

## 計算問題 —小テスト集—

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## 0.1 はじめに

本書は小テスト用のプリント集です。

学校では知識を与えることが主になりがちですが、計算能力等の能力が年々落ちてきているように感じています。数学では計算の遅さや不正確さが原因で話の理解が遅れていくこともあり、計算能力の不足は学習面に悪影響があります。そこで、脳を鍛え計算能力を身に付けさせるために行っていた小テストをまとめたものが本書です。

本書の目的は次のようになります。

### 目的 脳を鍛えること

元々は学校での授業で行う目的で作成していますが、塾での勉強や自習学習での利用、企業での研修、高齢者施設などでのボケ防止等でも利用出来るかもしれません。

PDF は GitHub でも公開中です。 [https://github.com/tyoji/math\\_workbook](https://github.com/tyoji/math_workbook)

## 0.2 使い方

次のような使い方を想定していますが、学習者の状況に応じて自由に利用してもらえればと思います。

- (1) 問題のページを印刷する
- (2) 黒板に 1 問目の解き方を解説し答えを書いておく
- (3) 解き方や解答等について質問を受け付ける
- (4) 質問がなくなれば、プリントを配布し解かせる
- (5) わからなければ、黒板を見て 1 問目を丸写しし、2 問目以降は数字を変えて同じように解く
- (6) 時間 (5 分から 10 分程度) 経過したら終了
- (7) 隣の人とプリントを交換し採点をさせる
- (8) 採点が終われば回収し、点数の集計をする

### 注意事項

- 生徒の能力を教師側が決めつけない。
- 小テストとして行うため、解いている時間は一人で静かに解かせる。一人で考えることこそが脳を鍛える。
- 問題を解くことを強要しない。
- 解けていなくてもあれこれ言わない。

プリントを配布前に第 1 問目の解説をし解答を書いておきます。これは、解き方がわからない人や勘違いしている人が小テストの時間を無駄に過ごさないためです。全員が一定数解けるようになったのであれば、省略してもいいです。

質問を受け付けるのも小テスト中に静かに取り組ませるためです。わからないなりに自力で考えることは脳の発達を促します。

小テストはわからない人が黒板を見ながら考えることに意味が有るので、邪魔になるような行為をするのであれば注意をします。

小テストの時間は解き終わらないぐらいの時間設定をします。具体的には教師が必死に解いてギリギリ解き終わるぐらいからその2倍ぐらいの時間を設定します。脳を鍛えることが目的の為、限界まで脳を使わせるように簡単に解き終わらないようにします。

採点は別の人のプリントを行います。これは、他人がどのように解いているかを知り自分の解き方を見直すきっかけにする為と、読めない文字を書いている事を指摘し他人に読めるように書くことを促す為です。学力の低い生徒が定期テストで読めない文字を書くことが多く、早めに指摘しておく意味を持ちます。

また、解答を提示する方法はいくつかあります。

- 口頭で読み上げる
- 黒板に書く
- 印刷して配布する

上にある方法がより教師生徒に負担が大きく、下にある方法が負担は軽いです。特に口頭で解答を読み上げると生徒は「聞く → 確認 → 丸付け」という事を繰り返します。聞く能力も鍛えられますし、採点時間が全員同じぐらいですから時間配分もしやすくなります。解答を読み上げる側も大変ではありますが。

答案を集めて集計をし、どの程度計算ができるかを確認します。

これを何度も繰り返すと多くの生徒が解き終わるようになります。

## 0.2.1 仕組み

人は猿が進化した動物なので、群れを作る性質を持ちます。具体的には、みんなと同じであろうとすることと、人よりも優れた存在になろうとすることです。

そこで、全員に同じ問題を何度も解かせるとサボっている生徒が出来なくてそれ以外の生徒が点数が伸びてきます。出来ていない生徒が少数になれば、その事を伝えてあげると周りにおいていかれることを恐れて取り組むようになります。

採点を隣の人と交換して行うもの同様の意味を持ちます。自分が出来ていないことを自分以外の人を知ることになるので、何とかしようと取り組むようになります。

こういった生徒のフォローの為に、テスト配布前に問題解説と質問受付を行います。

知識や技術の習得には本人の努力が不可欠です。他人から勉強をするように言われても本人の意思がなければ身につけません。そこで、勉強を促すのは自分自身やクラスメイト等からの緩やかなプレッシャーに任せます。教師側からの強要することは無いようにすることで、過度のプレッシャーが掛からないようになります。

目的は脳を鍛えることなので、静かに考える事が出来れば目的は達成です。考えた答えが正答でも誤答でも構いません。問題解説と小テストを繰り返す内に少しずつ正答に近づいていきます。この効果はどんなに勉強が出来ない生徒でもあらわれます。根気強く繰り返す必要があるかもしれませんが、全員同じように脳に負荷をかけて下さい。

# 第 1 章

# 数

## 1.1 自然数

### 1.1.1 加法

自然数の和を求める問題。

|                 |                |                |                 |
|-----------------|----------------|----------------|-----------------|
| (1) $46 + 24$   | (31) $31 + 91$ | (61) $67 + 85$ | (91) $44 + 95$  |
| (2) $26 + 33$   | (32) $43 + 35$ | (62) $98 + 64$ | (92) $48 + 99$  |
| (3) $87 + 8$    | (33) $94 + 89$ | (63) $75 + 49$ | (93) $43 + 83$  |
| (4) $71 + 91$   | (34) $22 + 23$ | (64) $86 + 59$ | (94) $42 + 23$  |
| (5) $7 + 50$    | (35) $58 + 20$ | (65) $47 + 71$ | (95) $19 + 29$  |
| (6) $76 + 36$   | (36) $7 + 60$  | (66) $45 + 61$ | (96) $11 + 72$  |
| (7) $26 + 46$   | (37) $67 + 72$ | (67) $40 + 63$ | (97) $59 + 21$  |
| (8) $94 + 96$   | (38) $15 + 3$  | (68) $81 + 59$ | (98) $59 + 66$  |
| (9) $96 + 40$   | (39) $2 + 82$  | (69) $9 + 56$  | (99) $24 + 43$  |
| (10) $13 + 23$  | (40) $74 + 66$ | (70) $37 + 36$ | (100) $85 + 82$ |
| (11) $51 + 75$  | (41) $46 + 0$  | (71) $61 + 89$ | (101) $26 + 89$ |
| (12) $84 + 77$  | (42) $18 + 92$ | (72) $36 + 54$ | (102) $83 + 67$ |
| (13) $97 + 26$  | (43) $98 + 82$ | (73) $4 + 38$  | (103) $41 + 91$ |
| (14) $39 + 17$  | (44) $37 + 85$ | (74) $83 + 1$  | (104) $54 + 69$ |
| (15) $24 + 80$  | (45) $0 + 59$  | (75) $17 + 48$ | (105) $82 + 99$ |
| (16) $64 + 97$  | (46) $9 + 87$  | (76) $52 + 80$ | (106) $5 + 10$  |
| (17) $3 + 47$   | (47) $16 + 22$ | (77) $34 + 43$ | (107) $75 + 24$ |
| (18) $52 + 87$  | (48) $52 + 18$ | (78) $69 + 73$ | (108) $41 + 12$ |
| (19) $100 + 92$ | (49) $49 + 74$ | (79) $49 + 26$ | (109) $8 + 56$  |
| (20) $82 + 68$  | (50) $47 + 4$  | (80) $6 + 66$  | (110) $34 + 52$ |
| (21) $18 + 12$  | (51) $70 + 22$ | (81) $93 + 39$ | (111) $91 + 78$ |
| (22) $70 + 94$  | (52) $36 + 93$ | (82) $20 + 35$ | (112) $30 + 57$ |
| (23) $84 + 89$  | (53) $94 + 21$ | (83) $41 + 6$  | (113) $55 + 10$ |
| (24) $78 + 98$  | (54) $2 + 73$  | (84) $30 + 13$ | (114) $81 + 95$ |
| (25) $97 + 27$  | (55) $50 + 55$ | (85) $71 + 47$ | (115) $93 + 69$ |
| (26) $29 + 42$  | (56) $67 + 38$ | (86) $26 + 17$ | (116) $47 + 49$ |
| (27) $67 + 8$   | (57) $66 + 88$ | (87) $17 + 14$ | (117) $66 + 6$  |
| (28) $26 + 55$  | (58) $58 + 91$ | (88) $32 + 65$ | (118) $44 + 52$ |
| (29) $30 + 5$   | (59) $48 + 99$ | (89) $96 + 26$ | (119) $75 + 79$ |
| (30) $80 + 4$   | (60) $74 + 91$ | (90) $24 + 47$ | (120) $41 + 58$ |

|                |                 |                 |                 |
|----------------|-----------------|-----------------|-----------------|
| (1) $1 + 74$   | (31) $67 + 7$   | (61) $55 + 34$  | (91) $10 + 10$  |
| (2) $12 + 46$  | (32) $36 + 69$  | (62) $80 + 69$  | (92) $38 + 58$  |
| (3) $18 + 38$  | (33) $48 + 29$  | (63) $88 + 51$  | (93) $26 + 16$  |
| (4) $42 + 7$   | (34) $87 + 18$  | (64) $85 + 100$ | (94) $16 + 3$   |
| (5) $44 + 62$  | (35) $89 + 92$  | (65) $5 + 23$   | (95) $32 + 24$  |
| (6) $66 + 94$  | (36) $23 + 51$  | (66) $66 + 83$  | (96) $68 + 96$  |
| (7) $96 + 69$  | (37) $62 + 95$  | (67) $63 + 19$  | (97) $10 + 42$  |
| (8) $20 + 100$ | (38) $22 + 3$   | (68) $15 + 23$  | (98) $83 + 54$  |
| (9) $70 + 15$  | (39) $31 + 22$  | (69) $48 + 44$  | (99) $100 + 48$ |
| (10) $14 + 35$ | (40) $95 + 98$  | (70) $87 + 41$  | (100) $59 + 38$ |
| (11) $90 + 42$ | (41) $48 + 96$  | (71) $70 + 76$  | (101) $17 + 58$ |
| (12) $57 + 76$ | (42) $42 + 66$  | (72) $21 + 25$  | (102) $6 + 65$  |
| (13) $82 + 88$ | (43) $58 + 15$  | (73) $95 + 19$  | (103) $63 + 15$ |
| (14) $48 + 74$ | (44) $4 + 96$   | (74) $57 + 34$  | (104) $9 + 63$  |
| (15) $23 + 84$ | (45) $60 + 98$  | (75) $13 + 1$   | (105) $89 + 42$ |
| (16) $77 + 52$ | (46) $27 + 82$  | (76) $22 + 10$  | (106) $95 + 22$ |
| (17) $72 + 54$ | (47) $82 + 12$  | (77) $30 + 8$   | (107) $73 + 85$ |
| (18) $17 + 65$ | (48) $85 + 60$  | (78) $78 + 49$  | (108) $0 + 99$  |
| (19) $62 + 71$ | (49) $39 + 57$  | (79) $88 + 6$   | (109) $53 + 60$ |
| (20) $74 + 79$ | (50) $49 + 98$  | (80) $48 + 93$  | (110) $16 + 6$  |
| (21) $89 + 89$ | (51) $1 + 4$    | (81) $42 + 59$  | (111) $49 + 22$ |
| (22) $26 + 36$ | (52) $85 + 90$  | (82) $58 + 91$  | (112) $30 + 28$ |
| (23) $62 + 67$ | (53) $100 + 23$ | (83) $1 + 31$   | (113) $20 + 7$  |
| (24) $55 + 17$ | (54) $53 + 97$  | (84) $62 + 92$  | (114) $16 + 40$ |
| (25) $77 + 68$ | (55) $21 + 11$  | (85) $57 + 100$ | (115) $38 + 54$ |
| (26) $2 + 100$ | (56) $18 + 53$  | (86) $85 + 20$  | (116) $32 + 23$ |
| (27) $47 + 92$ | (57) $13 + 16$  | (87) $30 + 31$  | (117) $41 + 93$ |
| (28) $6 + 49$  | (58) $65 + 34$  | (88) $57 + 60$  | (118) $95 + 1$  |
| (29) $62 + 89$ | (59) $70 + 8$   | (89) $3 + 30$   | (119) $29 + 7$  |
| (30) $43 + 93$ | (60) $63 + 17$  | (90) $83 + 38$  | (120) $5 + 28$  |



|                |                 |                |                  |
|----------------|-----------------|----------------|------------------|
| (1) $86 + 70$  | (31) $61 + 81$  | (61) $93 + 57$ | (91) $6 + 3$     |
| (2) $3 + 98$   | (32) $61 + 56$  | (62) $7 + 41$  | (92) $100 + 57$  |
| (3) $66 + 54$  | (33) $53 + 13$  | (63) $93 + 47$ | (93) $95 + 5$    |
| (4) $97 + 60$  | (34) $52 + 81$  | (64) $18 + 59$ | (94) $95 + 40$   |
| (5) $45 + 91$  | (35) $25 + 24$  | (65) $78 + 35$ | (95) $52 + 39$   |
| (6) $0 + 100$  | (36) $47 + 17$  | (66) $80 + 28$ | (96) $2 + 97$    |
| (7) $73 + 76$  | (37) $86 + 28$  | (67) $11 + 32$ | (97) $75 + 19$   |
| (8) $40 + 17$  | (38) $41 + 59$  | (68) $8 + 85$  | (98) $61 + 6$    |
| (9) $100 + 49$ | (39) $97 + 41$  | (69) $13 + 76$ | (99) $87 + 60$   |
| (10) $61 + 25$ | (40) $1 + 24$   | (70) $37 + 31$ | (100) $56 + 75$  |
| (11) $45 + 62$ | (41) $15 + 72$  | (71) $82 + 98$ | (101) $84 + 40$  |
| (12) $2 + 29$  | (42) $19 + 36$  | (72) $57 + 38$ | (102) $51 + 59$  |
| (13) $69 + 2$  | (43) $48 + 69$  | (73) $72 + 3$  | (103) $56 + 82$  |
| (14) $96 + 15$ | (44) $55 + 100$ | (74) $45 + 2$  | (104) $7 + 95$   |
| (15) $20 + 38$ | (45) $50 + 92$  | (75) $88 + 69$ | (105) $2 + 63$   |
| (16) $54 + 26$ | (46) $66 + 11$  | (76) $3 + 33$  | (106) $42 + 41$  |
| (17) $25 + 17$ | (47) $22 + 84$  | (77) $38 + 19$ | (107) $3 + 28$   |
| (18) $38 + 72$ | (48) $28 + 41$  | (78) $19 + 73$ | (108) $27 + 38$  |
| (19) $40 + 56$ | (49) $54 + 40$  | (79) $32 + 57$ | (109) $21 + 12$  |
| (20) $9 + 58$  | (50) $65 + 20$  | (80) $88 + 86$ | (110) $23 + 82$  |
| (21) $56 + 84$ | (51) $48 + 52$  | (81) $77 + 42$ | (111) $35 + 42$  |
| (22) $27 + 50$ | (52) $11 + 8$   | (82) $80 + 77$ | (112) $23 + 79$  |
| (23) $94 + 58$ | (53) $46 + 46$  | (83) $100 + 2$ | (113) $77 + 2$   |
| (24) $99 + 12$ | (54) $45 + 79$  | (84) $100 + 6$ | (114) $92 + 82$  |
| (25) $4 + 78$  | (55) $16 + 57$  | (85) $57 + 47$ | (115) $100 + 82$ |
| (26) $40 + 16$ | (56) $85 + 74$  | (86) $10 + 34$ | (116) $39 + 57$  |
| (27) $95 + 67$ | (57) $56 + 38$  | (87) $74 + 64$ | (117) $32 + 90$  |
| (28) $11 + 77$ | (58) $11 + 54$  | (88) $2 + 44$  | (118) $16 + 42$  |
| (29) $48 + 54$ | (59) $46 + 42$  | (89) $33 + 51$ | (119) $49 + 24$  |
| (30) $2 + 1$   | (60) $26 + 58$  | (90) $61 + 76$ | (120) $23 + 100$ |

|                 |                |                 |                 |
|-----------------|----------------|-----------------|-----------------|
| (1) $77 + 46$   | (31) $95 + 76$ | (61) $11 + 100$ | (91) $18 + 15$  |
| (2) $27 + 7$    | (32) $29 + 35$ | (62) $75 + 33$  | (92) $68 + 5$   |
| (3) $51 + 30$   | (33) $51 + 86$ | (63) $56 + 43$  | (93) $18 + 85$  |
| (4) $31 + 12$   | (34) $55 + 45$ | (64) $37 + 23$  | (94) $36 + 51$  |
| (5) $84 + 30$   | (35) $79 + 51$ | (65) $31 + 50$  | (95) $49 + 33$  |
| (6) $41 + 59$   | (36) $93 + 56$ | (66) $21 + 88$  | (96) $81 + 15$  |
| (7) $87 + 67$   | (37) $25 + 54$ | (67) $79 + 92$  | (97) $91 + 6$   |
| (8) $28 + 71$   | (38) $35 + 15$ | (68) $85 + 71$  | (98) $76 + 45$  |
| (9) $52 + 68$   | (39) $78 + 83$ | (69) $72 + 3$   | (99) $46 + 50$  |
| (10) $24 + 18$  | (40) $87 + 87$ | (70) $85 + 82$  | (100) $8 + 51$  |
| (11) $75 + 82$  | (41) $49 + 60$ | (71) $48 + 48$  | (101) $27 + 17$ |
| (12) $97 + 1$   | (42) $97 + 58$ | (72) $84 + 89$  | (102) $45 + 73$ |
| (13) $35 + 20$  | (43) $46 + 51$ | (73) $99 + 26$  | (103) $89 + 15$ |
| (14) $94 + 43$  | (44) $42 + 37$ | (74) $20 + 82$  | (104) $36 + 59$ |
| (15) $28 + 98$  | (45) $70 + 79$ | (75) $80 + 85$  | (105) $96 + 86$ |
| (16) $100 + 29$ | (46) $28 + 65$ | (76) $88 + 72$  | (106) $73 + 2$  |
| (17) $94 + 1$   | (47) $12 + 3$  | (77) $68 + 39$  | (107) $44 + 16$ |
| (18) $18 + 54$  | (48) $32 + 5$  | (78) $79 + 64$  | (108) $29 + 59$ |
| (19) $65 + 87$  | (49) $72 + 91$ | (79) $68 + 29$  | (109) $35 + 46$ |
| (20) $89 + 69$  | (50) $66 + 54$ | (80) $54 + 78$  | (110) $10 + 69$ |
| (21) $60 + 97$  | (51) $93 + 75$ | (81) $38 + 49$  | (111) $18 + 83$ |
| (22) $83 + 15$  | (52) $48 + 12$ | (82) $38 + 90$  | (112) $14 + 72$ |
| (23) $76 + 17$  | (53) $79 + 38$ | (83) $40 + 8$   | (113) $51 + 7$  |
| (24) $74 + 99$  | (54) $81 + 22$ | (84) $52 + 82$  | (114) $19 + 6$  |
| (25) $35 + 88$  | (55) $17 + 16$ | (85) $45 + 83$  | (115) $31 + 39$ |
| (26) $30 + 93$  | (56) $1 + 64$  | (86) $62 + 91$  | (116) $21 + 27$ |
| (27) $88 + 62$  | (57) $25 + 54$ | (87) $39 + 48$  | (117) $14 + 92$ |
| (28) $75 + 21$  | (58) $5 + 35$  | (88) $95 + 28$  | (118) $33 + 70$ |
| (29) $95 + 74$  | (59) $47 + 83$ | (89) $54 + 62$  | (119) $34 + 60$ |
| (30) $84 + 45$  | (60) $88 + 4$  | (90) $3 + 83$   | (120) $38 + 43$ |

|                |                |                |                  |
|----------------|----------------|----------------|------------------|
| (1) $76 + 55$  | (31) $91 + 56$ | (61) $7 + 33$  | (91) $100 + 31$  |
| (2) $1 + 79$   | (32) $68 + 15$ | (62) $97 + 78$ | (92) $23 + 4$    |
| (3) $24 + 32$  | (33) $58 + 92$ | (63) $37 + 64$ | (93) $30 + 34$   |
| (4) $61 + 48$  | (34) $77 + 66$ | (64) $69 + 51$ | (94) $86 + 75$   |
| (5) $10 + 89$  | (35) $99 + 24$ | (65) $21 + 43$ | (95) $35 + 57$   |
| (6) $35 + 75$  | (36) $23 + 84$ | (66) $4 + 78$  | (96) $20 + 51$   |
| (7) $39 + 98$  | (37) $70 + 74$ | (67) $69 + 97$ | (97) $96 + 65$   |
| (8) $91 + 69$  | (38) $9 + 38$  | (68) $14 + 97$ | (98) $28 + 53$   |
| (9) $26 + 55$  | (39) $95 + 61$ | (69) $68 + 68$ | (99) $94 + 62$   |
| (10) $47 + 78$ | (40) $4 + 4$   | (70) $36 + 14$ | (100) $9 + 97$   |
| (11) $40 + 5$  | (41) $89 + 5$  | (71) $88 + 2$  | (101) $20 + 58$  |
| (12) $8 + 63$  | (42) $8 + 88$  | (72) $41 + 20$ | (102) $13 + 22$  |
| (13) $98 + 51$ | (43) $48 + 62$ | (73) $80 + 37$ | (103) $71 + 30$  |
| (14) $30 + 46$ | (44) $0 + 62$  | (74) $40 + 56$ | (104) $25 + 100$ |
| (15) $17 + 80$ | (45) $73 + 43$ | (75) $73 + 90$ | (105) $98 + 54$  |
| (16) $25 + 21$ | (46) $71 + 72$ | (76) $91 + 33$ | (106) $37 + 32$  |
| (17) $25 + 85$ | (47) $87 + 4$  | (77) $28 + 95$ | (107) $56 + 36$  |
| (18) $26 + 4$  | (48) $61 + 86$ | (78) $97 + 0$  | (108) $57 + 71$  |
| (19) $67 + 24$ | (49) $49 + 91$ | (79) $64 + 42$ | (109) $58 + 86$  |
| (20) $93 + 51$ | (50) $46 + 92$ | (80) $91 + 87$ | (110) $32 + 11$  |
| (21) $79 + 7$  | (51) $99 + 85$ | (81) $61 + 65$ | (111) $42 + 86$  |
| (22) $69 + 30$ | (52) $13 + 6$  | (82) $61 + 15$ | (112) $37 + 66$  |
| (23) $82 + 10$ | (53) $28 + 49$ | (83) $95 + 40$ | (113) $90 + 33$  |
| (24) $61 + 51$ | (54) $29 + 46$ | (84) $9 + 3$   | (114) $7 + 47$   |
| (25) $9 + 91$  | (55) $56 + 80$ | (85) $15 + 60$ | (115) $46 + 53$  |
| (26) $77 + 33$ | (56) $69 + 50$ | (86) $49 + 49$ | (116) $12 + 2$   |
| (27) $7 + 69$  | (57) $81 + 66$ | (87) $10 + 45$ | (117) $90 + 100$ |
| (28) $44 + 32$ | (58) $24 + 38$ | (88) $76 + 88$ | (118) $35 + 66$  |
| (29) $88 + 11$ | (59) $47 + 41$ | (89) $44 + 88$ | (119) $72 + 45$  |
| (30) $79 + 33$ | (60) $33 + 20$ | (90) $15 + 85$ | (120) $55 + 92$  |

|                 |                 |                 |                  |
|-----------------|-----------------|-----------------|------------------|
| (1) $30 + 63$   | (31) $87 + 80$  | (61) $29 + 75$  | (91) $53 + 99$   |
| (2) $21 + 74$   | (32) $55 + 30$  | (62) $69 + 78$  | (92) $76 + 24$   |
| (3) $97 + 27$   | (33) $36 + 4$   | (63) $57 + 43$  | (93) $96 + 71$   |
| (4) $28 + 6$    | (34) $3 + 52$   | (64) $40 + 1$   | (94) $81 + 55$   |
| (5) $2 + 65$    | (35) $16 + 99$  | (65) $61 + 47$  | (95) $58 + 81$   |
| (6) $88 + 55$   | (36) $9 + 84$   | (66) $27 + 27$  | (96) $97 + 9$    |
| (7) $23 + 83$   | (37) $83 + 6$   | (67) $43 + 43$  | (97) $0 + 46$    |
| (8) $7 + 32$    | (38) $96 + 46$  | (68) $100 + 37$ | (98) $44 + 87$   |
| (9) $87 + 33$   | (39) $26 + 37$  | (69) $69 + 44$  | (99) $50 + 83$   |
| (10) $29 + 30$  | (40) $34 + 92$  | (70) $52 + 55$  | (100) $24 + 97$  |
| (11) $76 + 8$   | (41) $63 + 30$  | (71) $12 + 85$  | (101) $85 + 40$  |
| (12) $48 + 65$  | (42) $60 + 20$  | (72) $15 + 100$ | (102) $71 + 96$  |
| (13) $27 + 12$  | (43) $73 + 63$  | (73) $88 + 45$  | (103) $88 + 19$  |
| (14) $35 + 43$  | (44) $94 + 11$  | (74) $23 + 58$  | (104) $62 + 45$  |
| (15) $56 + 78$  | (45) $85 + 1$   | (75) $17 + 83$  | (105) $11 + 100$ |
| (16) $19 + 57$  | (46) $72 + 20$  | (76) $56 + 95$  | (106) $73 + 67$  |
| (17) $45 + 46$  | (47) $98 + 79$  | (77) $96 + 15$  | (107) $32 + 17$  |
| (18) $56 + 72$  | (48) $8 + 24$   | (78) $96 + 24$  | (108) $32 + 95$  |
| (19) $51 + 54$  | (49) $65 + 26$  | (79) $84 + 95$  | (109) $18 + 14$  |
| (20) $46 + 13$  | (50) $100 + 33$ | (80) $48 + 7$   | (110) $59 + 85$  |
| (21) $12 + 7$   | (51) $11 + 9$   | (81) $52 + 45$  | (111) $23 + 32$  |
| (22) $31 + 100$ | (52) $16 + 72$  | (82) $48 + 83$  | (112) $38 + 100$ |
| (23) $24 + 26$  | (53) $84 + 74$  | (83) $95 + 61$  | (113) $10 + 88$  |
| (24) $56 + 96$  | (54) $49 + 90$  | (84) $97 + 29$  | (114) $87 + 61$  |
| (25) $97 + 20$  | (55) $80 + 89$  | (85) $34 + 82$  | (115) $41 + 31$  |
| (26) $55 + 57$  | (56) $51 + 16$  | (86) $91 + 99$  | (116) $37 + 51$  |
| (27) $40 + 93$  | (57) $47 + 57$  | (87) $64 + 83$  | (117) $79 + 17$  |
| (28) $63 + 26$  | (58) $95 + 35$  | (88) $13 + 88$  | (118) $62 + 5$   |
| (29) $54 + 71$  | (59) $70 + 96$  | (89) $12 + 48$  | (119) $87 + 25$  |
| (30) $78 + 15$  | (60) $97 + 1$   | (90) $98 + 63$  | (120) $20 + 25$  |

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|----------------|----------------|-----------------|-----------------|
| (1) $91 + 29$  | (31) $88 + 78$ | (61) $7 + 45$   | (91) $90 + 57$  |
| (2) $13 + 15$  | (32) $78 + 77$ | (62) $70 + 73$  | (92) $27 + 83$  |
| (3) $92 + 59$  | (33) $53 + 78$ | (63) $8 + 88$   | (93) $51 + 33$  |
| (4) $1 + 46$   | (34) $73 + 40$ | (64) $20 + 70$  | (94) $68 + 26$  |
| (5) $14 + 90$  | (35) $51 + 96$ | (65) $73 + 51$  | (95) $79 + 81$  |
| (6) $79 + 97$  | (36) $71 + 25$ | (66) $98 + 38$  | (96) $6 + 16$   |
| (7) $82 + 5$   | (37) $72 + 14$ | (67) $12 + 97$  | (97) $33 + 71$  |
| (8) $46 + 80$  | (38) $62 + 7$  | (68) $32 + 63$  | (98) $64 + 32$  |
| (9) $56 + 34$  | (39) $92 + 48$ | (69) $73 + 28$  | (99) $34 + 12$  |
| (10) $18 + 50$ | (40) $2 + 7$   | (70) $90 + 23$  | (100) $33 + 77$ |
| (11) $97 + 69$ | (41) $17 + 35$ | (71) $34 + 73$  | (101) $65 + 18$ |
| (12) $17 + 6$  | (42) $13 + 56$ | (72) $6 + 87$   | (102) $19 + 68$ |
| (13) $38 + 16$ | (43) $94 + 3$  | (73) $53 + 24$  | (103) $28 + 90$ |
| (14) $13 + 45$ | (44) $60 + 57$ | (74) $21 + 61$  | (104) $68 + 40$ |
| (15) $40 + 20$ | (45) $15 + 28$ | (75) $57 + 46$  | (105) $70 + 48$ |
| (16) $87 + 68$ | (46) $83 + 70$ | (76) $33 + 15$  | (106) $66 + 67$ |
| (17) $4 + 34$  | (47) $14 + 84$ | (77) $31 + 0$   | (107) $43 + 90$ |
| (18) $25 + 4$  | (48) $1 + 74$  | (78) $72 + 78$  | (108) $84 + 2$  |
| (19) $42 + 22$ | (49) $57 + 72$ | (79) $88 + 28$  | (109) $26 + 48$ |
| (20) $24 + 38$ | (50) $84 + 87$ | (80) $53 + 50$  | (110) $78 + 15$ |
| (21) $61 + 38$ | (51) $39 + 71$ | (81) $39 + 9$   | (111) $95 + 32$ |
| (22) $26 + 97$ | (52) $77 + 67$ | (82) $63 + 95$  | (112) $76 + 88$ |
| (23) $96 + 75$ | (53) $79 + 85$ | (83) $73 + 73$  | (113) $68 + 32$ |
| (24) $71 + 47$ | (54) $76 + 87$ | (84) $64 + 3$   | (114) $24 + 39$ |
| (25) $54 + 45$ | (55) $83 + 46$ | (85) $71 + 100$ | (115) $99 + 43$ |
| (26) $78 + 72$ | (56) $20 + 39$ | (86) $46 + 43$  | (116) $22 + 2$  |
| (27) $14 + 13$ | (57) $35 + 68$ | (87) $11 + 86$  | (117) $94 + 63$ |
| (28) $54 + 29$ | (58) $72 + 1$  | (88) $39 + 83$  | (118) $40 + 49$ |
| (29) $50 + 34$ | (59) $98 + 87$ | (89) $25 + 15$  | (119) $29 + 88$ |
| (30) $60 + 76$ | (60) $3 + 39$  | (90) $85 + 69$  | (120) $96 + 79$ |

|                |                |                 |                 |
|----------------|----------------|-----------------|-----------------|
| (1) $61 + 16$  | (31) $77 + 30$ | (61) $64 + 90$  | (91) $72 + 78$  |
| (2) $11 + 13$  | (32) $20 + 54$ | (62) $53 + 80$  | (92) $36 + 33$  |
| (3) $26 + 73$  | (33) $83 + 81$ | (63) $78 + 99$  | (93) $81 + 85$  |
| (4) $97 + 96$  | (34) $86 + 51$ | (64) $43 + 80$  | (94) $75 + 54$  |
| (5) $75 + 6$   | (35) $25 + 44$ | (65) $40 + 19$  | (95) $70 + 32$  |
| (6) $93 + 40$  | (36) $33 + 76$ | (66) $76 + 40$  | (96) $89 + 38$  |
| (7) $70 + 74$  | (37) $55 + 83$ | (67) $37 + 83$  | (97) $100 + 35$ |
| (8) $31 + 38$  | (38) $10 + 32$ | (68) $36 + 91$  | (98) $71 + 6$   |
| (9) $71 + 29$  | (39) $36 + 98$ | (69) $38 + 87$  | (99) $90 + 4$   |
| (10) $57 + 58$ | (40) $87 + 63$ | (70) $94 + 12$  | (100) $18 + 70$ |
| (11) $18 + 16$ | (41) $76 + 75$ | (71) $49 + 99$  | (101) $82 + 64$ |
| (12) $48 + 92$ | (42) $48 + 32$ | (72) $100 + 59$ | (102) $56 + 9$  |
| (13) $74 + 23$ | (43) $43 + 68$ | (73) $79 + 2$   | (103) $53 + 63$ |
| (14) $33 + 86$ | (44) $8 + 51$  | (74) $85 + 100$ | (104) $45 + 6$  |
| (15) $65 + 46$ | (45) $91 + 97$ | (75) $66 + 33$  | (105) $89 + 33$ |
| (16) $89 + 45$ | (46) $47 + 97$ | (76) $84 + 39$  | (106) $10 + 7$  |
| (17) $85 + 88$ | (47) $96 + 98$ | (77) $83 + 60$  | (107) $87 + 64$ |
| (18) $43 + 18$ | (48) $55 + 10$ | (78) $32 + 22$  | (108) $41 + 78$ |
| (19) $56 + 2$  | (49) $2 + 93$  | (79) $80 + 20$  | (109) $15 + 48$ |
| (20) $81 + 95$ | (50) $57 + 70$ | (80) $86 + 9$   | (110) $57 + 43$ |
| (21) $72 + 55$ | (51) $56 + 33$ | (81) $59 + 81$  | (111) $81 + 78$ |
| (22) $12 + 30$ | (52) $47 + 33$ | (82) $24 + 38$  | (112) $0 + 53$  |
| (23) $11 + 50$ | (53) $46 + 60$ | (83) $34 + 39$  | (113) $98 + 55$ |
| (24) $52 + 11$ | (54) $44 + 99$ | (84) $20 + 74$  | (114) $5 + 32$  |
| (25) $14 + 86$ | (55) $7 + 18$  | (85) $55 + 77$  | (115) $77 + 37$ |
| (26) $59 + 14$ | (56) $94 + 46$ | (86) $51 + 95$  | (116) $0 + 39$  |
| (27) $33 + 70$ | (57) $87 + 78$ | (87) $53 + 76$  | (117) $89 + 2$  |
| (28) $37 + 95$ | (58) $56 + 24$ | (88) $4 + 39$   | (118) $9 + 10$  |
| (29) $95 + 28$ | (59) $90 + 66$ | (89) $17 + 42$  | (119) $89 + 49$ |
| (30) $79 + 11$ | (60) $48 + 66$ | (90) $51 + 17$  | (120) $57 + 60$ |

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|-----------------|-----------------|-----------------|-----------------|
| (1) $1 + 13$    | (31) $74 + 83$  | (61) $37 + 45$  | (91) $11 + 82$  |
| (2) $54 + 60$   | (32) $17 + 94$  | (62) $6 + 48$   | (92) $14 + 84$  |
| (3) $84 + 99$   | (33) $91 + 4$   | (63) $16 + 59$  | (93) $8 + 16$   |
| (4) $96 + 96$   | (34) $76 + 32$  | (64) $47 + 96$  | (94) $29 + 17$  |
| (5) $69 + 60$   | (35) $39 + 44$  | (65) $48 + 35$  | (95) $46 + 19$  |
| (6) $94 + 61$   | (36) $73 + 59$  | (66) $23 + 100$ | (96) $61 + 61$  |
| (7) $7 + 65$    | (37) $84 + 21$  | (67) $20 + 52$  | (97) $40 + 42$  |
| (8) $32 + 36$   | (38) $68 + 14$  | (68) $62 + 62$  | (98) $10 + 12$  |
| (9) $51 + 49$   | (39) $79 + 68$  | (69) $90 + 22$  | (99) $41 + 98$  |
| (10) $2 + 94$   | (40) $16 + 46$  | (70) $38 + 39$  | (100) $54 + 5$  |
| (11) $76 + 18$  | (41) $91 + 11$  | (71) $61 + 41$  | (101) $81 + 78$ |
| (12) $59 + 79$  | (42) $23 + 100$ | (72) $3 + 96$   | (102) $9 + 72$  |
| (13) $13 + 36$  | (43) $27 + 79$  | (73) $30 + 92$  | (103) $14 + 46$ |
| (14) $93 + 69$  | (44) $70 + 55$  | (74) $33 + 75$  | (104) $31 + 15$ |
| (15) $60 + 66$  | (45) $41 + 54$  | (75) $82 + 42$  | (105) $61 + 66$ |
| (16) $28 + 100$ | (46) $57 + 9$   | (76) $85 + 53$  | (106) $55 + 45$ |
| (17) $70 + 91$  | (47) $50 + 96$  | (77) $85 + 11$  | (107) $45 + 60$ |
| (18) $72 + 27$  | (48) $8 + 56$   | (78) $80 + 15$  | (108) $55 + 68$ |
| (19) $81 + 29$  | (49) $7 + 73$   | (79) $87 + 37$  | (109) $9 + 88$  |
| (20) $39 + 45$  | (50) $28 + 61$  | (80) $58 + 94$  | (110) $93 + 40$ |
| (21) $74 + 3$   | (51) $2 + 71$   | (81) $4 + 15$   | (111) $56 + 27$ |
| (22) $53 + 92$  | (52) $94 + 99$  | (82) $70 + 33$  | (112) $28 + 46$ |
| (23) $35 + 81$  | (53) $15 + 51$  | (83) $59 + 62$  | (113) $44 + 52$ |
| (24) $24 + 93$  | (54) $33 + 57$  | (84) $16 + 33$  | (114) $61 + 46$ |
| (25) $92 + 17$  | (55) $64 + 27$  | (85) $73 + 41$  | (115) $36 + 73$ |
| (26) $56 + 10$  | (56) $82 + 65$  | (86) $47 + 86$  | (116) $49 + 45$ |
| (27) $53 + 43$  | (57) $20 + 60$  | (87) $44 + 89$  | (117) $15 + 34$ |
| (28) $94 + 28$  | (58) $96 + 88$  | (88) $9 + 23$   | (118) $35 + 90$ |
| (29) $77 + 44$  | (59) $99 + 65$  | (89) $72 + 32$  | (119) $60 + 9$  |
| (30) $23 + 62$  | (60) $88 + 82$  | (90) $98 + 5$   | (120) $67 + 37$ |

|                |                |                 |                 |
|----------------|----------------|-----------------|-----------------|
| (1) $31 + 76$  | (31) $29 + 29$ | (61) $54 + 35$  | (91) $86 + 68$  |
| (2) $37 + 59$  | (32) $8 + 68$  | (62) $17 + 56$  | (92) $88 + 9$   |
| (3) $43 + 72$  | (33) $93 + 59$ | (63) $48 + 100$ | (93) $1 + 35$   |
| (4) $80 + 42$  | (34) $32 + 28$ | (64) $9 + 99$   | (94) $22 + 3$   |
| (5) $20 + 73$  | (35) $77 + 84$ | (65) $78 + 60$  | (95) $84 + 92$  |
| (6) $100 + 46$ | (36) $59 + 5$  | (66) $14 + 25$  | (96) $42 + 74$  |
| (7) $5 + 48$   | (37) $18 + 62$ | (67) $24 + 71$  | (97) $15 + 74$  |
| (8) $78 + 13$  | (38) $16 + 23$ | (68) $87 + 67$  | (98) $36 + 41$  |
| (9) $43 + 37$  | (39) $27 + 91$ | (69) $70 + 51$  | (99) $0 + 13$   |
| (10) $17 + 67$ | (40) $2 + 95$  | (70) $83 + 84$  | (100) $25 + 32$ |
| (11) $67 + 1$  | (41) $7 + 69$  | (71) $23 + 78$  | (101) $17 + 28$ |
| (12) $33 + 88$ | (42) $6 + 91$  | (72) $82 + 76$  | (102) $71 + 83$ |
| (13) $5 + 87$  | (43) $86 + 87$ | (73) $83 + 85$  | (103) $7 + 69$  |
| (14) $71 + 63$ | (44) $89 + 92$ | (74) $99 + 98$  | (104) $12 + 64$ |
| (15) $45 + 92$ | (45) $35 + 68$ | (75) $25 + 42$  | (105) $43 + 89$ |
| (16) $67 + 68$ | (46) $49 + 75$ | (76) $1 + 20$   | (106) $68 + 14$ |
| (17) $96 + 6$  | (47) $82 + 19$ | (77) $54 + 1$   | (107) $76 + 98$ |
| (18) $30 + 56$ | (48) $84 + 44$ | (78) $86 + 38$  | (108) $20 + 87$ |
| (19) $61 + 16$ | (49) $15 + 37$ | (79) $58 + 9$   | (109) $95 + 71$ |
| (20) $65 + 50$ | (50) $80 + 12$ | (80) $81 + 97$  | (110) $65 + 97$ |
| (21) $27 + 63$ | (51) $35 + 85$ | (81) $15 + 86$  | (111) $18 + 71$ |
| (22) $23 + 79$ | (52) $4 + 21$  | (82) $21 + 28$  | (112) $8 + 18$  |
| (23) $33 + 9$  | (53) $55 + 73$ | (83) $93 + 50$  | (113) $83 + 59$ |
| (24) $59 + 83$ | (54) $16 + 67$ | (84) $36 + 13$  | (114) $34 + 44$ |
| (25) $49 + 68$ | (55) $63 + 15$ | (85) $55 + 79$  | (115) $89 + 92$ |
| (26) $50 + 60$ | (56) $58 + 97$ | (86) $52 + 40$  | (116) $7 + 31$  |
| (27) $92 + 64$ | (57) $36 + 52$ | (87) $62 + 55$  | (117) $27 + 38$ |
| (28) $55 + 56$ | (58) $42 + 58$ | (88) $37 + 56$  | (118) $53 + 90$ |
| (29) $58 + 74$ | (59) $45 + 99$ | (89) $12 + 76$  | (119) $82 + 36$ |
| (30) $22 + 23$ | (60) $1 + 70$  | (90) $73 + 61$  | (120) $88 + 8$  |



|                |                |                 |                 |
|----------------|----------------|-----------------|-----------------|
| (1) $94 + 95$  | (31) $71 + 26$ | (61) $71 + 91$  | (91) $76 + 97$  |
| (2) $56 + 48$  | (32) $46 + 47$ | (62) $60 + 64$  | (92) $50 + 41$  |
| (3) $21 + 18$  | (33) $30 + 83$ | (63) $69 + 3$   | (93) $27 + 32$  |
| (4) $98 + 48$  | (34) $99 + 65$ | (64) $38 + 63$  | (94) $17 + 76$  |
| (5) $29 + 94$  | (35) $98 + 77$ | (65) $2 + 77$   | (95) $7 + 26$   |
| (6) $7 + 71$   | (36) $64 + 79$ | (66) $12 + 4$   | (96) $60 + 94$  |
| (7) $55 + 40$  | (37) $12 + 54$ | (67) $35 + 72$  | (97) $41 + 50$  |
| (8) $0 + 99$   | (38) $35 + 45$ | (68) $71 + 15$  | (98) $77 + 12$  |
| (9) $64 + 63$  | (39) $14 + 44$ | (69) $68 + 38$  | (99) $19 + 32$  |
| (10) $96 + 96$ | (40) $88 + 92$ | (70) $87 + 49$  | (100) $79 + 94$ |
| (11) $51 + 79$ | (41) $76 + 93$ | (71) $93 + 100$ | (101) $32 + 84$ |
| (12) $34 + 34$ | (42) $3 + 36$  | (72) $94 + 2$   | (102) $36 + 18$ |
| (13) $14 + 90$ | (43) $83 + 26$ | (73) $66 + 80$  | (103) $37 + 25$ |
| (14) $80 + 69$ | (44) $59 + 90$ | (74) $39 + 30$  | (104) $75 + 36$ |
| (15) $32 + 4$  | (45) $38 + 60$ | (75) $63 + 29$  | (105) $92 + 45$ |
| (16) $19 + 20$ | (46) $61 + 3$  | (76) $96 + 45$  | (106) $19 + 50$ |
| (17) $74 + 97$ | (47) $55 + 46$ | (77) $10 + 6$   | (107) $52 + 95$ |
| (18) $96 + 78$ | (48) $72 + 86$ | (78) $47 + 91$  | (108) $30 + 27$ |
| (19) $14 + 60$ | (49) $45 + 71$ | (79) $97 + 43$  | (109) $23 + 88$ |
| (20) $14 + 13$ | (50) $78 + 98$ | (80) $31 + 40$  | (110) $54 + 16$ |
| (21) $2 + 78$  | (51) $70 + 13$ | (81) $52 + 48$  | (111) $60 + 84$ |
| (22) $20 + 53$ | (52) $80 + 11$ | (82) $61 + 55$  | (112) $74 + 55$ |
| (23) $14 + 37$ | (53) $2 + 22$  | (83) $16 + 11$  | (113) $52 + 13$ |
| (24) $22 + 14$ | (54) $15 + 99$ | (84) $20 + 84$  | (114) $22 + 52$ |
| (25) $28 + 18$ | (55) $36 + 85$ | (85) $50 + 69$  | (115) $79 + 76$ |
| (26) $19 + 74$ | (56) $14 + 2$  | (86) $1 + 31$   | (116) $28 + 90$ |
| (27) $35 + 77$ | (57) $91 + 41$ | (87) $32 + 78$  | (117) $26 + 7$  |
| (28) $58 + 5$  | (58) $74 + 84$ | (88) $27 + 5$   | (118) $47 + 67$ |
| (29) $62 + 30$ | (59) $19 + 51$ | (89) $28 + 1$   | (119) $77 + 37$ |
| (30) $52 + 29$ | (60) $26 + 44$ | (90) $8 + 3$    | (120) $35 + 14$ |

|                |                 |                |                 |
|----------------|-----------------|----------------|-----------------|
| (1) $11 + 68$  | (31) $51 + 42$  | (61) $64 + 67$ | (91) $22 + 33$  |
| (2) $45 + 1$   | (32) $12 + 93$  | (62) $86 + 30$ | (92) $100 + 38$ |
| (3) $66 + 69$  | (33) $62 + 90$  | (63) $42 + 52$ | (93) $100 + 28$ |
| (4) $12 + 18$  | (34) $45 + 6$   | (64) $26 + 88$ | (94) $43 + 97$  |
| (5) $80 + 81$  | (35) $68 + 89$  | (65) $17 + 54$ | (95) $92 + 50$  |
| (6) $66 + 51$  | (36) $54 + 86$  | (66) $74 + 13$ | (96) $44 + 15$  |
| (7) $60 + 19$  | (37) $6 + 93$   | (67) $0 + 79$  | (97) $66 + 37$  |
| (8) $62 + 29$  | (38) $22 + 5$   | (68) $50 + 60$ | (98) $92 + 100$ |
| (9) $45 + 1$   | (39) $99 + 77$  | (69) $72 + 78$ | (99) $61 + 81$  |
| (10) $95 + 43$ | (40) $20 + 55$  | (70) $59 + 27$ | (100) $69 + 13$ |
| (11) $99 + 61$ | (41) $26 + 67$  | (71) $94 + 98$ | (101) $4 + 59$  |
| (12) $43 + 87$ | (42) $64 + 78$  | (72) $95 + 1$  | (102) $22 + 93$ |
| (13) $8 + 11$  | (43) $88 + 47$  | (73) $12 + 70$ | (103) $81 + 77$ |
| (14) $56 + 40$ | (44) $22 + 17$  | (74) $32 + 47$ | (104) $34 + 78$ |
| (15) $86 + 50$ | (45) $44 + 83$  | (75) $73 + 94$ | (105) $77 + 67$ |
| (16) $96 + 59$ | (46) $48 + 28$  | (76) $47 + 47$ | (106) $14 + 32$ |
| (17) $94 + 49$ | (47) $25 + 65$  | (77) $26 + 3$  | (107) $53 + 48$ |
| (18) $23 + 27$ | (48) $74 + 56$  | (78) $17 + 76$ | (108) $20 + 53$ |
| (19) $82 + 3$  | (49) $96 + 88$  | (79) $56 + 86$ | (109) $86 + 60$ |
| (20) $18 + 89$ | (50) $62 + 100$ | (80) $12 + 97$ | (110) $24 + 47$ |
| (21) $86 + 25$ | (51) $0 + 74$   | (81) $56 + 5$  | (111) $27 + 27$ |
| (22) $24 + 35$ | (52) $35 + 49$  | (82) $87 + 14$ | (112) $15 + 19$ |
| (23) $6 + 85$  | (53) $69 + 42$  | (83) $12 + 3$  | (113) $3 + 24$  |
| (24) $89 + 42$ | (54) $4 + 79$   | (84) $82 + 38$ | (114) $100 + 4$ |
| (25) $9 + 42$  | (55) $25 + 10$  | (85) $30 + 32$ | (115) $61 + 18$ |
| (26) $48 + 58$ | (56) $54 + 10$  | (86) $67 + 88$ | (116) $73 + 94$ |
| (27) $69 + 65$ | (57) $86 + 29$  | (87) $39 + 86$ | (117) $9 + 36$  |
| (28) $66 + 8$  | (58) $80 + 27$  | (88) $47 + 46$ | (118) $89 + 81$ |
| (29) $1 + 49$  | (59) $18 + 6$   | (89) $20 + 28$ | (119) $85 + 34$ |
| (30) $23 + 55$ | (60) $70 + 55$  | (90) $20 + 46$ | (120) $33 + 56$ |

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|----------------|----------------|-----------------|-----------------|
| (1) $56 + 60$  | (31) $97 + 18$ | (61) $91 + 25$  | (91) $3 + 33$   |
| (2) $38 + 13$  | (32) $46 + 38$ | (62) $93 + 36$  | (92) $22 + 8$   |
| (3) $43 + 58$  | (33) $85 + 81$ | (63) $100 + 63$ | (93) $65 + 56$  |
| (4) $67 + 37$  | (34) $4 + 79$  | (64) $76 + 37$  | (94) $19 + 81$  |
| (5) $69 + 50$  | (35) $44 + 39$ | (65) $59 + 57$  | (95) $70 + 2$   |
| (6) $71 + 28$  | (36) $66 + 72$ | (66) $85 + 56$  | (96) $5 + 64$   |
| (7) $10 + 76$  | (37) $70 + 88$ | (67) $66 + 75$  | (97) $15 + 54$  |
| (8) $100 + 42$ | (38) $55 + 24$ | (68) $66 + 83$  | (98) $4 + 61$   |
| (9) $8 + 93$   | (39) $98 + 65$ | (69) $72 + 41$  | (99) $82 + 97$  |
| (10) $55 + 41$ | (40) $95 + 82$ | (70) $59 + 85$  | (100) $10 + 13$ |
| (11) $88 + 36$ | (41) $39 + 66$ | (71) $4 + 88$   | (101) $65 + 2$  |
| (12) $66 + 3$  | (42) $32 + 16$ | (72) $99 + 36$  | (102) $4 + 83$  |
| (13) $83 + 68$ | (43) $58 + 24$ | (73) $57 + 71$  | (103) $51 + 88$ |
| (14) $66 + 17$ | (44) $30 + 6$  | (74) $56 + 100$ | (104) $43 + 32$ |
| (15) $58 + 64$ | (45) $55 + 77$ | (75) $73 + 94$  | (105) $33 + 14$ |
| (16) $9 + 81$  | (46) $98 + 26$ | (76) $79 + 70$  | (106) $90 + 90$ |
| (17) $34 + 62$ | (47) $92 + 77$ | (77) $20 + 90$  | (107) $19 + 26$ |
| (18) $15 + 88$ | (48) $13 + 73$ | (78) $83 + 14$  | (108) $52 + 54$ |
| (19) $39 + 60$ | (49) $67 + 21$ | (79) $79 + 7$   | (109) $27 + 72$ |
| (20) $52 + 68$ | (50) $98 + 95$ | (80) $26 + 86$  | (110) $54 + 2$  |
| (21) $54 + 75$ | (51) $21 + 15$ | (81) $13 + 19$  | (111) $56 + 89$ |
| (22) $14 + 57$ | (52) $14 + 36$ | (82) $73 + 61$  | (112) $80 + 4$  |
| (23) $81 + 95$ | (53) $88 + 43$ | (83) $6 + 39$   | (113) $4 + 28$  |
| (24) $52 + 40$ | (54) $46 + 3$  | (84) $36 + 45$  | (114) $74 + 71$ |
| (25) $56 + 22$ | (55) $32 + 64$ | (85) $39 + 95$  | (115) $72 + 15$ |
| (26) $33 + 66$ | (56) $77 + 76$ | (86) $27 + 57$  | (116) $3 + 19$  |
| (27) $68 + 29$ | (57) $23 + 26$ | (87) $99 + 81$  | (117) $95 + 76$ |
| (28) $53 + 19$ | (58) $24 + 7$  | (88) $12 + 95$  | (118) $43 + 83$ |
| (29) $11 + 32$ | (59) $54 + 91$ | (89) $100 + 46$ | (119) $19 + 74$ |
| (30) $19 + 14$ | (60) $32 + 37$ | (90) $70 + 51$  | (120) $30 + 83$ |

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|----------------|----------------|----------------|-----------------|
| (1) $82 + 65$  | (31) $69 + 65$ | (61) $31 + 57$ | (91) $61 + 15$  |
| (2) $22 + 65$  | (32) $13 + 68$ | (62) $48 + 27$ | (92) $58 + 39$  |
| (3) $61 + 46$  | (33) $74 + 65$ | (63) $76 + 98$ | (93) $23 + 73$  |
| (4) $94 + 45$  | (34) $58 + 81$ | (64) $77 + 59$ | (94) $47 + 65$  |
| (5) $54 + 68$  | (35) $12 + 17$ | (65) $1 + 4$   | (95) $92 + 45$  |
| (6) $9 + 100$  | (36) $43 + 99$ | (66) $54 + 83$ | (96) $20 + 21$  |
| (7) $82 + 82$  | (37) $75 + 9$  | (67) $35 + 65$ | (97) $39 + 27$  |
| (8) $23 + 67$  | (38) $17 + 13$ | (68) $53 + 54$ | (98) $79 + 74$  |
| (9) $3 + 5$    | (39) $28 + 90$ | (69) $17 + 91$ | (99) $39 + 62$  |
| (10) $96 + 61$ | (40) $76 + 88$ | (70) $51 + 25$ | (100) $5 + 31$  |
| (11) $28 + 41$ | (41) $11 + 21$ | (71) $100 + 4$ | (101) $27 + 62$ |
| (12) $80 + 38$ | (42) $63 + 19$ | (72) $39 + 30$ | (102) $89 + 45$ |
| (13) $83 + 49$ | (43) $32 + 34$ | (73) $94 + 9$  | (103) $59 + 76$ |
| (14) $49 + 61$ | (44) $30 + 92$ | (74) $33 + 65$ | (104) $87 + 7$  |
| (15) $68 + 21$ | (45) $39 + 8$  | (75) $53 + 28$ | (105) $19 + 54$ |
| (16) $34 + 80$ | (46) $72 + 60$ | (76) $38 + 59$ | (106) $23 + 37$ |
| (17) $75 + 69$ | (47) $88 + 1$  | (77) $5 + 61$  | (107) $91 + 35$ |
| (18) $87 + 30$ | (48) $52 + 90$ | (78) $24 + 61$ | (108) $28 + 43$ |
| (19) $2 + 25$  | (49) $58 + 25$ | (79) $3 + 75$  | (109) $12 + 89$ |
| (20) $91 + 53$ | (50) $89 + 88$ | (80) $81 + 50$ | (110) $12 + 47$ |
| (21) $67 + 11$ | (51) $15 + 46$ | (81) $9 + 89$  | (111) $78 + 53$ |
| (22) $46 + 45$ | (52) $35 + 92$ | (82) $66 + 93$ | (112) $20 + 64$ |
| (23) $92 + 12$ | (53) $57 + 55$ | (83) $9 + 26$  | (113) $58 + 17$ |
| (24) $63 + 64$ | (54) $9 + 60$  | (84) $33 + 70$ | (114) $30 + 57$ |
| (25) $98 + 85$ | (55) $10 + 14$ | (85) $13 + 64$ | (115) $12 + 88$ |
| (26) $52 + 30$ | (56) $0 + 48$  | (86) $37 + 79$ | (116) $25 + 72$ |
| (27) $57 + 28$ | (57) $13 + 73$ | (87) $87 + 29$ | (117) $65 + 95$ |
| (28) $76 + 87$ | (58) $57 + 38$ | (88) $71 + 72$ | (118) $79 + 13$ |
| (29) $66 + 13$ | (59) $26 + 37$ | (89) $5 + 60$  | (119) $16 + 17$ |
| (30) $37 + 50$ | (60) $81 + 66$ | (90) $34 + 42$ | (120) $29 + 63$ |

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|-----------------|-----------------|----------------|-----------------|
| (1) $69 + 55$   | (31) $71 + 48$  | (61) $14 + 30$ | (91) $9 + 74$   |
| (2) $75 + 63$   | (32) $13 + 88$  | (62) $61 + 95$ | (92) $83 + 55$  |
| (3) $39 + 86$   | (33) $48 + 57$  | (63) $32 + 72$ | (93) $52 + 41$  |
| (4) $54 + 40$   | (34) $36 + 56$  | (64) $23 + 26$ | (94) $98 + 27$  |
| (5) $57 + 12$   | (35) $100 + 94$ | (65) $20 + 40$ | (95) $68 + 93$  |
| (6) $9 + 34$    | (36) $67 + 97$  | (66) $10 + 57$ | (96) $29 + 24$  |
| (7) $63 + 29$   | (37) $23 + 68$  | (67) $78 + 96$ | (97) $36 + 89$  |
| (8) $23 + 48$   | (38) $87 + 69$  | (68) $55 + 83$ | (98) $11 + 43$  |
| (9) $49 + 98$   | (39) $80 + 81$  | (69) $5 + 80$  | (99) $56 + 51$  |
| (10) $90 + 73$  | (40) $10 + 86$  | (70) $92 + 13$ | (100) $13 + 89$ |
| (11) $63 + 31$  | (41) $12 + 16$  | (71) $37 + 35$ | (101) $66 + 19$ |
| (12) $50 + 46$  | (42) $85 + 49$  | (72) $26 + 64$ | (102) $94 + 34$ |
| (13) $32 + 81$  | (43) $6 + 94$   | (73) $70 + 63$ | (103) $84 + 96$ |
| (14) $70 + 78$  | (44) $48 + 25$  | (74) $29 + 28$ | (104) $71 + 70$ |
| (15) $68 + 85$  | (45) $25 + 33$  | (75) $62 + 33$ | (105) $81 + 54$ |
| (16) $31 + 86$  | (46) $83 + 89$  | (76) $10 + 16$ | (106) $92 + 77$ |
| (17) $75 + 23$  | (47) $70 + 66$  | (77) $87 + 54$ | (107) $46 + 26$ |
| (18) $100 + 39$ | (48) $16 + 16$  | (78) $9 + 41$  | (108) $86 + 20$ |
| (19) $3 + 6$    | (49) $29 + 68$  | (79) $66 + 45$ | (109) $52 + 42$ |
| (20) $70 + 8$   | (50) $33 + 42$  | (80) $70 + 9$  | (110) $73 + 2$  |
| (21) $37 + 14$  | (51) $87 + 95$  | (81) $84 + 91$ | (111) $90 + 11$ |
| (22) $36 + 43$  | (52) $1 + 96$   | (82) $77 + 8$  | (112) $48 + 78$ |
| (23) $92 + 35$  | (53) $41 + 1$   | (83) $26 + 75$ | (113) $87 + 58$ |
| (24) $29 + 50$  | (54) $78 + 41$  | (84) $62 + 25$ | (114) $44 + 52$ |
| (25) $45 + 3$   | (55) $50 + 8$   | (85) $67 + 50$ | (115) $59 + 15$ |
| (26) $64 + 2$   | (56) $2 + 84$   | (86) $40 + 22$ | (116) $20 + 67$ |
| (27) $94 + 62$  | (57) $60 + 100$ | (87) $23 + 17$ | (117) $54 + 40$ |
| (28) $85 + 29$  | (58) $16 + 5$   | (88) $42 + 1$  | (118) $3 + 95$  |
| (29) $91 + 79$  | (59) $47 + 0$   | (89) $30 + 97$ | (119) $59 + 97$ |
| (30) $69 + 88$  | (60) $59 + 91$  | (90) $52 + 12$ | (120) $23 + 36$ |

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|----------------|----------------|----------------|-----------------|
| (1) $83 + 62$  | (31) $4 + 21$  | (61) $36 + 52$ | (91) $64 + 90$  |
| (2) $86 + 11$  | (32) $51 + 23$ | (62) $31 + 7$  | (92) $36 + 14$  |
| (3) $90 + 80$  | (33) $61 + 42$ | (63) $7 + 36$  | (93) $74 + 22$  |
| (4) $16 + 55$  | (34) $45 + 94$ | (64) $91 + 57$ | (94) $38 + 62$  |
| (5) $76 + 99$  | (35) $48 + 46$ | (65) $61 + 93$ | (95) $50 + 81$  |
| (6) $36 + 93$  | (36) $96 + 80$ | (66) $14 + 42$ | (96) $9 + 96$   |
| (7) $9 + 88$   | (37) $63 + 72$ | (67) $28 + 3$  | (97) $83 + 21$  |
| (8) $68 + 16$  | (38) $4 + 64$  | (68) $17 + 45$ | (98) $69 + 17$  |
| (9) $93 + 35$  | (39) $98 + 37$ | (69) $44 + 61$ | (99) $90 + 33$  |
| (10) $93 + 37$ | (40) $70 + 97$ | (70) $39 + 20$ | (100) $96 + 65$ |
| (11) $87 + 65$ | (41) $52 + 85$ | (71) $59 + 59$ | (101) $48 + 30$ |
| (12) $38 + 23$ | (42) $1 + 98$  | (72) $3 + 88$  | (102) $40 + 66$ |
| (13) $72 + 56$ | (43) $65 + 45$ | (73) $32 + 72$ | (103) $81 + 77$ |
| (14) $40 + 30$ | (44) $79 + 8$  | (74) $60 + 54$ | (104) $60 + 97$ |
| (15) $94 + 11$ | (45) $49 + 82$ | (75) $90 + 8$  | (105) $19 + 96$ |
| (16) $55 + 80$ | (46) $65 + 38$ | (76) $15 + 45$ | (106) $25 + 51$ |
| (17) $2 + 24$  | (47) $59 + 61$ | (77) $21 + 28$ | (107) $25 + 38$ |
| (18) $47 + 18$ | (48) $51 + 66$ | (78) $8 + 60$  | (108) $78 + 63$ |
| (19) $69 + 35$ | (49) $85 + 11$ | (79) $44 + 70$ | (109) $76 + 72$ |
| (20) $35 + 99$ | (50) $22 + 6$  | (80) $53 + 6$  | (110) $75 + 64$ |
| (21) $26 + 82$ | (51) $40 + 77$ | (81) $68 + 4$  | (111) $85 + 69$ |
| (22) $57 + 30$ | (52) $11 + 25$ | (82) $29 + 22$ | (112) $95 + 4$  |
| (23) $10 + 55$ | (53) $39 + 49$ | (83) $24 + 35$ | (113) $78 + 55$ |
| (24) $40 + 89$ | (54) $11 + 76$ | (84) $47 + 31$ | (114) $40 + 63$ |
| (25) $79 + 72$ | (55) $35 + 66$ | (85) $56 + 88$ | (115) $91 + 82$ |
| (26) $76 + 48$ | (56) $86 + 92$ | (86) $45 + 44$ | (116) $83 + 37$ |
| (27) $88 + 47$ | (57) $10 + 70$ | (87) $56 + 84$ | (117) $41 + 33$ |
| (28) $72 + 75$ | (58) $78 + 92$ | (88) $87 + 80$ | (118) $1 + 66$  |
| (29) $38 + 21$ | (59) $51 + 23$ | (89) $98 + 73$ | (119) $29 + 19$ |
| (30) $48 + 52$ | (60) $62 + 28$ | (90) $12 + 56$ | (120) $83 + 41$ |

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|----------------|----------------|----------------|-----------------|
| (1) $26 + 27$  | (31) $18 + 99$ | (61) $17 + 58$ | (91) $41 + 57$  |
| (2) $83 + 48$  | (32) $30 + 67$ | (62) $97 + 55$ | (92) $32 + 63$  |
| (3) $1 + 55$   | (33) $15 + 51$ | (63) $67 + 43$ | (93) $16 + 51$  |
| (4) $69 + 71$  | (34) $25 + 42$ | (64) $50 + 99$ | (94) $31 + 33$  |
| (5) $1 + 72$   | (35) $75 + 98$ | (65) $58 + 69$ | (95) $2 + 100$  |
| (6) $94 + 70$  | (36) $81 + 4$  | (66) $90 + 41$ | (96) $47 + 51$  |
| (7) $43 + 64$  | (37) $96 + 73$ | (67) $90 + 20$ | (97) $49 + 61$  |
| (8) $82 + 85$  | (38) $27 + 35$ | (68) $7 + 46$  | (98) $48 + 7$   |
| (9) $47 + 73$  | (39) $49 + 94$ | (69) $80 + 30$ | (99) $75 + 59$  |
| (10) $55 + 43$ | (40) $36 + 75$ | (70) $33 + 95$ | (100) $27 + 95$ |
| (11) $4 + 53$  | (41) $15 + 22$ | (71) $95 + 73$ | (101) $39 + 69$ |
| (12) $27 + 28$ | (42) $76 + 44$ | (72) $11 + 29$ | (102) $52 + 70$ |
| (13) $92 + 6$  | (43) $73 + 3$  | (73) $54 + 22$ | (103) $23 + 43$ |
| (14) $25 + 3$  | (44) $22 + 35$ | (74) $67 + 79$ | (104) $31 + 41$ |
| (15) $29 + 43$ | (45) $21 + 66$ | (75) $64 + 83$ | (105) $99 + 11$ |
| (16) $99 + 75$ | (46) $57 + 91$ | (76) $49 + 86$ | (106) $44 + 53$ |
| (17) $9 + 17$  | (47) $88 + 10$ | (77) $22 + 26$ | (107) $30 + 4$  |
| (18) $18 + 94$ | (48) $42 + 43$ | (78) $14 + 56$ | (108) $48 + 44$ |
| (19) $70 + 51$ | (49) $31 + 78$ | (79) $71 + 53$ | (109) $3 + 24$  |
| (20) $72 + 50$ | (50) $27 + 2$  | (80) $62 + 3$  | (110) $48 + 73$ |
| (21) $8 + 27$  | (51) $51 + 73$ | (81) $26 + 32$ | (111) $75 + 37$ |
| (22) $9 + 35$  | (52) $90 + 86$ | (82) $86 + 57$ | (112) $28 + 71$ |
| (23) $49 + 56$ | (53) $35 + 98$ | (83) $78 + 58$ | (113) $14 + 16$ |
| (24) $69 + 75$ | (54) $9 + 79$  | (84) $25 + 52$ | (114) $28 + 13$ |
| (25) $72 + 5$  | (55) $87 + 78$ | (85) $22 + 60$ | (115) $75 + 89$ |
| (26) $43 + 86$ | (56) $55 + 83$ | (86) $6 + 20$  | (116) $2 + 74$  |
| (27) $52 + 55$ | (57) $24 + 54$ | (87) $15 + 37$ | (117) $98 + 99$ |
| (28) $19 + 91$ | (58) $93 + 93$ | (88) $2 + 92$  | (118) $96 + 62$ |
| (29) $47 + 55$ | (59) $19 + 9$  | (89) $16 + 37$ | (119) $43 + 73$ |
| (30) $54 + 38$ | (60) $5 + 96$  | (90) $8 + 69$  | (120) $38 + 4$  |

|                |                |                |                  |
|----------------|----------------|----------------|------------------|
| (1) $58 + 6$   | (31) $36 + 45$ | (61) $43 + 17$ | (91) $86 + 54$   |
| (2) $80 + 84$  | (32) $25 + 14$ | (62) $94 + 30$ | (92) $11 + 50$   |
| (3) $46 + 16$  | (33) $96 + 94$ | (63) $3 + 96$  | (93) $55 + 22$   |
| (4) $15 + 96$  | (34) $52 + 56$ | (64) $62 + 12$ | (94) $2 + 14$    |
| (5) $2 + 12$   | (35) $62 + 97$ | (65) $84 + 50$ | (95) $74 + 29$   |
| (6) $39 + 82$  | (36) $38 + 94$ | (66) $85 + 63$ | (96) $47 + 32$   |
| (7) $44 + 84$  | (37) $44 + 59$ | (67) $69 + 58$ | (97) $31 + 92$   |
| (8) $70 + 24$  | (38) $82 + 76$ | (68) $18 + 11$ | (98) $44 + 14$   |
| (9) $81 + 54$  | (39) $6 + 57$  | (69) $34 + 98$ | (99) $16 + 60$   |
| (10) $69 + 3$  | (40) $87 + 70$ | (70) $32 + 70$ | (100) $67 + 42$  |
| (11) $21 + 95$ | (41) $66 + 56$ | (71) $39 + 69$ | (101) $39 + 5$   |
| (12) $7 + 61$  | (42) $69 + 97$ | (72) $16 + 38$ | (102) $73 + 38$  |
| (13) $89 + 75$ | (43) $61 + 80$ | (73) $27 + 32$ | (103) $30 + 17$  |
| (14) $49 + 99$ | (44) $89 + 30$ | (74) $32 + 5$  | (104) $47 + 74$  |
| (15) $54 + 19$ | (45) $20 + 13$ | (75) $84 + 35$ | (105) $57 + 100$ |
| (16) $55 + 75$ | (46) $99 + 88$ | (76) $72 + 95$ | (106) $86 + 35$  |
| (17) $23 + 34$ | (47) $37 + 51$ | (77) $91 + 83$ | (107) $86 + 78$  |
| (18) $48 + 97$ | (48) $35 + 50$ | (78) $49 + 27$ | (108) $96 + 53$  |
| (19) $87 + 2$  | (49) $58 + 80$ | (79) $55 + 98$ | (109) $56 + 47$  |
| (20) $85 + 29$ | (50) $78 + 66$ | (80) $8 + 3$   | (110) $74 + 18$  |
| (21) $53 + 42$ | (51) $75 + 41$ | (81) $23 + 10$ | (111) $52 + 5$   |
| (22) $20 + 76$ | (52) $3 + 3$   | (82) $67 + 78$ | (112) $49 + 74$  |
| (23) $48 + 57$ | (53) $32 + 1$  | (83) $71 + 83$ | (113) $68 + 58$  |
| (24) $3 + 20$  | (54) $58 + 64$ | (84) $27 + 62$ | (114) $58 + 55$  |
| (25) $29 + 16$ | (55) $37 + 71$ | (85) $83 + 57$ | (115) $85 + 55$  |
| (26) $84 + 91$ | (56) $18 + 50$ | (86) $47 + 1$  | (116) $65 + 4$   |
| (27) $65 + 59$ | (57) $52 + 54$ | (87) $83 + 38$ | (117) $21 + 87$  |
| (28) $55 + 90$ | (58) $41 + 72$ | (88) $53 + 38$ | (118) $73 + 23$  |
| (29) $85 + 12$ | (59) $76 + 66$ | (89) $79 + 40$ | (119) $77 + 96$  |
| (30) $69 + 58$ | (60) $14 + 66$ | (90) $6 + 93$  | (120) $27 + 8$   |



|                 |                 |                |                 |
|-----------------|-----------------|----------------|-----------------|
| (1) $26 + 97$   | (31) $16 + 46$  | (61) $98 + 92$ | (91) $66 + 11$  |
| (2) $76 + 7$    | (32) $16 + 83$  | (62) $99 + 16$ | (92) $95 + 66$  |
| (3) $13 + 49$   | (33) $79 + 14$  | (63) $37 + 53$ | (93) $81 + 21$  |
| (4) $76 + 69$   | (34) $57 + 19$  | (64) $93 + 16$ | (94) $11 + 69$  |
| (5) $98 + 32$   | (35) $74 + 55$  | (65) $71 + 45$ | (95) $40 + 75$  |
| (6) $85 + 100$  | (36) $52 + 4$   | (66) $6 + 83$  | (96) $65 + 18$  |
| (7) $17 + 45$   | (37) $67 + 20$  | (67) $92 + 23$ | (97) $83 + 74$  |
| (8) $43 + 46$   | (38) $43 + 84$  | (68) $37 + 98$ | (98) $87 + 50$  |
| (9) $63 + 43$   | (39) $100 + 16$ | (69) $79 + 45$ | (99) $49 + 56$  |
| (10) $72 + 28$  | (40) $71 + 6$   | (70) $7 + 18$  | (100) $9 + 64$  |
| (11) $85 + 82$  | (41) $25 + 69$  | (71) $37 + 84$ | (101) $31 + 6$  |
| (12) $35 + 10$  | (42) $80 + 81$  | (72) $65 + 27$ | (102) $32 + 13$ |
| (13) $41 + 94$  | (43) $18 + 89$  | (73) $25 + 52$ | (103) $9 + 69$  |
| (14) $19 + 74$  | (44) $33 + 89$  | (74) $2 + 82$  | (104) $46 + 14$ |
| (15) $44 + 9$   | (45) $80 + 69$  | (75) $7 + 7$   | (105) $76 + 20$ |
| (16) $93 + 53$  | (46) $33 + 98$  | (76) $17 + 27$ | (106) $76 + 78$ |
| (17) $12 + 92$  | (47) $7 + 62$   | (77) $92 + 90$ | (107) $50 + 31$ |
| (18) $16 + 45$  | (48) $48 + 42$  | (78) $92 + 59$ | (108) $18 + 23$ |
| (19) $100 + 21$ | (49) $28 + 54$  | (79) $55 + 9$  | (109) $28 + 19$ |
| (20) $24 + 88$  | (50) $44 + 32$  | (80) $19 + 20$ | (110) $68 + 59$ |
| (21) $56 + 87$  | (51) $28 + 60$  | (81) $9 + 30$  | (111) $54 + 89$ |
| (22) $5 + 67$   | (52) $18 + 62$  | (82) $67 + 63$ | (112) $43 + 31$ |
| (23) $85 + 33$  | (53) $73 + 48$  | (83) $32 + 33$ | (113) $80 + 14$ |
| (24) $6 + 81$   | (54) $2 + 44$   | (84) $65 + 70$ | (114) $52 + 57$ |
| (25) $17 + 55$  | (55) $98 + 18$  | (85) $16 + 39$ | (115) $65 + 70$ |
| (26) $64 + 24$  | (56) $70 + 96$  | (86) $74 + 47$ | (116) $69 + 28$ |
| (27) $85 + 40$  | (57) $3 + 84$   | (87) $63 + 43$ | (117) $74 + 30$ |
| (28) $6 + 28$   | (58) $28 + 59$  | (88) $31 + 42$ | (118) $71 + 97$ |
| (29) $31 + 16$  | (59) $24 + 20$  | (89) $37 + 2$  | (119) $15 + 69$ |
| (30) $74 + 17$  | (60) $67 + 65$  | (90) $8 + 20$  | (120) $73 + 84$ |

|                |                |                |                 |
|----------------|----------------|----------------|-----------------|
| (1) $17 + 33$  | (31) $90 + 10$ | (61) $1 + 81$  | (91) $34 + 16$  |
| (2) $35 + 61$  | (32) $5 + 34$  | (62) $94 + 99$ | (92) $32 + 72$  |
| (3) $60 + 27$  | (33) $39 + 49$ | (63) $59 + 7$  | (93) $14 + 53$  |
| (4) $27 + 68$  | (34) $10 + 47$ | (64) $0 + 23$  | (94) $84 + 3$   |
| (5) $8 + 20$   | (35) $7 + 41$  | (65) $60 + 60$ | (95) $63 + 56$  |
| (6) $89 + 77$  | (36) $65 + 9$  | (66) $96 + 18$ | (96) $96 + 51$  |
| (7) $21 + 94$  | (37) $0 + 69$  | (67) $0 + 86$  | (97) $55 + 79$  |
| (8) $83 + 52$  | (38) $39 + 46$ | (68) $84 + 86$ | (98) $96 + 99$  |
| (9) $37 + 99$  | (39) $49 + 47$ | (69) $58 + 62$ | (99) $19 + 77$  |
| (10) $19 + 59$ | (40) $42 + 80$ | (70) $23 + 17$ | (100) $24 + 6$  |
| (11) $41 + 32$ | (41) $83 + 52$ | (71) $86 + 17$ | (101) $64 + 66$ |
| (12) $89 + 42$ | (42) $42 + 14$ | (72) $92 + 57$ | (102) $99 + 55$ |
| (13) $8 + 94$  | (43) $24 + 70$ | (73) $13 + 77$ | (103) $18 + 37$ |
| (14) $76 + 29$ | (44) $98 + 43$ | (74) $58 + 25$ | (104) $67 + 55$ |
| (15) $41 + 71$ | (45) $82 + 47$ | (75) $62 + 38$ | (105) $20 + 90$ |
| (16) $40 + 10$ | (46) $2 + 39$  | (76) $70 + 68$ | (106) $61 + 55$ |
| (17) $1 + 89$  | (47) $36 + 87$ | (77) $43 + 89$ | (107) $44 + 93$ |
| (18) $62 + 93$ | (48) $41 + 43$ | (78) $98 + 61$ | (108) $39 + 21$ |
| (19) $22 + 4$  | (49) $13 + 15$ | (79) $1 + 19$  | (109) $39 + 6$  |
| (20) $98 + 90$ | (50) $66 + 0$  | (80) $90 + 24$ | (110) $77 + 88$ |
| (21) $56 + 39$ | (51) $54 + 3$  | (81) $97 + 48$ | (111) $11 + 58$ |
| (22) $55 + 88$ | (52) $57 + 51$ | (82) $56 + 18$ | (112) $28 + 60$ |
| (23) $43 + 11$ | (53) $65 + 15$ | (83) $0 + 23$  | (113) $79 + 32$ |
| (24) $7 + 6$   | (54) $72 + 18$ | (84) $81 + 95$ | (114) $86 + 8$  |
| (25) $14 + 54$ | (55) $31 + 15$ | (85) $65 + 83$ | (115) $92 + 37$ |
| (26) $65 + 90$ | (56) $33 + 3$  | (86) $36 + 46$ | (116) $44 + 68$ |
| (27) $25 + 85$ | (57) $66 + 91$ | (87) $98 + 10$ | (117) $56 + 47$ |
| (28) $99 + 52$ | (58) $45 + 38$ | (88) $34 + 58$ | (118) $10 + 95$ |
| (29) $88 + 44$ | (59) $29 + 24$ | (89) $39 + 1$  | (119) $7 + 77$  |
| (30) $64 + 51$ | (60) $7 + 76$  | (90) $8 + 7$   | (120) $48 + 10$ |

|                       |                      |                      |                       |
|-----------------------|----------------------|----------------------|-----------------------|
| (1) $46 + 24 = 70$    | (31) $31 + 91 = 122$ | (61) $67 + 85 = 152$ | (91) $44 + 95 = 139$  |
| (2) $26 + 33 = 59$    | (32) $43 + 35 = 78$  | (62) $98 + 64 = 162$ | (92) $48 + 99 = 147$  |
| (3) $87 + 8 = 95$     | (33) $94 + 89 = 183$ | (63) $75 + 49 = 124$ | (93) $43 + 83 = 126$  |
| (4) $71 + 91 = 162$   | (34) $22 + 23 = 45$  | (64) $86 + 59 = 145$ | (94) $42 + 23 = 65$   |
| (5) $7 + 50 = 57$     | (35) $58 + 20 = 78$  | (65) $47 + 71 = 118$ | (95) $19 + 29 = 48$   |
| (6) $76 + 36 = 112$   | (36) $7 + 60 = 67$   | (66) $45 + 61 = 106$ | (96) $11 + 72 = 83$   |
| (7) $26 + 46 = 72$    | (37) $67 + 72 = 139$ | (67) $40 + 63 = 103$ | (97) $59 + 21 = 80$   |
| (8) $94 + 96 = 190$   | (38) $15 + 3 = 18$   | (68) $81 + 59 = 140$ | (98) $59 + 66 = 125$  |
| (9) $96 + 40 = 136$   | (39) $2 + 82 = 84$   | (69) $9 + 56 = 65$   | (99) $24 + 43 = 67$   |
| (10) $13 + 23 = 36$   | (40) $74 + 66 = 140$ | (70) $37 + 36 = 73$  | (100) $85 + 82 = 167$ |
| (11) $51 + 75 = 126$  | (41) $46 + 0 = 46$   | (71) $61 + 89 = 150$ | (101) $26 + 89 = 115$ |
| (12) $84 + 77 = 161$  | (42) $18 + 92 = 110$ | (72) $36 + 54 = 90$  | (102) $83 + 67 = 150$ |
| (13) $97 + 26 = 123$  | (43) $98 + 82 = 180$ | (73) $4 + 38 = 42$   | (103) $41 + 91 = 132$ |
| (14) $39 + 17 = 56$   | (44) $37 + 85 = 122$ | (74) $83 + 1 = 84$   | (104) $54 + 69 = 123$ |
| (15) $24 + 80 = 104$  | (45) $0 + 59 = 59$   | (75) $17 + 48 = 65$  | (105) $82 + 99 = 181$ |
| (16) $64 + 97 = 161$  | (46) $9 + 87 = 96$   | (76) $52 + 80 = 132$ | (106) $5 + 10 = 15$   |
| (17) $3 + 47 = 50$    | (47) $16 + 22 = 38$  | (77) $34 + 43 = 77$  | (107) $75 + 24 = 99$  |
| (18) $52 + 87 = 139$  | (48) $52 + 18 = 70$  | (78) $69 + 73 = 142$ | (108) $41 + 12 = 53$  |
| (19) $100 + 92 = 192$ | (49) $49 + 74 = 123$ | (79) $49 + 26 = 75$  | (109) $8 + 56 = 64$   |
| (20) $82 + 68 = 150$  | (50) $47 + 4 = 51$   | (80) $6 + 66 = 72$   | (110) $34 + 52 = 86$  |
| (21) $18 + 12 = 30$   | (51) $70 + 22 = 92$  | (81) $93 + 39 = 132$ | (111) $91 + 78 = 169$ |
| (22) $70 + 94 = 164$  | (52) $36 + 93 = 129$ | (82) $20 + 35 = 55$  | (112) $30 + 57 = 87$  |
| (23) $84 + 89 = 173$  | (53) $94 + 21 = 115$ | (83) $41 + 6 = 47$   | (113) $55 + 10 = 65$  |
| (24) $78 + 98 = 176$  | (54) $2 + 73 = 75$   | (84) $30 + 13 = 43$  | (114) $81 + 95 = 176$ |
| (25) $97 + 27 = 124$  | (55) $50 + 55 = 105$ | (85) $71 + 47 = 118$ | (115) $93 + 69 = 162$ |
| (26) $29 + 42 = 71$   | (56) $67 + 38 = 105$ | (86) $26 + 17 = 43$  | (116) $47 + 49 = 96$  |
| (27) $67 + 8 = 75$    | (57) $66 + 88 = 154$ | (87) $17 + 14 = 31$  | (117) $66 + 6 = 72$   |
| (28) $26 + 55 = 81$   | (58) $58 + 91 = 149$ | (88) $32 + 65 = 97$  | (118) $44 + 52 = 96$  |
| (29) $30 + 5 = 35$    | (59) $48 + 99 = 147$ | (89) $96 + 26 = 122$ | (119) $75 + 79 = 154$ |
| (30) $80 + 4 = 84$    | (60) $74 + 91 = 165$ | (90) $24 + 47 = 71$  | (120) $41 + 58 = 99$  |

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|----------------------|-----------------------|-----------------------|-----------------------|
| (1) $1 + 74 = 75$    | (31) $67 + 7 = 74$    | (61) $55 + 34 = 89$   | (91) $10 + 10 = 20$   |
| (2) $12 + 46 = 58$   | (32) $36 + 69 = 105$  | (62) $80 + 69 = 149$  | (92) $38 + 58 = 96$   |
| (3) $18 + 38 = 56$   | (33) $48 + 29 = 77$   | (63) $88 + 51 = 139$  | (93) $26 + 16 = 42$   |
| (4) $42 + 7 = 49$    | (34) $87 + 18 = 105$  | (64) $85 + 100 = 185$ | (94) $16 + 3 = 19$    |
| (5) $44 + 62 = 106$  | (35) $89 + 92 = 181$  | (65) $5 + 23 = 28$    | (95) $32 + 24 = 56$   |
| (6) $66 + 94 = 160$  | (36) $23 + 51 = 74$   | (66) $66 + 83 = 149$  | (96) $68 + 96 = 164$  |
| (7) $96 + 69 = 165$  | (37) $62 + 95 = 157$  | (67) $63 + 19 = 82$   | (97) $10 + 42 = 52$   |
| (8) $20 + 100 = 120$ | (38) $22 + 3 = 25$    | (68) $15 + 23 = 38$   | (98) $83 + 54 = 137$  |
| (9) $70 + 15 = 85$   | (39) $31 + 22 = 53$   | (69) $48 + 44 = 92$   | (99) $100 + 48 = 148$ |
| (10) $14 + 35 = 49$  | (40) $95 + 98 = 193$  | (70) $87 + 41 = 128$  | (100) $59 + 38 = 97$  |
| (11) $90 + 42 = 132$ | (41) $48 + 96 = 144$  | (71) $70 + 76 = 146$  | (101) $17 + 58 = 75$  |
| (12) $57 + 76 = 133$ | (42) $42 + 66 = 108$  | (72) $21 + 25 = 46$   | (102) $6 + 65 = 71$   |
| (13) $82 + 88 = 170$ | (43) $58 + 15 = 73$   | (73) $95 + 19 = 114$  | (103) $63 + 15 = 78$  |
| (14) $48 + 74 = 122$ | (44) $4 + 96 = 100$   | (74) $57 + 34 = 91$   | (104) $9 + 63 = 72$   |
| (15) $23 + 84 = 107$ | (45) $60 + 98 = 158$  | (75) $13 + 1 = 14$    | (105) $89 + 42 = 131$ |
| (16) $77 + 52 = 129$ | (46) $27 + 82 = 109$  | (76) $22 + 10 = 32$   | (106) $95 + 22 = 117$ |
| (17) $72 + 54 = 126$ | (47) $82 + 12 = 94$   | (77) $30 + 8 = 38$    | (107) $73 + 85 = 158$ |
| (18) $17 + 65 = 82$  | (48) $85 + 60 = 145$  | (78) $78 + 49 = 127$  | (108) $0 + 99 = 99$   |
| (19) $62 + 71 = 133$ | (49) $39 + 57 = 96$   | (79) $88 + 6 = 94$    | (109) $53 + 60 = 113$ |
| (20) $74 + 79 = 153$ | (50) $49 + 98 = 147$  | (80) $48 + 93 = 141$  | (110) $16 + 6 = 22$   |
| (21) $89 + 89 = 178$ | (51) $1 + 4 = 5$      | (81) $42 + 59 = 101$  | (111) $49 + 22 = 71$  |
| (22) $26 + 36 = 62$  | (52) $85 + 90 = 175$  | (82) $58 + 91 = 149$  | (112) $30 + 28 = 58$  |
| (23) $62 + 67 = 129$ | (53) $100 + 23 = 123$ | (83) $1 + 31 = 32$    | (113) $20 + 7 = 27$   |
| (24) $55 + 17 = 72$  | (54) $53 + 97 = 150$  | (84) $62 + 92 = 154$  | (114) $16 + 40 = 56$  |
| (25) $77 + 68 = 145$ | (55) $21 + 11 = 32$   | (85) $57 + 100 = 157$ | (115) $38 + 54 = 92$  |
| (26) $2 + 100 = 102$ | (56) $18 + 53 = 71$   | (86) $85 + 20 = 105$  | (116) $32 + 23 = 55$  |
| (27) $47 + 92 = 139$ | (57) $13 + 16 = 29$   | (87) $30 + 31 = 61$   | (117) $41 + 93 = 134$ |
| (28) $6 + 49 = 55$   | (58) $65 + 34 = 99$   | (88) $57 + 60 = 117$  | (118) $95 + 1 = 96$   |
| (29) $62 + 89 = 151$ | (59) $70 + 8 = 78$    | (89) $3 + 30 = 33$    | (119) $29 + 7 = 36$   |
| (30) $43 + 93 = 136$ | (60) $63 + 17 = 80$   | (90) $83 + 38 = 121$  | (120) $5 + 28 = 33$   |

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|----------------------|-----------------------|----------------------|------------------------|
| (1) $86 + 70 = 156$  | (31) $61 + 81 = 142$  | (61) $93 + 57 = 150$ | (91) $6 + 3 = 9$       |
| (2) $3 + 98 = 101$   | (32) $61 + 56 = 117$  | (62) $7 + 41 = 48$   | (92) $100 + 57 = 157$  |
| (3) $66 + 54 = 120$  | (33) $53 + 13 = 66$   | (63) $93 + 47 = 140$ | (93) $95 + 5 = 100$    |
| (4) $97 + 60 = 157$  | (34) $52 + 81 = 133$  | (64) $18 + 59 = 77$  | (94) $95 + 40 = 135$   |
| (5) $45 + 91 = 136$  | (35) $25 + 24 = 49$   | (65) $78 + 35 = 113$ | (95) $52 + 39 = 91$    |
| (6) $0 + 100 = 100$  | (36) $47 + 17 = 64$   | (66) $80 + 28 = 108$ | (96) $2 + 97 = 99$     |
| (7) $73 + 76 = 149$  | (37) $86 + 28 = 114$  | (67) $11 + 32 = 43$  | (97) $75 + 19 = 94$    |
| (8) $40 + 17 = 57$   | (38) $41 + 59 = 100$  | (68) $8 + 85 = 93$   | (98) $61 + 6 = 67$     |
| (9) $100 + 49 = 149$ | (39) $97 + 41 = 138$  | (69) $13 + 76 = 89$  | (99) $87 + 60 = 147$   |
| (10) $61 + 25 = 86$  | (40) $1 + 24 = 25$    | (70) $37 + 31 = 68$  | (100) $56 + 75 = 131$  |
| (11) $45 + 62 = 107$ | (41) $15 + 72 = 87$   | (71) $82 + 98 = 180$ | (101) $84 + 40 = 124$  |
| (12) $2 + 29 = 31$   | (42) $19 + 36 = 55$   | (72) $57 + 38 = 95$  | (102) $51 + 59 = 110$  |
| (13) $69 + 2 = 71$   | (43) $48 + 69 = 117$  | (73) $72 + 3 = 75$   | (103) $56 + 82 = 138$  |
| (14) $96 + 15 = 111$ | (44) $55 + 100 = 155$ | (74) $45 + 2 = 47$   | (104) $7 + 95 = 102$   |
| (15) $20 + 38 = 58$  | (45) $50 + 92 = 142$  | (75) $88 + 69 = 157$ | (105) $2 + 63 = 65$    |
| (16) $54 + 26 = 80$  | (46) $66 + 11 = 77$   | (76) $3 + 33 = 36$   | (106) $42 + 41 = 83$   |
| (17) $25 + 17 = 42$  | (47) $22 + 84 = 106$  | (77) $38 + 19 = 57$  | (107) $3 + 28 = 31$    |
| (18) $38 + 72 = 110$ | (48) $28 + 41 = 69$   | (78) $19 + 73 = 92$  | (108) $27 + 38 = 65$   |
| (19) $40 + 56 = 96$  | (49) $54 + 40 = 94$   | (79) $32 + 57 = 89$  | (109) $21 + 12 = 33$   |
| (20) $9 + 58 = 67$   | (50) $65 + 20 = 85$   | (80) $88 + 86 = 174$ | (110) $23 + 82 = 105$  |
| (21) $56 + 84 = 140$ | (51) $48 + 52 = 100$  | (81) $77 + 42 = 119$ | (111) $35 + 42 = 77$   |
| (22) $27 + 50 = 77$  | (52) $11 + 8 = 19$    | (82) $80 + 77 = 157$ | (112) $23 + 79 = 102$  |
| (23) $94 + 58 = 152$ | (53) $46 + 46 = 92$   | (83) $100 + 2 = 102$ | (113) $77 + 2 = 79$    |
| (24) $99 + 12 = 111$ | (54) $45 + 79 = 124$  | (84) $100 + 6 = 106$ | (114) $92 + 82 = 174$  |
| (25) $4 + 78 = 82$   | (55) $16 + 57 = 73$   | (85) $57 + 47 = 104$ | (115) $100 + 82 = 182$ |
| (26) $40 + 16 = 56$  | (56) $85 + 74 = 159$  | (86) $10 + 34 = 44$  | (116) $39 + 57 = 96$   |
| (27) $95 + 67 = 162$ | (57) $56 + 38 = 94$   | (87) $74 + 64 = 138$ | (117) $32 + 90 = 122$  |
| (28) $11 + 77 = 88$  | (58) $11 + 54 = 65$   | (88) $2 + 44 = 46$   | (118) $16 + 42 = 58$   |
| (29) $48 + 54 = 102$ | (59) $46 + 42 = 88$   | (89) $33 + 51 = 84$  | (119) $49 + 24 = 73$   |
| (30) $2 + 1 = 3$     | (60) $26 + 58 = 84$   | (90) $61 + 76 = 137$ | (120) $23 + 100 = 123$ |

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|-----------------------|----------------------|-----------------------|-----------------------|
| (1) $77 + 46 = 123$   | (31) $95 + 76 = 171$ | (61) $11 + 100 = 111$ | (91) $18 + 15 = 33$   |
| (2) $27 + 7 = 34$     | (32) $29 + 35 = 64$  | (62) $75 + 33 = 108$  | (92) $68 + 5 = 73$    |
| (3) $51 + 30 = 81$    | (33) $51 + 86 = 137$ | (63) $56 + 43 = 99$   | (93) $18 + 85 = 103$  |
| (4) $31 + 12 = 43$    | (34) $55 + 45 = 100$ | (64) $37 + 23 = 60$   | (94) $36 + 51 = 87$   |
| (5) $84 + 30 = 114$   | (35) $79 + 51 = 130$ | (65) $31 + 50 = 81$   | (95) $49 + 33 = 82$   |
| (6) $41 + 59 = 100$   | (36) $93 + 56 = 149$ | (66) $21 + 88 = 109$  | (96) $81 + 15 = 96$   |
| (7) $87 + 67 = 154$   | (37) $25 + 54 = 79$  | (67) $79 + 92 = 171$  | (97) $91 + 6 = 97$    |
| (8) $28 + 71 = 99$    | (38) $35 + 15 = 50$  | (68) $85 + 71 = 156$  | (98) $76 + 45 = 121$  |
| (9) $52 + 68 = 120$   | (39) $78 + 83 = 161$ | (69) $72 + 3 = 75$    | (99) $46 + 50 = 96$   |
| (10) $24 + 18 = 42$   | (40) $87 + 87 = 174$ | (70) $85 + 82 = 167$  | (100) $8 + 51 = 59$   |
| (11) $75 + 82 = 157$  | (41) $49 + 60 = 109$ | (71) $48 + 48 = 96$   | (101) $27 + 17 = 44$  |
| (12) $97 + 1 = 98$    | (42) $97 + 58 = 155$ | (72) $84 + 89 = 173$  | (102) $45 + 73 = 118$ |
| (13) $35 + 20 = 55$   | (43) $46 + 51 = 97$  | (73) $99 + 26 = 125$  | (103) $89 + 15 = 104$ |
| (14) $94 + 43 = 137$  | (44) $42 + 37 = 79$  | (74) $20 + 82 = 102$  | (104) $36 + 59 = 95$  |
| (15) $28 + 98 = 126$  | (45) $70 + 79 = 149$ | (75) $80 + 85 = 165$  | (105) $96 + 86 = 182$ |
| (16) $100 + 29 = 129$ | (46) $28 + 65 = 93$  | (76) $88 + 72 = 160$  | (106) $73 + 2 = 75$   |
| (17) $94 + 1 = 95$    | (47) $12 + 3 = 15$   | (77) $68 + 39 = 107$  | (107) $44 + 16 = 60$  |
| (18) $18 + 54 = 72$   | (48) $32 + 5 = 37$   | (78) $79 + 64 = 143$  | (108) $29 + 59 = 88$  |
| (19) $65 + 87 = 152$  | (49) $72 + 91 = 163$ | (79) $68 + 29 = 97$   | (109) $35 + 46 = 81$  |
| (20) $89 + 69 = 158$  | (50) $66 + 54 = 120$ | (80) $54 + 78 = 132$  | (110) $10 + 69 = 79$  |
| (21) $60 + 97 = 157$  | (51) $93 + 75 = 168$ | (81) $38 + 49 = 87$   | (111) $18 + 83 = 101$ |
| (22) $83 + 15 = 98$   | (52) $48 + 12 = 60$  | (82) $38 + 90 = 128$  | (112) $14 + 72 = 86$  |
| (23) $76 + 17 = 93$   | (53) $79 + 38 = 117$ | (83) $40 + 8 = 48$    | (113) $51 + 7 = 58$   |
| (24) $74 + 99 = 173$  | (54) $81 + 22 = 103$ | (84) $52 + 82 = 134$  | (114) $19 + 6 = 25$   |
| (25) $35 + 88 = 123$  | (55) $17 + 16 = 33$  | (85) $45 + 83 = 128$  | (115) $31 + 39 = 70$  |
| (26) $30 + 93 = 123$  | (56) $1 + 64 = 65$   | (86) $62 + 91 = 153$  | (116) $21 + 27 = 48$  |
| (27) $88 + 62 = 150$  | (57) $25 + 54 = 79$  | (87) $39 + 48 = 87$   | (117) $14 + 92 = 106$ |
| (28) $75 + 21 = 96$   | (58) $5 + 35 = 40$   | (88) $95 + 28 = 123$  | (118) $33 + 70 = 103$ |
| (29) $95 + 74 = 169$  | (59) $47 + 83 = 130$ | (89) $54 + 62 = 116$  | (119) $34 + 60 = 94$  |
| (30) $84 + 45 = 129$  | (60) $88 + 4 = 92$   | (90) $3 + 83 = 86$    | (120) $38 + 43 = 81$  |

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|----------------------|----------------------|----------------------|------------------------|
| (1) $76 + 55 = 131$  | (31) $91 + 56 = 147$ | (61) $7 + 33 = 40$   | (91) $100 + 31 = 131$  |
| (2) $1 + 79 = 80$    | (32) $68 + 15 = 83$  | (62) $97 + 78 = 175$ | (92) $23 + 4 = 27$     |
| (3) $24 + 32 = 56$   | (33) $58 + 92 = 150$ | (63) $37 + 64 = 101$ | (93) $30 + 34 = 64$    |
| (4) $61 + 48 = 109$  | (34) $77 + 66 = 143$ | (64) $69 + 51 = 120$ | (94) $86 + 75 = 161$   |
| (5) $10 + 89 = 99$   | (35) $99 + 24 = 123$ | (65) $21 + 43 = 64$  | (95) $35 + 57 = 92$    |
| (6) $35 + 75 = 110$  | (36) $23 + 84 = 107$ | (66) $4 + 78 = 82$   | (96) $20 + 51 = 71$    |
| (7) $39 + 98 = 137$  | (37) $70 + 74 = 144$ | (67) $69 + 97 = 166$ | (97) $96 + 65 = 161$   |
| (8) $91 + 69 = 160$  | (38) $9 + 38 = 47$   | (68) $14 + 97 = 111$ | (98) $28 + 53 = 81$    |
| (9) $26 + 55 = 81$   | (39) $95 + 61 = 156$ | (69) $68 + 68 = 136$ | (99) $94 + 62 = 156$   |
| (10) $47 + 78 = 125$ | (40) $4 + 4 = 8$     | (70) $36 + 14 = 50$  | (100) $9 + 97 = 106$   |
| (11) $40 + 5 = 45$   | (41) $89 + 5 = 94$   | (71) $88 + 2 = 90$   | (101) $20 + 58 = 78$   |
| (12) $8 + 63 = 71$   | (42) $8 + 88 = 96$   | (72) $41 + 20 = 61$  | (102) $13 + 22 = 35$   |
| (13) $98 + 51 = 149$ | (43) $48 + 62 = 110$ | (73) $80 + 37 = 117$ | (103) $71 + 30 = 101$  |
| (14) $30 + 46 = 76$  | (44) $0 + 62 = 62$   | (74) $40 + 56 = 96$  | (104) $25 + 100 = 125$ |
| (15) $17 + 80 = 97$  | (45) $73 + 43 = 116$ | (75) $73 + 90 = 163$ | (105) $98 + 54 = 152$  |
| (16) $25 + 21 = 46$  | (46) $71 + 72 = 143$ | (76) $91 + 33 = 124$ | (106) $37 + 32 = 69$   |
| (17) $25 + 85 = 110$ | (47) $87 + 4 = 91$   | (77) $28 + 95 = 123$ | (107) $56 + 36 = 92$   |
| (18) $26 + 4 = 30$   | (48) $61 + 86 = 147$ | (78) $97 + 0 = 97$   | (108) $57 + 71 = 128$  |
| (19) $67 + 24 = 91$  | (49) $49 + 91 = 140$ | (79) $64 + 42 = 106$ | (109) $58 + 86 = 144$  |
| (20) $93 + 51 = 144$ | (50) $46 + 92 = 138$ | (80) $91 + 87 = 178$ | (110) $32 + 11 = 43$   |
| (21) $79 + 7 = 86$   | (51) $99 + 85 = 184$ | (81) $61 + 65 = 126$ | (111) $42 + 86 = 128$  |
| (22) $69 + 30 = 99$  | (52) $13 + 6 = 19$   | (82) $61 + 15 = 76$  | (112) $37 + 66 = 103$  |
| (23) $82 + 10 = 92$  | (53) $28 + 49 = 77$  | (83) $95 + 40 = 135$ | (113) $90 + 33 = 123$  |
| (24) $61 + 51 = 112$ | (54) $29 + 46 = 75$  | (84) $9 + 3 = 12$    | (114) $7 + 47 = 54$    |
| (25) $9 + 91 = 100$  | (55) $56 + 80 = 136$ | (85) $15 + 60 = 75$  | (115) $46 + 53 = 99$   |
| (26) $77 + 33 = 110$ | (56) $69 + 50 = 119$ | (86) $49 + 49 = 98$  | (116) $12 + 2 = 14$    |
| (27) $7 + 69 = 76$   | (57) $81 + 66 = 147$ | (87) $10 + 45 = 55$  | (117) $90 + 100 = 190$ |
| (28) $44 + 32 = 76$  | (58) $24 + 38 = 62$  | (88) $76 + 88 = 164$ | (118) $35 + 66 = 101$  |
| (29) $88 + 11 = 99$  | (59) $47 + 41 = 88$  | (89) $44 + 88 = 132$ | (119) $72 + 45 = 117$  |
| (30) $79 + 33 = 112$ | (60) $33 + 20 = 53$  | (90) $15 + 85 = 100$ | (120) $55 + 92 = 147$  |

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|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $30 + 63 = 93$    | (31) $87 + 80 = 167$  | (61) $29 + 75 = 104$  | (91) $53 + 99 = 152$   |
| (2) $21 + 74 = 95$    | (32) $55 + 30 = 85$   | (62) $69 + 78 = 147$  | (92) $76 + 24 = 100$   |
| (3) $97 + 27 = 124$   | (33) $36 + 4 = 40$    | (63) $57 + 43 = 100$  | (93) $96 + 71 = 167$   |
| (4) $28 + 6 = 34$     | (34) $3 + 52 = 55$    | (64) $40 + 1 = 41$    | (94) $81 + 55 = 136$   |
| (5) $2 + 65 = 67$     | (35) $16 + 99 = 115$  | (65) $61 + 47 = 108$  | (95) $58 + 81 = 139$   |
| (6) $88 + 55 = 143$   | (36) $9 + 84 = 93$    | (66) $27 + 27 = 54$   | (96) $97 + 9 = 106$    |
| (7) $23 + 83 = 106$   | (37) $83 + 6 = 89$    | (67) $43 + 43 = 86$   | (97) $0 + 46 = 46$     |
| (8) $7 + 32 = 39$     | (38) $96 + 46 = 142$  | (68) $100 + 37 = 137$ | (98) $44 + 87 = 131$   |
| (9) $87 + 33 = 120$   | (39) $26 + 37 = 63$   | (69) $69 + 44 = 113$  | (99) $50 + 83 = 133$   |
| (10) $29 + 30 = 59$   | (40) $34 + 92 = 126$  | (70) $52 + 55 = 107$  | (100) $24 + 97 = 121$  |
| (11) $76 + 8 = 84$    | (41) $63 + 30 = 93$   | (71) $12 + 85 = 97$   | (101) $85 + 40 = 125$  |
| (12) $48 + 65 = 113$  | (42) $60 + 20 = 80$   | (72) $15 + 100 = 115$ | (102) $71 + 96 = 167$  |
| (13) $27 + 12 = 39$   | (43) $73 + 63 = 136$  | (73) $88 + 45 = 133$  | (103) $88 + 19 = 107$  |
| (14) $35 + 43 = 78$   | (44) $94 + 11 = 105$  | (74) $23 + 58 = 81$   | (104) $62 + 45 = 107$  |
| (15) $56 + 78 = 134$  | (45) $85 + 1 = 86$    | (75) $17 + 83 = 100$  | (105) $11 + 100 = 111$ |
| (16) $19 + 57 = 76$   | (46) $72 + 20 = 92$   | (76) $56 + 95 = 151$  | (106) $73 + 67 = 140$  |
| (17) $45 + 46 = 91$   | (47) $98 + 79 = 177$  | (77) $96 + 15 = 111$  | (107) $32 + 17 = 49$   |
| (18) $56 + 72 = 128$  | (48) $8 + 24 = 32$    | (78) $96 + 24 = 120$  | (108) $32 + 95 = 127$  |
| (19) $51 + 54 = 105$  | (49) $65 + 26 = 91$   | (79) $84 + 95 = 179$  | (109) $18 + 14 = 32$   |
| (20) $46 + 13 = 59$   | (50) $100 + 33 = 133$ | (80) $48 + 7 = 55$    | (110) $59 + 85 = 144$  |
| (21) $12 + 7 = 19$    | (51) $11 + 9 = 20$    | (81) $52 + 45 = 97$   | (111) $23 + 32 = 55$   |
| (22) $31 + 100 = 131$ | (52) $16 + 72 = 88$   | (82) $48 + 83 = 131$  | (112) $38 + 100 = 138$ |
| (23) $24 + 26 = 50$   | (53) $84 + 74 = 158$  | (83) $95 + 61 = 156$  | (113) $10 + 88 = 98$   |
| (24) $56 + 96 = 152$  | (54) $49 + 90 = 139$  | (84) $97 + 29 = 126$  | (114) $87 + 61 = 148$  |
| (25) $97 + 20 = 117$  | (55) $80 + 89 = 169$  | (85) $34 + 82 = 116$  | (115) $41 + 31 = 72$   |
| (26) $55 + 57 = 112$  | (56) $51 + 16 = 67$   | (86) $91 + 99 = 190$  | (116) $37 + 51 = 88$   |
| (27) $40 + 93 = 133$  | (57) $47 + 57 = 104$  | (87) $64 + 83 = 147$  | (117) $79 + 17 = 96$   |
| (28) $63 + 26 = 89$   | (58) $95 + 35 = 130$  | (88) $13 + 88 = 101$  | (118) $62 + 5 = 67$    |
| (29) $54 + 71 = 125$  | (59) $70 + 96 = 166$  | (89) $12 + 48 = 60$   | (119) $87 + 25 = 112$  |
| (30) $78 + 15 = 93$   | (60) $97 + 1 = 98$    | (90) $98 + 63 = 161$  | (120) $20 + 25 = 45$   |



|                      |                      |                       |                       |
|----------------------|----------------------|-----------------------|-----------------------|
| (1) $91 + 29 = 120$  | (31) $88 + 78 = 166$ | (61) $7 + 45 = 52$    | (91) $90 + 57 = 147$  |
| (2) $13 + 15 = 28$   | (32) $78 + 77 = 155$ | (62) $70 + 73 = 143$  | (92) $27 + 83 = 110$  |
| (3) $92 + 59 = 151$  | (33) $53 + 78 = 131$ | (63) $8 + 88 = 96$    | (93) $51 + 33 = 84$   |
| (4) $1 + 46 = 47$    | (34) $73 + 40 = 113$ | (64) $20 + 70 = 90$   | (94) $68 + 26 = 94$   |
| (5) $14 + 90 = 104$  | (35) $51 + 96 = 147$ | (65) $73 + 51 = 124$  | (95) $79 + 81 = 160$  |
| (6) $79 + 97 = 176$  | (36) $71 + 25 = 96$  | (66) $98 + 38 = 136$  | (96) $6 + 16 = 22$    |
| (7) $82 + 5 = 87$    | (37) $72 + 14 = 86$  | (67) $12 + 97 = 109$  | (97) $33 + 71 = 104$  |
| (8) $46 + 80 = 126$  | (38) $62 + 7 = 69$   | (68) $32 + 63 = 95$   | (98) $64 + 32 = 96$   |
| (9) $56 + 34 = 90$   | (39) $92 + 48 = 140$ | (69) $73 + 28 = 101$  | (99) $34 + 12 = 46$   |
| (10) $18 + 50 = 68$  | (40) $2 + 7 = 9$     | (70) $90 + 23 = 113$  | (100) $33 + 77 = 110$ |
| (11) $97 + 69 = 166$ | (41) $17 + 35 = 52$  | (71) $34 + 73 = 107$  | (101) $65 + 18 = 83$  |
| (12) $17 + 6 = 23$   | (42) $13 + 56 = 69$  | (72) $6 + 87 = 93$    | (102) $19 + 68 = 87$  |
| (13) $38 + 16 = 54$  | (43) $94 + 3 = 97$   | (73) $53 + 24 = 77$   | (103) $28 + 90 = 118$ |
| (14) $13 + 45 = 58$  | (44) $60 + 57 = 117$ | (74) $21 + 61 = 82$   | (104) $68 + 40 = 108$ |
| (15) $40 + 20 = 60$  | (45) $15 + 28 = 43$  | (75) $57 + 46 = 103$  | (105) $70 + 48 = 118$ |
| (16) $87 + 68 = 155$ | (46) $83 + 70 = 153$ | (76) $33 + 15 = 48$   | (106) $66 + 67 = 133$ |
| (17) $4 + 34 = 38$   | (47) $14 + 84 = 98$  | (77) $31 + 0 = 31$    | (107) $43 + 90 = 133$ |
| (18) $25 + 4 = 29$   | (48) $1 + 74 = 75$   | (78) $72 + 78 = 150$  | (108) $84 + 2 = 86$   |
| (19) $42 + 22 = 64$  | (49) $57 + 72 = 129$ | (79) $88 + 28 = 116$  | (109) $26 + 48 = 74$  |
| (20) $24 + 38 = 62$  | (50) $84 + 87 = 171$ | (80) $53 + 50 = 103$  | (110) $78 + 15 = 93$  |
| (21) $61 + 38 = 99$  | (51) $39 + 71 = 110$ | (81) $39 + 9 = 48$    | (111) $95 + 32 = 127$ |
| (22) $26 + 97 = 123$ | (52) $77 + 67 = 144$ | (82) $63 + 95 = 158$  | (112) $76 + 88 = 164$ |
| (23) $96 + 75 = 171$ | (53) $79 + 85 = 164$ | (83) $73 + 73 = 146$  | (113) $68 + 32 = 100$ |
| (24) $71 + 47 = 118$ | (54) $76 + 87 = 163$ | (84) $64 + 3 = 67$    | (114) $24 + 39 = 63$  |
| (25) $54 + 45 = 99$  | (55) $83 + 46 = 129$ | (85) $71 + 100 = 171$ | (115) $99 + 43 = 142$ |
| (26) $78 + 72 = 150$ | (56) $20 + 39 = 59$  | (86) $46 + 43 = 89$   | (116) $22 + 2 = 24$   |
| (27) $14 + 13 = 27$  | (57) $35 + 68 = 103$ | (87) $11 + 86 = 97$   | (117) $94 + 63 = 157$ |
| (28) $54 + 29 = 83$  | (58) $72 + 1 = 73$   | (88) $39 + 83 = 122$  | (118) $40 + 49 = 89$  |
| (29) $50 + 34 = 84$  | (59) $98 + 87 = 185$ | (89) $25 + 15 = 40$   | (119) $29 + 88 = 117$ |
| (30) $60 + 76 = 136$ | (60) $3 + 39 = 42$   | (90) $85 + 69 = 154$  | (120) $96 + 79 = 175$ |

|                      |                      |                       |                       |
|----------------------|----------------------|-----------------------|-----------------------|
| (1) $61 + 16 = 77$   | (31) $77 + 30 = 107$ | (61) $64 + 90 = 154$  | (91) $72 + 78 = 150$  |
| (2) $11 + 13 = 24$   | (32) $20 + 54 = 74$  | (62) $53 + 80 = 133$  | (92) $36 + 33 = 69$   |
| (3) $26 + 73 = 99$   | (33) $83 + 81 = 164$ | (63) $78 + 99 = 177$  | (93) $81 + 85 = 166$  |
| (4) $97 + 96 = 193$  | (34) $86 + 51 = 137$ | (64) $43 + 80 = 123$  | (94) $75 + 54 = 129$  |
| (5) $75 + 6 = 81$    | (35) $25 + 44 = 69$  | (65) $40 + 19 = 59$   | (95) $70 + 32 = 102$  |
| (6) $93 + 40 = 133$  | (36) $33 + 76 = 109$ | (66) $76 + 40 = 116$  | (96) $89 + 38 = 127$  |
| (7) $70 + 74 = 144$  | (37) $55 + 83 = 138$ | (67) $37 + 83 = 120$  | (97) $100 + 35 = 135$ |
| (8) $31 + 38 = 69$   | (38) $10 + 32 = 42$  | (68) $36 + 91 = 127$  | (98) $71 + 6 = 77$    |
| (9) $71 + 29 = 100$  | (39) $36 + 98 = 134$ | (69) $38 + 87 = 125$  | (99) $90 + 4 = 94$    |
| (10) $57 + 58 = 115$ | (40) $87 + 63 = 150$ | (70) $94 + 12 = 106$  | (100) $18 + 70 = 88$  |
| (11) $18 + 16 = 34$  | (41) $76 + 75 = 151$ | (71) $49 + 99 = 148$  | (101) $82 + 64 = 146$ |
| (12) $48 + 92 = 140$ | (42) $48 + 32 = 80$  | (72) $100 + 59 = 159$ | (102) $56 + 9 = 65$   |
| (13) $74 + 23 = 97$  | (43) $43 + 68 = 111$ | (73) $79 + 2 = 81$    | (103) $53 + 63 = 116$ |
| (14) $33 + 86 = 119$ | (44) $8 + 51 = 59$   | (74) $85 + 100 = 185$ | (104) $45 + 6 = 51$   |
| (15) $65 + 46 = 111$ | (45) $91 + 97 = 188$ | (75) $66 + 33 = 99$   | (105) $89 + 33 = 122$ |
| (16) $89 + 45 = 134$ | (46) $47 + 97 = 144$ | (76) $84 + 39 = 123$  | (106) $10 + 7 = 17$   |
| (17) $85 + 88 = 173$ | (47) $96 + 98 = 194$ | (77) $83 + 60 = 143$  | (107) $87 + 64 = 151$ |
| (18) $43 + 18 = 61$  | (48) $55 + 10 = 65$  | (78) $32 + 22 = 54$   | (108) $41 + 78 = 119$ |
| (19) $56 + 2 = 58$   | (49) $2 + 93 = 95$   | (79) $80 + 20 = 100$  | (109) $15 + 48 = 63$  |
| (20) $81 + 95 = 176$ | (50) $57 + 70 = 127$ | (80) $86 + 9 = 95$    | (110) $57 + 43 = 100$ |
| (21) $72 + 55 = 127$ | (51) $56 + 33 = 89$  | (81) $59 + 81 = 140$  | (111) $81 + 78 = 159$ |
| (22) $12 + 30 = 42$  | (52) $47 + 33 = 80$  | (82) $24 + 38 = 62$   | (112) $0 + 53 = 53$   |
| (23) $11 + 50 = 61$  | (53) $46 + 60 = 106$ | (83) $34 + 39 = 73$   | (113) $98 + 55 = 153$ |
| (24) $52 + 11 = 63$  | (54) $44 + 99 = 143$ | (84) $20 + 74 = 94$   | (114) $5 + 32 = 37$   |
| (25) $14 + 86 = 100$ | (55) $7 + 18 = 25$   | (85) $55 + 77 = 132$  | (115) $77 + 37 = 114$ |
| (26) $59 + 14 = 73$  | (56) $94 + 46 = 140$ | (86) $51 + 95 = 146$  | (116) $0 + 39 = 39$   |
| (27) $33 + 70 = 103$ | (57) $87 + 78 = 165$ | (87) $53 + 76 = 129$  | (117) $89 + 2 = 91$   |
| (28) $37 + 95 = 132$ | (58) $56 + 24 = 80$  | (88) $4 + 39 = 43$    | (118) $9 + 10 = 19$   |
| (29) $95 + 28 = 123$ | (59) $90 + 66 = 156$ | (89) $17 + 42 = 59$   | (119) $89 + 49 = 138$ |
| (30) $79 + 11 = 90$  | (60) $48 + 66 = 114$ | (90) $51 + 17 = 68$   | (120) $57 + 60 = 117$ |

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|-----------------------|-----------------------|-----------------------|-----------------------|
| (1) $1 + 13 = 14$     | (31) $74 + 83 = 157$  | (61) $37 + 45 = 82$   | (91) $11 + 82 = 93$   |
| (2) $54 + 60 = 114$   | (32) $17 + 94 = 111$  | (62) $6 + 48 = 54$    | (92) $14 + 84 = 98$   |
| (3) $84 + 99 = 183$   | (33) $91 + 4 = 95$    | (63) $16 + 59 = 75$   | (93) $8 + 16 = 24$    |
| (4) $96 + 96 = 192$   | (34) $76 + 32 = 108$  | (64) $47 + 96 = 143$  | (94) $29 + 17 = 46$   |
| (5) $69 + 60 = 129$   | (35) $39 + 44 = 83$   | (65) $48 + 35 = 83$   | (95) $46 + 19 = 65$   |
| (6) $94 + 61 = 155$   | (36) $73 + 59 = 132$  | (66) $23 + 100 = 123$ | (96) $61 + 61 = 122$  |
| (7) $7 + 65 = 72$     | (37) $84 + 21 = 105$  | (67) $20 + 52 = 72$   | (97) $40 + 42 = 82$   |
| (8) $32 + 36 = 68$    | (38) $68 + 14 = 82$   | (68) $62 + 62 = 124$  | (98) $10 + 12 = 22$   |
| (9) $51 + 49 = 100$   | (39) $79 + 68 = 147$  | (69) $90 + 22 = 112$  | (99) $41 + 98 = 139$  |
| (10) $2 + 94 = 96$    | (40) $16 + 46 = 62$   | (70) $38 + 39 = 77$   | (100) $54 + 5 = 59$   |
| (11) $76 + 18 = 94$   | (41) $91 + 11 = 102$  | (71) $61 + 41 = 102$  | (101) $81 + 78 = 159$ |
| (12) $59 + 79 = 138$  | (42) $23 + 100 = 123$ | (72) $3 + 96 = 99$    | (102) $9 + 72 = 81$   |
| (13) $13 + 36 = 49$   | (43) $27 + 79 = 106$  | (73) $30 + 92 = 122$  | (103) $14 + 46 = 60$  |
| (14) $93 + 69 = 162$  | (44) $70 + 55 = 125$  | (74) $33 + 75 = 108$  | (104) $31 + 15 = 46$  |
| (15) $60 + 66 = 126$  | (45) $41 + 54 = 95$   | (75) $82 + 42 = 124$  | (105) $61 + 66 = 127$ |
| (16) $28 + 100 = 128$ | (46) $57 + 9 = 66$    | (76) $85 + 53 = 138$  | (106) $55 + 45 = 100$ |
| (17) $70 + 91 = 161$  | (47) $50 + 96 = 146$  | (77) $85 + 11 = 96$   | (107) $45 + 60 = 105$ |
| (18) $72 + 27 = 99$   | (48) $8 + 56 = 64$    | (78) $80 + 15 = 95$   | (108) $55 + 68 = 123$ |
| (19) $81 + 29 = 110$  | (49) $7 + 73 = 80$    | (79) $87 + 37 = 124$  | (109) $9 + 88 = 97$   |
| (20) $39 + 45 = 84$   | (50) $28 + 61 = 89$   | (80) $58 + 94 = 152$  | (110) $93 + 40 = 133$ |
| (21) $74 + 3 = 77$    | (51) $2 + 71 = 73$    | (81) $4 + 15 = 19$    | (111) $56 + 27 = 83$  |
| (22) $53 + 92 = 145$  | (52) $94 + 99 = 193$  | (82) $70 + 33 = 103$  | (112) $28 + 46 = 74$  |
| (23) $35 + 81 = 116$  | (53) $15 + 51 = 66$   | (83) $59 + 62 = 121$  | (113) $44 + 52 = 96$  |
| (24) $24 + 93 = 117$  | (54) $33 + 57 = 90$   | (84) $16 + 33 = 49$   | (114) $61 + 46 = 107$ |
| (25) $92 + 17 = 109$  | (55) $64 + 27 = 91$   | (85) $73 + 41 = 114$  | (115) $36 + 73 = 109$ |
| (26) $56 + 10 = 66$   | (56) $82 + 65 = 147$  | (86) $47 + 86 = 133$  | (116) $49 + 45 = 94$  |
| (27) $53 + 43 = 96$   | (57) $20 + 60 = 80$   | (87) $44 + 89 = 133$  | (117) $15 + 34 = 49$  |
| (28) $94 + 28 = 122$  | (58) $96 + 88 = 184$  | (88) $9 + 23 = 32$    | (118) $35 + 90 = 125$ |
| (29) $77 + 44 = 121$  | (59) $99 + 65 = 164$  | (89) $72 + 32 = 104$  | (119) $60 + 9 = 69$   |
| (30) $23 + 62 = 85$   | (60) $88 + 82 = 170$  | (90) $98 + 5 = 103$   | (120) $67 + 37 = 104$ |

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|----------------------|----------------------|-----------------------|-----------------------|
| (1) $31 + 76 = 107$  | (31) $29 + 29 = 58$  | (61) $54 + 35 = 89$   | (91) $86 + 68 = 154$  |
| (2) $37 + 59 = 96$   | (32) $8 + 68 = 76$   | (62) $17 + 56 = 73$   | (92) $88 + 9 = 97$    |
| (3) $43 + 72 = 115$  | (33) $93 + 59 = 152$ | (63) $48 + 100 = 148$ | (93) $1 + 35 = 36$    |
| (4) $80 + 42 = 122$  | (34) $32 + 28 = 60$  | (64) $9 + 99 = 108$   | (94) $22 + 3 = 25$    |
| (5) $20 + 73 = 93$   | (35) $77 + 84 = 161$ | (65) $78 + 60 = 138$  | (95) $84 + 92 = 176$  |
| (6) $100 + 46 = 146$ | (36) $59 + 5 = 64$   | (66) $14 + 25 = 39$   | (96) $42 + 74 = 116$  |
| (7) $5 + 48 = 53$    | (37) $18 + 62 = 80$  | (67) $24 + 71 = 95$   | (97) $15 + 74 = 89$   |
| (8) $78 + 13 = 91$   | (38) $16 + 23 = 39$  | (68) $87 + 67 = 154$  | (98) $36 + 41 = 77$   |
| (9) $43 + 37 = 80$   | (39) $27 + 91 = 118$ | (69) $70 + 51 = 121$  | (99) $0 + 13 = 13$    |
| (10) $17 + 67 = 84$  | (40) $2 + 95 = 97$   | (70) $83 + 84 = 167$  | (100) $25 + 32 = 57$  |
| (11) $67 + 1 = 68$   | (41) $7 + 69 = 76$   | (71) $23 + 78 = 101$  | (101) $17 + 28 = 45$  |
| (12) $33 + 88 = 121$ | (42) $6 + 91 = 97$   | (72) $82 + 76 = 158$  | (102) $71 + 83 = 154$ |
| (13) $5 + 87 = 92$   | (43) $86 + 87 = 173$ | (73) $83 + 85 = 168$  | (103) $7 + 69 = 76$   |
| (14) $71 + 63 = 134$ | (44) $89 + 92 = 181$ | (74) $99 + 98 = 197$  | (104) $12 + 64 = 76$  |
| (15) $45 + 92 = 137$ | (45) $35 + 68 = 103$ | (75) $25 + 42 = 67$   | (105) $43 + 89 = 132$ |
| (16) $67 + 68 = 135$ | (46) $49 + 75 = 124$ | (76) $1 + 20 = 21$    | (106) $68 + 14 = 82$  |
| (17) $96 + 6 = 102$  | (47) $82 + 19 = 101$ | (77) $54 + 1 = 55$    | (107) $76 + 98 = 174$ |
| (18) $30 + 56 = 86$  | (48) $84 + 44 = 128$ | (78) $86 + 38 = 124$  | (108) $20 + 87 = 107$ |
| (19) $61 + 16 = 77$  | (49) $15 + 37 = 52$  | (79) $58 + 9 = 67$    | (109) $95 + 71 = 166$ |
| (20) $65 + 50 = 115$ | (50) $80 + 12 = 92$  | (80) $81 + 97 = 178$  | (110) $65 + 97 = 162$ |
| (21) $27 + 63 = 90$  | (51) $35 + 85 = 120$ | (81) $15 + 86 = 101$  | (111) $18 + 71 = 89$  |
| (22) $23 + 79 = 102$ | (52) $4 + 21 = 25$   | (82) $21 + 28 = 49$   | (112) $8 + 18 = 26$   |
| (23) $33 + 9 = 42$   | (53) $55 + 73 = 128$ | (83) $93 + 50 = 143$  | (113) $83 + 59 = 142$ |
| (24) $59 + 83 = 142$ | (54) $16 + 67 = 83$  | (84) $36 + 13 = 49$   | (114) $34 + 44 = 78$  |
| (25) $49 + 68 = 117$ | (55) $63 + 15 = 78$  | (85) $55 + 79 = 134$  | (115) $89 + 92 = 181$ |
| (26) $50 + 60 = 110$ | (56) $58 + 97 = 155$ | (86) $52 + 40 = 92$   | (116) $7 + 31 = 38$   |
| (27) $92 + 64 = 156$ | (57) $36 + 52 = 88$  | (87) $62 + 55 = 117$  | (117) $27 + 38 = 65$  |
| (28) $55 + 56 = 111$ | (58) $42 + 58 = 100$ | (88) $37 + 56 = 93$   | (118) $53 + 90 = 143$ |
| (29) $58 + 74 = 132$ | (59) $45 + 99 = 144$ | (89) $12 + 76 = 88$   | (119) $82 + 36 = 118$ |
| (30) $22 + 23 = 45$  | (60) $1 + 70 = 71$   | (90) $73 + 61 = 134$  | (120) $88 + 8 = 96$   |

|                      |                      |                       |                       |
|----------------------|----------------------|-----------------------|-----------------------|
| (1) $94 + 95 = 189$  | (31) $71 + 26 = 97$  | (61) $71 + 91 = 162$  | (91) $76 + 97 = 173$  |
| (2) $56 + 48 = 104$  | (32) $46 + 47 = 93$  | (62) $60 + 64 = 124$  | (92) $50 + 41 = 91$   |
| (3) $21 + 18 = 39$   | (33) $30 + 83 = 113$ | (63) $69 + 3 = 72$    | (93) $27 + 32 = 59$   |
| (4) $98 + 48 = 146$  | (34) $99 + 65 = 164$ | (64) $38 + 63 = 101$  | (94) $17 + 76 = 93$   |
| (5) $29 + 94 = 123$  | (35) $98 + 77 = 175$ | (65) $2 + 77 = 79$    | (95) $7 + 26 = 33$    |
| (6) $7 + 71 = 78$    | (36) $64 + 79 = 143$ | (66) $12 + 4 = 16$    | (96) $60 + 94 = 154$  |
| (7) $55 + 40 = 95$   | (37) $12 + 54 = 66$  | (67) $35 + 72 = 107$  | (97) $41 + 50 = 91$   |
| (8) $0 + 99 = 99$    | (38) $35 + 45 = 80$  | (68) $71 + 15 = 86$   | (98) $77 + 12 = 89$   |
| (9) $64 + 63 = 127$  | (39) $14 + 44 = 58$  | (69) $68 + 38 = 106$  | (99) $19 + 32 = 51$   |
| (10) $96 + 96 = 192$ | (40) $88 + 92 = 180$ | (70) $87 + 49 = 136$  | (100) $79 + 94 = 173$ |
| (11) $51 + 79 = 130$ | (41) $76 + 93 = 169$ | (71) $93 + 100 = 193$ | (101) $32 + 84 = 116$ |
| (12) $34 + 34 = 68$  | (42) $3 + 36 = 39$   | (72) $94 + 2 = 96$    | (102) $36 + 18 = 54$  |
| (13) $14 + 90 = 104$ | (43) $83 + 26 = 109$ | (73) $66 + 80 = 146$  | (103) $37 + 25 = 62$  |
| (14) $80 + 69 = 149$ | (44) $59 + 90 = 149$ | (74) $39 + 30 = 69$   | (104) $75 + 36 = 111$ |
| (15) $32 + 4 = 36$   | (45) $38 + 60 = 98$  | (75) $63 + 29 = 92$   | (105) $92 + 45 = 137$ |
| (16) $19 + 20 = 39$  | (46) $61 + 3 = 64$   | (76) $96 + 45 = 141$  | (106) $19 + 50 = 69$  |
| (17) $74 + 97 = 171$ | (47) $55 + 46 = 101$ | (77) $10 + 6 = 16$    | (107) $52 + 95 = 147$ |
| (18) $96 + 78 = 174$ | (48) $72 + 86 = 158$ | (78) $47 + 91 = 138$  | (108) $30 + 27 = 57$  |
| (19) $14 + 60 = 74$  | (49) $45 + 71 = 116$ | (79) $97 + 43 = 140$  | (109) $23 + 88 = 111$ |
| (20) $14 + 13 = 27$  | (50) $78 + 98 = 176$ | (80) $31 + 40 = 71$   | (110) $54 + 16 = 70$  |
| (21) $2 + 78 = 80$   | (51) $70 + 13 = 83$  | (81) $52 + 48 = 100$  | (111) $60 + 84 = 144$ |
| (22) $20 + 53 = 73$  | (52) $80 + 11 = 91$  | (82) $61 + 55 = 116$  | (112) $74 + 55 = 129$ |
| (23) $14 + 37 = 51$  | (53) $2 + 22 = 24$   | (83) $16 + 11 = 27$   | (113) $52 + 13 = 65$  |
| (24) $22 + 14 = 36$  | (54) $15 + 99 = 114$ | (84) $20 + 84 = 104$  | (114) $22 + 52 = 74$  |
| (25) $28 + 18 = 46$  | (55) $36 + 85 = 121$ | (85) $50 + 69 = 119$  | (115) $79 + 76 = 155$ |
| (26) $19 + 74 = 93$  | (56) $14 + 2 = 16$   | (86) $1 + 31 = 32$    | (116) $28 + 90 = 118$ |
| (27) $35 + 77 = 112$ | (57) $91 + 41 = 132$ | (87) $32 + 78 = 110$  | (117) $26 + 7 = 33$   |
| (28) $58 + 5 = 63$   | (58) $74 + 84 = 158$ | (88) $27 + 5 = 32$    | (118) $47 + 67 = 114$ |
| (29) $62 + 30 = 92$  | (59) $19 + 51 = 70$  | (89) $28 + 1 = 29$    | (119) $77 + 37 = 114$ |
| (30) $52 + 29 = 81$  | (60) $26 + 44 = 70$  | (90) $8 + 3 = 11$     | (120) $35 + 14 = 49$  |

|                      |                       |                      |                       |
|----------------------|-----------------------|----------------------|-----------------------|
| (1) $11 + 68 = 79$   | (31) $51 + 42 = 93$   | (61) $64 + 67 = 131$ | (91) $22 + 33 = 55$   |
| (2) $45 + 1 = 46$    | (32) $12 + 93 = 105$  | (62) $86 + 30 = 116$ | (92) $100 + 38 = 138$ |
| (3) $66 + 69 = 135$  | (33) $62 + 90 = 152$  | (63) $42 + 52 = 94$  | (93) $100 + 28 = 128$ |
| (4) $12 + 18 = 30$   | (34) $45 + 6 = 51$    | (64) $26 + 88 = 114$ | (94) $43 + 97 = 140$  |
| (5) $80 + 81 = 161$  | (35) $68 + 89 = 157$  | (65) $17 + 54 = 71$  | (95) $92 + 50 = 142$  |
| (6) $66 + 51 = 117$  | (36) $54 + 86 = 140$  | (66) $74 + 13 = 87$  | (96) $44 + 15 = 59$   |
| (7) $60 + 19 = 79$   | (37) $6 + 93 = 99$    | (67) $0 + 79 = 79$   | (97) $66 + 37 = 103$  |
| (8) $62 + 29 = 91$   | (38) $22 + 5 = 27$    | (68) $50 + 60 = 110$ | (98) $92 + 100 = 192$ |
| (9) $45 + 1 = 46$    | (39) $99 + 77 = 176$  | (69) $72 + 78 = 150$ | (99) $61 + 81 = 142$  |
| (10) $95 + 43 = 138$ | (40) $20 + 55 = 75$   | (70) $59 + 27 = 86$  | (100) $69 + 13 = 82$  |
| (11) $99 + 61 = 160$ | (41) $26 + 67 = 93$   | (71) $94 + 98 = 192$ | (101) $4 + 59 = 63$   |
| (12) $43 + 87 = 130$ | (42) $64 + 78 = 142$  | (72) $95 + 1 = 96$   | (102) $22 + 93 = 115$ |
| (13) $8 + 11 = 19$   | (43) $88 + 47 = 135$  | (73) $12 + 70 = 82$  | (103) $81 + 77 = 158$ |
| (14) $56 + 40 = 96$  | (44) $22 + 17 = 39$   | (74) $32 + 47 = 79$  | (104) $34 + 78 = 112$ |
| (15) $86 + 50 = 136$ | (45) $44 + 83 = 127$  | (75) $73 + 94 = 167$ | (105) $77 + 67 = 144$ |
| (16) $96 + 59 = 155$ | (46) $48 + 28 = 76$   | (76) $47 + 47 = 94$  | (106) $14 + 32 = 46$  |
| (17) $94 + 49 = 143$ | (47) $25 + 65 = 90$   | (77) $26 + 3 = 29$   | (107) $53 + 48 = 101$ |
| (18) $23 + 27 = 50$  | (48) $74 + 56 = 130$  | (78) $17 + 76 = 93$  | (108) $20 + 53 = 73$  |
| (19) $82 + 3 = 85$   | (49) $96 + 88 = 184$  | (79) $56 + 86 = 142$ | (109) $86 + 60 = 146$ |
| (20) $18 + 89 = 107$ | (50) $62 + 100 = 162$ | (80) $12 + 97 = 109$ | (110) $24 + 47 = 71$  |
| (21) $86 + 25 = 111$ | (51) $0 + 74 = 74$    | (81) $56 + 5 = 61$   | (111) $27 + 27 = 54$  |
| (22) $24 + 35 = 59$  | (52) $35 + 49 = 84$   | (82) $87 + 14 = 101$ | (112) $15 + 19 = 34$  |
| (23) $6 + 85 = 91$   | (53) $69 + 42 = 111$  | (83) $12 + 3 = 15$   | (113) $3 + 24 = 27$   |
| (24) $89 + 42 = 131$ | (54) $4 + 79 = 83$    | (84) $82 + 38 = 120$ | (114) $100 + 4 = 104$ |
| (25) $9 + 42 = 51$   | (55) $25 + 10 = 35$   | (85) $30 + 32 = 62$  | (115) $61 + 18 = 79$  |
| (26) $48 + 58 = 106$ | (56) $54 + 10 = 64$   | (86) $67 + 88 = 155$ | (116) $73 + 94 = 167$ |
| (27) $69 + 65 = 134$ | (57) $86 + 29 = 115$  | (87) $39 + 86 = 125$ | (117) $9 + 36 = 45$   |
| (28) $66 + 8 = 74$   | (58) $80 + 27 = 107$  | (88) $47 + 46 = 93$  | (118) $89 + 81 = 170$ |
| (29) $1 + 49 = 50$   | (59) $18 + 6 = 24$    | (89) $20 + 28 = 48$  | (119) $85 + 34 = 119$ |
| (30) $23 + 55 = 78$  | (60) $70 + 55 = 125$  | (90) $20 + 46 = 66$  | (120) $33 + 56 = 89$  |

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|----------------------|----------------------|-----------------------|-----------------------|
| (1) $56 + 60 = 116$  | (31) $97 + 18 = 115$ | (61) $91 + 25 = 116$  | (91) $3 + 33 = 36$    |
| (2) $38 + 13 = 51$   | (32) $46 + 38 = 84$  | (62) $93 + 36 = 129$  | (92) $22 + 8 = 30$    |
| (3) $43 + 58 = 101$  | (33) $85 + 81 = 166$ | (63) $100 + 63 = 163$ | (93) $65 + 56 = 121$  |
| (4) $67 + 37 = 104$  | (34) $4 + 79 = 83$   | (64) $76 + 37 = 113$  | (94) $19 + 81 = 100$  |
| (5) $69 + 50 = 119$  | (35) $44 + 39 = 83$  | (65) $59 + 57 = 116$  | (95) $70 + 2 = 72$    |
| (6) $71 + 28 = 99$   | (36) $66 + 72 = 138$ | (66) $85 + 56 = 141$  | (96) $5 + 64 = 69$    |
| (7) $10 + 76 = 86$   | (37) $70 + 88 = 158$ | (67) $66 + 75 = 141$  | (97) $15 + 54 = 69$   |
| (8) $100 + 42 = 142$ | (38) $55 + 24 = 79$  | (68) $66 + 83 = 149$  | (98) $4 + 61 = 65$    |
| (9) $8 + 93 = 101$   | (39) $98 + 65 = 163$ | (69) $72 + 41 = 113$  | (99) $82 + 97 = 179$  |
| (10) $55 + 41 = 96$  | (40) $95 + 82 = 177$ | (70) $59 + 85 = 144$  | (100) $10 + 13 = 23$  |
| (11) $88 + 36 = 124$ | (41) $39 + 66 = 105$ | (71) $4 + 88 = 92$    | (101) $65 + 2 = 67$   |
| (12) $66 + 3 = 69$   | (42) $32 + 16 = 48$  | (72) $99 + 36 = 135$  | (102) $4 + 83 = 87$   |
| (13) $83 + 68 = 151$ | (43) $58 + 24 = 82$  | (73) $57 + 71 = 128$  | (103) $51 + 88 = 139$ |
| (14) $66 + 17 = 83$  | (44) $30 + 6 = 36$   | (74) $56 + 100 = 156$ | (104) $43 + 32 = 75$  |
| (15) $58 + 64 = 122$ | (45) $55 + 77 = 132$ | (75) $73 + 94 = 167$  | (105) $33 + 14 = 47$  |
| (16) $9 + 81 = 90$   | (46) $98 + 26 = 124$ | (76) $79 + 70 = 149$  | (106) $90 + 90 = 180$ |
| (17) $34 + 62 = 96$  | (47) $92 + 77 = 169$ | (77) $20 + 90 = 110$  | (107) $19 + 26 = 45$  |
| (18) $15 + 88 = 103$ | (48) $13 + 73 = 86$  | (78) $83 + 14 = 97$   | (108) $52 + 54 = 106$ |
| (19) $39 + 60 = 99$  | (49) $67 + 21 = 88$  | (79) $79 + 7 = 86$    | (109) $27 + 72 = 99$  |
| (20) $52 + 68 = 120$ | (50) $98 + 95 = 193$ | (80) $26 + 86 = 112$  | (110) $54 + 2 = 56$   |
| (21) $54 + 75 = 129$ | (51) $21 + 15 = 36$  | (81) $13 + 19 = 32$   | (111) $56 + 89 = 145$ |
| (22) $14 + 57 = 71$  | (52) $14 + 36 = 50$  | (82) $73 + 61 = 134$  | (112) $80 + 4 = 84$   |
| (23) $81 + 95 = 176$ | (53) $88 + 43 = 131$ | (83) $6 + 39 = 45$    | (113) $4 + 28 = 32$   |
| (24) $52 + 40 = 92$  | (54) $46 + 3 = 49$   | (84) $36 + 45 = 81$   | (114) $74 + 71 = 145$ |
| (25) $56 + 22 = 78$  | (55) $32 + 64 = 96$  | (85) $39 + 95 = 134$  | (115) $72 + 15 = 87$  |
| (26) $33 + 66 = 99$  | (56) $77 + 76 = 153$ | (86) $27 + 57 = 84$   | (116) $3 + 19 = 22$   |
| (27) $68 + 29 = 97$  | (57) $23 + 26 = 49$  | (87) $99 + 81 = 180$  | (117) $95 + 76 = 171$ |
| (28) $53 + 19 = 72$  | (58) $24 + 7 = 31$   | (88) $12 + 95 = 107$  | (118) $43 + 83 = 126$ |
| (29) $11 + 32 = 43$  | (59) $54 + 91 = 145$ | (89) $100 + 46 = 146$ | (119) $19 + 74 = 93$  |
| (30) $19 + 14 = 33$  | (60) $32 + 37 = 69$  | (90) $70 + 51 = 121$  | (120) $30 + 83 = 113$ |

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|----------------------|----------------------|----------------------|-----------------------|
| (1) $82 + 65 = 147$  | (31) $69 + 65 = 134$ | (61) $31 + 57 = 88$  | (91) $61 + 15 = 76$   |
| (2) $22 + 65 = 87$   | (32) $13 + 68 = 81$  | (62) $48 + 27 = 75$  | (92) $58 + 39 = 97$   |
| (3) $61 + 46 = 107$  | (33) $74 + 65 = 139$ | (63) $76 + 98 = 174$ | (93) $23 + 73 = 96$   |
| (4) $94 + 45 = 139$  | (34) $58 + 81 = 139$ | (64) $77 + 59 = 136$ | (94) $47 + 65 = 112$  |
| (5) $54 + 68 = 122$  | (35) $12 + 17 = 29$  | (65) $1 + 4 = 5$     | (95) $92 + 45 = 137$  |
| (6) $9 + 100 = 109$  | (36) $43 + 99 = 142$ | (66) $54 + 83 = 137$ | (96) $20 + 21 = 41$   |
| (7) $82 + 82 = 164$  | (37) $75 + 9 = 84$   | (67) $35 + 65 = 100$ | (97) $39 + 27 = 66$   |
| (8) $23 + 67 = 90$   | (38) $17 + 13 = 30$  | (68) $53 + 54 = 107$ | (98) $79 + 74 = 153$  |
| (9) $3 + 5 = 8$      | (39) $28 + 90 = 118$ | (69) $17 + 91 = 108$ | (99) $39 + 62 = 101$  |
| (10) $96 + 61 = 157$ | (40) $76 + 88 = 164$ | (70) $51 + 25 = 76$  | (100) $5 + 31 = 36$   |
| (11) $28 + 41 = 69$  | (41) $11 + 21 = 32$  | (71) $100 + 4 = 104$ | (101) $27 + 62 = 89$  |
| (12) $80 + 38 = 118$ | (42) $63 + 19 = 82$  | (72) $39 + 30 = 69$  | (102) $89 + 45 = 134$ |
| (13) $83 + 49 = 132$ | (43) $32 + 34 = 66$  | (73) $94 + 9 = 103$  | (103) $59 + 76 = 135$ |
| (14) $49 + 61 = 110$ | (44) $30 + 92 = 122$ | (74) $33 + 65 = 98$  | (104) $87 + 7 = 94$   |
| (15) $68 + 21 = 89$  | (45) $39 + 8 = 47$   | (75) $53 + 28 = 81$  | (105) $19 + 54 = 73$  |
| (16) $34 + 80 = 114$ | (46) $72 + 60 = 132$ | (76) $38 + 59 = 97$  | (106) $23 + 37 = 60$  |
| (17) $75 + 69 = 144$ | (47) $88 + 1 = 89$   | (77) $5 + 61 = 66$   | (107) $91 + 35 = 126$ |
| (18) $87 + 30 = 117$ | (48) $52 + 90 = 142$ | (78) $24 + 61 = 85$  | (108) $28 + 43 = 71$  |
| (19) $2 + 25 = 27$   | (49) $58 + 25 = 83$  | (79) $3 + 75 = 78$   | (109) $12 + 89 = 101$ |
| (20) $91 + 53 = 144$ | (50) $89 + 88 = 177$ | (80) $81 + 50 = 131$ | (110) $12 + 47 = 59$  |
| (21) $67 + 11 = 78$  | (51) $15 + 46 = 61$  | (81) $9 + 89 = 98$   | (111) $78 + 53 = 131$ |
| (22) $46 + 45 = 91$  | (52) $35 + 92 = 127$ | (82) $66 + 93 = 159$ | (112) $20 + 64 = 84$  |
| (23) $92 + 12 = 104$ | (53) $57 + 55 = 112$ | (83) $9 + 26 = 35$   | (113) $58 + 17 = 75$  |
| (24) $63 + 64 = 127$ | (54) $9 + 60 = 69$   | (84) $33 + 70 = 103$ | (114) $30 + 57 = 87$  |
| (25) $98 + 85 = 183$ | (55) $10 + 14 = 24$  | (85) $13 + 64 = 77$  | (115) $12 + 88 = 100$ |
| (26) $52 + 30 = 82$  | (56) $0 + 48 = 48$   | (86) $37 + 79 = 116$ | (116) $25 + 72 = 97$  |
| (27) $57 + 28 = 85$  | (57) $13 + 73 = 86$  | (87) $87 + 29 = 116$ | (117) $65 + 95 = 160$ |
| (28) $76 + 87 = 163$ | (58) $57 + 38 = 95$  | (88) $71 + 72 = 143$ | (118) $79 + 13 = 92$  |
| (29) $66 + 13 = 79$  | (59) $26 + 37 = 63$  | (89) $5 + 60 = 65$   | (119) $16 + 17 = 33$  |
| (30) $37 + 50 = 87$  | (60) $81 + 66 = 147$ | (90) $34 + 42 = 76$  | (120) $29 + 63 = 92$  |



|                       |                       |                      |                       |
|-----------------------|-----------------------|----------------------|-----------------------|
| (1) $69 + 55 = 124$   | (31) $71 + 48 = 119$  | (61) $14 + 30 = 44$  | (91) $9 + 74 = 83$    |
| (2) $75 + 63 = 138$   | (32) $13 + 88 = 101$  | (62) $61 + 95 = 156$ | (92) $83 + 55 = 138$  |
| (3) $39 + 86 = 125$   | (33) $48 + 57 = 105$  | (63) $32 + 72 = 104$ | (93) $52 + 41 = 93$   |
| (4) $54 + 40 = 94$    | (34) $36 + 56 = 92$   | (64) $23 + 26 = 49$  | (94) $98 + 27 = 125$  |
| (5) $57 + 12 = 69$    | (35) $100 + 94 = 194$ | (65) $20 + 40 = 60$  | (95) $68 + 93 = 161$  |
| (6) $9 + 34 = 43$     | (36) $67 + 97 = 164$  | (66) $10 + 57 = 67$  | (96) $29 + 24 = 53$   |
| (7) $63 + 29 = 92$    | (37) $23 + 68 = 91$   | (67) $78 + 96 = 174$ | (97) $36 + 89 = 125$  |
| (8) $23 + 48 = 71$    | (38) $87 + 69 = 156$  | (68) $55 + 83 = 138$ | (98) $11 + 43 = 54$   |
| (9) $49 + 98 = 147$   | (39) $80 + 81 = 161$  | (69) $5 + 80 = 85$   | (99) $56 + 51 = 107$  |
| (10) $90 + 73 = 163$  | (40) $10 + 86 = 96$   | (70) $92 + 13 = 105$ | (100) $13 + 89 = 102$ |
| (11) $63 + 31 = 94$   | (41) $12 + 16 = 28$   | (71) $37 + 35 = 72$  | (101) $66 + 19 = 85$  |
| (12) $50 + 46 = 96$   | (42) $85 + 49 = 134$  | (72) $26 + 64 = 90$  | (102) $94 + 34 = 128$ |
| (13) $32 + 81 = 113$  | (43) $6 + 94 = 100$   | (73) $70 + 63 = 133$ | (103) $84 + 96 = 180$ |
| (14) $70 + 78 = 148$  | (44) $48 + 25 = 73$   | (74) $29 + 28 = 57$  | (104) $71 + 70 = 141$ |
| (15) $68 + 85 = 153$  | (45) $25 + 33 = 58$   | (75) $62 + 33 = 95$  | (105) $81 + 54 = 135$ |
| (16) $31 + 86 = 117$  | (46) $83 + 89 = 172$  | (76) $10 + 16 = 26$  | (106) $92 + 77 = 169$ |
| (17) $75 + 23 = 98$   | (47) $70 + 66 = 136$  | (77) $87 + 54 = 141$ | (107) $46 + 26 = 72$  |
| (18) $100 + 39 = 139$ | (48) $16 + 16 = 32$   | (78) $9 + 41 = 50$   | (108) $86 + 20 = 106$ |
| (19) $3 + 6 = 9$      | (49) $29 + 68 = 97$   | (79) $66 + 45 = 111$ | (109) $52 + 42 = 94$  |
| (20) $70 + 8 = 78$    | (50) $33 + 42 = 75$   | (80) $70 + 9 = 79$   | (110) $73 + 2 = 75$   |
| (21) $37 + 14 = 51$   | (51) $87 + 95 = 182$  | (81) $84 + 91 = 175$ | (111) $90 + 11 = 101$ |
| (22) $36 + 43 = 79$   | (52) $1 + 96 = 97$    | (82) $77 + 8 = 85$   | (112) $48 + 78 = 126$ |
| (23) $92 + 35 = 127$  | (53) $41 + 1 = 42$    | (83) $26 + 75 = 101$ | (113) $87 + 58 = 145$ |
| (24) $29 + 50 = 79$   | (54) $78 + 41 = 119$  | (84) $62 + 25 = 87$  | (114) $44 + 52 = 96$  |
| (25) $45 + 3 = 48$    | (55) $50 + 8 = 58$    | (85) $67 + 50 = 117$ | (115) $59 + 15 = 74$  |
| (26) $64 + 2 = 66$    | (56) $2 + 84 = 86$    | (86) $40 + 22 = 62$  | (116) $20 + 67 = 87$  |
| (27) $94 + 62 = 156$  | (57) $60 + 100 = 160$ | (87) $23 + 17 = 40$  | (117) $54 + 40 = 94$  |
| (28) $85 + 29 = 114$  | (58) $16 + 5 = 21$    | (88) $42 + 1 = 43$   | (118) $3 + 95 = 98$   |
| (29) $91 + 79 = 170$  | (59) $47 + 0 = 47$    | (89) $30 + 97 = 127$ | (119) $59 + 97 = 156$ |
| (30) $69 + 88 = 157$  | (60) $59 + 91 = 150$  | (90) $52 + 12 = 64$  | (120) $23 + 36 = 59$  |

|                      |                      |                      |                       |
|----------------------|----------------------|----------------------|-----------------------|
| (1) $83 + 62 = 145$  | (31) $4 + 21 = 25$   | (61) $36 + 52 = 88$  | (91) $64 + 90 = 154$  |
| (2) $86 + 11 = 97$   | (32) $51 + 23 = 74$  | (62) $31 + 7 = 38$   | (92) $36 + 14 = 50$   |
| (3) $90 + 80 = 170$  | (33) $61 + 42 = 103$ | (63) $7 + 36 = 43$   | (93) $74 + 22 = 96$   |
| (4) $16 + 55 = 71$   | (34) $45 + 94 = 139$ | (64) $91 + 57 = 148$ | (94) $38 + 62 = 100$  |
| (5) $76 + 99 = 175$  | (35) $48 + 46 = 94$  | (65) $61 + 93 = 154$ | (95) $50 + 81 = 131$  |
| (6) $36 + 93 = 129$  | (36) $96 + 80 = 176$ | (66) $14 + 42 = 56$  | (96) $9 + 96 = 105$   |
| (7) $9 + 88 = 97$    | (37) $63 + 72 = 135$ | (67) $28 + 3 = 31$   | (97) $83 + 21 = 104$  |
| (8) $68 + 16 = 84$   | (38) $4 + 64 = 68$   | (68) $17 + 45 = 62$  | (98) $69 + 17 = 86$   |
| (9) $93 + 35 = 128$  | (39) $98 + 37 = 135$ | (69) $44 + 61 = 105$ | (99) $90 + 33 = 123$  |
| (10) $93 + 37 = 130$ | (40) $70 + 97 = 167$ | (70) $39 + 20 = 59$  | (100) $96 + 65 = 161$ |
| (11) $87 + 65 = 152$ | (41) $52 + 85 = 137$ | (71) $59 + 59 = 118$ | (101) $48 + 30 = 78$  |
| (12) $38 + 23 = 61$  | (42) $1 + 98 = 99$   | (72) $3 + 88 = 91$   | (102) $40 + 66 = 106$ |
| (13) $72 + 56 = 128$ | (43) $65 + 45 = 110$ | (73) $32 + 72 = 104$ | (103) $81 + 77 = 158$ |
| (14) $40 + 30 = 70$  | (44) $79 + 8 = 87$   | (74) $60 + 54 = 114$ | (104) $60 + 97 = 157$ |
| (15) $94 + 11 = 105$ | (45) $49 + 82 = 131$ | (75) $90 + 8 = 98$   | (105) $19 + 96 = 115$ |
| (16) $55 + 80 = 135$ | (46) $65 + 38 = 103$ | (76) $15 + 45 = 60$  | (106) $25 + 51 = 76$  |
| (17) $2 + 24 = 26$   | (47) $59 + 61 = 120$ | (77) $21 + 28 = 49$  | (107) $25 + 38 = 63$  |
| (18) $47 + 18 = 65$  | (48) $51 + 66 = 117$ | (78) $8 + 60 = 68$   | (108) $78 + 63 = 141$ |
| (19) $69 + 35 = 104$ | (49) $85 + 11 = 96$  | (79) $44 + 70 = 114$ | (109) $76 + 72 = 148$ |
| (20) $35 + 99 = 134$ | (50) $22 + 6 = 28$   | (80) $53 + 6 = 59$   | (110) $75 + 64 = 139$ |
| (21) $26 + 82 = 108$ | (51) $40 + 77 = 117$ | (81) $68 + 4 = 72$   | (111) $85 + 69 = 154$ |
| (22) $57 + 30 = 87$  | (52) $11 + 25 = 36$  | (82) $29 + 22 = 51$  | (112) $95 + 4 = 99$   |
| (23) $10 + 55 = 65$  | (53) $39 + 49 = 88$  | (83) $24 + 35 = 59$  | (113) $78 + 55 = 133$ |
| (24) $40 + 89 = 129$ | (54) $11 + 76 = 87$  | (84) $47 + 31 = 78$  | (114) $40 + 63 = 103$ |
| (25) $79 + 72 = 151$ | (55) $35 + 66 = 101$ | (85) $56 + 88 = 144$ | (115) $91 + 82 = 173$ |
| (26) $76 + 48 = 124$ | (56) $86 + 92 = 178$ | (86) $45 + 44 = 89$  | (116) $83 + 37 = 120$ |
| (27) $88 + 47 = 135$ | (57) $10 + 70 = 80$  | (87) $56 + 84 = 140$ | (117) $41 + 33 = 74$  |
| (28) $72 + 75 = 147$ | (58) $78 + 92 = 170$ | (88) $87 + 80 = 167$ | (118) $1 + 66 = 67$   |
| (29) $38 + 21 = 59$  | (59) $51 + 23 = 74$  | (89) $98 + 73 = 171$ | (119) $29 + 19 = 48$  |
| (30) $48 + 52 = 100$ | (60) $62 + 28 = 90$  | (90) $12 + 56 = 68$  | (120) $83 + 41 = 124$ |

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|----------------------|----------------------|----------------------|-----------------------|
| (1) $26 + 27 = 53$   | (31) $18 + 99 = 117$ | (61) $17 + 58 = 75$  | (91) $41 + 57 = 98$   |
| (2) $83 + 48 = 131$  | (32) $30 + 67 = 97$  | (62) $97 + 55 = 152$ | (92) $32 + 63 = 95$   |
| (3) $1 + 55 = 56$    | (33) $15 + 51 = 66$  | (63) $67 + 43 = 110$ | (93) $16 + 51 = 67$   |
| (4) $69 + 71 = 140$  | (34) $25 + 42 = 67$  | (64) $50 + 99 = 149$ | (94) $31 + 33 = 64$   |
| (5) $1 + 72 = 73$    | (35) $75 + 98 = 173$ | (65) $58 + 69 = 127$ | (95) $2 + 100 = 102$  |
| (6) $94 + 70 = 164$  | (36) $81 + 4 = 85$   | (66) $90 + 41 = 131$ | (96) $47 + 51 = 98$   |
| (7) $43 + 64 = 107$  | (37) $96 + 73 = 169$ | (67) $90 + 20 = 110$ | (97) $49 + 61 = 110$  |
| (8) $82 + 85 = 167$  | (38) $27 + 35 = 62$  | (68) $7 + 46 = 53$   | (98) $48 + 7 = 55$    |
| (9) $47 + 73 = 120$  | (39) $49 + 94 = 143$ | (69) $80 + 30 = 110$ | (99) $75 + 59 = 134$  |
| (10) $55 + 43 = 98$  | (40) $36 + 75 = 111$ | (70) $33 + 95 = 128$ | (100) $27 + 95 = 122$ |
| (11) $4 + 53 = 57$   | (41) $15 + 22 = 37$  | (71) $95 + 73 = 168$ | (101) $39 + 69 = 108$ |
| (12) $27 + 28 = 55$  | (42) $76 + 44 = 120$ | (72) $11 + 29 = 40$  | (102) $52 + 70 = 122$ |
| (13) $92 + 6 = 98$   | (43) $73 + 3 = 76$   | (73) $54 + 22 = 76$  | (103) $23 + 43 = 66$  |
| (14) $25 + 3 = 28$   | (44) $22 + 35 = 57$  | (74) $67 + 79 = 146$ | (104) $31 + 41 = 72$  |
| (15) $29 + 43 = 72$  | (45) $21 + 66 = 87$  | (75) $64 + 83 = 147$ | (105) $99 + 11 = 110$ |
| (16) $99 + 75 = 174$ | (46) $57 + 91 = 148$ | (76) $49 + 86 = 135$ | (106) $44 + 53 = 97$  |
| (17) $9 + 17 = 26$   | (47) $88 + 10 = 98$  | (77) $22 + 26 = 48$  | (107) $30 + 4 = 34$   |
| (18) $18 + 94 = 112$ | (48) $42 + 43 = 85$  | (78) $14 + 56 = 70$  | (108) $48 + 44 = 92$  |
| (19) $70 + 51 = 121$ | (49) $31 + 78 = 109$ | (79) $71 + 53 = 124$ | (109) $3 + 24 = 27$   |
| (20) $72 + 50 = 122$ | (50) $27 + 2 = 29$   | (80) $62 + 3 = 65$   | (110) $48 + 73 = 121$ |
| (21) $8 + 27 = 35$   | (51) $51 + 73 = 124$ | (81) $26 + 32 = 58$  | (111) $75 + 37 = 112$ |
| (22) $9 + 35 = 44$   | (52) $90 + 86 = 176$ | (82) $86 + 57 = 143$ | (112) $28 + 71 = 99$  |
| (23) $49 + 56 = 105$ | (53) $35 + 98 = 133$ | (83) $78 + 58 = 136$ | (113) $14 + 16 = 30$  |
| (24) $69 + 75 = 144$ | (54) $9 + 79 = 88$   | (84) $25 + 52 = 77$  | (114) $28 + 13 = 41$  |
| (25) $72 + 5 = 77$   | (55) $87 + 78 = 165$ | (85) $22 + 60 = 82$  | (115) $75 + 89 = 164$ |
| (26) $43 + 86 = 129$ | (56) $55 + 83 = 138$ | (86) $6 + 20 = 26$   | (116) $2 + 74 = 76$   |
| (27) $52 + 55 = 107$ | (57) $24 + 54 = 78$  | (87) $15 + 37 = 52$  | (117) $98 + 99 = 197$ |
| (28) $19 + 91 = 110$ | (58) $93 + 93 = 186$ | (88) $2 + 92 = 94$   | (118) $96 + 62 = 158$ |
| (29) $47 + 55 = 102$ | (59) $19 + 9 = 28$   | (89) $16 + 37 = 53$  | (119) $43 + 73 = 116$ |
| (30) $54 + 38 = 92$  | (60) $5 + 96 = 101$  | (90) $8 + 69 = 77$   | (120) $38 + 4 = 42$   |

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|----------------------|----------------------|----------------------|------------------------|
| (1) $58 + 6 = 64$    | (31) $36 + 45 = 81$  | (61) $43 + 17 = 60$  | (91) $86 + 54 = 140$   |
| (2) $80 + 84 = 164$  | (32) $25 + 14 = 39$  | (62) $94 + 30 = 124$ | (92) $11 + 50 = 61$    |
| (3) $46 + 16 = 62$   | (33) $96 + 94 = 190$ | (63) $3 + 96 = 99$   | (93) $55 + 22 = 77$    |
| (4) $15 + 96 = 111$  | (34) $52 + 56 = 108$ | (64) $62 + 12 = 74$  | (94) $2 + 14 = 16$     |
| (5) $2 + 12 = 14$    | (35) $62 + 97 = 159$ | (65) $84 + 50 = 134$ | (95) $74 + 29 = 103$   |
| (6) $39 + 82 = 121$  | (36) $38 + 94 = 132$ | (66) $85 + 63 = 148$ | (96) $47 + 32 = 79$    |
| (7) $44 + 84 = 128$  | (37) $44 + 59 = 103$ | (67) $69 + 58 = 127$ | (97) $31 + 92 = 123$   |
| (8) $70 + 24 = 94$   | (38) $82 + 76 = 158$ | (68) $18 + 11 = 29$  | (98) $44 + 14 = 58$    |
| (9) $81 + 54 = 135$  | (39) $6 + 57 = 63$   | (69) $34 + 98 = 132$ | (99) $16 + 60 = 76$    |
| (10) $69 + 3 = 72$   | (40) $87 + 70 = 157$ | (70) $32 + 70 = 102$ | (100) $67 + 42 = 109$  |
| (11) $21 + 95 = 116$ | (41) $66 + 56 = 122$ | (71) $39 + 69 = 108$ | (101) $39 + 5 = 44$    |
| (12) $7 + 61 = 68$   | (42) $69 + 97 = 166$ | (72) $16 + 38 = 54$  | (102) $73 + 38 = 111$  |
| (13) $89 + 75 = 164$ | (43) $61 + 80 = 141$ | (73) $27 + 32 = 59$  | (103) $30 + 17 = 47$   |
| (14) $49 + 99 = 148$ | (44) $89 + 30 = 119$ | (74) $32 + 5 = 37$   | (104) $47 + 74 = 121$  |
| (15) $54 + 19 = 73$  | (45) $20 + 13 = 33$  | (75) $84 + 35 = 119$ | (105) $57 + 100 = 157$ |
| (16) $55 + 75 = 130$ | (46) $99 + 88 = 187$ | (76) $72 + 95 = 167$ | (106) $86 + 35 = 121$  |
| (17) $23 + 34 = 57$  | (47) $37 + 51 = 88$  | (77) $91 + 83 = 174$ | (107) $86 + 78 = 164$  |
| (18) $48 + 97 = 145$ | (48) $35 + 50 = 85$  | (78) $49 + 27 = 76$  | (108) $96 + 53 = 149$  |
| (19) $87 + 2 = 89$   | (49) $58 + 80 = 138$ | (79) $55 + 98 = 153$ | (109) $56 + 47 = 103$  |
| (20) $85 + 29 = 114$ | (50) $78 + 66 = 144$ | (80) $8 + 3 = 11$    | (110) $74 + 18 = 92$   |
| (21) $53 + 42 = 95$  | (51) $75 + 41 = 116$ | (81) $23 + 10 = 33$  | (111) $52 + 5 = 57$    |
| (22) $20 + 76 = 96$  | (52) $3 + 3 = 6$     | (82) $67 + 78 = 145$ | (112) $49 + 74 = 123$  |
| (23) $48 + 57 = 105$ | (53) $32 + 1 = 33$   | (83) $71 + 83 = 154$ | (113) $68 + 58 = 126$  |
| (24) $3 + 20 = 23$   | (54) $58 + 64 = 122$ | (84) $27 + 62 = 89$  | (114) $58 + 55 = 113$  |
| (25) $29 + 16 = 45$  | (55) $37 + 71 = 108$ | (85) $83 + 57 = 140$ | (115) $85 + 55 = 140$  |
| (26) $84 + 91 = 175$ | (56) $18 + 50 = 68$  | (86) $47 + 1 = 48$   | (116) $65 + 4 = 69$    |
| (27) $65 + 59 = 124$ | (57) $52 + 54 = 106$ | (87) $83 + 38 = 121$ | (117) $21 + 87 = 108$  |
| (28) $55 + 90 = 145$ | (58) $41 + 72 = 113$ | (88) $53 + 38 = 91$  | (118) $73 + 23 = 96$   |
| (29) $85 + 12 = 97$  | (59) $76 + 66 = 142$ | (89) $79 + 40 = 119$ | (119) $77 + 96 = 173$  |
| (30) $69 + 58 = 127$ | (60) $14 + 66 = 80$  | (90) $6 + 93 = 99$   | (120) $27 + 8 = 35$    |

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|-----------------------|-----------------------|----------------------|-----------------------|
| (1) $26 + 97 = 123$   | (31) $16 + 46 = 62$   | (61) $98 + 92 = 190$ | (91) $66 + 11 = 77$   |
| (2) $76 + 7 = 83$     | (32) $16 + 83 = 99$   | (62) $99 + 16 = 115$ | (92) $95 + 66 = 161$  |
| (3) $13 + 49 = 62$    | (33) $79 + 14 = 93$   | (63) $37 + 53 = 90$  | (93) $81 + 21 = 102$  |
| (4) $76 + 69 = 145$   | (34) $57 + 19 = 76$   | (64) $93 + 16 = 109$ | (94) $11 + 69 = 80$   |
| (5) $98 + 32 = 130$   | (35) $74 + 55 = 129$  | (65) $71 + 45 = 116$ | (95) $40 + 75 = 115$  |
| (6) $85 + 100 = 185$  | (36) $52 + 4 = 56$    | (66) $6 + 83 = 89$   | (96) $65 + 18 = 83$   |
| (7) $17 + 45 = 62$    | (37) $67 + 20 = 87$   | (67) $92 + 23 = 115$ | (97) $83 + 74 = 157$  |
| (8) $43 + 46 = 89$    | (38) $43 + 84 = 127$  | (68) $37 + 98 = 135$ | (98) $87 + 50 = 137$  |
| (9) $63 + 43 = 106$   | (39) $100 + 16 = 116$ | (69) $79 + 45 = 124$ | (99) $49 + 56 = 105$  |
| (10) $72 + 28 = 100$  | (40) $71 + 6 = 77$    | (70) $7 + 18 = 25$   | (100) $9 + 64 = 73$   |
| (11) $85 + 82 = 167$  | (41) $25 + 69 = 94$   | (71) $37 + 84 = 121$ | (101) $31 + 6 = 37$   |
| (12) $35 + 10 = 45$   | (42) $80 + 81 = 161$  | (72) $65 + 27 = 92$  | (102) $32 + 13 = 45$  |
| (13) $41 + 94 = 135$  | (43) $18 + 89 = 107$  | (73) $25 + 52 = 77$  | (103) $9 + 69 = 78$   |
| (14) $19 + 74 = 93$   | (44) $33 + 89 = 122$  | (74) $2 + 82 = 84$   | (104) $46 + 14 = 60$  |
| (15) $44 + 9 = 53$    | (45) $80 + 69 = 149$  | (75) $7 + 7 = 14$    | (105) $76 + 20 = 96$  |
| (16) $93 + 53 = 146$  | (46) $33 + 98 = 131$  | (76) $17 + 27 = 44$  | (106) $76 + 78 = 154$ |
| (17) $12 + 92 = 104$  | (47) $7 + 62 = 69$    | (77) $92 + 90 = 182$ | (107) $50 + 31 = 81$  |
| (18) $16 + 45 = 61$   | (48) $48 + 42 = 90$   | (78) $92 + 59 = 151$ | (108) $18 + 23 = 41$  |
| (19) $100 + 21 = 121$ | (49) $28 + 54 = 82$   | (79) $55 + 9 = 64$   | (109) $28 + 19 = 47$  |
| (20) $24 + 88 = 112$  | (50) $44 + 32 = 76$   | (80) $19 + 20 = 39$  | (110) $68 + 59 = 127$ |
| (21) $56 + 87 = 143$  | (51) $28 + 60 = 88$   | (81) $9 + 30 = 39$   | (111) $54 + 89 = 143$ |
| (22) $5 + 67 = 72$    | (52) $18 + 62 = 80$   | (82) $67 + 63 = 130$ | (112) $43 + 31 = 74$  |
| (23) $85 + 33 = 118$  | (53) $73 + 48 = 121$  | (83) $32 + 33 = 65$  | (113) $80 + 14 = 94$  |
| (24) $6 + 81 = 87$    | (54) $2 + 44 = 46$    | (84) $65 + 70 = 135$ | (114) $52 + 57 = 109$ |
| (25) $17 + 55 = 72$   | (55) $98 + 18 = 116$  | (85) $16 + 39 = 55$  | (115) $65 + 70 = 135$ |
| (26) $64 + 24 = 88$   | (56) $70 + 96 = 166$  | (86) $74 + 47 = 121$ | (116) $69 + 28 = 97$  |
| (27) $85 + 40 = 125$  | (57) $3 + 84 = 87$    | (87) $63 + 43 = 106$ | (117) $74 + 30 = 104$ |
| (28) $6 + 28 = 34$    | (58) $28 + 59 = 87$   | (88) $31 + 42 = 73$  | (118) $71 + 97 = 168$ |
| (29) $31 + 16 = 47$   | (59) $24 + 20 = 44$   | (89) $37 + 2 = 39$   | (119) $15 + 69 = 84$  |
| (30) $74 + 17 = 91$   | (60) $67 + 65 = 132$  | (90) $8 + 20 = 28$   | (120) $73 + 84 = 157$ |

|                      |                      |                      |                       |
|----------------------|----------------------|----------------------|-----------------------|
| (1) $17 + 33 = 50$   | (31) $90 + 10 = 100$ | (61) $1 + 81 = 82$   | (91) $34 + 16 = 50$   |
| (2) $35 + 61 = 96$   | (32) $5 + 34 = 39$   | (62) $94 + 99 = 193$ | (92) $32 + 72 = 104$  |
| (3) $60 + 27 = 87$   | (33) $39 + 49 = 88$  | (63) $59 + 7 = 66$   | (93) $14 + 53 = 67$   |
| (4) $27 + 68 = 95$   | (34) $10 + 47 = 57$  | (64) $0 + 23 = 23$   | (94) $84 + 3 = 87$    |
| (5) $8 + 20 = 28$    | (35) $7 + 41 = 48$   | (65) $60 + 60 = 120$ | (95) $63 + 56 = 119$  |
| (6) $89 + 77 = 166$  | (36) $65 + 9 = 74$   | (66) $96 + 18 = 114$ | (96) $96 + 51 = 147$  |
| (7) $21 + 94 = 115$  | (37) $0 + 69 = 69$   | (67) $0 + 86 = 86$   | (97) $55 + 79 = 134$  |
| (8) $83 + 52 = 135$  | (38) $39 + 46 = 85$  | (68) $84 + 86 = 170$ | (98) $96 + 99 = 195$  |
| (9) $37 + 99 = 136$  | (39) $49 + 47 = 96$  | (69) $58 + 62 = 120$ | (99) $19 + 77 = 96$   |
| (10) $19 + 59 = 78$  | (40) $42 + 80 = 122$ | (70) $23 + 17 = 40$  | (100) $24 + 6 = 30$   |
| (11) $41 + 32 = 73$  | (41) $83 + 52 = 135$ | (71) $86 + 17 = 103$ | (101) $64 + 66 = 130$ |
| (12) $89 + 42 = 131$ | (42) $42 + 14 = 56$  | (72) $92 + 57 = 149$ | (102) $99 + 55 = 154$ |
| (13) $8 + 94 = 102$  | (43) $24 + 70 = 94$  | (73) $13 + 77 = 90$  | (103) $18 + 37 = 55$  |
| (14) $76 + 29 = 105$ | (44) $98 + 43 = 141$ | (74) $58 + 25 = 83$  | (104) $67 + 55 = 122$ |
| (15) $41 + 71 = 112$ | (45) $82 + 47 = 129$ | (75) $62 + 38 = 100$ | (105) $20 + 90 = 110$ |
| (16) $40 + 10 = 50$  | (46) $2 + 39 = 41$   | (76) $70 + 68 = 138$ | (106) $61 + 55 = 116$ |
| (17) $1 + 89 = 90$   | (47) $36 + 87 = 123$ | (77) $43 + 89 = 132$ | (107) $44 + 93 = 137$ |
| (18) $62 + 93 = 155$ | (48) $41 + 43 = 84$  | (78) $98 + 61 = 159$ | (108) $39 + 21 = 60$  |
| (19) $22 + 4 = 26$   | (49) $13 + 15 = 28$  | (79) $1 + 19 = 20$   | (109) $39 + 6 = 45$   |
| (20) $98 + 90 = 188$ | (50) $66 + 0 = 66$   | (80) $90 + 24 = 114$ | (110) $77 + 88 = 165$ |
| (21) $56 + 39 = 95$  | (51) $54 + 3 = 57$   | (81) $97 + 48 = 145$ | (111) $11 + 58 = 69$  |
| (22) $55 + 88 = 143$ | (52) $57 + 51 = 108$ | (82) $56 + 18 = 74$  | (112) $28 + 60 = 88$  |
| (23) $43 + 11 = 54$  | (53) $65 + 15 = 80$  | (83) $0 + 23 = 23$   | (113) $79 + 32 = 111$ |
| (24) $7 + 6 = 13$    | (54) $72 + 18 = 90$  | (84) $81 + 95 = 176$ | (114) $86 + 8 = 94$   |
| (25) $14 + 54 = 68$  | (55) $31 + 15 = 46$  | (85) $65 + 83 = 148$ | (115) $92 + 37 = 129$ |
| (26) $65 + 90 = 155$ | (56) $33 + 3 = 36$   | (86) $36 + 46 = 82$  | (116) $44 + 68 = 112$ |
| (27) $25 + 85 = 110$ | (57) $66 + 91 = 157$ | (87) $98 + 10 = 108$ | (117) $56 + 47 = 103$ |
| (28) $99 + 52 = 151$ | (58) $45 + 38 = 83$  | (88) $34 + 58 = 92$  | (118) $10 + 95 = 105$ |
| (29) $88 + 44 = 132$ | (59) $29 + 24 = 53$  | (89) $39 + 1 = 40$   | (119) $7 + 77 = 84$   |
| (30) $64 + 51 = 115$ | (60) $7 + 76 = 83$   | (90) $8 + 7 = 15$    | (120) $48 + 10 = 58$  |

### 1.1.2 減法

自然数の差を求める問題。

|              |               |              |               |
|--------------|---------------|--------------|---------------|
| (1) 48 - 16  | (31) 40 - 24  | (61) 79 - 54 | (91) 38 - 32  |
| (2) 28 - 3   | (32) 79 - 66  | (62) 60 - 46 | (92) 72 - 58  |
| (3) 49 - 3   | (33) 74 - 26  | (63) 17 - 15 | (93) 94 - 47  |
| (4) 57 - 54  | (34) 25 - 4   | (64) 98 - 18 | (94) 13 - 0   |
| (5) 93 - 6   | (35) 86 - 51  | (65) 58 - 25 | (95) 75 - 66  |
| (6) 60 - 36  | (36) 76 - 65  | (66) 67 - 15 | (96) 84 - 27  |
| (7) 82 - 43  | (37) 97 - 23  | (67) 65 - 22 | (97) 85 - 31  |
| (8) 38 - 8   | (38) 56 - 52  | (68) 16 - 3  | (98) 96 - 18  |
| (9) 37 - 35  | (39) 9 - 3    | (69) 85 - 74 | (99) 66 - 48  |
| (10) 83 - 52 | (40) 80 - 31  | (70) 78 - 29 | (100) 80 - 67 |
| (11) 37 - 33 | (41) 54 - 53  | (71) 98 - 57 | (101) 48 - 36 |
| (12) 77 - 5  | (42) 96 - 33  | (72) 76 - 33 | (102) 16 - 13 |
| (13) 95 - 33 | (43) 91 - 47  | (73) 82 - 10 | (103) 79 - 66 |
| (14) 97 - 83 | (44) 64 - 19  | (74) 83 - 7  | (104) 67 - 50 |
| (15) 89 - 42 | (45) 39 - 12  | (75) 70 - 21 | (105) 90 - 70 |
| (16) 49 - 6  | (46) 47 - 29  | (76) 74 - 2  | (106) 62 - 46 |
| (17) 66 - 2  | (47) 70 - 4   | (77) 29 - 22 | (107) 80 - 71 |
| (18) 98 - 66 | (48) 4 - 0    | (78) 55 - 37 | (108) 76 - 15 |
| (19) 70 - 31 | (49) 61 - 43  | (79) 74 - 10 | (109) 34 - 25 |
| (20) 72 - 1  | (50) 94 - 24  | (80) 25 - 13 | (110) 69 - 46 |
| (21) 78 - 55 | (51) 66 - 61  | (81) 58 - 50 | (111) 72 - 22 |
| (22) 79 - 48 | (52) 41 - 36  | (82) 57 - 23 | (112) 46 - 1  |
| (23) 42 - 37 | (53) 78 - 13  | (83) 67 - 39 | (113) 56 - 42 |
| (24) 74 - 2  | (54) 99 - 98  | (84) 87 - 43 | (114) 90 - 81 |
| (25) 92 - 39 | (55) 44 - 7   | (85) 77 - 34 | (115) 60 - 55 |
| (26) 88 - 67 | (56) 100 - 73 | (86) 68 - 26 | (116) 76 - 4  |
| (27) 78 - 4  | (57) 43 - 20  | (87) 29 - 11 | (117) 40 - 2  |
| (28) 28 - 20 | (58) 76 - 28  | (88) 98 - 72 | (118) 65 - 30 |
| (29) 95 - 91 | (59) 98 - 24  | (89) 60 - 8  | (119) 80 - 56 |
| (30) 32 - 25 | (60) 66 - 40  | (90) 84 - 58 | (120) 94 - 71 |



|              |              |              |                |
|--------------|--------------|--------------|----------------|
| (1) 96 - 85  | (31) 76 - 47 | (61) 31 - 12 | (91) 91 - 56   |
| (2) 66 - 26  | (32) 30 - 7  | (62) 80 - 40 | (92) 100 - 55  |
| (3) 91 - 60  | (33) 62 - 14 | (63) 52 - 47 | (93) 24 - 8    |
| (4) 93 - 13  | (34) 68 - 3  | (64) 72 - 3  | (94) 95 - 71   |
| (5) 26 - 1   | (35) 78 - 11 | (65) 28 - 3  | (95) 24 - 13   |
| (6) 76 - 36  | (36) 90 - 66 | (66) 54 - 12 | (96) 33 - 17   |
| (7) 85 - 53  | (37) 98 - 90 | (67) 75 - 22 | (97) 52 - 35   |
| (8) 63 - 57  | (38) 60 - 33 | (68) 89 - 27 | (98) 98 - 46   |
| (9) 87 - 54  | (39) 96 - 45 | (69) 90 - 9  | (99) 76 - 47   |
| (10) 26 - 23 | (40) 43 - 31 | (70) 89 - 73 | (100) 94 - 62  |
| (11) 33 - 4  | (41) 36 - 5  | (71) 75 - 66 | (101) 91 - 45  |
| (12) 85 - 58 | (42) 69 - 56 | (72) 94 - 58 | (102) 55 - 18  |
| (13) 38 - 5  | (43) 63 - 3  | (73) 90 - 5  | (103) 23 - 3   |
| (14) 48 - 23 | (44) 68 - 50 | (74) 36 - 32 | (104) 43 - 18  |
| (15) 96 - 60 | (45) 46 - 25 | (75) 79 - 22 | (105) 82 - 33  |
| (16) 46 - 9  | (46) 26 - 5  | (76) 26 - 23 | (106) 97 - 66  |
| (17) 99 - 50 | (47) 24 - 5  | (77) 95 - 86 | (107) 50 - 26  |
| (18) 35 - 1  | (48) 86 - 63 | (78) 78 - 40 | (108) 77 - 22  |
| (19) 41 - 25 | (49) 59 - 31 | (79) 23 - 2  | (109) 98 - 32  |
| (20) 55 - 21 | (50) 89 - 22 | (80) 91 - 32 | (110) 96 - 71  |
| (21) 46 - 27 | (51) 47 - 11 | (81) 63 - 54 | (111) 79 - 76  |
| (22) 71 - 28 | (52) 58 - 9  | (82) 82 - 7  | (112) 87 - 40  |
| (23) 73 - 66 | (53) 53 - 22 | (83) 41 - 13 | (113) 57 - 48  |
| (24) 83 - 77 | (54) 19 - 5  | (84) 95 - 27 | (114) 64 - 46  |
| (25) 67 - 32 | (55) 64 - 30 | (85) 36 - 10 | (115) 50 - 3   |
| (26) 37 - 21 | (56) 32 - 29 | (86) 65 - 44 | (116) 100 - 98 |
| (27) 92 - 56 | (57) 91 - 57 | (87) 84 - 79 | (117) 48 - 6   |
| (28) 62 - 39 | (58) 73 - 4  | (88) 52 - 15 | (118) 63 - 23  |
| (29) 91 - 59 | (59) 69 - 51 | (89) 26 - 21 | (119) 98 - 1   |
| (30) 36 - 34 | (60) 74 - 5  | (90) 31 - 10 | (120) 87 - 37  |

|              |               |              |               |
|--------------|---------------|--------------|---------------|
| (1) 64 - 50  | (31) 95 - 52  | (61) 75 - 25 | (91) 79 - 18  |
| (2) 42 - 15  | (32) 77 - 57  | (62) 77 - 48 | (92) 33 - 19  |
| (3) 61 - 27  | (33) 86 - 20  | (63) 83 - 57 | (93) 64 - 31  |
| (4) 46 - 17  | (34) 66 - 38  | (64) 89 - 56 | (94) 49 - 39  |
| (5) 35 - 21  | (35) 95 - 89  | (65) 23 - 13 | (95) 68 - 55  |
| (6) 77 - 44  | (36) 85 - 13  | (66) 75 - 36 | (96) 93 - 70  |
| (7) 53 - 12  | (37) 52 - 49  | (67) 79 - 37 | (97) 56 - 40  |
| (8) 63 - 7   | (38) 29 - 21  | (68) 75 - 70 | (98) 81 - 81  |
| (9) 57 - 17  | (39) 54 - 51  | (69) 65 - 49 | (99) 93 - 52  |
| (10) 72 - 63 | (40) 92 - 57  | (70) 28 - 20 | (100) 43 - 36 |
| (11) 99 - 2  | (41) 40 - 14  | (71) 92 - 81 | (101) 90 - 29 |
| (12) 73 - 32 | (42) 50 - 2   | (72) 31 - 15 | (102) 89 - 9  |
| (13) 42 - 24 | (43) 100 - 73 | (73) 87 - 83 | (103) 36 - 0  |
| (14) 50 - 28 | (44) 37 - 26  | (74) 86 - 14 | (104) 34 - 29 |
| (15) 55 - 39 | (45) 65 - 25  | (75) 30 - 3  | (105) 50 - 33 |
| (16) 88 - 31 | (46) 82 - 38  | (76) 91 - 33 | (106) 82 - 39 |
| (17) 72 - 30 | (47) 42 - 8   | (77) 59 - 14 | (107) 55 - 30 |
| (18) 54 - 7  | (48) 58 - 32  | (78) 94 - 52 | (108) 82 - 9  |
| (19) 73 - 26 | (49) 97 - 91  | (79) 87 - 56 | (109) 90 - 9  |
| (20) 13 - 3  | (50) 97 - 44  | (80) 93 - 88 | (110) 64 - 40 |
| (21) 74 - 73 | (51) 40 - 22  | (81) 89 - 50 | (111) 38 - 33 |
| (22) 60 - 13 | (52) 87 - 0   | (82) 55 - 36 | (112) 57 - 33 |
| (23) 61 - 54 | (53) 41 - 38  | (83) 65 - 22 | (113) 82 - 69 |
| (24) 30 - 22 | (54) 17 - 5   | (84) 87 - 55 | (114) 51 - 19 |
| (25) 65 - 52 | (55) 88 - 19  | (85) 63 - 37 | (115) 95 - 89 |
| (26) 43 - 25 | (56) 96 - 47  | (86) 62 - 9  | (116) 62 - 51 |
| (27) 46 - 43 | (57) 68 - 51  | (87) 91 - 38 | (117) 85 - 77 |
| (28) 64 - 26 | (58) 84 - 47  | (88) 41 - 20 | (118) 12 - 10 |
| (29) 56 - 36 | (59) 81 - 67  | (89) 79 - 66 | (119) 88 - 52 |
| (30) 90 - 85 | (60) 22 - 4   | (90) 30 - 26 | (120) 76 - 14 |

|               |               |              |                |
|---------------|---------------|--------------|----------------|
| (1) 60 - 17   | (31) 82 - 45  | (61) 73 - 36 | (91) 64 - 59   |
| (2) 86 - 59   | (32) 79 - 10  | (62) 62 - 26 | (92) 66 - 7    |
| (3) 70 - 20   | (33) 77 - 66  | (63) 17 - 9  | (93) 55 - 53   |
| (4) 72 - 51   | (34) 72 - 4   | (64) 91 - 73 | (94) 81 - 36   |
| (5) 44 - 2    | (35) 89 - 1   | (65) 84 - 8  | (95) 93 - 69   |
| (6) 77 - 47   | (36) 19 - 11  | (66) 59 - 33 | (96) 31 - 29   |
| (7) 35 - 30   | (37) 64 - 29  | (67) 67 - 5  | (97) 86 - 27   |
| (8) 27 - 4    | (38) 21 - 19  | (68) 65 - 48 | (98) 50 - 28   |
| (9) 86 - 11   | (39) 98 - 12  | (69) 88 - 67 | (99) 92 - 63   |
| (10) 89 - 59  | (40) 97 - 89  | (70) 2 - 1   | (100) 100 - 24 |
| (11) 91 - 62  | (41) 96 - 11  | (71) 38 - 28 | (101) 70 - 67  |
| (12) 25 - 7   | (42) 90 - 50  | (72) 61 - 54 | (102) 77 - 63  |
| (13) 92 - 80  | (43) 68 - 42  | (73) 21 - 8  | (103) 71 - 56  |
| (14) 72 - 62  | (44) 98 - 93  | (74) 52 - 21 | (104) 44 - 18  |
| (15) 92 - 48  | (45) 66 - 12  | (75) 63 - 11 | (105) 16 - 13  |
| (16) 58 - 34  | (46) 82 - 73  | (76) 85 - 3  | (106) 40 - 12  |
| (17) 83 - 81  | (47) 59 - 16  | (77) 40 - 4  | (107) 30 - 26  |
| (18) 34 - 19  | (48) 42 - 20  | (78) 59 - 13 | (108) 78 - 66  |
| (19) 90 - 68  | (49) 72 - 59  | (79) 81 - 41 | (109) 28 - 17  |
| (20) 68 - 8   | (50) 100 - 30 | (80) 97 - 71 | (110) 77 - 32  |
| (21) 100 - 75 | (51) 45 - 24  | (81) 91 - 44 | (111) 74 - 5   |
| (22) 45 - 1   | (52) 93 - 10  | (82) 67 - 62 | (112) 76 - 32  |
| (23) 91 - 5   | (53) 72 - 57  | (83) 75 - 12 | (113) 74 - 58  |
| (24) 77 - 45  | (54) 54 - 7   | (84) 67 - 30 | (114) 88 - 40  |
| (25) 25 - 9   | (55) 91 - 68  | (85) 60 - 17 | (115) 61 - 4   |
| (26) 55 - 35  | (56) 78 - 27  | (86) 83 - 74 | (116) 14 - 4   |
| (27) 87 - 58  | (57) 38 - 38  | (87) 82 - 29 | (117) 52 - 24  |
| (28) 68 - 26  | (58) 77 - 58  | (88) 46 - 40 | (118) 99 - 37  |
| (29) 75 - 15  | (59) 80 - 56  | (89) 78 - 72 | (119) 30 - 28  |
| (30) 42 - 36  | (60) 45 - 45  | (90) 90 - 11 | (120) 68 - 64  |

|               |              |              |               |
|---------------|--------------|--------------|---------------|
| (1) 93 - 20   | (31) 54 - 5  | (61) 41 - 25 | (91) 94 - 62  |
| (2) 92 - 39   | (32) 58 - 27 | (62) 81 - 4  | (92) 43 - 18  |
| (3) 43 - 20   | (33) 78 - 42 | (63) 11 - 11 | (93) 91 - 52  |
| (4) 46 - 33   | (34) 72 - 29 | (64) 89 - 24 | (94) 100 - 35 |
| (5) 61 - 44   | (35) 80 - 73 | (65) 85 - 1  | (95) 74 - 71  |
| (6) 86 - 55   | (36) 94 - 9  | (66) 73 - 30 | (96) 80 - 6   |
| (7) 66 - 36   | (37) 92 - 33 | (67) 48 - 27 | (97) 36 - 32  |
| (8) 80 - 63   | (38) 94 - 41 | (68) 54 - 52 | (98) 71 - 19  |
| (9) 83 - 48   | (39) 66 - 48 | (69) 62 - 1  | (99) 95 - 86  |
| (10) 19 - 11  | (40) 51 - 5  | (70) 45 - 3  | (100) 73 - 56 |
| (11) 67 - 31  | (41) 68 - 12 | (71) 91 - 44 | (101) 48 - 19 |
| (12) 43 - 28  | (42) 63 - 14 | (72) 77 - 5  | (102) 80 - 78 |
| (13) 82 - 57  | (43) 89 - 11 | (73) 82 - 9  | (103) 22 - 3  |
| (14) 77 - 12  | (44) 66 - 17 | (74) 81 - 73 | (104) 34 - 13 |
| (15) 80 - 49  | (45) 85 - 47 | (75) 20 - 18 | (105) 75 - 32 |
| (16) 3 - 1    | (46) 69 - 20 | (76) 27 - 14 | (106) 71 - 49 |
| (17) 83 - 20  | (47) 76 - 75 | (77) 20 - 7  | (107) 75 - 11 |
| (18) 67 - 4   | (48) 13 - 3  | (78) 88 - 53 | (108) 91 - 78 |
| (19) 60 - 19  | (49) 23 - 22 | (79) 29 - 13 | (109) 29 - 9  |
| (20) 2 - 0    | (50) 57 - 24 | (80) 95 - 32 | (110) 92 - 19 |
| (21) 52 - 4   | (51) 49 - 11 | (81) 17 - 8  | (111) 34 - 22 |
| (22) 34 - 7   | (52) 91 - 90 | (82) 7 - 4   | (112) 91 - 77 |
| (23) 98 - 31  | (53) 39 - 4  | (83) 96 - 88 | (113) 90 - 68 |
| (24) 66 - 56  | (54) 99 - 32 | (84) 95 - 71 | (114) 73 - 0  |
| (25) 42 - 25  | (55) 94 - 45 | (85) 40 - 11 | (115) 88 - 57 |
| (26) 91 - 56  | (56) 74 - 67 | (86) 63 - 44 | (116) 31 - 27 |
| (27) 100 - 79 | (57) 90 - 60 | (87) 69 - 16 | (117) 87 - 36 |
| (28) 59 - 13  | (58) 5 - 4   | (88) 84 - 21 | (118) 45 - 29 |
| (29) 97 - 56  | (59) 76 - 12 | (89) 58 - 31 | (119) 77 - 29 |
| (30) 80 - 59  | (60) 69 - 31 | (90) 39 - 28 | (120) 50 - 18 |

|              |              |              |               |
|--------------|--------------|--------------|---------------|
| (1) 28 - 14  | (31) 79 - 18 | (61) 18 - 12 | (91) 70 - 14  |
| (2) 6 - 5    | (32) 64 - 10 | (62) 89 - 44 | (92) 35 - 31  |
| (3) 34 - 14  | (33) 90 - 2  | (63) 58 - 54 | (93) 89 - 24  |
| (4) 38 - 7   | (34) 62 - 44 | (64) 98 - 73 | (94) 88 - 23  |
| (5) 45 - 3   | (35) 10 - 10 | (65) 88 - 19 | (95) 91 - 64  |
| (6) 83 - 4   | (36) 3 - 3   | (66) 44 - 33 | (96) 88 - 58  |
| (7) 88 - 49  | (37) 84 - 71 | (67) 68 - 28 | (97) 97 - 52  |
| (8) 45 - 24  | (38) 71 - 36 | (68) 97 - 81 | (98) 66 - 26  |
| (9) 72 - 46  | (39) 63 - 11 | (69) 94 - 41 | (99) 96 - 20  |
| (10) 69 - 53 | (40) 67 - 48 | (70) 46 - 14 | (100) 73 - 7  |
| (11) 54 - 8  | (41) 55 - 3  | (71) 82 - 47 | (101) 66 - 42 |
| (12) 49 - 11 | (42) 98 - 32 | (72) 71 - 16 | (102) 98 - 32 |
| (13) 43 - 29 | (43) 99 - 84 | (73) 83 - 4  | (103) 56 - 27 |
| (14) 44 - 21 | (44) 94 - 76 | (74) 98 - 43 | (104) 62 - 25 |
| (15) 40 - 20 | (45) 92 - 18 | (75) 74 - 62 | (105) 89 - 38 |
| (16) 47 - 26 | (46) 61 - 3  | (76) 31 - 29 | (106) 83 - 18 |
| (17) 57 - 43 | (47) 100 - 3 | (77) 91 - 9  | (107) 64 - 45 |
| (18) 18 - 12 | (48) 77 - 36 | (78) 79 - 35 | (108) 76 - 65 |
| (19) 92 - 37 | (49) 92 - 31 | (79) 32 - 25 | (109) 78 - 70 |
| (20) 45 - 28 | (50) 43 - 39 | (80) 57 - 1  | (110) 96 - 21 |
| (21) 45 - 38 | (51) 67 - 56 | (81) 97 - 6  | (111) 75 - 35 |
| (22) 98 - 27 | (52) 77 - 41 | (82) 27 - 12 | (112) 78 - 17 |
| (23) 94 - 78 | (53) 49 - 1  | (83) 72 - 71 | (113) 84 - 48 |
| (24) 17 - 12 | (54) 30 - 4  | (84) 93 - 5  | (114) 83 - 41 |
| (25) 34 - 10 | (55) 94 - 40 | (85) 32 - 23 | (115) 73 - 46 |
| (26) 91 - 23 | (56) 65 - 36 | (86) 81 - 3  | (116) 16 - 4  |
| (27) 82 - 30 | (57) 20 - 18 | (87) 49 - 40 | (117) 59 - 7  |
| (28) 70 - 4  | (58) 44 - 2  | (88) 94 - 2  | (118) 83 - 46 |
| (29) 63 - 4  | (59) 60 - 5  | (89) 96 - 65 | (119) 36 - 34 |
| (30) 90 - 31 | (60) 93 - 79 | (90) 64 - 33 | (120) 94 - 70 |

|              |               |              |                |
|--------------|---------------|--------------|----------------|
| (1) 29 - 18  | (31) 79 - 42  | (61) 45 - 7  | (91) 17 - 0    |
| (2) 91 - 76  | (32) 89 - 44  | (62) 60 - 41 | (92) 68 - 44   |
| (3) 74 - 28  | (33) 89 - 45  | (63) 67 - 7  | (93) 59 - 39   |
| (4) 86 - 64  | (34) 77 - 35  | (64) 67 - 18 | (94) 97 - 58   |
| (5) 72 - 37  | (35) 86 - 44  | (65) 29 - 3  | (95) 32 - 7    |
| (6) 84 - 16  | (36) 58 - 9   | (66) 40 - 29 | (96) 25 - 3    |
| (7) 74 - 11  | (37) 73 - 55  | (67) 98 - 1  | (97) 49 - 12   |
| (8) 81 - 77  | (38) 77 - 5   | (68) 8 - 2   | (98) 100 - 22  |
| (9) 46 - 17  | (39) 63 - 38  | (69) 78 - 58 | (99) 69 - 55   |
| (10) 42 - 24 | (40) 24 - 22  | (70) 84 - 65 | (100) 100 - 94 |
| (11) 93 - 43 | (41) 78 - 21  | (71) 60 - 44 | (101) 77 - 65  |
| (12) 94 - 68 | (42) 67 - 15  | (72) 24 - 20 | (102) 12 - 7   |
| (13) 89 - 72 | (43) 98 - 69  | (73) 70 - 69 | (103) 80 - 59  |
| (14) 84 - 26 | (44) 81 - 28  | (74) 75 - 30 | (104) 95 - 92  |
| (15) 39 - 25 | (45) 87 - 59  | (75) 95 - 93 | (105) 84 - 67  |
| (16) 88 - 72 | (46) 48 - 6   | (76) 65 - 60 | (106) 95 - 36  |
| (17) 82 - 7  | (47) 100 - 46 | (77) 78 - 65 | (107) 79 - 52  |
| (18) 22 - 13 | (48) 65 - 19  | (78) 62 - 16 | (108) 80 - 50  |
| (19) 6 - 4   | (49) 98 - 53  | (79) 66 - 8  | (109) 79 - 21  |
| (20) 94 - 52 | (50) 20 - 12  | (80) 3 - 0   | (110) 58 - 56  |
| (21) 60 - 2  | (51) 69 - 58  | (81) 26 - 21 | (111) 71 - 17  |
| (22) 68 - 66 | (52) 53 - 33  | (82) 64 - 20 | (112) 16 - 3   |
| (23) 72 - 48 | (53) 60 - 48  | (83) 76 - 40 | (113) 88 - 12  |
| (24) 68 - 27 | (54) 63 - 51  | (84) 26 - 14 | (114) 66 - 57  |
| (25) 71 - 69 | (55) 70 - 47  | (85) 99 - 84 | (115) 50 - 8   |
| (26) 7 - 5   | (56) 80 - 37  | (86) 53 - 36 | (116) 37 - 16  |
| (27) 50 - 30 | (57) 8 - 5    | (87) 94 - 81 | (117) 76 - 24  |
| (28) 90 - 88 | (58) 49 - 18  | (88) 58 - 23 | (118) 99 - 19  |
| (29) 50 - 18 | (59) 84 - 37  | (89) 45 - 31 | (119) 81 - 0   |
| (30) 65 - 6  | (60) 81 - 65  | (90) 19 - 16 | (120) 98 - 18  |

|               |              |              |               |
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| (1) 37 - 12   | (31) 85 - 3  | (61) 74 - 1  | (91) 64 - 58  |
| (2) 51 - 15   | (32) 99 - 46 | (62) 13 - 11 | (92) 72 - 67  |
| (3) 96 - 94   | (33) 27 - 10 | (63) 25 - 18 | (93) 79 - 64  |
| (4) 98 - 67   | (34) 91 - 59 | (64) 29 - 17 | (94) 97 - 82  |
| (5) 78 - 53   | (35) 74 - 65 | (65) 24 - 16 | (95) 38 - 16  |
| (6) 99 - 93   | (36) 42 - 29 | (66) 75 - 1  | (96) 79 - 64  |
| (7) 79 - 37   | (37) 59 - 45 | (67) 69 - 20 | (97) 58 - 32  |
| (8) 52 - 38   | (38) 92 - 52 | (68) 64 - 58 | (98) 23 - 2   |
| (9) 70 - 36   | (39) 55 - 11 | (69) 58 - 22 | (99) 84 - 31  |
| (10) 76 - 58  | (40) 85 - 71 | (70) 85 - 35 | (100) 88 - 33 |
| (11) 48 - 18  | (41) 99 - 50 | (71) 78 - 8  | (101) 20 - 8  |
| (12) 68 - 39  | (42) 94 - 12 | (72) 76 - 9  | (102) 70 - 58 |
| (13) 30 - 22  | (43) 71 - 1  | (73) 86 - 15 | (103) 39 - 27 |
| (14) 85 - 39  | (44) 56 - 8  | (74) 85 - 51 | (104) 72 - 4  |
| (15) 79 - 1   | (45) 51 - 7  | (75) 54 - 21 | (105) 85 - 62 |
| (16) 67 - 14  | (46) 39 - 13 | (76) 77 - 9  | (106) 73 - 21 |
| (17) 94 - 32  | (47) 91 - 14 | (77) 47 - 2  | (107) 67 - 15 |
| (18) 89 - 88  | (48) 53 - 31 | (78) 37 - 24 | (108) 53 - 10 |
| (19) 47 - 43  | (49) 91 - 84 | (79) 88 - 26 | (109) 40 - 15 |
| (20) 100 - 65 | (50) 66 - 43 | (80) 75 - 75 | (110) 69 - 60 |
| (21) 91 - 42  | (51) 81 - 9  | (81) 98 - 26 | (111) 98 - 39 |
| (22) 55 - 8   | (52) 30 - 11 | (82) 76 - 4  | (112) 77 - 54 |
| (23) 50 - 38  | (53) 99 - 89 | (83) 88 - 80 | (113) 89 - 10 |
| (24) 56 - 11  | (54) 62 - 26 | (84) 25 - 20 | (114) 74 - 32 |
| (25) 91 - 54  | (55) 78 - 66 | (85) 64 - 47 | (115) 65 - 39 |
| (26) 45 - 30  | (56) 86 - 8  | (86) 38 - 26 | (116) 91 - 82 |
| (27) 48 - 41  | (57) 40 - 13 | (87) 43 - 40 | (117) 81 - 17 |
| (28) 15 - 11  | (58) 57 - 36 | (88) 51 - 41 | (118) 73 - 43 |
| (29) 31 - 11  | (59) 90 - 86 | (89) 50 - 0  | (119) 1 - 0   |
| (30) 57 - 31  | (60) 75 - 71 | (90) 41 - 18 | (120) 88 - 40 |

|              |               |              |               |
|--------------|---------------|--------------|---------------|
| (1) 32 - 29  | (31) 34 - 15  | (61) 94 - 0  | (91) 94 - 20  |
| (2) 50 - 48  | (32) 36 - 19  | (62) 70 - 3  | (92) 95 - 83  |
| (3) 82 - 59  | (33) 85 - 25  | (63) 81 - 20 | (93) 95 - 32  |
| (4) 38 - 29  | (34) 36 - 26  | (64) 57 - 32 | (94) 81 - 37  |
| (5) 19 - 18  | (35) 86 - 6   | (65) 94 - 45 | (95) 64 - 21  |
| (6) 47 - 3   | (36) 93 - 62  | (66) 78 - 61 | (96) 99 - 1   |
| (7) 95 - 46  | (37) 54 - 40  | (67) 84 - 52 | (97) 98 - 24  |
| (8) 49 - 22  | (38) 62 - 4   | (68) 64 - 31 | (98) 91 - 77  |
| (9) 53 - 11  | (39) 86 - 77  | (69) 98 - 23 | (99) 86 - 46  |
| (10) 72 - 62 | (40) 92 - 58  | (70) 40 - 40 | (100) 81 - 46 |
| (11) 77 - 31 | (41) 78 - 49  | (71) 47 - 7  | (101) 67 - 44 |
| (12) 98 - 91 | (42) 92 - 31  | (72) 43 - 33 | (102) 46 - 39 |
| (13) 79 - 20 | (43) 99 - 15  | (73) 99 - 73 | (103) 70 - 16 |
| (14) 63 - 5  | (44) 64 - 9   | (74) 40 - 39 | (104) 86 - 79 |
| (15) 53 - 27 | (45) 50 - 12  | (75) 61 - 14 | (105) 54 - 26 |
| (16) 59 - 33 | (46) 33 - 3   | (76) 69 - 58 | (106) 83 - 25 |
| (17) 50 - 36 | (47) 71 - 41  | (77) 93 - 21 | (107) 87 - 29 |
| (18) 69 - 31 | (48) 92 - 23  | (78) 22 - 11 | (108) 78 - 37 |
| (19) 73 - 15 | (49) 76 - 13  | (79) 85 - 68 | (109) 76 - 29 |
| (20) 68 - 39 | (50) 82 - 58  | (80) 74 - 58 | (110) 75 - 35 |
| (21) 43 - 35 | (51) 90 - 35  | (81) 51 - 34 | (111) 92 - 5  |
| (22) 73 - 15 | (52) 7 - 5    | (82) 42 - 21 | (112) 83 - 83 |
| (23) 66 - 58 | (53) 83 - 8   | (83) 51 - 18 | (113) 95 - 4  |
| (24) 52 - 21 | (54) 64 - 35  | (84) 20 - 8  | (114) 8 - 6   |
| (25) 28 - 13 | (55) 91 - 3   | (85) 46 - 13 | (115) 58 - 53 |
| (26) 63 - 37 | (56) 49 - 44  | (86) 31 - 15 | (116) 69 - 62 |
| (27) 37 - 16 | (57) 85 - 71  | (87) 92 - 47 | (117) 10 - 9  |
| (28) 39 - 9  | (58) 49 - 26  | (88) 45 - 34 | (118) 35 - 33 |
| (29) 48 - 10 | (59) 100 - 90 | (89) 73 - 39 | (119) 27 - 8  |
| (30) 57 - 31 | (60) 98 - 81  | (90) 86 - 11 | (120) 50 - 11 |



|               |               |               |                |
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| (1) 63 - 38   | (31) 63 - 60  | (61) 92 - 0   | (91) 58 - 33   |
| (2) 12 - 5    | (32) 67 - 60  | (62) 88 - 52  | (92) 49 - 45   |
| (3) 54 - 5    | (33) 94 - 72  | (63) 91 - 27  | (93) 70 - 58   |
| (4) 90 - 72   | (34) 67 - 36  | (64) 52 - 7   | (94) 43 - 19   |
| (5) 61 - 56   | (35) 83 - 50  | (65) 91 - 83  | (95) 67 - 17   |
| (6) 69 - 41   | (36) 100 - 83 | (66) 96 - 1   | (96) 56 - 42   |
| (7) 98 - 48   | (37) 42 - 22  | (67) 60 - 25  | (97) 79 - 26   |
| (8) 36 - 23   | (38) 65 - 33  | (68) 96 - 50  | (98) 85 - 76   |
| (9) 30 - 16   | (39) 38 - 0   | (69) 75 - 72  | (99) 72 - 32   |
| (10) 76 - 58  | (40) 96 - 37  | (70) 39 - 2   | (100) 43 - 21  |
| (11) 32 - 25  | (41) 77 - 47  | (71) 15 - 8   | (101) 94 - 31  |
| (12) 88 - 26  | (42) 33 - 8   | (72) 46 - 25  | (102) 100 - 68 |
| (13) 65 - 10  | (43) 39 - 33  | (73) 66 - 63  | (103) 84 - 47  |
| (14) 91 - 89  | (44) 50 - 0   | (74) 63 - 32  | (104) 76 - 31  |
| (15) 78 - 51  | (45) 72 - 28  | (75) 53 - 2   | (105) 41 - 2   |
| (16) 94 - 85  | (46) 87 - 38  | (76) 74 - 32  | (106) 100 - 19 |
| (17) 48 - 25  | (47) 27 - 9   | (77) 76 - 67  | (107) 89 - 9   |
| (18) 93 - 21  | (48) 95 - 67  | (78) 63 - 43  | (108) 51 - 27  |
| (19) 72 - 47  | (49) 27 - 25  | (79) 29 - 23  | (109) 74 - 53  |
| (20) 86 - 15  | (50) 18 - 1   | (80) 40 - 19  | (110) 51 - 50  |
| (21) 54 - 45  | (51) 48 - 18  | (81) 48 - 15  | (111) 82 - 60  |
| (22) 81 - 11  | (52) 54 - 46  | (82) 72 - 26  | (112) 73 - 0   |
| (23) 70 - 64  | (53) 84 - 29  | (83) 100 - 57 | (113) 73 - 25  |
| (24) 18 - 13  | (54) 77 - 57  | (84) 97 - 63  | (114) 98 - 86  |
| (25) 69 - 43  | (55) 56 - 26  | (85) 92 - 58  | (115) 9 - 7    |
| (26) 64 - 47  | (56) 46 - 6   | (86) 79 - 32  | (116) 96 - 82  |
| (27) 21 - 1   | (57) 98 - 91  | (87) 6 - 1    | (117) 52 - 21  |
| (28) 25 - 22  | (58) 93 - 58  | (88) 91 - 15  | (118) 73 - 57  |
| (29) 100 - 40 | (59) 69 - 10  | (89) 97 - 96  | (119) 98 - 66  |
| (30) 50 - 19  | (60) 96 - 40  | (90) 76 - 57  | (120) 80 - 53  |

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| (1) 76 - 8   | (31) 45 - 23  | (61) 34 - 11 | (91) 56 - 45  |
| (2) 93 - 92  | (32) 38 - 24  | (62) 20 - 11 | (92) 71 - 6   |
| (3) 96 - 60  | (33) 31 - 24  | (63) 81 - 70 | (93) 83 - 72  |
| (4) 82 - 10  | (34) 34 - 5   | (64) 84 - 25 | (94) 42 - 31  |
| (5) 49 - 22  | (35) 67 - 67  | (65) 98 - 3  | (95) 71 - 60  |
| (6) 69 - 2   | (36) 98 - 24  | (66) 35 - 29 | (96) 98 - 22  |
| (7) 78 - 22  | (37) 71 - 34  | (67) 70 - 57 | (97) 42 - 18  |
| (8) 61 - 6   | (38) 97 - 58  | (68) 85 - 83 | (98) 84 - 41  |
| (9) 54 - 40  | (39) 38 - 17  | (69) 40 - 8  | (99) 98 - 19  |
| (10) 86 - 13 | (40) 73 - 33  | (70) 22 - 15 | (100) 96 - 49 |
| (11) 88 - 49 | (41) 85 - 14  | (71) 39 - 34 | (101) 13 - 11 |
| (12) 68 - 15 | (42) 23 - 2   | (72) 53 - 51 | (102) 40 - 6  |
| (13) 79 - 10 | (43) 99 - 51  | (73) 43 - 4  | (103) 25 - 2  |
| (14) 87 - 32 | (44) 77 - 17  | (74) 70 - 27 | (104) 35 - 20 |
| (15) 92 - 76 | (45) 94 - 72  | (75) 45 - 2  | (105) 77 - 48 |
| (16) 94 - 29 | (46) 96 - 9   | (76) 33 - 22 | (106) 25 - 11 |
| (17) 88 - 23 | (47) 38 - 35  | (77) 90 - 59 | (107) 77 - 46 |
| (18) 65 - 60 | (48) 65 - 22  | (78) 86 - 50 | (108) 96 - 69 |
| (19) 45 - 28 | (49) 49 - 46  | (79) 50 - 48 | (109) 30 - 21 |
| (20) 37 - 8  | (50) 100 - 77 | (80) 97 - 87 | (110) 61 - 33 |
| (21) 64 - 45 | (51) 74 - 51  | (81) 9 - 3   | (111) 30 - 18 |
| (22) 88 - 67 | (52) 86 - 68  | (82) 81 - 56 | (112) 36 - 21 |
| (23) 15 - 13 | (53) 33 - 20  | (83) 81 - 53 | (113) 20 - 10 |
| (24) 69 - 69 | (54) 42 - 4   | (84) 77 - 20 | (114) 30 - 18 |
| (25) 48 - 28 | (55) 35 - 17  | (85) 88 - 87 | (115) 96 - 88 |
| (26) 84 - 79 | (56) 88 - 34  | (86) 75 - 74 | (116) 46 - 18 |
| (27) 85 - 37 | (57) 63 - 40  | (87) 58 - 16 | (117) 44 - 17 |
| (28) 87 - 1  | (58) 28 - 25  | (88) 71 - 19 | (118) 26 - 7  |
| (29) 59 - 27 | (59) 51 - 5   | (89) 57 - 32 | (119) 13 - 6  |
| (30) 75 - 11 | (60) 63 - 25  | (90) 14 - 3  | (120) 74 - 65 |

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| (1) 15 - 14   | (31) 96 - 13 | (61) 25 - 15 | (91) 16 - 6    |
| (2) 75 - 14   | (32) 96 - 75 | (62) 30 - 8  | (92) 58 - 11   |
| (3) 31 - 20   | (33) 39 - 14 | (63) 91 - 7  | (93) 88 - 15   |
| (4) 93 - 46   | (34) 58 - 17 | (64) 97 - 8  | (94) 51 - 50   |
| (5) 75 - 38   | (35) 85 - 51 | (65) 72 - 17 | (95) 3 - 0     |
| (6) 65 - 36   | (36) 42 - 42 | (66) 71 - 68 | (96) 95 - 38   |
| (7) 30 - 19   | (37) 85 - 6  | (67) 75 - 38 | (97) 95 - 67   |
| (8) 90 - 87   | (38) 46 - 9  | (68) 48 - 22 | (98) 71 - 19   |
| (9) 45 - 4    | (39) 54 - 21 | (69) 73 - 21 | (99) 63 - 41   |
| (10) 76 - 40  | (40) 35 - 9  | (70) 81 - 17 | (100) 66 - 33  |
| (11) 94 - 30  | (41) 76 - 38 | (71) 98 - 10 | (101) 63 - 47  |
| (12) 99 - 60  | (42) 63 - 62 | (72) 66 - 6  | (102) 52 - 26  |
| (13) 67 - 6   | (43) 83 - 31 | (73) 66 - 46 | (103) 98 - 78  |
| (14) 99 - 57  | (44) 79 - 47 | (74) 30 - 2  | (104) 90 - 61  |
| (15) 89 - 22  | (45) 57 - 22 | (75) 99 - 31 | (105) 22 - 21  |
| (16) 91 - 45  | (46) 35 - 31 | (76) 90 - 33 | (106) 23 - 9   |
| (17) 97 - 29  | (47) 77 - 55 | (77) 96 - 68 | (107) 41 - 17  |
| (18) 19 - 7   | (48) 97 - 91 | (78) 93 - 67 | (108) 92 - 6   |
| (19) 72 - 18  | (49) 39 - 32 | (79) 91 - 75 | (109) 85 - 25  |
| (20) 93 - 18  | (50) 34 - 10 | (80) 72 - 15 | (110) 93 - 4   |
| (21) 28 - 28  | (51) 60 - 33 | (81) 60 - 10 | (111) 12 - 7   |
| (22) 89 - 83  | (52) 92 - 48 | (82) 82 - 14 | (112) 62 - 18  |
| (23) 92 - 10  | (53) 26 - 21 | (83) 97 - 84 | (113) 73 - 57  |
| (24) 78 - 47  | (54) 45 - 37 | (84) 98 - 36 | (114) 53 - 46  |
| (25) 84 - 14  | (55) 91 - 28 | (85) 95 - 50 | (115) 56 - 47  |
| (26) 21 - 7   | (56) 70 - 41 | (86) 84 - 38 | (116) 100 - 62 |
| (27) 100 - 57 | (57) 21 - 7  | (87) 97 - 7  | (117) 79 - 74  |
| (28) 57 - 55  | (58) 40 - 8  | (88) 51 - 2  | (118) 13 - 10  |
| (29) 99 - 63  | (59) 76 - 27 | (89) 77 - 13 | (119) 91 - 41  |
| (30) 13 - 11  | (60) 14 - 14 | (90) 90 - 83 | (120) 83 - 50  |

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| (1) 52 - 8   | (31) 31 - 12 | (61) 74 - 56  | (91) 39 - 16  |
| (2) 88 - 15  | (32) 81 - 46 | (62) 91 - 66  | (92) 82 - 68  |
| (3) 49 - 41  | (33) 90 - 14 | (63) 97 - 41  | (93) 55 - 27  |
| (4) 94 - 22  | (34) 92 - 91 | (64) 40 - 36  | (94) 100 - 6  |
| (5) 22 - 18  | (35) 91 - 12 | (65) 58 - 3   | (95) 83 - 7   |
| (6) 58 - 6   | (36) 41 - 28 | (66) 44 - 27  | (96) 27 - 4   |
| (7) 8 - 1    | (37) 20 - 12 | (67) 100 - 83 | (97) 39 - 22  |
| (8) 99 - 1   | (38) 66 - 8  | (68) 76 - 22  | (98) 98 - 97  |
| (9) 91 - 27  | (39) 74 - 65 | (69) 97 - 74  | (99) 73 - 28  |
| (10) 79 - 59 | (40) 75 - 61 | (70) 86 - 17  | (100) 21 - 11 |
| (11) 92 - 48 | (41) 54 - 36 | (71) 84 - 35  | (101) 93 - 68 |
| (12) 64 - 47 | (42) 25 - 17 | (72) 56 - 11  | (102) 88 - 77 |
| (13) 38 - 10 | (43) 69 - 66 | (73) 99 - 66  | (103) 51 - 18 |
| (14) 75 - 0  | (44) 100 - 8 | (74) 81 - 53  | (104) 85 - 18 |
| (15) 90 - 25 | (45) 84 - 48 | (75) 93 - 38  | (105) 91 - 70 |
| (16) 75 - 21 | (46) 99 - 13 | (76) 56 - 19  | (106) 53 - 40 |
| (17) 84 - 41 | (47) 30 - 16 | (77) 61 - 34  | (107) 96 - 62 |
| (18) 62 - 21 | (48) 15 - 0  | (78) 73 - 23  | (108) 82 - 6  |
| (19) 95 - 11 | (49) 71 - 12 | (79) 93 - 15  | (109) 73 - 31 |
| (20) 89 - 46 | (50) 38 - 1  | (80) 60 - 9   | (110) 92 - 55 |
| (21) 80 - 38 | (51) 70 - 24 | (81) 38 - 7   | (111) 85 - 21 |
| (22) 9 - 5   | (52) 39 - 35 | (82) 73 - 68  | (112) 23 - 5  |
| (23) 53 - 17 | (53) 72 - 62 | (83) 98 - 64  | (113) 73 - 15 |
| (24) 89 - 67 | (54) 7 - 7   | (84) 99 - 9   | (114) 89 - 27 |
| (25) 58 - 45 | (55) 50 - 24 | (85) 23 - 18  | (115) 38 - 23 |
| (26) 84 - 32 | (56) 71 - 3  | (86) 18 - 5   | (116) 92 - 67 |
| (27) 96 - 17 | (57) 41 - 34 | (87) 96 - 25  | (117) 88 - 18 |
| (28) 78 - 53 | (58) 23 - 14 | (88) 39 - 27  | (118) 9 - 7   |
| (29) 84 - 38 | (59) 74 - 3  | (89) 67 - 31  | (119) 83 - 41 |
| (30) 95 - 17 | (60) 32 - 28 | (90) 68 - 7   | (120) 8 - 8   |

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| (1) 99 - 36   | (31) 41 - 34 | (61) 67 - 64 | (91) 82 - 6   |
| (2) 81 - 47   | (32) 94 - 39 | (62) 13 - 7  | (92) 90 - 45  |
| (3) 68 - 11   | (33) 46 - 15 | (63) 46 - 3  | (93) 33 - 31  |
| (4) 33 - 31   | (34) 80 - 45 | (64) 79 - 78 | (94) 88 - 72  |
| (5) 47 - 1    | (35) 39 - 26 | (65) 96 - 74 | (95) 70 - 22  |
| (6) 52 - 52   | (36) 70 - 53 | (66) 94 - 13 | (96) 46 - 10  |
| (7) 97 - 70   | (37) 63 - 35 | (67) 74 - 41 | (97) 48 - 13  |
| (8) 39 - 23   | (38) 94 - 61 | (68) 58 - 45 | (98) 66 - 31  |
| (9) 57 - 40   | (39) 82 - 9  | (69) 98 - 4  | (99) 78 - 2   |
| (10) 76 - 74  | (40) 49 - 44 | (70) 77 - 44 | (100) 63 - 43 |
| (11) 88 - 39  | (41) 60 - 18 | (71) 81 - 34 | (101) 45 - 15 |
| (12) 41 - 31  | (42) 83 - 5  | (72) 75 - 72 | (102) 58 - 29 |
| (13) 82 - 8   | (43) 53 - 19 | (73) 41 - 30 | (103) 43 - 22 |
| (14) 94 - 10  | (44) 16 - 6  | (74) 40 - 38 | (104) 57 - 8  |
| (15) 69 - 16  | (45) 73 - 14 | (75) 94 - 11 | (105) 82 - 33 |
| (16) 6 - 0    | (46) 53 - 28 | (76) 26 - 26 | (106) 30 - 28 |
| (17) 84 - 39  | (47) 75 - 43 | (77) 83 - 17 | (107) 97 - 17 |
| (18) 96 - 44  | (48) 96 - 1  | (78) 66 - 32 | (108) 67 - 22 |
| (19) 20 - 4   | (49) 68 - 43 | (79) 48 - 24 | (109) 94 - 11 |
| (20) 100 - 91 | (50) 99 - 68 | (80) 69 - 44 | (110) 80 - 61 |
| (21) 67 - 59  | (51) 68 - 34 | (81) 93 - 41 | (111) 75 - 48 |
| (22) 30 - 30  | (52) 76 - 26 | (82) 80 - 57 | (112) 75 - 22 |
| (23) 98 - 69  | (53) 87 - 3  | (83) 45 - 44 | (113) 85 - 19 |
| (24) 93 - 55  | (54) 87 - 58 | (84) 47 - 29 | (114) 32 - 24 |
| (25) 32 - 12  | (55) 53 - 14 | (85) 72 - 52 | (115) 36 - 26 |
| (26) 62 - 47  | (56) 98 - 6  | (86) 93 - 51 | (116) 55 - 49 |
| (27) 79 - 60  | (57) 65 - 16 | (87) 99 - 70 | (117) 66 - 62 |
| (28) 93 - 52  | (58) 85 - 53 | (88) 99 - 6  | (118) 73 - 41 |
| (29) 51 - 35  | (59) 64 - 26 | (89) 18 - 17 | (119) 85 - 79 |
| (30) 34 - 11  | (60) 87 - 76 | (90) 52 - 0  | (120) 39 - 15 |

|               |               |              |               |
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| (1) 90 - 84   | (31) 96 - 52  | (61) 74 - 20 | (91) 98 - 1   |
| (2) 45 - 20   | (32) 70 - 4   | (62) 88 - 65 | (92) 26 - 5   |
| (3) 76 - 6    | (33) 99 - 30  | (63) 55 - 10 | (93) 98 - 84  |
| (4) 56 - 5    | (34) 65 - 21  | (64) 88 - 80 | (94) 59 - 45  |
| (5) 90 - 84   | (35) 78 - 76  | (65) 70 - 58 | (95) 76 - 66  |
| (6) 80 - 49   | (36) 79 - 3   | (66) 81 - 72 | (96) 80 - 71  |
| (7) 5 - 3     | (37) 93 - 8   | (67) 83 - 19 | (97) 75 - 56  |
| (8) 85 - 30   | (38) 35 - 25  | (68) 97 - 86 | (98) 90 - 33  |
| (9) 87 - 76   | (39) 81 - 60  | (69) 84 - 64 | (99) 77 - 1   |
| (10) 75 - 22  | (40) 42 - 17  | (70) 72 - 16 | (100) 45 - 25 |
| (11) 100 - 29 | (41) 78 - 42  | (71) 86 - 77 | (101) 27 - 27 |
| (12) 92 - 5   | (42) 22 - 15  | (72) 58 - 11 | (102) 41 - 14 |
| (13) 61 - 45  | (43) 31 - 17  | (73) 80 - 64 | (103) 96 - 37 |
| (14) 94 - 40  | (44) 95 - 25  | (74) 81 - 44 | (104) 42 - 16 |
| (15) 5 - 5    | (45) 55 - 29  | (75) 89 - 20 | (105) 99 - 98 |
| (16) 86 - 70  | (46) 86 - 63  | (76) 80 - 15 | (106) 69 - 11 |
| (17) 13 - 8   | (47) 100 - 88 | (77) 99 - 2  | (107) 70 - 12 |
| (18) 65 - 20  | (48) 56 - 48  | (78) 72 - 43 | (108) 95 - 0  |
| (19) 82 - 37  | (49) 98 - 94  | (79) 79 - 52 | (109) 41 - 29 |
| (20) 91 - 60  | (50) 86 - 59  | (80) 78 - 41 | (110) 73 - 27 |
| (21) 82 - 5   | (51) 79 - 9   | (81) 92 - 78 | (111) 68 - 11 |
| (22) 8 - 4    | (52) 60 - 4   | (82) 50 - 40 | (112) 77 - 45 |
| (23) 94 - 63  | (53) 88 - 42  | (83) 50 - 23 | (113) 77 - 17 |
| (24) 63 - 51  | (54) 64 - 22  | (84) 18 - 11 | (114) 77 - 57 |
| (25) 90 - 77  | (55) 68 - 47  | (85) 65 - 22 | (115) 45 - 15 |
| (26) 64 - 21  | (56) 92 - 40  | (86) 28 - 5  | (116) 74 - 66 |
| (27) 83 - 45  | (57) 75 - 14  | (87) 96 - 2  | (117) 44 - 40 |
| (28) 56 - 51  | (58) 72 - 10  | (88) 97 - 23 | (118) 14 - 1  |
| (29) 55 - 6   | (59) 55 - 18  | (89) 94 - 4  | (119) 73 - 37 |
| (30) 29 - 2   | (60) 41 - 15  | (90) 92 - 53 | (120) 81 - 44 |

|              |              |               |                |
|--------------|--------------|---------------|----------------|
| (1) 85 - 77  | (31) 68 - 26 | (61) 79 - 24  | (91) 47 - 25   |
| (2) 54 - 50  | (32) 21 - 6  | (62) 100 - 58 | (92) 53 - 37   |
| (3) 30 - 2   | (33) 56 - 42 | (63) 68 - 24  | (93) 73 - 20   |
| (4) 99 - 81  | (34) 75 - 35 | (64) 88 - 61  | (94) 95 - 59   |
| (5) 59 - 31  | (35) 65 - 1  | (65) 45 - 9   | (95) 82 - 38   |
| (6) 67 - 15  | (36) 68 - 26 | (66) 20 - 13  | (96) 84 - 69   |
| (7) 72 - 18  | (37) 57 - 43 | (67) 100 - 4  | (97) 70 - 44   |
| (8) 36 - 0   | (38) 14 - 12 | (68) 96 - 24  | (98) 33 - 27   |
| (9) 81 - 6   | (39) 78 - 71 | (69) 62 - 58  | (99) 53 - 6    |
| (10) 68 - 32 | (40) 99 - 61 | (70) 54 - 4   | (100) 88 - 56  |
| (11) 78 - 60 | (41) 38 - 5  | (71) 57 - 13  | (101) 85 - 48  |
| (12) 96 - 54 | (42) 93 - 23 | (72) 97 - 50  | (102) 83 - 27  |
| (13) 6 - 6   | (43) 69 - 32 | (73) 81 - 78  | (103) 87 - 16  |
| (14) 94 - 69 | (44) 75 - 54 | (74) 75 - 67  | (104) 76 - 62  |
| (15) 17 - 15 | (45) 90 - 36 | (75) 91 - 72  | (105) 52 - 42  |
| (16) 19 - 10 | (46) 52 - 45 | (76) 82 - 43  | (106) 78 - 35  |
| (17) 89 - 78 | (47) 56 - 34 | (77) 90 - 67  | (107) 81 - 54  |
| (18) 65 - 49 | (48) 95 - 83 | (78) 78 - 28  | (108) 69 - 60  |
| (19) 98 - 24 | (49) 49 - 33 | (79) 78 - 47  | (109) 58 - 26  |
| (20) 79 - 71 | (50) 68 - 31 | (80) 67 - 37  | (110) 35 - 8   |
| (21) 88 - 21 | (51) 79 - 61 | (81) 90 - 62  | (111) 85 - 54  |
| (22) 64 - 26 | (52) 61 - 31 | (82) 55 - 29  | (112) 3 - 1    |
| (23) 48 - 0  | (53) 62 - 59 | (83) 94 - 68  | (113) 44 - 14  |
| (24) 65 - 27 | (54) 66 - 26 | (84) 43 - 40  | (114) 82 - 4   |
| (25) 63 - 33 | (55) 97 - 35 | (85) 31 - 24  | (115) 52 - 40  |
| (26) 95 - 43 | (56) 45 - 12 | (86) 50 - 38  | (116) 34 - 5   |
| (27) 73 - 3  | (57) 97 - 38 | (87) 49 - 45  | (117) 90 - 8   |
| (28) 22 - 5  | (58) 15 - 9  | (88) 73 - 38  | (118) 48 - 42  |
| (29) 86 - 33 | (59) 96 - 68 | (89) 100 - 17 | (119) 81 - 1   |
| (30) 85 - 48 | (60) 54 - 31 | (90) 77 - 56  | (120) 100 - 13 |

|               |              |               |                |
|---------------|--------------|---------------|----------------|
| (1) 21 - 6    | (31) 89 - 65 | (61) 87 - 71  | (91) 94 - 50   |
| (2) 67 - 38   | (32) 56 - 37 | (62) 100 - 32 | (92) 46 - 40   |
| (3) 77 - 20   | (33) 95 - 41 | (63) 98 - 30  | (93) 13 - 13   |
| (4) 80 - 32   | (34) 51 - 6  | (64) 99 - 15  | (94) 54 - 17   |
| (5) 79 - 1    | (35) 66 - 45 | (65) 96 - 56  | (95) 47 - 46   |
| (6) 75 - 27   | (36) 96 - 82 | (66) 51 - 49  | (96) 85 - 69   |
| (7) 61 - 27   | (37) 68 - 13 | (67) 18 - 3   | (97) 74 - 6    |
| (8) 98 - 72   | (38) 82 - 71 | (68) 69 - 14  | (98) 70 - 12   |
| (9) 61 - 18   | (39) 60 - 40 | (69) 94 - 64  | (99) 82 - 33   |
| (10) 88 - 4   | (40) 82 - 61 | (70) 69 - 24  | (100) 96 - 9   |
| (11) 73 - 32  | (41) 39 - 39 | (71) 59 - 0   | (101) 73 - 2   |
| (12) 75 - 68  | (42) 74 - 56 | (72) 51 - 18  | (102) 56 - 11  |
| (13) 100 - 71 | (43) 92 - 59 | (73) 55 - 13  | (103) 88 - 2   |
| (14) 32 - 20  | (44) 65 - 45 | (74) 62 - 47  | (104) 57 - 18  |
| (15) 70 - 38  | (45) 17 - 12 | (75) 71 - 71  | (105) 74 - 8   |
| (16) 71 - 63  | (46) 68 - 58 | (76) 10 - 3   | (106) 60 - 19  |
| (17) 76 - 76  | (47) 91 - 43 | (77) 99 - 74  | (107) 26 - 26  |
| (18) 25 - 20  | (48) 71 - 1  | (78) 67 - 22  | (108) 71 - 33  |
| (19) 98 - 26  | (49) 69 - 63 | (79) 99 - 24  | (109) 100 - 37 |
| (20) 93 - 73  | (50) 37 - 21 | (80) 100 - 37 | (110) 62 - 0   |
| (21) 89 - 38  | (51) 28 - 20 | (81) 60 - 18  | (111) 60 - 39  |
| (22) 87 - 39  | (52) 61 - 10 | (82) 17 - 8   | (112) 58 - 40  |
| (23) 96 - 86  | (53) 70 - 41 | (83) 86 - 69  | (113) 52 - 52  |
| (24) 95 - 6   | (54) 67 - 37 | (84) 53 - 9   | (114) 77 - 66  |
| (25) 80 - 47  | (55) 62 - 46 | (85) 62 - 47  | (115) 60 - 60  |
| (26) 68 - 10  | (56) 35 - 20 | (86) 98 - 76  | (116) 82 - 26  |
| (27) 75 - 31  | (57) 33 - 20 | (87) 85 - 44  | (117) 77 - 29  |
| (28) 92 - 70  | (58) 68 - 42 | (88) 66 - 45  | (118) 87 - 78  |
| (29) 71 - 48  | (59) 86 - 42 | (89) 42 - 14  | (119) 49 - 21  |
| (30) 100 - 24 | (60) 44 - 3  | (90) 54 - 19  | (120) 84 - 48  |



|              |               |               |                |
|--------------|---------------|---------------|----------------|
| (1) 76 - 52  | (31) 70 - 18  | (61) 53 - 40  | (91) 58 - 29   |
| (2) 56 - 9   | (32) 68 - 58  | (62) 65 - 3   | (92) 100 - 94  |
| (3) 43 - 5   | (33) 75 - 59  | (63) 41 - 11  | (93) 65 - 20   |
| (4) 80 - 38  | (34) 97 - 24  | (64) 91 - 79  | (94) 83 - 37   |
| (5) 56 - 33  | (35) 84 - 30  | (65) 56 - 36  | (95) 88 - 51   |
| (6) 72 - 11  | (36) 47 - 18  | (66) 69 - 54  | (96) 62 - 51   |
| (7) 100 - 35 | (37) 70 - 41  | (67) 85 - 31  | (97) 94 - 67   |
| (8) 99 - 84  | (38) 86 - 20  | (68) 87 - 66  | (98) 30 - 24   |
| (9) 39 - 35  | (39) 95 - 33  | (69) 90 - 2   | (99) 77 - 6    |
| (10) 90 - 11 | (40) 49 - 6   | (70) 70 - 18  | (100) 67 - 36  |
| (11) 67 - 44 | (41) 76 - 15  | (71) 40 - 15  | (101) 98 - 73  |
| (12) 53 - 52 | (42) 88 - 58  | (72) 47 - 42  | (102) 80 - 54  |
| (13) 63 - 11 | (43) 89 - 1   | (73) 98 - 17  | (103) 74 - 66  |
| (14) 95 - 89 | (44) 52 - 32  | (74) 46 - 44  | (104) 88 - 42  |
| (15) 67 - 41 | (45) 86 - 10  | (75) 44 - 23  | (105) 67 - 47  |
| (16) 40 - 38 | (46) 95 - 73  | (76) 83 - 10  | (106) 23 - 8   |
| (17) 26 - 21 | (47) 100 - 88 | (77) 100 - 56 | (107) 55 - 11  |
| (18) 62 - 26 | (48) 68 - 11  | (78) 23 - 7   | (108) 43 - 29  |
| (19) 96 - 3  | (49) 73 - 62  | (79) 83 - 22  | (109) 55 - 6   |
| (20) 63 - 35 | (50) 91 - 10  | (80) 78 - 36  | (110) 85 - 59  |
| (21) 47 - 31 | (51) 96 - 15  | (81) 75 - 18  | (111) 34 - 3   |
| (22) 91 - 68 | (52) 68 - 10  | (82) 77 - 63  | (112) 83 - 19  |
| (23) 98 - 47 | (53) 19 - 7   | (83) 65 - 21  | (113) 89 - 17  |
| (24) 72 - 2  | (54) 97 - 86  | (84) 74 - 11  | (114) 70 - 39  |
| (25) 98 - 19 | (55) 71 - 8   | (85) 99 - 10  | (115) 100 - 35 |
| (26) 86 - 57 | (56) 96 - 54  | (86) 67 - 20  | (116) 48 - 1   |
| (27) 55 - 13 | (57) 27 - 17  | (87) 61 - 40  | (117) 78 - 40  |
| (28) 92 - 87 | (58) 95 - 10  | (88) 12 - 5   | (118) 83 - 78  |
| (29) 83 - 23 | (59) 95 - 67  | (89) 80 - 19  | (119) 97 - 84  |
| (30) 81 - 2  | (60) 81 - 60  | (90) 99 - 26  | (120) 18 - 2   |

|              |              |              |               |
|--------------|--------------|--------------|---------------|
| (1) 51 - 29  | (31) 80 - 8  | (61) 75 - 47 | (91) 79 - 68  |
| (2) 63 - 5   | (32) 67 - 6  | (62) 80 - 25 | (92) 87 - 32  |
| (3) 63 - 49  | (33) 76 - 45 | (63) 97 - 23 | (93) 78 - 63  |
| (4) 68 - 20  | (34) 81 - 54 | (64) 88 - 15 | (94) 46 - 3   |
| (5) 94 - 57  | (35) 52 - 46 | (65) 60 - 48 | (95) 97 - 48  |
| (6) 81 - 57  | (36) 77 - 39 | (66) 34 - 28 | (96) 61 - 46  |
| (7) 97 - 85  | (37) 49 - 48 | (67) 84 - 39 | (97) 95 - 84  |
| (8) 35 - 15  | (38) 95 - 38 | (68) 68 - 5  | (98) 93 - 28  |
| (9) 36 - 12  | (39) 74 - 55 | (69) 76 - 29 | (99) 86 - 10  |
| (10) 68 - 51 | (40) 37 - 18 | (70) 58 - 20 | (100) 83 - 17 |
| (11) 82 - 23 | (41) 80 - 29 | (71) 85 - 21 | (101) 10 - 1  |
| (12) 26 - 10 | (42) 98 - 87 | (72) 60 - 56 | (102) 5 - 4   |
| (13) 89 - 35 | (43) 50 - 47 | (73) 26 - 10 | (103) 30 - 7  |
| (14) 39 - 25 | (44) 78 - 71 | (74) 90 - 8  | (104) 89 - 30 |
| (15) 80 - 66 | (45) 48 - 47 | (75) 51 - 50 | (105) 98 - 20 |
| (16) 30 - 22 | (46) 94 - 42 | (76) 97 - 34 | (106) 86 - 83 |
| (17) 89 - 15 | (47) 93 - 10 | (77) 45 - 45 | (107) 18 - 12 |
| (18) 61 - 6  | (48) 47 - 42 | (78) 93 - 67 | (108) 70 - 39 |
| (19) 63 - 1  | (49) 52 - 42 | (79) 26 - 20 | (109) 39 - 9  |
| (20) 40 - 39 | (50) 55 - 35 | (80) 68 - 19 | (110) 60 - 37 |
| (21) 29 - 6  | (51) 54 - 8  | (81) 99 - 3  | (111) 48 - 46 |
| (22) 36 - 13 | (52) 76 - 8  | (82) 79 - 74 | (112) 80 - 6  |
| (23) 95 - 41 | (53) 91 - 42 | (83) 87 - 7  | (113) 39 - 34 |
| (24) 71 - 44 | (54) 47 - 12 | (84) 67 - 41 | (114) 48 - 41 |
| (25) 96 - 42 | (55) 12 - 4  | (85) 58 - 22 | (115) 98 - 87 |
| (26) 97 - 55 | (56) 59 - 9  | (86) 73 - 70 | (116) 90 - 31 |
| (27) 24 - 23 | (57) 54 - 41 | (87) 74 - 50 | (117) 87 - 73 |
| (28) 84 - 48 | (58) 76 - 26 | (88) 95 - 70 | (118) 88 - 48 |
| (29) 98 - 67 | (59) 39 - 4  | (89) 65 - 35 | (119) 18 - 12 |
| (30) 91 - 13 | (60) 66 - 41 | (90) 97 - 6  | (120) 75 - 46 |

|              |               |              |               |
|--------------|---------------|--------------|---------------|
| (1) 63 - 22  | (31) 72 - 24  | (61) 97 - 88 | (91) 90 - 23  |
| (2) 40 - 39  | (32) 62 - 51  | (62) 41 - 19 | (92) 94 - 64  |
| (3) 45 - 39  | (33) 18 - 11  | (63) 75 - 57 | (93) 18 - 16  |
| (4) 71 - 12  | (34) 63 - 44  | (64) 33 - 11 | (94) 59 - 30  |
| (5) 83 - 56  | (35) 63 - 57  | (65) 40 - 15 | (95) 82 - 13  |
| (6) 51 - 5   | (36) 84 - 63  | (66) 96 - 25 | (96) 83 - 26  |
| (7) 78 - 73  | (37) 99 - 25  | (67) 65 - 64 | (97) 22 - 18  |
| (8) 86 - 9   | (38) 98 - 53  | (68) 74 - 52 | (98) 96 - 73  |
| (9) 59 - 39  | (39) 85 - 40  | (69) 99 - 8  | (99) 58 - 16  |
| (10) 90 - 61 | (40) 69 - 33  | (70) 60 - 43 | (100) 48 - 7  |
| (11) 85 - 81 | (41) 59 - 54  | (71) 54 - 12 | (101) 50 - 27 |
| (12) 84 - 73 | (42) 62 - 50  | (72) 53 - 36 | (102) 16 - 14 |
| (13) 92 - 87 | (43) 29 - 25  | (73) 99 - 4  | (103) 59 - 29 |
| (14) 93 - 8  | (44) 100 - 62 | (74) 29 - 24 | (104) 47 - 17 |
| (15) 50 - 37 | (45) 98 - 84  | (75) 70 - 64 | (105) 93 - 86 |
| (16) 67 - 57 | (46) 47 - 4   | (76) 39 - 11 | (106) 74 - 9  |
| (17) 98 - 30 | (47) 89 - 51  | (77) 12 - 5  | (107) 44 - 32 |
| (18) 52 - 35 | (48) 97 - 33  | (78) 45 - 14 | (108) 47 - 31 |
| (19) 65 - 19 | (49) 83 - 70  | (79) 54 - 18 | (109) 83 - 11 |
| (20) 64 - 46 | (50) 67 - 19  | (80) 9 - 1   | (110) 73 - 4  |
| (21) 76 - 48 | (51) 94 - 11  | (81) 90 - 83 | (111) 82 - 52 |
| (22) 91 - 59 | (52) 64 - 49  | (82) 71 - 7  | (112) 97 - 72 |
| (23) 68 - 42 | (53) 92 - 80  | (83) 93 - 38 | (113) 85 - 82 |
| (24) 30 - 6  | (54) 65 - 49  | (84) 60 - 22 | (114) 70 - 65 |
| (25) 78 - 46 | (55) 93 - 42  | (85) 88 - 75 | (115) 64 - 49 |
| (26) 60 - 13 | (56) 92 - 23  | (86) 68 - 7  | (116) 85 - 16 |
| (27) 89 - 38 | (57) 78 - 33  | (87) 83 - 34 | (117) 70 - 27 |
| (28) 97 - 47 | (58) 86 - 27  | (88) 70 - 16 | (118) 25 - 9  |
| (29) 58 - 1  | (59) 71 - 66  | (89) 98 - 24 | (119) 44 - 18 |
| (30) 76 - 11 | (60) 77 - 71  | (90) 26 - 3  | (120) 80 - 48 |

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|---------------------|----------------------|---------------------|----------------------|
| (1) $48 - 16 = 32$  | (31) $40 - 24 = 16$  | (61) $79 - 54 = 25$ | (91) $38 - 32 = 6$   |
| (2) $28 - 3 = 25$   | (32) $79 - 66 = 13$  | (62) $60 - 46 = 14$ | (92) $72 - 58 = 14$  |
| (3) $49 - 3 = 46$   | (33) $74 - 26 = 48$  | (63) $17 - 15 = 2$  | (93) $94 - 47 = 47$  |
| (4) $57 - 54 = 3$   | (34) $25 - 4 = 21$   | (64) $98 - 18 = 80$ | (94) $13 - 0 = 13$   |
| (5) $93 - 6 = 87$   | (35) $86 - 51 = 35$  | (65) $58 - 25 = 33$ | (95) $75 - 66 = 9$   |
| (6) $60 - 36 = 24$  | (36) $76 - 65 = 11$  | (66) $67 - 15 = 52$ | (96) $84 - 27 = 57$  |
| (7) $82 - 43 = 39$  | (37) $97 - 23 = 74$  | (67) $65 - 22 = 43$ | (97) $85 - 31 = 54$  |
| (8) $38 - 8 = 30$   | (38) $56 - 52 = 4$   | (68) $16 - 3 = 13$  | (98) $96 - 18 = 78$  |
| (9) $37 - 35 = 2$   | (39) $9 - 3 = 6$     | (69) $85 - 74 = 11$ | (99) $66 - 48 = 18$  |
| (10) $83 - 52 = 31$ | (40) $80 - 31 = 49$  | (70) $78 - 29 = 49$ | (100) $80 - 67 = 13$ |
| (11) $37 - 33 = 4$  | (41) $54 - 53 = 1$   | (71) $98 - 57 = 41$ | (101) $48 - 36 = 12$ |
| (12) $77 - 5 = 72$  | (42) $96 - 33 = 63$  | (72) $76 - 33 = 43$ | (102) $16 - 13 = 3$  |
| (13) $95 - 33 = 62$ | (43) $91 - 47 = 44$  | (73) $82 - 10 = 72$ | (103) $79 - 66 = 13$ |
| (14) $97 - 83 = 14$ | (44) $64 - 19 = 45$  | (74) $83 - 7 = 76$  | (104) $67 - 50 = 17$ |
| (15) $89 - 42 = 47$ | (45) $39 - 12 = 27$  | (75) $70 - 21 = 49$ | (105) $90 - 70 = 20$ |
| (16) $49 - 6 = 43$  | (46) $47 - 29 = 18$  | (76) $74 - 2 = 72$  | (106) $62 - 46 = 16$ |
| (17) $66 - 2 = 64$  | (47) $70 - 4 = 66$   | (77) $29 - 22 = 7$  | (107) $80 - 71 = 9$  |
| (18) $98 - 66 = 32$ | (48) $4 - 0 = 4$     | (78) $55 - 37 = 18$ | (108) $76 - 15 = 61$ |
| (19) $70 - 31 = 39$ | (49) $61 - 43 = 18$  | (79) $74 - 10 = 64$ | (109) $34 - 25 = 9$  |
| (20) $72 - 1 = 71$  | (50) $94 - 24 = 70$  | (80) $25 - 13 = 12$ | (110) $69 - 46 = 23$ |
| (21) $78 - 55 = 23$ | (51) $66 - 61 = 5$   | (81) $58 - 50 = 8$  | (111) $72 - 22 = 50$ |
| (22) $79 - 48 = 31$ | (52) $41 - 36 = 5$   | (82) $57 - 23 = 34$ | (112) $46 - 1 = 45$  |
| (23) $42 - 37 = 5$  | (53) $78 - 13 = 65$  | (83) $67 - 39 = 28$ | (113) $56 - 42 = 14$ |
| (24) $74 - 2 = 72$  | (54) $99 - 98 = 1$   | (84) $87 - 43 = 44$ | (114) $90 - 81 = 9$  |
| (25) $92 - 39 = 53$ | (55) $44 - 7 = 37$   | (85) $77 - 34 = 43$ | (115) $60 - 55 = 5$  |
| (26) $88 - 67 = 21$ | (56) $100 - 73 = 27$ | (86) $68 - 26 = 42$ | (116) $76 - 4 = 72$  |
| (27) $78 - 4 = 74$  | (57) $43 - 20 = 23$  | (87) $29 - 11 = 18$ | (117) $40 - 2 = 38$  |
| (28) $28 - 20 = 8$  | (58) $76 - 28 = 48$  | (88) $98 - 72 = 26$ | (118) $65 - 30 = 35$ |
| (29) $95 - 91 = 4$  | (59) $98 - 24 = 74$  | (89) $60 - 8 = 52$  | (119) $80 - 56 = 24$ |
| (30) $32 - 25 = 7$  | (60) $66 - 40 = 26$  | (90) $84 - 58 = 26$ | (120) $94 - 71 = 23$ |

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|---------------------|---------------------|---------------------|----------------------|
| (1) $96 - 85 = 11$  | (31) $76 - 47 = 29$ | (61) $31 - 12 = 19$ | (91) $91 - 56 = 35$  |
| (2) $66 - 26 = 40$  | (32) $30 - 7 = 23$  | (62) $80 - 40 = 40$ | (92) $100 - 55 = 45$ |
| (3) $91 - 60 = 31$  | (33) $62 - 14 = 48$ | (63) $52 - 47 = 5$  | (93) $24 - 8 = 16$   |
| (4) $93 - 13 = 80$  | (34) $68 - 3 = 65$  | (64) $72 - 3 = 69$  | (94) $95 - 71 = 24$  |
| (5) $26 - 1 = 25$   | (35) $78 - 11 = 67$ | (65) $28 - 3 = 25$  | (95) $24 - 13 = 11$  |
| (6) $76 - 36 = 40$  | (36) $90 - 66 = 24$ | (66) $54 - 12 = 42$ | (96) $33 - 17 = 16$  |
| (7) $85 - 53 = 32$  | (37) $98 - 90 = 8$  | (67) $75 - 22 = 53$ | (97) $52 - 35 = 17$  |
| (8) $63 - 57 = 6$   | (38) $60 - 33 = 27$ | (68) $89 - 27 = 62$ | (98) $98 - 46 = 52$  |
| (9) $87 - 54 = 33$  | (39) $96 - 45 = 51$ | (69) $90 - 9 = 81$  | (99) $76 - 47 = 29$  |
| (10) $26 - 23 = 3$  | (40) $43 - 31 = 12$ | (70) $89 - 73 = 16$ | (100) $94 - 62 = 32$ |
| (11) $33 - 4 = 29$  | (41) $36 - 5 = 31$  | (71) $75 - 66 = 9$  | (101) $91 - 45 = 46$ |
| (12) $85 - 58 = 27$ | (42) $69 - 56 = 13$ | (72) $94 - 58 = 36$ | (102) $55 - 18 = 37$ |
| (13) $38 - 5 = 33$  | (43) $63 - 3 = 60$  | (73) $90 - 5 = 85$  | (103) $23 - 3 = 20$  |
| (14) $48 - 23 = 25$ | (44) $68 - 50 = 18$ | (74) $36 - 32 = 4$  | (104) $43 - 18 = 25$ |
| (15) $96 - 60 = 36$ | (45) $46 - 25 = 21$ | (75) $79 - 22 = 57$ | (105) $82 - 33 = 49$ |
| (16) $46 - 9 = 37$  | (46) $26 - 5 = 21$  | (76) $26 - 23 = 3$  | (106) $97 - 66 = 31$ |
| (17) $99 - 50 = 49$ | (47) $24 - 5 = 19$  | (77) $95 - 86 = 9$  | (107) $50 - 26 = 24$ |
| (18) $35 - 1 = 34$  | (48) $86 - 63 = 23$ | (78) $78 - 40 = 38$ | (108) $77 - 22 = 55$ |
| (19) $41 - 25 = 16$ | (49) $59 - 31 = 28$ | (79) $23 - 2 = 21$  | (109) $98 - 32 = 66$ |
| (20) $55 - 21 = 34$ | (50) $89 - 22 = 67$ | (80) $91 - 32 = 59$ | (110) $96 - 71 = 25$ |
| (21) $46 - 27 = 19$ | (51) $47 - 11 = 36$ | (81) $63 - 54 = 9$  | (111) $79 - 76 = 3$  |
| (22) $71 - 28 = 43$ | (52) $58 - 9 = 49$  | (82) $82 - 7 = 75$  | (112) $87 - 40 = 47$ |
| (23) $73 - 66 = 7$  | (53) $53 - 22 = 31$ | (83) $41 - 13 = 28$ | (113) $57 - 48 = 9$  |
| (24) $83 - 77 = 6$  | (54) $19 - 5 = 14$  | (84) $95 - 27 = 68$ | (114) $64 - 46 = 18$ |
| (25) $67 - 32 = 35$ | (55) $64 - 30 = 34$ | (85) $36 - 10 = 26$ | (115) $50 - 3 = 47$  |
| (26) $37 - 21 = 16$ | (56) $32 - 29 = 3$  | (86) $65 - 44 = 21$ | (116) $100 - 98 = 2$ |
| (27) $92 - 56 = 36$ | (57) $91 - 57 = 34$ | (87) $84 - 79 = 5$  | (117) $48 - 6 = 42$  |
| (28) $62 - 39 = 23$ | (58) $73 - 4 = 69$  | (88) $52 - 15 = 37$ | (118) $63 - 23 = 40$ |
| (29) $91 - 59 = 32$ | (59) $69 - 51 = 18$ | (89) $26 - 21 = 5$  | (119) $98 - 1 = 97$  |
| (30) $36 - 34 = 2$  | (60) $74 - 5 = 69$  | (90) $31 - 10 = 21$ | (120) $87 - 37 = 50$ |

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|---------------------|----------------------|---------------------|----------------------|
| (1) $64 - 50 = 14$  | (31) $95 - 52 = 43$  | (61) $75 - 25 = 50$ | (91) $79 - 18 = 61$  |
| (2) $42 - 15 = 27$  | (32) $77 - 57 = 20$  | (62) $77 - 48 = 29$ | (92) $33 - 19 = 14$  |
| (3) $61 - 27 = 34$  | (33) $86 - 20 = 66$  | (63) $83 - 57 = 26$ | (93) $64 - 31 = 33$  |
| (4) $46 - 17 = 29$  | (34) $66 - 38 = 28$  | (64) $89 - 56 = 33$ | (94) $49 - 39 = 10$  |
| (5) $35 - 21 = 14$  | (35) $95 - 89 = 6$   | (65) $23 - 13 = 10$ | (95) $68 - 55 = 13$  |
| (6) $77 - 44 = 33$  | (36) $85 - 13 = 72$  | (66) $75 - 36 = 39$ | (96) $93 - 70 = 23$  |
| (7) $53 - 12 = 41$  | (37) $52 - 49 = 3$   | (67) $79 - 37 = 42$ | (97) $56 - 40 = 16$  |
| (8) $63 - 7 = 56$   | (38) $29 - 21 = 8$   | (68) $75 - 70 = 5$  | (98) $81 - 81 = 0$   |
| (9) $57 - 17 = 40$  | (39) $54 - 51 = 3$   | (69) $65 - 49 = 16$ | (99) $93 - 52 = 41$  |
| (10) $72 - 63 = 9$  | (40) $92 - 57 = 35$  | (70) $28 - 20 = 8$  | (100) $43 - 36 = 7$  |
| (11) $99 - 2 = 97$  | (41) $40 - 14 = 26$  | (71) $92 - 81 = 11$ | (101) $90 - 29 = 61$ |
| (12) $73 - 32 = 41$ | (42) $50 - 2 = 48$   | (72) $31 - 15 = 16$ | (102) $89 - 9 = 80$  |
| (13) $42 - 24 = 18$ | (43) $100 - 73 = 27$ | (73) $87 - 83 = 4$  | (103) $36 - 0 = 36$  |
| (14) $50 - 28 = 22$ | (44) $37 - 26 = 11$  | (74) $86 - 14 = 72$ | (104) $34 - 29 = 5$  |
| (15) $55 - 39 = 16$ | (45) $65 - 25 = 40$  | (75) $30 - 3 = 27$  | (105) $50 - 33 = 17$ |
| (16) $88 - 31 = 57$ | (46) $82 - 38 = 44$  | (76) $91 - 33 = 58$ | (106) $82 - 39 = 43$ |
| (17) $72 - 30 = 42$ | (47) $42 - 8 = 34$   | (77) $59 - 14 = 45$ | (107) $55 - 30 = 25$ |
| (18) $54 - 7 = 47$  | (48) $58 - 32 = 26$  | (78) $94 - 52 = 42$ | (108) $82 - 9 = 73$  |
| (19) $73 - 26 = 47$ | (49) $97 - 91 = 6$   | (79) $87 - 56 = 31$ | (109) $90 - 9 = 81$  |
| (20) $13 - 3 = 10$  | (50) $97 - 44 = 53$  | (80) $93 - 88 = 5$  | (110) $64 - 40 = 24$ |
| (21) $74 - 73 = 1$  | (51) $40 - 22 = 18$  | (81) $89 - 50 = 39$ | (111) $38 - 33 = 5$  |
| (22) $60 - 13 = 47$ | (52) $87 - 0 = 87$   | (82) $55 - 36 = 19$ | (112) $57 - 33 = 24$ |
| (23) $61 - 54 = 7$  | (53) $41 - 38 = 3$   | (83) $65 - 22 = 43$ | (113) $82 - 69 = 13$ |
| (24) $30 - 22 = 8$  | (54) $17 - 5 = 12$   | (84) $87 - 55 = 32$ | (114) $51 - 19 = 32$ |
| (25) $65 - 52 = 13$ | (55) $88 - 19 = 69$  | (85) $63 - 37 = 26$ | (115) $95 - 89 = 6$  |
| (26) $43 - 25 = 18$ | (56) $96 - 47 = 49$  | (86) $62 - 9 = 53$  | (116) $62 - 51 = 11$ |
| (27) $46 - 43 = 3$  | (57) $68 - 51 = 17$  | (87) $91 - 38 = 53$ | (117) $85 - 77 = 8$  |
| (28) $64 - 26 = 38$ | (58) $84 - 47 = 37$  | (88) $41 - 20 = 21$ | (118) $12 - 10 = 2$  |
| (29) $56 - 36 = 20$ | (59) $81 - 67 = 14$  | (89) $79 - 66 = 13$ | (119) $88 - 52 = 36$ |
| (30) $90 - 85 = 5$  | (60) $22 - 4 = 18$   | (90) $30 - 26 = 4$  | (120) $76 - 14 = 62$ |

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|----------------------|----------------------|---------------------|-----------------------|
| (1) $60 - 17 = 43$   | (31) $82 - 45 = 37$  | (61) $73 - 36 = 37$ | (91) $64 - 59 = 5$    |
| (2) $86 - 59 = 27$   | (32) $79 - 10 = 69$  | (62) $62 - 26 = 36$ | (92) $66 - 7 = 59$    |
| (3) $70 - 20 = 50$   | (33) $77 - 66 = 11$  | (63) $17 - 9 = 8$   | (93) $55 - 53 = 2$    |
| (4) $72 - 51 = 21$   | (34) $72 - 4 = 68$   | (64) $91 - 73 = 18$ | (94) $81 - 36 = 45$   |
| (5) $44 - 2 = 42$    | (35) $89 - 1 = 88$   | (65) $84 - 8 = 76$  | (95) $93 - 69 = 24$   |
| (6) $77 - 47 = 30$   | (36) $19 - 11 = 8$   | (66) $59 - 33 = 26$ | (96) $31 - 29 = 2$    |
| (7) $35 - 30 = 5$    | (37) $64 - 29 = 35$  | (67) $67 - 5 = 62$  | (97) $86 - 27 = 59$   |
| (8) $27 - 4 = 23$    | (38) $21 - 19 = 2$   | (68) $65 - 48 = 17$ | (98) $50 - 28 = 22$   |
| (9) $86 - 11 = 75$   | (39) $98 - 12 = 86$  | (69) $88 - 67 = 21$ | (99) $92 - 63 = 29$   |
| (10) $89 - 59 = 30$  | (40) $97 - 89 = 8$   | (70) $2 - 1 = 1$    | (100) $100 - 24 = 76$ |
| (11) $91 - 62 = 29$  | (41) $96 - 11 = 85$  | (71) $38 - 28 = 10$ | (101) $70 - 67 = 3$   |
| (12) $25 - 7 = 18$   | (42) $90 - 50 = 40$  | (72) $61 - 54 = 7$  | (102) $77 - 63 = 14$  |
| (13) $92 - 80 = 12$  | (43) $68 - 42 = 26$  | (73) $21 - 8 = 13$  | (103) $71 - 56 = 15$  |
| (14) $72 - 62 = 10$  | (44) $98 - 93 = 5$   | (74) $52 - 21 = 31$ | (104) $44 - 18 = 26$  |
| (15) $92 - 48 = 44$  | (45) $66 - 12 = 54$  | (75) $63 - 11 = 52$ | (105) $16 - 13 = 3$   |
| (16) $58 - 34 = 24$  | (46) $82 - 73 = 9$   | (76) $85 - 3 = 82$  | (106) $40 - 12 = 28$  |
| (17) $83 - 81 = 2$   | (47) $59 - 16 = 43$  | (77) $40 - 4 = 36$  | (107) $30 - 26 = 4$   |
| (18) $34 - 19 = 15$  | (48) $42 - 20 = 22$  | (78) $59 - 13 = 46$ | (108) $78 - 66 = 12$  |
| (19) $90 - 68 = 22$  | (49) $72 - 59 = 13$  | (79) $81 - 41 = 40$ | (109) $28 - 17 = 11$  |
| (20) $68 - 8 = 60$   | (50) $100 - 30 = 70$ | (80) $97 - 71 = 26$ | (110) $77 - 32 = 45$  |
| (21) $100 - 75 = 25$ | (51) $45 - 24 = 21$  | (81) $91 - 44 = 47$ | (111) $74 - 5 = 69$   |
| (22) $45 - 1 = 44$   | (52) $93 - 10 = 83$  | (82) $67 - 62 = 5$  | (112) $76 - 32 = 44$  |
| (23) $91 - 5 = 86$   | (53) $72 - 57 = 15$  | (83) $75 - 12 = 63$ | (113) $74 - 58 = 16$  |
| (24) $77 - 45 = 32$  | (54) $54 - 7 = 47$   | (84) $67 - 30 = 37$ | (114) $88 - 40 = 48$  |
| (25) $25 - 9 = 16$   | (55) $91 - 68 = 23$  | (85) $60 - 17 = 43$ | (115) $61 - 4 = 57$   |
| (26) $55 - 35 = 20$  | (56) $78 - 27 = 51$  | (86) $83 - 74 = 9$  | (116) $14 - 4 = 10$   |
| (27) $87 - 58 = 29$  | (57) $38 - 38 = 0$   | (87) $82 - 29 = 53$ | (117) $52 - 24 = 28$  |
| (28) $68 - 26 = 42$  | (58) $77 - 58 = 19$  | (88) $46 - 40 = 6$  | (118) $99 - 37 = 62$  |
| (29) $75 - 15 = 60$  | (59) $80 - 56 = 24$  | (89) $78 - 72 = 6$  | (119) $30 - 28 = 2$   |
| (30) $42 - 36 = 6$   | (60) $45 - 45 = 0$   | (90) $90 - 11 = 79$ | (120) $68 - 64 = 4$   |

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|----------------------|---------------------|---------------------|----------------------|
| (1) $93 - 20 = 73$   | (31) $54 - 5 = 49$  | (61) $41 - 25 = 16$ | (91) $94 - 62 = 32$  |
| (2) $92 - 39 = 53$   | (32) $58 - 27 = 31$ | (62) $81 - 4 = 77$  | (92) $43 - 18 = 25$  |
| (3) $43 - 20 = 23$   | (33) $78 - 42 = 36$ | (63) $11 - 11 = 0$  | (93) $91 - 52 = 39$  |
| (4) $46 - 33 = 13$   | (34) $72 - 29 = 43$ | (64) $89 - 24 = 65$ | (94) $100 - 35 = 65$ |
| (5) $61 - 44 = 17$   | (35) $80 - 73 = 7$  | (65) $85 - 1 = 84$  | (95) $74 - 71 = 3$   |
| (6) $86 - 55 = 31$   | (36) $94 - 9 = 85$  | (66) $73 - 30 = 43$ | (96) $80 - 6 = 74$   |
| (7) $66 - 36 = 30$   | (37) $92 - 33 = 59$ | (67) $48 - 27 = 21$ | (97) $36 - 32 = 4$   |
| (8) $80 - 63 = 17$   | (38) $94 - 41 = 53$ | (68) $54 - 52 = 2$  | (98) $71 - 19 = 52$  |
| (9) $83 - 48 = 35$   | (39) $66 - 48 = 18$ | (69) $62 - 1 = 61$  | (99) $95 - 86 = 9$   |
| (10) $19 - 11 = 8$   | (40) $51 - 5 = 46$  | (70) $45 - 3 = 42$  | (100) $73 - 56 = 17$ |
| (11) $67 - 31 = 36$  | (41) $68 - 12 = 56$ | (71) $91 - 44 = 47$ | (101) $48 - 19 = 29$ |
| (12) $43 - 28 = 15$  | (42) $63 - 14 = 49$ | (72) $77 - 5 = 72$  | (102) $80 - 78 = 2$  |
| (13) $82 - 57 = 25$  | (43) $89 - 11 = 78$ | (73) $82 - 9 = 73$  | (103) $22 - 3 = 19$  |
| (14) $77 - 12 = 65$  | (44) $66 - 17 = 49$ | (74) $81 - 73 = 8$  | (104) $34 - 13 = 21$ |
| (15) $80 - 49 = 31$  | (45) $85 - 47 = 38$ | (75) $20 - 18 = 2$  | (105) $75 - 32 = 43$ |
| (16) $3 - 1 = 2$     | (46) $69 - 20 = 49$ | (76) $27 - 14 = 13$ | (106) $71 - 49 = 22$ |
| (17) $83 - 20 = 63$  | (47) $76 - 75 = 1$  | (77) $20 - 7 = 13$  | (107) $75 - 11 = 64$ |
| (18) $67 - 4 = 63$   | (48) $13 - 3 = 10$  | (78) $88 - 53 = 35$ | (108) $91 - 78 = 13$ |
| (19) $60 - 19 = 41$  | (49) $23 - 22 = 1$  | (79) $29 - 13 = 16$ | (109) $29 - 9 = 20$  |
| (20) $2 - 0 = 2$     | (50) $57 - 24 = 33$ | (80) $95 - 32 = 63$ | (110) $92 - 19 = 73$ |
| (21) $52 - 4 = 48$   | (51) $49 - 11 = 38$ | (81) $17 - 8 = 9$   | (111) $34 - 22 = 12$ |
| (22) $34 - 7 = 27$   | (52) $91 - 90 = 1$  | (82) $7 - 4 = 3$    | (112) $91 - 77 = 14$ |
| (23) $98 - 31 = 67$  | (53) $39 - 4 = 35$  | (83) $96 - 88 = 8$  | (113) $90 - 68 = 22$ |
| (24) $66 - 56 = 10$  | (54) $99 - 32 = 67$ | (84) $95 - 71 = 24$ | (114) $73 - 0 = 73$  |
| (25) $42 - 25 = 17$  | (55) $94 - 45 = 49$ | (85) $40 - 11 = 29$ | (115) $88 - 57 = 31$ |
| (26) $91 - 56 = 35$  | (56) $74 - 67 = 7$  | (86) $63 - 44 = 19$ | (116) $31 - 27 = 4$  |
| (27) $100 - 79 = 21$ | (57) $90 - 60 = 30$ | (87) $69 - 16 = 53$ | (117) $87 - 36 = 51$ |
| (28) $59 - 13 = 46$  | (58) $5 - 4 = 1$    | (88) $84 - 21 = 63$ | (118) $45 - 29 = 16$ |
| (29) $97 - 56 = 41$  | (59) $76 - 12 = 64$ | (89) $58 - 31 = 27$ | (119) $77 - 29 = 48$ |
| (30) $80 - 59 = 21$  | (60) $69 - 31 = 38$ | (90) $39 - 28 = 11$ | (120) $50 - 18 = 32$ |



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|---------------------|---------------------|---------------------|----------------------|
| (1) $28 - 14 = 14$  | (31) $79 - 18 = 61$ | (61) $18 - 12 = 6$  | (91) $70 - 14 = 56$  |
| (2) $6 - 5 = 1$     | (32) $64 - 10 = 54$ | (62) $89 - 44 = 45$ | (92) $35 - 31 = 4$   |
| (3) $34 - 14 = 20$  | (33) $90 - 2 = 88$  | (63) $58 - 54 = 4$  | (93) $89 - 24 = 65$  |
| (4) $38 - 7 = 31$   | (34) $62 - 44 = 18$ | (64) $98 - 73 = 25$ | (94) $88 - 23 = 65$  |
| (5) $45 - 3 = 42$   | (35) $10 - 10 = 0$  | (65) $88 - 19 = 69$ | (95) $91 - 64 = 27$  |
| (6) $83 - 4 = 79$   | (36) $3 - 3 = 0$    | (66) $44 - 33 = 11$ | (96) $88 - 58 = 30$  |
| (7) $88 - 49 = 39$  | (37) $84 - 71 = 13$ | (67) $68 - 28 = 40$ | (97) $97 - 52 = 45$  |
| (8) $45 - 24 = 21$  | (38) $71 - 36 = 35$ | (68) $97 - 81 = 16$ | (98) $66 - 26 = 40$  |
| (9) $72 - 46 = 26$  | (39) $63 - 11 = 52$ | (69) $94 - 41 = 53$ | (99) $96 - 20 = 76$  |
| (10) $69 - 53 = 16$ | (40) $67 - 48 = 19$ | (70) $46 - 14 = 32$ | (100) $73 - 7 = 66$  |
| (11) $54 - 8 = 46$  | (41) $55 - 3 = 52$  | (71) $82 - 47 = 35$ | (101) $66 - 42 = 24$ |
| (12) $49 - 11 = 38$ | (42) $98 - 32 = 66$ | (72) $71 - 16 = 55$ | (102) $98 - 32 = 66$ |
| (13) $43 - 29 = 14$ | (43) $99 - 84 = 15$ | (73) $83 - 4 = 79$  | (103) $56 - 27 = 29$ |
| (14) $44 - 21 = 23$ | (44) $94 - 76 = 18$ | (74) $98 - 43 = 55$ | (104) $62 - 25 = 37$ |
| (15) $40 - 20 = 20$ | (45) $92 - 18 = 74$ | (75) $74 - 62 = 12$ | (105) $89 - 38 = 51$ |
| (16) $47 - 26 = 21$ | (46) $61 - 3 = 58$  | (76) $31 - 29 = 2$  | (106) $83 - 18 = 65$ |
| (17) $57 - 43 = 14$ | (47) $100 - 3 = 97$ | (77) $91 - 9 = 82$  | (107) $64 - 45 = 19$ |
| (18) $18 - 12 = 6$  | (48) $77 - 36 = 41$ | (78) $79 - 35 = 44$ | (108) $76 - 65 = 11$ |
| (19) $92 - 37 = 55$ | (49) $92 - 31 = 61$ | (79) $32 - 25 = 7$  | (109) $78 - 70 = 8$  |
| (20) $45 - 28 = 17$ | (50) $43 - 39 = 4$  | (80) $57 - 1 = 56$  | (110) $96 - 21 = 75$ |
| (21) $45 - 38 = 7$  | (51) $67 - 56 = 11$ | (81) $97 - 6 = 91$  | (111) $75 - 35 = 40$ |
| (22) $98 - 27 = 71$ | (52) $77 - 41 = 36$ | (82) $27 - 12 = 15$ | (112) $78 - 17 = 61$ |
| (23) $94 - 78 = 16$ | (53) $49 - 1 = 48$  | (83) $72 - 71 = 1$  | (113) $84 - 48 = 36$ |
| (24) $17 - 12 = 5$  | (54) $30 - 4 = 26$  | (84) $93 - 5 = 88$  | (114) $83 - 41 = 42$ |
| (25) $34 - 10 = 24$ | (55) $94 - 40 = 54$ | (85) $32 - 23 = 9$  | (115) $73 - 46 = 27$ |
| (26) $91 - 23 = 68$ | (56) $65 - 36 = 29$ | (86) $81 - 3 = 78$  | (116) $16 - 4 = 12$  |
| (27) $82 - 30 = 52$ | (57) $20 - 18 = 2$  | (87) $49 - 40 = 9$  | (117) $59 - 7 = 52$  |
| (28) $70 - 4 = 66$  | (58) $44 - 2 = 42$  | (88) $94 - 2 = 92$  | (118) $83 - 46 = 37$ |
| (29) $63 - 4 = 59$  | (59) $60 - 5 = 55$  | (89) $96 - 65 = 31$ | (119) $36 - 34 = 2$  |
| (30) $90 - 31 = 59$ | (60) $93 - 79 = 14$ | (90) $64 - 33 = 31$ | (120) $94 - 70 = 24$ |

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|---------------------|----------------------|---------------------|----------------------|
| (1) $29 - 18 = 11$  | (31) $79 - 42 = 37$  | (61) $45 - 7 = 38$  | (91) $17 - 0 = 17$   |
| (2) $91 - 76 = 15$  | (32) $89 - 44 = 45$  | (62) $60 - 41 = 19$ | (92) $68 - 44 = 24$  |
| (3) $74 - 28 = 46$  | (33) $89 - 45 = 44$  | (63) $67 - 7 = 60$  | (93) $59 - 39 = 20$  |
| (4) $86 - 64 = 22$  | (34) $77 - 35 = 42$  | (64) $67 - 18 = 49$ | (94) $97 - 58 = 39$  |
| (5) $72 - 37 = 35$  | (35) $86 - 44 = 42$  | (65) $29 - 3 = 26$  | (95) $32 - 7 = 25$   |
| (6) $84 - 16 = 68$  | (36) $58 - 9 = 49$   | (66) $40 - 29 = 11$ | (96) $25 - 3 = 22$   |
| (7) $74 - 11 = 63$  | (37) $73 - 55 = 18$  | (67) $98 - 1 = 97$  | (97) $49 - 12 = 37$  |
| (8) $81 - 77 = 4$   | (38) $77 - 5 = 72$   | (68) $8 - 2 = 6$    | (98) $100 - 22 = 78$ |
| (9) $46 - 17 = 29$  | (39) $63 - 38 = 25$  | (69) $78 - 58 = 20$ | (99) $69 - 55 = 14$  |
| (10) $42 - 24 = 18$ | (40) $24 - 22 = 2$   | (70) $84 - 65 = 19$ | (100) $100 - 94 = 6$ |
| (11) $93 - 43 = 50$ | (41) $78 - 21 = 57$  | (71) $60 - 44 = 16$ | (101) $77 - 65 = 12$ |
| (12) $94 - 68 = 26$ | (42) $67 - 15 = 52$  | (72) $24 - 20 = 4$  | (102) $12 - 7 = 5$   |
| (13) $89 - 72 = 17$ | (43) $98 - 69 = 29$  | (73) $70 - 69 = 1$  | (103) $80 - 59 = 21$ |
| (14) $84 - 26 = 58$ | (44) $81 - 28 = 53$  | (74) $75 - 30 = 45$ | (104) $95 - 92 = 3$  |
| (15) $39 - 25 = 14$ | (45) $87 - 59 = 28$  | (75) $95 - 93 = 2$  | (105) $84 - 67 = 17$ |
| (16) $88 - 72 = 16$ | (46) $48 - 6 = 42$   | (76) $65 - 60 = 5$  | (106) $95 - 36 = 59$ |
| (17) $82 - 7 = 75$  | (47) $100 - 46 = 54$ | (77) $78 - 65 = 13$ | (107) $79 - 52 = 27$ |
| (18) $22 - 13 = 9$  | (48) $65 - 19 = 46$  | (78) $62 - 16 = 46$ | (108) $80 - 50 = 30$ |
| (19) $6 - 4 = 2$    | (49) $98 - 53 = 45$  | (79) $66 - 8 = 58$  | (109) $79 - 21 = 58$ |
| (20) $94 - 52 = 42$ | (50) $20 - 12 = 8$   | (80) $3 - 0 = 3$    | (110) $58 - 56 = 2$  |
| (21) $60 - 2 = 58$  | (51) $69 - 58 = 11$  | (81) $26 - 21 = 5$  | (111) $71 - 17 = 54$ |
| (22) $68 - 66 = 2$  | (52) $53 - 33 = 20$  | (82) $64 - 20 = 44$ | (112) $16 - 3 = 13$  |
| (23) $72 - 48 = 24$ | (53) $60 - 48 = 12$  | (83) $76 - 40 = 36$ | (113) $88 - 12 = 76$ |
| (24) $68 - 27 = 41$ | (54) $63 - 51 = 12$  | (84) $26 - 14 = 12$ | (114) $66 - 57 = 9$  |
| (25) $71 - 69 = 2$  | (55) $70 - 47 = 23$  | (85) $99 - 84 = 15$ | (115) $50 - 8 = 42$  |
| (26) $7 - 5 = 2$    | (56) $80 - 37 = 43$  | (86) $53 - 36 = 17$ | (116) $37 - 16 = 21$ |
| (27) $50 - 30 = 20$ | (57) $8 - 5 = 3$     | (87) $94 - 81 = 13$ | (117) $76 - 24 = 52$ |
| (28) $90 - 88 = 2$  | (58) $49 - 18 = 31$  | (88) $58 - 23 = 35$ | (118) $99 - 19 = 80$ |
| (29) $50 - 18 = 32$ | (59) $84 - 37 = 47$  | (89) $45 - 31 = 14$ | (119) $81 - 0 = 81$  |
| (30) $65 - 6 = 59$  | (60) $81 - 65 = 16$  | (90) $19 - 16 = 3$  | (120) $98 - 18 = 80$ |

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|----------------------|---------------------|---------------------|----------------------|
| (1) $37 - 12 = 25$   | (31) $85 - 3 = 82$  | (61) $74 - 1 = 73$  | (91) $64 - 58 = 6$   |
| (2) $51 - 15 = 36$   | (32) $99 - 46 = 53$ | (62) $13 - 11 = 2$  | (92) $72 - 67 = 5$   |
| (3) $96 - 94 = 2$    | (33) $27 - 10 = 17$ | (63) $25 - 18 = 7$  | (93) $79 - 64 = 15$  |
| (4) $98 - 67 = 31$   | (34) $91 - 59 = 32$ | (64) $29 - 17 = 12$ | (94) $97 - 82 = 15$  |
| (5) $78 - 53 = 25$   | (35) $74 - 65 = 9$  | (65) $24 - 16 = 8$  | (95) $38 - 16 = 22$  |
| (6) $99 - 93 = 6$    | (36) $42 - 29 = 13$ | (66) $75 - 1 = 74$  | (96) $79 - 64 = 15$  |
| (7) $79 - 37 = 42$   | (37) $59 - 45 = 14$ | (67) $69 - 20 = 49$ | (97) $58 - 32 = 26$  |
| (8) $52 - 38 = 14$   | (38) $92 - 52 = 40$ | (68) $64 - 58 = 6$  | (98) $23 - 2 = 21$   |
| (9) $70 - 36 = 34$   | (39) $55 - 11 = 44$ | (69) $58 - 22 = 36$ | (99) $84 - 31 = 53$  |
| (10) $76 - 58 = 18$  | (40) $85 - 71 = 14$ | (70) $85 - 35 = 50$ | (100) $88 - 33 = 55$ |
| (11) $48 - 18 = 30$  | (41) $99 - 50 = 49$ | (71) $78 - 8 = 70$  | (101) $20 - 8 = 12$  |
| (12) $68 - 39 = 29$  | (42) $94 - 12 = 82$ | (72) $76 - 9 = 67$  | (102) $70 - 58 = 12$ |
| (13) $30 - 22 = 8$   | (43) $71 - 1 = 70$  | (73) $86 - 15 = 71$ | (103) $39 - 27 = 12$ |
| (14) $85 - 39 = 46$  | (44) $56 - 8 = 48$  | (74) $85 - 51 = 34$ | (104) $72 - 4 = 68$  |
| (15) $79 - 1 = 78$   | (45) $51 - 7 = 44$  | (75) $54 - 21 = 33$ | (105) $85 - 62 = 23$ |
| (16) $67 - 14 = 53$  | (46) $39 - 13 = 26$ | (76) $77 - 9 = 68$  | (106) $73 - 21 = 52$ |
| (17) $94 - 32 = 62$  | (47) $91 - 14 = 77$ | (77) $47 - 2 = 45$  | (107) $67 - 15 = 52$ |
| (18) $89 - 88 = 1$   | (48) $53 - 31 = 22$ | (78) $37 - 24 = 13$ | (108) $53 - 10 = 43$ |
| (19) $47 - 43 = 4$   | (49) $91 - 84 = 7$  | (79) $88 - 26 = 62$ | (109) $40 - 15 = 25$ |
| (20) $100 - 65 = 35$ | (50) $66 - 43 = 23$ | (80) $75 - 75 = 0$  | (110) $69 - 60 = 9$  |
| (21) $91 - 42 = 49$  | (51) $81 - 9 = 72$  | (81) $98 - 26 = 72$ | (111) $98 - 39 = 59$ |
| (22) $55 - 8 = 47$   | (52) $30 - 11 = 19$ | (82) $76 - 4 = 72$  | (112) $77 - 54 = 23$ |
| (23) $50 - 38 = 12$  | (53) $99 - 89 = 10$ | (83) $88 - 80 = 8$  | (113) $89 - 10 = 79$ |
| (24) $56 - 11 = 45$  | (54) $62 - 26 = 36$ | (84) $25 - 20 = 5$  | (114) $74 - 32 = 42$ |
| (25) $91 - 54 = 37$  | (55) $78 - 66 = 12$ | (85) $64 - 47 = 17$ | (115) $65 - 39 = 26$ |
| (26) $45 - 30 = 15$  | (56) $86 - 8 = 78$  | (86) $38 - 26 = 12$ | (116) $91 - 82 = 9$  |
| (27) $48 - 41 = 7$   | (57) $40 - 13 = 27$ | (87) $43 - 40 = 3$  | (117) $81 - 17 = 64$ |
| (28) $15 - 11 = 4$   | (58) $57 - 36 = 21$ | (88) $51 - 41 = 10$ | (118) $73 - 43 = 30$ |
| (29) $31 - 11 = 20$  | (59) $90 - 86 = 4$  | (89) $50 - 0 = 50$  | (119) $1 - 0 = 1$    |
| (30) $57 - 31 = 26$  | (60) $75 - 71 = 4$  | (90) $41 - 18 = 23$ | (120) $88 - 40 = 48$ |

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|---------------------|----------------------|---------------------|----------------------|
| (1) $32 - 29 = 3$   | (31) $34 - 15 = 19$  | (61) $94 - 0 = 94$  | (91) $94 - 20 = 74$  |
| (2) $50 - 48 = 2$   | (32) $36 - 19 = 17$  | (62) $70 - 3 = 67$  | (92) $95 - 83 = 12$  |
| (3) $82 - 59 = 23$  | (33) $85 - 25 = 60$  | (63) $81 - 20 = 61$ | (93) $95 - 32 = 63$  |
| (4) $38 - 29 = 9$   | (34) $36 - 26 = 10$  | (64) $57 - 32 = 25$ | (94) $81 - 37 = 44$  |
| (5) $19 - 18 = 1$   | (35) $86 - 6 = 80$   | (65) $94 - 45 = 49$ | (95) $64 - 21 = 43$  |
| (6) $47 - 3 = 44$   | (36) $93 - 62 = 31$  | (66) $78 - 61 = 17$ | (96) $99 - 1 = 98$   |
| (7) $95 - 46 = 49$  | (37) $54 - 40 = 14$  | (67) $84 - 52 = 32$ | (97) $98 - 24 = 74$  |
| (8) $49 - 22 = 27$  | (38) $62 - 4 = 58$   | (68) $64 - 31 = 33$ | (98) $91 - 77 = 14$  |
| (9) $53 - 11 = 42$  | (39) $86 - 77 = 9$   | (69) $98 - 23 = 75$ | (99) $86 - 46 = 40$  |
| (10) $72 - 62 = 10$ | (40) $92 - 58 = 34$  | (70) $40 - 40 = 0$  | (100) $81 - 46 = 35$ |
| (11) $77 - 31 = 46$ | (41) $78 - 49 = 29$  | (71) $47 - 7 = 40$  | (101) $67 - 44 = 23$ |
| (12) $98 - 91 = 7$  | (42) $92 - 31 = 61$  | (72) $43 - 33 = 10$ | (102) $46 - 39 = 7$  |
| (13) $79 - 20 = 59$ | (43) $99 - 15 = 84$  | (73) $99 - 73 = 26$ | (103) $70 - 16 = 54$ |
| (14) $63 - 5 = 58$  | (44) $64 - 9 = 55$   | (74) $40 - 39 = 1$  | (104) $86 - 79 = 7$  |
| (15) $53 - 27 = 26$ | (45) $50 - 12 = 38$  | (75) $61 - 14 = 47$ | (105) $54 - 26 = 28$ |
| (16) $59 - 33 = 26$ | (46) $33 - 3 = 30$   | (76) $69 - 58 = 11$ | (106) $83 - 25 = 58$ |
| (17) $50 - 36 = 14$ | (47) $71 - 41 = 30$  | (77) $93 - 21 = 72$ | (107) $87 - 29 = 58$ |
| (18) $69 - 31 = 38$ | (48) $92 - 23 = 69$  | (78) $22 - 11 = 11$ | (108) $78 - 37 = 41$ |
| (19) $73 - 15 = 58$ | (49) $76 - 13 = 63$  | (79) $85 - 68 = 17$ | (109) $76 - 29 = 47$ |
| (20) $68 - 39 = 29$ | (50) $82 - 58 = 24$  | (80) $74 - 58 = 16$ | (110) $75 - 35 = 40$ |
| (21) $43 - 35 = 8$  | (51) $90 - 35 = 55$  | (81) $51 - 34 = 17$ | (111) $92 - 5 = 87$  |
| (22) $73 - 15 = 58$ | (52) $7 - 5 = 2$     | (82) $42 - 21 = 21$ | (112) $83 - 83 = 0$  |
| (23) $66 - 58 = 8$  | (53) $83 - 8 = 75$   | (83) $51 - 18 = 33$ | (113) $95 - 4 = 91$  |
| (24) $52 - 21 = 31$ | (54) $64 - 35 = 29$  | (84) $20 - 8 = 12$  | (114) $8 - 6 = 2$    |
| (25) $28 - 13 = 15$ | (55) $91 - 3 = 88$   | (85) $46 - 13 = 33$ | (115) $58 - 53 = 5$  |
| (26) $63 - 37 = 26$ | (56) $49 - 44 = 5$   | (86) $31 - 15 = 16$ | (116) $69 - 62 = 7$  |
| (27) $37 - 16 = 21$ | (57) $85 - 71 = 14$  | (87) $92 - 47 = 45$ | (117) $10 - 9 = 1$   |
| (28) $39 - 9 = 30$  | (58) $49 - 26 = 23$  | (88) $45 - 34 = 11$ | (118) $35 - 33 = 2$  |
| (29) $48 - 10 = 38$ | (59) $100 - 90 = 10$ | (89) $73 - 39 = 34$ | (119) $27 - 8 = 19$  |
| (30) $57 - 31 = 26$ | (60) $98 - 81 = 17$  | (90) $86 - 11 = 75$ | (120) $50 - 11 = 39$ |

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|----------------------|----------------------|----------------------|-----------------------|
| (1) $63 - 38 = 25$   | (31) $63 - 60 = 3$   | (61) $92 - 0 = 92$   | (91) $58 - 33 = 25$   |
| (2) $12 - 5 = 7$     | (32) $67 - 60 = 7$   | (62) $88 - 52 = 36$  | (92) $49 - 45 = 4$    |
| (3) $54 - 5 = 49$    | (33) $94 - 72 = 22$  | (63) $91 - 27 = 64$  | (93) $70 - 58 = 12$   |
| (4) $90 - 72 = 18$   | (34) $67 - 36 = 31$  | (64) $52 - 7 = 45$   | (94) $43 - 19 = 24$   |
| (5) $61 - 56 = 5$    | (35) $83 - 50 = 33$  | (65) $91 - 83 = 8$   | (95) $67 - 17 = 50$   |
| (6) $69 - 41 = 28$   | (36) $100 - 83 = 17$ | (66) $96 - 1 = 95$   | (96) $56 - 42 = 14$   |
| (7) $98 - 48 = 50$   | (37) $42 - 22 = 20$  | (67) $60 - 25 = 35$  | (97) $79 - 26 = 53$   |
| (8) $36 - 23 = 13$   | (38) $65 - 33 = 32$  | (68) $96 - 50 = 46$  | (98) $85 - 76 = 9$    |
| (9) $30 - 16 = 14$   | (39) $38 - 0 = 38$   | (69) $75 - 72 = 3$   | (99) $72 - 32 = 40$   |
| (10) $76 - 58 = 18$  | (40) $96 - 37 = 59$  | (70) $39 - 2 = 37$   | (100) $43 - 21 = 22$  |
| (11) $32 - 25 = 7$   | (41) $77 - 47 = 30$  | (71) $15 - 8 = 7$    | (101) $94 - 31 = 63$  |
| (12) $88 - 26 = 62$  | (42) $33 - 8 = 25$   | (72) $46 - 25 = 21$  | (102) $100 - 68 = 32$ |
| (13) $65 - 10 = 55$  | (43) $39 - 33 = 6$   | (73) $66 - 63 = 3$   | (103) $84 - 47 = 37$  |
| (14) $91 - 89 = 2$   | (44) $50 - 0 = 50$   | (74) $63 - 32 = 31$  | (104) $76 - 31 = 45$  |
| (15) $78 - 51 = 27$  | (45) $72 - 28 = 44$  | (75) $53 - 2 = 51$   | (105) $41 - 2 = 39$   |
| (16) $94 - 85 = 9$   | (46) $87 - 38 = 49$  | (76) $74 - 32 = 42$  | (106) $100 - 19 = 81$ |
| (17) $48 - 25 = 23$  | (47) $27 - 9 = 18$   | (77) $76 - 67 = 9$   | (107) $89 - 9 = 80$   |
| (18) $93 - 21 = 72$  | (48) $95 - 67 = 28$  | (78) $63 - 43 = 20$  | (108) $51 - 27 = 24$  |
| (19) $72 - 47 = 25$  | (49) $27 - 25 = 2$   | (79) $29 - 23 = 6$   | (109) $74 - 53 = 21$  |
| (20) $86 - 15 = 71$  | (50) $18 - 1 = 17$   | (80) $40 - 19 = 21$  | (110) $51 - 50 = 1$   |
| (21) $54 - 45 = 9$   | (51) $48 - 18 = 30$  | (81) $48 - 15 = 33$  | (111) $82 - 60 = 22$  |
| (22) $81 - 11 = 70$  | (52) $54 - 46 = 8$   | (82) $72 - 26 = 46$  | (112) $73 - 0 = 73$   |
| (23) $70 - 64 = 6$   | (53) $84 - 29 = 55$  | (83) $100 - 57 = 43$ | (113) $73 - 25 = 48$  |
| (24) $18 - 13 = 5$   | (54) $77 - 57 = 20$  | (84) $97 - 63 = 34$  | (114) $98 - 86 = 12$  |
| (25) $69 - 43 = 26$  | (55) $56 - 26 = 30$  | (85) $92 - 58 = 34$  | (115) $9 - 7 = 2$     |
| (26) $64 - 47 = 17$  | (56) $46 - 6 = 40$   | (86) $79 - 32 = 47$  | (116) $96 - 82 = 14$  |
| (27) $21 - 1 = 20$   | (57) $98 - 91 = 7$   | (87) $6 - 1 = 5$     | (117) $52 - 21 = 31$  |
| (28) $25 - 22 = 3$   | (58) $93 - 58 = 35$  | (88) $91 - 15 = 76$  | (118) $73 - 57 = 16$  |
| (29) $100 - 40 = 60$ | (59) $69 - 10 = 59$  | (89) $97 - 96 = 1$   | (119) $98 - 66 = 32$  |
| (30) $50 - 19 = 31$  | (60) $96 - 40 = 56$  | (90) $76 - 57 = 19$  | (120) $80 - 53 = 27$  |

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|---------------------|----------------------|---------------------|----------------------|
| (1) $76 - 8 = 68$   | (31) $45 - 23 = 22$  | (61) $34 - 11 = 23$ | (91) $56 - 45 = 11$  |
| (2) $93 - 92 = 1$   | (32) $38 - 24 = 14$  | (62) $20 - 11 = 9$  | (92) $71 - 6 = 65$   |
| (3) $96 - 60 = 36$  | (33) $31 - 24 = 7$   | (63) $81 - 70 = 11$ | (93) $83 - 72 = 11$  |
| (4) $82 - 10 = 72$  | (34) $34 - 5 = 29$   | (64) $84 - 25 = 59$ | (94) $42 - 31 = 11$  |
| (5) $49 - 22 = 27$  | (35) $67 - 67 = 0$   | (65) $98 - 3 = 95$  | (95) $71 - 60 = 11$  |
| (6) $69 - 2 = 67$   | (36) $98 - 24 = 74$  | (66) $35 - 29 = 6$  | (96) $98 - 22 = 76$  |
| (7) $78 - 22 = 56$  | (37) $71 - 34 = 37$  | (67) $70 - 57 = 13$ | (97) $42 - 18 = 24$  |
| (8) $61 - 6 = 55$   | (38) $97 - 58 = 39$  | (68) $85 - 83 = 2$  | (98) $84 - 41 = 43$  |
| (9) $54 - 40 = 14$  | (39) $38 - 17 = 21$  | (69) $40 - 8 = 32$  | (99) $98 - 19 = 79$  |
| (10) $86 - 13 = 73$ | (40) $73 - 33 = 40$  | (70) $22 - 15 = 7$  | (100) $96 - 49 = 47$ |
| (11) $88 - 49 = 39$ | (41) $85 - 14 = 71$  | (71) $39 - 34 = 5$  | (101) $13 - 11 = 2$  |
| (12) $68 - 15 = 53$ | (42) $23 - 2 = 21$   | (72) $53 - 51 = 2$  | (102) $40 - 6 = 34$  |
| (13) $79 - 10 = 69$ | (43) $99 - 51 = 48$  | (73) $43 - 4 = 39$  | (103) $25 - 2 = 23$  |
| (14) $87 - 32 = 55$ | (44) $77 - 17 = 60$  | (74) $70 - 27 = 43$ | (104) $35 - 20 = 15$ |
| (15) $92 - 76 = 16$ | (45) $94 - 72 = 22$  | (75) $45 - 2 = 43$  | (105) $77 - 48 = 29$ |
| (16) $94 - 29 = 65$ | (46) $96 - 9 = 87$   | (76) $33 - 22 = 11$ | (106) $25 - 11 = 14$ |
| (17) $88 - 23 = 65$ | (47) $38 - 35 = 3$   | (77) $90 - 59 = 31$ | (107) $77 - 46 = 31$ |
| (18) $65 - 60 = 5$  | (48) $65 - 22 = 43$  | (78) $86 - 50 = 36$ | (108) $96 - 69 = 27$ |
| (19) $45 - 28 = 17$ | (49) $49 - 46 = 3$   | (79) $50 - 48 = 2$  | (109) $30 - 21 = 9$  |
| (20) $37 - 8 = 29$  | (50) $100 - 77 = 23$ | (80) $97 - 87 = 10$ | (110) $61 - 33 = 28$ |
| (21) $64 - 45 = 19$ | (51) $74 - 51 = 23$  | (81) $9 - 3 = 6$    | (111) $30 - 18 = 12$ |
| (22) $88 - 67 = 21$ | (52) $86 - 68 = 18$  | (82) $81 - 56 = 25$ | (112) $36 - 21 = 15$ |
| (23) $15 - 13 = 2$  | (53) $33 - 20 = 13$  | (83) $81 - 53 = 28$ | (113) $20 - 10 = 10$ |
| (24) $69 - 69 = 0$  | (54) $42 - 4 = 38$   | (84) $77 - 20 = 57$ | (114) $30 - 18 = 12$ |
| (25) $48 - 28 = 20$ | (55) $35 - 17 = 18$  | (85) $88 - 87 = 1$  | (115) $96 - 88 = 8$  |
| (26) $84 - 79 = 5$  | (56) $88 - 34 = 54$  | (86) $75 - 74 = 1$  | (116) $46 - 18 = 28$ |
| (27) $85 - 37 = 48$ | (57) $63 - 40 = 23$  | (87) $58 - 16 = 42$ | (117) $44 - 17 = 27$ |
| (28) $87 - 1 = 86$  | (58) $28 - 25 = 3$   | (88) $71 - 19 = 52$ | (118) $26 - 7 = 19$  |
| (29) $59 - 27 = 32$ | (59) $51 - 5 = 46$   | (89) $57 - 32 = 25$ | (119) $13 - 6 = 7$   |
| (30) $75 - 11 = 64$ | (60) $63 - 25 = 38$  | (90) $14 - 3 = 11$  | (120) $74 - 65 = 9$  |

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|----------------------|---------------------|---------------------|-----------------------|
| (1) $15 - 14 = 1$    | (31) $96 - 13 = 83$ | (61) $25 - 15 = 10$ | (91) $16 - 6 = 10$    |
| (2) $75 - 14 = 61$   | (32) $96 - 75 = 21$ | (62) $30 - 8 = 22$  | (92) $58 - 11 = 47$   |
| (3) $31 - 20 = 11$   | (33) $39 - 14 = 25$ | (63) $91 - 7 = 84$  | (93) $88 - 15 = 73$   |
| (4) $93 - 46 = 47$   | (34) $58 - 17 = 41$ | (64) $97 - 8 = 89$  | (94) $51 - 50 = 1$    |
| (5) $75 - 38 = 37$   | (35) $85 - 51 = 34$ | (65) $72 - 17 = 55$ | (95) $3 - 0 = 3$      |
| (6) $65 - 36 = 29$   | (36) $42 - 42 = 0$  | (66) $71 - 68 = 3$  | (96) $95 - 38 = 57$   |
| (7) $30 - 19 = 11$   | (37) $85 - 6 = 79$  | (67) $75 - 38 = 37$ | (97) $95 - 67 = 28$   |
| (8) $90 - 87 = 3$    | (38) $46 - 9 = 37$  | (68) $48 - 22 = 26$ | (98) $71 - 19 = 52$   |
| (9) $45 - 4 = 41$    | (39) $54 - 21 = 33$ | (69) $73 - 21 = 52$ | (99) $63 - 41 = 22$   |
| (10) $76 - 40 = 36$  | (40) $35 - 9 = 26$  | (70) $81 - 17 = 64$ | (100) $66 - 33 = 33$  |
| (11) $94 - 30 = 64$  | (41) $76 - 38 = 38$ | (71) $98 - 10 = 88$ | (101) $63 - 47 = 16$  |
| (12) $99 - 60 = 39$  | (42) $63 - 62 = 1$  | (72) $66 - 6 = 60$  | (102) $52 - 26 = 26$  |
| (13) $67 - 6 = 61$   | (43) $83 - 31 = 52$ | (73) $66 - 46 = 20$ | (103) $98 - 78 = 20$  |
| (14) $99 - 57 = 42$  | (44) $79 - 47 = 32$ | (74) $30 - 2 = 28$  | (104) $90 - 61 = 29$  |
| (15) $89 - 22 = 67$  | (45) $57 - 22 = 35$ | (75) $99 - 31 = 68$ | (105) $22 - 21 = 1$   |
| (16) $91 - 45 = 46$  | (46) $35 - 31 = 4$  | (76) $90 - 33 = 57$ | (106) $23 - 9 = 14$   |
| (17) $97 - 29 = 68$  | (47) $77 - 55 = 22$ | (77) $96 - 68 = 28$ | (107) $41 - 17 = 24$  |
| (18) $19 - 7 = 12$   | (48) $97 - 91 = 6$  | (78) $93 - 67 = 26$ | (108) $92 - 6 = 86$   |
| (19) $72 - 18 = 54$  | (49) $39 - 32 = 7$  | (79) $91 - 75 = 16$ | (109) $85 - 25 = 60$  |
| (20) $93 - 18 = 75$  | (50) $34 - 10 = 24$ | (80) $72 - 15 = 57$ | (110) $93 - 4 = 89$   |
| (21) $28 - 28 = 0$   | (51) $60 - 33 = 27$ | (81) $60 - 10 = 50$ | (111) $12 - 7 = 5$    |
| (22) $89 - 83 = 6$   | (52) $92 - 48 = 44$ | (82) $82 - 14 = 68$ | (112) $62 - 18 = 44$  |
| (23) $92 - 10 = 82$  | (53) $26 - 21 = 5$  | (83) $97 - 84 = 13$ | (113) $73 - 57 = 16$  |
| (24) $78 - 47 = 31$  | (54) $45 - 37 = 8$  | (84) $98 - 36 = 62$ | (114) $53 - 46 = 7$   |
| (25) $84 - 14 = 70$  | (55) $91 - 28 = 63$ | (85) $95 - 50 = 45$ | (115) $56 - 47 = 9$   |
| (26) $21 - 7 = 14$   | (56) $70 - 41 = 29$ | (86) $84 - 38 = 46$ | (116) $100 - 62 = 38$ |
| (27) $100 - 57 = 43$ | (57) $21 - 7 = 14$  | (87) $97 - 7 = 90$  | (117) $79 - 74 = 5$   |
| (28) $57 - 55 = 2$   | (58) $40 - 8 = 32$  | (88) $51 - 2 = 49$  | (118) $13 - 10 = 3$   |
| (29) $99 - 63 = 36$  | (59) $76 - 27 = 49$ | (89) $77 - 13 = 64$ | (119) $91 - 41 = 50$  |
| (30) $13 - 11 = 2$   | (60) $14 - 14 = 0$  | (90) $90 - 83 = 7$  | (120) $83 - 50 = 33$  |

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|---------------------|---------------------|----------------------|----------------------|
| (1) $52 - 8 = 44$   | (31) $31 - 12 = 19$ | (61) $74 - 56 = 18$  | (91) $39 - 16 = 23$  |
| (2) $88 - 15 = 73$  | (32) $81 - 46 = 35$ | (62) $91 - 66 = 25$  | (92) $82 - 68 = 14$  |
| (3) $49 - 41 = 8$   | (33) $90 - 14 = 76$ | (63) $97 - 41 = 56$  | (93) $55 - 27 = 28$  |
| (4) $94 - 22 = 72$  | (34) $92 - 91 = 1$  | (64) $40 - 36 = 4$   | (94) $100 - 6 = 94$  |
| (5) $22 - 18 = 4$   | (35) $91 - 12 = 79$ | (65) $58 - 3 = 55$   | (95) $83 - 7 = 76$   |
| (6) $58 - 6 = 52$   | (36) $41 - 28 = 13$ | (66) $44 - 27 = 17$  | (96) $27 - 4 = 23$   |
| (7) $8 - 1 = 7$     | (37) $20 - 12 = 8$  | (67) $100 - 83 = 17$ | (97) $39 - 22 = 17$  |
| (8) $99 - 1 = 98$   | (38) $66 - 8 = 58$  | (68) $76 - 22 = 54$  | (98) $98 - 97 = 1$   |
| (9) $91 - 27 = 64$  | (39) $74 - 65 = 9$  | (69) $97 - 74 = 23$  | (99) $73 - 28 = 45$  |
| (10) $79 - 59 = 20$ | (40) $75 - 61 = 14$ | (70) $86 - 17 = 69$  | (100) $21 - 11 = 10$ |
| (11) $92 - 48 = 44$ | (41) $54 - 36 = 18$ | (71) $84 - 35 = 49$  | (101) $93 - 68 = 25$ |
| (12) $64 - 47 = 17$ | (42) $25 - 17 = 8$  | (72) $56 - 11 = 45$  | (102) $88 - 77 = 11$ |
| (13) $38 - 10 = 28$ | (43) $69 - 66 = 3$  | (73) $99 - 66 = 33$  | (103) $51 - 18 = 33$ |
| (14) $75 - 0 = 75$  | (44) $100 - 8 = 92$ | (74) $81 - 53 = 28$  | (104) $85 - 18 = 67$ |
| (15) $90 - 25 = 65$ | (45) $84 - 48 = 36$ | (75) $93 - 38 = 55$  | (105) $91 - 70 = 21$ |
| (16) $75 - 21 = 54$ | (46) $99 - 13 = 86$ | (76) $56 - 19 = 37$  | (106) $53 - 40 = 13$ |
| (17) $84 - 41 = 43$ | (47) $30 - 16 = 14$ | (77) $61 - 34 = 27$  | (107) $96 - 62 = 34$ |
| (18) $62 - 21 = 41$ | (48) $15 - 0 = 15$  | (78) $73 - 23 = 50$  | (108) $82 - 6 = 76$  |
| (19) $95 - 11 = 84$ | (49) $71 - 12 = 59$ | (79) $93 - 15 = 78$  | (109) $73 - 31 = 42$ |
| (20) $89 - 46 = 43$ | (50) $38 - 1 = 37$  | (80) $60 - 9 = 51$   | (110) $92 - 55 = 37$ |
| (21) $80 - 38 = 42$ | (51) $70 - 24 = 46$ | (81) $38 - 7 = 31$   | (111) $85 - 21 = 64$ |
| (22) $9 - 5 = 4$    | (52) $39 - 35 = 4$  | (82) $73 - 68 = 5$   | (112) $23 - 5 = 18$  |
| (23) $53 - 17 = 36$ | (53) $72 - 62 = 10$ | (83) $98 - 64 = 34$  | (113) $73 - 15 = 58$ |
| (24) $89 - 67 = 22$ | (54) $7 - 7 = 0$    | (84) $99 - 9 = 90$   | (114) $89 - 27 = 62$ |
| (25) $58 - 45 = 13$ | (55) $50 - 24 = 26$ | (85) $23 - 18 = 5$   | (115) $38 - 23 = 15$ |
| (26) $84 - 32 = 52$ | (56) $71 - 3 = 68$  | (86) $18 - 5 = 13$   | (116) $92 - 67 = 25$ |
| (27) $96 - 17 = 79$ | (57) $41 - 34 = 7$  | (87) $96 - 25 = 71$  | (117) $88 - 18 = 70$ |
| (28) $78 - 53 = 25$ | (58) $23 - 14 = 9$  | (88) $39 - 27 = 12$  | (118) $9 - 7 = 2$    |
| (29) $84 - 38 = 46$ | (59) $74 - 3 = 71$  | (89) $67 - 31 = 36$  | (119) $83 - 41 = 42$ |
| (30) $95 - 17 = 78$ | (60) $32 - 28 = 4$  | (90) $68 - 7 = 61$   | (120) $8 - 8 = 0$    |



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|---------------------|---------------------|---------------------|----------------------|
| (1) $99 - 36 = 63$  | (31) $41 - 34 = 7$  | (61) $67 - 64 = 3$  | (91) $82 - 6 = 76$   |
| (2) $81 - 47 = 34$  | (32) $94 - 39 = 55$ | (62) $13 - 7 = 6$   | (92) $90 - 45 = 45$  |
| (3) $68 - 11 = 57$  | (33) $46 - 15 = 31$ | (63) $46 - 3 = 43$  | (93) $33 - 31 = 2$   |
| (4) $33 - 31 = 2$   | (34) $80 - 45 = 35$ | (64) $79 - 78 = 1$  | (94) $88 - 72 = 16$  |
| (5) $47 - 1 = 46$   | (35) $39 - 26 = 13$ | (65) $96 - 74 = 22$ | (95) $70 - 22 = 48$  |
| (6) $52 - 52 = 0$   | (36) $70 - 53 = 17$ | (66) $94 - 13 = 81$ | (96) $46 - 10 = 36$  |
| (7) $97 - 70 = 27$  | (37) $63 - 35 = 28$ | (67) $74 - 41 = 33$ | (97) $48 - 13 = 35$  |
| (8) $39 - 23 = 16$  | (38) $94 - 61 = 33$ | (68) $58 - 45 = 13$ | (98) $66 - 31 = 35$  |
| (9) $57 - 40 = 17$  | (39) $82 - 9 = 73$  | (69) $98 - 4 = 94$  | (99) $78 - 2 = 76$   |
| (10) $76 - 74 = 2$  | (40) $49 - 44 = 5$  | (70) $77 - 44 = 33$ | (100) $63 - 43 = 20$ |
| (11) $88 - 39 = 49$ | (41) $60 - 18 = 42$ | (71) $81 - 34 = 47$ | (101) $45 - 15 = 30$ |
| (12) $41 - 31 = 10$ | (42) $83 - 5 = 78$  | (72) $75 - 72 = 3$  | (102) $58 - 29 = 29$ |
| (13) $82 - 8 = 74$  | (43) $53 - 19 = 34$ | (73) $41 - 30 = 11$ | (103) $43 - 22 = 21$ |
| (14) $94 - 10 = 84$ | (44) $16 - 6 = 10$  | (74) $40 - 38 = 2$  | (104) $57 - 8 = 49$  |
| (15) $69 - 16 = 53$ | (45) $73 - 14 = 59$ | (75) $94 - 11 = 83$ | (105) $82 - 33 = 49$ |
| (16) $6 - 0 = 6$    | (46) $53 - 28 = 25$ | (76) $26 - 26 = 0$  | (106) $30 - 28 = 2$  |
| (17) $84 - 39 = 45$ | (47) $75 - 43 = 32$ | (77) $83 - 17 = 66$ | (107) $97 - 17 = 80$ |
| (18) $96 - 44 = 52$ | (48) $96 - 1 = 95$  | (78) $66 - 32 = 34$ | (108) $67 - 22 = 45$ |
| (19) $20 - 4 = 16$  | (49) $68 - 43 = 25$ | (79) $48 - 24 = 24$ | (109) $94 - 11 = 83$ |
| (20) $100 - 91 = 9$ | (50) $99 - 68 = 31$ | (80) $69 - 44 = 25$ | (110) $80 - 61 = 19$ |
| (21) $67 - 59 = 8$  | (51) $68 - 34 = 34$ | (81) $93 - 41 = 52$ | (111) $75 - 48 = 27$ |
| (22) $30 - 30 = 0$  | (52) $76 - 26 = 50$ | (82) $80 - 57 = 23$ | (112) $75 - 22 = 53$ |
| (23) $98 - 69 = 29$ | (53) $87 - 3 = 84$  | (83) $45 - 44 = 1$  | (113) $85 - 19 = 66$ |
| (24) $93 - 55 = 38$ | (54) $87 - 58 = 29$ | (84) $47 - 29 = 18$ | (114) $32 - 24 = 8$  |
| (25) $32 - 12 = 20$ | (55) $53 - 14 = 39$ | (85) $72 - 52 = 20$ | (115) $36 - 26 = 10$ |
| (26) $62 - 47 = 15$ | (56) $98 - 6 = 92$  | (86) $93 - 51 = 42$ | (116) $55 - 49 = 6$  |
| (27) $79 - 60 = 19$ | (57) $65 - 16 = 49$ | (87) $99 - 70 = 29$ | (117) $66 - 62 = 4$  |
| (28) $93 - 52 = 41$ | (58) $85 - 53 = 32$ | (88) $99 - 6 = 93$  | (118) $73 - 41 = 32$ |
| (29) $51 - 35 = 16$ | (59) $64 - 26 = 38$ | (89) $18 - 17 = 1$  | (119) $85 - 79 = 6$  |
| (30) $34 - 11 = 23$ | (60) $87 - 76 = 11$ | (90) $52 - 0 = 52$  | (120) $39 - 15 = 24$ |

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|----------------------|----------------------|---------------------|----------------------|
| (1) $90 - 84 = 6$    | (31) $96 - 52 = 44$  | (61) $74 - 20 = 54$ | (91) $98 - 1 = 97$   |
| (2) $45 - 20 = 25$   | (32) $70 - 4 = 66$   | (62) $88 - 65 = 23$ | (92) $26 - 5 = 21$   |
| (3) $76 - 6 = 70$    | (33) $99 - 30 = 69$  | (63) $55 - 10 = 45$ | (93) $98 - 84 = 14$  |
| (4) $56 - 5 = 51$    | (34) $65 - 21 = 44$  | (64) $88 - 80 = 8$  | (94) $59 - 45 = 14$  |
| (5) $90 - 84 = 6$    | (35) $78 - 76 = 2$   | (65) $70 - 58 = 12$ | (95) $76 - 66 = 10$  |
| (6) $80 - 49 = 31$   | (36) $79 - 3 = 76$   | (66) $81 - 72 = 9$  | (96) $80 - 71 = 9$   |
| (7) $5 - 3 = 2$      | (37) $93 - 8 = 85$   | (67) $83 - 19 = 64$ | (97) $75 - 56 = 19$  |
| (8) $85 - 30 = 55$   | (38) $35 - 25 = 10$  | (68) $97 - 86 = 11$ | (98) $90 - 33 = 57$  |
| (9) $87 - 76 = 11$   | (39) $81 - 60 = 21$  | (69) $84 - 64 = 20$ | (99) $77 - 1 = 76$   |
| (10) $75 - 22 = 53$  | (40) $42 - 17 = 25$  | (70) $72 - 16 = 56$ | (100) $45 - 25 = 20$ |
| (11) $100 - 29 = 71$ | (41) $78 - 42 = 36$  | (71) $86 - 77 = 9$  | (101) $27 - 27 = 0$  |
| (12) $92 - 5 = 87$   | (42) $22 - 15 = 7$   | (72) $58 - 11 = 47$ | (102) $41 - 14 = 27$ |
| (13) $61 - 45 = 16$  | (43) $31 - 17 = 14$  | (73) $80 - 64 = 16$ | (103) $96 - 37 = 59$ |
| (14) $94 - 40 = 54$  | (44) $95 - 25 = 70$  | (74) $81 - 44 = 37$ | (104) $42 - 16 = 26$ |
| (15) $5 - 5 = 0$     | (45) $55 - 29 = 26$  | (75) $89 - 20 = 69$ | (105) $99 - 98 = 1$  |
| (16) $86 - 70 = 16$  | (46) $86 - 63 = 23$  | (76) $80 - 15 = 65$ | (106) $69 - 11 = 58$ |
| (17) $13 - 8 = 5$    | (47) $100 - 88 = 12$ | (77) $99 - 2 = 97$  | (107) $70 - 12 = 58$ |
| (18) $65 - 20 = 45$  | (48) $56 - 48 = 8$   | (78) $72 - 43 = 29$ | (108) $95 - 0 = 95$  |
| (19) $82 - 37 = 45$  | (49) $98 - 94 = 4$   | (79) $79 - 52 = 27$ | (109) $41 - 29 = 12$ |
| (20) $91 - 60 = 31$  | (50) $86 - 59 = 27$  | (80) $78 - 41 = 37$ | (110) $73 - 27 = 46$ |
| (21) $82 - 5 = 77$   | (51) $79 - 9 = 70$   | (81) $92 - 78 = 14$ | (111) $68 - 11 = 57$ |
| (22) $8 - 4 = 4$     | (52) $60 - 4 = 56$   | (82) $50 - 40 = 10$ | (112) $77 - 45 = 32$ |
| (23) $94 - 63 = 31$  | (53) $88 - 42 = 46$  | (83) $50 - 23 = 27$ | (113) $77 - 17 = 60$ |
| (24) $63 - 51 = 12$  | (54) $64 - 22 = 42$  | (84) $18 - 11 = 7$  | (114) $77 - 57 = 20$ |
| (25) $90 - 77 = 13$  | (55) $68 - 47 = 21$  | (85) $65 - 22 = 43$ | (115) $45 - 15 = 30$ |
| (26) $64 - 21 = 43$  | (56) $92 - 40 = 52$  | (86) $28 - 5 = 23$  | (116) $74 - 66 = 8$  |
| (27) $83 - 45 = 38$  | (57) $75 - 14 = 61$  | (87) $96 - 2 = 94$  | (117) $44 - 40 = 4$  |
| (28) $56 - 51 = 5$   | (58) $72 - 10 = 62$  | (88) $97 - 23 = 74$ | (118) $14 - 1 = 13$  |
| (29) $55 - 6 = 49$   | (59) $55 - 18 = 37$  | (89) $94 - 4 = 90$  | (119) $73 - 37 = 36$ |
| (30) $29 - 2 = 27$   | (60) $41 - 15 = 26$  | (90) $92 - 53 = 39$ | (120) $81 - 44 = 37$ |

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|---------------------|---------------------|----------------------|-----------------------|
| (1) $85 - 77 = 8$   | (31) $68 - 26 = 42$ | (61) $79 - 24 = 55$  | (91) $47 - 25 = 22$   |
| (2) $54 - 50 = 4$   | (32) $21 - 6 = 15$  | (62) $100 - 58 = 42$ | (92) $53 - 37 = 16$   |
| (3) $30 - 2 = 28$   | (33) $56 - 42 = 14$ | (63) $68 - 24 = 44$  | (93) $73 - 20 = 53$   |
| (4) $99 - 81 = 18$  | (34) $75 - 35 = 40$ | (64) $88 - 61 = 27$  | (94) $95 - 59 = 36$   |
| (5) $59 - 31 = 28$  | (35) $65 - 1 = 64$  | (65) $45 - 9 = 36$   | (95) $82 - 38 = 44$   |
| (6) $67 - 15 = 52$  | (36) $68 - 26 = 42$ | (66) $20 - 13 = 7$   | (96) $84 - 69 = 15$   |
| (7) $72 - 18 = 54$  | (37) $57 - 43 = 14$ | (67) $100 - 4 = 96$  | (97) $70 - 44 = 26$   |
| (8) $36 - 0 = 36$   | (38) $14 - 12 = 2$  | (68) $96 - 24 = 72$  | (98) $33 - 27 = 6$    |
| (9) $81 - 6 = 75$   | (39) $78 - 71 = 7$  | (69) $62 - 58 = 4$   | (99) $53 - 6 = 47$    |
| (10) $68 - 32 = 36$ | (40) $99 - 61 = 38$ | (70) $54 - 4 = 50$   | (100) $88 - 56 = 32$  |
| (11) $78 - 60 = 18$ | (41) $38 - 5 = 33$  | (71) $57 - 13 = 44$  | (101) $85 - 48 = 37$  |
| (12) $96 - 54 = 42$ | (42) $93 - 23 = 70$ | (72) $97 - 50 = 47$  | (102) $83 - 27 = 56$  |
| (13) $6 - 6 = 0$    | (43) $69 - 32 = 37$ | (73) $81 - 78 = 3$   | (103) $87 - 16 = 71$  |
| (14) $94 - 69 = 25$ | (44) $75 - 54 = 21$ | (74) $75 - 67 = 8$   | (104) $76 - 62 = 14$  |
| (15) $17 - 15 = 2$  | (45) $90 - 36 = 54$ | (75) $91 - 72 = 19$  | (105) $52 - 42 = 10$  |
| (16) $19 - 10 = 9$  | (46) $52 - 45 = 7$  | (76) $82 - 43 = 39$  | (106) $78 - 35 = 43$  |
| (17) $89 - 78 = 11$ | (47) $56 - 34 = 22$ | (77) $90 - 67 = 23$  | (107) $81 - 54 = 27$  |
| (18) $65 - 49 = 16$ | (48) $95 - 83 = 12$ | (78) $78 - 28 = 50$  | (108) $69 - 60 = 9$   |
| (19) $98 - 24 = 74$ | (49) $49 - 33 = 16$ | (79) $78 - 47 = 31$  | (109) $58 - 26 = 32$  |
| (20) $79 - 71 = 8$  | (50) $68 - 31 = 37$ | (80) $67 - 37 = 30$  | (110) $35 - 8 = 27$   |
| (21) $88 - 21 = 67$ | (51) $79 - 61 = 18$ | (81) $90 - 62 = 28$  | (111) $85 - 54 = 31$  |
| (22) $64 - 26 = 38$ | (52) $61 - 31 = 30$ | (82) $55 - 29 = 26$  | (112) $3 - 1 = 2$     |
| (23) $48 - 0 = 48$  | (53) $62 - 59 = 3$  | (83) $94 - 68 = 26$  | (113) $44 - 14 = 30$  |
| (24) $65 - 27 = 38$ | (54) $66 - 26 = 40$ | (84) $43 - 40 = 3$   | (114) $82 - 4 = 78$   |
| (25) $63 - 33 = 30$ | (55) $97 - 35 = 62$ | (85) $31 - 24 = 7$   | (115) $52 - 40 = 12$  |
| (26) $95 - 43 = 52$ | (56) $45 - 12 = 33$ | (86) $50 - 38 = 12$  | (116) $34 - 5 = 29$   |
| (27) $73 - 3 = 70$  | (57) $97 - 38 = 59$ | (87) $49 - 45 = 4$   | (117) $90 - 8 = 82$   |
| (28) $22 - 5 = 17$  | (58) $15 - 9 = 6$   | (88) $73 - 38 = 35$  | (118) $48 - 42 = 6$   |
| (29) $86 - 33 = 53$ | (59) $96 - 68 = 28$ | (89) $100 - 17 = 83$ | (119) $81 - 1 = 80$   |
| (30) $85 - 48 = 37$ | (60) $54 - 31 = 23$ | (90) $77 - 56 = 21$  | (120) $100 - 13 = 87$ |

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|----------------------|---------------------|----------------------|-----------------------|
| (1) $21 - 6 = 15$    | (31) $89 - 65 = 24$ | (61) $87 - 71 = 16$  | (91) $94 - 50 = 44$   |
| (2) $67 - 38 = 29$   | (32) $56 - 37 = 19$ | (62) $100 - 32 = 68$ | (92) $46 - 40 = 6$    |
| (3) $77 - 20 = 57$   | (33) $95 - 41 = 54$ | (63) $98 - 30 = 68$  | (93) $13 - 13 = 0$    |
| (4) $80 - 32 = 48$   | (34) $51 - 6 = 45$  | (64) $99 - 15 = 84$  | (94) $54 - 17 = 37$   |
| (5) $79 - 1 = 78$    | (35) $66 - 45 = 21$ | (65) $96 - 56 = 40$  | (95) $47 - 46 = 1$    |
| (6) $75 - 27 = 48$   | (36) $96 - 82 = 14$ | (66) $51 - 49 = 2$   | (96) $85 - 69 = 16$   |
| (7) $61 - 27 = 34$   | (37) $68 - 13 = 55$ | (67) $18 - 3 = 15$   | (97) $74 - 6 = 68$    |
| (8) $98 - 72 = 26$   | (38) $82 - 71 = 11$ | (68) $69 - 14 = 55$  | (98) $70 - 12 = 58$   |
| (9) $61 - 18 = 43$   | (39) $60 - 40 = 20$ | (69) $94 - 64 = 30$  | (99) $82 - 33 = 49$   |
| (10) $88 - 4 = 84$   | (40) $82 - 61 = 21$ | (70) $69 - 24 = 45$  | (100) $96 - 9 = 87$   |
| (11) $73 - 32 = 41$  | (41) $39 - 39 = 0$  | (71) $59 - 0 = 59$   | (101) $73 - 2 = 71$   |
| (12) $75 - 68 = 7$   | (42) $74 - 56 = 18$ | (72) $51 - 18 = 33$  | (102) $56 - 11 = 45$  |
| (13) $100 - 71 = 29$ | (43) $92 - 59 = 33$ | (73) $55 - 13 = 42$  | (103) $88 - 2 = 86$   |
| (14) $32 - 20 = 12$  | (44) $65 - 45 = 20$ | (74) $62 - 47 = 15$  | (104) $57 - 18 = 39$  |
| (15) $70 - 38 = 32$  | (45) $17 - 12 = 5$  | (75) $71 - 71 = 0$   | (105) $74 - 8 = 66$   |
| (16) $71 - 63 = 8$   | (46) $68 - 58 = 10$ | (76) $10 - 3 = 7$    | (106) $60 - 19 = 41$  |
| (17) $76 - 76 = 0$   | (47) $91 - 43 = 48$ | (77) $99 - 74 = 25$  | (107) $26 - 26 = 0$   |
| (18) $25 - 20 = 5$   | (48) $71 - 1 = 70$  | (78) $67 - 22 = 45$  | (108) $71 - 33 = 38$  |
| (19) $98 - 26 = 72$  | (49) $69 - 63 = 6$  | (79) $99 - 24 = 75$  | (109) $100 - 37 = 63$ |
| (20) $93 - 73 = 20$  | (50) $37 - 21 = 16$ | (80) $100 - 37 = 63$ | (110) $62 - 0 = 62$   |
| (21) $89 - 38 = 51$  | (51) $28 - 20 = 8$  | (81) $60 - 18 = 42$  | (111) $60 - 39 = 21$  |
| (22) $87 - 39 = 48$  | (52) $61 - 10 = 51$ | (82) $17 - 8 = 9$    | (112) $58 - 40 = 18$  |
| (23) $96 - 86 = 10$  | (53) $70 - 41 = 29$ | (83) $86 - 69 = 17$  | (113) $52 - 52 = 0$   |
| (24) $95 - 6 = 89$   | (54) $67 - 37 = 30$ | (84) $53 - 9 = 44$   | (114) $77 - 66 = 11$  |
| (25) $80 - 47 = 33$  | (55) $62 - 46 = 16$ | (85) $62 - 47 = 15$  | (115) $60 - 60 = 0$   |
| (26) $68 - 10 = 58$  | (56) $35 - 20 = 15$ | (86) $98 - 76 = 22$  | (116) $82 - 26 = 56$  |
| (27) $75 - 31 = 44$  | (57) $33 - 20 = 13$ | (87) $85 - 44 = 41$  | (117) $77 - 29 = 48$  |
| (28) $92 - 70 = 22$  | (58) $68 - 42 = 26$ | (88) $66 - 45 = 21$  | (118) $87 - 78 = 9$   |
| (29) $71 - 48 = 23$  | (59) $86 - 42 = 44$ | (89) $42 - 14 = 28$  | (119) $49 - 21 = 28$  |
| (30) $100 - 24 = 76$ | (60) $44 - 3 = 41$  | (90) $54 - 19 = 35$  | (120) $84 - 48 = 36$  |

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|---------------------|----------------------|----------------------|-----------------------|
| (1) $76 - 52 = 24$  | (31) $70 - 18 = 52$  | (61) $53 - 40 = 13$  | (91) $58 - 29 = 29$   |
| (2) $56 - 9 = 47$   | (32) $68 - 58 = 10$  | (62) $65 - 3 = 62$   | (92) $100 - 94 = 6$   |
| (3) $43 - 5 = 38$   | (33) $75 - 59 = 16$  | (63) $41 - 11 = 30$  | (93) $65 - 20 = 45$   |
| (4) $80 - 38 = 42$  | (34) $97 - 24 = 73$  | (64) $91 - 79 = 12$  | (94) $83 - 37 = 46$   |
| (5) $56 - 33 = 23$  | (35) $84 - 30 = 54$  | (65) $56 - 36 = 20$  | (95) $88 - 51 = 37$   |
| (6) $72 - 11 = 61$  | (36) $47 - 18 = 29$  | (66) $69 - 54 = 15$  | (96) $62 - 51 = 11$   |
| (7) $100 - 35 = 65$ | (37) $70 - 41 = 29$  | (67) $85 - 31 = 54$  | (97) $94 - 67 = 27$   |
| (8) $99 - 84 = 15$  | (38) $86 - 20 = 66$  | (68) $87 - 66 = 21$  | (98) $30 - 24 = 6$    |
| (9) $39 - 35 = 4$   | (39) $95 - 33 = 62$  | (69) $90 - 2 = 88$   | (99) $77 - 6 = 71$    |
| (10) $90 - 11 = 79$ | (40) $49 - 6 = 43$   | (70) $70 - 18 = 52$  | (100) $67 - 36 = 31$  |
| (11) $67 - 44 = 23$ | (41) $76 - 15 = 61$  | (71) $40 - 15 = 25$  | (101) $98 - 73 = 25$  |
| (12) $53 - 52 = 1$  | (42) $88 - 58 = 30$  | (72) $47 - 42 = 5$   | (102) $80 - 54 = 26$  |
| (13) $63 - 11 = 52$ | (43) $89 - 1 = 88$   | (73) $98 - 17 = 81$  | (103) $74 - 66 = 8$   |
| (14) $95 - 89 = 6$  | (44) $52 - 32 = 20$  | (74) $46 - 44 = 2$   | (104) $88 - 42 = 46$  |
| (15) $67 - 41 = 26$ | (45) $86 - 10 = 76$  | (75) $44 - 23 = 21$  | (105) $67 - 47 = 20$  |
| (16) $40 - 38 = 2$  | (46) $95 - 73 = 22$  | (76) $83 - 10 = 73$  | (106) $23 - 8 = 15$   |
| (17) $26 - 21 = 5$  | (47) $100 - 88 = 12$ | (77) $100 - 56 = 44$ | (107) $55 - 11 = 44$  |
| (18) $62 - 26 = 36$ | (48) $68 - 11 = 57$  | (78) $23 - 7 = 16$   | (108) $43 - 29 = 14$  |
| (19) $96 - 3 = 93$  | (49) $73 - 62 = 11$  | (79) $83 - 22 = 61$  | (109) $55 - 6 = 49$   |
| (20) $63 - 35 = 28$ | (50) $91 - 10 = 81$  | (80) $78 - 36 = 42$  | (110) $85 - 59 = 26$  |
| (21) $47 - 31 = 16$ | (51) $96 - 15 = 81$  | (81) $75 - 18 = 57$  | (111) $34 - 3 = 31$   |
| (22) $91 - 68 = 23$ | (52) $68 - 10 = 58$  | (82) $77 - 63 = 14$  | (112) $83 - 19 = 64$  |
| (23) $98 - 47 = 51$ | (53) $19 - 7 = 12$   | (83) $65 - 21 = 44$  | (113) $89 - 17 = 72$  |
| (24) $72 - 2 = 70$  | (54) $97 - 86 = 11$  | (84) $74 - 11 = 63$  | (114) $70 - 39 = 31$  |
| (25) $98 - 19 = 79$ | (55) $71 - 8 = 63$   | (85) $99 - 10 = 89$  | (115) $100 - 35 = 65$ |
| (26) $86 - 57 = 29$ | (56) $96 - 54 = 42$  | (86) $67 - 20 = 47$  | (116) $48 - 1 = 47$   |
| (27) $55 - 13 = 42$ | (57) $27 - 17 = 10$  | (87) $61 - 40 = 21$  | (117) $78 - 40 = 38$  |
| (28) $92 - 87 = 5$  | (58) $95 - 10 = 85$  | (88) $12 - 5 = 7$    | (118) $83 - 78 = 5$   |
| (29) $83 - 23 = 60$ | (59) $95 - 67 = 28$  | (89) $80 - 19 = 61$  | (119) $97 - 84 = 13$  |
| (30) $81 - 2 = 79$  | (60) $81 - 60 = 21$  | (90) $99 - 26 = 73$  | (120) $18 - 2 = 16$   |

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|---------------------|---------------------|---------------------|----------------------|
| (1) $51 - 29 = 22$  | (31) $80 - 8 = 72$  | (61) $75 - 47 = 28$ | (91) $79 - 68 = 11$  |
| (2) $63 - 5 = 58$   | (32) $67 - 6 = 61$  | (62) $80 - 25 = 55$ | (92) $87 - 32 = 55$  |
| (3) $63 - 49 = 14$  | (33) $76 - 45 = 31$ | (63) $97 - 23 = 74$ | (93) $78 - 63 = 15$  |
| (4) $68 - 20 = 48$  | (34) $81 - 54 = 27$ | (64) $88 - 15 = 73$ | (94) $46 - 3 = 43$   |
| (5) $94 - 57 = 37$  | (35) $52 - 46 = 6$  | (65) $60 - 48 = 12$ | (95) $97 - 48 = 49$  |
| (6) $81 - 57 = 24$  | (36) $77 - 39 = 38$ | (66) $34 - 28 = 6$  | (96) $61 - 46 = 15$  |
| (7) $97 - 85 = 12$  | (37) $49 - 48 = 1$  | (67) $84 - 39 = 45$ | (97) $95 - 84 = 11$  |
| (8) $35 - 15 = 20$  | (38) $95 - 38 = 57$ | (68) $68 - 5 = 63$  | (98) $93 - 28 = 65$  |
| (9) $36 - 12 = 24$  | (39) $74 - 55 = 19$ | (69) $76 - 29 = 47$ | (99) $86 - 10 = 76$  |
| (10) $68 - 51 = 17$ | (40) $37 - 18 = 19$ | (70) $58 - 20 = 38$ | (100) $83 - 17 = 66$ |
| (11) $82 - 23 = 59$ | (41) $80 - 29 = 51$ | (71) $85 - 21 = 64$ | (101) $10 - 1 = 9$   |
| (12) $26 - 10 = 16$ | (42) $98 - 87 = 11$ | (72) $60 - 56 = 4$  | (102) $5 - 4 = 1$    |
| (13) $89 - 35 = 54$ | (43) $50 - 47 = 3$  | (73) $26 - 10 = 16$ | (103) $30 - 7 = 23$  |
| (14) $39 - 25 = 14$ | (44) $78 - 71 = 7$  | (74) $90 - 8 = 82$  | (104) $89 - 30 = 59$ |
| (15) $80 - 66 = 14$ | (45) $48 - 47 = 1$  | (75) $51 - 50 = 1$  | (105) $98 - 20 = 78$ |
| (16) $30 - 22 = 8$  | (46) $94 - 42 = 52$ | (76) $97 - 34 = 63$ | (106) $86 - 83 = 3$  |
| (17) $89 - 15 = 74$ | (47) $93 - 10 = 83$ | (77) $45 - 45 = 0$  | (107) $18 - 12 = 6$  |
| (18) $61 - 6 = 55$  | (48) $47 - 42 = 5$  | (78) $93 - 67 = 26$ | (108) $70 - 39 = 31$ |
| (19) $63 - 1 = 62$  | (49) $52 - 42 = 10$ | (79) $26 - 20 = 6$  | (109) $39 - 9 = 30$  |
| (20) $40 - 39 = 1$  | (50) $55 - 35 = 20$ | (80) $68 - 19 = 49$ | (110) $60 - 37 = 23$ |
| (21) $29 - 6 = 23$  | (51) $54 - 8 = 46$  | (81) $99 - 3 = 96$  | (111) $48 - 46 = 2$  |
| (22) $36 - 13 = 23$ | (52) $76 - 8 = 68$  | (82) $79 - 74 = 5$  | (112) $80 - 6 = 74$  |
| (23) $95 - 41 = 54$ | (53) $91 - 42 = 49$ | (83) $87 - 7 = 80$  | (113) $39 - 34 = 5$  |
| (24) $71 - 44 = 27$ | (54) $47 - 12 = 35$ | (84) $67 - 41 = 26$ | (114) $48 - 41 = 7$  |
| (25) $96 - 42 = 54$ | (55) $12 - 4 = 8$   | (85) $58 - 22 = 36$ | (115) $98 - 87 = 11$ |
| (26) $97 - 55 = 42$ | (56) $59 - 9 = 50$  | (86) $73 - 70 = 3$  | (116) $90 - 31 = 59$ |
| (27) $24 - 23 = 1$  | (57) $54 - 41 = 13$ | (87) $74 - 50 = 24$ | (117) $87 - 73 = 14$ |
| (28) $84 - 48 = 36$ | (58) $76 - 26 = 50$ | (88) $95 - 70 = 25$ | (118) $88 - 48 = 40$ |
| (29) $98 - 67 = 31$ | (59) $39 - 4 = 35$  | (89) $65 - 35 = 30$ | (119) $18 - 12 = 6$  |
| (30) $91 - 13 = 78$ | (60) $66 - 41 = 25$ | (90) $97 - 6 = 91$  | (120) $75 - 46 = 29$ |

- |                     |                      |                     |                      |
|---------------------|----------------------|---------------------|----------------------|
| (1) $63 - 22 = 41$  | (31) $72 - 24 = 48$  | (61) $97 - 88 = 9$  | (91) $90 - 23 = 67$  |
| (2) $40 - 39 = 1$   | (32) $62 - 51 = 11$  | (62) $41 - 19 = 22$ | (92) $94 - 64 = 30$  |
| (3) $45 - 39 = 6$   | (33) $18 - 11 = 7$   | (63) $75 - 57 = 18$ | (93) $18 - 16 = 2$   |
| (4) $71 - 12 = 59$  | (34) $63 - 44 = 19$  | (64) $33 - 11 = 22$ | (94) $59 - 30 = 29$  |
| (5) $83 - 56 = 27$  | (35) $63 - 57 = 6$   | (65) $40 - 15 = 25$ | (95) $82 - 13 = 69$  |
| (6) $51 - 5 = 46$   | (36) $84 - 63 = 21$  | (66) $96 - 25 = 71$ | (96) $83 - 26 = 57$  |
| (7) $78 - 73 = 5$   | (37) $99 - 25 = 74$  | (67) $65 - 64 = 1$  | (97) $22 - 18 = 4$   |
| (8) $86 - 9 = 77$   | (38) $98 - 53 = 45$  | (68) $74 - 52 = 22$ | (98) $96 - 73 = 23$  |
| (9) $59 - 39 = 20$  | (39) $85 - 40 = 45$  | (69) $99 - 8 = 91$  | (99) $58 - 16 = 42$  |
| (10) $90 - 61 = 29$ | (40) $69 - 33 = 36$  | (70) $60 - 43 = 17$ | (100) $48 - 7 = 41$  |
| (11) $85 - 81 = 4$  | (41) $59 - 54 = 5$   | (71) $54 - 12 = 42$ | (101) $50 - 27 = 23$ |
| (12) $84 - 73 = 11$ | (42) $62 - 50 = 12$  | (72) $53 - 36 = 17$ | (102) $16 - 14 = 2$  |
| (13) $92 - 87 = 5$  | (43) $29 - 25 = 4$   | (73) $99 - 4 = 95$  | (103) $59 - 29 = 30$ |
| (14) $93 - 8 = 85$  | (44) $100 - 62 = 38$ | (74) $29 - 24 = 5$  | (104) $47 - 17 = 30$ |
| (15) $50 - 37 = 13$ | (45) $98 - 84 = 14$  | (75) $70 - 64 = 6$  | (105) $93 - 86 = 7$  |
| (16) $67 - 57 = 10$ | (46) $47 - 4 = 43$   | (76) $39 - 11 = 28$ | (106) $74 - 9 = 65$  |
| (17) $98 - 30 = 68$ | (47) $89 - 51 = 38$  | (77) $12 - 5 = 7$   | (107) $44 - 32 = 12$ |
| (18) $52 - 35 = 17$ | (48) $97 - 33 = 64$  | (78) $45 - 14 = 31$ | (108) $47 - 31 = 16$ |
| (19) $65 - 19 = 46$ | (49) $83 - 70 = 13$  | (79) $54 - 18 = 36$ | (109) $83 - 11 = 72$ |
| (20) $64 - 46 = 18$ | (50) $67 - 19 = 48$  | (80) $9 - 1 = 8$    | (110) $73 - 4 = 69$  |
| (21) $76 - 48 = 28$ | (51) $94 - 11 = 83$  | (81) $90 - 83 = 7$  | (111) $82 - 52 = 30$ |
| (22) $91 - 59 = 32$ | (52) $64 - 49 = 15$  | (82) $71 - 7 = 64$  | (112) $97 - 72 = 25$ |
| (23) $68 - 42 = 26$ | (53) $92 - 80 = 12$  | (83) $93 - 38 = 55$ | (113) $85 - 82 = 3$  |
| (24) $30 - 6 = 24$  | (54) $65 - 49 = 16$  | (84) $60 - 22 = 38$ | (114) $70 - 65 = 5$  |
| (25) $78 - 46 = 32$ | (55) $93 - 42 = 51$  | (85) $88 - 75 = 13$ | (115) $64 - 49 = 15$ |
| (26) $60 - 13 = 47$ | (56) $92 - 23 = 69$  | (86) $68 - 7 = 61$  | (116) $85 - 16 = 69$ |
| (27) $89 - 38 = 51$ | (57) $78 - 33 = 45$  | (87) $83 - 34 = 49$ | (117) $70 - 27 = 43$ |
| (28) $97 - 47 = 50$ | (58) $86 - 27 = 59$  | (88) $70 - 16 = 54$ | (118) $25 - 9 = 16$  |
| (29) $58 - 1 = 57$  | (59) $71 - 66 = 5$   | (89) $98 - 24 = 74$ | (119) $44 - 18 = 26$ |
| (30) $76 - 11 = 65$ | (60) $77 - 71 = 6$   | (90) $26 - 3 = 23$  | (120) $80 - 48 = 32$ |

### 1.1.3 乗法

自然数の積を求める問題。



|                     |                    |                     |                     |
|---------------------|--------------------|---------------------|---------------------|
| (1) $9 \times 6$    | (31) $1 \times 5$  | (61) $6 \times 4$   | (91) $4 \times 9$   |
| (2) $4 \times 5$    | (32) $1 \times 3$  | (62) $6 \times 8$   | (92) $2 \times 3$   |
| (3) $5 \times 1$    | (33) $5 \times 3$  | (63) $4 \times 8$   | (93) $4 \times 7$   |
| (4) $2 \times 8$    | (34) $10 \times 6$ | (64) $8 \times 5$   | (94) $8 \times 1$   |
| (5) $6 \times 0$    | (35) $1 \times 8$  | (65) $6 \times 2$   | (95) $7 \times 3$   |
| (6) $10 \times 1$   | (36) $5 \times 5$  | (66) $1 \times 1$   | (96) $7 \times 4$   |
| (7) $7 \times 4$    | (37) $8 \times 4$  | (67) $10 \times 10$ | (97) $4 \times 4$   |
| (8) $5 \times 6$    | (38) $4 \times 8$  | (68) $3 \times 5$   | (98) $3 \times 2$   |
| (9) $10 \times 10$  | (39) $0 \times 3$  | (69) $6 \times 9$   | (99) $4 \times 1$   |
| (10) $1 \times 5$   | (40) $5 \times 3$  | (70) $7 \times 6$   | (100) $4 \times 10$ |
| (11) $5 \times 6$   | (41) $7 \times 10$ | (71) $4 \times 2$   | (101) $9 \times 5$  |
| (12) $4 \times 10$  | (42) $6 \times 10$ | (72) $6 \times 1$   | (102) $5 \times 10$ |
| (13) $5 \times 5$   | (43) $7 \times 6$  | (73) $6 \times 10$  | (103) $10 \times 5$ |
| (14) $6 \times 4$   | (44) $9 \times 5$  | (74) $6 \times 5$   | (104) $2 \times 8$  |
| (15) $2 \times 2$   | (45) $6 \times 5$  | (75) $1 \times 10$  | (105) $1 \times 4$  |
| (16) $9 \times 2$   | (46) $2 \times 6$  | (76) $4 \times 0$   | (106) $7 \times 4$  |
| (17) $5 \times 1$   | (47) $10 \times 7$ | (77) $2 \times 7$   | (107) $6 \times 6$  |
| (18) $10 \times 10$ | (48) $9 \times 3$  | (78) $2 \times 3$   | (108) $6 \times 5$  |
| (19) $8 \times 9$   | (49) $8 \times 7$  | (79) $2 \times 5$   | (109) $3 \times 10$ |
| (20) $5 \times 5$   | (50) $2 \times 5$  | (80) $3 \times 8$   | (110) $5 \times 10$ |
| (21) $2 \times 1$   | (51) $2 \times 7$  | (81) $6 \times 7$   | (111) $1 \times 3$  |
| (22) $10 \times 2$  | (52) $1 \times 10$ | (82) $2 \times 6$   | (112) $5 \times 6$  |
| (23) $6 \times 5$   | (53) $4 \times 7$  | (83) $8 \times 8$   | (113) $6 \times 8$  |
| (24) $9 \times 4$   | (54) $3 \times 9$  | (84) $7 \times 5$   | (114) $2 \times 3$  |
| (25) $3 \times 10$  | (55) $1 \times 6$  | (85) $2 \times 7$   | (115) $1 \times 0$  |
| (26) $8 \times 6$   | (56) $5 \times 5$  | (86) $3 \times 0$   | (116) $1 \times 9$  |
| (27) $10 \times 9$  | (57) $8 \times 9$  | (87) $4 \times 9$   | (117) $8 \times 0$  |
| (28) $8 \times 3$   | (58) $5 \times 8$  | (88) $5 \times 10$  | (118) $5 \times 2$  |
| (29) $8 \times 7$   | (59) $4 \times 6$  | (89) $10 \times 1$  | (119) $2 \times 0$  |
| (30) $5 \times 3$   | (60) $2 \times 5$  | (90) $1 \times 2$   | (120) $6 \times 4$  |

|                    |                    |                    |                     |
|--------------------|--------------------|--------------------|---------------------|
| (1) $2 \times 7$   | (31) $2 \times 0$  | (61) $8 \times 4$  | (91) $8 \times 9$   |
| (2) $10 \times 4$  | (32) $2 \times 4$  | (62) $6 \times 3$  | (92) $0 \times 10$  |
| (3) $3 \times 7$   | (33) $1 \times 10$ | (63) $4 \times 6$  | (93) $8 \times 2$   |
| (4) $7 \times 1$   | (34) $1 \times 4$  | (64) $1 \times 8$  | (94) $8 \times 5$   |
| (5) $5 \times 4$   | (35) $2 \times 8$  | (65) $7 \times 2$  | (95) $1 \times 5$   |
| (6) $5 \times 7$   | (36) $3 \times 4$  | (66) $7 \times 7$  | (96) $1 \times 8$   |
| (7) $7 \times 3$   | (37) $0 \times 4$  | (67) $5 \times 10$ | (97) $9 \times 2$   |
| (8) $8 \times 1$   | (38) $4 \times 9$  | (68) $3 \times 4$  | (98) $3 \times 0$   |
| (9) $6 \times 6$   | (39) $1 \times 3$  | (69) $6 \times 1$  | (99) $8 \times 9$   |
| (10) $2 \times 10$ | (40) $9 \times 8$  | (70) $7 \times 2$  | (100) $5 \times 4$  |
| (11) $10 \times 7$ | (41) $7 \times 0$  | (71) $7 \times 3$  | (101) $1 \times 8$  |
| (12) $0 \times 10$ | (42) $2 \times 3$  | (72) $6 \times 8$  | (102) $1 \times 10$ |
| (13) $1 \times 6$  | (43) $10 \times 1$ | (73) $6 \times 2$  | (103) $1 \times 2$  |
| (14) $10 \times 7$ | (44) $4 \times 3$  | (74) $4 \times 6$  | (104) $4 \times 2$  |
| (15) $8 \times 5$  | (45) $10 \times 7$ | (75) $10 \times 5$ | (105) $5 \times 0$  |
| (16) $10 \times 8$ | (46) $9 \times 6$  | (76) $5 \times 3$  | (106) $7 \times 3$  |
| (17) $7 \times 10$ | (47) $3 \times 0$  | (77) $3 \times 8$  | (107) $9 \times 5$  |
| (18) $7 \times 9$  | (48) $2 \times 0$  | (78) $10 \times 2$ | (108) $0 \times 2$  |
| (19) $9 \times 0$  | (49) $7 \times 6$  | (79) $1 \times 8$  | (109) $7 \times 4$  |
| (20) $8 \times 2$  | (50) $4 \times 5$  | (80) $9 \times 7$  | (110) $8 \times 5$  |
| (21) $6 \times 4$  | (51) $5 \times 10$ | (81) $3 \times 9$  | (111) $1 \times 8$  |
| (22) $2 \times 0$  | (52) $1 \times 3$  | (82) $8 \times 7$  | (112) $6 \times 8$  |
| (23) $8 \times 10$ | (53) $7 \times 8$  | (83) $4 \times 5$  | (113) $7 \times 4$  |
| (24) $9 \times 5$  | (54) $4 \times 6$  | (84) $9 \times 3$  | (114) $5 \times 9$  |
| (25) $3 \times 9$  | (55) $1 \times 9$  | (85) $4 \times 5$  | (115) $2 \times 10$ |
| (26) $4 \times 10$ | (56) $2 \times 8$  | (86) $4 \times 0$  | (116) $2 \times 7$  |
| (27) $6 \times 9$  | (57) $1 \times 7$  | (87) $7 \times 2$  | (117) $9 \times 2$  |
| (28) $9 \times 10$ | (58) $3 \times 2$  | (88) $6 \times 7$  | (118) $9 \times 7$  |
| (29) $9 \times 9$  | (59) $10 \times 5$ | (89) $10 \times 9$ | (119) $6 \times 4$  |
| (30) $9 \times 2$  | (60) $0 \times 6$  | (90) $7 \times 2$  | (120) $1 \times 9$  |

|                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|
| (1) $4 \times 9$   | (31) $1 \times 4$  | (61) $10 \times 6$ | (91) $2 \times 2$  |
| (2) $1 \times 3$   | (32) $1 \times 6$  | (62) $1 \times 8$  | (92) $2 \times 3$  |
| (3) $7 \times 10$  | (33) $2 \times 10$ | (63) $7 \times 10$ | (93) $6 \times 8$  |
| (4) $7 \times 7$   | (34) $9 \times 10$ | (64) $4 \times 3$  | (94) $5 \times 4$  |
| (5) $2 \times 2$   | (35) $5 \times 4$  | (65) $4 \times 9$  | (95) $5 \times 8$  |
| (6) $6 \times 10$  | (36) $6 \times 2$  | (66) $8 \times 1$  | (96) $6 \times 3$  |
| (7) $2 \times 8$   | (37) $4 \times 7$  | (67) $6 \times 3$  | (97) $2 \times 1$  |
| (8) $1 \times 6$   | (38) $6 \times 1$  | (68) $10 \times 4$ | (98) $5 \times 2$  |
| (9) $2 \times 8$   | (39) $7 \times 2$  | (69) $6 \times 7$  | (99) $2 \times 4$  |
| (10) $4 \times 8$  | (40) $5 \times 9$  | (70) $1 \times 1$  | (100) $4 \times 4$ |
| (11) $8 \times 5$  | (41) $0 \times 7$  | (71) $2 \times 8$  | (101) $7 \times 0$ |
| (12) $8 \times 4$  | (42) $9 \times 2$  | (72) $1 \times 2$  | (102) $5 \times 3$ |
| (13) $3 \times 4$  | (43) $2 \times 8$  | (73) $9 \times 0$  | (103) $5 \times 2$ |
| (14) $6 \times 10$ | (44) $10 \times 7$ | (74) $10 \times 9$ | (104) $7 \times 6$ |
| (15) $6 \times 5$  | (45) $5 \times 1$  | (75) $7 \times 1$  | (105) $9 \times 1$ |
| (16) $7 \times 6$  | (46) $1 \times 4$  | (76) $9 \times 10$ | (106) $4 \times 7$ |
| (17) $8 \times 8$  | (47) $8 \times 4$  | (77) $2 \times 9$  | (107) $6 \times 2$ |
| (18) $0 \times 9$  | (48) $2 \times 6$  | (78) $8 \times 8$  | (108) $3 \times 3$ |
| (19) $0 \times 5$  | (49) $7 \times 9$  | (79) $2 \times 5$  | (109) $2 \times 7$ |
| (20) $2 \times 9$  | (50) $6 \times 7$  | (80) $9 \times 1$  | (110) $8 \times 3$ |
| (21) $0 \times 6$  | (51) $2 \times 8$  | (81) $10 \times 7$ | (111) $4 \times 1$ |
| (22) $10 \times 8$ | (52) $6 \times 5$  | (82) $1 \times 2$  | (112) $6 \times 7$ |
| (23) $5 \times 5$  | (53) $10 \times 6$ | (83) $4 \times 6$  | (113) $9 \times 1$ |
| (24) $3 \times 6$  | (54) $8 \times 7$  | (84) $2 \times 9$  | (114) $2 \times 3$ |
| (25) $0 \times 9$  | (55) $7 \times 2$  | (85) $3 \times 3$  | (115) $4 \times 0$ |
| (26) $0 \times 10$ | (56) $2 \times 9$  | (86) $10 \times 5$ | (116) $1 \times 5$ |
| (27) $10 \times 8$ | (57) $3 \times 3$  | (87) $8 \times 9$  | (117) $1 \times 1$ |
| (28) $2 \times 2$  | (58) $9 \times 2$  | (88) $8 \times 9$  | (118) $7 \times 7$ |
| (29) $2 \times 2$  | (59) $6 \times 4$  | (89) $3 \times 8$  | (119) $3 \times 3$ |
| (30) $0 \times 1$  | (60) $1 \times 9$  | (90) $7 \times 6$  | (120) $1 \times 6$ |

|                    |                    |                    |                      |
|--------------------|--------------------|--------------------|----------------------|
| (1) $3 \times 8$   | (31) $5 \times 0$  | (61) $2 \times 7$  | (91) $7 \times 2$    |
| (2) $9 \times 4$   | (32) $1 \times 3$  | (62) $4 \times 2$  | (92) $10 \times 8$   |
| (3) $6 \times 8$   | (33) $3 \times 0$  | (63) $2 \times 9$  | (93) $8 \times 2$    |
| (4) $5 \times 1$   | (34) $2 \times 8$  | (64) $4 \times 3$  | (94) $3 \times 5$    |
| (5) $10 \times 5$  | (35) $1 \times 8$  | (65) $3 \times 7$  | (95) $1 \times 8$    |
| (6) $8 \times 8$   | (36) $2 \times 2$  | (66) $1 \times 3$  | (96) $3 \times 8$    |
| (7) $3 \times 1$   | (37) $10 \times 3$ | (67) $6 \times 3$  | (97) $8 \times 1$    |
| (8) $5 \times 1$   | (38) $5 \times 8$  | (68) $7 \times 6$  | (98) $3 \times 3$    |
| (9) $0 \times 3$   | (39) $8 \times 5$  | (69) $1 \times 10$ | (99) $7 \times 1$    |
| (10) $9 \times 8$  | (40) $7 \times 7$  | (70) $6 \times 9$  | (100) $10 \times 1$  |
| (11) $9 \times 2$  | (41) $10 \times 1$ | (71) $2 \times 8$  | (101) $1 \times 1$   |
| (12) $0 \times 10$ | (42) $4 \times 6$  | (72) $10 \times 3$ | (102) $10 \times 10$ |
| (13) $9 \times 8$  | (43) $8 \times 1$  | (73) $3 \times 4$  | (103) $4 \times 5$   |
| (14) $6 \times 1$  | (44) $8 \times 7$  | (74) $8 \times 7$  | (104) $10 \times 7$  |
| (15) $8 \times 3$  | (45) $3 \times 5$  | (75) $6 \times 10$ | (105) $5 \times 3$   |
| (16) $10 \times 7$ | (46) $1 \times 2$  | (76) $6 \times 4$  | (106) $9 \times 8$   |
| (17) $5 \times 7$  | (47) $3 \times 10$ | (77) $7 \times 1$  | (107) $9 \times 1$   |
| (18) $10 \times 5$ | (48) $2 \times 2$  | (78) $3 \times 1$  | (108) $5 \times 4$   |
| (19) $3 \times 6$  | (49) $4 \times 0$  | (79) $0 \times 7$  | (109) $0 \times 1$   |
| (20) $6 \times 5$  | (50) $7 \times 7$  | (80) $5 \times 1$  | (110) $3 \times 2$   |
| (21) $6 \times 9$  | (51) $8 \times 5$  | (81) $3 \times 9$  | (111) $9 \times 5$   |
| (22) $5 \times 9$  | (52) $5 \times 4$  | (82) $6 \times 8$  | (112) $5 \times 7$   |
| (23) $2 \times 3$  | (53) $9 \times 9$  | (83) $8 \times 5$  | (113) $0 \times 0$   |
| (24) $1 \times 2$  | (54) $1 \times 7$  | (84) $6 \times 8$  | (114) $6 \times 10$  |
| (25) $5 \times 8$  | (55) $8 \times 8$  | (85) $5 \times 7$  | (115) $7 \times 1$   |
| (26) $10 \times 2$ | (56) $9 \times 0$  | (86) $3 \times 1$  | (116) $1 \times 3$   |
| (27) $7 \times 0$  | (57) $1 \times 3$  | (87) $3 \times 1$  | (117) $1 \times 5$   |
| (28) $0 \times 4$  | (58) $6 \times 5$  | (88) $1 \times 8$  | (118) $8 \times 5$   |
| (29) $1 \times 1$  | (59) $9 \times 0$  | (89) $7 \times 3$  | (119) $0 \times 4$   |
| (30) $0 \times 2$  | (60) $8 \times 3$  | (90) $8 \times 2$  | (120) $3 \times 7$   |

|                    |                    |                     |                     |
|--------------------|--------------------|---------------------|---------------------|
| (1) $2 \times 1$   | (31) $7 \times 0$  | (61) $8 \times 4$   | (91) $1 \times 1$   |
| (2) $9 \times 2$   | (32) $9 \times 4$  | (62) $3 \times 1$   | (92) $9 \times 2$   |
| (3) $0 \times 4$   | (33) $4 \times 7$  | (63) $5 \times 4$   | (93) $8 \times 3$   |
| (4) $3 \times 1$   | (34) $3 \times 7$  | (64) $10 \times 9$  | (94) $6 \times 2$   |
| (5) $8 \times 9$   | (35) $7 \times 9$  | (65) $3 \times 6$   | (95) $9 \times 6$   |
| (6) $6 \times 10$  | (36) $2 \times 1$  | (66) $10 \times 10$ | (96) $10 \times 0$  |
| (7) $2 \times 7$   | (37) $3 \times 10$ | (67) $0 \times 9$   | (97) $7 \times 3$   |
| (8) $7 \times 10$  | (38) $2 \times 8$  | (68) $9 \times 4$   | (98) $9 \times 5$   |
| (9) $7 \times 0$   | (39) $5 \times 2$  | (69) $2 \times 3$   | (99) $7 \times 9$   |
| (10) $4 \times 8$  | (40) $2 \times 2$  | (70) $5 \times 8$   | (100) $6 \times 0$  |
| (11) $6 \times 0$  | (41) $9 \times 8$  | (71) $3 \times 6$   | (101) $0 \times 7$  |
| (12) $9 \times 5$  | (42) $2 \times 1$  | (72) $10 \times 6$  | (102) $8 \times 1$  |
| (13) $6 \times 4$  | (43) $6 \times 7$  | (73) $0 \times 9$   | (103) $0 \times 6$  |
| (14) $6 \times 1$  | (44) $8 \times 4$  | (74) $5 \times 9$   | (104) $0 \times 9$  |
| (15) $3 \times 7$  | (45) $3 \times 5$  | (75) $5 \times 0$   | (105) $6 \times 2$  |
| (16) $3 \times 5$  | (46) $5 \times 1$  | (76) $5 \times 2$   | (106) $0 \times 9$  |
| (17) $10 \times 8$ | (47) $2 \times 7$  | (77) $2 \times 2$   | (107) $7 \times 3$  |
| (18) $9 \times 9$  | (48) $5 \times 4$  | (78) $1 \times 6$   | (108) $8 \times 9$  |
| (19) $5 \times 7$  | (49) $3 \times 9$  | (79) $6 \times 10$  | (109) $8 \times 1$  |
| (20) $0 \times 3$  | (50) $3 \times 7$  | (80) $10 \times 6$  | (110) $1 \times 9$  |
| (21) $10 \times 3$ | (51) $1 \times 4$  | (81) $9 \times 8$   | (111) $1 \times 4$  |
| (22) $10 \times 0$ | (52) $9 \times 9$  | (82) $4 \times 5$   | (112) $7 \times 0$  |
| (23) $5 \times 4$  | (53) $2 \times 2$  | (83) $5 \times 10$  | (113) $0 \times 4$  |
| (24) $7 \times 4$  | (54) $6 \times 2$  | (84) $8 \times 7$   | (114) $9 \times 10$ |
| (25) $2 \times 9$  | (55) $5 \times 2$  | (85) $6 \times 9$   | (115) $1 \times 4$  |
| (26) $7 \times 5$  | (56) $10 \times 8$ | (86) $10 \times 3$  | (116) $1 \times 10$ |
| (27) $4 \times 0$  | (57) $0 \times 4$  | (87) $4 \times 0$   | (117) $10 \times 1$ |
| (28) $9 \times 8$  | (58) $0 \times 9$  | (88) $1 \times 5$   | (118) $4 \times 4$  |
| (29) $7 \times 9$  | (59) $9 \times 3$  | (89) $10 \times 1$  | (119) $0 \times 8$  |
| (30) $9 \times 1$  | (60) $10 \times 5$ | (90) $8 \times 3$   | (120) $4 \times 1$  |

|                    |                    |                    |                     |
|--------------------|--------------------|--------------------|---------------------|
| (1) $3 \times 5$   | (31) $1 \times 5$  | (61) $8 \times 7$  | (91) $9 \times 0$   |
| (2) $0 \times 7$   | (32) $2 \times 8$  | (62) $10 \times 9$ | (92) $9 \times 8$   |
| (3) $9 \times 7$   | (33) $3 \times 0$  | (63) $5 \times 5$  | (93) $2 \times 10$  |
| (4) $3 \times 0$   | (34) $3 \times 6$  | (64) $2 \times 10$ | (94) $1 \times 7$   |
| (5) $9 \times 1$   | (35) $4 \times 5$  | (65) $4 \times 1$  | (95) $1 \times 4$   |
| (6) $3 \times 9$   | (36) $8 \times 2$  | (66) $6 \times 10$ | (96) $7 \times 10$  |
| (7) $7 \times 9$   | (37) $8 \times 8$  | (67) $3 \times 7$  | (97) $2 \times 1$   |
| (8) $5 \times 3$   | (38) $7 \times 10$ | (68) $0 \times 1$  | (98) $7 \times 7$   |
| (9) $9 \times 10$  | (39) $6 \times 5$  | (69) $5 \times 2$  | (99) $10 \times 6$  |
| (10) $10 \times 9$ | (40) $10 \times 3$ | (70) $1 \times 5$  | (100) $0 \times 9$  |
| (11) $8 \times 2$  | (41) $0 \times 5$  | (71) $4 \times 4$  | (101) $3 \times 3$  |
| (12) $6 \times 7$  | (42) $5 \times 6$  | (72) $4 \times 5$  | (102) $8 \times 3$  |
| (13) $6 \times 9$  | (43) $10 \times 2$ | (73) $4 \times 9$  | (103) $8 \times 8$  |
| (14) $1 \times 6$  | (44) $8 \times 5$  | (74) $2 \times 3$  | (104) $10 \times 1$ |
| (15) $10 \times 2$ | (45) $6 \times 5$  | (75) $3 \times 4$  | (105) $3 \times 7$  |
| (16) $1 \times 8$  | (46) $5 \times 7$  | (76) $7 \times 8$  | (106) $3 \times 5$  |
| (17) $4 \times 5$  | (47) $2 \times 2$  | (77) $10 \times 3$ | (107) $10 \times 8$ |
| (18) $8 \times 9$  | (48) $5 \times 7$  | (78) $2 \times 4$  | (108) $1 \times 3$  |
| (19) $2 \times 5$  | (49) $9 \times 2$  | (79) $3 \times 6$  | (109) $1 \times 8$  |
| (20) $1 \times 2$  | (50) $4 \times 7$  | (80) $6 \times 3$  | (110) $1 \times 5$  |
| (21) $0 \times 9$  | (51) $10 \times 0$ | (81) $10 \times 1$ | (111) $6 \times 5$  |
| (22) $10 \times 1$ | (52) $7 \times 6$  | (82) $1 \times 6$  | (112) $1 \times 1$  |
| (23) $1 \times 3$  | (53) $6 \times 3$  | (83) $9 \times 6$  | (113) $6 \times 7$  |
| (24) $4 \times 6$  | (54) $10 \times 8$ | (84) $0 \times 5$  | (114) $1 \times 10$ |
| (25) $8 \times 9$  | (55) $1 \times 3$  | (85) $9 \times 4$  | (115) $2 \times 10$ |
| (26) $6 \times 0$  | (56) $9 \times 9$  | (86) $3 \times 8$  | (116) $8 \times 6$  |
| (27) $9 \times 8$  | (57) $4 \times 1$  | (87) $6 \times 10$ | (117) $9 \times 8$  |
| (28) $6 \times 6$  | (58) $6 \times 2$  | (88) $7 \times 3$  | (118) $1 \times 3$  |
| (29) $9 \times 7$  | (59) $6 \times 0$  | (89) $1 \times 3$  | (119) $7 \times 2$  |
| (30) $8 \times 5$  | (60) $9 \times 8$  | (90) $5 \times 3$  | (120) $8 \times 0$  |

|                     |                    |                    |                     |
|---------------------|--------------------|--------------------|---------------------|
| (1) $2 \times 10$   | (31) $4 \times 8$  | (61) $9 \times 5$  | (91) $6 \times 8$   |
| (2) $7 \times 2$    | (32) $7 \times 3$  | (62) $7 \times 10$ | (92) $4 \times 9$   |
| (3) $2 \times 4$    | (33) $5 \times 1$  | (63) $10 \times 7$ | (93) $8 \times 1$   |
| (4) $6 \times 7$    | (34) $2 \times 9$  | (64) $2 \times 5$  | (94) $4 \times 1$   |
| (5) $1 \times 6$    | (35) $4 \times 5$  | (65) $3 \times 4$  | (95) $8 \times 5$   |
| (6) $2 \times 5$    | (36) $6 \times 6$  | (66) $6 \times 1$  | (96) $7 \times 0$   |
| (7) $10 \times 9$   | (37) $7 \times 5$  | (67) $3 \times 9$  | (97) $3 \times 1$   |
| (8) $6 \times 10$   | (38) $8 \times 9$  | (68) $10 \times 8$ | (98) $9 \times 8$   |
| (9) $1 \times 5$    | (39) $2 \times 0$  | (69) $6 \times 5$  | (99) $5 \times 4$   |
| (10) $2 \times 6$   | (40) $7 \times 2$  | (70) $0 \times 0$  | (100) $2 \times 1$  |
| (11) $8 \times 2$   | (41) $7 \times 3$  | (71) $4 \times 4$  | (101) $3 \times 10$ |
| (12) $9 \times 2$   | (42) $0 \times 4$  | (72) $2 \times 3$  | (102) $5 \times 4$  |
| (13) $10 \times 9$  | (43) $2 \times 8$  | (73) $2 \times 3$  | (103) $0 \times 9$  |
| (14) $7 \times 6$   | (44) $3 \times 5$  | (74) $4 \times 1$  | (104) $6 \times 1$  |
| (15) $0 \times 9$   | (45) $7 \times 2$  | (75) $2 \times 9$  | (105) $8 \times 10$ |
| (16) $10 \times 3$  | (46) $1 \times 8$  | (76) $2 \times 6$  | (106) $9 \times 8$  |
| (17) $2 \times 6$   | (47) $6 \times 2$  | (77) $4 \times 3$  | (107) $3 \times 8$  |
| (18) $5 \times 6$   | (48) $4 \times 7$  | (78) $9 \times 9$  | (108) $6 \times 9$  |
| (19) $9 \times 6$   | (49) $3 \times 9$  | (79) $0 \times 6$  | (109) $10 \times 6$ |
| (20) $9 \times 8$   | (50) $2 \times 5$  | (80) $9 \times 4$  | (110) $3 \times 6$  |
| (21) $9 \times 9$   | (51) $9 \times 4$  | (81) $4 \times 1$  | (111) $1 \times 4$  |
| (22) $1 \times 5$   | (52) $10 \times 7$ | (82) $9 \times 3$  | (112) $6 \times 6$  |
| (23) $6 \times 0$   | (53) $7 \times 2$  | (83) $6 \times 8$  | (113) $10 \times 0$ |
| (24) $6 \times 1$   | (54) $9 \times 1$  | (84) $8 \times 0$  | (114) $8 \times 8$  |
| (25) $10 \times 10$ | (55) $7 \times 2$  | (85) $6 \times 1$  | (115) $0 \times 9$  |
| (26) $9 \times 7$   | (56) $4 \times 9$  | (86) $6 \times 6$  | (116) $1 \times 1$  |
| (27) $10 \times 2$  | (57) $4 \times 0$  | (87) $9 \times 1$  | (117) $1 \times 5$  |
| (28) $9 \times 10$  | (58) $8 \times 4$  | (88) $1 \times 1$  | (118) $1 \times 9$  |
| (29) $4 \times 5$   | (59) $4 \times 8$  | (89) $2 \times 4$  | (119) $0 \times 0$  |
| (30) $6 \times 6$   | (60) $2 \times 4$  | (90) $7 \times 7$  | (120) $10 \times 3$ |

|                    |                     |                     |                     |
|--------------------|---------------------|---------------------|---------------------|
| (1) $4 \times 6$   | (31) $9 \times 5$   | (61) $10 \times 7$  | (91) $0 \times 1$   |
| (2) $9 \times 7$   | (32) $9 \times 6$   | (62) $2 \times 0$   | (92) $7 \times 2$   |
| (3) $5 \times 9$   | (33) $1 \times 4$   | (63) $5 \times 3$   | (93) $3 \times 8$   |
| (4) $10 \times 10$ | (34) $8 \times 0$   | (64) $10 \times 6$  | (94) $7 \times 3$   |
| (5) $2 \times 2$   | (35) $10 \times 8$  | (65) $8 \times 8$   | (95) $0 \times 7$   |
| (6) $2 \times 6$   | (36) $2 \times 7$   | (66) $10 \times 8$  | (96) $6 \times 9$   |
| (7) $2 \times 9$   | (37) $4 \times 2$   | (67) $5 \times 6$   | (97) $1 \times 1$   |
| (8) $2 \times 0$   | (38) $5 \times 3$   | (68) $2 \times 2$   | (98) $2 \times 5$   |
| (9) $4 \times 4$   | (39) $3 \times 2$   | (69) $7 \times 7$   | (99) $7 \times 8$   |
| (10) $10 \times 9$ | (40) $10 \times 8$  | (70) $6 \times 7$   | (100) $7 \times 1$  |
| (11) $6 \times 4$  | (41) $10 \times 2$  | (71) $1 \times 0$   | (101) $10 \times 6$ |
| (12) $6 \times 1$  | (42) $4 \times 2$   | (72) $10 \times 2$  | (102) $8 \times 6$  |
| (13) $10 \times 8$ | (43) $10 \times 10$ | (73) $7 \times 8$   | (103) $2 \times 6$  |
| (14) $3 \times 4$  | (44) $5 \times 4$   | (74) $4 \times 7$   | (104) $6 \times 9$  |
| (15) $8 \times 0$  | (45) $3 \times 7$   | (75) $4 \times 7$   | (105) $7 \times 6$  |
| (16) $10 \times 4$ | (46) $2 \times 2$   | (76) $2 \times 1$   | (106) $2 \times 10$ |
| (17) $0 \times 8$  | (47) $9 \times 0$   | (77) $7 \times 2$   | (107) $9 \times 8$  |
| (18) $8 \times 3$  | (48) $2 \times 3$   | (78) $1 \times 1$   | (108) $6 \times 6$  |
| (19) $9 \times 3$  | (49) $5 \times 7$   | (79) $7 \times 6$   | (109) $6 \times 5$  |
| (20) $6 \times 7$  | (50) $4 \times 5$   | (80) $8 \times 7$   | (110) $7 \times 8$  |
| (21) $9 \times 2$  | (51) $4 \times 1$   | (81) $8 \times 8$   | (111) $6 \times 7$  |
| (22) $10 \times 3$ | (52) $0 \times 9$   | (82) $4 \times 8$   | (112) $3 \times 5$  |
| (23) $9 \times 6$  | (53) $1 \times 8$   | (83) $9 \times 9$   | (113) $4 \times 4$  |
| (24) $10 \times 1$ | (54) $1 \times 0$   | (84) $1 \times 2$   | (114) $9 \times 9$  |
| (25) $6 \times 0$  | (55) $7 \times 4$   | (85) $10 \times 10$ | (115) $7 \times 10$ |
| (26) $3 \times 3$  | (56) $4 \times 6$   | (86) $1 \times 9$   | (116) $6 \times 7$  |
| (27) $3 \times 10$ | (57) $3 \times 6$   | (87) $9 \times 6$   | (117) $6 \times 9$  |
| (28) $10 \times 1$ | (58) $3 \times 6$   | (88) $1 \times 10$  | (118) $1 \times 7$  |
| (29) $10 \times 1$ | (59) $9 \times 4$   | (89) $1 \times 2$   | (119) $1 \times 9$  |
| (30) $5 \times 5$  | (60) $9 \times 2$   | (90) $7 \times 4$   | (120) $10 \times 3$ |



|                    |                    |                    |                     |
|--------------------|--------------------|--------------------|---------------------|
| (1) $6 \times 4$   | (31) $6 \times 2$  | (61) $2 \times 9$  | (91) $0 \times 1$   |
| (2) $5 \times 2$   | (32) $1 \times 8$  | (62) $7 \times 7$  | (92) $5 \times 10$  |
| (3) $8 \times 8$   | (33) $5 \times 5$  | (63) $9 \times 4$  | (93) $6 \times 7$   |
| (4) $2 \times 9$   | (34) $7 \times 10$ | (64) $0 \times 9$  | (94) $3 \times 6$   |
| (5) $6 \times 8$   | (35) $3 \times 6$  | (65) $8 \times 5$  | (95) $10 \times 3$  |
| (6) $1 \times 1$   | (36) $3 \times 5$  | (66) $9 \times 1$  | (96) $3 \times 2$   |
| (7) $4 \times 1$   | (37) $7 \times 2$  | (67) $8 \times 9$  | (97) $2 \times 4$   |
| (8) $2 \times 6$   | (38) $4 \times 3$  | (68) $5 \times 9$  | (98) $1 \times 2$   |
| (9) $4 \times 10$  | (39) $5 \times 7$  | (69) $5 \times 10$ | (99) $3 \times 8$   |
| (10) $7 \times 6$  | (40) $1 \times 6$  | (70) $3 \times 7$  | (100) $3 \times 6$  |
| (11) $2 \times 5$  | (41) $6 \times 9$  | (71) $2 \times 1$  | (101) $10 \times 0$ |
| (12) $8 \times 4$  | (42) $2 \times 5$  | (72) $6 \times 6$  | (102) $8 \times 4$  |
| (13) $2 \times 5$  | (43) $2 \times 4$  | (73) $5 \times 3$  | (103) $4 \times 1$  |
| (14) $2 \times 9$  | (44) $9 \times 0$  | (74) $1 \times 5$  | (104) $9 \times 6$  |
| (15) $8 \times 9$  | (45) $9 \times 10$ | (75) $0 \times 5$  | (105) $5 \times 10$ |
| (16) $10 \times 9$ | (46) $2 \times 9$  | (76) $8 \times 8$  | (106) $4 \times 2$  |
| (17) $1 \times 9$  | (47) $4 \times 9$  | (77) $4 \times 8$  | (107) $0 \times 1$  |
| (18) $6 \times 6$  | (48) $1 \times 5$  | (78) $7 \times 8$  | (108) $9 \times 10$ |
| (19) $7 \times 8$  | (49) $4 \times 9$  | (79) $1 \times 3$  | (109) $2 \times 4$  |
| (20) $4 \times 2$  | (50) $1 \times 1$  | (80) $2 \times 7$  | (110) $5 \times 6$  |
| (21) $5 \times 7$  | (51) $9 \times 8$  | (81) $9 \times 1$  | (111) $6 \times 10$ |
| (22) $9 \times 2$  | (52) $10 \times 6$ | (82) $8 \times 6$  | (112) $3 \times 6$  |
| (23) $2 \times 6$  | (53) $2 \times 8$  | (83) $3 \times 8$  | (113) $10 \times 9$ |
| (24) $1 \times 9$  | (54) $3 \times 4$  | (84) $6 \times 9$  | (114) $5 \times 0$  |
| (25) $3 \times 4$  | (55) $1 \times 6$  | (85) $1 \times 7$  | (115) $6 \times 1$  |
| (26) $1 \times 4$  | (56) $8 \times 9$  | (86) $0 \times 2$  | (116) $3 \times 2$  |
| (27) $7 \times 2$  | (57) $0 \times 3$  | (87) $2 \times 5$  | (117) $9 \times 1$  |
| (28) $3 \times 1$  | (58) $8 \times 9$  | (88) $10 \times 8$ | (118) $7 \times 7$  |
| (29) $8 \times 2$  | (59) $4 \times 6$  | (89) $6 \times 0$  | (119) $2 \times 3$  |
| (30) $3 \times 5$  | (60) $0 \times 4$  | (90) $9 \times 7$  | (120) $2 \times 2$  |

|                    |                    |                    |                     |
|--------------------|--------------------|--------------------|---------------------|
| (1) $6 \times 3$   | (31) $4 \times 4$  | (61) $1 \times 1$  | (91) $2 \times 7$   |
| (2) $7 \times 9$   | (32) $7 \times 8$  | (62) $5 \times 4$  | (92) $3 \times 3$   |
| (3) $3 \times 5$   | (33) $4 \times 9$  | (63) $7 \times 4$  | (93) $3 \times 1$   |
| (4) $5 \times 1$   | (34) $7 \times 10$ | (64) $10 \times 7$ | (94) $5 \times 9$   |
| (5) $9 \times 5$   | (35) $1 \times 5$  | (65) $4 \times 4$  | (95) $7 \times 8$   |
| (6) $1 \times 3$   | (36) $4 \times 6$  | (66) $2 \times 2$  | (96) $5 \times 1$   |
| (7) $4 \times 5$   | (37) $9 \times 2$  | (67) $2 \times 3$  | (97) $10 \times 7$  |
| (8) $9 \times 6$   | (38) $6 \times 0$  | (68) $8 \times 9$  | (98) $2 \times 2$   |
| (9) $0 \times 0$   | (39) $2 \times 5$  | (69) $4 \times 7$  | (99) $7 \times 6$   |
| (10) $10 \times 1$ | (40) $5 \times 4$  | (70) $0 \times 6$  | (100) $10 \times 0$ |
| (11) $7 \times 5$  | (41) $3 \times 4$  | (71) $5 \times 4$  | (101) $10 \times 5$ |
| (12) $4 \times 10$ | (42) $0 \times 10$ | (72) $1 \times 2$  | (102) $9 \times 8$  |
| (13) $0 \times 3$  | (43) $9 \times 1$  | (73) $5 \times 4$  | (103) $1 \times 8$  |
| (14) $5 \times 5$  | (44) $6 \times 6$  | (74) $5 \times 9$  | (104) $3 \times 1$  |
| (15) $3 \times 7$  | (45) $5 \times 3$  | (75) $4 \times 2$  | (105) $1 \times 6$  |
| (16) $4 \times 4$  | (46) $0 \times 10$ | (76) $8 \times 6$  | (106) $5 \times 2$  |
| (17) $10 \times 7$ | (47) $9 \times 3$  | (77) $5 \times 6$  | (107) $2 \times 9$  |
| (18) $1 \times 9$  | (48) $8 \times 1$  | (78) $0 \times 8$  | (108) $9 \times 3$  |
| (19) $1 \times 8$  | (49) $3 \times 6$  | (79) $10 \times 3$ | (109) $4 \times 10$ |
| (20) $8 \times 4$  | (50) $10 \times 8$ | (80) $8 \times 7$  | (110) $1 \times 2$  |
| (21) $3 \times 3$  | (51) $2 \times 5$  | (81) $0 \times 6$  | (111) $3 \times 5$  |
| (22) $7 \times 1$  | (52) $8 \times 2$  | (82) $8 \times 7$  | (112) $2 \times 7$  |
| (23) $8 \times 9$  | (53) $8 \times 9$  | (83) $8 \times 7$  | (113) $6 \times 1$  |
| (24) $10 \times 3$ | (54) $1 \times 2$  | (84) $1 \times 6$  | (114) $9 \times 2$  |
| (25) $10 \times 1$ | (55) $6 \times 10$ | (85) $0 \times 8$  | (115) $4 \times 5$  |
| (26) $5 \times 5$  | (56) $2 \times 9$  | (86) $7 \times 4$  | (116) $4 \times 8$  |
| (27) $8 \times 6$  | (57) $9 \times 3$  | (87) $9 \times 9$  | (117) $7 \times 6$  |
| (28) $6 \times 1$  | (58) $4 \times 2$  | (88) $9 \times 9$  | (118) $7 \times 1$  |
| (29) $4 \times 2$  | (59) $5 \times 10$ | (89) $10 \times 9$ | (119) $2 \times 10$ |
| (30) $10 \times 2$ | (60) $9 \times 10$ | (90) $8 \times 3$  | (120) $6 \times 8$  |

|                     |                     |                    |                     |
|---------------------|---------------------|--------------------|---------------------|
| (1) $4 \times 4$    | (31) $7 \times 7$   | (61) $8 \times 7$  | (91) $9 \times 5$   |
| (2) $4 \times 9$    | (32) $8 \times 1$   | (62) $8 \times 1$  | (92) $4 \times 1$   |
| (3) $0 \times 4$    | (33) $0 \times 4$   | (63) $5 \times 8$  | (93) $7 \times 10$  |
| (4) $8 \times 9$    | (34) $6 \times 4$   | (64) $2 \times 5$  | (94) $1 \times 2$   |
| (5) $0 \times 10$   | (35) $0 \times 2$   | (65) $0 \times 0$  | (95) $1 \times 0$   |
| (6) $6 \times 6$    | (36) $6 \times 0$   | (66) $3 \times 3$  | (96) $2 \times 5$   |
| (7) $8 \times 9$    | (37) $10 \times 7$  | (67) $9 \times 9$  | (97) $0 \times 7$   |
| (8) $5 \times 4$    | (38) $5 \times 3$   | (68) $9 \times 4$  | (98) $6 \times 2$   |
| (9) $3 \times 10$   | (39) $5 \times 7$   | (69) $4 \times 10$ | (99) $1 \times 9$   |
| (10) $3 \times 10$  | (40) $5 \times 2$   | (70) $6 \times 7$  | (100) $7 \times 9$  |
| (11) $0 \times 8$   | (41) $10 \times 1$  | (71) $2 \times 4$  | (101) $10 \times 3$ |
| (12) $7 \times 10$  | (42) $6 \times 9$   | (72) $10 \times 6$ | (102) $0 \times 10$ |
| (13) $7 \times 9$   | (43) $8 \times 1$   | (73) $5 \times 7$  | (103) $7 \times 7$  |
| (14) $3 \times 2$   | (44) $7 \times 6$   | (74) $10 \times 6$ | (104) $1 \times 6$  |
| (15) $9 \times 0$   | (45) $0 \times 9$   | (75) $10 \times 1$ | (105) $4 \times 5$  |
| (16) $7 \times 4$   | (46) $7 \times 7$   | (76) $2 \times 7$  | (106) $7 \times 1$  |
| (17) $4 \times 2$   | (47) $10 \times 7$  | (77) $3 \times 2$  | (107) $6 \times 6$  |
| (18) $10 \times 10$ | (48) $1 \times 3$   | (78) $0 \times 9$  | (108) $5 \times 9$  |
| (19) $6 \times 10$  | (49) $10 \times 7$  | (79) $8 \times 10$ | (109) $1 \times 4$  |
| (20) $1 \times 9$   | (50) $2 \times 6$   | (80) $5 \times 4$  | (110) $5 \times 8$  |
| (21) $10 \times 2$  | (51) $7 \times 6$   | (81) $8 \times 9$  | (111) $1 \times 8$  |
| (22) $3 \times 1$   | (52) $3 \times 8$   | (82) $9 \times 4$  | (112) $7 \times 3$  |
| (23) $6 \times 7$   | (53) $5 \times 1$   | (83) $6 \times 8$  | (113) $9 \times 9$  |
| (24) $3 \times 2$   | (54) $7 \times 4$   | (84) $9 \times 7$  | (114) $9 \times 2$  |
| (25) $9 \times 5$   | (55) $1 \times 1$   | (85) $1 \times 1$  | (115) $4 \times 10$ |
| (26) $4 \times 3$   | (56) $10 \times 10$ | (86) $9 \times 8$  | (116) $10 \times 6$ |
| (27) $2 \times 7$   | (57) $8 \times 1$   | (87) $7 \times 9$  | (117) $10 \times 3$ |
| (28) $9 \times 1$   | (58) $7 \times 3$   | (88) $4 \times 9$  | (118) $8 \times 2$  |
| (29) $7 \times 8$   | (59) $3 \times 7$   | (89) $1 \times 7$  | (119) $3 \times 0$  |
| (30) $10 \times 8$  | (60) $8 \times 8$   | (90) $7 \times 3$  | (120) $2 \times 9$  |

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|--------------------|--------------------|---------------------|---------------------|
| (1) $10 \times 4$  | (31) $10 \times 1$ | (61) $7 \times 6$   | (91) $3 \times 3$   |
| (2) $4 \times 1$   | (32) $7 \times 9$  | (62) $8 \times 3$   | (92) $3 \times 8$   |
| (3) $9 \times 9$   | (33) $4 \times 8$  | (63) $8 \times 7$   | (93) $10 \times 7$  |
| (4) $10 \times 4$  | (34) $4 \times 2$  | (64) $10 \times 9$  | (94) $3 \times 7$   |
| (5) $3 \times 2$   | (35) $3 \times 5$  | (65) $10 \times 6$  | (95) $3 \times 5$   |
| (6) $5 \times 6$   | (36) $1 \times 3$  | (66) $8 \times 10$  | (96) $7 \times 10$  |
| (7) $1 \times 5$   | (37) $7 \times 7$  | (67) $9 \times 10$  | (97) $8 \times 7$   |
| (8) $7 \times 0$   | (38) $8 \times 2$  | (68) $8 \times 1$   | (98) $10 \times 6$  |
| (9) $6 \times 6$   | (39) $10 \times 3$ | (69) $5 \times 10$  | (99) $3 \times 5$   |
| (10) $9 \times 3$  | (40) $6 \times 6$  | (70) $5 \times 9$   | (100) $9 \times 1$  |
| (11) $2 \times 8$  | (41) $6 \times 1$  | (71) $8 \times 5$   | (101) $3 \times 9$  |
| (12) $6 \times 1$  | (42) $1 \times 8$  | (72) $1 \times 6$   | (102) $7 \times 6$  |
| (13) $9 \times 0$  | (43) $3 \times 1$  | (73) $2 \times 3$   | (103) $10 \times 7$ |
| (14) $9 \times 6$  | (44) $2 \times 0$  | (74) $4 \times 7$   | (104) $0 \times 4$  |
| (15) $1 \times 10$ | (45) $2 \times 3$  | (75) $4 \times 3$   | (105) $8 \times 2$  |
| (16) $2 \times 9$  | (46) $10 \times 8$ | (76) $8 \times 3$   | (106) $2 \times 10$ |
| (17) $3 \times 9$  | (47) $0 \times 3$  | (77) $2 \times 5$   | (107) $2 \times 1$  |
| (18) $6 \times 4$  | (48) $10 \times 9$ | (78) $4 \times 7$   | (108) $10 \times 0$ |
| (19) $6 \times 9$  | (49) $1 \times 5$  | (79) $9 \times 9$   | (109) $10 \times 1$ |
| (20) $5 \times 4$  | (50) $8 \times 7$  | (80) $0 \times 7$   | (110) $5 \times 2$  |
| (21) $5 \times 2$  | (51) $5 \times 6$  | (81) $0 \times 7$   | (111) $1 \times 7$  |
| (22) $7 \times 2$  | (52) $8 \times 4$  | (82) $4 \times 9$   | (112) $9 \times 1$  |
| (23) $2 \times 10$ | (53) $4 \times 5$  | (83) $10 \times 10$ | (113) $0 \times 6$  |
| (24) $10 \times 3$ | (54) $10 \times 3$ | (84) $7 \times 8$   | (114) $7 \times 10$ |
| (25) $5 \times 3$  | (55) $3 \times 1$  | (85) $4 \times 1$   | (115) $4 \times 10$ |
| (26) $2 \times 2$  | (56) $1 \times 2$  | (86) $2 \times 1$   | (116) $5 \times 10$ |
| (27) $2 \times 4$  | (57) $4 \times 0$  | (87) $5 \times 9$   | (117) $9 \times 9$  |
| (28) $8 \times 4$  | (58) $3 \times 3$  | (88) $8 \times 9$   | (118) $9 \times 2$  |
| (29) $9 \times 4$  | (59) $1 \times 5$  | (89) $4 \times 3$   | (119) $10 \times 0$ |
| (30) $4 \times 9$  | (60) $6 \times 3$  | (90) $10 \times 8$  | (120) $9 \times 6$  |

|                    |                    |                    |                     |
|--------------------|--------------------|--------------------|---------------------|
| (1) $2 \times 8$   | (31) $6 \times 2$  | (61) $0 \times 1$  | (91) $9 \times 5$   |
| (2) $2 \times 10$  | (32) $3 \times 1$  | (62) $2 \times 1$  | (92) $3 \times 5$   |
| (3) $6 \times 8$   | (33) $7 \times 7$  | (63) $4 \times 8$  | (93) $1 \times 0$   |
| (4) $10 \times 10$ | (34) $9 \times 3$  | (64) $4 \times 1$  | (94) $2 \times 3$   |
| (5) $7 \times 1$   | (35) $9 \times 0$  | (65) $3 \times 7$  | (95) $3 \times 10$  |
| (6) $6 \times 5$   | (36) $1 \times 6$  | (66) $2 \times 3$  | (96) $9 \times 10$  |
| (7) $8 \times 8$   | (37) $1 \times 6$  | (67) $0 \times 7$  | (97) $8 \times 1$   |
| (8) $9 \times 5$   | (38) $10 \times 0$ | (68) $5 \times 2$  | (98) $1 \times 4$   |
| (9) $8 \times 7$   | (39) $3 \times 2$  | (69) $6 \times 2$  | (99) $2 \times 4$   |
| (10) $9 \times 2$  | (40) $2 \times 6$  | (70) $1 \times 7$  | (100) $6 \times 4$  |
| (11) $3 \times 0$  | (41) $5 \times 4$  | (71) $2 \times 2$  | (101) $4 \times 4$  |
| (12) $4 \times 4$  | (42) $3 \times 6$  | (72) $2 \times 3$  | (102) $1 \times 6$  |
| (13) $7 \times 1$  | (43) $2 \times 8$  | (73) $6 \times 5$  | (103) $3 \times 4$  |
| (14) $5 \times 0$  | (44) $8 \times 10$ | (74) $6 \times 8$  | (104) $6 \times 3$  |
| (15) $5 \times 9$  | (45) $8 \times 4$  | (75) $9 \times 2$  | (105) $5 \times 3$  |
| (16) $1 \times 0$  | (46) $8 \times 5$  | (76) $6 \times 1$  | (106) $10 \times 4$ |
| (17) $3 \times 7$  | (47) $4 \times 7$  | (77) $1 \times 6$  | (107) $8 \times 4$  |
| (18) $6 \times 7$  | (48) $6 \times 6$  | (78) $1 \times 8$  | (108) $7 \times 3$  |
| (19) $6 \times 1$  | (49) $7 \times 9$  | (79) $8 \times 0$  | (109) $8 \times 3$  |
| (20) $2 \times 3$  | (50) $10 \times 9$ | (80) $7 \times 1$  | (110) $4 \times 4$  |
| (21) $0 \times 5$  | (51) $1 \times 3$  | (81) $8 \times 2$  | (111) $3 \times 8$  |
| (22) $7 \times 9$  | (52) $0 \times 6$  | (82) $7 \times 9$  | (112) $2 \times 8$  |
| (23) $10 \times 9$ | (53) $1 \times 10$ | (83) $0 \times 9$  | (113) $1 \times 1$  |
| (24) $4 \times 8$  | (54) $10 \times 4$ | (84) $4 \times 1$  | (114) $5 \times 1$  |
| (25) $10 \times 2$ | (55) $8 \times 8$  | (85) $7 \times 1$  | (115) $9 \times 1$  |
| (26) $6 \times 3$  | (56) $8 \times 4$  | (86) $2 \times 6$  | (116) $2 \times 1$  |
| (27) $8 \times 4$  | (57) $1 \times 8$  | (87) $4 \times 1$  | (117) $2 \times 1$  |
| (28) $1 \times 1$  | (58) $6 \times 5$  | (88) $3 \times 4$  | (118) $2 \times 8$  |
| (29) $0 \times 6$  | (59) $10 \times 6$ | (89) $9 \times 10$ | (119) $0 \times 2$  |
| (30) $4 \times 10$ | (60) $0 \times 7$  | (90) $9 \times 10$ | (120) $4 \times 4$  |

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|---------------------|--------------------|--------------------|---------------------|
| (1) $1 \times 4$    | (31) $10 \times 8$ | (61) $7 \times 8$  | (91) $9 \times 5$   |
| (2) $2 \times 9$    | (32) $5 \times 0$  | (62) $8 \times 8$  | (92) $4 \times 4$   |
| (3) $2 \times 2$    | (33) $9 \times 4$  | (63) $1 \times 3$  | (93) $10 \times 9$  |
| (4) $10 \times 8$   | (34) $4 \times 3$  | (64) $0 \times 1$  | (94) $2 \times 2$   |
| (5) $5 \times 6$    | (35) $7 \times 5$  | (65) $2 \times 3$  | (95) $1 \times 1$   |
| (6) $6 \times 6$    | (36) $1 \times 3$  | (66) $2 \times 2$  | (96) $8 \times 5$   |
| (7) $7 \times 2$    | (37) $7 \times 9$  | (67) $1 \times 3$  | (97) $7 \times 5$   |
| (8) $0 \times 7$    | (38) $4 \times 2$  | (68) $9 \times 4$  | (98) $0 \times 2$   |
| (9) $2 \times 5$    | (39) $7 \times 9$  | (69) $3 \times 6$  | (99) $5 \times 10$  |
| (10) $8 \times 7$   | (40) $7 \times 4$  | (70) $9 \times 1$  | (100) $1 \times 0$  |
| (11) $4 \times 6$   | (41) $3 \times 6$  | (71) $7 \times 8$  | (101) $0 \times 6$  |
| (12) $7 \times 4$   | (42) $4 \times 3$  | (72) $2 \times 5$  | (102) $8 \times 1$  |
| (13) $6 \times 10$  | (43) $0 \times 4$  | (73) $10 \times 8$ | (103) $4 \times 1$  |
| (14) $8 \times 8$   | (44) $7 \times 1$  | (74) $8 \times 6$  | (104) $4 \times 10$ |
| (15) $10 \times 10$ | (45) $3 \times 8$  | (75) $7 \times 6$  | (105) $2 \times 10$ |
| (16) $5 \times 0$   | (46) $4 \times 7$  | (76) $5 \times 2$  | (106) $7 \times 5$  |
| (17) $4 \times 0$   | (47) $6 \times 10$ | (77) $8 \times 4$  | (107) $0 \times 0$  |
| (18) $4 \times 7$   | (48) $10 \times 3$ | (78) $5 \times 8$  | (108) $10 \times 3$ |
| (19) $6 \times 2$   | (49) $8 \times 6$  | (79) $1 \times 7$  | (109) $2 \times 2$  |
| (20) $4 \times 1$   | (50) $7 \times 5$  | (80) $9 \times 4$  | (110) $5 \times 6$  |
| (21) $4 \times 8$   | (51) $6 \times 1$  | (81) $10 \times 5$ | (111) $5 \times 8$  |
| (22) $6 \times 10$  | (52) $4 \times 0$  | (82) $10 \times 1$ | (112) $2 \times 4$  |
| (23) $9 \times 9$   | (53) $5 \times 2$  | (83) $6 \times 3$  | (113) $8 \times 0$  |
| (24) $4 \times 6$   | (54) $3 \times 9$  | (84) $10 \times 2$ | (114) $10 \times 3$ |
| (25) $3 \times 1$   | (55) $7 \times 4$  | (85) $8 \times 8$  | (115) $2 \times 4$  |
| (26) $3 \times 4$   | (56) $1 \times 9$  | (86) $5 \times 10$ | (116) $2 \times 9$  |
| (27) $9 \times 10$  | (57) $7 \times 7$  | (87) $4 \times 10$ | (117) $3 \times 7$  |
| (28) $10 \times 2$  | (58) $6 \times 7$  | (88) $8 \times 4$  | (118) $9 \times 10$ |
| (29) $4 \times 9$   | (59) $9 \times 10$ | (89) $9 \times 8$  | (119) $3 \times 2$  |
| (30) $7 \times 9$   | (60) $4 \times 6$  | (90) $0 \times 7$  | (120) $6 \times 6$  |

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|---------------------|--------------------|--------------------|---------------------|
| (1) $8 \times 9$    | (31) $10 \times 2$ | (61) $0 \times 9$  | (91) $5 \times 2$   |
| (2) $5 \times 9$    | (32) $3 \times 2$  | (62) $9 \times 2$  | (92) $6 \times 4$   |
| (3) $7 \times 10$   | (33) $5 \times 8$  | (63) $10 \times 1$ | (93) $1 \times 4$   |
| (4) $4 \times 8$    | (34) $2 \times 3$  | (64) $8 \times 1$  | (94) $4 \times 6$   |
| (5) $4 \times 1$    | (35) $4 \times 5$  | (65) $8 \times 3$  | (95) $9 \times 9$   |
| (6) $7 \times 8$    | (36) $8 \times 10$ | (66) $3 \times 5$  | (96) $4 \times 4$   |
| (7) $5 \times 9$    | (37) $8 \times 6$  | (67) $10 \times 8$ | (97) $6 \times 5$   |
| (8) $9 \times 7$    | (38) $9 \times 9$  | (68) $1 \times 10$ | (98) $4 \times 9$   |
| (9) $3 \times 4$    | (39) $6 \times 9$  | (69) $1 \times 1$  | (99) $6 \times 10$  |
| (10) $4 \times 10$  | (40) $1 \times 3$  | (70) $10 \times 8$ | (100) $0 \times 7$  |
| (11) $4 \times 9$   | (41) $4 \times 5$  | (71) $8 \times 4$  | (101) $4 \times 9$  |
| (12) $6 \times 9$   | (42) $3 \times 5$  | (72) $9 \times 3$  | (102) $10 \times 3$ |
| (13) $1 \times 8$   | (43) $4 \times 5$  | (73) $3 \times 10$ | (103) $7 \times 1$  |
| (14) $10 \times 5$  | (44) $2 \times 3$  | (74) $2 \times 9$  | (104) $5 \times 10$ |
| (15) $1 \times 10$  | (45) $4 \times 10$ | (75) $9 \times 8$  | (105) $9 \times 9$  |
| (16) $0 \times 2$   | (46) $3 \times 8$  | (76) $7 \times 7$  | (106) $4 \times 1$  |
| (17) $8 \times 6$   | (47) $1 \times 7$  | (77) $6 \times 9$  | (107) $2 \times 8$  |
| (18) $10 \times 5$  | (48) $0 \times 1$  | (78) $8 \times 5$  | (108) $2 \times 0$  |
| (19) $10 \times 10$ | (49) $8 \times 2$  | (79) $6 \times 8$  | (109) $6 \times 10$ |
| (20) $2 \times 6$   | (50) $7 \times 7$  | (80) $10 \times 6$ | (110) $10 \times 5$ |
| (21) $9 \times 6$   | (51) $0 \times 3$  | (81) $2 \times 10$ | (111) $6 \times 4$  |
| (22) $1 \times 10$  | (52) $9 \times 10$ | (82) $1 \times 8$  | (112) $6 \times 10$ |
| (23) $10 \times 2$  | (53) $5 \times 7$  | (83) $2 \times 1$  | (113) $0 \times 6$  |
| (24) $2 \times 9$   | (54) $8 \times 6$  | (84) $5 \times 0$  | (114) $3 \times 4$  |
| (25) $10 \times 2$  | (55) $7 \times 2$  | (85) $3 \times 0$  | (115) $7 \times 4$  |
| (26) $6 \times 9$   | (56) $10 \times 6$ | (86) $8 \times 8$  | (116) $0 \times 10$ |
| (27) $1 \times 4$   | (57) $4 \times 4$  | (87) $7 \times 5$  | (117) $3 \times 9$  |
| (28) $4 \times 1$   | (58) $5 \times 1$  | (88) $5 \times 9$  | (118) $4 \times 6$  |
| (29) $6 \times 9$   | (59) $10 \times 9$ | (89) $5 \times 9$  | (119) $5 \times 2$  |
| (30) $0 \times 7$   | (60) $3 \times 4$  | (90) $6 \times 8$  | (120) $3 \times 1$  |

|                     |                    |                    |                     |
|---------------------|--------------------|--------------------|---------------------|
| (1) $1 \times 9$    | (31) $9 \times 3$  | (61) $4 \times 1$  | (91) $1 \times 3$   |
| (2) $8 \times 6$    | (32) $5 \times 3$  | (62) $3 \times 2$  | (92) $7 \times 4$   |
| (3) $5 \times 3$    | (33) $5 \times 2$  | (63) $8 \times 7$  | (93) $9 \times 7$   |
| (4) $5 \times 9$    | (34) $7 \times 2$  | (64) $7 \times 1$  | (94) $10 \times 0$  |
| (5) $8 \times 10$   | (35) $10 \times 4$ | (65) $7 \times 6$  | (95) $6 \times 0$   |
| (6) $1 \times 3$    | (36) $7 \times 8$  | (66) $0 \times 8$  | (96) $3 \times 2$   |
| (7) $4 \times 5$    | (37) $10 \times 7$ | (67) $3 \times 5$  | (97) $8 \times 2$   |
| (8) $9 \times 4$    | (38) $8 \times 7$  | (68) $7 \times 9$  | (98) $3 \times 8$   |
| (9) $5 \times 3$    | (39) $4 \times 0$  | (69) $1 \times 2$  | (99) $10 \times 7$  |
| (10) $7 \times 9$   | (40) $2 \times 8$  | (70) $1 \times 9$  | (100) $10 \times 6$ |
| (11) $10 \times 5$  | (41) $9 \times 10$ | (71) $6 \times 8$  | (101) $3 \times 2$  |
| (12) $2 \times 6$   | (42) $10 \times 6$ | (72) $9 \times 10$ | (102) $0 \times 9$  |
| (13) $6 \times 0$   | (43) $8 \times 2$  | (73) $4 \times 9$  | (103) $4 \times 10$ |
| (14) $3 \times 4$   | (44) $2 \times 10$ | (74) $10 \times 2$ | (104) $8 \times 7$  |
| (15) $1 \times 9$   | (45) $5 \times 0$  | (75) $4 \times 7$  | (105) $8 \times 1$  |
| (16) $2 \times 3$   | (46) $10 \times 6$ | (76) $9 \times 2$  | (106) $9 \times 7$  |
| (17) $10 \times 2$  | (47) $4 \times 7$  | (77) $5 \times 8$  | (107) $3 \times 8$  |
| (18) $10 \times 3$  | (48) $6 \times 1$  | (78) $10 \times 2$ | (108) $4 \times 3$  |
| (19) $6 \times 1$   | (49) $5 \times 2$  | (79) $6 \times 5$  | (109) $5 \times 3$  |
| (20) $8 \times 1$   | (50) $3 \times 9$  | (80) $10 \times 2$ | (110) $7 \times 5$  |
| (21) $9 \times 9$   | (51) $6 \times 8$  | (81) $0 \times 5$  | (111) $3 \times 4$  |
| (22) $3 \times 2$   | (52) $5 \times 2$  | (82) $2 \times 8$  | (112) $6 \times 6$  |
| (23) $7 \times 6$   | (53) $1 \times 9$  | (83) $6 \times 4$  | (113) $2 \times 4$  |
| (24) $7 \times 3$   | (54) $0 \times 8$  | (84) $6 \times 6$  | (114) $1 \times 9$  |
| (25) $2 \times 3$   | (55) $1 \times 7$  | (85) $3 \times 6$  | (115) $9 \times 5$  |
| (26) $10 \times 3$  | (56) $3 \times 8$  | (86) $5 \times 1$  | (116) $2 \times 7$  |
| (27) $10 \times 10$ | (57) $9 \times 5$  | (87) $0 \times 2$  | (117) $7 \times 6$  |
| (28) $6 \times 7$   | (58) $2 \times 5$  | (88) $5 \times 1$  | (118) $6 \times 5$  |
| (29) $8 \times 4$   | (59) $0 \times 4$  | (89) $2 \times 10$ | (119) $5 \times 7$  |
| (30) $6 \times 5$   | (60) $4 \times 7$  | (90) $7 \times 7$  | (120) $4 \times 4$  |



|                     |                    |                     |                     |
|---------------------|--------------------|---------------------|---------------------|
| (1) $7 \times 10$   | (31) $7 \times 6$  | (61) $2 \times 3$   | (91) $2 \times 5$   |
| (2) $7 \times 6$    | (32) $4 \times 4$  | (62) $5 \times 2$   | (92) $10 \times 8$  |
| (3) $0 \times 0$    | (33) $1 \times 6$  | (63) $6 \times 2$   | (93) $4 \times 5$   |
| (4) $2 \times 9$    | (34) $5 \times 8$  | (64) $3 \times 2$   | (94) $2 \times 6$   |
| (5) $10 \times 3$   | (35) $3 \times 7$  | (65) $3 \times 2$   | (95) $6 \times 9$   |
| (6) $3 \times 8$    | (36) $4 \times 5$  | (66) $9 \times 10$  | (96) $9 \times 4$   |
| (7) $7 \times 9$    | (37) $4 \times 10$ | (67) $1 \times 4$   | (97) $10 \times 6$  |
| (8) $9 \times 0$    | (38) $5 \times 6$  | (68) $9 \times 3$   | (98) $6 \times 0$   |
| (9) $2 \times 1$    | (39) $6 \times 8$  | (69) $4 \times 8$   | (99) $2 \times 9$   |
| (10) $2 \times 10$  | (40) $2 \times 9$  | (70) $5 \times 8$   | (100) $1 \times 2$  |
| (11) $2 \times 3$   | (41) $0 \times 1$  | (71) $2 \times 6$   | (101) $3 \times 8$  |
| (12) $10 \times 3$  | (42) $8 \times 10$ | (72) $9 \times 3$   | (102) $6 \times 5$  |
| (13) $10 \times 7$  | (43) $2 \times 5$  | (73) $5 \times 4$   | (103) $9 \times 10$ |
| (14) $3 \times 8$   | (44) $7 \times 4$  | (74) $0 \times 4$   | (104) $4 \times 8$  |
| (15) $10 \times 7$  | (45) $6 \times 3$  | (75) $1 \times 1$   | (105) $3 \times 3$  |
| (16) $10 \times 10$ | (46) $6 \times 3$  | (76) $4 \times 2$   | (106) $10 \times 8$ |
| (17) $7 \times 3$   | (47) $9 \times 7$  | (77) $10 \times 10$ | (107) $5 \times 1$  |
| (18) $9 \times 2$   | (48) $8 \times 5$  | (78) $4 \times 6$   | (108) $7 \times 8$  |
| (19) $1 \times 3$   | (49) $8 \times 6$  | (79) $7 \times 1$   | (109) $4 \times 10$ |
| (20) $8 \times 4$   | (50) $2 \times 9$  | (80) $9 \times 6$   | (110) $1 \times 3$  |
| (21) $9 \times 5$   | (51) $10 \times 1$ | (81) $2 \times 9$   | (111) $10 \times 7$ |
| (22) $6 \times 2$   | (52) $10 \times 9$ | (82) $10 \times 10$ | (112) $2 \times 6$  |
| (23) $1 \times 1$   | (53) $4 \times 5$  | (83) $1 \times 5$   | (113) $10 \times 4$ |
| (24) $6 \times 6$   | (54) $9 \times 3$  | (84) $7 \times 6$   | (114) $4 \times 8$  |
| (25) $3 \times 10$  | (55) $4 \times 7$  | (85) $4 \times 6$   | (115) $7 \times 9$  |
| (26) $7 \times 4$   | (56) $8 \times 3$  | (86) $3 \times 0$   | (116) $5 \times 0$  |
| (27) $9 \times 4$   | (57) $2 \times 2$  | (87) $6 \times 2$   | (117) $4 \times 2$  |
| (28) $2 \times 1$   | (58) $2 \times 6$  | (88) $7 \times 2$   | (118) $1 \times 8$  |
| (29) $0 \times 3$   | (59) $1 \times 10$ | (89) $4 \times 5$   | (119) $8 \times 2$  |
| (30) $10 \times 3$  | (60) $1 \times 2$  | (90) $9 \times 6$   | (120) $5 \times 7$  |

|                    |                     |                    |                     |
|--------------------|---------------------|--------------------|---------------------|
| (1) $3 \times 5$   | (31) $2 \times 5$   | (61) $2 \times 3$  | (91) $3 \times 4$   |
| (2) $5 \times 1$   | (32) $7 \times 10$  | (62) $9 \times 6$  | (92) $3 \times 4$   |
| (3) $2 \times 3$   | (33) $0 \times 1$   | (63) $5 \times 3$  | (93) $1 \times 10$  |
| (4) $7 \times 6$   | (34) $10 \times 1$  | (64) $5 \times 5$  | (94) $10 \times 1$  |
| (5) $5 \times 7$   | (35) $5 \times 3$   | (65) $9 \times 1$  | (95) $0 \times 5$   |
| (6) $5 \times 9$   | (36) $2 \times 7$   | (66) $2 \times 3$  | (96) $6 \times 8$   |
| (7) $7 \times 5$   | (37) $7 \times 8$   | (67) $10 \times 4$ | (97) $3 \times 5$   |
| (8) $3 \times 2$   | (38) $4 \times 6$   | (68) $4 \times 2$  | (98) $6 \times 6$   |
| (9) $5 \times 2$   | (39) $5 \times 8$   | (69) $0 \times 1$  | (99) $7 \times 4$   |
| (10) $7 \times 4$  | (40) $1 \times 3$   | (70) $3 \times 2$  | (100) $3 \times 8$  |
| (11) $3 \times 4$  | (41) $6 \times 1$   | (71) $4 \times 5$  | (101) $7 \times 2$  |
| (12) $5 \times 1$  | (42) $10 \times 7$  | (72) $0 \times 1$  | (102) $9 \times 3$  |
| (13) $7 \times 4$  | (43) $8 \times 7$   | (73) $7 \times 10$ | (103) $4 \times 5$  |
| (14) $6 \times 3$  | (44) $2 \times 6$   | (74) $10 \times 5$ | (104) $7 \times 0$  |
| (15) $10 \times 3$ | (45) $10 \times 7$  | (75) $8 \times 10$ | (105) $0 \times 6$  |
| (16) $6 \times 2$  | (46) $8 \times 4$   | (76) $3 \times 4$  | (106) $9 \times 9$  |
| (17) $6 \times 6$  | (47) $1 \times 7$   | (77) $5 \times 8$  | (107) $2 \times 2$  |
| (18) $0 \times 1$  | (48) $5 \times 2$   | (78) $6 \times 7$  | (108) $7 \times 2$  |
| (19) $7 \times 8$  | (49) $8 \times 9$   | (79) $0 \times 0$  | (109) $6 \times 4$  |
| (20) $9 \times 4$  | (50) $3 \times 8$   | (80) $1 \times 3$  | (110) $9 \times 2$  |
| (21) $2 \times 4$  | (51) $7 \times 2$   | (81) $7 \times 1$  | (111) $2 \times 9$  |
| (22) $1 \times 9$  | (52) $7 \times 7$   | (82) $3 \times 9$  | (112) $3 \times 1$  |
| (23) $2 \times 4$  | (53) $3 \times 0$   | (83) $8 \times 10$ | (113) $1 \times 9$  |
| (24) $2 \times 4$  | (54) $10 \times 1$  | (84) $6 \times 9$  | (114) $9 \times 7$  |
| (25) $9 \times 8$  | (55) $3 \times 2$   | (85) $2 \times 4$  | (115) $4 \times 7$  |
| (26) $10 \times 1$ | (56) $10 \times 10$ | (86) $0 \times 4$  | (116) $1 \times 6$  |
| (27) $5 \times 1$  | (57) $7 \times 7$   | (87) $3 \times 8$  | (117) $10 \times 7$ |
| (28) $2 \times 8$  | (58) $1 \times 5$   | (88) $8 \times 4$  | (118) $5 \times 2$  |
| (29) $1 \times 8$  | (59) $8 \times 6$   | (89) $7 \times 8$  | (119) $1 \times 8$  |
| (30) $7 \times 6$  | (60) $5 \times 6$   | (90) $4 \times 1$  | (120) $6 \times 7$  |

|                    |                    |                    |                     |
|--------------------|--------------------|--------------------|---------------------|
| (1) $2 \times 7$   | (31) $3 \times 2$  | (61) $5 \times 5$  | (91) $6 \times 6$   |
| (2) $3 \times 1$   | (32) $3 \times 6$  | (62) $3 \times 4$  | (92) $8 \times 7$   |
| (3) $9 \times 10$  | (33) $1 \times 7$  | (63) $6 \times 5$  | (93) $10 \times 7$  |
| (4) $1 \times 3$   | (34) $9 \times 6$  | (64) $8 \times 10$ | (94) $3 \times 6$   |
| (5) $1 \times 6$   | (35) $7 \times 6$  | (65) $6 \times 1$  | (95) $1 \times 5$   |
| (6) $1 \times 10$  | (36) $6 \times 0$  | (66) $0 \times 10$ | (96) $6 \times 6$   |
| (7) $5 \times 2$   | (37) $3 \times 2$  | (67) $1 \times 8$  | (97) $1 \times 2$   |
| (8) $1 \times 10$  | (38) $3 \times 1$  | (68) $8 \times 4$  | (98) $3 \times 5$   |
| (9) $8 \times 4$   | (39) $4 \times 1$  | (69) $1 \times 8$  | (99) $8 \times 3$   |
| (10) $9 \times 6$  | (40) $7 \times 7$  | (70) $6 \times 3$  | (100) $10 \times 3$ |
| (11) $10 \times 3$ | (41) $5 \times 10$ | (71) $2 \times 6$  | (101) $1 \times 7$  |
| (12) $9 \times 7$  | (42) $4 \times 5$  | (72) $1 \times 10$ | (102) $8 \times 7$  |
| (13) $6 \times 8$  | (43) $3 \times 7$  | (73) $8 \times 9$  | (103) $4 \times 9$  |
| (14) $7 \times 2$  | (44) $9 \times 10$ | (74) $8 \times 9$  | (104) $10 \times 2$ |
| (15) $0 \times 8$  | (45) $3 \times 8$  | (75) $9 \times 2$  | (105) $4 \times 1$  |
| (16) $5 \times 9$  | (46) $1 \times 5$  | (76) $7 \times 0$  | (106) $7 \times 3$  |
| (17) $10 \times 2$ | (47) $8 \times 4$  | (77) $0 \times 9$  | (107) $10 \times 5$ |
| (18) $5 \times 8$  | (48) $10 \times 7$ | (78) $2 \times 1$  | (108) $3 \times 9$  |
| (19) $1 \times 4$  | (49) $10 \times 8$ | (79) $1 \times 1$  | (109) $6 \times 7$  |
| (20) $5 \times 6$  | (50) $1 \times 10$ | (80) $3 \times 8$  | (110) $9 \times 5$  |
| (21) $1 \times 7$  | (51) $4 \times 2$  | (81) $9 \times 1$  | (111) $4 \times 0$  |
| (22) $3 \times 8$  | (52) $4 \times 9$  | (82) $2 \times 3$  | (112) $0 \times 10$ |
| (23) $6 \times 10$ | (53) $0 \times 7$  | (83) $8 \times 9$  | (113) $7 \times 8$  |
| (24) $9 \times 2$  | (54) $2 \times 6$  | (84) $3 \times 3$  | (114) $9 \times 3$  |
| (25) $6 \times 0$  | (55) $1 \times 0$  | (85) $9 \times 9$  | (115) $3 \times 2$  |
| (26) $10 \times 6$ | (56) $10 \times 2$ | (86) $3 \times 7$  | (116) $2 \times 8$  |
| (27) $5 \times 2$  | (57) $10 \times 5$ | (87) $6 \times 2$  | (117) $4 \times 1$  |
| (28) $9 \times 4$  | (58) $7 \times 3$  | (88) $3 \times 4$  | (118) $0 \times 2$  |
| (29) $2 \times 10$ | (59) $5 \times 2$  | (89) $7 \times 4$  | (119) $1 \times 9$  |
| (30) $5 \times 1$  | (60) $4 \times 3$  | (90) $9 \times 6$  | (120) $2 \times 3$  |

|                    |                    |                    |                     |
|--------------------|--------------------|--------------------|---------------------|
| (1) $3 \times 3$   | (31) $6 \times 2$  | (61) $8 \times 5$  | (91) $7 \times 8$   |
| (2) $9 \times 4$   | (32) $10 \times 9$ | (62) $8 \times 10$ | (92) $4 \times 3$   |
| (3) $7 \times 1$   | (33) $4 \times 8$  | (63) $5 \times 10$ | (93) $4 \times 2$   |
| (4) $2 \times 1$   | (34) $6 \times 6$  | (64) $0 \times 7$  | (94) $9 \times 6$   |
| (5) $3 \times 3$   | (35) $6 \times 6$  | (65) $10 \times 8$ | (95) $3 \times 2$   |
| (6) $2 \times 2$   | (36) $4 \times 0$  | (66) $5 \times 7$  | (96) $4 \times 3$   |
| (7) $1 \times 9$   | (37) $4 \times 3$  | (67) $5 \times 7$  | (97) $3 \times 3$   |
| (8) $6 \times 8$   | (38) $7 \times 2$  | (68) $5 \times 7$  | (98) $6 \times 9$   |
| (9) $9 \times 6$   | (39) $9 \times 1$  | (69) $7 \times 7$  | (99) $0 \times 3$   |
| (10) $7 \times 2$  | (40) $6 \times 6$  | (70) $3 \times 4$  | (100) $9 \times 1$  |
| (11) $4 \times 9$  | (41) $8 \times 9$  | (71) $3 \times 3$  | (101) $3 \times 0$  |
| (12) $9 \times 9$  | (42) $6 \times 10$ | (72) $9 \times 5$  | (102) $8 \times 9$  |
| (13) $6 \times 6$  | (43) $7 \times 2$  | (73) $5 \times 4$  | (103) $1 \times 8$  |
| (14) $1 \times 0$  | (44) $8 \times 1$  | (74) $8 \times 0$  | (104) $9 \times 3$  |
| (15) $6 \times 6$  | (45) $2 \times 7$  | (75) $1 \times 9$  | (105) $0 \times 7$  |
| (16) $7 \times 3$  | (46) $2 \times 10$ | (76) $4 \times 3$  | (106) $3 \times 2$  |
| (17) $4 \times 3$  | (47) $2 \times 9$  | (77) $4 \times 1$  | (107) $10 \times 8$ |
| (18) $0 \times 8$  | (48) $8 \times 8$  | (78) $3 \times 9$  | (108) $7 \times 3$  |
| (19) $6 \times 3$  | (49) $7 \times 3$  | (79) $7 \times 7$  | (109) $0 \times 5$  |
| (20) $5 \times 2$  | (50) $2 \times 5$  | (80) $10 \times 8$ | (110) $7 \times 7$  |
| (21) $5 \times 6$  | (51) $8 \times 0$  | (81) $9 \times 8$  | (111) $5 \times 5$  |
| (22) $7 \times 7$  | (52) $7 \times 9$  | (82) $7 \times 1$  | (112) $9 \times 1$  |
| (23) $9 \times 10$ | (53) $6 \times 4$  | (83) $3 \times 4$  | (113) $8 \times 1$  |
| (24) $1 \times 3$  | (54) $9 \times 10$ | (84) $2 \times 5$  | (114) $8 \times 7$  |
| (25) $7 \times 8$  | (55) $3 \times 8$  | (85) $1 \times 4$  | (115) $7 \times 10$ |
| (26) $8 \times 5$  | (56) $5 \times 2$  | (86) $2 \times 4$  | (116) $6 \times 7$  |
| (27) $7 \times 3$  | (57) $6 \times 3$  | (87) $4 \times 3$  | (117) $3 \times 8$  |
| (28) $5 \times 10$ | (58) $3 \times 4$  | (88) $1 \times 8$  | (118) $2 \times 3$  |
| (29) $7 \times 5$  | (59) $5 \times 5$  | (89) $10 \times 6$ | (119) $4 \times 2$  |
| (30) $1 \times 10$ | (60) $10 \times 7$ | (90) $3 \times 1$  | (120) $3 \times 5$  |

|                           |                         |                           |                          |
|---------------------------|-------------------------|---------------------------|--------------------------|
| (1) $9 \times 6 = 54$     | (31) $1 \times 5 = 5$   | (61) $6 \times 4 = 24$    | (91) $4 \times 9 = 36$   |
| (2) $4 \times 5 = 20$     | (32) $1 \times 3 = 3$   | (62) $6 \times 8 = 48$    | (92) $2 \times 3 = 6$    |
| (3) $5 \times 1 = 5$      | (33) $5 \times 3 = 15$  | (63) $4 \times 8 = 32$    | (93) $4 \times 7 = 28$   |
| (4) $2 \times 8 = 16$     | (34) $10 \times 6 = 60$ | (64) $8 \times 5 = 40$    | (94) $8 \times 1 = 8$    |
| (5) $6 \times 0 = 0$      | (35) $1 \times 8 = 8$   | (65) $6 \times 2 = 12$    | (95) $7 \times 3 = 21$   |
| (6) $10 \times 1 = 10$    | (36) $5 \times 5 = 25$  | (66) $1 \times 1 = 1$     | (96) $7 \times 4 = 28$   |
| (7) $7 \times 4 = 28$     | (37) $8 \times 4 = 32$  | (67) $10 \times 10 = 100$ | (97) $4 \times 4 = 16$   |
| (8) $5 \times 6 = 30$     | (38) $4 \times 8 = 32$  | (68) $3 \times 5 = 15$    | (98) $3 \times 2 = 6$    |
| (9) $10 \times 10 = 100$  | (39) $0 \times 3 = 0$   | (69) $6 \times 9 = 54$    | (99) $4 \times 1 = 4$    |
| (10) $1 \times 5 = 5$     | (40) $5 \times 3 = 15$  | (70) $7 \times 6 = 42$    | (100) $4 \times 10 = 40$ |
| (11) $5 \times 6 = 30$    | (41) $7 \times 10 = 70$ | (71) $4 \times 2 = 8$     | (101) $9 \times 5 = 45$  |
| (12) $4 \times 10 = 40$   | (42) $6 \times 10 = 60$ | (72) $6 \times 1 = 6$     | (102) $5 \times 10 = 50$ |
| (13) $5 \times 5 = 25$    | (43) $7 \times 6 = 42$  | (73) $6 \times 10 = 60$   | (103) $10 \times 5 = 50$ |
| (14) $6 \times 4 = 24$    | (44) $9 \times 5 = 45$  | (74) $6 \times 5 = 30$    | (104) $2 \times 8 = 16$  |
| (15) $2 \times 2 = 4$     | (45) $6 \times 5 = 30$  | (75) $1 \times 10 = 10$   | (105) $1 \times 4 = 4$   |
| (16) $9 \times 2 = 18$    | (46) $2 \times 6 = 12$  | (76) $4 \times 0 = 0$     | (106) $7 \times 4 = 28$  |
| (17) $5 \times 1 = 5$     | (47) $10 \times 7 = 70$ | (77) $2 \times 7 = 14$    | (107) $6 \times 6 = 36$  |
| (18) $10 \times 10 = 100$ | (48) $9 \times 3 = 27$  | (78) $2 \times 3 = 6$     | (108) $6 \times 5 = 30$  |
| (19) $8 \times 9 = 72$    | (49) $8 \times 7 = 56$  | (79) $2 \times 5 = 10$    | (109) $3 \times 10 = 30$ |
| (20) $5 \times 5 = 25$    | (50) $2 \times 5 = 10$  | (80) $3 \times 8 = 24$    | (110) $5 \times 10 = 50$ |
| (21) $2 \times 1 = 2$     | (51) $2 \times 7 = 14$  | (81) $6 \times 7 = 42$    | (111) $1 \times 3 = 3$   |
| (22) $10 \times 2 = 20$   | (52) $1 \times 10 = 10$ | (82) $2 \times 6 = 12$    | (112) $5 \times 6 = 30$  |
| (23) $6 \times 5 = 30$    | (53) $4 \times 7 = 28$  | (83) $8 \times 8 = 64$    | (113) $6 \times 8 = 48$  |
| (24) $9 \times 4 = 36$    | (54) $3 \times 9 = 27$  | (84) $7 \times 5 = 35$    | (114) $2 \times 3 = 6$   |
| (25) $3 \times 10 = 30$   | (55) $1 \times 6 = 6$   | (85) $2 \times 7 = 14$    | (115) $1 \times 0 = 0$   |
| (26) $8 \times 6 = 48$    | (56) $5 \times 5 = 25$  | (86) $3 \times 0 = 0$     | (116) $1 \times 9 = 9$   |
| (27) $10 \times 9 = 90$   | (57) $8 \times 9 = 72$  | (87) $4 \times 9 = 36$    | (117) $8 \times 0 = 0$   |
| (28) $8 \times 3 = 24$    | (58) $5 \times 8 = 40$  | (88) $5 \times 10 = 50$   | (118) $5 \times 2 = 10$  |
| (29) $8 \times 7 = 56$    | (59) $4 \times 6 = 24$  | (89) $10 \times 1 = 10$   | (119) $2 \times 0 = 0$   |
| (30) $5 \times 3 = 15$    | (60) $2 \times 5 = 10$  | (90) $1 \times 2 = 2$     | (120) $6 \times 4 = 24$  |

|                         |                         |                         |                          |
|-------------------------|-------------------------|-------------------------|--------------------------|
| (1) $2 \times 7 = 14$   | (31) $2 \times 0 = 0$   | (61) $8 \times 4 = 32$  | (91) $8 \times 9 = 72$   |
| (2) $10 \times 4 = 40$  | (32) $2 \times 4 = 8$   | (62) $6 \times 3 = 18$  | (92) $0 \times 10 = 0$   |
| (3) $3 \times 7 = 21$   | (33) $1 \times 10 = 10$ | (63) $4 \times 6 = 24$  | (93) $8 \times 2 = 16$   |
| (4) $7 \times 1 = 7$    | (34) $1 \times 4 = 4$   | (64) $1 \times 8 = 8$   | (94) $8 \times 5 = 40$   |
| (5) $5 \times 4 = 20$   | (35) $2 \times 8 = 16$  | (65) $7 \times 2 = 14$  | (95) $1 \times 5 = 5$    |
| (6) $5 \times 7 = 35$   | (36) $3 \times 4 = 12$  | (66) $7 \times 7 = 49$  | (96) $1 \times 8 = 8$    |
| (7) $7 \times 3 = 21$   | (37) $0 \times 4 = 0$   | (67) $5 \times 10 = 50$ | (97) $9 \times 2 = 18$   |
| (8) $8 \times 1 = 8$    | (38) $4 \times 9 = 36$  | (68) $3 \times 4 = 12$  | (98) $3 \times 0 = 0$    |
| (9) $6 \times 6 = 36$   | (39) $1 \times 3 = 3$   | (69) $6 \times 1 = 6$   | (99) $8 \times 9 = 72$   |
| (10) $2 \times 10 = 20$ | (40) $9 \times 8 = 72$  | (70) $7 \times 2 = 14$  | (100) $5 \times 4 = 20$  |
| (11) $10 \times 7 = 70$ | (41) $7 \times 0 = 0$   | (71) $7 \times 3 = 21$  | (101) $1 \times 8 = 8$   |
| (12) $0 \times 10 = 0$  | (42) $2 \times 3 = 6$   | (72) $6 \times 8 = 48$  | (102) $1 \times 10 = 10$ |
| (13) $1 \times 6 = 6$   | (43) $10 \times 1 = 10$ | (73) $6 \times 2 = 12$  | (103) $1 \times 2 = 2$   |
| (14) $10 \times 7 = 70$ | (44) $4 \times 3 = 12$  | (74) $4 \times 6 = 24$  | (104) $4 \times 2 = 8$   |
| (15) $8 \times 5 = 40$  | (45) $10 \times 7 = 70$ | (75) $10 \times 5 = 50$ | (105) $5 \times 0 = 0$   |
| (16) $10 \times 8 = 80$ | (46) $9 \times 6 = 54$  | (76) $5 \times 3 = 15$  | (106) $7 \times 3 = 21$  |
| (17) $7 \times 10 = 70$ | (47) $3 \times 0 = 0$   | (77) $3 \times 8 = 24$  | (107) $9 \times 5 = 45$  |
| (18) $7 \times 9 = 63$  | (48) $2 \times 0 = 0$   | (78) $10 \times 2 = 20$ | (108) $0 \times 2 = 0$   |
| (19) $9 \times 0 = 0$   | (49) $7 \times 6 = 42$  | (79) $1 \times 8 = 8$   | (109) $7 \times 4 = 28$  |
| (20) $8 \times 2 = 16$  | (50) $4 \times 5 = 20$  | (80) $9 \times 7 = 63$  | (110) $8 \times 5 = 40$  |
| (21) $6 \times 4 = 24$  | (51) $5 \times 10 = 50$ | (81) $3 \times 9 = 27$  | (111) $1 \times 8 = 8$   |
| (22) $2 \times 0 = 0$   | (52) $1 \times 3 = 3$   | (82) $8 \times 7 = 56$  | (112) $6 \times 8 = 48$  |
| (23) $8 \times 10 = 80$ | (53) $7 \times 8 = 56$  | (83) $4 \times 5 = 20$  | (113) $7 \times 4 = 28$  |
| (24) $9 \times 5 = 45$  | (54) $4 \times 6 = 24$  | (84) $9 \times 3 = 27$  | (114) $5 \times 9 = 45$  |
| (25) $3 \times 9 = 27$  | (55) $1 \times 9 = 9$   | (85) $4 \times 5 = 20$  | (115) $2 \times 10 = 20$ |
| (26) $4 \times 10 = 40$ | (56) $2 \times 8 = 16$  | (86) $4 \times 0 = 0$   | (116) $2 \times 7 = 14$  |
| (27) $6 \times 9 = 54$  | (57) $1 \times 7 = 7$   | (87) $7 \times 2 = 14$  | (117) $9 \times 2 = 18$  |
| (28) $9 \times 10 = 90$ | (58) $3 \times 2 = 6$   | (88) $6 \times 7 = 42$  | (118) $9 \times 7 = 63$  |
| (29) $9 \times 9 = 81$  | (59) $10 \times 5 = 50$ | (89) $10 \times 9 = 90$ | (119) $6 \times 4 = 24$  |
| (30) $9 \times 2 = 18$  | (60) $0 \times 6 = 0$   | (90) $7 \times 2 = 14$  | (120) $1 \times 9 = 9$   |

- |                         |                         |                         |                         |
|-------------------------|-------------------------|-------------------------|-------------------------|
| (1) $4 \times 9 = 36$   | (31) $1 \times 4 = 4$   | (61) $10 \times 6 = 60$ | (91) $2 \times 2 = 4$   |
| (2) $1 \times 3 = 3$    | (32) $1 \times 6 = 6$   | (62) $1 \times 8 = 8$   | (92) $2 \times 3 = 6$   |
| (3) $7 \times 10 = 70$  | (33) $2 \times 10 = 20$ | (63) $7 \times 10 = 70$ | (93) $6 \times 8 = 48$  |
| (4) $7 \times 7 = 49$   | (34) $9 \times 10 = 90$ | (64) $4 \times 3 = 12$  | (94) $5 \times 4 = 20$  |
| (5) $2 \times 2 = 4$    | (35) $5 \times 4 = 20$  | (65) $4 \times 9 = 36$  | (95) $5 \times 8 = 40$  |
| (6) $6 \times 10 = 60$  | (36) $6 \times 2 = 12$  | (66) $8 \times 1 = 8$   | (96) $6 \times 3 = 18$  |
| (7) $2 \times 8 = 16$   | (37) $4 \times 7 = 28$  | (67) $6 \times 3 = 18$  | (97) $2 \times 1 = 2$   |
| (8) $1 \times 6 = 6$    | (38) $6 \times 1 = 6$   | (68) $10 \times 4 = 40$ | (98) $5 \times 2 = 10$  |
| (9) $2 \times 8 = 16$   | (39) $7 \times 2 = 14$  | (69) $6 \times 7 = 42$  | (99) $2 \times 4 = 8$   |
| (10) $4 \times 8 = 32$  | (40) $5 \times 9 = 45$  | (70) $1 \times 1 = 1$   | (100) $4 \times 4 = 16$ |
| (11) $8 \times 5 = 40$  | (41) $0 \times 7 = 0$   | (71) $2 \times 8 = 16$  | (101) $7 \times 0 = 0$  |
| (12) $8 \times 4 = 32$  | (42) $9 \times 2 = 18$  | (72) $1 \times 2 = 2$   | (102) $5 \times 3 = 15$ |
| (13) $3 \times 4 = 12$  | (43) $2 \times 8 = 16$  | (73) $9 \times 0 = 0$   | (103) $5 \times 2 = 10$ |
| (14) $6 \times 10 = 60$ | (44) $10 \times 7 = 70$ | (74) $10 \times 9 = 90$ | (104) $7 \times 6 = 42$ |
| (15) $6 \times 5 = 30$  | (45) $5 \times 1 = 5$   | (75) $7 \times 1 = 7$   | (105) $9 \times 1 = 9$  |
| (16) $7 \times 6 = 42$  | (46) $1 \times 4 = 4$   | (76) $9 \times 10 = 90$ | (106) $4 \times 7 = 28$ |
| (17) $8 \times 8 = 64$  | (47) $8 \times 4 = 32$  | (77) $2 \times 9 = 18$  | (107) $6 \times 2 = 12$ |
| (18) $0 \times 9 = 0$   | (48) $2 \times 6 = 12$  | (78) $8 \times 8 = 64$  | (108) $3 \times 3 = 9$  |
| (19) $0 \times 5 = 0$   | (49) $7 \times 9 = 63$  | (79) $2 \times 5 = 10$  | (109) $2 \times 7 = 14$ |
| (20) $2 \times 9 = 18$  | (50) $6 \times 7 = 42$  | (80) $9 \times 1 = 9$   | (110) $8 \times 3 = 24$ |
| (21) $0 \times 6 = 0$   | (51) $2 \times 8 = 16$  | (81) $10 \times 7 = 70$ | (111) $4 \times 1 = 4$  |
| (22) $10 \times 8 = 80$ | (52) $6 \times 5 = 30$  | (82) $1 \times 2 = 2$   | (112) $6 \times 7 = 42$ |
| (23) $5 \times 5 = 25$  | (53) $10 \times 6 = 60$ | (83) $4 \times 6 = 24$  | (113) $9 \times 1 = 9$  |
| (24) $3 \times 6 = 18$  | (54) $8 \times 7 = 56$  | (84) $2 \times 9 = 18$  | (114) $2 \times 3 = 6$  |
| (25) $0 \times 9 = 0$   | (55) $7 \times 2 = 14$  | (85) $3 \times 3 = 9$   | (115) $4 \times 0 = 0$  |
| (26) $0 \times 10 = 0$  | (56) $2 \times 9 = 18$  | (86) $10 \times 5 = 50$ | (116) $1 \times 5 = 5$  |
| (27) $10 \times 8 = 80$ | (57) $3 \times 3 = 9$   | (87) $8 \times 9 = 72$  | (117) $1 \times 1 = 1$  |
| (28) $2 \times 2 = 4$   | (58) $9 \times 2 = 18$  | (88) $8 \times 9 = 72$  | (118) $7 \times 7 = 49$ |
| (29) $2 \times 2 = 4$   | (59) $6 \times 4 = 24$  | (89) $3 \times 8 = 24$  | (119) $3 \times 3 = 9$  |
| (30) $0 \times 1 = 0$   | (60) $1 \times 9 = 9$   | (90) $7 \times 6 = 42$  | (120) $1 \times 6 = 6$  |

|                         |                         |                         |                            |
|-------------------------|-------------------------|-------------------------|----------------------------|
| (1) $3 \times 8 = 24$   | (31) $5 \times 0 = 0$   | (61) $2 \times 7 = 14$  | (91) $7 \times 2 = 14$     |
| (2) $9 \times 4 = 36$   | (32) $1 \times 3 = 3$   | (62) $4 \times 2 = 8$   | (92) $10 \times 8 = 80$    |
| (3) $6 \times 8 = 48$   | (33) $3 \times 0 = 0$   | (63) $2 \times 9 = 18$  | (93) $8 \times 2 = 16$     |
| (4) $5 \times 1 = 5$    | (34) $2 \times 8 = 16$  | (64) $4 \times 3 = 12$  | (94) $3 \times 5 = 15$     |
| (5) $10 \times 5 = 50$  | (35) $1 \times 8 = 8$   | (65) $3 \times 7 = 21$  | (95) $1 \times 8 = 8$      |
| (6) $8 \times 8 = 64$   | (36) $2 \times 2 = 4$   | (66) $1 \times 3 = 3$   | (96) $3 \times 8 = 24$     |
| (7) $3 \times 1 = 3$    | (37) $10 \times 3 = 30$ | (67) $6 \times 3 = 18$  | (97) $8 \times 1 = 8$      |
| (8) $5 \times 1 = 5$    | (38) $5 \times 8 = 40$  | (68) $7 \times 6 = 42$  | (98) $3 \times 3 = 9$      |
| (9) $0 \times 3 = 0$    | (39) $8 \times 5 = 40$  | (69) $1 \times 10 = 10$ | (99) $7 \times 1 = 7$      |
| (10) $9 \times 8 = 72$  | (40) $7 \times 7 = 49$  | (70) $6 \times 9 = 54$  | (100) $10 \times 1 = 10$   |
| (11) $9 \times 2 = 18$  | (41) $10 \times 1 = 10$ | (71) $2 \times 8 = 16$  | (101) $1 \times 1 = 1$     |
| (12) $0 \times 10 = 0$  | (42) $4 \times 6 = 24$  | (72) $10 \times 3 = 30$ | (102) $10 \times 10 = 100$ |
| (13) $9 \times 8 = 72$  | (43) $8 \times 1 = 8$   | (73) $3 \times 4 = 12$  | (103) $4 \times 5 = 20$    |
| (14) $6 \times 1 = 6$   | (44) $8 \times 7 = 56$  | (74) $8 \times 7 = 56$  | (104) $10 \times 7 = 70$   |
| (15) $8 \times 3 = 24$  | (45) $3 \times 5 = 15$  | (75) $6 \times 10 = 60$ | (105) $5 \times 3 = 15$    |
| (16) $10 \times 7 = 70$ | (46) $1 \times 2 = 2$   | (76) $6 \times 4 = 24$  | (106) $9 \times 8 = 72$    |
| (17) $5 \times 7 = 35$  | (47) $3 \times 10 = 30$ | (77) $7 \times 1 = 7$   | (107) $9 \times 1 = 9$     |
| (18) $10 \times 5 = 50$ | (48) $2 \times 2 = 4$   | (78) $3 \times 1 = 3$   | (108) $5 \times 4 = 20$    |
| (19) $3 \times 6 = 18$  | (49) $4 \times 0 = 0$   | (79) $0 \times 7 = 0$   | (109) $0 \times 1 = 0$     |
| (20) $6 \times 5 = 30$  | (50) $7 \times 7 = 49$  | (80) $5 \times 1 = 5$   | (110) $3 \times 2 = 6$     |
| (21) $6 \times 9 = 54$  | (51) $8 \times 5 = 40$  | (81) $3 \times 9 = 27$  | (111) $9 \times 5 = 45$    |
| (22) $5 \times 9 = 45$  | (52) $5 \times 4 = 20$  | (82) $6 \times 8 = 48$  | (112) $5 \times 7 = 35$    |
| (23) $2 \times 3 = 6$   | (53) $9 \times 9 = 81$  | (83) $8 \times 5 = 40$  | (113) $0 \times 0 = 0$     |
| (24) $1 \times 2 = 2$   | (54) $1 \times 7 = 7$   | (84) $6 \times 8 = 48$  | (114) $6 \times 10 = 60$   |
| (25) $5 \times 8 = 40$  | (55) $8 \times 8 = 64$  | (85) $5 \times 7 = 35$  | (115) $7 \times 1 = 7$     |
| (26) $10 \times 2 = 20$ | (56) $9 \times 0 = 0$   | (86) $3 \times 1 = 3$   | (116) $1 \times 3 = 3$     |
| (27) $7 \times 0 = 0$   | (57) $1 \times 3 = 3$   | (87) $3 \times 1 = 3$   | (117) $1 \times 5 = 5$     |
| (28) $0 \times 4 = 0$   | (58) $6 \times 5 = 30$  | (88) $1 \times 8 = 8$   | (118) $8 \times 5 = 40$    |
| (29) $1 \times 1 = 1$   | (59) $9 \times 0 = 0$   | (89) $7 \times 3 = 21$  | (119) $0 \times 4 = 0$     |
| (30) $0 \times 2 = 0$   | (60) $8 \times 3 = 24$  | (90) $8 \times 2 = 16$  | (120) $3 \times 7 = 21$    |



- |                         |                         |                           |                          |
|-------------------------|-------------------------|---------------------------|--------------------------|
| (1) $2 \times 1 = 2$    | (31) $7 \times 0 = 0$   | (61) $8 \times 4 = 32$    | (91) $1 \times 1 = 1$    |
| (2) $9 \times 2 = 18$   | (32) $9 \times 4 = 36$  | (62) $3 \times 1 = 3$     | (92) $9 \times 2 = 18$   |
| (3) $0 \times 4 = 0$    | (33) $4 \times 7 = 28$  | (63) $5 \times 4 = 20$    | (93) $8 \times 3 = 24$   |
| (4) $3 \times 1 = 3$    | (34) $3 \times 7 = 21$  | (64) $10 \times 9 = 90$   | (94) $6 \times 2 = 12$   |
| (5) $8 \times 9 = 72$   | (35) $7 \times 9 = 63$  | (65) $3 \times 6 = 18$    | (95) $9 \times 6 = 54$   |
| (6) $6 \times 10 = 60$  | (36) $2 \times 1 = 2$   | (66) $10 \times 10 = 100$ | (96) $10 \times 0 = 0$   |
| (7) $2 \times 7 = 14$   | (37) $3 \times 10 = 30$ | (67) $0 \times 9 = 0$     | (97) $7 \times 3 = 21$   |
| (8) $7 \times 10 = 70$  | (38) $2 \times 8 = 16$  | (68) $9 \times 4 = 36$    | (98) $9 \times 5 = 45$   |
| (9) $7 \times 0 = 0$    | (39) $5 \times 2 = 10$  | (69) $2 \times 3 = 6$     | (99) $7 \times 9 = 63$   |
| (10) $4 \times 8 = 32$  | (40) $2 \times 2 = 4$   | (70) $5 \times 8 = 40$    | (100) $6 \times 0 = 0$   |
| (11) $6 \times 0 = 0$   | (41) $9 \times 8 = 72$  | (71) $3 \times 6 = 18$    | (101) $0 \times 7 = 0$   |
| (12) $9 \times 5 = 45$  | (42) $2 \times 1 = 2$   | (72) $10 \times 6 = 60$   | (102) $8 \times 1 = 8$   |
| (13) $6 \times 4 = 24$  | (43) $6 \times 7 = 42$  | (73) $0 \times 9 = 0$     | (103) $0 \times 6 = 0$   |
| (14) $6 \times 1 = 6$   | (44) $8 \times 4 = 32$  | (74) $5 \times 9 = 45$    | (104) $0 \times 9 = 0$   |
| (15) $3 \times 7 = 21$  | (45) $3 \times 5 = 15$  | (75) $5 \times 0 = 0$     | (105) $6 \times 2 = 12$  |
| (16) $3 \times 5 = 15$  | (46) $5 \times 1 = 5$   | (76) $5 \times 2 = 10$    | (106) $0 \times 9 = 0$   |
| (17) $10 \times 8 = 80$ | (47) $2 \times 7 = 14$  | (77) $2 \times 2 = 4$     | (107) $7 \times 3 = 21$  |
| (18) $9 \times 9 = 81$  | (48) $5 \times 4 = 20$  | (78) $1 \times 6 = 6$     | (108) $8 \times 9 = 72$  |
| (19) $5 \times 7 = 35$  | (49) $3 \times 9 = 27$  | (79) $6 \times 10 = 60$   | (109) $8 \times 1 = 8$   |
| (20) $0 \times 3 = 0$   | (50) $3 \times 7 = 21$  | (80) $10 \times 6 = 60$   | (110) $1 \times 9 = 9$   |
| (21) $10 \times 3 = 30$ | (51) $1 \times 4 = 4$   | (81) $9 \times 8 = 72$    | (111) $1 \times 4 = 4$   |
| (22) $10 \times 0 = 0$  | (52) $9 \times 9 = 81$  | (82) $4 \times 5 = 20$    | (112) $7 \times 0 = 0$   |
| (23) $5 \times 4 = 20$  | (53) $2 \times 2 = 4$   | (83) $5 \times 10 = 50$   | (113) $0 \times 4 = 0$   |
| (24) $7 \times 4 = 28$  | (54) $6 \times 2 = 12$  | (84) $8 \times 7 = 56$    | (114) $9 \times 10 = 90$ |
| (25) $2 \times 9 = 18$  | (55) $5 \times 2 = 10$  | (85) $6 \times 9 = 54$    | (115) $1 \times 4 = 4$   |
| (26) $7 \times 5 = 35$  | (56) $10 \times 8 = 80$ | (86) $10 \times 3 = 30$   | (116) $1 \times 10 = 10$ |
| (27) $4 \times 0 = 0$   | (57) $0 \times 4 = 0$   | (87) $4 \times 0 = 0$     | (117) $10 \times 1 = 10$ |
| (28) $9 \times 8 = 72$  | (58) $0 \times 9 = 0$   | (88) $1 \times 5 = 5$     | (118) $4 \times 4 = 16$  |
| (29) $7 \times 9 = 63$  | (59) $9 \times 3 = 27$  | (89) $10 \times 1 = 10$   | (119) $0 \times 8 = 0$   |
| (30) $9 \times 1 = 9$   | (60) $10 \times 5 = 50$ | (90) $8 \times 3 = 24$    | (120) $4 \times 1 = 4$   |

- |                         |                         |                         |                          |
|-------------------------|-------------------------|-------------------------|--------------------------|
| (1) $3 \times 5 = 15$   | (31) $1 \times 5 = 5$   | (61) $8 \times 7 = 56$  | (91) $9 \times 0 = 0$    |
| (2) $0 \times 7 = 0$    | (32) $2 \times 8 = 16$  | (62) $10 \times 9 = 90$ | (92) $9 \times 8 = 72$   |
| (3) $9 \times 7 = 63$   | (33) $3 \times 0 = 0$   | (63) $5 \times 5 = 25$  | (93) $2 \times 10 = 20$  |
| (4) $3 \times 0 = 0$    | (34) $3 \times 6 = 18$  | (64) $2 \times 10 = 20$ | (94) $1 \times 7 = 7$    |
| (5) $9 \times 1 = 9$    | (35) $4 \times 5 = 20$  | (65) $4 \times 1 = 4$   | (95) $1 \times 4 = 4$    |
| (6) $3 \times 9 = 27$   | (36) $8 \times 2 = 16$  | (66) $6 \times 10 = 60$ | (96) $7 \times 10 = 70$  |
| (7) $7 \times 9 = 63$   | (37) $8 \times 8 = 64$  | (67) $3 \times 7 = 21$  | (97) $2 \times 1 = 2$    |
| (8) $5 \times 3 = 15$   | (38) $7 \times 10 = 70$ | (68) $0 \times 1 = 0$   | (98) $7 \times 7 = 49$   |
| (9) $9 \times 10 = 90$  | (39) $6 \times 5 = 30$  | (69) $5 \times 2 = 10$  | (99) $10 \times 6 = 60$  |
| (10) $10 \times 9 = 90$ | (40) $10 \times 3 = 30$ | (70) $1 \times 5 = 5$   | (100) $0 \times 9 = 0$   |
| (11) $8 \times 2 = 16$  | (41) $0 \times 5 = 0$   | (71) $4 \times 4 = 16$  | (101) $3 \times 3 = 9$   |
| (12) $6 \times 7 = 42$  | (42) $5 \times 6 = 30$  | (72) $4 \times 5 = 20$  | (102) $8 \times 3 = 24$  |
| (13) $6 \times 9 = 54$  | (43) $10 \times 2 = 20$ | (73) $4 \times 9 = 36$  | (103) $8 \times 8 = 64$  |
| (14) $1 \times 6 = 6$   | (44) $8 \times 5 = 40$  | (74) $2 \times 3 = 6$   | (104) $10 \times 1 = 10$ |
| (15) $10 \times 2 = 20$ | (45) $6 \times 5 = 30$  | (75) $3 \times 4 = 12$  | (105) $3 \times 7 = 21$  |
| (16) $1 \times 8 = 8$   | (46) $5 \times 7 = 35$  | (76) $7 \times 8 = 56$  | (106) $3 \times 5 = 15$  |
| (17) $4 \times 5 = 20$  | (47) $2 \times 2 = 4$   | (77) $10 \times 3 = 30$ | (107) $10 \times 8 = 80$ |
| (18) $8 \times 9 = 72$  | (48) $5 \times 7 = 35$  | (78) $2 \times 4 = 8$   | (108) $1 \times 3 = 3$   |
| (19) $2 \times 5 = 10$  | (49) $9 \times 2 = 18$  | (79) $3 \times 6 = 18$  | (109) $1 \times 8 = 8$   |
| (20) $1 \times 2 = 2$   | (50) $4 \times 7 = 28$  | (80) $6 \times 3 = 18$  | (110) $1 \times 5 = 5$   |
| (21) $0 \times 9 = 0$   | (51) $10 \times 0 = 0$  | (81) $10 \times 1 = 10$ | (111) $6 \times 5 = 30$  |
| (22) $10 \times 1 = 10$ | (52) $7 \times 6 = 42$  | (82) $1 \times 6 = 6$   | (112) $1 \times 1 = 1$   |
| (23) $1 \times 3 = 3$   | (53) $6 \times 3 = 18$  | (83) $9 \times 6 = 54$  | (113) $6 \times 7 = 42$  |
| (24) $4 \times 6 = 24$  | (54) $10 \times 8 = 80$ | (84) $0 \times 5 = 0$   | (114) $1 \times 10 = 10$ |
| (25) $8 \times 9 = 72$  | (55) $1 \times 3 = 3$   | (85) $9 \times 4 = 36$  | (115) $2 \times 10 = 20$ |
| (26) $6 \times 0 = 0$   | (56) $9 \times 9 = 81$  | (86) $3 \times 8 = 24$  | (116) $8 \times 6 = 48$  |
| (27) $9 \times 8 = 72$  | (57) $4 \times 1 = 4$   | (87) $6 \times 10 = 60$ | (117) $9 \times 8 = 72$  |
| (28) $6 \times 6 = 36$  | (58) $6 \times 2 = 12$  | (88) $7 \times 3 = 21$  | (118) $1 \times 3 = 3$   |
| (29) $9 \times 7 = 63$  | (59) $6 \times 0 = 0$   | (89) $1 \times 3 = 3$   | (119) $7 \times 2 = 14$  |
| (30) $8 \times 5 = 40$  | (60) $9 \times 8 = 72$  | (90) $5 \times 3 = 15$  | (120) $8 \times 0 = 0$   |

|                           |                         |                         |                          |
|---------------------------|-------------------------|-------------------------|--------------------------|
| (1) $2 \times 10 = 20$    | (31) $4 \times 8 = 32$  | (61) $9 \times 5 = 45$  | (91) $6 \times 8 = 48$   |
| (2) $7 \times 2 = 14$     | (32) $7 \times 3 = 21$  | (62) $7 \times 10 = 70$ | (92) $4 \times 9 = 36$   |
| (3) $2 \times 4 = 8$      | (33) $5 \times 1 = 5$   | (63) $10 \times 7 = 70$ | (93) $8 \times 1 = 8$    |
| (4) $6 \times 7 = 42$     | (34) $2 \times 9 = 18$  | (64) $2 \times 5 = 10$  | (94) $4 \times 1 = 4$    |
| (5) $1 \times 6 = 6$      | (35) $4 \times 5 = 20$  | (65) $3 \times 4 = 12$  | (95) $8 \times 5 = 40$   |
| (6) $2 \times 5 = 10$     | (36) $6 \times 6 = 36$  | (66) $6 \times 1 = 6$   | (96) $7 \times 0 = 0$    |
| (7) $10 \times 9 = 90$    | (37) $7 \times 5 = 35$  | (67) $3 \times 9 = 27$  | (97) $3 \times 1 = 3$    |
| (8) $6 \times 10 = 60$    | (38) $8 \times 9 = 72$  | (68) $10 \times 8 = 80$ | (98) $9 \times 8 = 72$   |
| (9) $1 \times 5 = 5$      | (39) $2 \times 0 = 0$   | (69) $6 \times 5 = 30$  | (99) $5 \times 4 = 20$   |
| (10) $2 \times 6 = 12$    | (40) $7 \times 2 = 14$  | (70) $0 \times 0 = 0$   | (100) $2 \times 1 = 2$   |
| (11) $8 \times 2 = 16$    | (41) $7 \times 3 = 21$  | (71) $4 \times 4 = 16$  | (101) $3 \times 10 = 30$ |
| (12) $9 \times 2 = 18$    | (42) $0 \times 4 = 0$   | (72) $2 \times 3 = 6$   | (102) $5 \times 4 = 20$  |
| (13) $10 \times 9 = 90$   | (43) $2 \times 8 = 16$  | (73) $2 \times 3 = 6$   | (103) $0 \times 9 = 0$   |
| (14) $7 \times 6 = 42$    | (44) $3 \times 5 = 15$  | (74) $4 \times 1 = 4$   | (104) $6 \times 1 = 6$   |
| (15) $0 \times 9 = 0$     | (45) $7 \times 2 = 14$  | (75) $2 \times 9 = 18$  | (105) $8 \times 10 = 80$ |
| (16) $10 \times 3 = 30$   | (46) $1 \times 8 = 8$   | (76) $2 \times 6 = 12$  | (106) $9 \times 8 = 72$  |
| (17) $2 \times 6 = 12$    | (47) $6 \times 2 = 12$  | (77) $4 \times 3 = 12$  | (107) $3 \times 8 = 24$  |
| (18) $5 \times 6 = 30$    | (48) $4 \times 7 = 28$  | (78) $9 \times 9 = 81$  | (108) $6 \times 9 = 54$  |
| (19) $9 \times 6 = 54$    | (49) $3 \times 9 = 27$  | (79) $0 \times 6 = 0$   | (109) $10 \times 6 = 60$ |
| (20) $9 \times 8 = 72$    | (50) $2 \times 5 = 10$  | (80) $9 \times 4 = 36$  | (110) $3 \times 6 = 18$  |
| (21) $9 \times 9 = 81$    | (51) $9 \times 4 = 36$  | (81) $4 \times 1 = 4$   | (111) $1 \times 4 = 4$   |
| (22) $1 \times 5 = 5$     | (52) $10 \times 7 = 70$ | (82) $9 \times 3 = 27$  | (112) $6 \times 6 = 36$  |
| (23) $6 \times 0 = 0$     | (53) $7 \times 2 = 14$  | (83) $6 \times 8 = 48$  | (113) $10 \times 0 = 0$  |
| (24) $6 \times 1 = 6$     | (54) $9 \times 1 = 9$   | (84) $8 \times 0 = 0$   | (114) $8 \times 8 = 64$  |
| (25) $10 \times 10 = 100$ | (55) $7 \times 2 = 14$  | (85) $6 \times 1 = 6$   | (115) $0 \times 9 = 0$   |
| (26) $9 \times 7 = 63$    | (56) $4 \times 9 = 36$  | (86) $6 \times 6 = 36$  | (116) $1 \times 1 = 1$   |
| (27) $10 \times 2 = 20$   | (57) $4 \times 0 = 0$   | (87) $9 \times 1 = 9$   | (117) $1 \times 5 = 5$   |
| (28) $9 \times 10 = 90$   | (58) $8 \times 4 = 32$  | (88) $1 \times 1 = 1$   | (118) $1 \times 9 = 9$   |
| (29) $4 \times 5 = 20$    | (59) $4 \times 8 = 32$  | (89) $2 \times 4 = 8$   | (119) $0 \times 0 = 0$   |
| (30) $6 \times 6 = 36$    | (60) $2 \times 4 = 8$   | (90) $7 \times 7 = 49$  | (120) $10 \times 3 = 30$ |

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|--------------------------|---------------------------|---------------------------|--------------------------|
| (1) $4 \times 6 = 24$    | (31) $9 \times 5 = 45$    | (61) $10 \times 7 = 70$   | (91) $0 \times 1 = 0$    |
| (2) $9 \times 7 = 63$    | (32) $9 \times 6 = 54$    | (62) $2 \times 0 = 0$     | (92) $7 \times 2 = 14$   |
| (3) $5 \times 9 = 45$    | (33) $1 \times 4 = 4$     | (63) $5 \times 3 = 15$    | (93) $3 \times 8 = 24$   |
| (4) $10 \times 10 = 100$ | (34) $8 \times 0 = 0$     | (64) $10 \times 6 = 60$   | (94) $7 \times 3 = 21$   |
| (5) $2 \times 2 = 4$     | (35) $10 \times 8 = 80$   | (65) $8 \times 8 = 64$    | (95) $0 \times 7 = 0$    |
| (6) $2 \times 6 = 12$    | (36) $2 \times 7 = 14$    | (66) $10 \times 8 = 80$   | (96) $6 \times 9 = 54$   |
| (7) $2 \times 9 = 18$    | (37) $4 \times 2 = 8$     | (67) $5 \times 6 = 30$    | (97) $1 \times 1 = 1$    |
| (8) $2 \times 0 = 0$     | (38) $5 \times 3 = 15$    | (68) $2 \times 2 = 4$     | (98) $2 \times 5 = 10$   |
| (9) $4 \times 4 = 16$    | (39) $3 \times 2 = 6$     | (69) $7 \times 7 = 49$    | (99) $7 \times 8 = 56$   |
| (10) $10 \times 9 = 90$  | (40) $10 \times 8 = 80$   | (70) $6 \times 7 = 42$    | (100) $7 \times 1 = 7$   |
| (11) $6 \times 4 = 24$   | (41) $10 \times 2 = 20$   | (71) $1 \times 0 = 0$     | (101) $10 \times 6 = 60$ |
| (12) $6 \times 1 = 6$    | (42) $4 \times 2 = 8$     | (72) $10 \times 2 = 20$   | (102) $8 \times 6 = 48$  |
| (13) $10 \times 8 = 80$  | (43) $10 \times 10 = 100$ | (73) $7 \times 8 = 56$    | (103) $2 \times 6 = 12$  |
| (14) $3 \times 4 = 12$   | (44) $5 \times 4 = 20$    | (74) $4 \times 7 = 28$    | (104) $6 \times 9 = 54$  |
| (15) $8 \times 0 = 0$    | (45) $3 \times 7 = 21$    | (75) $4 \times 7 = 28$    | (105) $7 \times 6 = 42$  |
| (16) $10 \times 4 = 40$  | (46) $2 \times 2 = 4$     | (76) $2 \times 1 = 2$     | (106) $2 \times 10 = 20$ |
| (17) $0 \times 8 = 0$    | (47) $9 \times 0 = 0$     | (77) $7 \times 2 = 14$    | (107) $9 \times 8 = 72$  |
| (18) $8 \times 3 = 24$   | (48) $2 \times 3 = 6$     | (78) $1 \times 1 = 1$     | (108) $6 \times 6 = 36$  |
| (19) $9 \times 3 = 27$   | (49) $5 \times 7 = 35$    | (79) $7 \times 6 = 42$    | (109) $6 \times 5 = 30$  |
| (20) $6 \times 7 = 42$   | (50) $4 \times 5 = 20$    | (80) $8 \times 7 = 56$    | (110) $7 \times 8 = 56$  |
| (21) $9 \times 2 = 18$   | (51) $4 \times 1 = 4$     | (81) $8 \times 8 = 64$    | (111) $6 \times 7 = 42$  |
| (22) $10 \times 3 = 30$  | (52) $0 \times 9 = 0$     | (82) $4 \times 8 = 32$    | (112) $3 \times 5 = 15$  |
| (23) $9 \times 6 = 54$   | (53) $1 \times 8 = 8$     | (83) $9 \times 9 = 81$    | (113) $4 \times 4 = 16$  |
| (24) $10 \times 1 = 10$  | (54) $1 \times 0 = 0$     | (84) $1 \times 2 = 2$     | (114) $9 \times 9 = 81$  |
| (25) $6 \times 0 = 0$    | (55) $7 \times 4 = 28$    | (85) $10 \times 10 = 100$ | (115) $7 \times 10 = 70$ |
| (26) $3 \times 3 = 9$    | (56) $4 \times 6 = 24$    | (86) $1 \times 9 = 9$     | (116) $6 \times 7 = 42$  |
| (27) $3 \times 10 = 30$  | (57) $3 \times 6 = 18$    | (87) $9 \times 6 = 54$    | (117) $6 \times 9 = 54$  |
| (28) $10 \times 1 = 10$  | (58) $3 \times 6 = 18$    | (88) $1 \times 10 = 10$   | (118) $1 \times 7 = 7$   |
| (29) $10 \times 1 = 10$  | (59) $9 \times 4 = 36$    | (89) $1 \times 2 = 2$     | (119) $1 \times 9 = 9$   |
| (30) $5 \times 5 = 25$   | (60) $9 \times 2 = 18$    | (90) $7 \times 4 = 28$    | (120) $10 \times 3 = 30$ |

|                         |                         |                         |                          |
|-------------------------|-------------------------|-------------------------|--------------------------|
| (1) $6 \times 4 = 24$   | (31) $6 \times 2 = 12$  | (61) $2 \times 9 = 18$  | (91) $0 \times 1 = 0$    |
| (2) $5 \times 2 = 10$   | (32) $1 \times 8 = 8$   | (62) $7 \times 7 = 49$  | (92) $5 \times 10 = 50$  |
| (3) $8 \times 8 = 64$   | (33) $5 \times 5 = 25$  | (63) $9 \times 4 = 36$  | (93) $6 \times 7 = 42$   |
| (4) $2 \times 9 = 18$   | (34) $7 \times 10 = 70$ | (64) $0 \times 9 = 0$   | (94) $3 \times 6 = 18$   |
| (5) $6 \times 8 = 48$   | (35) $3 \times 6 = 18$  | (65) $8 \times 5 = 40$  | (95) $10 \times 3 = 30$  |
| (6) $1 \times 1 = 1$    | (36) $3 \times 5 = 15$  | (66) $9 \times 1 = 9$   | (96) $3 \times 2 = 6$    |
| (7) $4 \times 1 = 4$    | (37) $7 \times 2 = 14$  | (67) $8 \times 9 = 72$  | (97) $2 \times 4 = 8$    |
| (8) $2 \times 6 = 12$   | (38) $4 \times 3 = 12$  | (68) $5 \times 9 = 45$  | (98) $1 \times 2 = 2$    |
| (9) $4 \times 10 = 40$  | (39) $5 \times 7 = 35$  | (69) $5 \times 10 = 50$ | (99) $3 \times 8 = 24$   |
| (10) $7 \times 6 = 42$  | (40) $1 \times 6 = 6$   | (70) $3 \times 7 = 21$  | (100) $3 \times 6 = 18$  |
| (11) $2 \times 5 = 10$  | (41) $6 \times 9 = 54$  | (71) $2 \times 1 = 2$   | (101) $10 \times 0 = 0$  |
| (12) $8 \times 4 = 32$  | (42) $2 \times 5 = 10$  | (72) $6 \times 6 = 36$  | (102) $8 \times 4 = 32$  |
| (13) $2 \times 5 = 10$  | (43) $2 \times 4 = 8$   | (73) $5 \times 3 = 15$  | (103) $4 \times 1 = 4$   |
| (14) $2 \times 9 = 18$  | (44) $9 \times 0 = 0$   | (74) $1 \times 5 = 5$   | (104) $9 \times 6 = 54$  |
| (15) $8 \times 9 = 72$  | (45) $9 \times 10 = 90$ | (75) $0 \times 5 = 0$   | (105) $5 \times 10 = 50$ |
| (16) $10 \times 9 = 90$ | (46) $2 \times 9 = 18$  | (76) $8 \times 8 = 64$  | (106) $4 \times 2 = 8$   |
| (17) $1 \times 9 = 9$   | (47) $4 \times 9 = 36$  | (77) $4 \times 8 = 32$  | (107) $0 \times 1 = 0$   |
| (18) $6 \times 6 = 36$  | (48) $1 \times 5 = 5$   | (78) $7 \times 8 = 56$  | (108) $9 \times 10 = 90$ |
| (19) $7 \times 8 = 56$  | (49) $4 \times 9 = 36$  | (79) $1 \times 3 = 3$   | (109) $2 \times 4 = 8$   |
| (20) $4 \times 2 = 8$   | (50) $1 \times 1 = 1$   | (80) $2 \times 7 = 14$  | (110) $5 \times 6 = 30$  |
| (21) $5 \times 7 = 35$  | (51) $9 \times 8 = 72$  | (81) $9 \times 1 = 9$   | (111) $6 \times 10 = 60$ |
| (22) $9 \times 2 = 18$  | (52) $10 \times 6 = 60$ | (82) $8 \times 6 = 48$  | (112) $3 \times 6 = 18$  |
| (23) $2 \times 6 = 12$  | (53) $2 \times 8 = 16$  | (83) $3 \times 8 = 24$  | (113) $10 \times 9 = 90$ |
| (24) $1 \times 9 = 9$   | (54) $3 \times 4 = 12$  | (84) $6 \times 9 = 54$  | (114) $5 \times 0 = 0$   |
| (25) $3 \times 4 = 12$  | (55) $1 \times 6 = 6$   | (85) $1 \times 7 = 7$   | (115) $6 \times 1 = 6$   |
| (26) $1 \times 4 = 4$   | (56) $8 \times 9 = 72$  | (86) $0 \times 2 = 0$   | (116) $3 \times 2 = 6$   |
| (27) $7 \times 2 = 14$  | (57) $0 \times 3 = 0$   | (87) $2 \times 5 = 10$  | (117) $9 \times 1 = 9$   |
| (28) $3 \times 1 = 3$   | (58) $8 \times 9 = 72$  | (88) $10 \times 8 = 80$ | (118) $7 \times 7 = 49$  |
| (29) $8 \times 2 = 16$  | (59) $4 \times 6 = 24$  | (89) $6 \times 0 = 0$   | (119) $2 \times 3 = 6$   |
| (30) $3 \times 5 = 15$  | (60) $0 \times 4 = 0$   | (90) $9 \times 7 = 63$  | (120) $2 \times 2 = 4$   |

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|-------------------------|-------------------------|-------------------------|--------------------------|
| (1) $6 \times 3 = 18$   | (31) $4 \times 4 = 16$  | (61) $1 \times 1 = 1$   | (91) $2 \times 7 = 14$   |
| (2) $7 \times 9 = 63$   | (32) $7 \times 8 = 56$  | (62) $5 \times 4 = 20$  | (92) $3 \times 3 = 9$    |
| (3) $3 \times 5 = 15$   | (33) $4 \times 9 = 36$  | (63) $7 \times 4 = 28$  | (93) $3 \times 1 = 3$    |
| (4) $5 \times 1 = 5$    | (34) $7 \times 10 = 70$ | (64) $10 \times 7 = 70$ | (94) $5 \times 9 = 45$   |
| (5) $9 \times 5 = 45$   | (35) $1 \times 5 = 5$   | (65) $4 \times 4 = 16$  | (95) $7 \times 8 = 56$   |
| (6) $1 \times 3 = 3$    | (36) $4 \times 6 = 24$  | (66) $2 \times 2 = 4$   | (96) $5 \times 1 = 5$    |
| (7) $4 \times 5 = 20$   | (37) $9 \times 2 = 18$  | (67) $2 \times 3 = 6$   | (97) $10 \times 7 = 70$  |
| (8) $9 \times 6 = 54$   | (38) $6 \times 0 = 0$   | (68) $8 \times 9 = 72$  | (98) $2 \times 2 = 4$    |
| (9) $0 \times 0 = 0$    | (39) $2 \times 5 = 10$  | (69) $4 \times 7 = 28$  | (99) $7 \times 6 = 42$   |
| (10) $10 \times 1 = 10$ | (40) $5 \times 4 = 20$  | (70) $0 \times 6 = 0$   | (100) $10 \times 0 = 0$  |
| (11) $7 \times 5 = 35$  | (41) $3 \times 4 = 12$  | (71) $5 \times 4 = 20$  | (101) $10 \times 5 = 50$ |
| (12) $4 \times 10 = 40$ | (42) $0 \times 10 = 0$  | (72) $1 \times 2 = 2$   | (102) $9 \times 8 = 72$  |
| (13) $0 \times 3 = 0$   | (43) $9 \times 1 = 9$   | (73) $5 \times 4 = 20$  | (103) $1 \times 8 = 8$   |
| (14) $5 \times 5 = 25$  | (44) $6 \times 6 = 36$  | (74) $5 \times 9 = 45$  | (104) $3 \times 1 = 3$   |
| (15) $3 \times 7 = 21$  | (45) $5 \times 3 = 15$  | (75) $4 \times 2 = 8$   | (105) $1 \times 6 = 6$   |
| (16) $4 \times 4 = 16$  | (46) $0 \times 10 = 0$  | (76) $8 \times 6 = 48$  | (106) $5 \times 2 = 10$  |
| (17) $10 \times 7 = 70$ | (47) $9 \times 3 = 27$  | (77) $5 \times 6 = 30$  | (107) $2 \times 9 = 18$  |
| (18) $1 \times 9 = 9$   | (48) $8 \times 1 = 8$   | (78) $0 \times 8 = 0$   | (108) $9 \times 3 = 27$  |
| (19) $1 \times 8 = 8$   | (49) $3 \times 6 = 18$  | (79) $10 \times 3 = 30$ | (109) $4 \times 10 = 40$ |
| (20) $8 \times 4 = 32$  | (50) $10 \times 8 = 80$ | (80) $8 \times 7 = 56$  | (110) $1 \times 2 = 2$   |
| (21) $3 \times 3 = 9$   | (51) $2 \times 5 = 10$  | (81) $0 \times 6 = 0$   | (111) $3 \times 5 = 15$  |
| (22) $7 \times 1 = 7$   | (52) $8 \times 2 = 16$  | (82) $8 \times 7 = 56$  | (112) $2 \times 7 = 14$  |
| (23) $8 \times 9 = 72$  | (53) $8 \times 9 = 72$  | (83) $8 \times 7 = 56$  | (113) $6 \times 1 = 6$   |
| (24) $10 \times 3 = 30$ | (54) $1 \times 2 = 2$   | (84) $1 \times 6 = 6$   | (114) $9 \times 2 = 18$  |
| (25) $10 \times 1 = 10$ | (55) $6 \times 10 = 60$ | (85) $0 \times 8 = 0$   | (115) $4 \times 5 = 20$  |
| (26) $5 \times 5 = 25$  | (56) $2 \times 9 = 18$  | (86) $7 \times 4 = 28$  | (116) $4 \times 8 = 32$  |
| (27) $8 \times 6 = 48$  | (57) $9 \times 3 = 27$  | (87) $9 \times 9 = 81$  | (117) $7 \times 6 = 42$  |
| (28) $6 \times 1 = 6$   | (58) $4 \times 2 = 8$   | (88) $9 \times 9 = 81$  | (118) $7 \times 1 = 7$   |
| (29) $4 \times 2 = 8$   | (59) $5 \times 10 = 50$ | (89) $10 \times 9 = 90$ | (119) $2 \times 10 = 20$ |
| (30) $10 \times 2 = 20$ | (60) $9 \times 10 = 90$ | (90) $8 \times 3 = 24$  | (120) $6 \times 8 = 48$  |

|                           |                           |                         |                          |
|---------------------------|---------------------------|-------------------------|--------------------------|
| (1) $4 \times 4 = 16$     | (31) $7 \times 7 = 49$    | (61) $8 \times 7 = 56$  | (91) $9 \times 5 = 45$   |
| (2) $4 \times 9 = 36$     | (32) $8 \times 1 = 8$     | (62) $8 \times 1 = 8$   | (92) $4 \times 1 = 4$    |
| (3) $0 \times 4 = 0$      | (33) $0 \times 4 = 0$     | (63) $5 \times 8 = 40$  | (93) $7 \times 10 = 70$  |
| (4) $8 \times 9 = 72$     | (34) $6 \times 4 = 24$    | (64) $2 \times 5 = 10$  | (94) $1 \times 2 = 2$    |
| (5) $0 \times 10 = 0$     | (35) $0 \times 2 = 0$     | (65) $0 \times 0 = 0$   | (95) $1 \times 0 = 0$    |
| (6) $6 \times 6 = 36$     | (36) $6 \times 0 = 0$     | (66) $3 \times 3 = 9$   | (96) $2 \times 5 = 10$   |
| (7) $8 \times 9 = 72$     | (37) $10 \times 7 = 70$   | (67) $9 \times 9 = 81$  | (97) $0 \times 7 = 0$    |
| (8) $5 \times 4 = 20$     | (38) $5 \times 3 = 15$    | (68) $9 \times 4 = 36$  | (98) $6 \times 2 = 12$   |
| (9) $3 \times 10 = 30$    | (39) $5 \times 7 = 35$    | (69) $4 \times 10 = 40$ | (99) $1 \times 9 = 9$    |
| (10) $3 \times 10 = 30$   | (40) $5 \times 2 = 10$    | (70) $6 \times 7 = 42$  | (100) $7 \times 9 = 63$  |
| (11) $0 \times 8 = 0$     | (41) $10 \times 1 = 10$   | (71) $2 \times 4 = 8$   | (101) $10 \times 3 = 30$ |
| (12) $7 \times 10 = 70$   | (42) $6 \times 9 = 54$    | (72) $10 \times 6 = 60$ | (102) $0 \times 10 = 0$  |
| (13) $7 \times 9 = 63$    | (43) $8 \times 1 = 8$     | (73) $5 \times 7 = 35$  | (103) $7 \times 7 = 49$  |
| (14) $3 \times 2 = 6$     | (44) $7 \times 6 = 42$    | (74) $10 \times 6 = 60$ | (104) $1 \times 6 = 6$   |
| (15) $9 \times 0 = 0$     | (45) $0 \times 9 = 0$     | (75) $10 \times 1 = 10$ | (105) $4 \times 5 = 20$  |
| (16) $7 \times 4 = 28$    | (46) $7 \times 7 = 49$    | (76) $2 \times 7 = 14$  | (106) $7 \times 1 = 7$   |
| (17) $4 \times 2 = 8$     | (47) $10 \times 7 = 70$   | (77) $3 \times 2 = 6$   | (107) $6 \times 6 = 36$  |
| (18) $10 \times 10 = 100$ | (48) $1 \times 3 = 3$     | (78) $0 \times 9 = 0$   | (108) $5 \times 9 = 45$  |
| (19) $6 \times 10 = 60$   | (49) $10 \times 7 = 70$   | (79) $8 \times 10 = 80$ | (109) $1 \times 4 = 4$   |
| (20) $1 \times 9 = 9$     | (50) $2 \times 6 = 12$    | (80) $5 \times 4 = 20$  | (110) $5 \times 8 = 40$  |
| (21) $10 \times 2 = 20$   | (51) $7 \times 6 = 42$    | (81) $8 \times 9 = 72$  | (111) $1 \times 8 = 8$   |
| (22) $3 \times 1 = 3$     | (52) $3 \times 8 = 24$    | (82) $9 \times 4 = 36$  | (112) $7 \times 3 = 21$  |
| (23) $6 \times 7 = 42$    | (53) $5 \times 1 = 5$     | (83) $6 \times 8 = 48$  | (113) $9 \times 9 = 81$  |
| (24) $3 \times 2 = 6$     | (54) $7 \times 4 = 28$    | (84) $9 \times 7 = 63$  | (114) $9 \times 2 = 18$  |
| (25) $9 \times 5 = 45$    | (55) $1 \times 1 = 1$     | (85) $1 \times 1 = 1$   | (115) $4 \times 10 = 40$ |
| (26) $4 \times 3 = 12$    | (56) $10 \times 10 = 100$ | (86) $9 \times 8 = 72$  | (116) $10 \times 6 = 60$ |
| (27) $2 \times 7 = 14$    | (57) $8 \times 1 = 8$     | (87) $7 \times 9 = 63$  | (117) $10 \times 3 = 30$ |
| (28) $9 \times 1 = 9$     | (58) $7 \times 3 = 21$    | (88) $4 \times 9 = 36$  | (118) $8 \times 2 = 16$  |
| (29) $7 \times 8 = 56$    | (59) $3 \times 7 = 21$    | (89) $1 \times 7 = 7$   | (119) $3 \times 0 = 0$   |
| (30) $10 \times 8 = 80$   | (60) $8 \times 8 = 64$    | (90) $7 \times 3 = 21$  | (120) $2 \times 9 = 18$  |

- |                         |                         |                           |                          |
|-------------------------|-------------------------|---------------------------|--------------------------|
| (1) $10 \times 4 = 40$  | (31) $10 \times 1 = 10$ | (61) $7 \times 6 = 42$    | (91) $3 \times 3 = 9$    |
| (2) $4 \times 1 = 4$    | (32) $7 \times 9 = 63$  | (62) $8 \times 3 = 24$    | (92) $3 \times 8 = 24$   |
| (3) $9 \times 9 = 81$   | (33) $4 \times 8 = 32$  | (63) $8 \times 7 = 56$    | (93) $10 \times 7 = 70$  |
| (4) $10 \times 4 = 40$  | (34) $4 \times 2 = 8$   | (64) $10 \times 9 = 90$   | (94) $3 \times 7 = 21$   |
| (5) $3 \times 2 = 6$    | (35) $3 \times 5 = 15$  | (65) $10 \times 6 = 60$   | (95) $3 \times 5 = 15$   |
| (6) $5 \times 6 = 30$   | (36) $1 \times 3 = 3$   | (66) $8 \times 10 = 80$   | (96) $7 \times 10 = 70$  |
| (7) $1 \times 5 = 5$    | (37) $7 \times 7 = 49$  | (67) $9 \times 10 = 90$   | (97) $8 \times 7 = 56$   |
| (8) $7 \times 0 = 0$    | (38) $8 \times 2 = 16$  | (68) $8 \times 1 = 8$     | (98) $10 \times 6 = 60$  |
| (9) $6 \times 6 = 36$   | (39) $10 \times 3 = 30$ | (69) $5 \times 10 = 50$   | (99) $3 \times 5 = 15$   |
| (10) $9 \times 3 = 27$  | (40) $6 \times 6 = 36$  | (70) $5 \times 9 = 45$    | (100) $9 \times 1 = 9$   |
| (11) $2 \times 8 = 16$  | (41) $6 \times 1 = 6$   | (71) $8 \times 5 = 40$    | (101) $3 \times 9 = 27$  |
| (12) $6 \times 1 = 6$   | (42) $1 \times 8 = 8$   | (72) $1 \times 6 = 6$     | (102) $7 \times 6 = 42$  |
| (13) $9 \times 0 = 0$   | (43) $3 \times 1 = 3$   | (73) $2 \times 3 = 6$     | (103) $10 \times 7 = 70$ |
| (14) $9 \times 6 = 54$  | (44) $2 \times 0 = 0$   | (74) $4 \times 7 = 28$    | (104) $0 \times 4 = 0$   |
| (15) $1 \times 10 = 10$ | (45) $2 \times 3 = 6$   | (75) $4 \times 3 = 12$    | (105) $8 \times 2 = 16$  |
| (16) $2 \times 9 = 18$  | (46) $10 \times 8 = 80$ | (76) $8 \times 3 = 24$    | (106) $2 \times 10 = 20$ |
| (17) $3 \times 9 = 27$  | (47) $0 \times 3 = 0$   | (77) $2 \times 5 = 10$    | (107) $2 \times 1 = 2$   |
| (18) $6 \times 4 = 24$  | (48) $10 \times 9 = 90$ | (78) $4 \times 7 = 28$    | (108) $10 \times 0 = 0$  |
| (19) $6 \times 9 = 54$  | (49) $1 \times 5 = 5$   | (79) $9 \times 9 = 81$    | (109) $10 \times 1 = 10$ |
| (20) $5 \times 4 = 20$  | (50) $8 \times 7 = 56$  | (80) $0 \times 7 = 0$     | (110) $5 \times 2 = 10$  |
| (21) $5 \times 2 = 10$  | (51) $5 \times 6 = 30$  | (81) $0 \times 7 = 0$     | (111) $1 \times 7 = 7$   |
| (22) $7 \times 2 = 14$  | (52) $8 \times 4 = 32$  | (82) $4 \times 9 = 36$    | (112) $9 \times 1 = 9$   |
| (23) $2 \times 10 = 20$ | (53) $4 \times 5 = 20$  | (83) $10 \times 10 = 100$ | (113) $0 \times 6 = 0$   |
| (24) $10 \times 3 = 30$ | (54) $10 \times 3 = 30$ | (84) $7 \times 8 = 56$    | (114) $7 \times 10 = 70$ |
| (25) $5 \times 3 = 15$  | (55) $3 \times 1 = 3$   | (85) $4 \times 1 = 4$     | (115) $4 \times 10 = 40$ |
| (26) $2 \times 2 = 4$   | (56) $1 \times 2 = 2$   | (86) $2 \times 1 = 2$     | (116) $5 \times 10 = 50$ |
| (27) $2 \times 4 = 8$   | (57) $4 \times 0 = 0$   | (87) $5 \times 9 = 45$    | (117) $9 \times 9 = 81$  |
| (28) $8 \times 4 = 32$  | (58) $3 \times 3 = 9$   | (88) $8 \times 9 = 72$    | (118) $9 \times 2 = 18$  |
| (29) $9 \times 4 = 36$  | (59) $1 \times 5 = 5$   | (89) $4 \times 3 = 12$    | (119) $10 \times 0 = 0$  |
| (30) $4 \times 9 = 36$  | (60) $6 \times 3 = 18$  | (90) $10 \times 8 = 80$   | (120) $9 \times 6 = 54$  |



|                          |                         |                         |                          |
|--------------------------|-------------------------|-------------------------|--------------------------|
| (1) $2 \times 8 = 16$    | (31) $6 \times 2 = 12$  | (61) $0 \times 1 = 0$   | (91) $9 \times 5 = 45$   |
| (2) $2 \times 10 = 20$   | (32) $3 \times 1 = 3$   | (62) $2 \times 1 = 2$   | (92) $3 \times 5 = 15$   |
| (3) $6 \times 8 = 48$    | (33) $7 \times 7 = 49$  | (63) $4 \times 8 = 32$  | (93) $1 \times 0 = 0$    |
| (4) $10 \times 10 = 100$ | (34) $9 \times 3 = 27$  | (64) $4 \times 1 = 4$   | (94) $2 \times 3 = 6$    |
| (5) $7 \times 1 = 7$     | (35) $9 \times 0 = 0$   | (65) $3 \times 7 = 21$  | (95) $3 \times 10 = 30$  |
| (6) $6 \times 5 = 30$    | (36) $1 \times 6 = 6$   | (66) $2 \times 3 = 6$   | (96) $9 \times 10 = 90$  |
| (7) $8 \times 8 = 64$    | (37) $1 \times 6 = 6$   | (67) $0 \times 7 = 0$   | (97) $8 \times 1 = 8$    |
| (8) $9 \times 5 = 45$    | (38) $10 \times 0 = 0$  | (68) $5 \times 2 = 10$  | (98) $1 \times 4 = 4$    |
| (9) $8 \times 7 = 56$    | (39) $3 \times 2 = 6$   | (69) $6 \times 2 = 12$  | (99) $2 \times 4 = 8$    |
| (10) $9 \times 2 = 18$   | (40) $2 \times 6 = 12$  | (70) $1 \times 7 = 7$   | (100) $6 \times 4 = 24$  |
| (11) $3 \times 0 = 0$    | (41) $5 \times 4 = 20$  | (71) $2 \times 2 = 4$   | (101) $4 \times 4 = 16$  |
| (12) $4 \times 4 = 16$   | (42) $3 \times 6 = 18$  | (72) $2 \times 3 = 6$   | (102) $1 \times 6 = 6$   |
| (13) $7 \times 1 = 7$    | (43) $2 \times 8 = 16$  | (73) $6 \times 5 = 30$  | (103) $3 \times 4 = 12$  |
| (14) $5 \times 0 = 0$    | (44) $8 \times 10 = 80$ | (74) $6 \times 8 = 48$  | (104) $6 \times 3 = 18$  |
| (15) $5 \times 9 = 45$   | (45) $8 \times 4 = 32$  | (75) $9 \times 2 = 18$  | (105) $5 \times 3 = 15$  |
| (16) $1 \times 0 = 0$    | (46) $8 \times 5 = 40$  | (76) $6 \times 1 = 6$   | (106) $10 \times 4 = 40$ |
| (17) $3 \times 7 = 21$   | (47) $4 \times 7 = 28$  | (77) $1 \times 6 = 6$   | (107) $8 \times 4 = 32$  |
| (18) $6 \times 7 = 42$   | (48) $6 \times 6 = 36$  | (78) $1 \times 8 = 8$   | (108) $7 \times 3 = 21$  |
| (19) $6 \times 1 = 6$    | (49) $7 \times 9 = 63$  | (79) $8 \times 0 = 0$   | (109) $8 \times 3 = 24$  |
| (20) $2 \times 3 = 6$    | (50) $10 \times 9 = 90$ | (80) $7 \times 1 = 7$   | (110) $4 \times 4 = 16$  |
| (21) $0 \times 5 = 0$    | (51) $1 \times 3 = 3$   | (81) $8 \times 2 = 16$  | (111) $3 \times 8 = 24$  |
| (22) $7 \times 9 = 63$   | (52) $0 \times 6 = 0$   | (82) $7 \times 9 = 63$  | (112) $2 \times 8 = 16$  |
| (23) $10 \times 9 = 90$  | (53) $1 \times 10 = 10$ | (83) $0 \times 9 = 0$   | (113) $1 \times 1 = 1$   |
| (24) $4 \times 8 = 32$   | (54) $10 \times 4 = 40$ | (84) $4 \times 1 = 4$   | (114) $5 \times 1 = 5$   |
| (25) $10 \times 2 = 20$  | (55) $8 \times 8 = 64$  | (85) $7 \times 1 = 7$   | (115) $9 \times 1 = 9$   |
| (26) $6 \times 3 = 18$   | (56) $8 \times 4 = 32$  | (86) $2 \times 6 = 12$  | (116) $2 \times 1 = 2$   |
| (27) $8 \times 4 = 32$   | (57) $1 \times 8 = 8$   | (87) $4 \times 1 = 4$   | (117) $2 \times 1 = 2$   |
| (28) $1 \times 1 = 1$    | (58) $6 \times 5 = 30$  | (88) $3 \times 4 = 12$  | (118) $2 \times 8 = 16$  |
| (29) $0 \times 6 = 0$    | (59) $10 \times 6 = 60$ | (89) $9 \times 10 = 90$ | (119) $0 \times 2 = 0$   |
| (30) $4 \times 10 = 40$  | (60) $0 \times 7 = 0$   | (90) $9 \times 10 = 90$ | (120) $4 \times 4 = 16$  |

- |                           |                         |                         |                          |
|---------------------------|-------------------------|-------------------------|--------------------------|
| (1) $1 \times 4 = 4$      | (31) $10 \times 8 = 80$ | (61) $7 \times 8 = 56$  | (91) $9 \times 5 = 45$   |
| (2) $2 \times 9 = 18$     | (32) $5 \times 0 = 0$   | (62) $8 \times 8 = 64$  | (92) $4 \times 4 = 16$   |
| (3) $2 \times 2 = 4$      | (33) $9 \times 4 = 36$  | (63) $1 \times 3 = 3$   | (93) $10 \times 9 = 90$  |
| (4) $10 \times 8 = 80$    | (34) $4 \times 3 = 12$  | (64) $0 \times 1 = 0$   | (94) $2 \times 2 = 4$    |
| (5) $5 \times 6 = 30$     | (35) $7 \times 5 = 35$  | (65) $2 \times 3 = 6$   | (95) $1 \times 1 = 1$    |
| (6) $6 \times 6 = 36$     | (36) $1 \times 3 = 3$   | (66) $2 \times 2 = 4$   | (96) $8 \times 5 = 40$   |
| (7) $7 \times 2 = 14$     | (37) $7 \times 9 = 63$  | (67) $1 \times 3 = 3$   | (97) $7 \times 5 = 35$   |
| (8) $0 \times 7 = 0$      | (38) $4 \times 2 = 8$   | (68) $9 \times 4 = 36$  | (98) $0 \times 2 = 0$    |
| (9) $2 \times 5 = 10$     | (39) $7 \times 9 = 63$  | (69) $3 \times 6 = 18$  | (99) $5 \times 10 = 50$  |
| (10) $8 \times 7 = 56$    | (40) $7 \times 4 = 28$  | (70) $9 \times 1 = 9$   | (100) $1 \times 0 = 0$   |
| (11) $4 \times 6 = 24$    | (41) $3 \times 6 = 18$  | (71) $7 \times 8 = 56$  | (101) $0 \times 6 = 0$   |
| (12) $7 \times 4 = 28$    | (42) $4 \times 3 = 12$  | (72) $2 \times 5 = 10$  | (102) $8 \times 1 = 8$   |
| (13) $6 \times 10 = 60$   | (43) $0 \times 4 = 0$   | (73) $10 \times 8 = 80$ | (103) $4 \times 1 = 4$   |
| (14) $8 \times 8 = 64$    | (44) $7 \times 1 = 7$   | (74) $8 \times 6 = 48$  | (104) $4 \times 10 = 40$ |
| (15) $10 \times 10 = 100$ | (45) $3 \times 8 = 24$  | (75) $7 \times 6 = 42$  | (105) $2 \times 10 = 20$ |
| (16) $5 \times 0 = 0$     | (46) $4 \times 7 = 28$  | (76) $5 \times 2 = 10$  | (106) $7 \times 5 = 35$  |
| (17) $4 \times 0 = 0$     | (47) $6 \times 10 = 60$ | (77) $8 \times 4 = 32$  | (107) $0 \times 0 = 0$   |
| (18) $4 \times 7 = 28$    | (48) $10 \times 3 = 30$ | (78) $5 \times 8 = 40$  | (108) $10 \times 3 = 30$ |
| (19) $6 \times 2 = 12$    | (49) $8 \times 6 = 48$  | (79) $1 \times 7 = 7$   | (109) $2 \times 2 = 4$   |
| (20) $4 \times 1 = 4$     | (50) $7 \times 5 = 35$  | (80) $9 \times 4 = 36$  | (110) $5 \times 6 = 30$  |
| (21) $4 \times 8 = 32$    | (51) $6 \times 1 = 6$   | (81) $10 \times 5 = 50$ | (111) $5 \times 8 = 40$  |
| (22) $6 \times 10 = 60$   | (52) $4 \times 0 = 0$   | (82) $10 \times 1 = 10$ | (112) $2 \times 4 = 8$   |
| (23) $9 \times 9 = 81$    | (53) $5 \times 2 = 10$  | (83) $6 \times 3 = 18$  | (113) $8 \times 0 = 0$   |
| (24) $4 \times 6 = 24$    | (54) $3 \times 9 = 27$  | (84) $10 \times 2 = 20$ | (114) $10 \times 3 = 30$ |
| (25) $3 \times 1 = 3$     | (55) $7 \times 4 = 28$  | (85) $8 \times 8 = 64$  | (115) $2 \times 4 = 8$   |
| (26) $3 \times 4 = 12$    | (56) $1 \times 9 = 9$   | (86) $5 \times 10 = 50$ | (116) $2 \times 9 = 18$  |
| (27) $9 \times 10 = 90$   | (57) $7 \times 7 = 49$  | (87) $4 \times 10 = 40$ | (117) $3 \times 7 = 21$  |
| (28) $10 \times 2 = 20$   | (58) $6 \times 7 = 42$  | (88) $8 \times 4 = 32$  | (118) $9 \times 10 = 90$ |
| (29) $4 \times 9 = 36$    | (59) $9 \times 10 = 90$ | (89) $9 \times 8 = 72$  | (119) $3 \times 2 = 6$   |
| (30) $7 \times 9 = 63$    | (60) $4 \times 6 = 24$  | (90) $0 \times 7 = 0$   | (120) $6 \times 6 = 36$  |

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|---------------------------|-------------------------|-------------------------|--------------------------|
| (1) $8 \times 9 = 72$     | (31) $10 \times 2 = 20$ | (61) $0 \times 9 = 0$   | (91) $5 \times 2 = 10$   |
| (2) $5 \times 9 = 45$     | (32) $3 \times 2 = 6$   | (62) $9 \times 2 = 18$  | (92) $6 \times 4 = 24$   |
| (3) $7 \times 10 = 70$    | (33) $5 \times 8 = 40$  | (63) $10 \times 1 = 10$ | (93) $1 \times 4 = 4$    |
| (4) $4 \times 8 = 32$     | (34) $2 \times 3 = 6$   | (64) $8 \times 1 = 8$   | (94) $4 \times 6 = 24$   |
| (5) $4 \times 1 = 4$      | (35) $4 \times 5 = 20$  | (65) $8 \times 3 = 24$  | (95) $9 \times 9 = 81$   |
| (6) $7 \times 8 = 56$     | (36) $8 \times 10 = 80$ | (66) $3 \times 5 = 15$  | (96) $4 \times 4 = 16$   |
| (7) $5 \times 9 = 45$     | (37) $8 \times 6 = 48$  | (67) $10 \times 8 = 80$ | (97) $6 \times 5 = 30$   |
| (8) $9 \times 7 = 63$     | (38) $9 \times 9 = 81$  | (68) $1 \times 10 = 10$ | (98) $4 \times 9 = 36$   |
| (9) $3 \times 4 = 12$     | (39) $6 \times 9 = 54$  | (69) $1 \times 1 = 1$   | (99) $6 \times 10 = 60$  |
| (10) $4 \times 10 = 40$   | (40) $1 \times 3 = 3$   | (70) $10 \times 8 = 80$ | (100) $0 \times 7 = 0$   |
| (11) $4 \times 9 = 36$    | (41) $4 \times 5 = 20$  | (71) $8 \times 4 = 32$  | (101) $4 \times 9 = 36$  |
| (12) $6 \times 9 = 54$    | (42) $3 \times 5 = 15$  | (72) $9 \times 3 = 27$  | (102) $10 \times 3 = 30$ |
| (13) $1 \times 8 = 8$     | (43) $4 \times 5 = 20$  | (73) $3 \times 10 = 30$ | (103) $7 \times 1 = 7$   |
| (14) $10 \times 5 = 50$   | (44) $2 \times 3 = 6$   | (74) $2 \times 9 = 18$  | (104) $5 \times 10 = 50$ |
| (15) $1 \times 10 = 10$   | (45) $4 \times 10 = 40$ | (75) $9 \times 8 = 72$  | (105) $9 \times 9 = 81$  |
| (16) $0 \times 2 = 0$     | (46) $3 \times 8 = 24$  | (76) $7 \times 7 = 49$  | (106) $4 \times 1 = 4$   |
| (17) $8 \times 6 = 48$    | (47) $1 \times 7 = 7$   | (77) $6 \times 9 = 54$  | (107) $2 \times 8 = 16$  |
| (18) $10 \times 5 = 50$   | (48) $0 \times 1 = 0$   | (78) $8 \times 5 = 40$  | (108) $2 \times 0 = 0$   |
| (19) $10 \times 10 = 100$ | (49) $8 \times 2 = 16$  | (79) $6 \times 8 = 48$  | (109) $6 \times 10 = 60$ |
| (20) $2 \times 6 = 12$    | (50) $7 \times 7 = 49$  | (80) $10 \times 6 = 60$ | (110) $10 \times 5 = 50$ |
| (21) $9 \times 6 = 54$    | (51) $0 \times 3 = 0$   | (81) $2 \times 10 = 20$ | (111) $6 \times 4 = 24$  |
| (22) $1 \times 10 = 10$   | (52) $9 \times 10 = 90$ | (82) $1 \times 8 = 8$   | (112) $6 \times 10 = 60$ |
| (23) $10 \times 2 = 20$   | (53) $5 \times 7 = 35$  | (83) $2 \times 1 = 2$   | (113) $0 \times 6 = 0$   |
| (24) $2 \times 9 = 18$    | (54) $8 \times 6 = 48$  | (84) $5 \times 0 = 0$   | (114) $3 \times 4 = 12$  |
| (25) $10 \times 2 = 20$   | (55) $7 \times 2 = 14$  | (85) $3 \times 0 = 0$   | (115) $7 \times 4 = 28$  |
| (26) $6 \times 9 = 54$    | (56) $10 \times 6 = 60$ | (86) $8 \times 8 = 64$  | (116) $0 \times 10 = 0$  |
| (27) $1 \times 4 = 4$     | (57) $4 \times 4 = 16$  | (87) $7 \times 5 = 35$  | (117) $3 \times 9 = 27$  |
| (28) $4 \times 1 = 4$     | (58) $5 \times 1 = 5$   | (88) $5 \times 9 = 45$  | (118) $4 \times 6 = 24$  |
| (29) $6 \times 9 = 54$    | (59) $10 \times 9 = 90$ | (89) $5 \times 9 = 45$  | (119) $5 \times 2 = 10$  |
| (30) $0 \times 7 = 0$     | (60) $3 \times 4 = 12$  | (90) $6 \times 8 = 48$  | (120) $3 \times 1 = 3$   |

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|---------------------------|-------------------------|-------------------------|--------------------------|
| (1) $1 \times 9 = 9$      | (31) $9 \times 3 = 27$  | (61) $4 \times 1 = 4$   | (91) $1 \times 3 = 3$    |
| (2) $8 \times 6 = 48$     | (32) $5 \times 3 = 15$  | (62) $3 \times 2 = 6$   | (92) $7 \times 4 = 28$   |
| (3) $5 \times 3 = 15$     | (33) $5 \times 2 = 10$  | (63) $8 \times 7 = 56$  | (93) $9 \times 7 = 63$   |
| (4) $5 \times 9 = 45$     | (34) $7 \times 2 = 14$  | (64) $7 \times 1 = 7$   | (94) $10 \times 0 = 0$   |
| (5) $8 \times 10 = 80$    | (35) $10 \times 4 = 40$ | (65) $7 \times 6 = 42$  | (95) $6 \times 0 = 0$    |
| (6) $1 \times 3 = 3$      | (36) $7 \times 8 = 56$  | (66) $0 \times 8 = 0$   | (96) $3 \times 2 = 6$    |
| (7) $4 \times 5 = 20$     | (37) $10 \times 7 = 70$ | (67) $3 \times 5 = 15$  | (97) $8 \times 2 = 16$   |
| (8) $9 \times 4 = 36$     | (38) $8 \times 7 = 56$  | (68) $7 \times 9 = 63$  | (98) $3 \times 8 = 24$   |
| (9) $5 \times 3 = 15$     | (39) $4 \times 0 = 0$   | (69) $1 \times 2 = 2$   | (99) $10 \times 7 = 70$  |
| (10) $7 \times 9 = 63$    | (40) $2 \times 8 = 16$  | (70) $1 \times 9 = 9$   | (100) $10 \times 6 = 60$ |
| (11) $10 \times 5 = 50$   | (41) $9 \times 10 = 90$ | (71) $6 \times 8 = 48$  | (101) $3 \times 2 = 6$   |
| (12) $2 \times 6 = 12$    | (42) $10 \times 6 = 60$ | (72) $9 \times 10 = 90$ | (102) $0 \times 9 = 0$   |
| (13) $6 \times 0 = 0$     | (43) $8 \times 2 = 16$  | (73) $4 \times 9 = 36$  | (103) $4 \times 10 = 40$ |
| (14) $3 \times 4 = 12$    | (44) $2 \times 10 = 20$ | (74) $10 \times 2 = 20$ | (104) $8 \times 7 = 56$  |
| (15) $1 \times 9 = 9$     | (45) $5 \times 0 = 0$   | (75) $4 \times 7 = 28$  | (105) $8 \times 1 = 8$   |
| (16) $2 \times 3 = 6$     | (46) $10 \times 6 = 60$ | (76) $9 \times 2 = 18$  | (106) $9 \times 7 = 63$  |
| (17) $10 \times 2 = 20$   | (47) $4 \times 7 = 28$  | (77) $5 \times 8 = 40$  | (107) $3 \times 8 = 24$  |
| (18) $10 \times 3 = 30$   | (48) $6 \times 1 = 6$   | (78) $10 \times 2 = 20$ | (108) $4 \times 3 = 12$  |
| (19) $6 \times 1 = 6$     | (49) $5 \times 2 = 10$  | (79) $6 \times 5 = 30$  | (109) $5 \times 3 = 15$  |
| (20) $8 \times 1 = 8$     | (50) $3 \times 9 = 27$  | (80) $10 \times 2 = 20$ | (110) $7 \times 5 = 35$  |
| (21) $9 \times 9 = 81$    | (51) $6 \times 8 = 48$  | (81) $0 \times 5 = 0$   | (111) $3 \times 4 = 12$  |
| (22) $3 \times 2 = 6$     | (52) $5 \times 2 = 10$  | (82) $2 \times 8 = 16$  | (112) $6 \times 6 = 36$  |
| (23) $7 \times 6 = 42$    | (53) $1 \times 9 = 9$   | (83) $6 \times 4 = 24$  | (113) $2 \times 4 = 8$   |
| (24) $7 \times 3 = 21$    | (54) $0 \times 8 = 0$   | (84) $6 \times 6 = 36$  | (114) $1 \times 9 = 9$   |
| (25) $2 \times 3 = 6$     | (55) $1 \times 7 = 7$   | (85) $3 \times 6 = 18$  | (115) $9 \times 5 = 45$  |
| (26) $10 \times 3 = 30$   | (56) $3 \times 8 = 24$  | (86) $5 \times 1 = 5$   | (116) $2 \times 7 = 14$  |
| (27) $10 \times 10 = 100$ | (57) $9 \times 5 = 45$  | (87) $0 \times 2 = 0$   | (117) $7 \times 6 = 42$  |
| (28) $6 \times 7 = 42$    | (58) $2 \times 5 = 10$  | (88) $5 \times 1 = 5$   | (118) $6 \times 5 = 30$  |
| (29) $8 \times 4 = 32$    | (59) $0 \times 4 = 0$   | (89) $2 \times 10 = 20$ | (119) $5 \times 7 = 35$  |
| (30) $6 \times 5 = 30$    | (60) $4 \times 7 = 28$  | (90) $7 \times 7 = 49$  | (120) $4 \times 4 = 16$  |

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|---------------------------|-------------------------|---------------------------|--------------------------|
| (1) $7 \times 10 = 70$    | (31) $7 \times 6 = 42$  | (61) $2 \times 3 = 6$     | (91) $2 \times 5 = 10$   |
| (2) $7 \times 6 = 42$     | (32) $4 \times 4 = 16$  | (62) $5 \times 2 = 10$    | (92) $10 \times 8 = 80$  |
| (3) $0 \times 0 = 0$      | (33) $1 \times 6 = 6$   | (63) $6 \times 2 = 12$    | (93) $4 \times 5 = 20$   |
| (4) $2 \times 9 = 18$     | (34) $5 \times 8 = 40$  | (64) $3 \times 2 = 6$     | (94) $2 \times 6 = 12$   |
| (5) $10 \times 3 = 30$    | (35) $3 \times 7 = 21$  | (65) $3 \times 2 = 6$     | (95) $6 \times 9 = 54$   |
| (6) $3 \times 8 = 24$     | (36) $4 \times 5 = 20$  | (66) $9 \times 10 = 90$   | (96) $9 \times 4 = 36$   |
| (7) $7 \times 9 = 63$     | (37) $4 \times 10 = 40$ | (67) $1 \times 4 = 4$     | (97) $10 \times 6 = 60$  |
| (8) $9 \times 0 = 0$      | (38) $5 \times 6 = 30$  | (68) $9 \times 3 = 27$    | (98) $6 \times 0 = 0$    |
| (9) $2 \times 1 = 2$      | (39) $6 \times 8 = 48$  | (69) $4 \times 8 = 32$    | (99) $2 \times 9 = 18$   |
| (10) $2 \times 10 = 20$   | (40) $2 \times 9 = 18$  | (70) $5 \times 8 = 40$    | (100) $1 \times 2 = 2$   |
| (11) $2 \times 3 = 6$     | (41) $0 \times 1 = 0$   | (71) $2 \times 6 = 12$    | (101) $3 \times 8 = 24$  |
| (12) $10 \times 3 = 30$   | (42) $8 \times 10 = 80$ | (72) $9 \times 3 = 27$    | (102) $6 \times 5 = 30$  |
| (13) $10 \times 7 = 70$   | (43) $2 \times 5 = 10$  | (73) $5 \times 4 = 20$    | (103) $9 \times 10 = 90$ |
| (14) $3 \times 8 = 24$    | (44) $7 \times 4 = 28$  | (74) $0 \times 4 = 0$     | (104) $4 \times 8 = 32$  |
| (15) $10 \times 7 = 70$   | (45) $6 \times 3 = 18$  | (75) $1 \times 1 = 1$     | (105) $3 \times 3 = 9$   |
| (16) $10 \times 10 = 100$ | (46) $6 \times 3 = 18$  | (76) $4 \times 2 = 8$     | (106) $10 \times 8 = 80$ |
| (17) $7 \times 3 = 21$    | (47) $9 \times 7 = 63$  | (77) $10 \times 10 = 100$ | (107) $5 \times 1 = 5$   |
| (18) $9 \times 2 = 18$    | (48) $8 \times 5 = 40$  | (78) $4 \times 6 = 24$    | (108) $7 \times 8 = 56$  |
| (19) $1 \times 3 = 3$     | (49) $8 \times 6 = 48$  | (79) $7 \times 1 = 7$     | (109) $4 \times 10 = 40$ |
| (20) $8 \times 4 = 32$    | (50) $2 \times 9 = 18$  | (80) $9 \times 6 = 54$    | (110) $1 \times 3 = 3$   |
| (21) $9 \times 5 = 45$    | (51) $10 \times 1 = 10$ | (81) $2 \times 9 = 18$    | (111) $10 \times 7 = 70$ |
| (22) $6 \times 2 = 12$    | (52) $10 \times 9 = 90$ | (82) $10 \times 10 = 100$ | (112) $2 \times 6 = 12$  |
| (23) $1 \times 1 = 1$     | (53) $4 \times 5 = 20$  | (83) $1 \times 5 = 5$     | (113) $10 \times 4 = 40$ |
| (24) $6 \times 6 = 36$    | (54) $9 \times 3 = 27$  | (84) $7 \times 6 = 42$    | (114) $4 \times 8 = 32$  |
| (25) $3 \times 10 = 30$   | (55) $4 \times 7 = 28$  | (85) $4 \times 6 = 24$    | (115) $7 \times 9 = 63$  |
| (26) $7 \times 4 = 28$    | (56) $8 \times 3 = 24$  | (86) $3 \times 0 = 0$     | (116) $5 \times 0 = 0$   |
| (27) $9 \times 4 = 36$    | (57) $2 \times 2 = 4$   | (87) $6 \times 2 = 12$    | (117) $4 \times 2 = 8$   |
| (28) $2 \times 1 = 2$     | (58) $2 \times 6 = 12$  | (88) $7 \times 2 = 14$    | (118) $1 \times 8 = 8$   |
| (29) $0 \times 3 = 0$     | (59) $1 \times 10 = 10$ | (89) $4 \times 5 = 20$    | (119) $8 \times 2 = 16$  |
| (30) $10 \times 3 = 30$   | (60) $1 \times 2 = 2$   | (90) $9 \times 6 = 54$    | (120) $5 \times 7 = 35$  |

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|-------------------------|---------------------------|-------------------------|--------------------------|
| (1) $3 \times 5 = 15$   | (31) $2 \times 5 = 10$    | (61) $2 \times 3 = 6$   | (91) $3 \times 4 = 12$   |
| (2) $5 \times 1 = 5$    | (32) $7 \times 10 = 70$   | (62) $9 \times 6 = 54$  | (92) $3 \times 4 = 12$   |
| (3) $2 \times 3 = 6$    | (33) $0 \times 1 = 0$     | (63) $5 \times 3 = 15$  | (93) $1 \times 10 = 10$  |
| (4) $7 \times 6 = 42$   | (34) $10 \times 1 = 10$   | (64) $5 \times 5 = 25$  | (94) $10 \times 1 = 10$  |
| (5) $5 \times 7 = 35$   | (35) $5 \times 3 = 15$    | (65) $9 \times 1 = 9$   | (95) $0 \times 5 = 0$    |
| (6) $5 \times 9 = 45$   | (36) $2 \times 7 = 14$    | (66) $2 \times 3 = 6$   | (96) $6 \times 8 = 48$   |
| (7) $7 \times 5 = 35$   | (37) $7 \times 8 = 56$    | (67) $10 \times 4 = 40$ | (97) $3 \times 5 = 15$   |
| (8) $3 \times 2 = 6$    | (38) $4 \times 6 = 24$    | (68) $4 \times 2 = 8$   | (98) $6 \times 6 = 36$   |
| (9) $5 \times 2 = 10$   | (39) $5 \times 8 = 40$    | (69) $0 \times 1 = 0$   | (99) $7 \times 4 = 28$   |
| (10) $7 \times 4 = 28$  | (40) $1 \times 3 = 3$     | (70) $3 \times 2 = 6$   | (100) $3 \times 8 = 24$  |
| (11) $3 \times 4 = 12$  | (41) $6 \times 1 = 6$     | (71) $4 \times 5 = 20$  | (101) $7 \times 2 = 14$  |
| (12) $5 \times 1 = 5$   | (42) $10 \times 7 = 70$   | (72) $0 \times 1 = 0$   | (102) $9 \times 3 = 27$  |
| (13) $7 \times 4 = 28$  | (43) $8 \times 7 = 56$    | (73) $7 \times 10 = 70$ | (103) $4 \times 5 = 20$  |
| (14) $6 \times 3 = 18$  | (44) $2 \times 6 = 12$    | (74) $10 \times 5 = 50$ | (104) $7 \times 0 = 0$   |
| (15) $10 \times 3 = 30$ | (45) $10 \times 7 = 70$   | (75) $8 \times 10 = 80$ | (105) $0 \times 6 = 0$   |
| (16) $6 \times 2 = 12$  | (46) $8 \times 4 = 32$    | (76) $3 \times 4 = 12$  | (106) $9 \times 9 = 81$  |
| (17) $6 \times 6 = 36$  | (47) $1 \times 7 = 7$     | (77) $5 \times 8 = 40$  | (107) $2 \times 2 = 4$   |
| (18) $0 \times 1 = 0$   | (48) $5 \times 2 = 10$    | (78) $6 \times 7 = 42$  | (108) $7 \times 2 = 14$  |
| (19) $7 \times 8 = 56$  | (49) $8 \times 9 = 72$    | (79) $0 \times 0 = 0$   | (109) $6 \times 4 = 24$  |
| (20) $9 \times 4 = 36$  | (50) $3 \times 8 = 24$    | (80) $1 \times 3 = 3$   | (110) $9 \times 2 = 18$  |
| (21) $2 \times 4 = 8$   | (51) $7 \times 2 = 14$    | (81) $7 \times 1 = 7$   | (111) $2 \times 9 = 18$  |
| (22) $1 \times 9 = 9$   | (52) $7 \times 7 = 49$    | (82) $3 \times 9 = 27$  | (112) $3 \times 1 = 3$   |
| (23) $2 \times 4 = 8$   | (53) $3 \times 0 = 0$     | (83) $8 \times 10 = 80$ | (113) $1 \times 9 = 9$   |
| (24) $2 \times 4 = 8$   | (54) $10 \times 1 = 10$   | (84) $6 \times 9 = 54$  | (114) $9 \times 7 = 63$  |
| (25) $9 \times 8 = 72$  | (55) $3 \times 2 = 6$     | (85) $2 \times 4 = 8$   | (115) $4 \times 7 = 28$  |
| (26) $10 \times 1 = 10$ | (56) $10 \times 10 = 100$ | (86) $0 \times 4 = 0$   | (116) $1 \times 6 = 6$   |
| (27) $5 \times 1 = 5$   | (57) $7 \times 7 = 49$    | (87) $3 \times 8 = 24$  | (117) $10 \times 7 = 70$ |
| (28) $2 \times 8 = 16$  | (58) $1 \times 5 = 5$     | (88) $8 \times 4 = 32$  | (118) $5 \times 2 = 10$  |
| (29) $1 \times 8 = 8$   | (59) $8 \times 6 = 48$    | (89) $7 \times 8 = 56$  | (119) $1 \times 8 = 8$   |
| (30) $7 \times 6 = 42$  | (60) $5 \times 6 = 30$    | (90) $4 \times 1 = 4$   | (120) $6 \times 7 = 42$  |

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|-------------------------|-------------------------|-------------------------|--------------------------|
| (1) $2 \times 7 = 14$   | (31) $3 \times 2 = 6$   | (61) $5 \times 5 = 25$  | (91) $6 \times 6 = 36$   |
| (2) $3 \times 1 = 3$    | (32) $3 \times 6 = 18$  | (62) $3 \times 4 = 12$  | (92) $8 \times 7 = 56$   |
| (3) $9 \times 10 = 90$  | (33) $1 \times 7 = 7$   | (63) $6 \times 5 = 30$  | (93) $10 \times 7 = 70$  |
| (4) $1 \times 3 = 3$    | (34) $9 \times 6 = 54$  | (64) $8 \times 10 = 80$ | (94) $3 \times 6 = 18$   |
| (5) $1 \times 6 = 6$    | (35) $7 \times 6 = 42$  | (65) $6 \times 1 = 6$   | (95) $1 \times 5 = 5$    |
| (6) $1 \times 10 = 10$  | (36) $6 \times 0 = 0$   | (66) $0 \times 10 = 0$  | (96) $6 \times 6 = 36$   |
| (7) $5 \times 2 = 10$   | (37) $3 \times 2 = 6$   | (67) $1 \times 8 = 8$   | (97) $1 \times 2 = 2$    |
| (8) $1 \times 10 = 10$  | (38) $3 \times 1 = 3$   | (68) $8 \times 4 = 32$  | (98) $3 \times 5 = 15$   |
| (9) $8 \times 4 = 32$   | (39) $4 \times 1 = 4$   | (69) $1 \times 8 = 8$   | (99) $8 \times 3 = 24$   |
| (10) $9 \times 6 = 54$  | (40) $7 \times 7 = 49$  | (70) $6 \times 3 = 18$  | (100) $10 \times 3 = 30$ |
| (11) $10 \times 3 = 30$ | (41) $5 \times 10 = 50$ | (71) $2 \times 6 = 12$  | (101) $1 \times 7 = 7$   |
| (12) $9 \times 7 = 63$  | (42) $4 \times 5 = 20$  | (72) $1 \times 10 = 10$ | (102) $8 \times 7 = 56$  |
| (13) $6 \times 8 = 48$  | (43) $3 \times 7 = 21$  | (73) $8 \times 9 = 72$  | (103) $4 \times 9 = 36$  |
| (14) $7 \times 2 = 14$  | (44) $9 \times 10 = 90$ | (74) $8 \times 9 = 72$  | (104) $10 \times 2 = 20$ |
| (15) $0 \times 8 = 0$   | (45) $3 \times 8 = 24$  | (75) $9 \times 2 = 18$  | (105) $4 \times 1 = 4$   |
| (16) $5 \times 9 = 45$  | (46) $1 \times 5 = 5$   | (76) $7 \times 0 = 0$   | (106) $7 \times 3 = 21$  |
| (17) $10 \times 2 = 20$ | (47) $8 \times 4 = 32$  | (77) $0 \times 9 = 0$   | (107) $10 \times 5 = 50$ |
| (18) $5 \times 8 = 40$  | (48) $10 \times 7 = 70$ | (78) $2 \times 1 = 2$   | (108) $3 \times 9 = 27$  |
| (19) $1 \times 4 = 4$   | (49) $10 \times 8 = 80$ | (79) $1 \times 1 = 1$   | (109) $6 \times 7 = 42$  |
| (20) $5 \times 6 = 30$  | (50) $1 \times 10 = 10$ | (80) $3 \times 8 = 24$  | (110) $9 \times 5 = 45$  |
| (21) $1 \times 7 = 7$   | (51) $4 \times 2 = 8$   | (81) $9 \times 1 = 9$   | (111) $4 \times 0 = 0$   |
| (22) $3 \times 8 = 24$  | (52) $4 \times 9 = 36$  | (82) $2 \times 3 = 6$   | (112) $0 \times 10 = 0$  |
| (23) $6 \times 10 = 60$ | (53) $0 \times 7 = 0$   | (83) $8 \times 9 = 72$  | (113) $7 \times 8 = 56$  |
| (24) $9 \times 2 = 18$  | (54) $2 \times 6 = 12$  | (84) $3 \times 3 = 9$   | (114) $9 \times 3 = 27$  |
| (25) $6 \times 0 = 0$   | (55) $1 \times 0 = 0$   | (85) $9 \times 9 = 81$  | (115) $3 \times 2 = 6$   |
| (26) $10 \times 6 = 60$ | (56) $10 \times 2 = 20$ | (86) $3 \times 7 = 21$  | (116) $2 \times 8 = 16$  |
| (27) $5 \times 2 = 10$  | (57) $10 \times 5 = 50$ | (87) $6 \times 2 = 12$  | (117) $4 \times 1 = 4$   |
| (28) $9 \times 4 = 36$  | (58) $7 \times 3 = 21$  | (88) $3 \times 4 = 12$  | (118) $0 \times 2 = 0$   |
| (29) $2 \times 10 = 20$ | (59) $5 \times 2 = 10$  | (89) $7 \times 4 = 28$  | (119) $1 \times 9 = 9$   |
| (30) $5 \times 1 = 5$   | (60) $4 \times 3 = 12$  | (90) $9 \times 6 = 54$  | (120) $2 \times 3 = 6$   |

- |                         |                         |                         |                          |
|-------------------------|-------------------------|-------------------------|--------------------------|
| (1) $3 \times 3 = 9$    | (31) $6 \times 2 = 12$  | (61) $8 \times 5 = 40$  | (91) $7 \times 8 = 56$   |
| (2) $9 \times 4 = 36$   | (32) $10 \times 9 = 90$ | (62) $8 \times 10 = 80$ | (92) $4 \times 3 = 12$   |
| (3) $7 \times 1 = 7$    | (33) $4 \times 8 = 32$  | (63) $5 \times 10 = 50$ | (93) $4 \times 2 = 8$    |
| (4) $2 \times 1 = 2$    | (34) $6 \times 6 = 36$  | (64) $0 \times 7 = 0$   | (94) $9 \times 6 = 54$   |
| (5) $3 \times 3 = 9$    | (35) $6 \times 6 = 36$  | (65) $10 \times 8 = 80$ | (95) $3 \times 2 = 6$    |
| (6) $2 \times 2 = 4$    | (36) $4 \times 0 = 0$   | (66) $5 \times 7 = 35$  | (96) $4 \times 3 = 12$   |
| (7) $1 \times 9 = 9$    | (37) $4 \times 3 = 12$  | (67) $5 \times 7 = 35$  | (97) $3 \times 3 = 9$    |
| (8) $6 \times 8 = 48$   | (38) $7 \times 2 = 14$  | (68) $5 \times 7 = 35$  | (98) $6 \times 9 = 54$   |
| (9) $9 \times 6 = 54$   | (39) $9 \times 1 = 9$   | (69) $7 \times 7 = 49$  | (99) $0 \times 3 = 0$    |
| (10) $7 \times 2 = 14$  | (40) $6 \times 6 = 36$  | (70) $3 \times 4 = 12$  | (100) $9 \times 1 = 9$   |
| (11) $4 \times 9 = 36$  | (41) $8 \times 9 = 72$  | (71) $3 \times 3 = 9$   | (101) $3 \times 0 = 0$   |
| (12) $9 \times 9 = 81$  | (42) $6 \times 10 = 60$ | (72) $9 \times 5 = 45$  | (102) $8 \times 9 = 72$  |
| (13) $6 \times 6 = 36$  | (43) $7 \times 2 = 14$  | (73) $5 \times 4 = 20$  | (103) $1 \times 8 = 8$   |
| (14) $1 \times 0 = 0$   | (44) $8 \times 1 = 8$   | (74) $8 \times 0 = 0$   | (104) $9 \times 3 = 27$  |
| (15) $6 \times 6 = 36$  | (45) $2 \times 7 = 14$  | (75) $1 \times 9 = 9$   | (105) $0 \times 7 = 0$   |
| (16) $7 \times 3 = 21$  | (46) $2 \times 10 = 20$ | (76) $4 \times 3 = 12$  | (106) $3 \times 2 = 6$   |
| (17) $4 \times 3 = 12$  | (47) $2 \times 9 = 18$  | (77) $4 \times 1 = 4$   | (107) $10 \times 8 = 80$ |
| (18) $0 \times 8 = 0$   | (48) $8 \times 8 = 64$  | (78) $3 \times 9 = 27$  | (108) $7 \times 3 = 21$  |
| (19) $6 \times 3 = 18$  | (49) $7 \times 3 = 21$  | (79) $7 \times 7 = 49$  | (109) $0 \times 5 = 0$   |
| (20) $5 \times 2 = 10$  | (50) $2 \times 5 = 10$  | (80) $10 \times 8 = 80$ | (110) $7 \times 7 = 49$  |
| (21) $5 \times 6 = 30$  | (51) $8 \times 0 = 0$   | (81) $9 \times 8 = 72$  | (111) $5 \times 5 = 25$  |
| (22) $7 \times 7 = 49$  | (52) $7 \times 9 = 63$  | (82) $7 \times 1 = 7$   | (112) $9 \times 1 = 9$   |
| (23) $9 \times 10 = 90$ | (53) $6 \times 4 = 24$  | (83) $3 \times 4 = 12$  | (113) $8 \times 1 = 8$   |
| (24) $1 \times 3 = 3$   | (54) $9 \times 10 = 90$ | (84) $2 \times 5 = 10$  | (114) $8 \times 7 = 56$  |
| (25) $7 \times 8 = 56$  | (55) $3 \times 8 = 24$  | (85) $1 \times 4 = 4$   | (115) $7 \times 10 = 70$ |
| (26) $8 \times 5 = 40$  | (56) $5 \times 2 = 10$  | (86) $2 \times 4 = 8$   | (116) $6 \times 7 = 42$  |
| (27) $7 \times 3 = 21$  | (57) $6 \times 3 = 18$  | (87) $4 \times 3 = 12$  | (117) $3 \times 8 = 24$  |
| (28) $5 \times 10 = 50$ | (58) $3 \times 4 = 12$  | (88) $1 \times 8 = 8$   | (118) $2 \times 3 = 6$   |
| (29) $7 \times 5 = 35$  | (59) $5 \times 5 = 25$  | (89) $10 \times 6 = 60$ | (119) $4 \times 2 = 8$   |
| (30) $1 \times 10 = 10$ | (60) $10 \times 7 = 70$ | (90) $3 \times 1 = 3$   | (120) $3 \times 5 = 15$  |



#### 1.1.4 除法

自然数の商と剰余を求める問題。

(1)  $59 \div 3$

(19)  $64 \div 7$

(37)  $68 \div 6$

(55)  $33 \div 1$

(2)  $74 \div 4$

(20)  $70 \div 1$

(38)  $81 \div 2$

(56)  $70 \div 10$

(3)  $62 \div 4$

(21)  $86 \div 9$

(39)  $27 \div 7$

(57)  $32 \div 8$

(4)  $84 \div 5$

(22)  $62 \div 9$

(40)  $37 \div 10$

(58)  $33 \div 2$

(5)  $76 \div 6$

(23)  $1 \div 10$

(41)  $62 \div 8$

(59)  $12 \div 5$

(6)  $1 \div 2$

(24)  $86 \div 4$

(42)  $36 \div 5$

(60)  $19 \div 8$

(7)  $48 \div 9$

(25)  $73 \div 7$

(43)  $22 \div 7$

(61)  $37 \div 1$

(8)  $53 \div 6$

(26)  $59 \div 6$

(44)  $22 \div 5$

(62)  $75 \div 10$

(9)  $85 \div 10$

(27)  $45 \div 1$

(45)  $47 \div 3$

(63)  $34 \div 4$

(10)  $21 \div 3$

(28)  $24 \div 8$

(46)  $89 \div 2$

(64)  $18 \div 5$

(11)  $71 \div 9$

(29)  $7 \div 3$

(47)  $87 \div 2$

(65)  $60 \div 3$

(12)  $24 \div 1$

(30)  $48 \div 9$

(48)  $14 \div 4$

(66)  $22 \div 2$

(13)  $22 \div 1$

(31)  $39 \div 9$

(49)  $2 \div 8$

(67)  $40 \div 4$

(14)  $22 \div 2$

(32)  $79 \div 10$

(50)  $1 \div 2$

(68)  $91 \div 1$

(15)  $18 \div 2$

(33)  $4 \div 6$

(51)  $49 \div 5$

(69)  $89 \div 10$

(16)  $70 \div 7$

(34)  $56 \div 2$

(52)  $34 \div 9$

(70)  $1 \div 3$

(17)  $9 \div 5$

(35)  $50 \div 2$

(53)  $70 \div 2$

(71)  $86 \div 6$

(18)  $38 \div 1$

(36)  $58 \div 1$

(54)  $76 \div 2$

(72)  $46 \div 9$

(1)  $50 \div 5$

(19)  $25 \div 1$

(37)  $100 \div 6$

(55)  $89 \div 8$

(2)  $100 \div 8$

(20)  $87 \div 10$

(38)  $2 \div 6$

(56)  $66 \div 4$

(3)  $65 \div 8$

(21)  $46 \div 8$

(39)  $12 \div 7$

(57)  $55 \div 4$

(4)  $25 \div 9$

(22)  $72 \div 6$

(40)  $32 \div 10$

(58)  $64 \div 1$

(5)  $88 \div 4$

(23)  $42 \div 6$

(41)  $76 \div 9$

(59)  $38 \div 3$

(6)  $23 \div 4$

(24)  $47 \div 8$

(42)  $44 \div 6$

(60)  $25 \div 5$

(7)  $18 \div 9$

(25)  $38 \div 2$

(43)  $52 \div 10$

(61)  $77 \div 1$

(8)  $12 \div 6$

(26)  $54 \div 6$

(44)  $24 \div 6$

(62)  $21 \div 3$

(9)  $60 \div 4$

(27)  $55 \div 9$

(45)  $47 \div 8$

(63)  $67 \div 4$

(10)  $58 \div 9$

(28)  $57 \div 2$

(46)  $96 \div 10$

(64)  $89 \div 10$

(11)  $31 \div 6$

(29)  $64 \div 1$

(47)  $76 \div 8$

(65)  $81 \div 7$

(12)  $53 \div 1$

(30)  $24 \div 4$

(48)  $10 \div 10$

(66)  $30 \div 3$

(13)  $79 \div 6$

(31)  $98 \div 1$

(49)  $38 \div 3$

(67)  $26 \div 5$

(14)  $57 \div 1$

(32)  $9 \div 4$

(50)  $75 \div 4$

(68)  $40 \div 3$

(15)  $87 \div 4$

(33)  $58 \div 3$

(51)  $67 \div 4$

(69)  $83 \div 7$

(16)  $35 \div 7$

(34)  $35 \div 4$

(52)  $4 \div 3$

(70)  $20 \div 10$

(17)  $31 \div 1$

(35)  $38 \div 6$

(53)  $22 \div 8$

(71)  $10 \div 5$

(18)  $86 \div 3$

(36)  $33 \div 8$

(54)  $8 \div 2$

(72)  $47 \div 1$

- |                   |                   |                   |                   |
|-------------------|-------------------|-------------------|-------------------|
| (1) $86 \div 5$   | (19) $71 \div 4$  | (37) $8 \div 2$   | (55) $78 \div 9$  |
| (2) $30 \div 5$   | (20) $84 \div 7$  | (38) $71 \div 4$  | (56) $29 \div 6$  |
| (3) $99 \div 1$   | (21) $79 \div 10$ | (39) $78 \div 8$  | (57) $46 \div 4$  |
| (4) $1 \div 4$    | (22) $46 \div 8$  | (40) $74 \div 9$  | (58) $34 \div 9$  |
| (5) $52 \div 8$   | (23) $52 \div 4$  | (41) $66 \div 10$ | (59) $4 \div 7$   |
| (6) $4 \div 9$    | (24) $43 \div 5$  | (42) $66 \div 6$  | (60) $9 \div 5$   |
| (7) $8 \div 6$    | (25) $54 \div 1$  | (43) $75 \div 8$  | (61) $38 \div 3$  |
| (8) $100 \div 1$  | (26) $50 \div 5$  | (44) $84 \div 1$  | (62) $50 \div 7$  |
| (9) $42 \div 1$   | (27) $7 \div 7$   | (45) $54 \div 1$  | (63) $16 \div 1$  |
| (10) $56 \div 8$  | (28) $86 \div 9$  | (46) $79 \div 4$  | (64) $12 \div 9$  |
| (11) $9 \div 3$   | (29) $75 \div 3$  | (47) $55 \div 4$  | (65) $66 \div 6$  |
| (12) $61 \div 4$  | (30) $26 \div 1$  | (48) $99 \div 6$  | (66) $88 \div 6$  |
| (13) $62 \div 4$  | (31) $50 \div 10$ | (49) $99 \div 10$ | (67) $9 \div 1$   |
| (14) $71 \div 3$  | (32) $58 \div 9$  | (50) $83 \div 3$  | (68) $40 \div 8$  |
| (15) $44 \div 4$  | (33) $18 \div 7$  | (51) $63 \div 5$  | (69) $93 \div 1$  |
| (16) $97 \div 4$  | (34) $19 \div 1$  | (52) $62 \div 8$  | (70) $41 \div 5$  |
| (17) $41 \div 10$ | (35) $92 \div 6$  | (53) $42 \div 2$  | (71) $20 \div 10$ |
| (18) $76 \div 6$  | (36) $46 \div 5$  | (54) $26 \div 4$  | (72) $1 \div 1$   |

(1)

$33 \div 7$

(19)

$98 \div 8$

(37)

$41 \div 6$

(55)

$12 \div 4$

(2)

$34 \div 6$

(20)

$88 \div 2$

(38)

$83 \div 10$

(56)

$85 \div 10$

(3)

$63 \div 8$

(21)

$90 \div 9$

(39)

$58 \div 4$

(57)

$29 \div 3$

(4)

$94 \div 3$

(22)

$51 \div 9$

(40)

$51 \div 10$

(58)

$87 \div 9$

(5)

$21 \div 5$

(23)

$65 \div 1$

(41)

$28 \div 1$

(59)

$2 \div 5$

(6)

$30 \div 8$

(24)

$61 \div 2$

(42)

$11 \div 5$

(60)

$50 \div 1$

(7)

$21 \div 8$

(25)

$27 \div 8$

(43)

$30 \div 6$

(61)

$47 \div 10$

(8)

$0 \div 8$

(26)

$35 \div 4$

(44)

$17 \div 1$

(62)

$21 \div 2$

(9)

$88 \div 6$

(27)

$90 \div 6$

(45)

$93 \div 1$

(63)

$83 \div 10$

(10)

$5 \div 1$

(28)

$52 \div 3$

(46)

$38 \div 5$

(64)

$34 \div 9$

(11)

$38 \div 6$

(29)

$71 \div 10$

(47)

$27 \div 9$

(65)

$12 \div 7$

(12)

$50 \div 4$

(30)

$34 \div 6$

(48)

$57 \div 8$

(66)

$16 \div 5$

(13)

$2 \div 10$

(31)

$20 \div 7$

(49)

$83 \div 3$

(67)

$99 \div 5$

(14)

$85 \div 1$

(32)

$1 \div 5$

(50)

$69 \div 7$

(68)

$55 \div 1$

(15)

$83 \div 5$

(33)

$84 \div 4$

(51)

$59 \div 2$

(69)

$18 \div 6$

(16)

$31 \div 8$

(34)

$30 \div 6$

(52)

$44 \div 2$

(70)

$26 \div 5$

(17)

$42 \div 10$

(35)

$4 \div 7$

(53)

$6 \div 8$

(71)

$50 \div 9$

(18)

$78 \div 4$

(36)

$15 \div 4$

(54)

$14 \div 7$

(72)

$0 \div 9$

|                  |                   |                   |                  |
|------------------|-------------------|-------------------|------------------|
| (1) $6 \div 1$   | (19) $48 \div 10$ | (37) $5 \div 1$   | (55) $23 \div 9$ |
| (2) $19 \div 4$  | (20) $84 \div 7$  | (38) $74 \div 10$ | (56) $87 \div 2$ |
| (3) $91 \div 6$  | (21) $1 \div 4$   | (39) $46 \div 3$  | (57) $83 \div 2$ |
| (4) $90 \div 3$  | (22) $52 \div 2$  | (40) $96 \div 4$  | (58) $58 \div 2$ |
| (5) $79 \div 8$  | (23) $46 \div 5$  | (41) $54 \div 8$  | (59) $75 \div 8$ |
| (6) $52 \div 9$  | (24) $63 \div 10$ | (42) $33 \div 9$  | (60) $28 \div 1$ |
| (7) $51 \div 7$  | (25) $67 \div 6$  | (43) $77 \div 4$  | (61) $36 \div 5$ |
| (8) $100 \div 2$ | (26) $27 \div 10$ | (44) $37 \div 6$  | (62) $48 \div 2$ |
| (9) $79 \div 3$  | (27) $62 \div 9$  | (45) $37 \div 9$  | (63) $36 \div 3$ |
| (10) $7 \div 9$  | (28) $39 \div 1$  | (46) $64 \div 7$  | (64) $66 \div 3$ |
| (11) $30 \div 5$ | (29) $87 \div 9$  | (47) $77 \div 4$  | (65) $21 \div 6$ |
| (12) $97 \div 8$ | (30) $34 \div 1$  | (48) $33 \div 2$  | (66) $89 \div 3$ |
| (13) $55 \div 9$ | (31) $17 \div 2$  | (49) $70 \div 9$  | (67) $47 \div 7$ |
| (14) $7 \div 10$ | (32) $42 \div 2$  | (50) $84 \div 5$  | (68) $38 \div 8$ |
| (15) $63 \div 8$ | (33) $60 \div 9$  | (51) $32 \div 2$  | (69) $7 \div 10$ |
| (16) $3 \div 10$ | (34) $9 \div 1$   | (52) $12 \div 1$  | (70) $34 \div 2$ |
| (17) $6 \div 8$  | (35) $41 \div 10$ | (53) $36 \div 10$ | (71) $75 \div 3$ |
| (18) $66 \div 5$ | (36) $89 \div 2$  | (54) $79 \div 2$  | (72) $78 \div 4$ |

(1)  $51 \div 10$

(19)  $78 \div 1$

(37)  $52 \div 7$

(55)  $27 \div 8$

(2)  $21 \div 5$

(20)  $30 \div 9$

(38)  $34 \div 6$

(56)  $92 \div 3$

(3)  $97 \div 5$

(21)  $36 \div 9$

(39)  $52 \div 2$

(57)  $38 \div 9$

(4)  $54 \div 10$

(22)  $93 \div 7$

(40)  $12 \div 10$

(58)  $12 \div 9$

(5)  $32 \div 4$

(23)  $16 \div 9$

(41)  $91 \div 10$

(59)  $80 \div 6$

(6)  $92 \div 1$

(24)  $99 \div 9$

(42)  $60 \div 8$

(60)  $7 \div 9$

(7)  $59 \div 9$

(25)  $75 \div 7$

(43)  $71 \div 6$

(61)  $96 \div 10$

(8)  $12 \div 8$

(26)  $3 \div 7$

(44)  $5 \div 5$

(62)  $0 \div 9$

(9)  $84 \div 6$

(27)  $88 \div 10$

(45)  $17 \div 8$

(63)  $6 \div 4$

(10)  $72 \div 5$

(28)  $7 \div 3$

(46)  $9 \div 5$

(64)  $88 \div 10$

(11)  $52 \div 5$

(29)  $27 \div 5$

(47)  $41 \div 4$

(65)  $85 \div 8$

(12)  $99 \div 5$

(30)  $35 \div 3$

(48)  $49 \div 7$

(66)  $31 \div 3$

(13)  $63 \div 5$

(31)  $33 \div 9$

(49)  $73 \div 3$

(67)  $48 \div 8$

(14)  $18 \div 10$

(32)  $53 \div 9$

(50)  $26 \div 9$

(68)  $62 \div 7$

(15)  $50 \div 1$

(33)  $64 \div 9$

(51)  $74 \div 3$

(69)  $100 \div 3$

(16)  $57 \div 1$

(34)  $69 \div 6$

(52)  $4 \div 10$

(70)  $56 \div 7$

(17)  $96 \div 3$

(35)  $39 \div 5$

(53)  $38 \div 5$

(71)  $69 \div 5$

(18)  $36 \div 8$

(36)  $16 \div 3$

(54)  $58 \div 4$

(72)  $10 \div 3$

|                   |                   |                  |                  |
|-------------------|-------------------|------------------|------------------|
| (1) $80 \div 10$  | (19) $20 \div 8$  | (37) $55 \div 5$ | (55) $75 \div 6$ |
| (2) $99 \div 7$   | (20) $58 \div 7$  | (38) $81 \div 3$ | (56) $67 \div 1$ |
| (3) $12 \div 1$   | (21) $13 \div 10$ | (39) $5 \div 8$  | (57) $15 \div 4$ |
| (4) $17 \div 3$   | (22) $14 \div 6$  | (40) $89 \div 5$ | (58) $73 \div 4$ |
| (5) $38 \div 9$   | (23) $26 \div 10$ | (41) $72 \div 3$ | (59) $59 \div 7$ |
| (6) $22 \div 4$   | (24) $55 \div 8$  | (42) $18 \div 8$ | (60) $78 \div 5$ |
| (7) $4 \div 6$    | (25) $13 \div 8$  | (43) $82 \div 8$ | (61) $31 \div 7$ |
| (8) $75 \div 1$   | (26) $78 \div 10$ | (44) $71 \div 3$ | (62) $65 \div 8$ |
| (9) $38 \div 10$  | (27) $97 \div 1$  | (45) $82 \div 4$ | (63) $42 \div 5$ |
| (10) $93 \div 9$  | (28) $27 \div 10$ | (46) $48 \div 4$ | (64) $50 \div 3$ |
| (11) $73 \div 3$  | (29) $0 \div 2$   | (47) $74 \div 1$ | (65) $5 \div 3$  |
| (12) $74 \div 6$  | (30) $50 \div 7$  | (48) $15 \div 8$ | (66) $77 \div 6$ |
| (13) $21 \div 10$ | (31) $83 \div 8$  | (49) $29 \div 7$ | (67) $28 \div 1$ |
| (14) $27 \div 10$ | (32) $98 \div 8$  | (50) $96 \div 5$ | (68) $24 \div 7$ |
| (15) $55 \div 2$  | (33) $72 \div 7$  | (51) $64 \div 1$ | (69) $22 \div 1$ |
| (16) $21 \div 8$  | (34) $40 \div 7$  | (52) $74 \div 9$ | (70) $12 \div 3$ |
| (17) $8 \div 10$  | (35) $83 \div 1$  | (53) $54 \div 8$ | (71) $89 \div 3$ |
| (18) $29 \div 1$  | (36) $57 \div 3$  | (54) $9 \div 3$  | (72) $13 \div 9$ |



- |                   |                   |                  |                   |
|-------------------|-------------------|------------------|-------------------|
| (1) $38 \div 1$   | (19) $91 \div 6$  | (37) $65 \div 7$ | (55) $67 \div 9$  |
| (2) $59 \div 5$   | (20) $51 \div 4$  | (38) $25 \div 7$ | (56) $95 \div 7$  |
| (3) $33 \div 5$   | (21) $0 \div 5$   | (39) $68 \div 3$ | (57) $66 \div 6$  |
| (4) $89 \div 8$   | (22) $49 \div 10$ | (40) $97 \div 2$ | (58) $62 \div 10$ |
| (5) $70 \div 7$   | (23) $92 \div 6$  | (41) $41 \div 9$ | (59) $84 \div 8$  |
| (6) $39 \div 3$   | (24) $1 \div 6$   | (42) $41 \div 9$ | (60) $50 \div 2$  |
| (7) $22 \div 1$   | (25) $12 \div 2$  | (43) $97 \div 8$ | (61) $81 \div 8$  |
| (8) $76 \div 7$   | (26) $37 \div 2$  | (44) $39 \div 3$ | (62) $91 \div 4$  |
| (9) $59 \div 6$   | (27) $77 \div 8$  | (45) $21 \div 8$ | (63) $61 \div 7$  |
| (10) $79 \div 1$  | (28) $29 \div 5$  | (46) $68 \div 3$ | (64) $14 \div 8$  |
| (11) $42 \div 1$  | (29) $75 \div 10$ | (47) $77 \div 1$ | (65) $67 \div 3$  |
| (12) $33 \div 10$ | (30) $11 \div 10$ | (48) $21 \div 4$ | (66) $98 \div 4$  |
| (13) $58 \div 6$  | (31) $68 \div 1$  | (49) $80 \div 1$ | (67) $52 \div 9$  |
| (14) $20 \div 1$  | (32) $19 \div 3$  | (50) $35 \div 4$ | (68) $70 \div 6$  |
| (15) $81 \div 8$  | (33) $8 \div 9$   | (51) $77 \div 2$ | (69) $86 \div 6$  |
| (16) $61 \div 2$  | (34) $84 \div 7$  | (52) $97 \div 4$ | (70) $23 \div 8$  |
| (17) $9 \div 10$  | (35) $37 \div 4$  | (53) $78 \div 2$ | (71) $48 \div 8$  |
| (18) $43 \div 5$  | (36) $78 \div 10$ | (54) $25 \div 4$ | (72) $63 \div 6$  |

(1)  $42 \div 8$

(19)  $32 \div 2$

(37)  $68 \div 4$

(55)  $92 \div 1$

(2)  $1 \div 5$

(20)  $97 \div 4$

(38)  $57 \div 2$

(56)  $50 \div 9$

(3)  $44 \div 3$

(21)  $2 \div 10$

(39)  $52 \div 8$

(57)  $6 \div 1$

(4)  $12 \div 2$

(22)  $4 \div 8$

(40)  $48 \div 2$

(58)  $86 \div 1$

(5)  $58 \div 5$

(23)  $96 \div 4$

(41)  $85 \div 3$

(59)  $46 \div 1$

(6)  $7 \div 10$

(24)  $9 \div 3$

(42)  $69 \div 7$

(60)  $87 \div 4$

(7)  $75 \div 2$

(25)  $76 \div 3$

(43)  $70 \div 5$

(61)  $79 \div 10$

(8)  $5 \div 1$

(26)  $91 \div 3$

(44)  $45 \div 3$

(62)  $100 \div 5$

(9)  $87 \div 2$

(27)  $72 \div 8$

(45)  $100 \div 2$

(63)  $69 \div 7$

(10)  $30 \div 6$

(28)  $35 \div 2$

(46)  $85 \div 4$

(64)  $19 \div 4$

(11)  $22 \div 10$

(29)  $57 \div 1$

(47)  $7 \div 10$

(65)  $11 \div 5$

(12)  $48 \div 1$

(30)  $58 \div 9$

(48)  $4 \div 1$

(66)  $83 \div 6$

(13)  $77 \div 9$

(31)  $40 \div 4$

(49)  $11 \div 6$

(67)  $71 \div 5$

(14)  $55 \div 10$

(32)  $21 \div 6$

(50)  $21 \div 7$

(68)  $33 \div 1$

(15)  $7 \div 1$

(33)  $16 \div 10$

(51)  $79 \div 6$

(69)  $41 \div 3$

(16)  $74 \div 3$

(34)  $10 \div 4$

(52)  $82 \div 3$

(70)  $25 \div 5$

(17)  $65 \div 5$

(35)  $47 \div 5$

(53)  $53 \div 4$

(71)  $95 \div 8$

(18)  $82 \div 3$

(36)  $5 \div 5$

(54)  $39 \div 6$

(72)  $36 \div 1$

(1)  $41 \div 4$

(19)  $67 \div 4$

(37)  $18 \div 4$

(55)  $4 \div 10$

(2)  $25 \div 8$

(20)  $78 \div 6$

(38)  $4 \div 2$

(56)  $90 \div 9$

(3)  $99 \div 7$

(21)  $46 \div 6$

(39)  $64 \div 5$

(57)  $66 \div 6$

(4)  $86 \div 8$

(22)  $27 \div 2$

(40)  $10 \div 5$

(58)  $24 \div 5$

(5)  $36 \div 9$

(23)  $92 \div 3$

(41)  $47 \div 7$

(59)  $6 \div 3$

(6)  $12 \div 6$

(24)  $37 \div 5$

(42)  $52 \div 3$

(60)  $3 \div 8$

(7)  $95 \div 4$

(25)  $35 \div 8$

(43)  $11 \div 4$

(61)  $7 \div 5$

(8)  $20 \div 9$

(26)  $95 \div 6$

(44)  $65 \div 5$

(62)  $13 \div 10$

(9)  $10 \div 5$

(27)  $43 \div 2$

(45)  $95 \div 4$

(63)  $27 \div 3$

(10)  $67 \div 4$

(28)  $4 \div 7$

(46)  $42 \div 5$

(64)  $17 \div 4$

(11)  $8 \div 5$

(29)  $89 \div 7$

(47)  $67 \div 7$

(65)  $52 \div 1$

(12)  $66 \div 1$

(30)  $61 \div 4$

(48)  $12 \div 2$

(66)  $83 \div 2$

(13)  $65 \div 5$

(31)  $17 \div 1$

(49)  $29 \div 4$

(67)  $88 \div 1$

(14)  $9 \div 4$

(32)  $70 \div 7$

(50)  $25 \div 4$

(68)  $98 \div 4$

(15)  $58 \div 2$

(33)  $8 \div 3$

(51)  $39 \div 6$

(69)  $96 \div 8$

(16)  $2 \div 8$

(34)  $63 \div 10$

(52)  $82 \div 1$

(70)  $17 \div 8$

(17)  $48 \div 9$

(35)  $28 \div 2$

(53)  $79 \div 5$

(71)  $36 \div 3$

(18)  $77 \div 8$

(36)  $23 \div 1$

(54)  $46 \div 4$

(72)  $44 \div 8$

(1)  $51 \div 7$

(19)  $97 \div 10$

(37)  $37 \div 10$

(55)  $80 \div 2$

(2)  $98 \div 8$

(20)  $77 \div 3$

(38)  $26 \div 1$

(56)  $59 \div 8$

(3)  $20 \div 1$

(21)  $18 \div 10$

(39)  $44 \div 10$

(57)  $38 \div 5$

(4)  $12 \div 8$

(22)  $41 \div 3$

(40)  $84 \div 6$

(58)  $88 \div 6$

(5)  $86 \div 4$

(23)  $31 \div 2$

(41)  $51 \div 9$

(59)  $95 \div 7$

(6)  $99 \div 9$

(24)  $66 \div 6$

(42)  $56 \div 3$

(60)  $16 \div 2$

(7)  $3 \div 1$

(25)  $73 \div 4$

(43)  $45 \div 3$

(61)  $35 \div 8$

(8)  $28 \div 7$

(26)  $63 \div 5$

(44)  $82 \div 10$

(62)  $100 \div 8$

(9)  $97 \div 8$

(27)  $11 \div 5$

(45)  $58 \div 4$

(63)  $85 \div 4$

(10)  $28 \div 6$

(28)  $65 \div 4$

(46)  $40 \div 10$

(64)  $78 \div 5$

(11)  $40 \div 8$

(29)  $70 \div 6$

(47)  $41 \div 5$

(65)  $13 \div 3$

(12)  $26 \div 5$

(30)  $53 \div 6$

(48)  $98 \div 2$

(66)  $15 \div 2$

(13)  $52 \div 1$

(31)  $29 \div 4$

(49)  $65 \div 9$

(67)  $50 \div 5$

(14)  $7 \div 9$

(32)  $52 \div 5$

(50)  $11 \div 9$

(68)  $79 \div 10$

(15)  $2 \div 7$

(33)  $40 \div 9$

(51)  $64 \div 7$

(69)  $52 \div 5$

(16)  $65 \div 1$

(34)  $15 \div 5$

(52)  $82 \div 6$

(70)  $54 \div 9$

(17)  $40 \div 9$

(35)  $100 \div 8$

(53)  $87 \div 5$

(71)  $5 \div 8$

(18)  $2 \div 6$

(36)  $8 \div 10$

(54)  $51 \div 2$

(72)  $11 \div 3$

(1)  $9 \div 6$

(19)  $25 \div 5$

(37)  $39 \div 3$

(55)  $21 \div 6$

(2)  $64 \div 6$

(20)  $81 \div 3$

(38)  $7 \div 3$

(56)  $6 \div 7$

(3)  $22 \div 5$

(21)  $23 \div 8$

(39)  $79 \div 10$

(57)  $3 \div 3$

(4)  $35 \div 10$

(22)  $96 \div 10$

(40)  $14 \div 10$

(58)  $45 \div 8$

(5)  $85 \div 3$

(23)  $24 \div 10$

(41)  $82 \div 7$

(59)  $82 \div 2$

(6)  $99 \div 9$

(24)  $17 \div 3$

(42)  $52 \div 7$

(60)  $9 \div 6$

(7)  $17 \div 10$

(25)  $34 \div 10$

(43)  $19 \div 10$

(61)  $82 \div 10$

(8)  $76 \div 7$

(26)  $6 \div 8$

(44)  $68 \div 3$

(62)  $12 \div 8$

(9)  $97 \div 6$

(27)  $59 \div 5$

(45)  $90 \div 1$

(63)  $99 \div 1$

(10)  $1 \div 7$

(28)  $14 \div 4$

(46)  $84 \div 8$

(64)  $4 \div 8$

(11)  $6 \div 8$

(29)  $54 \div 1$

(47)  $81 \div 4$

(65)  $7 \div 2$

(12)  $77 \div 5$

(30)  $97 \div 6$

(48)  $5 \div 7$

(66)  $40 \div 8$

(13)  $27 \div 8$

(31)  $30 \div 7$

(49)  $37 \div 6$

(67)  $35 \div 3$

(14)  $65 \div 10$

(32)  $49 \div 7$

(50)  $83 \div 6$

(68)  $24 \div 10$

(15)  $92 \div 6$

(33)  $16 \div 8$

(51)  $6 \div 6$

(69)  $4 \div 4$

(16)  $85 \div 6$

(34)  $70 \div 4$

(52)  $40 \div 8$

(70)  $38 \div 8$

(17)  $64 \div 2$

(35)  $67 \div 9$

(53)  $40 \div 9$

(71)  $79 \div 1$

(18)  $86 \div 3$

(36)  $16 \div 8$

(54)  $92 \div 4$

(72)  $100 \div 8$

(1)  $2 \div 6$

(19)  $77 \div 2$

(37)  $15 \div 8$

(55)  $86 \div 4$

(2)  $38 \div 10$

(20)  $83 \div 2$

(38)  $48 \div 3$

(56)  $28 \div 9$

(3)  $92 \div 8$

(21)  $71 \div 9$

(39)  $67 \div 5$

(57)  $75 \div 5$

(4)  $46 \div 6$

(22)  $32 \div 2$

(40)  $75 \div 7$

(58)  $15 \div 5$

(5)  $90 \div 1$

(23)  $95 \div 6$

(41)  $27 \div 5$

(59)  $99 \div 8$

(6)  $5 \div 1$

(24)  $46 \div 4$

(42)  $37 \div 1$

(60)  $56 \div 4$

(7)  $56 \div 10$

(25)  $61 \div 9$

(43)  $2 \div 7$

(61)  $89 \div 6$

(8)  $44 \div 9$

(26)  $93 \div 10$

(44)  $70 \div 3$

(62)  $18 \div 5$

(9)  $2 \div 8$

(27)  $56 \div 10$

(45)  $4 \div 7$

(63)  $68 \div 10$

(10)  $78 \div 4$

(28)  $69 \div 2$

(46)  $19 \div 5$

(64)  $53 \div 10$

(11)  $66 \div 1$

(29)  $94 \div 5$

(47)  $26 \div 10$

(65)  $33 \div 5$

(12)  $29 \div 10$

(30)  $93 \div 7$

(48)  $22 \div 2$

(66)  $30 \div 1$

(13)  $48 \div 8$

(31)  $91 \div 9$

(49)  $40 \div 3$

(67)  $92 \div 8$

(14)  $16 \div 4$

(32)  $59 \div 9$

(50)  $69 \div 10$

(68)  $100 \div 7$

(15)  $75 \div 7$

(33)  $65 \div 1$

(51)  $6 \div 2$

(69)  $58 \div 8$

(16)  $46 \div 9$

(34)  $12 \div 7$

(52)  $74 \div 4$

(70)  $16 \div 4$

(17)  $58 \div 5$

(35)  $45 \div 3$

(53)  $59 \div 1$

(71)  $37 \div 4$

(18)  $33 \div 7$

(36)  $54 \div 4$

(54)  $73 \div 9$

(72)  $89 \div 8$

- |                  |                   |                   |                  |
|------------------|-------------------|-------------------|------------------|
| (1) $100 \div 4$ | (19) $79 \div 8$  | (37) $82 \div 9$  | (55) $21 \div 1$ |
| (2) $75 \div 10$ | (20) $53 \div 7$  | (38) $80 \div 3$  | (56) $71 \div 7$ |
| (3) $29 \div 5$  | (21) $1 \div 10$  | (39) $3 \div 3$   | (57) $66 \div 3$ |
| (4) $82 \div 6$  | (22) $27 \div 5$  | (40) $66 \div 7$  | (58) $94 \div 8$ |
| (5) $30 \div 10$ | (23) $16 \div 2$  | (41) $67 \div 9$  | (59) $68 \div 9$ |
| (6) $81 \div 3$  | (24) $55 \div 8$  | (42) $81 \div 8$  | (60) $51 \div 1$ |
| (7) $58 \div 4$  | (25) $25 \div 2$  | (43) $42 \div 3$  | (61) $74 \div 6$ |
| (8) $8 \div 8$   | (26) $52 \div 7$  | (44) $93 \div 5$  | (62) $23 \div 5$ |
| (9) $58 \div 1$  | (27) $27 \div 3$  | (45) $51 \div 6$  | (63) $79 \div 1$ |
| (10) $38 \div 2$ | (28) $29 \div 10$ | (46) $77 \div 5$  | (64) $66 \div 9$ |
| (11) $58 \div 6$ | (29) $1 \div 8$   | (47) $71 \div 5$  | (65) $28 \div 5$ |
| (12) $70 \div 1$ | (30) $23 \div 9$  | (48) $56 \div 8$  | (66) $7 \div 10$ |
| (13) $74 \div 9$ | (31) $41 \div 3$  | (49) $4 \div 10$  | (67) $82 \div 5$ |
| (14) $58 \div 9$ | (32) $69 \div 4$  | (50) $81 \div 8$  | (68) $11 \div 9$ |
| (15) $73 \div 1$ | (33) $8 \div 1$   | (51) $69 \div 10$ | (69) $14 \div 8$ |
| (16) $5 \div 10$ | (34) $42 \div 4$  | (52) $29 \div 9$  | (70) $95 \div 2$ |
| (17) $17 \div 5$ | (35) $42 \div 1$  | (53) $80 \div 8$  | (71) $49 \div 1$ |
| (18) $37 \div 7$ | (36) $93 \div 7$  | (54) $42 \div 9$  | (72) $79 \div 3$ |

- |                   |                   |                   |                   |
|-------------------|-------------------|-------------------|-------------------|
| (1) $74 \div 6$   | (19) $97 \div 3$  | (37) $74 \div 8$  | (55) $81 \div 9$  |
| (2) $3 \div 7$    | (20) $71 \div 10$ | (38) $41 \div 8$  | (56) $40 \div 5$  |
| (3) $63 \div 5$   | (21) $62 \div 3$  | (39) $66 \div 8$  | (57) $10 \div 5$  |
| (4) $21 \div 6$   | (22) $37 \div 3$  | (40) $47 \div 3$  | (58) $16 \div 3$  |
| (5) $99 \div 10$  | (23) $85 \div 4$  | (41) $13 \div 2$  | (59) $37 \div 1$  |
| (6) $26 \div 1$   | (24) $11 \div 4$  | (42) $2 \div 5$   | (60) $78 \div 6$  |
| (7) $95 \div 4$   | (25) $32 \div 5$  | (43) $68 \div 7$  | (61) $46 \div 1$  |
| (8) $15 \div 7$   | (26) $38 \div 1$  | (44) $15 \div 1$  | (62) $74 \div 7$  |
| (9) $20 \div 3$   | (27) $85 \div 2$  | (45) $98 \div 8$  | (63) $81 \div 7$  |
| (10) $45 \div 6$  | (28) $17 \div 8$  | (46) $82 \div 8$  | (64) $66 \div 4$  |
| (11) $20 \div 1$  | (29) $58 \div 6$  | (47) $62 \div 3$  | (65) $18 \div 2$  |
| (12) $77 \div 8$  | (30) $20 \div 4$  | (48) $39 \div 10$ | (66) $59 \div 10$ |
| (13) $89 \div 2$  | (31) $94 \div 4$  | (49) $13 \div 2$  | (67) $48 \div 10$ |
| (14) $77 \div 7$  | (32) $12 \div 10$ | (50) $9 \div 2$   | (68) $35 \div 4$  |
| (15) $22 \div 2$  | (33) $78 \div 10$ | (51) $10 \div 4$  | (69) $20 \div 3$  |
| (16) $76 \div 10$ | (34) $84 \div 6$  | (52) $75 \div 1$  | (70) $89 \div 9$  |
| (17) $3 \div 2$   | (35) $71 \div 2$  | (53) $35 \div 5$  | (71) $87 \div 7$  |
| (18) $30 \div 3$  | (36) $61 \div 2$  | (54) $15 \div 5$  | (72) $40 \div 4$  |



(1)  $47 \div 4$

(19)  $96 \div 4$

(37)  $84 \div 2$

(55)  $1 \div 7$

(2)  $30 \div 9$

(20)  $81 \div 5$

(38)  $77 \div 8$

(56)  $23 \div 5$

(3)  $80 \div 8$

(21)  $17 \div 10$

(39)  $87 \div 5$

(57)  $55 \div 7$

(4)  $38 \div 6$

(22)  $57 \div 10$

(40)  $74 \div 7$

(58)  $72 \div 8$

(5)  $11 \div 8$

(23)  $85 \div 5$

(41)  $52 \div 9$

(59)  $61 \div 6$

(6)  $51 \div 6$

(24)  $43 \div 10$

(42)  $69 \div 4$

(60)  $75 \div 10$

(7)  $88 \div 7$

(25)  $90 \div 10$

(43)  $1 \div 9$

(61)  $23 \div 1$

(8)  $48 \div 3$

(26)  $84 \div 5$

(44)  $38 \div 4$

(62)  $50 \div 1$

(9)  $24 \div 6$

(27)  $76 \div 4$

(45)  $28 \div 9$

(63)  $20 \div 7$

(10)  $85 \div 5$

(28)  $8 \div 7$

(46)  $96 \div 6$

(64)  $75 \div 7$

(11)  $19 \div 5$

(29)  $90 \div 4$

(47)  $97 \div 1$

(65)  $72 \div 5$

(12)  $40 \div 6$

(30)  $77 \div 9$

(48)  $1 \div 7$

(66)  $64 \div 9$

(13)  $100 \div 10$

(31)  $26 \div 5$

(49)  $31 \div 7$

(67)  $29 \div 5$

(14)  $7 \div 7$

(32)  $81 \div 10$

(50)  $99 \div 5$

(68)  $91 \div 3$

(15)  $10 \div 6$

(33)  $89 \div 9$

(51)  $13 \div 4$

(69)  $79 \div 9$

(16)  $68 \div 6$

(34)  $29 \div 4$

(52)  $26 \div 8$

(70)  $67 \div 9$

(17)  $68 \div 10$

(35)  $49 \div 10$

(53)  $10 \div 1$

(71)  $9 \div 4$

(18)  $76 \div 2$

(36)  $65 \div 4$

(54)  $95 \div 9$

(72)  $45 \div 9$

|                   |                   |                   |                   |
|-------------------|-------------------|-------------------|-------------------|
| (1) $92 \div 1$   | (19) $46 \div 2$  | (37) $9 \div 4$   | (55) $72 \div 9$  |
| (2) $19 \div 8$   | (20) $70 \div 10$ | (38) $84 \div 5$  | (56) $17 \div 3$  |
| (3) $25 \div 3$   | (21) $3 \div 10$  | (39) $14 \div 1$  | (57) $65 \div 10$ |
| (4) $41 \div 7$   | (22) $65 \div 4$  | (40) $5 \div 2$   | (58) $14 \div 6$  |
| (5) $96 \div 3$   | (23) $90 \div 1$  | (41) $23 \div 5$  | (59) $17 \div 3$  |
| (6) $79 \div 5$   | (24) $97 \div 8$  | (42) $18 \div 8$  | (60) $43 \div 4$  |
| (7) $64 \div 6$   | (25) $68 \div 3$  | (43) $99 \div 8$  | (61) $12 \div 7$  |
| (8) $75 \div 9$   | (26) $18 \div 2$  | (44) $77 \div 9$  | (62) $72 \div 2$  |
| (9) $54 \div 8$   | (27) $41 \div 7$  | (45) $90 \div 1$  | (63) $68 \div 5$  |
| (10) $45 \div 10$ | (28) $70 \div 9$  | (46) $81 \div 8$  | (64) $41 \div 8$  |
| (11) $68 \div 1$  | (29) $95 \div 8$  | (47) $100 \div 2$ | (65) $16 \div 7$  |
| (12) $37 \div 4$  | (30) $50 \div 10$ | (48) $38 \div 5$  | (66) $73 \div 7$  |
| (13) $34 \div 10$ | (31) $49 \div 6$  | (49) $30 \div 3$  | (67) $65 \div 9$  |
| (14) $98 \div 7$  | (32) $0 \div 9$   | (50) $93 \div 4$  | (68) $51 \div 4$  |
| (15) $59 \div 6$  | (33) $67 \div 3$  | (51) $42 \div 9$  | (69) $41 \div 8$  |
| (16) $56 \div 6$  | (34) $76 \div 3$  | (52) $44 \div 4$  | (70) $85 \div 5$  |
| (17) $47 \div 10$ | (35) $53 \div 10$ | (53) $13 \div 5$  | (71) $53 \div 5$  |
| (18) $99 \div 6$  | (36) $52 \div 2$  | (54) $73 \div 8$  | (72) $88 \div 2$  |

|                   |                   |                   |                   |
|-------------------|-------------------|-------------------|-------------------|
| (1) $26 \div 10$  | (19) $39 \div 5$  | (37) $19 \div 7$  | (55) $43 \div 8$  |
| (2) $43 \div 2$   | (20) $73 \div 7$  | (38) $17 \div 10$ | (56) $10 \div 7$  |
| (3) $61 \div 7$   | (21) $66 \div 1$  | (39) $80 \div 8$  | (57) $92 \div 6$  |
| (4) $81 \div 2$   | (22) $9 \div 6$   | (40) $67 \div 3$  | (58) $26 \div 6$  |
| (5) $21 \div 8$   | (23) $45 \div 4$  | (41) $39 \div 5$  | (59) $95 \div 6$  |
| (6) $69 \div 1$   | (24) $72 \div 1$  | (42) $18 \div 5$  | (60) $77 \div 7$  |
| (7) $11 \div 1$   | (25) $2 \div 6$   | (43) $46 \div 9$  | (61) $31 \div 7$  |
| (8) $60 \div 4$   | (26) $4 \div 1$   | (44) $59 \div 8$  | (62) $86 \div 3$  |
| (9) $1 \div 7$    | (27) $91 \div 2$  | (45) $68 \div 10$ | (63) $6 \div 5$   |
| (10) $69 \div 9$  | (28) $67 \div 4$  | (46) $78 \div 9$  | (64) $61 \div 4$  |
| (11) $66 \div 3$  | (29) $90 \div 6$  | (47) $82 \div 3$  | (65) $57 \div 9$  |
| (12) $1 \div 10$  | (30) $84 \div 5$  | (48) $20 \div 6$  | (66) $32 \div 5$  |
| (13) $87 \div 10$ | (31) $31 \div 5$  | (49) $73 \div 7$  | (67) $67 \div 4$  |
| (14) $67 \div 2$  | (32) $39 \div 3$  | (50) $81 \div 5$  | (68) $5 \div 6$   |
| (15) $13 \div 3$  | (33) $42 \div 6$  | (51) $3 \div 7$   | (69) $44 \div 10$ |
| (16) $84 \div 8$  | (34) $28 \div 10$ | (52) $18 \div 10$ | (70) $82 \div 9$  |
| (17) $43 \div 9$  | (35) $23 \div 3$  | (53) $92 \div 3$  | (71) $81 \div 8$  |
| (18) $70 \div 6$  | (36) $77 \div 4$  | (54) $92 \div 3$  | (72) $55 \div 9$  |

(1)

$2 \div 3$

(19)

$32 \div 9$

(37)

$24 \div 3$

(55)

$29 \div 8$

(2)

$7 \div 7$

(20)

$42 \div 3$

(38)

$82 \div 7$

(56)

$76 \div 9$

(3)

$29 \div 6$

(21)

$9 \div 2$

(39)

$10 \div 8$

(57)

$81 \div 4$

(4)

$78 \div 3$

(22)

$50 \div 10$

(40)

$76 \div 6$

(58)

$10 \div 8$

(5)

$14 \div 6$

(23)

$7 \div 6$

(41)

$51 \div 3$

(59)

$49 \div 7$

(6)

$16 \div 4$

(24)

$65 \div 8$

(42)

$78 \div 3$

(60)

$15 \div 5$

(7)

$13 \div 2$

(25)

$36 \div 8$

(43)

$88 \div 6$

(61)

$6 \div 8$

(8)

$44 \div 1$

(26)

$87 \div 8$

(44)

$66 \div 8$

(62)

$13 \div 9$

(9)

$17 \div 1$

(27)

$24 \div 2$

(45)

$21 \div 1$

(63)

$40 \div 7$

(10)

$92 \div 5$

(28)

$81 \div 10$

(46)

$66 \div 2$

(64)

$38 \div 4$

(11)

$2 \div 1$

(29)

$36 \div 1$

(47)

$39 \div 8$

(65)

$69 \div 5$

(12)

$21 \div 6$

(30)

$27 \div 1$

(48)

$67 \div 10$

(66)

$80 \div 2$

(13)

$70 \div 9$

(31)

$8 \div 10$

(49)

$30 \div 7$

(67)

$58 \div 4$

(14)

$81 \div 6$

(32)

$98 \div 4$

(50)

$76 \div 2$

(68)

$85 \div 4$

(15)

$45 \div 6$

(33)

$79 \div 3$

(51)

$100 \div 3$

(69)

$81 \div 10$

(16)

$23 \div 1$

(34)

$27 \div 4$

(52)

$62 \div 9$

(70)

$64 \div 6$

(17)

$60 \div 8$

(35)

$56 \div 4$

(53)

$86 \div 8$

(71)

$78 \div 8$

(18)

$7 \div 3$

(36)

$68 \div 3$

(54)

$60 \div 1$

(72)

$97 \div 6$

(1)  $5 \div 6$

(19)  $24 \div 5$

(37)  $65 \div 2$

(55)  $67 \div 2$

(2)  $76 \div 2$

(20)  $95 \div 2$

(38)  $54 \div 10$

(56)  $33 \div 7$

(3)  $57 \div 6$

(21)  $94 \div 10$

(39)  $33 \div 3$

(57)  $34 \div 10$

(4)  $1 \div 9$

(22)  $65 \div 8$

(40)  $73 \div 2$

(58)  $96 \div 1$

(5)  $41 \div 2$

(23)  $64 \div 9$

(41)  $53 \div 4$

(59)  $95 \div 10$

(6)  $75 \div 1$

(24)  $10 \div 1$

(42)  $87 \div 9$

(60)  $81 \div 6$

(7)  $30 \div 8$

(25)  $9 \div 2$

(43)  $23 \div 1$

(61)  $43 \div 10$

(8)  $52 \div 3$

(26)  $85 \div 5$

(44)  $32 \div 6$

(62)  $13 \div 4$

(9)  $86 \div 7$

(27)  $23 \div 3$

(45)  $21 \div 2$

(63)  $48 \div 1$

(10)  $59 \div 9$

(28)  $47 \div 10$

(46)  $28 \div 4$

(64)  $100 \div 4$

(11)  $75 \div 3$

(29)  $52 \div 4$

(47)  $33 \div 4$

(65)  $50 \div 10$

(12)  $45 \div 2$

(30)  $57 \div 5$

(48)  $36 \div 9$

(66)  $66 \div 10$

(13)  $91 \div 9$

(31)  $98 \div 5$

(49)  $15 \div 3$

(67)  $46 \div 6$

(14)  $21 \div 9$

(32)  $84 \div 3$

(50)  $0 \div 2$

(68)  $70 \div 4$

(15)  $84 \div 3$

(33)  $86 \div 2$

(51)  $93 \div 3$

(69)  $82 \div 8$

(16)  $51 \div 7$

(34)  $27 \div 5$

(52)  $69 \div 3$

(70)  $55 \div 2$

(17)  $68 \div 6$

(35)  $78 \div 3$

(53)  $65 \div 5$

(71)  $16 \div 1$

(18)  $37 \div 6$

(36)  $58 \div 8$

(54)  $96 \div 5$

(72)  $50 \div 7$

- |                                |                                |                                |                                |
|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $59 \div 3$<br>商 19, 余り 2  | (19) $64 \div 7$<br>商 9, 余り 1  | (37) $68 \div 6$<br>商 11, 余り 2 | (55) $33 \div 1$<br>商 33, 余り 0 |
| (2) $74 \div 4$<br>商 18, 余り 2  | (20) $70 \div 1$<br>商 70, 余り 0 | (38) $81 \div 2$<br>商 40, 余り 1 | (56) $70 \div 10$<br>商 7, 余り 0 |
| (3) $62 \div 4$<br>商 15, 余り 2  | (21) $86 \div 9$<br>商 9, 余り 5  | (39) $27 \div 7$<br>商 3, 余り 6  | (57) $32 \div 8$<br>商 4, 余り 0  |
| (4) $84 \div 5$<br>商 16, 余り 4  | (22) $62 \div 9$<br>商 6, 余り 8  | (40) $37 \div 10$<br>商 3, 余り 7 | (58) $33 \div 2$<br>商 16, 余り 1 |
| (5) $76 \div 6$<br>商 12, 余り 4  | (23) $1 \div 10$<br>商 0, 余り 1  | (41) $62 \div 8$<br>商 7, 余り 6  | (59) $12 \div 5$<br>商 2, 余り 2  |
| (6) $1 \div 2$<br>商 0, 余り 1    | (24) $86 \div 4$<br>商 21, 余り 2 | (42) $36 \div 5$<br>商 7, 余り 1  | (60) $19 \div 8$<br>商 2, 余り 3  |
| (7) $48 \div 9$<br>商 5, 余り 3   | (25) $73 \div 7$<br>商 10, 余り 3 | (43) $22 \div 7$<br>商 3, 余り 1  | (61) $37 \div 1$<br>商 37, 余り 0 |
| (8) $53 \div 6$<br>商 8, 余り 5   | (26) $59 \div 6$<br>商 9, 余り 5  | (44) $22 \div 5$<br>商 4, 余り 2  | (62) $75 \div 10$<br>商 7, 余り 5 |
| (9) $85 \div 10$<br>商 8, 余り 5  | (27) $45 \div 1$<br>商 45, 余り 0 | (45) $47 \div 3$<br>商 15, 余り 2 | (63) $34 \div 4$<br>商 8, 余り 2  |
| (10) $21 \div 3$<br>商 7, 余り 0  | (28) $24 \div 8$<br>商 3, 余り 0  | (46) $89 \div 2$<br>商 44, 余り 1 | (64) $18 \div 5$<br>商 3, 余り 3  |
| (11) $71 \div 9$<br>商 7, 余り 8  | (29) $7 \div 3$<br>商 2, 余り 1   | (47) $87 \div 2$<br>商 43, 余り 1 | (65) $60 \div 3$<br>商 20, 余り 0 |
| (12) $24 \div 1$<br>商 24, 余り 0 | (30) $48 \div 9$<br>商 5, 余り 3  | (48) $14 \div 4$<br>商 3, 余り 2  | (66) $22 \div 2$<br>商 11, 余り 0 |
| (13) $22 \div 1$<br>商 22, 余り 0 | (31) $39 \div 9$<br>商 4, 余り 3  | (49) $2 \div 8$<br>商 0, 余り 2   | (67) $40 \div 4$<br>商 10, 余り 0 |
| (14) $22 \div 2$<br>商 11, 余り 0 | (32) $79 \div 10$<br>商 7, 余り 9 | (50) $1 \div 2$<br>商 0, 余り 1   | (68) $91 \div 1$<br>商 91, 余り 0 |
| (15) $18 \div 2$<br>商 9, 余り 0  | (33) $4 \div 6$<br>商 0, 余り 4   | (51) $49 \div 5$<br>商 9, 余り 4  | (69) $89 \div 10$<br>商 8, 余り 9 |
| (16) $70 \div 7$<br>商 10, 余り 0 | (34) $56 \div 2$<br>商 28, 余り 0 | (52) $34 \div 9$<br>商 3, 余り 7  | (70) $1 \div 3$<br>商 0, 余り 1   |
| (17) $9 \div 5$<br>商 1, 余り 4   | (35) $50 \div 2$<br>商 25, 余り 0 | (53) $70 \div 2$<br>商 35, 余り 0 | (71) $86 \div 6$<br>商 14, 余り 2 |
| (18) $38 \div 1$<br>商 38, 余り 0 | (36) $58 \div 1$<br>商 58, 余り 0 | (54) $76 \div 2$<br>商 38, 余り 0 | (72) $46 \div 9$<br>商 5, 余り 1  |

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|--------------------------------|--------------------------------|---------------------------------|--------------------------------|
| (1) $50 \div 5$<br>商 10, 余り 0  | (19) $25 \div 1$<br>商 25, 余り 0 | (37) $100 \div 6$<br>商 16, 余り 4 | (55) $89 \div 8$<br>商 11, 余り 1 |
| (2) $100 \div 8$<br>商 12, 余り 4 | (20) $87 \div 10$<br>商 8, 余り 7 | (38) $2 \div 6$<br>商 0, 余り 2    | (56) $66 \div 4$<br>商 16, 余り 2 |
| (3) $65 \div 8$<br>商 8, 余り 1   | (21) $46 \div 8$<br>商 5, 余り 6  | (39) $12 \div 7$<br>商 1, 余り 5   | (57) $55 \div 4$<br>商 13, 余り 3 |
| (4) $25 \div 9$<br>商 2, 余り 7   | (22) $72 \div 6$<br>商 12, 余り 0 | (40) $32 \div 10$<br>商 3, 余り 2  | (58) $64 \div 1$<br>商 64, 余り 0 |
| (5) $88 \div 4$<br>商 22, 余り 0  | (23) $42 \div 6$<br>商 7, 余り 0  | (41) $76 \div 9$<br>商 8, 余り 4   | (59) $38 \div 3$<br>商 12, 余り 2 |
| (6) $23 \div 4$<br>商 5, 余り 3   | (24) $47 \div 8$<br>商 5, 余り 7  | (42) $44 \div 6$<br>商 7, 余り 2   | (60) $25 \div 5$<br>商 5, 余り 0  |
| (7) $18 \div 9$<br>商 2, 余り 0   | (25) $38 \div 2$<br>商 19, 余り 0 | (43) $52 \div 10$<br>商 5, 余り 2  | (61) $77 \div 1$<br>商 77, 余り 0 |
| (8) $12 \div 6$<br>商 2, 余り 0   | (26) $54 \div 6$<br>商 9, 余り 0  | (44) $24 \div 6$<br>商 4, 余り 0   | (62) $21 \div 3$<br>商 7, 余り 0  |
| (9) $60 \div 4$<br>商 15, 余り 0  | (27) $55 \div 9$<br>商 6, 余り 1  | (45) $47 \div 8$<br>商 5, 余り 7   | (63) $67 \div 4$<br>商 16, 余り 3 |
| (10) $58 \div 9$<br>商 6, 余り 4  | (28) $57 \div 2$<br>商 28, 余り 1 | (46) $96 \div 10$<br>商 9, 余り 6  | (64) $89 \div 10$<br>商 8, 余り 9 |
| (11) $31 \div 6$<br>商 5, 余り 1  | (29) $64 \div 1$<br>商 64, 余り 0 | (47) $76 \div 8$<br>商 9, 余り 4   | (65) $81 \div 7$<br>商 11, 余り 4 |
| (12) $53 \div 1$<br>商 53, 余り 0 | (30) $24 \div 4$<br>商 6, 余り 0  | (48) $10 \div 10$<br>商 1, 余り 0  | (66) $30 \div 3$<br>商 10, 余り 0 |
| (13) $79 \div 6$<br>商 13, 余り 1 | (31) $98 \div 1$<br>商 98, 余り 0 | (49) $38 \div 3$<br>商 12, 余り 2  | (67) $26 \div 5$<br>商 5, 余り 1  |
| (14) $57 \div 1$<br>商 57, 余り 0 | (32) $9 \div 4$<br>商 2, 余り 1   | (50) $75 \div 4$<br>商 18, 余り 3  | (68) $40 \div 3$<br>商 13, 余り 1 |
| (15) $87 \div 4$<br>商 21, 余り 3 | (33) $58 \div 3$<br>商 19, 余り 1 | (51) $67 \div 4$<br>商 16, 余り 3  | (69) $83 \div 7$<br>商 11, 余り 6 |
| (16) $35 \div 7$<br>商 5, 余り 0  | (34) $35 \div 4$<br>商 8, 余り 3  | (52) $4 \div 3$<br>商 1, 余り 1    | (70) $20 \div 10$<br>商 2, 余り 0 |
| (17) $31 \div 1$<br>商 31, 余り 0 | (35) $38 \div 6$<br>商 6, 余り 2  | (53) $22 \div 8$<br>商 2, 余り 6   | (71) $10 \div 5$<br>商 2, 余り 0  |
| (18) $86 \div 3$<br>商 28, 余り 2 | (36) $33 \div 8$<br>商 4, 余り 1  | (54) $8 \div 2$<br>商 4, 余り 0    | (72) $47 \div 1$<br>商 47, 余り 0 |

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|---------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $86 \div 5$<br>商 17, 余り 1   | (19) $71 \div 4$<br>商 17, 余り 3 | (37) $8 \div 2$<br>商 4, 余り 0   | (55) $78 \div 9$<br>商 8, 余り 6  |
| (2) $30 \div 5$<br>商 6, 余り 0    | (20) $84 \div 7$<br>商 12, 余り 0 | (38) $71 \div 4$<br>商 17, 余り 3 | (56) $29 \div 6$<br>商 4, 余り 5  |
| (3) $99 \div 1$<br>商 99, 余り 0   | (21) $79 \div 10$<br>商 7, 余り 9 | (39) $78 \div 8$<br>商 9, 余り 6  | (57) $46 \div 4$<br>商 11, 余り 2 |
| (4) $1 \div 4$<br>商 0, 余り 1     | (22) $46 \div 8$<br>商 5, 余り 6  | (40) $74 \div 9$<br>商 8, 余り 2  | (58) $34 \div 9$<br>商 3, 余り 7  |
| (5) $52 \div 8$<br>商 6, 余り 4    | (23) $52 \div 4$<br>商 13, 余り 0 | (41) $66 \div 10$<br>商 6, 余り 6 | (59) $4 \div 7$<br>商 0, 余り 4   |
| (6) $4 \div 9$<br>商 0, 余り 4     | (24) $43 \div 5$<br>商 8, 余り 3  | (42) $66 \div 6$<br>商 11, 余り 0 | (60) $9 \div 5$<br>商 1, 余り 4   |
| (7) $8 \div 6$<br>商 1, 余り 2     | (25) $54 \div 1$<br>商 54, 余り 0 | (43) $75 \div 8$<br>商 9, 余り 3  | (61) $38 \div 3$<br>商 12, 余り 2 |
| (8) $100 \div 1$<br>商 100, 余り 0 | (26) $50 \div 5$<br>商 10, 余り 0 | (44) $84 \div 1$<br>商 84, 余り 0 | (62) $50 \div 7$<br>商 7, 余り 1  |
| (9) $42 \div 1$<br>商 42, 余り 0   | (27) $7 \div 7$<br>商 1, 余り 0   | (45) $54 \div 1$<br>商 54, 余り 0 | (63) $16 \div 1$<br>商 16, 余り 0 |
| (10) $56 \div 8$<br>商 7, 余り 0   | (28) $86 \div 9$<br>商 9, 余り 5  | (46) $79 \div 4$<br>商 19, 余り 3 | (64) $12 \div 9$<br>商 1, 余り 3  |
| (11) $9 \div 3$<br>商 3, 余り 0    | (29) $75 \div 3$<br>商 25, 余り 0 | (47) $55 \div 4$<br>商 13, 余り 3 | (65) $66 \div 6$<br>商 11, 余り 0 |
| (12) $61 \div 4$<br>商 15, 余り 1  | (30) $26 \div 1$<br>商 26, 余り 0 | (48) $99 \div 6$<br>商 16, 余り 3 | (66) $88 \div 6$<br>商 14, 余り 4 |
| (13) $62 \div 4$<br>商 15, 余り 2  | (31) $50 \div 10$<br>商 5, 余り 0 | (49) $99 \div 10$<br>商 9, 余り 9 | (67) $9 \div 1$<br>商 9, 余り 0   |
| (14) $71 \div 3$<br>商 23, 余り 2  | (32) $58 \div 9$<br>商 6, 余り 4  | (50) $83 \div 3$<br>商 27, 余り 2 | (68) $40 \div 8$<br>商 5, 余り 0  |
| (15) $44 \div 4$<br>商 11, 余り 0  | (33) $18 \div 7$<br>商 2, 余り 4  | (51) $63 \div 5$<br>商 12, 余り 3 | (69) $93 \div 1$<br>商 93, 余り 0 |
| (16) $97 \div 4$<br>商 24, 余り 1  | (34) $19 \div 1$<br>商 19, 余り 0 | (52) $62 \div 8$<br>商 7, 余り 6  | (70) $41 \div 5$<br>商 8, 余り 1  |
| (17) $41 \div 10$<br>商 4, 余り 1  | (35) $92 \div 6$<br>商 15, 余り 2 | (53) $42 \div 2$<br>商 21, 余り 0 | (71) $20 \div 10$<br>商 2, 余り 0 |
| (18) $76 \div 6$<br>商 12, 余り 4  | (36) $46 \div 5$<br>商 9, 余り 1  | (54) $26 \div 4$<br>商 6, 余り 2  | (72) $1 \div 1$<br>商 1, 余り 0   |



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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $33 \div 7$<br>商 4, 余り 5   | (19) $98 \div 8$<br>商 12, 余り 2 | (37) $41 \div 6$<br>商 6, 余り 5  | (55) $12 \div 4$<br>商 3, 余り 0  |
| (2) $34 \div 6$<br>商 5, 余り 4   | (20) $88 \div 2$<br>商 44, 余り 0 | (38) $83 \div 10$<br>商 8, 余り 3 | (56) $85 \div 10$<br>商 8, 余り 5 |
| (3) $63 \div 8$<br>商 7, 余り 7   | (21) $90 \div 9$<br>商 10, 余り 0 | (39) $58 \div 4$<br>商 14, 余り 2 | (57) $29 \div 3$<br>商 9, 余り 2  |
| (4) $94 \div 3$<br>商 31, 余り 1  | (22) $51 \div 9$<br>商 5, 余り 6  | (40) $51 \div 10$<br>商 5, 余り 1 | (58) $87 \div 9$<br>商 9, 余り 6  |
| (5) $21 \div 5$<br>商 4, 余り 1   | (23) $65 \div 1$<br>商 65, 余り 0 | (41) $28 \div 1$<br>商 28, 余り 0 | (59) $2 \div 5$<br>商 0, 余り 2   |
| (6) $30 \div 8$<br>商 3, 余り 6   | (24) $61 \div 2$<br>商 30, 余り 1 | (42) $11 \div 5$<br>商 2, 余り 1  | (60) $50 \div 1$<br>商 50, 余り 0 |
| (7) $21 \div 8$<br>商 2, 余り 5   | (25) $27 \div 8$<br>商 3, 余り 3  | (43) $30 \div 6$<br>商 5, 余り 0  | (61) $47 \div 10$<br>商 4, 余り 7 |
| (8) $0 \div 8$<br>商 0, 余り 0    | (26) $35 \div 4$<br>商 8, 余り 3  | (44) $17 \div 1$<br>商 17, 余り 0 | (62) $21 \div 2$<br>商 10, 余り 1 |
| (9) $88 \div 6$<br>商 14, 余り 4  | (27) $90 \div 6$<br>商 15, 余り 0 | (45) $93 \div 1$<br>商 93, 余り 0 | (63) $83 \div 10$<br>商 8, 余り 3 |
| (10) $5 \div 1$<br>商 5, 余り 0   | (28) $52 \div 3$<br>商 17, 余り 1 | (46) $38 \div 5$<br>商 7, 余り 3  | (64) $34 \div 9$<br>商 3, 余り 7  |
| (11) $38 \div 6$<br>商 6, 余り 2  | (29) $71 \div 10$<br>商 7, 余り 1 | (47) $27 \div 9$<br>商 3, 余り 0  | (65) $12 \div 7$<br>商 1, 余り 5  |
| (12) $50 \div 4$<br>商 12, 余り 2 | (30) $34 \div 6$<br>商 5, 余り 4  | (48) $57 \div 8$<br>商 7, 余り 1  | (66) $16 \div 5$<br>商 3, 余り 1  |
| (13) $2 \div 10$<br>商 0, 余り 2  | (31) $20 \div 7$<br>商 2, 余り 6  | (49) $83 \div 3$<br>商 27, 余り 2 | (67) $99 \div 5$<br>商 19, 余り 4 |
| (14) $85 \div 1$<br>商 85, 余り 0 | (32) $1 \div 5$<br>商 0, 余り 1   | (50) $69 \div 7$<br>商 9, 余り 6  | (68) $55 \div 1$<br>商 55, 余り 0 |
| (15) $83 \div 5$<br>商 16, 余り 3 | (33) $84 \div 4$<br>商 21, 余り 0 | (51) $59 \div 2$<br>商 29, 余り 1 | (69) $18 \div 6$<br>商 3, 余り 0  |
| (16) $31 \div 8$<br>商 3, 余り 7  | (34) $30 \div 6$<br>商 5, 余り 0  | (52) $44 \div 2$<br>商 22, 余り 0 | (70) $26 \div 5$<br>商 5, 余り 1  |
| (17) $42 \div 10$<br>商 4, 余り 2 | (35) $4 \div 7$<br>商 0, 余り 4   | (53) $6 \div 8$<br>商 0, 余り 6   | (71) $50 \div 9$<br>商 5, 余り 5  |
| (18) $78 \div 4$<br>商 19, 余り 2 | (36) $15 \div 4$<br>商 3, 余り 3  | (54) $14 \div 7$<br>商 2, 余り 0  | (72) $0 \div 9$<br>商 0, 余り 0   |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $6 \div 1$<br>商 6, 余り 0    | (19) $48 \div 10$<br>商 4, 余り 8 | (37) $5 \div 1$<br>商 5, 余り 0   | (55) $23 \div 9$<br>商 2, 余り 5  |
| (2) $19 \div 4$<br>商 4, 余り 3   | (20) $84 \div 7$<br>商 12, 余り 0 | (38) $74 \div 10$<br>商 7, 余り 4 | (56) $87 \div 2$<br>商 43, 余り 1 |
| (3) $91 \div 6$<br>商 15, 余り 1  | (21) $1 \div 4$<br>商 0, 余り 1   | (39) $46 \div 3$<br>商 15, 余り 1 | (57) $83 \div 2$<br>商 41, 余り 1 |
| (4) $90 \div 3$<br>商 30, 余り 0  | (22) $52 \div 2$<br>商 26, 余り 0 | (40) $96 \div 4$<br>商 24, 余り 0 | (58) $58 \div 2$<br>商 29, 余り 0 |
| (5) $79 \div 8$<br>商 9, 余り 7   | (23) $46 \div 5$<br>商 9, 余り 1  | (41) $54 \div 8$<br>商 6, 余り 6  | (59) $75 \div 8$<br>商 9, 余り 3  |
| (6) $52 \div 9$<br>商 5, 余り 7   | (24) $63 \div 10$<br>商 6, 余り 3 | (42) $33 \div 9$<br>商 3, 余り 6  | (60) $28 \div 1$<br>商 28, 余り 0 |
| (7) $51 \div 7$<br>商 7, 余り 2   | (25) $67 \div 6$<br>商 11, 余り 1 | (43) $77 \div 4$<br>商 19, 余り 1 | (61) $36 \div 5$<br>商 7, 余り 1  |
| (8) $100 \div 2$<br>商 50, 余り 0 | (26) $27 \div 10$<br>商 2, 余り 7 | (44) $37 \div 6$<br>商 6, 余り 1  | (62) $48 \div 2$<br>商 24, 余り 0 |
| (9) $79 \div 3$<br>商 26, 余り 1  | (27) $62 \div 9$<br>商 6, 余り 8  | (45) $37 \div 9$<br>商 4, 余り 1  | (63) $36 \div 3$<br>商 12, 余り 0 |
| (10) $7 \div 9$<br>商 0, 余り 7   | (28) $39 \div 1$<br>商 39, 余り 0 | (46) $64 \div 7$<br>商 9, 余り 1  | (64) $66 \div 3$<br>商 22, 余り 0 |
| (11) $30 \div 5$<br>商 6, 余り 0  | (29) $87 \div 9$<br>商 9, 余り 6  | (47) $77 \div 4$<br>商 19, 余り 1 | (65) $21 \div 6$<br>商 3, 余り 3  |
| (12) $97 \div 8$<br>商 12, 余り 1 | (30) $34 \div 1$<br>商 34, 余り 0 | (48) $33 \div 2$<br>商 16, 余り 1 | (66) $89 \div 3$<br>商 29, 余り 2 |
| (13) $55 \div 9$<br>商 6, 余り 1  | (31) $17 \div 2$<br>商 8, 余り 1  | (49) $70 \div 9$<br>商 7, 余り 7  | (67) $47 \div 7$<br>商 6, 余り 5  |
| (14) $7 \div 10$<br>商 0, 余り 7  | (32) $42 \div 2$<br>商 21, 余り 0 | (50) $84 \div 5$<br>商 16, 余り 4 | (68) $38 \div 8$<br>商 4, 余り 6  |
| (15) $63 \div 8$<br>商 7, 余り 7  | (33) $60 \div 9$<br>商 6, 余り 6  | (51) $32 \div 2$<br>商 16, 余り 0 | (69) $7 \div 10$<br>商 0, 余り 7  |
| (16) $3 \div 10$<br>商 0, 余り 3  | (34) $9 \div 1$<br>商 9, 余り 0   | (52) $12 \div 1$<br>商 12, 余り 0 | (70) $34 \div 2$<br>商 17, 余り 0 |
| (17) $6 \div 8$<br>商 0, 余り 6   | (35) $41 \div 10$<br>商 4, 余り 1 | (53) $36 \div 10$<br>商 3, 余り 6 | (71) $75 \div 3$<br>商 25, 余り 0 |
| (18) $66 \div 5$<br>商 13, 余り 1 | (36) $89 \div 2$<br>商 44, 余り 1 | (54) $79 \div 2$<br>商 39, 余り 1 | (72) $78 \div 4$<br>商 19, 余り 2 |

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|--------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $51 \div 10$<br>商 5, 余り 1  | (19) $78 \div 1$<br>商 78, 余り 0 | (37) $52 \div 7$<br>商 7, 余り 3  | (55) $27 \div 8$<br>商 3, 余り 3   |
| (2) $21 \div 5$<br>商 4, 余り 1   | (20) $30 \div 9$<br>商 3, 余り 3  | (38) $34 \div 6$<br>商 5, 余り 4  | (56) $92 \div 3$<br>商 30, 余り 2  |
| (3) $97 \div 5$<br>商 19, 余り 2  | (21) $36 \div 9$<br>商 4, 余り 0  | (39) $52 \div 2$<br>商 26, 余り 0 | (57) $38 \div 9$<br>商 4, 余り 2   |
| (4) $54 \div 10$<br>商 5, 余り 4  | (22) $93 \div 7$<br>商 13, 余り 2 | (40) $12 \div 10$<br>商 1, 余り 2 | (58) $12 \div 9$<br>商 1, 余り 3   |
| (5) $32 \div 4$<br>商 8, 余り 0   | (23) $16 \div 9$<br>商 1, 余り 7  | (41) $91 \div 10$<br>商 9, 余り 1 | (59) $80 \div 6$<br>商 13, 余り 2  |
| (6) $92 \div 1$<br>商 92, 余り 0  | (24) $99 \div 9$<br>商 11, 余り 0 | (42) $60 \div 8$<br>商 7, 余り 4  | (60) $7 \div 9$<br>商 0, 余り 7    |
| (7) $59 \div 9$<br>商 6, 余り 5   | (25) $75 \div 7$<br>商 10, 余り 5 | (43) $71 \div 6$<br>商 11, 余り 5 | (61) $96 \div 10$<br>商 9, 余り 6  |
| (8) $12 \div 8$<br>商 1, 余り 4   | (26) $3 \div 7$<br>商 0, 余り 3   | (44) $5 \div 5$<br>商 1, 余り 0   | (62) $0 \div 9$<br>商 0, 余り 0    |
| (9) $84 \div 6$<br>商 14, 余り 0  | (27) $88 \div 10$<br>商 8, 余り 8 | (45) $17 \div 8$<br>商 2, 余り 1  | (63) $6 \div 4$<br>商 1, 余り 2    |
| (10) $72 \div 5$<br>商 14, 余り 2 | (28) $7 \div 3$<br>商 2, 余り 1   | (46) $9 \div 5$<br>商 1, 余り 4   | (64) $88 \div 10$<br>商 8, 余り 8  |
| (11) $52 \div 5$<br>商 10, 余り 2 | (29) $27 \div 5$<br>商 5, 余り 2  | (47) $41 \div 4$<br>商 10, 余り 1 | (65) $85 \div 8$<br>商 10, 余り 5  |
| (12) $99 \div 5$<br>商 19, 余り 4 | (30) $35 \div 3$<br>商 11, 余り 2 | (48) $49 \div 7$<br>商 7, 余り 0  | (66) $31 \div 3$<br>商 10, 余り 1  |
| (13) $63 \div 5$<br>商 12, 余り 3 | (31) $33 \div 9$<br>商 3, 余り 6  | (49) $73 \div 3$<br>商 24, 余り 1 | (67) $48 \div 8$<br>商 6, 余り 0   |
| (14) $18 \div 10$<br>商 1, 余り 8 | (32) $53 \div 9$<br>商 5, 余り 8  | (50) $26 \div 9$<br>商 2, 余り 8  | (68) $62 \div 7$<br>商 8, 余り 6   |
| (15) $50 \div 1$<br>商 50, 余り 0 | (33) $64 \div 9$<br>商 7, 余り 1  | (51) $74 \div 3$<br>商 24, 余り 2 | (69) $100 \div 3$<br>商 33, 余り 1 |
| (16) $57 \div 1$<br>商 57, 余り 0 | (34) $69 \div 6$<br>商 11, 余り 3 | (52) $4 \div 10$<br>商 0, 余り 4  | (70) $56 \div 7$<br>商 8, 余り 0   |
| (17) $96 \div 3$<br>商 32, 余り 0 | (35) $39 \div 5$<br>商 7, 余り 4  | (53) $38 \div 5$<br>商 7, 余り 3  | (71) $69 \div 5$<br>商 13, 余り 4  |
| (18) $36 \div 8$<br>商 4, 余り 4  | (36) $16 \div 3$<br>商 5, 余り 1  | (54) $58 \div 4$<br>商 14, 余り 2 | (72) $10 \div 3$<br>商 3, 余り 1   |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $80 \div 10$<br>商 8, 余り 0  | (19) $20 \div 8$<br>商 2, 余り 4  | (37) $55 \div 5$<br>商 11, 余り 0 | (55) $75 \div 6$<br>商 12, 余り 3 |
| (2) $99 \div 7$<br>商 14, 余り 1  | (20) $58 \div 7$<br>商 8, 余り 2  | (38) $81 \div 3$<br>商 27, 余り 0 | (56) $67 \div 1$<br>商 67, 余り 0 |
| (3) $12 \div 1$<br>商 12, 余り 0  | (21) $13 \div 10$<br>商 1, 余り 3 | (39) $5 \div 8$<br>商 0, 余り 5   | (57) $15 \div 4$<br>商 3, 余り 3  |
| (4) $17 \div 3$<br>商 5, 余り 2   | (22) $14 \div 6$<br>商 2, 余り 2  | (40) $89 \div 5$<br>商 17, 余り 4 | (58) $73 \div 4$<br>商 18, 余り 1 |
| (5) $38 \div 9$<br>商 4, 余り 2   | (23) $26 \div 10$<br>商 2, 余り 6 | (41) $72 \div 3$<br>商 24, 余り 0 | (59) $59 \div 7$<br>商 8, 余り 3  |
| (6) $22 \div 4$<br>商 5, 余り 2   | (24) $55 \div 8$<br>商 6, 余り 7  | (42) $18 \div 8$<br>商 2, 余り 2  | (60) $78 \div 5$<br>商 15, 余り 3 |
| (7) $4 \div 6$<br>商 0, 余り 4    | (25) $13 \div 8$<br>商 1, 余り 5  | (43) $82 \div 8$<br>商 10, 余り 2 | (61) $31 \div 7$<br>商 4, 余り 3  |
| (8) $75 \div 1$<br>商 75, 余り 0  | (26) $78 \div 10$<br>商 7, 余り 8 | (44) $71 \div 3$<br>商 23, 余り 2 | (62) $65 \div 8$<br>商 8, 余り 1  |
| (9) $38 \div 10$<br>商 3, 余り 8  | (27) $97 \div 1$<br>商 97, 余り 0 | (45) $82 \div 4$<br>商 20, 余り 2 | (63) $42 \div 5$<br>商 8, 余り 2  |
| (10) $93 \div 9$<br>商 10, 余り 3 | (28) $27 \div 10$<br>商 2, 余り 7 | (46) $48 \div 4$<br>商 12, 余り 0 | (64) $50 \div 3$<br>商 16, 余り 2 |
| (11) $73 \div 3$<br>商 24, 余り 1 | (29) $0 \div 2$<br>商 0, 余り 0   | (47) $74 \div 1$<br>商 74, 余り 0 | (65) $5 \div 3$<br>商 1, 余り 2   |
| (12) $74 \div 6$<br>商 12, 余り 2 | (30) $50 \div 7$<br>商 7, 余り 1  | (48) $15 \div 8$<br>商 1, 余り 7  | (66) $77 \div 6$<br>商 12, 余り 5 |
| (13) $21 \div 10$<br>商 2, 余り 1 | (31) $83 \div 8$<br>商 10, 余り 3 | (49) $29 \div 7$<br>商 4, 余り 1  | (67) $28 \div 1$<br>商 28, 余り 0 |
| (14) $27 \div 10$<br>商 2, 余り 7 | (32) $98 \div 8$<br>商 12, 余り 2 | (50) $96 \div 5$<br>商 19, 余り 1 | (68) $24 \div 7$<br>商 3, 余り 3  |
| (15) $55 \div 2$<br>商 27, 余り 1 | (33) $72 \div 7$<br>商 10, 余り 2 | (51) $64 \div 1$<br>商 64, 余り 0 | (69) $22 \div 1$<br>商 22, 余り 0 |
| (16) $21 \div 8$<br>商 2, 余り 5  | (34) $40 \div 7$<br>商 5, 余り 5  | (52) $74 \div 9$<br>商 8, 余り 2  | (70) $12 \div 3$<br>商 4, 余り 0  |
| (17) $8 \div 10$<br>商 0, 余り 8  | (35) $83 \div 1$<br>商 83, 余り 0 | (53) $54 \div 8$<br>商 6, 余り 6  | (71) $89 \div 3$<br>商 29, 余り 2 |
| (18) $29 \div 1$<br>商 29, 余り 0 | (36) $57 \div 3$<br>商 19, 余り 0 | (54) $9 \div 3$<br>商 3, 余り 0   | (72) $13 \div 9$<br>商 1, 余り 4  |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $38 \div 1$<br>商 38, 余り 0  | (19) $91 \div 6$<br>商 15, 余り 1 | (37) $65 \div 7$<br>商 9, 余り 2  | (55) $67 \div 9$<br>商 7, 余り 4  |
| (2) $59 \div 5$<br>商 11, 余り 4  | (20) $51 \div 4$<br>商 12, 余り 3 | (38) $25 \div 7$<br>商 3, 余り 4  | (56) $95 \div 7$<br>商 13, 余り 4 |
| (3) $33 \div 5$<br>商 6, 余り 3   | (21) $0 \div 5$<br>商 0, 余り 0   | (39) $68 \div 3$<br>商 22, 余り 2 | (57) $66 \div 6$<br>商 11, 余り 0 |
| (4) $89 \div 8$<br>商 11, 余り 1  | (22) $49 \div 10$<br>商 4, 余り 9 | (40) $97 \div 2$<br>商 48, 余り 1 | (58) $62 \div 10$<br>商 6, 余り 2 |
| (5) $70 \div 7$<br>商 10, 余り 0  | (23) $92 \div 6$<br>商 15, 余り 2 | (41) $41 \div 9$<br>商 4, 余り 5  | (59) $84 \div 8$<br>商 10, 余り 4 |
| (6) $39 \div 3$<br>商 13, 余り 0  | (24) $1 \div 6$<br>商 0, 余り 1   | (42) $41 \div 9$<br>商 4, 余り 5  | (60) $50 \div 2$<br>商 25, 余り 0 |
| (7) $22 \div 1$<br>商 22, 余り 0  | (25) $12 \div 2$<br>商 6, 余り 0  | (43) $97 \div 8$<br>商 12, 余り 1 | (61) $81 \div 8$<br>商 10, 余り 1 |
| (8) $76 \div 7$<br>商 10, 余り 6  | (26) $37 \div 2$<br>商 18, 余り 1 | (44) $39 \div 3$<br>商 13, 余り 0 | (62) $91 \div 4$<br>商 22, 余り 3 |
| (9) $59 \div 6$<br>商 9, 余り 5   | (27) $77 \div 8$<br>商 9, 余り 5  | (45) $21 \div 8$<br>商 2, 余り 5  | (63) $61 \div 7$<br>商 8, 余り 5  |
| (10) $79 \div 1$<br>商 79, 余り 0 | (28) $29 \div 5$<br>商 5, 余り 4  | (46) $68 \div 3$<br>商 22, 余り 2 | (64) $14 \div 8$<br>商 1, 余り 6  |
| (11) $42 \div 1$<br>商 42, 余り 0 | (29) $75 \div 10$<br>商 7, 余り 5 | (47) $77 \div 1$<br>商 77, 余り 0 | (65) $67 \div 3$<br>商 22, 余り 1 |
| (12) $33 \div 10$<br>商 3, 余り 3 | (30) $11 \div 10$<br>商 1, 余り 1 | (48) $21 \div 4$<br>商 5, 余り 1  | (66) $98 \div 4$<br>商 24, 余り 2 |
| (13) $58 \div 6$<br>商 9, 余り 4  | (31) $68 \div 1$<br>商 68, 余り 0 | (49) $80 \div 1$<br>商 80, 余り 0 | (67) $52 \div 9$<br>商 5, 余り 7  |
| (14) $20 \div 1$<br>商 20, 余り 0 | (32) $19 \div 3$<br>商 6, 余り 1  | (50) $35 \div 4$<br>商 8, 余り 3  | (68) $70 \div 6$<br>商 11, 余り 4 |
| (15) $81 \div 8$<br>商 10, 余り 1 | (33) $8 \div 9$<br>商 0, 余り 8   | (51) $77 \div 2$<br>商 38, 余り 1 | (69) $86 \div 6$<br>商 14, 余り 2 |
| (16) $61 \div 2$<br>商 30, 余り 1 | (34) $84 \div 7$<br>商 12, 余り 0 | (52) $97 \div 4$<br>商 24, 余り 1 | (70) $23 \div 8$<br>商 2, 余り 7  |
| (17) $9 \div 10$<br>商 0, 余り 9  | (35) $37 \div 4$<br>商 9, 余り 1  | (53) $78 \div 2$<br>商 39, 余り 0 | (71) $48 \div 8$<br>商 6, 余り 0  |
| (18) $43 \div 5$<br>商 8, 余り 3  | (36) $78 \div 10$<br>商 7, 余り 8 | (54) $25 \div 4$<br>商 6, 余り 1  | (72) $63 \div 6$<br>商 10, 余り 3 |

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|--------------------------------|--------------------------------|---------------------------------|---------------------------------|
| (1) $42 \div 8$<br>商 5, 余り 2   | (19) $32 \div 2$<br>商 16, 余り 0 | (37) $68 \div 4$<br>商 17, 余り 0  | (55) $92 \div 1$<br>商 92, 余り 0  |
| (2) $1 \div 5$<br>商 0, 余り 1    | (20) $97 \div 4$<br>商 24, 余り 1 | (38) $57 \div 2$<br>商 28, 余り 1  | (56) $50 \div 9$<br>商 5, 余り 5   |
| (3) $44 \div 3$<br>商 14, 余り 2  | (21) $2 \div 10$<br>商 0, 余り 2  | (39) $52 \div 8$<br>商 6, 余り 4   | (57) $6 \div 1$<br>商 6, 余り 0    |
| (4) $12 \div 2$<br>商 6, 余り 0   | (22) $4 \div 8$<br>商 0, 余り 4   | (40) $48 \div 2$<br>商 24, 余り 0  | (58) $86 \div 1$<br>商 86, 余り 0  |
| (5) $58 \div 5$<br>商 11, 余り 3  | (23) $96 \div 4$<br>商 24, 余り 0 | (41) $85 \div 3$<br>商 28, 余り 1  | (59) $46 \div 1$<br>商 46, 余り 0  |
| (6) $7 \div 10$<br>商 0, 余り 7   | (24) $9 \div 3$<br>商 3, 余り 0   | (42) $69 \div 7$<br>商 9, 余り 6   | (60) $87 \div 4$<br>商 21, 余り 3  |
| (7) $75 \div 2$<br>商 37, 余り 1  | (25) $76 \div 3$<br>商 25, 余り 1 | (43) $70 \div 5$<br>商 14, 余り 0  | (61) $79 \div 10$<br>商 7, 余り 9  |
| (8) $5 \div 1$<br>商 5, 余り 0    | (26) $91 \div 3$<br>商 30, 余り 1 | (44) $45 \div 3$<br>商 15, 余り 0  | (62) $100 \div 5$<br>商 20, 余り 0 |
| (9) $87 \div 2$<br>商 43, 余り 1  | (27) $72 \div 8$<br>商 9, 余り 0  | (45) $100 \div 2$<br>商 50, 余り 0 | (63) $69 \div 7$<br>商 9, 余り 6   |
| (10) $30 \div 6$<br>商 5, 余り 0  | (28) $35 \div 2$<br>商 17, 余り 1 | (46) $85 \div 4$<br>商 21, 余り 1  | (64) $19 \div 4$<br>商 4, 余り 3   |
| (11) $22 \div 10$<br>商 2, 余り 2 | (29) $57 \div 1$<br>商 57, 余り 0 | (47) $7 \div 10$<br>商 0, 余り 7   | (65) $11 \div 5$<br>商 2, 余り 1   |
| (12) $48 \div 1$<br>商 48, 余り 0 | (30) $58 \div 9$<br>商 6, 余り 4  | (48) $4 \div 1$<br>商 4, 余り 0    | (66) $83 \div 6$<br>商 13, 余り 5  |
| (13) $77 \div 9$<br>商 8, 余り 5  | (31) $40 \div 4$<br>商 10, 余り 0 | (49) $11 \div 6$<br>商 1, 余り 5   | (67) $71 \div 5$<br>商 14, 余り 1  |
| (14) $55 \div 10$<br>商 5, 余り 5 | (32) $21 \div 6$<br>商 3, 余り 3  | (50) $21 \div 7$<br>商 3, 余り 0   | (68) $33 \div 1$<br>商 33, 余り 0  |
| (15) $7 \div 1$<br>商 7, 余り 0   | (33) $16 \div 10$<br>商 1, 余り 6 | (51) $79 \div 6$<br>商 13, 余り 1  | (69) $41 \div 3$<br>商 13, 余り 2  |
| (16) $74 \div 3$<br>商 24, 余り 2 | (34) $10 \div 4$<br>商 2, 余り 2  | (52) $82 \div 3$<br>商 27, 余り 1  | (70) $25 \div 5$<br>商 5, 余り 0   |
| (17) $65 \div 5$<br>商 13, 余り 0 | (35) $47 \div 5$<br>商 9, 余り 2  | (53) $53 \div 4$<br>商 13, 余り 1  | (71) $95 \div 8$<br>商 11, 余り 7  |
| (18) $82 \div 3$<br>商 27, 余り 1 | (36) $5 \div 5$<br>商 1, 余り 0   | (54) $39 \div 6$<br>商 6, 余り 3   | (72) $36 \div 1$<br>商 36, 余り 0  |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $41 \div 4$<br>商 10, 余り 1  | (19) $67 \div 4$<br>商 16, 余り 3 | (37) $18 \div 4$<br>商 4, 余り 2  | (55) $4 \div 10$<br>商 0, 余り 4  |
| (2) $25 \div 8$<br>商 3, 余り 1   | (20) $78 \div 6$<br>商 13, 余り 0 | (38) $4 \div 2$<br>商 2, 余り 0   | (56) $90 \div 9$<br>商 10, 余り 0 |
| (3) $99 \div 7$<br>商 14, 余り 1  | (21) $46 \div 6$<br>商 7, 余り 4  | (39) $64 \div 5$<br>商 12, 余り 4 | (57) $66 \div 6$<br>商 11, 余り 0 |
| (4) $86 \div 8$<br>商 10, 余り 6  | (22) $27 \div 2$<br>商 13, 余り 1 | (40) $10 \div 5$<br>商 2, 余り 0  | (58) $24 \div 5$<br>商 4, 余り 4  |
| (5) $36 \div 9$<br>商 4, 余り 0   | (23) $92 \div 3$<br>商 30, 余り 2 | (41) $47 \div 7$<br>商 6, 余り 5  | (59) $6 \div 3$<br>商 2, 余り 0   |
| (6) $12 \div 6$<br>商 2, 余り 0   | (24) $37 \div 5$<br>商 7, 余り 2  | (42) $52 \div 3$<br>商 17, 余り 1 | (60) $3 \div 8$<br>商 0, 余り 3   |
| (7) $95 \div 4$<br>商 23, 余り 3  | (25) $35 \div 8$<br>商 4, 余り 3  | (43) $11 \div 4$<br>商 2, 余り 3  | (61) $7 \div 5$<br>商 1, 余り 2   |
| (8) $20 \div 9$<br>商 2, 余り 2   | (26) $95 \div 6$<br>商 15, 余り 5 | (44) $65 \div 5$<br>商 13, 余り 0 | (62) $13 \div 10$<br>商 1, 余り 3 |
| (9) $10 \div 5$<br>商 2, 余り 0   | (27) $43 \div 2$<br>商 21, 余り 1 | (45) $95 \div 4$<br>商 23, 余り 3 | (63) $27 \div 3$<br>商 9, 余り 0  |
| (10) $67 \div 4$<br>商 16, 余り 3 | (28) $4 \div 7$<br>商 0, 余り 4   | (46) $42 \div 5$<br>商 8, 余り 2  | (64) $17 \div 4$<br>商 4, 余り 1  |
| (11) $8 \div 5$<br>商 1, 余り 3   | (29) $89 \div 7$<br>商 12, 余り 5 | (47) $67 \div 7$<br>商 9, 余り 4  | (65) $52 \div 1$<br>商 52, 余り 0 |
| (12) $66 \div 1$<br>商 66, 余り 0 | (30) $61 \div 4$<br>商 15, 余り 1 | (48) $12 \div 2$<br>商 6, 余り 0  | (66) $83 \div 2$<br>商 41, 余り 1 |
| (13) $65 \div 5$<br>商 13, 余り 0 | (31) $17 \div 1$<br>商 17, 余り 0 | (49) $29 \div 4$<br>商 7, 余り 1  | (67) $88 \div 1$<br>商 88, 余り 0 |
| (14) $9 \div 4$<br>商 2, 余り 1   | (32) $70 \div 7$<br>商 10, 余り 0 | (50) $25 \div 4$<br>商 6, 余り 1  | (68) $98 \div 4$<br>商 24, 余り 2 |
| (15) $58 \div 2$<br>商 29, 余り 0 | (33) $8 \div 3$<br>商 2, 余り 2   | (51) $39 \div 6$<br>商 6, 余り 3  | (69) $96 \div 8$<br>商 12, 余り 0 |
| (16) $2 \div 8$<br>商 0, 余り 2   | (34) $63 \div 10$<br>商 6, 余り 3 | (52) $82 \div 1$<br>商 82, 余り 0 | (70) $17 \div 8$<br>商 2, 余り 1  |
| (17) $48 \div 9$<br>商 5, 余り 3  | (35) $28 \div 2$<br>商 14, 余り 0 | (53) $79 \div 5$<br>商 15, 余り 4 | (71) $36 \div 3$<br>商 12, 余り 0 |
| (18) $77 \div 8$<br>商 9, 余り 5  | (36) $23 \div 1$<br>商 23, 余り 0 | (54) $46 \div 4$<br>商 11, 余り 2 | (72) $44 \div 8$<br>商 5, 余り 4  |

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|--------------------------------|---------------------------------|--------------------------------|---------------------------------|
| (1) $51 \div 7$<br>商 7, 余り 2   | (19) $97 \div 10$<br>商 9, 余り 7  | (37) $37 \div 10$<br>商 3, 余り 7 | (55) $80 \div 2$<br>商 40, 余り 0  |
| (2) $98 \div 8$<br>商 12, 余り 2  | (20) $77 \div 3$<br>商 25, 余り 2  | (38) $26 \div 1$<br>商 26, 余り 0 | (56) $59 \div 8$<br>商 7, 余り 3   |
| (3) $20 \div 1$<br>商 20, 余り 0  | (21) $18 \div 10$<br>商 1, 余り 8  | (39) $44 \div 10$<br>商 4, 余り 4 | (57) $38 \div 5$<br>商 7, 余り 3   |
| (4) $12 \div 8$<br>商 1, 余り 4   | (22) $41 \div 3$<br>商 13, 余り 2  | (40) $84 \div 6$<br>商 14, 余り 0 | (58) $88 \div 6$<br>商 14, 余り 4  |
| (5) $86 \div 4$<br>商 21, 余り 2  | (23) $31 \div 2$<br>商 15, 余り 1  | (41) $51 \div 9$<br>商 5, 余り 6  | (59) $95 \div 7$<br>商 13, 余り 4  |
| (6) $99 \div 9$<br>商 11, 余り 0  | (24) $66 \div 6$<br>商 11, 余り 0  | (42) $56 \div 3$<br>商 18, 余り 2 | (60) $16 \div 2$<br>商 8, 余り 0   |
| (7) $3 \div 1$<br>商 3, 余り 0    | (25) $73 \div 4$<br>商 18, 余り 1  | (43) $45 \div 3$<br>商 15, 余り 0 | (61) $35 \div 8$<br>商 4, 余り 3   |
| (8) $28 \div 7$<br>商 4, 余り 0   | (26) $63 \div 5$<br>商 12, 余り 3  | (44) $82 \div 10$<br>商 8, 余り 2 | (62) $100 \div 8$<br>商 12, 余り 4 |
| (9) $97 \div 8$<br>商 12, 余り 1  | (27) $11 \div 5$<br>商 2, 余り 1   | (45) $58 \div 4$<br>商 14, 余り 2 | (63) $85 \div 4$<br>商 21, 余り 1  |
| (10) $28 \div 6$<br>商 4, 余り 4  | (28) $65 \div 4$<br>商 16, 余り 1  | (46) $40 \div 10$<br>商 4, 余り 0 | (64) $78 \div 5$<br>商 15, 余り 3  |
| (11) $40 \div 8$<br>商 5, 余り 0  | (29) $70 \div 6$<br>商 11, 余り 4  | (47) $41 \div 5$<br>商 8, 余り 1  | (65) $13 \div 3$<br>商 4, 余り 1   |
| (12) $26 \div 5$<br>商 5, 余り 1  | (30) $53 \div 6$<br>商 8, 余り 5   | (48) $98 \div 2$<br>商 49, 余り 0 | (66) $15 \div 2$<br>商 7, 余り 1   |
| (13) $52 \div 1$<br>商 52, 余り 0 | (31) $29 \div 4$<br>商 7, 余り 1   | (49) $65 \div 9$<br>商 7, 余り 2  | (67) $50 \div 5$<br>商 10, 余り 0  |
| (14) $7 \div 9$<br>商 0, 余り 7   | (32) $52 \div 5$<br>商 10, 余り 2  | (50) $11 \div 9$<br>商 1, 余り 2  | (68) $79 \div 10$<br>商 7, 余り 9  |
| (15) $2 \div 7$<br>商 0, 余り 2   | (33) $40 \div 9$<br>商 4, 余り 4   | (51) $64 \div 7$<br>商 9, 余り 1  | (69) $52 \div 5$<br>商 10, 余り 2  |
| (16) $65 \div 1$<br>商 65, 余り 0 | (34) $15 \div 5$<br>商 3, 余り 0   | (52) $82 \div 6$<br>商 13, 余り 4 | (70) $54 \div 9$<br>商 6, 余り 0   |
| (17) $40 \div 9$<br>商 4, 余り 4  | (35) $100 \div 8$<br>商 12, 余り 4 | (53) $87 \div 5$<br>商 17, 余り 2 | (71) $5 \div 8$<br>商 0, 余り 5    |
| (18) $2 \div 6$<br>商 0, 余り 2   | (36) $8 \div 10$<br>商 0, 余り 8   | (54) $51 \div 2$<br>商 25, 余り 1 | (72) $11 \div 3$<br>商 3, 余り 2   |



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|--------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $9 \div 6$<br>商 1, 余り 3    | (19) $25 \div 5$<br>商 5, 余り 0  | (37) $39 \div 3$<br>商 13, 余り 0 | (55) $21 \div 6$<br>商 3, 余り 3   |
| (2) $64 \div 6$<br>商 10, 余り 4  | (20) $81 \div 3$<br>商 27, 余り 0 | (38) $7 \div 3$<br>商 2, 余り 1   | (56) $6 \div 7$<br>商 0, 余り 6    |
| (3) $22 \div 5$<br>商 4, 余り 2   | (21) $23 \div 8$<br>商 2, 余り 7  | (39) $79 \div 10$<br>商 7, 余り 9 | (57) $3 \div 3$<br>商 1, 余り 0    |
| (4) $35 \div 10$<br>商 3, 余り 5  | (22) $96 \div 10$<br>商 9, 余り 6 | (40) $14 \div 10$<br>商 1, 余り 4 | (58) $45 \div 8$<br>商 5, 余り 5   |
| (5) $85 \div 3$<br>商 28, 余り 1  | (23) $24 \div 10$<br>商 2, 余り 4 | (41) $82 \div 7$<br>商 11, 余り 5 | (59) $82 \div 2$<br>商 41, 余り 0  |
| (6) $99 \div 9$<br>商 11, 余り 0  | (24) $17 \div 3$<br>商 5, 余り 2  | (42) $52 \div 7$<br>商 7, 余り 3  | (60) $9 \div 6$<br>商 1, 余り 3    |
| (7) $17 \div 10$<br>商 1, 余り 7  | (25) $34 \div 10$<br>商 3, 余り 4 | (43) $19 \div 10$<br>商 1, 余り 9 | (61) $82 \div 10$<br>商 8, 余り 2  |
| (8) $76 \div 7$<br>商 10, 余り 6  | (26) $6 \div 8$<br>商 0, 余り 6   | (44) $68 \div 3$<br>商 22, 余り 2 | (62) $12 \div 8$<br>商 1, 余り 4   |
| (9) $97 \div 6$<br>商 16, 余り 1  | (27) $59 \div 5$<br>商 11, 余り 4 | (45) $90 \div 1$<br>商 90, 余り 0 | (63) $99 \div 1$<br>商 99, 余り 0  |
| (10) $1 \div 7$<br>商 0, 余り 1   | (28) $14 \div 4$<br>商 3, 余り 2  | (46) $84 \div 8$<br>商 10, 余り 4 | (64) $4 \div 8$<br>商 0, 余り 4    |
| (11) $6 \div 8$<br>商 0, 余り 6   | (29) $54 \div 1$<br>商 54, 余り 0 | (47) $81 \div 4$<br>商 20, 余り 1 | (65) $7 \div 2$<br>商 3, 余り 1    |
| (12) $77 \div 5$<br>商 15, 余り 2 | (30) $97 \div 6$<br>商 16, 余り 1 | (48) $5 \div 7$<br>商 0, 余り 5   | (66) $40 \div 8$<br>商 5, 余り 0   |
| (13) $27 \div 8$<br>商 3, 余り 3  | (31) $30 \div 7$<br>商 4, 余り 2  | (49) $37 \div 6$<br>商 6, 余り 1  | (67) $35 \div 3$<br>商 11, 余り 2  |
| (14) $65 \div 10$<br>商 6, 余り 5 | (32) $49 \div 7$<br>商 7, 余り 0  | (50) $83 \div 6$<br>商 13, 余り 5 | (68) $24 \div 10$<br>商 2, 余り 4  |
| (15) $92 \div 6$<br>商 15, 余り 2 | (33) $16 \div 8$<br>商 2, 余り 0  | (51) $6 \div 6$<br>商 1, 余り 0   | (69) $4 \div 4$<br>商 1, 余り 0    |
| (16) $85 \div 6$<br>商 14, 余り 1 | (34) $70 \div 4$<br>商 17, 余り 2 | (52) $40 \div 8$<br>商 5, 余り 0  | (70) $38 \div 8$<br>商 4, 余り 6   |
| (17) $64 \div 2$<br>商 32, 余り 0 | (35) $67 \div 9$<br>商 7, 余り 4  | (53) $40 \div 9$<br>商 4, 余り 4  | (71) $79 \div 1$<br>商 79, 余り 0  |
| (18) $86 \div 3$<br>商 28, 余り 2 | (36) $16 \div 8$<br>商 2, 余り 0  | (54) $92 \div 4$<br>商 23, 余り 0 | (72) $100 \div 8$<br>商 12, 余り 4 |

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|--------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $2 \div 6$<br>商 0, 余り 2    | (19) $77 \div 2$<br>商 38, 余り 1 | (37) $15 \div 8$<br>商 1, 余り 7  | (55) $86 \div 4$<br>商 21, 余り 2  |
| (2) $38 \div 10$<br>商 3, 余り 8  | (20) $83 \div 2$<br>商 41, 余り 1 | (38) $48 \div 3$<br>商 16, 余り 0 | (56) $28 \div 9$<br>商 3, 余り 1   |
| (3) $92 \div 8$<br>商 11, 余り 4  | (21) $71 \div 9$<br>商 7, 余り 8  | (39) $67 \div 5$<br>商 13, 余り 2 | (57) $75 \div 5$<br>商 15, 余り 0  |
| (4) $46 \div 6$<br>商 7, 余り 4   | (22) $32 \div 2$<br>商 16, 余り 0 | (40) $75 \div 7$<br>商 10, 余り 5 | (58) $15 \div 5$<br>商 3, 余り 0   |
| (5) $90 \div 1$<br>商 90, 余り 0  | (23) $95 \div 6$<br>商 15, 余り 5 | (41) $27 \div 5$<br>商 5, 余り 2  | (59) $99 \div 8$<br>商 12, 余り 3  |
| (6) $5 \div 1$<br>商 5, 余り 0    | (24) $46 \div 4$<br>商 11, 余り 2 | (42) $37 \div 1$<br>商 37, 余り 0 | (60) $56 \div 4$<br>商 14, 余り 0  |
| (7) $56 \div 10$<br>商 5, 余り 6  | (25) $61 \div 9$<br>商 6, 余り 7  | (43) $2 \div 7$<br>商 0, 余り 2   | (61) $89 \div 6$<br>商 14, 余り 5  |
| (8) $44 \div 9$<br>商 4, 余り 8   | (26) $93 \div 10$<br>商 9, 余り 3 | (44) $70 \div 3$<br>商 23, 余り 1 | (62) $18 \div 5$<br>商 3, 余り 3   |
| (9) $2 \div 8$<br>商 0, 余り 2    | (27) $56 \div 10$<br>商 5, 余り 6 | (45) $4 \div 7$<br>商 0, 余り 4   | (63) $68 \div 10$<br>商 6, 余り 8  |
| (10) $78 \div 4$<br>商 19, 余り 2 | (28) $69 \div 2$<br>商 34, 余り 1 | (46) $19 \div 5$<br>商 3, 余り 4  | (64) $53 \div 10$<br>商 5, 余り 3  |
| (11) $66 \div 1$<br>商 66, 余り 0 | (29) $94 \div 5$<br>商 18, 余り 4 | (47) $26 \div 10$<br>商 2, 余り 6 | (65) $33 \div 5$<br>商 6, 余り 3   |
| (12) $29 \div 10$<br>商 2, 余り 9 | (30) $93 \div 7$<br>商 13, 余り 2 | (48) $22 \div 2$<br>商 11, 余り 0 | (66) $30 \div 1$<br>商 30, 余り 0  |
| (13) $48 \div 8$<br>商 6, 余り 0  | (31) $91 \div 9$<br>商 10, 余り 1 | (49) $40 \div 3$<br>商 13, 余り 1 | (67) $92 \div 8$<br>商 11, 余り 4  |
| (14) $16 \div 4$<br>商 4, 余り 0  | (32) $59 \div 9$<br>商 6, 余り 5  | (50) $69 \div 10$<br>商 6, 余り 9 | (68) $100 \div 7$<br>商 14, 余り 2 |
| (15) $75 \div 7$<br>商 10, 余り 5 | (33) $65 \div 1$<br>商 65, 余り 0 | (51) $6 \div 2$<br>商 3, 余り 0   | (69) $58 \div 8$<br>商 7, 余り 2   |
| (16) $46 \div 9$<br>商 5, 余り 1  | (34) $12 \div 7$<br>商 1, 余り 5  | (52) $74 \div 4$<br>商 18, 余り 2 | (70) $16 \div 4$<br>商 4, 余り 0   |
| (17) $58 \div 5$<br>商 11, 余り 3 | (35) $45 \div 3$<br>商 15, 余り 0 | (53) $59 \div 1$<br>商 59, 余り 0 | (71) $37 \div 4$<br>商 9, 余り 1   |
| (18) $33 \div 7$<br>商 4, 余り 5  | (36) $54 \div 4$<br>商 13, 余り 2 | (54) $73 \div 9$<br>商 8, 余り 1  | (72) $89 \div 8$<br>商 11, 余り 1  |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $100 \div 4$<br>商 25, 余り 0 | (19) $79 \div 8$<br>商 9, 余り 7  | (37) $82 \div 9$<br>商 9, 余り 1  | (55) $21 \div 1$<br>商 21, 余り 0 |
| (2) $75 \div 10$<br>商 7, 余り 5  | (20) $53 \div 7$<br>商 7, 余り 4  | (38) $80 \div 3$<br>商 26, 余り 2 | (56) $71 \div 7$<br>商 10, 余り 1 |
| (3) $29 \div 5$<br>商 5, 余り 4   | (21) $1 \div 10$<br>商 0, 余り 1  | (39) $3 \div 3$<br>商 1, 余り 0   | (57) $66 \div 3$<br>商 22, 余り 0 |
| (4) $82 \div 6$<br>商 13, 余り 4  | (22) $27 \div 5$<br>商 5, 余り 2  | (40) $66 \div 7$<br>商 9, 余り 3  | (58) $94 \div 8$<br>商 11, 余り 6 |
| (5) $30 \div 10$<br>商 3, 余り 0  | (23) $16 \div 2$<br>商 8, 余り 0  | (41) $67 \div 9$<br>商 7, 余り 4  | (59) $68 \div 9$<br>商 7, 余り 5  |
| (6) $81 \div 3$<br>商 27, 余り 0  | (24) $55 \div 8$<br>商 6, 余り 7  | (42) $81 \div 8$<br>商 10, 余り 1 | (60) $51 \div 1$<br>商 51, 余り 0 |
| (7) $58 \div 4$<br>商 14, 余り 2  | (25) $25 \div 2$<br>商 12, 余り 1 | (43) $42 \div 3$<br>商 14, 余り 0 | (61) $74 \div 6$<br>商 12, 余り 2 |
| (8) $8 \div 8$<br>商 1, 余り 0    | (26) $52 \div 7$<br>商 7, 余り 3  | (44) $93 \div 5$<br>商 18, 余り 3 | (62) $23 \div 5$<br>商 4, 余り 3  |
| (9) $58 \div 1$<br>商 58, 余り 0  | (27) $27 \div 3$<br>商 9, 余り 0  | (45) $51 \div 6$<br>商 8, 余り 3  | (63) $79 \div 1$<br>商 79, 余り 0 |
| (10) $38 \div 2$<br>商 19, 余り 0 | (28) $29 \div 10$<br>商 2, 余り 9 | (46) $77 \div 5$<br>商 15, 余り 2 | (64) $66 \div 9$<br>商 7, 余り 3  |
| (11) $58 \div 6$<br>商 9, 余り 4  | (29) $1 \div 8$<br>商 0, 余り 1   | (47) $71 \div 5$<br>商 14, 余り 1 | (65) $28 \div 5$<br>商 5, 余り 3  |
| (12) $70 \div 1$<br>商 70, 余り 0 | (30) $23 \div 9$<br>商 2, 余り 5  | (48) $56 \div 8$<br>商 7, 余り 0  | (66) $7 \div 10$<br>商 0, 余り 7  |
| (13) $74 \div 9$<br>商 8, 余り 2  | (31) $41 \div 3$<br>商 13, 余り 2 | (49) $4 \div 10$<br>商 0, 余り 4  | (67) $82 \div 5$<br>商 16, 余り 2 |
| (14) $58 \div 9$<br>商 6, 余り 4  | (32) $69 \div 4$<br>商 17, 余り 1 | (50) $81 \div 8$<br>商 10, 余り 1 | (68) $11 \div 9$<br>商 1, 余り 2  |
| (15) $73 \div 1$<br>商 73, 余り 0 | (33) $8 \div 1$<br>商 8, 余り 0   | (51) $69 \div 10$<br>商 6, 余り 9 | (69) $14 \div 8$<br>商 1, 余り 6  |
| (16) $5 \div 10$<br>商 0, 余り 5  | (34) $42 \div 4$<br>商 10, 余り 2 | (52) $29 \div 9$<br>商 3, 余り 2  | (70) $95 \div 2$<br>商 47, 余り 1 |
| (17) $17 \div 5$<br>商 3, 余り 2  | (35) $42 \div 1$<br>商 42, 余り 0 | (53) $80 \div 8$<br>商 10, 余り 0 | (71) $49 \div 1$<br>商 49, 余り 0 |
| (18) $37 \div 7$<br>商 5, 余り 2  | (36) $93 \div 7$<br>商 13, 余り 2 | (54) $42 \div 9$<br>商 4, 余り 6  | (72) $79 \div 3$<br>商 26, 余り 1 |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $74 \div 6$<br>商 12, 余り 2  | (19) $97 \div 3$<br>商 32, 余り 1 | (37) $74 \div 8$<br>商 9, 余り 2  | (55) $81 \div 9$<br>商 9, 余り 0  |
| (2) $3 \div 7$<br>商 0, 余り 3    | (20) $71 \div 10$<br>商 7, 余り 1 | (38) $41 \div 8$<br>商 5, 余り 1  | (56) $40 \div 5$<br>商 8, 余り 0  |
| (3) $63 \div 5$<br>商 12, 余り 3  | (21) $62 \div 3$<br>商 20, 余り 2 | (39) $66 \div 8$<br>商 8, 余り 2  | (57) $10 \div 5$<br>商 2, 余り 0  |
| (4) $21 \div 6$<br>商 3, 余り 3   | (22) $37 \div 3$<br>商 12, 余り 1 | (40) $47 \div 3$<br>商 15, 余り 2 | (58) $16 \div 3$<br>商 5, 余り 1  |
| (5) $99 \div 10$<br>商 9, 余り 9  | (23) $85 \div 4$<br>商 21, 余り 1 | (41) $13 \div 2$<br>商 6, 余り 1  | (59) $37 \div 1$<br>商 37, 余り 0 |
| (6) $26 \div 1$<br>商 26, 余り 0  | (24) $11 \div 4$<br>商 2, 余り 3  | (42) $2 \div 5$<br>商 0, 余り 2   | (60) $78 \div 6$<br>商 13, 余り 0 |
| (7) $95 \div 4$<br>商 23, 余り 3  | (25) $32 \div 5$<br>商 6, 余り 2  | (43) $68 \div 7$<br>商 9, 余り 5  | (61) $46 \div 1$<br>商 46, 余り 0 |
| (8) $15 \div 7$<br>商 2, 余り 1   | (26) $38 \div 1$<br>商 38, 余り 0 | (44) $15 \div 1$<br>商 15, 余り 0 | (62) $74 \div 7$<br>商 10, 余り 4 |
| (9) $20 \div 3$<br>商 6, 余り 2   | (27) $85 \div 2$<br>商 42, 余り 1 | (45) $98 \div 8$<br>商 12, 余り 2 | (63) $81 \div 7$<br>商 11, 余り 4 |
| (10) $45 \div 6$<br>商 7, 余り 3  | (28) $17 \div 8$<br>商 2, 余り 1  | (46) $82 \div 8$<br>商 10, 余り 2 | (64) $66 \div 4$<br>商 16, 余り 2 |
| (11) $20 \div 1$<br>商 20, 余り 0 | (29) $58 \div 6$<br>商 9, 余り 4  | (47) $62 \div 3$<br>商 20, 余り 2 | (65) $18 \div 2$<br>商 9, 余り 0  |
| (12) $77 \div 8$<br>商 9, 余り 5  | (30) $20 \div 4$<br>商 5, 余り 0  | (48) $39 \div 10$<br>商 3, 余り 9 | (66) $59 \div 10$<br>商 5, 余り 9 |
| (13) $89 \div 2$<br>商 44, 余り 1 | (31) $94 \div 4$<br>商 23, 余り 2 | (49) $13 \div 2$<br>商 6, 余り 1  | (67) $48 \div 10$<br>商 4, 余り 8 |
| (14) $77 \div 7$<br>商 11, 余り 0 | (32) $12 \div 10$<br>商 1, 余り 2 | (50) $9 \div 2$<br>商 4, 余り 1   | (68) $35 \div 4$<br>商 8, 余り 3  |
| (15) $22 \div 2$<br>商 11, 余り 0 | (33) $78 \div 10$<br>商 7, 余り 8 | (51) $10 \div 4$<br>商 2, 余り 2  | (69) $20 \div 3$<br>商 6, 余り 2  |
| (16) $76 \div 10$<br>商 7, 余り 6 | (34) $84 \div 6$<br>商 14, 余り 0 | (52) $75 \div 1$<br>商 75, 余り 0 | (70) $89 \div 9$<br>商 9, 余り 8  |
| (17) $3 \div 2$<br>商 1, 余り 1   | (35) $71 \div 2$<br>商 35, 余り 1 | (53) $35 \div 5$<br>商 7, 余り 0  | (71) $87 \div 7$<br>商 12, 余り 3 |
| (18) $30 \div 3$<br>商 10, 余り 0 | (36) $61 \div 2$<br>商 30, 余り 1 | (54) $15 \div 5$<br>商 3, 余り 0  | (72) $40 \div 4$<br>商 10, 余り 0 |

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|----------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $47 \div 4$<br>商 11, 余り 3    | (19) $96 \div 4$<br>商 24, 余り 0 | (37) $84 \div 2$<br>商 42, 余り 0 | (55) $1 \div 7$<br>商 0, 余り 1   |
| (2) $30 \div 9$<br>商 3, 余り 3     | (20) $81 \div 5$<br>商 16, 余り 1 | (38) $77 \div 8$<br>商 9, 余り 5  | (56) $23 \div 5$<br>商 4, 余り 3  |
| (3) $80 \div 8$<br>商 10, 余り 0    | (21) $17 \div 10$<br>商 1, 余り 7 | (39) $87 \div 5$<br>商 17, 余り 2 | (57) $55 \div 7$<br>商 7, 余り 6  |
| (4) $38 \div 6$<br>商 6, 余り 2     | (22) $57 \div 10$<br>商 5, 余り 7 | (40) $74 \div 7$<br>商 10, 余り 4 | (58) $72 \div 8$<br>商 9, 余り 0  |
| (5) $11 \div 8$<br>商 1, 余り 3     | (23) $85 \div 5$<br>商 17, 余り 0 | (41) $52 \div 9$<br>商 5, 余り 7  | (59) $61 \div 6$<br>商 10, 余り 1 |
| (6) $51 \div 6$<br>商 8, 余り 3     | (24) $43 \div 10$<br>商 4, 余り 3 | (42) $69 \div 4$<br>商 17, 余り 1 | (60) $75 \div 10$<br>商 7, 余り 5 |
| (7) $88 \div 7$<br>商 12, 余り 4    | (25) $90 \div 10$<br>商 9, 余り 0 | (43) $1 \div 9$<br>商 0, 余り 1   | (61) $23 \div 1$<br>商 23, 余り 0 |
| (8) $48 \div 3$<br>商 16, 余り 0    | (26) $84 \div 5$<br>商 16, 余り 4 | (44) $38 \div 4$<br>商 9, 余り 2  | (62) $50 \div 1$<br>商 50, 余り 0 |
| (9) $24 \div 6$<br>商 4, 余り 0     | (27) $76 \div 4$<br>商 19, 余り 0 | (45) $28 \div 9$<br>商 3, 余り 1  | (63) $20 \div 7$<br>商 2, 余り 6  |
| (10) $85 \div 5$<br>商 17, 余り 0   | (28) $8 \div 7$<br>商 1, 余り 1   | (46) $96 \div 6$<br>商 16, 余り 0 | (64) $75 \div 7$<br>商 10, 余り 5 |
| (11) $19 \div 5$<br>商 3, 余り 4    | (29) $90 \div 4$<br>商 22, 余り 2 | (47) $97 \div 1$<br>商 97, 余り 0 | (65) $72 \div 5$<br>商 14, 余り 2 |
| (12) $40 \div 6$<br>商 6, 余り 4    | (30) $77 \div 9$<br>商 8, 余り 5  | (48) $1 \div 7$<br>商 0, 余り 1   | (66) $64 \div 9$<br>商 7, 余り 1  |
| (13) $100 \div 10$<br>商 10, 余り 0 | (31) $26 \div 5$<br>商 5, 余り 1  | (49) $31 \div 7$<br>商 4, 余り 3  | (67) $29 \div 5$<br>商 5, 余り 4  |
| (14) $7 \div 7$<br>商 1, 余り 0     | (32) $81 \div 10$<br>商 8, 余り 1 | (50) $99 \div 5$<br>商 19, 余り 4 | (68) $91 \div 3$<br>商 30, 余り 1 |
| (15) $10 \div 6$<br>商 1, 余り 4    | (33) $89 \div 9$<br>商 9, 余り 8  | (51) $13 \div 4$<br>商 3, 余り 1  | (69) $79 \div 9$<br>商 8, 余り 7  |
| (16) $68 \div 6$<br>商 11, 余り 2   | (34) $29 \div 4$<br>商 7, 余り 1  | (52) $26 \div 8$<br>商 3, 余り 2  | (70) $67 \div 9$<br>商 7, 余り 4  |
| (17) $68 \div 10$<br>商 6, 余り 8   | (35) $49 \div 10$<br>商 4, 余り 9 | (53) $10 \div 1$<br>商 10, 余り 0 | (71) $9 \div 4$<br>商 2, 余り 1   |
| (18) $76 \div 2$<br>商 38, 余り 0   | (36) $65 \div 4$<br>商 16, 余り 1 | (54) $95 \div 9$<br>商 10, 余り 5 | (72) $45 \div 9$<br>商 5, 余り 0  |

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| (1) $92 \div 1$<br>商 92, 余り 0  | (19) $46 \div 2$<br>商 23, 余り 0 | (37) $9 \div 4$<br>商 2, 余り 1    | (55) $72 \div 9$<br>商 8, 余り 0  |
| (2) $19 \div 8$<br>商 2, 余り 3   | (20) $70 \div 10$<br>商 7, 余り 0 | (38) $84 \div 5$<br>商 16, 余り 4  | (56) $17 \div 3$<br>商 5, 余り 2  |
| (3) $25 \div 3$<br>商 8, 余り 1   | (21) $3 \div 10$<br>商 0, 余り 3  | (39) $14 \div 1$<br>商 14, 余り 0  | (57) $65 \div 10$<br>商 6, 余り 5 |
| (4) $41 \div 7$<br>商 5, 余り 6   | (22) $65 \div 4$<br>商 16, 余り 1 | (40) $5 \div 2$<br>商 2, 余り 1    | (58) $14 \div 6$<br>商 2, 余り 2  |
| (5) $96 \div 3$<br>商 32, 余り 0  | (23) $90 \div 1$<br>商 90, 余り 0 | (41) $23 \div 5$<br>商 4, 余り 3   | (59) $17 \div 3$<br>商 5, 余り 2  |
| (6) $79 \div 5$<br>商 15, 余り 4  | (24) $97 \div 8$<br>商 12, 余り 1 | (42) $18 \div 8$<br>商 2, 余り 2   | (60) $43 \div 4$<br>商 10, 余り 3 |
| (7) $64 \div 6$<br>商 10, 余り 4  | (25) $68 \div 3$<br>商 22, 余り 2 | (43) $99 \div 8$<br>商 12, 余り 3  | (61) $12 \div 7$<br>商 1, 余り 5  |
| (8) $75 \div 9$<br>商 8, 余り 3   | (26) $18 \div 2$<br>商 9, 余り 0  | (44) $77 \div 9$<br>商 8, 余り 5   | (62) $72 \div 2$<br>商 36, 余り 0 |
| (9) $54 \div 8$<br>商 6, 余り 6   | (27) $41 \div 7$<br>商 5, 余り 6  | (45) $90 \div 1$<br>商 90, 余り 0  | (63) $68 \div 5$<br>商 13, 余り 3 |
| (10) $45 \div 10$<br>商 4, 余り 5 | (28) $70 \div 9$<br>商 7, 余り 7  | (46) $81 \div 8$<br>商 10, 余り 1  | (64) $41 \div 8$<br>商 5, 余り 1  |
| (11) $68 \div 1$<br>商 68, 余り 0 | (29) $95 \div 8$<br>商 11, 余り 7 | (47) $100 \div 2$<br>商 50, 余り 0 | (65) $16 \div 7$<br>商 2, 余り 2  |
| (12) $37 \div 4$<br>商 9, 余り 1  | (30) $50 \div 10$<br>商 5, 余り 0 | (48) $38 \div 5$<br>商 7, 余り 3   | (66) $73 \div 7$<br>商 10, 余り 3 |
| (13) $34 \div 10$<br>商 3, 余り 4 | (31) $49 \div 6$<br>商 8, 余り 1  | (49) $30 \div 3$<br>商 10, 余り 0  | (67) $65 \div 9$<br>商 7, 余り 2  |
| (14) $98 \div 7$<br>商 14, 余り 0 | (32) $0 \div 9$<br>商 0, 余り 0   | (50) $93 \div 4$<br>商 23, 余り 1  | (68) $51 \div 4$<br>商 12, 余り 3 |
| (15) $59 \div 6$<br>商 9, 余り 5  | (33) $67 \div 3$<br>商 22, 余り 1 | (51) $42 \div 9$<br>商 4, 余り 6   | (69) $41 \div 8$<br>商 5, 余り 1  |
| (16) $56 \div 6$<br>商 9, 余り 2  | (34) $76 \div 3$<br>商 25, 余り 1 | (52) $44 \div 4$<br>商 11, 余り 0  | (70) $85 \div 5$<br>商 17, 余り 0 |
| (17) $47 \div 10$<br>商 4, 余り 7 | (35) $53 \div 10$<br>商 5, 余り 3 | (53) $13 \div 5$<br>商 2, 余り 3   | (71) $53 \div 5$<br>商 10, 余り 3 |
| (18) $99 \div 6$<br>商 16, 余り 3 | (36) $52 \div 2$<br>商 26, 余り 0 | (54) $73 \div 8$<br>商 9, 余り 1   | (72) $88 \div 2$<br>商 44, 余り 0 |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $26 \div 10$<br>商 2, 余り 6  | (19) $39 \div 5$<br>商 7, 余り 4  | (37) $19 \div 7$<br>商 2, 余り 5  | (55) $43 \div 8$<br>商 5, 余り 3  |
| (2) $43 \div 2$<br>商 21, 余り 1  | (20) $73 \div 7$<br>商 10, 余り 3 | (38) $17 \div 10$<br>商 1, 余り 7 | (56) $10 \div 7$<br>商 1, 余り 3  |
| (3) $61 \div 7$<br>商 8, 余り 5   | (21) $66 \div 1$<br>商 66, 余り 0 | (39) $80 \div 8$<br>商 10, 余り 0 | (57) $92 \div 6$<br>商 15, 余り 2 |
| (4) $81 \div 2$<br>商 40, 余り 1  | (22) $9 \div 6$<br>商 1, 余り 3   | (40) $67 \div 3$<br>商 22, 余り 1 | (58) $26 \div 6$<br>商 4, 余り 2  |
| (5) $21 \div 8$<br>商 2, 余り 5   | (23) $45 \div 4$<br>商 11, 余り 1 | (41) $39 \div 5$<br>商 7, 余り 4  | (59) $95 \div 6$<br>商 15, 余り 5 |
| (6) $69 \div 1$<br>商 69, 余り 0  | (24) $72 \div 1$<br>商 72, 余り 0 | (42) $18 \div 5$<br>商 3, 余り 3  | (60) $77 \div 7$<br>商 11, 余り 0 |
| (7) $11 \div 1$<br>商 11, 余り 0  | (25) $2 \div 6$<br>商 0, 余り 2   | (43) $46 \div 9$<br>商 5, 余り 1  | (61) $31 \div 7$<br>商 4, 余り 3  |
| (8) $60 \div 4$<br>商 15, 余り 0  | (26) $4 \div 1$<br>商 4, 余り 0   | (44) $59 \div 8$<br>商 7, 余り 3  | (62) $86 \div 3$<br>商 28, 余り 2 |
| (9) $1 \div 7$<br>商 0, 余り 1    | (27) $91 \div 2$<br>商 45, 余り 1 | (45) $68 \div 10$<br>商 6, 余り 8 | (63) $6 \div 5$<br>商 1, 余り 1   |
| (10) $69 \div 9$<br>商 7, 余り 6  | (28) $67 \div 4$<br>商 16, 余り 3 | (46) $78 \div 9$<br>商 8, 余り 6  | (64) $61 \div 4$<br>商 15, 余り 1 |
| (11) $66 \div 3$<br>商 22, 余り 0 | (29) $90 \div 6$<br>商 15, 余り 0 | (47) $82 \div 3$<br>商 27, 余り 1 | (65) $57 \div 9$<br>商 6, 余り 3  |
| (12) $1 \div 10$<br>商 0, 余り 1  | (30) $84 \div 5$<br>商 16, 余り 4 | (48) $20 \div 6$<br>商 3, 余り 2  | (66) $32 \div 5$<br>商 6, 余り 2  |
| (13) $87 \div 10$<br>商 8, 余り 7 | (31) $31 \div 5$<br>商 6, 余り 1  | (49) $73 \div 7$<br>商 10, 余り 3 | (67) $67 \div 4$<br>商 16, 余り 3 |
| (14) $67 \div 2$<br>商 33, 余り 1 | (32) $39 \div 3$<br>商 13, 余り 0 | (50) $81 \div 5$<br>商 16, 余り 1 | (68) $5 \div 6$<br>商 0, 余り 5   |
| (15) $13 \div 3$<br>商 4, 余り 1  | (33) $42 \div 6$<br>商 7, 余り 0  | (51) $3 \div 7$<br>商 0, 余り 3   | (69) $44 \div 10$<br>商 4, 余り 4 |
| (16) $84 \div 8$<br>商 10, 余り 4 | (34) $28 \div 10$<br>商 2, 余り 8 | (52) $18 \div 10$<br>商 1, 余り 8 | (70) $82 \div 9$<br>商 9, 余り 1  |
| (17) $43 \div 9$<br>商 4, 余り 7  | (35) $23 \div 3$<br>商 7, 余り 2  | (53) $92 \div 3$<br>商 30, 余り 2 | (71) $81 \div 8$<br>商 10, 余り 1 |
| (18) $70 \div 6$<br>商 11, 余り 4 | (36) $77 \div 4$<br>商 19, 余り 1 | (54) $92 \div 3$<br>商 30, 余り 2 | (72) $55 \div 9$<br>商 6, 余り 1  |

- |                                |                                |                                 |                                |
|--------------------------------|--------------------------------|---------------------------------|--------------------------------|
| (1) $2 \div 3$<br>商 0, 余り 2    | (19) $32 \div 9$<br>商 3, 余り 5  | (37) $24 \div 3$<br>商 8, 余り 0   | (55) $29 \div 8$<br>商 3, 余り 5  |
| (2) $7 \div 7$<br>商 1, 余り 0    | (20) $42 \div 3$<br>商 14, 余り 0 | (38) $82 \div 7$<br>商 11, 余り 5  | (56) $76 \div 9$<br>商 8, 余り 4  |
| (3) $29 \div 6$<br>商 4, 余り 5   | (21) $9 \div 2$<br>商 4, 余り 1   | (39) $10 \div 8$<br>商 1, 余り 2   | (57) $81 \div 4$<br>商 20, 余り 1 |
| (4) $78 \div 3$<br>商 26, 余り 0  | (22) $50 \div 10$<br>商 5, 余り 0 | (40) $76 \div 6$<br>商 12, 余り 4  | (58) $10 \div 8$<br>商 1, 余り 2  |
| (5) $14 \div 6$<br>商 2, 余り 2   | (23) $7 \div 6$<br>商 1, 余り 1   | (41) $51 \div 3$<br>商 17, 余り 0  | (59) $49 \div 7$<br>商 7, 余り 0  |
| (6) $16 \div 4$<br>商 4, 余り 0   | (24) $65 \div 8$<br>商 8, 余り 1  | (42) $78 \div 3$<br>商 26, 余り 0  | (60) $15 \div 5$<br>商 3, 余り 0  |
| (7) $13 \div 2$<br>商 6, 余り 1   | (25) $36 \div 8$<br>商 4, 余り 4  | (43) $88 \div 6$<br>商 14, 余り 4  | (61) $6 \div 8$<br>商 0, 余り 6   |
| (8) $44 \div 1$<br>商 44, 余り 0  | (26) $87 \div 8$<br>商 10, 余り 7 | (44) $66 \div 8$<br>商 8, 余り 2   | (62) $13 \div 9$<br>商 1, 余り 4  |
| (9) $17 \div 1$<br>商 17, 余り 0  | (27) $24 \div 2$<br>商 12, 余り 0 | (45) $21 \div 1$<br>商 21, 余り 0  | (63) $40 \div 7$<br>商 5, 余り 5  |
| (10) $92 \div 5$<br>商 18, 余り 2 | (28) $81 \div 10$<br>商 8, 余り 1 | (46) $66 \div 2$<br>商 33, 余り 0  | (64) $38 \div 4$<br>商 9, 余り 2  |
| (11) $2 \div 1$<br>商 2, 余り 0   | (29) $36 \div 1$<br>商 36, 余り 0 | (47) $39 \div 8$<br>商 4, 余り 7   | (65) $69 \div 5$<br>商 13, 余り 4 |
| (12) $21 \div 6$<br>商 3, 余り 3  | (30) $27 \div 1$<br>商 27, 余り 0 | (48) $67 \div 10$<br>商 6, 余り 7  | (66) $80 \div 2$<br>商 40, 余り 0 |
| (13) $70 \div 9$<br>商 7, 余り 7  | (31) $8 \div 10$<br>商 0, 余り 8  | (49) $30 \div 7$<br>商 4, 余り 2   | (67) $58 \div 4$<br>商 14, 余り 2 |
| (14) $81 \div 6$<br>商 13, 余り 3 | (32) $98 \div 4$<br>商 24, 余り 2 | (50) $76 \div 2$<br>商 38, 余り 0  | (68) $85 \div 4$<br>商 21, 余り 1 |
| (15) $45 \div 6$<br>商 7, 余り 3  | (33) $79 \div 3$<br>商 26, 余り 1 | (51) $100 \div 3$<br>商 33, 余り 1 | (69) $81 \div 10$<br>商 8, 余り 1 |
| (16) $23 \div 1$<br>商 23, 余り 0 | (34) $27 \div 4$<br>商 6, 余り 3  | (52) $62 \div 9$<br>商 6, 余り 8   | (70) $64 \div 6$<br>商 10, 余り 4 |
| (17) $60 \div 8$<br>商 7, 余り 4  | (35) $56 \div 4$<br>商 14, 余り 0 | (53) $86 \div 8$<br>商 10, 余り 6  | (71) $78 \div 8$<br>商 9, 余り 6  |
| (18) $7 \div 3$<br>商 2, 余り 1   | (36) $68 \div 3$<br>商 22, 余り 2 | (54) $60 \div 1$<br>商 60, 余り 0  | (72) $97 \div 6$<br>商 16, 余り 1 |



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|--------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $5 \div 6$<br>商 0, 余り 5    | (19) $24 \div 5$<br>商 4, 余り 4  | (37) $65 \div 2$<br>商 32, 余り 1 | (55) $67 \div 2$<br>商 33, 余り 1  |
| (2) $76 \div 2$<br>商 38, 余り 0  | (20) $95 \div 2$<br>商 47, 余り 1 | (38) $54 \div 10$<br>商 5, 余り 4 | (56) $33 \div 7$<br>商 4, 余り 5   |
| (3) $57 \div 6$<br>商 9, 余り 3   | (21) $94 \div 10$<br>商 9, 余り 4 | (39) $33 \div 3$<br>商 11, 余り 0 | (57) $34 \div 10$<br>商 3, 余り 4  |
| (4) $1 \div 9$<br>商 0, 余り 1    | (22) $65 \div 8$<br>商 8, 余り 1  | (40) $73 \div 2$<br>商 36, 余り 1 | (58) $96 \div 1$<br>商 96, 余り 0  |
| (5) $41 \div 2$<br>商 20, 余り 1  | (23) $64 \div 9$<br>商 7, 余り 1  | (41) $53 \div 4$<br>商 13, 余り 1 | (59) $95 \div 10$<br>商 9, 余り 5  |
| (6) $75 \div 1$<br>商 75, 余り 0  | (24) $10 \div 1$<br>商 10, 余り 0 | (42) $87 \div 9$<br>商 9, 余り 6  | (60) $81 \div 6$<br>商 13, 余り 3  |
| (7) $30 \div 8$<br>商 3, 余り 6   | (25) $9 \div 2$<br>商 4, 余り 1   | (43) $23 \div 1$<br>商 23, 余り 0 | (61) $43 \div 10$<br>商 4, 余り 3  |
| (8) $52 \div 3$<br>商 17, 余り 1  | (26) $85 \div 5$<br>商 17, 余り 0 | (44) $32 \div 6$<br>商 5, 余り 2  | (62) $13 \div 4$<br>商 3, 余り 1   |
| (9) $86 \div 7$<br>商 12, 余り 2  | (27) $23 \div 3$<br>商 7, 余り 2  | (45) $21 \div 2$<br>商 10, 余り 1 | (63) $48 \div 1$<br>商 48, 余り 0  |
| (10) $59 \div 9$<br>商 6, 余り 5  | (28) $47 \div 10$<br>商 4, 余り 7 | (46) $28 \div 4$<br>商 7, 余り 0  | (64) $100 \div 4$<br>商 25, 余り 0 |
| (11) $75 \div 3$<br>商 25, 余り 0 | (29) $52 \div 4$<br>商 13, 余り 0 | (47) $33 \div 4$<br>商 8, 余り 1  | (65) $50 \div 10$<br>商 5, 余り 0  |
| (12) $45 \div 2$<br>商 22, 余り 1 | (30) $57 \div 5$<br>商 11, 余り 2 | (48) $36 \div 9$<br>商 4, 余り 0  | (66) $66 \div 10$<br>商 6, 余り 6  |
| (13) $91 \div 9$<br>商 10, 余り 1 | (31) $98 \div 5$<br>商 19, 余り 3 | (49) $15 \div 3$<br>商 5, 余り 0  | (67) $46 \div 6$<br>商 7, 余り 4   |
| (14) $21 \div 9$<br>商 2, 余り 3  | (32) $84 \div 3$<br>商 28, 余り 0 | (50) $0 \div 2$<br>商 0, 余り 0   | (68) $70 \div 4$<br>商 17, 余り 2  |
| (15) $84 \div 3$<br>商 28, 余り 0 | (33) $86 \div 2$<br>商 43, 余り 0 | (51) $93 \div 3$<br>商 31, 余り 0 | (69) $82 \div 8$<br>商 10, 余り 2  |
| (16) $51 \div 7$<br>商 7, 余り 2  | (34) $27 \div 5$<br>商 5, 余り 2  | (52) $69 \div 3$<br>商 23, 余り 0 | (70) $55 \div 2$<br>商 27, 余り 1  |
| (17) $68 \div 6$<br>商 11, 余り 2 | (35) $78 \div 3$<br>商 26, 余り 0 | (53) $65 \div 5$<br>商 13, 余り 0 | (71) $16 \div 1$<br>商 16, 余り 0  |
| (18) $37 \div 6$<br>商 6, 余り 1  | (36) $58 \div 8$<br>商 7, 余り 2  | (54) $96 \div 5$<br>商 19, 余り 1 | (72) $50 \div 7$<br>商 7, 余り 1   |

## 1.2 整数

### 1.2.1 加減法

整数の和や差を求める問題。

- |                    |                    |                    |                     |
|--------------------|--------------------|--------------------|---------------------|
| (1) $-27 - 61$     | (20) $67 + (-8)$   | (39) $-60 - 8$     | (58) $28 + 93$      |
| (2) $-19 - (-83)$  | (21) $-8 - (-63)$  | (40) $-82 + 26$    | (59) $-42 - (-84)$  |
| (3) $-84 + (-79)$  | (22) $-78 + (-39)$ | (41) $17 - 38$     | (60) $66 - 36$      |
| (4) $-65 - 46$     | (23) $-36 - 51$    | (42) $-70 + (-43)$ | (61) $-96 - (-95)$  |
| (5) $83 - 66$      | (24) $88 + 26$     | (43) $40 - (-73)$  | (62) $-100 + (-23)$ |
| (6) $-77 + (-95)$  | (25) $-84 - 18$    | (44) $26 + 45$     | (63) $-85 + (-69)$  |
| (7) $-8 - (-18)$   | (26) $-8 + 42$     | (45) $11 + (-86)$  | (64) $-34 + 89$     |
| (8) $5 + (-73)$    | (27) $-51 + 41$    | (46) $38 + 88$     | (65) $-78 + 45$     |
| (9) $-17 + 37$     | (28) $50 + 86$     | (47) $67 - 35$     | (66) $6 - (-56)$    |
| (10) $-95 - (-11)$ | (29) $16 - 60$     | (48) $-79 - (-58)$ | (67) $6 - (-40)$    |
| (11) $64 + 37$     | (30) $-1 + (-16)$  | (49) $-15 - 68$    | (68) $67 + (-39)$   |
| (12) $-56 + 76$    | (31) $69 + (-30)$  | (50) $-3 - 59$     | (69) $89 - (-90)$   |
| (13) $-79 - 46$    | (32) $-51 + 84$    | (51) $-86 - 26$    | (70) $-45 + (-34)$  |
| (14) $-10 - (-1)$  | (33) $-99 - (-79)$ | (52) $72 + (-66)$  | (71) $-38 + 47$     |
| (15) $-30 - (-27)$ | (34) $86 + (-24)$  | (53) $72 - 79$     | (72) $36 - 34$      |
| (16) $7 - 16$      | (35) $-52 + 41$    | (54) $-36 - 70$    | (73) $64 + 88$      |
| (17) $-35 + 3$     | (36) $-5 + 47$     | (55) $61 + (-25)$  | (74) $-38 + 32$     |
| (18) $-13 + 80$    | (37) $29 - 15$     | (56) $-53 - (-66)$ | (75) $9 + (-87)$    |
| (19) $-79 + (-38)$ | (38) $29 + (-40)$  | (57) $-83 + (-98)$ | (76) $59 + 45$      |

|                     |                    |                    |                    |
|---------------------|--------------------|--------------------|--------------------|
| (1) $-3 + (-46)$    | (20) $10 + (-59)$  | (39) $-40 - (-40)$ | (58) $-18 + 28$    |
| (2) $2 - 21$        | (21) $-53 + 37$    | (40) $74 - (-33)$  | (59) $-69 + (-10)$ |
| (3) $64 - (-42)$    | (22) $-95 - (-44)$ | (41) $68 - (-57)$  | (60) $70 + (-22)$  |
| (4) $31 - 77$       | (23) $28 + (-13)$  | (42) $-68 - 4$     | (61) $82 - 39$     |
| (5) $-7 - 20$       | (24) $-13 + 98$    | (43) $-36 + (-11)$ | (62) $33 + 88$     |
| (6) $-41 + (-44)$   | (25) $-85 - 75$    | (44) $-75 + 42$    | (63) $-54 - 41$    |
| (7) $-27 - (-96)$   | (26) $-42 - (-72)$ | (45) $49 + (-4)$   | (64) $7 + (-70)$   |
| (8) $-77 - (-6)$    | (27) $51 + (-56)$  | (46) $61 - 47$     | (65) $-38 + 81$    |
| (9) $46 + (-32)$    | (28) $36 - (-34)$  | (47) $-26 - 21$    | (66) $50 + 37$     |
| (10) $-33 + 99$     | (29) $-81 + (-79)$ | (48) $-74 + 29$    | (67) $88 + 94$     |
| (11) $-65 + (-37)$  | (30) $-31 + 4$     | (49) $-71 - (-90)$ | (68) $82 - (-41)$  |
| (12) $95 + 41$      | (31) $-83 - (-63)$ | (50) $95 - (-37)$  | (69) $-62 + 90$    |
| (13) $-100 + (-11)$ | (32) $34 + 48$     | (51) $66 - 61$     | (70) $85 - (-18)$  |
| (14) $33 - (-81)$   | (33) $-78 - 73$    | (52) $96 + (-67)$  | (71) $96 - 25$     |
| (15) $90 - 17$      | (34) $-34 + (-65)$ | (53) $58 + 18$     | (72) $-10 - 35$    |
| (16) $28 + 13$      | (35) $-90 + 24$    | (54) $64 - (-16)$  | (73) $88 - 100$    |
| (17) $77 - 7$       | (36) $29 - 27$     | (55) $-7 + (-17)$  | (74) $-12 + 96$    |
| (18) $-67 + 89$     | (37) $81 - (-65)$  | (56) $22 + (-57)$  | (75) $64 - 54$     |
| (19) $-84 + 13$     | (38) $-88 + (-33)$ | (57) $14 + 14$     | (76) $45 + 41$     |

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|--------------------|---------------------|--------------------|---------------------|
| (1) $-92 + (-6)$   | (20) $-8 - 82$      | (39) $-73 - (-91)$ | (58) $-11 - (-71)$  |
| (2) $-63 + (-87)$  | (21) $78 + (-24)$   | (40) $-61 + (-77)$ | (59) $89 + (-4)$    |
| (3) $-33 + (-36)$  | (22) $50 - 56$      | (41) $-68 - (-14)$ | (60) $-71 - 23$     |
| (4) $-100 + 14$    | (23) $61 - (-93)$   | (42) $48 + (-77)$  | (61) $-100 + (-26)$ |
| (5) $-70 - 82$     | (24) $-54 - 73$     | (43) $-7 + (-50)$  | (62) $92 + 29$      |
| (6) $-31 - 58$     | (25) $39 - 74$      | (44) $70 + 19$     | (63) $-70 - (-36)$  |
| (7) $-65 + 33$     | (26) $-53 - 46$     | (45) $24 + 4$      | (64) $-65 - 70$     |
| (8) $41 - 41$      | (27) $-68 - (-100)$ | (46) $-63 - 46$    | (65) $41 - (-99)$   |
| (9) $77 + 95$      | (28) $-21 - (-20)$  | (47) $-74 - 3$     | (66) $-41 - 45$     |
| (10) $81 + 4$      | (29) $-23 - (-4)$   | (48) $37 - (-91)$  | (67) $-52 + 82$     |
| (11) $-29 + (-92)$ | (30) $89 + 100$     | (49) $-39 - 73$    | (68) $51 + (-61)$   |
| (12) $42 + 20$     | (31) $10 - 13$      | (50) $-92 - 29$    | (69) $-21 - (-49)$  |
| (13) $-87 - 13$    | (32) $-86 - (-91)$  | (51) $89 + (-53)$  | (70) $57 + 37$      |
| (14) $94 + (-56)$  | (33) $14 + 59$      | (52) $99 - 75$     | (71) $59 - 50$      |
| (15) $75 - 65$     | (34) $86 - 66$      | (53) $-74 + (-13)$ | (72) $-35 + (-81)$  |
| (16) $82 + 78$     | (35) $-4 - (-82)$   | (54) $-37 + (-80)$ | (73) $-72 - 91$     |
| (17) $-25 + 87$    | (36) $20 - 73$      | (55) $-95 - 81$    | (74) $10 - (-26)$   |
| (18) $-14 + 44$    | (37) $11 - (-67)$   | (56) $76 + (-98)$  | (75) $35 + (-59)$   |
| (19) $-50 + (-45)$ | (38) $-22 - 20$     | (57) $-37 - (-3)$  | (76) $100 - (-79)$  |

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|--------------------|--------------------|---------------------|--------------------|
| (1) $88 + 26$      | (20) $-94 + (-14)$ | (39) $73 - 60$      | (58) $-81 - (-32)$ |
| (2) $-99 - (-95)$  | (21) $-22 - (-35)$ | (40) $-69 + (-51)$  | (59) $61 - (-20)$  |
| (3) $18 - 92$      | (22) $-25 + 52$    | (41) $83 - (-44)$   | (60) $53 - 89$     |
| (4) $-41 - (-93)$  | (23) $-94 - (-8)$  | (42) $-100 + (-97)$ | (61) $84 + 35$     |
| (5) $0 + 82$       | (24) $-72 - (-67)$ | (43) $9 + 37$       | (62) $74 + (-49)$  |
| (6) $-31 - 87$     | (25) $34 + 43$     | (44) $73 + 96$      | (63) $-69 - 65$    |
| (7) $-48 - 35$     | (26) $80 - 52$     | (45) $-48 - 16$     | (64) $-12 + (-64)$ |
| (8) $15 + 82$      | (27) $-32 + (-95)$ | (46) $2 + 71$       | (65) $43 + 61$     |
| (9) $-54 + 68$     | (28) $11 - (-72)$  | (47) $-6 - (-88)$   | (66) $19 - 49$     |
| (10) $-98 + (-45)$ | (29) $2 + 96$      | (48) $-89 + (-86)$  | (67) $-9 - (-30)$  |
| (11) $19 + (-97)$  | (30) $-16 - 56$    | (49) $-16 + 75$     | (68) $-66 + (-58)$ |
| (12) $-6 - (-26)$  | (31) $20 + (-32)$  | (50) $-93 - (-94)$  | (69) $64 - (-89)$  |
| (13) $-24 + 95$    | (32) $84 - (-12)$  | (51) $-79 - 57$     | (70) $-52 - (-8)$  |
| (14) $-81 + (-88)$ | (33) $-98 - (-38)$ | (52) $-72 - 79$     | (71) $-68 - (-68)$ |
| (15) $62 - 4$      | (34) $91 + 85$     | (53) $20 + 93$      | (72) $14 + 31$     |
| (16) $83 - (-8)$   | (35) $-95 - (-12)$ | (54) $42 - (-95)$   | (73) $92 + 86$     |
| (17) $11 + 25$     | (36) $69 - (-90)$  | (55) $-80 + (-58)$  | (74) $-97 - 68$    |
| (18) $91 + (-58)$  | (37) $-93 - 28$    | (56) $12 - (-74)$   | (75) $57 - 15$     |
| (19) $37 - (-80)$  | (38) $-79 - (-32)$ | (57) $98 - (-23)$   | (76) $-31 - (-13)$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $-55 - (-76)$  | (20) $20 - (-8)$   | (39) $-92 - 64$    | (58) $33 - (-23)$  |
| (2) $4 + (-15)$    | (21) $57 - (-64)$  | (40) $93 - 45$     | (59) $20 - (-47)$  |
| (3) $-90 + 55$     | (22) $-82 + (-59)$ | (41) $64 - 91$     | (60) $48 - (-11)$  |
| (4) $27 - (-51)$   | (23) $-34 - 4$     | (42) $43 - 10$     | (61) $79 + (-46)$  |
| (5) $92 + 14$      | (24) $-73 + 64$    | (43) $-37 - (-76)$ | (62) $-81 - (-18)$ |
| (6) $60 + 57$      | (25) $-86 + (-92)$ | (44) $-90 + (-45)$ | (63) $82 + (-3)$   |
| (7) $100 + 26$     | (26) $-41 - 39$    | (45) $-26 + (-10)$ | (64) $-46 - (-52)$ |
| (8) $71 - (-46)$   | (27) $-41 - (-74)$ | (46) $-43 + 59$    | (65) $66 - (-3)$   |
| (9) $88 + 62$      | (28) $-83 + 59$    | (47) $-35 - (-1)$  | (66) $71 + (-24)$  |
| (10) $-2 + (-25)$  | (29) $91 - 55$     | (48) $15 + (-14)$  | (67) $-77 + (-90)$ |
| (11) $17 - (-87)$  | (30) $45 + 79$     | (49) $13 + 29$     | (68) $11 - 80$     |
| (12) $47 - 7$      | (31) $65 + 42$     | (50) $-40 - (-19)$ | (69) $-63 - (-12)$ |
| (13) $49 - 38$     | (32) $61 + (-35)$  | (51) $-82 + (-58)$ | (70) $-98 + 19$    |
| (14) $45 + (-92)$  | (33) $-90 - 75$    | (52) $-77 - (-81)$ | (71) $66 + (-42)$  |
| (15) $72 - (-55)$  | (34) $78 + (-56)$  | (53) $50 - (-47)$  | (72) $-78 - (-25)$ |
| (16) $-42 - (-61)$ | (35) $-72 - (-93)$ | (54) $-49 - 97$    | (73) $64 - 15$     |
| (17) $57 - 89$     | (36) $2 - (-65)$   | (55) $-55 - (-57)$ | (74) $71 - (-57)$  |
| (18) $-62 - 90$    | (37) $-95 + (-53)$ | (56) $85 - (-12)$  | (75) $57 + 91$     |
| (19) $-46 - 100$   | (38) $80 + 50$     | (57) $-73 - (-29)$ | (76) $57 - 13$     |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $-72 - 43$     | (20) $-16 + (-77)$ | (39) $-52 + 8$     | (58) $73 - 6$      |
| (2) $-36 + (-20)$  | (21) $11 + (-27)$  | (40) $57 + (-11)$  | (59) $68 - (-6)$   |
| (3) $64 + (-17)$   | (22) $-45 + (-28)$ | (41) $16 - 12$     | (60) $-62 - (-9)$  |
| (4) $94 - (-56)$   | (23) $-31 - 39$    | (42) $37 + 52$     | (61) $60 + (-90)$  |
| (5) $-96 - 10$     | (24) $-1 - 67$     | (43) $59 - 61$     | (62) $49 - (-77)$  |
| (6) $-34 - 2$      | (25) $-84 + 72$    | (44) $-1 - 4$      | (63) $11 - (-15)$  |
| (7) $-32 - 29$     | (26) $15 + (-76)$  | (45) $-68 - 31$    | (64) $41 + (-76)$  |
| (8) $42 - (-67)$   | (27) $-30 - (-4)$  | (46) $96 - (-20)$  | (65) $96 + (-37)$  |
| (9) $-38 - 97$     | (28) $-6 - (-61)$  | (47) $46 - 45$     | (66) $-51 + 32$    |
| (10) $46 - 12$     | (29) $13 + 31$     | (48) $-67 - (-35)$ | (67) $96 + 64$     |
| (11) $10 + (-47)$  | (30) $8 + (-10)$   | (49) $27 - (-76)$  | (68) $-44 - 57$    |
| (12) $80 - (-42)$  | (31) $66 - 98$     | (50) $62 - (-81)$  | (69) $-95 + (-38)$ |
| (13) $-68 + 48$    | (32) $-41 - 66$    | (51) $-42 + (-13)$ | (70) $-82 - (-59)$ |
| (14) $-78 + 32$    | (33) $-99 + (-77)$ | (52) $-19 + (-62)$ | (71) $-3 + (-25)$  |
| (15) $93 - 84$     | (34) $-5 - 92$     | (53) $85 + 100$    | (72) $90 + (-55)$  |
| (16) $66 - 86$     | (35) $98 - (-53)$  | (54) $85 - 35$     | (73) $63 - (-55)$  |
| (17) $-92 + 50$    | (36) $-48 + (-17)$ | (55) $42 - 5$      | (74) $25 + 15$     |
| (18) $-65 - (-49)$ | (37) $95 - 0$      | (56) $54 + (-100)$ | (75) $54 - (-74)$  |
| (19) $-97 - (-65)$ | (38) $44 - 48$     | (57) $-1 - (-90)$  | (76) $27 + 47$     |



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|--------------------|--------------------|--------------------|--------------------|
| (1) $81 - (-85)$   | (20) $4 + 13$      | (39) $77 + (-32)$  | (58) $97 + (-15)$  |
| (2) $38 - 85$      | (21) $19 - 83$     | (40) $-69 + 0$     | (59) $56 + 55$     |
| (3) $-4 - (-26)$   | (22) $87 - (-12)$  | (41) $1 - 60$      | (60) $-92 - (-93)$ |
| (4) $11 + 95$      | (23) $-7 - (-6)$   | (42) $-65 - (-64)$ | (61) $-83 + (-25)$ |
| (5) $68 - (-27)$   | (24) $-51 + 31$    | (43) $-37 + 100$   | (62) $-36 - (-39)$ |
| (6) $59 - 38$      | (25) $65 - 76$     | (44) $-97 + 75$    | (63) $-10 - (-11)$ |
| (7) $-35 + (-35)$  | (26) $17 - (-96)$  | (45) $46 + 28$     | (64) $80 + 58$     |
| (8) $35 + (-50)$   | (27) $67 + 5$      | (46) $-2 + 10$     | (65) $-91 + (-16)$ |
| (9) $82 - 56$      | (28) $76 + (-80)$  | (47) $2 - 37$      | (66) $15 - 13$     |
| (10) $-22 - (-73)$ | (29) $-94 + 48$    | (48) $15 + 63$     | (67) $-71 - (-81)$ |
| (11) $89 - 5$      | (30) $48 - 40$     | (49) $-42 - 3$     | (68) $96 + (-19)$  |
| (12) $-82 - (-24)$ | (31) $-24 - (-52)$ | (50) $78 - (-79)$  | (69) $6 - (-67)$   |
| (13) $14 - (-20)$  | (32) $-66 - (-11)$ | (51) $16 - 60$     | (70) $85 + (-78)$  |
| (14) $74 - 24$     | (33) $-60 + 79$    | (52) $52 - (-41)$  | (71) $-6 + 85$     |
| (15) $-34 - 93$    | (34) $-96 + 68$    | (53) $82 + (-12)$  | (72) $-22 + 42$    |
| (16) $-67 - (-44)$ | (35) $-33 - (-61)$ | (54) $-45 - 98$    | (73) $-41 + (-77)$ |
| (17) $-95 - (-14)$ | (36) $95 - 43$     | (55) $77 - 8$      | (74) $-38 - 49$    |
| (18) $92 + 44$     | (37) $13 + (-39)$  | (56) $-62 + 94$    | (75) $-66 - (-71)$ |
| (19) $86 + (-50)$  | (38) $-100 + 57$   | (57) $-51 - (-93)$ | (76) $-69 - (-3)$  |

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|--------------------|--------------------|---------------------|--------------------|
| (1) $48 + 30$      | (20) $14 + (-84)$  | (39) $97 + (-7)$    | (58) $-41 - (-49)$ |
| (2) $15 + 3$       | (21) $-80 + (-1)$  | (40) $-72 + 20$     | (59) $-20 + 99$    |
| (3) $-35 + 27$     | (22) $-89 - (-99)$ | (41) $-52 + (-56)$  | (60) $-59 + 92$    |
| (4) $37 + (-21)$   | (23) $-55 + (-49)$ | (42) $60 + 12$      | (61) $80 + 4$      |
| (5) $7 - (-58)$    | (24) $74 - (-49)$  | (43) $-50 + (-53)$  | (62) $-95 - 12$    |
| (6) $-70 - 37$     | (25) $-70 + (-15)$ | (44) $-46 - 4$      | (63) $69 + (-21)$  |
| (7) $-31 - (-25)$  | (26) $-71 + (-94)$ | (45) $94 - (-72)$   | (64) $-50 - (-40)$ |
| (8) $-92 + (-12)$  | (27) $-30 + 99$    | (46) $-49 - (-59)$  | (65) $85 + 26$     |
| (9) $7 - 78$       | (28) $-98 + (-99)$ | (47) $20 + 94$      | (66) $94 - 58$     |
| (10) $-41 - 25$    | (29) $-64 - 95$    | (48) $-80 + 16$     | (67) $-83 + (-43)$ |
| (11) $-47 - (-53)$ | (30) $-62 - (-73)$ | (49) $1 + (-23)$    | (68) $-7 + (-97)$  |
| (12) $-77 + (-32)$ | (31) $48 - 82$     | (50) $15 - (-29)$   | (69) $83 + 18$     |
| (13) $-74 + 19$    | (32) $-1 - (-86)$  | (51) $3 + 31$       | (70) $66 - 47$     |
| (14) $-38 + (-48)$ | (33) $-1 + 64$     | (52) $-12 - 52$     | (71) $17 + 98$     |
| (15) $57 - (-25)$  | (34) $-55 - 93$    | (53) $-84 + 38$     | (72) $4 - (-26)$   |
| (16) $51 + (-10)$  | (35) $77 + 44$     | (54) $57 - 97$      | (73) $24 + (-1)$   |
| (17) $25 + 61$     | (36) $-9 - (-24)$  | (55) $91 - (-63)$   | (74) $-80 - 97$    |
| (18) $52 + 10$     | (37) $-76 + (-87)$ | (56) $-100 + (-30)$ | (75) $49 + 9$      |
| (19) $96 - 27$     | (38) $-8 - (-2)$   | (57) $46 - (-15)$   | (76) $-84 + 73$    |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $-88 - (-94)$  | (20) $84 + (-61)$  | (39) $31 + 46$     | (58) $14 - (-37)$  |
| (2) $91 + (-30)$   | (21) $21 - 70$     | (40) $16 + (-67)$  | (59) $-3 - (-40)$  |
| (3) $-64 - 96$     | (22) $-59 - 96$    | (41) $-89 - (-65)$ | (60) $-21 + (-39)$ |
| (4) $-63 - 69$     | (23) $15 - (-98)$  | (42) $-78 - (-80)$ | (61) $-96 + 44$    |
| (5) $-72 - (-28)$  | (24) $8 - 82$      | (43) $68 - 13$     | (62) $6 + 29$      |
| (6) $-87 + 75$     | (25) $6 + 79$      | (44) $21 - (-35)$  | (63) $-19 + (-9)$  |
| (7) $-99 + (-29)$  | (26) $-44 + (-53)$ | (45) $24 + 40$     | (64) $42 + 64$     |
| (8) $81 - (-87)$   | (27) $-46 - (-71)$ | (46) $36 + (-9)$   | (65) $84 + 58$     |
| (9) $-8 - (-84)$   | (28) $-34 - (-25)$ | (47) $-46 + 63$    | (66) $-90 - 96$    |
| (10) $89 - 44$     | (29) $-48 - 92$    | (48) $46 + 41$     | (67) $-95 - (-34)$ |
| (11) $78 + 20$     | (30) $-13 - 7$     | (49) $-76 - (-29)$ | (68) $-80 - (-33)$ |
| (12) $83 + 99$     | (31) $32 + (-89)$  | (50) $-39 + (-58)$ | (69) $6 + 8$       |
| (13) $-12 - (-39)$ | (32) $-41 - (-56)$ | (51) $50 + 45$     | (70) $57 + (-95)$  |
| (14) $79 + (-60)$  | (33) $44 - (-72)$  | (52) $65 + 76$     | (71) $-40 - 35$    |
| (15) $-16 - 82$    | (34) $42 + 5$      | (53) $14 - 88$     | (72) $2 - (-99)$   |
| (16) $9 + (-48)$   | (35) $37 - 91$     | (54) $1 - (-39)$   | (73) $-95 + (-25)$ |
| (17) $52 - (-2)$   | (36) $-52 + 3$     | (55) $-20 + 35$    | (74) $7 - (-82)$   |
| (18) $-28 - 15$    | (37) $-51 + (-3)$  | (56) $49 - 11$     | (75) $-29 + 5$     |
| (19) $79 + 19$     | (38) $97 + (-53)$  | (57) $-73 - (-57)$ | (76) $12 + 37$     |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $70 - 41$      | (20) $-75 + 11$    | (39) $-34 - 63$    | (58) $70 + (-17)$  |
| (2) $75 - 69$      | (21) $87 - (-11)$  | (40) $30 - 65$     | (59) $-51 + 9$     |
| (3) $-5 - (-13)$   | (22) $38 - 62$     | (41) $17 - (-27)$  | (60) $18 + 75$     |
| (4) $49 + 68$      | (23) $7 - (-68)$   | (42) $61 + (-55)$  | (61) $-10 + (-17)$ |
| (5) $-78 + (-12)$  | (24) $-6 - 3$      | (43) $93 - (-45)$  | (62) $-22 + 24$    |
| (6) $69 + (-90)$   | (25) $59 - (-51)$  | (44) $42 + (-14)$  | (63) $-28 - 27$    |
| (7) $73 + 57$      | (26) $89 - 96$     | (45) $-61 - (-18)$ | (64) $80 + (-6)$   |
| (8) $68 + 83$      | (27) $73 + 83$     | (46) $-7 + (-64)$  | (65) $-67 + (-54)$ |
| (9) $49 + (-85)$   | (28) $70 - 11$     | (47) $-91 + 53$    | (66) $58 + (-37)$  |
| (10) $-60 - (-6)$  | (29) $27 + 19$     | (48) $23 + 7$      | (67) $-22 + (-75)$ |
| (11) $-98 - (-37)$ | (30) $32 - 54$     | (49) $-5 + (-7)$   | (68) $3 - 87$      |
| (12) $-77 + (-5)$  | (31) $-56 - 15$    | (50) $-21 - (-56)$ | (69) $-47 - (-3)$  |
| (13) $-13 + 53$    | (32) $-99 + (-50)$ | (51) $-17 + (-27)$ | (70) $54 - 21$     |
| (14) $-28 + 17$    | (33) $-31 + 33$    | (52) $84 - (-44)$  | (71) $-63 + (-34)$ |
| (15) $14 + (-9)$   | (34) $70 + 27$     | (53) $-4 - 19$     | (72) $-67 + (-54)$ |
| (16) $32 - 0$      | (35) $30 - (-69)$  | (54) $64 - 51$     | (73) $-21 - 39$    |
| (17) $63 + (-17)$  | (36) $89 + 61$     | (55) $-59 - (-50)$ | (74) $47 - (-93)$  |
| (18) $83 - (-55)$  | (37) $-85 + 9$     | (56) $19 - 25$     | (75) $88 - 18$     |
| (19) $-17 + 59$    | (38) $53 - 85$     | (57) $99 + (-69)$  | (76) $-21 - (-59)$ |

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|--------------------|--------------------|--------------------|---------------------|
| (1) $88 - 51$      | (20) $-84 - 42$    | (39) $-21 - 9$     | (58) $43 + 91$      |
| (2) $15 + 85$      | (21) $36 - (-63)$  | (40) $-63 - (-36)$ | (59) $-19 + (-47)$  |
| (3) $-91 - 32$     | (22) $52 + 11$     | (41) $1 - (-12)$   | (60) $-96 - (-94)$  |
| (4) $-65 - (-36)$  | (23) $100 - 53$    | (42) $0 - 25$      | (61) $-92 - 41$     |
| (5) $-70 - (-39)$  | (24) $78 - (-2)$   | (43) $48 - (-41)$  | (62) $-19 - (-44)$  |
| (6) $-58 + 10$     | (25) $-42 - (-68)$ | (44) $-69 + (-43)$ | (63) $-69 - (-74)$  |
| (7) $49 - (-36)$   | (26) $-66 + (-81)$ | (45) $-23 + (-72)$ | (64) $-86 + (-48)$  |
| (8) $59 - 12$      | (27) $22 + (-7)$   | (46) $-42 + 86$    | (65) $-85 - (-46)$  |
| (9) $70 - (-24)$   | (28) $-1 + (-65)$  | (47) $-52 + 43$    | (66) $42 + 76$      |
| (10) $-29 - 42$    | (29) $67 + 50$     | (48) $-56 + 39$    | (67) $25 + 45$      |
| (11) $100 + (-46)$ | (30) $74 + 33$     | (49) $-62 + 93$    | (68) $42 - 70$      |
| (12) $54 - 74$     | (31) $0 + (-39)$   | (50) $34 - (-7)$   | (69) $-100 + (-87)$ |
| (13) $36 - 70$     | (32) $96 + (-6)$   | (51) $44 - 67$     | (70) $84 + 45$      |
| (14) $51 + 92$     | (33) $-54 - (-97)$ | (52) $-55 + 46$    | (71) $-68 + (-45)$  |
| (15) $-43 - 33$    | (34) $92 + 75$     | (53) $72 - 15$     | (72) $70 + (-97)$   |
| (16) $73 - 1$      | (35) $6 - 96$      | (54) $44 - 36$     | (73) $39 - (-6)$    |
| (17) $-18 + (-19)$ | (36) $16 - (-49)$  | (55) $27 + (-30)$  | (74) $-59 - (-73)$  |
| (18) $31 - (-12)$  | (37) $53 - (-60)$  | (56) $1 + (-65)$   | (75) $-77 + (-46)$  |
| (19) $49 - 68$     | (38) $-16 - 73$    | (57) $53 + 14$     | (76) $1 - (-38)$    |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $-8 + 27$      | (20) $-44 + (-2)$  | (39) $96 + 8$      | (58) $74 - (-56)$  |
| (2) $-87 + (-97)$  | (21) $86 - 11$     | (40) $57 - (-60)$  | (59) $-69 + 92$    |
| (3) $-49 + 72$     | (22) $93 - (-7)$   | (41) $11 - 88$     | (60) $98 - 58$     |
| (4) $-3 - 92$      | (23) $52 - 70$     | (42) $34 - 65$     | (61) $-82 - 90$    |
| (5) $1 + (-32)$    | (24) $-21 + (-22)$ | (43) $21 - (-71)$  | (62) $71 - 1$      |
| (6) $0 - (-74)$    | (25) $-20 + 67$    | (44) $-74 + 44$    | (63) $-44 + 57$    |
| (7) $-38 + (-32)$  | (26) $44 - 19$     | (45) $-50 - 78$    | (64) $-95 - (-75)$ |
| (8) $95 - (-20)$   | (27) $-13 + 94$    | (46) $-21 + 95$    | (65) $96 + 78$     |
| (9) $78 - 26$      | (28) $-90 - 49$    | (47) $71 + (-99)$  | (66) $6 - 62$      |
| (10) $99 - (-29)$  | (29) $7 - (-99)$   | (48) $53 + 72$     | (67) $24 + (-54)$  |
| (11) $-26 - (-84)$ | (30) $-93 + 5$     | (49) $11 - (-70)$  | (68) $-62 - 52$    |
| (12) $14 + 63$     | (31) $-52 + 92$    | (50) $3 - (-30)$   | (69) $-28 - (-49)$ |
| (13) $44 - (-46)$  | (32) $-66 - (-52)$ | (51) $8 - 94$      | (70) $-89 + 12$    |
| (14) $-87 + (-72)$ | (33) $58 - (-60)$  | (52) $-70 + (-84)$ | (71) $-35 - 59$    |
| (15) $-31 + 47$    | (34) $59 - 38$     | (53) $65 + (-44)$  | (72) $-95 + 90$    |
| (16) $-84 + 21$    | (35) $-22 - (-90)$ | (54) $30 - 43$     | (73) $85 - (-9)$   |
| (17) $100 - (-88)$ | (36) $27 - (-7)$   | (55) $86 - 22$     | (74) $-11 - (-45)$ |
| (18) $11 - 18$     | (37) $45 - (-62)$  | (56) $-56 + (-68)$ | (75) $-19 + (-45)$ |
| (19) $-71 + 99$    | (38) $-34 - (-94)$ | (57) $85 + 72$     | (76) $-72 + (-31)$ |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $84 - (-98)$   | (20) $35 + (-3)$   | (39) $-48 + 6$     | (58) $-58 + 23$    |
| (2) $17 + 13$      | (21) $-15 + 45$    | (40) $-97 + 70$    | (59) $-53 - (-57)$ |
| (3) $-5 - (-45)$   | (22) $-11 - (-95)$ | (41) $-25 - 61$    | (60) $88 + 71$     |
| (4) $-52 + 58$     | (23) $-17 - (-78)$ | (42) $81 + 86$     | (61) $56 - (-95)$  |
| (5) $18 - 59$      | (24) $41 - (-36)$  | (43) $-2 + 55$     | (62) $-46 - 35$    |
| (6) $-40 - (-98)$  | (25) $95 - 26$     | (44) $-18 + (-72)$ | (63) $5 - (-15)$   |
| (7) $-87 + (-82)$  | (26) $8 + 45$      | (45) $73 - 69$     | (64) $-41 - (-73)$ |
| (8) $-18 - (-68)$  | (27) $-42 - (-70)$ | (46) $-98 + 89$    | (65) $14 + 88$     |
| (9) $-75 + 12$     | (28) $61 + (-27)$  | (47) $-82 - 9$     | (66) $2 + (-37)$   |
| (10) $-56 + 10$    | (29) $71 - 48$     | (48) $78 + (-1)$   | (67) $52 + (-97)$  |
| (11) $43 + (-100)$ | (30) $-18 - 68$    | (49) $44 + 12$     | (68) $-28 + (-87)$ |
| (12) $-43 - 25$    | (31) $-55 - (-32)$ | (50) $-97 + (-42)$ | (69) $-45 + 4$     |
| (13) $24 - 71$     | (32) $76 - 10$     | (51) $6 + 23$      | (70) $-82 - (-22)$ |
| (14) $13 - 75$     | (33) $-32 + (-77)$ | (52) $9 + (-2)$    | (71) $-34 - 31$    |
| (15) $-21 + (-35)$ | (34) $23 - (-22)$  | (53) $69 - 82$     | (72) $-13 + (-78)$ |
| (16) $65 - (-71)$  | (35) $45 - 75$     | (54) $39 + (-21)$  | (73) $47 + (-56)$  |
| (17) $88 + (-42)$  | (36) $-40 + 88$    | (55) $-43 + (-75)$ | (74) $87 - (-1)$   |
| (18) $91 + (-50)$  | (37) $83 - 22$     | (56) $41 - (-12)$  | (75) $-31 + (-11)$ |
| (19) $38 + (-75)$  | (38) $76 - 44$     | (57) $65 + (-72)$  | (76) $-29 - 48$    |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $12 + 41$      | (20) $63 + (-86)$  | (39) $-60 - 18$    | (58) $35 + (-44)$  |
| (2) $-22 - (-18)$  | (21) $39 + 68$     | (40) $-51 + (-88)$ | (59) $-92 - (-90)$ |
| (3) $-16 + (-28)$  | (22) $-4 - 52$     | (41) $-65 - (-48)$ | (60) $-78 - (-39)$ |
| (4) $15 + (-72)$   | (23) $-31 + 68$    | (42) $-63 + (-79)$ | (61) $64 - (-51)$  |
| (5) $-64 + 3$      | (24) $34 + (-8)$   | (43) $-91 - (-20)$ | (62) $19 - 84$     |
| (6) $0 - 27$       | (25) $-78 + 78$    | (44) $79 - (-46)$  | (63) $34 - (-7)$   |
| (7) $-28 - 51$     | (26) $89 - (-54)$  | (45) $-11 - (-6)$  | (64) $3 - (-65)$   |
| (8) $-34 + (-97)$  | (27) $-36 + (-1)$  | (46) $34 - 98$     | (65) $48 - (-10)$  |
| (9) $45 + 93$      | (28) $53 + (-54)$  | (47) $-46 + 57$    | (66) $-96 - (-51)$ |
| (10) $61 + 42$     | (29) $23 - 31$     | (48) $-72 - 53$    | (67) $45 - (-22)$  |
| (11) $75 + 98$     | (30) $-37 - (-76)$ | (49) $-71 - 38$    | (68) $-28 - 80$    |
| (12) $47 - 90$     | (31) $-59 - 98$    | (50) $-84 - (-40)$ | (69) $-26 + (-41)$ |
| (13) $-96 - (-35)$ | (32) $1 - (-27)$   | (51) $-18 + (-89)$ | (70) $-91 - 84$    |
| (14) $-30 - (-89)$ | (33) $-98 - 35$    | (52) $91 + 83$     | (71) $-79 + 58$    |
| (15) $15 + 55$     | (34) $14 - 35$     | (53) $-66 + 23$    | (72) $37 - (-79)$  |
| (16) $57 - (-87)$  | (35) $-33 + (-6)$  | (54) $27 - (-31)$  | (73) $48 + (-9)$   |
| (17) $-38 - 11$    | (36) $-35 - (-23)$ | (55) $44 + 100$    | (74) $-68 + (-22)$ |
| (18) $6 - 36$      | (37) $-57 + 10$    | (56) $-89 - (-26)$ | (75) $12 - (-25)$  |
| (19) $57 - 83$     | (38) $66 - 55$     | (57) $-16 + 60$    | (76) $-13 + (-43)$ |



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|--------------------|---------------------|--------------------|--------------------|
| (1) $-89 - (-40)$  | (20) $-18 - (-35)$  | (39) $43 - 69$     | (58) $47 - (-16)$  |
| (2) $25 - 84$      | (21) $80 + (-45)$   | (40) $-66 + 6$     | (59) $65 + (-20)$  |
| (3) $77 - (-64)$   | (22) $-25 + 58$     | (41) $-92 - (-11)$ | (60) $34 + 44$     |
| (4) $-98 - (-22)$  | (23) $42 + (-72)$   | (42) $-35 + (-6)$  | (61) $44 - 32$     |
| (5) $50 - (-34)$   | (24) $2 - (-3)$     | (43) $63 - 3$      | (62) $-86 - (-25)$ |
| (6) $53 + 100$     | (25) $-27 + 97$     | (44) $-78 + (-32)$ | (63) $-56 - 82$    |
| (7) $-94 + 20$     | (26) $-38 + (-85)$  | (45) $80 - (-53)$  | (64) $-56 + (-20)$ |
| (8) $-13 + 90$     | (27) $57 + 23$      | (46) $-91 - (-48)$ | (65) $10 + (-58)$  |
| (9) $-19 + (-40)$  | (28) $34 + 64$      | (47) $39 + (-54)$  | (66) $-56 - (-73)$ |
| (10) $73 + 31$     | (29) $19 - (-30)$   | (48) $26 + 26$     | (67) $-88 + (-89)$ |
| (11) $45 + 11$     | (30) $-82 + 24$     | (49) $-95 - 96$    | (68) $-32 + 55$    |
| (12) $61 + (-77)$  | (31) $29 + 35$      | (50) $-76 - (-13)$ | (69) $49 - 62$     |
| (13) $-32 + (-92)$ | (32) $79 - 13$      | (51) $-85 + 59$    | (70) $6 + (-34)$   |
| (14) $-72 + (-22)$ | (33) $-83 - 45$     | (52) $44 + 65$     | (71) $5 + 53$      |
| (15) $-47 - 98$    | (34) $-100 - (-70)$ | (53) $52 - 24$     | (72) $32 - (-70)$  |
| (16) $53 + (-24)$  | (35) $-5 - 89$      | (54) $-58 - (-89)$ | (73) $12 - (-30)$  |
| (17) $80 + (-8)$   | (36) $-6 + 96$      | (55) $24 - (-75)$  | (74) $-82 - (-72)$ |
| (18) $-86 - (-77)$ | (37) $-31 + (-82)$  | (56) $44 - 79$     | (75) $-27 - 14$    |
| (19) $-92 + (-31)$ | (38) $-28 + 55$     | (57) $-88 - (-99)$ | (76) $28 - 74$     |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $-17 - 18$     | (20) $1 + (-58)$   | (39) $84 - (-98)$  | (58) $55 + (-13)$  |
| (2) $-8 + 80$      | (21) $38 + (-54)$  | (40) $-16 - (-79)$ | (59) $-58 + 1$     |
| (3) $-60 + 84$     | (22) $90 + 2$      | (41) $-48 - (-23)$ | (60) $-74 + (-35)$ |
| (4) $-1 - (-74)$   | (23) $92 + 46$     | (42) $-25 - 50$    | (61) $-76 + 29$    |
| (5) $-88 - 77$     | (24) $-91 - (-66)$ | (43) $61 + 40$     | (62) $85 - (-3)$   |
| (6) $85 + (-36)$   | (25) $-16 - 53$    | (44) $-27 - 89$    | (63) $-42 - 70$    |
| (7) $92 - 14$      | (26) $-65 + (-27)$ | (45) $-48 - (-4)$  | (64) $-24 + 100$   |
| (8) $-100 + 69$    | (27) $-92 + 22$    | (46) $-29 - (-16)$ | (65) $-7 + 96$     |
| (9) $39 + 68$      | (28) $-40 - 36$    | (47) $-67 - (-21)$ | (66) $-36 - (-14)$ |
| (10) $5 - (-66)$   | (29) $5 + 55$      | (48) $-34 - (-47)$ | (67) $-91 - 34$    |
| (11) $-82 + 66$    | (30) $-26 + (-13)$ | (49) $-99 + 84$    | (68) $22 + 5$      |
| (12) $44 + 98$     | (31) $69 + 1$      | (50) $-67 + 13$    | (69) $33 - (-76)$  |
| (13) $-3 + (-54)$  | (32) $35 - 54$     | (51) $-3 - 50$     | (70) $34 - (-32)$  |
| (14) $-65 - (-12)$ | (33) $-73 - 74$    | (52) $38 - 41$     | (71) $-72 + 7$     |
| (15) $-100 - 93$   | (34) $-67 + (-80)$ | (53) $41 + (-39)$  | (72) $47 - 53$     |
| (16) $-39 - (-69)$ | (35) $65 + (-51)$  | (54) $-19 + (-42)$ | (73) $63 + (-84)$  |
| (17) $-9 - (-62)$  | (36) $6 - 50$      | (55) $24 - (-9)$   | (74) $-51 + (-15)$ |
| (18) $30 + (-32)$  | (37) $-63 + 54$    | (56) $-58 - (-22)$ | (75) $-26 + 73$    |
| (19) $-88 + 49$    | (38) $-55 + 30$    | (57) $-15 - (-49)$ | (76) $-76 + 76$    |

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|--------------------|---------------------|--------------------|---------------------|
| (1) $-79 - 58$     | (20) $31 + (-53)$   | (39) $-93 + (-19)$ | (58) $84 + 74$      |
| (2) $70 + 45$      | (21) $-12 + 50$     | (40) $-20 - (-32)$ | (59) $-9 + (-92)$   |
| (3) $-60 + 78$     | (22) $86 - (-69)$   | (41) $-70 - 15$    | (60) $-5 - 78$      |
| (4) $2 - (-53)$    | (23) $72 - (-96)$   | (42) $5 - 51$      | (61) $52 - 57$      |
| (5) $-29 + (-4)$   | (24) $80 + (-53)$   | (43) $78 - 2$      | (62) $-92 - (-35)$  |
| (6) $-75 - 97$     | (25) $-58 + 20$     | (44) $90 + 57$     | (63) $14 + (-89)$   |
| (7) $-80 + (-32)$  | (26) $-100 + (-20)$ | (45) $-3 - (-24)$  | (64) $83 + 69$      |
| (8) $66 + (-5)$    | (27) $-35 + (-54)$  | (46) $-95 - 88$    | (65) $77 + (-50)$   |
| (9) $-22 - (-63)$  | (28) $-43 + (-44)$  | (47) $-45 - (-95)$ | (66) $38 + (-59)$   |
| (10) $95 + 64$     | (29) $-70 + 81$     | (48) $57 + (-36)$  | (67) $90 + (-21)$   |
| (11) $-16 + (-24)$ | (30) $18 - (-53)$   | (49) $-98 + (-72)$ | (68) $15 - 12$      |
| (12) $-8 - (-14)$  | (31) $76 + 100$     | (50) $46 + (-79)$  | (69) $71 + 68$      |
| (13) $16 - 86$     | (32) $-58 + (-12)$  | (51) $-29 + 7$     | (70) $-100 + (-17)$ |
| (14) $-77 - 97$    | (33) $50 - 72$      | (52) $-85 - (-30)$ | (71) $-61 + (-94)$  |
| (15) $-71 + 14$    | (34) $0 - (-35)$    | (53) $69 + (-33)$  | (72) $-74 - (-89)$  |
| (16) $-14 - 66$    | (35) $14 - 82$      | (54) $29 - 39$     | (73) $64 + 100$     |
| (17) $-9 - (-60)$  | (36) $67 + 10$      | (55) $-94 + 95$    | (74) $-91 - 78$     |
| (18) $-92 + (-48)$ | (37) $-75 + 17$     | (56) $-50 + 0$     | (75) $84 + (-62)$   |
| (19) $27 + (-69)$  | (38) $-69 - 38$     | (57) $-1 + 31$     | (76) $-74 + (-9)$   |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $65 + 16$      | (20) $11 - 79$     | (39) $56 + 87$     | (58) $-28 - 58$    |
| (2) $-3 - (-10)$   | (21) $19 - 56$     | (40) $-13 - 84$    | (59) $-6 - (-8)$   |
| (3) $-17 - 64$     | (22) $93 - (-95)$  | (41) $-36 + (-15)$ | (60) $-97 + (-28)$ |
| (4) $-95 + (-14)$  | (23) $-12 + (-31)$ | (42) $-56 + 40$    | (61) $33 + 63$     |
| (5) $86 - (-58)$   | (24) $42 - 100$    | (43) $-15 - (-92)$ | (62) $43 - 0$      |
| (6) $-13 + 65$     | (25) $-64 - (-8)$  | (44) $30 - 84$     | (63) $73 + 18$     |
| (7) $91 + 62$      | (26) $-74 - 58$    | (45) $-30 + (-52)$ | (64) $100 + (-78)$ |
| (8) $-11 + (-98)$  | (27) $-88 - 66$    | (46) $-89 + 53$    | (65) $68 + 65$     |
| (9) $3 + 63$       | (28) $77 + 73$     | (47) $-76 + 76$    | (66) $44 - (-100)$ |
| (10) $-25 - (-92)$ | (29) $82 - (-57)$  | (48) $19 - (-65)$  | (67) $19 - (-92)$  |
| (11) $-51 + (-24)$ | (30) $91 - (-100)$ | (49) $-61 + 4$     | (68) $72 - (-22)$  |
| (12) $69 - 86$     | (31) $-20 - (-12)$ | (50) $-93 + (-3)$  | (69) $-54 + 93$    |
| (13) $-56 + 92$    | (32) $-51 + 43$    | (51) $-48 + (-26)$ | (70) $-90 - 39$    |
| (14) $61 - (-52)$  | (33) $-9 + 10$     | (52) $26 + (-19)$  | (71) $35 - (-90)$  |
| (15) $-39 - 95$    | (34) $44 - (-54)$  | (53) $52 - 28$     | (72) $65 + (-55)$  |
| (16) $-68 - 100$   | (35) $-53 + (-54)$ | (54) $14 - 18$     | (73) $-40 - 96$    |
| (17) $-91 - 100$   | (36) $21 + (-88)$  | (55) $76 - (-31)$  | (74) $-61 + 11$    |
| (18) $72 - 12$     | (37) $-7 + 59$     | (56) $-73 - 69$    | (75) $70 + (-42)$  |
| (19) $-71 - (-66)$ | (38) $96 + 60$     | (57) $-1 - (-18)$  | (76) $-68 + 1$     |

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|--------------------|--------------------|--------------------|--------------------|
| (1) $85 - 87$      | (20) $99 + (-38)$  | (39) $-55 + 69$    | (58) $26 - (-57)$  |
| (2) $-29 + (-62)$  | (21) $-56 - (-53)$ | (40) $61 - (-39)$  | (59) $-59 - (-73)$ |
| (3) $-91 - (-32)$  | (22) $46 + 16$     | (41) $-28 + 80$    | (60) $-68 + (-49)$ |
| (4) $-45 + (-76)$  | (23) $-42 + 22$    | (42) $22 + (-44)$  | (61) $-100 - 0$    |
| (5) $99 + 100$     | (24) $39 + 18$     | (43) $64 + 47$     | (62) $-18 + 64$    |
| (6) $22 - (-68)$   | (25) $27 - 2$      | (44) $54 - (-87)$  | (63) $8 + (-20)$   |
| (7) $24 - (-5)$    | (26) $82 - (-65)$  | (45) $-100 + 22$   | (64) $-38 - (-24)$ |
| (8) $19 + 35$      | (27) $90 - 56$     | (46) $14 - 70$     | (65) $-50 + 24$    |
| (9) $92 + 53$      | (28) $72 + 14$     | (47) $-94 - 61$    | (66) $-78 - 45$    |
| (10) $-26 - 21$    | (29) $2 - 82$      | (48) $57 - (-88)$  | (67) $1 + (-15)$   |
| (11) $36 - 16$     | (30) $-39 + (-39)$ | (49) $0 + (-29)$   | (68) $46 + 57$     |
| (12) $-33 + (-77)$ | (31) $13 - 98$     | (50) $-21 + 18$    | (69) $-18 - 85$    |
| (13) $-9 - (-9)$   | (32) $85 + 54$     | (51) $-37 + (-1)$  | (70) $-12 + (-3)$  |
| (14) $-82 + 91$    | (33) $-98 + (-68)$ | (52) $-22 - (-73)$ | (71) $-37 + (-44)$ |
| (15) $89 + 45$     | (34) $-4 + (-11)$  | (53) $-69 - (-15)$ | (72) $-72 - (-89)$ |
| (16) $79 - (-64)$  | (35) $39 - (-86)$  | (54) $-78 - (-93)$ | (73) $-78 + 76$    |
| (17) $9 + (-41)$   | (36) $-38 + (-68)$ | (55) $44 + 8$      | (74) $-84 - (-65)$ |
| (18) $-55 + (-68)$ | (37) $-44 + 36$    | (56) $-65 + 47$    | (75) $-25 + (-24)$ |
| (19) $5 + 47$      | (38) $29 + (-34)$  | (57) $-83 + (-16)$ | (76) $42 - (-55)$  |

(1)  $-49 + 2$

(20)  $-69 - 33$

(39)  $-7 + 84$

(58)  $-73 - 57$

(2)  $47 - (-49)$

(21)  $5 + 45$

(40)  $-59 - (-70)$

(59)  $58 + (-18)$

(3)  $-65 - (-71)$

(22)  $-78 + 1$

(41)  $9 - 29$

(60)  $19 + (-9)$

(4)  $86 - 39$

(23)  $-52 - (-11)$

(42)  $45 - (-75)$

(61)  $52 + 22$

(5)  $32 + 38$

(24)  $-46 + 31$

(43)  $-25 + (-10)$

(62)  $-33 - (-88)$

(6)  $4 - 66$

(25)  $-48 - 46$

(44)  $-78 - 18$

(63)  $67 + (-75)$

(7)  $-26 - 27$

(26)  $-18 - (-46)$

(45)  $-75 - 35$

(64)  $-16 - (-20)$

(8)  $-46 - (-15)$

(27)  $-28 - 55$

(46)  $-58 + 70$

(65)  $-84 - 52$

(9)  $61 - 41$

(28)  $55 + (-72)$

(47)  $-67 + 16$

(66)  $-3 + (-96)$

(10)  $-54 - 3$

(29)  $38 - 34$

(48)  $66 - (-19)$

(67)  $16 - (-35)$

(11)  $79 - (-36)$

(30)  $-63 - 80$

(49)  $-18 + 15$

(68)  $98 - 46$

(12)  $9 + (-17)$

(31)  $-72 - (-44)$

(50)  $-25 + 42$

(69)  $10 - (-47)$

(13)  $-14 - (-24)$

(32)  $75 - 74$

(51)  $-82 - 61$

(70)  $-46 - (-79)$

(14)  $5 - (-20)$

(33)  $66 - (-63)$

(52)  $18 + 61$

(71)  $83 - 76$

(15)  $-6 - 47$

(34)  $-69 - 15$

(53)  $-54 - 43$

(72)  $25 + (-45)$

(16)  $57 - (-79)$

(35)  $15 - (-9)$

(54)  $74 + (-99)$

(73)  $-23 + 97$

(17)  $8 + 58$

(36)  $-88 + 82$

(55)  $-54 + 97$

(74)  $48 - (-6)$

(18)  $-12 + (-98)$

(37)  $72 + 91$

(56)  $-66 + 39$

(75)  $23 - 18$

(19)  $-82 - (-12)$

(38)  $83 + 34$

(57)  $-36 - 64$

(76)  $20 - (-83)$

- |                                |                                |                                |                                 |
|--------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $-27 - 61$<br>$= -88$      | (20) $67 + (-8)$<br>$= 59$     | (39) $-60 - 8$<br>$= -68$      | (58) $28 + 93$<br>$= 121$       |
| (2) $-19 - (-83)$<br>$= 64$    | (21) $-8 - (-63)$<br>$= 55$    | (40) $-82 + 26$<br>$= -56$     | (59) $-42 - (-84)$<br>$= 42$    |
| (3) $-84 + (-79)$<br>$= -163$  | (22) $-78 + (-39)$<br>$= -117$ | (41) $17 - 38$<br>$= -21$      | (60) $66 - 36$<br>$= 30$        |
| (4) $-65 - 46$<br>$= -111$     | (23) $-36 - 51$<br>$= -87$     | (42) $-70 + (-43)$<br>$= -113$ | (61) $-96 - (-95)$<br>$= -1$    |
| (5) $83 - 66$<br>$= 17$        | (24) $88 + 26$<br>$= 114$      | (43) $40 - (-73)$<br>$= 113$   | (62) $-100 + (-23)$<br>$= -123$ |
| (6) $-77 + (-95)$<br>$= -172$  | (25) $-84 - 18$<br>$= -102$    | (44) $26 + 45$<br>$= 71$       | (63) $-85 + (-69)$<br>$= -154$  |
| (7) $-8 - (-18)$<br>$= 10$     | (26) $-8 + 42$<br>$= 34$       | (45) $11 + (-86)$<br>$= -75$   | (64) $-34 + 89$<br>$= 55$       |
| (8) $5 + (-73)$<br>$= -68$     | (27) $-51 + 41$<br>$= -10$     | (46) $38 + 88$<br>$= 126$      | (65) $-78 + 45$<br>$= -33$      |
| (9) $-17 + 37$<br>$= 20$       | (28) $50 + 86$<br>$= 136$      | (47) $67 - 35$<br>$= 32$       | (66) $6 - (-56)$<br>$= 62$      |
| (10) $-95 - (-11)$<br>$= -84$  | (29) $16 - 60$<br>$= -44$      | (48) $-79 - (-58)$<br>$= -21$  | (67) $6 - (-40)$<br>$= 46$      |
| (11) $64 + 37$<br>$= 101$      | (30) $-1 + (-16)$<br>$= -17$   | (49) $-15 - 68$<br>$= -83$     | (68) $67 + (-39)$<br>$= 28$     |
| (12) $-56 + 76$<br>$= 20$      | (31) $69 + (-30)$<br>$= 39$    | (50) $-3 - 59$<br>$= -62$      | (69) $89 - (-90)$<br>$= 179$    |
| (13) $-79 - 46$<br>$= -125$    | (32) $-51 + 84$<br>$= 33$      | (51) $-86 - 26$<br>$= -112$    | (70) $-45 + (-34)$<br>$= -79$   |
| (14) $-10 - (-1)$<br>$= -9$    | (33) $-99 - (-79)$<br>$= -20$  | (52) $72 + (-66)$<br>$= 6$     | (71) $-38 + 47$<br>$= 9$        |
| (15) $-30 - (-27)$<br>$= -3$   | (34) $86 + (-24)$<br>$= 62$    | (53) $72 - 79$<br>$= -7$       | (72) $36 - 34$<br>$= 2$         |
| (16) $7 - 16$<br>$= -9$        | (35) $-52 + 41$<br>$= -11$     | (54) $-36 - 70$<br>$= -106$    | (73) $64 + 88$<br>$= 152$       |
| (17) $-35 + 3$<br>$= -32$      | (36) $-5 + 47$<br>$= 42$       | (55) $61 + (-25)$<br>$= 36$    | (74) $-38 + 32$<br>$= -6$       |
| (18) $-13 + 80$<br>$= 67$      | (37) $29 - 15$<br>$= 14$       | (56) $-53 - (-66)$<br>$= 13$   | (75) $9 + (-87)$<br>$= -78$     |
| (19) $-79 + (-38)$<br>$= -117$ | (38) $29 + (-40)$<br>$= -11$   | (57) $-83 + (-98)$<br>$= -181$ | (76) $59 + 45$<br>$= 104$       |

|                                 |                                |                               |                               |
|---------------------------------|--------------------------------|-------------------------------|-------------------------------|
| (1) $-3 + (-46)$<br>$= -49$     | (20) $10 + (-59)$<br>$= -49$   | (39) $-40 - (-40)$<br>$= 0$   | (58) $-18 + 28$<br>$= 10$     |
| (2) $2 - 21$<br>$= -19$         | (21) $-53 + 37$<br>$= -16$     | (40) $74 - (-33)$<br>$= 107$  | (59) $-69 + (-10)$<br>$= -79$ |
| (3) $64 - (-42)$<br>$= 106$     | (22) $-95 - (-44)$<br>$= -51$  | (41) $68 - (-57)$<br>$= 125$  | (60) $70 + (-22)$<br>$= 48$   |
| (4) $31 - 77$<br>$= -46$        | (23) $28 + (-13)$<br>$= 15$    | (42) $-68 - 4$<br>$= -72$     | (61) $82 - 39$<br>$= 43$      |
| (5) $-7 - 20$<br>$= -27$        | (24) $-13 + 98$<br>$= 85$      | (43) $-36 + (-11)$<br>$= -47$ | (62) $33 + 88$<br>$= 121$     |
| (6) $-41 + (-44)$<br>$= -85$    | (25) $-85 - 75$<br>$= -160$    | (44) $-75 + 42$<br>$= -33$    | (63) $-54 - 41$<br>$= -95$    |
| (7) $-27 - (-96)$<br>$= 69$     | (26) $-42 - (-72)$<br>$= 30$   | (45) $49 + (-4)$<br>$= 45$    | (64) $7 + (-70)$<br>$= -63$   |
| (8) $-77 - (-6)$<br>$= -71$     | (27) $51 + (-56)$<br>$= -5$    | (46) $61 - 47$<br>$= 14$      | (65) $-38 + 81$<br>$= 43$     |
| (9) $46 + (-32)$<br>$= 14$      | (28) $36 - (-34)$<br>$= 70$    | (47) $-26 - 21$<br>$= -47$    | (66) $50 + 37$<br>$= 87$      |
| (10) $-33 + 99$<br>$= 66$       | (29) $-81 + (-79)$<br>$= -160$ | (48) $-74 + 29$<br>$= -45$    | (67) $88 + 94$<br>$= 182$     |
| (11) $-65 + (-37)$<br>$= -102$  | (30) $-31 + 4$<br>$= -27$      | (49) $-71 - (-90)$<br>$= 19$  | (68) $82 - (-41)$<br>$= 123$  |
| (12) $95 + 41$<br>$= 136$       | (31) $-83 - (-63)$<br>$= -20$  | (50) $95 - (-37)$<br>$= 132$  | (69) $-62 + 90$<br>$= 28$     |
| (13) $-100 + (-11)$<br>$= -111$ | (32) $34 + 48$<br>$= 82$       | (51) $66 - 61$<br>$= 5$       | (70) $85 - (-18)$<br>$= 103$  |
| (14) $33 - (-81)$<br>$= 114$    | (33) $-78 - 73$<br>$= -151$    | (52) $96 + (-67)$<br>$= 29$   | (71) $96 - 25$<br>$= 71$      |
| (15) $90 - 17$<br>$= 73$        | (34) $-34 + (-65)$<br>$= -99$  | (53) $58 + 18$<br>$= 76$      | (72) $-10 - 35$<br>$= -45$    |
| (16) $28 + 13$<br>$= 41$        | (35) $-90 + 24$<br>$= -66$     | (54) $64 - (-16)$<br>$= 80$   | (73) $88 - 100$<br>$= -12$    |
| (17) $77 - 7$<br>$= 70$         | (36) $29 - 27$<br>$= 2$        | (55) $-7 + (-17)$<br>$= -24$  | (74) $-12 + 96$<br>$= 84$     |
| (18) $-67 + 89$<br>$= 22$       | (37) $81 - (-65)$<br>$= 146$   | (56) $22 + (-57)$<br>$= -35$  | (75) $64 - 54$<br>$= 10$      |
| (19) $-84 + 13$<br>$= -71$      | (38) $-88 + (-33)$<br>$= -121$ | (57) $14 + 14$<br>$= 28$      | (76) $45 + 41$<br>$= 86$      |



- |                              |                             |                              |                               |
|------------------------------|-----------------------------|------------------------------|-------------------------------|
| (1) $-92 + (-6)$<br>= -98    | (20) $-8 - 82$<br>= -90     | (39) $-73 - (-91)$<br>= 18   | (58) $-11 - (-71)$<br>= 60    |
| (2) $-63 + (-87)$<br>= -150  | (21) $78 + (-24)$<br>= 54   | (40) $-61 + (-77)$<br>= -138 | (59) $89 + (-4)$<br>= 85      |
| (3) $-33 + (-36)$<br>= -69   | (22) $50 - 56$<br>= -6      | (41) $-68 - (-14)$<br>= -54  | (60) $-71 - 23$<br>= -94      |
| (4) $-100 + 14$<br>= -86     | (23) $61 - (-93)$<br>= 154  | (42) $48 + (-77)$<br>= -29   | (61) $-100 + (-26)$<br>= -126 |
| (5) $-70 - 82$<br>= -152     | (24) $-54 - 73$<br>= -127   | (43) $-7 + (-50)$<br>= -57   | (62) $92 + 29$<br>= 121       |
| (6) $-31 - 58$<br>= -89      | (25) $39 - 74$<br>= -35     | (44) $70 + 19$<br>= 89       | (63) $-70 - (-36)$<br>= -34   |
| (7) $-65 + 33$<br>= -32      | (26) $-53 - 46$<br>= -99    | (45) $24 + 4$<br>= 28        | (64) $-65 - 70$<br>= -135     |
| (8) $41 - 41$<br>= 0         | (27) $-68 - (-100)$<br>= 32 | (46) $-63 - 46$<br>= -109    | (65) $41 - (-99)$<br>= 140    |
| (9) $77 + 95$<br>= 172       | (28) $-21 - (-20)$<br>= -1  | (47) $-74 - 3$<br>= -77      | (66) $-41 - 45$<br>= -86      |
| (10) $81 + 4$<br>= 85        | (29) $-23 - (-4)$<br>= -19  | (48) $37 - (-91)$<br>= 128   | (67) $-52 + 82$<br>= 30       |
| (11) $-29 + (-92)$<br>= -121 | (30) $89 + 100$<br>= 189    | (49) $-39 - 73$<br>= -112    | (68) $51 + (-61)$<br>= -10    |
| (12) $42 + 20$<br>= 62       | (31) $10 - 13$<br>= -3      | (50) $-92 - 29$<br>= -121    | (69) $-21 - (-49)$<br>= 28    |
| (13) $-87 - 13$<br>= -100    | (32) $-86 - (-91)$<br>= 5   | (51) $89 + (-53)$<br>= 36    | (70) $57 + 37$<br>= 94        |
| (14) $94 + (-56)$<br>= 38    | (33) $14 + 59$<br>= 73      | (52) $99 - 75$<br>= 24       | (71) $59 - 50$<br>= 9         |
| (15) $75 - 65$<br>= 10       | (34) $86 - 66$<br>= 20      | (53) $-74 + (-13)$<br>= -87  | (72) $-35 + (-81)$<br>= -116  |
| (16) $82 + 78$<br>= 160      | (35) $-4 - (-82)$<br>= 78   | (54) $-37 + (-80)$<br>= -117 | (73) $-72 - 91$<br>= -163     |
| (17) $-25 + 87$<br>= 62      | (36) $20 - 73$<br>= -53     | (55) $-95 - 81$<br>= -176    | (74) $10 - (-26)$<br>= 36     |
| (18) $-14 + 44$<br>= 30      | (37) $11 - (-67)$<br>= 78   | (56) $76 + (-98)$<br>= -22   | (75) $35 + (-59)$<br>= -24    |
| (19) $-50 + (-45)$<br>= -95  | (38) $-22 - 20$<br>= -42    | (57) $-37 - (-3)$<br>= -34   | (76) $100 - (-79)$<br>= 179   |

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|------------------------------|------------------------------|-------------------------------|------------------------------|
| (1) $88 + 26$<br>= 114       | (20) $-94 + (-14)$<br>= -108 | (39) $73 - 60$<br>= 13        | (58) $-81 - (-32)$<br>= -49  |
| (2) $-99 - (-95)$<br>= -4    | (21) $-22 - (-35)$<br>= 13   | (40) $-69 + (-51)$<br>= -120  | (59) $61 - (-20)$<br>= 81    |
| (3) $18 - 92$<br>= -74       | (22) $-25 + 52$<br>= 27      | (41) $83 - (-44)$<br>= 127    | (60) $53 - 89$<br>= -36      |
| (4) $-41 - (-93)$<br>= 52    | (23) $-94 - (-8)$<br>= -86   | (42) $-100 + (-97)$<br>= -197 | (61) $84 + 35$<br>= 119      |
| (5) $0 + 82$<br>= 82         | (24) $-72 - (-67)$<br>= -5   | (43) $9 + 37$<br>= 46         | (62) $74 + (-49)$<br>= 25    |
| (6) $-31 - 87$<br>= -118     | (25) $34 + 43$<br>= 77       | (44) $73 + 96$<br>= 169       | (63) $-69 - 65$<br>= -134    |
| (7) $-48 - 35$<br>= -83      | (26) $80 - 52$<br>= 28       | (45) $-48 - 16$<br>= -64      | (64) $-12 + (-64)$<br>= -76  |
| (8) $15 + 82$<br>= 97        | (27) $-32 + (-95)$<br>= -127 | (46) $2 + 71$<br>= 73         | (65) $43 + 61$<br>= 104      |
| (9) $-54 + 68$<br>= 14       | (28) $11 - (-72)$<br>= 83    | (47) $-6 - (-88)$<br>= 82     | (66) $19 - 49$<br>= -30      |
| (10) $-98 + (-45)$<br>= -143 | (29) $2 + 96$<br>= 98        | (48) $-89 + (-86)$<br>= -175  | (67) $-9 - (-30)$<br>= 21    |
| (11) $19 + (-97)$<br>= -78   | (30) $-16 - 56$<br>= -72     | (49) $-16 + 75$<br>= 59       | (68) $-66 + (-58)$<br>= -124 |
| (12) $-6 - (-26)$<br>= 20    | (31) $20 + (-32)$<br>= -12   | (50) $-93 - (-94)$<br>= 1     | (69) $64 - (-89)$<br>= 153   |
| (13) $-24 + 95$<br>= 71      | (32) $84 - (-12)$<br>= 96    | (51) $-79 - 57$<br>= -136     | (70) $-52 - (-8)$<br>= -44   |
| (14) $-81 + (-88)$<br>= -169 | (33) $-98 - (-38)$<br>= -60  | (52) $-72 - 79$<br>= -151     | (71) $-68 - (-68)$<br>= 0    |
| (15) $62 - 4$<br>= 58        | (34) $91 + 85$<br>= 176      | (53) $20 + 93$<br>= 113       | (72) $14 + 31$<br>= 45       |
| (16) $83 - (-8)$<br>= 91     | (35) $-95 - (-12)$<br>= -83  | (54) $42 - (-95)$<br>= 137    | (73) $92 + 86$<br>= 178      |
| (17) $11 + 25$<br>= 36       | (36) $69 - (-90)$<br>= 159   | (55) $-80 + (-58)$<br>= -138  | (74) $-97 - 68$<br>= -165    |
| (18) $91 + (-58)$<br>= 33    | (37) $-93 - 28$<br>= -121    | (56) $12 - (-74)$<br>= 86     | (75) $57 - 15$<br>= 42       |
| (19) $37 - (-80)$<br>= 117   | (38) $-79 - (-32)$<br>= -47  | (57) $98 - (-23)$<br>= 121    | (76) $-31 - (-13)$<br>= -18  |

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|----------------------------|------------------------------|------------------------------|------------------------------|
| (1) $-55 - (-76)$<br>= 21  | (20) $20 - (-8)$<br>= 28     | (39) $-92 - 64$<br>= -156    | (58) $33 - (-23)$<br>= 56    |
| (2) $4 + (-15)$<br>= -11   | (21) $57 - (-64)$<br>= 121   | (40) $93 - 45$<br>= 48       | (59) $20 - (-47)$<br>= 67    |
| (3) $-90 + 55$<br>= -35    | (22) $-82 + (-59)$<br>= -141 | (41) $64 - 91$<br>= -27      | (60) $48 - (-11)$<br>= 59    |
| (4) $27 - (-51)$<br>= 78   | (23) $-34 - 4$<br>= -38      | (42) $43 - 10$<br>= 33       | (61) $79 + (-46)$<br>= 33    |
| (5) $92 + 14$<br>= 106     | (24) $-73 + 64$<br>= -9      | (43) $-37 - (-76)$<br>= 39   | (62) $-81 - (-18)$<br>= -63  |
| (6) $60 + 57$<br>= 117     | (25) $-86 + (-92)$<br>= -178 | (44) $-90 + (-45)$<br>= -135 | (63) $82 + (-3)$<br>= 79     |
| (7) $100 + 26$<br>= 126    | (26) $-41 - 39$<br>= -80     | (45) $-26 + (-10)$<br>= -36  | (64) $-46 - (-52)$<br>= 6    |
| (8) $71 - (-46)$<br>= 117  | (27) $-41 - (-74)$<br>= 33   | (46) $-43 + 59$<br>= 16      | (65) $66 - (-3)$<br>= 69     |
| (9) $88 + 62$<br>= 150     | (28) $-83 + 59$<br>= -24     | (47) $-35 - (-1)$<br>= -34   | (66) $71 + (-24)$<br>= 47    |
| (10) $-2 + (-25)$<br>= -27 | (29) $91 - 55$<br>= 36       | (48) $15 + (-14)$<br>= 1     | (67) $-77 + (-90)$<br>= -167 |
| (11) $17 - (-87)$<br>= 104 | (30) $45 + 79$<br>= 124      | (49) $13 + 29$<br>= 42       | (68) $11 - 80$<br>= -69      |
| (12) $47 - 7$<br>= 40      | (31) $65 + 42$<br>= 107      | (50) $-40 - (-19)$<br>= -21  | (69) $-63 - (-12)$<br>= -51  |
| (13) $49 - 38$<br>= 11     | (32) $61 + (-35)$<br>= 26    | (51) $-82 + (-58)$<br>= -140 | (70) $-98 + 19$<br>= -79     |
| (14) $45 + (-92)$<br>= -47 | (33) $-90 - 75$<br>= -165    | (52) $-77 - (-81)$<br>= 4    | (71) $66 + (-42)$<br>= 24    |
| (15) $72 - (-55)$<br>= 127 | (34) $78 + (-56)$<br>= 22    | (53) $50 - (-47)$<br>= 97    | (72) $-78 - (-25)$<br>= -53  |
| (16) $-42 - (-61)$<br>= 19 | (35) $-72 - (-93)$<br>= 21   | (54) $-49 - 97$<br>= -146    | (73) $64 - 15$<br>= 49       |
| (17) $57 - 89$<br>= -32    | (36) $2 - (-65)$<br>= 67     | (55) $-55 - (-57)$<br>= 2    | (74) $71 - (-57)$<br>= 128   |
| (18) $-62 - 90$<br>= -152  | (37) $-95 + (-53)$<br>= -148 | (56) $85 - (-12)$<br>= 97    | (75) $57 + 91$<br>= 148      |
| (19) $-46 - 100$<br>= -146 | (38) $80 + 50$<br>= 130      | (57) $-73 - (-29)$<br>= -44  | (76) $57 - 13$<br>= 44       |

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|-------------------------------|--------------------------------|-------------------------------|--------------------------------|
| (1) $-72 - 43$<br>$= -115$    | (20) $-16 + (-77)$<br>$= -93$  | (39) $-52 + 8$<br>$= -44$     | (58) $73 - 6$<br>$= 67$        |
| (2) $-36 + (-20)$<br>$= -56$  | (21) $11 + (-27)$<br>$= -16$   | (40) $57 + (-11)$<br>$= 46$   | (59) $68 - (-6)$<br>$= 74$     |
| (3) $64 + (-17)$<br>$= 47$    | (22) $-45 + (-28)$<br>$= -73$  | (41) $16 - 12$<br>$= 4$       | (60) $-62 - (-9)$<br>$= -53$   |
| (4) $94 - (-56)$<br>$= 150$   | (23) $-31 - 39$<br>$= -70$     | (42) $37 + 52$<br>$= 89$      | (61) $60 + (-90)$<br>$= -30$   |
| (5) $-96 - 10$<br>$= -106$    | (24) $-1 - 67$<br>$= -68$      | (43) $59 - 61$<br>$= -2$      | (62) $49 - (-77)$<br>$= 126$   |
| (6) $-34 - 2$<br>$= -36$      | (25) $-84 + 72$<br>$= -12$     | (44) $-1 - 4$<br>$= -5$       | (63) $11 - (-15)$<br>$= 26$    |
| (7) $-32 - 29$<br>$= -61$     | (26) $15 + (-76)$<br>$= -61$   | (45) $-68 - 31$<br>$= -99$    | (64) $41 + (-76)$<br>$= -35$   |
| (8) $42 - (-67)$<br>$= 109$   | (27) $-30 - (-4)$<br>$= -26$   | (46) $96 - (-20)$<br>$= 116$  | (65) $96 + (-37)$<br>$= 59$    |
| (9) $-38 - 97$<br>$= -135$    | (28) $-6 - (-61)$<br>$= 55$    | (47) $46 - 45$<br>$= 1$       | (66) $-51 + 32$<br>$= -19$     |
| (10) $46 - 12$<br>$= 34$      | (29) $13 + 31$<br>$= 44$       | (48) $-67 - (-35)$<br>$= -32$ | (67) $96 + 64$<br>$= 160$      |
| (11) $10 + (-47)$<br>$= -37$  | (30) $8 + (-10)$<br>$= -2$     | (49) $27 - (-76)$<br>$= 103$  | (68) $-44 - 57$<br>$= -101$    |
| (12) $80 - (-42)$<br>$= 122$  | (31) $66 - 98$<br>$= -32$      | (50) $62 - (-81)$<br>$= 143$  | (69) $-95 + (-38)$<br>$= -133$ |
| (13) $-68 + 48$<br>$= -20$    | (32) $-41 - 66$<br>$= -107$    | (51) $-42 + (-13)$<br>$= -55$ | (70) $-82 - (-59)$<br>$= -23$  |
| (14) $-78 + 32$<br>$= -46$    | (33) $-99 + (-77)$<br>$= -176$ | (52) $-19 + (-62)$<br>$= -81$ | (71) $-3 + (-25)$<br>$= -28$   |
| (15) $93 - 84$<br>$= 9$       | (34) $-5 - 92$<br>$= -97$      | (53) $85 + 100$<br>$= 185$    | (72) $90 + (-55)$<br>$= 35$    |
| (16) $66 - 86$<br>$= -20$     | (35) $98 - (-53)$<br>$= 151$   | (54) $85 - 35$<br>$= 50$      | (73) $63 - (-55)$<br>$= 118$   |
| (17) $-92 + 50$<br>$= -42$    | (36) $-48 + (-17)$<br>$= -65$  | (55) $42 - 5$<br>$= 37$       | (74) $25 + 15$<br>$= 40$       |
| (18) $-65 - (-49)$<br>$= -16$ | (37) $95 - 0$<br>$= 95$        | (56) $54 + (-100)$<br>$= -46$ | (75) $54 - (-74)$<br>$= 128$   |
| (19) $-97 - (-65)$<br>$= -32$ | (38) $44 - 48$<br>$= -4$       | (57) $-1 - (-90)$<br>$= 89$   | (76) $27 + 47$<br>$= 74$       |

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|-------------------------------|-------------------------------|------------------------------|--------------------------------|
| (1) $81 - (-85)$<br>$= 166$   | (20) $4 + 13$<br>$= 17$       | (39) $77 + (-32)$<br>$= 45$  | (58) $97 + (-15)$<br>$= 82$    |
| (2) $38 - 85$<br>$= -47$      | (21) $19 - 83$<br>$= -64$     | (40) $-69 + 0$<br>$= -69$    | (59) $56 + 55$<br>$= 111$      |
| (3) $-4 - (-26)$<br>$= 22$    | (22) $87 - (-12)$<br>$= 99$   | (41) $1 - 60$<br>$= -59$     | (60) $-92 - (-93)$<br>$= 1$    |
| (4) $11 + 95$<br>$= 106$      | (23) $-7 - (-6)$<br>$= -1$    | (42) $-65 - (-64)$<br>$= -1$ | (61) $-83 + (-25)$<br>$= -108$ |
| (5) $68 - (-27)$<br>$= 95$    | (24) $-51 + 31$<br>$= -20$    | (43) $-37 + 100$<br>$= 63$   | (62) $-36 - (-39)$<br>$= 3$    |
| (6) $59 - 38$<br>$= 21$       | (25) $65 - 76$<br>$= -11$     | (44) $-97 + 75$<br>$= -22$   | (63) $-10 - (-11)$<br>$= 1$    |
| (7) $-35 + (-35)$<br>$= -70$  | (26) $17 - (-96)$<br>$= 113$  | (45) $46 + 28$<br>$= 74$     | (64) $80 + 58$<br>$= 138$      |
| (8) $35 + (-50)$<br>$= -15$   | (27) $67 + 5$<br>$= 72$       | (46) $-2 + 10$<br>$= 8$      | (65) $-91 + (-16)$<br>$= -107$ |
| (9) $82 - 56$<br>$= 26$       | (28) $76 + (-80)$<br>$= -4$   | (47) $2 - 37$<br>$= -35$     | (66) $15 - 13$<br>$= 2$        |
| (10) $-22 - (-73)$<br>$= 51$  | (29) $-94 + 48$<br>$= -46$    | (48) $15 + 63$<br>$= 78$     | (67) $-71 - (-81)$<br>$= 10$   |
| (11) $89 - 5$<br>$= 84$       | (30) $48 - 40$<br>$= 8$       | (49) $-42 - 3$<br>$= -45$    | (68) $96 + (-19)$<br>$= 77$    |
| (12) $-82 - (-24)$<br>$= -58$ | (31) $-24 - (-52)$<br>$= 28$  | (50) $78 - (-79)$<br>$= 157$ | (69) $6 - (-67)$<br>$= 73$     |
| (13) $14 - (-20)$<br>$= 34$   | (32) $-66 - (-11)$<br>$= -55$ | (51) $16 - 60$<br>$= -44$    | (70) $85 + (-78)$<br>$= 7$     |
| (14) $74 - 24$<br>$= 50$      | (33) $-60 + 79$<br>$= 19$     | (52) $52 - (-41)$<br>$= 93$  | (71) $-6 + 85$<br>$= 79$       |
| (15) $-34 - 93$<br>$= -127$   | (34) $-96 + 68$<br>$= -28$    | (53) $82 + (-12)$<br>$= 70$  | (72) $-22 + 42$<br>$= 20$      |
| (16) $-67 - (-44)$<br>$= -23$ | (35) $-33 - (-61)$<br>$= 28$  | (54) $-45 - 98$<br>$= -143$  | (73) $-41 + (-77)$<br>$= -118$ |
| (17) $-95 - (-14)$<br>$= -81$ | (36) $95 - 43$<br>$= 52$      | (55) $77 - 8$<br>$= 69$      | (74) $-38 - 49$<br>$= -87$     |
| (18) $92 + 44$<br>$= 136$     | (37) $13 + (-39)$<br>$= -26$  | (56) $-62 + 94$<br>$= 32$    | (75) $-66 - (-71)$<br>$= 5$    |
| (19) $86 + (-50)$<br>$= 36$   | (38) $-100 + 57$<br>$= -43$   | (57) $-51 - (-93)$<br>$= 42$ | (76) $-69 - (-3)$<br>$= -66$   |

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|--------------------------------|--------------------------------|---------------------------------|--------------------------------|
| (1) $48 + 30$<br>$= 78$        | (20) $14 + (-84)$<br>$= -70$   | (39) $97 + (-7)$<br>$= 90$      | (58) $-41 - (-49)$<br>$= 8$    |
| (2) $15 + 3$<br>$= 18$         | (21) $-80 + (-1)$<br>$= -81$   | (40) $-72 + 20$<br>$= -52$      | (59) $-20 + 99$<br>$= 79$      |
| (3) $-35 + 27$<br>$= -8$       | (22) $-89 - (-99)$<br>$= 10$   | (41) $-52 + (-56)$<br>$= -108$  | (60) $-59 + 92$<br>$= 33$      |
| (4) $37 + (-21)$<br>$= 16$     | (23) $-55 + (-49)$<br>$= -104$ | (42) $60 + 12$<br>$= 72$        | (61) $80 + 4$<br>$= 84$        |
| (5) $7 - (-58)$<br>$= 65$      | (24) $74 - (-49)$<br>$= 123$   | (43) $-50 + (-53)$<br>$= -103$  | (62) $-95 - 12$<br>$= -107$    |
| (6) $-70 - 37$<br>$= -107$     | (25) $-70 + (-15)$<br>$= -85$  | (44) $-46 - 4$<br>$= -50$       | (63) $69 + (-21)$<br>$= 48$    |
| (7) $-31 - (-25)$<br>$= -6$    | (26) $-71 + (-94)$<br>$= -165$ | (45) $94 - (-72)$<br>$= 166$    | (64) $-50 - (-40)$<br>$= -10$  |
| (8) $-92 + (-12)$<br>$= -104$  | (27) $-30 + 99$<br>$= 69$      | (46) $-49 - (-59)$<br>$= 10$    | (65) $85 + 26$<br>$= 111$      |
| (9) $7 - 78$<br>$= -71$        | (28) $-98 + (-99)$<br>$= -197$ | (47) $20 + 94$<br>$= 114$       | (66) $94 - 58$<br>$= 36$       |
| (10) $-41 - 25$<br>$= -66$     | (29) $-64 - 95$<br>$= -159$    | (48) $-80 + 16$<br>$= -64$      | (67) $-83 + (-43)$<br>$= -126$ |
| (11) $-47 - (-53)$<br>$= 6$    | (30) $-62 - (-73)$<br>$= 11$   | (49) $1 + (-23)$<br>$= -22$     | (68) $-7 + (-97)$<br>$= -104$  |
| (12) $-77 + (-32)$<br>$= -109$ | (31) $48 - 82$<br>$= -34$      | (50) $15 - (-29)$<br>$= 44$     | (69) $83 + 18$<br>$= 101$      |
| (13) $-74 + 19$<br>$= -55$     | (32) $-1 - (-86)$<br>$= 85$    | (51) $3 + 31$<br>$= 34$         | (70) $66 - 47$<br>$= 19$       |
| (14) $-38 + (-48)$<br>$= -86$  | (33) $-1 + 64$<br>$= 63$       | (52) $-12 - 52$<br>$= -64$      | (71) $17 + 98$<br>$= 115$      |
| (15) $57 - (-25)$<br>$= 82$    | (34) $-55 - 93$<br>$= -148$    | (53) $-84 + 38$<br>$= -46$      | (72) $4 - (-26)$<br>$= 30$     |
| (16) $51 + (-10)$<br>$= 41$    | (35) $77 + 44$<br>$= 121$      | (54) $57 - 97$<br>$= -40$       | (73) $24 + (-1)$<br>$= 23$     |
| (17) $25 + 61$<br>$= 86$       | (36) $-9 - (-24)$<br>$= 15$    | (55) $91 - (-63)$<br>$= 154$    | (74) $-80 - 97$<br>$= -177$    |
| (18) $52 + 10$<br>$= 62$       | (37) $-76 + (-87)$<br>$= -163$ | (56) $-100 + (-30)$<br>$= -130$ | (75) $49 + 9$<br>$= 58$        |
| (19) $96 - 27$<br>$= 69$       | (38) $-8 - (-2)$<br>$= -6$     | (57) $46 - (-15)$<br>$= 61$     | (76) $-84 + 73$<br>$= -11$     |

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|-----------------------------|-----------------------------|-----------------------------|------------------------------|
| (1) $-88 - (-94)$<br>= 6    | (20) $84 + (-61)$<br>= 23   | (39) $31 + 46$<br>= 77      | (58) $14 - (-37)$<br>= 51    |
| (2) $91 + (-30)$<br>= 61    | (21) $21 - 70$<br>= -49     | (40) $16 + (-67)$<br>= -51  | (59) $-3 - (-40)$<br>= 37    |
| (3) $-64 - 96$<br>= -160    | (22) $-59 - 96$<br>= -155   | (41) $-89 - (-65)$<br>= -24 | (60) $-21 + (-39)$<br>= -60  |
| (4) $-63 - 69$<br>= -132    | (23) $15 - (-98)$<br>= 113  | (42) $-78 - (-80)$<br>= 2   | (61) $-96 + 44$<br>= -52     |
| (5) $-72 - (-28)$<br>= -44  | (24) $8 - 82$<br>= -74      | (43) $68 - 13$<br>= 55      | (62) $6 + 29$<br>= 35        |
| (6) $-87 + 75$<br>= -12     | (25) $6 + 79$<br>= 85       | (44) $21 - (-35)$<br>= 56   | (63) $-19 + (-9)$<br>= -28   |
| (7) $-99 + (-29)$<br>= -128 | (26) $-44 + (-53)$<br>= -97 | (45) $24 + 40$<br>= 64      | (64) $42 + 64$<br>= 106      |
| (8) $81 - (-87)$<br>= 168   | (27) $-46 - (-71)$<br>= 25  | (46) $36 + (-9)$<br>= 27    | (65) $84 + 58$<br>= 142      |
| (9) $-8 - (-84)$<br>= 76    | (28) $-34 - (-25)$<br>= -9  | (47) $-46 + 63$<br>= 17     | (66) $-90 - 96$<br>= -186    |
| (10) $89 - 44$<br>= 45      | (29) $-48 - 92$<br>= -140   | (48) $46 + 41$<br>= 87      | (67) $-95 - (-34)$<br>= -61  |
| (11) $78 + 20$<br>= 98      | (30) $-13 - 7$<br>= -20     | (49) $-76 - (-29)$<br>= -47 | (68) $-80 - (-33)$<br>= -47  |
| (12) $83 + 99$<br>= 182     | (31) $32 + (-89)$<br>= -57  | (50) $-39 + (-58)$<br>= -97 | (69) $6 + 8$<br>= 14         |
| (13) $-12 - (-39)$<br>= 27  | (32) $-41 - (-56)$<br>= 15  | (51) $50 + 45$<br>= 95      | (70) $57 + (-95)$<br>= -38   |
| (14) $79 + (-60)$<br>= 19   | (33) $44 - (-72)$<br>= 116  | (52) $65 + 76$<br>= 141     | (71) $-40 - 35$<br>= -75     |
| (15) $-16 - 82$<br>= -98    | (34) $42 + 5$<br>= 47       | (53) $14 - 88$<br>= -74     | (72) $2 - (-99)$<br>= 101    |
| (16) $9 + (-48)$<br>= -39   | (35) $37 - 91$<br>= -54     | (54) $1 - (-39)$<br>= 40    | (73) $-95 + (-25)$<br>= -120 |
| (17) $52 - (-2)$<br>= 54    | (36) $-52 + 3$<br>= -49     | (55) $-20 + 35$<br>= 15     | (74) $7 - (-82)$<br>= 89     |
| (18) $-28 - 15$<br>= -43    | (37) $-51 + (-3)$<br>= -54  | (56) $49 - 11$<br>= 38      | (75) $-29 + 5$<br>= -24      |
| (19) $79 + 19$<br>= 98      | (38) $97 + (-53)$<br>= 44   | (57) $-73 - (-57)$<br>= -16 | (76) $12 + 37$<br>= 49       |

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| (1) $70 - 41$<br>$= 29$       | (20) $-75 + 11$<br>$= -64$     | (39) $-34 - 63$<br>$= -97$    | (58) $70 + (-17)$<br>$= 53$    |
| (2) $75 - 69$<br>$= 6$        | (21) $87 - (-11)$<br>$= 98$    | (40) $30 - 65$<br>$= -35$     | (59) $-51 + 9$<br>$= -42$      |
| (3) $-5 - (-13)$<br>$= 8$     | (22) $38 - 62$<br>$= -24$      | (41) $17 - (-27)$<br>$= 44$   | (60) $18 + 75$<br>$= 93$       |
| (4) $49 + 68$<br>$= 117$      | (23) $7 - (-68)$<br>$= 75$     | (42) $61 + (-55)$<br>$= 6$    | (61) $-10 + (-17)$<br>$= -27$  |
| (5) $-78 + (-12)$<br>$= -90$  | (24) $-6 - 3$<br>$= -9$        | (43) $93 - (-45)$<br>$= 138$  | (62) $-22 + 24$<br>$= 2$       |
| (6) $69 + (-90)$<br>$= -21$   | (25) $59 - (-51)$<br>$= 110$   | (44) $42 + (-14)$<br>$= 28$   | (63) $-28 - 27$<br>$= -55$     |
| (7) $73 + 57$<br>$= 130$      | (26) $89 - 96$<br>$= -7$       | (45) $-61 - (-18)$<br>$= -43$ | (64) $80 + (-6)$<br>$= 74$     |
| (8) $68 + 83$<br>$= 151$      | (27) $73 + 83$<br>$= 156$      | (46) $-7 + (-64)$<br>$= -71$  | (65) $-67 + (-54)$<br>$= -121$ |
| (9) $49 + (-85)$<br>$= -36$   | (28) $70 - 11$<br>$= 59$       | (47) $-91 + 53$<br>$= -38$    | (66) $58 + (-37)$<br>$= 21$    |
| (10) $-60 - (-6)$<br>$= -54$  | (29) $27 + 19$<br>$= 46$       | (48) $23 + 7$<br>$= 30$       | (67) $-22 + (-75)$<br>$= -97$  |
| (11) $-98 - (-37)$<br>$= -61$ | (30) $32 - 54$<br>$= -22$      | (49) $-5 + (-7)$<br>$= -12$   | (68) $3 - 87$<br>$= -84$       |
| (12) $-77 + (-5)$<br>$= -82$  | (31) $-56 - 15$<br>$= -71$     | (50) $-21 - (-56)$<br>$= 35$  | (69) $-47 - (-3)$<br>$= -44$   |
| (13) $-13 + 53$<br>$= 40$     | (32) $-99 + (-50)$<br>$= -149$ | (51) $-17 + (-27)$<br>$= -44$ | (70) $54 - 21$<br>$= 33$       |
| (14) $-28 + 17$<br>$= -11$    | (33) $-31 + 33$<br>$= 2$       | (52) $84 - (-44)$<br>$= 128$  | (71) $-63 + (-34)$<br>$= -97$  |
| (15) $14 + (-9)$<br>$= 5$     | (34) $70 + 27$<br>$= 97$       | (53) $-4 - 19$<br>$= -23$     | (72) $-67 + (-54)$<br>$= -121$ |
| (16) $32 - 0$<br>$= 32$       | (35) $30 - (-69)$<br>$= 99$    | (54) $64 - 51$<br>$= 13$      | (73) $-21 - 39$<br>$= -60$     |
| (17) $63 + (-17)$<br>$= 46$   | (36) $89 + 61$<br>$= 150$      | (55) $-59 - (-50)$<br>$= -9$  | (74) $47 - (-93)$<br>$= 140$   |
| (18) $83 - (-55)$<br>$= 138$  | (37) $-85 + 9$<br>$= -76$      | (56) $19 - 25$<br>$= -6$      | (75) $88 - 18$<br>$= 70$       |
| (19) $-17 + 59$<br>$= 42$     | (38) $53 - 85$<br>$= -32$      | (57) $99 + (-69)$<br>$= 30$   | (76) $-21 - (-59)$<br>$= 38$   |



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|-------------------------------|--------------------------------|--------------------------------|---------------------------------|
| (1) $88 - 51$<br>$= 37$       | (20) $-84 - 42$<br>$= -126$    | (39) $-21 - 9$<br>$= -30$      | (58) $43 + 91$<br>$= 134$       |
| (2) $15 + 85$<br>$= 100$      | (21) $36 - (-63)$<br>$= 99$    | (40) $-63 - (-36)$<br>$= -27$  | (59) $-19 + (-47)$<br>$= -66$   |
| (3) $-91 - 32$<br>$= -123$    | (22) $52 + 11$<br>$= 63$       | (41) $1 - (-12)$<br>$= 13$     | (60) $-96 - (-94)$<br>$= -2$    |
| (4) $-65 - (-36)$<br>$= -29$  | (23) $100 - 53$<br>$= 47$      | (42) $0 - 25$<br>$= -25$       | (61) $-92 - 41$<br>$= -133$     |
| (5) $-70 - (-39)$<br>$= -31$  | (24) $78 - (-2)$<br>$= 80$     | (43) $48 - (-41)$<br>$= 89$    | (62) $-19 - (-44)$<br>$= 25$    |
| (6) $-58 + 10$<br>$= -48$     | (25) $-42 - (-68)$<br>$= 26$   | (44) $-69 + (-43)$<br>$= -112$ | (63) $-69 - (-74)$<br>$= 5$     |
| (7) $49 - (-36)$<br>$= 85$    | (26) $-66 + (-81)$<br>$= -147$ | (45) $-23 + (-72)$<br>$= -95$  | (64) $-86 + (-48)$<br>$= -134$  |
| (8) $59 - 12$<br>$= 47$       | (27) $22 + (-7)$<br>$= 15$     | (46) $-42 + 86$<br>$= 44$      | (65) $-85 - (-46)$<br>$= -39$   |
| (9) $70 - (-24)$<br>$= 94$    | (28) $-1 + (-65)$<br>$= -66$   | (47) $-52 + 43$<br>$= -9$      | (66) $42 + 76$<br>$= 118$       |
| (10) $-29 - 42$<br>$= -71$    | (29) $67 + 50$<br>$= 117$      | (48) $-56 + 39$<br>$= -17$     | (67) $25 + 45$<br>$= 70$        |
| (11) $100 + (-46)$<br>$= 54$  | (30) $74 + 33$<br>$= 107$      | (49) $-62 + 93$<br>$= 31$      | (68) $42 - 70$<br>$= -28$       |
| (12) $54 - 74$<br>$= -20$     | (31) $0 + (-39)$<br>$= -39$    | (50) $34 - (-7)$<br>$= 41$     | (69) $-100 + (-87)$<br>$= -187$ |
| (13) $36 - 70$<br>$= -34$     | (32) $96 + (-6)$<br>$= 90$     | (51) $44 - 67$<br>$= -23$      | (70) $84 + 45$<br>$= 129$       |
| (14) $51 + 92$<br>$= 143$     | (33) $-54 - (-97)$<br>$= 43$   | (52) $-55 + 46$<br>$= -9$      | (71) $-68 + (-45)$<br>$= -113$  |
| (15) $-43 - 33$<br>$= -76$    | (34) $92 + 75$<br>$= 167$      | (53) $72 - 15$<br>$= 57$       | (72) $70 + (-97)$<br>$= -27$    |
| (16) $73 - 1$<br>$= 72$       | (35) $6 - 96$<br>$= -90$       | (54) $44 - 36$<br>$= 8$        | (73) $39 - (-6)$<br>$= 45$      |
| (17) $-18 + (-19)$<br>$= -37$ | (36) $16 - (-49)$<br>$= 65$    | (55) $27 + (-30)$<br>$= -3$    | (74) $-59 - (-73)$<br>$= 14$    |
| (18) $31 - (-12)$<br>$= 43$   | (37) $53 - (-60)$<br>$= 113$   | (56) $1 + (-65)$<br>$= -64$    | (75) $-77 + (-46)$<br>$= -123$  |
| (19) $49 - 68$<br>$= -19$     | (38) $-16 - 73$<br>$= -89$     | (57) $53 + 14$<br>$= 67$       | (76) $1 - (-38)$<br>$= 39$      |

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| (1) $-8 + 27$<br>$= 19$        | (20) $-44 + (-2)$<br>$= -46$  | (39) $96 + 8$<br>$= 104$       | (58) $74 - (-56)$<br>$= 130$   |
| (2) $-87 + (-97)$<br>$= -184$  | (21) $86 - 11$<br>$= 75$      | (40) $57 - (-60)$<br>$= 117$   | (59) $-69 + 92$<br>$= 23$      |
| (3) $-49 + 72$<br>$= 23$       | (22) $93 - (-7)$<br>$= 100$   | (41) $11 - 88$<br>$= -77$      | (60) $98 - 58$<br>$= 40$       |
| (4) $-3 - 92$<br>$= -95$       | (23) $52 - 70$<br>$= -18$     | (42) $34 - 65$<br>$= -31$      | (61) $-82 - 90$<br>$= -172$    |
| (5) $1 + (-32)$<br>$= -31$     | (24) $-21 + (-22)$<br>$= -43$ | (43) $21 - (-71)$<br>$= 92$    | (62) $71 - 1$<br>$= 70$        |
| (6) $0 - (-74)$<br>$= 74$      | (25) $-20 + 67$<br>$= 47$     | (44) $-74 + 44$<br>$= -30$     | (63) $-44 + 57$<br>$= 13$      |
| (7) $-38 + (-32)$<br>$= -70$   | (26) $44 - 19$<br>$= 25$      | (45) $-50 - 78$<br>$= -128$    | (64) $-95 - (-75)$<br>$= -20$  |
| (8) $95 - (-20)$<br>$= 115$    | (27) $-13 + 94$<br>$= 81$     | (46) $-21 + 95$<br>$= 74$      | (65) $96 + 78$<br>$= 174$      |
| (9) $78 - 26$<br>$= 52$        | (28) $-90 - 49$<br>$= -139$   | (47) $71 + (-99)$<br>$= -28$   | (66) $6 - 62$<br>$= -56$       |
| (10) $99 - (-29)$<br>$= 128$   | (29) $7 - (-99)$<br>$= 106$   | (48) $53 + 72$<br>$= 125$      | (67) $24 + (-54)$<br>$= -30$   |
| (11) $-26 - (-84)$<br>$= 58$   | (30) $-93 + 5$<br>$= -88$     | (49) $11 - (-70)$<br>$= 81$    | (68) $-62 - 52$<br>$= -114$    |
| (12) $14 + 63$<br>$= 77$       | (31) $-52 + 92$<br>$= 40$     | (50) $3 - (-30)$<br>$= 33$     | (69) $-28 - (-49)$<br>$= 21$   |
| (13) $44 - (-46)$<br>$= 90$    | (32) $-66 - (-52)$<br>$= -14$ | (51) $8 - 94$<br>$= -86$       | (70) $-89 + 12$<br>$= -77$     |
| (14) $-87 + (-72)$<br>$= -159$ | (33) $58 - (-60)$<br>$= 118$  | (52) $-70 + (-84)$<br>$= -154$ | (71) $-35 - 59$<br>$= -94$     |
| (15) $-31 + 47$<br>$= 16$      | (34) $59 - 38$<br>$= 21$      | (53) $65 + (-44)$<br>$= 21$    | (72) $-95 + 90$<br>$= -5$      |
| (16) $-84 + 21$<br>$= -63$     | (35) $-22 - (-90)$<br>$= 68$  | (54) $30 - 43$<br>$= -13$      | (73) $85 - (-9)$<br>$= 94$     |
| (17) $100 - (-88)$<br>$= 188$  | (36) $27 - (-7)$<br>$= 34$    | (55) $86 - 22$<br>$= 64$       | (74) $-11 - (-45)$<br>$= 34$   |
| (18) $11 - 18$<br>$= -7$       | (37) $45 - (-62)$<br>$= 107$  | (56) $-56 + (-68)$<br>$= -124$ | (75) $-19 + (-45)$<br>$= -64$  |
| (19) $-71 + 99$<br>$= 28$      | (38) $-34 - (-94)$<br>$= 60$  | (57) $85 + 72$<br>$= 157$      | (76) $-72 + (-31)$<br>$= -103$ |

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| (1) $84 - (-98)$<br>$= 182$   | (20) $35 + (-3)$<br>$= 32$     | (39) $-48 + 6$<br>$= -42$      | (58) $-58 + 23$<br>$= -35$     |
| (2) $17 + 13$<br>$= 30$       | (21) $-15 + 45$<br>$= 30$      | (40) $-97 + 70$<br>$= -27$     | (59) $-53 - (-57)$<br>$= 4$    |
| (3) $-5 - (-45)$<br>$= 40$    | (22) $-11 - (-95)$<br>$= 84$   | (41) $-25 - 61$<br>$= -86$     | (60) $88 + 71$<br>$= 159$      |
| (4) $-52 + 58$<br>$= 6$       | (23) $-17 - (-78)$<br>$= 61$   | (42) $81 + 86$<br>$= 167$      | (61) $56 - (-95)$<br>$= 151$   |
| (5) $18 - 59$<br>$= -41$      | (24) $41 - (-36)$<br>$= 77$    | (43) $-2 + 55$<br>$= 53$       | (62) $-46 - 35$<br>$= -81$     |
| (6) $-40 - (-98)$<br>$= 58$   | (25) $95 - 26$<br>$= 69$       | (44) $-18 + (-72)$<br>$= -90$  | (63) $5 - (-15)$<br>$= 20$     |
| (7) $-87 + (-82)$<br>$= -169$ | (26) $8 + 45$<br>$= 53$        | (45) $73 - 69$<br>$= 4$        | (64) $-41 - (-73)$<br>$= 32$   |
| (8) $-18 - (-68)$<br>$= 50$   | (27) $-42 - (-70)$<br>$= 28$   | (46) $-98 + 89$<br>$= -9$      | (65) $14 + 88$<br>$= 102$      |
| (9) $-75 + 12$<br>$= -63$     | (28) $61 + (-27)$<br>$= 34$    | (47) $-82 - 9$<br>$= -91$      | (66) $2 + (-37)$<br>$= -35$    |
| (10) $-56 + 10$<br>$= -46$    | (29) $71 - 48$<br>$= 23$       | (48) $78 + (-1)$<br>$= 77$     | (67) $52 + (-97)$<br>$= -45$   |
| (11) $43 + (-100)$<br>$= -57$ | (30) $-18 - 68$<br>$= -86$     | (49) $44 + 12$<br>$= 56$       | (68) $-28 + (-87)$<br>$= -115$ |
| (12) $-43 - 25$<br>$= -68$    | (31) $-55 - (-32)$<br>$= -23$  | (50) $-97 + (-42)$<br>$= -139$ | (69) $-45 + 4$<br>$= -41$      |
| (13) $24 - 71$<br>$= -47$     | (32) $76 - 10$<br>$= 66$       | (51) $6 + 23$<br>$= 29$        | (70) $-82 - (-22)$<br>$= -60$  |
| (14) $13 - 75$<br>$= -62$     | (33) $-32 + (-77)$<br>$= -109$ | (52) $9 + (-2)$<br>$= 7$       | (71) $-34 - 31$<br>$= -65$     |
| (15) $-21 + (-35)$<br>$= -56$ | (34) $23 - (-22)$<br>$= 45$    | (53) $69 - 82$<br>$= -13$      | (72) $-13 + (-78)$<br>$= -91$  |
| (16) $65 - (-71)$<br>$= 136$  | (35) $45 - 75$<br>$= -30$      | (54) $39 + (-21)$<br>$= 18$    | (73) $47 + (-56)$<br>$= -9$    |
| (17) $88 + (-42)$<br>$= 46$   | (36) $-40 + 88$<br>$= 48$      | (55) $-43 + (-75)$<br>$= -118$ | (74) $87 - (-1)$<br>$= 88$     |
| (18) $91 + (-50)$<br>$= 41$   | (37) $83 - 22$<br>$= 61$       | (56) $41 - (-12)$<br>$= 53$    | (75) $-31 + (-11)$<br>$= -42$  |
| (19) $38 + (-75)$<br>$= -37$  | (38) $76 - 44$<br>$= 32$       | (57) $65 + (-72)$<br>$= -7$    | (76) $-29 - 48$<br>$= -77$     |

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| (1) $12 + 41$<br>$= 53$       | (20) $63 + (-86)$<br>$= -23$  | (39) $-60 - 18$<br>$= -78$     | (58) $35 + (-44)$<br>$= -9$   |
| (2) $-22 - (-18)$<br>$= -4$   | (21) $39 + 68$<br>$= 107$     | (40) $-51 + (-88)$<br>$= -139$ | (59) $-92 - (-90)$<br>$= -2$  |
| (3) $-16 + (-28)$<br>$= -44$  | (22) $-4 - 52$<br>$= -56$     | (41) $-65 - (-48)$<br>$= -17$  | (60) $-78 - (-39)$<br>$= -39$ |
| (4) $15 + (-72)$<br>$= -57$   | (23) $-31 + 68$<br>$= 37$     | (42) $-63 + (-79)$<br>$= -142$ | (61) $64 - (-51)$<br>$= 115$  |
| (5) $-64 + 3$<br>$= -61$      | (24) $34 + (-8)$<br>$= 26$    | (43) $-91 - (-20)$<br>$= -71$  | (62) $19 - 84$<br>$= -65$     |
| (6) $0 - 27$<br>$= -27$       | (25) $-78 + 78$<br>$= 0$      | (44) $79 - (-46)$<br>$= 125$   | (63) $34 - (-7)$<br>$= 41$    |
| (7) $-28 - 51$<br>$= -79$     | (26) $89 - (-54)$<br>$= 143$  | (45) $-11 - (-6)$<br>$= -5$    | (64) $3 - (-65)$<br>$= 68$    |
| (8) $-34 + (-97)$<br>$= -131$ | (27) $-36 + (-1)$<br>$= -37$  | (46) $34 - 98$<br>$= -64$      | (65) $48 - (-10)$<br>$= 58$   |
| (9) $45 + 93$<br>$= 138$      | (28) $53 + (-54)$<br>$= -1$   | (47) $-46 + 57$<br>$= 11$      | (66) $-96 - (-51)$<br>$= -45$ |
| (10) $61 + 42$<br>$= 103$     | (29) $23 - 31$<br>$= -8$      | (48) $-72 - 53$<br>$= -125$    | (67) $45 - (-22)$<br>$= 67$   |
| (11) $75 + 98$<br>$= 173$     | (30) $-37 - (-76)$<br>$= 39$  | (49) $-71 - 38$<br>$= -109$    | (68) $-28 - 80$<br>$= -108$   |
| (12) $47 - 90$<br>$= -43$     | (31) $-59 - 98$<br>$= -157$   | (50) $-84 - (-40)$<br>$= -44$  | (69) $-26 + (-41)$<br>$= -67$ |
| (13) $-96 - (-35)$<br>$= -61$ | (32) $1 - (-27)$<br>$= 28$    | (51) $-18 + (-89)$<br>$= -107$ | (70) $-91 - 84$<br>$= -175$   |
| (14) $-30 - (-89)$<br>$= 59$  | (33) $-98 - 35$<br>$= -133$   | (52) $91 + 83$<br>$= 174$      | (71) $-79 + 58$<br>$= -21$    |
| (15) $15 + 55$<br>$= 70$      | (34) $14 - 35$<br>$= -21$     | (53) $-66 + 23$<br>$= -43$     | (72) $37 - (-79)$<br>$= 116$  |
| (16) $57 - (-87)$<br>$= 144$  | (35) $-33 + (-6)$<br>$= -39$  | (54) $27 - (-31)$<br>$= 58$    | (73) $48 + (-9)$<br>$= 39$    |
| (17) $-38 - 11$<br>$= -49$    | (36) $-35 - (-23)$<br>$= -12$ | (55) $44 + 100$<br>$= 144$     | (74) $-68 + (-22)$<br>$= -90$ |
| (18) $6 - 36$<br>$= -30$      | (37) $-57 + 10$<br>$= -47$    | (56) $-89 - (-26)$<br>$= -63$  | (75) $12 - (-25)$<br>$= 37$   |
| (19) $57 - 83$<br>$= -26$     | (38) $66 - 55$<br>$= 11$      | (57) $-16 + 60$<br>$= 44$      | (76) $-13 + (-43)$<br>$= -56$ |

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|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| (1) $-89 - (-40)$<br>$= -49$   | (20) $-18 - (-35)$<br>$= 17$   | (39) $43 - 69$<br>$= -26$      | (58) $47 - (-16)$<br>$= 63$    |
| (2) $25 - 84$<br>$= -59$       | (21) $80 + (-45)$<br>$= 35$    | (40) $-66 + 6$<br>$= -60$      | (59) $65 + (-20)$<br>$= 45$    |
| (3) $77 - (-64)$<br>$= 141$    | (22) $-25 + 58$<br>$= 33$      | (41) $-92 - (-11)$<br>$= -81$  | (60) $34 + 44$<br>$= 78$       |
| (4) $-98 - (-22)$<br>$= -76$   | (23) $42 + (-72)$<br>$= -30$   | (42) $-35 + (-6)$<br>$= -41$   | (61) $44 - 32$<br>$= 12$       |
| (5) $50 - (-34)$<br>$= 84$     | (24) $2 - (-3)$<br>$= 5$       | (43) $63 - 3$<br>$= 60$        | (62) $-86 - (-25)$<br>$= -61$  |
| (6) $53 + 100$<br>$= 153$      | (25) $-27 + 97$<br>$= 70$      | (44) $-78 + (-32)$<br>$= -110$ | (63) $-56 - 82$<br>$= -138$    |
| (7) $-94 + 20$<br>$= -74$      | (26) $-38 + (-85)$<br>$= -123$ | (45) $80 - (-53)$<br>$= 133$   | (64) $-56 + (-20)$<br>$= -76$  |
| (8) $-13 + 90$<br>$= 77$       | (27) $57 + 23$<br>$= 80$       | (46) $-91 - (-48)$<br>$= -43$  | (65) $10 + (-58)$<br>$= -48$   |
| (9) $-19 + (-40)$<br>$= -59$   | (28) $34 + 64$<br>$= 98$       | (47) $39 + (-54)$<br>$= -15$   | (66) $-56 - (-73)$<br>$= 17$   |
| (10) $73 + 31$<br>$= 104$      | (29) $19 - (-30)$<br>$= 49$    | (48) $26 + 26$<br>$= 52$       | (67) $-88 + (-89)$<br>$= -177$ |
| (11) $45 + 11$<br>$= 56$       | (30) $-82 + 24$<br>$= -58$     | (49) $-95 - 96$<br>$= -191$    | (68) $-32 + 55$<br>$= 23$      |
| (12) $61 + (-77)$<br>$= -16$   | (31) $29 + 35$<br>$= 64$       | (50) $-76 - (-13)$<br>$= -63$  | (69) $49 - 62$<br>$= -13$      |
| (13) $-32 + (-92)$<br>$= -124$ | (32) $79 - 13$<br>$= 66$       | (51) $-85 + 59$<br>$= -26$     | (70) $6 + (-34)$<br>$= -28$    |
| (14) $-72 + (-22)$<br>$= -94$  | (33) $-83 - 45$<br>$= -128$    | (52) $44 + 65$<br>$= 109$      | (71) $5 + 53$<br>$= 58$        |
| (15) $-47 - 98$<br>$= -145$    | (34) $-100 - (-70)$<br>$= -30$ | (53) $52 - 24$<br>$= 28$       | (72) $32 - (-70)$<br>$= 102$   |
| (16) $53 + (-24)$<br>$= 29$    | (35) $-5 - 89$<br>$= -94$      | (54) $-58 - (-89)$<br>$= 31$   | (73) $12 - (-30)$<br>$= 42$    |
| (17) $80 + (-8)$<br>$= 72$     | (36) $-6 + 96$<br>$= 90$       | (55) $24 - (-75)$<br>$= 99$    | (74) $-82 - (-72)$<br>$= -10$  |
| (18) $-86 - (-77)$<br>$= -9$   | (37) $-31 + (-82)$<br>$= -113$ | (56) $44 - 79$<br>$= -35$      | (75) $-27 - 14$<br>$= -41$     |
| (19) $-92 + (-31)$<br>$= -123$ | (38) $-28 + 55$<br>$= 27$      | (57) $-88 - (-99)$<br>$= 11$   | (76) $28 - 74$<br>$= -46$      |

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| (1) $-17 - 18$<br>$= -35$     | (20) $1 + (-58)$<br>$= -57$    | (39) $84 - (-98)$<br>$= 182$  | (58) $55 + (-13)$<br>$= 42$    |
| (2) $-8 + 80$<br>$= 72$       | (21) $38 + (-54)$<br>$= -16$   | (40) $-16 - (-79)$<br>$= 63$  | (59) $-58 + 1$<br>$= -57$      |
| (3) $-60 + 84$<br>$= 24$      | (22) $90 + 2$<br>$= 92$        | (41) $-48 - (-23)$<br>$= -25$ | (60) $-74 + (-35)$<br>$= -109$ |
| (4) $-1 - (-74)$<br>$= 73$    | (23) $92 + 46$<br>$= 138$      | (42) $-25 - 50$<br>$= -75$    | (61) $-76 + 29$<br>$= -47$     |
| (5) $-88 - 77$<br>$= -165$    | (24) $-91 - (-66)$<br>$= -25$  | (43) $61 + 40$<br>$= 101$     | (62) $85 - (-3)$<br>$= 88$     |
| (6) $85 + (-36)$<br>$= 49$    | (25) $-16 - 53$<br>$= -69$     | (44) $-27 - 89$<br>$= -116$   | (63) $-42 - 70$<br>$= -112$    |
| (7) $92 - 14$<br>$= 78$       | (26) $-65 + (-27)$<br>$= -92$  | (45) $-48 - (-4)$<br>$= -44$  | (64) $-24 + 100$<br>$= 76$     |
| (8) $-100 + 69$<br>$= -31$    | (27) $-92 + 22$<br>$= -70$     | (46) $-29 - (-16)$<br>$= -13$ | (65) $-7 + 96$<br>$= 89$       |
| (9) $39 + 68$<br>$= 107$      | (28) $-40 - 36$<br>$= -76$     | (47) $-67 - (-21)$<br>$= -46$ | (66) $-36 - (-14)$<br>$= -22$  |
| (10) $5 - (-66)$<br>$= 71$    | (29) $5 + 55$<br>$= 60$        | (48) $-34 - (-47)$<br>$= 13$  | (67) $-91 - 34$<br>$= -125$    |
| (11) $-82 + 66$<br>$= -16$    | (30) $-26 + (-13)$<br>$= -39$  | (49) $-99 + 84$<br>$= -15$    | (68) $22 + 5$<br>$= 27$        |
| (12) $44 + 98$<br>$= 142$     | (31) $69 + 1$<br>$= 70$        | (50) $-67 + 13$<br>$= -54$    | (69) $33 - (-76)$<br>$= 109$   |
| (13) $-3 + (-54)$<br>$= -57$  | (32) $35 - 54$<br>$= -19$      | (51) $-3 - 50$<br>$= -53$     | (70) $34 - (-32)$<br>$= 66$    |
| (14) $-65 - (-12)$<br>$= -53$ | (33) $-73 - 74$<br>$= -147$    | (52) $38 - 41$<br>$= -3$      | (71) $-72 + 7$<br>$= -65$      |
| (15) $-100 - 93$<br>$= -193$  | (34) $-67 + (-80)$<br>$= -147$ | (53) $41 + (-39)$<br>$= 2$    | (72) $47 - 53$<br>$= -6$       |
| (16) $-39 - (-69)$<br>$= 30$  | (35) $65 + (-51)$<br>$= 14$    | (54) $-19 + (-42)$<br>$= -61$ | (73) $63 + (-84)$<br>$= -21$   |
| (17) $-9 - (-62)$<br>$= 53$   | (36) $6 - 50$<br>$= -44$       | (55) $24 - (-9)$<br>$= 33$    | (74) $-51 + (-15)$<br>$= -66$  |
| (18) $30 + (-32)$<br>$= -2$   | (37) $-63 + 54$<br>$= -9$      | (56) $-58 - (-22)$<br>$= -36$ | (75) $-26 + 73$<br>$= 47$      |
| (19) $-88 + 49$<br>$= -39$    | (38) $-55 + 30$<br>$= -25$     | (57) $-15 - (-49)$<br>$= 34$  | (76) $-76 + 76$<br>$= 0$       |

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| (1) $-79 - 58$<br>$= -137$     | (20) $31 + (-53)$<br>$= -22$    | (39) $-93 + (-19)$<br>$= -112$ | (58) $84 + 74$<br>$= 158$       |
| (2) $70 + 45$<br>$= 115$       | (21) $-12 + 50$<br>$= 38$       | (40) $-20 - (-32)$<br>$= 12$   | (59) $-9 + (-92)$<br>$= -101$   |
| (3) $-60 + 78$<br>$= 18$       | (22) $86 - (-69)$<br>$= 155$    | (41) $-70 - 15$<br>$= -85$     | (60) $-5 - 78$<br>$= -83$       |
| (4) $2 - (-53)$<br>$= 55$      | (23) $72 - (-96)$<br>$= 168$    | (42) $5 - 51$<br>$= -46$       | (61) $52 - 57$<br>$= -5$        |
| (5) $-29 + (-4)$<br>$= -33$    | (24) $80 + (-53)$<br>$= 27$     | (43) $78 - 2$<br>$= 76$        | (62) $-92 - (-35)$<br>$= -57$   |
| (6) $-75 - 97$<br>$= -172$     | (25) $-58 + 20$<br>$= -38$      | (44) $90 + 57$<br>$= 147$      | (63) $14 + (-89)$<br>$= -75$    |
| (7) $-80 + (-32)$<br>$= -112$  | (26) $-100 + (-20)$<br>$= -120$ | (45) $-3 - (-24)$<br>$= 21$    | (64) $83 + 69$<br>$= 152$       |
| (8) $66 + (-5)$<br>$= 61$      | (27) $-35 + (-54)$<br>$= -89$   | (46) $-95 - 88$<br>$= -183$    | (65) $77 + (-50)$<br>$= 27$     |
| (9) $-22 - (-63)$<br>$= 41$    | (28) $-43 + (-44)$<br>$= -87$   | (47) $-45 - (-95)$<br>$= 50$   | (66) $38 + (-59)$<br>$= -21$    |
| (10) $95 + 64$<br>$= 159$      | (29) $-70 + 81$<br>$= 11$       | (48) $57 + (-36)$<br>$= 21$    | (67) $90 + (-21)$<br>$= 69$     |
| (11) $-16 + (-24)$<br>$= -40$  | (30) $18 - (-53)$<br>$= 71$     | (49) $-98 + (-72)$<br>$= -170$ | (68) $15 - 12$<br>$= 3$         |
| (12) $-8 - (-14)$<br>$= 6$     | (31) $76 + 100$<br>$= 176$      | (50) $46 + (-79)$<br>$= -33$   | (69) $71 + 68$<br>$= 139$       |
| (13) $16 - 86$<br>$= -70$      | (32) $-58 + (-12)$<br>$= -70$   | (51) $-29 + 7$<br>$= -22$      | (70) $-100 + (-17)$<br>$= -117$ |
| (14) $-77 - 97$<br>$= -174$    | (33) $50 - 72$<br>$= -22$       | (52) $-85 - (-30)$<br>$= -55$  | (71) $-61 + (-94)$<br>$= -155$  |
| (15) $-71 + 14$<br>$= -57$     | (34) $0 - (-35)$<br>$= 35$      | (53) $69 + (-33)$<br>$= 36$    | (72) $-74 - (-89)$<br>$= 15$    |
| (16) $-14 - 66$<br>$= -80$     | (35) $14 - 82$<br>$= -68$       | (54) $29 - 39$<br>$= -10$      | (73) $64 + 100$<br>$= 164$      |
| (17) $-9 - (-60)$<br>$= 51$    | (36) $67 + 10$<br>$= 77$        | (55) $-94 + 95$<br>$= 1$       | (74) $-91 - 78$<br>$= -169$     |
| (18) $-92 + (-48)$<br>$= -140$ | (37) $-75 + 17$<br>$= -58$      | (56) $-50 + 0$<br>$= -50$      | (75) $84 + (-62)$<br>$= 22$     |
| (19) $27 + (-69)$<br>$= -42$   | (38) $-69 - 38$<br>$= -107$     | (57) $-1 + 31$<br>$= 30$       | (76) $-74 + (-9)$<br>$= -83$    |

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| (1) $65 + 16$<br>$= 81$       | (20) $11 - 79$<br>$= -68$      | (39) $56 + 87$<br>$= 143$     | (58) $-28 - 58$<br>$= -86$     |
| (2) $-3 - (-10)$<br>$= 7$     | (21) $19 - 56$<br>$= -37$      | (40) $-13 - 84$<br>$= -97$    | (59) $-6 - (-8)$<br>$= 2$      |
| (3) $-17 - 64$<br>$= -81$     | (22) $93 - (-95)$<br>$= 188$   | (41) $-36 + (-15)$<br>$= -51$ | (60) $-97 + (-28)$<br>$= -125$ |
| (4) $-95 + (-14)$<br>$= -109$ | (23) $-12 + (-31)$<br>$= -43$  | (42) $-56 + 40$<br>$= -16$    | (61) $33 + 63$<br>$= 96$       |
| (5) $86 - (-58)$<br>$= 144$   | (24) $42 - 100$<br>$= -58$     | (43) $-15 - (-92)$<br>$= 77$  | (62) $43 - 0$<br>$= 43$        |
| (6) $-13 + 65$<br>$= 52$      | (25) $-64 - (-8)$<br>$= -56$   | (44) $30 - 84$<br>$= -54$     | (63) $73 + 18$<br>$= 91$       |
| (7) $91 + 62$<br>$= 153$      | (26) $-74 - 58$<br>$= -132$    | (45) $-30 + (-52)$<br>$= -82$ | (64) $100 + (-78)$<br>$= 22$   |
| (8) $-11 + (-98)$<br>$= -109$ | (27) $-88 - 66$<br>$= -154$    | (46) $-89 + 53$<br>$= -36$    | (65) $68 + 65$<br>$= 133$      |
| (9) $3 + 63$<br>$= 66$        | (28) $77 + 73$<br>$= 150$      | (47) $-76 + 76$<br>$= 0$      | (66) $44 - (-100)$<br>$= 144$  |
| (10) $-25 - (-92)$<br>$= 67$  | (29) $82 - (-57)$<br>$= 139$   | (48) $19 - (-65)$<br>$= 84$   | (67) $19 - (-92)$<br>$= 111$   |
| (11) $-51 + (-24)$<br>$= -75$ | (30) $91 - (-100)$<br>$= 191$  | (49) $-61 + 4$<br>$= -57$     | (68) $72 - (-22)$<br>$= 94$    |
| (12) $69 - 86$<br>$= -17$     | (31) $-20 - (-12)$<br>$= -8$   | (50) $-93 + (-3)$<br>$= -96$  | (69) $-54 + 93$<br>$= 39$      |
| (13) $-56 + 92$<br>$= 36$     | (32) $-51 + 43$<br>$= -8$      | (51) $-48 + (-26)$<br>$= -74$ | (70) $-90 - 39$<br>$= -129$    |
| (14) $61 - (-52)$<br>$= 113$  | (33) $-9 + 10$<br>$= 1$        | (52) $26 + (-19)$<br>$= 7$    | (71) $35 - (-90)$<br>$= 125$   |
| (15) $-39 - 95$<br>$= -134$   | (34) $44 - (-54)$<br>$= 98$    | (53) $52 - 28$<br>$= 24$      | (72) $65 + (-55)$<br>$= 10$    |
| (16) $-68 - 100$<br>$= -168$  | (35) $-53 + (-54)$<br>$= -107$ | (54) $14 - 18$<br>$= -4$      | (73) $-40 - 96$<br>$= -136$    |
| (17) $-91 - 100$<br>$= -191$  | (36) $21 + (-88)$<br>$= -67$   | (55) $76 - (-31)$<br>$= 107$  | (74) $-61 + 11$<br>$= -50$     |
| (18) $72 - 12$<br>$= 60$      | (37) $-7 + 59$<br>$= 52$       | (56) $-73 - 69$<br>$= -142$   | (75) $70 + (-42)$<br>$= 28$    |
| (19) $-71 - (-66)$<br>$= -5$  | (38) $96 + 60$<br>$= 156$      | (57) $-1 - (-18)$<br>$= 17$   | (76) $-68 + 1$<br>$= -67$      |



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| (1) $85 - 87$<br>$= -2$        | (20) $99 + (-38)$<br>$= 61$    | (39) $-55 + 69$<br>$= 14$     | (58) $26 - (-57)$<br>$= 83$    |
| (2) $-29 + (-62)$<br>$= -91$   | (21) $-56 - (-53)$<br>$= -3$   | (40) $61 - (-39)$<br>$= 100$  | (59) $-59 - (-73)$<br>$= 14$   |
| (3) $-91 - (-32)$<br>$= -59$   | (22) $46 + 16$<br>$= 62$       | (41) $-28 + 80$<br>$= 52$     | (60) $-68 + (-49)$<br>$= -117$ |
| (4) $-45 + (-76)$<br>$= -121$  | (23) $-42 + 22$<br>$= -20$     | (42) $22 + (-44)$<br>$= -22$  | (61) $-100 - 0$<br>$= -100$    |
| (5) $99 + 100$<br>$= 199$      | (24) $39 + 18$<br>$= 57$       | (43) $64 + 47$<br>$= 111$     | (62) $-18 + 64$<br>$= 46$      |
| (6) $22 - (-68)$<br>$= 90$     | (25) $27 - 2$<br>$= 25$        | (44) $54 - (-87)$<br>$= 141$  | (63) $8 + (-20)$<br>$= -12$    |
| (7) $24 - (-5)$<br>$= 29$      | (26) $82 - (-65)$<br>$= 147$   | (45) $-100 + 22$<br>$= -78$   | (64) $-38 - (-24)$<br>$= -14$  |
| (8) $19 + 35$<br>$= 54$        | (27) $90 - 56$<br>$= 34$       | (46) $14 - 70$<br>$= -56$     | (65) $-50 + 24$<br>$= -26$     |
| (9) $92 + 53$<br>$= 145$       | (28) $72 + 14$<br>$= 86$       | (47) $-94 - 61$<br>$= -155$   | (66) $-78 - 45$<br>$= -123$    |
| (10) $-26 - 21$<br>$= -47$     | (29) $2 - 82$<br>$= -80$       | (48) $57 - (-88)$<br>$= 145$  | (67) $1 + (-15)$<br>$= -14$    |
| (11) $36 - 16$<br>$= 20$       | (30) $-39 + (-39)$<br>$= -78$  | (49) $0 + (-29)$<br>$= -29$   | (68) $46 + 57$<br>$= 103$      |
| (12) $-33 + (-77)$<br>$= -110$ | (31) $13 - 98$<br>$= -85$      | (50) $-21 + 18$<br>$= -3$     | (69) $-18 - 85$<br>$= -103$    |
| (13) $-9 - (-9)$<br>$= 0$      | (32) $85 + 54$<br>$= 139$      | (51) $-37 + (-1)$<br>$= -38$  | (70) $-12 + (-3)$<br>$= -15$   |
| (14) $-82 + 91$<br>$= 9$       | (33) $-98 + (-68)$<br>$= -166$ | (52) $-22 - (-73)$<br>$= 51$  | (71) $-37 + (-44)$<br>$= -81$  |
| (15) $89 + 45$<br>$= 134$      | (34) $-4 + (-11)$<br>$= -15$   | (53) $-69 - (-15)$<br>$= -54$ | (72) $-72 - (-89)$<br>$= 17$   |
| (16) $79 - (-64)$<br>$= 143$   | (35) $39 - (-86)$<br>$= 125$   | (54) $-78 - (-93)$<br>$= 15$  | (73) $-78 + 76$<br>$= -2$      |
| (17) $9 + (-41)$<br>$= -32$    | (36) $-38 + (-68)$<br>$= -106$ | (55) $44 + 8$<br>$= 52$       | (74) $-84 - (-65)$<br>$= -19$  |
| (18) $-55 + (-68)$<br>$= -123$ | (37) $-44 + 36$<br>$= -8$      | (56) $-65 + 47$<br>$= -18$    | (75) $-25 + (-24)$<br>$= -49$  |
| (19) $5 + 47$<br>$= 52$        | (38) $29 + (-34)$<br>$= -5$    | (57) $-83 + (-16)$<br>$= -99$ | (76) $42 - (-55)$<br>$= 97$    |

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| (1) $-49 + 2$<br>$= -47$       | (20) $-69 - 33$<br>$= -102$   | (39) $-7 + 84$<br>$= 77$      | (58) $-73 - 57$<br>$= -130$  |
| (2) $47 - (-49)$<br>$= 96$     | (21) $5 + 45$<br>$= 50$       | (40) $-59 - (-70)$<br>$= 11$  | (59) $58 + (-18)$<br>$= 40$  |
| (3) $-65 - (-71)$<br>$= 6$     | (22) $-78 + 1$<br>$= -77$     | (41) $9 - 29$<br>$= -20$      | (60) $19 + (-9)$<br>$= 10$   |
| (4) $86 - 39$<br>$= 47$        | (23) $-52 - (-11)$<br>$= -41$ | (42) $45 - (-75)$<br>$= 120$  | (61) $52 + 22$<br>$= 74$     |
| (5) $32 + 38$<br>$= 70$        | (24) $-46 + 31$<br>$= -15$    | (43) $-25 + (-10)$<br>$= -35$ | (62) $-33 - (-88)$<br>$= 55$ |
| (6) $4 - 66$<br>$= -62$        | (25) $-48 - 46$<br>$= -94$    | (44) $-78 - 18$<br>$= -96$    | (63) $67 + (-75)$<br>$= -8$  |
| (7) $-26 - 27$<br>$= -53$      | (26) $-18 - (-46)$<br>$= 28$  | (45) $-75 - 35$<br>$= -110$   | (64) $-16 - (-20)$<br>$= 4$  |
| (8) $-46 - (-15)$<br>$= -31$   | (27) $-28 - 55$<br>$= -83$    | (46) $-58 + 70$<br>$= 12$     | (65) $-84 - 52$<br>$= -136$  |
| (9) $61 - 41$<br>$= 20$        | (28) $55 + (-72)$<br>$= -17$  | (47) $-67 + 16$<br>$= -51$    | (66) $-3 + (-96)$<br>$= -99$ |
| (10) $-54 - 3$<br>$= -57$      | (29) $38 - 34$<br>$= 4$       | (48) $66 - (-19)$<br>$= 85$   | (67) $16 - (-35)$<br>$= 51$  |
| (11) $79 - (-36)$<br>$= 115$   | (30) $-63 - 80$<br>$= -143$   | (49) $-18 + 15$<br>$= -3$     | (68) $98 - 46$<br>$= 52$     |
| (12) $9 + (-17)$<br>$= -8$     | (31) $-72 - (-44)$<br>$= -28$ | (50) $-25 + 42$<br>$= 17$     | (69) $10 - (-47)$<br>$= 57$  |
| (13) $-14 - (-24)$<br>$= 10$   | (32) $75 - 74$<br>$= 1$       | (51) $-82 - 61$<br>$= -143$   | (70) $-46 - (-79)$<br>$= 33$ |
| (14) $5 - (-20)$<br>$= 25$     | (33) $66 - (-63)$<br>$= 129$  | (52) $18 + 61$<br>$= 79$      | (71) $83 - 76$<br>$= 7$      |
| (15) $-6 - 47$<br>$= -53$      | (34) $-69 - 15$<br>$= -84$    | (53) $-54 - 43$<br>$= -97$    | (72) $25 + (-45)$<br>$= -20$ |
| (16) $57 - (-79)$<br>$= 136$   | (35) $15 - (-9)$<br>$= 24$    | (54) $74 + (-99)$<br>$= -25$  | (73) $-23 + 97$<br>$= 74$    |
| (17) $8 + 58$<br>$= 66$        | (36) $-88 + 82$<br>$= -6$     | (55) $-54 + 97$<br>$= 43$     | (74) $48 - (-6)$<br>$= 54$   |
| (18) $-12 + (-98)$<br>$= -110$ | (37) $72 + 91$<br>$= 163$     | (56) $-66 + 39$<br>$= -27$    | (75) $23 - 18$<br>$= 5$      |
| (19) $-82 - (-12)$<br>$= -70$  | (38) $83 + 34$<br>$= 117$     | (57) $-36 - 64$<br>$= -100$   | (76) $20 - (-83)$<br>$= 103$ |

### 1.2.2 乗法

整数の積を求める問題。

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-7 \times (-7)$  | (31) $-8 \times (-2)$ | (61) $-8 \times (-3)$ | (91) $5 \times (-7)$   |
| (2) $7 \times (-1)$   | (32) $7 \times 3$     | (62) $7 \times 2$     | (92) $-7 \times (-9)$  |
| (3) $8 \times (-6)$   | (33) $9 \times (-9)$  | (63) $8 \times (-3)$  | (93) $-2 \times (-3)$  |
| (4) $-7 \times 5$     | (34) $5 \times 3$     | (64) $-1 \times (-9)$ | (94) $-7 \times 0$     |
| (5) $-5 \times 6$     | (35) $7 \times (-9)$  | (65) $5 \times 2$     | (95) $3 \times (-2)$   |
| (6) $8 \times (-8)$   | (36) $-7 \times 4$    | (66) $-8 \times 6$    | (96) $-2 \times (-1)$  |
| (7) $7 \times (-2)$   | (37) $-2 \times 9$    | (67) $-3 \times 8$    | (97) $-9 \times (-2)$  |
| (8) $0 \times 6$      | (38) $4 \times (-7)$  | (68) $0 \times 7$     | (98) $4 \times (-9)$   |
| (9) $-9 \times 8$     | (39) $-9 \times 1$    | (69) $-1 \times (-8)$ | (99) $-1 \times 5$     |
| (10) $-2 \times 5$    | (40) $-6 \times (-6)$ | (70) $4 \times 9$     | (100) $-1 \times (-1)$ |
| (11) $6 \times 7$     | (41) $7 \times 6$     | (71) $-6 \times 1$    | (101) $3 \times 4$     |
| (12) $2 \times (-9)$  | (42) $-3 \times 1$    | (72) $-7 \times (-1)$ | (102) $-8 \times (-3)$ |
| (13) $3 \times 2$     | (43) $7 \times (-2)$  | (73) $-4 \times 0$    | (103) $7 \times (-4)$  |
| (14) $-3 \times 1$    | (44) $-5 \times (-8)$ | (74) $4 \times (-6)$  | (104) $-5 \times 2$    |
| (15) $1 \times 4$     | (45) $-9 \times 7$    | (75) $5 \times (-2)$  | (105) $1 \times 9$     |
| (16) $-6 \times 6$    | (46) $-3 \times 6$    | (76) $9 \times 3$     | (106) $-3 \times 2$    |
| (17) $-7 \times 3$    | (47) $7 \times 9$     | (77) $-7 \times 4$    | (107) $9 \times (-8)$  |
| (18) $4 \times 1$     | (48) $-9 \times (-1)$ | (78) $4 \times (-1)$  | (108) $-1 \times 4$    |
| (19) $3 \times 2$     | (49) $0 \times (-1)$  | (79) $-9 \times 3$    | (109) $-6 \times 1$    |
| (20) $4 \times (-1)$  | (50) $8 \times (-1)$  | (80) $-4 \times 2$    | (110) $-1 \times (-6)$ |
| (21) $-6 \times 0$    | (51) $-9 \times 4$    | (81) $5 \times 8$     | (111) $6 \times (-4)$  |
| (22) $8 \times (-9)$  | (52) $-7 \times 3$    | (82) $2 \times (-3)$  | (112) $-8 \times (-7)$ |
| (23) $-8 \times (-9)$ | (53) $-8 \times (-2)$ | (83) $-5 \times 9$    | (113) $-9 \times (-7)$ |
| (24) $0 \times 4$     | (54) $-1 \times 7$    | (84) $-9 \times 5$    | (114) $-8 \times 7$    |
| (25) $7 \times (-3)$  | (55) $-5 \times (-9)$ | (85) $0 \times (-9)$  | (115) $1 \times (-3)$  |
| (26) $0 \times 1$     | (56) $-9 \times (-8)$ | (86) $4 \times (-6)$  | (116) $2 \times 5$     |
| (27) $-8 \times (-5)$ | (57) $-6 \times (-3)$ | (87) $-6 \times (-5)$ | (117) $4 \times (-9)$  |
| (28) $-9 \times 8$    | (58) $1 \times (-2)$  | (88) $5 \times (-5)$  | (118) $1 \times 9$     |
| (29) $-9 \times (-6)$ | (59) $-1 \times (-3)$ | (89) $-8 \times 5$    | (119) $6 \times (-3)$  |
| (30) $4 \times (-8)$  | (60) $-5 \times 8$    | (90) $3 \times 2$     | (120) $9 \times 6$     |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $8 \times 9$      | (31) $0 \times 8$     | (61) $-2 \times 0$    | (91) $5 \times 4$      |
| (2) $-4 \times (-8)$  | (32) $3 \times (-9)$  | (62) $6 \times 6$     | (92) $6 \times 2$      |
| (3) $-6 \times 6$     | (33) $8 \times 4$     | (63) $4 \times (-9)$  | (93) $7 \times 6$      |
| (4) $1 \times (-4)$   | (34) $5 \times 2$     | (64) $-7 \times (-2)$ | (94) $4 \times (-6)$   |
| (5) $-5 \times (-3)$  | (35) $-4 \times (-8)$ | (65) $-5 \times (-8)$ | (95) $-8 \times 4$     |
| (6) $0 \times 7$      | (36) $-3 \times (-8)$ | (66) $6 \times 7$     | (96) $3 \times (-3)$   |
| (7) $-7 \times 3$     | (37) $2 \times (-1)$  | (67) $5 \times 1$     | (97) $2 \times (-2)$   |
| (8) $-9 \times (-9)$  | (38) $-6 \times 8$    | (68) $-1 \times (-6)$ | (98) $-4 \times 8$     |
| (9) $-1 \times 3$     | (39) $0 \times (-7)$  | (69) $-2 \times (-2)$ | (99) $8 \times (-7)$   |
| (10) $-1 \times 9$    | (40) $7 \times 2$     | (70) $1 \times 2$     | (100) $6 \times (-6)$  |
| (11) $-7 \times 0$    | (41) $-3 \times 3$    | (71) $7 \times 2$     | (101) $1 \times 5$     |
| (12) $2 \times 8$     | (42) $5 \times (-4)$  | (72) $-8 \times 1$    | (102) $-2 \times 9$    |
| (13) $8 \times 8$     | (43) $-6 \times 7$    | (73) $-1 \times 2$    | (103) $-3 \times 5$    |
| (14) $7 \times 3$     | (44) $1 \times 2$     | (74) $0 \times (-7)$  | (104) $5 \times 0$     |
| (15) $-8 \times (-5)$ | (45) $9 \times 6$     | (75) $-5 \times (-3)$ | (105) $-2 \times (-2)$ |
| (16) $0 \times (-7)$  | (46) $-7 \times 3$    | (76) $-2 \times 1$    | (106) $7 \times (-9)$  |
| (17) $3 \times 5$     | (47) $-2 \times 8$    | (77) $-2 \times (-1)$ | (107) $7 \times (-3)$  |
| (18) $0 \times 8$     | (48) $-1 \times 2$    | (78) $7 \times (-6)$  | (108) $4 \times 3$     |
| (19) $-6 \times (-7)$ | (49) $-2 \times (-1)$ | (79) $-1 \times (-6)$ | (109) $-1 \times 9$    |
| (20) $-3 \times 8$    | (50) $-7 \times (-2)$ | (80) $-5 \times 9$    | (110) $-8 \times 2$    |
| (21) $-4 \times (-2)$ | (51) $3 \times (-5)$  | (81) $-5 \times 5$    | (111) $3 \times (-8)$  |
| (22) $-8 \times 7$    | (52) $-4 \times (-8)$ | (82) $0 \times 1$     | (112) $-6 \times 8$    |
| (23) $-9 \times (-4)$ | (53) $5 \times 1$     | (83) $8 \times (-3)$  | (113) $-2 \times (-4)$ |
| (24) $-9 \times 5$    | (54) $-9 \times 8$    | (84) $3 \times (-6)$  | (114) $-7 \times 2$    |
| (25) $-1 \times 0$    | (55) $4 \times 9$     | (85) $3 \times 1$     | (115) $3 \times 3$     |
| (26) $-2 \times 4$    | (56) $0 \times 4$     | (86) $-9 \times 1$    | (116) $-7 \times 1$    |
| (27) $-7 \times (-8)$ | (57) $6 \times 5$     | (87) $-2 \times (-1)$ | (117) $1 \times (-6)$  |
| (28) $2 \times 4$     | (58) $5 \times 7$     | (88) $8 \times (-6)$  | (118) $7 \times 4$     |
| (29) $-3 \times 2$    | (59) $8 \times 8$     | (89) $-9 \times 0$    | (119) $6 \times 6$     |
| (30) $8 \times (-4)$  | (60) $5 \times (-9)$  | (90) $9 \times 1$     | (120) $0 \times (-1)$  |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $0 \times 9$      | (31) $-4 \times 8$    | (61) $7 \times 5$     | (91) $-7 \times 0$     |
| (2) $6 \times (-7)$   | (32) $7 \times (-8)$  | (62) $8 \times (-2)$  | (92) $-2 \times 4$     |
| (3) $-6 \times (-8)$  | (33) $-9 \times 6$    | (63) $-5 \times (-7)$ | (93) $2 \times 5$      |
| (4) $-6 \times (-7)$  | (34) $0 \times (-3)$  | (64) $2 \times (-3)$  | (94) $-9 \times (-4)$  |
| (5) $-9 \times (-1)$  | (35) $-5 \times 4$    | (65) $2 \times (-4)$  | (95) $-8 \times (-7)$  |
| (6) $-3 \times 4$     | (36) $2 \times 0$     | (66) $-2 \times 5$    | (96) $-9 \times 7$     |
| (7) $-8 \times 7$     | (37) $7 \times (-4)$  | (67) $-9 \times (-1)$ | (97) $6 \times 6$      |
| (8) $-4 \times (-1)$  | (38) $-6 \times (-1)$ | (68) $-5 \times 2$    | (98) $5 \times (-3)$   |
| (9) $-8 \times 0$     | (39) $-5 \times (-4)$ | (69) $3 \times (-9)$  | (99) $4 \times (-7)$   |
| (10) $-4 \times (-4)$ | (40) $-5 \times (-2)$ | (70) $7 \times 0$     | (100) $8 \times (-8)$  |
| (11) $-7 \times (-3)$ | (41) $-7 \times (-7)$ | (71) $-3 \times (-6)$ | (101) $8 \times (-4)$  |
| (12) $5 \times 3$     | (42) $-9 \times (-6)$ | (72) $-7 \times 7$    | (102) $-3 \times (-2)$ |
| (13) $-1 \times 7$    | (43) $-1 \times (-8)$ | (73) $-7 \times 9$    | (103) $-5 \times 0$    |
| (14) $-7 \times (-3)$ | (44) $1 \times 6$     | (74) $-9 \times (-4)$ | (104) $4 \times 8$     |
| (15) $7 \times 4$     | (45) $4 \times 7$     | (75) $7 \times (-6)$  | (105) $0 \times (-5)$  |
| (16) $4 \times (-2)$  | (46) $5 \times 7$     | (76) $-7 \times (-9)$ | (106) $-8 \times (-6)$ |
| (17) $-3 \times (-1)$ | (47) $9 \times (-1)$  | (77) $-7 \times 7$    | (107) $0 \times 9$     |
| (18) $-9 \times 9$    | (48) $0 \times (-9)$  | (78) $-1 \times (-6)$ | (108) $1 \times (-1)$  |
| (19) $9 \times 5$     | (49) $8 \times 4$     | (79) $-5 \times (-9)$ | (109) $7 \times 2$     |
| (20) $3 \times 9$     | (50) $-4 \times 5$    | (80) $2 \times (-4)$  | (110) $8 \times 6$     |
| (21) $2 \times 4$     | (51) $-9 \times (-1)$ | (81) $-8 \times 9$    | (111) $0 \times 1$     |
| (22) $5 \times 4$     | (52) $-4 \times (-2)$ | (82) $-5 \times 3$    | (112) $-9 \times (-3)$ |
| (23) $-5 \times 5$    | (53) $3 \times (-8)$  | (83) $1 \times (-3)$  | (113) $7 \times (-1)$  |
| (24) $-8 \times (-1)$ | (54) $1 \times 9$     | (84) $-1 \times (-1)$ | (114) $5 \times (-8)$  |
| (25) $-3 \times (-4)$ | (55) $-2 \times 4$    | (85) $-7 \times (-7)$ | (115) $-4 \times 2$    |
| (26) $-9 \times (-8)$ | (56) $-9 \times 6$    | (86) $-4 \times 3$    | (116) $-2 \times 0$    |
| (27) $7 \times (-2)$  | (57) $-2 \times 9$    | (87) $7 \times (-6)$  | (117) $1 \times 8$     |
| (28) $0 \times (-2)$  | (58) $5 \times 6$     | (88) $2 \times 5$     | (118) $-5 \times 3$    |
| (29) $6 \times 9$     | (59) $-9 \times 0$    | (89) $7 \times 3$     | (119) $2 \times (-7)$  |
| (30) $4 \times 1$     | (60) $-6 \times (-7)$ | (90) $-3 \times 8$    | (120) $-9 \times 5$    |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-9 \times 0$     | (31) $4 \times 7$     | (61) $-6 \times (-9)$ | (91) $-3 \times 8$     |
| (2) $-7 \times (-1)$  | (32) $7 \times 4$     | (62) $1 \times 7$     | (92) $-9 \times 0$     |
| (3) $-5 \times 8$     | (33) $3 \times (-2)$  | (63) $5 \times 3$     | (93) $1 \times 7$      |
| (4) $-9 \times 0$     | (34) $4 \times 8$     | (64) $2 \times 6$     | (94) $9 \times 7$      |
| (5) $3 \times 1$      | (35) $4 \times 2$     | (65) $4 \times 9$     | (95) $-4 \times 4$     |
| (6) $-3 \times (-3)$  | (36) $2 \times 8$     | (66) $2 \times (-7)$  | (96) $-9 \times 7$     |
| (7) $-5 \times 0$     | (37) $8 \times 0$     | (67) $4 \times 7$     | (97) $-5 \times (-7)$  |
| (8) $-1 \times 5$     | (38) $8 \times (-4)$  | (68) $-3 \times 8$    | (98) $2 \times 0$      |
| (9) $-5 \times 1$     | (39) $-3 \times (-9)$ | (69) $-1 \times 0$    | (99) $4 \times 9$      |
| (10) $-4 \times 7$    | (40) $-3 \times 1$    | (70) $9 \times (-8)$  | (100) $7 \times (-6)$  |
| (11) $2 \times (-8)$  | (41) $-4 \times (-2)$ | (71) $6 \times 4$     | (101) $1 \times 5$     |
| (12) $2 \times (-9)$  | (42) $-7 \times 0$    | (72) $-5 \times 7$    | (102) $-4 \times 4$    |
| (13) $-1 \times 0$    | (43) $4 \times 2$     | (73) $-7 \times 9$    | (103) $5 \times 2$     |
| (14) $2 \times 7$     | (44) $-3 \times 9$    | (74) $-9 \times 7$    | (104) $-5 \times (-8)$ |
| (15) $-2 \times (-7)$ | (45) $9 \times 3$     | (75) $-1 \times 6$    | (105) $9 \times (-1)$  |
| (16) $8 \times (-2)$  | (46) $-1 \times 8$    | (76) $-9 \times 9$    | (106) $6 \times 0$     |
| (17) $9 \times (-6)$  | (47) $-1 \times 3$    | (77) $0 \times 8$     | (107) $1 \times 4$     |
| (18) $8 \times 5$     | (48) $6 \times 3$     | (78) $3 \times (-8)$  | (108) $7 \times (-3)$  |
| (19) $8 \times (-3)$  | (49) $-3 \times 1$    | (79) $-9 \times (-9)$ | (109) $0 \times 1$     |
| (20) $6 \times 0$     | (50) $-8 \times (-9)$ | (80) $-5 \times 7$    | (110) $-6 \times 5$    |
| (21) $-9 \times (-7)$ | (51) $-6 \times (-2)$ | (81) $7 \times (-2)$  | (111) $9 \times 6$     |
| (22) $5 \times (-4)$  | (52) $7 \times (-5)$  | (82) $0 \times (-6)$  | (112) $-5 \times 8$    |
| (23) $-1 \times 2$    | (53) $-7 \times 6$    | (83) $3 \times 2$     | (113) $8 \times 3$     |
| (24) $8 \times (-1)$  | (54) $-7 \times 2$    | (84) $-5 \times 0$    | (114) $-3 \times 9$    |
| (25) $4 \times (-1)$  | (55) $6 \times 9$     | (85) $-2 \times 5$    | (115) $-3 \times 2$    |
| (26) $-7 \times 5$    | (56) $1 \times (-3)$  | (86) $2 \times (-4)$  | (116) $-7 \times 0$    |
| (27) $1 \times (-7)$  | (57) $-2 \times 2$    | (87) $-4 \times 0$    | (117) $4 \times (-9)$  |
| (28) $9 \times 8$     | (58) $7 \times (-4)$  | (88) $6 \times 9$     | (118) $0 \times (-4)$  |
| (29) $-2 \times 1$    | (59) $6 \times (-9)$  | (89) $-4 \times (-7)$ | (119) $-9 \times (-3)$ |
| (30) $-7 \times (-1)$ | (60) $-4 \times (-9)$ | (90) $-4 \times (-1)$ | (120) $-5 \times (-7)$ |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $2 \times (-3)$   | (31) $-1 \times (-4)$ | (61) $9 \times 0$     | (91) $4 \times 7$      |
| (2) $7 \times (-9)$   | (32) $-6 \times 2$    | (62) $-5 \times (-5)$ | (92) $-1 \times (-5)$  |
| (3) $2 \times 6$      | (33) $8 \times 0$     | (63) $6 \times (-1)$  | (93) $8 \times (-9)$   |
| (4) $1 \times 9$      | (34) $-3 \times (-4)$ | (64) $3 \times 2$     | (94) $4 \times (-1)$   |
| (5) $9 \times (-2)$   | (35) $-4 \times 3$    | (65) $-1 \times 5$    | (95) $7 \times (-5)$   |
| (6) $-2 \times 8$     | (36) $6 \times (-9)$  | (66) $-5 \times 6$    | (96) $-2 \times (-8)$  |
| (7) $-2 \times 0$     | (37) $-2 \times 8$    | (67) $-7 \times (-6)$ | (97) $2 \times 3$      |
| (8) $-6 \times (-4)$  | (38) $7 \times 9$     | (68) $-7 \times 8$    | (98) $8 \times 0$      |
| (9) $0 \times (-1)$   | (39) $8 \times (-6)$  | (69) $-3 \times (-3)$ | (99) $-2 \times 1$     |
| (10) $5 \times (-3)$  | (40) $-7 \times (-5)$ | (70) $2 \times (-1)$  | (100) $0 \times 0$     |
| (11) $-3 \times 5$    | (41) $1 \times (-4)$  | (71) $-2 \times (-3)$ | (101) $-7 \times (-7)$ |
| (12) $1 \times 5$     | (42) $4 \times 1$     | (72) $4 \times (-7)$  | (102) $-8 \times 9$    |
| (13) $-8 \times 9$    | (43) $3 \times (-6)$  | (73) $7 \times 5$     | (103) $8 \times 8$     |
| (14) $-5 \times 1$    | (44) $8 \times (-4)$  | (74) $2 \times 3$     | (104) $1 \times 8$     |
| (15) $8 \times (-5)$  | (45) $-7 \times 8$    | (75) $1 \times 3$     | (105) $-6 \times (-9)$ |
| (16) $-1 \times (-5)$ | (46) $-2 \times 6$    | (76) $-1 \times 6$    | (106) $-5 \times 3$    |
| (17) $0 \times 5$     | (47) $8 \times (-7)$  | (77) $4 \times 4$     | (107) $-5 \times 2$    |
| (18) $-9 \times 2$    | (48) $-6 \times (-5)$ | (78) $-3 \times (-7)$ | (108) $-5 \times (-2)$ |
| (19) $-4 \times (-1)$ | (49) $4 \times 8$     | (79) $9 \times (-1)$  | (109) $-8 \times (-8)$ |
| (20) $7 \times (-6)$  | (50) $8 \times 5$     | (80) $1 \times (-3)$  | (110) $5 \times 0$     |
| (21) $6 \times (-6)$  | (51) $7 \times (-9)$  | (81) $8 \times 9$     | (111) $0 \times (-3)$  |
| (22) $9 \times (-6)$  | (52) $0 \times 1$     | (82) $3 \times (-9)$  | (112) $0 \times 6$     |
| (23) $0 \times 1$     | (53) $-4 \times (-6)$ | (83) $-5 \times 9$    | (113) $8 \times (-1)$  |
| (24) $-2 \times (-6)$ | (54) $3 \times (-4)$  | (84) $7 \times (-3)$  | (114) $5 \times 4$     |
| (25) $2 \times (-6)$  | (55) $8 \times (-5)$  | (85) $-5 \times 0$    | (115) $7 \times 1$     |
| (26) $-9 \times 9$    | (56) $-1 \times (-8)$ | (86) $3 \times (-8)$  | (116) $8 \times (-2)$  |
| (27) $-3 \times 6$    | (57) $-7 \times (-4)$ | (87) $1 \times (-6)$  | (117) $-1 \times 1$    |
| (28) $6 \times 0$     | (58) $-6 \times 3$    | (88) $-3 \times (-2)$ | (118) $3 \times 7$     |
| (29) $-5 \times 9$    | (59) $6 \times 8$     | (89) $3 \times (-2)$  | (119) $6 \times (-4)$  |
| (30) $-9 \times (-5)$ | (60) $-7 \times (-8)$ | (90) $3 \times (-6)$  | (120) $-4 \times (-7)$ |



- |                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-4 \times (-4)$  | (31) $-2 \times (-6)$ | (61) $-6 \times (-2)$ | (91) $-3 \times 8$     |
| (2) $6 \times 8$      | (32) $-6 \times 0$    | (62) $6 \times (-1)$  | (92) $-7 \times 1$     |
| (3) $-6 \times 8$     | (33) $3 \times 6$     | (63) $-8 \times (-9)$ | (93) $-8 \times 0$     |
| (4) $1 \times (-9)$   | (34) $6 \times 7$     | (64) $0 \times (-5)$  | (94) $6 \times 1$      |
| (5) $-6 \times 2$     | (35) $-6 \times (-6)$ | (65) $-7 \times (-5)$ | (95) $0 \times (-5)$   |
| (6) $5 \times (-3)$   | (36) $-5 \times (-1)$ | (66) $2 \times (-6)$  | (96) $7 \times 9$      |
| (7) $7 \times (-1)$   | (37) $-2 \times 8$    | (67) $9 \times 8$     | (97) $6 \times 9$      |
| (8) $3 \times 7$      | (38) $9 \times 5$     | (68) $2 \times (-5)$  | (98) $9 \times 6$      |
| (9) $7 \times 0$      | (39) $-3 \times 4$    | (69) $2 \times 9$     | (99) $-2 \times 6$     |
| (10) $-1 \times (-6)$ | (40) $4 \times 2$     | (70) $4 \times 7$     | (100) $-6 \times (-2)$ |
| (11) $-1 \times 7$    | (41) $9 \times (-3)$  | (71) $-7 \times (-8)$ | (101) $9 \times 8$     |
| (12) $2 \times (-4)$  | (42) $-9 \times 8$    | (72) $8 \times (-8)$  | (102) $-6 \times 1$    |
| (13) $2 \times (-6)$  | (43) $1 \times (-3)$  | (73) $-3 \times 5$    | (103) $1 \times (-6)$  |
| (14) $5 \times (-1)$  | (44) $9 \times 4$     | (74) $-2 \times 9$    | (104) $-8 \times 3$    |
| (15) $0 \times 0$     | (45) $1 \times (-2)$  | (75) $7 \times (-3)$  | (105) $1 \times 0$     |
| (16) $-1 \times 1$    | (46) $0 \times 0$     | (76) $-4 \times (-9)$ | (106) $7 \times 8$     |
| (17) $7 \times (-6)$  | (47) $-6 \times 5$    | (77) $-9 \times (-8)$ | (107) $9 \times 7$     |
| (18) $2 \times 5$     | (48) $7 \times (-1)$  | (78) $5 \times (-6)$  | (108) $-4 \times 6$    |
| (19) $-6 \times (-2)$ | (49) $-8 \times 6$    | (79) $0 \times (-5)$  | (109) $2 \times 7$     |
| (20) $5 \times 7$     | (50) $-7 \times 7$    | (80) $4 \times 2$     | (110) $3 \times 1$     |
| (21) $5 \times (-4)$  | (51) $-3 \times 5$    | (81) $5 \times (-4)$  | (111) $0 \times (-5)$  |
| (22) $1 \times 2$     | (52) $3 \times 0$     | (82) $-9 \times (-1)$ | (112) $-1 \times (-1)$ |
| (23) $-5 \times 8$    | (53) $1 \times 4$     | (83) $-1 \times 8$    | (113) $-1 \times (-6)$ |
| (24) $4 \times (-8)$  | (54) $1 \times (-9)$  | (84) $-8 \times (-2)$ | (114) $9 \times 8$     |
| (25) $-2 \times (-6)$ | (55) $3 \times 9$     | (85) $5 \times (-4)$  | (115) $4 \times (-3)$  |
| (26) $3 \times 0$     | (56) $-9 \times 0$    | (86) $-6 \times (-6)$ | (116) $-9 \times 7$    |
| (27) $-6 \times (-9)$ | (57) $-9 \times (-7)$ | (87) $-5 \times 3$    | (117) $-7 \times 0$    |
| (28) $6 \times 0$     | (58) $7 \times 0$     | (88) $-6 \times (-3)$ | (118) $3 \times (-8)$  |
| (29) $-4 \times (-2)$ | (59) $6 \times (-8)$  | (89) $-9 \times 0$    | (119) $1 \times 3$     |
| (30) $-9 \times (-3)$ | (60) $-8 \times (-9)$ | (90) $-9 \times 6$    | (120) $1 \times (-9)$  |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $4 \times (-3)$   | (31) $-7 \times (-3)$ | (61) $-9 \times (-7)$ | (91) $0 \times (-2)$   |
| (2) $-4 \times (-6)$  | (32) $-4 \times 0$    | (62) $5 \times 9$     | (92) $-7 \times (-9)$  |
| (3) $-2 \times (-6)$  | (33) $9 \times (-8)$  | (63) $6 \times 0$     | (93) $5 \times 9$      |
| (4) $9 \times (-8)$   | (34) $-4 \times (-9)$ | (64) $3 \times (-9)$  | (94) $6 \times (-5)$   |
| (5) $3 \times 5$      | (35) $2 \times (-4)$  | (65) $7 \times 1$     | (95) $-6 \times 7$     |
| (6) $5 \times (-9)$   | (36) $0 \times 8$     | (66) $3 \times 4$     | (96) $6 \times 9$      |
| (7) $-4 \times 0$     | (37) $-5 \times (-6)$ | (67) $-5 \times 7$    | (97) $2 \times (-7)$   |
| (8) $4 \times (-6)$   | (38) $-6 \times (-9)$ | (68) $-5 \times 0$    | (98) $2 \times 6$      |
| (9) $2 \times 0$      | (39) $-1 \times (-1)$ | (69) $-4 \times (-7)$ | (99) $6 \times 9$      |
| (10) $-1 \times (-8)$ | (40) $9 \times (-8)$  | (70) $-7 \times (-2)$ | (100) $1 \times 8$     |
| (11) $-9 \times 8$    | (41) $-1 \times 7$    | (71) $7 \times 6$     | (101) $8 \times (-3)$  |
| (12) $4 \times 4$     | (42) $-7 \times 6$    | (72) $3 \times (-9)$  | (102) $-5 \times (-9)$ |
| (13) $-5 \times (-8)$ | (43) $2 \times (-9)$  | (73) $7 \times 5$     | (103) $-5 \times 5$    |
| (14) $-3 \times 4$    | (44) $6 \times (-3)$  | (74) $-1 \times 4$    | (104) $-9 \times 5$    |
| (15) $3 \times (-8)$  | (45) $-9 \times 8$    | (75) $8 \times (-5)$  | (105) $9 \times 5$     |
| (16) $3 \times 0$     | (46) $3 \times 1$     | (76) $-7 \times (-6)$ | (106) $8 \times 1$     |
| (17) $-6 \times 7$    | (47) $8 \times 8$     | (77) $5 \times (-9)$  | (107) $8 \times (-7)$  |
| (18) $-3 \times (-9)$ | (48) $6 \times (-5)$  | (78) $0 \times (-7)$  | (108) $4 \times 9$     |
| (19) $-5 \times (-7)$ | (49) $-5 \times (-1)$ | (79) $-2 \times (-2)$ | (109) $-2 \times 2$    |
| (20) $-3 \times (-4)$ | (50) $-2 \times 3$    | (80) $-5 \times 1$    | (110) $-7 \times (-7)$ |
| (21) $5 \times 6$     | (51) $5 \times 8$     | (81) $-7 \times (-5)$ | (111) $2 \times 6$     |
| (22) $4 \times (-5)$  | (52) $4 \times 3$     | (82) $-5 \times (-2)$ | (112) $-3 \times 1$    |
| (23) $8 \times (-9)$  | (53) $3 \times 3$     | (83) $-3 \times (-7)$ | (113) $2 \times (-8)$  |
| (24) $-4 \times 2$    | (54) $-9 \times (-1)$ | (84) $-8 \times 4$    | (114) $-6 \times 7$    |
| (25) $-4 \times 5$    | (55) $8 \times 0$     | (85) $-1 \times 1$    | (115) $9 \times 8$     |
| (26) $9 \times 4$     | (56) $8 \times (-8)$  | (86) $-8 \times 5$    | (116) $9 \times 1$     |
| (27) $3 \times 7$     | (57) $4 \times 8$     | (87) $-4 \times 5$    | (117) $-7 \times 4$    |
| (28) $-5 \times (-8)$ | (58) $-1 \times (-5)$ | (88) $-6 \times 4$    | (118) $9 \times 8$     |
| (29) $-4 \times 4$    | (59) $-4 \times 2$    | (89) $-8 \times (-6)$ | (119) $-6 \times (-8)$ |
| (30) $8 \times 2$     | (60) $8 \times 6$     | (90) $2 \times 2$     | (120) $-6 \times 3$    |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-1 \times (-5)$  | (31) $1 \times (-6)$  | (61) $1 \times 1$     | (91) $-8 \times (-3)$  |
| (2) $-5 \times 7$     | (32) $-1 \times (-1)$ | (62) $-8 \times (-7)$ | (92) $1 \times 4$      |
| (3) $2 \times 6$      | (33) $-9 \times 1$    | (63) $-3 \times 5$    | (93) $5 \times 1$      |
| (4) $-6 \times 8$     | (34) $-5 \times (-1)$ | (64) $5 \times (-8)$  | (94) $-6 \times (-4)$  |
| (5) $6 \times (-9)$   | (35) $-3 \times 4$    | (65) $-5 \times (-7)$ | (95) $0 \times (-8)$   |
| (6) $1 \times (-1)$   | (36) $-3 \times 5$    | (66) $-5 \times 4$    | (96) $2 \times (-4)$   |
| (7) $2 \times (-3)$   | (37) $0 \times 5$     | (67) $-9 \times (-4)$ | (97) $7 \times 2$      |
| (8) $-1 \times (-3)$  | (38) $2 \times 5$     | (68) $4 \times 6$     | (98) $1 \times 7$      |
| (9) $-7 \times (-7)$  | (39) $-9 \times (-5)$ | (69) $-5 \times (-8)$ | (99) $7 \times (-3)$   |
| (10) $-5 \times (-9)$ | (40) $9 \times 6$     | (70) $-9 \times 7$    | (100) $3 \times (-1)$  |
| (11) $-7 \times 2$    | (41) $-7 \times 0$    | (71) $-1 \times 7$    | (101) $2 \times 1$     |
| (12) $4 \times (-9)$  | (42) $-8 \times (-7)$ | (72) $8 \times 5$     | (102) $3 \times 3$     |
| (13) $-9 \times 3$    | (43) $-3 \times 0$    | (73) $-7 \times 7$    | (103) $-3 \times (-6)$ |
| (14) $-8 \times (-4)$ | (44) $2 \times (-1)$  | (74) $-4 \times (-9)$ | (104) $3 \times 9$     |
| (15) $-7 \times 2$    | (45) $-2 \times (-3)$ | (75) $2 \times 1$     | (105) $0 \times 2$     |
| (16) $-5 \times 1$    | (46) $6 \times 6$     | (76) $-3 \times 0$    | (106) $4 \times 0$     |
| (17) $-6 \times 7$    | (47) $-1 \times (-4)$ | (77) $7 \times (-1)$  | (107) $0 \times 0$     |
| (18) $1 \times 2$     | (48) $0 \times 6$     | (78) $9 \times (-2)$  | (108) $-4 \times (-2)$ |
| (19) $2 \times 1$     | (49) $9 \times (-7)$  | (79) $8 \times (-7)$  | (109) $-2 \times 4$    |
| (20) $2 \times 7$     | (50) $0 \times 5$     | (80) $9 \times 5$     | (110) $4 \times (-1)$  |
| (21) $2 \times 0$     | (51) $8 \times (-5)$  | (81) $6 \times 5$     | (111) $-9 \times (-2)$ |
| (22) $8 \times (-9)$  | (52) $-1 \times (-5)$ | (82) $0 \times 6$     | (112) $-2 \times 2$    |
| (23) $-7 \times (-4)$ | (53) $3 \times 1$     | (83) $-2 \times (-4)$ | (113) $6 \times (-5)$  |
| (24) $7 \times 7$     | (54) $2 \times (-5)$  | (84) $-9 \times 3$    | (114) $-8 \times (-8)$ |
| (25) $-1 \times 9$    | (55) $-6 \times (-9)$ | (85) $2 \times (-5)$  | (115) $-8 \times (-5)$ |
| (26) $7 \times (-3)$  | (56) $1 \times 0$     | (86) $-7 \times 6$    | (116) $-9 \times (-1)$ |
| (27) $-3 \times 5$    | (57) $2 \times (-6)$  | (87) $1 \times (-8)$  | (117) $2 \times (-6)$  |
| (28) $6 \times (-9)$  | (58) $2 \times (-9)$  | (88) $-6 \times (-6)$ | (118) $1 \times (-5)$  |
| (29) $0 \times 4$     | (59) $-1 \times 9$    | (89) $3 \times 7$     | (119) $7 \times (-7)$  |
| (30) $8 \times (-7)$  | (60) $-8 \times 4$    | (90) $8 \times (-3)$  | (120) $5 \times (-6)$  |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-3 \times 8$     | (31) $7 \times 7$     | (61) $7 \times (-9)$  | (91) $2 \times (-9)$   |
| (2) $-5 \times 0$     | (32) $9 \times 4$     | (62) $-5 \times (-2)$ | (92) $-7 \times 6$     |
| (3) $6 \times (-6)$   | (33) $2 \times 9$     | (63) $-7 \times (-4)$ | (93) $0 \times (-8)$   |
| (4) $-8 \times 2$     | (34) $-6 \times (-5)$ | (64) $-7 \times 0$    | (94) $-1 \times 8$     |
| (5) $-3 \times 9$     | (35) $4 \times (-3)$  | (65) $-2 \times 4$    | (95) $-8 \times (-5)$  |
| (6) $-5 \times 7$     | (36) $-1 \times (-3)$ | (66) $4 \times (-5)$  | (96) $8 \times 3$      |
| (7) $4 \times (-8)$   | (37) $1 \times (-6)$  | (67) $-6 \times 8$    | (97) $-3 \times 0$     |
| (8) $1 \times (-1)$   | (38) $0 \times 4$     | (68) $1 \times 6$     | (98) $4 \times (-8)$   |
| (9) $1 \times 4$      | (39) $-9 \times 0$    | (69) $6 \times 9$     | (99) $9 \times 4$      |
| (10) $9 \times (-4)$  | (40) $-2 \times (-7)$ | (70) $4 \times (-6)$  | (100) $4 \times (-4)$  |
| (11) $1 \times 6$     | (41) $-4 \times 9$    | (71) $5 \times 1$     | (101) $-1 \times (-7)$ |
| (12) $-9 \times 2$    | (42) $3 \times 4$     | (72) $2 \times 2$     | (102) $3 \times (-4)$  |
| (13) $-3 \times 3$    | (43) $-3 \times (-2)$ | (73) $4 \times (-4)$  | (103) $6 \times (-4)$  |
| (14) $-2 \times (-5)$ | (44) $9 \times (-1)$  | (74) $-4 \times 9$    | (104) $-2 \times 5$    |
| (15) $-1 \times (-8)$ | (45) $6 \times 2$     | (75) $7 \times 6$     | (105) $5 \times (-1)$  |
| (16) $8 \times (-4)$  | (46) $-4 \times 9$    | (76) $8 \times (-4)$  | (106) $-1 \times (-8)$ |
| (17) $-3 \times (-5)$ | (47) $8 \times (-4)$  | (77) $-3 \times (-8)$ | (107) $8 \times 5$     |
| (18) $-6 \times 4$    | (48) $-2 \times (-4)$ | (78) $-2 \times (-7)$ | (108) $4 \times 8$     |
| (19) $-5 \times (-6)$ | (49) $0 \times (-7)$  | (79) $1 \times (-2)$  | (109) $-1 \times 3$    |
| (20) $0 \times 3$     | (50) $2 \times (-8)$  | (80) $3 \times 8$     | (110) $-8 \times (-3)$ |
| (21) $-1 \times (-6)$ | (51) $4 \times 4$     | (81) $7 \times 2$     | (111) $-3 \times (-6)$ |
| (22) $9 \times (-2)$  | (52) $-9 \times (-6)$ | (82) $-2 \times 8$    | (112) $9 \times 1$     |
| (23) $-9 \times 4$    | (53) $2 \times 6$     | (83) $-9 \times (-6)$ | (113) $2 \times 6$     |
| (24) $2 \times 1$     | (54) $-8 \times (-5)$ | (84) $8 \times 3$     | (114) $-5 \times (-1)$ |
| (25) $8 \times 9$     | (55) $-1 \times (-5)$ | (85) $6 \times 5$     | (115) $-5 \times (-3)$ |
| (26) $7 \times 7$     | (56) $4 \times 1$     | (86) $7 \times 8$     | (116) $-9 \times (-1)$ |
| (27) $6 \times 7$     | (57) $7 \times 9$     | (87) $-1 \times (-7)$ | (117) $9 \times 8$     |
| (28) $-7 \times 8$    | (58) $6 \times (-7)$  | (88) $8 \times (-9)$  | (118) $-4 \times (-4)$ |
| (29) $-9 \times (-5)$ | (59) $7 \times (-8)$  | (89) $-2 \times 4$    | (119) $-8 \times 4$    |
| (30) $9 \times 1$     | (60) $0 \times (-4)$  | (90) $-2 \times 9$    | (120) $-4 \times 6$    |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-5 \times 2$     | (31) $8 \times (-1)$  | (61) $-6 \times 1$    | (91) $-2 \times 4$     |
| (2) $-7 \times 6$     | (32) $-5 \times (-3)$ | (62) $-5 \times 7$    | (92) $6 \times (-7)$   |
| (3) $1 \times (-4)$   | (33) $8 \times (-3)$  | (63) $6 \times (-6)$  | (93) $-2 \times 0$     |
| (4) $6 \times 9$      | (34) $-6 \times 4$    | (64) $0 \times (-1)$  | (94) $-2 \times 2$     |
| (5) $4 \times 9$      | (35) $0 \times 9$     | (65) $2 \times 4$     | (95) $-5 \times 7$     |
| (6) $7 \times (-2)$   | (36) $-4 \times (-8)$ | (66) $4 \times 7$     | (96) $-5 \times 7$     |
| (7) $8 \times 3$      | (37) $7 \times (-9)$  | (67) $-4 \times (-9)$ | (97) $7 \times 0$      |
| (8) $3 \times 2$      | (38) $-8 \times (-3)$ | (68) $-2 \times 8$    | (98) $2 \times (-2)$   |
| (9) $-4 \times 9$     | (39) $-3 \times (-2)$ | (69) $-4 \times 7$    | (99) $-3 \times 7$     |
| (10) $8 \times (-8)$  | (40) $9 \times 5$     | (70) $0 \times 5$     | (100) $5 \times (-6)$  |
| (11) $-1 \times (-4)$ | (41) $9 \times 1$     | (71) $-8 \times 6$    | (101) $-6 \times 4$    |
| (12) $-3 \times (-2)$ | (42) $8 \times (-6)$  | (72) $-2 \times 1$    | (102) $8 \times 7$     |
| (13) $-7 \times (-2)$ | (43) $-1 \times (-7)$ | (73) $1 \times (-9)$  | (103) $4 \times (-2)$  |
| (14) $2 \times 7$     | (44) $9 \times (-7)$  | (74) $-1 \times 6$    | (104) $-3 \times 7$    |
| (15) $-6 \times 9$    | (45) $-4 \times (-3)$ | (75) $3 \times (-7)$  | (105) $1 \times 8$     |
| (16) $1 \times 0$     | (46) $2 \times 0$     | (76) $-5 \times (-3)$ | (106) $8 \times (-2)$  |
| (17) $5 \times (-9)$  | (47) $1 \times 9$     | (77) $9 \times 8$     | (107) $-6 \times (-5)$ |
| (18) $9 \times (-9)$  | (48) $6 \times 2$     | (78) $1 \times (-7)$  | (108) $-4 \times (-6)$ |
| (19) $7 \times 6$     | (49) $-1 \times 0$    | (79) $-1 \times 2$    | (109) $9 \times 6$     |
| (20) $-6 \times 3$    | (50) $-1 \times (-1)$ | (80) $-2 \times 9$    | (110) $4 \times (-1)$  |
| (21) $-3 \times 4$    | (51) $-2 \times 2$    | (81) $-1 \times 4$    | (111) $-3 \times (-2)$ |
| (22) $-1 \times 6$    | (52) $4 \times (-2)$  | (82) $2 \times (-2)$  | (112) $9 \times (-9)$  |
| (23) $-4 \times (-8)$ | (53) $-7 \times 3$    | (83) $-9 \times 1$    | (113) $-8 \times (-5)$ |
| (24) $-4 \times (-8)$ | (54) $0 \times (-7)$  | (84) $-4 \times (-3)$ | (114) $-4 \times 5$    |
| (25) $-3 \times 0$    | (55) $-3 \times (-8)$ | (85) $-1 \times (-2)$ | (115) $-6 \times (-3)$ |
| (26) $8 \times 5$     | (56) $-9 \times (-2)$ | (86) $4 \times (-6)$  | (116) $2 \times (-6)$  |
| (27) $-9 \times 8$    | (57) $4 \times (-3)$  | (87) $6 \times 1$     | (117) $-8 \times 5$    |
| (28) $-4 \times (-2)$ | (58) $0 \times 5$     | (88) $4 \times (-9)$  | (118) $3 \times (-1)$  |
| (29) $6 \times (-6)$  | (59) $-3 \times 1$    | (89) $-8 \times (-6)$ | (119) $8 \times 6$     |
| (30) $-2 \times 4$    | (60) $0 \times (-4)$  | (90) $-8 \times (-3)$ | (120) $1 \times 0$     |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-2 \times 9$     | (31) $1 \times (-2)$  | (61) $3 \times 7$     | (91) $1 \times (-5)$   |
| (2) $8 \times 7$      | (32) $-5 \times 4$    | (62) $-8 \times (-8)$ | (92) $-2 \times (-6)$  |
| (3) $6 \times (-2)$   | (33) $2 \times (-3)$  | (63) $-4 \times (-5)$ | (93) $-6 \times 8$     |
| (4) $6 \times (-8)$   | (34) $-8 \times 9$    | (64) $9 \times (-2)$  | (94) $2 \times 6$      |
| (5) $7 \times 4$      | (35) $-7 \times (-7)$ | (65) $-4 \times (-4)$ | (95) $2 \times (-9)$   |
| (6) $-1 \times (-2)$  | (36) $8 \times (-5)$  | (66) $6 \times (-4)$  | (96) $-8 \times 9$     |
| (7) $-4 \times (-3)$  | (37) $-8 \times (-5)$ | (67) $-3 \times (-2)$ | (97) $9 \times (-8)$   |
| (8) $3 \times (-2)$   | (38) $8 \times 0$     | (68) $4 \times 8$     | (98) $7 \times (-8)$   |
| (9) $-4 \times 6$     | (39) $4 \times 0$     | (69) $1 \times (-2)$  | (99) $-6 \times (-6)$  |
| (10) $5 \times (-3)$  | (40) $-1 \times (-9)$ | (70) $-4 \times (-5)$ | (100) $8 \times (-4)$  |
| (11) $4 \times (-3)$  | (41) $8 \times 1$     | (71) $-3 \times 4$    | (101) $-1 \times 0$    |
| (12) $3 \times (-3)$  | (42) $6 \times 9$     | (72) $8 \times 2$     | (102) $-2 \times (-1)$ |
| (13) $-4 \times (-4)$ | (43) $1 \times 7$     | (73) $-7 \times (-7)$ | (103) $3 \times (-8)$  |
| (14) $0 \times 4$     | (44) $-9 \times (-2)$ | (74) $-6 \times (-4)$ | (104) $-8 \times 2$    |
| (15) $8 \times 3$     | (45) $-1 \times 7$    | (75) $-3 \times 5$    | (105) $-1 \times 2$    |
| (16) $-2 \times 3$    | (46) $5 \times (-3)$  | (76) $7 \times 2$     | (106) $-1 \times 9$    |
| (17) $6 \times 2$     | (47) $-9 \times (-2)$ | (77) $6 \times (-2)$  | (107) $5 \times 5$     |
| (18) $-2 \times 9$    | (48) $7 \times 1$     | (78) $-2 \times (-7)$ | (108) $5 \times 4$     |
| (19) $0 \times (-1)$  | (49) $-8 \times 8$    | (79) $-1 \times (-9)$ | (109) $4 \times (-7)$  |
| (20) $0 \times (-6)$  | (50) $-1 \times 7$    | (80) $-2 \times (-7)$ | (110) $-2 \times (-4)$ |
| (21) $7 \times 0$     | (51) $-7 \times 8$    | (81) $-2 \times 9$    | (111) $4 \times (-7)$  |
| (22) $-3 \times 8$    | (52) $-1 \times (-7)$ | (82) $3 \times (-8)$  | (112) $9 \times (-3)$  |
| (23) $-5 \times 1$    | (53) $2 \times (-4)$  | (83) $2 \times 3$     | (113) $-4 \times (-7)$ |
| (24) $-4 \times (-7)$ | (54) $6 \times 6$     | (84) $5 \times 2$     | (114) $-8 \times (-6)$ |
| (25) $-9 \times 2$    | (55) $-6 \times 6$    | (85) $1 \times 1$     | (115) $-4 \times 6$    |
| (26) $5 \times (-9)$  | (56) $-9 \times 8$    | (86) $5 \times 4$     | (116) $7 \times (-1)$  |
| (27) $8 \times (-4)$  | (57) $7 \times 9$     | (87) $3 \times (-7)$  | (117) $-8 \times (-3)$ |
| (28) $6 \times 3$     | (58) $3 \times (-6)$  | (88) $8 \times 7$     | (118) $-1 \times 8$    |
| (29) $0 \times 2$     | (59) $9 \times (-5)$  | (89) $-6 \times (-9)$ | (119) $-3 \times (-4)$ |
| (30) $-3 \times (-8)$ | (60) $5 \times (-8)$  | (90) $-5 \times 5$    | (120) $4 \times (-4)$  |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $0 \times (-7)$   | (31) $8 \times 8$     | (61) $-5 \times 6$    | (91) $4 \times 1$      |
| (2) $2 \times (-2)$   | (32) $-4 \times 0$    | (62) $9 \times 5$     | (92) $-2 \times (-5)$  |
| (3) $-7 \times (-2)$  | (33) $9 \times 1$     | (63) $3 \times 5$     | (93) $5 \times (-8)$   |
| (4) $4 \times 2$      | (34) $-4 \times (-3)$ | (64) $7 \times 8$     | (94) $8 \times 3$      |
| (5) $0 \times (-1)$   | (35) $5 \times 3$     | (65) $8 \times (-7)$  | (95) $1 \times (-7)$   |
| (6) $-8 \times 1$     | (36) $-4 \times (-9)$ | (66) $-1 \times 2$    | (96) $0 \times (-7)$   |
| (7) $4 \times (-1)$   | (37) $4 \times (-9)$  | (67) $-8 \times (-4)$ | (97) $6 \times (-4)$   |
| (8) $-3 \times 9$     | (38) $9 \times 3$     | (68) $9 \times 7$     | (98) $2 \times 3$      |
| (9) $2 \times (-4)$   | (39) $-9 \times 3$    | (69) $-5 \times (-1)$ | (99) $0 \times (-6)$   |
| (10) $-3 \times (-6)$ | (40) $-7 \times (-7)$ | (70) $4 \times (-2)$  | (100) $8 \times 0$     |
| (11) $8 \times 5$     | (41) $-3 \times 0$    | (71) $-2 \times (-7)$ | (101) $9 \times (-4)$  |
| (12) $-5 \times (-9)$ | (42) $-2 \times (-4)$ | (72) $-4 \times 7$    | (102) $0 \times 4$     |
| (13) $-6 \times (-2)$ | (43) $-6 \times 5$    | (73) $0 \times 5$     | (103) $-4 \times 5$    |
| (14) $-8 \times (-8)$ | (44) $5 \times (-7)$  | (74) $5 \times (-2)$  | (104) $3 \times 5$     |
| (15) $0 \times 6$     | (45) $-5 \times 1$    | (75) $-4 \times 0$    | (105) $9 \times 3$     |
| (16) $8 \times 8$     | (46) $6 \times 2$     | (76) $-4 \times (-1)$ | (106) $6 \times 0$     |
| (17) $-5 \times 2$    | (47) $1 \times 8$     | (77) $-1 \times (-6)$ | (107) $2 \times 2$     |
| (18) $3 \times (-7)$  | (48) $-1 \times (-1)$ | (78) $-5 \times 4$    | (108) $-1 \times 7$    |
| (19) $5 \times (-6)$  | (49) $-9 \times (-7)$ | (79) $-5 \times 4$    | (109) $-4 \times 5$    |
| (20) $5 \times 9$     | (50) $-4 \times 6$    | (80) $-4 \times (-4)$ | (110) $7 \times (-1)$  |
| (21) $0 \times 0$     | (51) $7 \times 5$     | (81) $-6 \times (-4)$ | (111) $3 \times (-3)$  |
| (22) $-3 \times 4$    | (52) $-2 \times 7$    | (82) $8 \times (-2)$  | (112) $8 \times 7$     |
| (23) $5 \times 5$     | (53) $7 \times 9$     | (83) $-2 \times (-2)$ | (113) $-2 \times 9$    |
| (24) $-5 \times 4$    | (54) $0 \times (-7)$  | (84) $-9 \times 8$    | (114) $9 \times (-1)$  |
| (25) $-1 \times (-1)$ | (55) $-7 \times (-8)$ | (85) $-5 \times (-5)$ | (115) $3 \times (-4)$  |
| (26) $-6 \times 0$    | (56) $9 \times 2$     | (86) $-7 \times (-7)$ | (116) $2 \times 3$     |
| (27) $-9 \times 9$    | (57) $3 \times (-8)$  | (87) $4 \times (-6)$  | (117) $1 \times (-7)$  |
| (28) $-2 \times 6$    | (58) $2 \times (-2)$  | (88) $5 \times 8$     | (118) $-6 \times (-9)$ |
| (29) $-4 \times (-3)$ | (59) $-2 \times (-5)$ | (89) $5 \times (-5)$  | (119) $5 \times 2$     |
| (30) $5 \times 3$     | (60) $-6 \times (-8)$ | (90) $8 \times 4$     | (120) $-9 \times 6$    |

- |                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-4 \times (-7)$  | (31) $-9 \times (-5)$ | (61) $-8 \times 8$    | (91) $-6 \times (-6)$  |
| (2) $6 \times 9$      | (32) $-3 \times 8$    | (62) $4 \times (-7)$  | (92) $-7 \times 3$     |
| (3) $-7 \times 3$     | (33) $4 \times (-3)$  | (63) $8 \times 0$     | (93) $0 \times 7$      |
| (4) $7 \times 4$      | (34) $-6 \times (-8)$ | (64) $3 \times (-3)$  | (94) $-6 \times 1$     |
| (5) $2 \times 8$      | (35) $6 \times 5$     | (65) $4 \times (-3)$  | (95) $4 \times 6$      |
| (6) $4 \times (-7)$   | (36) $8 \times 8$     | (66) $9 \times (-9)$  | (96) $0 \times 4$      |
| (7) $-5 \times 3$     | (37) $2 \times 1$     | (67) $2 \times 9$     | (97) $-9 \times (-5)$  |
| (8) $-8 \times 0$     | (38) $-6 \times (-5)$ | (68) $9 \times 3$     | (98) $4 \times (-8)$   |
| (9) $-3 \times 8$     | (39) $6 \times (-8)$  | (69) $-2 \times 4$    | (99) $9 \times 2$      |
| (10) $-4 \times (-2)$ | (40) $-6 \times (-3)$ | (70) $-7 \times (-1)$ | (100) $9 \times (-7)$  |
| (11) $1 \times 7$     | (41) $-7 \times (-9)$ | (71) $-1 \times 6$    | (101) $7 \times 8$     |
| (12) $2 \times 8$     | (42) $0 \times (-4)$  | (72) $1 \times 1$     | (102) $-3 \times 6$    |
| (13) $2 \times (-8)$  | (43) $1 \times (-5)$  | (73) $-8 \times (-9)$ | (103) $-4 \times (-7)$ |
| (14) $-8 \times 4$    | (44) $1 \times 4$     | (74) $5 \times 7$     | (104) $-3 \times 5$    |
| (15) $0 \times 7$     | (45) $-6 \times 4$    | (75) $-3 \times 8$    | (105) $2 \times 7$     |
| (16) $2 \times 9$     | (46) $4 \times 6$     | (76) $7 \times (-4)$  | (106) $-6 \times (-6)$ |
| (17) $-3 \times (-2)$ | (47) $4 \times (-5)$  | (77) $-8 \times (-4)$ | (107) $3 \times (-9)$  |
| (18) $3 \times 2$     | (48) $7 \times 8$     | (78) $-1 \times 4$    | (108) $-6 \times (-9)$ |
| (19) $-4 \times 7$    | (49) $-5 \times (-5)$ | (79) $6 \times (-4)$  | (109) $9 \times 3$     |
| (20) $-3 \times (-1)$ | (50) $2 \times 6$     | (80) $6 \times (-6)$  | (110) $7 \times 7$     |
| (21) $1 \times 3$     | (51) $-5 \times (-8)$ | (81) $0 \times (-3)$  | (111) $7 \times 6$     |
| (22) $-1 \times (-5)$ | (52) $9 \times 7$     | (82) $7 \times (-7)$  | (112) $2 \times 7$     |
| (23) $-2 \times 6$    | (53) $-4 \times (-6)$ | (83) $-5 \times 2$    | (113) $-2 \times (-3)$ |
| (24) $-4 \times 2$    | (54) $-9 \times 1$    | (84) $7 \times (-6)$  | (114) $1 \times 4$     |
| (25) $1 \times 0$     | (55) $0 \times 8$     | (85) $9 \times 6$     | (115) $2 \times 8$     |
| (26) $5 \times 5$     | (56) $-4 \times (-3)$ | (86) $7 \times 6$     | (116) $5 \times 8$     |
| (27) $2 \times (-4)$  | (57) $2 \times 6$     | (87) $6 \times 3$     | (117) $6 \times 1$     |
| (28) $-6 \times 8$    | (58) $0 \times 4$     | (88) $5 \times 8$     | (118) $8 \times 8$     |
| (29) $-7 \times (-9)$ | (59) $-5 \times 6$    | (89) $-6 \times (-2)$ | (119) $8 \times 6$     |
| (30) $-5 \times 2$    | (60) $-4 \times (-1)$ | (90) $-6 \times (-1)$ | (120) $3 \times 5$     |



|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $4 \times (-7)$   | (31) $0 \times 7$     | (61) $-6 \times 7$    | (91) $3 \times (-3)$   |
| (2) $-8 \times (-7)$  | (32) $-1 \times (-6)$ | (62) $-7 \times 5$    | (92) $9 \times 1$      |
| (3) $-4 \times (-1)$  | (33) $6 \times 5$     | (63) $7 \times (-9)$  | (93) $-3 \times (-8)$  |
| (4) $-8 \times 5$     | (34) $6 \times 5$     | (64) $7 \times (-3)$  | (94) $9 \times 6$      |
| (5) $-2 \times (-3)$  | (35) $6 \times (-8)$  | (65) $4 \times (-8)$  | (95) $-9 \times 1$     |
| (6) $-9 \times (-4)$  | (36) $5 \times 7$     | (66) $2 \times (-5)$  | (96) $6 \times 7$      |
| (7) $8 \times (-7)$   | (37) $2 \times (-5)$  | (67) $2 \times (-7)$  | (97) $-6 \times 9$     |
| (8) $-1 \times 2$     | (38) $1 \times (-9)$  | (68) $7 \times 6$     | (98) $5 \times (-2)$   |
| (9) $9 \times (-9)$   | (39) $6 \times 9$     | (69) $8 \times 1$     | (99) $1 \times 6$      |
| (10) $7 \times (-7)$  | (40) $4 \times (-9)$  | (70) $5 \times 0$     | (100) $4 \times 3$     |
| (11) $6 \times 7$     | (41) $9 \times 3$     | (71) $2 \times 2$     | (101) $3 \times 9$     |
| (12) $2 \times (-9)$  | (42) $5 \times (-5)$  | (72) $2 \times (-8)$  | (102) $-5 \times (-6)$ |
| (13) $-1 \times (-8)$ | (43) $-9 \times 7$    | (73) $-1 \times 5$    | (103) $-3 \times (-3)$ |
| (14) $-8 \times 8$    | (44) $-9 \times 6$    | (74) $5 \times (-1)$  | (104) $9 \times 0$     |
| (15) $5 \times (-8)$  | (45) $-2 \times (-1)$ | (75) $8 \times 6$     | (105) $9 \times (-3)$  |
| (16) $-2 \times 6$    | (46) $8 \times (-1)$  | (76) $5 \times 0$     | (106) $-2 \times (-8)$ |
| (17) $1 \times (-8)$  | (47) $-9 \times (-4)$ | (77) $-9 \times 0$    | (107) $-6 \times 0$    |
| (18) $3 \times 7$     | (48) $8 \times (-3)$  | (78) $4 \times 2$     | (108) $6 \times (-4)$  |
| (19) $-3 \times (-2)$ | (49) $6 \times (-9)$  | (79) $4 \times (-2)$  | (109) $-9 \times (-1)$ |
| (20) $8 \times (-1)$  | (50) $6 \times (-6)$  | (80) $5 \times 4$     | (110) $3 \times (-7)$  |
| (21) $-7 \times 9$    | (51) $-9 \times (-7)$ | (81) $8 \times 5$     | (111) $-8 \times (-4)$ |
| (22) $-1 \times (-3)$ | (52) $6 \times 2$     | (82) $2 \times 1$     | (112) $-3 \times (-7)$ |
| (23) $9 \times (-1)$  | (53) $-6 \times 2$    | (83) $-7 \times 0$    | (113) $-6 \times 2$    |
| (24) $8 \times 5$     | (54) $-7 \times 0$    | (84) $0 \times 3$     | (114) $3 \times (-6)$  |
| (25) $-6 \times (-9)$ | (55) $8 \times (-7)$  | (85) $6 \times (-1)$  | (115) $6 \times 3$     |
| (26) $6 \times (-5)$  | (56) $8 \times 0$     | (86) $-8 \times 0$    | (116) $0 \times 8$     |
| (27) $0 \times (-4)$  | (57) $3 \times (-4)$  | (87) $-7 \times (-8)$ | (117) $8 \times (-9)$  |
| (28) $-1 \times (-8)$ | (58) $-4 \times 7$    | (88) $-8 \times 8$    | (118) $5 \times (-1)$  |
| (29) $1 \times 4$     | (59) $-3 \times 8$    | (89) $-2 \times (-7)$ | (119) $-6 \times (-2)$ |
| (30) $-5 \times (-8)$ | (60) $-2 \times (-9)$ | (90) $5 \times 9$     | (120) $7 \times 8$     |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $7 \times (-2)$   | (31) $-9 \times 7$    | (61) $-3 \times 8$    | (91) $-3 \times 0$     |
| (2) $-7 \times 0$     | (32) $-7 \times (-3)$ | (62) $-5 \times (-5)$ | (92) $3 \times 0$      |
| (3) $3 \times 9$      | (33) $8 \times (-8)$  | (63) $-5 \times (-7)$ | (93) $3 \times 1$      |
| (4) $-3 \times 5$     | (34) $2 \times (-4)$  | (64) $-7 \times (-9)$ | (94) $-9 \times (-6)$  |
| (5) $-1 \times (-3)$  | (35) $-8 \times 5$    | (65) $2 \times 5$     | (95) $3 \times 5$      |
| (6) $6 \times 0$      | (36) $1 \times (-3)$  | (66) $-5 \times 9$    | (96) $-1 \times 3$     |
| (7) $1 \times 1$      | (37) $0 \times (-4)$  | (67) $6 \times 1$     | (97) $-7 \times (-7)$  |
| (8) $-1 \times 4$     | (38) $1 \times (-3)$  | (68) $1 \times 9$     | (98) $-1 \times 8$     |
| (9) $-1 \times (-8)$  | (39) $-7 \times 2$    | (69) $-9 \times (-8)$ | (99) $-3 \times (-9)$  |
| (10) $0 \times (-4)$  | (40) $3 \times (-7)$  | (70) $-3 \times (-8)$ | (100) $-8 \times (-6)$ |
| (11) $-2 \times 7$    | (41) $5 \times (-1)$  | (71) $9 \times 9$     | (101) $-6 \times 6$    |
| (12) $3 \times (-9)$  | (42) $7 \times 1$     | (72) $8 \times 7$     | (102) $-4 \times (-6)$ |
| (13) $-2 \times (-6)$ | (43) $8 \times 2$     | (73) $3 \times 4$     | (103) $9 \times 8$     |
| (14) $-4 \times 1$    | (44) $9 \times (-7)$  | (74) $-8 \times (-1)$ | (104) $8 \times 4$     |
| (15) $3 \times (-3)$  | (45) $5 \times 4$     | (75) $7 \times (-9)$  | (105) $7 \times 6$     |
| (16) $4 \times (-2)$  | (46) $-1 \times (-8)$ | (76) $-8 \times 2$    | (106) $8 \times 2$     |
| (17) $3 \times 9$     | (47) $-9 \times (-3)$ | (77) $8 \times (-7)$  | (107) $3 \times (-7)$  |
| (18) $-9 \times 6$    | (48) $-1 \times (-1)$ | (78) $-8 \times (-4)$ | (108) $2 \times 9$     |
| (19) $6 \times 5$     | (49) $4 \times 9$     | (79) $-7 \times (-9)$ | (109) $6 \times (-9)$  |
| (20) $-9 \times 5$    | (50) $1 \times (-1)$  | (80) $-1 \times (-7)$ | (110) $-7 \times 7$    |
| (21) $6 \times 5$     | (51) $1 \times 5$     | (81) $-3 \times 9$    | (111) $1 \times (-3)$  |
| (22) $9 \times (-3)$  | (52) $6 \times 0$     | (82) $4 \times 3$     | (112) $-7 \times (-3)$ |
| (23) $6 \times (-7)$  | (53) $-1 \times 8$    | (83) $-2 \times (-9)$ | (113) $8 \times 4$     |
| (24) $-8 \times (-9)$ | (54) $-2 \times 3$    | (84) $-9 \times (-7)$ | (114) $-5 \times (-7)$ |
| (25) $-1 \times 2$    | (55) $9 \times 9$     | (85) $5 \times 4$     | (115) $-1 \times 4$    |
| (26) $1 \times 8$     | (56) $0 \times 5$     | (86) $-2 \times 8$    | (116) $9 \times 7$     |
| (27) $-2 \times (-5)$ | (57) $-8 \times (-1)$ | (87) $-2 \times (-4)$ | (117) $-6 \times (-3)$ |
| (28) $0 \times (-9)$  | (58) $-2 \times (-3)$ | (88) $6 \times (-4)$  | (118) $-5 \times 3$    |
| (29) $8 \times (-2)$  | (59) $-5 \times 0$    | (89) $-9 \times 2$    | (119) $9 \times (-1)$  |
| (30) $-7 \times (-6)$ | (60) $4 \times (-3)$  | (90) $5 \times (-1)$  | (120) $-7 \times 5$    |

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|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-2 \times 6$     | (31) $1 \times (-8)$  | (61) $-5 \times 0$    | (91) $5 \times (-4)$   |
| (2) $9 \times 9$      | (32) $-3 \times (-2)$ | (62) $8 \times 5$     | (92) $-4 \times (-7)$  |
| (3) $0 \times 2$      | (33) $-7 \times (-5)$ | (63) $3 \times 0$     | (93) $4 \times 3$      |
| (4) $-7 \times 1$     | (34) $-7 \times (-1)$ | (64) $-1 \times 0$    | (94) $-9 \times 8$     |
| (5) $-1 \times 9$     | (35) $8 \times (-3)$  | (65) $6 \times (-7)$  | (95) $-7 \times 4$     |
| (6) $-4 \times 3$     | (36) $-4 \times 9$    | (66) $-8 \times 0$    | (96) $-2 \times (-7)$  |
| (7) $6 \times 9$      | (37) $4 \times 1$     | (67) $3 \times (-3)$  | (97) $8 \times (-9)$   |
| (8) $2 \times 9$      | (38) $-7 \times (-1)$ | (68) $-5 \times (-5)$ | (98) $-7 \times 7$     |
| (9) $9 \times (-8)$   | (39) $0 \times (-5)$  | (69) $-3 \times 9$    | (99) $0 \times 0$      |
| (10) $-8 \times (-4)$ | (40) $-1 \times (-7)$ | (70) $-3 \times (-5)$ | (100) $5 \times (-5)$  |
| (11) $-6 \times 4$    | (41) $6 \times 6$     | (71) $2 \times 2$     | (101) $-7 \times 2$    |
| (12) $-8 \times (-8)$ | (42) $0 \times 4$     | (72) $-7 \times 2$    | (102) $-8 \times 8$    |
| (13) $-9 \times 7$    | (43) $1 \times (-1)$  | (73) $3 \times (-4)$  | (103) $-9 \times 2$    |
| (14) $9 \times 6$     | (44) $8 \times (-7)$  | (74) $-7 \times 5$    | (104) $-6 \times 9$    |
| (15) $-3 \times (-3)$ | (45) $-1 \times 0$    | (75) $-7 \times 3$    | (105) $-2 \times 9$    |
| (16) $7 \times (-2)$  | (46) $7 \times (-4)$  | (76) $4 \times 8$     | (106) $-6 \times (-4)$ |
| (17) $2 \times (-3)$  | (47) $1 \times 3$     | (77) $7 \times (-9)$  | (107) $4 \times 2$     |
| (18) $3 \times 5$     | (48) $9 \times (-3)$  | (78) $1 \times (-3)$  | (108) $3 \times 0$     |
| (19) $-4 \times 3$    | (49) $-2 \times (-2)$ | (79) $-2 \times 5$    | (109) $-4 \times 7$    |
| (20) $-6 \times 9$    | (50) $6 \times 3$     | (80) $-7 \times (-5)$ | (110) $-4 \times (-8)$ |
| (21) $7 \times 4$     | (51) $-9 \times 7$    | (81) $-6 \times (-5)$ | (111) $-6 \times 3$    |
| (22) $3 \times 0$     | (52) $-5 \times 4$    | (82) $4 \times (-3)$  | (112) $3 \times (-6)$  |
| (23) $5 \times (-6)$  | (53) $-5 \times 9$    | (83) $6 \times 4$     | (113) $5 \times (-8)$  |
| (24) $-8 \times (-5)$ | (54) $6 \times 1$     | (84) $9 \times (-3)$  | (114) $2 \times 9$     |
| (25) $-4 \times (-4)$ | (55) $-7 \times (-4)$ | (85) $2 \times (-8)$  | (115) $-9 \times (-3)$ |
| (26) $-4 \times (-9)$ | (56) $-1 \times (-4)$ | (86) $9 \times (-7)$  | (116) $7 \times 4$     |
| (27) $9 \times 5$     | (57) $-2 \times (-5)$ | (87) $0 \times (-4)$  | (117) $-7 \times (-9)$ |
| (28) $-7 \times 8$    | (58) $2 \times (-3)$  | (88) $-6 \times 3$    | (118) $-9 \times (-2)$ |
| (29) $9 \times (-6)$  | (59) $6 \times (-4)$  | (89) $-6 \times (-8)$ | (119) $3 \times 0$     |
| (30) $0 \times 8$     | (60) $-2 \times 2$    | (90) $-4 \times 6$    | (120) $-8 \times 2$    |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $4 \times (-4)$   | (31) $2 \times (-8)$  | (61) $-9 \times 3$    | (91) $0 \times 0$      |
| (2) $-5 \times (-5)$  | (32) $-1 \times 3$    | (62) $-4 \times 3$    | (92) $3 \times 0$      |
| (3) $4 \times 4$      | (33) $1 \times (-4)$  | (63) $-2 \times (-3)$ | (93) $-4 \times (-7)$  |
| (4) $9 \times (-4)$   | (34) $5 \times 1$     | (64) $0 \times (-4)$  | (94) $4 \times 2$      |
| (5) $-4 \times (-5)$  | (35) $-3 \times (-4)$ | (65) $-2 \times 4$    | (95) $1 \times 5$      |
| (6) $5 \times 0$      | (36) $1 \times 7$     | (66) $-4 \times 0$    | (96) $9 \times 3$      |
| (7) $-7 \times 8$     | (37) $-4 \times (-9)$ | (67) $4 \times 2$     | (97) $6 \times 8$      |
| (8) $9 \times (-2)$   | (38) $4 \times (-2)$  | (68) $9 \times 0$     | (98) $-1 \times 2$     |
| (9) $-2 \times 9$     | (39) $-2 \times (-6)$ | (69) $-1 \times (-4)$ | (99) $5 \times (-7)$   |
| (10) $-7 \times 1$    | (40) $2 \times 3$     | (70) $9 \times (-2)$  | (100) $3 \times 9$     |
| (11) $-4 \times 4$    | (41) $-9 \times (-1)$ | (71) $2 \times 2$     | (101) $6 \times (-1)$  |
| (12) $-8 \times (-1)$ | (42) $-8 \times (-2)$ | (72) $-5 \times (-1)$ | (102) $7 \times 8$     |
| (13) $5 \times 5$     | (43) $-8 \times (-3)$ | (73) $4 \times 4$     | (103) $-4 \times (-5)$ |
| (14) $-4 \times 7$    | (44) $-3 \times (-8)$ | (74) $-1 \times (-6)$ | (104) $2 \times (-9)$  |
| (15) $-6 \times (-6)$ | (45) $-6 \times (-1)$ | (75) $9 \times 1$     | (105) $-2 \times 1$    |
| (16) $9 \times (-6)$  | (46) $2 \times 5$     | (76) $6 \times (-2)$  | (106) $-3 \times (-2)$ |
| (17) $2 \times 2$     | (47) $-6 \times (-6)$ | (77) $-6 \times (-6)$ | (107) $6 \times (-2)$  |
| (18) $-1 \times (-5)$ | (48) $-5 \times (-8)$ | (78) $-9 \times (-2)$ | (108) $6 \times (-4)$  |
| (19) $-7 \times 8$    | (49) $-1 \times (-1)$ | (79) $-7 \times 3$    | (109) $-4 \times (-9)$ |
| (20) $7 \times 8$     | (50) $2 \times (-1)$  | (80) $7 \times 8$     | (110) $7 \times (-9)$  |
| (21) $-8 \times (-8)$ | (51) $-4 \times 0$    | (81) $-8 \times 8$    | (111) $2 \times 5$     |
| (22) $9 \times 7$     | (52) $-1 \times 9$    | (82) $2 \times (-8)$  | (112) $-6 \times 0$    |
| (23) $5 \times 3$     | (53) $-4 \times (-7)$ | (83) $-8 \times 1$    | (113) $7 \times 9$     |
| (24) $6 \times 8$     | (54) $-8 \times 0$    | (84) $6 \times 9$     | (114) $-2 \times 7$    |
| (25) $8 \times 9$     | (55) $6 \times 1$     | (85) $3 \times 3$     | (115) $0 \times 7$     |
| (26) $8 \times (-5)$  | (56) $6 \times 0$     | (86) $-5 \times 9$    | (116) $-8 \times (-4)$ |
| (27) $3 \times (-1)$  | (57) $7 \times (-7)$  | (87) $1 \times 5$     | (117) $6 \times 9$     |
| (28) $-7 \times (-2)$ | (58) $5 \times 1$     | (88) $7 \times 1$     | (118) $0 \times 8$     |
| (29) $3 \times (-8)$  | (59) $-9 \times (-1)$ | (89) $2 \times (-2)$  | (119) $1 \times (-4)$  |
| (30) $-3 \times (-7)$ | (60) $1 \times (-6)$  | (90) $1 \times (-1)$  | (120) $7 \times (-9)$  |

- |                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $8 \times (-9)$   | (31) $3 \times 4$     | (61) $3 \times (-5)$  | (91) $0 \times 6$      |
| (2) $-4 \times 6$     | (32) $2 \times 6$     | (62) $1 \times (-6)$  | (92) $0 \times 8$      |
| (3) $-6 \times 6$     | (33) $-3 \times 0$    | (63) $7 \times (-9)$  | (93) $3 \times 4$      |
| (4) $7 \times (-7)$   | (34) $-1 \times (-9)$ | (64) $-1 \times 4$    | (94) $2 \times (-8)$   |
| (5) $7 \times (-3)$   | (35) $7 \times (-2)$  | (65) $-1 \times (-9)$ | (95) $-6 \times (-3)$  |
| (6) $-4 \times 3$     | (36) $-1 \times 1$    | (66) $7 \times 5$     | (96) $1 \times 7$      |
| (7) $-9 \times (-2)$  | (37) $4 \times 9$     | (67) $3 \times 7$     | (97) $0 \times 8$      |
| (8) $2 \times 4$      | (38) $8 \times 5$     | (68) $6 \times 5$     | (98) $-7 \times (-5)$  |
| (9) $1 \times 8$      | (39) $8 \times 1$     | (69) $8 \times (-4)$  | (99) $4 \times 4$      |
| (10) $0 \times 9$     | (40) $-3 \times 1$    | (70) $-8 \times (-9)$ | (100) $-8 \times (-4)$ |
| (11) $-6 \times 2$    | (41) $8 \times (-2)$  | (71) $1 \times 3$     | (101) $9 \times (-2)$  |
| (12) $7 \times (-5)$  | (42) $4 \times 9$     | (72) $-2 \times 9$    | (102) $3 \times 9$     |
| (13) $-6 \times 9$    | (43) $-3 \times 9$    | (73) $-7 \times 9$    | (103) $-7 \times 1$    |
| (14) $2 \times 8$     | (44) $7 \times (-2)$  | (74) $-1 \times 4$    | (104) $9 \times (-5)$  |
| (15) $2 \times 7$     | (45) $-3 \times (-6)$ | (75) $6 \times (-5)$  | (105) $-1 \times 5$    |
| (16) $-8 \times 9$    | (46) $-4 \times 9$    | (76) $6 \times (-3)$  | (106) $8 \times 5$     |
| (17) $2 \times (-6)$  | (47) $7 \times 2$     | (77) $-5 \times (-9)$ | (107) $6 \times (-1)$  |
| (18) $-1 \times 2$    | (48) $-7 \times 8$    | (78) $9 \times 3$     | (108) $0 \times 9$     |
| (19) $0 \times (-7)$  | (49) $-5 \times (-8)$ | (79) $3 \times 8$     | (109) $1 \times (-6)$  |
| (20) $-2 \times 1$    | (50) $-1 \times (-4)$ | (80) $4 \times 9$     | (110) $-1 \times (-7)$ |
| (21) $5 \times (-2)$  | (51) $-4 \times (-1)$ | (81) $-4 \times 4$    | (111) $7 \times 3$     |
| (22) $-1 \times 5$    | (52) $9 \times 5$     | (82) $5 \times (-4)$  | (112) $-6 \times 7$    |
| (23) $0 \times 7$     | (53) $9 \times (-5)$  | (83) $-1 \times (-9)$ | (113) $-8 \times 7$    |
| (24) $-6 \times (-5)$ | (54) $3 \times (-7)$  | (84) $-4 \times 4$    | (114) $-6 \times (-8)$ |
| (25) $1 \times (-8)$  | (55) $-7 \times 5$    | (85) $-8 \times (-6)$ | (115) $-7 \times (-8)$ |
| (26) $7 \times (-6)$  | (56) $-4 \times 6$    | (86) $-2 \times (-7)$ | (116) $0 \times (-5)$  |
| (27) $-8 \times (-1)$ | (57) $5 \times 3$     | (87) $-5 \times (-8)$ | (117) $4 \times 4$     |
| (28) $7 \times 2$     | (58) $-9 \times (-1)$ | (88) $-9 \times 2$    | (118) $-1 \times 0$    |
| (29) $1 \times (-6)$  | (59) $5 \times (-2)$  | (89) $2 \times (-5)$  | (119) $-7 \times (-1)$ |
| (30) $1 \times 6$     | (60) $-3 \times 6$    | (90) $-2 \times 9$    | (120) $-5 \times 7$    |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $-7 \times (-6)$  | (31) $4 \times (-9)$  | (61) $5 \times 5$     | (91) $3 \times (-1)$   |
| (2) $6 \times 9$      | (32) $0 \times 8$     | (62) $2 \times 8$     | (92) $5 \times 6$      |
| (3) $9 \times (-5)$   | (33) $7 \times 5$     | (63) $-7 \times (-2)$ | (93) $-2 \times 8$     |
| (4) $4 \times (-7)$   | (34) $-3 \times 1$    | (64) $3 \times (-3)$  | (94) $5 \times 3$      |
| (5) $-8 \times 8$     | (35) $1 \times (-7)$  | (65) $-9 \times (-6)$ | (95) $5 \times (-2)$   |
| (6) $-4 \times 6$     | (36) $-1 \times 5$    | (66) $-3 \times (-1)$ | (96) $4 \times (-9)$   |
| (7) $-5 \times 5$     | (37) $7 \times 2$     | (67) $-6 \times 7$    | (97) $5 \times (-7)$   |
| (8) $-5 \times (-7)$  | (38) $2 \times (-9)$  | (68) $-5 \times (-2)$ | (98) $-7 \times 2$     |
| (9) $-7 \times (-5)$  | (39) $4 \times 2$     | (69) $6 \times 1$     | (99) $0 \times 3$      |
| (10) $9 \times 9$     | (40) $8 \times 6$     | (70) $9 \times 7$     | (100) $-8 \times (-1)$ |
| (11) $-6 \times (-3)$ | (41) $0 \times 9$     | (71) $-2 \times 1$    | (101) $7 \times 0$     |
| (12) $-5 \times 5$    | (42) $-7 \times (-3)$ | (72) $0 \times 7$     | (102) $1 \times 0$     |
| (13) $-1 \times (-7)$ | (43) $2 \times (-5)$  | (73) $3 \times (-6)$  | (103) $9 \times (-3)$  |
| (14) $-3 \times 1$    | (44) $0 \times 9$     | (74) $-5 \times 4$    | (104) $-3 \times (-8)$ |
| (15) $-6 \times 6$    | (45) $1 \times (-9)$  | (75) $-2 \times (-2)$ | (105) $8 \times (-2)$  |
| (16) $4 \times (-2)$  | (46) $2 \times 7$     | (76) $2 \times (-4)$  | (106) $-8 \times 5$    |
| (17) $-3 \times (-6)$ | (47) $4 \times 0$     | (77) $-8 \times 4$    | (107) $-3 \times 3$    |
| (18) $5 \times (-3)$  | (48) $9 \times (-5)$  | (78) $-4 \times (-8)$ | (108) $0 \times 8$     |
| (19) $-9 \times (-1)$ | (49) $1 \times (-8)$  | (79) $-7 \times 6$    | (109) $4 \times (-6)$  |
| (20) $-1 \times (-2)$ | (50) $-1 \times 4$    | (80) $6 \times (-4)$  | (110) $8 \times (-4)$  |
| (21) $0 \times (-2)$  | (51) $5 \times (-1)$  | (81) $9 \times 9$     | (111) $7 \times 2$     |
| (22) $2 \times (-7)$  | (52) $0 \times 5$     | (82) $-6 \times 4$    | (112) $3 \times (-5)$  |
| (23) $-1 \times 8$    | (53) $7 \times (-9)$  | (83) $-5 \times 8$    | (113) $2 \times 1$     |
| (24) $-7 \times 5$    | (54) $4 \times (-8)$  | (84) $-4 \times 7$    | (114) $8 \times 2$     |
| (25) $8 \times 5$     | (55) $7 \times (-4)$  | (85) $-8 \times 9$    | (115) $-7 \times 7$    |
| (26) $-3 \times (-5)$ | (56) $-5 \times 9$    | (86) $-1 \times (-6)$ | (116) $-3 \times 1$    |
| (27) $-1 \times 3$    | (57) $-1 \times 7$    | (87) $-3 \times (-3)$ | (117) $-8 \times 4$    |
| (28) $8 \times (-9)$  | (58) $6 \times 5$     | (88) $9 \times (-3)$  | (118) $0 \times 1$     |
| (29) $-6 \times (-1)$ | (59) $-8 \times 2$    | (89) $7 \times 3$     | (119) $5 \times (-1)$  |
| (30) $-4 \times 8$    | (60) $-1 \times 6$    | (90) $6 \times 5$     | (120) $-5 \times (-2)$ |

|                       |                       |                       |                        |
|-----------------------|-----------------------|-----------------------|------------------------|
| (1) $7 \times 7$      | (31) $3 \times (-3)$  | (61) $3 \times (-4)$  | (91) $3 \times 9$      |
| (2) $-6 \times (-6)$  | (32) $6 \times (-7)$  | (62) $-4 \times 0$    | (92) $7 \times 1$      |
| (3) $6 \times 3$      | (33) $-3 \times 0$    | (63) $-3 \times (-7)$ | (93) $8 \times (-6)$   |
| (4) $-7 \times (-1)$  | (34) $-1 \times (-1)$ | (64) $-8 \times 7$    | (94) $7 \times (-3)$   |
| (5) $-4 \times 8$     | (35) $-5 \times (-5)$ | (65) $6 \times (-7)$  | (95) $-2 \times 8$     |
| (6) $1 \times (-3)$   | (36) $-1 \times (-7)$ | (66) $-9 \times (-6)$ | (96) $-5 \times (-2)$  |
| (7) $9 \times 2$      | (37) $-3 \times (-6)$ | (67) $-9 \times (-3)$ | (97) $0 \times 8$      |
| (8) $8 \times 8$      | (38) $-5 \times (-4)$ | (68) $-2 \times 2$    | (98) $-4 \times 0$     |
| (9) $-1 \times (-3)$  | (39) $2 \times 7$     | (69) $-8 \times 8$    | (99) $1 \times (-2)$   |
| (10) $-2 \times (-7)$ | (40) $-7 \times (-3)$ | (70) $7 \times 9$     | (100) $6 \times (-7)$  |
| (11) $5 \times 7$     | (41) $2 \times 2$     | (71) $-3 \times 9$    | (101) $4 \times (-4)$  |
| (12) $7 \times 9$     | (42) $-1 \times 1$    | (72) $3 \times (-4)$  | (102) $5 \times 6$     |
| (13) $6 \times (-1)$  | (43) $-1 \times 3$    | (73) $9 \times 8$     | (103) $4 \times 4$     |
| (14) $9 \times 9$     | (44) $-6 \times (-3)$ | (74) $-7 \times (-4)$ | (104) $2 \times (-3)$  |
| (15) $6 \times 2$     | (45) $-9 \times 9$    | (75) $8 \times 6$     | (105) $1 \times (-9)$  |
| (16) $5 \times (-2)$  | (46) $-6 \times (-8)$ | (76) $-3 \times 8$    | (106) $0 \times (-1)$  |
| (17) $5 \times 2$     | (47) $-8 \times (-8)$ | (77) $-8 \times (-7)$ | (107) $-3 \times 3$    |
| (18) $-5 \times 7$    | (48) $2 \times (-8)$  | (78) $-2 \times 1$    | (108) $3 \times (-7)$  |
| (19) $-6 \times 9$    | (49) $-8 \times (-9)$ | (79) $-9 \times 5$    | (109) $-1 \times (-4)$ |
| (20) $5 \times (-8)$  | (50) $5 \times (-9)$  | (80) $-1 \times (-3)$ | (110) $-1 \times 7$    |
| (21) $-4 \times 8$    | (51) $-6 \times (-5)$ | (81) $-7 \times (-1)$ | (111) $7 \times 5$     |
| (22) $-9 \times 8$    | (52) $-2 \times 5$    | (82) $-8 \times 8$    | (112) $-4 \times 5$    |
| (23) $6 \times 2$     | (53) $8 \times 1$     | (83) $-4 \times (-9)$ | (113) $-3 \times 3$    |
| (24) $9 \times (-3)$  | (54) $9 \times (-3)$  | (84) $-3 \times (-5)$ | (114) $-1 \times (-9)$ |
| (25) $-5 \times 1$    | (55) $2 \times (-6)$  | (85) $0 \times 1$     | (115) $-3 \times 4$    |
| (26) $5 \times (-7)$  | (56) $-4 \times 0$    | (86) $2 \times 1$     | (116) $7 \times (-2)$  |
| (27) $0 \times (-6)$  | (57) $-3 \times 9$    | (87) $-3 \times (-9)$ | (117) $-6 \times (-6)$ |
| (28) $7 \times 7$     | (58) $-2 \times 9$    | (88) $-3 \times (-8)$ | (118) $5 \times (-2)$  |
| (29) $8 \times 0$     | (59) $2 \times 0$     | (89) $-2 \times 8$    | (119) $-2 \times 4$    |
| (30) $-8 \times 2$    | (60) $1 \times 3$     | (90) $-9 \times (-4)$ | (120) $3 \times 0$     |

- |                            |                            |                            |                             |
|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-7 \times (-7) = 49$  | (31) $-8 \times (-2) = 16$ | (61) $-8 \times (-3) = 24$ | (91) $5 \times (-7) = -35$  |
| (2) $7 \times (-1) = -7$   | (32) $7 \times 3 = 21$     | (62) $7 \times 2 = 14$     | (92) $-7 \times (-9) = 63$  |
| (3) $8 \times (-6) = -48$  | (33) $9 \times (-9) = -81$ | (63) $8 \times (-3) = -24$ | (93) $-2 \times (-3) = 6$   |
| (4) $-7 \times 5 = -35$    | (34) $5 \times 3 = 15$     | (64) $-1 \times (-9) = 9$  | (94) $-7 \times 0 = 0$      |
| (5) $-5 \times 6 = -30$    | (35) $7 \times (-9) = -63$ | (65) $5 \times 2 = 10$     | (95) $3 \times (-2) = -6$   |
| (6) $8 \times (-8) = -64$  | (36) $-7 \times 4 = -28$   | (66) $-8 \times 6 = -48$   | (96) $-2 \times (-1) = 2$   |
| (7) $7 \times (-2) = -14$  | (37) $-2 \times 9 = -18$   | (67) $-3 \times 8 = -24$   | (97) $-9 \times (-2) = 18$  |
| (8) $0 \times 6 = 0$       | (38) $4 \times (-7) = -28$ | (68) $0 \times 7 = 0$      | (98) $4 \times (-9) = -36$  |
| (9) $-9 \times 8 = -72$    | (39) $-9 \times 1 = -9$    | (69) $-1 \times (-8) = 8$  | (99) $-1 \times 5 = -5$     |
| (10) $-2 \times 5 = -10$   | (40) $-6 \times (-6) = 36$ | (70) $4 \times 9 = 36$     | (100) $-1 \times (-1) = 1$  |
| (11) $6 \times 7 = 42$     | (41) $7 \times 6 = 42$     | (71) $-6 \times 1 = -6$    | (101) $3 \times 4 = 12$     |
| (12) $2 \times (-9) = -18$ | (42) $-3 \times 1 = -3$    | (72) $-7 \times (-1) = 7$  | (102) $-8 \times (-3) = 24$ |
| (13) $3 \times 2 = 6$      | (43) $7 \times (-2) = -14$ | (73) $-4 \times 0 = 0$     | (103) $7 \times (-4) = -28$ |
| (14) $-3 \times 1 = -3$    | (44) $-5 \times (-8) = 40$ | (74) $4 \times (-6) = -24$ | (104) $-5 \times 2 = -10$   |
| (15) $1 \times 4 = 4$      | (45) $-9 \times 7 = -63$   | (75) $5 \times (-2) = -10$ | (105) $1 \times 9 = 9$      |
| (16) $-6 \times 6 = -36$   | (46) $-3 \times 6 = -18$   | (76) $9 \times 3 = 27$     | (106) $-3 \times 2 = -6$    |
| (17) $-7 \times 3 = -21$   | (47) $7 \times 9 = 63$     | (77) $-7 \times 4 = -28$   | (107) $9 \times (-8) = -72$ |
| (18) $4 \times 1 = 4$      | (48) $-9 \times (-1) = 9$  | (78) $4 \times (-1) = -4$  | (108) $-1 \times 4 = -4$    |
| (19) $3 \times 2 = 6$      | (49) $0 \times (-1) = 0$   | (79) $-9 \times 3 = -27$   | (109) $-6 \times 1 = -6$    |
| (20) $4 \times (-1) = -4$  | (50) $8 \times (-1) = -8$  | (80) $-4 \times 2 = -8$    | (110) $-1 \times (-6) = 6$  |
| (21) $-6 \times 0 = 0$     | (51) $-9 \times 4 = -36$   | (81) $5 \times 8 = 40$     | (111) $6 \times (-4) = -24$ |
| (22) $8 \times (-9) = -72$ | (52) $-7 \times 3 = -21$   | (82) $2 \times (-3) = -6$  | (112) $-8 \times (-7) = 56$ |
| (23) $-8 \times (-9) = 72$ | (53) $-8 \times (-2) = 16$ | (83) $-5 \times 9 = -45$   | (113) $-9 \times (-7) = 63$ |
| (24) $0 \times 4 = 0$      | (54) $-1 \times 7 = -7$    | (84) $-9 \times 5 = -45$   | (114) $-8 \times 7 = -56$   |
| (25) $7 \times (-3) = -21$ | (55) $-5 \times (-9) = 45$ | (85) $0 \times (-9) = 0$   | (115) $1 \times (-3) = -3$  |
| (26) $0 \times 1 = 0$      | (56) $-9 \times (-8) = 72$ | (86) $4 \times (-6) = -24$ | (116) $2 \times 5 = 10$     |
| (27) $-8 \times (-5) = 40$ | (57) $-6 \times (-3) = 18$ | (87) $-6 \times (-5) = 30$ | (117) $4 \times (-9) = -36$ |
| (28) $-9 \times 8 = -72$   | (58) $1 \times (-2) = -2$  | (88) $5 \times (-5) = -25$ | (118) $1 \times 9 = 9$      |
| (29) $-9 \times (-6) = 54$ | (59) $-1 \times (-3) = 3$  | (89) $-8 \times 5 = -40$   | (119) $6 \times (-3) = -18$ |
| (30) $4 \times (-8) = -32$ | (60) $-5 \times 8 = -40$   | (90) $3 \times 2 = 6$      | (120) $9 \times 6 = 54$     |



- |                            |                            |                            |                             |
|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $8 \times 9 = 72$      | (31) $0 \times 8 = 0$      | (61) $-2 \times 0 = 0$     | (91) $5 \times 4 = 20$      |
| (2) $-4 \times (-8) = 32$  | (32) $3 \times (-9) = -27$ | (62) $6 \times 6 = 36$     | (92) $6 \times 2 = 12$      |
| (3) $-6 \times 6 = -36$    | (33) $8 \times 4 = 32$     | (63) $4 \times (-9) = -36$ | (93) $7 \times 6 = 42$      |
| (4) $1 \times (-4) = -4$   | (34) $5 \times 2 = 10$     | (64) $-7 \times (-2) = 14$ | (94) $4 \times (-6) = -24$  |
| (5) $-5 \times (-3) = 15$  | (35) $-4 \times (-8) = 32$ | (65) $-5 \times (-8) = 40$ | (95) $-8 \times 4 = -32$    |
| (6) $0 \times 7 = 0$       | (36) $-3 \times (-8) = 24$ | (66) $6 \times 7 = 42$     | (96) $3 \times (-3) = -9$   |
| (7) $-7 \times 3 = -21$    | (37) $2 \times (-1) = -2$  | (67) $5 \times 1 = 5$      | (97) $2 \times (-2) = -4$   |
| (8) $-9 \times (-9) = 81$  | (38) $-6 \times 8 = -48$   | (68) $-1 \times (-6) = 6$  | (98) $-4 \times 8 = -32$    |
| (9) $-1 \times 3 = -3$     | (39) $0 \times (-7) = 0$   | (69) $-2 \times (-2) = 4$  | (99) $8 \times (-7) = -56$  |
| (10) $-1 \times 9 = -9$    | (40) $7 \times 2 = 14$     | (70) $1 \times 2 = 2$      | (100) $6 \times (-6) = -36$ |
| (11) $-7 \times 0 = 0$     | (41) $-3 \times 3 = -9$    | (71) $7 \times 2 = 14$     | (101) $1 \times 5 = 5$      |
| (12) $2 \times 8 = 16$     | (42) $5 \times (-4) = -20$ | (72) $-8 \times 1 = -8$    | (102) $-2 \times 9 = -18$   |
| (13) $8 \times 8 = 64$     | (43) $-6 \times 7 = -42$   | (73) $-1 \times 2 = -2$    | (103) $-3 \times 5 = -15$   |
| (14) $7 \times 3 = 21$     | (44) $1 \times 2 = 2$      | (74) $0 \times (-7) = 0$   | (104) $5 \times 0 = 0$      |
| (15) $-8 \times (-5) = 40$ | (45) $9 \times 6 = 54$     | (75) $-5 \times (-3) = 15$ | (105) $-2 \times (-2) = 4$  |
| (16) $0 \times (-7) = 0$   | (46) $-7 \times 3 = -21$   | (76) $-2 \times 1 = -2$    | (106) $7 \times (-9) = -63$ |
| (17) $3 \times 5 = 15$     | (47) $-2 \times 8 = -16$   | (77) $-2 \times (-1) = 2$  | (107) $7 \times (-3) = -21$ |
| (18) $0 \times 8 = 0$      | (48) $-1 \times 2 = -2$    | (78) $7 \times (-6) = -42$ | (108) $4 \times 3 = 12$     |
| (19) $-6 \times (-7) = 42$ | (49) $-2 \times (-1) = 2$  | (79) $-1 \times (-6) = 6$  | (109) $-1 \times 9 = -9$    |
| (20) $-3 \times 8 = -24$   | (50) $-7 \times (-2) = 14$ | (80) $-5 \times 9 = -45$   | (110) $-8 \times 2 = -16$   |
| (21) $-4 \times (-2) = 8$  | (51) $3 \times (-5) = -15$ | (81) $-5 \times 5 = -25$   | (111) $3 \times (-8) = -24$ |
| (22) $-8 \times 7 = -56$   | (52) $-4 \times (-8) = 32$ | (82) $0 \times 1 = 0$      | (112) $-6 \times 8 = -48$   |
| (23) $-9 \times (-4) = 36$ | (53) $5 \times 1 = 5$      | (83) $8 \times (-3) = -24$ | (113) $-2 \times (-4) = 8$  |
| (24) $-9 \times 5 = -45$   | (54) $-9 \times 8 = -72$   | (84) $3 \times (-6) = -18$ | (114) $-7 \times 2 = -14$   |
| (25) $-1 \times 0 = 0$     | (55) $4 \times 9 = 36$     | (85) $3 \times 1 = 3$      | (115) $3 \times 3 = 9$      |
| (26) $-2 \times 4 = -8$    | (56) $0 \times 4 = 0$      | (86) $-9 \times 1 = -9$    | (116) $-7 \times 1 = -7$    |
| (27) $-7 \times (-8) = 56$ | (57) $6 \times 5 = 30$     | (87) $-2 \times (-1) = 2$  | (117) $1 \times (-6) = -6$  |
| (28) $2 \times 4 = 8$      | (58) $5 \times 7 = 35$     | (88) $8 \times (-6) = -48$ | (118) $7 \times 4 = 28$     |
| (29) $-3 \times 2 = -6$    | (59) $8 \times 8 = 64$     | (89) $-9 \times 0 = 0$     | (119) $6 \times 6 = 36$     |
| (30) $8 \times (-4) = -32$ | (60) $5 \times (-9) = -45$ | (90) $9 \times 1 = 9$      | (120) $0 \times (-1) = 0$   |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $0 \times 9 = 0$       | (31) $-4 \times 8 = -32$   | (61) $7 \times 5 = 35$     | (91) $-7 \times 0 = 0$      |
| (2) $6 \times (-7) = -42$  | (32) $7 \times (-8) = -56$ | (62) $8 \times (-2) = -16$ | (92) $-2 \times 4 = -8$     |
| (3) $-6 \times (-8) = 48$  | (33) $-9 \times 6 = -54$   | (63) $-5 \times (-7) = 35$ | (93) $2 \times 5 = 10$      |
| (4) $-6 \times (-7) = 42$  | (34) $0 \times (-3) = 0$   | (64) $2 \times (-3) = -6$  | (94) $-9 \times (-4) = 36$  |
| (5) $-9 \times (-1) = 9$   | (35) $-5 \times 4 = -20$   | (65) $2 \times (-4) = -8$  | (95) $-8 \times (-7) = 56$  |
| (6) $-3 \times 4 = -12$    | (36) $2 \times 0 = 0$      | (66) $-2 \times 5 = -10$   | (96) $-9 \times 7 = -63$    |
| (7) $-8 \times 7 = -56$    | (37) $7 \times (-4) = -28$ | (67) $-9 \times (-1) = 9$  | (97) $6 \times 6 = 36$      |
| (8) $-4 \times (-1) = 4$   | (38) $-6 \times (-1) = 6$  | (68) $-5 \times 2 = -10$   | (98) $5 \times (-3) = -15$  |
| (9) $-8 \times 0 = 0$      | (39) $-5 \times (-4) = 20$ | (69) $3 \times (-9) = -27$ | (99) $4 \times (-7) = -28$  |
| (10) $-4 \times (-4) = 16$ | (40) $-5 \times (-2) = 10$ | (70) $7 \times 0 = 0$      | (100) $8 \times (-8) = -64$ |
| (11) $-7 \times (-3) = 21$ | (41) $-7 \times (-7) = 49$ | (71) $-3 \times (-6) = 18$ | (101) $8 \times (-4) = -32$ |
| (12) $5 \times 3 = 15$     | (42) $-9 \times (-6) = 54$ | (72) $-7 \times 7 = -49$   | (102) $-3 \times (-2) = 6$  |
| (13) $-1 \times 7 = -7$    | (43) $-1 \times (-8) = 8$  | (73) $-7 \times 9 = -63$   | (103) $-5 \times 0 = 0$     |
| (14) $-7 \times (-3) = 21$ | (44) $1 \times 6 = 6$      | (74) $-9 \times (-4) = 36$ | (104) $4 \times 8 = 32$     |
| (15) $7 \times 4 = 28$     | (45) $4 \times 7 = 28$     | (75) $7 \times (-6) = -42$ | (105) $0 \times (-5) = 0$   |
| (16) $4 \times (-2) = -8$  | (46) $5 \times 7 = 35$     | (76) $-7 \times (-9) = 63$ | (106) $-8 \times (-6) = 48$ |
| (17) $-3 \times (-1) = 3$  | (47) $9 \times (-1) = -9$  | (77) $-7 \times 7 = -49$   | (107) $0 \times 9 = 0$      |
| (18) $-9 \times 9 = -81$   | (48) $0 \times (-9) = 0$   | (78) $-1 \times (-6) = 6$  | (108) $1 \times (-1) = -1$  |
| (19) $9 \times 5 = 45$     | (49) $8 \times 4 = 32$     | (79) $-5 \times (-9) = 45$ | (109) $7 \times 2 = 14$     |
| (20) $3 \times 9 = 27$     | (50) $-4 \times 5 = -20$   | (80) $2 \times (-4) = -8$  | (110) $8 \times 6 = 48$     |
| (21) $2 \times 4 = 8$      | (51) $-9 \times (-1) = 9$  | (81) $-8 \times 9 = -72$   | (111) $0 \times 1 = 0$      |
| (22) $5 \times 4 = 20$     | (52) $-4 \times (-2) = 8$  | (82) $-5 \times 3 = -15$   | (112) $-9 \times (-3) = 27$ |
| (23) $-5 \times 5 = -25$   | (53) $3 \times (-8) = -24$ | (83) $1 \times (-3) = -3$  | (113) $7 \times (-1) = -7$  |
| (24) $-8 \times (-1) = 8$  | (54) $1 \times 9 = 9$      | (84) $-1 \times (-1) = 1$  | (114) $5 \times (-8) = -40$ |
| (25) $-3 \times (-4) = 12$ | (55) $-2 \times 4 = -8$    | (85) $-7 \times (-7) = 49$ | (115) $-4 \times 2 = -8$    |
| (26) $-9 \times (-8) = 72$ | (56) $-9 \times 6 = -54$   | (86) $-4 \times 3 = -12$   | (116) $-2 \times 0 = 0$     |
| (27) $7 \times (-2) = -14$ | (57) $-2 \times 9 = -18$   | (87) $7 \times (-6) = -42$ | (117) $1 \times 8 = 8$      |
| (28) $0 \times (-2) = 0$   | (58) $5 \times 6 = 30$     | (88) $2 \times 5 = 10$     | (118) $-5 \times 3 = -15$   |
| (29) $6 \times 9 = 54$     | (59) $-9 \times 0 = 0$     | (89) $7 \times 3 = 21$     | (119) $2 \times (-7) = -14$ |
| (30) $4 \times 1 = 4$      | (60) $-6 \times (-7) = 42$ | (90) $-3 \times 8 = -24$   | (120) $-9 \times 5 = -45$   |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-9 \times 0 = 0$      | (31) $4 \times 7 = 28$     | (61) $-6 \times (-9) = 54$ | (91) $-3 \times 8 = -24$    |
| (2) $-7 \times (-1) = 7$   | (32) $7 \times 4 = 28$     | (62) $1 \times 7 = 7$      | (92) $-9 \times 0 = 0$      |
| (3) $-5 \times 8 = -40$    | (33) $3 \times (-2) = -6$  | (63) $5 \times 3 = 15$     | (93) $1 \times 7 = 7$       |
| (4) $-9 \times 0 = 0$      | (34) $4 \times 8 = 32$     | (64) $2 \times 6 = 12$     | (94) $9 \times 7 = 63$      |
| (5) $3 \times 1 = 3$       | (35) $4 \times 2 = 8$      | (65) $4 \times 9 = 36$     | (95) $-4 \times 4 = -16$    |
| (6) $-3 \times (-3) = 9$   | (36) $2 \times 8 = 16$     | (66) $2 \times (-7) = -14$ | (96) $-9 \times 7 = -63$    |
| (7) $-5 \times 0 = 0$      | (37) $8 \times 0 = 0$      | (67) $4 \times 7 = 28$     | (97) $-5 \times (-7) = 35$  |
| (8) $-1 \times 5 = -5$     | (38) $8 \times (-4) = -32$ | (68) $-3 \times 8 = -24$   | (98) $2 \times 0 = 0$       |
| (9) $-5 \times 1 = -5$     | (39) $-3 \times (-9) = 27$ | (69) $-1 \times 0 = 0$     | (99) $4 \times 9 = 36$      |
| (10) $-4 \times 7 = -28$   | (40) $-3 \times 1 = -3$    | (70) $9 \times (-8) = -72$ | (100) $7 \times (-6) = -42$ |
| (11) $2 \times (-8) = -16$ | (41) $-4 \times (-2) = 8$  | (71) $6 \times 4 = 24$     | (101) $1 \times 5 = 5$      |
| (12) $2 \times (-9) = -18$ | (42) $-7 \times 0 = 0$     | (72) $-5 \times 7 = -35$   | (102) $-4 \times 4 = -16$   |
| (13) $-1 \times 0 = 0$     | (43) $4 \times 2 = 8$      | (73) $-7 \times 9 = -63$   | (103) $5 \times 2 = 10$     |
| (14) $2 \times 7 = 14$     | (44) $-3 \times 9 = -27$   | (74) $-9 \times 7 = -63$   | (104) $-5 \times (-8) = 40$ |
| (15) $-2 \times (-7) = 14$ | (45) $9 \times 3 = 27$     | (75) $-1 \times 6 = -6$    | (105) $9 \times (-1) = -9$  |
| (16) $8 \times (-2) = -16$ | (46) $-1 \times 8 = -8$    | (76) $-9 \times 9 = -81$   | (106) $6 \times 0 = 0$      |
| (17) $9 \times (-6) = -54$ | (47) $-1 \times 3 = -3$    | (77) $0 \times 8 = 0$      | (107) $1 \times 4 = 4$      |
| (18) $8 \times 5 = 40$     | (48) $6 \times 3 = 18$     | (78) $3 \times (-8) = -24$ | (108) $7 \times (-3) = -21$ |
| (19) $8 \times (-3) = -24$ | (49) $-3 \times 1 = -3$    | (79) $-9 \times (-9) = 81$ | (109) $0 \times 1 = 0$      |
| (20) $6 \times 0 = 0$      | (50) $-8 \times (-9) = 72$ | (80) $-5 \times 7 = -35$   | (110) $-6 \times 5 = -30$   |
| (21) $-9 \times (-7) = 63$ | (51) $-6 \times (-2) = 12$ | (81) $7 \times (-2) = -14$ | (111) $9 \times 6 = 54$     |
| (22) $5 \times (-4) = -20$ | (52) $7 \times (-5) = -35$ | (82) $0 \times (-6) = 0$   | (112) $-5 \times 8 = -40$   |
| (23) $-1 \times 2 = -2$    | (53) $-7 \times 6 = -42$   | (83) $3 \times 2 = 6$      | (113) $8 \times 3 = 24$     |
| (24) $8 \times (-1) = -8$  | (54) $-7 \times 2 = -14$   | (84) $-5 \times 0 = 0$     | (114) $-3 \times 9 = -27$   |
| (25) $4 \times (-1) = -4$  | (55) $6 \times 9 = 54$     | (85) $-2 \times 5 = -10$   | (115) $-3 \times 2 = -6$    |
| (26) $-7 \times 5 = -35$   | (56) $1 \times (-3) = -3$  | (86) $2 \times (-4) = -8$  | (116) $-7 \times 0 = 0$     |
| (27) $1 \times (-7) = -7$  | (57) $-2 \times 2 = -4$    | (87) $-4 \times 0 = 0$     | (117) $4 \times (-9) = -36$ |
| (28) $9 \times 8 = 72$     | (58) $7 \times (-4) = -28$ | (88) $6 \times 9 = 54$     | (118) $0 \times (-4) = 0$   |
| (29) $-2 \times 1 = -2$    | (59) $6 \times (-9) = -54$ | (89) $-4 \times (-7) = 28$ | (119) $-9 \times (-3) = 27$ |
| (30) $-7 \times (-1) = 7$  | (60) $-4 \times (-9) = 36$ | (90) $-4 \times (-1) = 4$  | (120) $-5 \times (-7) = 35$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $2 \times (-3) = -6$   | (31) $-1 \times (-4) = 4$  | (61) $9 \times 0 = 0$      | (91) $4 \times 7 = 28$      |
| (2) $7 \times (-9) = -63$  | (32) $-6 \times 2 = -12$   | (62) $-5 \times (-5) = 25$ | (92) $-1 \times (-5) = 5$   |
| (3) $2 \times 6 = 12$      | (33) $8 \times 0 = 0$      | (63) $6 \times (-1) = -6$  | (93) $8 \times (-9) = -72$  |
| (4) $1 \times 9 = 9$       | (34) $-3 \times (-4) = 12$ | (64) $3 \times 2 = 6$      | (94) $4 \times (-1) = -4$   |
| (5) $9 \times (-2) = -18$  | (35) $-4 \times 3 = -12$   | (65) $-1 \times 5 = -5$    | (95) $7 \times (-5) = -35$  |
| (6) $-2 \times 8 = -16$    | (36) $6 \times (-9) = -54$ | (66) $-5 \times 6 = -30$   | (96) $-2 \times (-8) = 16$  |
| (7) $-2 \times 0 = 0$      | (37) $-2 \times 8 = -16$   | (67) $-7 \times (-6) = 42$ | (97) $2 \times 3 = 6$       |
| (8) $-6 \times (-4) = 24$  | (38) $7 \times 9 = 63$     | (68) $-7 \times 8 = -56$   | (98) $8 \times 0 = 0$       |
| (9) $0 \times (-1) = 0$    | (39) $8 \times (-6) = -48$ | (69) $-3 \times (-3) = 9$  | (99) $-2 \times 1 = -2$     |
| (10) $5 \times (-3) = -15$ | (40) $-7 \times (-5) = 35$ | (70) $2 \times (-1) = -2$  | (100) $0 \times 0 = 0$      |
| (11) $-3 \times 5 = -15$   | (41) $1 \times (-4) = -4$  | (71) $-2 \times (-3) = 6$  | (101) $-7 \times (-7) = 49$ |
| (12) $1 \times 5 = 5$      | (42) $4 \times 1 = 4$      | (72) $4 \times (-7) = -28$ | (102) $-8 \times 9 = -72$   |
| (13) $-8 \times 9 = -72$   | (43) $3 \times (-6) = -18$ | (73) $7 \times 5 = 35$     | (103) $8 \times 8 = 64$     |
| (14) $-5 \times 1 = -5$    | (44) $8 \times (-4) = -32$ | (74) $2 \times 3 = 6$      | (104) $1 \times 8 = 8$      |
| (15) $8 \times (-5) = -40$ | (45) $-7 \times 8 = -56$   | (75) $1 \times 3 = 3$      | (105) $-6 \times (-9) = 54$ |
| (16) $-1 \times (-5) = 5$  | (46) $-2 \times 6 = -12$   | (76) $-1 \times 6 = -6$    | (106) $-5 \times 3 = -15$   |
| (17) $0 \times 5 = 0$      | (47) $8 \times (-7) = -56$ | (77) $4 \times 4 = 16$     | (107) $-5 \times 2 = -10$   |
| (18) $-9 \times 2 = -18$   | (48) $-6 \times (-5) = 30$ | (78) $-3 \times (-7) = 21$ | (108) $-5 \times (-2) = 10$ |
| (19) $-4 \times (-1) = 4$  | (49) $4 \times 8 = 32$     | (79) $9 \times (-1) = -9$  | (109) $-8 \times (-8) = 64$ |
| (20) $7 \times (-6) = -42$ | (50) $8 \times 5 = 40$     | (80) $1 \times (-3) = -3$  | (110) $5 \times 0 = 0$      |
| (21) $6 \times (-6) = -36$ | (51) $7 \times (-9) = -63$ | (81) $8 \times 9 = 72$     | (111) $0 \times (-3) = 0$   |
| (22) $9 \times (-6) = -54$ | (52) $0 \times 1 = 0$      | (82) $3 \times (-9) = -27$ | (112) $0 \times 6 = 0$      |
| (23) $0 \times 1 = 0$      | (53) $-4 \times (-6) = 24$ | (83) $-5 \times 9 = -45$   | (113) $8 \times (-1) = -8$  |
| (24) $-2 \times (-6) = 12$ | (54) $3 \times (-4) = -12$ | (84) $7 \times (-3) = -21$ | (114) $5 \times 4 = 20$     |
| (25) $2 \times (-6) = -12$ | (55) $8 \times (-5) = -40$ | (85) $-5 \times 0 = 0$     | (115) $7 \times 1 = 7$      |
| (26) $-9 \times 9 = -81$   | (56) $-1 \times (-8) = 8$  | (86) $3 \times (-8) = -24$ | (116) $8 \times (-2) = -16$ |
| (27) $-3 \times 6 = -18$   | (57) $-7 \times (-4) = 28$ | (87) $1 \times (-6) = -6$  | (117) $-1 \times 1 = -1$    |
| (28) $6 \times 0 = 0$      | (58) $-6 \times 3 = -18$   | (88) $-3 \times (-2) = 6$  | (118) $3 \times 7 = 21$     |
| (29) $-5 \times 9 = -45$   | (59) $6 \times 8 = 48$     | (89) $3 \times (-2) = -6$  | (119) $6 \times (-4) = -24$ |
| (30) $-9 \times (-5) = 45$ | (60) $-7 \times (-8) = 56$ | (90) $3 \times (-6) = -18$ | (120) $-4 \times (-7) = 28$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-4 \times (-4) = 16$  | (31) $-2 \times (-6) = 12$ | (61) $-6 \times (-2) = 12$ | (91) $-3 \times 8 = -24$    |
| (2) $6 \times 8 = 48$      | (32) $-6 \times 0 = 0$     | (62) $6 \times (-1) = -6$  | (92) $-7 \times 1 = -7$     |
| (3) $-6 \times 8 = -48$    | (33) $3 \times 6 = 18$     | (63) $-8 \times (-9) = 72$ | (93) $-8 \times 0 = 0$      |
| (4) $1 \times (-9) = -9$   | (34) $6 \times 7 = 42$     | (64) $0 \times (-5) = 0$   | (94) $6 \times 1 = 6$       |
| (5) $-6 \times 2 = -12$    | (35) $-6 \times (-6) = 36$ | (65) $-7 \times (-5) = 35$ | (95) $0 \times (-5) = 0$    |
| (6) $5 \times (-3) = -15$  | (36) $-5 \times (-1) = 5$  | (66) $2 \times (-6) = -12$ | (96) $7 \times 9 = 63$      |
| (7) $7 \times (-1) = -7$   | (37) $-2 \times 8 = -16$   | (67) $9 \times 8 = 72$     | (97) $6 \times 9 = 54$      |
| (8) $3 \times 7 = 21$      | (38) $9 \times 5 = 45$     | (68) $2 \times (-5) = -10$ | (98) $9 \times 6 = 54$      |
| (9) $7 \times 0 = 0$       | (39) $-3 \times 4 = -12$   | (69) $2 \times 9 = 18$     | (99) $-2 \times 6 = -12$    |
| (10) $-1 \times (-6) = 6$  | (40) $4 \times 2 = 8$      | (70) $4 \times 7 = 28$     | (100) $-6 \times (-2) = 12$ |
| (11) $-1 \times 7 = -7$    | (41) $9 \times (-3) = -27$ | (71) $-7 \times (-8) = 56$ | (101) $9 \times 8 = 72$     |
| (12) $2 \times (-4) = -8$  | (42) $-9 \times 8 = -72$   | (72) $8 \times (-8) = -64$ | (102) $-6 \times 1 = -6$    |
| (13) $2 \times (-6) = -12$ | (43) $1 \times (-3) = -3$  | (73) $-3 \times 5 = -15$   | (103) $1 \times (-6) = -6$  |
| (14) $5 \times (-1) = -5$  | (44) $9 \times 4 = 36$     | (74) $-2 \times 9 = -18$   | (104) $-8 \times 3 = -24$   |
| (15) $0 \times 0 = 0$      | (45) $1 \times (-2) = -2$  | (75) $7 \times (-3) = -21$ | (105) $1 \times 0 = 0$      |
| (16) $-1 \times 1 = -1$    | (46) $0 \times 0 = 0$      | (76) $-4 \times (-9) = 36$ | (106) $7 \times 8 = 56$     |
| (17) $7 \times (-6) = -42$ | (47) $-6 \times 5 = -30$   | (77) $-9 \times (-8) = 72$ | (107) $9 \times 7 = 63$     |
| (18) $2 \times 5 = 10$     | (48) $7 \times (-1) = -7$  | (78) $5 \times (-6) = -30$ | (108) $-4 \times 6 = -24$   |
| (19) $-6 \times (-2) = 12$ | (49) $-8 \times 6 = -48$   | (79) $0 \times (-5) = 0$   | (109) $2 \times 7 = 14$     |
| (20) $5 \times 7 = 35$     | (50) $-7 \times 7 = -49$   | (80) $4 \times 2 = 8$      | (110) $3 \times 1 = 3$      |
| (21) $5 \times (-4) = -20$ | (51) $-3 \times 5 = -15$   | (81) $5 \times (-4) = -20$ | (111) $0 \times (-5) = 0$   |
| (22) $1 \times 2 = 2$      | (52) $3 \times 0 = 0$      | (82) $-9 \times (-1) = 9$  | (112) $-1 \times (-1) = 1$  |
| (23) $-5 \times 8 = -40$   | (53) $1 \times 4 = 4$      | (83) $-1 \times 8 = -8$    | (113) $-1 \times (-6) = 6$  |
| (24) $4 \times (-8) = -32$ | (54) $1 \times (-9) = -9$  | (84) $-8 \times (-2) = 16$ | (114) $9 \times 8 = 72$     |
| (25) $-2 \times (-6) = 12$ | (55) $3 \times 9 = 27$     | (85) $5 \times (-4) = -20$ | (115) $4 \times (-3) = -12$ |
| (26) $3 \times 0 = 0$      | (56) $-9 \times 0 = 0$     | (86) $-6 \times (-6) = 36$ | (116) $-9 \times 7 = -63$   |
| (27) $-6 \times (-9) = 54$ | (57) $-9 \times (-7) = 63$ | (87) $-5 \times 3 = -15$   | (117) $-7 \times 0 = 0$     |
| (28) $6 \times 0 = 0$      | (58) $7 \times 0 = 0$      | (88) $-6 \times (-3) = 18$ | (118) $3 \times (-8) = -24$ |
| (29) $-4 \times (-2) = 8$  | (59) $6 \times (-8) = -48$ | (89) $-9 \times 0 = 0$     | (119) $1 \times 3 = 3$      |
| (30) $-9 \times (-3) = 27$ | (60) $-8 \times (-9) = 72$ | (90) $-9 \times 6 = -54$   | (120) $1 \times (-9) = -9$  |

- |                            |                            |                            |                             |
|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $4 \times (-3) = -12$  | (31) $-7 \times (-3) = 21$ | (61) $-9 \times (-7) = 63$ | (91) $0 \times (-2) = 0$    |
| (2) $-4 \times (-6) = 24$  | (32) $-4 \times 0 = 0$     | (62) $5 \times 9 = 45$     | (92) $-7 \times (-9) = 63$  |
| (3) $-2 \times (-6) = 12$  | (33) $9 \times (-8) = -72$ | (63) $6 \times 0 = 0$      | (93) $5 \times 9 = 45$      |
| (4) $9 \times (-8) = -72$  | (34) $-4 \times (-9) = 36$ | (64) $3 \times (-9) = -27$ | (94) $6 \times (-5) = -30$  |
| (5) $3 \times 5 = 15$      | (35) $2 \times (-4) = -8$  | (65) $7 \times 1 = 7$      | (95) $-6 \times 7 = -42$    |
| (6) $5 \times (-9) = -45$  | (36) $0 \times 8 = 0$      | (66) $3 \times 4 = 12$     | (96) $6 \times 9 = 54$      |
| (7) $-4 \times 0 = 0$      | (37) $-5 \times (-6) = 30$ | (67) $-5 \times 7 = -35$   | (97) $2 \times (-7) = -14$  |
| (8) $4 \times (-6) = -24$  | (38) $-6 \times (-9) = 54$ | (68) $-5 \times 0 = 0$     | (98) $2 \times 6 = 12$      |
| (9) $2 \times 0 = 0$       | (39) $-1 \times (-1) = 1$  | (69) $-4 \times (-7) = 28$ | (99) $6 \times 9 = 54$      |
| (10) $-1 \times (-8) = 8$  | (40) $9 \times (-8) = -72$ | (70) $-7 \times (-2) = 14$ | (100) $1 \times 8 = 8$      |
| (11) $-9 \times 8 = -72$   | (41) $-1 \times 7 = -7$    | (71) $7 \times 6 = 42$     | (101) $8 \times (-3) = -24$ |
| (12) $4 \times 4 = 16$     | (42) $-7 \times 6 = -42$   | (72) $3 \times (-9) = -27$ | (102) $-5 \times (-9) = 45$ |
| (13) $-5 \times (-8) = 40$ | (43) $2 \times (-9) = -18$ | (73) $7 \times 5 = 35$     | (103) $-5 \times 5 = -25$   |
| (14) $-3 \times 4 = -12$   | (44) $6 \times (-3) = -18$ | (74) $-1 \times 4 = -4$    | (104) $-9 \times 5 = -45$   |
| (15) $3 \times (-8) = -24$ | (45) $-9 \times 8 = -72$   | (75) $8 \times (-5) = -40$ | (105) $9 \times 5 = 45$     |
| (16) $3 \times 0 = 0$      | (46) $3 \times 1 = 3$      | (76) $-7 \times (-6) = 42$ | (106) $8 \times 1 = 8$      |
| (17) $-6 \times 7 = -42$   | (47) $8 \times 8 = 64$     | (77) $5 \times (-9) = -45$ | (107) $8 \times (-7) = -56$ |
| (18) $-3 \times (-9) = 27$ | (48) $6 \times (-5) = -30$ | (78) $0 \times (-7) = 0$   | (108) $4 \times 9 = 36$     |
| (19) $-5 \times (-7) = 35$ | (49) $-5 \times (-1) = 5$  | (79) $-2 \times (-2) = 4$  | (109) $-2 \times 2 = -4$    |
| (20) $-3 \times (-4) = 12$ | (50) $-2 \times 3 = -6$    | (80) $-5 \times 1 = -5$    | (110) $-7 \times (-7) = 49$ |
| (21) $5 \times 6 = 30$     | (51) $5 \times 8 = 40$     | (81) $-7 \times (-5) = 35$ | (111) $2 \times 6 = 12$     |
| (22) $4 \times (-5) = -20$ | (52) $4 \times 3 = 12$     | (82) $-5 \times (-2) = 10$ | (112) $-3 \times 1 = -3$    |
| (23) $8 \times (-9) = -72$ | (53) $3 \times 3 = 9$      | (83) $-3 \times (-7) = 21$ | (113) $2 \times (-8) = -16$ |
| (24) $-4 \times 2 = -8$    | (54) $-9 \times (-1) = 9$  | (84) $-8 \times 4 = -32$   | (114) $-6 \times 7 = -42$   |
| (25) $-4 \times 5 = -20$   | (55) $8 \times 0 = 0$      | (85) $-1 \times 1 = -1$    | (115) $9 \times 8 = 72$     |
| (26) $9 \times 4 = 36$     | (56) $8 \times (-8) = -64$ | (86) $-8 \times 5 = -40$   | (116) $9 \times 1 = 9$      |
| (27) $3 \times 7 = 21$     | (57) $4 \times 8 = 32$     | (87) $-4 \times 5 = -20$   | (117) $-7 \times 4 = -28$   |
| (28) $-5 \times (-8) = 40$ | (58) $-1 \times (-5) = 5$  | (88) $-6 \times 4 = -24$   | (118) $9 \times 8 = 72$     |
| (29) $-4 \times 4 = -16$   | (59) $-4 \times 2 = -8$    | (89) $-8 \times (-6) = 48$ | (119) $-6 \times (-8) = 48$ |
| (30) $8 \times 2 = 16$     | (60) $8 \times 6 = 48$     | (90) $2 \times 2 = 4$      | (120) $-6 \times 3 = -18$   |

- |                            |                            |                            |                             |
|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-1 \times (-5) = 5$   | (31) $1 \times (-6) = -6$  | (61) $1 \times 1 = 1$      | (91) $-8 \times (-3) = 24$  |
| (2) $-5 \times 7 = -35$    | (32) $-1 \times (-1) = 1$  | (62) $-8 \times (-7) = 56$ | (92) $1 \times 4 = 4$       |
| (3) $2 \times 6 = 12$      | (33) $-9 \times 1 = -9$    | (63) $-3 \times 5 = -15$   | (93) $5 \times 1 = 5$       |
| (4) $-6 \times 8 = -48$    | (34) $-5 \times (-1) = 5$  | (64) $5 \times (-8) = -40$ | (94) $-6 \times (-4) = 24$  |
| (5) $6 \times (-9) = -54$  | (35) $-3 \times 4 = -12$   | (65) $-5 \times (-7) = 35$ | (95) $0 \times (-8) = 0$    |
| (6) $1 \times (-1) = -1$   | (36) $-3 \times 5 = -15$   | (66) $-5 \times 4 = -20$   | (96) $2 \times (-4) = -8$   |
| (7) $2 \times (-3) = -6$   | (37) $0 \times 5 = 0$      | (67) $-9 \times (-4) = 36$ | (97) $7 \times 2 = 14$      |
| (8) $-1 \times (-3) = 3$   | (38) $2 \times 5 = 10$     | (68) $4 \times 6 = 24$     | (98) $1 \times 7 = 7$       |
| (9) $-7 \times (-7) = 49$  | (39) $-9 \times (-5) = 45$ | (69) $-5 \times (-8) = 40$ | (99) $7 \times (-3) = -21$  |
| (10) $-5 \times (-9) = 45$ | (40) $9 \times 6 = 54$     | (70) $-9 \times 7 = -63$   | (100) $3 \times (-1) = -3$  |
| (11) $-7 \times 2 = -14$   | (41) $-7 \times 0 = 0$     | (71) $-1 \times 7 = -7$    | (101) $2 \times 1 = 2$      |
| (12) $4 \times (-9) = -36$ | (42) $-8 \times (-7) = 56$ | (72) $8 \times 5 = 40$     | (102) $3 \times 3 = 9$      |
| (13) $-9 \times 3 = -27$   | (43) $-3 \times 0 = 0$     | (73) $-7 \times 7 = -49$   | (103) $-3 \times (-6) = 18$ |
| (14) $-8 \times (-4) = 32$ | (44) $2 \times (-1) = -2$  | (74) $-4 \times (-9) = 36$ | (104) $3 \times 9 = 27$     |
| (15) $-7 \times 2 = -14$   | (45) $-2 \times (-3) = 6$  | (75) $2 \times 1 = 2$      | (105) $0 \times 2 = 0$      |
| (16) $-5 \times 1 = -5$    | (46) $6 \times 6 = 36$     | (76) $-3 \times 0 = 0$     | (106) $4 \times 0 = 0$      |
| (17) $-6 \times 7 = -42$   | (47) $-1 \times (-4) = 4$  | (77) $7 \times (-1) = -7$  | (107) $0 \times 0 = 0$      |
| (18) $1 \times 2 = 2$      | (48) $0 \times 6 = 0$      | (78) $9 \times (-2) = -18$ | (108) $-4 \times (-2) = 8$  |
| (19) $2 \times 1 = 2$      | (49) $9 \times (-7) = -63$ | (79) $8 \times (-7) = -56$ | (109) $-2 \times 4 = -8$    |
| (20) $2 \times 7 = 14$     | (50) $0 \times 5 = 0$      | (80) $9 \times 5 = 45$     | (110) $4 \times (-1) = -4$  |
| (21) $2 \times 0 = 0$      | (51) $8 \times (-5) = -40$ | (81) $6 \times 5 = 30$     | (111) $-9 \times (-2) = 18$ |
| (22) $8 \times (-9) = -72$ | (52) $-1 \times (-5) = 5$  | (82) $0 \times 6 = 0$      | (112) $-2 \times 2 = -4$    |
| (23) $-7 \times (-4) = 28$ | (53) $3 \times 1 = 3$      | (83) $-2 \times (-4) = 8$  | (113) $6 \times (-5) = -30$ |
| (24) $7 \times 7 = 49$     | (54) $2 \times (-5) = -10$ | (84) $-9 \times 3 = -27$   | (114) $-8 \times (-8) = 64$ |
| (25) $-1 \times 9 = -9$    | (55) $-6 \times (-9) = 54$ | (85) $2 \times (-5) = -10$ | (115) $-8 \times (-5) = 40$ |
| (26) $7 \times (-3) = -21$ | (56) $1 \times 0 = 0$      | (86) $-7 \times 6 = -42$   | (116) $-9 \times (-1) = 9$  |
| (27) $-3 \times 5 = -15$   | (57) $2 \times (-6) = -12$ | (87) $1 \times (-8) = -8$  | (117) $2 \times (-6) = -12$ |
| (28) $6 \times (-9) = -54$ | (58) $2 \times (-9) = -18$ | (88) $-6 \times (-6) = 36$ | (118) $1 \times (-5) = -5$  |
| (29) $0 \times 4 = 0$      | (59) $-1 \times 9 = -9$    | (89) $3 \times 7 = 21$     | (119) $7 \times (-7) = -49$ |
| (30) $8 \times (-7) = -56$ | (60) $-8 \times 4 = -32$   | (90) $8 \times (-3) = -24$ | (120) $5 \times (-6) = -30$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-3 \times 8 = -24$    | (31) $7 \times 7 = 49$     | (61) $7 \times (-9) = -63$ | (91) $2 \times (-9) = -18$  |
| (2) $-5 \times 0 = 0$      | (32) $9 \times 4 = 36$     | (62) $-5 \times (-2) = 10$ | (92) $-7 \times 6 = -42$    |
| (3) $6 \times (-6) = -36$  | (33) $2 \times 9 = 18$     | (63) $-7 \times (-4) = 28$ | (93) $0 \times (-8) = 0$    |
| (4) $-8 \times 2 = -16$    | (34) $-6 \times (-5) = 30$ | (64) $-7 \times 0 = 0$     | (94) $-1 \times 8 = -8$     |
| (5) $-3 \times 9 = -27$    | (35) $4 \times (-3) = -12$ | (65) $-2 \times 4 = -8$    | (95) $-8 \times (-5) = 40$  |
| (6) $-5 \times 7 = -35$    | (36) $-1 \times (-3) = 3$  | (66) $4 \times (-5) = -20$ | (96) $8 \times 3 = 24$      |
| (7) $4 \times (-8) = -32$  | (37) $1 \times (-6) = -6$  | (67) $-6 \times 8 = -48$   | (97) $-3 \times 0 = 0$      |
| (8) $1 \times (-1) = -1$   | (38) $0 \times 4 = 0$      | (68) $1 \times 6 = 6$      | (98) $4 \times (-8) = -32$  |
| (9) $1 \times 4 = 4$       | (39) $-9 \times 0 = 0$     | (69) $6 \times 9 = 54$     | (99) $9 \times 4 = 36$      |
| (10) $9 \times (-4) = -36$ | (40) $-2 \times (-7) = 14$ | (70) $4 \times (-6) = -24$ | (100) $4 \times (-4) = -16$ |
| (11) $1 \times 6 = 6$      | (41) $-4 \times 9 = -36$   | (71) $5 \times 1 = 5$      | (101) $-1 \times (-7) = 7$  |
| (12) $-9 \times 2 = -18$   | (42) $3 \times 4 = 12$     | (72) $2 \times 2 = 4$      | (102) $3 \times (-4) = -12$ |
| (13) $-3 \times 3 = -9$    | (43) $-3 \times (-2) = 6$  | (73) $4 \times (-4) = -16$ | (103) $6 \times (-4) = -24$ |
| (14) $-2 \times (-5) = 10$ | (44) $9 \times (-1) = -9$  | (74) $-4 \times 9 = -36$   | (104) $-2 \times 5 = -10$   |
| (15) $-1 \times (-8) = 8$  | (45) $6 \times 2 = 12$     | (75) $7 \times 6 = 42$     | (105) $5 \times (-1) = -5$  |
| (16) $8 \times (-4) = -32$ | (46) $-4 \times 9 = -36$   | (76) $8 \times (-4) = -32$ | (106) $-1 \times (-8) = 8$  |
| (17) $-3 \times (-5) = 15$ | (47) $8 \times (-4) = -32$ | (77) $-3 \times (-8) = 24$ | (107) $8 \times 5 = 40$     |
| (18) $-6 \times 4 = -24$   | (48) $-2 \times (-4) = 8$  | (78) $-2 \times (-7) = 14$ | (108) $4 \times 8 = 32$     |
| (19) $-5 \times (-6) = 30$ | (49) $0 \times (-7) = 0$   | (79) $1 \times (-2) = -2$  | (109) $-1 \times 3 = -3$    |
| (20) $0 \times 3 = 0$      | (50) $2 \times (-8) = -16$ | (80) $3 \times 8 = 24$     | (110) $-8 \times (-3) = 24$ |
| (21) $-1 \times (-6) = 6$  | (51) $4 \times 4 = 16$     | (81) $7 \times 2 = 14$     | (111) $-3 \times (-6) = 18$ |
| (22) $9 \times (-2) = -18$ | (52) $-9 \times (-6) = 54$ | (82) $-2 \times 8 = -16$   | (112) $9 \times 1 = 9$      |
| (23) $-9 \times 4 = -36$   | (53) $2 \times 6 = 12$     | (83) $-9 \times (-6) = 54$ | (113) $2 \times 6 = 12$     |
| (24) $2 \times 1 = 2$      | (54) $-8 \times (-5) = 40$ | (84) $8 \times 3 = 24$     | (114) $-5 \times (-1) = 5$  |
| (25) $8 \times 9 = 72$     | (55) $-1 \times (-5) = 5$  | (85) $6 \times 5 = 30$     | (115) $-5 \times (-3) = 15$ |
| (26) $7 \times 7 = 49$     | (56) $4 \times 1 = 4$      | (86) $7 \times 8 = 56$     | (116) $-9 \times (-1) = 9$  |
| (27) $6 \times 7 = 42$     | (57) $7 \times 9 = 63$     | (87) $-1 \times (-7) = 7$  | (117) $9 \times 8 = 72$     |
| (28) $-7 \times 8 = -56$   | (58) $6 \times (-7) = -42$ | (88) $8 \times (-9) = -72$ | (118) $-4 \times (-4) = 16$ |
| (29) $-9 \times (-5) = 45$ | (59) $7 \times (-8) = -56$ | (89) $-2 \times 4 = -8$    | (119) $-8 \times 4 = -32$   |
| (30) $9 \times 1 = 9$      | (60) $0 \times (-4) = 0$   | (90) $-2 \times 9 = -18$   | (120) $-4 \times 6 = -24$   |



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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-5 \times 2 = -10$    | (31) $8 \times (-1) = -8$  | (61) $-6 \times 1 = -6$    | (91) $-2 \times 4 = -8$     |
| (2) $-7 \times 6 = -42$    | (32) $-5 \times (-3) = 15$ | (62) $-5 \times 7 = -35$   | (92) $6 \times (-7) = -42$  |
| (3) $1 \times (-4) = -4$   | (33) $8 \times (-3) = -24$ | (63) $6 \times (-6) = -36$ | (93) $-2 \times 0 = 0$      |
| (4) $6 \times 9 = 54$      | (34) $-6 \times 4 = -24$   | (64) $0 \times (-1) = 0$   | (94) $-2 \times 2 = -4$     |
| (5) $4 \times 9 = 36$      | (35) $0 \times 9 = 0$      | (65) $2 \times 4 = 8$      | (95) $-5 \times 7 = -35$    |
| (6) $7 \times (-2) = -14$  | (36) $-4 \times (-8) = 32$ | (66) $4 \times 7 = 28$     | (96) $-5 \times 7 = -35$    |
| (7) $8 \times 3 = 24$      | (37) $7 \times (-9) = -63$ | (67) $-4 \times (-9) = 36$ | (97) $7 \times 0 = 0$       |
| (8) $3 \times 2 = 6$       | (38) $-8 \times (-3) = 24$ | (68) $-2 \times 8 = -16$   | (98) $2 \times (-2) = -4$   |
| (9) $-4 \times 9 = -36$    | (39) $-3 \times (-2) = 6$  | (69) $-4 \times 7 = -28$   | (99) $-3 \times 7 = -21$    |
| (10) $8 \times (-8) = -64$ | (40) $9 \times 5 = 45$     | (70) $0 \times 5 = 0$      | (100) $5 \times (-6) = -30$ |
| (11) $-1 \times (-4) = 4$  | (41) $9 \times 1 = 9$      | (71) $-8 \times 6 = -48$   | (101) $-6 \times 4 = -24$   |
| (12) $-3 \times (-2) = 6$  | (42) $8 \times (-6) = -48$ | (72) $-2 \times 1 = -2$    | (102) $8 \times 7 = 56$     |
| (13) $-7 \times (-2) = 14$ | (43) $-1 \times (-7) = 7$  | (73) $1 \times (-9) = -9$  | (103) $4 \times (-2) = -8$  |
| (14) $2 \times 7 = 14$     | (44) $9 \times (-7) = -63$ | (74) $-1 \times 6 = -6$    | (104) $-3 \times 7 = -21$   |
| (15) $-6 \times 9 = -54$   | (45) $-4 \times (-3) = 12$ | (75) $3 \times (-7) = -21$ | (105) $1 \times 8 = 8$      |
| (16) $1 \times 0 = 0$      | (46) $2 \times 0 = 0$      | (76) $-5 \times (-3) = 15$ | (106) $8 \times (-2) = -16$ |
| (17) $5 \times (-9) = -45$ | (47) $1 \times 9 = 9$      | (77) $9 \times 8 = 72$     | (107) $-6 \times (-5) = 30$ |
| (18) $9 \times (-9) = -81$ | (48) $6 \times 2 = 12$     | (78) $1 \times (-7) = -7$  | (108) $-4 \times (-6) = 24$ |
| (19) $7 \times 6 = 42$     | (49) $-1 \times 0 = 0$     | (79) $-1 \times 2 = -2$    | (109) $9 \times 6 = 54$     |
| (20) $-6 \times 3 = -18$   | (50) $-1 \times (-1) = 1$  | (80) $-2 \times 9 = -18$   | (110) $4 \times (-1) = -4$  |
| (21) $-3 \times 4 = -12$   | (51) $-2 \times 2 = -4$    | (81) $-1 \times 4 = -4$    | (111) $-3 \times (-2) = 6$  |
| (22) $-1 \times 6 = -6$    | (52) $4 \times (-2) = -8$  | (82) $2 \times (-2) = -4$  | (112) $9 \times (-9) = -81$ |
| (23) $-4 \times (-8) = 32$ | (53) $-7 \times 3 = -21$   | (83) $-9 \times 1 = -9$    | (113) $-8 \times (-5) = 40$ |
| (24) $-4 \times (-8) = 32$ | (54) $0 \times (-7) = 0$   | (84) $-4 \times (-3) = 12$ | (114) $-4 \times 5 = -20$   |
| (25) $-3 \times 0 = 0$     | (55) $-3 \times (-8) = 24$ | (85) $-1 \times (-2) = 2$  | (115) $-6 \times (-3) = 18$ |
| (26) $8 \times 5 = 40$     | (56) $-9 \times (-2) = 18$ | (86) $4 \times (-6) = -24$ | (116) $2 \times (-6) = -12$ |
| (27) $-9 \times 8 = -72$   | (57) $4 \times (-3) = -12$ | (87) $6 \times 1 = 6$      | (117) $-8 \times 5 = -40$   |
| (28) $-4 \times (-2) = 8$  | (58) $0 \times 5 = 0$      | (88) $4 \times (-9) = -36$ | (118) $3 \times (-1) = -3$  |
| (29) $6 \times (-6) = -36$ | (59) $-3 \times 1 = -3$    | (89) $-8 \times (-6) = 48$ | (119) $8 \times 6 = 48$     |
| (30) $-2 \times 4 = -8$    | (60) $0 \times (-4) = 0$   | (90) $-8 \times (-3) = 24$ | (120) $1 \times 0 = 0$      |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-2 \times 9 = -18$    | (31) $1 \times (-2) = -2$  | (61) $3 \times 7 = 21$     | (91) $1 \times (-5) = -5$   |
| (2) $8 \times 7 = 56$      | (32) $-5 \times 4 = -20$   | (62) $-8 \times (-8) = 64$ | (92) $-2 \times (-6) = 12$  |
| (3) $6 \times (-2) = -12$  | (33) $2 \times (-3) = -6$  | (63) $-4 \times (-5) = 20$ | (93) $-6 \times 8 = -48$    |
| (4) $6 \times (-8) = -48$  | (34) $-8 \times 9 = -72$   | (64) $9 \times (-2) = -18$ | (94) $2 \times 6 = 12$      |
| (5) $7 \times 4 = 28$      | (35) $-7 \times (-7) = 49$ | (65) $-4 \times (-4) = 16$ | (95) $2 \times (-9) = -18$  |
| (6) $-1 \times (-2) = 2$   | (36) $8 \times (-5) = -40$ | (66) $6 \times (-4) = -24$ | (96) $-8 \times 9 = -72$    |
| (7) $-4 \times (-3) = 12$  | (37) $-8 \times (-5) = 40$ | (67) $-3 \times (-2) = 6$  | (97) $9 \times (-8) = -72$  |
| (8) $3 \times (-2) = -6$   | (38) $8 \times 0 = 0$      | (68) $4 \times 8 = 32$     | (98) $7 \times (-8) = -56$  |
| (9) $-4 \times 6 = -24$    | (39) $4 \times 0 = 0$      | (69) $1 \times (-2) = -2$  | (99) $-6 \times (-6) = 36$  |
| (10) $5 \times (-3) = -15$ | (40) $-1 \times (-9) = 9$  | (70) $-4 \times (-5) = 20$ | (100) $8 \times (-4) = -32$ |
| (11) $4 \times (-3) = -12$ | (41) $8 \times 1 = 8$      | (71) $-3 \times 4 = -12$   | (101) $-1 \times 0 = 0$     |
| (12) $3 \times (-3) = -9$  | (42) $6 \times 9 = 54$     | (72) $8 \times 2 = 16$     | (102) $-2 \times (-1) = 2$  |
| (13) $-4 \times (-4) = 16$ | (43) $1 \times 7 = 7$      | (73) $-7 \times (-7) = 49$ | (103) $3 \times (-8) = -24$ |
| (14) $0 \times 4 = 0$      | (44) $-9 \times (-2) = 18$ | (74) $-6 \times (-4) = 24$ | (104) $-8 \times 2 = -16$   |
| (15) $8 \times 3 = 24$     | (45) $-1 \times 7 = -7$    | (75) $-3 \times 5 = -15$   | (105) $-1 \times 2 = -2$    |
| (16) $-2 \times 3 = -6$    | (46) $5 \times (-3) = -15$ | (76) $7 \times 2 = 14$     | (106) $-1 \times 9 = -9$    |
| (17) $6 \times 2 = 12$     | (47) $-9 \times (-2) = 18$ | (77) $6 \times (-2) = -12$ | (107) $5 \times 5 = 25$     |
| (18) $-2 \times 9 = -18$   | (48) $7 \times 1 = 7$      | (78) $-2 \times (-7) = 14$ | (108) $5 \times 4 = 20$     |
| (19) $0 \times (-1) = 0$   | (49) $-8 \times 8 = -64$   | (79) $-1 \times (-9) = 9$  | (109) $4 \times (-7) = -28$ |
| (20) $0 \times (-6) = 0$   | (50) $-1 \times 7 = -7$    | (80) $-2 \times (-7) = 14$ | (110) $-2 \times (-4) = 8$  |
| (21) $7 \times 0 = 0$      | (51) $-7 \times 8 = -56$   | (81) $-2 \times 9 = -18$   | (111) $4 \times (-7) = -28$ |
| (22) $-3 \times 8 = -24$   | (52) $-1 \times (-7) = 7$  | (82) $3 \times (-8) = -24$ | (112) $9 \times (-3) = -27$ |
| (23) $-5 \times 1 = -5$    | (53) $2 \times (-4) = -8$  | (83) $2 \times 3 = 6$      | (113) $-4 \times (-7) = 28$ |
| (24) $-4 \times (-7) = 28$ | (54) $6 \times 6 = 36$     | (84) $5 \times 2 = 10$     | (114) $-8 \times (-6) = 48$ |
| (25) $-9 \times 2 = -18$   | (55) $-6 \times 6 = -36$   | (85) $1 \times 1 = 1$      | (115) $-4 \times 6 = -24$   |
| (26) $5 \times (-9) = -45$ | (56) $-9 \times 8 = -72$   | (86) $5 \times 4 = 20$     | (116) $7 \times (-1) = -7$  |
| (27) $8 \times (-4) = -32$ | (57) $7 \times 9 = 63$     | (87) $3 \times (-7) = -21$ | (117) $-8 \times (-3) = 24$ |
| (28) $6 \times 3 = 18$     | (58) $3 \times (-6) = -18$ | (88) $8 \times 7 = 56$     | (118) $-1 \times 8 = -8$    |
| (29) $0 \times 2 = 0$      | (59) $9 \times (-5) = -45$ | (89) $-6 \times (-9) = 54$ | (119) $-3 \times (-4) = 12$ |
| (30) $-3 \times (-8) = 24$ | (60) $5 \times (-8) = -40$ | (90) $-5 \times 5 = -25$   | (120) $4 \times (-4) = -16$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $0 \times (-7) = 0$    | (31) $8 \times 8 = 64$     | (61) $-5 \times 6 = -30$   | (91) $4 \times 1 = 4$       |
| (2) $2 \times (-2) = -4$   | (32) $-4 \times 0 = 0$     | (62) $9 \times 5 = 45$     | (92) $-2 \times (-5) = 10$  |
| (3) $-7 \times (-2) = 14$  | (33) $9 \times 1 = 9$      | (63) $3 \times 5 = 15$     | (93) $5 \times (-8) = -40$  |
| (4) $4 \times 2 = 8$       | (34) $-4 \times (-3) = 12$ | (64) $7 \times 8 = 56$     | (94) $8 \times 3 = 24$      |
| (5) $0 \times (-1) = 0$    | (35) $5 \times 3 = 15$     | (65) $8 \times (-7) = -56$ | (95) $1 \times (-7) = -7$   |
| (6) $-8 \times 1 = -8$     | (36) $-4 \times (-9) = 36$ | (66) $-1 \times 2 = -2$    | (96) $0 \times (-7) = 0$    |
| (7) $4 \times (-1) = -4$   | (37) $4 \times (-9) = -36$ | (67) $-8 \times (-4) = 32$ | (97) $6 \times (-4) = -24$  |
| (8) $-3 \times 9 = -27$    | (38) $9 \times 3 = 27$     | (68) $9 \times 7 = 63$     | (98) $2 \times 3 = 6$       |
| (9) $2 \times (-4) = -8$   | (39) $-9 \times 3 = -27$   | (69) $-5 \times (-1) = 5$  | (99) $0 \times (-6) = 0$    |
| (10) $-3 \times (-6) = 18$ | (40) $-7 \times (-7) = 49$ | (70) $4 \times (-2) = -8$  | (100) $8 \times 0 = 0$      |
| (11) $8 \times 5 = 40$     | (41) $-3 \times 0 = 0$     | (71) $-2 \times (-7) = 14$ | (101) $9 \times (-4) = -36$ |
| (12) $-5 \times (-9) = 45$ | (42) $-2 \times (-4) = 8$  | (72) $-4 \times 7 = -28$   | (102) $0 \times 4 = 0$      |
| (13) $-6 \times (-2) = 12$ | (43) $-6 \times 5 = -30$   | (73) $0 \times 5 = 0$      | (103) $-4 \times 5 = -20$   |
| (14) $-8 \times (-8) = 64$ | (44) $5 \times (-7) = -35$ | (74) $5 \times (-2) = -10$ | (104) $3 \times 5 = 15$     |
| (15) $0 \times 6 = 0$      | (45) $-5 \times 1 = -5$    | (75) $-4 \times 0 = 0$     | (105) $9 \times 3 = 27$     |
| (16) $8 \times 8 = 64$     | (46) $6 \times 2 = 12$     | (76) $-4 \times (-1) = 4$  | (106) $6 \times 0 = 0$      |
| (17) $-5 \times 2 = -10$   | (47) $1 \times 8 = 8$      | (77) $-1 \times (-6) = 6$  | (107) $2 \times 2 = 4$      |
| (18) $3 \times (-7) = -21$ | (48) $-1 \times (-1) = 1$  | (78) $-5 \times 4 = -20$   | (108) $-1 \times 7 = -7$    |
| (19) $5 \times (-6) = -30$ | (49) $-9 \times (-7) = 63$ | (79) $-5 \times 4 = -20$   | (109) $-4 \times 5 = -20$   |
| (20) $5 \times 9 = 45$     | (50) $-4 \times 6 = -24$   | (80) $-4 \times (-4) = 16$ | (110) $7 \times (-1) = -7$  |
| (21) $0 \times 0 = 0$      | (51) $7 \times 5 = 35$     | (81) $-6 \times (-4) = 24$ | (111) $3 \times (-3) = -9$  |
| (22) $-3 \times 4 = -12$   | (52) $-2 \times 7 = -14$   | (82) $8 \times (-2) = -16$ | (112) $8 \times 7 = 56$     |
| (23) $5 \times 5 = 25$     | (53) $7 \times 9 = 63$     | (83) $-2 \times (-2) = 4$  | (113) $-2 \times 9 = -18$   |
| (24) $-5 \times 4 = -20$   | (54) $0 \times (-7) = 0$   | (84) $-9 \times 8 = -72$   | (114) $9 \times (-1) = -9$  |
| (25) $-1 \times (-1) = 1$  | (55) $-7 \times (-8) = 56$ | (85) $-5 \times (-5) = 25$ | (115) $3 \times (-4) = -12$ |
| (26) $-6 \times 0 = 0$     | (56) $9 \times 2 = 18$     | (86) $-7 \times (-7) = 49$ | (116) $2 \times 3 = 6$      |
| (27) $-9 \times 9 = -81$   | (57) $3 \times (-8) = -24$ | (87) $4 \times (-6) = -24$ | (117) $1 \times (-7) = -7$  |
| (28) $-2 \times 6 = -12$   | (58) $2 \times (-2) = -4$  | (88) $5 \times 8 = 40$     | (118) $-6 \times (-9) = 54$ |
| (29) $-4 \times (-3) = 12$ | (59) $-2 \times (-5) = 10$ | (89) $5 \times (-5) = -25$ | (119) $5 \times 2 = 10$     |
| (30) $5 \times 3 = 15$     | (60) $-6 \times (-8) = 48$ | (90) $8 \times 4 = 32$     | (120) $-9 \times 6 = -54$   |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-4 \times (-7) = 28$  | (31) $-9 \times (-5) = 45$ | (61) $-8 \times 8 = -64$   | (91) $-6 \times (-6) = 36$  |
| (2) $6 \times 9 = 54$      | (32) $-3 \times 8 = -24$   | (62) $4 \times (-7) = -28$ | (92) $-7 \times 3 = -21$    |
| (3) $-7 \times 3 = -21$    | (33) $4 \times (-3) = -12$ | (63) $8 \times 0 = 0$      | (93) $0 \times 7 = 0$       |
| (4) $7 \times 4 = 28$      | (34) $-6 \times (-8) = 48$ | (64) $3 \times (-3) = -9$  | (94) $-6 \times 1 = -6$     |
| (5) $2 \times 8 = 16$      | (35) $6 \times 5 = 30$     | (65) $4 \times (-3) = -12$ | (95) $4 \times 6 = 24$      |
| (6) $4 \times (-7) = -28$  | (36) $8 \times 8 = 64$     | (66) $9 \times (-9) = -81$ | (96) $0 \times 4 = 0$       |
| (7) $-5 \times 3 = -15$    | (37) $2 \times 1 = 2$      | (67) $2 \times 9 = 18$     | (97) $-9 \times (-5) = 45$  |
| (8) $-8 \times 0 = 0$      | (38) $-6 \times (-5) = 30$ | (68) $9 \times 3 = 27$     | (98) $4 \times (-8) = -32$  |
| (9) $-3 \times 8 = -24$    | (39) $6 \times (-8) = -48$ | (69) $-2 \times 4 = -8$    | (99) $9 \times 2 = 18$      |
| (10) $-4 \times (-2) = 8$  | (40) $-6 \times (-3) = 18$ | (70) $-7 \times (-1) = 7$  | (100) $9 \times (-7) = -63$ |
| (11) $1 \times 7 = 7$      | (41) $-7 \times (-9) = 63$ | (71) $-1 \times 6 = -6$    | (101) $7 \times 8 = 56$     |
| (12) $2 \times 8 = 16$     | (42) $0 \times (-4) = 0$   | (72) $1 \times 1 = 1$      | (102) $-3 \times 6 = -18$   |
| (13) $2 \times (-8) = -16$ | (43) $1 \times (-5) = -5$  | (73) $-8 \times (-9) = 72$ | (103) $-4 \times (-7) = 28$ |
| (14) $-8 \times 4 = -32$   | (44) $1 \times 4 = 4$      | (74) $5 \times 7 = 35$     | (104) $-3 \times 5 = -15$   |
| (15) $0 \times 7 = 0$      | (45) $-6 \times 4 = -24$   | (75) $-3 \times 8 = -24$   | (105) $2 \times 7 = 14$     |
| (16) $2 \times 9 = 18$     | (46) $4 \times 6 = 24$     | (76) $7 \times (-4) = -28$ | (106) $-6 \times (-6) = 36$ |
| (17) $-3 \times (-2) = 6$  | (47) $4 \times (-5) = -20$ | (77) $-8 \times (-4) = 32$ | (107) $3 \times (-9) = -27$ |
| (18) $3 \times 2 = 6$      | (48) $7 \times 8 = 56$     | (78) $-1 \times 4 = -4$    | (108) $-6 \times (-9) = 54$ |
| (19) $-4 \times 7 = -28$   | (49) $-5 \times (-5) = 25$ | (79) $6 \times (-4) = -24$ | (109) $9 \times 3 = 27$     |
| (20) $-3 \times (-1) = 3$  | (50) $2 \times 6 = 12$     | (80) $6 \times (-6) = -36$ | (110) $7 \times 7 = 49$     |
| (21) $1 \times 3 = 3$      | (51) $-5 \times (-8) = 40$ | (81) $0 \times (-3) = 0$   | (111) $7 \times 6 = 42$     |
| (22) $-1 \times (-5) = 5$  | (52) $9 \times 7 = 63$     | (82) $7 \times (-7) = -49$ | (112) $2 \times 7 = 14$     |
| (23) $-2 \times 6 = -12$   | (53) $-4 \times (-6) = 24$ | (83) $-5 \times 2 = -10$   | (113) $-2 \times (-3) = 6$  |
| (24) $-4 \times 2 = -8$    | (54) $-9 \times 1 = -9$    | (84) $7 \times (-6) = -42$ | (114) $1 \times 4 = 4$      |
| (25) $1 \times 0 = 0$      | (55) $0 \times 8 = 0$      | (85) $9 \times 6 = 54$     | (115) $2 \times 8 = 16$     |
| (26) $5 \times 5 = 25$     | (56) $-4 \times (-3) = 12$ | (86) $7 \times 6 = 42$     | (116) $5 \times 8 = 40$     |
| (27) $2 \times (-4) = -8$  | (57) $2 \times 6 = 12$     | (87) $6 \times 3 = 18$     | (117) $6 \times 1 = 6$      |
| (28) $-6 \times 8 = -48$   | (58) $0 \times 4 = 0$      | (88) $5 \times 8 = 40$     | (118) $8 \times 8 = 64$     |
| (29) $-7 \times (-9) = 63$ | (59) $-5 \times 6 = -30$   | (89) $-6 \times (-2) = 12$ | (119) $8 \times 6 = 48$     |
| (30) $-5 \times 2 = -10$   | (60) $-4 \times (-1) = 4$  | (90) $-6 \times (-1) = 6$  | (120) $3 \times 5 = 15$     |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $4 \times (-7) = -28$  | (31) $0 \times 7 = 0$      | (61) $-6 \times 7 = -42$   | (91) $3 \times (-3) = -9$   |
| (2) $-8 \times (-7) = 56$  | (32) $-1 \times (-6) = 6$  | (62) $-7 \times 5 = -35$   | (92) $9 \times 1 = 9$       |
| (3) $-4 \times (-1) = 4$   | (33) $6 \times 5 = 30$     | (63) $7 \times (-9) = -63$ | (93) $-3 \times (-8) = 24$  |
| (4) $-8 \times 5 = -40$    | (34) $6 \times 5 = 30$     | (64) $7 \times (-3) = -21$ | (94) $9 \times 6 = 54$      |
| (5) $-2 \times (-3) = 6$   | (35) $6 \times (-8) = -48$ | (65) $4 \times (-8) = -32$ | (95) $-9 \times 1 = -9$     |
| (6) $-9 \times (-4) = 36$  | (36) $5 \times 7 = 35$     | (66) $2 \times (-5) = -10$ | (96) $6 \times 7 = 42$      |
| (7) $8 \times (-7) = -56$  | (37) $2 \times (-5) = -10$ | (67) $2 \times (-7) = -14$ | (97) $-6 \times 9 = -54$    |
| (8) $-1 \times 2 = -2$     | (38) $1 \times (-9) = -9$  | (68) $7 \times 6 = 42$     | (98) $5 \times (-2) = -10$  |
| (9) $9 \times (-9) = -81$  | (39) $6 \times 9 = 54$     | (69) $8 \times 1 = 8$      | (99) $1 \times 6 = 6$       |
| (10) $7 \times (-7) = -49$ | (40) $4 \times (-9) = -36$ | (70) $5 \times 0 = 0$      | (100) $4 \times 3 = 12$     |
| (11) $6 \times 7 = 42$     | (41) $9 \times 3 = 27$     | (71) $2 \times 2 = 4$      | (101) $3 \times 9 = 27$     |
| (12) $2 \times (-9) = -18$ | (42) $5 \times (-5) = -25$ | (72) $2 \times (-8) = -16$ | (102) $-5 \times (-6) = 30$ |
| (13) $-1 \times (-8) = 8$  | (43) $-9 \times 7 = -63$   | (73) $-1 \times 5 = -5$    | (103) $-3 \times (-3) = 9$  |
| (14) $-8 \times 8 = -64$   | (44) $-9 \times 6 = -54$   | (74) $5 \times (-1) = -5$  | (104) $9 \times 0 = 0$      |
| (15) $5 \times (-8) = -40$ | (45) $-2 \times (-1) = 2$  | (75) $8 \times 6 = 48$     | (105) $9 \times (-3) = -27$ |
| (16) $-2 \times 6 = -12$   | (46) $8 \times (-1) = -8$  | (76) $5 \times 0 = 0$      | (106) $-2 \times (-8) = 16$ |
| (17) $1 \times (-8) = -8$  | (47) $-9 \times (-4) = 36$ | (77) $-9 \times 0 = 0$     | (107) $-6 \times 0 = 0$     |
| (18) $3 \times 7 = 21$     | (48) $8 \times (-3) = -24$ | (78) $4 \times 2 = 8$      | (108) $6 \times (-4) = -24$ |
| (19) $-3 \times (-2) = 6$  | (49) $6 \times (-9) = -54$ | (79) $4 \times (-2) = -8$  | (109) $-9 \times (-1) = 9$  |
| (20) $8 \times (-1) = -8$  | (50) $6 \times (-6) = -36$ | (80) $5 \times 4 = 20$     | (110) $3 \times (-7) = -21$ |
| (21) $-7 \times 9 = -63$   | (51) $-9 \times (-7) = 63$ | (81) $8 \times 5 = 40$     | (111) $-8 \times (-4) = 32$ |
| (22) $-1 \times (-3) = 3$  | (52) $6 \times 2 = 12$     | (82) $2 \times 1 = 2$      | (112) $-3 \times (-7) = 21$ |
| (23) $9 \times (-1) = -9$  | (53) $-6 \times 2 = -12$   | (83) $-7 \times 0 = 0$     | (113) $-6 \times 2 = -12$   |
| (24) $8 \times 5 = 40$     | (54) $-7 \times 0 = 0$     | (84) $0 \times 3 = 0$      | (114) $3 \times (-6) = -18$ |
| (25) $-6 \times (-9) = 54$ | (55) $8 \times (-7) = -56$ | (85) $6 \times (-1) = -6$  | (115) $6 \times 3 = 18$     |
| (26) $6 \times (-5) = -30$ | (56) $8 \times 0 = 0$      | (86) $-8 \times 0 = 0$     | (116) $0 \times 8 = 0$      |
| (27) $0 \times (-4) = 0$   | (57) $3 \times (-4) = -12$ | (87) $-7 \times (-8) = 56$ | (117) $8 \times (-9) = -72$ |
| (28) $-1 \times (-8) = 8$  | (58) $-4 \times 7 = -28$   | (88) $-8 \times 8 = -64$   | (118) $5 \times (-1) = -5$  |
| (29) $1 \times 4 = 4$      | (59) $-3 \times 8 = -24$   | (89) $-2 \times (-7) = 14$ | (119) $-6 \times (-2) = 12$ |
| (30) $-5 \times (-8) = 40$ | (60) $-2 \times (-9) = 18$ | (90) $5 \times 9 = 45$     | (120) $7 \times 8 = 56$     |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $7 \times (-2) = -14$  | (31) $-9 \times 7 = -63$   | (61) $-3 \times 8 = -24$   | (91) $-3 \times 0 = 0$      |
| (2) $-7 \times 0 = 0$      | (32) $-7 \times (-3) = 21$ | (62) $-5 \times (-5) = 25$ | (92) $3 \times 0 = 0$       |
| (3) $3 \times 9 = 27$      | (33) $8 \times (-8) = -64$ | (63) $-5 \times (-7) = 35$ | (93) $3 \times 1 = 3$       |
| (4) $-3 \times 5 = -15$    | (34) $2 \times (-4) = -8$  | (64) $-7 \times (-9) = 63$ | (94) $-9 \times (-6) = 54$  |
| (5) $-1 \times (-3) = 3$   | (35) $-8 \times 5 = -40$   | (65) $2 \times 5 = 10$     | (95) $3 \times 5 = 15$      |
| (6) $6 \times 0 = 0$       | (36) $1 \times (-3) = -3$  | (66) $-5 \times 9 = -45$   | (96) $-1 \times 3 = -3$     |
| (7) $1 \times 1 = 1$       | (37) $0 \times (-4) = 0$   | (67) $6 \times 1 = 6$      | (97) $-7 \times (-7) = 49$  |
| (8) $-1 \times 4 = -4$     | (38) $1 \times (-3) = -3$  | (68) $1 \times 9 = 9$      | (98) $-1 \times 8 = -8$     |
| (9) $-1 \times (-8) = 8$   | (39) $-7 \times 2 = -14$   | (69) $-9 \times (-8) = 72$ | (99) $-3 \times (-9) = 27$  |
| (10) $0 \times (-4) = 0$   | (40) $3 \times (-7) = -21$ | (70) $-3 \times (-8) = 24$ | (100) $-8 \times (-6) = 48$ |
| (11) $-2 \times 7 = -14$   | (41) $5 \times (-1) = -5$  | (71) $9 \times 9 = 81$     | (101) $-6 \times 6 = -36$   |
| (12) $3 \times (-9) = -27$ | (42) $7 \times 1 = 7$      | (72) $8 \times 7 = 56$     | (102) $-4 \times (-6) = 24$ |
| (13) $-2 \times (-6) = 12$ | (43) $8 \times 2 = 16$     | (73) $3 \times 4 = 12$     | (103) $9 \times 8 = 72$     |
| (14) $-4 \times 1 = -4$    | (44) $9 \times (-7) = -63$ | (74) $-8 \times (-1) = 8$  | (104) $8 \times 4 = 32$     |
| (15) $3 \times (-3) = -9$  | (45) $5 \times 4 = 20$     | (75) $7 \times (-9) = -63$ | (105) $7 \times 6 = 42$     |
| (16) $4 \times (-2) = -8$  | (46) $-1 \times (-8) = 8$  | (76) $-8 \times 2 = -16$   | (106) $8 \times 2 = 16$     |
| (17) $3 \times 9 = 27$     | (47) $-9 \times (-3) = 27$ | (77) $8 \times (-7) = -56$ | (107) $3 \times (-7) = -21$ |
| (18) $-9 \times 6 = -54$   | (48) $-1 \times (-1) = 1$  | (78) $-8 \times (-4) = 32$ | (108) $2 \times 9 = 18$     |
| (19) $6 \times 5 = 30$     | (49) $4 \times 9 = 36$     | (79) $-7 \times (-9) = 63$ | (109) $6 \times (-9) = -54$ |
| (20) $-9 \times 5 = -45$   | (50) $1 \times (-1) = -1$  | (80) $-1 \times (-7) = 7$  | (110) $-7 \times 7 = -49$   |
| (21) $6 \times 5 = 30$     | (51) $1 \times 5 = 5$      | (81) $-3 \times 9 = -27$   | (111) $1 \times (-3) = -3$  |
| (22) $9 \times (-3) = -27$ | (52) $6 \times 0 = 0$      | (82) $4 \times 3 = 12$     | (112) $-7 \times (-3) = 21$ |
| (23) $6 \times (-7) = -42$ | (53) $-1 \times 8 = -8$    | (83) $-2 \times (-9) = 18$ | (113) $8 \times 4 = 32$     |
| (24) $-8 \times (-9) = 72$ | (54) $-2 \times 3 = -6$    | (84) $-9 \times (-7) = 63$ | (114) $-5 \times (-7) = 35$ |
| (25) $-1 \times 2 = -2$    | (55) $9 \times 9 = 81$     | (85) $5 \times 4 = 20$     | (115) $-1 \times 4 = -4$    |
| (26) $1 \times 8 = 8$      | (56) $0 \times 5 = 0$      | (86) $-2 \times 8 = -16$   | (116) $9 \times 7 = 63$     |
| (27) $-2 \times (-5) = 10$ | (57) $-8 \times (-1) = 8$  | (87) $-2 \times (-4) = 8$  | (117) $-6 \times (-3) = 18$ |
| (28) $0 \times (-9) = 0$   | (58) $-2 \times (-3) = 6$  | (88) $6 \times (-4) = -24$ | (118) $-5 \times 3 = -15$   |
| (29) $8 \times (-2) = -16$ | (59) $-5 \times 0 = 0$     | (89) $-9 \times 2 = -18$   | (119) $9 \times (-1) = -9$  |
| (30) $-7 \times (-6) = 42$ | (60) $4 \times (-3) = -12$ | (90) $5 \times (-1) = -5$  | (120) $-7 \times 5 = -35$   |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-2 \times 6 = -12$    | (31) $1 \times (-8) = -8$  | (61) $-5 \times 0 = 0$     | (91) $5 \times (-4) = -20$  |
| (2) $9 \times 9 = 81$      | (32) $-3 \times (-2) = 6$  | (62) $8 \times 5 = 40$     | (92) $-4 \times (-7) = 28$  |
| (3) $0 \times 2 = 0$       | (33) $-7 \times (-5) = 35$ | (63) $3 \times 0 = 0$      | (93) $4 \times 3 = 12$      |
| (4) $-7 \times 1 = -7$     | (34) $-7 \times (-1) = 7$  | (64) $-1 \times 0 = 0$     | (94) $-9 \times 8 = -72$    |
| (5) $-1 \times 9 = -9$     | (35) $8 \times (-3) = -24$ | (65) $6 \times (-7) = -42$ | (95) $-7 \times 4 = -28$    |
| (6) $-4 \times 3 = -12$    | (36) $-4 \times 9 = -36$   | (66) $-8 \times 0 = 0$     | (96) $-2 \times (-7) = 14$  |
| (7) $6 \times 9 = 54$      | (37) $4 \times 1 = 4$      | (67) $3 \times (-3) = -9$  | (97) $8 \times (-9) = -72$  |
| (8) $2 \times 9 = 18$      | (38) $-7 \times (-1) = 7$  | (68) $-5 \times (-5) = 25$ | (98) $-7 \times 7 = -49$    |
| (9) $9 \times (-8) = -72$  | (39) $0 \times (-5) = 0$   | (69) $-3 \times 9 = -27$   | (99) $0 \times 0 = 0$       |
| (10) $-8 \times (-4) = 32$ | (40) $-1 \times (-7) = 7$  | (70) $-3 \times (-5) = 15$ | (100) $5 \times (-5) = -25$ |
| (11) $-6 \times 4 = -24$   | (41) $6 \times 6 = 36$     | (71) $2 \times 2 = 4$      | (101) $-7 \times 2 = -14$   |
| (12) $-8 \times (-8) = 64$ | (42) $0 \times 4 = 0$      | (72) $-7 \times 2 = -14$   | (102) $-8 \times 8 = -64$   |
| (13) $-9 \times 7 = -63$   | (43) $1 \times (-1) = -1$  | (73) $3 \times (-4) = -12$ | (103) $-9 \times 2 = -18$   |
| (14) $9 \times 6 = 54$     | (44) $8 \times (-7) = -56$ | (74) $-7 \times 5 = -35$   | (104) $-6 \times 9 = -54$   |
| (15) $-3 \times (-3) = 9$  | (45) $-1 \times 0 = 0$     | (75) $-7 \times 3 = -21$   | (105) $-2 \times 9 = -18$   |
| (16) $7 \times (-2) = -14$ | (46) $7 \times (-4) = -28$ | (76) $4 \times 8 = 32$     | (106) $-6 \times (-4) = 24$ |
| (17) $2 \times (-3) = -6$  | (47) $1 \times 3 = 3$      | (77) $7 \times (-9) = -63$ | (107) $4 \times 2 = 8$      |
| (18) $3 \times 5 = 15$     | (48) $9 \times (-3) = -27$ | (78) $1 \times (-3) = -3$  | (108) $3 \times 0 = 0$      |
| (19) $-4 \times 3 = -12$   | (49) $-2 \times (-2) = 4$  | (79) $-2 \times 5 = -10$   | (109) $-4 \times 7 = -28$   |
| (20) $-6 \times 9 = -54$   | (50) $6 \times 3 = 18$     | (80) $-7 \times (-5) = 35$ | (110) $-4 \times (-8) = 32$ |
| (21) $7 \times 4 = 28$     | (51) $-9 \times 7 = -63$   | (81) $-6 \times (-5) = 30$ | (111) $-6 \times 3 = -18$   |
| (22) $3 \times 0 = 0$      | (52) $-5 \times 4 = -20$   | (82) $4 \times (-3) = -12$ | (112) $3 \times (-6) = -18$ |
| (23) $5 \times (-6) = -30$ | (53) $-5 \times 9 = -45$   | (83) $6 \times 4 = 24$     | (113) $5 \times (-8) = -40$ |
| (24) $-8 \times (-5) = 40$ | (54) $6 \times 1 = 6$      | (84) $9 \times (-3) = -27$ | (114) $2 \times 9 = 18$     |
| (25) $-4 \times (-4) = 16$ | (55) $-7 \times (-4) = 28$ | (85) $2 \times (-8) = -16$ | (115) $-9 \times (-3) = 27$ |
| (26) $-4 \times (-9) = 36$ | (56) $-1 \times (-4) = 4$  | (86) $9 \times (-7) = -63$ | (116) $7 \times 4 = 28$     |
| (27) $9 \times 5 = 45$     | (57) $-2 \times (-5) = 10$ | (87) $0 \times (-4) = 0$   | (117) $-7 \times (-9) = 63$ |
| (28) $-7 \times 8 = -56$   | (58) $2 \times (-3) = -6$  | (88) $-6 \times 3 = -18$   | (118) $-9 \times (-2) = 18$ |
| (29) $9 \times (-6) = -54$ | (59) $6 \times (-4) = -24$ | (89) $-6 \times (-8) = 48$ | (119) $3 \times 0 = 0$      |
| (30) $0 \times 8 = 0$      | (60) $-2 \times 2 = -4$    | (90) $-4 \times 6 = -24$   | (120) $-8 \times 2 = -16$   |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $4 \times (-4) = -16$  | (31) $2 \times (-8) = -16$ | (61) $-9 \times 3 = -27$   | (91) $0 \times 0 = 0$       |
| (2) $-5 \times (-5) = 25$  | (32) $-1 \times 3 = -3$    | (62) $-4 \times 3 = -12$   | (92) $3 \times 0 = 0$       |
| (3) $4 \times 4 = 16$      | (33) $1 \times (-4) = -4$  | (63) $-2 \times (-3) = 6$  | (93) $-4 \times (-7) = 28$  |
| (4) $9 \times (-4) = -36$  | (34) $5 \times 1 = 5$      | (64) $0 \times (-4) = 0$   | (94) $4 \times 2 = 8$       |
| (5) $-4 \times (-5) = 20$  | (35) $-3 \times (-4) = 12$ | (65) $-2 \times 4 = -8$    | (95) $1 \times 5 = 5$       |
| (6) $5 \times 0 = 0$       | (36) $1 \times 7 = 7$      | (66) $-4 \times 0 = 0$     | (96) $9 \times 3 = 27$      |
| (7) $-7 \times 8 = -56$    | (37) $-4 \times (-9) = 36$ | (67) $4 \times 2 = 8$      | (97) $6 \times 8 = 48$      |
| (8) $9 \times (-2) = -18$  | (38) $4 \times (-2) = -8$  | (68) $9 \times 0 = 0$      | (98) $-1 \times 2 = -2$     |
| (9) $-2 \times 9 = -18$    | (39) $-2 \times (-6) = 12$ | (69) $-1 \times (-4) = 4$  | (99) $5 \times (-7) = -35$  |
| (10) $-7 \times 1 = -7$    | (40) $2 \times 3 = 6$      | (70) $9 \times (-2) = -18$ | (100) $3 \times 9 = 27$     |
| (11) $-4 \times 4 = -16$   | (41) $-9 \times (-1) = 9$  | (71) $2 \times 2 = 4$      | (101) $6 \times (-1) = -6$  |
| (12) $-8 \times (-1) = 8$  | (42) $-8 \times (-2) = 16$ | (72) $-5 \times (-1) = 5$  | (102) $7 \times 8 = 56$     |
| (13) $5 \times 5 = 25$     | (43) $-8 \times (-3) = 24$ | (73) $4 \times 4 = 16$     | (103) $-4 \times (-5) = 20$ |
| (14) $-4 \times 7 = -28$   | (44) $-3 \times (-8) = 24$ | (74) $-1 \times (-6) = 6$  | (104) $2 \times (-9) = -18$ |
| (15) $-6 \times (-6) = 36$ | (45) $-6 \times (-1) = 6$  | (75) $9 \times 1 = 9$      | (105) $-2 \times 1 = -2$    |
| (16) $9 \times (-6) = -54$ | (46) $2 \times 5 = 10$     | (76) $6 \times (-2) = -12$ | (106) $-3 \times (-2) = 6$  |
| (17) $2 \times 2 = 4$      | (47) $-6 \times (-6) = 36$ | (77) $-6 \times (-6) = 36$ | (107) $6 \times (-2) = -12$ |
| (18) $-1 \times (-5) = 5$  | (48) $-5 \times (-8) = 40$ | (78) $-9 \times (-2) = 18$ | (108) $6 \times (-4) = -24$ |
| (19) $-7 \times 8 = -56$   | (49) $-1 \times (-1) = 1$  | (79) $-7 \times 3 = -21$   | (109) $-4 \times (-9) = 36$ |
| (20) $7 \times 8 = 56$     | (50) $2 \times (-1) = -2$  | (80) $7 \times 8 = 56$     | (110) $7 \times (-9) = -63$ |
| (21) $-8 \times (-8) = 64$ | (51) $-4 \times 0 = 0$     | (81) $-8 \times 8 = -64$   | (111) $2 \times 5 = 10$     |
| (22) $9 \times 7 = 63$     | (52) $-1 \times 9 = -9$    | (82) $2 \times (-8) = -16$ | (112) $-6 \times 0 = 0$     |
| (23) $5 \times 3 = 15$     | (53) $-4 \times (-7) = 28$ | (83) $-8 \times 1 = -8$    | (113) $7 \times 9 = 63$     |
| (24) $6 \times 8 = 48$     | (54) $-8 \times 0 = 0$     | (84) $6 \times 9 = 54$     | (114) $-2 \times 7 = -14$   |
| (25) $8 \times 9 = 72$     | (55) $6 \times 1 = 6$      | (85) $3 \times 3 = 9$      | (115) $0 \times 7 = 0$      |
| (26) $8 \times (-5) = -40$ | (56) $6 \times 0 = 0$      | (86) $-5 \times 9 = -45$   | (116) $-8 \times (-4) = 32$ |
| (27) $3 \times (-1) = -3$  | (57) $7 \times (-7) = -49$ | (87) $1 \times 5 = 5$      | (117) $6 \times 9 = 54$     |
| (28) $-7 \times (-2) = 14$ | (58) $5 \times 1 = 5$      | (88) $7 \times 1 = 7$      | (118) $0 \times 8 = 0$      |
| (29) $3 \times (-8) = -24$ | (59) $-9 \times (-1) = 9$  | (89) $2 \times (-2) = -4$  | (119) $1 \times (-4) = -4$  |
| (30) $-3 \times (-7) = 21$ | (60) $1 \times (-6) = -6$  | (90) $1 \times (-1) = -1$  | (120) $7 \times (-9) = -63$ |



- |                            |                            |                            |                             |
|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $8 \times (-9) = -72$  | (31) $3 \times 4 = 12$     | (61) $3 \times (-5) = -15$ | (91) $0 \times 6 = 0$       |
| (2) $-4 \times 6 = -24$    | (32) $2 \times 6 = 12$     | (62) $1 \times (-6) = -6$  | (92) $0 \times 8 = 0$       |
| (3) $-6 \times 6 = -36$    | (33) $-3 \times 0 = 0$     | (63) $7 \times (-9) = -63$ | (93) $3 \times 4 = 12$      |
| (4) $7 \times (-7) = -49$  | (34) $-1 \times (-9) = 9$  | (64) $-1 \times 4 = -4$    | (94) $2 \times (-8) = -16$  |
| (5) $7 \times (-3) = -21$  | (35) $7 \times (-2) = -14$ | (65) $-1 \times (-9) = 9$  | (95) $-6 \times (-3) = 18$  |
| (6) $-4 \times 3 = -12$    | (36) $-1 \times 1 = -1$    | (66) $7 \times 5 = 35$     | (96) $1 \times 7 = 7$       |
| (7) $-9 \times (-2) = 18$  | (37) $4 \times 9 = 36$     | (67) $3 \times 7 = 21$     | (97) $0 \times 8 = 0$       |
| (8) $2 \times 4 = 8$       | (38) $8 \times 5 = 40$     | (68) $6 \times 5 = 30$     | (98) $-7 \times (-5) = 35$  |
| (9) $1 \times 8 = 8$       | (39) $8 \times 1 = 8$      | (69) $8 \times (-4) = -32$ | (99) $4 \times 4 = 16$      |
| (10) $0 \times 9 = 0$      | (40) $-3 \times 1 = -3$    | (70) $-8 \times (-9) = 72$ | (100) $-8 \times (-4) = 32$ |
| (11) $-6 \times 2 = -12$   | (41) $8 \times (-2) = -16$ | (71) $1 \times 3 = 3$      | (101) $9 \times (-2) = -18$ |
| (12) $7 \times (-5) = -35$ | (42) $4 \times 9 = 36$     | (72) $-2 \times 9 = -18$   | (102) $3 \times 9 = 27$     |
| (13) $-6 \times 9 = -54$   | (43) $-3 \times 9 = -27$   | (73) $-7 \times 9 = -63$   | (103) $-7 \times 1 = -7$    |
| (14) $2 \times 8 = 16$     | (44) $7 \times (-2) = -14$ | (74) $-1 \times 4 = -4$    | (104) $9 \times (-5) = -45$ |
| (15) $2 \times 7 = 14$     | (45) $-3 \times (-6) = 18$ | (75) $6 \times (-5) = -30$ | (105) $-1 \times 5 = -5$    |
| (16) $-8 \times 9 = -72$   | (46) $-4 \times 9 = -36$   | (76) $6 \times (-3) = -18$ | (106) $8 \times 5 = 40$     |
| (17) $2 \times (-6) = -12$ | (47) $7 \times 2 = 14$     | (77) $-5 \times (-9) = 45$ | (107) $6 \times (-1) = -6$  |
| (18) $-1 \times 2 = -2$    | (48) $-7 \times 8 = -56$   | (78) $9 \times 3 = 27$     | (108) $0 \times 9 = 0$      |
| (19) $0 \times (-7) = 0$   | (49) $-5 \times (-8) = 40$ | (79) $3 \times 8 = 24$     | (109) $1 \times (-6) = -6$  |
| (20) $-2 \times 1 = -2$    | (50) $-1 \times (-4) = 4$  | (80) $4 \times 9 = 36$     | (110) $-1 \times (-7) = 7$  |
| (21) $5 \times (-2) = -10$ | (51) $-4 \times (-1) = 4$  | (81) $-4 \times 4 = -16$   | (111) $7 \times 3 = 21$     |
| (22) $-1 \times 5 = -5$    | (52) $9 \times 5 = 45$     | (82) $5 \times (-4) = -20$ | (112) $-6 \times 7 = -42$   |
| (23) $0 \times 7 = 0$      | (53) $9 \times (-5) = -45$ | (83) $-1 \times (-9) = 9$  | (113) $-8 \times 7 = -56$   |
| (24) $-6 \times (-5) = 30$ | (54) $3 \times (-7) = -21$ | (84) $-4 \times 4 = -16$   | (114) $-6 \times (-8) = 48$ |
| (25) $1 \times (-8) = -8$  | (55) $-7 \times 5 = -35$   | (85) $-8 \times (-6) = 48$ | (115) $-7 \times (-8) = 56$ |
| (26) $7 \times (-6) = -42$ | (56) $-4 \times 6 = -24$   | (86) $-2 \times (-7) = 14$ | (116) $0 \times (-5) = 0$   |
| (27) $-8 \times (-1) = 8$  | (57) $5 \times 3 = 15$     | (87) $-5 \times (-8) = 40$ | (117) $4 \times 4 = 16$     |
| (28) $7 \times 2 = 14$     | (58) $-9 \times (-1) = 9$  | (88) $-9 \times 2 = -18$   | (118) $-1 \times 0 = 0$     |
| (29) $1 \times (-6) = -6$  | (59) $5 \times (-2) = -10$ | (89) $2 \times (-5) = -10$ | (119) $-7 \times (-1) = 7$  |
| (30) $1 \times 6 = 6$      | (60) $-3 \times 6 = -18$   | (90) $-2 \times 9 = -18$   | (120) $-5 \times 7 = -35$   |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $-7 \times (-6) = 42$  | (31) $4 \times (-9) = -36$ | (61) $5 \times 5 = 25$     | (91) $3 \times (-1) = -3$   |
| (2) $6 \times 9 = 54$      | (32) $0 \times 8 = 0$      | (62) $2 \times 8 = 16$     | (92) $5 \times 6 = 30$      |
| (3) $9 \times (-5) = -45$  | (33) $7 \times 5 = 35$     | (63) $-7 \times (-2) = 14$ | (93) $-2 \times 8 = -16$    |
| (4) $4 \times (-7) = -28$  | (34) $-3 \times 1 = -3$    | (64) $3 \times (-3) = -9$  | (94) $5 \times 3 = 15$      |
| (5) $-8 \times 8 = -64$    | (35) $1 \times (-7) = -7$  | (65) $-9 \times (-6) = 54$ | (95) $5 \times (-2) = -10$  |
| (6) $-4 \times 6 = -24$    | (36) $-1 \times 5 = -5$    | (66) $-3 \times (-1) = 3$  | (96) $4 \times (-9) = -36$  |
| (7) $-5 \times 5 = -25$    | (37) $7 \times 2 = 14$     | (67) $-6 \times 7 = -42$   | (97) $5 \times (-7) = -35$  |
| (8) $-5 \times (-7) = 35$  | (38) $2 \times (-9) = -18$ | (68) $-5 \times (-2) = 10$ | (98) $-7 \times 2 = -14$    |
| (9) $-7 \times (-5) = 35$  | (39) $4 \times 2 = 8$      | (69) $6 \times 1 = 6$      | (99) $0 \times 3 = 0$       |
| (10) $9 \times 9 = 81$     | (40) $8 \times 6 = 48$     | (70) $9 \times 7 = 63$     | (100) $-8 \times (-1) = 8$  |
| (11) $-6 \times (-3) = 18$ | (41) $0 \times 9 = 0$      | (71) $-2 \times 1 = -2$    | (101) $7 \times 0 = 0$      |
| (12) $-5 \times 5 = -25$   | (42) $-7 \times (-3) = 21$ | (72) $0 \times 7 = 0$      | (102) $1 \times 0 = 0$      |
| (13) $-1 \times (-7) = 7$  | (43) $2 \times (-5) = -10$ | (73) $3 \times (-6) = -18$ | (103) $9 \times (-3) = -27$ |
| (14) $-3 \times 1 = -3$    | (44) $0 \times 9 = 0$      | (74) $-5 \times 4 = -20$   | (104) $-3 \times (-8) = 24$ |
| (15) $-6 \times 6 = -36$   | (45) $1 \times (-9) = -9$  | (75) $-2 \times (-2) = 4$  | (105) $8 \times (-2) = -16$ |
| (16) $4 \times (-2) = -8$  | (46) $2 \times 7 = 14$     | (76) $2 \times (-4) = -8$  | (106) $-8 \times 5 = -40$   |
| (17) $-3 \times (-6) = 18$ | (47) $4 \times 0 = 0$      | (77) $-8 \times 4 = -32$   | (107) $-3 \times 3 = -9$    |
| (18) $5 \times (-3) = -15$ | (48) $9 \times (-5) = -45$ | (78) $-4 \times (-8) = 32$ | (108) $0 \times 8 = 0$      |
| (19) $-9 \times (-1) = 9$  | (49) $1 \times (-8) = -8$  | (79) $-7 \times 6 = -42$   | (109) $4 \times (-6) = -24$ |
| (20) $-1 \times (-2) = 2$  | (50) $-1 \times 4 = -4$    | (80) $6 \times (-4) = -24$ | (110) $8 \times (-4) = -32$ |
| (21) $0 \times (-2) = 0$   | (51) $5 \times (-1) = -5$  | (81) $9 \times 9 = 81$     | (111) $7 \times 2 = 14$     |
| (22) $2 \times (-7) = -14$ | (52) $0 \times 5 = 0$      | (82) $-6 \times 4 = -24$   | (112) $3 \times (-5) = -15$ |
| (23) $-1 \times 8 = -8$    | (53) $7 \times (-9) = -63$ | (83) $-5 \times 8 = -40$   | (113) $2 \times 1 = 2$      |
| (24) $-7 \times 5 = -35$   | (54) $4 \times (-8) = -32$ | (84) $-4 \times 7 = -28$   | (114) $8 \times 2 = 16$     |
| (25) $8 \times 5 = 40$     | (55) $7 \times (-4) = -28$ | (85) $-8 \times 9 = -72$   | (115) $-7 \times 7 = -49$   |
| (26) $-3 \times (-5) = 15$ | (56) $-5 \times 9 = -45$   | (86) $-1 \times (-6) = 6$  | (116) $-3 \times 1 = -3$    |
| (27) $-1 \times 3 = -3$    | (57) $-1 \times 7 = -7$    | (87) $-3 \times (-3) = 9$  | (117) $-8 \times 4 = -32$   |
| (28) $8 \times (-9) = -72$ | (58) $6 \times 5 = 30$     | (88) $9 \times (-3) = -27$ | (118) $0 \times 1 = 0$      |
| (29) $-6 \times (-1) = 6$  | (59) $-8 \times 2 = -16$   | (89) $7 \times 3 = 21$     | (119) $5 \times (-1) = -5$  |
| (30) $-4 \times 8 = -32$   | (60) $-1 \times 6 = -6$    | (90) $6 \times 5 = 30$     | (120) $-5 \times (-2) = 10$ |

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|----------------------------|----------------------------|----------------------------|-----------------------------|
| (1) $7 \times 7 = 49$      | (31) $3 \times (-3) = -9$  | (61) $3 \times (-4) = -12$ | (91) $3 \times 9 = 27$      |
| (2) $-6 \times (-6) = 36$  | (32) $6 \times (-7) = -42$ | (62) $-4 \times 0 = 0$     | (92) $7 \times 1 = 7$       |
| (3) $6 \times 3 = 18$      | (33) $-3 \times 0 = 0$     | (63) $-3 \times (-7) = 21$ | (93) $8 \times (-6) = -48$  |
| (4) $-7 \times (-1) = 7$   | (34) $-1 \times (-1) = 1$  | (64) $-8 \times 7 = -56$   | (94) $7 \times (-3) = -21$  |
| (5) $-4 \times 8 = -32$    | (35) $-5 \times (-5) = 25$ | (65) $6 \times (-7) = -42$ | (95) $-2 \times 8 = -16$    |
| (6) $1 \times (-3) = -3$   | (36) $-1 \times (-7) = 7$  | (66) $-9 \times (-6) = 54$ | (96) $-5 \times (-2) = 10$  |
| (7) $9 \times 2 = 18$      | (37) $-3 \times (-6) = 18$ | (67) $-9 \times (-3) = 27$ | (97) $0 \times 8 = 0$       |
| (8) $8 \times 8 = 64$      | (38) $-5 \times (-4) = 20$ | (68) $-2 \times 2 = -4$    | (98) $-4 \times 0 = 0$      |
| (9) $-1 \times (-3) = 3$   | (39) $2 \times 7 = 14$     | (69) $-8 \times 8 = -64$   | (99) $1 \times (-2) = -2$   |
| (10) $-2 \times (-7) = 14$ | (40) $-7 \times (-3) = 21$ | (70) $7 \times 9 = 63$     | (100) $6 \times (-7) = -42$ |
| (11) $5 \times 7 = 35$     | (41) $2 \times 2 = 4$      | (71) $-3 \times 9 = -27$   | (101) $4 \times (-4) = -16$ |
| (12) $7 \times 9 = 63$     | (42) $-1 \times 1 = -1$    | (72) $3 \times (-4) = -12$ | (102) $5 \times 6 = 30$     |
| (13) $6 \times (-1) = -6$  | (43) $-1 \times 3 = -3$    | (73) $9 \times 8 = 72$     | (103) $4 \times 4 = 16$     |
| (14) $9 \times 9 = 81$     | (44) $-6 \times (-3) = 18$ | (74) $-7 \times (-4) = 28$ | (104) $2 \times (-3) = -6$  |
| (15) $6 \times 2 = 12$     | (45) $-9 \times 9 = -81$   | (75) $8 \times 6 = 48$     | (105) $1 \times (-9) = -9$  |
| (16) $5 \times (-2) = -10$ | (46) $-6 \times (-8) = 48$ | (76) $-3 \times 8 = -24$   | (106) $0 \times (-1) = 0$   |
| (17) $5 \times 2 = 10$     | (47) $-8 \times (-8) = 64$ | (77) $-8 \times (-7) = 56$ | (107) $-3 \times 3 = -9$    |
| (18) $-5 \times 7 = -35$   | (48) $2 \times (-8) = -16$ | (78) $-2 \times 1 = -2$    | (108) $3 \times (-7) = -21$ |
| (19) $-6 \times 9 = -54$   | (49) $-8 \times (-9) = 72$ | (79) $-9 \times 5 = -45$   | (109) $-1 \times (-4) = 4$  |
| (20) $5 \times (-8) = -40$ | (50) $5 \times (-9) = -45$ | (80) $-1 \times (-3) = 3$  | (110) $-1 \times 7 = -7$    |
| (21) $-4 \times 8 = -32$   | (51) $-6 \times (-5) = 30$ | (81) $-7 \times (-1) = 7$  | (111) $7 \times 5 = 35$     |
| (22) $-9 \times 8 = -72$   | (52) $-2 \times 5 = -10$   | (82) $-8 \times 8 = -64$   | (112) $-4 \times 5 = -20$   |
| (23) $6 \times 2 = 12$     | (53) $8 \times 1 = 8$      | (83) $-4 \times (-9) = 36$ | (113) $-3 \times 3 = -9$    |
| (24) $9 \times (-3) = -27$ | (54) $9 \times (-3) = -27$ | (84) $-3 \times (-5) = 15$ | (114) $-1 \times (-9) = 9$  |
| (25) $-5 \times 1 = -5$    | (55) $2 \times (-6) = -12$ | (85) $0 \times 1 = 0$      | (115) $-3 \times 4 = -12$   |
| (26) $5 \times (-7) = -35$ | (56) $-4 \times 0 = 0$     | (86) $2 \times 1 = 2$      | (116) $7 \times (-2) = -14$ |
| (27) $0 \times (-6) = 0$   | (57) $-3 \times 9 = -27$   | (87) $-3 \times (-9) = 27$ | (117) $-6 \times (-6) = 36$ |
| (28) $7 \times 7 = 49$     | (58) $-2 \times 9 = -18$   | (88) $-3 \times (-8) = 24$ | (118) $5 \times (-2) = -10$ |
| (29) $8 \times 0 = 0$      | (59) $2 \times 0 = 0$      | (89) $-2 \times 8 = -16$   | (119) $-2 \times 4 = -8$    |
| (30) $-8 \times 2 = -16$   | (60) $1 \times 3 = 3$      | (90) $-9 \times (-4) = 36$ | (120) $3 \times 0 = 0$      |

### 1.2.3 除法

整数の商と剰余を求める問題。

(1)  $-8 \div 5$

(19)  $-74 \div 7$

(37)  $90 \div 8$

(55)  $96 \div 6$

(2)  $-54 \div 8$

(20)  $-47 \div 6$

(38)  $-50 \div 7$

(56)  $78 \div 3$

(3)  $-28 \div 7$

(21)  $-15 \div 1$

(39)  $76 \div 1$

(57)  $-10 \div 3$

(4)  $80 \div 4$

(22)  $-55 \div 2$

(40)  $-92 \div 9$

(58)  $58 \div 6$

(5)  $93 \div 3$

(23)  $-30 \div 2$

(41)  $-66 \div 6$

(59)  $65 \div 2$

(6)  $80 \div 10$

(24)  $-95 \div 3$

(42)  $-41 \div 9$

(60)  $60 \div 3$

(7)  $-99 \div 4$

(25)  $7 \div 2$

(43)  $-1 \div 1$

(61)  $5 \div 9$

(8)  $73 \div 6$

(26)  $74 \div 4$

(44)  $33 \div 10$

(62)  $-64 \div 7$

(9)  $65 \div 9$

(27)  $-3 \div 6$

(45)  $60 \div 8$

(63)  $-67 \div 8$

(10)  $48 \div 3$

(28)  $20 \div 2$

(46)  $42 \div 8$

(64)  $45 \div 8$

(11)  $-6 \div 7$

(29)  $-47 \div 10$

(47)  $-90 \div 7$

(65)  $-33 \div 6$

(12)  $32 \div 9$

(30)  $74 \div 9$

(48)  $-69 \div 3$

(66)  $-91 \div 8$

(13)  $72 \div 6$

(31)  $96 \div 5$

(49)  $37 \div 1$

(67)  $-73 \div 2$

(14)  $-40 \div 6$

(32)  $21 \div 6$

(50)  $98 \div 10$

(68)  $92 \div 5$

(15)  $-31 \div 1$

(33)  $24 \div 8$

(51)  $78 \div 2$

(69)  $27 \div 5$

(16)  $-47 \div 10$

(34)  $-68 \div 6$

(52)  $76 \div 2$

(70)  $-67 \div 5$

(17)  $60 \div 3$

(35)  $10 \div 8$

(53)  $-76 \div 7$

(71)  $-93 \div 1$

(18)  $61 \div 4$

(36)  $-89 \div 8$

(54)  $15 \div 6$

(72)  $-97 \div 3$

|                    |                    |                   |                   |
|--------------------|--------------------|-------------------|-------------------|
| (1) $19 \div 5$    | (19) $-52 \div 5$  | (37) $91 \div 3$  | (55) $-94 \div 6$ |
| (2) $63 \div 4$    | (20) $81 \div 8$   | (38) $98 \div 3$  | (56) $-99 \div 6$ |
| (3) $-71 \div 7$   | (21) $-91 \div 3$  | (39) $-6 \div 3$  | (57) $-27 \div 5$ |
| (4) $61 \div 10$   | (22) $-64 \div 7$  | (40) $-86 \div 5$ | (58) $82 \div 5$  |
| (5) $-24 \div 5$   | (23) $-34 \div 7$  | (41) $-70 \div 6$ | (59) $64 \div 1$  |
| (6) $28 \div 3$    | (24) $31 \div 6$   | (42) $71 \div 10$ | (60) $65 \div 5$  |
| (7) $0 \div 9$     | (25) $84 \div 6$   | (43) $64 \div 2$  | (61) $-99 \div 3$ |
| (8) $63 \div 9$    | (26) $88 \div 8$   | (44) $-23 \div 7$ | (62) $100 \div 8$ |
| (9) $55 \div 9$    | (27) $54 \div 2$   | (45) $-6 \div 4$  | (63) $-78 \div 3$ |
| (10) $-20 \div 3$  | (28) $58 \div 5$   | (46) $35 \div 3$  | (64) $-96 \div 7$ |
| (11) $85 \div 10$  | (29) $28 \div 3$   | (47) $88 \div 7$  | (65) $-93 \div 4$ |
| (12) $-4 \div 3$   | (30) $39 \div 5$   | (48) $-58 \div 7$ | (66) $-25 \div 3$ |
| (13) $26 \div 5$   | (31) $-87 \div 6$  | (49) $-43 \div 3$ | (67) $-56 \div 8$ |
| (14) $-24 \div 6$  | (32) $-100 \div 2$ | (50) $-82 \div 9$ | (68) $91 \div 6$  |
| (15) $-89 \div 2$  | (33) $-87 \div 3$  | (51) $18 \div 8$  | (69) $-94 \div 3$ |
| (16) $22 \div 8$   | (34) $99 \div 1$   | (52) $83 \div 7$  | (70) $1 \div 8$   |
| (17) $-41 \div 10$ | (35) $62 \div 4$   | (53) $-65 \div 6$ | (71) $-43 \div 3$ |
| (18) $-51 \div 8$  | (36) $-71 \div 7$  | (54) $72 \div 8$  | (72) $84 \div 8$  |

- |                   |                    |                    |                   |
|-------------------|--------------------|--------------------|-------------------|
| (1) $-96 \div 5$  | (19) $4 \div 3$    | (37) $52 \div 8$   | (55) $26 \div 4$  |
| (2) $-96 \div 9$  | (20) $56 \div 2$   | (38) $-16 \div 8$  | (56) $65 \div 9$  |
| (3) $15 \div 7$   | (21) $14 \div 9$   | (39) $72 \div 7$   | (57) $-33 \div 2$ |
| (4) $-11 \div 4$  | (22) $-16 \div 1$  | (40) $22 \div 6$   | (58) $-62 \div 7$ |
| (5) $-40 \div 1$  | (23) $68 \div 5$   | (41) $-3 \div 10$  | (59) $-12 \div 3$ |
| (6) $-61 \div 4$  | (24) $-58 \div 2$  | (42) $-86 \div 10$ | (60) $54 \div 1$  |
| (7) $41 \div 6$   | (25) $3 \div 3$    | (43) $45 \div 10$  | (61) $-74 \div 9$ |
| (8) $-93 \div 1$  | (26) $64 \div 7$   | (44) $66 \div 6$   | (62) $-8 \div 3$  |
| (9) $90 \div 3$   | (27) $-93 \div 9$  | (45) $49 \div 3$   | (63) $-68 \div 2$ |
| (10) $-21 \div 3$ | (28) $32 \div 5$   | (46) $47 \div 6$   | (64) $-67 \div 2$ |
| (11) $71 \div 5$  | (29) $89 \div 7$   | (47) $53 \div 5$   | (65) $72 \div 1$  |
| (12) $79 \div 8$  | (30) $-11 \div 4$  | (48) $-63 \div 8$  | (66) $71 \div 10$ |
| (13) $68 \div 7$  | (31) $34 \div 3$   | (49) $66 \div 8$   | (67) $92 \div 8$  |
| (14) $27 \div 4$  | (32) $-26 \div 9$  | (50) $-36 \div 6$  | (68) $-25 \div 2$ |
| (15) $24 \div 7$  | (33) $-58 \div 10$ | (51) $89 \div 4$   | (69) $-93 \div 3$ |
| (16) $-5 \div 8$  | (34) $-99 \div 7$  | (52) $26 \div 8$   | (70) $31 \div 10$ |
| (17) $95 \div 7$  | (35) $-19 \div 9$  | (53) $-23 \div 9$  | (71) $90 \div 2$  |
| (18) $-16 \div 5$ | (36) $-26 \div 4$  | (54) $-40 \div 10$ | (72) $49 \div 5$  |

- |                   |                    |                   |                   |
|-------------------|--------------------|-------------------|-------------------|
| (1) $34 \div 3$   | (19) $21 \div 2$   | (37) $-20 \div 3$ | (55) $95 \div 2$  |
| (2) $37 \div 3$   | (20) $39 \div 1$   | (38) $-24 \div 5$ | (56) $73 \div 9$  |
| (3) $-8 \div 1$   | (21) $19 \div 8$   | (39) $7 \div 5$   | (57) $67 \div 7$  |
| (4) $-6 \div 7$   | (22) $-47 \div 10$ | (40) $-45 \div 2$ | (58) $56 \div 5$  |
| (5) $57 \div 5$   | (23) $85 \div 1$   | (41) $4 \div 10$  | (59) $-73 \div 5$ |
| (6) $31 \div 1$   | (24) $-69 \div 8$  | (42) $14 \div 6$  | (60) $50 \div 3$  |
| (7) $28 \div 6$   | (25) $55 \div 2$   | (43) $24 \div 9$  | (61) $-52 \div 8$ |
| (8) $64 \div 10$  | (26) $-40 \div 2$  | (44) $89 \div 2$  | (62) $35 \div 3$  |
| (9) $84 \div 9$   | (27) $-56 \div 9$  | (45) $31 \div 3$  | (63) $-80 \div 9$ |
| (10) $0 \div 6$   | (28) $35 \div 1$   | (46) $-38 \div 3$ | (64) $64 \div 7$  |
| (11) $-18 \div 4$ | (29) $72 \div 9$   | (47) $-37 \div 2$ | (65) $12 \div 1$  |
| (12) $0 \div 6$   | (30) $-60 \div 10$ | (48) $32 \div 5$  | (66) $-31 \div 4$ |
| (13) $-39 \div 5$ | (31) $-25 \div 5$  | (49) $74 \div 1$  | (67) $83 \div 8$  |
| (14) $-67 \div 1$ | (32) $89 \div 2$   | (50) $-82 \div 5$ | (68) $-98 \div 2$ |
| (15) $31 \div 2$  | (33) $88 \div 2$   | (51) $-32 \div 8$ | (69) $36 \div 1$  |
| (16) $-68 \div 8$ | (34) $-55 \div 3$  | (52) $-73 \div 4$ | (70) $-26 \div 4$ |
| (17) $54 \div 2$  | (35) $-79 \div 7$  | (53) $33 \div 4$  | (71) $46 \div 6$  |
| (18) $77 \div 10$ | (36) $4 \div 8$    | (54) $-85 \div 5$ | (72) $-12 \div 8$ |



- |                   |                    |                   |                   |
|-------------------|--------------------|-------------------|-------------------|
| (1) $44 \div 10$  | (19) $2 \div 7$    | (37) $38 \div 10$ | (55) $77 \div 10$ |
| (2) $60 \div 5$   | (20) $71 \div 4$   | (38) $-69 \div 8$ | (56) $45 \div 7$  |
| (3) $23 \div 8$   | (21) $-46 \div 5$  | (39) $1 \div 7$   | (57) $7 \div 7$   |
| (4) $10 \div 10$  | (22) $91 \div 4$   | (40) $-87 \div 3$ | (58) $80 \div 8$  |
| (5) $55 \div 8$   | (23) $-43 \div 1$  | (41) $-64 \div 1$ | (59) $76 \div 8$  |
| (6) $56 \div 5$   | (24) $-34 \div 8$  | (42) $-21 \div 5$ | (60) $59 \div 5$  |
| (7) $-9 \div 6$   | (25) $-99 \div 7$  | (43) $-77 \div 2$ | (61) $74 \div 6$  |
| (8) $-49 \div 1$  | (26) $-23 \div 3$  | (44) $43 \div 1$  | (62) $-48 \div 6$ |
| (9) $32 \div 5$   | (27) $78 \div 6$   | (45) $13 \div 5$  | (63) $80 \div 6$  |
| (10) $-42 \div 2$ | (28) $-74 \div 7$  | (46) $-1 \div 3$  | (64) $-80 \div 2$ |
| (11) $-75 \div 9$ | (29) $-100 \div 4$ | (47) $-73 \div 1$ | (65) $-89 \div 8$ |
| (12) $64 \div 3$  | (30) $-25 \div 3$  | (48) $62 \div 6$  | (66) $12 \div 1$  |
| (13) $75 \div 5$  | (31) $-1 \div 1$   | (49) $-29 \div 5$ | (67) $-47 \div 3$ |
| (14) $-82 \div 6$ | (32) $66 \div 8$   | (50) $42 \div 6$  | (68) $-82 \div 2$ |
| (15) $26 \div 3$  | (33) $28 \div 7$   | (51) $44 \div 4$  | (69) $23 \div 4$  |
| (16) $64 \div 3$  | (34) $49 \div 4$   | (52) $32 \div 2$  | (70) $-43 \div 4$ |
| (17) $10 \div 1$  | (35) $-68 \div 3$  | (53) $-42 \div 5$ | (71) $84 \div 10$ |
| (18) $19 \div 5$  | (36) $85 \div 5$   | (54) $22 \div 10$ | (72) $94 \div 5$  |

- |                   |                   |                    |                   |
|-------------------|-------------------|--------------------|-------------------|
| (1) $54 \div 10$  | (19) $-64 \div 3$ | (37) $-43 \div 1$  | (55) $-22 \div 4$ |
| (2) $-94 \div 7$  | (20) $90 \div 2$  | (38) $10 \div 10$  | (56) $-76 \div 7$ |
| (3) $14 \div 2$   | (21) $-67 \div 4$ | (39) $-66 \div 2$  | (57) $59 \div 4$  |
| (4) $82 \div 3$   | (22) $-77 \div 4$ | (40) $26 \div 10$  | (58) $-32 \div 3$ |
| (5) $18 \div 9$   | (23) $43 \div 1$  | (41) $-26 \div 4$  | (59) $-14 \div 7$ |
| (6) $-16 \div 6$  | (24) $96 \div 7$  | (42) $48 \div 10$  | (60) $-53 \div 2$ |
| (7) $-43 \div 6$  | (25) $-88 \div 6$ | (43) $-32 \div 3$  | (61) $-45 \div 8$ |
| (8) $61 \div 10$  | (26) $-49 \div 1$ | (44) $-6 \div 5$   | (62) $36 \div 5$  |
| (9) $100 \div 10$ | (27) $48 \div 2$  | (45) $-30 \div 6$  | (63) $37 \div 9$  |
| (10) $79 \div 1$  | (28) $71 \div 10$ | (46) $-75 \div 2$  | (64) $-57 \div 1$ |
| (11) $-80 \div 6$ | (29) $-19 \div 4$ | (47) $-76 \div 10$ | (65) $58 \div 5$  |
| (12) $28 \div 1$  | (30) $40 \div 9$  | (48) $-82 \div 4$  | (66) $-48 \div 1$ |
| (13) $-14 \div 4$ | (31) $67 \div 9$  | (49) $-89 \div 8$  | (67) $27 \div 9$  |
| (14) $-75 \div 2$ | (32) $23 \div 3$  | (50) $27 \div 8$   | (68) $28 \div 6$  |
| (15) $-72 \div 7$ | (33) $56 \div 9$  | (51) $-88 \div 9$  | (69) $-77 \div 6$ |
| (16) $-68 \div 1$ | (34) $40 \div 8$  | (52) $-20 \div 3$  | (70) $40 \div 10$ |
| (17) $-26 \div 3$ | (35) $-22 \div 9$ | (53) $12 \div 9$   | (71) $-65 \div 4$ |
| (18) $25 \div 6$  | (36) $4 \div 3$   | (54) $15 \div 6$   | (72) $41 \div 5$  |

(1)  $29 \div 7$

(19)  $3 \div 10$

(37)  $-12 \div 2$

(55)  $83 \div 8$

(2)  $1 \div 9$

(20)  $66 \div 7$

(38)  $50 \div 7$

(56)  $30 \div 4$

(3)  $-81 \div 7$

(21)  $54 \div 9$

(39)  $22 \div 2$

(57)  $-39 \div 1$

(4)  $38 \div 3$

(22)  $-11 \div 1$

(40)  $59 \div 5$

(58)  $10 \div 7$

(5)  $-92 \div 9$

(23)  $83 \div 3$

(41)  $-9 \div 3$

(59)  $-91 \div 10$

(6)  $68 \div 5$

(24)  $-70 \div 9$

(42)  $-93 \div 6$

(60)  $-76 \div 4$

(7)  $36 \div 10$

(25)  $63 \div 7$

(43)  $-87 \div 2$

(61)  $35 \div 7$

(8)  $-63 \div 1$

(26)  $57 \div 5$

(44)  $-85 \div 9$

(62)  $59 \div 7$

(9)  $9 \div 6$

(27)  $-13 \div 8$

(45)  $93 \div 5$

(63)  $11 \div 2$

(10)  $-56 \div 8$

(28)  $28 \div 7$

(46)  $-73 \div 4$

(64)  $64 \div 1$

(11)  $56 \div 10$

(29)  $28 \div 7$

(47)  $-21 \div 3$

(65)  $31 \div 4$

(12)  $54 \div 10$

(30)  $84 \div 10$

(48)  $45 \div 8$

(66)  $73 \div 9$

(13)  $26 \div 10$

(31)  $24 \div 3$

(49)  $-1 \div 8$

(67)  $-29 \div 10$

(14)  $16 \div 1$

(32)  $-20 \div 8$

(50)  $-14 \div 8$

(68)  $-84 \div 6$

(15)  $59 \div 2$

(33)  $18 \div 7$

(51)  $68 \div 2$

(69)  $-6 \div 7$

(16)  $2 \div 2$

(34)  $-78 \div 3$

(52)  $61 \div 2$

(70)  $87 \div 3$

(17)  $13 \div 8$

(35)  $21 \div 5$

(53)  $86 \div 2$

(71)  $92 \div 10$

(18)  $61 \div 9$

(36)  $58 \div 8$

(54)  $-10 \div 3$

(72)  $56 \div 1$

- |                   |                    |                    |                    |
|-------------------|--------------------|--------------------|--------------------|
| (1) $17 \div 10$  | (19) $57 \div 3$   | (37) $24 \div 10$  | (55) $22 \div 4$   |
| (2) $-77 \div 2$  | (20) $45 \div 9$   | (38) $-73 \div 5$  | (56) $100 \div 9$  |
| (3) $31 \div 9$   | (21) $4 \div 8$    | (39) $84 \div 9$   | (57) $-52 \div 1$  |
| (4) $29 \div 1$   | (22) $79 \div 1$   | (40) $-96 \div 8$  | (58) $-14 \div 2$  |
| (5) $23 \div 1$   | (23) $59 \div 3$   | (41) $-33 \div 2$  | (59) $-2 \div 6$   |
| (6) $-18 \div 10$ | (24) $-80 \div 4$  | (42) $-38 \div 2$  | (60) $43 \div 1$   |
| (7) $-92 \div 10$ | (25) $-77 \div 9$  | (43) $100 \div 10$ | (61) $-82 \div 10$ |
| (8) $65 \div 10$  | (26) $-88 \div 7$  | (44) $-69 \div 2$  | (62) $-4 \div 10$  |
| (9) $-82 \div 2$  | (27) $-78 \div 6$  | (45) $43 \div 6$   | (63) $4 \div 9$    |
| (10) $14 \div 2$  | (28) $47 \div 10$  | (46) $59 \div 9$   | (64) $99 \div 10$  |
| (11) $-95 \div 7$ | (29) $27 \div 10$  | (47) $92 \div 7$   | (65) $-22 \div 7$  |
| (12) $-13 \div 9$ | (30) $-10 \div 5$  | (48) $19 \div 6$   | (66) $-38 \div 5$  |
| (13) $34 \div 7$  | (31) $-96 \div 10$ | (49) $-89 \div 5$  | (67) $-26 \div 5$  |
| (14) $69 \div 3$  | (32) $79 \div 10$  | (50) $-69 \div 5$  | (68) $56 \div 10$  |
| (15) $92 \div 1$  | (33) $52 \div 2$   | (51) $-27 \div 5$  | (69) $-6 \div 4$   |
| (16) $-30 \div 9$ | (34) $94 \div 7$   | (52) $81 \div 4$   | (70) $48 \div 2$   |
| (17) $30 \div 2$  | (35) $-24 \div 3$  | (53) $-35 \div 6$  | (71) $-36 \div 10$ |
| (18) $6 \div 9$   | (36) $78 \div 9$   | (54) $5 \div 7$    | (72) $-27 \div 10$ |

(1)  $-37 \div 9$

(19)  $-58 \div 10$

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(9)  $-5 \div 7$

(27)  $-81 \div 9$

(45)  $-79 \div 3$

(63)  $-9 \div 8$

(10)  $22 \div 1$

(28)  $-88 \div 4$

(46)  $-36 \div 4$

(64)  $-9 \div 2$

(11)  $96 \div 5$

(29)  $-91 \div 9$

(47)  $12 \div 1$

(65)  $-57 \div 10$

(12)  $-87 \div 2$

(30)  $100 \div 7$

(48)  $43 \div 9$

(66)  $-51 \div 10$

(13)  $-31 \div 10$

(31)  $-92 \div 3$

(49)  $-62 \div 2$

(67)  $-40 \div 8$

(14)  $46 \div 6$

(32)  $36 \div 4$

(50)  $56 \div 5$

(68)  $63 \div 8$

(15)  $62 \div 6$

(33)  $-73 \div 3$

(51)  $2 \div 4$

(69)  $-16 \div 5$

(16)  $47 \div 6$

(34)  $63 \div 1$

(52)  $-26 \div 10$

(70)  $73 \div 4$

(17)  $57 \div 2$

(35)  $15 \div 2$

(53)  $-65 \div 9$

(71)  $39 \div 1$

(18)  $21 \div 8$

(36)  $-39 \div 5$

(54)  $75 \div 9$

(72)  $98 \div 9$

(1)  $37 \div 7$

(19)  $81 \div 10$

(37)  $30 \div 1$

(55)  $-17 \div 4$

(2)  $71 \div 5$

(20)  $11 \div 10$

(38)  $-13 \div 2$

(56)  $7 \div 7$

(3)  $5 \div 2$

(21)  $-10 \div 2$

(39)  $-37 \div 7$

(57)  $9 \div 8$

(4)  $30 \div 4$

(22)  $74 \div 7$

(40)  $-8 \div 10$

(58)  $-100 \div 10$

(5)  $50 \div 9$

(23)  $-92 \div 4$

(41)  $97 \div 4$

(59)  $-28 \div 1$

(6)  $-78 \div 8$

(24)  $-50 \div 7$

(42)  $-62 \div 6$

(60)  $76 \div 1$

(7)  $97 \div 6$

(25)  $-68 \div 9$

(43)  $13 \div 4$

(61)  $-61 \div 3$

(8)  $10 \div 5$

(26)  $-17 \div 8$

(44)  $84 \div 7$

(62)  $99 \div 1$

(9)  $11 \div 1$

(27)  $70 \div 7$

(45)  $55 \div 5$

(63)  $-52 \div 4$

(10)  $33 \div 10$

(28)  $88 \div 1$

(46)  $-59 \div 10$

(64)  $30 \div 4$

(11)  $10 \div 4$

(29)  $97 \div 7$

(47)  $-79 \div 2$

(65)  $-86 \div 10$

(12)  $-70 \div 5$

(30)  $63 \div 4$

(48)  $78 \div 6$

(66)  $81 \div 4$

(13)  $72 \div 1$

(31)  $-71 \div 5$

(49)  $94 \div 7$

(67)  $-33 \div 4$

(14)  $-97 \div 10$

(32)  $32 \div 9$

(50)  $49 \div 5$

(68)  $-5 \div 1$

(15)  $57 \div 6$

(33)  $-15 \div 10$

(51)  $-19 \div 9$

(69)  $21 \div 6$

(16)  $-63 \div 5$

(34)  $23 \div 4$

(52)  $74 \div 1$

(70)  $41 \div 4$

(17)  $-28 \div 7$

(35)  $41 \div 3$

(53)  $-25 \div 9$

(71)  $47 \div 4$

(18)  $50 \div 8$

(36)  $58 \div 5$

(54)  $-82 \div 9$

(72)  $-19 \div 1$

(1)  $42 \div 3$

(19)  $-38 \div 3$

(37)  $-36 \div 7$

(55)  $-91 \div 10$

(2)  $-57 \div 5$

(20)  $61 \div 10$

(38)  $-42 \div 1$

(56)  $35 \div 7$

(3)  $-3 \div 9$

(21)  $88 \div 1$

(39)  $-5 \div 1$

(57)  $84 \div 2$

(4)  $-38 \div 1$

(22)  $55 \div 5$

(40)  $-36 \div 2$

(58)  $-97 \div 6$

(5)  $-32 \div 2$

(23)  $-9 \div 4$

(41)  $-84 \div 10$

(59)  $89 \div 6$

(6)  $-73 \div 8$

(24)  $56 \div 1$

(42)  $-7 \div 10$

(60)  $96 \div 3$

(7)  $87 \div 1$

(25)  $-11 \div 9$

(43)  $-41 \div 10$

(61)  $55 \div 1$

(8)  $-81 \div 9$

(26)  $-75 \div 10$

(44)  $95 \div 4$

(62)  $-92 \div 1$

(9)  $54 \div 7$

(27)  $-17 \div 9$

(45)  $-28 \div 10$

(63)  $-78 \div 10$

(10)  $-32 \div 7$

(28)  $92 \div 1$

(46)  $-78 \div 9$

(64)  $61 \div 7$

(11)  $99 \div 3$

(29)  $93 \div 8$

(47)  $-93 \div 8$

(65)  $40 \div 4$

(12)  $-55 \div 3$

(30)  $-5 \div 10$

(48)  $22 \div 5$

(66)  $59 \div 9$

(13)  $-17 \div 2$

(31)  $-25 \div 7$

(49)  $97 \div 4$

(67)  $-25 \div 8$

(14)  $-32 \div 2$

(32)  $-71 \div 1$

(50)  $-6 \div 2$

(68)  $85 \div 1$

(15)  $78 \div 8$

(33)  $56 \div 4$

(51)  $-79 \div 9$

(69)  $44 \div 5$

(16)  $53 \div 2$

(34)  $-65 \div 4$

(52)  $27 \div 3$

(70)  $64 \div 9$

(17)  $-78 \div 4$

(35)  $5 \div 2$

(53)  $-97 \div 5$

(71)  $96 \div 1$

(18)  $-19 \div 1$

(36)  $100 \div 2$

(54)  $-53 \div 4$

(72)  $-82 \div 10$



- |                                       |                                       |                                       |                                       |
|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| (1) $-8 \div 5$<br>商 $-2$ , 余り $2$    | (19) $-74 \div 7$<br>商 $-11$ , 余り $3$ | (37) $90 \div 8$<br>商 $11$ , 余り $2$   | (55) $96 \div 6$<br>商 $16$ , 余り $0$   |
| (2) $-54 \div 8$<br>商 $-7$ , 余り $2$   | (20) $-47 \div 6$<br>商 $-8$ , 余り $1$  | (38) $-50 \div 7$<br>商 $-8$ , 余り $6$  | (56) $78 \div 3$<br>商 $26$ , 余り $0$   |
| (3) $-28 \div 7$<br>商 $-4$ , 余り $0$   | (21) $-15 \div 1$<br>商 $-15$ , 余り $0$ | (39) $76 \div 1$<br>商 $76$ , 余り $0$   | (57) $-10 \div 3$<br>商 $-4$ , 余り $2$  |
| (4) $80 \div 4$<br>商 $20$ , 余り $0$    | (22) $-55 \div 2$<br>商 $-28$ , 余り $1$ | (40) $-92 \div 9$<br>商 $-11$ , 余り $7$ | (58) $58 \div 6$<br>商 $9$ , 余り $4$    |
| (5) $93 \div 3$<br>商 $31$ , 余り $0$    | (23) $-30 \div 2$<br>商 $-15$ , 余り $0$ | (41) $-66 \div 6$<br>商 $-11$ , 余り $0$ | (59) $65 \div 2$<br>商 $32$ , 余り $1$   |
| (6) $80 \div 10$<br>商 $8$ , 余り $0$    | (24) $-95 \div 3$<br>商 $-32$ , 余り $1$ | (42) $-41 \div 9$<br>商 $-5$ , 余り $4$  | (60) $60 \div 3$<br>商 $20$ , 余り $0$   |
| (7) $-99 \div 4$<br>商 $-25$ , 余り $1$  | (25) $7 \div 2$<br>商 $3$ , 余り $1$     | (43) $-1 \div 1$<br>商 $-1$ , 余り $0$   | (61) $5 \div 9$<br>商 $0$ , 余り $5$     |
| (8) $73 \div 6$<br>商 $12$ , 余り $1$    | (26) $74 \div 4$<br>商 $18$ , 余り $2$   | (44) $33 \div 10$<br>商 $3$ , 余り $3$   | (62) $-64 \div 7$<br>商 $-10$ , 余り $6$ |
| (9) $65 \div 9$<br>商 $7$ , 余り $2$     | (27) $-3 \div 6$<br>商 $-1$ , 余り $3$   | (45) $60 \div 8$<br>商 $7$ , 余り $4$    | (63) $-67 \div 8$<br>商 $-9$ , 余り $5$  |
| (10) $48 \div 3$<br>商 $16$ , 余り $0$   | (28) $20 \div 2$<br>商 $10$ , 余り $0$   | (46) $42 \div 8$<br>商 $5$ , 余り $2$    | (64) $45 \div 8$<br>商 $5$ , 余り $5$    |
| (11) $-6 \div 7$<br>商 $-1$ , 余り $1$   | (29) $-47 \div 10$<br>商 $-5$ , 余り $3$ | (47) $-90 \div 7$<br>商 $-13$ , 余り $1$ | (65) $-33 \div 6$<br>商 $-6$ , 余り $3$  |
| (12) $32 \div 9$<br>商 $3$ , 余り $5$    | (30) $74 \div 9$<br>商 $8$ , 余り $2$    | (48) $-69 \div 3$<br>商 $-23$ , 余り $0$ | (66) $-91 \div 8$<br>商 $-12$ , 余り $5$ |
| (13) $72 \div 6$<br>商 $12$ , 余り $0$   | (31) $96 \div 5$<br>商 $19$ , 余り $1$   | (49) $37 \div 1$<br>商 $37$ , 余り $0$   | (67) $-73 \div 2$<br>商 $-37$ , 余り $1$ |
| (14) $-40 \div 6$<br>商 $-7$ , 余り $2$  | (32) $21 \div 6$<br>商 $3$ , 余り $3$    | (50) $98 \div 10$<br>商 $9$ , 余り $8$   | (68) $92 \div 5$<br>商 $18$ , 余り $2$   |
| (15) $-31 \div 1$<br>商 $-31$ , 余り $0$ | (33) $24 \div 8$<br>商 $3$ , 余り $0$    | (51) $78 \div 2$<br>商 $39$ , 余り $0$   | (69) $27 \div 5$<br>商 $5$ , 余り $2$    |
| (16) $-47 \div 10$<br>商 $-5$ , 余り $3$ | (34) $-68 \div 6$<br>商 $-12$ , 余り $4$ | (52) $76 \div 2$<br>商 $38$ , 余り $0$   | (70) $-67 \div 5$<br>商 $-14$ , 余り $3$ |
| (17) $60 \div 3$<br>商 $20$ , 余り $0$   | (35) $10 \div 8$<br>商 $1$ , 余り $2$    | (53) $-76 \div 7$<br>商 $-11$ , 余り $1$ | (71) $-93 \div 1$<br>商 $-93$ , 余り $0$ |
| (18) $61 \div 4$<br>商 $15$ , 余り $1$   | (36) $-89 \div 8$<br>商 $-12$ , 余り $7$ | (54) $15 \div 6$<br>商 $2$ , 余り $3$    | (72) $-97 \div 3$<br>商 $-33$ , 余り $2$ |

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|----------------------------------|-----------------------------------|----------------------------------|----------------------------------|
| (1) $19 \div 5$<br>商 3, 余り 4     | (19) $-52 \div 5$<br>商 -11, 余り 3  | (37) $91 \div 3$<br>商 30, 余り 1   | (55) $-94 \div 6$<br>商 -16, 余り 2 |
| (2) $63 \div 4$<br>商 15, 余り 3    | (20) $81 \div 8$<br>商 10, 余り 1    | (38) $98 \div 3$<br>商 32, 余り 2   | (56) $-99 \div 6$<br>商 -17, 余り 3 |
| (3) $-71 \div 7$<br>商 -11, 余り 6  | (21) $-91 \div 3$<br>商 -31, 余り 2  | (39) $-6 \div 3$<br>商 -2, 余り 0   | (57) $-27 \div 5$<br>商 -6, 余り 3  |
| (4) $61 \div 10$<br>商 6, 余り 1    | (22) $-64 \div 7$<br>商 -10, 余り 6  | (40) $-86 \div 5$<br>商 -18, 余り 4 | (58) $82 \div 5$<br>商 16, 余り 2   |
| (5) $-24 \div 5$<br>商 -5, 余り 1   | (23) $-34 \div 7$<br>商 -5, 余り 1   | (41) $-70 \div 6$<br>商 -12, 余り 2 | (59) $64 \div 1$<br>商 64, 余り 0   |
| (6) $28 \div 3$<br>商 9, 余り 1     | (24) $31 \div 6$<br>商 5, 余り 1     | (42) $71 \div 10$<br>商 7, 余り 1   | (60) $65 \div 5$<br>商 13, 余り 0   |
| (7) $0 \div 9$<br>商 0, 余り 0      | (25) $84 \div 6$<br>商 14, 余り 0    | (43) $64 \div 2$<br>商 32, 余り 0   | (61) $-99 \div 3$<br>商 -33, 余り 0 |
| (8) $63 \div 9$<br>商 7, 余り 0     | (26) $88 \div 8$<br>商 11, 余り 0    | (44) $-23 \div 7$<br>商 -4, 余り 5  | (62) $100 \div 8$<br>商 12, 余り 4  |
| (9) $55 \div 9$<br>商 6, 余り 1     | (27) $54 \div 2$<br>商 27, 余り 0    | (45) $-6 \div 4$<br>商 -2, 余り 2   | (63) $-78 \div 3$<br>商 -26, 余り 0 |
| (10) $-20 \div 3$<br>商 -7, 余り 1  | (28) $58 \div 5$<br>商 11, 余り 3    | (46) $35 \div 3$<br>商 11, 余り 2   | (64) $-96 \div 7$<br>商 -14, 余り 2 |
| (11) $85 \div 10$<br>商 8, 余り 5   | (29) $28 \div 3$<br>商 9, 余り 1     | (47) $88 \div 7$<br>商 12, 余り 4   | (65) $-93 \div 4$<br>商 -24, 余り 3 |
| (12) $-4 \div 3$<br>商 -2, 余り 2   | (30) $39 \div 5$<br>商 7, 余り 4     | (48) $-58 \div 7$<br>商 -9, 余り 5  | (66) $-25 \div 3$<br>商 -9, 余り 2  |
| (13) $26 \div 5$<br>商 5, 余り 1    | (31) $-87 \div 6$<br>商 -15, 余り 3  | (49) $-43 \div 3$<br>商 -15, 余り 2 | (67) $-56 \div 8$<br>商 -7, 余り 0  |
| (14) $-24 \div 6$<br>商 -4, 余り 0  | (32) $-100 \div 2$<br>商 -50, 余り 0 | (50) $-82 \div 9$<br>商 -10, 余り 8 | (68) $91 \div 6$<br>商 15, 余り 1   |
| (15) $-89 \div 2$<br>商 -45, 余り 1 | (33) $-87 \div 3$<br>商 -29, 余り 0  | (51) $18 \div 8$<br>商 2, 余り 2    | (69) $-94 \div 3$<br>商 -32, 余り 2 |
| (16) $22 \div 8$<br>商 2, 余り 6    | (34) $99 \div 1$<br>商 99, 余り 0    | (52) $83 \div 7$<br>商 11, 余り 6   | (70) $1 \div 8$<br>商 0, 余り 1     |
| (17) $-41 \div 10$<br>商 -5, 余り 9 | (35) $62 \div 4$<br>商 15, 余り 2    | (53) $-65 \div 6$<br>商 -11, 余り 1 | (71) $-43 \div 3$<br>商 -15, 余り 2 |
| (18) $-51 \div 8$<br>商 -7, 余り 5  | (36) $-71 \div 7$<br>商 -11, 余り 6  | (54) $72 \div 8$<br>商 9, 余り 0    | (72) $84 \div 8$<br>商 10, 余り 4   |

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|--------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| (1) $-96 \div 5$<br>商 $-20$ , 余り $4$ | (19) $4 \div 3$<br>商 $1$ , 余り $1$     | (37) $52 \div 8$<br>商 $6$ , 余り $4$    | (55) $26 \div 4$<br>商 $6$ , 余り $2$    |
| (2) $-96 \div 9$<br>商 $-11$ , 余り $3$ | (20) $56 \div 2$<br>商 $28$ , 余り $0$   | (38) $-16 \div 8$<br>商 $-2$ , 余り $0$  | (56) $65 \div 9$<br>商 $7$ , 余り $2$    |
| (3) $15 \div 7$<br>商 $2$ , 余り $1$    | (21) $14 \div 9$<br>商 $1$ , 余り $5$    | (39) $72 \div 7$<br>商 $10$ , 余り $2$   | (57) $-33 \div 2$<br>商 $-17$ , 余り $1$ |
| (4) $-11 \div 4$<br>商 $-3$ , 余り $1$  | (22) $-16 \div 1$<br>商 $-16$ , 余り $0$ | (40) $22 \div 6$<br>商 $3$ , 余り $4$    | (58) $-62 \div 7$<br>商 $-9$ , 余り $1$  |
| (5) $-40 \div 1$<br>商 $-40$ , 余り $0$ | (23) $68 \div 5$<br>商 $13$ , 余り $3$   | (41) $-3 \div 10$<br>商 $-1$ , 余り $7$  | (59) $-12 \div 3$<br>商 $-4$ , 余り $0$  |
| (6) $-61 \div 4$<br>商 $-16$ , 余り $3$ | (24) $-58 \div 2$<br>商 $-29$ , 余り $0$ | (42) $-86 \div 10$<br>商 $-9$ , 余り $4$ | (60) $54 \div 1$<br>商 $54$ , 余り $0$   |
| (7) $41 \div 6$<br>商 $6$ , 余り $5$    | (25) $3 \div 3$<br>商 $1$ , 余り $0$     | (43) $45 \div 10$<br>商 $4$ , 余り $5$   | (61) $-74 \div 9$<br>商 $-9$ , 余り $7$  |
| (8) $-93 \div 1$<br>商 $-93$ , 余り $0$ | (26) $64 \div 7$<br>商 $9$ , 余り $1$    | (44) $66 \div 6$<br>商 $11$ , 余り $0$   | (62) $-8 \div 3$<br>商 $-3$ , 余り $1$   |
| (9) $90 \div 3$<br>商 $30$ , 余り $0$   | (27) $-93 \div 9$<br>商 $-11$ , 余り $6$ | (45) $49 \div 3$<br>商 $16$ , 余り $1$   | (63) $-68 \div 2$<br>商 $-34$ , 余り $0$ |
| (10) $-21 \div 3$<br>商 $-7$ , 余り $0$ | (28) $32 \div 5$<br>商 $6$ , 余り $2$    | (46) $47 \div 6$<br>商 $7$ , 余り $5$    | (64) $-67 \div 2$<br>商 $-34$ , 余り $1$ |
| (11) $71 \div 5$<br>商 $14$ , 余り $1$  | (29) $89 \div 7$<br>商 $12$ , 余り $5$   | (47) $53 \div 5$<br>商 $10$ , 余り $3$   | (65) $72 \div 1$<br>商 $72$ , 余り $0$   |
| (12) $79 \div 8$<br>商 $9$ , 余り $7$   | (30) $-11 \div 4$<br>商 $-3$ , 余り $1$  | (48) $-63 \div 8$<br>商 $-8$ , 余り $1$  | (66) $71 \div 10$<br>商 $7$ , 余り $1$   |
| (13) $68 \div 7$<br>商 $9$ , 余り $5$   | (31) $34 \div 3$<br>商 $11$ , 余り $1$   | (49) $66 \div 8$<br>商 $8$ , 余り $2$    | (67) $92 \div 8$<br>商 $11$ , 余り $4$   |
| (14) $27 \div 4$<br>商 $6$ , 余り $3$   | (32) $-26 \div 9$<br>商 $-3$ , 余り $1$  | (50) $-36 \div 6$<br>商 $-6$ , 余り $0$  | (68) $-25 \div 2$<br>商 $-13$ , 余り $1$ |
| (15) $24 \div 7$<br>商 $3$ , 余り $3$   | (33) $-58 \div 10$<br>商 $-6$ , 余り $2$ | (51) $89 \div 4$<br>商 $22$ , 余り $1$   | (69) $-93 \div 3$<br>商 $-31$ , 余り $0$ |
| (16) $-5 \div 8$<br>商 $-1$ , 余り $3$  | (34) $-99 \div 7$<br>商 $-15$ , 余り $6$ | (52) $26 \div 8$<br>商 $3$ , 余り $2$    | (70) $31 \div 10$<br>商 $3$ , 余り $1$   |
| (17) $95 \div 7$<br>商 $13$ , 余り $4$  | (35) $-19 \div 9$<br>商 $-3$ , 余り $8$  | (53) $-23 \div 9$<br>商 $-3$ , 余り $4$  | (71) $90 \div 2$<br>商 $45$ , 余り $0$   |
| (18) $-16 \div 5$<br>商 $-4$ , 余り $4$ | (36) $-26 \div 4$<br>商 $-7$ , 余り $2$  | (54) $-40 \div 10$<br>商 $-4$ , 余り $0$ | (72) $49 \div 5$<br>商 $9$ , 余り $4$    |

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|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| (1) $34 \div 3$<br>商 11, 余り 1    | (19) $21 \div 2$<br>商 10, 余り 1   | (37) $-20 \div 3$<br>商 -7, 余り 1  | (55) $95 \div 2$<br>商 47, 余り 1   |
| (2) $37 \div 3$<br>商 12, 余り 1    | (20) $39 \div 1$<br>商 39, 余り 0   | (38) $-24 \div 5$<br>商 -5, 余り 1  | (56) $73 \div 9$<br>商 8, 余り 1    |
| (3) $-8 \div 1$<br>商 -8, 余り 0    | (21) $19 \div 8$<br>商 2, 余り 3    | (39) $7 \div 5$<br>商 1, 余り 2     | (57) $67 \div 7$<br>商 9, 余り 4    |
| (4) $-6 \div 7$<br>商 -1, 余り 1    | (22) $-47 \div 10$<br>商 -5, 余り 3 | (40) $-45 \div 2$<br>商 -23, 余り 1 | (58) $56 \div 5$<br>商 11, 余り 1   |
| (5) $57 \div 5$<br>商 11, 余り 2    | (23) $85 \div 1$<br>商 85, 余り 0   | (41) $4 \div 10$<br>商 0, 余り 4    | (59) $-73 \div 5$<br>商 -15, 余り 2 |
| (6) $31 \div 1$<br>商 31, 余り 0    | (24) $-69 \div 8$<br>商 -9, 余り 3  | (42) $14 \div 6$<br>商 2, 余り 2    | (60) $50 \div 3$<br>商 16, 余り 2   |
| (7) $28 \div 6$<br>商 4, 余り 4     | (25) $55 \div 2$<br>商 27, 余り 1   | (43) $24 \div 9$<br>商 2, 余り 6    | (61) $-52 \div 8$<br>商 -7, 余り 4  |
| (8) $64 \div 10$<br>商 6, 余り 4    | (26) $-40 \div 2$<br>商 -20, 余り 0 | (44) $89 \div 2$<br>商 44, 余り 1   | (62) $35 \div 3$<br>商 11, 余り 2   |
| (9) $84 \div 9$<br>商 9, 余り 3     | (27) $-56 \div 9$<br>商 -7, 余り 7  | (45) $31 \div 3$<br>商 10, 余り 1   | (63) $-80 \div 9$<br>商 -9, 余り 1  |
| (10) $0 \div 6$<br>商 0, 余り 0     | (28) $35 \div 1$<br>商 35, 余り 0   | (46) $-38 \div 3$<br>商 -13, 余り 1 | (64) $64 \div 7$<br>商 9, 余り 1    |
| (11) $-18 \div 4$<br>商 -5, 余り 2  | (29) $72 \div 9$<br>商 8, 余り 0    | (47) $-37 \div 2$<br>商 -19, 余り 1 | (65) $12 \div 1$<br>商 12, 余り 0   |
| (12) $0 \div 6$<br>商 0, 余り 0     | (30) $-60 \div 10$<br>商 -6, 余り 0 | (48) $32 \div 5$<br>商 6, 余り 2    | (66) $-31 \div 4$<br>商 -8, 余り 1  |
| (13) $-39 \div 5$<br>商 -8, 余り 1  | (31) $-25 \div 5$<br>商 -5, 余り 0  | (49) $74 \div 1$<br>商 74, 余り 0   | (67) $83 \div 8$<br>商 10, 余り 3   |
| (14) $-67 \div 1$<br>商 -67, 余り 0 | (32) $89 \div 2$<br>商 44, 余り 1   | (50) $-82 \div 5$<br>商 -17, 余り 3 | (68) $-98 \div 2$<br>商 -49, 余り 0 |
| (15) $31 \div 2$<br>商 15, 余り 1   | (33) $88 \div 2$<br>商 44, 余り 0   | (51) $-32 \div 8$<br>商 -4, 余り 0  | (69) $36 \div 1$<br>商 36, 余り 0   |
| (16) $-68 \div 8$<br>商 -9, 余り 4  | (34) $-55 \div 3$<br>商 -19, 余り 2 | (52) $-73 \div 4$<br>商 -19, 余り 3 | (70) $-26 \div 4$<br>商 -7, 余り 2  |
| (17) $54 \div 2$<br>商 27, 余り 0   | (35) $-79 \div 7$<br>商 -12, 余り 5 | (53) $33 \div 4$<br>商 8, 余り 1    | (71) $46 \div 6$<br>商 7, 余り 4    |
| (18) $77 \div 10$<br>商 7, 余り 7   | (36) $4 \div 8$<br>商 0, 余り 4     | (54) $-85 \div 5$<br>商 -17, 余り 0 | (72) $-12 \div 8$<br>商 -2, 余り 4  |

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|----------------------------------|-----------------------------------|----------------------------------|----------------------------------|
| (1) $44 \div 10$<br>商 4, 余り 4    | (19) $2 \div 7$<br>商 0, 余り 2      | (37) $38 \div 10$<br>商 3, 余り 8   | (55) $77 \div 10$<br>商 7, 余り 7   |
| (2) $60 \div 5$<br>商 12, 余り 0    | (20) $71 \div 4$<br>商 17, 余り 3    | (38) $-69 \div 8$<br>商 -9, 余り 3  | (56) $45 \div 7$<br>商 6, 余り 3    |
| (3) $23 \div 8$<br>商 2, 余り 7     | (21) $-46 \div 5$<br>商 -10, 余り 4  | (39) $1 \div 7$<br>商 0, 余り 1     | (57) $7 \div 7$<br>商 1, 余り 0     |
| (4) $10 \div 10$<br>商 1, 余り 0    | (22) $91 \div 4$<br>商 22, 余り 3    | (40) $-87 \div 3$<br>商 -29, 余り 0 | (58) $80 \div 8$<br>商 10, 余り 0   |
| (5) $55 \div 8$<br>商 6, 余り 7     | (23) $-43 \div 1$<br>商 -43, 余り 0  | (41) $-64 \div 1$<br>商 -64, 余り 0 | (59) $76 \div 8$<br>商 9, 余り 4    |
| (6) $56 \div 5$<br>商 11, 余り 1    | (24) $-34 \div 8$<br>商 -5, 余り 6   | (42) $-21 \div 5$<br>商 -5, 余り 4  | (60) $59 \div 5$<br>商 11, 余り 4   |
| (7) $-9 \div 6$<br>商 -2, 余り 3    | (25) $-99 \div 7$<br>商 -15, 余り 6  | (43) $-77 \div 2$<br>商 -39, 余り 1 | (61) $74 \div 6$<br>商 12, 余り 2   |
| (8) $-49 \div 1$<br>商 -49, 余り 0  | (26) $-23 \div 3$<br>商 -8, 余り 1   | (44) $43 \div 1$<br>商 43, 余り 0   | (62) $-48 \div 6$<br>商 -8, 余り 0  |
| (9) $32 \div 5$<br>商 6, 余り 2     | (27) $78 \div 6$<br>商 13, 余り 0    | (45) $13 \div 5$<br>商 2, 余り 3    | (63) $80 \div 6$<br>商 13, 余り 2   |
| (10) $-42 \div 2$<br>商 -21, 余り 0 | (28) $-74 \div 7$<br>商 -11, 余り 3  | (46) $-1 \div 3$<br>商 -1, 余り 2   | (64) $-80 \div 2$<br>商 -40, 余り 0 |
| (11) $-75 \div 9$<br>商 -9, 余り 6  | (29) $-100 \div 4$<br>商 -25, 余り 0 | (47) $-73 \div 1$<br>商 -73, 余り 0 | (65) $-89 \div 8$<br>商 -12, 余り 7 |
| (12) $64 \div 3$<br>商 21, 余り 1   | (30) $-25 \div 3$<br>商 -9, 余り 2   | (48) $62 \div 6$<br>商 10, 余り 2   | (66) $12 \div 1$<br>商 12, 余り 0   |
| (13) $75 \div 5$<br>商 15, 余り 0   | (31) $-1 \div 1$<br>商 -1, 余り 0    | (49) $-29 \div 5$<br>商 -6, 余り 1  | (67) $-47 \div 3$<br>商 -16, 余り 1 |
| (14) $-82 \div 6$<br>商 -14, 余り 2 | (32) $66 \div 8$<br>商 8, 余り 2     | (50) $42 \div 6$<br>商 7, 余り 0    | (68) $-82 \div 2$<br>商 -41, 余り 0 |
| (15) $26 \div 3$<br>商 8, 余り 2    | (33) $28 \div 7$<br>商 4, 余り 0     | (51) $44 \div 4$<br>商 11, 余り 0   | (69) $23 \div 4$<br>商 5, 余り 3    |
| (16) $64 \div 3$<br>商 21, 余り 1   | (34) $49 \div 4$<br>商 12, 余り 1    | (52) $32 \div 2$<br>商 16, 余り 0   | (70) $-43 \div 4$<br>商 -11, 余り 1 |
| (17) $10 \div 1$<br>商 10, 余り 0   | (35) $-68 \div 3$<br>商 -23, 余り 1  | (53) $-42 \div 5$<br>商 -9, 余り 3  | (71) $84 \div 10$<br>商 8, 余り 4   |
| (18) $19 \div 5$<br>商 3, 余り 4    | (36) $85 \div 5$<br>商 17, 余り 0    | (54) $22 \div 10$<br>商 2, 余り 2   | (72) $94 \div 5$<br>商 18, 余り 4   |

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|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| (1) $54 \div 10$<br>商 5, 余り 4     | (19) $-64 \div 3$<br>商 - 22, 余り 2 | (37) $-43 \div 1$<br>商 - 43, 余り 0 | (55) $-22 \div 4$<br>商 - 6, 余り 2  |
| (2) $-94 \div 7$<br>商 - 14, 余り 4  | (20) $90 \div 2$<br>商 45, 余り 0    | (38) $10 \div 10$<br>商 1, 余り 0    | (56) $-76 \div 7$<br>商 - 11, 余り 1 |
| (3) $14 \div 2$<br>商 7, 余り 0      | (21) $-67 \div 4$<br>商 - 17, 余り 1 | (39) $-66 \div 2$<br>商 - 33, 余り 0 | (57) $59 \div 4$<br>商 14, 余り 3    |
| (4) $82 \div 3$<br>商 27, 余り 1     | (22) $-77 \div 4$<br>商 - 20, 余り 3 | (40) $26 \div 10$<br>商 2, 余り 6    | (58) $-32 \div 3$<br>商 - 11, 余り 1 |
| (5) $18 \div 9$<br>商 2, 余り 0      | (23) $43 \div 1$<br>商 43, 余り 0    | (41) $-26 \div 4$<br>商 - 7, 余り 2  | (59) $-14 \div 7$<br>商 - 2, 余り 0  |
| (6) $-16 \div 6$<br>商 - 3, 余り 2   | (24) $96 \div 7$<br>商 13, 余り 5    | (42) $48 \div 10$<br>商 4, 余り 8    | (60) $-53 \div 2$<br>商 - 27, 余り 1 |
| (7) $-43 \div 6$<br>商 - 8, 余り 5   | (25) $-88 \div 6$<br>商 - 15, 余り 2 | (43) $-32 \div 3$<br>商 - 11, 余り 1 | (61) $-45 \div 8$<br>商 - 6, 余り 3  |
| (8) $61 \div 10$<br>商 6, 余り 1     | (26) $-49 \div 1$<br>商 - 49, 余り 0 | (44) $-6 \div 5$<br>商 - 2, 余り 4   | (62) $36 \div 5$<br>商 7, 余り 1     |
| (9) $100 \div 10$<br>商 10, 余り 0   | (27) $48 \div 2$<br>商 24, 余り 0    | (45) $-30 \div 6$<br>商 - 5, 余り 0  | (63) $37 \div 9$<br>商 4, 余り 1     |
| (10) $79 \div 1$<br>商 79, 余り 0    | (28) $71 \div 10$<br>商 7, 余り 1    | (46) $-75 \div 2$<br>商 - 38, 余り 1 | (64) $-57 \div 1$<br>商 - 57, 余り 0 |
| (11) $-80 \div 6$<br>商 - 14, 余り 4 | (29) $-19 \div 4$<br>商 - 5, 余り 1  | (47) $-76 \div 10$<br>商 - 8, 余り 4 | (65) $58 \div 5$<br>商 11, 余り 3    |
| (12) $28 \div 1$<br>商 28, 余り 0    | (30) $40 \div 9$<br>商 4, 余り 4     | (48) $-82 \div 4$<br>商 - 21, 余り 2 | (66) $-48 \div 1$<br>商 - 48, 余り 0 |
| (13) $-14 \div 4$<br>商 - 4, 余り 2  | (31) $67 \div 9$<br>商 7, 余り 4     | (49) $-89 \div 8$<br>商 - 12, 余り 7 | (67) $27 \div 9$<br>商 3, 余り 0     |
| (14) $-75 \div 2$<br>商 - 38, 余り 1 | (32) $23 \div 3$<br>商 7, 余り 2     | (50) $27 \div 8$<br>商 3, 余り 3     | (68) $28 \div 6$<br>商 4, 余り 4     |
| (15) $-72 \div 7$<br>商 - 11, 余り 5 | (33) $56 \div 9$<br>商 6, 余り 2     | (51) $-88 \div 9$<br>商 - 10, 余り 2 | (69) $-77 \div 6$<br>商 - 13, 余り 1 |
| (16) $-68 \div 1$<br>商 - 68, 余り 0 | (34) $40 \div 8$<br>商 5, 余り 0     | (52) $-20 \div 3$<br>商 - 7, 余り 1  | (70) $40 \div 10$<br>商 4, 余り 0    |
| (17) $-26 \div 3$<br>商 - 9, 余り 1  | (35) $-22 \div 9$<br>商 - 3, 余り 5  | (53) $12 \div 9$<br>商 1, 余り 3     | (71) $-65 \div 4$<br>商 - 17, 余り 3 |
| (18) $25 \div 6$<br>商 4, 余り 1     | (36) $4 \div 3$<br>商 1, 余り 1      | (54) $15 \div 6$<br>商 2, 余り 3     | (72) $41 \div 5$<br>商 8, 余り 1     |

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|---------------------------------|----------------------------------|----------------------------------|-----------------------------------|
| (1) $29 \div 7$<br>商 4, 余り 1    | (19) $3 \div 10$<br>商 0, 余り 3    | (37) $-12 \div 2$<br>商 -6, 余り 0  | (55) $83 \div 8$<br>商 10, 余り 3    |
| (2) $1 \div 9$<br>商 0, 余り 1     | (20) $66 \div 7$<br>商 9, 余り 3    | (38) $50 \div 7$<br>商 7, 余り 1    | (56) $30 \div 4$<br>商 7, 余り 2     |
| (3) $-81 \div 7$<br>商 -12, 余り 3 | (21) $54 \div 9$<br>商 6, 余り 0    | (39) $22 \div 2$<br>商 11, 余り 0   | (57) $-39 \div 1$<br>商 -39, 余り 0  |
| (4) $38 \div 3$<br>商 12, 余り 2   | (22) $-11 \div 1$<br>商 -11, 余り 0 | (40) $59 \div 5$<br>商 11, 余り 4   | (58) $10 \div 7$<br>商 1, 余り 3     |
| (5) $-92 \div 9$<br>商 -11, 余り 7 | (23) $83 \div 3$<br>商 27, 余り 2   | (41) $-9 \div 3$<br>商 -3, 余り 0   | (59) $-91 \div 10$<br>商 -10, 余り 9 |
| (6) $68 \div 5$<br>商 13, 余り 3   | (24) $-70 \div 9$<br>商 -8, 余り 2  | (42) $-93 \div 6$<br>商 -16, 余り 3 | (60) $-76 \div 4$<br>商 -19, 余り 0  |
| (7) $36 \div 10$<br>商 3, 余り 6   | (25) $63 \div 7$<br>商 9, 余り 0    | (43) $-87 \div 2$<br>商 -44, 余り 1 | (61) $35 \div 7$<br>商 5, 余り 0     |
| (8) $-63 \div 1$<br>商 -63, 余り 0 | (26) $57 \div 5$<br>商 11, 余り 2   | (44) $-85 \div 9$<br>商 -10, 余り 5 | (62) $59 \div 7$<br>商 8, 余り 3     |
| (9) $9 \div 6$<br>商 1, 余り 3     | (27) $-13 \div 8$<br>商 -2, 余り 3  | (45) $93 \div 5$<br>商 18, 余り 3   | (63) $11 \div 2$<br>商 5, 余り 1     |
| (10) $-56 \div 8$<br>商 -7, 余り 0 | (28) $28 \div 7$<br>商 4, 余り 0    | (46) $-73 \div 4$<br>商 -19, 余り 3 | (64) $64 \div 1$<br>商 64, 余り 0    |
| (11) $56 \div 10$<br>商 5, 余り 6  | (29) $28 \div 7$<br>商 4, 余り 0    | (47) $-21 \div 3$<br>商 -7, 余り 0  | (65) $31 \div 4$<br>商 7, 余り 3     |
| (12) $54 \div 10$<br>商 5, 余り 4  | (30) $84 \div 10$<br>商 8, 余り 4   | (48) $45 \div 8$<br>商 5, 余り 5    | (66) $73 \div 9$<br>商 8, 余り 1     |
| (13) $26 \div 10$<br>商 2, 余り 6  | (31) $24 \div 3$<br>商 8, 余り 0    | (49) $-1 \div 8$<br>商 -1, 余り 7   | (67) $-29 \div 10$<br>商 -3, 余り 1  |
| (14) $16 \div 1$<br>商 16, 余り 0  | (32) $-20 \div 8$<br>商 -3, 余り 4  | (50) $-14 \div 8$<br>商 -2, 余り 2  | (68) $-84 \div 6$<br>商 -14, 余り 0  |
| (15) $59 \div 2$<br>商 29, 余り 1  | (33) $18 \div 7$<br>商 2, 余り 4    | (51) $68 \div 2$<br>商 34, 余り 0   | (69) $-6 \div 7$<br>商 -1, 余り 1    |
| (16) $2 \div 2$<br>商 1, 余り 0    | (34) $-78 \div 3$<br>商 -26, 余り 0 | (52) $61 \div 2$<br>商 30, 余り 1   | (70) $87 \div 3$<br>商 29, 余り 0    |
| (17) $13 \div 8$<br>商 1, 余り 5   | (35) $21 \div 5$<br>商 4, 余り 1    | (53) $86 \div 2$<br>商 43, 余り 0   | (71) $92 \div 10$<br>商 9, 余り 2    |
| (18) $61 \div 9$<br>商 6, 余り 7   | (36) $58 \div 8$<br>商 7, 余り 2    | (54) $-10 \div 3$<br>商 -4, 余り 2  | (72) $56 \div 1$<br>商 56, 余り 0    |

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|----------------------------------|-----------------------------------|----------------------------------|----------------------------------|
| (1) $17 \div 10$<br>商 1, 余り 7    | (19) $57 \div 3$<br>商 19, 余り 0    | (37) $24 \div 10$<br>商 2, 余り 4   | (55) $22 \div 4$<br>商 5, 余り 2    |
| (2) $-77 \div 2$<br>商 -39, 余り 1  | (20) $45 \div 9$<br>商 5, 余り 0     | (38) $-73 \div 5$<br>商 -15, 余り 2 | (56) $100 \div 9$<br>商 11, 余り 1  |
| (3) $31 \div 9$<br>商 3, 余り 4     | (21) $4 \div 8$<br>商 0, 余り 4      | (39) $84 \div 9$<br>商 9, 余り 3    | (57) $-52 \div 1$<br>商 -52, 余り 0 |
| (4) $29 \div 1$<br>商 29, 余り 0    | (22) $79 \div 1$<br>商 79, 余り 0    | (40) $-96 \div 8$<br>商 -12, 余り 0 | (58) $-14 \div 2$<br>商 -7, 余り 0  |
| (5) $23 \div 1$<br>商 23, 余り 0    | (23) $59 \div 3$<br>商 19, 余り 2    | (41) $-33 \div 2$<br>商 -17, 余り 1 | (59) $-2 \div 6$<br>商 -1, 余り 4   |
| (6) $-18 \div 10$<br>商 -2, 余り 2  | (24) $-80 \div 4$<br>商 -20, 余り 0  | (42) $-38 \div 2$<br>商 -19, 余り 0 | (60) $43 \div 1$<br>商 43, 余り 0   |
| (7) $-92 \div 10$<br>商 -10, 余り 8 | (25) $-77 \div 9$<br>商 -9, 余り 4   | (43) $100 \div 10$<br>商 10, 余り 0 | (61) $-82 \div 10$<br>商 -9, 余り 8 |
| (8) $65 \div 10$<br>商 6, 余り 5    | (26) $-88 \div 7$<br>商 -13, 余り 3  | (44) $-69 \div 2$<br>商 -35, 余り 1 | (62) $-4 \div 10$<br>商 -1, 余り 6  |
| (9) $-82 \div 2$<br>商 -41, 余り 0  | (27) $-78 \div 6$<br>商 -13, 余り 0  | (45) $43 \div 6$<br>商 7, 余り 1    | (63) $4 \div 9$<br>商 0, 余り 4     |
| (10) $14 \div 2$<br>商 7, 余り 0    | (28) $47 \div 10$<br>商 4, 余り 7    | (46) $59 \div 9$<br>商 6, 余り 5    | (64) $99 \div 10$<br>商 9, 余り 9   |
| (11) $-95 \div 7$<br>商 -14, 余り 3 | (29) $27 \div 10$<br>商 2, 余り 7    | (47) $92 \div 7$<br>商 13, 余り 1   | (65) $-22 \div 7$<br>商 -4, 余り 6  |
| (12) $-13 \div 9$<br>商 -2, 余り 5  | (30) $-10 \div 5$<br>商 -2, 余り 0   | (48) $19 \div 6$<br>商 3, 余り 1    | (66) $-38 \div 5$<br>商 -8, 余り 2  |
| (13) $34 \div 7$<br>商 4, 余り 6    | (31) $-96 \div 10$<br>商 -10, 余り 4 | (49) $-89 \div 5$<br>商 -18, 余り 1 | (67) $-26 \div 5$<br>商 -6, 余り 4  |
| (14) $69 \div 3$<br>商 23, 余り 0   | (32) $79 \div 10$<br>商 7, 余り 9    | (50) $-69 \div 5$<br>商 -14, 余り 1 | (68) $56 \div 10$<br>商 5, 余り 6   |
| (15) $92 \div 1$<br>商 92, 余り 0   | (33) $52 \div 2$<br>商 26, 余り 0    | (51) $-27 \div 5$<br>商 -6, 余り 3  | (69) $-6 \div 4$<br>商 -2, 余り 2   |
| (16) $-30 \div 9$<br>商 -4, 余り 6  | (34) $94 \div 7$<br>商 13, 余り 3    | (52) $81 \div 4$<br>商 20, 余り 1   | (70) $48 \div 2$<br>商 24, 余り 0   |
| (17) $30 \div 2$<br>商 15, 余り 0   | (35) $-24 \div 3$<br>商 -8, 余り 0   | (53) $-35 \div 6$<br>商 -6, 余り 1  | (71) $-36 \div 10$<br>商 -4, 余り 4 |
| (18) $6 \div 9$<br>商 0, 余り 6     | (36) $78 \div 9$<br>商 8, 余り 6     | (54) $5 \div 7$<br>商 0, 余り 5     | (72) $-27 \div 10$<br>商 -3, 余り 3 |



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|---------------------------------------|----------------------------------------|---------------------------------------|---------------------------------------|
| (1) $-37 \div 9$<br>商 $-5$ , 余り $8$   | (19) $-58 \div 10$<br>商 $-6$ , 余り $2$  | (37) $74 \div 10$<br>商 $7$ , 余り $4$   | (55) $-35 \div 3$<br>商 $-12$ , 余り $1$ |
| (2) $-75 \div 8$<br>商 $-10$ , 余り $5$  | (20) $78 \div 9$<br>商 $8$ , 余り $6$     | (38) $62 \div 1$<br>商 $62$ , 余り $0$   | (56) $87 \div 6$<br>商 $14$ , 余り $3$   |
| (3) $94 \div 7$<br>商 $13$ , 余り $3$    | (21) $-11 \div 8$<br>商 $-2$ , 余り $5$   | (39) $78 \div 8$<br>商 $9$ , 余り $6$    | (57) $-60 \div 6$<br>商 $-10$ , 余り $0$ |
| (4) $45 \div 9$<br>商 $5$ , 余り $0$     | (22) $33 \div 3$<br>商 $11$ , 余り $0$    | (40) $15 \div 6$<br>商 $2$ , 余り $3$    | (58) $-5 \div 1$<br>商 $-5$ , 余り $0$   |
| (5) $5 \div 2$<br>商 $2$ , 余り $1$      | (23) $16 \div 10$<br>商 $1$ , 余り $6$    | (41) $-50 \div 1$<br>商 $-50$ , 余り $0$ | (59) $31 \div 8$<br>商 $3$ , 余り $7$    |
| (6) $-19 \div 5$<br>商 $-4$ , 余り $1$   | (24) $-9 \div 6$<br>商 $-2$ , 余り $3$    | (42) $-60 \div 4$<br>商 $-15$ , 余り $0$ | (60) $-72 \div 7$<br>商 $-11$ , 余り $5$ |
| (7) $91 \div 8$<br>商 $11$ , 余り $3$    | (25) $66 \div 7$<br>商 $9$ , 余り $3$     | (43) $2 \div 3$<br>商 $0$ , 余り $2$     | (61) $58 \div 6$<br>商 $9$ , 余り $4$    |
| (8) $65 \div 4$<br>商 $16$ , 余り $1$    | (26) $-36 \div 4$<br>商 $-9$ , 余り $0$   | (44) $80 \div 2$<br>商 $40$ , 余り $0$   | (62) $-38 \div 2$<br>商 $-19$ , 余り $0$ |
| (9) $39 \div 2$<br>商 $19$ , 余り $1$    | (27) $-14 \div 10$<br>商 $-2$ , 余り $6$  | (45) $8 \div 1$<br>商 $8$ , 余り $0$     | (63) $89 \div 3$<br>商 $29$ , 余り $2$   |
| (10) $-45 \div 6$<br>商 $-8$ , 余り $3$  | (28) $-78 \div 10$<br>商 $-8$ , 余り $2$  | (46) $79 \div 1$<br>商 $79$ , 余り $0$   | (64) $-31 \div 1$<br>商 $-31$ , 余り $0$ |
| (11) $27 \div 8$<br>商 $3$ , 余り $3$    | (29) $-100 \div 7$<br>商 $-15$ , 余り $5$ | (47) $68 \div 1$<br>商 $68$ , 余り $0$   | (65) $-65 \div 5$<br>商 $-13$ , 余り $0$ |
| (12) $-87 \div 6$<br>商 $-15$ , 余り $3$ | (30) $-18 \div 7$<br>商 $-3$ , 余り $3$   | (48) $-41 \div 6$<br>商 $-7$ , 余り $1$  | (66) $16 \div 10$<br>商 $1$ , 余り $6$   |
| (13) $-12 \div 3$<br>商 $-4$ , 余り $0$  | (31) $85 \div 6$<br>商 $14$ , 余り $1$    | (49) $-17 \div 7$<br>商 $-3$ , 余り $4$  | (67) $-20 \div 4$<br>商 $-5$ , 余り $0$  |
| (14) $5 \div 3$<br>商 $1$ , 余り $2$     | (32) $85 \div 2$<br>商 $42$ , 余り $1$    | (50) $-85 \div 6$<br>商 $-15$ , 余り $5$ | (68) $39 \div 3$<br>商 $13$ , 余り $0$   |
| (15) $-61 \div 6$<br>商 $-11$ , 余り $5$ | (33) $-1 \div 7$<br>商 $-1$ , 余り $6$    | (51) $-25 \div 2$<br>商 $-13$ , 余り $1$ | (69) $-33 \div 3$<br>商 $-11$ , 余り $0$ |
| (16) $60 \div 7$<br>商 $8$ , 余り $4$    | (34) $-86 \div 9$<br>商 $-10$ , 余り $4$  | (52) $-2 \div 6$<br>商 $-1$ , 余り $4$   | (70) $41 \div 9$<br>商 $4$ , 余り $5$    |
| (17) $20 \div 1$<br>商 $20$ , 余り $0$   | (35) $82 \div 4$<br>商 $20$ , 余り $2$    | (53) $34 \div 5$<br>商 $6$ , 余り $4$    | (71) $-26 \div 8$<br>商 $-4$ , 余り $6$  |
| (18) $-61 \div 2$<br>商 $-31$ , 余り $1$ | (36) $-8 \div 6$<br>商 $-2$ , 余り $4$    | (54) $-23 \div 10$<br>商 $-3$ , 余り $7$ | (72) $76 \div 2$<br>商 $38$ , 余り $0$   |

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|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| (1) $-72 \div 5$<br>商 $-15$ , 余り $3$  | (19) $-17 \div 4$<br>商 $-5$ , 余り $3$  | (37) $-30 \div 3$<br>商 $-10$ , 余り $0$ | (55) $39 \div 6$<br>商 $6$ , 余り $3$    |
| (2) $96 \div 1$<br>商 $96$ , 余り $0$    | (20) $-3 \div 9$<br>商 $-1$ , 余り $6$   | (38) $89 \div 8$<br>商 $11$ , 余り $1$   | (56) $36 \div 10$<br>商 $3$ , 余り $6$   |
| (3) $68 \div 4$<br>商 $17$ , 余り $0$    | (21) $-66 \div 5$<br>商 $-14$ , 余り $4$ | (39) $-24 \div 8$<br>商 $-3$ , 余り $0$  | (57) $48 \div 2$<br>商 $24$ , 余り $0$   |
| (4) $44 \div 2$<br>商 $22$ , 余り $0$    | (22) $33 \div 6$<br>商 $5$ , 余り $3$    | (40) $-90 \div 9$<br>商 $-10$ , 余り $0$ | (58) $55 \div 7$<br>商 $7$ , 余り $6$    |
| (5) $7 \div 1$<br>商 $7$ , 余り $0$      | (23) $99 \div 5$<br>商 $19$ , 余り $4$   | (41) $-32 \div 2$<br>商 $-16$ , 余り $0$ | (59) $71 \div 1$<br>商 $71$ , 余り $0$   |
| (6) $22 \div 4$<br>商 $5$ , 余り $2$     | (24) $-70 \div 7$<br>商 $-10$ , 余り $0$ | (42) $-67 \div 7$<br>商 $-10$ , 余り $3$ | (60) $-90 \div 8$<br>商 $-12$ , 余り $6$ |
| (7) $-30 \div 4$<br>商 $-8$ , 余り $2$   | (25) $58 \div 9$<br>商 $6$ , 余り $4$    | (43) $20 \div 2$<br>商 $10$ , 余り $0$   | (61) $75 \div 2$<br>商 $37$ , 余り $1$   |
| (8) $-8 \div 6$<br>商 $-2$ , 余り $4$    | (26) $12 \div 8$<br>商 $1$ , 余り $4$    | (44) $43 \div 1$<br>商 $43$ , 余り $0$   | (62) $-20 \div 9$<br>商 $-3$ , 余り $7$  |
| (9) $86 \div 7$<br>商 $12$ , 余り $2$    | (27) $-66 \div 9$<br>商 $-8$ , 余り $6$  | (45) $-78 \div 1$<br>商 $-78$ , 余り $0$ | (63) $-83 \div 7$<br>商 $-12$ , 余り $1$ |
| (10) $-86 \div 10$<br>商 $-9$ , 余り $4$ | (28) $-75 \div 9$<br>商 $-9$ , 余り $6$  | (46) $12 \div 2$<br>商 $6$ , 余り $0$    | (64) $66 \div 10$<br>商 $6$ , 余り $6$   |
| (11) $-68 \div 6$<br>商 $-12$ , 余り $4$ | (29) $63 \div 8$<br>商 $7$ , 余り $7$    | (47) $74 \div 10$<br>商 $7$ , 余り $4$   | (65) $-12 \div 4$<br>商 $-3$ , 余り $0$  |
| (12) $-44 \div 9$<br>商 $-5$ , 余り $1$  | (30) $-15 \div 5$<br>商 $-3$ , 余り $0$  | (48) $10 \div 5$<br>商 $2$ , 余り $0$    | (66) $-84 \div 2$<br>商 $-42$ , 余り $0$ |
| (13) $63 \div 7$<br>商 $9$ , 余り $0$    | (31) $-4 \div 4$<br>商 $-1$ , 余り $0$   | (49) $44 \div 10$<br>商 $4$ , 余り $4$   | (67) $98 \div 10$<br>商 $9$ , 余り $8$   |
| (14) $67 \div 8$<br>商 $8$ , 余り $3$    | (32) $-54 \div 3$<br>商 $-18$ , 余り $0$ | (50) $-97 \div 4$<br>商 $-25$ , 余り $3$ | (68) $-20 \div 1$<br>商 $-20$ , 余り $0$ |
| (15) $-43 \div 8$<br>商 $-6$ , 余り $5$  | (33) $77 \div 10$<br>商 $7$ , 余り $7$   | (51) $-18 \div 2$<br>商 $-9$ , 余り $0$  | (69) $39 \div 1$<br>商 $39$ , 余り $0$   |
| (16) $25 \div 2$<br>商 $12$ , 余り $1$   | (34) $20 \div 6$<br>商 $3$ , 余り $2$    | (52) $50 \div 8$<br>商 $6$ , 余り $2$    | (70) $-78 \div 7$<br>商 $-12$ , 余り $6$ |
| (17) $4 \div 8$<br>商 $0$ , 余り $4$     | (35) $-4 \div 3$<br>商 $-2$ , 余り $2$   | (53) $39 \div 3$<br>商 $13$ , 余り $0$   | (71) $72 \div 1$<br>商 $72$ , 余り $0$   |
| (18) $-4 \div 8$<br>商 $-1$ , 余り $4$   | (36) $-33 \div 3$<br>商 $-11$ , 余り $0$ | (54) $87 \div 6$<br>商 $14$ , 余り $3$   | (72) $-45 \div 3$<br>商 $-15$ , 余り $0$ |

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|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| (1) $-56 \div 6$<br>商 $-10$ , 余り $4$  | (19) $-69 \div 10$<br>商 $-7$ , 余り $1$ | (37) $-70 \div 4$<br>商 $-18$ , 余り $2$ | (55) $-36 \div 9$<br>商 $-4$ , 余り $0$  |
| (2) $92 \div 1$<br>商 $92$ , 余り $0$    | (20) $-34 \div 3$<br>商 $-12$ , 余り $2$ | (38) $45 \div 8$<br>商 $5$ , 余り $5$    | (56) $73 \div 2$<br>商 $36$ , 余り $1$   |
| (3) $-16 \div 8$<br>商 $-2$ , 余り $0$   | (21) $34 \div 4$<br>商 $8$ , 余り $2$    | (39) $3 \div 2$<br>商 $1$ , 余り $1$     | (57) $-56 \div 5$<br>商 $-12$ , 余り $4$ |
| (4) $-34 \div 4$<br>商 $-9$ , 余り $2$   | (22) $59 \div 1$<br>商 $59$ , 余り $0$   | (40) $-40 \div 2$<br>商 $-20$ , 余り $0$ | (58) $-81 \div 3$<br>商 $-27$ , 余り $0$ |
| (5) $-76 \div 4$<br>商 $-19$ , 余り $0$  | (23) $-91 \div 5$<br>商 $-19$ , 余り $4$ | (41) $41 \div 5$<br>商 $8$ , 余り $1$    | (59) $93 \div 6$<br>商 $15$ , 余り $3$   |
| (6) $35 \div 8$<br>商 $4$ , 余り $3$     | (24) $-78 \div 10$<br>商 $-8$ , 余り $2$ | (42) $9 \div 1$<br>商 $9$ , 余り $0$     | (60) $-67 \div 3$<br>商 $-23$ , 余り $2$ |
| (7) $-82 \div 8$<br>商 $-11$ , 余り $6$  | (25) $71 \div 2$<br>商 $35$ , 余り $1$   | (43) $-6 \div 5$<br>商 $-2$ , 余り $4$   | (61) $-48 \div 2$<br>商 $-24$ , 余り $0$ |
| (8) $-33 \div 5$<br>商 $-7$ , 余り $2$   | (26) $-51 \div 4$<br>商 $-13$ , 余り $1$ | (44) $15 \div 10$<br>商 $1$ , 余り $5$   | (62) $0 \div 6$<br>商 $0$ , 余り $0$     |
| (9) $-96 \div 9$<br>商 $-11$ , 余り $3$  | (27) $-67 \div 1$<br>商 $-67$ , 余り $0$ | (45) $-71 \div 2$<br>商 $-36$ , 余り $1$ | (63) $47 \div 1$<br>商 $47$ , 余り $0$   |
| (10) $-92 \div 7$<br>商 $-14$ , 余り $6$ | (28) $71 \div 4$<br>商 $17$ , 余り $3$   | (46) $31 \div 5$<br>商 $6$ , 余り $1$    | (64) $-70 \div 8$<br>商 $-9$ , 余り $2$  |
| (11) $26 \div 5$<br>商 $5$ , 余り $1$    | (29) $-88 \div 4$<br>商 $-22$ , 余り $0$ | (47) $-73 \div 5$<br>商 $-15$ , 余り $2$ | (65) $12 \div 8$<br>商 $1$ , 余り $4$    |
| (12) $-45 \div 10$<br>商 $-5$ , 余り $5$ | (30) $-20 \div 3$<br>商 $-7$ , 余り $1$  | (48) $15 \div 7$<br>商 $2$ , 余り $1$    | (66) $-10 \div 7$<br>商 $-2$ , 余り $4$  |
| (13) $-60 \div 2$<br>商 $-30$ , 余り $0$ | (31) $-77 \div 3$<br>商 $-26$ , 余り $1$ | (49) $-92 \div 6$<br>商 $-16$ , 余り $4$ | (67) $-43 \div 4$<br>商 $-11$ , 余り $1$ |
| (14) $-70 \div 2$<br>商 $-35$ , 余り $0$ | (32) $-46 \div 3$<br>商 $-16$ , 余り $2$ | (50) $-94 \div 2$<br>商 $-47$ , 余り $0$ | (68) $-82 \div 6$<br>商 $-14$ , 余り $2$ |
| (15) $4 \div 1$<br>商 $4$ , 余り $0$     | (33) $57 \div 10$<br>商 $5$ , 余り $7$   | (51) $79 \div 1$<br>商 $79$ , 余り $0$   | (69) $51 \div 9$<br>商 $5$ , 余り $6$    |
| (16) $89 \div 9$<br>商 $9$ , 余り $8$    | (34) $23 \div 6$<br>商 $3$ , 余り $5$    | (52) $89 \div 4$<br>商 $22$ , 余り $1$   | (70) $100 \div 9$<br>商 $11$ , 余り $1$  |
| (17) $63 \div 6$<br>商 $10$ , 余り $3$   | (35) $-53 \div 9$<br>商 $-6$ , 余り $1$  | (53) $33 \div 5$<br>商 $6$ , 余り $3$    | (71) $29 \div 2$<br>商 $14$ , 余り $1$   |
| (18) $1 \div 5$<br>商 $0$ , 余り $1$     | (36) $30 \div 5$<br>商 $6$ , 余り $0$    | (54) $31 \div 2$<br>商 $15$ , 余り $1$   | (72) $75 \div 3$<br>商 $25$ , 余り $0$   |

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|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| (1) $48 \div 9$<br>商 5, 余り 3      | (19) $-66 \div 3$<br>商 - 22, 余り 0 | (37) $-16 \div 6$<br>商 - 3, 余り 2  | (55) $96 \div 5$<br>商 19, 余り 1    |
| (2) $-32 \div 10$<br>商 - 4, 余り 8  | (20) $71 \div 6$<br>商 11, 余り 5    | (38) $-63 \div 9$<br>商 - 7, 余り 0  | (56) $45 \div 2$<br>商 22, 余り 1    |
| (3) $-1 \div 7$<br>商 - 1, 余り 6    | (21) $88 \div 6$<br>商 14, 余り 4    | (39) $-55 \div 1$<br>商 - 55, 余り 0 | (57) $67 \div 1$<br>商 67, 余り 0    |
| (4) $77 \div 6$<br>商 12, 余り 5     | (22) $-31 \div 10$<br>商 - 4, 余り 9 | (40) $87 \div 5$<br>商 17, 余り 2    | (58) $4 \div 4$<br>商 1, 余り 0      |
| (5) $53 \div 7$<br>商 7, 余り 4      | (23) $76 \div 9$<br>商 8, 余り 4     | (41) $-12 \div 7$<br>商 - 2, 余り 2  | (59) $69 \div 8$<br>商 8, 余り 5     |
| (6) $42 \div 10$<br>商 4, 余り 2     | (24) $74 \div 1$<br>商 74, 余り 0    | (42) $-12 \div 9$<br>商 - 2, 余り 6  | (60) $70 \div 5$<br>商 14, 余り 0    |
| (7) $8 \div 5$<br>商 1, 余り 3       | (25) $13 \div 10$<br>商 1, 余り 3    | (43) $67 \div 8$<br>商 8, 余り 3     | (61) $31 \div 7$<br>商 4, 余り 3     |
| (8) $42 \div 6$<br>商 7, 余り 0      | (26) $-73 \div 5$<br>商 - 15, 余り 2 | (44) $-88 \div 3$<br>商 - 30, 余り 2 | (62) $-14 \div 2$<br>商 - 7, 余り 0  |
| (9) $-33 \div 3$<br>商 - 11, 余り 0  | (27) $73 \div 5$<br>商 14, 余り 3    | (45) $-61 \div 5$<br>商 - 13, 余り 4 | (63) $-63 \div 4$<br>商 - 16, 余り 1 |
| (10) $34 \div 7$<br>商 4, 余り 6     | (28) $-93 \div 6$<br>商 - 16, 余り 3 | (46) $82 \div 6$<br>商 13, 余り 4    | (64) $-66 \div 4$<br>商 - 17, 余り 2 |
| (11) $-59 \div 4$<br>商 - 15, 余り 1 | (29) $5 \div 9$<br>商 0, 余り 5      | (47) $-50 \div 7$<br>商 - 8, 余り 6  | (65) $30 \div 5$<br>商 6, 余り 0     |
| (12) $69 \div 10$<br>商 6, 余り 9    | (30) $68 \div 2$<br>商 34, 余り 0    | (48) $91 \div 1$<br>商 91, 余り 0    | (66) $8 \div 6$<br>商 1, 余り 2      |
| (13) $-75 \div 10$<br>商 - 8, 余り 5 | (31) $84 \div 7$<br>商 12, 余り 0    | (49) $-39 \div 5$<br>商 - 8, 余り 1  | (67) $-45 \div 2$<br>商 - 23, 余り 1 |
| (14) $4 \div 6$<br>商 0, 余り 4      | (32) $-86 \div 4$<br>商 - 22, 余り 2 | (50) $-67 \div 9$<br>商 - 8, 余り 5  | (68) $-20 \div 10$<br>商 - 2, 余り 0 |
| (15) $-22 \div 6$<br>商 - 4, 余り 2  | (33) $88 \div 10$<br>商 8, 余り 8    | (51) $0 \div 8$<br>商 0, 余り 0      | (69) $-99 \div 4$<br>商 - 25, 余り 1 |
| (16) $-55 \div 5$<br>商 - 11, 余り 0 | (34) $-15 \div 5$<br>商 - 3, 余り 0  | (52) $-32 \div 5$<br>商 - 7, 余り 3  | (70) $-83 \div 1$<br>商 - 83, 余り 0 |
| (17) $-33 \div 2$<br>商 - 17, 余り 1 | (35) $-22 \div 9$<br>商 - 3, 余り 5  | (53) $-22 \div 2$<br>商 - 11, 余り 0 | (71) $95 \div 2$<br>商 47, 余り 1    |
| (18) $-31 \div 9$<br>商 - 4, 余り 5  | (36) $-46 \div 4$<br>商 - 12, 余り 2 | (54) $-27 \div 6$<br>商 - 5, 余り 3  | (72) $95 \div 9$<br>商 10, 余り 5    |

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|---------------------------------------|-----------------------------------------|---------------------------------------|----------------------------------------|
| (1) $-62 \div 3$<br>商 $-21$ , 余り $1$  | (19) $-92 \div 1$<br>商 $-92$ , 余り $0$   | (37) $22 \div 10$<br>商 $2$ , 余り $2$   | (55) $93 \div 4$<br>商 $23$ , 余り $1$    |
| (2) $51 \div 4$<br>商 $12$ , 余り $3$    | (20) $36 \div 6$<br>商 $6$ , 余り $0$      | (38) $2 \div 6$<br>商 $0$ , 余り $2$     | (56) $88 \div 6$<br>商 $14$ , 余り $4$    |
| (3) $38 \div 5$<br>商 $7$ , 余り $3$     | (21) $62 \div 4$<br>商 $15$ , 余り $2$     | (39) $61 \div 2$<br>商 $30$ , 余り $1$   | (57) $62 \div 4$<br>商 $15$ , 余り $2$    |
| (4) $76 \div 8$<br>商 $9$ , 余り $4$     | (22) $-91 \div 4$<br>商 $-23$ , 余り $1$   | (40) $97 \div 1$<br>商 $97$ , 余り $0$   | (58) $-22 \div 1$<br>商 $-22$ , 余り $0$  |
| (5) $-53 \div 7$<br>商 $-8$ , 余り $3$   | (23) $3 \div 6$<br>商 $0$ , 余り $3$       | (41) $-59 \div 3$<br>商 $-20$ , 余り $1$ | (59) $38 \div 7$<br>商 $5$ , 余り $3$     |
| (6) $-49 \div 6$<br>商 $-9$ , 余り $5$   | (24) $67 \div 2$<br>商 $33$ , 余り $1$     | (42) $-15 \div 2$<br>商 $-8$ , 余り $1$  | (60) $-98 \div 5$<br>商 $-20$ , 余り $2$  |
| (7) $-62 \div 2$<br>商 $-31$ , 余り $0$  | (25) $68 \div 2$<br>商 $34$ , 余り $0$     | (43) $29 \div 1$<br>商 $29$ , 余り $0$   | (61) $-11 \div 5$<br>商 $-3$ , 余り $4$   |
| (8) $63 \div 3$<br>商 $21$ , 余り $0$    | (26) $87 \div 5$<br>商 $17$ , 余り $2$     | (44) $57 \div 5$<br>商 $11$ , 余り $2$   | (62) $-100 \div 4$<br>商 $-25$ , 余り $0$ |
| (9) $90 \div 5$<br>商 $18$ , 余り $0$    | (27) $89 \div 3$<br>商 $29$ , 余り $2$     | (45) $34 \div 7$<br>商 $4$ , 余り $6$    | (63) $3 \div 4$<br>商 $0$ , 余り $3$      |
| (10) $-56 \div 7$<br>商 $-8$ , 余り $0$  | (28) $3 \div 1$<br>商 $3$ , 余り $0$       | (46) $-15 \div 8$<br>商 $-2$ , 余り $1$  | (64) $93 \div 8$<br>商 $11$ , 余り $5$    |
| (11) $90 \div 7$<br>商 $12$ , 余り $6$   | (29) $-2 \div 8$<br>商 $-1$ , 余り $6$     | (47) $73 \div 4$<br>商 $18$ , 余り $1$   | (65) $-51 \div 7$<br>商 $-8$ , 余り $5$   |
| (12) $52 \div 5$<br>商 $10$ , 余り $2$   | (30) $-36 \div 9$<br>商 $-4$ , 余り $0$    | (48) $-99 \div 7$<br>商 $-15$ , 余り $6$ | (66) $-94 \div 8$<br>商 $-12$ , 余り $2$  |
| (13) $72 \div 7$<br>商 $10$ , 余り $2$   | (31) $-100 \div 1$<br>商 $-100$ , 余り $0$ | (49) $-38 \div 9$<br>商 $-5$ , 余り $7$  | (67) $43 \div 6$<br>商 $7$ , 余り $1$     |
| (14) $-77 \div 7$<br>商 $-11$ , 余り $0$ | (32) $6 \div 1$<br>商 $6$ , 余り $0$       | (50) $7 \div 1$<br>商 $7$ , 余り $0$     | (68) $-70 \div 8$<br>商 $-9$ , 余り $2$   |
| (15) $10 \div 1$<br>商 $10$ , 余り $0$   | (33) $-1 \div 2$<br>商 $-1$ , 余り $1$     | (51) $-49 \div 7$<br>商 $-7$ , 余り $0$  | (69) $-93 \div 7$<br>商 $-14$ , 余り $5$  |
| (16) $10 \div 4$<br>商 $2$ , 余り $2$    | (34) $59 \div 9$<br>商 $6$ , 余り $5$      | (52) $71 \div 5$<br>商 $14$ , 余り $1$   | (70) $15 \div 5$<br>商 $3$ , 余り $0$     |
| (17) $-51 \div 2$<br>商 $-26$ , 余り $1$ | (35) $-27 \div 7$<br>商 $-4$ , 余り $1$    | (53) $35 \div 8$<br>商 $4$ , 余り $3$    | (71) $-85 \div 7$<br>商 $-13$ , 余り $6$  |
| (18) $-24 \div 2$<br>商 $-12$ , 余り $0$ | (36) $-50 \div 8$<br>商 $-7$ , 余り $6$    | (54) $-92 \div 4$<br>商 $-23$ , 余り $0$ | (72) $-20 \div 8$<br>商 $-3$ , 余り $4$   |

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|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| (1) $-87 \div 8$<br>商 $-11$ , 余り $1$  | (19) $-15 \div 3$<br>商 $-5$ , 余り $0$  | (37) $-16 \div 8$<br>商 $-2$ , 余り $0$  | (55) $-61 \div 2$<br>商 $-31$ , 余り $1$ |
| (2) $50 \div 3$<br>商 $16$ , 余り $2$    | (20) $87 \div 4$<br>商 $21$ , 余り $3$   | (38) $-30 \div 7$<br>商 $-5$ , 余り $5$  | (56) $-61 \div 8$<br>商 $-8$ , 余り $3$  |
| (3) $-7 \div 4$<br>商 $-2$ , 余り $1$    | (21) $59 \div 6$<br>商 $9$ , 余り $5$    | (39) $64 \div 6$<br>商 $10$ , 余り $4$   | (57) $-93 \div 8$<br>商 $-12$ , 余り $3$ |
| (4) $71 \div 8$<br>商 $8$ , 余り $7$     | (22) $5 \div 10$<br>商 $0$ , 余り $5$    | (40) $-44 \div 9$<br>商 $-5$ , 余り $1$  | (58) $73 \div 3$<br>商 $24$ , 余り $1$   |
| (5) $-29 \div 6$<br>商 $-5$ , 余り $1$   | (23) $-16 \div 10$<br>商 $-2$ , 余り $4$ | (41) $15 \div 9$<br>商 $1$ , 余り $6$    | (59) $-17 \div 3$<br>商 $-6$ , 余り $1$  |
| (6) $29 \div 5$<br>商 $5$ , 余り $4$     | (24) $-82 \div 10$<br>商 $-9$ , 余り $8$ | (42) $58 \div 7$<br>商 $8$ , 余り $2$    | (60) $37 \div 7$<br>商 $5$ , 余り $2$    |
| (7) $56 \div 6$<br>商 $9$ , 余り $2$     | (25) $-16 \div 4$<br>商 $-4$ , 余り $0$  | (43) $-81 \div 9$<br>商 $-9$ , 余り $0$  | (61) $-74 \div 6$<br>商 $-13$ , 余り $4$ |
| (8) $-57 \div 6$<br>商 $-10$ , 余り $3$  | (26) $-46 \div 5$<br>商 $-10$ , 余り $4$ | (44) $-1 \div 6$<br>商 $-1$ , 余り $5$   | (62) $-82 \div 6$<br>商 $-14$ , 余り $2$ |
| (9) $96 \div 4$<br>商 $24$ , 余り $0$    | (27) $14 \div 2$<br>商 $7$ , 余り $0$    | (45) $78 \div 6$<br>商 $13$ , 余り $0$   | (63) $6 \div 2$<br>商 $3$ , 余り $0$     |
| (10) $-98 \div 4$<br>商 $-25$ , 余り $2$ | (28) $96 \div 9$<br>商 $10$ , 余り $6$   | (46) $-91 \div 7$<br>商 $-13$ , 余り $0$ | (64) $8 \div 7$<br>商 $1$ , 余り $1$     |
| (11) $2 \div 6$<br>商 $0$ , 余り $2$     | (29) $80 \div 5$<br>商 $16$ , 余り $0$   | (47) $70 \div 6$<br>商 $11$ , 余り $4$   | (65) $-91 \div 3$<br>商 $-31$ , 余り $2$ |
| (12) $92 \div 10$<br>商 $9$ , 余り $2$   | (30) $-56 \div 4$<br>商 $-14$ , 余り $0$ | (48) $-71 \div 8$<br>商 $-9$ , 余り $1$  | (66) $40 \div 1$<br>商 $40$ , 余り $0$   |
| (13) $57 \div 6$<br>商 $9$ , 余り $3$    | (31) $93 \div 4$<br>商 $23$ , 余り $1$   | (49) $-98 \div 8$<br>商 $-13$ , 余り $6$ | (67) $81 \div 2$<br>商 $40$ , 余り $1$   |
| (14) $-42 \div 3$<br>商 $-14$ , 余り $0$ | (32) $-96 \div 8$<br>商 $-12$ , 余り $0$ | (50) $42 \div 9$<br>商 $4$ , 余り $6$    | (68) $-31 \div 6$<br>商 $-6$ , 余り $5$  |
| (15) $83 \div 10$<br>商 $8$ , 余り $3$   | (33) $79 \div 3$<br>商 $26$ , 余り $1$   | (51) $-43 \div 7$<br>商 $-7$ , 余り $6$  | (69) $65 \div 6$<br>商 $10$ , 余り $5$   |
| (16) $-44 \div 6$<br>商 $-8$ , 余り $4$  | (34) $-55 \div 2$<br>商 $-28$ , 余り $1$ | (52) $-83 \div 6$<br>商 $-14$ , 余り $1$ | (70) $7 \div 4$<br>商 $1$ , 余り $3$     |
| (17) $-42 \div 9$<br>商 $-5$ , 余り $3$  | (35) $-90 \div 5$<br>商 $-18$ , 余り $0$ | (53) $-80 \div 5$<br>商 $-16$ , 余り $0$ | (71) $89 \div 2$<br>商 $44$ , 余り $1$   |
| (18) $9 \div 1$<br>商 $9$ , 余り $0$     | (36) $89 \div 8$<br>商 $11$ , 余り $1$   | (54) $-71 \div 5$<br>商 $-15$ , 余り $4$ | (72) $-37 \div 10$<br>商 $-4$ , 余り $3$ |

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|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| (1) $29 \div 1$<br>商 29, 余り 0    | (19) $-51 \div 6$<br>商 -9, 余り 3  | (37) $63 \div 10$<br>商 6, 余り 3   | (55) $82 \div 1$<br>商 82, 余り 0   |
| (2) $53 \div 10$<br>商 5, 余り 3    | (20) $79 \div 6$<br>商 13, 余り 1   | (38) $74 \div 2$<br>商 37, 余り 0   | (56) $-12 \div 4$<br>商 -3, 余り 0  |
| (3) $-98 \div 4$<br>商 -25, 余り 2  | (21) $97 \div 7$<br>商 13, 余り 6   | (39) $46 \div 5$<br>商 9, 余り 1    | (57) $93 \div 3$<br>商 31, 余り 0   |
| (4) $-99 \div 10$<br>商 -10, 余り 1 | (22) $99 \div 7$<br>商 14, 余り 1   | (40) $73 \div 5$<br>商 14, 余り 3   | (58) $-28 \div 2$<br>商 -14, 余り 0 |
| (5) $3 \div 2$<br>商 1, 余り 1      | (23) $63 \div 6$<br>商 10, 余り 3   | (41) $55 \div 9$<br>商 6, 余り 1    | (59) $-26 \div 3$<br>商 -9, 余り 1  |
| (6) $85 \div 7$<br>商 12, 余り 1    | (24) $50 \div 6$<br>商 8, 余り 2    | (42) $94 \div 4$<br>商 23, 余り 2   | (60) $-11 \div 1$<br>商 -11, 余り 0 |
| (7) $69 \div 2$<br>商 34, 余り 1    | (25) $-2 \div 9$<br>商 -1, 余り 7   | (43) $98 \div 2$<br>商 49, 余り 0   | (61) $31 \div 3$<br>商 10, 余り 1   |
| (8) $92 \div 9$<br>商 10, 余り 2    | (26) $59 \div 4$<br>商 14, 余り 3   | (44) $30 \div 1$<br>商 30, 余り 0   | (62) $8 \div 4$<br>商 2, 余り 0     |
| (9) $1 \div 10$<br>商 0, 余り 1     | (27) $-92 \div 6$<br>商 -16, 余り 4 | (45) $-14 \div 4$<br>商 -4, 余り 2  | (63) $-58 \div 1$<br>商 -58, 余り 0 |
| (10) $67 \div 8$<br>商 8, 余り 3    | (28) $5 \div 1$<br>商 5, 余り 0     | (46) $-96 \div 8$<br>商 -12, 余り 0 | (64) $-86 \div 6$<br>商 -15, 余り 4 |
| (11) $-36 \div 8$<br>商 -5, 余り 4  | (29) $-58 \div 6$<br>商 -10, 余り 2 | (47) $-74 \div 1$<br>商 -74, 余り 0 | (65) $-73 \div 8$<br>商 -10, 余り 7 |
| (12) $-71 \div 6$<br>商 -12, 余り 1 | (30) $-96 \div 7$<br>商 -14, 余り 2 | (48) $0 \div 6$<br>商 0, 余り 0     | (66) $-45 \div 6$<br>商 -8, 余り 3  |
| (13) $-75 \div 7$<br>商 -11, 余り 2 | (31) $9 \div 8$<br>商 1, 余り 1     | (49) $52 \div 10$<br>商 5, 余り 2   | (67) $-63 \div 3$<br>商 -21, 余り 0 |
| (14) $73 \div 4$<br>商 18, 余り 1   | (32) $64 \div 2$<br>商 32, 余り 0   | (50) $-86 \div 5$<br>商 -18, 余り 4 | (68) $-71 \div 4$<br>商 -18, 余り 1 |
| (15) $-6 \div 10$<br>商 -1, 余り 4  | (33) $82 \div 7$<br>商 11, 余り 5   | (51) $17 \div 4$<br>商 4, 余り 1    | (69) $49 \div 10$<br>商 4, 余り 9   |
| (16) $40 \div 4$<br>商 10, 余り 0   | (34) $51 \div 7$<br>商 7, 余り 2    | (52) $-69 \div 5$<br>商 -14, 余り 1 | (70) $-19 \div 10$<br>商 -2, 余り 1 |
| (17) $-1 \div 1$<br>商 -1, 余り 0   | (35) $83 \div 3$<br>商 27, 余り 2   | (53) $40 \div 3$<br>商 13, 余り 1   | (71) $69 \div 4$<br>商 17, 余り 1   |
| (18) $34 \div 6$<br>商 5, 余り 4    | (36) $95 \div 7$<br>商 13, 余り 4   | (54) $14 \div 6$<br>商 2, 余り 2    | (72) $-43 \div 10$<br>商 -5, 余り 7 |

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| (1) $86 \div 3$<br>商 28, 余り 2    | (19) $-99 \div 4$<br>商 -25, 余り 1 | (37) $40 \div 3$<br>商 13, 余り 1   | (55) $-29 \div 5$<br>商 -6, 余り 1  |
| (2) $71 \div 4$<br>商 17, 余り 3    | (20) $13 \div 4$<br>商 3, 余り 1    | (38) $59 \div 6$<br>商 9, 余り 5    | (56) $-6 \div 8$<br>商 -1, 余り 2   |
| (3) $35 \div 8$<br>商 4, 余り 3     | (21) $-37 \div 3$<br>商 -13, 余り 2 | (39) $-70 \div 2$<br>商 -35, 余り 0 | (57) $88 \div 7$<br>商 12, 余り 4   |
| (4) $78 \div 6$<br>商 13, 余り 0    | (22) $-92 \div 2$<br>商 -46, 余り 0 | (40) $18 \div 1$<br>商 18, 余り 0   | (58) $42 \div 10$<br>商 4, 余り 2   |
| (5) $-71 \div 9$<br>商 -8, 余り 1   | (23) $54 \div 2$<br>商 27, 余り 0   | (41) $-38 \div 9$<br>商 -5, 余り 7  | (59) $-29 \div 7$<br>商 -5, 余り 6  |
| (6) $-58 \div 2$<br>商 -29, 余り 0  | (24) $100 \div 4$<br>商 25, 余り 0  | (42) $-66 \div 2$<br>商 -33, 余り 0 | (60) $1 \div 6$<br>商 0, 余り 1     |
| (7) $58 \div 7$<br>商 8, 余り 2     | (25) $-79 \div 4$<br>商 -20, 余り 1 | (43) $-92 \div 7$<br>商 -14, 余り 6 | (61) $-45 \div 8$<br>商 -6, 余り 3  |
| (8) $-77 \div 7$<br>商 -11, 余り 0  | (26) $-18 \div 1$<br>商 -18, 余り 0 | (44) $47 \div 2$<br>商 23, 余り 1   | (62) $81 \div 6$<br>商 13, 余り 3   |
| (9) $3 \div 2$<br>商 1, 余り 1      | (27) $53 \div 7$<br>商 7, 余り 4    | (45) $-75 \div 9$<br>商 -9, 余り 6  | (63) $70 \div 9$<br>商 7, 余り 7    |
| (10) $97 \div 8$<br>商 12, 余り 1   | (28) $-98 \div 4$<br>商 -25, 余り 2 | (46) $-92 \div 6$<br>商 -16, 余り 4 | (64) $42 \div 6$<br>商 7, 余り 0    |
| (11) $-62 \div 7$<br>商 -9, 余り 1  | (29) $43 \div 1$<br>商 43, 余り 0   | (47) $92 \div 10$<br>商 9, 余り 2   | (65) $-99 \div 1$<br>商 -99, 余り 0 |
| (12) $-79 \div 7$<br>商 -12, 余り 5 | (30) $-90 \div 8$<br>商 -12, 余り 6 | (48) $-32 \div 3$<br>商 -11, 余り 1 | (66) $15 \div 3$<br>商 5, 余り 0    |
| (13) $-48 \div 2$<br>商 -24, 余り 0 | (31) $44 \div 1$<br>商 44, 余り 0   | (49) $9 \div 1$<br>商 9, 余り 0     | (67) $27 \div 5$<br>商 5, 余り 2    |
| (14) $90 \div 10$<br>商 9, 余り 0   | (32) $42 \div 2$<br>商 21, 余り 0   | (50) $32 \div 7$<br>商 4, 余り 4    | (68) $56 \div 1$<br>商 56, 余り 0   |
| (15) $76 \div 4$<br>商 19, 余り 0   | (33) $-68 \div 4$<br>商 -17, 余り 0 | (51) $-10 \div 3$<br>商 -4, 余り 2  | (69) $-45 \div 2$<br>商 -23, 余り 1 |
| (16) $-85 \div 1$<br>商 -85, 余り 0 | (34) $25 \div 4$<br>商 6, 余り 1    | (52) $88 \div 5$<br>商 17, 余り 3   | (70) $-17 \div 2$<br>商 -9, 余り 1  |
| (17) $-13 \div 10$<br>商 -2, 余り 7 | (35) $3 \div 6$<br>商 0, 余り 3     | (53) $-85 \div 4$<br>商 -22, 余り 3 | (71) $29 \div 4$<br>商 7, 余り 1    |
| (18) $12 \div 1$<br>商 12, 余り 0   | (36) $-60 \div 10$<br>商 -6, 余り 0 | (54) $92 \div 2$<br>商 46, 余り 0   | (72) $-24 \div 8$<br>商 -3, 余り 0  |



- |                                  |                                  |                                  |                                  |
|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| (1) $35 \div 10$<br>商 3, 余り 5    | (19) $-68 \div 5$<br>商 -14, 余り 2 | (37) $90 \div 3$<br>商 30, 余り 0   | (55) $-44 \div 5$<br>商 -9, 余り 1  |
| (2) $38 \div 10$<br>商 3, 余り 8    | (20) $-87 \div 8$<br>商 -11, 余り 1 | (38) $28 \div 4$<br>商 7, 余り 0    | (56) $41 \div 4$<br>商 10, 余り 1   |
| (3) $-62 \div 3$<br>商 -21, 余り 1  | (21) $82 \div 7$<br>商 11, 余り 5   | (39) $1 \div 4$<br>商 0, 余り 1     | (57) $-53 \div 3$<br>商 -18, 余り 1 |
| (4) $-51 \div 2$<br>商 -26, 余り 1  | (22) $-2 \div 4$<br>商 -1, 余り 2   | (40) $-35 \div 8$<br>商 -5, 余り 5  | (58) $-52 \div 1$<br>商 -52, 余り 0 |
| (5) $65 \div 5$<br>商 13, 余り 0    | (23) $-97 \div 1$<br>商 -97, 余り 0 | (41) $-97 \div 3$<br>商 -33, 余り 2 | (59) $18 \div 4$<br>商 4, 余り 2    |
| (6) $-87 \div 4$<br>商 -22, 余り 1  | (24) $-27 \div 10$<br>商 -3, 余り 3 | (42) $-32 \div 9$<br>商 -4, 余り 4  | (60) $-28 \div 9$<br>商 -4, 余り 8  |
| (7) $84 \div 5$<br>商 16, 余り 4    | (25) $92 \div 7$<br>商 13, 余り 1   | (43) $34 \div 10$<br>商 3, 余り 4   | (61) $72 \div 2$<br>商 36, 余り 0   |
| (8) $25 \div 5$<br>商 5, 余り 0     | (26) $-89 \div 9$<br>商 -10, 余り 1 | (44) $-10 \div 6$<br>商 -2, 余り 2  | (62) $-72 \div 10$<br>商 -8, 余り 8 |
| (9) $-87 \div 5$<br>商 -18, 余り 3  | (27) $2 \div 8$<br>商 0, 余り 2     | (45) $-52 \div 7$<br>商 -8, 余り 4  | (63) $-8 \div 7$<br>商 -2, 余り 6   |
| (10) $91 \div 5$<br>商 18, 余り 1   | (28) $-77 \div 5$<br>商 -16, 余り 3 | (46) $-55 \div 8$<br>商 -7, 余り 1  | (64) $9 \div 7$<br>商 1, 余り 2     |
| (11) $-82 \div 2$<br>商 -41, 余り 0 | (29) $-33 \div 8$<br>商 -5, 余り 7  | (47) $-74 \div 7$<br>商 -11, 余り 3 | (65) $-9 \div 4$<br>商 -3, 余り 3   |
| (12) $-82 \div 2$<br>商 -41, 余り 0 | (30) $50 \div 9$<br>商 5, 余り 5    | (48) $-92 \div 1$<br>商 -92, 余り 0 | (66) $-74 \div 1$<br>商 -74, 余り 0 |
| (13) $-66 \div 6$<br>商 -11, 余り 0 | (31) $-37 \div 9$<br>商 -5, 余り 8  | (49) $68 \div 6$<br>商 11, 余り 2   | (67) $-37 \div 10$<br>商 -4, 余り 3 |
| (14) $18 \div 6$<br>商 3, 余り 0    | (32) $-2 \div 8$<br>商 -1, 余り 6   | (50) $40 \div 10$<br>商 4, 余り 0   | (68) $-6 \div 5$<br>商 -2, 余り 4   |
| (15) $-86 \div 9$<br>商 -10, 余り 4 | (33) $83 \div 1$<br>商 83, 余り 0   | (51) $82 \div 3$<br>商 27, 余り 1   | (69) $44 \div 1$<br>商 44, 余り 0   |
| (16) $99 \div 5$<br>商 19, 余り 4   | (34) $92 \div 9$<br>商 10, 余り 2   | (52) $3 \div 1$<br>商 3, 余り 0     | (70) $41 \div 1$<br>商 41, 余り 0   |
| (17) $93 \div 10$<br>商 9, 余り 3   | (35) $43 \div 7$<br>商 6, 余り 1    | (53) $-77 \div 3$<br>商 -26, 余り 1 | (71) $35 \div 10$<br>商 3, 余り 5   |
| (18) $-34 \div 4$<br>商 -9, 余り 2  | (36) $-22 \div 4$<br>商 -6, 余り 2  | (54) $-81 \div 2$<br>商 -41, 余り 1 | (72) $47 \div 9$<br>商 5, 余り 2    |

- |                                  |                                  |                                  |                                  |
|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| (1) $84 \div 1$<br>商 84, 余り 0    | (19) $69 \div 6$<br>商 11, 余り 3   | (37) $42 \div 2$<br>商 21, 余り 0   | (55) $-17 \div 3$<br>商 -6, 余り 1  |
| (2) $-14 \div 3$<br>商 -5, 余り 1   | (20) $-20 \div 2$<br>商 -10, 余り 0 | (38) $37 \div 9$<br>商 4, 余り 1    | (56) $-75 \div 5$<br>商 -15, 余り 0 |
| (3) $86 \div 8$<br>商 10, 余り 6    | (21) $46 \div 8$<br>商 5, 余り 6    | (39) $-38 \div 4$<br>商 -10, 余り 2 | (57) $-2 \div 1$<br>商 -2, 余り 0   |
| (4) $-89 \div 6$<br>商 -15, 余り 1  | (22) $-86 \div 2$<br>商 -43, 余り 0 | (40) $-36 \div 4$<br>商 -9, 余り 0  | (58) $5 \div 5$<br>商 1, 余り 0     |
| (5) $5 \div 8$<br>商 0, 余り 5      | (23) $86 \div 9$<br>商 9, 余り 5    | (41) $15 \div 8$<br>商 1, 余り 7    | (59) $21 \div 7$<br>商 3, 余り 0    |
| (6) $-67 \div 4$<br>商 -17, 余り 1  | (24) $89 \div 9$<br>商 9, 余り 8    | (42) $56 \div 9$<br>商 6, 余り 2    | (60) $20 \div 7$<br>商 2, 余り 6    |
| (7) $89 \div 7$<br>商 12, 余り 5    | (25) $18 \div 7$<br>商 2, 余り 4    | (43) $-57 \div 10$<br>商 -6, 余り 3 | (61) $-1 \div 7$<br>商 -1, 余り 6   |
| (8) $82 \div 5$<br>商 16, 余り 2    | (26) $-56 \div 7$<br>商 -8, 余り 0  | (44) $-72 \div 6$<br>商 -12, 余り 0 | (62) $87 \div 10$<br>商 8, 余り 7   |
| (9) $-5 \div 7$<br>商 -1, 余り 2    | (27) $-81 \div 9$<br>商 -9, 余り 0  | (45) $-79 \div 3$<br>商 -27, 余り 2 | (63) $-9 \div 8$<br>商 -2, 余り 7   |
| (10) $22 \div 1$<br>商 22, 余り 0   | (28) $-88 \div 4$<br>商 -22, 余り 0 | (46) $-36 \div 4$<br>商 -9, 余り 0  | (64) $-9 \div 2$<br>商 -5, 余り 1   |
| (11) $96 \div 5$<br>商 19, 余り 1   | (29) $-91 \div 9$<br>商 -11, 余り 8 | (47) $12 \div 1$<br>商 12, 余り 0   | (65) $-57 \div 10$<br>商 -6, 余り 3 |
| (12) $-87 \div 2$<br>商 -44, 余り 1 | (30) $100 \div 7$<br>商 14, 余り 2  | (48) $43 \div 9$<br>商 4, 余り 7    | (66) $-51 \div 10$<br>商 -6, 余り 9 |
| (13) $-31 \div 10$<br>商 -4, 余り 9 | (31) $-92 \div 3$<br>商 -31, 余り 1 | (49) $-62 \div 2$<br>商 -31, 余り 0 | (67) $-40 \div 8$<br>商 -5, 余り 0  |
| (14) $46 \div 6$<br>商 7, 余り 4    | (32) $36 \div 4$<br>商 9, 余り 0    | (50) $56 \div 5$<br>商 11, 余り 1   | (68) $63 \div 8$<br>商 7, 余り 7    |
| (15) $62 \div 6$<br>商 10, 余り 2   | (33) $-73 \div 3$<br>商 -25, 余り 2 | (51) $2 \div 4$<br>商 0, 余り 2     | (69) $-16 \div 5$<br>商 -4, 余り 4  |
| (16) $47 \div 6$<br>商 7, 余り 5    | (34) $63 \div 1$<br>商 63, 余り 0   | (52) $-26 \div 10$<br>商 -3, 余り 4 | (70) $73 \div 4$<br>商 18, 余り 1   |
| (17) $57 \div 2$<br>商 28, 余り 1   | (35) $15 \div 2$<br>商 7, 余り 1    | (53) $-65 \div 9$<br>商 -8, 余り 7  | (71) $39 \div 1$<br>商 39, 余り 0   |
| (18) $21 \div 8$<br>商 2, 余り 5    | (36) $-39 \div 5$<br>商 -8, 余り 1  | (54) $75 \div 9$<br>商 8, 余り 3    | (72) $98 \div 9$<br>商 10, 余り 8   |

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|-----------------------------------|----------------------------------|----------------------------------|------------------------------------|
| (1) $37 \div 7$<br>商 5, 余り 2      | (19) $81 \div 10$<br>商 8, 余り 1   | (37) $30 \div 1$<br>商 30, 余り 0   | (55) $-17 \div 4$<br>商 -5, 余り 3    |
| (2) $71 \div 5$<br>商 14, 余り 1     | (20) $11 \div 10$<br>商 1, 余り 1   | (38) $-13 \div 2$<br>商 -7, 余り 1  | (56) $7 \div 7$<br>商 1, 余り 0       |
| (3) $5 \div 2$<br>商 2, 余り 1       | (21) $-10 \div 2$<br>商 -5, 余り 0  | (39) $-37 \div 7$<br>商 -6, 余り 5  | (57) $9 \div 8$<br>商 1, 余り 1       |
| (4) $30 \div 4$<br>商 7, 余り 2      | (22) $74 \div 7$<br>商 10, 余り 4   | (40) $-8 \div 10$<br>商 -1, 余り 2  | (58) $-100 \div 10$<br>商 -10, 余り 0 |
| (5) $50 \div 9$<br>商 5, 余り 5      | (23) $-92 \div 4$<br>商 -23, 余り 0 | (41) $97 \div 4$<br>商 24, 余り 1   | (59) $-28 \div 1$<br>商 -28, 余り 0   |
| (6) $-78 \div 8$<br>商 -10, 余り 2   | (24) $-50 \div 7$<br>商 -8, 余り 6  | (42) $-62 \div 6$<br>商 -11, 余り 4 | (60) $76 \div 1$<br>商 76, 余り 0     |
| (7) $97 \div 6$<br>商 16, 余り 1     | (25) $-68 \div 9$<br>商 -8, 余り 4  | (43) $13 \div 4$<br>商 3, 余り 1    | (61) $-61 \div 3$<br>商 -21, 余り 2   |
| (8) $10 \div 5$<br>商 2, 余り 0      | (26) $-17 \div 8$<br>商 -3, 余り 7  | (44) $84 \div 7$<br>商 12, 余り 0   | (62) $99 \div 1$<br>商 99, 余り 0     |
| (9) $11 \div 1$<br>商 11, 余り 0     | (27) $70 \div 7$<br>商 10, 余り 0   | (45) $55 \div 5$<br>商 11, 余り 0   | (63) $-52 \div 4$<br>商 -13, 余り 0   |
| (10) $33 \div 10$<br>商 3, 余り 3    | (28) $88 \div 1$<br>商 88, 余り 0   | (46) $-59 \div 10$<br>商 -6, 余り 1 | (64) $30 \div 4$<br>商 7, 余り 2      |
| (11) $10 \div 4$<br>商 2, 余り 2     | (29) $97 \div 7$<br>商 13, 余り 6   | (47) $-79 \div 2$<br>商 -40, 余り 1 | (65) $-86 \div 10$<br>商 -9, 余り 4   |
| (12) $-70 \div 5$<br>商 -14, 余り 0  | (30) $63 \div 4$<br>商 15, 余り 3   | (48) $78 \div 6$<br>商 13, 余り 0   | (66) $81 \div 4$<br>商 20, 余り 1     |
| (13) $72 \div 1$<br>商 72, 余り 0    | (31) $-71 \div 5$<br>商 -15, 余り 4 | (49) $94 \div 7$<br>商 13, 余り 3   | (67) $-33 \div 4$<br>商 -9, 余り 3    |
| (14) $-97 \div 10$<br>商 -10, 余り 3 | (32) $32 \div 9$<br>商 3, 余り 5    | (50) $49 \div 5$<br>商 9, 余り 4    | (68) $-5 \div 1$<br>商 -5, 余り 0     |
| (15) $57 \div 6$<br>商 9, 余り 3     | (33) $-15 \div 10$<br>商 -2, 余り 5 | (51) $-19 \div 9$<br>商 -3, 余り 8  | (69) $21 \div 6$<br>商 3, 余り 3      |
| (16) $-63 \div 5$<br>商 -13, 余り 2  | (34) $23 \div 4$<br>商 5, 余り 3    | (52) $74 \div 1$<br>商 74, 余り 0   | (70) $41 \div 4$<br>商 10, 余り 1     |
| (17) $-28 \div 7$<br>商 -4, 余り 0   | (35) $41 \div 3$<br>商 13, 余り 2   | (53) $-25 \div 9$<br>商 -3, 余り 2  | (71) $47 \div 4$<br>商 11, 余り 3     |
| (18) $50 \div 8$<br>商 6, 余り 2     | (36) $58 \div 5$<br>商 11, 余り 3   | (54) $-82 \div 9$<br>商 -10, 余り 8 | (72) $-19 \div 1$<br>商 -19, 余り 0   |

- |                                  |                                  |                                  |                                   |
|----------------------------------|----------------------------------|----------------------------------|-----------------------------------|
| (1) $42 \div 3$<br>商 14, 余り 0    | (19) $-38 \div 3$<br>商 -13, 余り 1 | (37) $-36 \div 7$<br>商 -6, 余り 6  | (55) $-91 \div 10$<br>商 -10, 余り 9 |
| (2) $-57 \div 5$<br>商 -12, 余り 3  | (20) $61 \div 10$<br>商 6, 余り 1   | (38) $-42 \div 1$<br>商 -42, 余り 0 | (56) $35 \div 7$<br>商 5, 余り 0     |
| (3) $-3 \div 9$<br>商 -1, 余り 6    | (21) $88 \div 1$<br>商 88, 余り 0   | (39) $-5 \div 1$<br>商 -5, 余り 0   | (57) $84 \div 2$<br>商 42, 余り 0    |
| (4) $-38 \div 1$<br>商 -38, 余り 0  | (22) $55 \div 5$<br>商 11, 余り 0   | (40) $-36 \div 2$<br>商 -18, 余り 0 | (58) $-97 \div 6$<br>商 -17, 余り 5  |
| (5) $-32 \div 2$<br>商 -16, 余り 0  | (23) $-9 \div 4$<br>商 -3, 余り 3   | (41) $-84 \div 10$<br>商 -9, 余り 6 | (59) $89 \div 6$<br>商 14, 余り 5    |
| (6) $-73 \div 8$<br>商 -10, 余り 7  | (24) $56 \div 1$<br>商 56, 余り 0   | (42) $-7 \div 10$<br>商 -1, 余り 3  | (60) $96 \div 3$<br>商 32, 余り 0    |
| (7) $87 \div 1$<br>商 87, 余り 0    | (25) $-11 \div 9$<br>商 -2, 余り 7  | (43) $-41 \div 10$<br>商 -5, 余り 9 | (61) $55 \div 1$<br>商 55, 余り 0    |
| (8) $-81 \div 9$<br>商 -9, 余り 0   | (26) $-75 \div 10$<br>商 -8, 余り 5 | (44) $95 \div 4$<br>商 23, 余り 3   | (62) $-92 \div 1$<br>商 -92, 余り 0  |
| (9) $54 \div 7$<br>商 7, 余り 5     | (27) $-17 \div 9$<br>商 -2, 余り 1  | (45) $-28 \div 10$<br>商 -3, 余り 2 | (63) $-78 \div 10$<br>商 -8, 余り 2  |
| (10) $-32 \div 7$<br>商 -5, 余り 3  | (28) $92 \div 1$<br>商 92, 余り 0   | (46) $-78 \div 9$<br>商 -9, 余り 3  | (64) $61 \div 7$<br>商 8, 余り 5     |
| (11) $99 \div 3$<br>商 33, 余り 0   | (29) $93 \div 8$<br>商 11, 余り 5   | (47) $-93 \div 8$<br>商 -12, 余り 3 | (65) $40 \div 4$<br>商 10, 余り 0    |
| (12) $-55 \div 3$<br>商 -19, 余り 2 | (30) $-5 \div 10$<br>商 -1, 余り 5  | (48) $22 \div 5$<br>商 4, 余り 2    | (66) $59 \div 9$<br>商 6, 余り 5     |
| (13) $-17 \div 2$<br>商 -9, 余り 1  | (31) $-25 \div 7$<br>商 -4, 余り 3  | (49) $97 \div 4$<br>商 24, 余り 1   | (67) $-25 \div 8$<br>商 -4, 余り 7   |
| (14) $-32 \div 2$<br>商 -16, 余り 0 | (32) $-71 \div 1$<br>商 -71, 余り 0 | (50) $-6 \div 2$<br>商 -3, 余り 0   | (68) $85 \div 1$<br>商 85, 余り 0    |
| (15) $78 \div 8$<br>商 9, 余り 6    | (33) $56 \div 4$<br>商 14, 余り 0   | (51) $-79 \div 9$<br>商 -9, 余り 2  | (69) $44 \div 5$<br>商 8, 余り 4     |
| (16) $53 \div 2$<br>商 26, 余り 1   | (34) $-65 \div 4$<br>商 -17, 余り 3 | (52) $27 \div 3$<br>商 9, 余り 0    | (70) $64 \div 9$<br>商 7, 余り 1     |
| (17) $-78 \div 4$<br>商 -20, 余り 2 | (35) $5 \div 2$<br>商 2, 余り 1     | (53) $-97 \div 5$<br>商 -20, 余り 3 | (71) $96 \div 1$<br>商 96, 余り 0    |
| (18) $-19 \div 1$<br>商 -19, 余り 0 | (36) $100 \div 2$<br>商 50, 余り 0  | (54) $-53 \div 4$<br>商 -14, 余り 3 | (72) $-82 \div 10$<br>商 -9, 余り 8  |

## 1.3 有理数

### 1.3.1 加減法

有理数の和や差を求める問題。

(1)  $-3 + \left(-\frac{3}{4}\right)$

(11)  $\frac{3}{4} - \frac{8}{3}$

(21)  $2 + 3$

(31)  $\frac{1}{8} + \left(-\frac{1}{10}\right)$

(2)  $-\frac{10}{9} + \left(-\frac{2}{3}\right)$

(12)  $-5 - \frac{7}{4}$

(22)  $1 + 1$

(32)  $-\frac{1}{7} - \frac{3}{2}$

(3)  $-\frac{4}{7} - \left(-\frac{1}{2}\right)$

(13)  $-6 + \left(-\frac{1}{3}\right)$

(23)  $-\frac{8}{3} + 3$

(33)  $1 + \frac{8}{3}$

(4)  $9 - \left(-\frac{8}{7}\right)$

(14)  $-\frac{7}{4} + \frac{8}{3}$

(24)  $\frac{2}{5} + \left(-\frac{5}{3}\right)$

(34)  $-3 + \frac{2}{3}$

(5)  $\frac{3}{5} + (-3)$

(15)  $-\frac{3}{7} - \left(-\frac{5}{4}\right)$

(25)  $\frac{3}{5} - \frac{4}{3}$

(35)  $-2 - \frac{3}{7}$

(6)  $-1 - \frac{3}{4}$

(16)  $-\frac{4}{5} + (-1)$

(26)  $2 + \frac{8}{3}$

(36)  $\frac{7}{3} + (-1)$

(7)  $\frac{3}{10} + (-5)$

(17)  $\frac{1}{5} + \left(-\frac{4}{5}\right)$

(27)  $-\frac{4}{9} + \left(-\frac{4}{3}\right)$

(37)  $-\frac{5}{3} - \frac{1}{9}$

(8)  $-\frac{8}{7} + \frac{4}{3}$

(18)  $-\frac{5}{2} - \left(-\frac{7}{8}\right)$

(28)  $-\frac{2}{5} + \left(-\frac{1}{3}\right)$

(38)  $\frac{4}{5} + 7$

(9)  $-5 - \frac{2}{3}$

(19)  $-\frac{2}{3} + \frac{9}{4}$

(29)  $\frac{7}{8} - \frac{8}{9}$

(39)  $-\frac{2}{3} - \left(-\frac{1}{3}\right)$

(10)  $\frac{1}{7} - \frac{4}{3}$

(20)  $-\frac{1}{3} + \frac{3}{2}$

(30)  $-\frac{5}{6} + (-3)$

(40)  $-\frac{9}{5} + 1$

(1)  $\frac{5}{3} - (-3)$

(11)  $-\frac{1}{8} - \left(-\frac{1}{3}\right)$

(21)  $\frac{1}{6} + \left(-\frac{1}{7}\right)$

(31)  $-\frac{1}{3} + \left(-\frac{1}{2}\right)$

(2)  $-\frac{4}{9} + 10$

(12)  $-\frac{8}{5} - \frac{3}{2}$

(22)  $\frac{4}{7} - \frac{2}{3}$

(32)  $\frac{9}{8} - \left(-\frac{5}{7}\right)$

(3)  $-\frac{9}{5} - \left(-\frac{1}{4}\right)$

(13)  $-\frac{1}{2} + \frac{3}{5}$

(23)  $3 - \frac{3}{5}$

(33)  $-3 + (-2)$

(4)  $\frac{7}{5} - \frac{1}{5}$

(14)  $-5 - \left(-\frac{1}{3}\right)$

(24)  $\frac{5}{2} + \left(-\frac{1}{2}\right)$

(34)  $-\frac{1}{4} - \frac{3}{2}$

(5)  $\frac{7}{3} + \left(-\frac{5}{8}\right)$

(15)  $-\frac{3}{7} + \frac{3}{5}$

(25)  $-\frac{3}{7} - \frac{8}{9}$

(35)  $9 - \frac{4}{9}$

(6)  $\frac{4}{9} + 5$

(16)  $-\frac{1}{2} + \left(-\frac{7}{5}\right)$

(26)  $\frac{2}{7} + (-8)$

(36)  $-\frac{5}{9} + \left(-\frac{5}{2}\right)$

(7)  $\frac{3}{2} + \frac{10}{7}$

(17)  $-2 - \left(-\frac{5}{3}\right)$

(27)  $-\frac{1}{2} + (-7)$

(37)  $-3 - \left(-\frac{2}{3}\right)$

(8)  $-\frac{1}{9} - (-2)$

(18)  $\frac{1}{8} + 6$

(28)  $-\frac{4}{5} + 6$

(38)  $\frac{1}{2} + \frac{6}{7}$

(9)  $\frac{1}{7} + \left(-\frac{1}{3}\right)$

(19)  $-\frac{3}{7} + \frac{4}{7}$

(29)  $\frac{10}{3} - \left(-\frac{6}{7}\right)$

(39)  $\frac{2}{3} + 2$

(10)  $-\frac{1}{8} - 5$

(20)  $\frac{1}{4} - \left(-\frac{3}{7}\right)$

(30)  $\frac{5}{3} + \frac{4}{3}$

(40)  $-6 - \frac{3}{2}$

(1)  $\frac{8}{7} - \frac{2}{7}$

(11)  $-2 + \frac{1}{4}$

(21)  $-6 - \frac{8}{9}$

(31)  $4 - 1$

(2)  $-\frac{3}{2} - \left(-\frac{6}{7}\right)$

(12)  $-\frac{8}{5} + \frac{1}{9}$

(22)  $-\frac{2}{5} + \left(-\frac{1}{2}\right)$

(32)  $\frac{2}{7} + \frac{1}{8}$

(3)  $-\frac{1}{6} + \frac{7}{8}$

(13)  $-\frac{9}{5} - \frac{8}{9}$

(23)  $-\frac{3}{4} + \frac{1}{4}$

(33)  $-\frac{5}{4} - \frac{1}{7}$

(4)  $\frac{2}{3} + \left(-\frac{1}{10}\right)$

(14)  $-1 + (-3)$

(24)  $1 - \left(-\frac{3}{2}\right)$

(34)  $\frac{2}{3} + \left(-\frac{4}{3}\right)$

(5)  $-1 + \left(-\frac{4}{9}\right)$

(15)  $\frac{1}{3} - (-8)$

(25)  $-1 + 3$

(35)  $-\frac{7}{10} - (-7)$

(6)  $\frac{1}{6} - \left(-\frac{5}{2}\right)$

(16)  $-10 + \frac{6}{5}$

(26)  $-\frac{8}{5} + (-2)$

(36)  $\frac{1}{2} - \frac{5}{6}$

(7)  $-2 - \frac{5}{7}$

(17)  $-\frac{7}{2} + 1$

(27)  $-\frac{10}{3} - \frac{9}{2}$

(37)  $\frac{3}{8} - \left(-\frac{7}{9}\right)$

(8)  $\frac{1}{2} - \left(-\frac{1}{3}\right)$

(18)  $-\frac{3}{7} - \frac{5}{7}$

(28)  $1 + \left(-\frac{4}{7}\right)$

(38)  $\frac{1}{3} - (-1)$

(9)  $1 - \left(-\frac{1}{10}\right)$

(19)  $\frac{7}{3} - 2$

(29)  $2 + \left(-\frac{9}{2}\right)$

(39)  $1 - \frac{2}{3}$

(10)  $-\frac{5}{4} + \left(-\frac{1}{3}\right)$

(20)  $\frac{1}{2} + (-8)$

(30)  $-\frac{2}{7} + \frac{7}{10}$

(40)  $\frac{7}{4} + 1$



(1)  $\frac{1}{3} - 2$

(11)  $10 - \left(-\frac{3}{2}\right)$

(21)  $-8 + (-2)$

(31)  $\frac{9}{4} + 1$

(2)  $\frac{1}{4} + 3$

(12)  $-\frac{5}{4} + \frac{7}{9}$

(22)  $\frac{7}{8} - 5$

(32)  $-5 - 1$

(3)  $\frac{3}{10} + \left(-\frac{5}{2}\right)$

(13)  $-1 + \left(-\frac{2}{3}\right)$

(23)  $\frac{3}{2} + \frac{2}{9}$

(33)  $-\frac{5}{2} - (-1)$

(4)  $\frac{3}{5} - \frac{4}{9}$

(14)  $\frac{3}{8} + 10$

(24)  $\frac{4}{7} - \frac{7}{5}$

(34)  $-\frac{7}{4} + \left(-\frac{1}{2}\right)$

(5)  $-\frac{4}{3} + \frac{3}{8}$

(15)  $-\frac{5}{4} + \left(-\frac{6}{5}\right)$

(25)  $-8 + 2$

(35)  $\frac{7}{9} - 1$

(6)  $-\frac{5}{8} + \left(-\frac{4}{7}\right)$

(16)  $-\frac{4}{7} - \left(-\frac{8}{3}\right)$

(26)  $\frac{4}{5} - 1$

(36)  $-5 - \left(-\frac{1}{2}\right)$

(7)  $-1 + \left(-\frac{1}{5}\right)$

(17)  $\frac{1}{2} + \frac{5}{8}$

(27)  $3 - \frac{4}{7}$

(37)  $-\frac{1}{4} + \left(-\frac{1}{7}\right)$

(8)  $\frac{7}{5} - \left(-\frac{5}{2}\right)$

(18)  $-\frac{1}{3} - \left(-\frac{1}{10}\right)$

(28)  $5 + \left(-\frac{1}{8}\right)$

(38)  $\frac{5}{2} - (-3)$

(9)  $-1 - (-1)$

(19)  $1 - \left(-\frac{1}{4}\right)$

(29)  $4 - 4$

(39)  $\frac{2}{3} + \frac{9}{8}$

(10)  $-\frac{5}{8} - \frac{1}{3}$

(20)  $-1 - (-1)$

(30)  $\frac{7}{4} - (-5)$

(40)  $\frac{7}{10} - \left(-\frac{9}{5}\right)$

(1)  $-5 - \left(-\frac{6}{5}\right)$

(11)  $\frac{8}{7} + \left(-\frac{8}{5}\right)$

(21)  $-\frac{3}{8} + \left(-\frac{4}{7}\right)$

(31)  $-1 - \frac{3}{5}$

(2)  $-1 - \left(-\frac{10}{7}\right)$

(12)  $\frac{6}{5} - \left(-\frac{1}{4}\right)$

(22)  $-\frac{9}{10} - (-1)$

(32)  $1 - \left(-\frac{5}{3}\right)$

(3)  $-\frac{3}{4} + \left(-\frac{1}{4}\right)$

(13)  $-9 - \left(-\frac{1}{2}\right)$

(23)  $-\frac{2}{5} + (-2)$

(33)  $\frac{9}{8} - (-3)$

(4)  $-5 - \frac{9}{10}$

(14)  $\frac{2}{5} - \left(-\frac{9}{2}\right)$

(24)  $-\frac{5}{8} - \frac{4}{9}$

(34)  $-\frac{1}{8} - 3$

(5)  $\frac{1}{9} + \frac{10}{7}$

(15)  $-5 - \frac{4}{7}$

(25)  $1 + \left(-\frac{3}{8}\right)$

(35)  $\frac{7}{9} + \frac{7}{3}$

(6)  $-\frac{10}{7} + \left(-\frac{7}{6}\right)$

(16)  $-\frac{1}{4} - \frac{2}{3}$

(26)  $1 - (-3)$

(36)  $-\frac{3}{4} + \frac{1}{2}$

(7)  $-5 + \left(-\frac{2}{7}\right)$

(17)  $\frac{4}{3} + \frac{1}{7}$

(27)  $-\frac{10}{3} - 2$

(37)  $-\frac{2}{3} - \frac{2}{3}$

(8)  $7 + \left(-\frac{8}{3}\right)$

(18)  $-\frac{5}{7} + 1$

(28)  $-\frac{1}{6} - \frac{1}{8}$

(38)  $-\frac{5}{9} - \frac{6}{5}$

(9)  $-3 + 1$

(19)  $\frac{5}{2} + \frac{3}{8}$

(29)  $\frac{10}{9} - \frac{5}{2}$

(39)  $\frac{3}{8} - \frac{1}{2}$

(10)  $2 + (-3)$

(20)  $\frac{2}{3} - \left(-\frac{3}{4}\right)$

(30)  $-1 - \frac{3}{5}$

(40)  $-\frac{1}{7} + \frac{5}{8}$

(1)  $-\frac{1}{2} - \frac{4}{3}$

(11)  $-\frac{8}{9} + \frac{1}{5}$

(21)  $-10 + (-4)$

(31)  $-\frac{1}{5} - \left(-\frac{1}{2}\right)$

(2)  $3 - (-1)$

(12)  $\frac{3}{5} - \frac{2}{5}$

(22)  $\frac{7}{3} - \left(-\frac{4}{7}\right)$

(32)  $-\frac{1}{4} + \left(-\frac{7}{2}\right)$

(3)  $\frac{1}{2} - 2$

(13)  $-\frac{5}{6} + \frac{9}{8}$

(23)  $-\frac{3}{8} + \frac{1}{7}$

(33)  $-\frac{3}{5} + 8$

(4)  $-\frac{1}{2} + \left(-\frac{4}{5}\right)$

(14)  $-\frac{5}{4} + \frac{4}{5}$

(24)  $-\frac{1}{3} - (-6)$

(34)  $\frac{7}{10} + \frac{4}{9}$

(5)  $-\frac{1}{2} + 1$

(15)  $-\frac{3}{10} + \frac{2}{7}$

(25)  $-\frac{2}{3} - \left(-\frac{4}{3}\right)$

(35)  $-\frac{1}{5} + \frac{5}{2}$

(6)  $-1 + \frac{5}{4}$

(16)  $\frac{3}{2} - \frac{5}{2}$

(26)  $\frac{3}{2} + \left(-\frac{3}{5}\right)$

(36)  $\frac{1}{5} + \left(-\frac{2}{3}\right)$

(7)  $-\frac{7}{9} + 4$

(17)  $\frac{5}{3} - 10$

(27)  $-3 + \frac{4}{3}$

(37)  $-\frac{7}{5} - \left(-\frac{4}{5}\right)$

(8)  $-8 - \left(-\frac{3}{4}\right)$

(18)  $\frac{7}{3} - \frac{10}{3}$

(28)  $3 - \frac{5}{6}$

(38)  $-7 + \left(-\frac{7}{2}\right)$

(9)  $-\frac{6}{7} + 4$

(19)  $\frac{1}{10} - \frac{1}{3}$

(29)  $-\frac{3}{2} - \left(-\frac{1}{8}\right)$

(39)  $-\frac{1}{4} - \left(-\frac{9}{10}\right)$

(10)  $1 + \frac{1}{2}$

(20)  $\frac{5}{6} + (-10)$

(30)  $-\frac{2}{9} - \left(-\frac{1}{9}\right)$

(40)  $\frac{6}{5} + (-1)$

(1)  $-\frac{5}{7} + \left(-\frac{7}{10}\right)$

(11)  $1 + 1$

(21)  $-1 + (-1)$

(31)  $-\frac{3}{4} - 1$

(2)  $-7 - \left(-\frac{2}{5}\right)$

(12)  $-\frac{7}{8} + (-3)$

(22)  $\frac{3}{5} - \frac{7}{2}$

(32)  $\frac{10}{3} + (-4)$

(3)  $-\frac{7}{2} + \frac{7}{6}$

(13)  $\frac{2}{7} + \left(-\frac{1}{3}\right)$

(23)  $-\frac{9}{2} - \frac{3}{4}$

(33)  $\frac{3}{2} - \frac{7}{3}$

(4)  $-\frac{5}{9} + \frac{3}{4}$

(14)  $-\frac{7}{5} + \frac{4}{3}$

(24)  $9 + 1$

(34)  $\frac{6}{7} + \left(-\frac{10}{7}\right)$

(5)  $-\frac{8}{5} - \left(-\frac{1}{10}\right)$

(15)  $-\frac{2}{7} - (-5)$

(25)  $\frac{1}{8} - \frac{5}{3}$

(35)  $\frac{5}{3} + \frac{4}{5}$

(6)  $-2 - \frac{8}{7}$

(16)  $-\frac{3}{10} - \left(-\frac{3}{5}\right)$

(26)  $\frac{5}{3} + \frac{3}{4}$

(36)  $-\frac{1}{2} - \left(-\frac{7}{9}\right)$

(7)  $-\frac{7}{9} + \left(-\frac{4}{9}\right)$

(17)  $-\frac{1}{3} + \frac{2}{5}$

(27)  $\frac{5}{6} - (-2)$

(37)  $-1 + \frac{2}{7}$

(28)  $-\frac{4}{3} + \frac{1}{2}$

(8)  $3 + \left(-\frac{5}{4}\right)$

(18)  $-\frac{1}{2} - 2$

(38)  $-\frac{7}{2} + (-1)$

(9)  $-\frac{3}{10} - \frac{1}{2}$

(19)  $\frac{7}{6} + (-1)$

(29)  $-\frac{8}{3} + \left(-\frac{1}{2}\right)$

(39)  $-\frac{8}{3} + \left(-\frac{4}{7}\right)$

(30)  $-\frac{9}{2} + \frac{9}{2}$

(20)  $-1 + (-2)$

(10)  $\frac{2}{3} - \frac{8}{3}$

(40)  $-\frac{1}{3} + 1$

(1)  $-\frac{3}{4} - \frac{9}{4}$

(11)  $-2 - \left(-\frac{5}{9}\right)$

(21)  $\frac{2}{5} - 5$

(31)  $\frac{1}{2} - \left(-\frac{7}{8}\right)$

(2)  $\frac{4}{3} + \left(-\frac{7}{2}\right)$

(12)  $-10 + \frac{10}{9}$

(22)  $-\frac{10}{3} + \left(-\frac{2}{7}\right)$

(32)  $-\frac{1}{3} + \left(-\frac{9}{8}\right)$

(3)  $\frac{10}{7} + \frac{5}{3}$

(13)  $-1 + (-1)$

(23)  $9 + \frac{2}{3}$

(33)  $\frac{9}{8} - \frac{7}{2}$

(4)  $-1 - \left(-\frac{6}{5}\right)$

(14)  $3 + \frac{1}{5}$

(24)  $2 + \left(-\frac{7}{9}\right)$

(34)  $-\frac{1}{3} - \frac{7}{10}$

(5)  $-\frac{4}{3} - (-2)$

(15)  $3 + \frac{5}{9}$

(25)  $-\frac{4}{3} - \frac{1}{3}$

(35)  $7 + \frac{5}{8}$

(6)  $\frac{2}{5} - \frac{5}{3}$

(16)  $\frac{5}{3} - \frac{9}{4}$

(26)  $-\frac{3}{8} - \left(-\frac{1}{2}\right)$

(36)  $-3 - \left(-\frac{1}{2}\right)$

(7)  $-\frac{3}{5} + \frac{2}{3}$

(17)  $10 + \left(-\frac{9}{4}\right)$

(27)  $2 + \left(-\frac{5}{3}\right)$

(37)  $-\frac{8}{5} - \left(-\frac{3}{10}\right)$

(8)  $\frac{3}{2} - \left(-\frac{5}{7}\right)$

(18)  $\frac{9}{5} + \left(-\frac{6}{5}\right)$

(28)  $4 + \left(-\frac{1}{4}\right)$

(38)  $\frac{4}{5} - 2$

(9)  $\frac{5}{2} + \frac{2}{5}$

(19)  $-\frac{1}{2} + 1$

(29)  $\frac{3}{2} + \frac{2}{3}$

(39)  $\frac{2}{7} + \frac{8}{7}$

(10)  $\frac{1}{10} + \left(-\frac{4}{5}\right)$

(20)  $\frac{3}{4} + \frac{1}{7}$

(30)  $\frac{7}{9} - (-1)$

(40)  $-\frac{9}{2} - 1$

(1)  $2 - \frac{3}{2}$

(11)  $\frac{3}{4} + (-1)$

(21)  $\frac{8}{3} + 1$

(31)  $-\frac{5}{3} + \frac{2}{7}$

(2)  $-\frac{6}{5} + \left(-\frac{9}{2}\right)$

(12)  $\frac{4}{9} - \frac{1}{9}$

(22)  $\frac{9}{2} - \left(-\frac{1}{9}\right)$

(32)  $-\frac{3}{2} + \frac{7}{8}$

(3)  $1 + \left(-\frac{7}{4}\right)$

(13)  $\frac{2}{5} + \left(-\frac{10}{7}\right)$

(23)  $-1 + \frac{1}{3}$

(33)  $-\frac{3}{7} - \frac{4}{5}$

(4)  $-5 + \frac{5}{7}$

(14)  $-4 + \left(-\frac{5}{4}\right)$

(24)  $-\frac{3}{2} - (-3)$

(34)  $2 - 2$

(5)  $-\frac{6}{5} + \frac{5}{2}$

(15)  $-\frac{5}{2} - \frac{4}{7}$

(25)  $\frac{4}{5} + (-1)$

(35)  $\frac{2}{3} - \left(-\frac{5}{3}\right)$

(6)  $-4 - 3$

(16)  $-1 - (-8)$

(26)  $\frac{5}{9} + \left(-\frac{9}{7}\right)$

(36)  $\frac{5}{4} + \left(-\frac{2}{3}\right)$

(7)  $-\frac{1}{2} - \left(-\frac{7}{4}\right)$

(17)  $\frac{3}{2} - \frac{7}{4}$

(27)  $-\frac{7}{10} + \frac{1}{6}$

(37)  $\frac{8}{7} + \frac{3}{2}$

(8)  $5 - \left(-\frac{3}{10}\right)$

(18)  $-\frac{9}{10} + \frac{7}{6}$

(28)  $-2 + \frac{5}{3}$

(38)  $-1 + \left(-\frac{1}{3}\right)$

(9)  $\frac{10}{9} + \frac{5}{6}$

(19)  $-\frac{1}{3} + \left(-\frac{5}{9}\right)$

(29)  $\frac{2}{5} + \left(-\frac{3}{4}\right)$

(39)  $2 + \frac{10}{3}$

(10)  $\frac{3}{10} + \frac{9}{5}$

(20)  $2 - \frac{5}{3}$

(30)  $-\frac{1}{2} + \left(-\frac{3}{2}\right)$

(40)  $-1 - (-6)$

(1)  $-\frac{1}{2} + \frac{5}{7}$

(11)  $-\frac{1}{6} + 1$

(21)  $\frac{3}{5} - (-1)$

(31)  $\frac{2}{5} + \left(-\frac{4}{3}\right)$

(2)  $\frac{1}{4} - \frac{9}{2}$

(12)  $\frac{4}{5} + \frac{2}{5}$

(22)  $-\frac{9}{8} + 1$

(32)  $\frac{4}{3} - \left(-\frac{3}{2}\right)$

(3)  $-3 - \left(-\frac{9}{2}\right)$

(13)  $-2 + \frac{1}{3}$

(23)  $-9 - \left(-\frac{9}{8}\right)$

(33)  $\frac{7}{5} + \left(-\frac{5}{3}\right)$

(4)  $\frac{5}{2} - \frac{1}{3}$

(14)  $-7 - \frac{7}{6}$

(24)  $-\frac{4}{7} + \frac{7}{9}$

(34)  $\frac{3}{5} + \frac{8}{9}$

(5)  $\frac{9}{2} + 4$

(15)  $1 - \frac{1}{3}$

(25)  $9 + 1$

(35)  $\frac{3}{4} + \left(-\frac{9}{10}\right)$

(6)  $-\frac{1}{5} - \frac{2}{3}$

(16)  $\frac{1}{8} - \frac{2}{3}$

(26)  $-\frac{6}{5} + \frac{6}{7}$

(36)  $-\frac{8}{7} + \frac{3}{2}$

(7)  $-\frac{10}{9} + \frac{9}{2}$

(17)  $\frac{5}{4} - \frac{5}{9}$

(27)  $-\frac{1}{2} + \frac{1}{10}$

(37)  $-\frac{6}{5} - (-10)$

(8)  $1 + \frac{1}{2}$

(18)  $\frac{5}{6} + 3$

(28)  $1 - \left(-\frac{9}{10}\right)$

(38)  $-\frac{9}{5} - \left(-\frac{1}{2}\right)$

(9)  $-\frac{9}{7} + \frac{1}{10}$

(19)  $\frac{1}{2} - \left(-\frac{9}{2}\right)$

(29)  $\frac{7}{8} - \left(-\frac{1}{10}\right)$

(39)  $\frac{5}{2} - \frac{9}{4}$

(10)  $\frac{4}{7} + \frac{1}{5}$

(20)  $-2 + \frac{2}{5}$

(30)  $-\frac{2}{5} - (-8)$

(40)  $-\frac{5}{7} + \left(-\frac{9}{2}\right)$

(1)  $-9 + \left(-\frac{9}{4}\right)$

(11)  $-\frac{1}{7} - \left(-\frac{1}{2}\right)$

(21)  $\frac{10}{3} - \frac{8}{5}$

(31)  $-\frac{1}{5} + (-10)$

(2)  $5 - 8$

(12)  $\frac{3}{7} - \frac{1}{4}$

(22)  $\frac{3}{7} - 1$

(32)  $-\frac{9}{10} + \left(-\frac{7}{3}\right)$

(3)  $\frac{2}{5} + \frac{3}{2}$

(13)  $-\frac{1}{2} + \left(-\frac{3}{5}\right)$

(23)  $-\frac{7}{6} - \left(-\frac{4}{5}\right)$

(33)  $\frac{8}{3} + (-6)$

(4)  $7 + \frac{2}{5}$

(14)  $\frac{2}{5} - \left(-\frac{7}{4}\right)$

(24)  $\frac{1}{2} + \frac{4}{7}$

(34)  $-\frac{6}{5} - \left(-\frac{1}{2}\right)$

(5)  $-\frac{5}{2} + \left(-\frac{9}{4}\right)$

(15)  $8 - 2$

(25)  $\frac{1}{7} - 5$

(35)  $\frac{3}{2} - \frac{1}{7}$

(6)  $-1 + \frac{4}{3}$

(16)  $2 + 1$

(26)  $-\frac{9}{8} - \left(-\frac{7}{6}\right)$

(36)  $\frac{3}{10} + \frac{2}{7}$

(7)  $\frac{10}{7} - \left(-\frac{7}{9}\right)$

(17)  $-\frac{8}{7} - \frac{5}{4}$

(27)  $\frac{2}{3} - \frac{8}{9}$

(37)  $\frac{6}{5} - (-1)$

(8)  $\frac{3}{2} - 1$

(18)  $\frac{4}{3} - \frac{1}{3}$

(28)  $\frac{1}{7} + \left(-\frac{9}{10}\right)$

(38)  $-\frac{4}{5} - \left(-\frac{9}{2}\right)$

(9)  $-1 - \frac{7}{3}$

(19)  $\frac{10}{3} - \left(-\frac{1}{2}\right)$

(29)  $\frac{5}{8} + \left(-\frac{1}{8}\right)$

(39)  $-\frac{2}{3} - (-2)$

(10)  $9 + \left(-\frac{1}{9}\right)$

(20)  $-2 + \left(-\frac{1}{8}\right)$

(30)  $\frac{5}{2} + \frac{3}{4}$

(40)  $-2 + \left(-\frac{7}{9}\right)$



(1)  $5 + (-3)$

(11)  $-\frac{4}{7} - \left(-\frac{4}{7}\right)$

(21)  $-\frac{3}{5} + \frac{5}{4}$

(31)  $\frac{7}{6} - \frac{2}{3}$

(2)  $-\frac{1}{5} - \frac{1}{4}$

(12)  $-1 - \frac{1}{5}$

(22)  $-\frac{2}{3} + \left(-\frac{5}{8}\right)$

(32)  $2 + \left(-\frac{1}{3}\right)$

(3)  $\frac{2}{5} + \left(-\frac{9}{4}\right)$

(13)  $\frac{1}{2} + \frac{3}{2}$

(23)  $\frac{4}{3} - \left(-\frac{8}{5}\right)$

(33)  $-\frac{5}{2} + \left(-\frac{1}{2}\right)$

(4)  $\frac{1}{3} - \frac{9}{4}$

(14)  $-\frac{3}{10} - 1$

(24)  $-\frac{1}{3} - \left(-\frac{5}{7}\right)$

(34)  $-\frac{5}{2} - \left(-\frac{5}{6}\right)$

(5)  $-\frac{1}{8} - \frac{5}{8}$

(15)  $1 - \frac{9}{7}$

(25)  $-\frac{1}{9} + \left(-\frac{7}{8}\right)$

(35)  $-5 + \left(-\frac{4}{5}\right)$

(6)  $-\frac{1}{9} - (-5)$

(16)  $-\frac{10}{7} + (-1)$

(26)  $\frac{2}{5} - (-10)$

(36)  $-\frac{3}{5} + \left(-\frac{9}{2}\right)$

(7)  $\frac{3}{10} + (-1)$

(17)  $\frac{1}{2} + \frac{5}{2}$

(27)  $-3 + 2$

(37)  $\frac{8}{5} - \frac{3}{5}$

(8)  $-\frac{8}{7} - 8$

(18)  $\frac{9}{4} - (-2)$

(28)  $\frac{7}{8} - \frac{4}{5}$

(38)  $\frac{5}{7} - \left(-\frac{9}{7}\right)$

(9)  $\frac{4}{5} - \frac{3}{5}$

(19)  $\frac{1}{6} - (-2)$

(29)  $-\frac{5}{4} + \frac{3}{10}$

(39)  $1 + \frac{1}{3}$

(10)  $-\frac{5}{9} - \frac{8}{9}$

(20)  $\frac{5}{3} + \frac{7}{3}$

(30)  $-1 - \frac{1}{2}$

(40)  $-\frac{6}{7} + 1$

(1)  $-\frac{5}{3} - (-3)$

(11)  $\frac{5}{9} - \left(-\frac{3}{5}\right)$

(21)  $1 + \frac{5}{3}$

(31)  $\frac{8}{5} + \frac{9}{7}$

(2)  $-\frac{3}{7} + \frac{5}{4}$

(12)  $\frac{5}{2} + 1$

(22)  $-1 - (-7)$

(32)  $\frac{5}{9} + \left(-\frac{5}{2}\right)$

(3)  $-1 - \left(-\frac{3}{7}\right)$

(13)  $-\frac{1}{2} - (-1)$

(23)  $\frac{10}{3} - \left(-\frac{7}{5}\right)$

(33)  $-\frac{9}{7} - \left(-\frac{9}{4}\right)$

(4)  $-\frac{9}{8} + \frac{4}{9}$

(14)  $-\frac{9}{5} + \left(-\frac{5}{3}\right)$

(24)  $-\frac{1}{4} + \left(-\frac{4}{3}\right)$

(34)  $-\frac{7}{10} + \frac{10}{3}$

(5)  $-2 - 1$

(15)  $-\frac{5}{3} - \left(-\frac{5}{4}\right)$

(25)  $-\frac{4}{9} + \left(-\frac{4}{9}\right)$

(35)  $-\frac{1}{10} - \left(-\frac{9}{10}\right)$

(6)  $-\frac{3}{4} - \frac{3}{10}$

(16)  $\frac{7}{6} + 10$

(26)  $\frac{10}{3} + \frac{3}{5}$

(36)  $-\frac{2}{9} - \left(-\frac{3}{10}\right)$

(7)  $\frac{10}{3} - \left(-\frac{3}{4}\right)$

(17)  $-\frac{9}{10} - 7$

(27)  $\frac{8}{5} + (-1)$

(37)  $-1 + \frac{1}{4}$

(8)  $-\frac{4}{9} + \left(-\frac{4}{5}\right)$

(18)  $\frac{10}{3} - \left(-\frac{9}{8}\right)$

(28)  $-\frac{1}{2} + \left(-\frac{7}{8}\right)$

(38)  $\frac{10}{7} - \frac{2}{9}$

(9)  $-6 - \left(-\frac{9}{4}\right)$

(19)  $-\frac{6}{7} - \left(-\frac{8}{9}\right)$

(29)  $-\frac{1}{7} - \frac{8}{3}$

(39)  $2 + \frac{4}{3}$

(10)  $-\frac{4}{7} + 10$

(20)  $\frac{1}{5} + (-5)$

(30)  $6 - \frac{1}{2}$

(40)  $-\frac{3}{8} + \left(-\frac{2}{7}\right)$

(1)  $\frac{5}{4} - \left(-\frac{5}{4}\right)$

(11)  $-5 + \left(-\frac{6}{7}\right)$

(21)  $\frac{7}{6} - (-1)$

(31)  $\frac{1}{2} - 5$

(2)  $1 - (-1)$

(12)  $-\frac{2}{5} + 1$

(22)  $-\frac{2}{5} - \left(-\frac{1}{6}\right)$

(32)  $-3 - \left(-\frac{2}{9}\right)$

(3)  $3 + \frac{10}{3}$

(13)  $-\frac{1}{3} - \frac{8}{3}$

(23)  $-\frac{6}{7} + \frac{1}{2}$

(33)  $\frac{1}{7} + 1$

(4)  $\frac{2}{5} + (-6)$

(14)  $\frac{5}{2} + 2$

(24)  $\frac{3}{4} + \left(-\frac{1}{7}\right)$

(34)  $\frac{2}{3} + \frac{4}{9}$

(5)  $\frac{1}{2} + \frac{1}{4}$

(15)  $-1 + \frac{3}{2}$

(25)  $\frac{1}{2} - \frac{1}{3}$

(35)  $-\frac{1}{4} - \frac{1}{7}$

(6)  $\frac{1}{3} - \frac{4}{7}$

(16)  $-\frac{5}{7} - \left(-\frac{4}{5}\right)$

(26)  $-5 - (-4)$

(36)  $\frac{5}{8} - \frac{6}{5}$

(7)  $-5 - \left(-\frac{4}{5}\right)$

(17)  $\frac{1}{2} + \frac{1}{2}$

(27)  $5 - 3$

(37)  $\frac{7}{6} + (-2)$

(8)  $\frac{7}{6} + \left(-\frac{8}{7}\right)$

(18)  $1 + \frac{9}{8}$

(28)  $\frac{8}{7} + (-1)$

(38)  $\frac{1}{7} - \left(-\frac{2}{5}\right)$

(9)  $-2 + \frac{3}{4}$

(19)  $\frac{8}{7} + \frac{7}{10}$

(29)  $1 - \frac{1}{3}$

(39)  $-\frac{10}{3} + \frac{3}{2}$

(10)  $4 - \left(-\frac{5}{6}\right)$

(20)  $-\frac{7}{8} - 1$

(30)  $\frac{2}{3} + \frac{9}{4}$

(40)  $\frac{4}{5} - \frac{9}{8}$

(1)  $\frac{3}{4} - (-2)$

(11)  $-\frac{2}{3} - \left(-\frac{3}{5}\right)$

(21)  $-\frac{1}{7} - (-9)$

(31)  $\frac{3}{8} - \left(-\frac{1}{5}\right)$

(2)  $\frac{2}{7} - \left(-\frac{8}{5}\right)$

(12)  $-\frac{5}{8} + \frac{3}{8}$

(22)  $-3 - \left(-\frac{4}{9}\right)$

(32)  $-3 - 1$

(3)  $-\frac{5}{7} - \left(-\frac{8}{7}\right)$

(13)  $1 - \frac{9}{10}$

(23)  $\frac{1}{2} + \frac{5}{4}$

(33)  $-4 - \left(-\frac{1}{10}\right)$

(4)  $-\frac{3}{2} + \frac{1}{3}$

(14)  $-\frac{1}{2} - \frac{8}{3}$

(24)  $-\frac{10}{9} - (-1)$

(34)  $\frac{5}{2} - \left(-\frac{10}{3}\right)$

(5)  $-\frac{2}{3} - \left(-\frac{7}{6}\right)$

(15)  $1 - \left(-\frac{3}{2}\right)$

(25)  $-\frac{3}{8} - 1$

(35)  $-8 - \left(-\frac{4}{7}\right)$

(6)  $-8 - \left(-\frac{5}{9}\right)$

(16)  $-1 + (-5)$

(26)  $7 + 1$

(36)  $-\frac{5}{4} - \left(-\frac{5}{4}\right)$

(7)  $2 - \left(-\frac{3}{4}\right)$

(17)  $\frac{1}{4} + \left(-\frac{5}{4}\right)$

(27)  $\frac{3}{7} - \frac{9}{5}$

(37)  $\frac{7}{3} - \frac{5}{4}$

(18)  $\frac{1}{10} - \left(-\frac{9}{8}\right)$

(28)  $-4 - \frac{3}{5}$

(38)  $4 - \frac{9}{4}$

(8)  $-\frac{3}{5} + \frac{9}{7}$

(19)  $\frac{7}{8} - (-7)$

(29)  $1 - 1$

(39)  $-\frac{7}{8} + (-2)$

(9)  $-\frac{2}{3} - \left(-\frac{5}{3}\right)$

(20)  $\frac{5}{2} - 8$

(30)  $\frac{2}{3} + 1$

(40)  $\frac{8}{9} - \frac{8}{9}$

(10)  $\frac{7}{10} + \left(-\frac{9}{4}\right)$

(1)  $-\frac{7}{5} + \left(-\frac{10}{9}\right)$

(11)  $-\frac{1}{2} + \frac{9}{4}$

(21)  $1 + \frac{1}{3}$

(31)  $-\frac{7}{5} + \left(-\frac{2}{3}\right)$

(2)  $-\frac{6}{5} - \frac{9}{7}$

(12)  $-9 - \left(-\frac{5}{4}\right)$

(22)  $\frac{5}{2} - (-1)$

(32)  $3 + \frac{3}{5}$

(3)  $-\frac{5}{2} - \frac{3}{4}$

(13)  $-\frac{9}{7} + \left(-\frac{8}{9}\right)$

(23)  $-5 - \left(-\frac{6}{7}\right)$

(33)  $\frac{4}{3} + 10$

(4)  $2 + \left(-\frac{4}{3}\right)$

(14)  $\frac{10}{7} + \frac{1}{2}$

(24)  $-\frac{1}{2} + \left(-\frac{1}{8}\right)$

(34)  $-\frac{5}{9} - 1$

(5)  $-1 - \left(-\frac{1}{3}\right)$

(15)  $-\frac{8}{3} + (-3)$

(25)  $\frac{3}{10} - \left(-\frac{4}{5}\right)$

(35)  $2 - \frac{1}{5}$

(6)  $\frac{5}{7} + \frac{9}{10}$

(16)  $\frac{5}{4} - 4$

(26)  $\frac{2}{3} - \left(-\frac{5}{8}\right)$

(36)  $\frac{5}{3} - (-4)$

(7)  $\frac{1}{4} + \left(-\frac{3}{8}\right)$

(17)  $1 + \frac{3}{7}$

(27)  $-\frac{1}{2} + \frac{5}{4}$

(37)  $-3 + 1$

(8)  $-\frac{3}{5} + \left(-\frac{2}{9}\right)$

(18)  $-\frac{3}{4} + (-2)$

(28)  $-\frac{10}{9} - \left(-\frac{4}{9}\right)$

(38)  $-2 + \left(-\frac{1}{3}\right)$

(9)  $\frac{6}{7} + 5$

(19)  $5 - \left(-\frac{3}{5}\right)$

(29)  $-\frac{2}{3} + \frac{9}{2}$

(39)  $2 - 7$

(40)  $\frac{1}{3} - \frac{3}{4}$

(10)  $\frac{5}{9} - (-6)$

(20)  $\frac{4}{9} + \left(-\frac{3}{8}\right)$

(30)  $1 - \left(-\frac{4}{5}\right)$

(1)  $-\frac{1}{3} + (-1)$

(11)  $1 - \left(-\frac{1}{5}\right)$

(21)  $-8 + \left(-\frac{5}{9}\right)$

(31)  $-\frac{4}{5} - \left(-\frac{3}{8}\right)$

(2)  $\frac{8}{3} - \left(-\frac{1}{2}\right)$

(12)  $-\frac{3}{2} - \frac{5}{3}$

(22)  $-4 - \frac{7}{8}$

(32)  $\frac{7}{3} - \left(-\frac{9}{5}\right)$

(3)  $\frac{3}{10} + \left(-\frac{1}{5}\right)$

(13)  $-\frac{5}{2} + \frac{9}{7}$

(23)  $\frac{3}{4} - \left(-\frac{1}{2}\right)$

(33)  $-\frac{9}{8} + \left(-\frac{1}{7}\right)$

(4)  $\frac{2}{5} - \left(-\frac{3}{2}\right)$

(14)  $-\frac{3}{10} - 10$

(24)  $\frac{1}{8} + \left(-\frac{4}{9}\right)$

(34)  $1 + \left(-\frac{4}{3}\right)$

(5)  $-1 + (-3)$

(15)  $\frac{4}{5} + \frac{5}{3}$

(25)  $-\frac{2}{3} - \frac{5}{4}$

(35)  $-2 - \frac{3}{2}$

(6)  $\frac{4}{3} + \frac{1}{7}$

(16)  $-\frac{7}{4} + \left(-\frac{8}{7}\right)$

(26)  $-2 - \left(-\frac{5}{4}\right)$

(36)  $-\frac{4}{3} + (-1)$

(7)  $-\frac{1}{2} - \frac{1}{4}$

(17)  $-\frac{5}{6} - \left(-\frac{7}{6}\right)$

(27)  $2 - \left(-\frac{2}{5}\right)$

(37)  $\frac{9}{4} - (-1)$

(8)  $1 + \left(-\frac{4}{9}\right)$

(18)  $2 - \left(-\frac{1}{9}\right)$

(28)  $-\frac{1}{3} - \left(-\frac{9}{4}\right)$

(38)  $\frac{1}{4} + \frac{1}{10}$

(9)  $-1 + \left(-\frac{5}{4}\right)$

(19)  $\frac{5}{4} - \left(-\frac{7}{3}\right)$

(29)  $1 + \frac{1}{2}$

(39)  $\frac{2}{7} - (-1)$

(10)  $-\frac{6}{5} - \left(-\frac{1}{10}\right)$

(20)  $\frac{5}{2} - 4$

(30)  $-2 - 4$

(40)  $-8 - \frac{10}{9}$

(1)  $3 - \frac{1}{4}$

(11)  $-\frac{9}{7} + \left(-\frac{9}{5}\right)$

(21)  $-\frac{1}{9} - \frac{5}{7}$

(31)  $\frac{1}{6} + \frac{7}{4}$

(2)  $-4 - 1$

(12)  $-\frac{7}{10} + (-1)$

(22)  $\frac{5}{7} - \frac{5}{3}$

(32)  $-1 + \left(-\frac{4}{9}\right)$

(3)  $-\frac{4}{5} - \left(-\frac{9}{7}\right)$

(13)  $\frac{6}{7} - \left(-\frac{4}{5}\right)$

(23)  $3 + \left(-\frac{1}{4}\right)$

(33)  $-\frac{2}{9} + \left(-\frac{2}{3}\right)$

(4)  $-\frac{2}{3} + \frac{1}{9}$

(14)  $\frac{1}{3} + (-2)$

(24)  $\frac{9}{5} + \left(-\frac{5}{7}\right)$

(34)  $\frac{4}{3} - 2$

(5)  $-5 + \frac{3}{7}$

(15)  $-5 - \frac{8}{9}$

(25)  $-\frac{2}{5} + \frac{10}{9}$

(35)  $-2 - (-1)$

(6)  $\frac{3}{2} - 1$

(16)  $-\frac{7}{4} + \left(-\frac{10}{7}\right)$

(26)  $\frac{10}{9} + \left(-\frac{1}{9}\right)$

(36)  $\frac{2}{7} + \frac{3}{4}$

(7)  $\frac{7}{2} + \frac{1}{2}$

(17)  $2 + \frac{1}{2}$

(27)  $-\frac{5}{6} + \left(-\frac{1}{2}\right)$

(37)  $-\frac{3}{2} - \frac{9}{2}$

(8)  $\frac{5}{6} - 4$

(18)  $\frac{1}{2} + \frac{2}{3}$

(28)  $-1 + \frac{7}{4}$

(38)  $\frac{7}{3} + \frac{6}{7}$

(9)  $\frac{9}{8} - (-1)$

(19)  $-\frac{1}{6} - \frac{3}{2}$

(29)  $-\frac{10}{7} - \left(-\frac{1}{2}\right)$

(39)  $-1 + \frac{5}{7}$

(10)  $\frac{6}{7} + \frac{3}{4}$

(20)  $-\frac{4}{7} + \frac{7}{5}$

(30)  $\frac{9}{10} + \frac{1}{2}$

(40)  $1 - \frac{3}{7}$

(1)  $9 - \frac{7}{3}$

(11)  $-\frac{5}{8} - 3$

(21)  $\frac{3}{5} - \frac{10}{9}$

(31)  $\frac{5}{7} - \left(-\frac{7}{4}\right)$

(2)  $-\frac{4}{5} - \left(-\frac{3}{10}\right)$

(12)  $-\frac{7}{3} + \left(-\frac{1}{9}\right)$

(22)  $\frac{9}{10} - \frac{1}{4}$

(32)  $1 + \frac{5}{4}$

(3)  $-\frac{3}{4} + \left(-\frac{7}{3}\right)$

(13)  $6 + \left(-\frac{1}{4}\right)$

(23)  $-\frac{5}{2} - \left(-\frac{5}{2}\right)$

(33)  $\frac{7}{2} - \frac{1}{3}$

(4)  $-\frac{3}{7} - \left(-\frac{3}{4}\right)$

(14)  $\frac{8}{3} + (-5)$

(24)  $-\frac{1}{2} + \left(-\frac{6}{5}\right)$

(34)  $-\frac{4}{3} - \frac{2}{9}$

(5)  $2 + (-10)$

(15)  $\frac{1}{6} + \left(-\frac{1}{7}\right)$

(25)  $-\frac{4}{3} + 2$

(35)  $-\frac{1}{5} + \frac{4}{5}$

(6)  $-\frac{5}{6} - \frac{1}{2}$

(16)  $2 - \frac{10}{7}$

(26)  $-3 + \left(-\frac{2}{5}\right)$

(36)  $\frac{5}{3} + \left(-\frac{5}{7}\right)$

(7)  $-\frac{3}{8} + \left(-\frac{7}{2}\right)$

(17)  $-\frac{6}{5} + \frac{2}{3}$

(27)  $-\frac{9}{5} + \left(-\frac{1}{2}\right)$

(37)  $\frac{3}{2} - \left(-\frac{8}{9}\right)$

(8)  $\frac{10}{7} + \frac{9}{7}$

(18)  $-\frac{6}{5} - \frac{3}{4}$

(28)  $-\frac{1}{5} + \frac{9}{2}$

(38)  $\frac{1}{5} - \left(-\frac{5}{3}\right)$

(9)  $-6 - 1$

(19)  $\frac{3}{5} - 1$

(29)  $-\frac{8}{9} - (-2)$

(39)  $-\frac{4}{5} - (-5)$

(10)  $\frac{1}{4} - (-2)$

(20)  $-\frac{1}{2} + \frac{1}{4}$

(30)  $-\frac{1}{4} - \left(-\frac{3}{5}\right)$

(40)  $\frac{6}{5} + \left(-\frac{1}{4}\right)$



(1)  $\frac{1}{3} - (-5)$

(11)  $\frac{7}{9} + (-1)$

(21)  $-1 + \frac{10}{3}$

(31)  $-\frac{7}{4} - \left(-\frac{4}{3}\right)$

(2)  $\frac{9}{4} + \left(-\frac{7}{6}\right)$

(12)  $-\frac{2}{5} - \frac{9}{4}$

(22)  $-4 + \frac{9}{8}$

(32)  $1 + \left(-\frac{1}{2}\right)$

(3)  $-3 - \frac{5}{9}$

(13)  $\frac{2}{5} + \left(-\frac{5}{3}\right)$

(23)  $-\frac{5}{3} - \frac{1}{3}$

(33)  $-3 + \frac{4}{7}$

(4)  $\frac{3}{2} - \left(-\frac{7}{5}\right)$

(14)  $-\frac{3}{5} - \frac{2}{7}$

(24)  $-\frac{3}{5} - \left(-\frac{6}{7}\right)$

(34)  $\frac{2}{3} + \frac{8}{3}$

(5)  $9 + \frac{3}{2}$

(15)  $\frac{4}{5} - \left(-\frac{2}{9}\right)$

(25)  $-\frac{1}{6} + \left(-\frac{3}{2}\right)$

(35)  $5 + \frac{1}{5}$

(6)  $\frac{1}{2} - \left(-\frac{4}{7}\right)$

(16)  $-\frac{7}{2} - \frac{2}{7}$

(26)  $-1 - \left(-\frac{7}{10}\right)$

(36)  $1 - (-2)$

(7)  $-\frac{3}{4} - \frac{7}{2}$

(17)  $\frac{9}{8} - \frac{1}{2}$

(27)  $-10 + (-4)$

(37)  $\frac{4}{5} + \left(-\frac{8}{5}\right)$

(8)  $\frac{5}{2} - \frac{1}{9}$

(18)  $-\frac{5}{3} - \left(-\frac{9}{5}\right)$

(28)  $-\frac{5}{9} + \frac{1}{9}$

(38)  $7 + \left(-\frac{5}{7}\right)$

(9)  $\frac{5}{8} + \frac{10}{3}$

(19)  $\frac{10}{7} + 3$

(29)  $-\frac{2}{3} + \left(-\frac{4}{5}\right)$

(39)  $\frac{5}{7} + \frac{3}{4}$

(10)  $\frac{5}{3} - \left(-\frac{7}{4}\right)$

(20)  $\frac{5}{7} - \frac{1}{4}$

(30)  $-2 - \left(-\frac{1}{9}\right)$

(40)  $1 + (-1)$

$$(1) -3 + \left(-\frac{3}{4}\right) \\ = -\frac{15}{4}$$

$$(2) -\frac{10}{9} + \left(-\frac{2}{3}\right) \\ = -\frac{16}{9}$$

$$(3) -\frac{4}{7} - \left(-\frac{1}{2}\right) \\ = -\frac{1}{14}$$

$$(4) 9 - \left(-\frac{8}{7}\right) \\ = \frac{71}{7}$$

$$(5) \frac{3}{5} + (-3) \\ = -\frac{12}{5}$$

$$(6) -1 - \frac{3}{4} \\ = -\frac{7}{4}$$

$$(7) \frac{3}{10} + (-5) \\ = -\frac{47}{10}$$

$$(8) -\frac{8}{7} + \frac{4}{3} \\ = \frac{4}{21}$$

$$(9) -5 - \frac{2}{3} \\ = -\frac{17}{3}$$

$$(10) \frac{1}{7} - \frac{4}{3} \\ = -\frac{25}{21}$$

$$(11) \frac{3}{4} - \frac{8}{3} \\ = -\frac{23}{12}$$

$$(12) -5 - \frac{7}{4} \\ = -\frac{27}{4}$$

$$(13) -6 + \left(-\frac{1}{3}\right) \\ = -\frac{19}{3}$$

$$(14) -\frac{7}{4} + \frac{8}{3} \\ = \frac{11}{12}$$

$$(15) -\frac{3}{7} - \left(-\frac{5}{4}\right) \\ = \frac{23}{28}$$

$$(16) -\frac{4}{5} + (-1) \\ = -\frac{9}{5}$$

$$(17) \frac{1}{5} + \left(-\frac{4}{5}\right) \\ = -\frac{3}{5}$$

$$(18) -\frac{5}{2} - \left(-\frac{7}{8}\right) \\ = -\frac{13}{8}$$

$$(19) -\frac{2}{3} + \frac{9}{4} \\ = \frac{19}{12}$$

$$(20) -\frac{1}{3} + \frac{3}{2} \\ = \frac{5}{6}$$

$$(21) 2 + 3 \\ = 5$$

$$(22) 1 + 1 \\ = 2$$

$$(23) -\frac{8}{3} + 3 \\ = \frac{1}{3}$$

$$(24) \frac{2}{5} + \left(-\frac{5}{3}\right) \\ = -\frac{19}{15}$$

$$(25) \frac{3}{5} - \frac{4}{3} \\ = -\frac{11}{15}$$

$$(26) 2 + \frac{8}{3} \\ = \frac{14}{3}$$

$$(27) -\frac{4}{9} + \left(-\frac{4}{3}\right) \\ = -\frac{16}{9}$$

$$(28) -\frac{2}{5} + \left(-\frac{1}{3}\right) \\ = -\frac{11}{15}$$

$$(29) \frac{7}{8} - \frac{8}{9} \\ = -\frac{1}{72}$$

$$(30) -\frac{5}{6} + (-3) \\ = -\frac{23}{6}$$

$$(31) \frac{1}{8} + \left(-\frac{1}{10}\right) \\ = \frac{1}{40}$$

$$(32) -\frac{1}{7} - \frac{3}{2} \\ = -\frac{23}{14}$$

$$(33) 1 + \frac{8}{3} \\ = \frac{11}{3}$$

$$(34) -3 + \frac{2}{3} \\ = -\frac{7}{3}$$

$$(35) -2 - \frac{3}{7} \\ = -\frac{17}{7}$$

$$(36) \frac{7}{3} + (-1) \\ = \frac{4}{3}$$

$$(37) -\frac{5}{3} - \frac{1}{9} \\ = -\frac{16}{9}$$

$$(38) \frac{4}{5} + 7 \\ = \frac{39}{5}$$

$$(39) -\frac{2}{3} - \left(-\frac{1}{3}\right) \\ = -\frac{1}{3}$$

$$(40) -\frac{9}{5} + 1 \\ = -\frac{4}{5}$$

$$(1) \frac{5}{3} - (-3) \\ = \frac{14}{3}$$

$$(11) -\frac{1}{8} - \left(-\frac{1}{3}\right) \\ = \frac{5}{24}$$

$$(21) \frac{1}{6} + \left(-\frac{1}{7}\right) \\ = \frac{1}{42}$$

$$(31) -\frac{1}{3} + \left(-\frac{1}{2}\right) \\ = -\frac{5}{6}$$

$$(2) -\frac{4}{9} + 10 \\ = \frac{86}{9}$$

$$(12) -\frac{8}{5} - \frac{3}{2} \\ = -\frac{31}{10}$$

$$(22) \frac{4}{7} - \frac{2}{3} \\ = -\frac{2}{21}$$

$$(32) \frac{9}{8} - \left(-\frac{5}{7}\right) \\ = \frac{103}{56}$$

$$(3) -\frac{9}{5} - \left(-\frac{1}{4}\right) \\ = -\frac{31}{20}$$

$$(13) -\frac{1}{2} + \frac{3}{5} \\ = \frac{1}{10}$$

$$(23) 3 - \frac{3}{5} \\ = \frac{12}{5}$$

$$(33) -3 + (-2) \\ = -5$$

$$(4) \frac{7}{5} - \frac{1}{5} \\ = \frac{6}{5}$$

$$(14) -5 - \left(-\frac{1}{3}\right) \\ = -\frac{14}{3}$$

$$(24) \frac{5}{2} + \left(-\frac{1}{2}\right) \\ = 2$$

$$(34) -\frac{1}{4} - \frac{3}{2} \\ = -\frac{7}{4}$$

$$(5) \frac{7}{3} + \left(-\frac{5}{8}\right) \\ = \frac{41}{24}$$

$$(15) -\frac{3}{7} + \frac{3}{5} \\ = \frac{3}{35}$$

$$(25) -\frac{3}{7} - \frac{8}{9} \\ = -\frac{83}{63}$$

$$(35) 9 - \frac{4}{9} \\ = \frac{77}{9}$$

$$(6) \frac{4}{9} + 5 \\ = \frac{49}{9}$$

$$(16) -\frac{1}{2} + \left(-\frac{7}{5}\right) \\ = -\frac{19}{10}$$

$$(26) \frac{2}{7} + (-8) \\ = -\frac{54}{7}$$

$$(36) -\frac{5}{9} + \left(-\frac{5}{2}\right) \\ = -\frac{55}{18}$$

$$(7) \frac{3}{2} + \frac{10}{7} \\ = \frac{41}{14}$$

$$(17) -2 - \left(-\frac{5}{3}\right) \\ = -\frac{1}{3}$$

$$(27) -\frac{1}{2} + (-7) \\ = -\frac{15}{2}$$

$$(37) -3 - \left(-\frac{2}{3}\right) \\ = -\frac{7}{3}$$

$$(8) -\frac{1}{9} - (-2) \\ = \frac{17}{9}$$

$$(18) \frac{1}{8} + 6 \\ = \frac{49}{8}$$

$$(28) -\frac{4}{5} + 6 \\ = \frac{26}{5}$$

$$(38) \frac{1}{2} + \frac{6}{7} \\ = \frac{19}{14}$$

$$(9) \frac{1}{7} + \left(-\frac{1}{3}\right) \\ = -\frac{4}{21}$$

$$(19) -\frac{3}{7} + \frac{4}{7} \\ = \frac{1}{7}$$

$$(29) \frac{10}{3} - \left(-\frac{6}{7}\right) \\ = \frac{88}{21}$$

$$(39) \frac{2}{3} + 2 \\ = \frac{8}{3}$$

$$(10) -\frac{1}{8} - 5 \\ = -\frac{41}{8}$$

$$(20) \frac{1}{4} - \left(-\frac{3}{7}\right) \\ = \frac{19}{28}$$

$$(30) \frac{5}{3} + \frac{4}{3} \\ = 3$$

$$(40) -6 - \frac{3}{2} \\ = -\frac{15}{2}$$

$$(1) \frac{8}{7} - \frac{2}{7} \\ = \frac{6}{7}$$

$$(11) -2 + \frac{1}{4} \\ = -\frac{7}{4}$$

$$(21) -6 - \frac{8}{9} \\ = -\frac{62}{9}$$

$$(31) 4 - 1 \\ = 3$$

$$(2) -\frac{3}{2} - \left(-\frac{6}{7}\right) \\ = -\frac{9}{14}$$

$$(12) -\frac{8}{5} + \frac{1}{9} \\ = -\frac{67}{45}$$

$$(22) -\frac{2}{5} + \left(-\frac{1}{2}\right) \\ = -\frac{9}{10}$$

$$(32) \frac{2}{7} + \frac{1}{8} \\ = \frac{23}{56}$$

$$(3) -\frac{1}{6} + \frac{7}{8} \\ = \frac{17}{24}$$

$$(13) -\frac{9}{5} - \frac{8}{9} \\ = -\frac{121}{45}$$

$$(23) -\frac{3}{4} + \frac{1}{4} \\ = -\frac{1}{2}$$

$$(33) -\frac{5}{4} - \frac{1}{7} \\ = -\frac{39}{28}$$

$$(4) \frac{2}{3} + \left(-\frac{1}{10}\right) \\ = \frac{17}{30}$$

$$(14) -1 + (-3) \\ = -4$$

$$(24) 1 - \left(-\frac{3}{2}\right) \\ = \frac{5}{2}$$

$$(34) \frac{2}{3} + \left(-\frac{4}{3}\right) \\ = -\frac{2}{3}$$

$$(5) -1 + \left(-\frac{4}{9}\right) \\ = -\frac{13}{9}$$

$$(15) \frac{1}{3} - (-8) \\ = \frac{25}{3}$$

$$(25) -1 + 3 \\ = 2$$

$$(35) -\frac{7}{10} - (-7) \\ = \frac{10}{10}$$

$$(6) \frac{1}{6} - \left(-\frac{5}{2}\right) \\ = \frac{8}{3}$$

$$(16) -10 + \frac{6}{5} \\ = -\frac{44}{5}$$

$$(26) -\frac{8}{5} + (-2) \\ = -\frac{18}{5}$$

$$(36) \frac{1}{2} - \frac{5}{6} \\ = -\frac{1}{3}$$

$$(7) -2 - \frac{5}{7} \\ = -\frac{19}{7}$$

$$(17) -\frac{7}{2} + 1 \\ = -\frac{5}{2}$$

$$(27) -\frac{10}{3} - \frac{9}{2} \\ = -\frac{47}{6}$$

$$(37) \frac{3}{8} - \left(-\frac{7}{9}\right) \\ = \frac{83}{72}$$

$$(8) \frac{1}{2} - \left(-\frac{1}{3}\right) \\ = \frac{5}{6}$$

$$(18) -\frac{3}{7} - \frac{5}{7} \\ = -\frac{8}{7}$$

$$(28) 1 + \left(-\frac{4}{7}\right) \\ = \frac{3}{7}$$

$$(38) \frac{1}{3} - (-1) \\ = \frac{4}{3}$$

$$(9) 1 - \left(-\frac{1}{10}\right) \\ = \frac{11}{10}$$

$$(19) \frac{7}{3} - 2 \\ = \frac{1}{3}$$

$$(29) 2 + \left(-\frac{9}{2}\right) \\ = -\frac{5}{2}$$

$$(39) 1 - \frac{2}{3} \\ = \frac{1}{3}$$

$$(10) -\frac{5}{4} + \left(-\frac{1}{3}\right) \\ = -\frac{19}{12}$$

$$(20) \frac{1}{2} + (-8) \\ = -\frac{15}{2}$$

$$(30) -\frac{2}{7} + \frac{7}{10} \\ = \frac{29}{70}$$

$$(40) \frac{7}{4} + 1 \\ = \frac{11}{4}$$

$$(1) \frac{1}{3} - 2 \\ = -\frac{5}{3}$$

$$(2) \frac{1}{4} + 3 \\ = \frac{13}{4}$$

$$(3) \frac{3}{10} + \left(-\frac{5}{2}\right) \\ = -\frac{11}{5}$$

$$(4) \frac{3}{5} - \frac{4}{9} \\ = \frac{7}{45}$$

$$(5) -\frac{4}{3} + \frac{3}{8} \\ = -\frac{23}{24}$$

$$(6) -\frac{5}{8} + \left(-\frac{4}{7}\right) \\ = -\frac{67}{56}$$

$$(7) -1 + \left(-\frac{1}{5}\right) \\ = -\frac{6}{5}$$

$$(8) \frac{7}{5} - \left(-\frac{5}{2}\right) \\ = \frac{39}{10}$$

$$(9) -1 - (-1) \\ = 0$$

$$(10) -\frac{5}{8} - \frac{1}{3} \\ = -\frac{23}{24}$$

$$(11) 10 - \left(-\frac{3}{2}\right) \\ = \frac{23}{2}$$

$$(12) -\frac{5}{4} + \frac{7}{9} \\ = -\frac{17}{36}$$

$$(13) -1 + \left(-\frac{2}{3}\right) \\ = -\frac{5}{3}$$

$$(14) \frac{3}{8} + 10 \\ = \frac{83}{8}$$

$$(15) -\frac{5}{4} + \left(-\frac{6}{5}\right) \\ = -\frac{49}{20}$$

$$(16) -\frac{4}{7} - \left(-\frac{8}{3}\right) \\ = \frac{44}{21}$$

$$(17) \frac{1}{2} + \frac{5}{8} \\ = \frac{9}{8}$$

$$(18) -\frac{1}{3} - \left(-\frac{1}{10}\right) \\ = -\frac{7}{30}$$

$$(19) 1 - \left(-\frac{1}{4}\right) \\ = \frac{5}{4}$$

$$(20) -1 - (-1) \\ = 0$$

$$(21) -8 + (-2) \\ = -10$$

$$(22) \frac{7}{8} - 5 \\ = -\frac{33}{8}$$

$$(23) \frac{3}{2} + \frac{2}{9} \\ = \frac{31}{18}$$

$$(24) \frac{4}{7} - \frac{7}{5} \\ = -\frac{29}{35}$$

$$(25) -8 + 2 \\ = -6$$

$$(26) \frac{4}{5} - 1 \\ = -\frac{1}{5}$$

$$(27) 3 - \frac{4}{7} \\ = \frac{17}{7}$$

$$(28) 5 + \left(-\frac{1}{8}\right) \\ = \frac{39}{8}$$

$$(29) 4 - 4 \\ = 0$$

$$(30) \frac{7}{4} - (-5) \\ = \frac{27}{4}$$

$$(31) \frac{9}{4} + 1 \\ = \frac{13}{4}$$

$$(32) -5 - 1 \\ = -6$$

$$(33) -\frac{5}{2} - (-1) \\ = -\frac{3}{2}$$

$$(34) -\frac{7}{4} + \left(-\frac{1}{2}\right) \\ = -\frac{9}{4}$$

$$(35) \frac{7}{9} - 1 \\ = -\frac{2}{9}$$

$$(36) -5 - \left(-\frac{1}{2}\right) \\ = -\frac{9}{2}$$

$$(37) -\frac{1}{4} + \left(-\frac{1}{7}\right) \\ = -\frac{11}{28}$$

$$(38) \frac{5}{2} - (-3) \\ = \frac{11}{2}$$

$$(39) \frac{2}{3} + \frac{9}{8} \\ = \frac{43}{24}$$

$$(40) \frac{7}{10} - \left(-\frac{9}{5}\right) \\ = \frac{5}{2}$$

$$(1) -5 - \left(-\frac{6}{5}\right) \\ = -\frac{19}{5}$$

$$(11) \frac{8}{7} + \left(-\frac{8}{5}\right) \\ = -\frac{16}{35}$$

$$(21) -\frac{3}{8} + \left(-\frac{4}{7}\right) \\ = -\frac{53}{56}$$

$$(31) -1 - \frac{3}{5} \\ = -\frac{8}{5}$$

$$(2) -1 - \left(-\frac{10}{7}\right) \\ = \frac{3}{7}$$

$$(12) \frac{6}{5} - \left(-\frac{1}{4}\right) \\ = \frac{29}{20}$$

$$(22) -\frac{9}{10} - (-1) \\ = \frac{1}{10}$$

$$(32) 1 - \left(-\frac{5}{3}\right) \\ = \frac{8}{3}$$

$$(3) -\frac{3}{4} + \left(-\frac{1}{4}\right) \\ = -1$$

$$(13) -9 - \left(-\frac{1}{2}\right) \\ = -\frac{17}{2}$$

$$(23) -\frac{2}{5} + (-2) \\ = -\frac{12}{5}$$

$$(33) \frac{9}{8} - (-3) \\ = \frac{33}{8}$$

$$(4) -5 - \frac{9}{10} \\ = -\frac{59}{10}$$

$$(14) \frac{2}{5} - \left(-\frac{9}{2}\right) \\ = \frac{49}{10}$$

$$(24) -\frac{5}{8} - \frac{4}{9} \\ = -\frac{77}{72}$$

$$(34) -\frac{1}{8} - 3 \\ = -\frac{25}{8}$$

$$(5) \frac{1}{9} + \frac{10}{7} \\ = \frac{97}{63}$$

$$(15) -5 - \frac{4}{7} \\ = -\frac{39}{7}$$

$$(25) 1 + \left(-\frac{3}{8}\right) \\ = \frac{5}{8}$$

$$(35) \frac{7}{9} + \frac{7}{3} \\ = \frac{28}{9}$$

$$(6) -\frac{10}{7} + \left(-\frac{7}{6}\right) \\ = -\frac{109}{42}$$

$$(16) -\frac{1}{4} - \frac{2}{3} \\ = -\frac{11}{12}$$

$$(26) 1 - (-3) \\ = 4$$

$$(36) -\frac{3}{4} + \frac{1}{2} \\ = -\frac{1}{4}$$

$$(7) -5 + \left(-\frac{2}{7}\right) \\ = -\frac{37}{7}$$

$$(17) \frac{4}{3} + \frac{1}{7} \\ = \frac{31}{21}$$

$$(27) -\frac{10}{3} - 2 \\ = -\frac{16}{3}$$

$$(37) -\frac{2}{3} - \frac{2}{3} \\ = -\frac{4}{3}$$

$$(8) 7 + \left(-\frac{8}{3}\right) \\ = \frac{13}{3}$$

$$(18) -\frac{5}{7} + 1 \\ = \frac{2}{7}$$

$$(28) -\frac{1}{6} - \frac{1}{8} \\ = -\frac{7}{24}$$

$$(38) -\frac{5}{9} - \frac{6}{5} \\ = -\frac{79}{45}$$

$$(9) -3 + 1 \\ = -2$$

$$(19) \frac{5}{2} + \frac{3}{8} \\ = \frac{23}{8}$$

$$(29) \frac{10}{9} - \frac{5}{2} \\ = -\frac{1}{18}$$

$$(39) \frac{3}{8} - \frac{1}{2} \\ = -\frac{1}{8}$$

$$(10) 2 + (-3) \\ = -1$$

$$(20) \frac{2}{3} - \left(-\frac{3}{4}\right) \\ = \frac{17}{12}$$

$$(30) -1 - \frac{3}{5} \\ = -\frac{8}{5}$$

$$(40) -\frac{1}{7} + \frac{5}{8} \\ = \frac{27}{56}$$

$$(1) -\frac{1}{2} - \frac{4}{3}$$

$$= -\frac{11}{6}$$

$$(2) 3 - (-1)$$

$$= 4$$

$$(3) \frac{1}{2} - 2$$

$$= -\frac{3}{2}$$

$$(4) -\frac{1}{2} + \left(-\frac{4}{5}\right)$$

$$= -\frac{13}{10}$$

$$(5) -\frac{1}{2} + 1$$

$$= \frac{1}{2}$$

$$(6) -1 + \frac{5}{4}$$

$$= \frac{1}{4}$$

$$(7) -\frac{7}{9} + 4$$

$$= \frac{29}{9}$$

$$(8) -8 - \left(-\frac{3}{4}\right)$$

$$= -\frac{29}{4}$$

$$(9) -\frac{6}{7} + 4$$

$$= \frac{22}{7}$$

$$(10) 1 + \frac{1}{2}$$

$$= \frac{3}{2}$$

$$(11) -\frac{8}{9} + \frac{1}{5}$$

$$= -\frac{31}{45}$$

$$(12) \frac{3}{5} - \frac{2}{5}$$

$$= \frac{1}{5}$$

$$(13) -\frac{5}{6} + \frac{9}{8}$$

$$= \frac{7}{24}$$

$$(14) -\frac{5}{4} + \frac{4}{5}$$

$$= -\frac{9}{20}$$

$$(15) -\frac{3}{10} + \frac{2}{7}$$

$$= -\frac{1}{70}$$

$$(16) \frac{3}{2} - \frac{5}{2}$$

$$= -1$$

$$(17) \frac{5}{3} - 10$$

$$= -\frac{25}{3}$$

$$(18) \frac{7}{3} - \frac{10}{3}$$

$$= -1$$

$$(19) \frac{1}{10} - \frac{1}{3}$$

$$= -\frac{7}{30}$$

$$(20) \frac{5}{6} + (-10)$$

$$= -\frac{55}{6}$$

$$(21) -10 + (-4)$$

$$= -14$$

$$(22) \frac{7}{3} - \left(-\frac{4}{7}\right)$$

$$= \frac{61}{21}$$

$$(23) -\frac{3}{8} + \frac{1}{7}$$

$$= -\frac{13}{56}$$

$$(24) -\frac{1}{3} - (-6)$$

$$= \frac{17}{3}$$

$$(25) -\frac{2}{3} - \left(-\frac{4}{3}\right)$$

$$= \frac{2}{3}$$

$$(26) \frac{3}{2} + \left(-\frac{3}{5}\right)$$

$$= \frac{9}{10}$$

$$(27) -3 + \frac{4}{3}$$

$$= -\frac{5}{3}$$

$$(28) 3 - \frac{5}{6}$$

$$= \frac{13}{6}$$

$$(29) -\frac{3}{2} - \left(-\frac{1}{8}\right)$$

$$= -\frac{11}{8}$$

$$(30) -\frac{2}{9} - \left(-\frac{1}{9}\right)$$

$$= -\frac{1}{9}$$

$$(31) -\frac{1}{5} - \left(-\frac{1}{2}\right)$$

$$= \frac{3}{10}$$

$$(32) -\frac{1}{4} + \left(-\frac{7}{2}\right)$$

$$= -\frac{15}{4}$$

$$(33) -\frac{3}{5} + 8$$

$$= \frac{37}{5}$$

$$(34) \frac{7}{10} + \frac{4}{9}$$

$$= \frac{103}{90}$$

$$(35) -\frac{1}{5} + \frac{5}{2}$$

$$= \frac{23}{10}$$

$$(36) \frac{1}{5} + \left(-\frac{2}{3}\right)$$

$$= -\frac{7}{15}$$

$$(37) -\frac{7}{5} - \left(-\frac{4}{5}\right)$$

$$= -\frac{3}{5}$$

$$(38) -7 + \left(-\frac{7}{2}\right)$$

$$= -\frac{21}{2}$$

$$(39) -\frac{1}{4} - \left(-\frac{9}{10}\right)$$

$$= \frac{13}{20}$$

$$(40) \frac{6}{5} + (-1)$$

$$= \frac{1}{5}$$

$$(1) -\frac{5}{7} + \left(-\frac{7}{10}\right) \\ = -\frac{99}{70}$$

$$(2) -7 - \left(-\frac{2}{5}\right) \\ = -\frac{33}{5}$$

$$(3) -\frac{7}{2} + \frac{7}{6} \\ = -\frac{7}{3}$$

$$(4) -\frac{5}{9} + \frac{3}{4} \\ = \frac{7}{36}$$

$$(5) -\frac{8}{5} - \left(-\frac{1}{10}\right) \\ = -\frac{3}{2}$$

$$(6) -2 - \frac{8}{7} \\ = -\frac{22}{7}$$

$$(7) -\frac{7}{9} + \left(-\frac{4}{9}\right) \\ = -\frac{11}{9}$$

$$(8) 3 + \left(-\frac{5}{4}\right) \\ = \frac{7}{4}$$

$$(9) -\frac{3}{10} - \frac{1}{2} \\ = -\frac{4}{5}$$

$$(10) \frac{2}{3} - \frac{8}{3} \\ = -2$$

$$(11) 1 + 1 \\ = 2$$

$$(12) -\frac{7}{8} + (-3) \\ = -\frac{31}{8}$$

$$(13) \frac{2}{7} + \left(-\frac{1}{3}\right) \\ = -\frac{1}{21}$$

$$(14) -\frac{7}{5} + \frac{4}{3} \\ = -\frac{1}{15}$$

$$(15) -\frac{2}{7} - (-5) \\ = \frac{33}{7}$$

$$(16) -\frac{3}{10} - \left(-\frac{3}{5}\right) \\ = \frac{3}{10}$$

$$(17) -\frac{1}{3} + \frac{2}{5} \\ = \frac{1}{15}$$

$$(18) -\frac{1}{2} - 2 \\ = -\frac{5}{2}$$

$$(19) \frac{7}{6} + (-1) \\ = \frac{1}{6}$$

$$(20) -1 + (-2) \\ = -3$$

$$(21) -1 + (-1) \\ = -2$$

$$(22) \frac{3}{5} - \frac{7}{2} \\ = -\frac{29}{10}$$

$$(23) -\frac{9}{2} - \frac{3}{4} \\ = -\frac{21}{4}$$

$$(24) 9 + 1 \\ = 10$$

$$(25) \frac{1}{8} - \frac{5}{3} \\ = -\frac{37}{24}$$

$$(26) \frac{5}{3} + \frac{3}{4} \\ = \frac{29}{12}$$

$$(27) \frac{5}{6} - (-2) \\ = \frac{17}{6}$$

$$(28) -\frac{4}{3} + \frac{1}{2} \\ = -\frac{5}{6}$$

$$(29) -\frac{8}{3} + \left(-\frac{1}{2}\right) \\ = -\frac{19}{6}$$

$$(30) -\frac{9}{2} + \frac{9}{2} \\ = 0$$

$$(31) -\frac{3}{4} - 1 \\ = -\frac{7}{4}$$

$$(32) \frac{10}{3} + (-4) \\ = -\frac{2}{3}$$

$$(33) \frac{3}{2} - \frac{7}{3} \\ = -\frac{5}{6}$$

$$(34) \frac{6}{7} + \left(-\frac{10}{7}\right) \\ = -\frac{4}{7}$$

$$(35) \frac{5}{3} + \frac{4}{5} \\ = \frac{37}{15}$$

$$(36) -\frac{1}{2} - \left(-\frac{7}{9}\right) \\ = \frac{5}{18}$$

$$(37) -1 + \frac{2}{7} \\ = -\frac{5}{7}$$

$$(38) -\frac{7}{2} + (-1) \\ = -\frac{9}{2}$$

$$(39) -\frac{8}{3} + \left(-\frac{4}{7}\right) \\ = -\frac{68}{21}$$

$$(40) -\frac{1}{3} + 1 \\ = \frac{2}{3}$$



$$(1) \quad -\frac{3}{4} - \frac{9}{4} \\ = -3$$

$$(11) \quad -2 - \left(-\frac{5}{9}\right) \\ = -\frac{13}{9}$$

$$(21) \quad \frac{2}{5} - 5 \\ = -\frac{23}{5}$$

$$(31) \quad \frac{1}{2} - \left(-\frac{7}{8}\right) \\ = \frac{11}{8}$$

$$(2) \quad \frac{4}{3} + \left(-\frac{7}{2}\right) \\ = -\frac{13}{6}$$

$$(12) \quad -10 + \frac{10}{9} \\ = -\frac{80}{9}$$

$$(22) \quad -\frac{10}{3} + \left(-\frac{2}{7}\right) \\ = -\frac{76}{21}$$

$$(32) \quad -\frac{1}{3} + \left(-\frac{9}{8}\right) \\ = -\frac{35}{24}$$

$$(3) \quad \frac{10}{7} + \frac{5}{3} \\ = \frac{65}{21}$$

$$(13) \quad -1 + (-1) \\ = -2$$

$$(23) \quad 9 + \frac{2}{3} \\ = \frac{29}{3}$$

$$(33) \quad \frac{9}{8} - \frac{7}{2} \\ = -\frac{19}{8}$$

$$(4) \quad -1 - \left(-\frac{6}{5}\right) \\ = \frac{1}{5}$$

$$(14) \quad 3 + \frac{1}{5} \\ = \frac{16}{5}$$

$$(24) \quad 2 + \left(-\frac{7}{9}\right) \\ = \frac{11}{9}$$

$$(34) \quad -\frac{1}{3} - \frac{7}{10} \\ = -\frac{31}{30}$$

$$(5) \quad -\frac{4}{3} - (-2) \\ = \frac{2}{3}$$

$$(15) \quad 3 + \frac{5}{9} \\ = \frac{32}{9}$$

$$(25) \quad -\frac{4}{3} - \frac{1}{3} \\ = -\frac{5}{3}$$

$$(35) \quad 7 + \frac{5}{8} \\ = \frac{61}{8}$$

$$(6) \quad \frac{2}{5} - \frac{5}{3} \\ = -\frac{19}{15}$$

$$(16) \quad \frac{5}{3} - \frac{9}{4} \\ = -\frac{7}{12}$$

$$(26) \quad -\frac{3}{8} - \left(-\frac{1}{2}\right) \\ = \frac{1}{8}$$

$$(36) \quad -3 - \left(-\frac{1}{2}\right) \\ = -\frac{5}{2}$$

$$(7) \quad -\frac{3}{5} + \frac{2}{3} \\ = \frac{1}{15}$$

$$(17) \quad 10 + \left(-\frac{9}{4}\right) \\ = \frac{31}{4}$$

$$(27) \quad 2 + \left(-\frac{5}{3}\right) \\ = \frac{1}{3}$$

$$(37) \quad -\frac{8}{5} - \left(-\frac{3}{10}\right) \\ = -\frac{13}{10}$$

$$(8) \quad \frac{3}{2} - \left(-\frac{5}{7}\right) \\ = \frac{31}{14}$$

$$(18) \quad \frac{9}{5} + \left(-\frac{6}{5}\right) \\ = \frac{3}{5}$$

$$(28) \quad 4 + \left(-\frac{1}{4}\right) \\ = \frac{15}{4}$$

$$(38) \quad \frac{4}{5} - 2 \\ = -\frac{6}{5}$$

$$(9) \quad \frac{5}{2} + \frac{2}{5} \\ = \frac{29}{10}$$

$$(19) \quad -\frac{1}{2} + 1 \\ = \frac{1}{2}$$

$$(29) \quad \frac{3}{2} + \frac{2}{3} \\ = \frac{13}{6}$$

$$(39) \quad \frac{2}{7} + \frac{8}{7} \\ = \frac{10}{7}$$

$$(10) \quad \frac{1}{10} + \left(-\frac{4}{5}\right) \\ = -\frac{7}{10}$$

$$(20) \quad \frac{3}{4} + \frac{1}{7} \\ = \frac{25}{28}$$

$$(30) \quad \frac{7}{9} - (-1) \\ = \frac{16}{9}$$

$$(40) \quad -\frac{9}{2} - 1 \\ = -\frac{11}{2}$$

$$(1) 2 - \frac{3}{2}$$

$$= \frac{1}{2}$$

$$(11) \frac{3}{4} + (-1)$$

$$= -\frac{1}{4}$$

$$(21) \frac{8}{3} + 1$$

$$= \frac{11}{3}$$

$$(31) -\frac{5}{3} + \frac{2}{7}$$

$$= -\frac{29}{21}$$

$$(2) -\frac{6}{5} + \left(-\frac{9}{2}\right)$$

$$= -\frac{57}{10}$$

$$(12) \frac{4}{9} - \frac{1}{9}$$

$$= \frac{1}{3}$$

$$(22) \frac{9}{2} - \left(-\frac{1}{9}\right)$$

$$= \frac{83}{18}$$

$$(32) -\frac{3}{2} + \frac{7}{8}$$

$$= -\frac{5}{8}$$

$$(3) 1 + \left(-\frac{7}{4}\right)$$

$$= -\frac{3}{4}$$

$$(13) \frac{2}{5} + \left(-\frac{10}{7}\right)$$

$$= -\frac{36}{35}$$

$$(23) -1 + \frac{1}{3}$$

$$= -\frac{2}{3}$$

$$(33) -\frac{3}{7} - \frac{4}{5}$$

$$= -\frac{43}{35}$$

$$(4) -5 + \frac{5}{7}$$

$$= -\frac{30}{7}$$

$$(14) -4 + \left(-\frac{5}{4}\right)$$

$$= -\frac{21}{4}$$

$$(24) -\frac{3}{2} - (-3)$$

$$= \frac{3}{2}$$

$$(34) 2 - 2$$

$$= 0$$

$$(5) -\frac{6}{5} + \frac{5}{2}$$

$$= \frac{13}{10}$$

$$(15) -\frac{5}{2} - \frac{4}{7}$$

$$= -\frac{43}{14}$$

$$(25) \frac{4}{5} + (-1)$$

$$= -\frac{1}{5}$$

$$(35) \frac{2}{3} - \left(-\frac{5}{3}\right)$$

$$= \frac{7}{3}$$

$$(6) -4 - 3$$

$$= -7$$

$$(16) -1 - (-8)$$

$$= 7$$

$$(26) \frac{5}{9} + \left(-\frac{9}{7}\right)$$

$$= -\frac{46}{63}$$

$$(36) \frac{5}{4} + \left(-\frac{2}{3}\right)$$

$$= \frac{7}{12}$$

$$(7) -\frac{1}{2} - \left(-\frac{7}{4}\right)$$

$$= \frac{5}{4}$$

$$(17) \frac{3}{2} - \frac{7}{4}$$

$$= -\frac{1}{4}$$

$$(27) -\frac{7}{10} + \frac{1}{6}$$

$$= -\frac{8}{15}$$

$$(37) \frac{8}{7} + \frac{3}{2}$$

$$= \frac{37}{14}$$

$$(8) 5 - \left(-\frac{3}{10}\right)$$

$$= \frac{53}{10}$$

$$(18) -\frac{9}{10} + \frac{7}{6}$$

$$= \frac{1}{15}$$

$$(28) -2 + \frac{5}{3}$$

$$= -\frac{1}{3}$$

$$(38) -1 + \left(-\frac{1}{3}\right)$$

$$= -\frac{4}{3}$$

$$(9) \frac{10}{9} + \frac{5}{6}$$

$$= \frac{35}{18}$$

$$(19) -\frac{1}{3} + \left(-\frac{5}{9}\right)$$

$$= -\frac{8}{9}$$

$$(29) \frac{2}{5} + \left(-\frac{3}{4}\right)$$

$$= -\frac{7}{20}$$

$$(39) 2 + \frac{10}{3}$$

$$= \frac{16}{3}$$

$$(10) \frac{3}{10} + \frac{9}{5}$$

$$= \frac{21}{10}$$

$$(20) 2 - \frac{5}{3}$$

$$= \frac{1}{3}$$

$$(30) -\frac{1}{2} + \left(-\frac{3}{2}\right)$$

$$= -2$$

$$(40) -1 - (-6)$$

$$= 5$$

$$(1) -\frac{1}{2} + \frac{5}{7}$$

$$= -\frac{3}{14}$$

$$(11) -\frac{1}{6} + 1$$

$$= \frac{5}{6}$$

$$(21) \frac{3}{5} - (-1)$$

$$= \frac{8}{5}$$

$$(31) \frac{2}{5} + \left(-\frac{4}{3}\right)$$

$$= -\frac{14}{15}$$

$$(2) \frac{1}{4} - \frac{9}{2}$$

$$= -\frac{17}{4}$$

$$(12) \frac{4}{5} + \frac{2}{5}$$

$$= \frac{6}{5}$$

$$(22) -\frac{9}{8} + 1$$

$$= -\frac{1}{8}$$

$$(32) \frac{4}{3} - \left(-\frac{3}{2}\right)$$

$$= \frac{17}{6}$$

$$(3) -3 - \left(-\frac{9}{2}\right)$$

$$= \frac{3}{2}$$

$$(13) -2 + \frac{1}{3}$$

$$= -\frac{5}{3}$$

$$(23) -9 - \left(-\frac{9}{8}\right)$$

$$= -\frac{63}{8}$$

$$(33) \frac{7}{5} + \left(-\frac{5}{3}\right)$$

$$= -\frac{4}{15}$$

$$(4) \frac{5}{2} - \frac{1}{3}$$

$$= \frac{13}{6}$$

$$(14) -7 - \frac{7}{6}$$

$$= -\frac{49}{6}$$

$$(24) -\frac{4}{7} + \frac{7}{9}$$

$$= \frac{13}{63}$$

$$(34) \frac{3}{5} + \frac{8}{9}$$

$$= \frac{67}{45}$$

$$(5) \frac{9}{2} + 4$$

$$= \frac{17}{2}$$

$$(15) 1 - \frac{1}{3}$$

$$= \frac{2}{3}$$

$$(25) 9 + 1$$

$$= 10$$

$$(35) \frac{3}{4} + \left(-\frac{9}{10}\right)$$

$$= -\frac{3}{20}$$

$$(6) -\frac{1}{5} - \frac{2}{3}$$

$$= -\frac{13}{15}$$

$$(16) \frac{1}{8} - \frac{2}{3}$$

$$= -\frac{13}{24}$$

$$(26) -\frac{6}{5} + \frac{6}{7}$$

$$= -\frac{12}{35}$$

$$(36) -\frac{8}{7} + \frac{3}{2}$$

$$= \frac{5}{14}$$

$$(7) -\frac{10}{9} + \frac{9}{2}$$

$$= \frac{61}{18}$$

$$(17) \frac{5}{4} - \frac{5}{9}$$

$$= \frac{25}{36}$$

$$(27) -\frac{1}{2} + \frac{1}{10}$$

$$= -\frac{2}{5}$$

$$(37) -\frac{6}{5} - (-10)$$

$$= \frac{44}{5}$$

$$(8) 1 + \frac{1}{2}$$

$$= \frac{3}{2}$$

$$(18) \frac{5}{6} + 3$$

$$= \frac{23}{6}$$

$$(28) 1 - \left(-\frac{9}{10}\right)$$

$$= \frac{19}{10}$$

$$(38) -\frac{9}{5} - \left(-\frac{1}{2}\right)$$

$$= -\frac{13}{10}$$

$$(9) -\frac{9}{7} + \frac{1}{10}$$

$$= -\frac{83}{70}$$

$$(19) \frac{1}{2} - \left(-\frac{9}{2}\right)$$

$$= 5$$

$$(29) \frac{7}{8} - \left(-\frac{1}{10}\right)$$

$$= \frac{39}{40}$$

$$(39) \frac{5}{2} - \frac{9}{4}$$

$$= \frac{1}{4}$$

$$(10) \frac{4}{7} + \frac{1}{5}$$

$$= \frac{27}{35}$$

$$(20) -2 + \frac{2}{5}$$

$$= -\frac{8}{5}$$

$$(30) -\frac{2}{5} - (-8)$$

$$= \frac{38}{5}$$

$$(40) -\frac{5}{7} + \left(-\frac{9}{2}\right)$$

$$= -\frac{73}{14}$$

$$(1) -9 + \left(-\frac{9}{4}\right) \\ = -\frac{45}{4}$$

$$(11) -\frac{1}{7} - \left(-\frac{1}{2}\right) \\ = \frac{5}{14}$$

$$(21) \frac{10}{3} - \frac{8}{5} \\ = \frac{26}{15}$$

$$(31) -\frac{1}{5} + (-10) \\ = -\frac{51}{5}$$

$$(2) 5 - 8 \\ = -3$$

$$(12) \frac{3}{7} - \frac{1}{4} \\ = \frac{5}{28}$$

$$(22) \frac{3}{7} - 1 \\ = -\frac{4}{7}$$

$$(32) -\frac{9}{10} + \left(-\frac{7}{3}\right) \\ = -\frac{97}{30}$$

$$(3) \frac{2}{5} + \frac{3}{2} \\ = \frac{19}{10}$$

$$(13) -\frac{1}{2} + \left(-\frac{3}{5}\right) \\ = -\frac{11}{10}$$

$$(23) -\frac{7}{6} - \left(-\frac{4}{5}\right) \\ = -\frac{11}{30}$$

$$(33) \frac{8}{3} + (-6) \\ = -\frac{10}{3}$$

$$(4) 7 + \frac{2}{5} \\ = \frac{37}{5}$$

$$(14) \frac{2}{5} - \left(-\frac{7}{4}\right) \\ = \frac{43}{20}$$

$$(24) \frac{1}{2} + \frac{4}{7} \\ = \frac{15}{14}$$

$$(34) -\frac{6}{5} - \left(-\frac{1}{2}\right) \\ = -\frac{7}{10}$$

$$(5) -\frac{5}{2} + \left(-\frac{9}{4}\right) \\ = -\frac{19}{4}$$

$$(15) 8 - 2 \\ = 6$$

$$(25) \frac{1}{7} - 5 \\ = -\frac{34}{7}$$

$$(35) \frac{3}{2} - \frac{1}{7} \\ = \frac{19}{14}$$

$$(6) -1 + \frac{4}{3} \\ = \frac{1}{3}$$

$$(16) 2 + 1 \\ = 3$$

$$(26) -\frac{9}{8} - \left(-\frac{7}{6}\right) \\ = \frac{1}{24}$$

$$(36) \frac{3}{10} + \frac{2}{7} \\ = \frac{41}{70}$$

$$(7) \frac{10}{7} - \left(-\frac{7}{9}\right) \\ = \frac{139}{63}$$

$$(17) -\frac{8}{7} - \frac{5}{4} \\ = -\frac{67}{28}$$

$$(27) \frac{2}{3} - \frac{8}{9} \\ = -\frac{2}{9}$$

$$(37) \frac{6}{5} - (-1) \\ = \frac{11}{5}$$

$$(8) \frac{3}{2} - 1 \\ = \frac{1}{2}$$

$$(18) \frac{4}{3} - \frac{1}{3} \\ = 1$$

$$(28) \frac{1}{7} + \left(-\frac{9}{10}\right) \\ = -\frac{53}{70}$$

$$(38) -\frac{4}{5} - \left(-\frac{9}{2}\right) \\ = \frac{37}{10}$$

$$(9) -1 - \frac{7}{3} \\ = -\frac{10}{3}$$

$$(19) \frac{10}{3} - \left(-\frac{1}{2}\right) \\ = \frac{23}{6}$$

$$(29) \frac{5}{8} + \left(-\frac{1}{8}\right) \\ = \frac{1}{2}$$

$$(39) -\frac{2}{3} - (-2) \\ = \frac{4}{3}$$

$$(10) 9 + \left(-\frac{1}{9}\right) \\ = \frac{80}{9}$$

$$(20) -2 + \left(-\frac{1}{8}\right) \\ = -\frac{17}{8}$$

$$(30) \frac{5}{2} + \frac{3}{4} \\ = \frac{13}{4}$$

$$(40) -2 + \left(-\frac{7}{9}\right) \\ = -\frac{25}{9}$$

$$(1) \ 5 + (-3) \\ = 2$$

$$(11) \ -\frac{4}{7} - \left(-\frac{4}{7}\right) \\ = 0$$

$$(21) \ -\frac{3}{5} + \frac{5}{4} \\ = \frac{13}{20}$$

$$(31) \ \frac{7}{6} - \frac{2}{3} \\ = \frac{1}{2}$$

$$(2) \ -\frac{1}{5} - \frac{1}{4} \\ = -\frac{9}{20}$$

$$(12) \ -1 - \frac{1}{5} \\ = -\frac{6}{5}$$

$$(22) \ -\frac{2}{3} + \left(-\frac{5}{8}\right) \\ = -\frac{31}{24}$$

$$(32) \ 2 + \left(-\frac{1}{3}\right) \\ = \frac{5}{3}$$

$$(3) \ \frac{2}{5} + \left(-\frac{9}{4}\right) \\ = -\frac{37}{20}$$

$$(13) \ \frac{1}{2} + \frac{3}{2} \\ = 2$$

$$(23) \ \frac{4}{3} - \left(-\frac{8}{5}\right) \\ = \frac{44}{15}$$

$$(33) \ -\frac{5}{2} + \left(-\frac{1}{2}\right) \\ = -3$$

$$(4) \ \frac{1}{3} - \frac{9}{4} \\ = -\frac{23}{12}$$

$$(14) \ -\frac{3}{10} - 1 \\ = -\frac{13}{10}$$

$$(24) \ -\frac{1}{3} - \left(-\frac{5}{7}\right) \\ = \frac{8}{21}$$

$$(34) \ -\frac{5}{2} - \left(-\frac{5}{6}\right) \\ = -\frac{5}{3}$$

$$(5) \ -\frac{1}{8} - \frac{5}{8} \\ = -\frac{3}{4}$$

$$(15) \ 1 - \frac{9}{7} \\ = -\frac{2}{7}$$

$$(25) \ -\frac{1}{9} + \left(-\frac{7}{8}\right) \\ = -\frac{71}{72}$$

$$(35) \ -5 + \left(-\frac{4}{5}\right) \\ = -\frac{29}{5}$$

$$(6) \ -\frac{1}{9} - (-5) \\ = \frac{44}{9}$$

$$(16) \ -\frac{10}{7} + (-1) \\ = -\frac{17}{7}$$

$$(26) \ \frac{2}{5} - (-10) \\ = \frac{52}{5}$$

$$(36) \ -\frac{3}{5} + \left(-\frac{9}{2}\right) \\ = -\frac{51}{10}$$

$$(7) \ \frac{3}{10} + (-1) \\ = -\frac{7}{10}$$

$$(17) \ \frac{1}{2} + \frac{5}{2} \\ = 3$$

$$(27) \ -3 + 2 \\ = -1$$

$$(37) \ \frac{8}{5} - \frac{3}{5} \\ = 1$$

$$(8) \ -\frac{8}{7} - 8 \\ = -\frac{64}{7}$$

$$(18) \ \frac{9}{4} - (-2) \\ = \frac{17}{4}$$

$$(28) \ \frac{7}{8} - \frac{4}{5} \\ = \frac{3}{40}$$

$$(38) \ \frac{5}{7} - \left(-\frac{9}{7}\right) \\ = 2$$

$$(9) \ \frac{4}{5} - \frac{3}{5} \\ = \frac{1}{5}$$

$$(19) \ \frac{1}{6} - (-2) \\ = \frac{13}{6}$$

$$(29) \ -\frac{5}{4} + \frac{3}{10} \\ = -\frac{19}{20}$$

$$(39) \ 1 + \frac{1}{3} \\ = \frac{4}{3}$$

$$(10) \ -\frac{5}{9} - \frac{8}{9} \\ = -\frac{13}{9}$$

$$(20) \ \frac{5}{3} + \frac{7}{3} \\ = 4$$

$$(30) \ -1 - \frac{1}{2} \\ = -\frac{3}{2}$$

$$(40) \ -\frac{6}{7} + 1 \\ = \frac{1}{7}$$

$$(1) -\frac{5}{3} - (-3) \\ = \frac{4}{3}$$

$$(11) \frac{5}{9} - \left(-\frac{3}{5}\right) \\ = \frac{52}{45}$$

$$(21) 1 + \frac{5}{3} \\ = \frac{8}{3}$$

$$(31) \frac{8}{5} + \frac{9}{7} \\ = \frac{101}{35}$$

$$(2) -\frac{3}{7} + \frac{5}{4} \\ = \frac{23}{28}$$

$$(12) \frac{5}{2} + 1 \\ = \frac{7}{2}$$

$$(22) -1 - (-7) \\ = 6$$

$$(32) \frac{5}{9} + \left(-\frac{5}{2}\right) \\ = -\frac{35}{18}$$

$$(3) -1 - \left(-\frac{3}{7}\right) \\ = -\frac{4}{7}$$

$$(13) -\frac{1}{2} - (-1) \\ = \frac{1}{2}$$

$$(23) \frac{10}{3} - \left(-\frac{7}{5}\right) \\ = \frac{71}{15}$$

$$(33) -\frac{9}{7} - \left(-\frac{9}{4}\right) \\ = \frac{27}{28}$$

$$(4) -\frac{9}{8} + \frac{4}{9} \\ = -\frac{49}{72}$$

$$(14) -\frac{9}{5} + \left(-\frac{5}{3}\right) \\ = -\frac{52}{15}$$

$$(24) -\frac{1}{4} + \left(-\frac{4}{3}\right) \\ = -\frac{19}{12}$$

$$(34) -\frac{7}{10} + \frac{10}{3} \\ = \frac{13}{30}$$

$$(5) -2 - 1 \\ = -3$$

$$(15) -\frac{5}{3} - \left(-\frac{5}{4}\right) \\ = -\frac{5}{12}$$

$$(25) -\frac{4}{9} + \left(-\frac{4}{9}\right) \\ = -\frac{8}{9}$$

$$(35) -\frac{1}{10} - \left(-\frac{9}{10}\right) \\ = \frac{4}{5}$$

$$(6) -\frac{3}{4} - \frac{3}{10} \\ = -\frac{21}{20}$$

$$(16) \frac{7}{6} + 10 \\ = \frac{67}{6}$$

$$(26) \frac{10}{3} + \frac{3}{5} \\ = \frac{59}{15}$$

$$(36) -\frac{2}{9} - \left(-\frac{3}{10}\right) \\ = \frac{7}{90}$$

$$(7) \frac{10}{3} - \left(-\frac{3}{4}\right) \\ = \frac{49}{12}$$

$$(17) -\frac{9}{10} - 7 \\ = -\frac{79}{10}$$

$$(27) \frac{8}{5} + (-1) \\ = \frac{3}{5}$$

$$(37) -1 + \frac{1}{4} \\ = -\frac{3}{4}$$

$$(8) -\frac{4}{9} + \left(-\frac{4}{5}\right) \\ = -\frac{56}{45}$$

$$(18) \frac{10}{3} - \left(-\frac{9}{8}\right) \\ = \frac{107}{24}$$

$$(28) -\frac{1}{2} + \left(-\frac{7}{8}\right) \\ = -\frac{11}{8}$$

$$(38) \frac{10}{7} - \frac{2}{9} \\ = \frac{76}{63}$$

$$(9) -6 - \left(-\frac{9}{4}\right) \\ = -\frac{15}{4}$$

$$(19) -\frac{6}{7} - \left(-\frac{8}{9}\right) \\ = \frac{2}{63}$$

$$(29) -\frac{1}{7} - \frac{8}{3} \\ = -\frac{59}{21}$$

$$(39) 2 + \frac{4}{3} \\ = \frac{10}{3}$$

$$(10) -\frac{4}{7} + 10 \\ = \frac{66}{7}$$

$$(20) \frac{1}{5} + (-5) \\ = -\frac{24}{5}$$

$$(30) 6 - \frac{1}{2} \\ = \frac{11}{2}$$

$$(40) -\frac{3}{8} + \left(-\frac{2}{7}\right) \\ = -\frac{37}{56}$$

$$(1) \frac{5}{4} - \left(-\frac{5}{4}\right) \\ = \frac{5}{2}$$

$$(11) -5 + \left(-\frac{6}{7}\right) \\ = -\frac{41}{7}$$

$$(21) \frac{7}{6} - (-1) \\ = \frac{13}{6}$$

$$(31) \frac{1}{2} - 5 \\ = -\frac{9}{2}$$

$$(2) 1 - (-1) \\ = 2$$

$$(12) -\frac{2}{5} + 1 \\ = \frac{3}{5}$$

$$(22) -\frac{2}{5} - \left(-\frac{1}{6}\right) \\ = -\frac{7}{30}$$

$$(32) -3 - \left(-\frac{2}{9}\right) \\ = -\frac{25}{9}$$

$$(3) 3 + \frac{10}{3} \\ = \frac{19}{3}$$

$$(13) -\frac{1}{3} - \frac{8}{3} \\ = -3$$

$$(23) -\frac{6}{7} + \frac{1}{2} \\ = -\frac{5}{14}$$

$$(33) \frac{1}{7} + 1 \\ = \frac{8}{7}$$

$$(4) \frac{2}{5} + (-6) \\ = -\frac{28}{5}$$

$$(14) \frac{5}{2} + 2 \\ = \frac{9}{2}$$

$$(24) \frac{3}{4} + \left(-\frac{1}{7}\right) \\ = \frac{17}{28}$$

$$(34) \frac{2}{3} + \frac{4}{9} \\ = \frac{10}{9}$$

$$(5) \frac{1}{2} + \frac{1}{4} \\ = \frac{3}{4}$$

$$(15) -1 + \frac{3}{2} \\ = \frac{1}{2}$$

$$(25) \frac{1}{2} - \frac{1}{3} \\ = \frac{1}{6}$$

$$(35) -\frac{1}{4} - \frac{1}{7} \\ = -\frac{11}{28}$$

$$(6) \frac{1}{3} - \frac{4}{7} \\ = -\frac{11}{21}$$

$$(16) -\frac{5}{7} - \left(-\frac{4}{5}\right) \\ = \frac{3}{35}$$

$$(26) -5 - (-4) \\ = -1$$

$$(36) \frac{5}{8} - \frac{6}{5} \\ = -\frac{23}{40}$$

$$(7) -5 - \left(-\frac{4}{5}\right) \\ = -\frac{21}{5}$$

$$(17) \frac{1}{2} + \frac{1}{2} \\ = 1$$

$$(27) 5 - 3 \\ = 2$$

$$(37) \frac{7}{6} + (-2) \\ = -\frac{5}{6}$$

$$(8) \frac{7}{6} + \left(-\frac{8}{7}\right) \\ = \frac{1}{42}$$

$$(18) 1 + \frac{9}{8} \\ = \frac{17}{8}$$

$$(28) \frac{8}{7} + (-1) \\ = \frac{1}{7}$$

$$(38) \frac{1}{7} - \left(-\frac{2}{5}\right) \\ = \frac{19}{35}$$

$$(9) -2 + \frac{3}{4} \\ = -\frac{5}{4}$$

$$(19) \frac{8}{7} + \frac{7}{10} \\ = \frac{129}{70}$$

$$(29) 1 - \frac{1}{3} \\ = \frac{2}{3}$$

$$(39) -\frac{10}{3} + \frac{3}{2} \\ = -\frac{11}{6}$$

$$(10) 4 - \left(-\frac{5}{6}\right) \\ = \frac{29}{6}$$

$$(20) -\frac{7}{8} - 1 \\ = -\frac{15}{8}$$

$$(30) \frac{2}{3} + \frac{9}{4} \\ = \frac{35}{12}$$

$$(40) \frac{4}{5} - \frac{9}{8} \\ = -\frac{13}{40}$$

$$(1) \frac{3}{4} - (-2) \\ = \frac{11}{4}$$

$$(11) -\frac{2}{3} - \left(-\frac{3}{5}\right) \\ = -\frac{1}{15}$$

$$(21) -\frac{1}{7} - (-9) \\ = \frac{62}{7}$$

$$(31) \frac{3}{8} - \left(-\frac{1}{5}\right) \\ = \frac{23}{40}$$

$$(2) \frac{2}{7} - \left(-\frac{8}{5}\right) \\ = \frac{66}{35}$$

$$(12) -\frac{5}{8} + \frac{3}{8} \\ = -\frac{1}{4}$$

$$(22) -3 - \left(-\frac{4}{9}\right) \\ = -\frac{23}{9}$$

$$(32) -3 - 1 \\ = -4$$

$$(3) -\frac{5}{7} - \left(-\frac{8}{7}\right) \\ = \frac{3}{7}$$

$$(13) 1 - \frac{9}{10} \\ = \frac{1}{10}$$

$$(23) \frac{1}{2} + \frac{5}{4} \\ = \frac{7}{4}$$

$$(33) -4 - \left(-\frac{1}{10}\right) \\ = -\frac{39}{10}$$

$$(4) -\frac{3}{2} + \frac{1}{3} \\ = -\frac{7}{6}$$

$$(14) -\frac{1}{2} - \frac{8}{3} \\ = -\frac{19}{6}$$

$$(24) -\frac{10}{9} - (-1) \\ = -\frac{1}{9}$$

$$(34) \frac{5}{2} - \left(-\frac{10}{3}\right) \\ = \frac{35}{6}$$

$$(5) -\frac{2}{3} - \left(-\frac{7}{6}\right) \\ = \frac{1}{2}$$

$$(15) 1 - \left(-\frac{3}{2}\right) \\ = \frac{5}{2}$$

$$(25) -\frac{3}{8} - 1 \\ = -\frac{11}{8}$$

$$(35) -8 - \left(-\frac{4}{7}\right) \\ = -\frac{52}{7}$$

$$(6) -8 - \left(-\frac{5}{9}\right) \\ = -\frac{67}{9}$$

$$(16) -1 + (-5) \\ = -6$$

$$(26) 7 + 1 \\ = 8$$

$$(36) -\frac{5}{4} - \left(-\frac{5}{4}\right) \\ = 0$$

$$(7) 2 - \left(-\frac{3}{4}\right) \\ = \frac{11}{4}$$

$$(17) \frac{1}{4} + \left(-\frac{5}{4}\right) \\ = -1$$

$$(27) \frac{3}{7} - \frac{9}{5} \\ = -\frac{48}{35}$$

$$(37) \frac{7}{3} - \frac{5}{4} \\ = \frac{13}{12}$$

$$(18) \frac{1}{10} - \left(-\frac{9}{8}\right) \\ = \frac{49}{40}$$

$$(28) -4 - \frac{3}{5} \\ = -\frac{23}{5}$$

$$(38) 4 - \frac{9}{4} \\ = \frac{7}{4}$$

$$(8) -\frac{3}{5} + \frac{9}{7} \\ = \frac{24}{35}$$

$$(19) \frac{7}{8} - (-7) \\ = \frac{63}{8}$$

$$(29) 1 - 1 \\ = 0$$

$$(39) -\frac{7}{8} + (-2) \\ = -\frac{23}{8}$$

$$(9) -\frac{2}{3} - \left(-\frac{5}{3}\right) \\ = 1$$

$$(20) \frac{5}{2} - 8 \\ = -\frac{11}{2}$$

$$(30) \frac{2}{3} + 1 \\ = \frac{5}{3}$$

$$(40) \frac{8}{9} - \frac{8}{9} \\ = 0$$

$$(10) \frac{7}{10} + \left(-\frac{9}{4}\right) \\ = -\frac{31}{20}$$



$$(1) -\frac{7}{5} + \left(-\frac{10}{9}\right) \\ = -\frac{113}{45}$$

$$(11) -\frac{1}{2} + \frac{9}{4} \\ = \frac{7}{4}$$

$$(21) 1 + \frac{1}{3} \\ = \frac{4}{3}$$

$$(31) -\frac{7}{5} + \left(-\frac{2}{3}\right) \\ = -\frac{31}{15}$$

$$(2) -\frac{6}{5} - \frac{9}{7} \\ = -\frac{87}{35}$$

$$(12) -9 - \left(-\frac{5}{4}\right) \\ = -\frac{31}{4}$$

$$(22) \frac{5}{2} - (-1) \\ = \frac{7}{2}$$

$$(32) 3 + \frac{3}{5} \\ = \frac{18}{5}$$

$$(3) -\frac{5}{2} - \frac{3}{4} \\ = -\frac{13}{4}$$

$$(13) -\frac{9}{7} + \left(-\frac{8}{9}\right) \\ = -\frac{137}{63}$$

$$(23) -5 - \left(-\frac{6}{7}\right) \\ = -\frac{29}{7}$$

$$(33) \frac{4}{3} + 10 \\ = \frac{34}{3}$$

$$(4) 2 + \left(-\frac{4}{3}\right) \\ = \frac{2}{3}$$

$$(14) \frac{10}{7} + \frac{1}{2} \\ = \frac{27}{14}$$

$$(24) -\frac{1}{2} + \left(-\frac{1}{8}\right) \\ = -\frac{5}{8}$$

$$(34) -\frac{5}{9} - 1 \\ = -\frac{14}{9}$$

$$(5) -1 - \left(-\frac{1}{3}\right) \\ = -\frac{2}{3}$$

$$(15) -\frac{8}{3} + (-3) \\ = -\frac{17}{3}$$

$$(25) \frac{3}{10} - \left(-\frac{4}{5}\right) \\ = \frac{11}{10}$$

$$(35) 2 - \frac{1}{5} \\ = \frac{9}{5}$$

$$(6) \frac{5}{7} + \frac{9}{10} \\ = \frac{113}{70}$$

$$(16) \frac{5}{4} - 4 \\ = -\frac{11}{4}$$

$$(26) \frac{2}{3} - \left(-\frac{5}{8}\right) \\ = \frac{31}{24}$$

$$(36) \frac{5}{3} - (-4) \\ = \frac{17}{3}$$

$$(7) \frac{1}{4} + \left(-\frac{3}{8}\right) \\ = -\frac{1}{8}$$

$$(17) 1 + \frac{3}{7} \\ = \frac{10}{7}$$

$$(27) -\frac{1}{2} + \frac{5}{4} \\ = \frac{3}{4}$$

$$(37) -3 + 1 \\ = -2$$

$$(8) -\frac{3}{5} + \left(-\frac{2}{9}\right) \\ = -\frac{37}{45}$$

$$(18) -\frac{3}{4} + (-2) \\ = -\frac{11}{4}$$

$$(28) -\frac{10}{9} - \left(-\frac{4}{9}\right) \\ = -\frac{2}{3}$$

$$(38) -2 + \left(-\frac{1}{3}\right) \\ = -\frac{7}{3}$$

$$(9) \frac{6}{7} + 5 \\ = \frac{41}{7}$$

$$(19) 5 - \left(-\frac{3}{5}\right) \\ = \frac{28}{5}$$

$$(29) -\frac{2}{3} + \frac{9}{2} \\ = \frac{23}{6}$$

$$(39) 2 - 7 \\ = -5$$

$$(10) \frac{5}{9} - (-6) \\ = \frac{59}{9}$$

$$(20) \frac{4}{9} + \left(-\frac{3}{8}\right) \\ = \frac{5}{72}$$

$$(30) 1 - \left(-\frac{4}{5}\right) \\ = \frac{9}{5}$$

$$(40) \frac{1}{3} - \frac{3}{5} \\ = -\frac{2}{15}$$

$$(1) -\frac{1}{3} + (-1) \\ = -\frac{4}{3}$$

$$(11) 1 - \left(-\frac{1}{5}\right) \\ = \frac{6}{5}$$

$$(21) -8 + \left(-\frac{5}{9}\right) \\ = -\frac{77}{9}$$

$$(31) -\frac{4}{5} - \left(-\frac{3}{8}\right) \\ = -\frac{17}{40}$$

$$(2) \frac{8}{3} - \left(-\frac{1}{2}\right) \\ = \frac{19}{6}$$

$$(12) -\frac{3}{2} - \frac{5}{3} \\ = -\frac{19}{6}$$

$$(22) -4 - \frac{7}{8} \\ = -\frac{39}{8}$$

$$(32) \frac{7}{3} - \left(-\frac{9}{5}\right) \\ = \frac{62}{15}$$

$$(3) \frac{3}{10} + \left(-\frac{1}{5}\right) \\ = \frac{1}{10}$$

$$(13) -\frac{5}{2} + \frac{9}{7} \\ = -\frac{17}{14}$$

$$(23) \frac{3}{4} - \left(-\frac{1}{2}\right) \\ = \frac{5}{4}$$

$$(33) -\frac{9}{8} + \left(-\frac{1}{7}\right) \\ = -\frac{71}{56}$$

$$(4) \frac{2}{5} - \left(-\frac{3}{2}\right) \\ = \frac{19}{10}$$

$$(14) -\frac{3}{10} - 10 \\ = -\frac{103}{10}$$

$$(24) \frac{1}{8} + \left(-\frac{4}{9}\right) \\ = -\frac{23}{72}$$

$$(34) 1 + \left(-\frac{4}{3}\right) \\ = -\frac{1}{3}$$

$$(5) -1 + (-3) \\ = -4$$

$$(15) \frac{4}{5} + \frac{5}{3} \\ = \frac{37}{15}$$

$$(25) -\frac{2}{3} - \frac{5}{4} \\ = -\frac{23}{12}$$

$$(35) -2 - \frac{3}{2} \\ = -\frac{7}{2}$$

$$(6) \frac{4}{3} + \frac{1}{7} \\ = \frac{31}{21}$$

$$(16) -\frac{7}{4} + \left(-\frac{8}{7}\right) \\ = -\frac{81}{28}$$

$$(26) -2 - \left(-\frac{5}{4}\right) \\ = -\frac{3}{4}$$

$$(36) -\frac{4}{3} + (-1) \\ = -\frac{7}{3}$$

$$(7) -\frac{1}{2} - \frac{1}{4} \\ = -\frac{3}{4}$$

$$(17) -\frac{5}{6} - \left(-\frac{7}{6}\right) \\ = \frac{1}{3}$$

$$(27) 2 - \left(-\frac{2}{5}\right) \\ = \frac{12}{5}$$

$$(37) \frac{9}{4} - (-1) \\ = \frac{13}{4}$$

$$(8) 1 + \left(-\frac{4}{9}\right) \\ = \frac{5}{9}$$

$$(18) 2 - \left(-\frac{1}{9}\right) \\ = \frac{19}{9}$$

$$(28) -\frac{1}{3} - \left(-\frac{9}{4}\right) \\ = \frac{23}{12}$$

$$(38) \frac{1}{4} + \frac{1}{10} \\ = \frac{7}{20}$$

$$(9) -1 + \left(-\frac{5}{4}\right) \\ = -\frac{9}{4}$$

$$(19) \frac{5}{4} - \left(-\frac{7}{3}\right) \\ = \frac{43}{12}$$

$$(29) 1 + \frac{1}{2} \\ = \frac{3}{2}$$

$$(39) \frac{2}{7} - (-1) \\ = \frac{9}{7}$$

$$(10) -\frac{6}{5} - \left(-\frac{1}{10}\right) \\ = -\frac{11}{10}$$

$$(20) \frac{5}{2} - 4 \\ = -\frac{3}{2}$$

$$(30) -2 - 4 \\ = -6$$

$$(40) -8 - \frac{10}{9} \\ = -\frac{82}{9}$$

$$(1) 3 - \frac{1}{4} \\ = \frac{11}{4}$$

$$(11) -\frac{9}{7} + \left(-\frac{9}{5}\right) \\ = -\frac{108}{35}$$

$$(21) -\frac{1}{9} - \frac{5}{7} \\ = -\frac{52}{63}$$

$$(31) \frac{1}{6} + \frac{7}{4} \\ = \frac{23}{12}$$

$$(2) -4 - 1 \\ = -5$$

$$(12) -\frac{7}{10} + (-1) \\ = -\frac{17}{10}$$

$$(22) \frac{5}{7} - \frac{5}{3} \\ = -\frac{20}{21}$$

$$(32) -1 + \left(-\frac{4}{9}\right) \\ = -\frac{13}{9}$$

$$(3) -\frac{4}{5} - \left(-\frac{9}{7}\right) \\ = \frac{17}{35}$$

$$(13) \frac{6}{7} - \left(-\frac{4}{5}\right) \\ = \frac{58}{35}$$

$$(23) 3 + \left(-\frac{1}{4}\right) \\ = \frac{11}{4}$$

$$(33) -\frac{2}{9} + \left(-\frac{2}{3}\right) \\ = -\frac{8}{9}$$

$$(4) -\frac{2}{3} + \frac{1}{9} \\ = -\frac{5}{9}$$

$$(14) \frac{1}{3} + (-2) \\ = -\frac{5}{3}$$

$$(24) \frac{9}{5} + \left(-\frac{5}{7}\right) \\ = \frac{38}{35}$$

$$(34) \frac{4}{3} - 2 \\ = -\frac{2}{3}$$

$$(5) -5 + \frac{3}{7} \\ = -\frac{32}{7}$$

$$(15) -5 - \frac{8}{9} \\ = -\frac{53}{9}$$

$$(25) -\frac{2}{5} + \frac{10}{9} \\ = \frac{32}{45}$$

$$(35) -2 - (-1) \\ = -1$$

$$(6) \frac{3}{2} - 1 \\ = \frac{1}{2}$$

$$(16) -\frac{7}{4} + \left(-\frac{10}{7}\right) \\ = -\frac{89}{28}$$

$$(26) \frac{10}{9} + \left(-\frac{1}{9}\right) \\ = 1$$

$$(36) \frac{2}{7} + \frac{3}{4} \\ = \frac{29}{28}$$

$$(7) \frac{7}{2} + \frac{1}{2} \\ = 4$$

$$(17) 2 + \frac{1}{2} \\ = \frac{5}{2}$$

$$(27) -\frac{5}{6} + \left(-\frac{1}{2}\right) \\ = -\frac{4}{3}$$

$$(37) -\frac{3}{2} - \frac{9}{2} \\ = -6$$

$$(8) \frac{5}{6} - 4 \\ = -\frac{19}{6}$$

$$(18) \frac{1}{2} + \frac{2}{3} \\ = \frac{7}{6}$$

$$(28) -1 + \frac{7}{4} \\ = \frac{3}{4}$$

$$(38) \frac{7}{3} + \frac{6}{7} \\ = \frac{67}{21}$$

$$(9) \frac{9}{8} - (-1) \\ = \frac{17}{8}$$

$$(19) -\frac{1}{6} - \frac{3}{2} \\ = -\frac{5}{3}$$

$$(29) -\frac{10}{7} - \left(-\frac{1}{2}\right) \\ = -\frac{13}{14}$$

$$(39) -1 + \frac{5}{7} \\ = -\frac{2}{7}$$

$$(10) \frac{6}{7} + \frac{3}{4} \\ = \frac{45}{28}$$

$$(20) -\frac{4}{7} + \frac{7}{5} \\ = \frac{29}{35}$$

$$(30) \frac{9}{10} + \frac{1}{2} \\ = \frac{7}{5}$$

$$(40) 1 - \frac{3}{7} \\ = \frac{4}{7}$$

$$(1) 9 - \frac{7}{3}$$

$$= \frac{20}{3}$$

$$(11) -\frac{5}{8} - 3$$

$$= -\frac{29}{8}$$

$$(21) \frac{3}{5} - \frac{10}{9}$$

$$= -\frac{23}{45}$$

$$(31) \frac{5}{7} - \left(-\frac{7}{4}\right)$$

$$= \frac{69}{28}$$

$$(2) -\frac{4}{5} - \left(-\frac{3}{10}\right)$$

$$= -\frac{1}{2}$$

$$(12) -\frac{7}{3} + \left(-\frac{1}{9}\right)$$

$$= -\frac{22}{9}$$

$$(22) \frac{9}{10} - \frac{1}{4}$$

$$= \frac{13}{20}$$

$$(32) 1 + \frac{5}{4}$$

$$= \frac{9}{4}$$

$$(3) -\frac{3}{4} + \left(-\frac{7}{3}\right)$$

$$= -\frac{37}{12}$$

$$(13) 6 + \left(-\frac{1}{4}\right)$$

$$= \frac{23}{4}$$

$$(23) -\frac{5}{2} - \left(-\frac{5}{2}\right)$$

$$= 0$$

$$(33) \frac{7}{2} - \frac{1}{3}$$

$$= \frac{19}{6}$$

$$(4) -\frac{3}{7} - \left(-\frac{3}{4}\right)$$

$$= \frac{9}{28}$$

$$(14) \frac{8}{3} + (-5)$$

$$= -\frac{7}{3}$$

$$(24) -\frac{1}{2} + \left(-\frac{6}{5}\right)$$

$$= -\frac{17}{10}$$

$$(34) -\frac{4}{3} - \frac{2}{9}$$

$$= -\frac{14}{9}$$

$$(5) 2 + (-10)$$

$$= -8$$

$$(15) \frac{1}{6} + \left(-\frac{1}{7}\right)$$

$$= \frac{1}{42}$$

$$(25) -\frac{4}{3} + 2$$

$$= \frac{2}{3}$$

$$(35) -\frac{1}{5} + \frac{4}{5}$$

$$= \frac{3}{5}$$

$$(6) -\frac{5}{6} - \frac{1}{2}$$

$$= -\frac{4}{3}$$

$$(16) 2 - \frac{10}{7}$$

$$= \frac{4}{7}$$

$$(26) -3 + \left(-\frac{2}{5}\right)$$

$$= -\frac{17}{5}$$

$$(36) \frac{5}{3} + \left(-\frac{5}{7}\right)$$

$$= \frac{20}{21}$$

$$(7) -\frac{3}{8} + \left(-\frac{7}{2}\right)$$

$$= -\frac{31}{8}$$

$$(17) -\frac{6}{5} + \frac{2}{3}$$

$$= -\frac{8}{15}$$

$$(27) -\frac{9}{5} + \left(-\frac{1}{2}\right)$$

$$= -\frac{23}{10}$$

$$(37) \frac{3}{2} - \left(-\frac{8}{9}\right)$$

$$= \frac{43}{18}$$

$$(8) \frac{10}{7} + \frac{9}{7}$$

$$= \frac{19}{7}$$

$$(18) -\frac{6}{5} - \frac{3}{4}$$

$$= -\frac{39}{20}$$

$$(28) -\frac{1}{5} + \frac{9}{2}$$

$$= \frac{43}{10}$$

$$(38) \frac{1}{5} - \left(-\frac{5}{3}\right)$$

$$= \frac{28}{15}$$

$$(9) -6 - 1$$

$$= -7$$

$$(19) \frac{3}{5} - 1$$

$$= -\frac{2}{5}$$

$$(29) -\frac{8}{9} - (-2)$$

$$= \frac{10}{9}$$

$$(39) -\frac{4}{5} - (-5)$$

$$= \frac{21}{5}$$

$$(10) \frac{1}{4} - (-2)$$

$$= \frac{9}{4}$$

$$(20) -\frac{1}{2} + \frac{1}{4}$$

$$= -\frac{1}{4}$$

$$(30) -\frac{1}{4} - \left(-\frac{3}{5}\right)$$

$$= \frac{7}{20}$$

$$(40) \frac{6}{5} + \left(-\frac{1}{4}\right)$$

$$= \frac{19}{20}$$

$$(1) \frac{1}{3} - (-5) \\ = \frac{16}{3}$$

$$(11) \frac{7}{9} + (-1) \\ = -\frac{2}{9}$$

$$(21) -1 + \frac{10}{3} \\ = \frac{7}{3}$$

$$(31) -\frac{7}{4} - \left(-\frac{4}{3}\right) \\ = -\frac{5}{12}$$

$$(2) \frac{9}{4} + \left(-\frac{7}{6}\right) \\ = \frac{13}{12}$$

$$(12) -\frac{2}{5} - \frac{9}{4} \\ = -\frac{53}{20}$$

$$(22) -4 + \frac{9}{8} \\ = -\frac{23}{8}$$

$$(32) 1 + \left(-\frac{1}{2}\right) \\ = \frac{1}{2}$$

$$(3) -3 - \frac{5}{9} \\ = -\frac{32}{9}$$

$$(13) \frac{2}{5} + \left(-\frac{5}{3}\right) \\ = -\frac{19}{15}$$

$$(23) -\frac{5}{3} - \frac{1}{3} \\ = -2$$

$$(33) -3 + \frac{4}{7} \\ = -\frac{17}{7}$$

$$(4) \frac{3}{2} - \left(-\frac{7}{5}\right) \\ = \frac{29}{10}$$

$$(14) -\frac{3}{5} - \frac{2}{7} \\ = -\frac{31}{35}$$

$$(24) -\frac{3}{5} - \left(-\frac{6}{7}\right) \\ = \frac{9}{35}$$

$$(34) \frac{2}{3} + \frac{8}{3} \\ = \frac{10}{3}$$

$$(5) 9 + \frac{3}{2} \\ = \frac{21}{2}$$

$$(15) \frac{4}{5} - \left(-\frac{2}{9}\right) \\ = \frac{46}{45}$$

$$(25) -\frac{1}{6} + \left(-\frac{3}{2}\right) \\ = -\frac{5}{3}$$

$$(35) 5 + \frac{1}{5} \\ = \frac{26}{5}$$

$$(6) \frac{1}{2} - \left(-\frac{4}{7}\right) \\ = \frac{15}{14}$$

$$(16) -\frac{7}{2} - \frac{2}{7} \\ = -\frac{53}{14}$$

$$(26) -1 - \left(-\frac{7}{10}\right) \\ = -\frac{3}{10}$$

$$(36) 1 - (-2) \\ = 3$$

$$(7) -\frac{3}{4} - \frac{7}{2} \\ = -\frac{17}{4}$$

$$(17) \frac{9}{8} - \frac{1}{2} \\ = \frac{5}{8}$$

$$(27) -10 + (-4) \\ = -14$$

$$(37) \frac{4}{5} + \left(-\frac{8}{5}\right) \\ = -\frac{4}{5}$$

$$(8) \frac{5}{2} - \frac{1}{9} \\ = \frac{43}{18}$$

$$(18) -\frac{5}{3} - \left(-\frac{9}{5}\right) \\ = \frac{2}{15}$$

$$(28) -\frac{5}{9} + \frac{1}{9} \\ = -\frac{4}{9}$$

$$(38) 7 + \left(-\frac{5}{7}\right) \\ = \frac{44}{7}$$

$$(9) \frac{5}{8} + \frac{10}{3} \\ = \frac{95}{24}$$

$$(19) \frac{10}{7} + 3 \\ = \frac{31}{7}$$

$$(29) -\frac{2}{3} + \left(-\frac{4}{5}\right) \\ = -\frac{22}{15}$$

$$(39) \frac{5}{7} + \frac{3}{4} \\ = \frac{41}{28}$$

$$(10) \frac{5}{3} - \left(-\frac{7}{4}\right) \\ = \frac{41}{12}$$

$$(20) \frac{5}{7} - \frac{1}{4} \\ = \frac{13}{28}$$

$$(30) -2 - \left(-\frac{1}{9}\right) \\ = -\frac{17}{9}$$

$$(40) 1 + (-1) \\ = 0$$

### 1.3.2 乗除法

有理数の積や商を求める問題。

$$(1) -\frac{9}{10} \times \frac{3}{10}$$

$$(11) \frac{1}{2} \times \left(-\frac{5}{3}\right)$$

$$(21) \frac{4}{5} \times 1$$

$$(31) -1 \times 1$$

$$(2) -8 \div \frac{8}{7}$$

$$(12) -\frac{7}{8} \times \frac{9}{4}$$

$$(22) -\frac{1}{5} \div \left(-\frac{5}{2}\right)$$

$$(32) \frac{5}{2} \div \frac{1}{2}$$

$$(3) -2 \times \left(-\frac{9}{7}\right)$$

$$(13) -4 \div \frac{1}{4}$$

$$(23) -\frac{6}{7} \times \frac{1}{2}$$

$$(33) -9 \times (-3)$$

$$(4) \frac{4}{5} \times (-2)$$

$$(14) -\frac{2}{3} \times \frac{1}{5}$$

$$(24) \frac{1}{6} \div 10$$

$$(34) -8 \div \frac{7}{9}$$

$$(5) \frac{9}{10} \div \left(-\frac{3}{2}\right)$$

$$(15) \frac{10}{9} \div 1$$

$$(25) \frac{2}{5} \div 4$$

$$(35) -\frac{5}{8} \times \left(-\frac{3}{4}\right)$$

$$(6) \frac{4}{5} \div \left(-\frac{2}{3}\right)$$

$$(16) \frac{2}{9} \times \frac{5}{2}$$

$$(26) \frac{1}{3} \times 1$$

$$(36) \frac{1}{2} \div 2$$

$$(7) -\frac{1}{6} \times \frac{7}{9}$$

$$(17) \frac{4}{3} \times \frac{1}{8}$$

$$(27) \frac{1}{4} \div \frac{3}{8}$$

$$(37) -\frac{9}{4} \div (-2)$$

$$(8) \frac{7}{3} \div 9$$

$$(18) -\frac{3}{2} \div \frac{2}{3}$$

$$(28) \frac{3}{2} \times \left(-\frac{2}{9}\right)$$

$$(38) \frac{5}{2} \div \frac{1}{7}$$

$$(9) -2 \times \frac{2}{3}$$

$$(19) \frac{9}{8} \div \frac{4}{5}$$

$$(29) -\frac{5}{4} \div \left(-\frac{1}{10}\right)$$

$$(39) -\frac{8}{3} \div \frac{7}{10}$$

$$(10) -\frac{4}{5} \times \frac{1}{2}$$

$$(20) \frac{3}{2} \div \frac{3}{4}$$

$$(30) -2 \times \frac{8}{9}$$

$$(40) -\frac{9}{10} \div \left(-\frac{5}{3}\right)$$

(1)  $-\frac{5}{3} \div \left(-\frac{5}{4}\right)$

(11)  $\frac{1}{5} \div \frac{2}{5}$

(21)  $1 \div 6$

(31)  $\frac{5}{3} \times \left(-\frac{6}{5}\right)$

(2)  $-\frac{7}{5} \div \left(-\frac{4}{3}\right)$

(12)  $\frac{3}{10} \times \frac{3}{4}$

(22)  $-\frac{5}{7} \div \left(-\frac{8}{3}\right)$

(32)  $1 \div \frac{2}{3}$

(3)  $-\frac{2}{3} \times \left(-\frac{5}{3}\right)$

(13)  $\frac{1}{5} \times \left(-\frac{10}{9}\right)$

(23)  $-\frac{5}{2} \div 3$

(33)  $-\frac{5}{2} \div \left(-\frac{2}{7}\right)$

(4)  $2 \div \left(-\frac{1}{3}\right)$

(14)  $-1 \times \left(-\frac{1}{2}\right)$

(24)  $-\frac{2}{5} \div (-8)$

(34)  $-\frac{1}{3} \div (-1)$

(5)  $5 \div \left(-\frac{1}{4}\right)$

(15)  $\frac{9}{7} \div (-1)$

(25)  $5 \times \frac{5}{8}$

(35)  $1 \times 1$

(6)  $\frac{3}{4} \div \left(-\frac{5}{2}\right)$

(16)  $2 \div 1$

(26)  $\frac{1}{9} \div \left(-\frac{3}{2}\right)$

(36)  $-\frac{7}{10} \div \left(-\frac{7}{8}\right)$

(7)  $-\frac{1}{4} \times \left(-\frac{7}{10}\right)$

(17)  $-1 \times \left(-\frac{2}{3}\right)$

(27)  $\frac{7}{5} \div \left(-\frac{1}{3}\right)$

(37)  $2 \times 3$

(8)  $\frac{1}{2} \div \left(-\frac{1}{9}\right)$

(18)  $-\frac{5}{9} \div \frac{5}{2}$

(28)  $-5 \div (-1)$

(38)  $\frac{1}{4} \times \frac{10}{9}$

(9)  $-\frac{7}{10} \times \left(-\frac{1}{2}\right)$

(19)  $-5 \times \left(-\frac{1}{3}\right)$

(29)  $2 \div 2$

(39)  $3 \times \frac{3}{7}$

(10)  $3 \div \frac{4}{9}$

(20)  $-\frac{8}{9} \div \left(-\frac{10}{9}\right)$

(30)  $1 \div \left(-\frac{1}{5}\right)$

(40)  $\frac{3}{4} \div \left(-\frac{1}{2}\right)$



(1)  $\frac{4}{7} \div \left(-\frac{1}{5}\right)$

(11)  $-1 \times (-2)$

(21)  $\frac{10}{9} \times 2$

(31)  $-\frac{3}{5} \times 3$

(2)  $-\frac{1}{5} \div \frac{1}{5}$

(12)  $-1 \div \left(-\frac{3}{8}\right)$

(22)  $-\frac{9}{8} \times \left(-\frac{4}{3}\right)$

(32)  $-1 \times \frac{2}{3}$

(3)  $-\frac{1}{3} \times \frac{5}{4}$

(13)  $\frac{3}{2} \div 4$

(23)  $-1 \times \left(-\frac{9}{4}\right)$

(33)  $1 \times \frac{1}{5}$

(4)  $-\frac{4}{3} \div (-2)$

(14)  $-\frac{1}{4} \times \left(-\frac{7}{3}\right)$

(24)  $\frac{6}{7} \times \left(-\frac{2}{7}\right)$

(34)  $1 \div \frac{1}{5}$

(5)  $2 \div \frac{9}{8}$

(15)  $\frac{5}{3} \div \left(-\frac{10}{7}\right)$

(25)  $-\frac{7}{3} \div (-3)$

(35)  $-\frac{2}{5} \times \left(-\frac{6}{5}\right)$

(6)  $-\frac{8}{7} \div \left(-\frac{6}{5}\right)$

(16)  $-\frac{1}{3} \times 2$

(26)  $-1 \div \left(-\frac{9}{5}\right)$

(36)  $-5 \times \frac{10}{9}$

(7)  $-\frac{8}{5} \times (-3)$

(17)  $\frac{9}{2} \div \left(-\frac{4}{7}\right)$

(27)  $-1 \div \frac{1}{2}$

(37)  $-\frac{1}{2} \div \left(-\frac{2}{3}\right)$

(8)  $4 \div \frac{1}{10}$

(18)  $\frac{10}{9} \times \left(-\frac{5}{7}\right)$

(28)  $8 \times \left(-\frac{5}{3}\right)$

(38)  $\frac{3}{4} \div \left(-\frac{3}{5}\right)$

(9)  $\frac{5}{4} \div \left(-\frac{8}{7}\right)$

(19)  $\frac{3}{10} \div \left(-\frac{1}{2}\right)$

(29)  $8 \div \left(-\frac{1}{2}\right)$

(39)  $-\frac{1}{5} \times (-2)$

(10)  $\frac{3}{2} \div \left(-\frac{3}{5}\right)$

(20)  $-\frac{4}{7} \times \left(-\frac{5}{3}\right)$

(30)  $-\frac{4}{3} \div \frac{2}{3}$

(40)  $\frac{2}{3} \times \left(-\frac{7}{3}\right)$

(1)  $\frac{4}{7} \times \left(-\frac{1}{2}\right)$

(11)  $-\frac{5}{8} \div \left(-\frac{2}{3}\right)$

(21)  $-\frac{3}{7} \times \left(-\frac{5}{8}\right)$

(31)  $1 \times \left(-\frac{8}{3}\right)$

(2)  $\frac{1}{3} \div \frac{10}{7}$

(12)  $\frac{2}{5} \times 4$

(22)  $-\frac{3}{2} \times \frac{9}{5}$

(32)  $\frac{1}{5} \div \left(-\frac{3}{2}\right)$

(3)  $\frac{9}{5} \div 3$

(13)  $-\frac{9}{5} \div \frac{9}{5}$

(23)  $\frac{8}{9} \times \left(-\frac{1}{5}\right)$

(33)  $3 \times 9$

(4)  $\frac{6}{5} \div \left(-\frac{3}{5}\right)$

(14)  $\frac{1}{7} \div \frac{6}{7}$

(24)  $-\frac{4}{5} \times 2$

(34)  $\frac{1}{5} \div (-5)$

(5)  $1 \div (-2)$

(15)  $-1 \div \frac{1}{4}$

(25)  $\frac{3}{2} \div \frac{1}{2}$

(35)  $\frac{7}{8} \div \left(-\frac{7}{4}\right)$

(6)  $\frac{9}{5} \times 4$

(16)  $-1 \times \left(-\frac{7}{9}\right)$

(26)  $-9 \times \left(-\frac{5}{2}\right)$

(36)  $-\frac{7}{2} \div (-4)$

(7)  $\frac{4}{9} \div 1$

(17)  $\frac{3}{5} \div \left(-\frac{2}{7}\right)$

(27)  $-\frac{1}{3} \div \frac{4}{9}$

(37)  $\frac{5}{8} \div \frac{3}{4}$

(8)  $-\frac{1}{6} \times \left(-\frac{1}{5}\right)$

(18)  $4 \div \frac{1}{9}$

(28)  $\frac{6}{7} \div \frac{2}{3}$

(38)  $-1 \div 3$

(9)  $-\frac{10}{3} \div (-3)$

(19)  $\frac{5}{8} \times \left(-\frac{1}{2}\right)$

(29)  $\frac{3}{8} \div \frac{3}{8}$

(39)  $-\frac{3}{2} \div \frac{1}{8}$

(10)  $-\frac{5}{6} \div (-3)$

(20)  $3 \div \frac{8}{5}$

(30)  $-\frac{5}{4} \div \frac{1}{3}$

(40)  $7 \times 1$

(1)  $\frac{4}{3} \times (-1)$

(11)  $1 \div \left(-\frac{1}{8}\right)$

(21)  $\frac{5}{3} \times \frac{7}{4}$

(31)  $-\frac{9}{2} \times \frac{9}{7}$

(2)  $\frac{3}{4} \div 1$

(12)  $1 \div \frac{2}{3}$

(22)  $-\frac{5}{2} \div \left(-\frac{3}{4}\right)$

(32)  $-\frac{8}{5} \div \left(-\frac{1}{4}\right)$

(3)  $\frac{5}{7} \times \left(-\frac{5}{3}\right)$

(13)  $\frac{1}{4} \div 1$

(23)  $-5 \div 1$

(33)  $\frac{9}{2} \times 1$

(4)  $-\frac{1}{2} \div \frac{3}{4}$

(14)  $\frac{5}{4} \times \left(-\frac{7}{2}\right)$

(24)  $-\frac{5}{4} \div \frac{3}{2}$

(34)  $1 \times \left(-\frac{6}{5}\right)$

(5)  $-1 \div \frac{4}{3}$

(15)  $\frac{5}{4} \div \left(-\frac{3}{5}\right)$

(25)  $-\frac{7}{10} \times \left(-\frac{3}{8}\right)$

(35)  $-\frac{4}{5} \times \left(-\frac{3}{5}\right)$

(6)  $\frac{5}{8} \times \frac{5}{2}$

(16)  $-\frac{2}{3} \div \frac{1}{8}$

(26)  $-2 \div \left(-\frac{1}{4}\right)$

(36)  $\frac{1}{2} \div \frac{7}{4}$

(7)  $4 \times \left(-\frac{6}{7}\right)$

(17)  $-\frac{1}{2} \times \frac{7}{2}$

(27)  $-\frac{1}{4} \times (-1)$

(37)  $-\frac{7}{6} \div \left(-\frac{5}{8}\right)$

(8)  $-\frac{9}{2} \div \frac{2}{5}$

(18)  $1 \times \left(-\frac{9}{7}\right)$

(28)  $\frac{5}{3} \times \left(-\frac{8}{5}\right)$

(38)  $\frac{4}{3} \times (-5)$

(9)  $9 \times 1$

(19)  $\frac{4}{5} \times \frac{6}{7}$

(29)  $-\frac{10}{3} \div (-3)$

(39)  $9 \div \frac{9}{8}$

(10)  $\frac{4}{3} \div 1$

(20)  $-1 \div (-2)$

(30)  $-\frac{3}{2} \div \frac{3}{5}$

(40)  $-\frac{1}{4} \times \frac{4}{9}$

(1)  $-\frac{9}{8} \times \left(-\frac{8}{3}\right)$

(11)  $-7 \div \left(-\frac{8}{7}\right)$

(21)  $-\frac{4}{5} \times 3$

(31)  $\frac{4}{3} \div \frac{10}{7}$

(2)  $7 \div 1$

(12)  $-\frac{5}{4} \times \frac{3}{5}$

(22)  $\frac{8}{7} \div \frac{3}{5}$

(32)  $\frac{10}{7} \div \frac{7}{10}$

(3)  $-1 \div 5$

(13)  $-\frac{5}{2} \times \frac{1}{2}$

(23)  $\frac{5}{8} \times \frac{1}{2}$

(33)  $1 \div \frac{9}{2}$

(4)  $-2 \times 1$

(14)  $\frac{7}{9} \div 3$

(24)  $-2 \div \frac{2}{5}$

(34)  $-9 \times \frac{3}{10}$

(5)  $-\frac{2}{3} \div \frac{7}{3}$

(15)  $\frac{5}{4} \times \frac{1}{3}$

(25)  $-\frac{9}{5} \div \frac{5}{2}$

(35)  $-\frac{5}{7} \div \frac{9}{10}$

(6)  $\frac{1}{6} \times \left(-\frac{3}{10}\right)$

(16)  $\frac{9}{10} \div \left(-\frac{1}{9}\right)$

(26)  $-8 \div \left(-\frac{1}{6}\right)$

(36)  $\frac{9}{4} \times \left(-\frac{9}{8}\right)$

(7)  $-\frac{7}{10} \times \frac{1}{5}$

(27)  $-\frac{4}{5} \div \left(-\frac{3}{5}\right)$

(37)  $\frac{9}{10} \div \left(-\frac{3}{8}\right)$

(8)  $\frac{5}{9} \times 1$

(17)  $-\frac{9}{8} \div 5$

(28)  $-\frac{1}{10} \times 2$

(38)  $-3 \div \frac{9}{5}$

(9)  $\frac{4}{5} \div (-3)$

(18)  $\frac{5}{7} \div 3$

(29)  $-\frac{7}{6} \div \left(-\frac{7}{4}\right)$

(39)  $\frac{5}{7} \div \left(-\frac{5}{3}\right)$

(10)  $2 \div \frac{4}{3}$

(19)  $-\frac{1}{10} \times \frac{3}{5}$

(30)  $-\frac{5}{2} \times \frac{5}{4}$

(40)  $-\frac{6}{5} \times \left(-\frac{10}{9}\right)$

(20)  $-2 \div \frac{1}{7}$

(1)  $-2 \div \frac{4}{3}$

(11)  $-2 \times \left(-\frac{8}{9}\right)$

(21)  $-7 \div \frac{5}{4}$

(31)  $2 \times \left(-\frac{7}{2}\right)$

(2)  $\frac{2}{9} \div 2$

(12)  $4 \times 4$

(22)  $-\frac{1}{6} \times (-1)$

(32)  $\frac{1}{3} \div \left(-\frac{1}{2}\right)$

(3)  $\frac{7}{2} \div \frac{5}{8}$

(13)  $\frac{4}{3} \div \left(-\frac{5}{6}\right)$

(23)  $\frac{1}{2} \times (-1)$

(33)  $-\frac{2}{5} \times \left(-\frac{7}{10}\right)$

(4)  $\frac{5}{6} \times \frac{5}{4}$

(14)  $\frac{7}{5} \div 2$

(24)  $8 \div \left(-\frac{1}{6}\right)$

(34)  $-\frac{5}{9} \times \left(-\frac{7}{4}\right)$

(5)  $3 \div (-6)$

(15)  $5 \times \frac{7}{3}$

(25)  $\frac{10}{9} \times 1$

(35)  $-\frac{8}{3} \times \frac{3}{5}$

(6)  $1 \times (-1)$

(16)  $\frac{1}{5} \times \frac{6}{7}$

(26)  $\frac{1}{5} \div \left(-\frac{9}{4}\right)$

(36)  $1 \div \left(-\frac{7}{3}\right)$

(7)  $\frac{3}{4} \times \frac{3}{8}$

(17)  $-\frac{7}{5} \times (-2)$

(27)  $-2 \div 4$

(37)  $-\frac{4}{9} \times \frac{5}{9}$

(8)  $-\frac{10}{9} \div \left(-\frac{4}{5}\right)$

(18)  $\frac{7}{9} \times \left(-\frac{9}{2}\right)$

(28)  $\frac{9}{4} \times 1$

(38)  $-\frac{2}{5} \div 2$

(9)  $-\frac{7}{2} \times \left(-\frac{4}{3}\right)$

(19)  $-\frac{4}{5} \times (-1)$

(29)  $-1 \times 3$

(39)  $\frac{8}{5} \div \left(-\frac{3}{8}\right)$

(10)  $1 \times \frac{1}{2}$

(20)  $-\frac{2}{9} \div \frac{4}{5}$

(30)  $-2 \div \left(-\frac{6}{7}\right)$

(40)  $\frac{4}{3} \div (-10)$

(1)  $-\frac{9}{10} \times 6$

(11)  $-\frac{7}{10} \div \frac{1}{5}$

(21)  $\frac{1}{9} \div (-3)$

(31)  $2 \div \frac{7}{6}$

(2)  $-\frac{7}{3} \times \frac{5}{4}$

(12)  $-\frac{6}{5} \div \frac{10}{9}$

(22)  $-\frac{1}{2} \times \left(-\frac{8}{3}\right)$

(32)  $\frac{1}{3} \div \frac{10}{9}$

(3)  $-\frac{9}{10} \times \frac{7}{5}$

(13)  $-\frac{4}{5} \times \left(-\frac{2}{3}\right)$

(23)  $\frac{8}{7} \times \left(-\frac{3}{10}\right)$

(33)  $1 \times \frac{7}{5}$

(4)  $\frac{2}{5} \div \frac{3}{2}$

(14)  $\frac{9}{4} \times \left(-\frac{1}{8}\right)$

(24)  $-\frac{1}{3} \times \frac{1}{5}$

(34)  $1 \times \frac{4}{5}$

(5)  $-\frac{10}{7} \div 2$

(15)  $\frac{10}{9} \times (-1)$

(25)  $1 \div \frac{3}{2}$

(35)  $-\frac{3}{4} \times \left(-\frac{1}{4}\right)$

(6)  $\frac{4}{3} \times \left(-\frac{1}{4}\right)$

(16)  $-2 \div \left(-\frac{9}{4}\right)$

(26)  $\frac{1}{5} \times \frac{1}{4}$

(36)  $\frac{2}{3} \times \left(-\frac{2}{9}\right)$

(7)  $-6 \div (-8)$

(17)  $\frac{7}{2} \div \frac{5}{6}$

(27)  $-\frac{1}{7} \div \frac{6}{7}$

(37)  $-1 \div \left(-\frac{10}{9}\right)$

(8)  $-\frac{5}{2} \div \frac{9}{2}$

(18)  $-\frac{9}{10} \times \frac{7}{2}$

(28)  $\frac{1}{3} \times \frac{4}{5}$

(38)  $-1 \times (-1)$

(9)  $\frac{7}{8} \div \frac{9}{7}$

(19)  $-2 \div \left(-\frac{8}{9}\right)$

(29)  $\frac{8}{9} \times (-4)$

(39)  $\frac{5}{8} \div 3$

(10)  $-\frac{2}{9} \times \left(-\frac{5}{4}\right)$

(20)  $-3 \div 5$

(30)  $\frac{7}{9} \div \frac{4}{5}$

(40)  $-\frac{9}{7} \div \frac{3}{7}$

(1)  $\frac{1}{4} \div 5$

(11)  $\frac{1}{2} \times \frac{1}{9}$

(21)  $\frac{4}{3} \div \left(-\frac{1}{2}\right)$

(31)  $-2 \times \left(-\frac{1}{2}\right)$

(2)  $-\frac{8}{3} \times 1$

(12)  $1 \div (-4)$

(22)  $2 \times \left(-\frac{8}{5}\right)$

(32)  $\frac{4}{7} \times \left(-\frac{2}{3}\right)$

(3)  $\frac{1}{2} \div 2$

(13)  $\frac{3}{8} \times \left(-\frac{8}{5}\right)$

(23)  $\frac{1}{9} \times \frac{8}{7}$

(33)  $-1 \div \left(-\frac{10}{9}\right)$

(4)  $-9 \times \left(-\frac{6}{7}\right)$

(14)  $2 \times \frac{9}{4}$

(24)  $-\frac{8}{9} \div \left(-\frac{7}{2}\right)$

(34)  $-\frac{2}{5} \div \left(-\frac{3}{4}\right)$

(5)  $3 \times \frac{1}{8}$

(15)  $\frac{3}{5} \div \frac{9}{4}$

(25)  $\frac{7}{8} \div \left(-\frac{4}{7}\right)$

(35)  $1 \times \left(-\frac{4}{3}\right)$

(6)  $\frac{7}{3} \div \frac{5}{7}$

(16)  $-\frac{3}{5} \div 4$

(26)  $-\frac{3}{5} \div 1$

(36)  $-\frac{8}{7} \times 2$

(7)  $-\frac{2}{3} \div \frac{7}{5}$

(17)  $-\frac{10}{7} \times \frac{2}{3}$

(27)  $\frac{4}{5} \div \frac{7}{6}$

(37)  $\frac{1}{3} \div \left(-\frac{7}{4}\right)$

(8)  $-1 \times \frac{1}{2}$

(18)  $\frac{1}{2} \div (-1)$

(28)  $-\frac{1}{6} \div \frac{9}{10}$

(38)  $\frac{4}{9} \times \left(-\frac{7}{10}\right)$

(9)  $\frac{2}{5} \div \frac{5}{8}$

(19)  $10 \times \left(-\frac{9}{8}\right)$

(29)  $\frac{3}{5} \div \left(-\frac{8}{7}\right)$

(39)  $-\frac{4}{7} \times 5$

(10)  $-\frac{3}{4} \times \frac{1}{9}$

(20)  $-\frac{1}{3} \times \left(-\frac{7}{10}\right)$

(30)  $-\frac{2}{3} \div 4$

(40)  $\frac{10}{7} \div \frac{7}{3}$

(1)  $-\frac{3}{2} \times 2$

(11)  $\frac{8}{5} \times 3$

(21)  $\frac{4}{5} \times \left(-\frac{7}{9}\right)$

(31)  $10 \div 10$

(2)  $-\frac{4}{3} \div \left(-\frac{1}{2}\right)$

(12)  $\frac{1}{3} \times \left(-\frac{9}{4}\right)$

(22)  $-\frac{3}{10} \times (-5)$

(32)  $-\frac{3}{10} \div \frac{5}{2}$

(3)  $\frac{8}{9} \times \left(-\frac{8}{5}\right)$

(13)  $\frac{4}{7} \div \frac{4}{3}$

(23)  $-1 \div \frac{3}{8}$

(33)  $\frac{5}{9} \times \left(-\frac{3}{2}\right)$

(4)  $-\frac{5}{7} \div (-6)$

(14)  $\frac{1}{5} \times (-4)$

(24)  $-\frac{1}{3} \times \left(-\frac{8}{3}\right)$

(34)  $-\frac{5}{7} \times \frac{9}{7}$

(5)  $-2 \div \frac{3}{2}$

(15)  $-4 \div \left(-\frac{3}{4}\right)$

(25)  $-6 \times \frac{1}{5}$

(35)  $-\frac{1}{5} \div 5$

(6)  $\frac{3}{2} \div \frac{2}{3}$

(16)  $-1 \div \frac{7}{10}$

(26)  $\frac{4}{3} \times \frac{5}{3}$

(36)  $-2 \div \left(-\frac{1}{4}\right)$

(7)  $-\frac{7}{10} \div (-8)$

(17)  $-\frac{6}{5} \div \left(-\frac{9}{10}\right)$

(27)  $10 \times \frac{4}{3}$

(37)  $-2 \div \left(-\frac{8}{3}\right)$

(8)  $-\frac{1}{7} \times \frac{7}{9}$

(18)  $\frac{2}{5} \div 1$

(28)  $\frac{4}{9} \times \frac{8}{3}$

(38)  $-\frac{1}{6} \div \left(-\frac{1}{5}\right)$

(9)  $-\frac{10}{7} \times \left(-\frac{1}{4}\right)$

(19)  $-\frac{7}{3} \times \left(-\frac{5}{7}\right)$

(29)  $-1 \times \left(-\frac{1}{10}\right)$

(39)  $-\frac{4}{5} \times \frac{5}{3}$

(10)  $\frac{5}{3} \times \frac{7}{8}$

(20)  $-\frac{1}{2} \times \frac{9}{4}$

(30)  $-\frac{6}{7} \div 8$

(40)  $-\frac{3}{7} \times \frac{7}{9}$



(1)  $-2 \div \frac{3}{2}$

(11)  $1 \div \left(-\frac{5}{3}\right)$

(21)  $1 \times \frac{9}{8}$

(31)  $\frac{1}{7} \div 1$

(2)  $\frac{3}{5} \div \frac{1}{7}$

(12)  $-\frac{3}{10} \div \left(-\frac{5}{2}\right)$

(22)  $\frac{4}{5} \times \frac{5}{4}$

(32)  $1 \div \frac{4}{3}$

(3)  $-\frac{3}{5} \times (-1)$

(13)  $7 \div \left(-\frac{3}{2}\right)$

(23)  $\frac{3}{2} \div \left(-\frac{5}{4}\right)$

(33)  $-\frac{9}{2} \times \frac{3}{8}$

(4)  $1 \div \left(-\frac{8}{3}\right)$

(14)  $-1 \times \left(-\frac{7}{5}\right)$

(24)  $4 \times \frac{5}{3}$

(34)  $-\frac{9}{5} \times (-2)$

(5)  $\frac{1}{4} \div \frac{1}{4}$

(15)  $-8 \div 1$

(25)  $2 \times 4$

(35)  $\frac{4}{9} \times \frac{1}{4}$

(6)  $-4 \times \left(-\frac{3}{4}\right)$

(16)  $3 \div 7$

(26)  $-\frac{2}{5} \times \left(-\frac{2}{5}\right)$

(36)  $\frac{3}{10} \div \frac{3}{5}$

(7)  $\frac{2}{7} \div \left(-\frac{3}{7}\right)$

(17)  $\frac{3}{5} \times \left(-\frac{7}{10}\right)$

(27)  $-\frac{4}{7} \div \left(-\frac{5}{3}\right)$

(37)  $-\frac{9}{10} \times 1$

(8)  $-\frac{6}{5} \times \frac{3}{4}$

(18)  $-\frac{7}{9} \times \frac{9}{5}$

(28)  $\frac{5}{3} \div \left(-\frac{2}{3}\right)$

(38)  $\frac{10}{7} \div \frac{1}{6}$

(9)  $\frac{1}{9} \times \frac{9}{5}$

(19)  $1 \times \frac{3}{2}$

(29)  $1 \div \frac{9}{10}$

(39)  $\frac{1}{3} \div \frac{1}{4}$

(10)  $-\frac{1}{2} \div \left(-\frac{3}{2}\right)$

(20)  $-\frac{1}{3} \times (-4)$

(30)  $2 \div \frac{4}{9}$

(40)  $\frac{3}{2} \times \left(-\frac{1}{5}\right)$

(1)  $-\frac{2}{3} \times (-2)$

(11)  $-\frac{7}{9} \times \left(-\frac{4}{7}\right)$

(21)  $-\frac{2}{9} \times \left(-\frac{3}{5}\right)$

(31)  $\frac{9}{10} \times \left(-\frac{1}{2}\right)$

(2)  $-\frac{5}{9} \div \left(-\frac{9}{5}\right)$

(12)  $-\frac{1}{2} \times \frac{1}{10}$

(22)  $-\frac{7}{6} \div \left(-\frac{9}{4}\right)$

(32)  $\frac{10}{3} \times \frac{9}{7}$

(3)  $\frac{7}{4} \times \left(-\frac{5}{4}\right)$

(13)  $\frac{1}{7} \div \frac{10}{3}$

(23)  $-\frac{8}{9} \times \left(-\frac{1}{5}\right)$

(33)  $-\frac{9}{4} \div \frac{8}{5}$

(4)  $2 \div \frac{5}{8}$

(14)  $5 \times \left(-\frac{5}{4}\right)$

(24)  $\frac{9}{8} \times \left(-\frac{4}{5}\right)$

(34)  $\frac{5}{3} \div \frac{1}{9}$

(5)  $-\frac{1}{2} \div (-4)$

(15)  $\frac{8}{3} \times \frac{3}{2}$

(25)  $4 \div (-1)$

(35)  $1 \times \left(-\frac{4}{7}\right)$

(6)  $\frac{1}{2} \div 4$

(16)  $1 \div \frac{4}{3}$

(26)  $-\frac{2}{3} \times 3$

(36)  $\frac{3}{8} \div \left(-\frac{7}{3}\right)$

(7)  $\frac{7}{5} \times \frac{3}{8}$

(17)  $-1 \times \left(-\frac{4}{5}\right)$

(27)  $-1 \div \left(-\frac{1}{9}\right)$

(37)  $-\frac{3}{5} \div (-7)$

(8)  $\frac{4}{3} \times \left(-\frac{4}{5}\right)$

(18)  $-\frac{3}{7} \times \left(-\frac{5}{4}\right)$

(28)  $\frac{3}{10} \div \left(-\frac{5}{9}\right)$

(38)  $-\frac{7}{2} \times \left(-\frac{9}{4}\right)$

(9)  $-\frac{3}{2} \div \left(-\frac{8}{7}\right)$

(19)  $-\frac{1}{6} \div \frac{2}{3}$

(29)  $3 \div \frac{10}{3}$

(39)  $3 \div (-1)$

(10)  $\frac{1}{2} \times \left(-\frac{3}{5}\right)$

(20)  $-\frac{1}{3} \div \left(-\frac{1}{2}\right)$

(30)  $-\frac{9}{5} \div \left(-\frac{2}{3}\right)$

(40)  $-9 \div \frac{1}{10}$

(1)  $\frac{4}{5} \div (-2)$

(11)  $-\frac{9}{7} \div \left(-\frac{7}{8}\right)$

(21)  $-\frac{5}{8} \times \left(-\frac{7}{4}\right)$

(31)  $\frac{1}{2} \times \frac{9}{2}$

(2)  $\frac{7}{9} \div \frac{5}{4}$

(12)  $-\frac{5}{8} \times \frac{3}{2}$

(22)  $-9 \times \left(-\frac{5}{9}\right)$

(32)  $\frac{1}{4} \times \left(-\frac{5}{3}\right)$

(3)  $-\frac{10}{3} \div \frac{2}{9}$

(13)  $5 \div \frac{3}{2}$

(23)  $-5 \times \left(-\frac{9}{8}\right)$

(33)  $-\frac{9}{8} \div 1$

(4)  $-\frac{4}{7} \div \left(-\frac{2}{7}\right)$

(14)  $\frac{5}{3} \div \left(-\frac{3}{4}\right)$

(24)  $-\frac{6}{5} \times \left(-\frac{9}{2}\right)$

(34)  $\frac{2}{5} \times \frac{8}{9}$

(5)  $\frac{7}{10} \div 1$

(15)  $-\frac{7}{6} \div \frac{2}{5}$

(25)  $\frac{3}{7} \div \frac{6}{7}$

(35)  $\frac{3}{7} \div \left(-\frac{1}{3}\right)$

(6)  $-\frac{1}{10} \div \left(-\frac{10}{9}\right)$

(16)  $\frac{5}{7} \div \frac{3}{5}$

(26)  $-\frac{3}{5} \div \left(-\frac{6}{7}\right)$

(36)  $\frac{4}{5} \times \frac{1}{2}$

(7)  $\frac{5}{2} \div \frac{2}{9}$

(17)  $\frac{3}{10} \div \left(-\frac{6}{5}\right)$

(27)  $-1 \div \frac{7}{8}$

(37)  $-2 \div \frac{3}{2}$

(8)  $\frac{10}{3} \times \left(-\frac{1}{3}\right)$

(18)  $\frac{8}{3} \times \left(-\frac{10}{9}\right)$

(28)  $1 \div \left(-\frac{6}{5}\right)$

(38)  $\frac{2}{5} \div \frac{1}{6}$

(9)  $-\frac{9}{8} \div \left(-\frac{2}{3}\right)$

(19)  $\frac{1}{2} \times \left(-\frac{3}{2}\right)$

(29)  $\frac{2}{3} \times \frac{1}{8}$

(39)  $-1 \times \frac{4}{3}$

(10)  $-2 \div 2$

(20)  $-\frac{1}{7} \div \left(-\frac{2}{9}\right)$

(30)  $\frac{5}{2} \times \frac{2}{3}$

(40)  $-\frac{1}{2} \div 1$

(1)  $\frac{3}{5} \times \left(-\frac{3}{10}\right)$

(11)  $\frac{3}{5} \times \left(-\frac{4}{7}\right)$

(21)  $\frac{5}{8} \times \left(-\frac{1}{3}\right)$

(31)  $10 \times \left(-\frac{2}{5}\right)$

(2)  $\frac{3}{4} \div \left(-\frac{5}{8}\right)$

(12)  $\frac{10}{7} \times 10$

(22)  $\frac{3}{8} \div \left(-\frac{3}{4}\right)$

(32)  $-\frac{4}{9} \times \frac{7}{6}$

(3)  $-\frac{9}{2} \times (-4)$

(13)  $\frac{5}{3} \div \left(-\frac{9}{8}\right)$

(23)  $-\frac{10}{3} \div \left(-\frac{9}{10}\right)$

(33)  $-\frac{9}{8} \div \frac{3}{2}$

(4)  $1 \times \left(-\frac{8}{5}\right)$

(14)  $\frac{4}{9} \div \left(-\frac{4}{3}\right)$

(24)  $\frac{3}{2} \times \frac{4}{5}$

(34)  $\frac{3}{5} \div \left(-\frac{1}{2}\right)$

(5)  $-3 \times (-2)$

(15)  $\frac{5}{9} \times \frac{5}{3}$

(25)  $\frac{7}{6} \div \frac{5}{2}$

(35)  $-1 \times \frac{7}{6}$

(6)  $-\frac{10}{7} \times \frac{9}{10}$

(16)  $-\frac{5}{9} \times \frac{1}{7}$

(26)  $\frac{1}{3} \div \frac{1}{5}$

(36)  $\frac{2}{7} \div \left(-\frac{5}{6}\right)$

(7)  $-\frac{8}{3} \div 1$

(17)  $\frac{4}{3} \times \frac{1}{2}$

(27)  $\frac{3}{5} \times \frac{1}{3}$

(37)  $\frac{1}{2} \div \frac{6}{7}$

(8)  $-4 \times 4$

(18)  $1 \div \left(-\frac{7}{6}\right)$

(28)  $-\frac{5}{2} \times (-1)$

(38)  $-\frac{9}{7} \div 1$

(9)  $-\frac{1}{3} \div 1$

(19)  $\frac{2}{5} \div 1$

(29)  $-6 \times \left(-\frac{5}{2}\right)$

(39)  $-\frac{6}{7} \div \frac{3}{4}$

(10)  $-\frac{1}{3} \times (-6)$

(20)  $-\frac{8}{9} \div \left(-\frac{5}{4}\right)$

(30)  $-\frac{7}{3} \times (-4)$

(40)  $\frac{3}{5} \div \frac{5}{8}$

(1)  $1 \times \left(-\frac{4}{9}\right)$

(11)  $-\frac{3}{5} \div (-3)$

(21)  $1 \times \frac{7}{4}$

(31)  $2 \times \frac{2}{3}$

(2)  $-\frac{4}{7} \div \frac{3}{7}$

(12)  $\frac{9}{8} \div 3$

(22)  $-\frac{4}{5} \times \frac{2}{5}$

(32)  $-9 \times \frac{5}{6}$

(3)  $-2 \times \frac{10}{7}$

(13)  $\frac{7}{3} \div \left(-\frac{9}{4}\right)$

(23)  $-1 \times \left(-\frac{1}{2}\right)$

(33)  $-\frac{2}{7} \times \frac{5}{9}$

(4)  $1 \times (-5)$

(14)  $-1 \times \left(-\frac{1}{2}\right)$

(24)  $-\frac{7}{8} \times (-3)$

(34)  $-\frac{3}{2} \times \left(-\frac{5}{9}\right)$

(5)  $1 \div \left(-\frac{9}{8}\right)$

(15)  $5 \div (-4)$

(25)  $\frac{10}{7} \times \frac{5}{2}$

(35)  $-\frac{5}{8} \times \frac{7}{8}$

(6)  $\frac{1}{5} \div \frac{4}{5}$

(16)  $\frac{7}{10} \div (-9)$

(26)  $-\frac{4}{9} \times \left(-\frac{1}{4}\right)$

(36)  $5 \times \frac{4}{5}$

(7)  $-\frac{9}{8} \times \frac{4}{7}$

(17)  $\frac{2}{7} \div \frac{4}{9}$

(27)  $-\frac{10}{9} \times (-1)$

(37)  $\frac{5}{2} \div \frac{3}{2}$

(8)  $-\frac{1}{4} \div \left(-\frac{3}{2}\right)$

(18)  $\frac{7}{9} \times 5$

(28)  $-\frac{8}{5} \times \frac{1}{3}$

(38)  $\frac{5}{2} \times \left(-\frac{3}{4}\right)$

(9)  $-\frac{3}{7} \div \left(-\frac{2}{9}\right)$

(19)  $\frac{1}{7} \times \frac{8}{3}$

(29)  $\frac{8}{7} \times \frac{1}{4}$

(39)  $\frac{3}{5} \div \left(-\frac{1}{6}\right)$

(10)  $-\frac{5}{4} \div \frac{9}{7}$

(20)  $\frac{1}{3} \div \left(-\frac{5}{8}\right)$

(30)  $-\frac{5}{3} \times \left(-\frac{5}{3}\right)$

(40)  $1 \div \left(-\frac{4}{5}\right)$

(1)  $\frac{4}{9} \times \left(-\frac{1}{8}\right)$

(11)  $-1 \div \left(-\frac{5}{4}\right)$

(21)  $\frac{3}{2} \times \left(-\frac{1}{2}\right)$

(31)  $7 \times \left(-\frac{1}{2}\right)$

(2)  $1 \times \left(-\frac{7}{3}\right)$

(12)  $1 \times \frac{3}{5}$

(22)  $-1 \div \frac{2}{5}$

(32)  $\frac{7}{8} \div \frac{8}{5}$

(3)  $9 \div \frac{8}{3}$

(13)  $\frac{7}{2} \div \left(-\frac{2}{5}\right)$

(23)  $-\frac{1}{3} \times 4$

(33)  $-2 \div \left(-\frac{9}{8}\right)$

(4)  $-\frac{7}{5} \div 3$

(14)  $-\frac{4}{5} \times 1$

(24)  $-\frac{7}{3} \div (-8)$

(34)  $-\frac{3}{5} \times \left(-\frac{8}{9}\right)$

(5)  $-\frac{5}{8} \div \frac{10}{3}$

(15)  $\frac{9}{8} \times \left(-\frac{9}{4}\right)$

(25)  $-2 \times \left(-\frac{1}{2}\right)$

(35)  $\frac{5}{4} \times \frac{1}{10}$

(6)  $-\frac{5}{4} \times \frac{8}{9}$

(16)  $-6 \div \left(-\frac{5}{2}\right)$

(26)  $1 \div \left(-\frac{7}{4}\right)$

(36)  $-\frac{1}{2} \times \left(-\frac{1}{9}\right)$

(7)  $1 \div 2$

(17)  $\frac{3}{10} \times \left(-\frac{7}{5}\right)$

(27)  $1 \times \left(-\frac{1}{2}\right)$

(37)  $\frac{3}{2} \div \frac{3}{8}$

(8)  $\frac{10}{3} \div (-6)$

(18)  $-\frac{9}{5} \div (-3)$

(28)  $-\frac{5}{9} \div \frac{4}{7}$

(38)  $\frac{2}{3} \times \left(-\frac{5}{4}\right)$

(9)  $-5 \div \left(-\frac{1}{9}\right)$

(19)  $-\frac{8}{5} \div (-2)$

(29)  $\frac{1}{6} \times \left(-\frac{7}{10}\right)$

(39)  $-\frac{4}{3} \times \left(-\frac{1}{5}\right)$

(10)  $-3 \times \frac{3}{7}$

(20)  $\frac{3}{4} \times \left(-\frac{7}{5}\right)$

(30)  $\frac{1}{5} \div \frac{3}{2}$

(40)  $-1 \div \frac{1}{5}$

(1)  $-\frac{10}{7} \times \frac{3}{5}$

(11)  $8 \times 2$

(21)  $-1 \times \left(-\frac{5}{3}\right)$

(31)  $-\frac{1}{4} \times \left(-\frac{9}{2}\right)$

(2)  $\frac{8}{3} \times \left(-\frac{3}{2}\right)$

(12)  $-\frac{2}{3} \times \left(-\frac{1}{7}\right)$

(22)  $-\frac{10}{7} \times \left(-\frac{10}{7}\right)$

(32)  $\frac{4}{5} \times \frac{7}{6}$

(3)  $-\frac{5}{3} \div (-1)$

(13)  $-\frac{2}{3} \div \left(-\frac{7}{10}\right)$

(23)  $-\frac{5}{9} \times \frac{2}{3}$

(33)  $1 \div \left(-\frac{3}{4}\right)$

(4)  $3 \div \left(-\frac{5}{2}\right)$

(14)  $-\frac{7}{2} \times \frac{7}{2}$

(24)  $-\frac{9}{4} \div 1$

(34)  $3 \div \left(-\frac{3}{5}\right)$

(5)  $2 \times \left(-\frac{9}{5}\right)$

(15)  $\frac{5}{2} \times (-3)$

(25)  $\frac{2}{9} \times \frac{1}{9}$

(35)  $1 \times \left(-\frac{8}{9}\right)$

(6)  $\frac{3}{8} \times \frac{10}{9}$

(16)  $\frac{1}{7} \div (-1)$

(26)  $\frac{1}{4} \times (-2)$

(36)  $-\frac{10}{3} \div (-1)$

(7)  $1 \div \frac{2}{3}$

(17)  $-\frac{2}{9} \times (-2)$

(27)  $\frac{8}{7} \div \frac{7}{6}$

(37)  $\frac{9}{4} \div \left(-\frac{9}{10}\right)$

(8)  $-1 \div \frac{3}{5}$

(18)  $\frac{1}{9} \times (-1)$

(28)  $\frac{5}{4} \div (-1)$

(38)  $-\frac{5}{2} \times \frac{5}{2}$

(9)  $-3 \div \frac{3}{2}$

(19)  $\frac{1}{7} \times \left(-\frac{3}{4}\right)$

(29)  $8 \div (-3)$

(39)  $4 \times \frac{1}{2}$

(10)  $-\frac{4}{5} \times \frac{3}{4}$

(20)  $\frac{5}{2} \div \frac{6}{7}$

(30)  $1 \times \left(-\frac{2}{7}\right)$

(40)  $\frac{7}{10} \times \frac{6}{5}$

(1)  $-\frac{9}{7} \div \frac{1}{9}$

(11)  $-3 \div \frac{2}{9}$

(21)  $\frac{5}{3} \div \frac{1}{2}$

(31)  $-\frac{2}{9} \times \frac{7}{9}$

(2)  $\frac{8}{9} \times 2$

(12)  $-\frac{3}{2} \div \frac{3}{10}$

(22)  $\frac{1}{8} \div \left(-\frac{5}{2}\right)$

(32)  $\frac{3}{5} \div \frac{10}{9}$

(3)  $\frac{3}{2} \div \left(-\frac{7}{2}\right)$

(13)  $\frac{1}{3} \times \left(-\frac{2}{3}\right)$

(23)  $-\frac{5}{3} \div \frac{10}{9}$

(33)  $1 \div \left(-\frac{1}{2}\right)$

(4)  $\frac{1}{2} \times \left(-\frac{2}{9}\right)$

(14)  $\frac{5}{9} \div \left(-\frac{4}{3}\right)$

(24)  $-\frac{5}{3} \times \frac{7}{4}$

(34)  $-\frac{1}{6} \div 3$

(5)  $-9 \times (-1)$

(15)  $-\frac{2}{5} \times (-3)$

(25)  $-1 \times \frac{1}{4}$

(35)  $-\frac{3}{5} \div \left(-\frac{1}{10}\right)$

(6)  $\frac{9}{10} \div \frac{2}{3}$

(16)  $-\frac{4}{5} \times (-3)$

(26)  $\frac{3}{4} \div \frac{9}{5}$

(36)  $-\frac{1}{5} \times \left(-\frac{10}{9}\right)$

(7)  $\frac{1}{7} \div 2$

(17)  $-\frac{8}{3} \div \frac{1}{9}$

(27)  $-\frac{1}{9} \times 9$

(37)  $\frac{4}{5} \times \left(-\frac{2}{7}\right)$

(8)  $-\frac{4}{5} \times 1$

(18)  $-2 \div (-3)$

(28)  $\frac{1}{8} \times \frac{2}{9}$

(38)  $\frac{1}{5} \times \left(-\frac{8}{9}\right)$

(9)  $\frac{3}{2} \times \frac{7}{9}$

(19)  $-\frac{5}{4} \div \left(-\frac{2}{3}\right)$

(29)  $\frac{7}{2} \div \frac{1}{5}$

(39)  $-\frac{7}{9} \times \left(-\frac{1}{2}\right)$

(10)  $-\frac{1}{2} \div \left(-\frac{1}{9}\right)$

(20)  $\frac{1}{2} \div (-1)$

(30)  $\frac{2}{5} \div (-4)$

(40)  $-\frac{9}{4} \div \left(-\frac{2}{5}\right)$



(1)  $1 \times \frac{5}{2}$

(11)  $-2 \div \frac{2}{3}$

(21)  $\frac{10}{7} \div (-1)$

(31)  $-\frac{1}{2} \div \left(-\frac{7}{10}\right)$

(2)  $10 \div \left(-\frac{10}{3}\right)$

(12)  $-6 \times \left(-\frac{1}{2}\right)$

(22)  $\frac{5}{7} \times \frac{5}{8}$

(32)  $-\frac{2}{3} \times \frac{1}{3}$

(3)  $\frac{3}{2} \div (-3)$

(13)  $-\frac{7}{5} \times \left(-\frac{9}{5}\right)$

(23)  $\frac{1}{3} \times \left(-\frac{6}{5}\right)$

(33)  $-\frac{10}{9} \div \frac{3}{10}$

(4)  $\frac{3}{4} \times \left(-\frac{1}{3}\right)$

(14)  $1 \times 5$

(24)  $4 \div \frac{2}{5}$

(34)  $\frac{4}{9} \times \left(-\frac{1}{4}\right)$

(5)  $-\frac{3}{5} \div \frac{7}{5}$

(15)  $-2 \div \left(-\frac{1}{5}\right)$

(25)  $-\frac{4}{7} \div (-6)$

(35)  $-\frac{5}{2} \div \left(-\frac{3}{5}\right)$

(16)  $-\frac{1}{2} \times (-3)$

(6)  $-6 \div \frac{2}{3}$

(17)  $-2 \div 5$

(26)  $-\frac{5}{3} \times 2$

(36)  $\frac{1}{4} \div \left(-\frac{1}{2}\right)$

(7)  $-\frac{1}{5} \times \frac{8}{7}$

(18)  $-\frac{9}{10} \div 2$

(27)  $-\frac{2}{7} \times \frac{3}{8}$

(37)  $1 \times \frac{10}{7}$

(8)  $-\frac{2}{7} \times \left(-\frac{1}{3}\right)$

(19)  $-\frac{5}{4} \div \frac{2}{3}$

(28)  $\frac{8}{3} \times (-1)$

(38)  $\frac{1}{4} \times \left(-\frac{3}{5}\right)$

(9)  $-\frac{2}{3} \times \left(-\frac{7}{10}\right)$

(20)  $-1 \times \left(-\frac{6}{7}\right)$

(29)  $-\frac{3}{4} \times (-1)$

(39)  $-\frac{5}{3} \div \frac{4}{5}$

(10)  $-\frac{6}{5} \times 1$

(30)  $4 \div \frac{4}{7}$

(40)  $1 \times \left(-\frac{7}{2}\right)$

(1)  $-\frac{7}{5} \div (-1)$

(11)  $-\frac{3}{2} \times 1$

(21)  $-\frac{1}{4} \div \left(-\frac{2}{5}\right)$

(31)  $\frac{1}{8} \div \frac{3}{8}$

(2)  $-\frac{1}{7} \times \frac{9}{5}$

(12)  $-\frac{2}{7} \times \left(-\frac{4}{9}\right)$

(22)  $\frac{1}{5} \div 1$

(32)  $-\frac{8}{3} \div \frac{4}{7}$

(3)  $\frac{5}{7} \div (-5)$

(13)  $-\frac{1}{10} \times \left(-\frac{5}{4}\right)$

(23)  $\frac{3}{2} \times \frac{5}{4}$

(33)  $\frac{10}{7} \div \frac{3}{2}$

(4)  $-\frac{2}{7} \times \frac{2}{3}$

(14)  $\frac{8}{7} \times \left(-\frac{3}{5}\right)$

(24)  $\frac{8}{3} \times 1$

(34)  $1 \times (-1)$

(5)  $-\frac{5}{7} \times 5$

(15)  $1 \times \left(-\frac{7}{4}\right)$

(25)  $-\frac{4}{9} \div \frac{5}{7}$

(35)  $-\frac{8}{5} \times (-3)$

(6)  $-\frac{4}{3} \div \frac{4}{5}$

(16)  $\frac{5}{4} \times \left(-\frac{9}{7}\right)$

(26)  $\frac{3}{2} \times \frac{5}{7}$

(36)  $-1 \div (-7)$

(7)  $2 \div \frac{7}{3}$

(17)  $-\frac{7}{6} \times \left(-\frac{3}{2}\right)$

(27)  $\frac{2}{3} \div (-1)$

(37)  $\frac{5}{3} \times (-1)$

(8)  $-\frac{6}{5} \times \frac{10}{9}$

(18)  $\frac{1}{5} \times \left(-\frac{5}{7}\right)$

(28)  $-\frac{2}{5} \times \left(-\frac{2}{5}\right)$

(38)  $-\frac{1}{10} \div \left(-\frac{8}{5}\right)$

(9)  $\frac{5}{2} \times \frac{4}{5}$

(19)  $\frac{3}{2} \div \frac{7}{4}$

(29)  $\frac{3}{5} \times 3$

(39)  $\frac{1}{4} \times (-2)$

(10)  $\frac{3}{2} \times \frac{1}{3}$

(20)  $\frac{4}{3} \div (-2)$

(30)  $\frac{7}{2} \div \frac{3}{5}$

(40)  $1 \times \left(-\frac{5}{8}\right)$

$$(1) -\frac{9}{10} \times \frac{3}{10} \\ = -\frac{27}{100}$$

$$(11) \frac{1}{2} \times \left(-\frac{5}{3}\right) \\ = -\frac{5}{6}$$

$$(21) \frac{4}{5} \times 1 \\ = \frac{4}{5}$$

$$(31) -1 \times 1 \\ = -1$$

$$(2) -8 \div \frac{8}{7} \\ = -7$$

$$(12) -\frac{7}{8} \times \frac{9}{4} \\ = -\frac{63}{32}$$

$$(22) -\frac{1}{5} \div \left(-\frac{5}{2}\right) \\ = \frac{2}{25}$$

$$(32) \frac{5}{2} \div \frac{1}{2} \\ = 5$$

$$(3) -2 \times \left(-\frac{9}{7}\right) \\ = \frac{18}{7}$$

$$(13) -4 \div \frac{1}{4} \\ = -16$$

$$(23) -\frac{6}{7} \times \frac{1}{2} \\ = -\frac{3}{7}$$

$$(33) -9 \times (-3) \\ = 27$$

$$(4) \frac{4}{5} \times (-2) \\ = -\frac{8}{5}$$

$$(14) -\frac{2}{3} \times \frac{1}{5} \\ = -\frac{2}{15}$$

$$(24) \frac{1}{6} \div 10 \\ = \frac{1}{60}$$

$$(34) -8 \div \frac{7}{9} \\ = -\frac{72}{7}$$

$$(5) \frac{9}{10} \div \left(-\frac{3}{2}\right) \\ = -\frac{3}{5}$$

$$(15) \frac{10}{9} \div 1 \\ = \frac{10}{9}$$

$$(25) \frac{2}{5} \div 4 \\ = \frac{1}{10}$$

$$(35) -\frac{5}{8} \times \left(-\frac{3}{4}\right) \\ = \frac{15}{32}$$

$$(6) \frac{4}{5} \div \left(-\frac{2}{3}\right) \\ = -\frac{6}{5}$$

$$(16) \frac{2}{9} \times \frac{5}{2} \\ = \frac{5}{9}$$

$$(26) \frac{1}{3} \times 1 \\ = \frac{1}{3}$$

$$(36) \frac{1}{2} \div 2 \\ = \frac{1}{4}$$

$$(7) -\frac{1}{6} \times \frac{7}{9} \\ = -\frac{7}{54}$$

$$(17) \frac{4}{3} \times \frac{1}{8} \\ = \frac{1}{6}$$

$$(27) \frac{1}{4} \div \frac{3}{8} \\ = \frac{2}{3}$$

$$(37) -\frac{9}{4} \div (-2) \\ = \frac{9}{8}$$

$$(8) \frac{7}{3} \div 9 \\ = \frac{7}{27}$$

$$(18) -\frac{3}{2} \div \frac{2}{3} \\ = -\frac{9}{4}$$

$$(28) \frac{3}{2} \times \left(-\frac{2}{9}\right) \\ = -\frac{1}{3}$$

$$(38) \frac{5}{2} \div \frac{1}{7} \\ = \frac{35}{2}$$

$$(9) -2 \times \frac{2}{3} \\ = -\frac{4}{3}$$

$$(19) \frac{9}{8} \div \frac{4}{5} \\ = \frac{45}{32}$$

$$(29) -\frac{5}{4} \div \left(-\frac{1}{10}\right) \\ = \frac{25}{2}$$

$$(39) -\frac{8}{3} \div \frac{7}{10} \\ = -\frac{80}{21}$$

$$(10) -\frac{4}{5} \times \frac{1}{2} \\ = -\frac{2}{5}$$

$$(20) \frac{3}{2} \div \frac{3}{4} \\ = 2$$

$$(30) -2 \times \frac{8}{9} \\ = -\frac{16}{9}$$

$$(40) -\frac{9}{10} \div \left(-\frac{5}{3}\right) \\ = \frac{27}{50}$$

$$(1) -\frac{5}{3} \div \left(-\frac{5}{4}\right) \\ = \frac{4}{3}$$

$$(11) \frac{1}{5} \div \frac{2}{5} \\ = \frac{1}{2}$$

$$(21) 1 \div 6 \\ = \frac{1}{6}$$

$$(31) \frac{5}{3} \times \left(-\frac{6}{5}\right) \\ = -2$$

$$(2) -\frac{7}{5} \div \left(-\frac{4}{3}\right) \\ = \frac{21}{20}$$

$$(12) \frac{3}{10} \times \frac{3}{4} \\ = \frac{9}{40}$$

$$(22) -\frac{5}{7} \div \left(-\frac{8}{3}\right) \\ = \frac{15}{56}$$

$$(32) 1 \div \frac{2}{3} \\ = \frac{3}{2}$$

$$(3) -\frac{2}{3} \times \left(-\frac{5}{3}\right) \\ = \frac{10}{9}$$

$$(13) \frac{1}{5} \times \left(-\frac{10}{9}\right) \\ = -\frac{2}{9}$$

$$(23) -\frac{5}{2} \div 3 \\ = -\frac{5}{6}$$

$$(33) -\frac{5}{2} \div \left(-\frac{2}{7}\right) \\ = \frac{35}{4}$$

$$(4) 2 \div \left(-\frac{1}{3}\right) \\ = -6$$

$$(14) -1 \times \left(-\frac{1}{2}\right) \\ = \frac{1}{2}$$

$$(24) -\frac{2}{5} \div (-8) \\ = \frac{1}{20}$$

$$(34) -\frac{1}{3} \div (-1) \\ = \frac{1}{3}$$

$$(5) 5 \div \left(-\frac{1}{4}\right) \\ = -20$$

$$(15) \frac{9}{7} \div (-1) \\ = -\frac{9}{7}$$

$$(25) 5 \times \frac{5}{8} \\ = \frac{25}{8}$$

$$(35) 1 \times 1 \\ = 1$$

$$(6) \frac{3}{4} \div \left(-\frac{5}{2}\right) \\ = -\frac{3}{10}$$

$$(16) 2 \div 1 \\ = 2$$

$$(26) \frac{1}{9} \div \left(-\frac{3}{2}\right) \\ = -\frac{2}{27}$$

$$(36) -\frac{7}{10} \div \left(-\frac{7}{8}\right) \\ = \frac{4}{5}$$

$$(7) -\frac{1}{4} \times \left(-\frac{7}{10}\right) \\ = \frac{7}{40}$$

$$(17) -1 \times \left(-\frac{2}{3}\right) \\ = \frac{2}{3}$$

$$(27) \frac{7}{5} \div \left(-\frac{1}{3}\right) \\ = -\frac{21}{5}$$

$$(37) 2 \times 3 \\ = 6$$

$$(8) \frac{1}{2} \div \left(-\frac{1}{9}\right) \\ = -\frac{9}{2}$$

$$(18) -\frac{5}{9} \div \frac{5}{2} \\ = -\frac{2}{9}$$

$$(28) -5 \div (-1) \\ = 5$$

$$(38) \frac{1}{4} \times \frac{10}{9} \\ = \frac{5}{18}$$

$$(9) -\frac{7}{10} \times \left(-\frac{1}{2}\right) \\ = \frac{7}{20}$$

$$(19) -5 \times \left(-\frac{1}{3}\right) \\ = \frac{5}{3}$$

$$(29) 2 \div 2 \\ = 1$$

$$(39) 3 \times \frac{3}{7} \\ = \frac{9}{7}$$

$$(10) 3 \div \frac{4}{9} \\ = \frac{27}{4}$$

$$(20) -\frac{8}{9} \div \left(-\frac{10}{9}\right) \\ = \frac{4}{5}$$

$$(30) 1 \div \left(-\frac{1}{5}\right) \\ = -5$$

$$(40) \frac{3}{4} \div \left(-\frac{1}{2}\right) \\ = -\frac{3}{2}$$

$$(1) \frac{4}{7} \div \left(-\frac{1}{5}\right) \\ = -\frac{20}{7}$$

$$(11) -1 \times (-2) \\ = 2$$

$$(21) \frac{10}{9} \times 2 \\ = \frac{20}{9}$$

$$(31) -\frac{3}{5} \times 3 \\ = -\frac{9}{5}$$

$$(2) -\frac{1}{5} \div \frac{1}{5} \\ = -1$$

$$(12) -1 \div \left(-\frac{3}{8}\right) \\ = \frac{8}{3}$$

$$(22) -\frac{9}{8} \times \left(-\frac{4}{3}\right) \\ = \frac{3}{2}$$

$$(32) -1 \times \frac{2}{3} \\ = -\frac{2}{3}$$

$$(3) -\frac{1}{3} \times \frac{5}{4} \\ = -\frac{5}{12}$$

$$(13) \frac{3}{2} \div 4 \\ = \frac{3}{8}$$

$$(23) -1 \times \left(-\frac{9}{4}\right) \\ = \frac{9}{4}$$

$$(33) 1 \times \frac{1}{5} \\ = \frac{1}{5}$$

$$(4) -\frac{4}{3} \div (-2) \\ = \frac{2}{3}$$

$$(14) -\frac{1}{4} \times \left(-\frac{7}{3}\right) \\ = \frac{7}{12}$$

$$(24) \frac{6}{7} \times \left(-\frac{2}{7}\right) \\ = -\frac{12}{49}$$

$$(34) 1 \div \frac{1}{5} \\ = 5$$

$$(5) 2 \div \frac{9}{8} \\ = \frac{16}{9}$$

$$(15) \frac{5}{3} \div \left(-\frac{10}{7}\right) \\ = -\frac{7}{6}$$

$$(25) -\frac{7}{3} \div (-3) \\ = \frac{7}{9}$$

$$(35) -\frac{2}{5} \times \left(-\frac{6}{5}\right) \\ = \frac{12}{25}$$

$$(6) -\frac{8}{7} \div \left(-\frac{6}{5}\right) \\ = \frac{20}{21}$$

$$(16) -\frac{1}{3} \times 2 \\ = -\frac{2}{3}$$

$$(26) -1 \div \left(-\frac{9}{5}\right) \\ = \frac{5}{9}$$

$$(36) -5 \times \frac{10}{9} \\ = -\frac{50}{9}$$

$$(7) -\frac{8}{5} \times (-3) \\ = \frac{24}{5}$$

$$(17) \frac{9}{2} \div \left(-\frac{4}{7}\right) \\ = -\frac{63}{8}$$

$$(27) -1 \div \frac{1}{2} \\ = -2$$

$$(37) -\frac{1}{2} \div \left(-\frac{2}{3}\right) \\ = \frac{3}{4}$$

$$(8) 4 \div \frac{1}{10} \\ = 40$$

$$(18) \frac{10}{9} \times \left(-\frac{5}{7}\right) \\ = -\frac{50}{63}$$

$$(28) 8 \times \left(-\frac{5}{3}\right) \\ = -\frac{40}{3}$$

$$(38) \frac{3}{4} \div \left(-\frac{3}{5}\right) \\ = -\frac{5}{4}$$

$$(9) \frac{5}{4} \div \left(-\frac{8}{7}\right) \\ = -\frac{35}{32}$$

$$(19) \frac{3}{10} \div \left(-\frac{1}{2}\right) \\ = -\frac{3}{5}$$

$$(29) 8 \div \left(-\frac{1}{2}\right) \\ = -16$$

$$(39) -\frac{1}{5} \times (-2) \\ = \frac{2}{5}$$

$$(10) \frac{3}{2} \div \left(-\frac{3}{5}\right) \\ = -\frac{5}{2}$$

$$(20) -\frac{4}{7} \times \left(-\frac{5}{3}\right) \\ = \frac{20}{21}$$

$$(30) -\frac{4}{3} \div \frac{2}{3} \\ = -2$$

$$(40) \frac{2}{3} \times \left(-\frac{7}{3}\right) \\ = -\frac{14}{9}$$

$$(1) \frac{4}{7} \times \left(-\frac{1}{2}\right) \\ = -\frac{2}{7}$$

$$(11) -\frac{5}{8} \div \left(-\frac{2}{3}\right) \\ = \frac{15}{16}$$

$$(21) -\frac{3}{7} \times \left(-\frac{5}{8}\right) \\ = \frac{15}{56}$$

$$(31) 1 \times \left(-\frac{8}{3}\right) \\ = -\frac{8}{3}$$

$$(2) \frac{1}{3} \div \frac{10}{7} \\ = \frac{7}{30}$$

$$(12) \frac{2}{5} \times 4 \\ = \frac{8}{5}$$

$$(22) -\frac{3}{2} \times \frac{9}{5} \\ = -\frac{27}{10}$$

$$(32) \frac{1}{5} \div \left(-\frac{3}{2}\right) \\ = -\frac{2}{15}$$

$$(3) \frac{9}{5} \div 3 \\ = \frac{3}{5}$$

$$(13) -\frac{9}{5} \div \frac{9}{5} \\ = -1$$

$$(23) \frac{8}{9} \times \left(-\frac{1}{5}\right) \\ = -\frac{8}{45}$$

$$(33) 3 \times 9 \\ = 27$$

$$(4) \frac{6}{5} \div \left(-\frac{3}{5}\right) \\ = -2$$

$$(14) \frac{1}{7} \div \frac{6}{7} \\ = \frac{1}{6}$$

$$(24) -\frac{4}{5} \times 2 \\ = -\frac{8}{5}$$

$$(34) \frac{1}{5} \div (-5) \\ = -\frac{1}{25}$$

$$(5) 1 \div (-2) \\ = -\frac{1}{2}$$

$$(15) -1 \div \frac{1}{4} \\ = -4$$

$$(25) \frac{3}{2} \div \frac{1}{2} \\ = 3$$

$$(35) \frac{7}{8} \div \left(-\frac{7}{4}\right) \\ = -\frac{1}{2}$$

$$(6) \frac{9}{5} \times 4 \\ = \frac{36}{5}$$

$$(16) -1 \times \left(-\frac{7}{9}\right) \\ = \frac{7}{9}$$

$$(26) -9 \times \left(-\frac{5}{2}\right) \\ = \frac{45}{2}$$

$$(36) -\frac{7}{2} \div (-4) \\ = \frac{7}{8}$$

$$(7) \frac{4}{9} \div 1 \\ = \frac{4}{9}$$

$$(17) \frac{3}{5} \div \left(-\frac{2}{7}\right) \\ = -\frac{21}{10}$$

$$(27) -\frac{1}{3} \div \frac{4}{9} \\ = -\frac{3}{4}$$

$$(37) \frac{5}{8} \div \frac{3}{4} \\ = \frac{5}{6}$$

$$(8) -\frac{1}{6} \times \left(-\frac{1}{5}\right) \\ = \frac{1}{30}$$

$$(18) 4 \div \frac{1}{9} \\ = 36$$

$$(28) \frac{6}{7} \div \frac{2}{3} \\ = \frac{9}{7}$$

$$(38) -1 \div 3 \\ = -\frac{1}{3}$$

$$(9) -\frac{10}{3} \div (-3) \\ = \frac{10}{9}$$

$$(19) \frac{5}{8} \times \left(-\frac{1}{2}\right) \\ = -\frac{5}{16}$$

$$(29) \frac{3}{8} \div \frac{3}{8} \\ = 1$$

$$(39) -\frac{3}{2} \div \frac{1}{8} \\ = -12$$

$$(10) -\frac{5}{6} \div (-3) \\ = \frac{5}{18}$$

$$(20) 3 \div \frac{8}{5} \\ = \frac{15}{8}$$

$$(30) -\frac{5}{4} \div \frac{1}{3} \\ = -\frac{15}{4}$$

$$(40) 7 \times 1 \\ = 7$$

$$(1) \frac{4}{3} \times (-1) \\ = -\frac{4}{3}$$

$$(11) 1 \div \left(-\frac{1}{8}\right) \\ = -8$$

$$(21) \frac{5}{3} \times \frac{7}{4} \\ = \frac{35}{12}$$

$$(31) -\frac{9}{2} \times \frac{9}{7} \\ = -\frac{81}{14}$$

$$(2) \frac{3}{4} \div 1 \\ = \frac{3}{4}$$

$$(12) 1 \div \frac{2}{3} \\ = \frac{3}{2}$$

$$(22) -\frac{5}{2} \div \left(-\frac{3}{4}\right) \\ = \frac{10}{3}$$

$$(32) -\frac{8}{5} \div \left(-\frac{1}{4}\right) \\ = \frac{32}{5}$$

$$(3) \frac{5}{7} \times \left(-\frac{5}{3}\right) \\ = -\frac{25}{21}$$

$$(13) \frac{1}{4} \div 1 \\ = \frac{1}{4}$$

$$(23) -5 \div 1 \\ = -5$$

$$(33) \frac{9}{2} \times 1 \\ = \frac{9}{2}$$

$$(4) -\frac{1}{2} \div \frac{3}{4} \\ = -\frac{2}{3}$$

$$(14) \frac{5}{4} \times \left(-\frac{7}{2}\right) \\ = -\frac{35}{8}$$

$$(24) -\frac{5}{4} \div \frac{3}{2} \\ = -\frac{5}{6}$$

$$(34) 1 \times \left(-\frac{6}{5}\right) \\ = -\frac{6}{5}$$

$$(5) -1 \div \frac{4}{3} \\ = -\frac{3}{4}$$

$$(15) \frac{5}{4} \div \left(-\frac{3}{5}\right) \\ = -\frac{25}{12}$$

$$(25) -\frac{7}{10} \times \left(-\frac{3}{8}\right) \\ = \frac{21}{80}$$

$$(35) -\frac{4}{5} \times \left(-\frac{3}{5}\right) \\ = \frac{12}{25}$$

$$(6) \frac{5}{8} \times \frac{5}{2} \\ = \frac{25}{16}$$

$$(16) -\frac{2}{3} \div \frac{1}{8} \\ = -\frac{16}{3}$$

$$(26) -2 \div \left(-\frac{1}{4}\right) \\ = 8$$

$$(36) \frac{1}{2} \div \frac{7}{4} \\ = \frac{2}{7}$$

$$(7) 4 \times \left(-\frac{6}{7}\right) \\ = -\frac{24}{7}$$

$$(17) -\frac{1}{2} \times \frac{7}{2} \\ = -\frac{7}{4}$$

$$(27) -\frac{1}{4} \times (-1) \\ = \frac{1}{4}$$

$$(37) -\frac{7}{6} \div \left(-\frac{5}{8}\right) \\ = \frac{28}{15}$$

$$(8) -\frac{9}{2} \div \frac{2}{5} \\ = -\frac{45}{4}$$

$$(18) 1 \times \left(-\frac{9}{7}\right) \\ = -\frac{9}{7}$$

$$(28) \frac{5}{3} \times \left(-\frac{8}{5}\right) \\ = -\frac{8}{3}$$

$$(38) \frac{4}{3} \times (-5) \\ = -\frac{20}{3}$$

$$(9) 9 \times 1 \\ = 9$$

$$(19) \frac{4}{5} \times \frac{6}{7} \\ = \frac{24}{35}$$

$$(29) -\frac{10}{3} \div (-3) \\ = \frac{10}{9}$$

$$(39) 9 \div \frac{9}{8} \\ = 8$$

$$(10) \frac{4}{3} \div 1 \\ = \frac{4}{3}$$

$$(20) -1 \div (-2) \\ = \frac{1}{2}$$

$$(30) -\frac{3}{2} \div \frac{3}{5} \\ = -\frac{5}{2}$$

$$(40) -\frac{1}{4} \times \frac{4}{9} \\ = -\frac{1}{9}$$

$$(1) -\frac{9}{8} \times \left(-\frac{8}{3}\right) \\ = 3$$

$$(11) -7 \div \left(-\frac{8}{7}\right) \\ = \frac{49}{8}$$

$$(21) -\frac{4}{5} \times 3 \\ = -\frac{12}{5}$$

$$(31) \frac{4}{3} \div \frac{10}{7} \\ = \frac{14}{15}$$

$$(2) 7 \div 1 \\ = 7$$

$$(12) -\frac{5}{4} \times \frac{3}{5} \\ = -\frac{3}{4}$$

$$(22) \frac{8}{7} \div \frac{3}{5} \\ = \frac{40}{21}$$

$$(32) \frac{10}{7} \div \frac{7}{10} \\ = \frac{100}{49}$$

$$(3) -1 \div 5 \\ = -\frac{1}{5}$$

$$(13) -\frac{5}{2} \times \frac{1}{2} \\ = -\frac{5}{4}$$

$$(23) \frac{5}{8} \times \frac{1}{2} \\ = \frac{5}{16}$$

$$(33) 1 \div \frac{9}{2} \\ = \frac{2}{9}$$

$$(4) -2 \times 1 \\ = -2$$

$$(14) \frac{7}{9} \div 3 \\ = \frac{7}{27}$$

$$(24) -2 \div \frac{2}{5} \\ = -5$$

$$(34) -9 \times \frac{3}{10} \\ = -\frac{27}{10}$$

$$(5) -\frac{2}{3} \div \frac{7}{3} \\ = -\frac{2}{7}$$

$$(15) \frac{5}{4} \times \frac{1}{3} \\ = \frac{5}{12}$$

$$(25) -\frac{9}{5} \div \frac{5}{2} \\ = -\frac{18}{25}$$

$$(35) -\frac{5}{7} \div \frac{9}{10} \\ = -\frac{50}{63}$$

$$(6) \frac{1}{6} \times \left(-\frac{3}{10}\right) \\ = -\frac{1}{20}$$

$$(16) \frac{9}{10} \div \left(-\frac{1}{9}\right) \\ = -\frac{81}{10}$$

$$(26) -8 \div \left(-\frac{1}{6}\right) \\ = 48$$

$$(36) \frac{9}{4} \times \left(-\frac{9}{8}\right) \\ = -\frac{81}{32}$$

$$(7) -\frac{7}{10} \times \frac{1}{5} \\ = -\frac{7}{50}$$

$$(17) -\frac{9}{8} \div 5 \\ = -\frac{9}{40}$$

$$(27) -\frac{4}{5} \div \left(-\frac{3}{5}\right) \\ = \frac{4}{3}$$

$$(37) \frac{9}{10} \div \left(-\frac{3}{8}\right) \\ = -\frac{12}{5}$$

$$(8) \frac{5}{9} \times 1 \\ = \frac{5}{9}$$

$$(18) \frac{5}{7} \div 3 \\ = \frac{5}{21}$$

$$(28) -\frac{1}{10} \times 2 \\ = -\frac{1}{5}$$

$$(38) -3 \div \frac{9}{5} \\ = -\frac{5}{3}$$

$$(9) \frac{4}{5} \div (-3) \\ = -\frac{4}{15}$$

$$(19) -\frac{1}{10} \times \frac{3}{5} \\ = -\frac{3}{50}$$

$$(29) -\frac{7}{6} \div \left(-\frac{7}{4}\right) \\ = \frac{2}{3}$$

$$(39) \frac{5}{7} \div \left(-\frac{5}{3}\right) \\ = -\frac{3}{7}$$

$$(10) 2 \div \frac{4}{3} \\ = \frac{3}{2}$$

$$(20) -2 \div \frac{1}{7} \\ = -14$$

$$(30) -\frac{5}{2} \times \frac{5}{4} \\ = -\frac{25}{8}$$

$$(40) -\frac{6}{5} \times \left(-\frac{10}{9}\right) \\ = \frac{4}{3}$$



$$(1) -2 \div \frac{4}{3} \\ = -\frac{3}{2}$$

$$(11) -2 \times \left(-\frac{8}{9}\right) \\ = \frac{16}{9}$$

$$(21) -7 \div \frac{5}{4} \\ = -\frac{28}{5}$$

$$(31) 2 \times \left(-\frac{7}{2}\right) \\ = -7$$

$$(2) \frac{2}{9} \div 2 \\ = \frac{1}{9}$$

$$(12) 4 \times 4 \\ = 16$$

$$(22) -\frac{1}{6} \times (-1) \\ = \frac{1}{6}$$

$$(32) \frac{1}{3} \div \left(-\frac{1}{2}\right) \\ = -\frac{2}{3}$$

$$(3) \frac{7}{2} \div \frac{5}{8} \\ = \frac{28}{5}$$

$$(13) \frac{4}{3} \div \left(-\frac{5}{6}\right) \\ = -\frac{8}{5}$$

$$(23) \frac{1}{2} \times (-1) \\ = -\frac{1}{2}$$

$$(33) -\frac{2}{5} \times \left(-\frac{7}{10}\right) \\ = \frac{7}{25}$$

$$(4) \frac{5}{6} \times \frac{5}{4} \\ = \frac{25}{24}$$

$$(14) \frac{7}{5} \div 2 \\ = \frac{7}{10}$$

$$(24) 8 \div \left(-\frac{1}{6}\right) \\ = -48$$

$$(34) -\frac{5}{9} \times \left(-\frac{7}{4}\right) \\ = \frac{35}{36}$$

$$(5) 3 \div (-6) \\ = -\frac{1}{2}$$

$$(15) 5 \times \frac{7}{3} \\ = \frac{35}{3}$$

$$(25) \frac{10}{9} \times 1 \\ = \frac{10}{9}$$

$$(35) -\frac{8}{3} \times \frac{3}{5} \\ = -\frac{8}{5}$$

$$(6) 1 \times (-1) \\ = -1$$

$$(16) \frac{1}{5} \times \frac{6}{7} \\ = \frac{6}{35}$$

$$(26) \frac{1}{5} \div \left(-\frac{9}{4}\right) \\ = -\frac{4}{45}$$

$$(36) 1 \div \left(-\frac{7}{3}\right) \\ = -\frac{3}{7}$$

$$(7) \frac{3}{4} \times \frac{3}{8} \\ = \frac{9}{32}$$

$$(17) -\frac{7}{5} \times (-2) \\ = \frac{14}{5}$$

$$(27) -2 \div 4 \\ = -\frac{1}{2}$$

$$(37) -\frac{4}{9} \times \frac{5}{9} \\ = -\frac{20}{81}$$

$$(8) -\frac{10}{9} \div \left(-\frac{4}{5}\right) \\ = \frac{25}{18}$$

$$(18) \frac{7}{9} \times \left(-\frac{9}{2}\right) \\ = -\frac{7}{2}$$

$$(28) \frac{9}{4} \times 1 \\ = \frac{9}{4}$$

$$(38) -\frac{2}{5} \div 2 \\ = -\frac{1}{5}$$

$$(9) -\frac{7}{2} \times \left(-\frac{4}{3}\right) \\ = \frac{14}{3}$$

$$(19) -\frac{4}{5} \times (-1) \\ = \frac{4}{5}$$

$$(29) -1 \times 3 \\ = -3$$

$$(39) \frac{8}{5} \div \left(-\frac{3}{8}\right) \\ = -\frac{64}{15}$$

$$(10) 1 \times \frac{1}{2} \\ = \frac{1}{2}$$

$$(20) -\frac{2}{9} \div \frac{4}{5} \\ = -\frac{5}{18}$$

$$(30) -2 \div \left(-\frac{6}{7}\right) \\ = \frac{7}{3}$$

$$(40) \frac{4}{3} \div (-10) \\ = -\frac{2}{15}$$

$$(1) -\frac{9}{10} \times 6$$

$$= -\frac{27}{5}$$

$$(11) -\frac{7}{10} \div \frac{1}{5}$$

$$= -\frac{7}{2}$$

$$(21) \frac{1}{9} \div (-3)$$

$$= -\frac{1}{27}$$

$$(31) 2 \div \frac{7}{6}$$

$$= \frac{12}{7}$$

$$(2) -\frac{7}{3} \times \frac{5}{4}$$

$$= -\frac{35}{12}$$

$$(12) -\frac{6}{5} \div \frac{10}{9}$$

$$= -\frac{27}{25}$$

$$(22) -\frac{1}{2} \times \left(-\frac{8}{3}\right)$$

$$= \frac{4}{3}$$

$$(32) \frac{1}{3} \div \frac{10}{9}$$

$$= \frac{3}{10}$$

$$(3) -\frac{9}{10} \times \frac{7}{5}$$

$$= -\frac{63}{50}$$

$$(13) -\frac{4}{5} \times \left(-\frac{2}{3}\right)$$

$$= \frac{8}{15}$$

$$(23) \frac{8}{7} \times \left(-\frac{3}{10}\right)$$

$$= -\frac{12}{35}$$

$$(33) 1 \times \frac{7}{5}$$

$$= \frac{7}{5}$$

$$(4) \frac{2}{5} \div \frac{3}{2}$$

$$= \frac{4}{15}$$

$$(14) \frac{9}{4} \times \left(-\frac{1}{8}\right)$$

$$= -\frac{9}{32}$$

$$(24) -\frac{1}{3} \times \frac{1}{5}$$

$$= -\frac{1}{15}$$

$$(34) 1 \times \frac{4}{5}$$

$$= \frac{4}{5}$$

$$(5) -\frac{10}{7} \div 2$$

$$= -\frac{5}{7}$$

$$(15) \frac{10}{9} \times (-1)$$

$$= -\frac{10}{9}$$

$$(25) 1 \div \frac{3}{2}$$

$$= \frac{2}{3}$$

$$(35) -\frac{3}{4} \times \left(-\frac{1}{4}\right)$$

$$= \frac{3}{16}$$

$$(6) \frac{4}{3} \times \left(-\frac{1}{4}\right)$$

$$= -\frac{1}{3}$$

$$(16) -2 \div \left(-\frac{9}{4}\right)$$

$$= \frac{8}{9}$$

$$(26) \frac{1}{5} \times \frac{1}{4}$$

$$= \frac{1}{20}$$

$$(36) \frac{2}{3} \times \left(-\frac{2}{9}\right)$$

$$= -\frac{4}{27}$$

$$(7) -6 \div (-8)$$

$$= \frac{3}{4}$$

$$(17) \frac{7}{2} \div \frac{5}{6}$$

$$= \frac{21}{5}$$

$$(27) -\frac{1}{7} \div \frac{6}{7}$$

$$= -\frac{1}{6}$$

$$(37) -1 \div \left(-\frac{10}{9}\right)$$

$$= \frac{9}{10}$$

$$(8) -\frac{5}{2} \div \frac{9}{2}$$

$$= -\frac{5}{9}$$

$$(18) -\frac{9}{10} \times \frac{7}{2}$$

$$= -\frac{63}{20}$$

$$(28) \frac{1}{3} \times \frac{4}{5}$$

$$= \frac{4}{15}$$

$$(38) -1 \times (-1)$$

$$= 1$$

$$(9) \frac{7}{8} \div \frac{9}{7}$$

$$= \frac{49}{72}$$

$$(19) -2 \div \left(-\frac{8}{9}\right)$$

$$= \frac{9}{4}$$

$$(29) \frac{8}{9} \times (-4)$$

$$= -\frac{32}{9}$$

$$(39) \frac{5}{8} \div 3$$

$$= \frac{5}{24}$$

$$(10) -\frac{2}{9} \times \left(-\frac{5}{4}\right)$$

$$= \frac{5}{18}$$

$$(20) -3 \div 5$$

$$= -\frac{3}{5}$$

$$(30) \frac{7}{9} \div \frac{4}{5}$$

$$= \frac{35}{36}$$

$$(40) -\frac{9}{7} \div \frac{3}{7}$$

$$= -3$$

$$(1) \frac{1}{4} \div 5 \\ = \frac{1}{20}$$

$$(11) \frac{1}{2} \times \frac{1}{9} \\ = \frac{1}{18}$$

$$(21) \frac{4}{3} \div \left(-\frac{1}{2}\right) \\ = -\frac{8}{3}$$

$$(31) -2 \times \left(-\frac{1}{2}\right) \\ = 1$$

$$(2) -\frac{8}{3} \times 1 \\ = -\frac{8}{3}$$

$$(12) 1 \div (-4) \\ = -\frac{1}{4}$$

$$(22) 2 \times \left(-\frac{8}{5}\right) \\ = -\frac{16}{5}$$

$$(32) \frac{4}{7} \times \left(-\frac{2}{3}\right) \\ = -\frac{8}{21}$$

$$(3) \frac{1}{2} \div 2 \\ = \frac{1}{4}$$

$$(13) \frac{3}{8} \times \left(-\frac{8}{5}\right) \\ = -\frac{3}{5}$$

$$(23) \frac{1}{9} \times \frac{8}{7} \\ = \frac{8}{63}$$

$$(33) -1 \div \left(-\frac{10}{9}\right) \\ = \frac{9}{10}$$

$$(4) -9 \times \left(-\frac{6}{7}\right) \\ = \frac{54}{7}$$

$$(14) 2 \times \frac{9}{4} \\ = \frac{9}{2}$$

$$(24) -\frac{8}{9} \div \left(-\frac{7}{2}\right) \\ = \frac{16}{63}$$

$$(34) -\frac{2}{5} \div \left(-\frac{3}{4}\right) \\ = \frac{8}{15}$$

$$(5) 3 \times \frac{1}{8} \\ = \frac{3}{8}$$

$$(15) \frac{3}{5} \div \frac{4}{4} \\ = \frac{4}{15}$$

$$(25) \frac{7}{8} \div \left(-\frac{4}{7}\right) \\ = -\frac{49}{32}$$

$$(35) 1 \times \left(-\frac{4}{3}\right) \\ = -\frac{4}{3}$$

$$(6) \frac{7}{3} \div \frac{5}{7} \\ = \frac{49}{15}$$

$$(16) -\frac{3}{5} \div 4 \\ = -\frac{3}{20}$$

$$(26) -\frac{3}{5} \div 1 \\ = -\frac{3}{5}$$

$$(36) -\frac{8}{7} \times 2 \\ = -\frac{16}{7}$$

$$(7) -\frac{2}{3} \div \frac{7}{5} \\ = -\frac{10}{21}$$

$$(17) -\frac{10}{7} \times \frac{2}{3} \\ = -\frac{20}{21}$$

$$(27) \frac{4}{5} \div \frac{7}{6} \\ = \frac{24}{35}$$

$$(37) \frac{1}{3} \div \left(-\frac{7}{4}\right) \\ = -\frac{4}{21}$$

$$(8) -1 \times \frac{1}{2} \\ = -\frac{1}{2}$$

$$(18) \frac{1}{2} \div (-1) \\ = -\frac{1}{2}$$

$$(28) -\frac{1}{6} \div \frac{9}{10} \\ = -\frac{5}{27}$$

$$(38) \frac{4}{9} \times \left(-\frac{7}{10}\right) \\ = -\frac{14}{45}$$

$$(9) \frac{2}{5} \div \frac{5}{8} \\ = \frac{16}{25}$$

$$(19) 10 \times \left(-\frac{9}{8}\right) \\ = -\frac{45}{4}$$

$$(29) \frac{3}{5} \div \left(-\frac{8}{7}\right) \\ = -\frac{21}{40}$$

$$(39) -\frac{4}{7} \times 5 \\ = -\frac{20}{7}$$

$$(10) -\frac{3}{4} \times \frac{1}{9} \\ = -\frac{1}{12}$$

$$(20) -\frac{1}{3} \times \left(-\frac{7}{10}\right) \\ = \frac{7}{30}$$

$$(30) -\frac{2}{3} \div 4 \\ = -\frac{1}{6}$$

$$(40) \frac{10}{7} \div \frac{7}{3} \\ = \frac{30}{49}$$

$$(1) -\frac{3}{2} \times 2 \\ = -3$$

$$(11) \frac{8}{5} \times 3 \\ = \frac{24}{5}$$

$$(21) \frac{4}{5} \times \left(-\frac{7}{9}\right) \\ = -\frac{28}{45}$$

$$(31) 10 \div 10 \\ = 1$$

$$(2) -\frac{4}{3} \div \left(-\frac{1}{2}\right) \\ = \frac{8}{3}$$

$$(12) \frac{1}{3} \times \left(-\frac{9}{4}\right) \\ = -\frac{3}{4}$$

$$(22) -\frac{3}{10} \times (-5) \\ = \frac{3}{2}$$

$$(32) -\frac{3}{10} \div \frac{5}{2} \\ = -\frac{3}{25}$$

$$(3) \frac{8}{9} \times \left(-\frac{8}{5}\right) \\ = -\frac{64}{45}$$

$$(13) \frac{4}{7} \div \frac{4}{3} \\ = \frac{3}{7}$$

$$(23) -1 \div \frac{3}{8} \\ = -\frac{8}{3}$$

$$(33) \frac{5}{9} \times \left(-\frac{3}{2}\right) \\ = -\frac{5}{6}$$

$$(4) -\frac{5}{7} \div (-6) \\ = \frac{5}{42}$$

$$(14) \frac{1}{5} \times (-4) \\ = -\frac{4}{5}$$

$$(24) -\frac{1}{3} \times \left(-\frac{8}{3}\right) \\ = \frac{8}{9}$$

$$(34) -\frac{5}{7} \times \frac{9}{7} \\ = -\frac{45}{49}$$

$$(5) -2 \div \frac{3}{2} \\ = -\frac{4}{3}$$

$$(15) -4 \div \left(-\frac{3}{4}\right) \\ = \frac{16}{3}$$

$$(25) -6 \times \frac{1}{5} \\ = -\frac{6}{5}$$

$$(35) -\frac{1}{5} \div 5 \\ = -\frac{1}{25}$$

$$(6) \frac{3}{2} \div \frac{2}{3} \\ = \frac{9}{4}$$

$$(16) -1 \div \frac{7}{10} \\ = -\frac{10}{7}$$

$$(26) \frac{4}{3} \times \frac{5}{3} \\ = \frac{20}{9}$$

$$(36) -2 \div \left(-\frac{1}{4}\right) \\ = 8$$

$$(7) -\frac{7}{10} \div (-8) \\ = \frac{7}{80}$$

$$(17) -\frac{6}{5} \div \left(-\frac{9}{10}\right) \\ = \frac{4}{3}$$

$$(27) 10 \times \frac{4}{3} \\ = \frac{40}{3}$$

$$(37) -2 \div \left(-\frac{8}{3}\right) \\ = \frac{3}{4}$$

$$(8) -\frac{1}{7} \times \frac{7}{9} \\ = -\frac{1}{9}$$

$$(18) \frac{2}{5} \div 1 \\ = \frac{2}{5}$$

$$(28) \frac{4}{9} \times \frac{8}{3} \\ = \frac{32}{27}$$

$$(38) -\frac{1}{6} \div \left(-\frac{1}{5}\right) \\ = \frac{5}{6}$$

$$(9) -\frac{10}{7} \times \left(-\frac{1}{4}\right) \\ = \frac{5}{14}$$

$$(19) -\frac{7}{3} \times \left(-\frac{5}{7}\right) \\ = \frac{5}{3}$$

$$(29) -1 \times \left(-\frac{1}{10}\right) \\ = \frac{1}{10}$$

$$(39) -\frac{4}{5} \times \frac{5}{3} \\ = -\frac{4}{3}$$

$$(10) \frac{5}{3} \times \frac{7}{8} \\ = \frac{35}{24}$$

$$(20) -\frac{1}{2} \times \frac{9}{4} \\ = -\frac{9}{8}$$

$$(30) -\frac{6}{7} \div 8 \\ = -\frac{3}{28}$$

$$(40) -\frac{3}{7} \times \frac{7}{9} \\ = -\frac{1}{3}$$

$$(1) -2 \div \frac{3}{2}$$

$$= -\frac{4}{3}$$

$$(11) 1 \div \left(-\frac{5}{3}\right)$$

$$= -\frac{3}{5}$$

$$(21) 1 \times \frac{9}{8}$$

$$= \frac{9}{8}$$

$$(31) \frac{1}{7} \div 1$$

$$= \frac{1}{7}$$

$$(2) \frac{3}{5} \div \frac{1}{7}$$

$$= \frac{21}{5}$$

$$(12) -\frac{3}{10} \div \left(-\frac{5}{2}\right)$$

$$= \frac{3}{25}$$

$$(22) \frac{4}{5} \times \frac{5}{4}$$

$$= 1$$

$$(32) 1 \div \frac{4}{3}$$

$$= \frac{3}{4}$$

$$(3) -\frac{3}{5} \times (-1)$$

$$= \frac{3}{5}$$

$$(13) 7 \div \left(-\frac{3}{2}\right)$$

$$= -\frac{14}{3}$$

$$(23) \frac{3}{2} \div \left(-\frac{5}{4}\right)$$

$$= -\frac{6}{5}$$

$$(33) -\frac{9}{2} \times \frac{3}{8}$$

$$= -\frac{27}{16}$$

$$(4) 1 \div \left(-\frac{8}{3}\right)$$

$$= -\frac{3}{8}$$

$$(14) -1 \times \left(-\frac{7}{5}\right)$$

$$= \frac{7}{5}$$

$$(24) 4 \times \frac{5}{3}$$

$$= \frac{20}{3}$$

$$(34) -\frac{9}{5} \times (-2)$$

$$= \frac{18}{5}$$

$$(5) \frac{1}{4} \div \frac{1}{4}$$

$$= 1$$

$$(15) -8 \div 1$$

$$= -8$$

$$(25) 2 \times 4$$

$$= 8$$

$$(35) \frac{4}{9} \times \frac{1}{4}$$

$$= \frac{1}{9}$$

$$(6) -4 \times \left(-\frac{3}{4}\right)$$

$$= 3$$

$$(16) 3 \div \frac{7}{3}$$

$$= \frac{9}{7}$$

$$(26) -\frac{2}{5} \times \left(-\frac{2}{5}\right)$$

$$= \frac{4}{25}$$

$$(36) \frac{3}{10} \div \frac{3}{5}$$

$$= \frac{1}{2}$$

$$(7) \frac{2}{7} \div \left(-\frac{3}{7}\right)$$

$$= -\frac{2}{3}$$

$$(17) \frac{3}{5} \times \left(-\frac{7}{10}\right)$$

$$= -\frac{21}{50}$$

$$(27) -\frac{4}{7} \div \left(-\frac{5}{3}\right)$$

$$= \frac{12}{35}$$

$$(37) -\frac{9}{10} \times 1$$

$$= -\frac{9}{10}$$

$$(8) -\frac{6}{5} \times \frac{3}{4}$$

$$= -\frac{9}{10}$$

$$(18) -\frac{7}{9} \times \frac{9}{5}$$

$$= -\frac{7}{5}$$

$$(28) \frac{5}{3} \div \left(-\frac{2}{3}\right)$$

$$= -\frac{5}{2}$$

$$(38) \frac{10}{7} \div \frac{1}{6}$$

$$= \frac{60}{7}$$

$$(9) \frac{1}{9} \times \frac{9}{5}$$

$$= \frac{1}{5}$$

$$(19) 1 \times \frac{3}{2}$$

$$= \frac{3}{2}$$

$$(29) 1 \div \frac{9}{10}$$

$$= \frac{10}{9}$$

$$(39) \frac{1}{3} \div \frac{1}{4}$$

$$= \frac{4}{3}$$

$$(10) -\frac{1}{2} \div \left(-\frac{3}{2}\right)$$

$$= \frac{1}{3}$$

$$(20) -\frac{1}{3} \times (-4)$$

$$= \frac{4}{3}$$

$$(30) 2 \div \frac{4}{9}$$

$$= \frac{9}{2}$$

$$(40) \frac{3}{2} \times \left(-\frac{1}{5}\right)$$

$$= -\frac{3}{10}$$

$$(1) -\frac{2}{3} \times (-2) \\ = \frac{4}{3}$$

$$(11) -\frac{7}{9} \times \left(-\frac{4}{7}\right) \\ = \frac{4}{9}$$

$$(21) -\frac{2}{9} \times \left(-\frac{3}{5}\right) \\ = \frac{2}{15}$$

$$(31) \frac{9}{10} \times \left(-\frac{1}{2}\right) \\ = -\frac{9}{20}$$

$$(2) -\frac{5}{9} \div \left(-\frac{9}{5}\right) \\ = \frac{25}{81}$$

$$(12) -\frac{1}{2} \times \frac{1}{10} \\ = -\frac{1}{20}$$

$$(22) -\frac{7}{6} \div \left(-\frac{9}{4}\right) \\ = \frac{14}{27}$$

$$(32) \frac{10}{3} \times \frac{9}{7} \\ = \frac{30}{7}$$

$$(3) \frac{7}{4} \times \left(-\frac{5}{4}\right) \\ = -\frac{35}{16}$$

$$(13) \frac{1}{7} \div \frac{10}{3} \\ = \frac{3}{70}$$

$$(23) -\frac{8}{9} \times \left(-\frac{1}{5}\right) \\ = \frac{8}{45}$$

$$(33) -\frac{9}{4} \div \frac{8}{5} \\ = -\frac{45}{32}$$

$$(4) 2 \div \frac{5}{8} \\ = \frac{16}{5}$$

$$(14) 5 \times \left(-\frac{5}{4}\right) \\ = -\frac{25}{4}$$

$$(24) \frac{9}{8} \times \left(-\frac{4}{5}\right) \\ = -\frac{9}{10}$$

$$(34) \frac{5}{3} \div \frac{1}{9} \\ = 15$$

$$(5) -\frac{1}{2} \div (-4) \\ = \frac{1}{8}$$

$$(15) \frac{8}{3} \times \frac{3}{2} \\ = 4$$

$$(25) 4 \div (-1) \\ = -4$$

$$(35) 1 \times \left(-\frac{4}{7}\right) \\ = -\frac{4}{7}$$

$$(6) \frac{1}{2} \div 4 \\ = \frac{1}{8}$$

$$(16) 1 \div \frac{4}{3} \\ = \frac{3}{4}$$

$$(26) -\frac{2}{3} \times 3 \\ = -2$$

$$(36) \frac{3}{8} \div \left(-\frac{7}{3}\right) \\ = -\frac{9}{56}$$

$$(7) \frac{7}{5} \times \frac{3}{8} \\ = \frac{21}{40}$$

$$(17) -1 \times \left(-\frac{4}{5}\right) \\ = \frac{4}{5}$$

$$(27) -1 \div \left(-\frac{1}{9}\right) \\ = 9$$

$$(37) -\frac{3}{5} \div (-7) \\ = \frac{3}{35}$$

$$(8) \frac{4}{3} \times \left(-\frac{4}{5}\right) \\ = -\frac{16}{15}$$

$$(18) -\frac{3}{7} \times \left(-\frac{5}{4}\right) \\ = \frac{15}{28}$$

$$(28) \frac{3}{10} \div \left(-\frac{5}{9}\right) \\ = -\frac{27}{50}$$

$$(38) -\frac{7}{2} \times \left(-\frac{9}{4}\right) \\ = \frac{63}{8}$$

$$(9) -\frac{3}{2} \div \left(-\frac{8}{7}\right) \\ = \frac{21}{16}$$

$$(19) -\frac{1}{6} \div \frac{2}{3} \\ = -\frac{1}{4}$$

$$(29) 3 \div \frac{10}{3} \\ = \frac{9}{10}$$

$$(39) 3 \div (-1) \\ = -3$$

$$(10) \frac{1}{2} \times \left(-\frac{3}{5}\right) \\ = -\frac{3}{10}$$

$$(20) -\frac{1}{3} \div \left(-\frac{1}{2}\right) \\ = \frac{2}{3}$$

$$(30) -\frac{9}{5} \div \left(-\frac{2}{3}\right) \\ = \frac{27}{10}$$

$$(40) -9 \div \frac{1}{10} \\ = -90$$

$$(1) \frac{4}{5} \div (-2) \\ = -\frac{2}{5}$$

$$(11) -\frac{9}{7} \div \left(-\frac{7}{8}\right) \\ = \frac{72}{49}$$

$$(21) -\frac{5}{8} \times \left(-\frac{7}{4}\right) \\ = \frac{35}{32}$$

$$(31) \frac{1}{2} \times \frac{9}{2} \\ = \frac{9}{4}$$

$$(2) \frac{7}{9} \div \frac{5}{4} \\ = \frac{28}{45}$$

$$(12) -\frac{5}{8} \times \frac{3}{2} \\ = -\frac{15}{16}$$

$$(22) -9 \times \left(-\frac{5}{9}\right) \\ = 5$$

$$(32) \frac{1}{4} \times \left(-\frac{5}{3}\right) \\ = -\frac{5}{12}$$

$$(3) -\frac{10}{3} \div \frac{2}{9} \\ = -15$$

$$(13) 5 \div \frac{3}{2} \\ = \frac{10}{3}$$

$$(23) -5 \times \left(-\frac{9}{8}\right) \\ = \frac{45}{8}$$

$$(33) -\frac{9}{8} \div 1 \\ = -\frac{9}{8}$$

$$(4) -\frac{4}{7} \div \left(-\frac{2}{7}\right) \\ = 2$$

$$(14) \frac{5}{3} \div \left(-\frac{3}{4}\right) \\ = -\frac{20}{9}$$

$$(24) -\frac{6}{5} \times \left(-\frac{9}{2}\right) \\ = \frac{27}{5}$$

$$(34) \frac{2}{5} \times \frac{8}{9} \\ = \frac{16}{45}$$

$$(5) \frac{7}{10} \div 1 \\ = \frac{7}{10}$$

$$(15) -\frac{7}{6} \div \frac{2}{5} \\ = -\frac{35}{12}$$

$$(25) \frac{3}{7} \div \frac{6}{7} \\ = \frac{1}{2}$$

$$(35) \frac{3}{7} \div \left(-\frac{1}{3}\right) \\ = -\frac{9}{7}$$

$$(6) -\frac{1}{10} \div \left(-\frac{10}{9}\right) \\ = \frac{9}{100}$$

$$(16) \frac{5}{7} \div \frac{3}{5} \\ = \frac{25}{21}$$

$$(26) -\frac{3}{5} \div \left(-\frac{6}{7}\right) \\ = \frac{7}{10}$$

$$(36) \frac{4}{5} \times \frac{1}{2} \\ = \frac{2}{5}$$

$$(7) \frac{5}{2} \div \frac{2}{9} \\ = \frac{45}{4}$$

$$(17) \frac{3}{10} \div \left(-\frac{6}{5}\right) \\ = -\frac{1}{4}$$

$$(27) -1 \div \frac{7}{8} \\ = -\frac{8}{7}$$

$$(37) -2 \div \frac{3}{2} \\ = -\frac{4}{3}$$

$$(8) \frac{10}{3} \times \left(-\frac{1}{3}\right) \\ = -\frac{10}{9}$$

$$(18) \frac{8}{3} \times \left(-\frac{10}{9}\right) \\ = -\frac{80}{27}$$

$$(28) 1 \div \left(-\frac{6}{5}\right) \\ = -\frac{5}{6}$$

$$(38) \frac{2}{5} \div \frac{1}{6} \\ = \frac{12}{5}$$

$$(9) -\frac{9}{8} \div \left(-\frac{2}{3}\right) \\ = \frac{27}{16}$$

$$(19) \frac{1}{2} \times \left(-\frac{3}{2}\right) \\ = -\frac{3}{4}$$

$$(29) \frac{2}{3} \times \frac{1}{8} \\ = \frac{1}{12}$$

$$(39) -1 \times \frac{4}{3} \\ = -\frac{4}{3}$$

$$(10) -2 \div 2 \\ = -1$$

$$(20) -\frac{1}{7} \div \left(-\frac{2}{9}\right) \\ = \frac{9}{14}$$

$$(30) \frac{5}{2} \times \frac{2}{3} \\ = \frac{5}{3}$$

$$(40) -\frac{1}{2} \div 1 \\ = -\frac{1}{2}$$

$$(1) \frac{3}{5} \times \left(-\frac{3}{10}\right) \\ = -\frac{9}{50}$$

$$(11) \frac{3}{5} \times \left(-\frac{4}{7}\right) \\ = -\frac{12}{35}$$

$$(21) \frac{5}{8} \times \left(-\frac{1}{3}\right) \\ = -\frac{5}{24}$$

$$(31) 10 \times \left(-\frac{2}{5}\right) \\ = -4$$

$$(2) \frac{3}{4} \div \left(-\frac{5}{8}\right) \\ = -\frac{6}{5}$$

$$(12) \frac{10}{7} \times 10 \\ = \frac{100}{7}$$

$$(22) \frac{3}{8} \div \left(-\frac{3}{4}\right) \\ = -\frac{1}{2}$$

$$(32) -\frac{4}{9} \times \frac{7}{6} \\ = -\frac{14}{27}$$

$$(3) -\frac{9}{2} \times (-4) \\ = 18$$

$$(13) \frac{5}{3} \div \left(-\frac{9}{8}\right) \\ = -\frac{40}{27}$$

$$(23) -\frac{10}{3} \div \left(-\frac{9}{10}\right) \\ = \frac{100}{27}$$

$$(33) -\frac{9}{8} \div \frac{3}{2} \\ = -\frac{3}{4}$$

$$(4) 1 \times \left(-\frac{8}{5}\right) \\ = -\frac{8}{5}$$

$$(14) \frac{4}{9} \div \left(-\frac{4}{3}\right) \\ = -\frac{1}{3}$$

$$(24) \frac{3}{2} \times \frac{4}{5} \\ = \frac{6}{5}$$

$$(34) \frac{3}{5} \div \left(-\frac{1}{2}\right) \\ = -\frac{6}{5}$$

$$(5) -3 \times (-2) \\ = 6$$

$$(15) \frac{5}{9} \times \frac{5}{3} \\ = \frac{25}{27}$$

$$(25) \frac{7}{6} \div \frac{5}{2} \\ = \frac{7}{15}$$

$$(35) -1 \times \frac{7}{6} \\ = -\frac{7}{6}$$

$$(6) -\frac{10}{7} \times \frac{9}{10} \\ = -\frac{9}{7}$$

$$(16) -\frac{5}{9} \times \frac{1}{7} \\ = -\frac{5}{63}$$

$$(26) \frac{1}{3} \div \frac{1}{5} \\ = \frac{5}{3}$$

$$(36) \frac{2}{7} \div \left(-\frac{5}{6}\right) \\ = -\frac{12}{35}$$

$$(7) -\frac{8}{3} \div 1 \\ = -\frac{8}{3}$$

$$(17) \frac{4}{3} \times \frac{1}{2} \\ = \frac{2}{3}$$

$$(27) \frac{3}{5} \times \frac{1}{3} \\ = \frac{1}{5}$$

$$(37) \frac{1}{2} \div \frac{6}{7} \\ = \frac{7}{12}$$

$$(8) -4 \times 4 \\ = -16$$

$$(18) 1 \div \left(-\frac{7}{6}\right) \\ = -\frac{6}{7}$$

$$(28) -\frac{5}{2} \times (-1) \\ = \frac{5}{2}$$

$$(38) -\frac{9}{7} \div 1 \\ = -\frac{9}{7}$$

$$(9) -\frac{1}{3} \div 1 \\ = -\frac{1}{3}$$

$$(19) \frac{2}{5} \div 1 \\ = \frac{2}{5}$$

$$(29) -6 \times \left(-\frac{5}{2}\right) \\ = 15$$

$$(39) -\frac{6}{7} \div \frac{3}{4} \\ = -\frac{8}{7}$$

$$(10) -\frac{1}{3} \times (-6) \\ = 2$$

$$(20) -\frac{8}{9} \div \left(-\frac{5}{4}\right) \\ = \frac{32}{45}$$

$$(30) -\frac{7}{3} \times (-4) \\ = \frac{28}{3}$$

$$(40) \frac{3}{5} \div \frac{5}{8} \\ = \frac{24}{25}$$



$$(1) 1 \times \left(-\frac{4}{9}\right) \\ = -\frac{4}{9}$$

$$(11) -\frac{3}{5} \div (-3) \\ = \frac{1}{5}$$

$$(21) 1 \times \frac{7}{4} \\ = \frac{7}{4}$$

$$(31) 2 \times \frac{2}{3} \\ = \frac{4}{3}$$

$$(2) -\frac{4}{7} \div \frac{3}{7} \\ = -\frac{4}{3}$$

$$(12) \frac{9}{8} \div 3 \\ = \frac{3}{8}$$

$$(22) -\frac{4}{5} \times \frac{2}{5} \\ = -\frac{8}{25}$$

$$(32) -9 \times \frac{5}{6} \\ = -\frac{15}{2}$$

$$(3) -2 \times \frac{10}{7} \\ = -\frac{20}{7}$$

$$(13) \frac{7}{3} \div \left(-\frac{9}{4}\right) \\ = -\frac{28}{27}$$

$$(23) -1 \times \left(-\frac{1}{2}\right) \\ = \frac{1}{2}$$

$$(33) -\frac{2}{7} \times \frac{5}{9} \\ = -\frac{10}{63}$$

$$(4) 1 \times (-5) \\ = -5$$

$$(14) -1 \times \left(-\frac{1}{2}\right) \\ = \frac{1}{2}$$

$$(24) -\frac{7}{8} \times (-3) \\ = \frac{21}{8}$$

$$(34) -\frac{3}{2} \times \left(-\frac{5}{9}\right) \\ = \frac{5}{6}$$

$$(5) 1 \div \left(-\frac{9}{8}\right) \\ = -\frac{8}{9}$$

$$(15) 5 \div (-4) \\ = -\frac{5}{4}$$

$$(25) \frac{10}{7} \times \frac{5}{2} \\ = \frac{25}{7}$$

$$(35) -\frac{5}{8} \times \frac{7}{8} \\ = -\frac{35}{64}$$

$$(6) \frac{1}{5} \div \frac{4}{5} \\ = \frac{1}{4}$$

$$(16) \frac{7}{10} \div (-9) \\ = -\frac{7}{90}$$

$$(26) -\frac{4}{9} \times \left(-\frac{1}{4}\right) \\ = \frac{1}{9}$$

$$(36) 5 \times \frac{4}{5} \\ = 4$$

$$(7) -\frac{9}{8} \times \frac{4}{7} \\ = -\frac{9}{14}$$

$$(17) \frac{2}{7} \div \frac{4}{9} \\ = \frac{9}{14}$$

$$(27) -\frac{10}{9} \times (-1) \\ = \frac{10}{9}$$

$$(37) \frac{5}{2} \div \frac{3}{2} \\ = \frac{5}{3}$$

$$(8) -\frac{1}{4} \div \left(-\frac{3}{2}\right) \\ = \frac{1}{6}$$

$$(18) \frac{7}{9} \times 5 \\ = \frac{35}{9}$$

$$(28) -\frac{8}{5} \times \frac{1}{3} \\ = -\frac{8}{15}$$

$$(38) \frac{5}{2} \times \left(-\frac{3}{4}\right) \\ = -\frac{15}{8}$$

$$(9) -\frac{3}{7} \div \left(-\frac{2}{9}\right) \\ = \frac{27}{14}$$

$$(19) \frac{1}{7} \times \frac{8}{3} \\ = \frac{8}{21}$$

$$(29) \frac{8}{7} \times \frac{1}{4} \\ = \frac{2}{7}$$

$$(39) \frac{3}{5} \div \left(-\frac{1}{6}\right) \\ = -\frac{18}{5}$$

$$(10) -\frac{5}{4} \div \frac{9}{7} \\ = -\frac{35}{36}$$

$$(20) \frac{1}{3} \div \left(-\frac{5}{8}\right) \\ = -\frac{8}{15}$$

$$(30) -\frac{5}{3} \times \left(-\frac{5}{3}\right) \\ = \frac{25}{9}$$

$$(40) 1 \div \left(-\frac{4}{5}\right) \\ = -\frac{5}{4}$$

$$(1) \frac{4}{9} \times \left(-\frac{1}{8}\right) \\ = -\frac{1}{18}$$

$$(11) -1 \div \left(-\frac{5}{4}\right) \\ = \frac{4}{5}$$

$$(21) \frac{3}{2} \times \left(-\frac{1}{2}\right) \\ = -\frac{3}{4}$$

$$(31) 7 \times \left(-\frac{1}{2}\right) \\ = -\frac{7}{2}$$

$$(2) 1 \times \left(-\frac{7}{3}\right) \\ = -\frac{7}{3}$$

$$(12) 1 \times \frac{3}{5} \\ = \frac{3}{5}$$

$$(22) -1 \div \frac{2}{5} \\ = -\frac{5}{2}$$

$$(32) \frac{7}{8} \div \frac{8}{5} \\ = \frac{35}{64}$$

$$(3) 9 \div \frac{8}{3} \\ = \frac{27}{8}$$

$$(13) \frac{7}{2} \div \left(-\frac{2}{5}\right) \\ = -\frac{35}{4}$$

$$(23) -\frac{1}{3} \times 4 \\ = -\frac{4}{3}$$

$$(33) -2 \div \left(-\frac{9}{8}\right) \\ = \frac{16}{9}$$

$$(4) -\frac{7}{5} \div 3 \\ = -\frac{7}{15}$$

$$(14) -\frac{4}{5} \times 1 \\ = -\frac{4}{5}$$

$$(24) -\frac{7}{3} \div (-8) \\ = \frac{7}{24}$$

$$(34) -\frac{3}{5} \times \left(-\frac{8}{9}\right) \\ = \frac{8}{15}$$

$$(5) -\frac{5}{8} \div \frac{10}{3} \\ = -\frac{15}{16}$$

$$(15) \frac{9}{8} \times \left(-\frac{9}{4}\right) \\ = -\frac{81}{32}$$

$$(25) -2 \times \left(-\frac{1}{2}\right) \\ = 1$$

$$(35) \frac{5}{4} \times \frac{1}{10} \\ = \frac{1}{8}$$

$$(6) -\frac{5}{4} \times \frac{8}{9} \\ = -\frac{10}{9}$$

$$(16) -6 \div \left(-\frac{5}{2}\right) \\ = \frac{12}{5}$$

$$(26) 1 \div \left(-\frac{7}{4}\right) \\ = -\frac{4}{7}$$

$$(36) -\frac{1}{2} \times \left(-\frac{1}{9}\right) \\ = \frac{1}{18}$$

$$(7) 1 \div 2 \\ = \frac{1}{2}$$

$$(17) \frac{3}{10} \times \left(-\frac{7}{5}\right) \\ = -\frac{21}{50}$$

$$(27) 1 \times \left(-\frac{1}{2}\right) \\ = -\frac{1}{2}$$

$$(37) \frac{3}{2} \div \frac{3}{8} \\ = 4$$

$$(8) \frac{10}{3} \div (-6) \\ = -\frac{5}{9}$$

$$(18) -\frac{9}{5} \div (-3) \\ = \frac{3}{5}$$

$$(28) -\frac{5}{9} \div \frac{4}{7} \\ = -\frac{35}{36}$$

$$(38) \frac{2}{3} \times \left(-\frac{5}{4}\right) \\ = -\frac{5}{6}$$

$$(9) -5 \div \left(-\frac{1}{9}\right) \\ = 45$$

$$(19) -\frac{8}{5} \div (-2) \\ = \frac{4}{5}$$

$$(29) \frac{1}{6} \times \left(-\frac{7}{10}\right) \\ = -\frac{7}{60}$$

$$(39) -\frac{4}{3} \times \left(-\frac{1}{5}\right) \\ = \frac{4}{15}$$

$$(10) -3 \times \frac{3}{7} \\ = -\frac{9}{7}$$

$$(20) \frac{3}{4} \times \left(-\frac{7}{5}\right) \\ = -\frac{21}{20}$$

$$(30) \frac{1}{5} \div \frac{3}{2} \\ = \frac{2}{15}$$

$$(40) -1 \div \frac{1}{5} \\ = -5$$

$$(1) -\frac{10}{7} \times \frac{3}{5} \\ = -\frac{6}{7}$$

$$(11) 8 \times 2 \\ = 16$$

$$(21) -1 \times \left(-\frac{5}{3}\right) \\ = \frac{5}{3}$$

$$(31) -\frac{1}{4} \times \left(-\frac{9}{2}\right) \\ = \frac{9}{8}$$

$$(2) \frac{8}{3} \times \left(-\frac{3}{2}\right) \\ = -4$$

$$(12) -\frac{2}{3} \times \left(-\frac{1}{7}\right) \\ = \frac{2}{21}$$

$$(22) -\frac{10}{7} \times \left(-\frac{10}{7}\right) \\ = \frac{100}{49}$$

$$(32) \frac{4}{5} \times \frac{7}{6} \\ = \frac{14}{15}$$

$$(3) -\frac{5}{3} \div (-1) \\ = \frac{5}{3}$$

$$(13) -\frac{2}{3} \div \left(-\frac{7}{10}\right) \\ = \frac{20}{21}$$

$$(23) -\frac{5}{9} \times \frac{2}{3} \\ = -\frac{10}{27}$$

$$(33) 1 \div \left(-\frac{3}{4}\right) \\ = -\frac{4}{3}$$

$$(4) 3 \div \left(-\frac{5}{2}\right) \\ = -\frac{6}{5}$$

$$(14) -\frac{7}{2} \times \frac{7}{2} \\ = -\frac{49}{4}$$

$$(24) -\frac{9}{4} \div 1 \\ = -\frac{9}{4}$$

$$(34) 3 \div \left(-\frac{3}{5}\right) \\ = -5$$

$$(5) 2 \times \left(-\frac{9}{5}\right) \\ = -\frac{18}{5}$$

$$(15) \frac{5}{2} \times (-3) \\ = -\frac{15}{2}$$

$$(25) \frac{2}{9} \times \frac{1}{9} \\ = \frac{2}{81}$$

$$(35) 1 \times \left(-\frac{8}{9}\right) \\ = -\frac{8}{9}$$

$$(6) \frac{3}{8} \times \frac{10}{9} \\ = \frac{5}{12}$$

$$(16) \frac{1}{7} \div (-1) \\ = -\frac{1}{7}$$

$$(26) \frac{1}{4} \times (-2) \\ = -\frac{1}{2}$$

$$(36) -\frac{10}{3} \div (-1) \\ = \frac{10}{3}$$

$$(7) 1 \div \frac{2}{3} \\ = \frac{3}{2}$$

$$(17) -\frac{2}{9} \times (-2) \\ = \frac{4}{9}$$

$$(27) \frac{8}{7} \div \frac{7}{6} \\ = \frac{48}{49}$$

$$(37) \frac{9}{4} \div \left(-\frac{9}{10}\right) \\ = -\frac{5}{2}$$

$$(8) -1 \div \frac{3}{5} \\ = -\frac{5}{3}$$

$$(18) \frac{1}{9} \times (-1) \\ = -\frac{1}{9}$$

$$(28) \frac{5}{4} \div (-1) \\ = -\frac{5}{4}$$

$$(38) -\frac{5}{2} \times \frac{5}{2} \\ = -\frac{25}{4}$$

$$(9) -3 \div \frac{3}{2} \\ = -2$$

$$(19) \frac{1}{7} \times \left(-\frac{3}{4}\right) \\ = -\frac{3}{28}$$

$$(29) 8 \div (-3) \\ = -\frac{8}{3}$$

$$(39) 4 \times \frac{1}{2} \\ = 2$$

$$(10) -\frac{4}{5} \times \frac{3}{4} \\ = -\frac{3}{5}$$

$$(20) \frac{5}{2} \div \frac{6}{7} \\ = \frac{35}{12}$$

$$(30) 1 \times \left(-\frac{2}{7}\right) \\ = -\frac{2}{7}$$

$$(40) \frac{7}{10} \times \frac{6}{5} \\ = \frac{21}{25}$$

$$(1) -\frac{9}{7} \div \frac{1}{9}$$

$$= -\frac{81}{7}$$

$$(11) -3 \div \frac{2}{9}$$

$$= -\frac{27}{2}$$

$$(21) \frac{5}{3} \div \frac{1}{2}$$

$$= \frac{10}{3}$$

$$(31) -\frac{2}{9} \times \frac{7}{9}$$

$$= -\frac{14}{81}$$

$$(2) \frac{8}{9} \times 2$$

$$= \frac{16}{9}$$

$$(12) -\frac{3}{2} \div \frac{3}{10}$$

$$= -5$$

$$(22) \frac{1}{8} \div \left(-\frac{5}{2}\right)$$

$$= -\frac{1}{20}$$

$$(32) \frac{3}{5} \div \frac{10}{9}$$

$$= \frac{27}{50}$$

$$(3) \frac{3}{2} \div \left(-\frac{7}{2}\right)$$

$$= -\frac{3}{7}$$

$$(13) \frac{1}{3} \times \left(-\frac{2}{3}\right)$$

$$= -\frac{2}{9}$$

$$(23) -\frac{5}{3} \div \frac{10}{9}$$

$$= -\frac{3}{2}$$

$$(33) 1 \div \left(-\frac{1}{2}\right)$$

$$= -2$$

$$(4) \frac{1}{2} \times \left(-\frac{2}{9}\right)$$

$$= -\frac{1}{9}$$

$$(14) \frac{5}{9} \div \left(-\frac{4}{3}\right)$$

$$= -\frac{5}{12}$$

$$(24) -\frac{5}{3} \times \frac{7}{4}$$

$$= -\frac{35}{12}$$

$$(34) -\frac{1}{6} \div 3$$

$$= -\frac{1}{18}$$

$$(5) -9 \times (-1)$$

$$= 9$$

$$(15) -\frac{2}{5} \times (-3)$$

$$= \frac{6}{5}$$

$$(25) -1 \times \frac{1}{4}$$

$$= -\frac{1}{4}$$

$$(35) -\frac{3}{5} \div \left(-\frac{1}{10}\right)$$

$$= 6$$

$$(6) \frac{9}{10} \div \frac{2}{3}$$

$$= \frac{27}{20}$$

$$(16) -\frac{4}{5} \times (-3)$$

$$= \frac{12}{5}$$

$$(26) \frac{3}{4} \div \frac{9}{5}$$

$$= \frac{5}{12}$$

$$(36) -\frac{1}{5} \times \left(-\frac{10}{9}\right)$$

$$= \frac{2}{9}$$

$$(7) \frac{1}{7} \div 2$$

$$= \frac{1}{14}$$

$$(17) -\frac{8}{3} \div \frac{1}{9}$$

$$= -24$$

$$(27) -\frac{1}{9} \times 9$$

$$= -1$$

$$(37) \frac{4}{5} \times \left(-\frac{2}{7}\right)$$

$$= -\frac{8}{35}$$

$$(8) -\frac{4}{5} \times 1$$

$$= -\frac{4}{5}$$

$$(18) -\frac{2}{2} \div (-3)$$

$$= \frac{3}{3}$$

$$(28) \frac{1}{8} \times \frac{2}{9}$$

$$= \frac{1}{36}$$

$$(38) \frac{1}{5} \times \left(-\frac{8}{9}\right)$$

$$= -\frac{8}{45}$$

$$(9) \frac{3}{2} \times \frac{7}{9}$$

$$= \frac{7}{6}$$

$$(19) -\frac{5}{4} \div \left(-\frac{2}{3}\right)$$

$$= \frac{15}{8}$$

$$(29) \frac{7}{2} \div \frac{1}{5}$$

$$= \frac{35}{2}$$

$$(39) -\frac{7}{9} \times \left(-\frac{1}{2}\right)$$

$$= \frac{7}{18}$$

$$(10) -\frac{1}{2} \div \left(-\frac{1}{9}\right)$$

$$= \frac{9}{2}$$

$$(20) \frac{1}{2} \div (-1)$$

$$= -\frac{1}{2}$$

$$(30) \frac{2}{5} \div (-4)$$

$$= -\frac{1}{10}$$

$$(40) -\frac{9}{4} \div \left(-\frac{2}{5}\right)$$

$$= \frac{45}{8}$$

$$(1) 1 \times \frac{5}{2} \\ = \frac{5}{2}$$

$$(2) 10 \div \left(-\frac{10}{3}\right) \\ = -3$$

$$(3) \frac{3}{2} \div (-3) \\ = -\frac{1}{2}$$

$$(4) \frac{3}{4} \times \left(-\frac{1}{3}\right) \\ = -\frac{1}{4}$$

$$(5) -\frac{3}{5} \div \frac{7}{5} \\ = -\frac{3}{7}$$

$$(6) -6 \div \frac{2}{3} \\ = -9$$

$$(7) -\frac{1}{5} \times \frac{8}{7} \\ = -\frac{8}{35}$$

$$(8) -\frac{2}{7} \times \left(-\frac{1}{3}\right) \\ = \frac{2}{21}$$

$$(9) -\frac{2}{3} \times \left(-\frac{7}{10}\right) \\ = \frac{7}{15}$$

$$(10) -\frac{6}{5} \times 1 \\ = -\frac{6}{5}$$

$$(11) -2 \div \frac{2}{3} \\ = -3$$

$$(12) -6 \times \left(-\frac{1}{2}\right) \\ = 3$$

$$(13) -\frac{7}{5} \times \left(-\frac{9}{5}\right) \\ = \frac{63}{25}$$

$$(14) 1 \times 5 \\ = 5$$

$$(15) -2 \div \left(-\frac{1}{5}\right) \\ = 10$$

$$(16) -\frac{1}{2} \times (-3) \\ = \frac{3}{2}$$

$$(17) -2 \div 5 \\ = -\frac{2}{5}$$

$$(18) -\frac{9}{10} \div 2 \\ = -\frac{9}{20}$$

$$(19) -\frac{5}{4} \div \frac{2}{3} \\ = -\frac{15}{8}$$

$$(20) -1 \times \left(-\frac{6}{7}\right) \\ = \frac{6}{7}$$

$$(21) \frac{10}{7} \div (-1) \\ = -\frac{10}{7}$$

$$(22) \frac{5}{7} \times \frac{5}{8} \\ = \frac{25}{56}$$

$$(23) \frac{1}{3} \times \left(-\frac{6}{5}\right) \\ = -\frac{2}{5}$$

$$(24) 4 \div \frac{2}{5} \\ = 10$$

$$(25) -\frac{4}{7} \div (-6) \\ = \frac{2}{21}$$

$$(26) -\frac{5}{3} \times 2 \\ = -\frac{10}{3}$$

$$(27) -\frac{2}{7} \times \frac{3}{8} \\ = -\frac{3}{28}$$

$$(28) \frac{8}{3} \times (-1) \\ = -\frac{8}{3}$$

$$(29) -\frac{3}{4} \times (-1) \\ = \frac{3}{4}$$

$$(30) 4 \div \frac{4}{7} \\ = 7$$

$$(31) -\frac{1}{2} \div \left(-\frac{7}{10}\right) \\ = \frac{5}{7}$$

$$(32) -\frac{2}{3} \times \frac{1}{3} \\ = -\frac{2}{9}$$

$$(33) -\frac{10}{9} \div \frac{3}{10} \\ = -\frac{100}{27}$$

$$(34) \frac{4}{9} \times \left(-\frac{1}{4}\right) \\ = -\frac{1}{9}$$

$$(35) -\frac{5}{2} \div \left(-\frac{3}{5}\right) \\ = \frac{25}{6}$$

$$(36) \frac{1}{4} \div \left(-\frac{1}{2}\right) \\ = -\frac{1}{2}$$

$$(37) 1 \times \frac{10}{7} \\ = \frac{10}{7}$$

$$(38) \frac{1}{4} \times \left(-\frac{3}{5}\right) \\ = -\frac{3}{20}$$

$$(39) -\frac{5}{3} \div \frac{4}{5} \\ = -\frac{25}{12}$$

$$(40) 1 \times \left(-\frac{7}{2}\right) \\ = -\frac{7}{2}$$

$$(1) -\frac{7}{5} \div (-1) \\ = \frac{7}{5}$$

$$(11) -\frac{3}{2} \times 1 \\ = -\frac{3}{2}$$

$$(21) -\frac{1}{4} \div \left(-\frac{2}{5}\right) \\ = \frac{5}{8}$$

$$(31) \frac{1}{8} \div \frac{3}{8} \\ = \frac{1}{3}$$

$$(2) -\frac{1}{7} \times \frac{9}{5} \\ = -\frac{9}{35}$$

$$(12) -\frac{2}{7} \times \left(-\frac{4}{9}\right) \\ = \frac{8}{63}$$

$$(22) \frac{1}{5} \div 1 \\ = \frac{1}{5}$$

$$(32) -\frac{8}{3} \div \frac{4}{7} \\ = -\frac{14}{3}$$

$$(3) \frac{5}{7} \div (-5) \\ = -\frac{1}{7}$$

$$(13) -\frac{1}{10} \times \left(-\frac{5}{4}\right) \\ = \frac{1}{8}$$

$$(23) \frac{3}{2} \times \frac{5}{4} \\ = \frac{15}{8}$$

$$(33) \frac{10}{7} \div \frac{3}{2} \\ = \frac{20}{21}$$

$$(4) -\frac{2}{7} \times \frac{2}{3} \\ = -\frac{4}{21}$$

$$(14) \frac{8}{7} \times \left(-\frac{3}{5}\right) \\ = -\frac{24}{35}$$

$$(24) \frac{8}{3} \times 1 \\ = \frac{8}{3}$$

$$(34) 1 \times (-1) \\ = -1$$

$$(5) -\frac{5}{7} \times 5 \\ = -\frac{25}{7}$$

$$(15) 1 \times \left(-\frac{7}{4}\right) \\ = -\frac{7}{4}$$

$$(25) -\frac{4}{9} \div \frac{5}{7} \\ = -\frac{28}{45}$$

$$(35) -\frac{8}{5} \times (-3) \\ = \frac{24}{5}$$

$$(6) -\frac{4}{3} \div \frac{4}{5} \\ = -\frac{5}{3}$$

$$(16) \frac{5}{4} \times \left(-\frac{9}{7}\right) \\ = -\frac{45}{28}$$

$$(26) \frac{3}{2} \times \frac{5}{7} \\ = \frac{15}{14}$$

$$(36) -1 \div (-7) \\ = \frac{1}{7}$$

$$(7) 2 \div \frac{7}{3} \\ = \frac{6}{7}$$

$$(17) -\frac{7}{6} \times \left(-\frac{3}{2}\right) \\ = \frac{7}{4}$$

$$(27) \frac{2}{3} \div (-1) \\ = -\frac{2}{3}$$

$$(37) \frac{5}{3} \times (-1) \\ = -\frac{5}{3}$$

$$(8) -\frac{6}{5} \times \frac{10}{9} \\ = -\frac{4}{3}$$

$$(18) \frac{1}{5} \times \left(-\frac{5}{7}\right) \\ = -\frac{1}{7}$$

$$(28) -\frac{2}{5} \times \left(-\frac{2}{5}\right) \\ = \frac{4}{25}$$

$$(38) -\frac{1}{10} \div \left(-\frac{8}{5}\right) \\ = \frac{1}{16}$$

$$(9) \frac{5}{2} \times \frac{4}{5} \\ = 2$$

$$(19) \frac{3}{2} \div \frac{7}{4} \\ = \frac{6}{7}$$

$$(29) \frac{3}{5} \times 3 \\ = \frac{9}{5}$$

$$(39) \frac{1}{4} \times (-2) \\ = -\frac{1}{2}$$

$$(10) \frac{3}{2} \times \frac{1}{3} \\ = \frac{1}{2}$$

$$(20) \frac{4}{3} \div (-2) \\ = -\frac{2}{3}$$

$$(30) \frac{7}{2} \div \frac{3}{5} \\ = \frac{35}{6}$$

$$(40) 1 \times \left(-\frac{5}{8}\right) \\ = -\frac{5}{8}$$

## 1.4 実数

### 1.4.1 分母の有理化

分母の有理化を行う問題。

$$(1) \frac{3\sqrt{2}+3}{-4\sqrt{2}-2}$$

$$(6) \frac{-4\sqrt{5}+2}{-2\sqrt{5}-3}$$

$$(11) \frac{-2-3\sqrt{3}}{-1+4\sqrt{3}}$$

$$(2) \frac{-5\sqrt{2}+\sqrt{5}}{\sqrt{2}+4\sqrt{5}}$$

$$(7) \frac{-\sqrt{3}+1}{3\sqrt{3}}$$

$$(12) \frac{-\sqrt{2}-2}{-2\sqrt{2}+1}$$

$$(3) \frac{-1+3\sqrt{5}}{-4-\sqrt{5}}$$

$$(8) \frac{2+2\sqrt{2}}{-3+2\sqrt{2}}$$

$$(13) \frac{-3\sqrt{3}-6}{-5\sqrt{3}+4}$$

$$(4) \frac{2\sqrt{5}+1}{-2\sqrt{5}+5}$$

$$(9) \frac{-2\sqrt{3}-\sqrt{5}}{-5\sqrt{3}+3\sqrt{5}}$$

$$(14) \frac{3\sqrt{2}-\sqrt{5}}{5\sqrt{2}-3\sqrt{5}}$$

$$(5) \frac{5\sqrt{5}-3}{3\sqrt{5}+1}$$

$$(10) \frac{2\sqrt{2}-1}{-2\sqrt{2}+5}$$

$$(15) \frac{-5+4\sqrt{3}}{5+5\sqrt{3}}$$



$$(1) \frac{2\sqrt{3}+5}{-2\sqrt{3}+1}$$

$$(6) \frac{-\sqrt{2}+3}{-4\sqrt{2}}$$

$$(11) \frac{-\sqrt{3}+\sqrt{5}}{2\sqrt{3}-\sqrt{5}}$$

$$(2) \frac{5}{4+\sqrt{5}}$$

$$(7) \frac{-4-\sqrt{3}}{3+2\sqrt{3}}$$

$$(12) \frac{-2+4\sqrt{3}}{-10+3\sqrt{3}}$$

$$(3) \frac{5\sqrt{5}+4\sqrt{2}}{-2\sqrt{5}-\sqrt{2}}$$

$$(8) \frac{-4\sqrt{5}-5\sqrt{2}}{-2\sqrt{5}-2\sqrt{2}}$$

$$(13) \frac{-3\sqrt{3}-2\sqrt{5}}{3\sqrt{3}+5\sqrt{5}}$$

$$(4) \frac{-5\sqrt{5}}{3\sqrt{5}-\sqrt{2}}$$

$$(9) \frac{3\sqrt{2}+10}{2\sqrt{2}+8}$$

$$(14) \frac{5\sqrt{5}+2}{4\sqrt{5}+10}$$

$$(5) \frac{3\sqrt{3}+10}{5\sqrt{3}+2}$$

$$(10) \frac{-3}{-1+4\sqrt{5}}$$

$$(15) \frac{-5\sqrt{5}}{2\sqrt{2}-4\sqrt{5}}$$

$$(1) \frac{8 - \sqrt{3}}{8 - 4\sqrt{3}}$$

$$(6) \frac{\sqrt{5} - 10}{2\sqrt{5}}$$

$$(11) \frac{\sqrt{3} - 6}{-5\sqrt{3}}$$

$$(2) \frac{-4 - 3\sqrt{3}}{4 - 5\sqrt{3}}$$

$$(7) \frac{\sqrt{3} - 2}{\sqrt{3} + 8}$$

$$(12) \frac{-4\sqrt{2} - 3}{-4\sqrt{2} - 4}$$

$$(3) \frac{2 + 3\sqrt{5}}{-10 + 2\sqrt{5}}$$

$$(8) \frac{-3\sqrt{3} + 3}{-5\sqrt{3}}$$

$$(13) \frac{5 + 3\sqrt{5}}{1 + 5\sqrt{5}}$$

$$(4) \frac{-5\sqrt{2} + 4\sqrt{5}}{-4\sqrt{5}}$$

$$(9) \frac{3\sqrt{3}}{5\sqrt{5} - 2\sqrt{3}}$$

$$(14) \frac{8 + 2\sqrt{5}}{-10 + \sqrt{5}}$$

$$(5) \frac{-5\sqrt{5} - 5}{5\sqrt{5} - 4}$$

$$(10) \frac{5\sqrt{2} - 4\sqrt{5}}{-5\sqrt{2} + 5\sqrt{5}}$$

$$(15) \frac{-2\sqrt{3} - 5}{-3\sqrt{3} + 5}$$

$$(1) \frac{-\sqrt{3}}{5\sqrt{3}-4}$$

$$(6) \frac{4-\sqrt{5}}{6+4\sqrt{5}}$$

$$(11) \frac{4+5\sqrt{2}}{-4-\sqrt{2}}$$

$$(2) \frac{-5\sqrt{5}-2\sqrt{2}}{5\sqrt{5}+4\sqrt{2}}$$

$$(7) \frac{3+\sqrt{2}}{-1+\sqrt{2}}$$

$$(12) \frac{-4\sqrt{3}+5\sqrt{2}}{-2\sqrt{3}+5\sqrt{2}}$$

$$(3) \frac{3\sqrt{5}+10}{-2\sqrt{5}-10}$$

$$(8) \frac{8}{-3\sqrt{3}+4}$$

$$(13) \frac{-4\sqrt{3}+4}{\sqrt{3}+4}$$

$$(4) \frac{2\sqrt{3}+4}{\sqrt{3}+2}$$

$$(9) \frac{-\sqrt{2}-2}{\sqrt{2}+4}$$

$$(14) \frac{4-3\sqrt{3}}{-10-3\sqrt{3}}$$

$$(5) \frac{5\sqrt{2}+3\sqrt{3}}{\sqrt{3}}$$

$$(10) \frac{-6+2\sqrt{5}}{2-\sqrt{5}}$$

$$(15) \frac{1}{-4\sqrt{3}+4}$$

$$(1) \frac{5 - 3\sqrt{2}}{\sqrt{2}}$$

$$(6) \frac{-2\sqrt{3} + 2}{-3\sqrt{3} + 2}$$

$$(11) \frac{-4\sqrt{2} + 2}{-5\sqrt{2} + 10}$$

$$(2) \frac{-\sqrt{5} - \sqrt{2}}{\sqrt{2}}$$

$$(7) \frac{-\sqrt{3} + 2\sqrt{2}}{2\sqrt{3} - \sqrt{2}}$$

$$(12) \frac{-2\sqrt{3} - 1}{-2\sqrt{3} - 2}$$

$$(3) \frac{-10 - 5\sqrt{2}}{2 + \sqrt{2}}$$

$$(8) \frac{-2}{2 + 5\sqrt{2}}$$

$$(13) \frac{-5\sqrt{2} - 3}{-\sqrt{2}}$$

$$(4) \frac{4\sqrt{3} - 5\sqrt{5}}{-\sqrt{3} - 3\sqrt{5}}$$

$$(9) \frac{-3\sqrt{3} + \sqrt{2}}{-5\sqrt{3} - 4\sqrt{2}}$$

$$(14) \frac{-\sqrt{3} - 10}{-2\sqrt{3} - 6}$$

$$(5) \frac{2\sqrt{3} - 5}{3\sqrt{3}}$$

$$(10) \frac{10 - 5\sqrt{3}}{2 - \sqrt{3}}$$

$$(15) \frac{-2 + \sqrt{5}}{8 - \sqrt{5}}$$

$$(1) \frac{-4\sqrt{3} + 3\sqrt{5}}{2\sqrt{3} + 4\sqrt{5}}$$

$$(6) \frac{\sqrt{2} - 3}{2\sqrt{2} - 4}$$

$$(11) \frac{-3\sqrt{2} + \sqrt{5}}{-4\sqrt{2} - 2\sqrt{5}}$$

$$(2) \frac{-5\sqrt{3} - 6}{-3\sqrt{3}}$$

$$(7) \frac{3\sqrt{3} - 6}{\sqrt{3} - 8}$$

$$(12) \frac{2}{10 + 5\sqrt{3}}$$

$$(3) \frac{-5 + 2\sqrt{3}}{3 + 5\sqrt{3}}$$

$$(8) \frac{4}{-8 + 3\sqrt{3}}$$

$$(13) \frac{\sqrt{2} - 1}{3\sqrt{2} + 4}$$

$$(4) \frac{3\sqrt{3} + 6}{3\sqrt{3} + 2}$$

$$(9) \frac{1 - \sqrt{2}}{3 - 4\sqrt{2}}$$

$$(14) \frac{5 + 2\sqrt{3}}{-1 + 2\sqrt{3}}$$

$$(5) \frac{2\sqrt{3} - 6}{-3\sqrt{3} + 2}$$

$$(10) \frac{-4\sqrt{2} - 2\sqrt{5}}{-3\sqrt{2}}$$

$$(15) \frac{\sqrt{2} - 1}{4\sqrt{2} - 4}$$

$$(1) \frac{4\sqrt{2} + \sqrt{5}}{\sqrt{2} - \sqrt{5}}$$

$$(6) \frac{2\sqrt{5}}{2\sqrt{5} - 5}$$

$$(11) \frac{-3\sqrt{5} - 1}{\sqrt{5} - 1}$$

$$(2) \frac{\sqrt{3}}{2\sqrt{2} - \sqrt{3}}$$

$$(7) \frac{2 + 5\sqrt{2}}{10 - 4\sqrt{2}}$$

$$(12) \frac{-2\sqrt{3} - 5\sqrt{5}}{3\sqrt{5}}$$

$$(3) \frac{4\sqrt{2} + 5\sqrt{5}}{3\sqrt{2} + 4\sqrt{5}}$$

$$(8) \frac{-\sqrt{5} - 2}{3\sqrt{5} - 4}$$

$$(13) \frac{2\sqrt{2} + 5\sqrt{3}}{-3\sqrt{2} + 4\sqrt{3}}$$

$$(4) \frac{5\sqrt{2} + 2\sqrt{5}}{-5\sqrt{2} + 2\sqrt{5}}$$

$$(9) \frac{5\sqrt{2} + 2\sqrt{3}}{-\sqrt{2} - 4\sqrt{3}}$$

$$(14) \frac{-10 + 4\sqrt{2}}{-3\sqrt{2}}$$

$$(5) \frac{2\sqrt{3}}{4\sqrt{3} + 5\sqrt{5}}$$

$$(10) \frac{-2 - \sqrt{2}}{-8 - 5\sqrt{2}}$$

$$(15) \frac{-\sqrt{2} - 4\sqrt{5}}{-3\sqrt{2} - 5\sqrt{5}}$$

$$(1) \frac{-\sqrt{5}-5\sqrt{2}}{4\sqrt{5}}$$

$$(6) \frac{-\sqrt{5}+2}{-2\sqrt{5}-4}$$

$$(11) \frac{-4\sqrt{3}-2}{-3\sqrt{3}-4}$$

$$(2) \frac{-1-\sqrt{2}}{-2-\sqrt{2}}$$

$$(7) \frac{-4\sqrt{5}-5}{-3\sqrt{5}}$$

$$(12) \frac{-1-3\sqrt{5}}{2-2\sqrt{5}}$$

$$(3) \frac{4-\sqrt{2}}{3+\sqrt{2}}$$

$$(8) \frac{6+4\sqrt{3}}{6+\sqrt{3}}$$

$$(13) \frac{2\sqrt{3}}{-\sqrt{3}+5\sqrt{2}}$$

$$(4) \frac{-2\sqrt{5}}{5+2\sqrt{5}}$$

$$(9) \frac{-\sqrt{2}+5\sqrt{3}}{-4\sqrt{2}-2\sqrt{3}}$$

$$(14) \frac{2\sqrt{2}+3\sqrt{3}}{-5\sqrt{2}+3\sqrt{3}}$$

$$(5) \frac{8-\sqrt{2}}{2+5\sqrt{2}}$$

$$(10) \frac{-4\sqrt{5}+3\sqrt{3}}{2\sqrt{3}}$$

$$(15) \frac{-2\sqrt{5}-4\sqrt{2}}{-3\sqrt{2}}$$

$$(1) \frac{-3\sqrt{2} + 5\sqrt{5}}{4\sqrt{5}}$$

$$(6) \frac{-1 - 5\sqrt{5}}{-2 + 3\sqrt{5}}$$

$$(11) \frac{-5\sqrt{3} - 3\sqrt{5}}{-3\sqrt{3} - 2\sqrt{5}}$$

$$(2) \frac{-3\sqrt{3}}{\sqrt{3} - 5\sqrt{5}}$$

$$(7) \frac{1}{2\sqrt{5} + 1}$$

$$(12) \frac{-6 - \sqrt{5}}{-6 - \sqrt{5}}$$

$$(3) \frac{-4\sqrt{5}}{-5\sqrt{5} - \sqrt{2}}$$

$$(8) \frac{2\sqrt{2} + 6}{5\sqrt{2} - 4}$$

$$(13) \frac{-4 + 4\sqrt{2}}{-1 - \sqrt{2}}$$

$$(4) \frac{3\sqrt{2} - 10}{-4\sqrt{2} - 6}$$

$$(9) \frac{2\sqrt{3} - \sqrt{5}}{-3\sqrt{3} - 5\sqrt{5}}$$

$$(14) \frac{3\sqrt{2} + 3\sqrt{5}}{3\sqrt{2} + 2\sqrt{5}}$$

$$(5) \frac{5\sqrt{2}}{4\sqrt{2} + 6}$$

$$(10) \frac{-4\sqrt{5}}{\sqrt{5} + \sqrt{3}}$$

$$(15) \frac{\sqrt{3} + 5}{2\sqrt{3} + 4}$$



$$(1) \frac{8 - 2\sqrt{5}}{8 + 5\sqrt{5}}$$

$$(6) \frac{4\sqrt{5} - 2\sqrt{2}}{-2\sqrt{5} - 5\sqrt{2}}$$

$$(11) \frac{2\sqrt{2} - 1}{-2\sqrt{2} - 2}$$

$$(2) \frac{2 + 2\sqrt{2}}{2 - 5\sqrt{2}}$$

$$(7) \frac{-4\sqrt{5} - 2\sqrt{3}}{\sqrt{5} - 5\sqrt{3}}$$

$$(12) \frac{-1 + \sqrt{5}}{-2 - 4\sqrt{5}}$$

$$(3) \frac{5\sqrt{5}}{5\sqrt{5} - 3}$$

$$(8) \frac{-4}{-\sqrt{3} - 2}$$

$$(13) \frac{\sqrt{5} + 8}{-4\sqrt{5}}$$

$$(4) \frac{-\sqrt{2} + 3}{2\sqrt{2} - 2}$$

$$(9) \frac{-4\sqrt{3} + 10}{-\sqrt{3} + 4}$$

$$(14) \frac{-4 - 5\sqrt{5}}{1 - 3\sqrt{5}}$$

$$(5) \frac{-1}{2\sqrt{3} - 4}$$

$$(10) \frac{3\sqrt{2}}{-5\sqrt{3} + 4\sqrt{2}}$$

$$(15) \frac{\sqrt{2} - 1}{\sqrt{2} - 3}$$

$$(1) \frac{5 - 2\sqrt{3}}{3 - \sqrt{3}}$$

$$(6) \frac{5\sqrt{5} - 5\sqrt{2}}{-\sqrt{5} - \sqrt{2}}$$

$$(11) \frac{4\sqrt{2} - 4\sqrt{5}}{-3\sqrt{2} + 3\sqrt{5}}$$

$$(2) \frac{-5}{-2\sqrt{5} + 4}$$

$$(7) \frac{3\sqrt{2} + 2\sqrt{3}}{-2\sqrt{2}}$$

$$(12) \frac{-3\sqrt{3} + 6}{2\sqrt{3} - 10}$$

$$(3) \frac{-4}{2\sqrt{5} - 1}$$

$$(8) \frac{\sqrt{5}}{-\sqrt{5} - \sqrt{3}}$$

$$(13) \frac{2\sqrt{2} + 3\sqrt{3}}{-4\sqrt{2} + \sqrt{3}}$$

$$(4) \frac{\sqrt{3} + 4}{-2\sqrt{3} + 2}$$

$$(9) \frac{-6 - \sqrt{3}}{-8 - 3\sqrt{3}}$$

$$(14) \frac{-3\sqrt{2} - 4}{-4\sqrt{2} - 2}$$

$$(5) \frac{\sqrt{5} + 2}{4\sqrt{5} - 4}$$

$$(10) \frac{1 - \sqrt{5}}{3 - 2\sqrt{5}}$$

$$(15) \frac{4\sqrt{2}}{\sqrt{3} + 3\sqrt{2}}$$

$$(1) \frac{\sqrt{5} - \sqrt{2}}{-\sqrt{2}}$$

$$(6) \frac{-\sqrt{3} + 2\sqrt{5}}{2\sqrt{3} + 2\sqrt{5}}$$

$$(11) \frac{\sqrt{3} + 4\sqrt{5}}{-4\sqrt{3} - \sqrt{5}}$$

$$(2) \frac{-\sqrt{3} + 5\sqrt{2}}{\sqrt{3} + 2\sqrt{2}}$$

$$(7) \frac{-2\sqrt{5} - 3\sqrt{2}}{\sqrt{5} - \sqrt{2}}$$

$$(12) \frac{\sqrt{5}}{2\sqrt{5} + 4}$$

$$(3) \frac{6 + 5\sqrt{5}}{10 + 3\sqrt{5}}$$

$$(8) \frac{-5\sqrt{2} + 8}{-\sqrt{2} + 2}$$

$$(13) \frac{2\sqrt{5} - 3\sqrt{2}}{-2\sqrt{5} + 2\sqrt{2}}$$

$$(4) \frac{-2 + 4\sqrt{3}}{8 - \sqrt{3}}$$

$$(9) \frac{-4\sqrt{5}}{-5\sqrt{5} - 3\sqrt{2}}$$

$$(14) \frac{-\sqrt{2} - 2}{-\sqrt{2} - 5}$$

$$(5) \frac{-3\sqrt{3}}{\sqrt{3} + 3\sqrt{5}}$$

$$(10) \frac{-2\sqrt{5}}{3\sqrt{3} - 3\sqrt{5}}$$

$$(15) \frac{5\sqrt{2} + 2\sqrt{3}}{5\sqrt{2} + \sqrt{3}}$$

$$(1) \frac{3 - \sqrt{2}}{4 + 3\sqrt{2}}$$

$$(6) \frac{\sqrt{5}}{\sqrt{5}}$$

$$(11) \frac{-4\sqrt{5} + 2\sqrt{3}}{3\sqrt{5} + 3\sqrt{3}}$$

$$(2) \frac{-\sqrt{3}}{-\sqrt{3} - \sqrt{5}}$$

$$(7) \frac{-2\sqrt{3}}{5\sqrt{3} + 5}$$

$$(12) \frac{4\sqrt{3} + 3\sqrt{5}}{-4\sqrt{5}}$$

$$(3) \frac{-2 - \sqrt{5}}{4 - 4\sqrt{5}}$$

$$(8) \frac{3}{3 + \sqrt{2}}$$

$$(13) \frac{\sqrt{5}}{2\sqrt{3} - \sqrt{5}}$$

$$(4) \frac{-3\sqrt{2} + 6}{4\sqrt{2} - 8}$$

$$(9) \frac{4\sqrt{2} + 2\sqrt{5}}{\sqrt{2} - 5\sqrt{5}}$$

$$(14) \frac{-3\sqrt{2} + \sqrt{3}}{4\sqrt{2} + 3\sqrt{3}}$$

$$(5) \frac{-4 + 5\sqrt{3}}{5 + 3\sqrt{3}}$$

$$(10) \frac{-6 - 5\sqrt{5}}{4 + 5\sqrt{5}}$$

$$(15) \frac{3}{\sqrt{3} + 1}$$

$$(1) \frac{-3\sqrt{3} - 4\sqrt{2}}{3\sqrt{3} - 2\sqrt{2}}$$

$$(6) \frac{6 + 3\sqrt{5}}{-2 + 4\sqrt{5}}$$

$$(11) \frac{3\sqrt{2} - 3\sqrt{3}}{-3\sqrt{2} + \sqrt{3}}$$

$$(2) \frac{5 - \sqrt{5}}{-4 + 4\sqrt{5}}$$

$$(7) \frac{-5\sqrt{5} - 5}{-3\sqrt{5} - 3}$$

$$(12) \frac{-3\sqrt{3} - 5\sqrt{2}}{-3\sqrt{3} - \sqrt{2}}$$

$$(3) \frac{6 + 2\sqrt{2}}{-4 - 3\sqrt{2}}$$

$$(8) \frac{-5\sqrt{2}}{-5\sqrt{3} + 4\sqrt{2}}$$

$$(13) \frac{-4}{-10 - 3\sqrt{2}}$$

$$(4) \frac{-2 - 4\sqrt{5}}{-1 + 3\sqrt{5}}$$

$$(9) \frac{-2\sqrt{3} - 3\sqrt{2}}{2\sqrt{3} + \sqrt{2}}$$

$$(14) \frac{\sqrt{2} - 4}{-\sqrt{2} + 2}$$

$$(5) \frac{\sqrt{2} - 5\sqrt{3}}{\sqrt{2} - 2\sqrt{3}}$$

$$(10) \frac{2\sqrt{3} - 3\sqrt{2}}{3\sqrt{3} + 2\sqrt{2}}$$

$$(15) \frac{-1 - 4\sqrt{5}}{4 + 2\sqrt{5}}$$

$$(1) \frac{-2}{\sqrt{2}-2}$$

$$(6) \frac{-\sqrt{2}+4\sqrt{5}}{\sqrt{5}}$$

$$(11) \frac{4\sqrt{3}+1}{\sqrt{3}-3}$$

$$(2) \frac{-2-4\sqrt{2}}{-2+5\sqrt{2}}$$

$$(7) \frac{-5}{-5-4\sqrt{2}}$$

$$(12) \frac{3\sqrt{3}+\sqrt{5}}{4\sqrt{3}-\sqrt{5}}$$

$$(3) \frac{3+4\sqrt{5}}{2\sqrt{5}}$$

$$(8) \frac{2+\sqrt{3}}{5\sqrt{3}}$$

$$(13) \frac{-5\sqrt{2}-2\sqrt{3}}{2\sqrt{2}+3\sqrt{3}}$$

$$(4) \frac{5\sqrt{3}-4}{-3\sqrt{3}-1}$$

$$(9) \frac{-5\sqrt{2}}{10-2\sqrt{2}}$$

$$(14) \frac{\sqrt{2}+2\sqrt{3}}{-\sqrt{2}}$$

$$(5) \frac{-1+3\sqrt{5}}{-4\sqrt{5}}$$

$$(10) \frac{-4\sqrt{3}+5\sqrt{2}}{5\sqrt{3}-4\sqrt{2}}$$

$$(15) \frac{-2-3\sqrt{5}}{8-4\sqrt{5}}$$

$$(1) \frac{2}{1+4\sqrt{3}}$$

$$(6) \frac{-4+4\sqrt{2}}{4-\sqrt{2}}$$

$$(11) \frac{-\sqrt{3}}{4\sqrt{3}}$$

$$(2) \frac{3\sqrt{5}-4\sqrt{2}}{-2\sqrt{5}+5\sqrt{2}}$$

$$(7) \frac{2\sqrt{2}}{-2\sqrt{5}-\sqrt{2}}$$

$$(12) \frac{\sqrt{5}+2\sqrt{2}}{2\sqrt{2}}$$

$$(3) \frac{-4\sqrt{5}}{3\sqrt{2}-3\sqrt{5}}$$

$$(8) \frac{\sqrt{2}+8}{-3\sqrt{2}-4}$$

$$(13) \frac{-4\sqrt{2}-3}{-\sqrt{2}+2}$$

$$(4) \frac{\sqrt{2}+2}{-\sqrt{2}+3}$$

$$(9) \frac{-2-3\sqrt{2}}{3-4\sqrt{2}}$$

$$(14) \frac{5-5\sqrt{5}}{4+5\sqrt{5}}$$

$$(5) \frac{\sqrt{3}+8}{-3\sqrt{3}-4}$$

$$(10) \frac{\sqrt{5}+4}{3\sqrt{5}+5}$$

$$(15) \frac{3\sqrt{5}+6}{5\sqrt{5}}$$

$$(1) \frac{3\sqrt{3} + 5\sqrt{2}}{-3\sqrt{3} + \sqrt{2}}$$

$$(6) \frac{-5 + \sqrt{5}}{-3 - 2\sqrt{5}}$$

$$(11) \frac{-1 - \sqrt{2}}{-4 + \sqrt{2}}$$

$$(2) \frac{-2}{-\sqrt{2} + 2}$$

$$(7) \frac{3\sqrt{3} + \sqrt{5}}{5\sqrt{3} + 2\sqrt{5}}$$

$$(12) \frac{\sqrt{3}}{-6 - 2\sqrt{3}}$$

$$(3) \frac{1}{-4 - \sqrt{5}}$$

$$(8) \frac{-4\sqrt{3} - 3\sqrt{2}}{-3\sqrt{2}}$$

$$(13) \frac{3\sqrt{5} - \sqrt{3}}{-2\sqrt{5} + 5\sqrt{3}}$$

$$(4) \frac{-6 + \sqrt{5}}{-6 + 2\sqrt{5}}$$

$$(9) \frac{5 + \sqrt{5}}{2 + 5\sqrt{5}}$$

$$(14) \frac{-5\sqrt{2} - 8}{-2\sqrt{2} - 4}$$

$$(5) \frac{2\sqrt{5} + 4}{-5\sqrt{5} + 3}$$

$$(10) \frac{-3\sqrt{5} + 5\sqrt{2}}{2\sqrt{5} + 3\sqrt{2}}$$

$$(15) \frac{5\sqrt{5} - 4\sqrt{3}}{-3\sqrt{5} - 3\sqrt{3}}$$



$$(1) \frac{\sqrt{2}}{-5\sqrt{2} - 3\sqrt{5}}$$

$$(6) \frac{-2 + \sqrt{3}}{-4\sqrt{3}}$$

$$(11) \frac{\sqrt{5} + 10}{-\sqrt{5} - 6}$$

$$(2) \frac{-2 + 2\sqrt{5}}{-1 + \sqrt{5}}$$

$$(7) \frac{4\sqrt{3} + 3\sqrt{5}}{-\sqrt{3} - 5\sqrt{5}}$$

$$(12) \frac{3\sqrt{5} - 10}{-4\sqrt{5} - 6}$$

$$(3) \frac{3\sqrt{5} + 3\sqrt{2}}{\sqrt{5} + \sqrt{2}}$$

$$(8) \frac{5 - 5\sqrt{2}}{2 + 5\sqrt{2}}$$

$$(13) \frac{4\sqrt{5} + 5\sqrt{3}}{\sqrt{5} - \sqrt{3}}$$

$$(4) \frac{-3\sqrt{5} - 2}{5\sqrt{5} - 3}$$

$$(9) \frac{-4\sqrt{5} + 5\sqrt{2}}{-2\sqrt{5} - 5\sqrt{2}}$$

$$(14) \frac{5 + 3\sqrt{2}}{-3 - 2\sqrt{2}}$$

$$(5) \frac{\sqrt{3} + \sqrt{5}}{-4\sqrt{3} + 4\sqrt{5}}$$

$$(10) \frac{4\sqrt{5} + 2\sqrt{3}}{3\sqrt{5} - 3\sqrt{3}}$$

$$(15) \frac{\sqrt{3} + 2}{-2\sqrt{3}}$$

$$(1) \frac{2\sqrt{5} - 3\sqrt{3}}{-4\sqrt{5} - 3\sqrt{3}}$$

$$(6) \frac{4\sqrt{3} + \sqrt{5}}{4\sqrt{5}}$$

$$(11) \frac{2 - 5\sqrt{5}}{3 - \sqrt{5}}$$

$$(2) \frac{2\sqrt{2} + 2\sqrt{5}}{-\sqrt{2} + \sqrt{5}}$$

$$(7) \frac{5 + 3\sqrt{2}}{-3 - 3\sqrt{2}}$$

$$(12) \frac{\sqrt{3} + 2\sqrt{5}}{-2\sqrt{3} + 2\sqrt{5}}$$

$$(3) \frac{4\sqrt{3} - 3\sqrt{2}}{4\sqrt{3} - 3\sqrt{2}}$$

$$(8) \frac{-2\sqrt{5} + 2\sqrt{2}}{5\sqrt{5} - 3\sqrt{2}}$$

$$(13) \frac{-4 + 5\sqrt{3}}{-1 - 3\sqrt{3}}$$

$$(4) \frac{-3\sqrt{2} - 8}{4\sqrt{2} - 6}$$

$$(9) \frac{3\sqrt{2} + 8}{-4\sqrt{2}}$$

$$(14) \frac{5 - 4\sqrt{3}}{-4 + 4\sqrt{3}}$$

$$(5) \frac{2\sqrt{5} + 2}{3\sqrt{5} - 4}$$

$$(10) \frac{-2 + 4\sqrt{2}}{3 + 3\sqrt{2}}$$

$$(15) \frac{3 - 2\sqrt{5}}{3 - 3\sqrt{5}}$$

$$(1) \frac{3\sqrt{3}-2}{\sqrt{3}}$$

$$(6) \frac{\sqrt{2}}{1+\sqrt{2}}$$

$$(11) \frac{-5\sqrt{3}-1}{-3\sqrt{3}-4}$$

$$(2) \frac{3\sqrt{2}}{3\sqrt{2}+4\sqrt{3}}$$

$$(7) \frac{4+\sqrt{5}}{1+2\sqrt{5}}$$

$$(12) \frac{-10+\sqrt{3}}{-6+3\sqrt{3}}$$

$$(3) \frac{5\sqrt{5}-\sqrt{3}}{\sqrt{5}+2\sqrt{3}}$$

$$(8) \frac{-3}{5\sqrt{5}+3}$$

$$(13) \frac{-2-4\sqrt{3}}{-1+5\sqrt{3}}$$

$$(4) \frac{-2\sqrt{2}+2}{2\sqrt{2}-1}$$

$$(9) \frac{2\sqrt{3}-1}{4\sqrt{3}+2}$$

$$(14) \frac{3\sqrt{2}}{4-3\sqrt{2}}$$

$$(5) \frac{2}{-\sqrt{2}+2}$$

$$(10) \frac{-\sqrt{5}-2}{-3\sqrt{5}+10}$$

$$(15) \frac{-4\sqrt{5}}{5\sqrt{5}}$$

$$\begin{aligned}(1) \quad & \frac{3\sqrt{2}+3}{-4\sqrt{2}-2} \\ &= \frac{-9-3\sqrt{2}}{14}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{-4\sqrt{5}+2}{-2\sqrt{5}-3} \\ &= \frac{46-16\sqrt{5}}{11}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-2-3\sqrt{3}}{-1+4\sqrt{3}} \\ &= \frac{-11\sqrt{3}-38}{47}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{-5\sqrt{2}+\sqrt{5}}{\sqrt{2}+4\sqrt{5}} \\ &= \frac{-7\sqrt{10}+10}{26}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{-\sqrt{3}+1}{3\sqrt{3}} \\ &= \frac{\sqrt{3}-3}{9}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-\sqrt{2}-2}{-2\sqrt{2}+1} \\ &= \frac{6+5\sqrt{2}}{7}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{-1+3\sqrt{5}}{-4-\sqrt{5}} \\ &= \frac{-13\sqrt{5}+19}{11}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{2+2\sqrt{2}}{-3+2\sqrt{2}} \\ &= -10\sqrt{2}-14\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-3\sqrt{3}-6}{-5\sqrt{3}+4} \\ &= \frac{69+42\sqrt{3}}{59}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{2\sqrt{5}+1}{-2\sqrt{5}+5} \\ &= \frac{25+12\sqrt{5}}{5}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{-2\sqrt{3}-\sqrt{5}}{-5\sqrt{3}+3\sqrt{5}} \\ &= \frac{45+11\sqrt{15}}{30}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{3\sqrt{2}-\sqrt{5}}{5\sqrt{2}-3\sqrt{5}} \\ &= \frac{4\sqrt{10}+15}{5}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{5\sqrt{5}-3}{3\sqrt{5}+1} \\ &= \frac{-7\sqrt{5}+39}{22}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{2\sqrt{2}-1}{-2\sqrt{2}+5} \\ &= \frac{8\sqrt{2}+3}{17}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{-5+4\sqrt{3}}{5+5\sqrt{3}} \\ &= \frac{-9\sqrt{3}+17}{10}\end{aligned}$$

$$(1) \frac{2\sqrt{3}+5}{-2\sqrt{3}+1} \\ = \frac{-12\sqrt{3}-17}{11}$$

$$(6) \frac{-\sqrt{2}+3}{-4\sqrt{2}} \\ = \frac{-3\sqrt{2}+2}{8}$$

$$(11) \frac{-\sqrt{3}+\sqrt{5}}{2\sqrt{3}-\sqrt{5}} \\ = \frac{\sqrt{15}-1}{7}$$

$$(2) \frac{5}{4+\sqrt{5}} \\ = \frac{-5\sqrt{5}+20}{11}$$

$$(7) \frac{-4-\sqrt{3}}{3+2\sqrt{3}} \\ = \frac{6-5\sqrt{3}}{3}$$

$$(12) \frac{-2+4\sqrt{3}}{-10+3\sqrt{3}} \\ = \frac{-16-34\sqrt{3}}{73}$$

$$(3) \frac{5\sqrt{5}+4\sqrt{2}}{-2\sqrt{5}-\sqrt{2}} \\ = \frac{-14-\sqrt{10}}{6}$$

$$(8) \frac{-4\sqrt{5}-5\sqrt{2}}{-2\sqrt{5}-2\sqrt{2}} \\ = \frac{\sqrt{10}+10}{6}$$

$$(13) \frac{-3\sqrt{3}-2\sqrt{5}}{3\sqrt{3}+5\sqrt{5}} \\ = \frac{-23-9\sqrt{15}}{98}$$

$$(4) \frac{-5\sqrt{5}}{3\sqrt{5}-\sqrt{2}} \\ = \frac{-5\sqrt{10}-75}{43}$$

$$(9) \frac{3\sqrt{2}+10}{2\sqrt{2}+8} \\ = \frac{17+\sqrt{2}}{14}$$

$$(14) \frac{5\sqrt{5}+2}{4\sqrt{5}+10} \\ = \frac{21\sqrt{5}-40}{10}$$

$$(5) \frac{3\sqrt{3}+10}{5\sqrt{3}+2} \\ = \frac{44\sqrt{3}+25}{71}$$

$$(10) \frac{-3}{-1+4\sqrt{5}} \\ = \frac{-3-12\sqrt{5}}{79}$$

$$(15) \frac{-5\sqrt{5}}{2\sqrt{2}-4\sqrt{5}} \\ = \frac{5\sqrt{10}+50}{36}$$

$$(1) \frac{8 - \sqrt{3}}{8 - 4\sqrt{3}} \\ = \frac{13 + 6\sqrt{3}}{4}$$

$$(6) \frac{\sqrt{5} - 10}{2\sqrt{5}} \\ = \frac{1 - 2\sqrt{5}}{2}$$

$$(11) \frac{\sqrt{3} - 6}{-5\sqrt{3}} \\ = \frac{2\sqrt{3} - 1}{5}$$

$$(2) \frac{-4 - 3\sqrt{3}}{4 - 5\sqrt{3}} \\ = \frac{61 + 32\sqrt{3}}{59}$$

$$(7) \frac{\sqrt{3} - 2}{\sqrt{3} + 8} \\ = \frac{10\sqrt{3} - 19}{61}$$

$$(12) \frac{-4\sqrt{2} - 3}{-4\sqrt{2} - 4} \\ = \frac{-\sqrt{2} + 5}{4}$$

$$(3) \frac{2 + 3\sqrt{5}}{-10 + 2\sqrt{5}} \\ = \frac{-25 - 17\sqrt{5}}{40}$$

$$(8) \frac{-3\sqrt{3} + 3}{-5\sqrt{3}} \\ = \frac{3 - \sqrt{3}}{5}$$

$$(13) \frac{5 + 3\sqrt{5}}{1 + 5\sqrt{5}} \\ = \frac{11\sqrt{5} + 35}{62}$$

$$(4) \frac{-5\sqrt{2} + 4\sqrt{5}}{-4\sqrt{5}} \\ = \frac{-4 + \sqrt{10}}{4}$$

$$(9) \frac{3\sqrt{3}}{5\sqrt{5} - 2\sqrt{3}} \\ = \frac{18 + 15\sqrt{15}}{113}$$

$$(14) \frac{8 + 2\sqrt{5}}{-10 + \sqrt{5}} \\ = \frac{-90 - 28\sqrt{5}}{95}$$

$$(5) \frac{-5\sqrt{5} - 5}{5\sqrt{5} - 4} \\ = \frac{-145 - 45\sqrt{5}}{109}$$

$$(10) \frac{5\sqrt{2} - 4\sqrt{5}}{-5\sqrt{2} + 5\sqrt{5}} \\ = \frac{\sqrt{10} - 10}{15}$$

$$(15) \frac{-2\sqrt{3} - 5}{-3\sqrt{3} + 5} \\ = \frac{25\sqrt{3} + 43}{2}$$

$$\begin{aligned}(1) \quad & \frac{-\sqrt{3}}{5\sqrt{3}-4} \\ &= \frac{-15-4\sqrt{3}}{59}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{4-\sqrt{5}}{6+4\sqrt{5}} \\ &= \frac{-2+\sqrt{5}}{2}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{4+5\sqrt{2}}{-4-\sqrt{2}} \\ &= \frac{-3-8\sqrt{2}}{7}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{-5\sqrt{5}-2\sqrt{2}}{5\sqrt{5}+4\sqrt{2}} \\ &= \frac{-109+10\sqrt{10}}{93}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{3+\sqrt{2}}{-1+\sqrt{2}} \\ &= 4\sqrt{2}+5\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-4\sqrt{3}+5\sqrt{2}}{-2\sqrt{3}+5\sqrt{2}} \\ &= \frac{13-5\sqrt{6}}{19}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{3\sqrt{5}+10}{-2\sqrt{5}-10} \\ &= \frac{-\sqrt{5}-7}{8}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{8}{-3\sqrt{3}+4} \\ &= \frac{-24\sqrt{3}-32}{11}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-4\sqrt{3}+4}{\sqrt{3}+4} \\ &= \frac{-20\sqrt{3}+28}{13}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{2\sqrt{3}+4}{\sqrt{3}+2} \\ &= 2\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{-\sqrt{2}-2}{\sqrt{2}+4} \\ &= \frac{-\sqrt{2}-3}{7}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{4-3\sqrt{3}}{-10-3\sqrt{3}} \\ &= \frac{-67+42\sqrt{3}}{73}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{5\sqrt{2}+3\sqrt{3}}{\sqrt{3}} \\ &= \frac{9+5\sqrt{6}}{3}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{-6+2\sqrt{5}}{2-\sqrt{5}} \\ &= 2\sqrt{5}+2\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{1}{-4\sqrt{3}+4} \\ &= \frac{-1-\sqrt{3}}{8}\end{aligned}$$

$$\begin{aligned}(1) \quad & \frac{5-3\sqrt{2}}{\sqrt{2}} \\ &= \frac{-6+5\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{-2\sqrt{3}+2}{-3\sqrt{3}+2} \\ &= \frac{-2\sqrt{3}+14}{23}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-4\sqrt{2}+2}{-5\sqrt{2}+10} \\ &= \frac{-2-3\sqrt{2}}{5}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{-\sqrt{5}-\sqrt{2}}{\sqrt{2}} \\ &= \frac{-2-\sqrt{10}}{2}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{-\sqrt{3}+2\sqrt{2}}{2\sqrt{3}-\sqrt{2}} \\ &= \frac{3\sqrt{6}-2}{10}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-2\sqrt{3}-1}{-2\sqrt{3}-2} \\ &= \frac{5-\sqrt{3}}{4}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{-10-5\sqrt{2}}{2+\sqrt{2}} \\ &= -5\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{-2}{2+5\sqrt{2}} \\ &= \frac{2-5\sqrt{2}}{23}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-5\sqrt{2}-3}{-\sqrt{2}} \\ &= \frac{10+3\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{4\sqrt{3}-5\sqrt{5}}{-\sqrt{3}-3\sqrt{5}} \\ &= \frac{87-17\sqrt{15}}{42}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{-3\sqrt{3}+\sqrt{2}}{-5\sqrt{3}-4\sqrt{2}} \\ &= \frac{53-17\sqrt{6}}{43}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{-\sqrt{3}-10}{-2\sqrt{3}-6} \\ &= \frac{27-7\sqrt{3}}{12}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{2\sqrt{3}-5}{3\sqrt{3}} \\ &= \frac{6-5\sqrt{3}}{9}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{10-5\sqrt{3}}{2-\sqrt{3}} \\ &= 5\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{-2+\sqrt{5}}{8-\sqrt{5}} \\ &= \frac{-11+6\sqrt{5}}{59}\end{aligned}$$



$$\begin{aligned}(1) \quad & \frac{-4\sqrt{3} + 3\sqrt{5}}{2\sqrt{3} + 4\sqrt{5}} \\ &= \frac{42 - 11\sqrt{15}}{34}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{\sqrt{2} - 3}{2\sqrt{2} - 4} \\ &= \frac{4 + \sqrt{2}}{4}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-3\sqrt{2} + \sqrt{5}}{-4\sqrt{2} - 2\sqrt{5}} \\ &= \frac{17 - 5\sqrt{10}}{6}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{-5\sqrt{3} - 6}{-3\sqrt{3}} \\ &= \frac{5 + 2\sqrt{3}}{3}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{3\sqrt{3} - 6}{\sqrt{3} - 8} \\ &= \frac{39 - 18\sqrt{3}}{61}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{2}{10 + 5\sqrt{3}} \\ &= \frac{-2\sqrt{3} + 4}{5}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{-5 + 2\sqrt{3}}{3 + 5\sqrt{3}} \\ &= \frac{-31\sqrt{3} + 45}{66}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{4}{-8 + 3\sqrt{3}} \\ &= \frac{-32 - 12\sqrt{3}}{37}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{\sqrt{2} - 1}{3\sqrt{2} + 4} \\ &= \frac{10 - 7\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{3\sqrt{3} + 6}{3\sqrt{3} + 2} \\ &= \frac{15 + 12\sqrt{3}}{23}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{1 - \sqrt{2}}{3 - 4\sqrt{2}} \\ &= \frac{5 - \sqrt{2}}{23}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{5 + 2\sqrt{3}}{-1 + 2\sqrt{3}} \\ &= \frac{17 + 12\sqrt{3}}{11}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{2\sqrt{3} - 6}{-3\sqrt{3} + 2} \\ &= \frac{14\sqrt{3} - 6}{23}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{-4\sqrt{2} - 2\sqrt{5}}{-3\sqrt{2}} \\ &= \frac{4 + \sqrt{10}}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{\sqrt{2} - 1}{4\sqrt{2} - 4} \\ &= \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(1) \quad & \frac{4\sqrt{2} + \sqrt{5}}{\sqrt{2} - \sqrt{5}} \\ &= \frac{-13 - 5\sqrt{10}}{3}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{2\sqrt{5}}{2\sqrt{5} - 5} \\ &= -2\sqrt{5} - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-3\sqrt{5} - 1}{\sqrt{5} - 1} \\ &= -\sqrt{5} - 4\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{\sqrt{3}}{2\sqrt{2} - \sqrt{3}} \\ &= \frac{3 + 2\sqrt{6}}{5}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{2 + 5\sqrt{2}}{10 - 4\sqrt{2}} \\ &= \frac{30 + 29\sqrt{2}}{34}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-2\sqrt{3} - 5\sqrt{5}}{3\sqrt{5}} \\ &= \frac{-25 - 2\sqrt{15}}{15}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{4\sqrt{2} + 5\sqrt{5}}{3\sqrt{2} + 4\sqrt{5}} \\ &= \frac{76 + \sqrt{10}}{62}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{-\sqrt{5} - 2}{3\sqrt{5} - 4} \\ &= \frac{-23 - 10\sqrt{5}}{29}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{2\sqrt{2} + 5\sqrt{3}}{-3\sqrt{2} + 4\sqrt{3}} \\ &= \frac{72 + 23\sqrt{6}}{30}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{5\sqrt{2} + 2\sqrt{5}}{-5\sqrt{2} + 2\sqrt{5}} \\ &= \frac{-7 - 2\sqrt{10}}{3}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{5\sqrt{2} + 2\sqrt{3}}{-\sqrt{2} - 4\sqrt{3}} \\ &= \frac{-7 - 9\sqrt{6}}{23}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{-10 + 4\sqrt{2}}{-3\sqrt{2}} \\ &= \frac{-4 + 5\sqrt{2}}{3}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{2\sqrt{3}}{4\sqrt{3} + 5\sqrt{5}} \\ &= \frac{10\sqrt{15} - 24}{77}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{-2 - \sqrt{2}}{-8 - 5\sqrt{2}} \\ &= \frac{-\sqrt{2} + 3}{7}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{-\sqrt{2} - 4\sqrt{5}}{-3\sqrt{2} - 5\sqrt{5}} \\ &= \frac{94 - 7\sqrt{10}}{107}\end{aligned}$$

$$(1) \frac{-\sqrt{5}-5\sqrt{2}}{4\sqrt{5}} \\ = \frac{-1-\sqrt{10}}{4}$$

$$(6) \frac{-\sqrt{5}+2}{-2\sqrt{5}-4} \\ = \frac{-4\sqrt{5}+9}{2}$$

$$(11) \frac{-4\sqrt{3}-2}{-3\sqrt{3}-4} \\ = \frac{-10\sqrt{3}+28}{11}$$

$$(2) \frac{-1-\sqrt{2}}{-2-\sqrt{2}} \\ = \frac{\sqrt{2}}{2}$$

$$(7) \frac{-4\sqrt{5}-5}{-3\sqrt{5}} \\ = \frac{4+\sqrt{5}}{3}$$

$$(12) \frac{-1-3\sqrt{5}}{2-2\sqrt{5}} \\ = \frac{\sqrt{5}+4}{2}$$

$$(3) \frac{4-\sqrt{2}}{3+\sqrt{2}} \\ = -\sqrt{2}+2$$

$$(8) \frac{6+4\sqrt{3}}{6+\sqrt{3}} \\ = \frac{6\sqrt{3}+8}{11}$$

$$(13) \frac{2\sqrt{3}}{-\sqrt{3}+5\sqrt{2}} \\ = \frac{6+10\sqrt{6}}{47}$$

$$(4) \frac{-2\sqrt{5}}{5+2\sqrt{5}} \\ = -2\sqrt{5}+4$$

$$(9) \frac{-\sqrt{2}+5\sqrt{3}}{-4\sqrt{2}-2\sqrt{3}} \\ = \frac{-11\sqrt{6}+19}{10}$$

$$(14) \frac{2\sqrt{2}+3\sqrt{3}}{-5\sqrt{2}+3\sqrt{3}} \\ = \frac{-21\sqrt{6}-47}{23}$$

$$(5) \frac{8-\sqrt{2}}{2+5\sqrt{2}} \\ = \frac{-13+21\sqrt{2}}{23}$$

$$(10) \frac{-4\sqrt{5}+3\sqrt{3}}{2\sqrt{3}} \\ = \frac{-4\sqrt{15}+9}{6}$$

$$(15) \frac{-2\sqrt{5}-4\sqrt{2}}{-3\sqrt{2}} \\ = \frac{4+\sqrt{10}}{3}$$

$$(1) \frac{-3\sqrt{2} + 5\sqrt{5}}{4\sqrt{5}} \\ = \frac{25 - 3\sqrt{10}}{20}$$

$$(6) \frac{-1 - 5\sqrt{5}}{-2 + 3\sqrt{5}} \\ = \frac{-13\sqrt{5} - 77}{41}$$

$$(11) \frac{-5\sqrt{3} - 3\sqrt{5}}{-3\sqrt{3} - 2\sqrt{5}} \\ = \frac{15 - \sqrt{15}}{7}$$

$$(2) \frac{-3\sqrt{3}}{\sqrt{3} - 5\sqrt{5}} \\ = \frac{9 + 15\sqrt{15}}{122}$$

$$(7) \frac{1}{2\sqrt{5} + 1} \\ = \frac{2\sqrt{5} - 1}{19}$$

$$(12) \frac{-6 - \sqrt{5}}{-6 - \sqrt{5}} \\ = 1$$

$$(3) \frac{-4\sqrt{5}}{-5\sqrt{5} - \sqrt{2}} \\ = \frac{100 - 4\sqrt{10}}{123}$$

$$(8) \frac{2\sqrt{2} + 6}{5\sqrt{2} - 4} \\ = \frac{19\sqrt{2} + 22}{17}$$

$$(13) \frac{-4 + 4\sqrt{2}}{-1 - \sqrt{2}} \\ = 8\sqrt{2} - 12$$

$$(4) \frac{3\sqrt{2} - 10}{-4\sqrt{2} - 6} \\ = \frac{42 - 29\sqrt{2}}{2}$$

$$(9) \frac{2\sqrt{3} - \sqrt{5}}{-3\sqrt{3} - 5\sqrt{5}} \\ = \frac{-13\sqrt{15} + 43}{98}$$

$$(14) \frac{3\sqrt{2} + 3\sqrt{5}}{3\sqrt{2} + 2\sqrt{5}} \\ = \frac{12 - 3\sqrt{10}}{2}$$

$$(5) \frac{5\sqrt{2}}{4\sqrt{2} + 6} \\ = \frac{15\sqrt{2} - 20}{2}$$

$$(10) \frac{-4\sqrt{5}}{\sqrt{5} + \sqrt{3}} \\ = -10 + 2\sqrt{15}$$

$$(15) \frac{\sqrt{3} + 5}{2\sqrt{3} + 4} \\ = \frac{7 - 3\sqrt{3}}{2}$$

$$\begin{aligned} (1) \quad & \frac{8 - 2\sqrt{5}}{8 + 5\sqrt{5}} \\ &= \frac{56\sqrt{5} - 114}{61} \end{aligned}$$

$$\begin{aligned} (6) \quad & \frac{4\sqrt{5} - 2\sqrt{2}}{-2\sqrt{5} - 5\sqrt{2}} \\ &= \frac{10 - 4\sqrt{10}}{5} \end{aligned}$$

$$\begin{aligned} (11) \quad & \frac{2\sqrt{2} - 1}{-2\sqrt{2} - 2} \\ &= \frac{-5 + 3\sqrt{2}}{2} \end{aligned}$$

$$\begin{aligned} (2) \quad & \frac{2 + 2\sqrt{2}}{2 - 5\sqrt{2}} \\ &= \frac{-12 - 7\sqrt{2}}{23} \end{aligned}$$

$$\begin{aligned} (7) \quad & \frac{-4\sqrt{5} - 2\sqrt{3}}{\sqrt{5} - 5\sqrt{3}} \\ &= \frac{11\sqrt{15} + 25}{35} \end{aligned}$$

$$\begin{aligned} (12) \quad & \frac{-1 + \sqrt{5}}{-2 - 4\sqrt{5}} \\ &= \frac{3\sqrt{5} - 11}{38} \end{aligned}$$

$$\begin{aligned} (3) \quad & \frac{5\sqrt{5}}{5\sqrt{5} - 3} \\ &= \frac{15\sqrt{5} + 125}{116} \end{aligned}$$

$$\begin{aligned} (8) \quad & \frac{-4}{-\sqrt{3} - 2} \\ &= -4\sqrt{3} + 8 \end{aligned}$$

$$\begin{aligned} (13) \quad & \frac{\sqrt{5} + 8}{-4\sqrt{5}} \\ &= \frac{-5 - 8\sqrt{5}}{20} \end{aligned}$$

$$\begin{aligned} (4) \quad & \frac{-\sqrt{2} + 3}{2\sqrt{2} - 2} \\ &= \frac{1 + 2\sqrt{2}}{2} \end{aligned}$$

$$\begin{aligned} (9) \quad & \frac{-4\sqrt{3} + 10}{-\sqrt{3} + 4} \\ &= \frac{28 - 6\sqrt{3}}{13} \end{aligned}$$

$$\begin{aligned} (14) \quad & \frac{-4 - 5\sqrt{5}}{1 - 3\sqrt{5}} \\ &= \frac{17\sqrt{5} + 79}{44} \end{aligned}$$

$$\begin{aligned} (5) \quad & \frac{-1}{2\sqrt{3} - 4} \\ &= \frac{\sqrt{3} + 2}{2} \end{aligned}$$

$$\begin{aligned} (10) \quad & \frac{3\sqrt{2}}{-5\sqrt{3} + 4\sqrt{2}} \\ &= \frac{-15\sqrt{6} - 24}{43} \end{aligned}$$

$$\begin{aligned} (15) \quad & \frac{\sqrt{2} - 1}{\sqrt{2} - 3} \\ &= \frac{1 - 2\sqrt{2}}{7} \end{aligned}$$

$$(1) \frac{5-2\sqrt{3}}{3-\sqrt{3}} \\ = \frac{-\sqrt{3}+9}{6}$$

$$(6) \frac{5\sqrt{5}-5\sqrt{2}}{-\sqrt{5}-\sqrt{2}} \\ = \frac{-35+10\sqrt{10}}{3}$$

$$(11) \frac{4\sqrt{2}-4\sqrt{5}}{-3\sqrt{2}+3\sqrt{5}} \\ = \frac{-4}{3}$$

$$(2) \frac{-5}{-2\sqrt{5}+4} \\ = \frac{10+5\sqrt{5}}{2}$$

$$(7) \frac{3\sqrt{2}+2\sqrt{3}}{-2\sqrt{2}} \\ = \frac{-\sqrt{6}-3}{2}$$

$$(12) \frac{-3\sqrt{3}+6}{2\sqrt{3}-10} \\ = \frac{9\sqrt{3}-21}{44}$$

$$(3) \frac{-4}{2\sqrt{5}-1} \\ = \frac{-4-8\sqrt{5}}{19}$$

$$(8) \frac{\sqrt{5}}{-\sqrt{5}-\sqrt{3}} \\ = \frac{\sqrt{15}-5}{2}$$

$$(13) \frac{2\sqrt{2}+3\sqrt{3}}{-4\sqrt{2}+\sqrt{3}} \\ = \frac{-14\sqrt{6}-25}{29}$$

$$(4) \frac{\sqrt{3}+4}{-2\sqrt{3}+2} \\ = \frac{-7-5\sqrt{3}}{4}$$

$$(9) \frac{-6-\sqrt{3}}{-8-3\sqrt{3}} \\ = \frac{39-10\sqrt{3}}{37}$$

$$(14) \frac{-3\sqrt{2}-4}{-4\sqrt{2}-2} \\ = \frac{8+5\sqrt{2}}{14}$$

$$(5) \frac{\sqrt{5}+2}{4\sqrt{5}-4} \\ = \frac{3\sqrt{5}+7}{16}$$

$$(10) \frac{1-\sqrt{5}}{3-2\sqrt{5}} \\ = \frac{7+\sqrt{5}}{11}$$

$$(15) \frac{4\sqrt{2}}{\sqrt{3}+3\sqrt{2}} \\ = \frac{-4\sqrt{6}+24}{15}$$

$$(1) \frac{\sqrt{5} - \sqrt{2}}{-\sqrt{2}} \\ = \frac{-\sqrt{10} + 2}{2}$$

$$(6) \frac{-\sqrt{3} + 2\sqrt{5}}{2\sqrt{3} + 2\sqrt{5}} \\ = \frac{13 - 3\sqrt{15}}{4}$$

$$(11) \frac{\sqrt{3} + 4\sqrt{5}}{-4\sqrt{3} - \sqrt{5}} \\ = \frac{-15\sqrt{15} + 8}{43}$$

$$(2) \frac{-\sqrt{3} + 5\sqrt{2}}{\sqrt{3} + 2\sqrt{2}} \\ = \frac{-7\sqrt{6} + 23}{5}$$

$$(7) \frac{-2\sqrt{5} - 3\sqrt{2}}{\sqrt{5} - \sqrt{2}} \\ = \frac{-5\sqrt{10} - 16}{3}$$

$$(12) \frac{\sqrt{5}}{2\sqrt{5} + 4} \\ = \frac{-2\sqrt{5} + 5}{2}$$

$$(3) \frac{6 + 5\sqrt{5}}{10 + 3\sqrt{5}} \\ = \frac{-15 + 32\sqrt{5}}{55}$$

$$(8) \frac{-5\sqrt{2} + 8}{-\sqrt{2} + 2} \\ = -\sqrt{2} + 3$$

$$(13) \frac{2\sqrt{5} - 3\sqrt{2}}{-2\sqrt{5} + 2\sqrt{2}} \\ = \frac{-4 + \sqrt{10}}{6}$$

$$(4) \frac{-2 + 4\sqrt{3}}{8 - \sqrt{3}} \\ = \frac{30\sqrt{3} - 4}{61}$$

$$(9) \frac{-4\sqrt{5}}{-5\sqrt{5} - 3\sqrt{2}} \\ = \frac{-12\sqrt{10} + 100}{107}$$

$$(14) \frac{-\sqrt{2} - 2}{-\sqrt{2} - 5} \\ = \frac{3\sqrt{2} + 8}{23}$$

$$(5) \frac{-3\sqrt{3}}{\sqrt{3} + 3\sqrt{5}} \\ = \frac{-3\sqrt{15} + 3}{14}$$

$$(10) \frac{-2\sqrt{5}}{3\sqrt{3} - 3\sqrt{5}} \\ = \frac{5 + \sqrt{15}}{3}$$

$$(15) \frac{5\sqrt{2} + 2\sqrt{3}}{5\sqrt{2} + \sqrt{3}} \\ = \frac{44 + 5\sqrt{6}}{47}$$

$$(1) \frac{3 - \sqrt{2}}{4 + 3\sqrt{2}} \\ = \frac{-18 + 13\sqrt{2}}{2}$$

$$(6) \frac{\sqrt{5}}{\sqrt{5}} \\ = 1$$

$$(11) \frac{-4\sqrt{5} + 2\sqrt{3}}{3\sqrt{5} + 3\sqrt{3}} \\ = \frac{-13 + 3\sqrt{15}}{3}$$

$$(2) \frac{-\sqrt{3}}{-\sqrt{3} - \sqrt{5}} \\ = \frac{\sqrt{15} - 3}{2}$$

$$(7) \frac{-2\sqrt{3}}{5\sqrt{3} + 5} \\ = \frac{\sqrt{3} - 3}{5}$$

$$(12) \frac{4\sqrt{3} + 3\sqrt{5}}{-4\sqrt{5}} \\ = \frac{-15 - 4\sqrt{15}}{20}$$

$$(3) \frac{-2 - \sqrt{5}}{4 - 4\sqrt{5}} \\ = \frac{3\sqrt{5} + 7}{16}$$

$$(8) \frac{3}{3 + \sqrt{2}} \\ = \frac{-3\sqrt{2} + 9}{7}$$

$$(13) \frac{\sqrt{5}}{2\sqrt{3} - \sqrt{5}} \\ = \frac{5 + 2\sqrt{15}}{7}$$

$$(4) \frac{-3\sqrt{2} + 6}{4\sqrt{2} - 8} \\ = \frac{-3}{4}$$

$$(9) \frac{4\sqrt{2} + 2\sqrt{5}}{\sqrt{2} - 5\sqrt{5}} \\ = \frac{-22\sqrt{10} - 58}{123}$$

$$(14) \frac{-3\sqrt{2} + \sqrt{3}}{4\sqrt{2} + 3\sqrt{3}} \\ = \frac{-33 + 13\sqrt{6}}{5}$$

$$(5) \frac{-4 + 5\sqrt{3}}{5 + 3\sqrt{3}} \\ = \frac{65 - 37\sqrt{3}}{2}$$

$$(10) \frac{-6 - 5\sqrt{5}}{4 + 5\sqrt{5}} \\ = \frac{-101 - 10\sqrt{5}}{109}$$

$$(15) \frac{3}{\sqrt{3} + 1} \\ = \frac{3\sqrt{3} - 3}{2}$$



$$\begin{aligned}(1) \quad & \frac{-3\sqrt{3} - 4\sqrt{2}}{3\sqrt{3} - 2\sqrt{2}} \\ &= \frac{-18\sqrt{6} - 43}{19}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{6 + 3\sqrt{5}}{-2 + 4\sqrt{5}} \\ &= \frac{15\sqrt{5} + 36}{38}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{3\sqrt{2} - 3\sqrt{3}}{-3\sqrt{2} + \sqrt{3}} \\ &= \frac{2\sqrt{6} - 3}{5}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{5 - \sqrt{5}}{-4 + 4\sqrt{5}} \\ &= \frac{\sqrt{5}}{4}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{-5\sqrt{5} - 5}{-3\sqrt{5} - 3} \\ &= \frac{5}{3}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-3\sqrt{3} - 5\sqrt{2}}{-3\sqrt{3} - \sqrt{2}} \\ &= \frac{12\sqrt{6} + 17}{25}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{6 + 2\sqrt{2}}{-4 - 3\sqrt{2}} \\ &= 6 - 5\sqrt{2}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{-5\sqrt{2}}{-5\sqrt{3} + 4\sqrt{2}} \\ &= \frac{40 + 25\sqrt{6}}{43}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-4}{-10 - 3\sqrt{2}} \\ &= \frac{20 - 6\sqrt{2}}{41}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{-2 - 4\sqrt{5}}{-1 + 3\sqrt{5}} \\ &= \frac{-5\sqrt{5} - 31}{22}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{-2\sqrt{3} - 3\sqrt{2}}{2\sqrt{3} + \sqrt{2}} \\ &= \frac{-3 - 2\sqrt{6}}{5}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{\sqrt{2} - 4}{-\sqrt{2} + 2} \\ &= -3 - \sqrt{2}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{\sqrt{2} - 5\sqrt{3}}{\sqrt{2} - 2\sqrt{3}} \\ &= \frac{28 + 3\sqrt{6}}{10}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{2\sqrt{3} - 3\sqrt{2}}{3\sqrt{3} + 2\sqrt{2}} \\ &= \frac{-13\sqrt{6} + 30}{19}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{-1 - 4\sqrt{5}}{4 + 2\sqrt{5}} \\ &= \frac{-18 + 7\sqrt{5}}{2}\end{aligned}$$

$$(1) \frac{-2}{\sqrt{2}-2} \\ = 2 + \sqrt{2}$$

$$(6) \frac{-\sqrt{2}+4\sqrt{5}}{\sqrt{5}} \\ = \frac{20-\sqrt{10}}{5}$$

$$(11) \frac{4\sqrt{3}+1}{\sqrt{3}-3} \\ = \frac{-15-13\sqrt{3}}{6}$$

$$(2) \frac{-2-4\sqrt{2}}{-2+5\sqrt{2}} \\ = \frac{-9\sqrt{2}-22}{23}$$

$$(7) \frac{-5}{-5-4\sqrt{2}} \\ = \frac{-25+20\sqrt{2}}{7}$$

$$(12) \frac{3\sqrt{3}+\sqrt{5}}{4\sqrt{3}-\sqrt{5}} \\ = \frac{7\sqrt{15}+41}{43}$$

$$(3) \frac{3+4\sqrt{5}}{2\sqrt{5}} \\ = \frac{3\sqrt{5}+20}{10}$$

$$(8) \frac{2+\sqrt{3}}{5\sqrt{3}} \\ = \frac{3+2\sqrt{3}}{15}$$

$$(13) \frac{-5\sqrt{2}-2\sqrt{3}}{2\sqrt{2}+3\sqrt{3}} \\ = \frac{-11\sqrt{6}+2}{19}$$

$$(4) \frac{5\sqrt{3}-4}{-3\sqrt{3}-1} \\ = \frac{-49+17\sqrt{3}}{26}$$

$$(9) \frac{-5\sqrt{2}}{10-2\sqrt{2}} \\ = \frac{-10-25\sqrt{2}}{46}$$

$$(14) \frac{\sqrt{2}+2\sqrt{3}}{-\sqrt{2}} \\ = -1-\sqrt{6}$$

$$(5) \frac{-1+3\sqrt{5}}{-4\sqrt{5}} \\ = \frac{-15+\sqrt{5}}{20}$$

$$(10) \frac{-4\sqrt{3}+5\sqrt{2}}{5\sqrt{3}-4\sqrt{2}} \\ = \frac{9\sqrt{6}-20}{43}$$

$$(15) \frac{-2-3\sqrt{5}}{8-4\sqrt{5}} \\ = \frac{19+8\sqrt{5}}{4}$$

$$\begin{aligned}(1) \quad & \frac{2}{1+4\sqrt{3}} \\ &= \frac{8\sqrt{3}-2}{47}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{-4+4\sqrt{2}}{4-\sqrt{2}} \\ &= \frac{-4+6\sqrt{2}}{7}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-\sqrt{3}}{4\sqrt{3}} \\ &= \frac{-1}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{3\sqrt{5}-4\sqrt{2}}{-2\sqrt{5}+5\sqrt{2}} \\ &= \frac{-10+7\sqrt{10}}{30}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{2\sqrt{2}}{-2\sqrt{5}-\sqrt{2}} \\ &= \frac{2-2\sqrt{10}}{9}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{\sqrt{5}+2\sqrt{2}}{2\sqrt{2}} \\ &= \frac{4+\sqrt{10}}{4}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{-4\sqrt{5}}{3\sqrt{2}-3\sqrt{5}} \\ &= \frac{20+4\sqrt{10}}{9}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{\sqrt{2}+8}{-3\sqrt{2}-4} \\ &= -10\sqrt{2}+13\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-4\sqrt{2}-3}{-\sqrt{2}+2} \\ &= \frac{-14-11\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{\sqrt{2}+2}{-\sqrt{2}+3} \\ &= \frac{5\sqrt{2}+8}{7}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{-2-3\sqrt{2}}{3-4\sqrt{2}} \\ &= \frac{30+17\sqrt{2}}{23}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{5-5\sqrt{5}}{4+5\sqrt{5}} \\ &= \frac{45\sqrt{5}-145}{109}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{\sqrt{3}+8}{-3\sqrt{3}-4} \\ &= \frac{23-20\sqrt{3}}{11}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{\sqrt{5}+4}{3\sqrt{5}+5} \\ &= \frac{-5+7\sqrt{5}}{20}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{3\sqrt{5}+6}{5\sqrt{5}} \\ &= \frac{15+6\sqrt{5}}{25}\end{aligned}$$

$$(1) \frac{3\sqrt{3} + 5\sqrt{2}}{-3\sqrt{3} + \sqrt{2}} \\ = \frac{-18\sqrt{6} - 37}{25}$$

$$(6) \frac{-5 + \sqrt{5}}{-3 - 2\sqrt{5}} \\ = \frac{-25 + 13\sqrt{5}}{11}$$

$$(11) \frac{-1 - \sqrt{2}}{-4 + \sqrt{2}} \\ = \frac{6 + 5\sqrt{2}}{14}$$

$$(2) \frac{-2}{-\sqrt{2} + 2} \\ = -2 - \sqrt{2}$$

$$(7) \frac{3\sqrt{3} + \sqrt{5}}{5\sqrt{3} + 2\sqrt{5}} \\ = \frac{35 - \sqrt{15}}{55}$$

$$(12) \frac{\sqrt{3}}{-6 - 2\sqrt{3}} \\ = \frac{-\sqrt{3} + 1}{4}$$

$$(3) \frac{1}{-4 - \sqrt{5}} \\ = \frac{\sqrt{5} - 4}{11}$$

$$(8) \frac{-4\sqrt{3} - 3\sqrt{2}}{-3\sqrt{2}} \\ = \frac{2\sqrt{6} + 3}{3}$$

$$(13) \frac{3\sqrt{5} - \sqrt{3}}{-2\sqrt{5} + 5\sqrt{3}} \\ = \frac{13\sqrt{15} + 15}{55}$$

$$(4) \frac{-6 + \sqrt{5}}{-6 + 2\sqrt{5}} \\ = \frac{3\sqrt{5} + 13}{8}$$

$$(9) \frac{5 + \sqrt{5}}{2 + 5\sqrt{5}} \\ = \frac{15 + 23\sqrt{5}}{121}$$

$$(14) \frac{-5\sqrt{2} - 8}{-2\sqrt{2} - 4} \\ = \frac{\sqrt{2} + 3}{2}$$

$$(5) \frac{2\sqrt{5} + 4}{-5\sqrt{5} + 3} \\ = \frac{-31 - 13\sqrt{5}}{58}$$

$$(10) \frac{-3\sqrt{5} + 5\sqrt{2}}{2\sqrt{5} + 3\sqrt{2}} \\ = \frac{-60 + 19\sqrt{10}}{2}$$

$$(15) \frac{5\sqrt{5} - 4\sqrt{3}}{-3\sqrt{5} - 3\sqrt{3}} \\ = \frac{-37 + 9\sqrt{15}}{6}$$

$$(1) \frac{\sqrt{2}}{-5\sqrt{2} - 3\sqrt{5}} \\ = \frac{3\sqrt{10} - 10}{5}$$

$$(6) \frac{-2 + \sqrt{3}}{-4\sqrt{3}} \\ = \frac{-3 + 2\sqrt{3}}{12}$$

$$(11) \frac{\sqrt{5} + 10}{-\sqrt{5} - 6} \\ = \frac{4\sqrt{5} - 55}{31}$$

$$(2) \frac{-2 + 2\sqrt{5}}{-1 + \sqrt{5}} \\ = 2$$

$$(7) \frac{4\sqrt{3} + 3\sqrt{5}}{-\sqrt{3} - 5\sqrt{5}} \\ = \frac{-17\sqrt{15} - 63}{122}$$

$$(12) \frac{3\sqrt{5} - 10}{-4\sqrt{5} - 6} \\ = \frac{-60 + 29\sqrt{5}}{22}$$

$$(3) \frac{3\sqrt{5} + 3\sqrt{2}}{\sqrt{5} + \sqrt{2}} \\ = 3$$

$$(8) \frac{5 - 5\sqrt{2}}{2 + 5\sqrt{2}} \\ = \frac{-60 + 35\sqrt{2}}{46}$$

$$(13) \frac{4\sqrt{5} + 5\sqrt{3}}{\sqrt{5} - \sqrt{3}} \\ = \frac{35 + 9\sqrt{15}}{2}$$

$$(4) \frac{-3\sqrt{5} - 2}{5\sqrt{5} - 3} \\ = \frac{-19\sqrt{5} - 81}{116}$$

$$(9) \frac{-4\sqrt{5} + 5\sqrt{2}}{-2\sqrt{5} - 5\sqrt{2}} \\ = -3 + \sqrt{10}$$

$$(14) \frac{5 + 3\sqrt{2}}{-3 - 2\sqrt{2}} \\ = -3 + \sqrt{2}$$

$$(5) \frac{\sqrt{3} + \sqrt{5}}{-4\sqrt{3} + 4\sqrt{5}} \\ = \frac{\sqrt{15} + 4}{4}$$

$$(10) \frac{4\sqrt{5} + 2\sqrt{3}}{3\sqrt{5} - 3\sqrt{3}} \\ = \frac{13 + 3\sqrt{15}}{3}$$

$$(15) \frac{\sqrt{3} + 2}{-2\sqrt{3}} \\ = \frac{-3 - 2\sqrt{3}}{6}$$

$$\begin{aligned}(1) \quad & \frac{2\sqrt{5} - 3\sqrt{3}}{-4\sqrt{5} - 3\sqrt{3}} \\ &= \frac{-67 + 18\sqrt{15}}{53}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{4\sqrt{3} + \sqrt{5}}{4\sqrt{5}} \\ &= \frac{5 + 4\sqrt{15}}{20}\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{2 - 5\sqrt{5}}{3 - \sqrt{5}} \\ &= \frac{-13\sqrt{5} - 19}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{2\sqrt{2} + 2\sqrt{5}}{-\sqrt{2} + \sqrt{5}} \\ &= \frac{4\sqrt{10} + 14}{3}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{5 + 3\sqrt{2}}{-3 - 3\sqrt{2}} \\ &= \frac{-1 - 2\sqrt{2}}{3}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{\sqrt{3} + 2\sqrt{5}}{-2\sqrt{3} + 2\sqrt{5}} \\ &= \frac{3\sqrt{15} + 13}{4}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{4\sqrt{3} - 3\sqrt{2}}{4\sqrt{3} - 3\sqrt{2}} \\ &= 1\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{-2\sqrt{5} + 2\sqrt{2}}{5\sqrt{5} - 3\sqrt{2}} \\ &= \frac{4\sqrt{10} - 38}{107}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-4 + 5\sqrt{3}}{-1 - 3\sqrt{3}} \\ &= \frac{-49 + 17\sqrt{3}}{26}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{-3\sqrt{2} - 8}{4\sqrt{2} - 6} \\ &= \frac{36 + 25\sqrt{2}}{2}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{3\sqrt{2} + 8}{-4\sqrt{2}} \\ &= \frac{-3 - 4\sqrt{2}}{4}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{5 - 4\sqrt{3}}{-4 + 4\sqrt{3}} \\ &= \frac{\sqrt{3} - 7}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{2\sqrt{5} + 2}{3\sqrt{5} - 4} \\ &= \frac{38 + 14\sqrt{5}}{29}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{-2 + 4\sqrt{2}}{3 + 3\sqrt{2}} \\ &= \frac{10 - 6\sqrt{2}}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{3 - 2\sqrt{5}}{3 - 3\sqrt{5}} \\ &= \frac{7 - \sqrt{5}}{12}\end{aligned}$$

$$\begin{aligned}(1) \quad & \frac{3\sqrt{3}-2}{\sqrt{3}} \\ &= \frac{9-2\sqrt{3}}{3}\end{aligned}$$

$$\begin{aligned}(6) \quad & \frac{\sqrt{2}}{1+\sqrt{2}} \\ &= -\sqrt{2}+2\end{aligned}$$

$$\begin{aligned}(11) \quad & \frac{-5\sqrt{3}-1}{-3\sqrt{3}-4} \\ &= \frac{-17\sqrt{3}+41}{11}\end{aligned}$$

$$\begin{aligned}(2) \quad & \frac{3\sqrt{2}}{3\sqrt{2}+4\sqrt{3}} \\ &= \frac{2\sqrt{6}-3}{5}\end{aligned}$$

$$\begin{aligned}(7) \quad & \frac{4+\sqrt{5}}{1+2\sqrt{5}} \\ &= \frac{6+7\sqrt{5}}{19}\end{aligned}$$

$$\begin{aligned}(12) \quad & \frac{-10+\sqrt{3}}{-6+3\sqrt{3}} \\ &= \frac{17+8\sqrt{3}}{3}\end{aligned}$$

$$\begin{aligned}(3) \quad & \frac{5\sqrt{5}-\sqrt{3}}{\sqrt{5}+2\sqrt{3}} \\ &= \frac{-31+11\sqrt{15}}{7}\end{aligned}$$

$$\begin{aligned}(8) \quad & \frac{-3}{5\sqrt{5}+3} \\ &= \frac{-15\sqrt{5}+9}{116}\end{aligned}$$

$$\begin{aligned}(13) \quad & \frac{-2-4\sqrt{3}}{-1+5\sqrt{3}} \\ &= \frac{-31-7\sqrt{3}}{37}\end{aligned}$$

$$\begin{aligned}(4) \quad & \frac{-2\sqrt{2}+2}{2\sqrt{2}-1} \\ &= \frac{-6+2\sqrt{2}}{7}\end{aligned}$$

$$\begin{aligned}(9) \quad & \frac{2\sqrt{3}-1}{4\sqrt{3}+2} \\ &= \frac{13-4\sqrt{3}}{22}\end{aligned}$$

$$\begin{aligned}(14) \quad & \frac{3\sqrt{2}}{4-3\sqrt{2}} \\ &= -9-6\sqrt{2}\end{aligned}$$

$$\begin{aligned}(5) \quad & \frac{2}{-\sqrt{2}+2} \\ &= 2+\sqrt{2}\end{aligned}$$

$$\begin{aligned}(10) \quad & \frac{-\sqrt{5}-2}{-3\sqrt{5}+10} \\ &= \frac{-16\sqrt{5}-35}{55}\end{aligned}$$

$$\begin{aligned}(15) \quad & \frac{-4\sqrt{5}}{5\sqrt{5}} \\ &= \frac{-4}{5}\end{aligned}$$

## 1.5 複素数

### 1.5.1 加減法

複素数の和や差を求める問題。



(1)  $(-10 - 2i) + (-5 - 4i)$

(20)  $(-5 - i) - (-9 - 3i)$

(39)  $(-3 + 4i) + (8 + 5i)$

(2)  $(2 + 7i) - (3 - 8i)$

(21)  $(2 + 5i) + (2 - 10i)$

(40)  $(3 + 5i) + (-6 - 3i)$

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(22)  $(-10 - 10i) + (-8 + 3i)$

(41)  $(7 - 8i) - (9 + 10i)$

(4)  $(-3 - 4i) + (-10 + 5i)$

(23)  $(1 - i) + (-10 - 5i)$

(42)  $(-9 - 9i) + (-5 - 7i)$

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(49)  $(4 - 9i) - (-4 - 2i)$

(12)  $(-9 + 6i) - (-7 - 9i)$

(31)  $(5 - 2i) - (5 - 10i)$

(50)  $(-10 - 9i) + (-7 - 5i)$

(13)  $(-1 - 2i) - (1 + 3i)$

(32)  $(-10 + 7i) - (3 + 2i)$

(51)  $(-9 - 8i) - (-10 - 5i)$

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| (5) $(1 + 7i) - (-10 - 4i)$  | (24) $(-7 - 2i) + (2 + 7i)$    | (43) $(-10 - 8i) - (6 + 6i)$  |
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| (8) $(8 + 10i) - (8 + 3i)$   | (27) $(-2 + 5i) + (10 - 2i)$   | (46) $(8 - 6i) + (-7 + 10i)$  |
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(26)  $(6 + 7i) - (-6 - 8i)$

(45)  $(5 + 3i) + (3 - 6i)$

(8)  $(-2 + 9i) + (10 + 3i)$

(27)  $(-3 - 3i) - (7 - 8i)$

(46)  $(5 - 8i) - (-4 + 10i)$

(9)  $(7 - 5i) + (-8 - 9i)$

(28)  $(-7 + 6i) - (3 - 5i)$

(47)  $(2 - 10i) + (-4 - 10i)$

(10)  $(10 - 7i) + (-7 + 8i)$

(29)  $(-1 + i) + (-9 - 4i)$

(48)  $(-5 - 8i) + (9 - 9i)$

(11)  $(5 + 3i) + (-8 + 2i)$

(30)  $(-5 + 7i) - (3 + 4i)$

(49)  $(4 + i) + (7 + 9i)$

(12)  $(7 + 6i) + (5 - 6i)$

(31)  $(-7 + 4i) + (-8 - 7i)$

(50)  $(8 - 9i) - (1 - 8i)$

(13)  $(-9 - i) - (-10 + 2i)$

(32)  $(10 + 7i) + (9 - i)$

(51)  $(-5 - 2i) - (6 - 6i)$

(14)  $(-7 - i) - (-1 - 8i)$

(33)  $(6 + 5i) + (-9 - 2i)$

(52)  $(9 - 6i) - (-1 + 2i)$

(15)  $(-10 - 7i) - (1 + 8i)$

(34)  $(-3 - 9i) - (2 - 9i)$

(53)  $(-5 - 9i) - (2 + 2i)$

(16)  $(-10 - i) + (8 - 9i)$

(35)  $(9 - 3i) + (2 - 4i)$

(54)  $(-9 - 8i) + (-1 + 5i)$

(17)  $(10 + 3i) + (-2 + i)$

(36)  $(9 + 3i) + (-5 - 2i)$

(55)  $(-8 - 6i) - (-9 + 3i)$

(18)  $(-2 - i) + (5 + 8i)$

(37)  $(8 + 6i) + (-10 + 5i)$

(56)  $(-8 - 7i) - (-4 + 7i)$

(19)  $(-8 - 2i) - (-5 - 6i)$

(38)  $(-7 - i) - (2 + 10i)$

(57)  $(5 - 10i) - (-8 + 9i)$

- |                                                |                                                |                                                |
|------------------------------------------------|------------------------------------------------|------------------------------------------------|
| (1) $(-10 - 2i) + (-5 - 4i)$<br>$= -15 - 6i$   | (20) $(-5 - i) - (-9 - 3i)$<br>$= 4 + 2i$      | (39) $(-3 + 4i) + (8 + 5i)$<br>$= 5 + 9i$      |
| (2) $(2 + 7i) - (3 - 8i)$<br>$= -1 + 15i$      | (21) $(2 + 5i) + (2 - 10i)$<br>$= 4 - 5i$      | (40) $(3 + 5i) + (-6 - 3i)$<br>$= -3 + 2i$     |
| (3) $(-10 + 2i) - (-9 + 8i)$<br>$= -1 - 6i$    | (22) $(-10 - 10i) + (-8 + 3i)$<br>$= -18 - 7i$ | (41) $(7 - 8i) - (9 + 10i)$<br>$= -2 - 18i$    |
| (4) $(-3 - 4i) + (-10 + 5i)$<br>$= -13 + i$    | (23) $(1 - i) + (-10 - 5i)$<br>$= -9 - 6i$     | (42) $(-9 - 9i) + (-5 - 7i)$<br>$= -14 - 16i$  |
| (5) $(4 - 6i) + (-6 - 9i)$<br>$= -2 - 15i$     | (24) $(-1 - 4i) - (-10 + 6i)$<br>$= 9 - 10i$   | (43) $(-9 + 6i) + (8 + 6i)$<br>$= -1 + 12i$    |
| (6) $(-4 + 4i) - (4 + 2i)$<br>$= -8 + 2i$      | (25) $(-10 - 6i) + (10 + 3i)$<br>$= -3i$       | (44) $(-4 - i) - (10 + i)$<br>$= -14 - 2i$     |
| (7) $(-1 + 3i) - (-4 - i)$<br>$= 3 + 4i$       | (26) $(2 + 4i) + (3 + 6i)$<br>$= 5 + 10i$      | (45) $(6 + 2i) - (-6 - i)$<br>$= 12 + 3i$      |
| (8) $(4 + 5i) + (2 - i)$<br>$= 6 + 4i$         | (27) $(4 + 8i) + (9 + 5i)$<br>$= 13 + 13i$     | (46) $(4 + 8i) + (-8 + i)$<br>$= -4 + 9i$      |
| (9) $(4 + 9i) + (5 + 4i)$<br>$= 9 + 13i$       | (28) $(-2 - 5i) - (-2 + 7i)$<br>$= -12i$       | (47) $(3 - i) + (2 - i)$<br>$= 5 - 2i$         |
| (10) $(-10 - 9i) + (-6 - 8i)$<br>$= -16 - 17i$ | (29) $(1 + 10i) + (-8 + 5i)$<br>$= -7 + 15i$   | (48) $(1 - 4i) - (-1 + 2i)$<br>$= 2 - 6i$      |
| (11) $(-1 + 4i) + (-4 - 4i)$<br>$= -5$         | (30) $(8 - 10i) - (-4 - 2i)$<br>$= 12 - 8i$    | (49) $(4 - 9i) - (-4 - 2i)$<br>$= 8 - 7i$      |
| (12) $(-9 + 6i) - (-7 - 9i)$<br>$= -2 + 15i$   | (31) $(5 - 2i) - (5 - 10i)$<br>$= 8i$          | (50) $(-10 - 9i) + (-7 - 5i)$<br>$= -17 - 14i$ |
| (13) $(-1 - 2i) - (1 + 3i)$<br>$= -2 - 5i$     | (32) $(-10 + 7i) - (3 + 2i)$<br>$= -13 + 5i$   | (51) $(-9 - 8i) - (-10 - 5i)$<br>$= 1 - 3i$    |
| (14) $(-1 - 2i) + (-8 + 9i)$<br>$= -9 + 7i$    | (33) $(-9 - 9i) - (3 + 8i)$<br>$= -12 - 17i$   | (52) $(9 - 10i) + (-10 - 2i)$<br>$= -1 - 12i$  |
| (15) $(5 - 6i) + (-4 + 3i)$<br>$= 1 - 3i$      | (34) $(10 + 10i) - (10 - 8i)$<br>$= 18i$       | (53) $(1 - 2i) + (7 + 3i)$<br>$= 8 + i$        |
| (16) $(-7 - i) + (-5 - i)$<br>$= -12 - 2i$     | (35) $(5 + i) + (6 - 3i)$<br>$= 11 - 2i$       | (54) $(6 + 5i) + (-4 - 10i)$<br>$= 2 - 5i$     |
| (17) $(-2 - 7i) + (2 - 2i)$<br>$= -9i$         | (36) $(-9 + 2i) - (1 - 4i)$<br>$= -10 + 6i$    | (55) $(8 - 6i) - (-10 - 2i)$<br>$= 18 - 4i$    |
| (18) $(-2 + 3i) - (-10 - 8i)$<br>$= 8 + 11i$   | (37) $(7 + 5i) + (-4 + 9i)$<br>$= 3 + 14i$     | (56) $(-7 - i) + (4 - 5i)$<br>$= -3 - 6i$      |
| (19) $(-9 - 4i) - (9 + 9i)$<br>$= -18 - 13i$   | (38) $(3 - i) + (-4 + 5i)$<br>$= -1 + 4i$      | (57) $(-1 + 8i) + (3 - 4i)$<br>$= 2 + 4i$      |

- |                                              |                                               |                                              |
|----------------------------------------------|-----------------------------------------------|----------------------------------------------|
| (1) $(4 - 6i) + (-2 - 7i)$<br>$= 2 - 13i$    | (20) $(3 - 9i) - (8 - 2i)$<br>$= -5 - 7i$     | (39) $(1 + i) - (-7 + 6i)$<br>$= 8 - 5i$     |
| (2) $(4 + 6i) + (-9 + 8i)$<br>$= -5 + 14i$   | (21) $(7 + 6i) + (1 - 5i)$<br>$= 8 + i$       | (40) $(1 + 2i) + (1 + 9i)$<br>$= 2 + 11i$    |
| (3) $(7 - 5i) - (-4 - 4i)$<br>$= 11 - i$     | (22) $(-6 - 10i) - (2 - i)$<br>$= -8 - 9i$    | (41) $(8 - 8i) - (4 - 5i)$<br>$= 4 - 3i$     |
| (4) $(-7 - i) + (-6 + 2i)$<br>$= -13 + i$    | (23) $(8 + 4i) + (5 + 4i)$<br>$= 13 + 8i$     | (42) $(-10 - 2i) + (6 + 8i)$<br>$= -4 + 6i$  |
| (5) $(3 + i) - (-7 + 4i)$<br>$= 10 - 3i$     | (24) $(3 - 5i) + (10 + 3i)$<br>$= 13 - 2i$    | (43) $(-8 - 8i) - (-6 + i)$<br>$= -2 - 9i$   |
| (6) $(-6 + 3i) - (6 + 2i)$<br>$= -12 + i$    | (25) $(-7 + 9i) + (1 - 9i)$<br>$= -6$         | (44) $(6 - i) + (-10 + 3i)$<br>$= -4 + 2i$   |
| (7) $(-1 - 9i) - (-6 - i)$<br>$= 5 - 8i$     | (26) $(-7 - 5i) + (-9 + i)$<br>$= -16 - 4i$   | (45) $(-2 + i) + (-4 + 4i)$<br>$= -6 + 5i$   |
| (8) $(-5 + 3i) + (3 - 8i)$<br>$= -2 - 5i$    | (27) $(-3 + 5i) + (-8 - 10i)$<br>$= -11 - 5i$ | (46) $(-5 + 5i) - (7 - 2i)$<br>$= -12 + 7i$  |
| (9) $(-4 + 2i) + (-1 + 10i)$<br>$= -5 + 12i$ | (28) $(1 + 7i) + (-6 + 10i)$<br>$= -5 + 17i$  | (47) $(-2 - 3i) - (7 + 3i)$<br>$= -9 - 6i$   |
| (10) $(6 + 6i) - (6 - 8i)$<br>$= 14i$        | (29) $(-2 - 3i) + (4 - 2i)$<br>$= 2 - 5i$     | (48) $(9 - 10i) - (9 + 2i)$<br>$= -12i$      |
| (11) $(8 + 4i) - (2 + 10i)$<br>$= 6 - 6i$    | (30) $(-9 + 9i) + (4 + 6i)$<br>$= -5 + 15i$   | (49) $(-7 + 9i) - (5 + 6i)$<br>$= -12 + 3i$  |
| (12) $(-9 - i) + (9 - 5i)$<br>$= -6i$        | (31) $(3 - 7i) - (2 - 5i)$<br>$= 1 - 2i$      | (50) $(-2 + 10i) + (3 + 5i)$<br>$= 1 + 15i$  |
| (13) $(-10 + 5i) + (-9 - 6i)$<br>$= -19 - i$ | (32) $(-7 + 5i) - (-8 + 4i)$<br>$= 1 + i$     | (51) $(9 + 4i) + (-8 + i)$<br>$= 1 + 5i$     |
| (14) $(4 + 3i) - (-1 + 2i)$<br>$= 5 + i$     | (33) $(5 + 5i) + (-7 + 2i)$<br>$= -2 + 7i$    | (52) $(10 - 6i) - (6 + 5i)$<br>$= 4 - 11i$   |
| (15) $(3 + i) + (-4 + i)$<br>$= -1 + 2i$     | (34) $(4 + 2i) + (-6 + 6i)$<br>$= -2 + 8i$    | (53) $(10 + i) + (1 - 9i)$<br>$= 11 - 8i$    |
| (16) $(-5 - 2i) - (-1 + 3i)$<br>$= -4 - 5i$  | (35) $(10 + 5i) + (5 + 4i)$<br>$= 15 + 9i$    | (54) $(2 - 8i) + (6 - 7i)$<br>$= 8 - 15i$    |
| (17) $(8 + 9i) - (-3 + 10i)$<br>$= 11 - i$   | (36) $(4 + 4i) - (-8 - 7i)$<br>$= 12 + 11i$   | (55) $(-5 - 10i) - (7 + i)$<br>$= -12 - 11i$ |
| (18) $(-4 - i) - (10 - 3i)$<br>$= -14 + 2i$  | (37) $(4 + 6i) + (-4 + 5i)$<br>$= 11i$        | (56) $(-7 - 2i) + (2 - 6i)$<br>$= -5 - 8i$   |
| (19) $(10 + 10i) - (10 + 7i)$<br>$= 3i$      | (38) $(3 - 7i) + (-10 - 10i)$<br>$= -7 - 17i$ | (57) $(7 + i) - (-7 - 3i)$<br>$= 14 + 4i$    |

- |                                              |                                              |                                              |
|----------------------------------------------|----------------------------------------------|----------------------------------------------|
| (1) $(10 + 7i) + (6 - 8i)$<br>$= 16 - i$     | (20) $(7 - i) + (5 + 3i)$<br>$= 12 + 2i$     | (39) $(10 - 8i) - (-3 + 7i)$<br>$= 13 - 15i$ |
| (2) $(2 + 9i) + (-8 + 4i)$<br>$= -6 + 13i$   | (21) $(-8 - 9i) + (-1 - 7i)$<br>$= -9 - 16i$ | (40) $(7 + 10i) - (-6 + 9i)$<br>$= 13 + i$   |
| (3) $(-4 - 2i) - (6 + 7i)$<br>$= -10 - 9i$   | (22) $(-8 + 4i) - (-2 - i)$<br>$= -6 + 5i$   | (41) $(-3 + 9i) - (2 - 8i)$<br>$= -5 + 17i$  |
| (4) $(-10 - 6i) - (3 - 5i)$<br>$= -13 - i$   | (23) $(-6 + 3i) + (-7 - i)$<br>$= -13 + 2i$  | (42) $(6 + 6i) - (10 - 5i)$<br>$= -4 + 11i$  |
| (5) $(-7 + i) + (-10 - 4i)$<br>$= -17 - 3i$  | (24) $(10 + 4i) - (9 - 3i)$<br>$= 1 + 7i$    | (43) $(2 + 8i) - (5 - i)$<br>$= -3 + 9i$     |
| (6) $(-7 + 7i) + (-5 + 7i)$<br>$= -12 + 14i$ | (25) $(-2 - 6i) + (6 - 6i)$<br>$= 4 - 12i$   | (44) $(4 + 5i) + (9 - 3i)$<br>$= 13 + 2i$    |
| (7) $(-6 - 6i) - (-3 + 4i)$<br>$= -3 - 10i$  | (26) $(1 - 2i) + (-9 - 7i)$<br>$= -8 - 9i$   | (45) $(5 - i) - (-1 + 6i)$<br>$= 6 - 7i$     |
| (8) $(7 + 4i) + (2 - 7i)$<br>$= 9 - 3i$      | (27) $(-5 + 3i) - (-6 - 5i)$<br>$= 1 + 8i$   | (46) $(5 - 3i) + (4 + 8i)$<br>$= 9 + 5i$     |
| (9) $(-4 - 4i) + (10 - 5i)$<br>$= 6 - 9i$    | (28) $(6 - 9i) - (4 - i)$<br>$= 2 - 8i$      | (47) $(-1 - 9i) + (7 - 4i)$<br>$= 6 - 13i$   |
| (10) $(5 + 2i) + (-1 + 3i)$<br>$= 4 + 5i$    | (29) $(-1 - 9i) - (1 + 7i)$<br>$= -2 - 16i$  | (48) $(-2 - 5i) - (-5 + 9i)$<br>$= 3 - 14i$  |
| (11) $(2 - 2i) + (-9 + i)$<br>$= -7 - i$     | (30) $(4 + i) - (1 + 10i)$<br>$= 3 - 9i$     | (49) $(7 + 4i) - (6 - 10i)$<br>$= 1 + 14i$   |
| (12) $(6 + i) + (2 - 4i)$<br>$= 8 - 3i$      | (31) $(4 + 3i) - (-4 + 2i)$<br>$= 8 + i$     | (50) $(10 + 10i) + (3 - 10i)$<br>$= 13$      |
| (13) $(10 - 7i) + (7 - 7i)$<br>$= 17 - 14i$  | (32) $(9 + 9i) - (-3 + 3i)$<br>$= 12 + 6i$   | (51) $(3 + 2i) - (-6 + 10i)$<br>$= 9 - 8i$   |
| (14) $(8 - 10i) + (-2 - 6i)$<br>$= 6 - 16i$  | (33) $(5 - 7i) + (3 - 4i)$<br>$= 8 - 11i$    | (52) $(-8 + i) + (-4 + 8i)$<br>$= -12 + 9i$  |
| (15) $(9 + 2i) - (8 + 5i)$<br>$= 1 - 3i$     | (34) $(4 - 4i) + (-6 - 9i)$<br>$= -2 - 13i$  | (53) $(-8 + 3i) + (-2 - 3i)$<br>$= -10$      |
| (16) $(9 - 4i) + (9 - i)$<br>$= 18 - 5i$     | (35) $(-2 + i) + (1 - 8i)$<br>$= -1 - 7i$    | (54) $(3 - 3i) - (4 - 10i)$<br>$= -1 + 7i$   |
| (17) $(-3 + 2i) - (4 + 7i)$<br>$= -7 - 5i$   | (36) $(5 + 7i) - (1 - 6i)$<br>$= 4 + 13i$    | (55) $(7 + 9i) + (10 + 4i)$<br>$= 17 + 13i$  |
| (18) $(-10 + 5i) + (7 - 6i)$<br>$= -3 - i$   | (37) $(4 + 6i) + (7 + 6i)$<br>$= 11 + 12i$   | (56) $(6 + 5i) + (-2 + 4i)$<br>$= 4 + 9i$    |
| (19) $(1 + 10i) - (-5 - 3i)$<br>$= 6 + 13i$  | (38) $(-9 + 9i) - (6 + 9i)$<br>$= -15$       | (57) $(9 - 10i) + (3 + 7i)$<br>$= 12 - 3i$   |

- |                                                |                                                |                                               |
|------------------------------------------------|------------------------------------------------|-----------------------------------------------|
| (1) $(10 - 6i) - (-9 - 8i)$<br>$= 19 + 2i$     | (20) $(7 + 7i) + (2 - i)$<br>$= 9 + 6i$        | (39) $(6 - 4i) - (9 - 2i)$<br>$= -3 - 2i$     |
| (2) $(-7 + 7i) + (-8 - 6i)$<br>$= -15 + i$     | (21) $(1 - 6i) + (8 - 8i)$<br>$= 9 - 14i$      | (40) $(7 - i) + (9 + 10i)$<br>$= 16 + 9i$     |
| (3) $(9 + i) - (-9 + 6i)$<br>$= 18 - 5i$       | (22) $(-8 - 5i) + (-9 + 6i)$<br>$= -17 + i$    | (41) $(-3 - 3i) - (7 + 5i)$<br>$= -10 - 8i$   |
| (4) $(5 + 5i) + (4 + 3i)$<br>$= 9 + 8i$        | (23) $(4 - 9i) + (5 - 6i)$<br>$= 9 - 15i$      | (42) $(-5 + 5i) - (5 + 7i)$<br>$= -10 - 2i$   |
| (5) $(-9 + 2i) + (-3 - 10i)$<br>$= -12 - 8i$   | (24) $(10 + 4i) - (-6 + 6i)$<br>$= 16 - 2i$    | (43) $(-8 + i) + (-3 + 7i)$<br>$= -11 + 8i$   |
| (6) $(-3 - 4i) - (3 - 8i)$<br>$= -6 + 4i$      | (25) $(8 + 2i) - (-1 - 7i)$<br>$= 9 + 9i$      | (44) $(-9 - 9i) - (6 + 10i)$<br>$= -15 - 19i$ |
| (7) $(6 + 3i) - (-8 + 10i)$<br>$= 14 - 7i$     | (26) $(10 - 6i) + (7 + 7i)$<br>$= 17 + i$      | (45) $(-10 + 6i) - (2 - 2i)$<br>$= -12 + 8i$  |
| (8) $(-9 - 2i) + (10 + 9i)$<br>$= 1 + 7i$      | (27) $(-1 - 3i) - (3 + 2i)$<br>$= -4 - 5i$     | (46) $(7 + 2i) + (-9 + 4i)$<br>$= -2 + 6i$    |
| (9) $(-4 - 4i) + (-10 + 8i)$<br>$= -14 + 4i$   | (28) $(-2 - 10i) - (-4 + 2i)$<br>$= 2 - 12i$   | (47) $(10 - i) + (6 + 8i)$<br>$= 16 + 7i$     |
| (10) $(1 - 10i) + (-1 - 2i)$<br>$= -12i$       | (29) $(-5 - 5i) - (3 + 7i)$<br>$= -8 - 12i$    | (48) $(5 + 8i) - (-9 + 5i)$<br>$= 14 + 3i$    |
| (11) $(-2 + 8i) + (-10 - 10i)$<br>$= -12 - 2i$ | (30) $(-3 + 4i) + (-7 + i)$<br>$= -10 + 5i$    | (49) $(-6 - 4i) - (6 - 8i)$<br>$= -12 + 4i$   |
| (12) $(9 - 3i) - (4 + 9i)$<br>$= 5 - 12i$      | (31) $(10 + 2i) - (-9 + 6i)$<br>$= 19 - 4i$    | (50) $(7 - i) + (-5 - 7i)$<br>$= 2 - 8i$      |
| (13) $(9 - i) - (-3 - i)$<br>$= 12$            | (32) $(5 + 5i) - (8 + i)$<br>$= -3 + 4i$       | (51) $(6 + 2i) - (-2 - 2i)$<br>$= 8 + 4i$     |
| (14) $(-6 - 9i) - (3 - 9i)$<br>$= -9$          | (33) $(-5 - 4i) - (-4 - 7i)$<br>$= -1 + 3i$    | (52) $(-1 + 6i) + (8 - i)$<br>$= 7 + 5i$      |
| (15) $(-4 + 4i) + (-9 + 2i)$<br>$= -13 + 6i$   | (34) $(-8 + 2i) - (4 - 2i)$<br>$= -12 + 4i$    | (53) $(-3 + 8i) + (-2 - 10i)$<br>$= -5 - 2i$  |
| (16) $(6 + 8i) + (-10 + 5i)$<br>$= -4 + 13i$   | (35) $(-1 - 3i) + (8 - i)$<br>$= 7 - 4i$       | (54) $(4 + 6i) - (10 + i)$<br>$= -6 + 5i$     |
| (17) $(7 - 7i) - (-8 + i)$<br>$= 15 - 8i$      | (36) $(-3 + 9i) + (-9 + 5i)$<br>$= -12 + 14i$  | (55) $(10 - 9i) + (1 + 4i)$<br>$= 11 - 5i$    |
| (18) $(-8 + 9i) - (-2 + 4i)$<br>$= -6 + 5i$    | (37) $(-9 - 6i) - (7 - 8i)$<br>$= -16 + 2i$    | (56) $(-5 + 10i) - (-4 - 6i)$<br>$= -1 + 16i$ |
| (19) $(-8 - 2i) - (-4 - 8i)$<br>$= -4 + 6i$    | (38) $(-1 + 7i) + (-10 - 10i)$<br>$= -11 - 3i$ | (57) $(-1 - 2i) - (-3 + 6i)$<br>$= 2 - 8i$    |



- |                                               |                                               |                                               |
|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| (1) $(10 - 3i) - (6 + 7i)$<br>$= 4 - 10i$     | (20) $(-8 - 10i) + (5 - 10i)$<br>$= -3 - 20i$ | (39) $(7 + i) - (-10 + 10i)$<br>$= 17 - 9i$   |
| (2) $(7 - 9i) - (-6 - 7i)$<br>$= 13 - 2i$     | (21) $(-9 + 8i) + (4 - 4i)$<br>$= -5 + 4i$    | (40) $(-9 - 10i) + (-3 + 4i)$<br>$= -12 - 6i$ |
| (3) $(-7 + 7i) - (-9 - i)$<br>$= 2 + 8i$      | (22) $(-1 + 5i) - (-8 + 7i)$<br>$= 7 - 2i$    | (41) $(1 - 2i) + (5 - 6i)$<br>$= 6 - 8i$      |
| (4) $(9 + 10i) - (-1 + 3i)$<br>$= 10 + 7i$    | (23) $(-5 + 6i) + (8 - 8i)$<br>$= 3 - 2i$     | (42) $(4 - 3i) - (1 + 9i)$<br>$= 3 - 12i$     |
| (5) $(-2 + 7i) - (-7 + i)$<br>$= 5 + 6i$      | (24) $(-2 + 4i) + (7 + 4i)$<br>$= 5 + 8i$     | (43) $(-5 + 9i) + (-8 + 4i)$<br>$= -13 + 13i$ |
| (6) $(-7 + 8i) - (10 - 4i)$<br>$= -17 + 12i$  | (25) $(-8 - 3i) - (-6 + 9i)$<br>$= -2 - 12i$  | (44) $(10 - i) + (-5 - 10i)$<br>$= 5 - 11i$   |
| (7) $(2 - 4i) - (-6 + 5i)$<br>$= 8 - 9i$      | (26) $(-3 - 10i) + (-5 - 4i)$<br>$= -8 - 14i$ | (45) $(9 + 6i) + (1 - 8i)$<br>$= 10 - 2i$     |
| (8) $(-4 - 7i) - (-10 + 6i)$<br>$= 6 - 13i$   | (27) $(-1 - 7i) - (2 - 5i)$<br>$= -3 - 2i$    | (46) $(8 - 5i) - (8 - 5i)$<br>$= 0$           |
| (9) $(3 + 7i) - (10 - 2i)$<br>$= -7 + 9i$     | (28) $(9 + 7i) - (-8 - 7i)$<br>$= 17 + 14i$   | (47) $(-2 - 6i) + (-4 + i)$<br>$= -6 - 5i$    |
| (10) $(1 - 7i) + (-10 - i)$<br>$= -9 - 8i$    | (29) $(-5 - 7i) + (-9 - 9i)$<br>$= -14 - 16i$ | (48) $(-6 + i) - (1 - 9i)$<br>$= -7 + 10i$    |
| (11) $(-5 - 9i) + (-8 - 6i)$<br>$= -13 - 15i$ | (30) $(9 + 10i) - (7 + i)$<br>$= 2 + 9i$      | (49) $(4 + i) + (-7 + 3i)$<br>$= -3 + 4i$     |
| (12) $(5 - 9i) + (7 + 10i)$<br>$= 12 + i$     | (31) $(4 + 7i) - (1 + 9i)$<br>$= 3 - 2i$      | (50) $(3 - 9i) - (10 + 9i)$<br>$= -7 - 18i$   |
| (13) $(4 - 9i) + (10 - 2i)$<br>$= 14 - 11i$   | (32) $(5 + 4i) - (8 + 3i)$<br>$= -3 + i$      | (51) $(-10 + 5i) - (4 - i)$<br>$= -14 + 6i$   |
| (14) $(5 - 4i) - (7 + 8i)$<br>$= -2 - 12i$    | (33) $(-7 + 10i) + (-8 - 2i)$<br>$= -15 + 8i$ | (52) $(-7 + 3i) - (-1 - 6i)$<br>$= -6 + 9i$   |
| (15) $(1 + 6i) - (1 - 7i)$<br>$= 13i$         | (34) $(-2 + 7i) - (3 + 6i)$<br>$= -5 + i$     | (53) $(-7 + 8i) + (8 + 7i)$<br>$= 1 + 15i$    |
| (16) $(-3 - 9i) - (1 + 7i)$<br>$= -4 - 16i$   | (35) $(1 + 10i) - (8 - 7i)$<br>$= -7 + 17i$   | (54) $(9 - 2i) + (2 - 3i)$<br>$= 11 - 5i$     |
| (17) $(5 - 3i) - (3 - 9i)$<br>$= 2 + 6i$      | (36) $(-5 + 2i) + (-10 - i)$<br>$= -15 + i$   | (55) $(1 + 7i) + (9 + 5i)$<br>$= 10 + 12i$    |
| (18) $(-7 + 6i) + (2 + 4i)$<br>$= -5 + 10i$   | (37) $(-3 + 2i) + (4 - 9i)$<br>$= 1 - 7i$     | (56) $(7 - 4i) - (5 + 2i)$<br>$= 2 - 6i$      |
| (19) $(3 + 8i) + (2 + 10i)$<br>$= 5 + 18i$    | (38) $(6 - 2i) - (6 - 3i)$<br>$= i$           | (57) $(4 + 8i) + (2 + 8i)$<br>$= 6 + 16i$     |

- |                                              |                                              |                                              |
|----------------------------------------------|----------------------------------------------|----------------------------------------------|
| (1) $(-3 - 3i) - (3 + i)$<br>$= -6 - 4i$     | (20) $(4 + 6i) - (1 + 2i)$<br>$= 3 + 4i$     | (39) $(2 - 3i) - (2 - 7i)$<br>$= 4i$         |
| (2) $(4 - 5i) + (8 - i)$<br>$= 12 - 6i$      | (21) $(-8 + 3i) + (-3 - i)$<br>$= -11 + 2i$  | (40) $(3 + 2i) + (10 - 5i)$<br>$= 13 - 3i$   |
| (3) $(3 + 6i) + (-7 + 10i)$<br>$= -4 + 16i$  | (22) $(1 - i) + (-10 - 3i)$<br>$= -9 - 4i$   | (41) $(2 + 4i) - (7 + 4i)$<br>$= -5$         |
| (4) $(1 + 10i) - (10 + 4i)$<br>$= -9 + 6i$   | (23) $(-4 + i) - (4 + 3i)$<br>$= -8 - 2i$    | (42) $(-1 - 5i) + (-2 - 3i)$<br>$= -3 - 8i$  |
| (5) $(-5 - 5i) + (-4 - 7i)$<br>$= -9 - 12i$  | (24) $(8 + 8i) - (-4 - 4i)$<br>$= 12 + 12i$  | (43) $(-6 + 9i) - (-10 - i)$<br>$= 4 + 10i$  |
| (6) $(-9 - 4i) + (-4 + 4i)$<br>$= -13$       | (25) $(5 + 10i) + (7 - i)$<br>$= 12 + 9i$    | (44) $(10 - 9i) - (-5 + 9i)$<br>$= 15 - 18i$ |
| (7) $(8 - 5i) - (-4 - 2i)$<br>$= 12 - 3i$    | (26) $(7 + 10i) + (9 - 3i)$<br>$= 16 + 7i$   | (45) $(-4 - i) + (10 - 8i)$<br>$= 6 - 9i$    |
| (8) $(1 + 3i) + (-8 - 9i)$<br>$= -7 - 6i$    | (27) $(3 - 10i) - (-8 - i)$<br>$= 11 - 9i$   | (46) $(2 + 8i) - (5 + 10i)$<br>$= -3 - 2i$   |
| (9) $(3 - 10i) + (-2 - 10i)$<br>$= 1 - 20i$  | (28) $(-6 + 3i) + (-1 - 10i)$<br>$= -7 - 7i$ | (47) $(9 + 8i) + (8 - 8i)$<br>$= 17$         |
| (10) $(-4 + 3i) + (7 + i)$<br>$= 3 + 4i$     | (29) $(4 + 6i) - (-9 + 8i)$<br>$= 13 - 2i$   | (48) $(-6 + 6i) - (-9 + 10i)$<br>$= 3 - 4i$  |
| (11) $(-8 - 6i) - (1 + i)$<br>$= -9 - 7i$    | (30) $(-3 - 7i) + (3 + 3i)$<br>$= -4i$       | (49) $(2 - 9i) - (-2 - 2i)$<br>$= 4 - 7i$    |
| (12) $(8 - 3i) + (-7 + 9i)$<br>$= 1 + 6i$    | (31) $(7 - i) + (-9 - 10i)$<br>$= -2 - 11i$  | (50) $(-7 + 9i) + (-6 - 7i)$<br>$= -13 + 2i$ |
| (13) $(-3 - 7i) + (-9 + 6i)$<br>$= -12 - i$  | (32) $(2 - i) - (10 - 7i)$<br>$= -8 + 6i$    | (51) $(-10 - 5i) - (-2 + 4i)$<br>$= -8 - 9i$ |
| (14) $(-7 + 10i) - (-1 + 5i)$<br>$= -6 + 5i$ | (33) $(2 - i) + (10 + 10i)$<br>$= 12 + 9i$   | (52) $(10 + 8i) - (1 - 4i)$<br>$= 9 + 12i$   |
| (15) $(-4 - 7i) - (8 - 10i)$<br>$= -12 + 3i$ | (34) $(10 + 7i) - (-2 - 6i)$<br>$= 12 + 13i$ | (53) $(-7 + 7i) - (2 + 8i)$<br>$= -9 - i$    |
| (16) $(-2 + 9i) + (-1 - 2i)$<br>$= -3 + 7i$  | (35) $(10 + 10i) + (-7 + i)$<br>$= 3 + 11i$  | (54) $(2 - i) + (9 + i)$<br>$= 11$           |
| (17) $(-7 + i) + (-3 - 5i)$<br>$= -10 - 4i$  | (36) $(1 + i) + (6 + 7i)$<br>$= 7 + 8i$      | (55) $(9 - 7i) + (7 - 8i)$<br>$= 16 - 15i$   |
| (18) $(-1 + 9i) - (-7 - 10i)$<br>$= 6 + 19i$ | (37) $(4 + 7i) - (1 - 10i)$<br>$= 3 + 17i$   | (56) $(-4 - 4i) + (4 - 7i)$<br>$= -11i$      |
| (19) $(4 - i) + (8 - 7i)$<br>$= 12 - 8i$     | (38) $(5 - 6i) + (-7 - 4i)$<br>$= -2 - 10i$  | (57) $(4 + 9i) - (-8 + 9i)$<br>$= 12$        |

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|----------------------------------------------|----------------------------------------------|------------------------------------------------|
| (1) $(-4 + 8i) + (1 + i)$<br>$= -3 + 9i$     | (20) $(-1 - 2i) - (5 + 3i)$<br>$= -6 - 5i$   | (39) $(1 - 8i) - (4 + 7i)$<br>$= -3 - 15i$     |
| (2) $(-1 - i) + (1 + 9i)$<br>$= 8i$          | (21) $(-4 - i) - (7 - 10i)$<br>$= -11 + 9i$  | (40) $(4 - 3i) + (-2 + 4i)$<br>$= 2 + i$       |
| (3) $(-2 - 7i) - (-8 - 3i)$<br>$= 6 - 4i$    | (22) $(2 + 7i) - (-7 - 2i)$<br>$= 9 + 9i$    | (41) $(-6 + 3i) + (-5 + 10i)$<br>$= -11 + 13i$ |
| (4) $(-9 - 6i) - (-9 - i)$<br>$= -5i$        | (23) $(7 - 4i) - (-10 + 7i)$<br>$= 17 - 11i$ | (42) $(-6 - 8i) - (-7 + 3i)$<br>$= 1 - 11i$    |
| (5) $(-1 - i) + (5 + 5i)$<br>$= 4 + 4i$      | (24) $(2 + 8i) + (3 + 2i)$<br>$= 5 + 10i$    | (43) $(-4 - 10i) - (4 + 2i)$<br>$= -8 - 12i$   |
| (6) $(5 - 5i) + (7 + 5i)$<br>$= 12$          | (25) $(-8 + 9i) - (-9 + 7i)$<br>$= 1 + 2i$   | (44) $(6 + 10i) - (8 - 4i)$<br>$= -2 + 14i$    |
| (7) $(-8 - 5i) + (8 - 7i)$<br>$= -12i$       | (26) $(-6 + 4i) + (-3 - 8i)$<br>$= -9 - 4i$  | (45) $(-9 + 2i) - (-8 + 8i)$<br>$= -1 - 6i$    |
| (8) $(2 + i) - (-5 + 3i)$<br>$= 7 - 2i$      | (27) $(-7 + 8i) + (1 - 10i)$<br>$= -6 - 2i$  | (46) $(-2 + 3i) - (-8 - 10i)$<br>$= 6 + 13i$   |
| (9) $(10 - i) + (-6 - 8i)$<br>$= 4 - 9i$     | (28) $(-7 - 8i) - (-10 - 10i)$<br>$= 3 + 2i$ | (47) $(-10 + 4i) + (7 - 8i)$<br>$= -3 - 4i$    |
| (10) $(-10 + 9i) + (1 - 2i)$<br>$= -9 + 7i$  | (29) $(-3 + 9i) - (3 + 10i)$<br>$= -6 - i$   | (48) $(-3 + 8i) - (7 + 10i)$<br>$= -10 - 2i$   |
| (11) $(-4 - 10i) - (8 - 3i)$<br>$= -12 - 7i$ | (30) $(4 - 7i) + (8 + 7i)$<br>$= 12$         | (49) $(-1 + 8i) - (-2 + 7i)$<br>$= 1 + i$      |
| (12) $(6 - 5i) - (10 + 5i)$<br>$= -4 - 10i$  | (31) $(-10 + 3i) + (-2 - 4i)$<br>$= -12 - i$ | (50) $(5 - 4i) - (5 - i)$<br>$= -3i$           |
| (13) $(-6 - 5i) - (-2 - 5i)$<br>$= -4$       | (32) $(10 - 2i) - (-1 + 6i)$<br>$= 11 - 8i$  | (51) $(1 - i) - (9 + 5i)$<br>$= -8 - 6i$       |
| (14) $(-5 - 8i) + (9 + 3i)$<br>$= 4 - 5i$    | (33) $(10 - 9i) - (-3 + 6i)$<br>$= 13 - 15i$ | (52) $(-6 + i) + (7 + 8i)$<br>$= 1 + 9i$       |
| (15) $(8 - 9i) - (8 - 2i)$<br>$= -7i$        | (34) $(-8 - 3i) - (-4 - 4i)$<br>$= -4 + i$   | (53) $(-3 + 3i) - (5 + 7i)$<br>$= -8 - 4i$     |
| (16) $(2 - 10i) - (-5 + 5i)$<br>$= 7 - 15i$  | (35) $(-1 - 4i) - (-5 - i)$<br>$= 4 - 3i$    | (54) $(8 - i) - (-9 - 9i)$<br>$= 17 + 8i$      |
| (17) $(1 + 2i) + (9 + 2i)$<br>$= 10 + 4i$    | (36) $(-3 + 7i) + (7 + 4i)$<br>$= 4 + 11i$   | (55) $(-6 + 6i) + (8 + 8i)$<br>$= 2 + 14i$     |
| (18) $(2 - 10i) - (-3 + 3i)$<br>$= 5 - 13i$  | (37) $(10 - 7i) - (-1 - 3i)$<br>$= 11 - 4i$  | (56) $(10 - 8i) + (7 - 9i)$<br>$= 17 - 17i$    |
| (19) $(-4 + 9i) + (1 - 10i)$<br>$= -3 - i$   | (38) $(-9 + 9i) + (9 + 4i)$<br>$= 13i$       | (57) $(1 - 7i) - (1 - 7i)$<br>$= 0$            |

- |                                                |                                               |                                                |
|------------------------------------------------|-----------------------------------------------|------------------------------------------------|
| (1) $(-7 - 10i) - (-8 + 7i)$<br>$= 1 - 17i$    | (20) $(-5 - 10i) - (9 - 7i)$<br>$= -14 - 3i$  | (39) $(-9 + i) - (-4 - 6i)$<br>$= -5 + 7i$     |
| (2) $(-2 + 7i) + (10 - 6i)$<br>$= 8 + i$       | (21) $(2 + 9i) - (9 + 3i)$<br>$= -7 + 6i$     | (40) $(4 - i) - (-9 - 6i)$<br>$= 13 + 5i$      |
| (3) $(4 + 9i) + (6 - 9i)$<br>$= 10$            | (22) $(-8 + i) + (2 - 7i)$<br>$= -6 - 6i$     | (41) $(-2 - 10i) + (3 - 3i)$<br>$= 1 - 13i$    |
| (4) $(-3 + 4i) - (8 + 6i)$<br>$= -11 - 2i$     | (23) $(1 - 7i) - (7 + 4i)$<br>$= -6 - 11i$    | (42) $(-10 - 8i) - (-7 + 6i)$<br>$= -3 - 14i$  |
| (5) $(3 + 8i) - (2 - 10i)$<br>$= 1 + 18i$      | (24) $(-10 + 3i) + (-10 - 2i)$<br>$= -20 + i$ | (43) $(-3 + 8i) + (7 - 8i)$<br>$= 4$           |
| (6) $(-9 - 9i) - (-7 + 9i)$<br>$= -2 - 18i$    | (25) $(9 + 4i) - (-6 + 2i)$<br>$= 15 + 2i$    | (44) $(9 - 7i) + (-1 + 5i)$<br>$= 8 - 2i$      |
| (7) $(3 + i) - (-5 + 3i)$<br>$= 8 - 2i$        | (26) $(3 - 10i) + (9 - 4i)$<br>$= 12 - 14i$   | (45) $(-5 - 9i) + (6 - i)$<br>$= 1 - 10i$      |
| (8) $(2 - 6i) - (2 + 3i)$<br>$= -9i$           | (27) $(-9 - i) - (6 - 8i)$<br>$= -15 + 7i$    | (46) $(-3 - 10i) + (-4 - 2i)$<br>$= -7 - 12i$  |
| (9) $(4 + 9i) + (-6 + 3i)$<br>$= -2 + 12i$     | (28) $(-6 - 7i) + (5 - 7i)$<br>$= -1 - 14i$   | (47) $(6 - 7i) + (6 - 5i)$<br>$= 12 - 12i$     |
| (10) $(-9 + 6i) - (1 + 8i)$<br>$= -10 - 2i$    | (29) $(5 - 3i) - (2 + 10i)$<br>$= 3 - 13i$    | (48) $(1 - 8i) + (6 - 2i)$<br>$= 7 - 10i$      |
| (11) $(3 + 6i) + (-8 - 5i)$<br>$= -5 + i$      | (30) $(-9 - 4i) - (-1 - 6i)$<br>$= -8 + 2i$   | (49) $(-9 + 10i) + (-8 + 3i)$<br>$= -17 + 13i$ |
| (12) $(-3 - 4i) - (-1 + 9i)$<br>$= -2 - 13i$   | (31) $(-3 + 7i) + (2 + 6i)$<br>$= -1 + 13i$   | (50) $(-4 + 3i) - (9 + 6i)$<br>$= -13 - 3i$    |
| (13) $(3 + 2i) + (-2 + 3i)$<br>$= 1 + 5i$      | (32) $(-1 - 10i) - (10 - 10i)$<br>$= -11$     | (51) $(-7 - 8i) - (-9 + 7i)$<br>$= 2 - 15i$    |
| (14) $(-9 - 10i) + (-9 - 4i)$<br>$= -18 - 14i$ | (33) $(-1 + i) + (-4 + 3i)$<br>$= -5 + 4i$    | (52) $(5 + 5i) - (-3 - 9i)$<br>$= 8 + 14i$     |
| (15) $(-3 - 10i) - (-4 + 7i)$<br>$= 1 - 17i$   | (34) $(-8 + 6i) + (2 - 4i)$<br>$= -6 + 2i$    | (53) $(-1 - i) - (4 + 4i)$<br>$= -5 - 5i$      |
| (16) $(10 - 7i) + (-4 + i)$<br>$= 6 - 6i$      | (35) $(-10 - 6i) - (-2 + 8i)$<br>$= -8 - 14i$ | (54) $(10 - 7i) + (-7 + i)$<br>$= 3 - 6i$      |
| (17) $(-10 - 5i) + (-4 - 9i)$<br>$= -14 - 14i$ | (36) $(-8 - 7i) - (10 + 8i)$<br>$= -18 - 15i$ | (55) $(3 + 4i) + (-6 + 8i)$<br>$= -3 + 12i$    |
| (18) $(7 - i) + (-7 - 3i)$<br>$= -4i$          | (37) $(-2 + 8i) - (5 - 2i)$<br>$= -7 + 10i$   | (56) $(10 + 2i) + (7 + i)$<br>$= 17 + 3i$      |
| (19) $(-6 - 9i) + (-1 - 5i)$<br>$= -7 - 14i$   | (38) $(-4 - 4i) - (-7 - 5i)$<br>$= 3 + i$     | (57) $(-7 + i) - (8 + 2i)$<br>$= -15 - i$      |

- |                                              |                                                |                                               |
|----------------------------------------------|------------------------------------------------|-----------------------------------------------|
| (1) $(-2 + i) - (-1 - 5i)$<br>$= -1 + 6i$    | (20) $(-8 - 4i) + (-8 - 9i)$<br>$= -16 - 13i$  | (39) $(5 - 8i) - (-2 - 3i)$<br>$= 7 - 5i$     |
| (2) $(-6 - 5i) - (8 + 8i)$<br>$= -14 - 13i$  | (21) $(-4 - 10i) + (-3 - 4i)$<br>$= -7 - 14i$  | (40) $(-7 + 5i) - (5 + 4i)$<br>$= -12 + i$    |
| (3) $(-6 + 10i) + (-1 - 8i)$<br>$= -7 + 2i$  | (22) $(3 + 10i) - (10 + 8i)$<br>$= -7 + 2i$    | (41) $(-8 - 6i) - (-8 - 9i)$<br>$= 3i$        |
| (4) $(7 - 10i) + (9 + i)$<br>$= 16 - 9i$     | (23) $(10 + 2i) - (1 - 7i)$<br>$= 9 + 9i$      | (42) $(3 + 8i) + (9 - 10i)$<br>$= 12 - 2i$    |
| (5) $(-6 - 6i) - (-9 - 5i)$<br>$= 3 - i$     | (24) $(-7 + 8i) + (-8 + 4i)$<br>$= -15 + 12i$  | (43) $(-3 - i) - (-9 - 8i)$<br>$= 6 + 7i$     |
| (6) $(-6 - 7i) - (-10 - 8i)$<br>$= 4 + i$    | (25) $(5 + 9i) + (3 - 2i)$<br>$= 8 + 7i$       | (44) $(5 + 9i) + (4 - 10i)$<br>$= 9 - i$      |
| (7) $(5 + i) + (4 - 6i)$<br>$= 9 - 5i$       | (26) $(3 - 4i) + (-10 + 9i)$<br>$= -7 + 5i$    | (45) $(-4 + 7i) - (-8 + 9i)$<br>$= 4 - 2i$    |
| (8) $(1 - 5i) + (3 - 10i)$<br>$= 4 - 15i$    | (27) $(6 + 3i) - (-6 + 9i)$<br>$= 12 - 6i$     | (46) $(5 - 4i) - (-2 + 4i)$<br>$= 7 - 8i$     |
| (9) $(3 + 5i) + (-2 + 3i)$<br>$= 1 + 8i$     | (28) $(2 + 2i) + (-9 - 9i)$<br>$= -7 - 7i$     | (47) $(-10 - 2i) + (2 - 10i)$<br>$= -8 - 12i$ |
| (10) $(2 - 7i) - (-7 + 4i)$<br>$= 9 - 11i$   | (29) $(-6 - i) - (4 + i)$<br>$= -10 - 2i$      | (48) $(3 + 4i) + (3 + 2i)$<br>$= 6 + 6i$      |
| (11) $(6 + 3i) - (10 - 6i)$<br>$= -4 + 9i$   | (30) $(-3 - 8i) + (-1 - 9i)$<br>$= -4 - 17i$   | (49) $(-9 - 6i) + (10 - 7i)$<br>$= 1 - 13i$   |
| (12) $(10 - 3i) + (-9 - 5i)$<br>$= 1 - 8i$   | (31) $(-9 + 7i) - (-2 + 2i)$<br>$= -7 + 5i$    | (50) $(-9 - 4i) - (4 - 6i)$<br>$= -13 + 2i$   |
| (13) $(-5 + 4i) + (-10 - i)$<br>$= -15 + 3i$ | (32) $(2 - 4i) + (6 + 9i)$<br>$= 8 + 5i$       | (51) $(-1 - 4i) + (-9 + 6i)$<br>$= -10 + 2i$  |
| (14) $(10 - 6i) - (-1 + 9i)$<br>$= 11 - 15i$ | (33) $(2 - 7i) + (3 - 10i)$<br>$= 5 - 17i$     | (52) $(-8 - 3i) - (-2 + 6i)$<br>$= -6 - 9i$   |
| (15) $(5 - 9i) - (6 - 4i)$<br>$= -1 - 5i$    | (34) $(-9 - 4i) + (5 + i)$<br>$= -4 - 3i$      | (53) $(-4 - 7i) + (9 + 8i)$<br>$= 5 + i$      |
| (16) $(-5 - i) - (-3 + 7i)$<br>$= -2 - 8i$   | (35) $(-3 - 10i) + (1 + 8i)$<br>$= -2 - 2i$    | (54) $(-1 + 4i) + (10 - 10i)$<br>$= 9 - 6i$   |
| (17) $(-8 + 4i) + (2 + 6i)$<br>$= -6 + 10i$  | (36) $(-6 + 9i) + (-10 + 8i)$<br>$= -16 + 17i$ | (55) $(8 + 5i) + (-2 - 10i)$<br>$= 6 - 5i$    |
| (18) $(2 + 10i) - (3 + 4i)$<br>$= -1 + 6i$   | (37) $(-6 + 2i) - (2 + 4i)$<br>$= -8 - 2i$     | (56) $(6 - 4i) - (2 - 8i)$<br>$= 4 + 4i$      |
| (19) $(3 - 4i) + (2 - 3i)$<br>$= 5 - 7i$     | (38) $(-7 - 3i) + (5 + 4i)$<br>$= -2 + i$      | (57) $(10 - 9i) + (-3 + 9i)$<br>$= 7$         |

- |                                              |                                                |                                               |
|----------------------------------------------|------------------------------------------------|-----------------------------------------------|
| (1) $(10 + 6i) - (9 - 9i)$<br>$= 1 + 15i$    | (20) $(2 - 3i) + (-4 - 8i)$<br>$= -2 - 11i$    | (39) $(6 + 3i) - (-3 + 9i)$<br>$= 9 - 6i$     |
| (2) $(-6 - 4i) - (9 + 2i)$<br>$= -15 - 6i$   | (21) $(6 + 6i) + (2 + 3i)$<br>$= 8 + 9i$       | (40) $(-10 - 3i) - (10 - 7i)$<br>$= -20 + 4i$ |
| (3) $(-4 + 5i) - (6 - 5i)$<br>$= -10 + 10i$  | (22) $(6 + 5i) - (-1 + 3i)$<br>$= 7 + 2i$      | (41) $(2 - 8i) - (3 - 6i)$<br>$= -1 - 2i$     |
| (4) $(-9 - 5i) - (-6 - 9i)$<br>$= -3 + 4i$   | (23) $(-10 - 3i) - (-3 + 10i)$<br>$= -7 - 13i$ | (42) $(-7 - i) + (4 + 10i)$<br>$= -3 + 9i$    |
| (5) $(10 + 5i) + (7 - 10i)$<br>$= 17 - 5i$   | (24) $(-2 + 7i) - (-1 + 2i)$<br>$= -1 + 5i$    | (43) $(-5 - 8i) + (-1 - 8i)$<br>$= -6 - 16i$  |
| (6) $(9 + 9i) + (8 + 7i)$<br>$= 17 + 16i$    | (25) $(-3 - 10i) - (3 - 7i)$<br>$= -6 - 3i$    | (44) $(10 + i) - (-3 - 2i)$<br>$= 13 + 3i$    |
| (7) $(-8 - 10i) - (-10 + 7i)$<br>$= 2 - 17i$ | (26) $(-10 + 2i) + (10 - 9i)$<br>$= -7i$       | (45) $(-9 - 7i) - (4 + 5i)$<br>$= -13 - 12i$  |
| (8) $(-1 + 3i) - (-6 - i)$<br>$= 5 + 4i$     | (27) $(9 + 7i) + (-6 - 6i)$<br>$= 3 + i$       | (46) $(-4 - 5i) + (-5 + 4i)$<br>$= -9 - i$    |
| (9) $(5 + i) + (-5 - 10i)$<br>$= -9i$        | (28) $(6 - 2i) - (3 + 10i)$<br>$= 3 - 12i$     | (47) $(7 - 9i) - (-4 + 9i)$<br>$= 11 - 18i$   |
| (10) $(-5 - 3i) - (1 + 3i)$<br>$= -6 - 6i$   | (29) $(1 + 6i) + (9 - 6i)$<br>$= 10$           | (48) $(5 - 5i) + (-2 + 2i)$<br>$= 3 - 3i$     |
| (11) $(4 - 6i) - (6 + 10i)$<br>$= -2 - 16i$  | (30) $(7 - 8i) + (3 + 10i)$<br>$= 10 + 2i$     | (49) $(-7 - 7i) + (4 - 3i)$<br>$= -3 - 10i$   |
| (12) $(3 - i) - (-9 + 6i)$<br>$= 12 - 7i$    | (31) $(-3 + 3i) + (-5 + 10i)$<br>$= -8 + 13i$  | (50) $(-4 - 7i) + (-2 + 3i)$<br>$= -6 - 4i$   |
| (13) $(4 + 9i) - (7 - 9i)$<br>$= -3 + 18i$   | (32) $(-8 - 5i) + (-1 + 8i)$<br>$= -9 + 3i$    | (51) $(3 - 6i) - (10 + i)$<br>$= -7 - 7i$     |
| (14) $(-4 + 8i) - (-7 + 5i)$<br>$= 3 + 3i$   | (33) $(-5 + 8i) - (-5 - 6i)$<br>$= 14i$        | (52) $(-7 - 4i) + (1 - 4i)$<br>$= -6 - 8i$    |
| (15) $(9 + 8i) - (8 - 10i)$<br>$= 1 + 18i$   | (34) $(-10 + 9i) - (7 - 6i)$<br>$= -17 + 15i$  | (53) $(9 + i) + (2 - 9i)$<br>$= 11 - 8i$      |
| (16) $(7 - i) - (9 - 2i)$<br>$= -2 + i$      | (35) $(1 - 7i) - (-8 - 6i)$<br>$= 9 - i$       | (54) $(3 + 7i) + (-6 + 2i)$<br>$= -3 + 9i$    |
| (17) $(-1 + 4i) + (-2 - 2i)$<br>$= -3 + 2i$  | (36) $(-7 + 8i) - (-8 + 6i)$<br>$= 1 + 2i$     | (55) $(1 - 10i) + (8 + 5i)$<br>$= 9 - 5i$     |
| (18) $(3 - 7i) + (-5 + 2i)$<br>$= -2 - 5i$   | (37) $(-6 + 7i) - (5 + 3i)$<br>$= -11 + 4i$    | (56) $(8 - 10i) + (8 + 3i)$<br>$= 16 - 7i$    |
| (19) $(-3 + 4i) - (-4 + 10i)$<br>$= 1 - 6i$  | (38) $(-7 + 2i) - (10 - 8i)$<br>$= -17 + 10i$  | (57) $(9 + 2i) - (4 + 7i)$<br>$= 5 - 5i$      |

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|----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| (1) $(1 - 10i) - (10 + 5i)$<br>$= -9 - 15i$  | (20) $(-3 + 2i) + (2 + 7i)$<br>$= -1 + 9i$    | (39) $(-2 + 8i) - (-9 + 10i)$<br>$= 7 - 2i$   |
| (2) $(3 + i) + (10 - 6i)$<br>$= 13 - 5i$     | (21) $(9 - i) - (-8 + 2i)$<br>$= 17 - 3i$     | (40) $(-6 + 9i) - (-9 + 10i)$<br>$= 3 - i$    |
| (3) $(-4 + 2i) - (8 + i)$<br>$= -12 + i$     | (22) $(-7 - 3i) + (5 - 6i)$<br>$= -2 - 9i$    | (41) $(-4 + 5i) - (-3 - 10i)$<br>$= -1 + 15i$ |
| (4) $(-4 + 10i) - (9 + 5i)$<br>$= -13 + 5i$  | (23) $(-10 - 5i) - (-1 + 6i)$<br>$= -9 - 11i$ | (42) $(6 + 8i) - (-1 + 8i)$<br>$= 7$          |
| (5) $(1 + 7i) - (7 - 9i)$<br>$= -6 + 16i$    | (24) $(-7 + 4i) + (-8 - 4i)$<br>$= -15$       | (43) $(-8 - 7i) + (-8 - 2i)$<br>$= -16 - 9i$  |
| (6) $(10 + 4i) + (-1 + 4i)$<br>$= 9 + 8i$    | (25) $(5 - 2i) - (8 + 3i)$<br>$= -3 - 5i$     | (44) $(4 - 6i) + (2 + 10i)$<br>$= 6 + 4i$     |
| (7) $(1 + 8i) - (2 - 2i)$<br>$= -1 + 10i$    | (26) $(-10 + 3i) - (-3 - 4i)$<br>$= -7 + 7i$  | (45) $(-3 - 2i) + (4 - 4i)$<br>$= 1 - 6i$     |
| (8) $(9 + 7i) - (8 - 7i)$<br>$= 1 + 14i$     | (27) $(2 + 10i) + (1 - 7i)$<br>$= 3 + 3i$     | (46) $(6 - 10i) + (-10 + 5i)$<br>$= -4 - 5i$  |
| (9) $(6 - i) + (-10 - 6i)$<br>$= -4 - 7i$    | (28) $(3 - 5i) - (7 - 9i)$<br>$= -4 + 4i$     | (47) $(-7 - 10i) - (-1 + 7i)$<br>$= -6 - 17i$ |
| (10) $(-2 + 5i) - (7 + 4i)$<br>$= -9 + i$    | (29) $(-7 - 6i) - (10 - i)$<br>$= -17 - 5i$   | (48) $(-1 - 4i) + (7 + 8i)$<br>$= 6 + 4i$     |
| (11) $(1 - 3i) - (10 - 4i)$<br>$= -9 + i$    | (30) $(10 - 7i) - (-9 - i)$<br>$= 19 - 6i$    | (49) $(6 - 10i) - (3 - 6i)$<br>$= 3 - 4i$     |
| (12) $(-7 - 8i) + (1 - 9i)$<br>$= -6 - 17i$  | (31) $(6 - i) + (-9 - 8i)$<br>$= -3 - 9i$     | (50) $(-8 + 9i) - (-4 - 8i)$<br>$= -4 + 17i$  |
| (13) $(2 - 5i) - (7 + 5i)$<br>$= -5 - 10i$   | (32) $(-10 - 6i) + (5 + 2i)$<br>$= -5 - 4i$   | (51) $(-1 - 3i) + (-4 + 7i)$<br>$= -5 + 4i$   |
| (14) $(7 + 4i) + (5 + i)$<br>$= 12 + 5i$     | (33) $(-2 - 3i) - (-4 + 10i)$<br>$= 2 - 13i$  | (52) $(8 + 10i) - (1 + 6i)$<br>$= 7 + 4i$     |
| (15) $(2 + i) - (-6 + 8i)$<br>$= 8 - 7i$     | (34) $(7 - 6i) + (10 + 9i)$<br>$= 17 + 3i$    | (53) $(4 + 6i) - (-1 + i)$<br>$= 5 + 5i$      |
| (16) $(3 - i) + (-8 + 5i)$<br>$= -5 + 4i$    | (35) $(4 - 10i) + (-10 + 2i)$<br>$= -6 - 8i$  | (54) $(9 + i) + (9 - 6i)$<br>$= 18 - 5i$      |
| (17) $(10 - 9i) - (-1 + 3i)$<br>$= 11 - 12i$ | (36) $(8 + 5i) - (7 - 10i)$<br>$= 1 + 15i$    | (55) $(10 - 4i) + (-8 - i)$<br>$= 2 - 5i$     |
| (18) $(-3 - 4i) + (-8 + 4i)$<br>$= -11$      | (37) $(-8 + 4i) + (5 - 4i)$<br>$= -3$         | (56) $(-7 - 2i) - (-6 - 5i)$<br>$= -1 + 3i$   |
| (19) $(6 - 7i) - (5 - 4i)$<br>$= 1 - 3i$     | (38) $(9 + 5i) - (8 + 5i)$<br>$= 1$           | (57) $(1 - 9i) + (9 + 7i)$<br>$= 10 - 2i$     |

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|-----------------------------------------------|----------------------------------------------|-----------------------------------------------|
| (1) $(-2 + 7i) - (8 - i)$<br>$= -10 + 8i$     | (20) $(4 + 9i) - (10 + 10i)$<br>$= -6 - i$   | (39) $(-3 + 7i) + (-10 + i)$<br>$= -13 + 8i$  |
| (2) $(1 - 10i) - (-1 + 3i)$<br>$= 2 - 13i$    | (21) $(-6 - 5i) - (-6 - 9i)$<br>$= 4i$       | (40) $(10 - 10i) - (-9 + 9i)$<br>$= 19 - 19i$ |
| (3) $(10 - 4i) + (-4 - 2i)$<br>$= 6 - 6i$     | (22) $(-3 - 8i) + (10 + 3i)$<br>$= 7 - 5i$   | (41) $(-7 + 10i) + (-9 - 10i)$<br>$= -16$     |
| (4) $(5 + 2i) + (9 - 8i)$<br>$= 14 - 6i$      | (23) $(-2 - 6i) + (-8 - i)$<br>$= -10 - 7i$  | (42) $(-7 + 10i) - (-2 + i)$<br>$= -5 + 9i$   |
| (5) $(3 + 8i) - (-3 + 6i)$<br>$= 6 + 2i$      | (24) $(-7 - 9i) - (1 + 8i)$<br>$= -8 - 17i$  | (43) $(-6 + 2i) + (4 - 6i)$<br>$= -2 - 4i$    |
| (6) $(4 - 6i) + (-7 + 8i)$<br>$= -3 + 2i$     | (25) $(5 - 5i) + (-5 + 5i)$<br>$= 0$         | (44) $(-3 - 10i) + (6 + 5i)$<br>$= 3 - 5i$    |
| (7) $(8 + 2i) - (-6 - 7i)$<br>$= 14 + 9i$     | (26) $(-1 + 5i) + (9 + 8i)$<br>$= 8 + 13i$   | (45) $(-7 + 6i) - (-6 + 4i)$<br>$= -1 + 2i$   |
| (8) $(-5 - 8i) - (-5 + 8i)$<br>$= -16i$       | (27) $(-2 + i) + (-9 + 2i)$<br>$= -11 + 3i$  | (46) $(1 - 4i) + (-7 + i)$<br>$= -6 - 3i$     |
| (9) $(7 + 5i) - (5 - 5i)$<br>$= 2 + 10i$      | (28) $(-2 + 2i) - (-10 - 5i)$<br>$= 8 + 7i$  | (47) $(-2 - 3i) - (-8 + 8i)$<br>$= 6 - 11i$   |
| (10) $(-1 + 8i) + (-6 + 10i)$<br>$= -7 + 18i$ | (29) $(-6 + 5i) - (-4 + 7i)$<br>$= -2 - 2i$  | (48) $(4 + i) - (-2 + 10i)$<br>$= 6 - 9i$     |
| (11) $(-4 + i) - (-10 + 4i)$<br>$= 6 - 3i$    | (30) $(-1 - 2i) - (2 - 10i)$<br>$= -3 + 8i$  | (49) $(-3 - 4i) - (-5 + i)$<br>$= 2 - 5i$     |
| (12) $(-5 - 3i) - (-10 - 10i)$<br>$= 5 + 7i$  | (31) $(-6 - 8i) - (-9 - 9i)$<br>$= 3 + i$    | (50) $(4 + 3i) - (-2 - 3i)$<br>$= 6 + 6i$     |
| (13) $(-1 - 6i) + (9 - 8i)$<br>$= 8 - 14i$    | (32) $(-3 + 9i) - (-3 + 2i)$<br>$= 7i$       | (51) $(8 - i) - (-2 + 4i)$<br>$= 10 - 5i$     |
| (14) $(8 - 4i) - (8 - 6i)$<br>$= 2i$          | (33) $(-10 + 5i) - (2 + 8i)$<br>$= -12 - 3i$ | (52) $(3 + 8i) - (-1 + 2i)$<br>$= 4 + 6i$     |
| (15) $(1 - 7i) - (10 - 10i)$<br>$= -9 + 3i$   | (34) $(-6 + 4i) - (-1 + 8i)$<br>$= -5 - 4i$  | (53) $(-6 - 5i) - (6 + 2i)$<br>$= -12 - 7i$   |
| (16) $(-2 + 4i) - (4 + 7i)$<br>$= -6 - 3i$    | (35) $(-2 - 10i) - (-7 + 3i)$<br>$= 5 - 13i$ | (54) $(5 - 10i) + (-6 + i)$<br>$= -1 - 9i$    |
| (17) $(8 - 5i) - (-4 + 3i)$<br>$= 12 - 8i$    | (36) $(7 + 3i) + (-6 - 4i)$<br>$= 1 - i$     | (55) $(-2 + 7i) - (-9 + 9i)$<br>$= 7 - 2i$    |
| (18) $(-4 + 2i) - (-1 - 8i)$<br>$= -3 + 10i$  | (37) $(-1 - 6i) + (4 - 6i)$<br>$= 3 - 12i$   | (56) $(-5 + 6i) + (5 + 9i)$<br>$= 15i$        |
| (19) $(4 - 10i) - (-4 - 9i)$<br>$= 8 - i$     | (38) $(-3 + 6i) - (1 + 8i)$<br>$= -4 - 2i$   | (57) $(-6 - 3i) - (-5 - 6i)$<br>$= -1 + 3i$   |



- |                                               |                                               |                                                |
|-----------------------------------------------|-----------------------------------------------|------------------------------------------------|
| (1) $(1 + 6i) + (2 + i)$<br>$= 3 + 7i$        | (20) $(-5 + 5i) + (9 - 7i)$<br>$= 4 - 2i$     | (39) $(3 + 8i) - (-7 + 2i)$<br>$= 10 + 6i$     |
| (2) $(-3 - 8i) + (4 + 2i)$<br>$= 1 - 6i$      | (21) $(10 + 6i) + (-4 + 5i)$<br>$= 6 + 11i$   | (40) $(-6 - 4i) - (10 + i)$<br>$= -16 - 5i$    |
| (3) $(-9 + 8i) - (8 - i)$<br>$= -17 + 9i$     | (22) $(2 - 5i) - (2 - 9i)$<br>$= 4i$          | (41) $(-1 - 3i) - (6 - 6i)$<br>$= -7 + 3i$     |
| (4) $(9 - 9i) - (-10 + 3i)$<br>$= 19 - 12i$   | (23) $(4 - i) - (10 + 9i)$<br>$= -6 - 10i$    | (42) $(-5 + 9i) - (3 + 9i)$<br>$= -8$          |
| (5) $(7 + 4i) + (-10 + i)$<br>$= -3 + 5i$     | (24) $(-10 + 7i) - (-1 + 9i)$<br>$= -9 - 2i$  | (43) $(-6 - 3i) - (3 - 10i)$<br>$= -9 + 7i$    |
| (6) $(-8 - 9i) - (5 + 5i)$<br>$= -13 - 14i$   | (25) $(7 - 4i) + (-8 - 9i)$<br>$= -1 - 13i$   | (44) $(2 - 10i) + (7 - i)$<br>$= 9 - 11i$      |
| (7) $(-2 - 3i) + (-3 + 2i)$<br>$= -5 - i$     | (26) $(3 - 9i) - (1 + 5i)$<br>$= 2 - 14i$     | (45) $(-3 + 5i) - (6 + 10i)$<br>$= -9 - 5i$    |
| (8) $(-5 - 4i) + (-7 - 10i)$<br>$= -12 - 14i$ | (27) $(-7 - 8i) - (-1 - 9i)$<br>$= -6 + i$    | (46) $(2 + 9i) - (1 - 5i)$<br>$= 1 + 14i$      |
| (9) $(3 + i) - (-5 + 10i)$<br>$= 8 - 9i$      | (28) $(8 + 7i) - (-5 + 8i)$<br>$= 13 - i$     | (47) $(-4 - 2i) + (-4 + 5i)$<br>$= -8 + 3i$    |
| (10) $(1 - 3i) + (6 - 3i)$<br>$= 7 - 6i$      | (29) $(-8 + 2i) + (-8 + 2i)$<br>$= -16 + 4i$  | (48) $(-1 + 5i) + (8 + 3i)$<br>$= 7 + 8i$      |
| (11) $(-8 - 3i) - (10 - 6i)$<br>$= -18 + 3i$  | (30) $(5 + i) + (5 + i)$<br>$= 10 + 2i$       | (49) $(9 - 9i) + (6 + 4i)$<br>$= 15 - 5i$      |
| (12) $(8 - 4i) - (9 + 9i)$<br>$= -1 - 13i$    | (31) $(-4 + 5i) + (-7 - 7i)$<br>$= -11 - 2i$  | (50) $(-10 - 3i) - (-4 + 10i)$<br>$= -6 - 13i$ |
| (13) $(8 + 2i) - (-6 - 3i)$<br>$= 14 + 5i$    | (32) $(-6 - 4i) - (3 - 4i)$<br>$= -9$         | (51) $(-8 + 6i) + (4 - 6i)$<br>$= -4$          |
| (14) $(2 + 5i) + (-1 - 6i)$<br>$= 1 - i$      | (33) $(4 - 9i) + (-3 + 9i)$<br>$= 1$          | (52) $(-2 + 8i) - (-6 + 9i)$<br>$= 4 - i$      |
| (15) $(-4 + 7i) + (6 - 4i)$<br>$= 2 + 3i$     | (34) $(6 - 10i) - (-1 + 10i)$<br>$= 7 - 20i$  | (53) $(4 + 5i) - (7 + 7i)$<br>$= -3 - 2i$      |
| (16) $(-9 + 5i) - (-6 + i)$<br>$= -3 + 4i$    | (35) $(-5 - 5i) - (10 + 7i)$<br>$= -15 - 12i$ | (54) $(-3 - 7i) - (-6 + 2i)$<br>$= 3 - 9i$     |
| (17) $(-4 + 10i) - (4 + 8i)$<br>$= -8 + 2i$   | (36) $(2 - 9i) + (-3 - 10i)$<br>$= -1 - 19i$  | (55) $(-4 + i) - (7 - 5i)$<br>$= -11 + 6i$     |
| (18) $(8 - 10i) + (-3 - 2i)$<br>$= 5 - 12i$   | (37) $(5 + 7i) - (-8 + 8i)$<br>$= 13 - i$     | (56) $(-2 - i) - (-9 + 2i)$<br>$= 7 - 3i$      |
| (19) $(-2 - 8i) - (-8 - 9i)$<br>$= 6 + i$     | (38) $(1 + 9i) - (-3 + 5i)$<br>$= 4 + 4i$     | (57) $(-9 - 8i) + (-8 - 2i)$<br>$= -17 - 10i$  |

- |                                                |                                              |                                              |
|------------------------------------------------|----------------------------------------------|----------------------------------------------|
| (1) $(5 + 6i) - (-7 + 10i)$<br>$= 12 - 4i$     | (20) $(-9 + 10i) - (-9 - 3i)$<br>$= 13i$     | (39) $(4 + 4i) - (-6 + 10i)$<br>$= 10 - 6i$  |
| (2) $(-8 - 10i) - (3 - 5i)$<br>$= -11 - 5i$    | (21) $(1 + i) - (6 + 6i)$<br>$= -5 - 5i$     | (40) $(3 - 8i) + (4 - 6i)$<br>$= 7 - 14i$    |
| (3) $(6 - 10i) - (-8 - 7i)$<br>$= 14 - 3i$     | (22) $(-3 - 2i) + (-8 - 6i)$<br>$= -11 - 8i$ | (41) $(-9 + 2i) - (-9 - 6i)$<br>$= 8i$       |
| (4) $(4 + 7i) + (-6 - 3i)$<br>$= -2 + 4i$      | (23) $(-10 + 8i) + (7 - 7i)$<br>$= -3 + i$   | (42) $(6 - i) + (-3 - 8i)$<br>$= 3 - 9i$     |
| (5) $(-3 - 3i) + (4 + 9i)$<br>$= 1 + 6i$       | (24) $(9 - 4i) - (-4 + 10i)$<br>$= 13 - 14i$ | (43) $(3 + 5i) + (4 + 6i)$<br>$= 7 + 11i$    |
| (6) $(9 + 3i) + (10 - 5i)$<br>$= 19 - 2i$      | (25) $(9 + 7i) + (10 + 2i)$<br>$= 19 + 9i$   | (44) $(5 - 6i) + (8 - 2i)$<br>$= 13 - 8i$    |
| (7) $(-4 + 6i) + (-5 - 9i)$<br>$= -9 - 3i$     | (26) $(3 - 10i) + (-5 + 5i)$<br>$= -2 - 5i$  | (45) $(-2 + 6i) + (-7 + 4i)$<br>$= -9 + 10i$ |
| (8) $(7 - 6i) + (5 + 10i)$<br>$= 12 + 4i$      | (27) $(-10 + i) - (2 + 5i)$<br>$= -12 - 4i$  | (46) $(5 - 5i) - (9 + 2i)$<br>$= -4 - 7i$    |
| (9) $(-8 - 4i) + (8 + 6i)$<br>$= 2i$           | (28) $(-8 - 5i) - (8 + 8i)$<br>$= -16 - 13i$ | (47) $(-1 + i) - (-1 + 9i)$<br>$= -8i$       |
| (10) $(2 + i) + (7 - 4i)$<br>$= 9 - 3i$        | (29) $(-4 + 10i) + (6 - 2i)$<br>$= 2 + 8i$   | (48) $(-3 - 6i) - (10 - 5i)$<br>$= -13 - i$  |
| (11) $(-5 + i) + (-3 + 8i)$<br>$= -8 + 9i$     | (30) $(6 + 8i) - (4 + 8i)$<br>$= 2$          | (49) $(9 + 2i) + (8 - 5i)$<br>$= 17 - 3i$    |
| (12) $(-8 + 10i) - (-9 - 9i)$<br>$= 1 + 19i$   | (31) $(2 + 6i) + (10 + 5i)$<br>$= 12 + 11i$  | (50) $(10 - 3i) + (-2 + 10i)$<br>$= 8 + 7i$  |
| (13) $(-7 + 6i) + (-5 - 9i)$<br>$= -12 - 3i$   | (32) $(3 + 7i) - (5 + 7i)$<br>$= -2$         | (51) $(6 + 8i) - (10 - 3i)$<br>$= -4 + 11i$  |
| (14) $(-4 - 10i) - (-6 - 8i)$<br>$= 2 - 2i$    | (33) $(-9 + 2i) + (-5 + 3i)$<br>$= -14 + 5i$ | (52) $(-10 + i) + (-5 + 3i)$<br>$= -15 + 4i$ |
| (15) $(-7 - 3i) + (-10 - 8i)$<br>$= -17 - 11i$ | (34) $(-4 + 2i) - (-3 + 6i)$<br>$= -1 - 4i$  | (53) $(-1 + 3i) + (-8 - 3i)$<br>$= -9$       |
| (16) $(1 - i) + (10 - 9i)$<br>$= 11 - 10i$     | (35) $(7 - 2i) + (-2 + 2i)$<br>$= 5$         | (54) $(2 - i) + (-3 + 2i)$<br>$= -1 + i$     |
| (17) $(9 + 9i) + (-10 - 2i)$<br>$= -1 + 7i$    | (36) $(9 + 8i) + (-4 - i)$<br>$= 5 + 7i$     | (55) $(7 - 8i) + (-8 - 5i)$<br>$= -1 - 13i$  |
| (18) $(-1 + 9i) + (-10 + 2i)$<br>$= -11 + 11i$ | (37) $(1 + 4i) - (1 + 3i)$<br>$= i$          | (56) $(5 - 2i) + (1 - 4i)$<br>$= 6 - 6i$     |
| (19) $(4 - 2i) - (3 - 9i)$<br>$= 1 + 7i$       | (38) $(-4 - 4i) + (4 - i)$<br>$= -5i$        | (57) $(1 + 6i) + (-8 - 7i)$<br>$= -7 - i$    |

- |                                                |                                               |                                                |
|------------------------------------------------|-----------------------------------------------|------------------------------------------------|
| (1) $(2 - 10i) - (-1 - 4i)$<br>$= 3 - 6i$      | (20) $(6 + 8i) + (4 - 8i)$<br>$= 10$          | (39) $(7 - 10i) + (-6 - 10i)$<br>$= 1 - 20i$   |
| (2) $(-1 - 8i) + (-3 + 7i)$<br>$= -4 - i$      | (21) $(-9 - 10i) + (-8 - i)$<br>$= -17 - 11i$ | (40) $(-6 - 9i) - (-9 - 5i)$<br>$= 3 - 4i$     |
| (3) $(6 - 6i) + (3 + 7i)$<br>$= 9 + i$         | (22) $(1 - 3i) - (-6 + 3i)$<br>$= 7 - 6i$     | (41) $(-10 - 8i) - (-5 + 8i)$<br>$= -5 - 16i$  |
| (4) $(-9 - i) + (-3 + 4i)$<br>$= -12 + 3i$     | (23) $(-5 - 7i) - (6 + 3i)$<br>$= -11 - 10i$  | (42) $(-3 + 6i) - (9 + 2i)$<br>$= -12 + 4i$    |
| (5) $(-10 + 7i) - (5 - 5i)$<br>$= -15 + 12i$   | (24) $(4 + 8i) + (-1 + 5i)$<br>$= 3 + 13i$    | (43) $(8 + 4i) + (-5 + 3i)$<br>$= 3 + 7i$      |
| (6) $(10 + 3i) + (-7 - 8i)$<br>$= 3 - 5i$      | (25) $(-9 - 10i) + (-3 + 8i)$<br>$= -12 - 2i$ | (44) $(8 - 6i) - (-7 - 4i)$<br>$= 15 - 2i$     |
| (7) $(-8 + 7i) - (-6 - 7i)$<br>$= -2 + 14i$    | (26) $(3 + 10i) + (-9 - 4i)$<br>$= -6 + 6i$   | (45) $(4 - 10i) + (-9 - 10i)$<br>$= -5 - 20i$  |
| (8) $(-2 + 8i) - (6 - i)$<br>$= -8 + 9i$       | (27) $(9 - 4i) - (-8 - 9i)$<br>$= 17 + 5i$    | (46) $(3 + 6i) - (-10 + 7i)$<br>$= 13 - i$     |
| (9) $(-5 + 5i) - (-4 + i)$<br>$= -1 + 4i$      | (28) $(1 - 3i) + (8 - 3i)$<br>$= 9 - 6i$      | (47) $(-9 + 10i) + (-8 - 7i)$<br>$= -17 + 3i$  |
| (10) $(-5 + 6i) - (2 - 3i)$<br>$= -7 + 9i$     | (29) $(-4 - 7i) - (9 + 2i)$<br>$= -13 - 9i$   | (48) $(1 + 6i) + (-8 - 8i)$<br>$= -7 - 2i$     |
| (11) $(6 + 2i) + (-4 + 8i)$<br>$= 2 + 10i$     | (30) $(8 + 7i) - (-1 + 3i)$<br>$= 9 + 4i$     | (49) $(7 + 6i) - (9 + 8i)$<br>$= -2 - 2i$      |
| (12) $(-10 + 6i) - (9 - 10i)$<br>$= -19 + 16i$ | (31) $(-3 + 2i) - (6 + 8i)$<br>$= -9 - 6i$    | (50) $(-8 + 10i) - (-7 - 10i)$<br>$= -1 + 20i$ |
| (13) $(6 + i) - (7 + 4i)$<br>$= -1 - 3i$       | (32) $(7 - 9i) - (-10 - 6i)$<br>$= 17 - 3i$   | (51) $(9 - 6i) - (3 + 10i)$<br>$= 6 - 16i$     |
| (14) $(-7 - 2i) + (10 + 7i)$<br>$= 3 + 5i$     | (33) $(-5 + 8i) + (4 - 6i)$<br>$= -1 + 2i$    | (52) $(-1 + 5i) + (8 + i)$<br>$= 7 + 6i$       |
| (15) $(8 - 6i) + (-7 - 6i)$<br>$= 1 - 12i$     | (34) $(7 - 5i) + (8 + 10i)$<br>$= 15 + 5i$    | (53) $(6 + i) + (-5 + 10i)$<br>$= 1 + 11i$     |
| (16) $(-1 + 5i) - (-2 - 10i)$<br>$= 1 + 15i$   | (35) $(10 + 2i) + (7 + 10i)$<br>$= 17 + 12i$  | (54) $(-6 + i) + (8 - 6i)$<br>$= 2 - 5i$       |
| (17) $(-6 + 10i) + (9 - 7i)$<br>$= 3 + 3i$     | (36) $(9 + 6i) + (7 - 2i)$<br>$= 16 + 4i$     | (55) $(6 - 5i) - (7 + 6i)$<br>$= -1 - 11i$     |
| (18) $(-6 + 3i) - (-7 + 3i)$<br>$= 1$          | (37) $(-7 - 8i) + (-4 + 7i)$<br>$= -11 - i$   | (56) $(2 - 6i) + (8 + 5i)$<br>$= 10 - i$       |
| (19) $(-5 - i) - (7 + 6i)$<br>$= -12 - 7i$     | (38) $(-2 + 4i) + (9 + 9i)$<br>$= 7 + 13i$    | (57) $(3 - 7i) - (7 - 6i)$<br>$= -4 - i$       |

- |                                              |                                                |                                                |
|----------------------------------------------|------------------------------------------------|------------------------------------------------|
| (1) $(9 - 3i) + (9 + 5i)$<br>$= 18 + 2i$     | (20) $(-3 + 7i) + (10 - 2i)$<br>$= 7 + 5i$     | (39) $(-6 + 6i) - (-8 + 4i)$<br>$= 2 + 2i$     |
| (2) $(-5 + 6i) - (-8 - i)$<br>$= 3 + 7i$     | (21) $(-10 - 7i) + (-1 + 7i)$<br>$= -11$       | (40) $(8 - 2i) - (2 - 7i)$<br>$= 6 + 5i$       |
| (3) $(-7 + 4i) - (-4 - 8i)$<br>$= -3 + 12i$  | (22) $(-3 - 10i) + (-8 + 2i)$<br>$= -11 - 8i$  | (41) $(-7 + 7i) + (8 + 9i)$<br>$= 1 + 16i$     |
| (4) $(-9 - 9i) + (1 - 4i)$<br>$= -8 - 13i$   | (23) $(3 + 2i) + (-3 + 7i)$<br>$= 9i$          | (42) $(10 - 2i) + (10 - 10i)$<br>$= 20 - 12i$  |
| (5) $(7 - i) + (6 - 9i)$<br>$= 13 - 10i$     | (24) $(-10 + 9i) - (-8 + 5i)$<br>$= -2 + 4i$   | (43) $(9 - 7i) - (10 - 5i)$<br>$= -1 - 2i$     |
| (6) $(5 + i) - (-2 + 4i)$<br>$= 7 - 3i$      | (25) $(8 + 10i) + (-10 - 4i)$<br>$= -2 + 6i$   | (44) $(-9 - 8i) - (-4 + 5i)$<br>$= -5 - 13i$   |
| (7) $(-9 - 10i) - (1 + 3i)$<br>$= -10 - 13i$ | (26) $(-8 + 3i) - (-3 - 7i)$<br>$= -5 + 10i$   | (45) $(5 + 7i) - (1 - 8i)$<br>$= 4 + 15i$      |
| (8) $(-2 + 10i) - (-5 - 5i)$<br>$= 3 + 15i$  | (27) $(6 + 7i) - (-1 - i)$<br>$= 7 + 8i$       | (46) $(6 + 6i) - (-10 - 4i)$<br>$= 16 + 10i$   |
| (9) $(-4 + 9i) - (-4 + 8i)$<br>$= i$         | (28) $(-8 + 10i) + (-3 - 9i)$<br>$= -11 + i$   | (47) $(1 - 8i) - (-10 + 2i)$<br>$= 11 - 10i$   |
| (10) $(-1 + 7i) - (-4 + 7i)$<br>$= 3$        | (29) $(4 - 9i) + (-10 + 6i)$<br>$= -6 - 3i$    | (48) $(-10 - 8i) + (6 - 6i)$<br>$= -4 - 14i$   |
| (11) $(-8 + 4i) - (3 - 8i)$<br>$= -11 + 12i$ | (30) $(-7 + 3i) + (1 + i)$<br>$= -6 + 4i$      | (49) $(-8 - 7i) + (4 + 3i)$<br>$= -4 - 4i$     |
| (12) $(9 - i) - (-8 + 4i)$<br>$= 17 - 5i$    | (31) $(-8 - 3i) - (7 - 7i)$<br>$= -15 + 4i$    | (50) $(5 - 10i) + (7 - 9i)$<br>$= 12 - 19i$    |
| (13) $(-9 + 2i) + (-9 + i)$<br>$= -18 + 3i$  | (32) $(3 + 7i) + (9 + 6i)$<br>$= 12 + 13i$     | (51) $(8 - 9i) + (1 - 3i)$<br>$= 9 - 12i$      |
| (14) $(-9 + 5i) - (6 - i)$<br>$= -15 + 6i$   | (33) $(-2 - i) + (-10 + 9i)$<br>$= -12 + 8i$   | (52) $(3 + 6i) - (6 + 6i)$<br>$= -3$           |
| (15) $(-9 - 2i) - (-5 - 10i)$<br>$= -4 + 8i$ | (34) $(-2 + 8i) - (9 - 5i)$<br>$= -11 + 13i$   | (53) $(-3 + 10i) - (10 + i)$<br>$= -13 + 9i$   |
| (16) $(8 - 4i) + (8 - 4i)$<br>$= 16 - 8i$    | (35) $(-4 + 3i) - (6 + 8i)$<br>$= -10 - 5i$    | (54) $(-3 + 8i) - (-5 - 10i)$<br>$= 2 + 18i$   |
| (17) $(-9 + 10i) - (-8 + 2i)$<br>$= -1 + 8i$ | (36) $(-10 + 8i) + (-3 + 6i)$<br>$= -13 + 14i$ | (55) $(6 - 7i) + (-3 + i)$<br>$= 3 - 6i$       |
| (18) $(8 + 10i) + (2 + 10i)$<br>$= 10 + 20i$ | (37) $(-4 + 7i) - (-2 + 9i)$<br>$= -2 - 2i$    | (56) $(9 - 7i) + (-7 + 10i)$<br>$= 2 + 3i$     |
| (19) $(-5 + 3i) + (-8 - 4i)$<br>$= -13 - i$  | (38) $(-1 - 6i) - (-4 + 2i)$<br>$= 3 - 8i$     | (57) $(-10 - 2i) + (-5 + 10i)$<br>$= -15 + 8i$ |

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|----------------------------------------------|----------------------------------------------|------------------------------------------------|
| (1) $(6 + 8i) + (-9 - 10i)$<br>$= -3 - 2i$   | (20) $(-6 - 6i) - (-2 - 8i)$<br>$= -4 + 2i$  | (39) $(3 + 8i) - (-4 - i)$<br>$= 7 + 9i$       |
| (2) $(-4 - 9i) + (4 - 2i)$<br>$= -11i$       | (21) $(10 - 6i) - (-8 + 9i)$<br>$= 18 - 15i$ | (40) $(-6 - i) + (-9 + 6i)$<br>$= -15 + 5i$    |
| (3) $(6 - 2i) + (-4 - 5i)$<br>$= 2 - 7i$     | (22) $(-9 + i) + (6 + 7i)$<br>$= -3 + 8i$    | (41) $(-2 - 7i) + (-3 + i)$<br>$= -5 - 6i$     |
| (4) $(5 + 8i) - (8 + i)$<br>$= -3 + 7i$      | (23) $(9 - 8i) - (7 + 5i)$<br>$= 2 - 13i$    | (42) $(-3 + i) + (1 + 2i)$<br>$= -2 + 3i$      |
| (5) $(-7 + 2i) - (10 - 8i)$<br>$= -17 + 10i$ | (24) $(7 - 6i) - (4 + 4i)$<br>$= 3 - 10i$    | (43) $(-6 + 8i) + (6 + 4i)$<br>$= 12i$         |
| (6) $(-7 - i) - (10 - 6i)$<br>$= -17 + 5i$   | (25) $(10 - 2i) - (9 + 7i)$<br>$= 1 - 9i$    | (44) $(3 + 5i) + (-7 - 6i)$<br>$= -4 - i$      |
| (7) $(5 + 8i) - (-10 + i)$<br>$= 15 + 7i$    | (26) $(-9 - 5i) + (3 - 8i)$<br>$= -6 - 13i$  | (45) $(-9 - 2i) + (-8 - 3i)$<br>$= -17 - 5i$   |
| (8) $(-9 + 3i) - (10 - 4i)$<br>$= -19 + 7i$  | (27) $(-3 + 6i) - (-7 - 7i)$<br>$= 4 + 13i$  | (46) $(7 + 7i) - (-5 + 8i)$<br>$= 12 - i$      |
| (9) $(4 + 8i) + (-3 + 5i)$<br>$= 1 + 13i$    | (28) $(5 - 9i) - (-8 + 8i)$<br>$= 13 - 17i$  | (47) $(-9 - 2i) + (8 - 5i)$<br>$= -1 - 7i$     |
| (10) $(-10 - 10i) + (5 + i)$<br>$= -5 - 9i$  | (29) $(4 + 2i) + (4 - 9i)$<br>$= 8 - 7i$     | (48) $(6 + 2i) + (-3 + i)$<br>$= 3 + 3i$       |
| (11) $(-2 + 10i) + (-9 - i)$<br>$= -11 + 9i$ | (30) $(1 - 6i) + (-4 + 3i)$<br>$= -3 - 3i$   | (49) $(-1 - 10i) - (-3 + 2i)$<br>$= 2 - 12i$   |
| (12) $(5 + 4i) + (10 - 8i)$<br>$= 15 - 4i$   | (31) $(9 + 7i) - (5 + 2i)$<br>$= 4 + 5i$     | (50) $(5 + 6i) + (-4 - 9i)$<br>$= 1 - 3i$      |
| (13) $(-1 + 5i) + (4 - i)$<br>$= 3 + 4i$     | (32) $(3 + 8i) - (-1 + 4i)$<br>$= 4 + 4i$    | (51) $(-10 + 6i) - (4 + 4i)$<br>$= -14 + 2i$   |
| (14) $(2 + 9i) - (-4 - i)$<br>$= 6 + 10i$    | (33) $(6 - 6i) + (4 + 5i)$<br>$= 10 - i$     | (52) $(8 + 6i) - (-1 + 10i)$<br>$= 9 - 4i$     |
| (15) $(-9 + 6i) - (-3 - 4i)$<br>$= -6 + 10i$ | (34) $(-3 + 8i) + (-3 - 7i)$<br>$= -6 + i$   | (53) $(-5 + 6i) - (-2 - 3i)$<br>$= -3 + 9i$    |
| (16) $(3 + 6i) - (2 - 4i)$<br>$= 1 + 10i$    | (35) $(1 + 6i) - (-9 + 4i)$<br>$= 10 + 2i$   | (54) $(-10 + 4i) + (-4 - 10i)$<br>$= -14 - 6i$ |
| (17) $(-5 - 3i) - (6 - 10i)$<br>$= -11 + 7i$ | (36) $(-9 + 5i) - (1 + 6i)$<br>$= -10 - i$   | (55) $(7 + i) - (-3 + 3i)$<br>$= 10 - 2i$      |
| (18) $(2 + 6i) - (4 + 2i)$<br>$= -2 + 4i$    | (37) $(3 + 8i) - (1 + 7i)$<br>$= 2 + i$      | (56) $(2 - 6i) - (8 + 4i)$<br>$= -6 - 10i$     |
| (19) $(-2 + 4i) - (-5 + 9i)$<br>$= 3 - 5i$   | (38) $(10 - 3i) + (-1 - 5i)$<br>$= 9 - 8i$   | (57) $(-6 + 3i) + (3 + 8i)$<br>$= -3 + 11i$    |

- |                                              |                                                |                                                |
|----------------------------------------------|------------------------------------------------|------------------------------------------------|
| (1) $(-7 + 4i) - (-8 - 10i)$<br>$= 1 + 14i$  | (20) $(-7 + 7i) + (-2 - 8i)$<br>$= -9 - i$     | (39) $(6 + 9i) + (7 + 5i)$<br>$= 13 + 14i$     |
| (2) $(-1 - i) - (-8 + 3i)$<br>$= 7 - 4i$     | (21) $(-6 - i) + (9 + 7i)$<br>$= 3 + 6i$       | (40) $(6 + 10i) - (7 - 5i)$<br>$= -1 + 15i$    |
| (3) $(-6 + 7i) + (-3 + i)$<br>$= -9 + 8i$    | (22) $(-5 - 4i) - (-3 - 7i)$<br>$= -2 + 3i$    | (41) $(9 - 4i) + (1 + 10i)$<br>$= 10 + 6i$     |
| (4) $(-3 - 2i) - (5 - 7i)$<br>$= -8 + 5i$    | (23) $(-5 + 2i) - (-5 - 2i)$<br>$= 4i$         | (42) $(-3 + 2i) + (-9 - 10i)$<br>$= -12 - 8i$  |
| (5) $(1 + 7i) - (-10 - 4i)$<br>$= 11 + 11i$  | (24) $(-7 - 2i) + (2 + 7i)$<br>$= -5 + 5i$     | (43) $(-10 - 8i) - (6 + 6i)$<br>$= -16 - 14i$  |
| (6) $(5 + i) + (-1 + 2i)$<br>$= 4 + 3i$      | (25) $(-6 - 3i) + (-9 - i)$<br>$= -15 - 4i$    | (44) $(-2 + 5i) + (-6 + 2i)$<br>$= -8 + 7i$    |
| (7) $(-3 - 10i) - (9 + 9i)$<br>$= -12 - 19i$ | (26) $(-3 + 3i) + (3 - 7i)$<br>$= -4i$         | (45) $(7 + 4i) + (-4 - 8i)$<br>$= 3 - 4i$      |
| (8) $(8 + 10i) - (8 + 3i)$<br>$= 7i$         | (27) $(-2 + 5i) + (10 - 2i)$<br>$= 8 + 3i$     | (46) $(8 - 6i) + (-7 + 10i)$<br>$= 1 + 4i$     |
| (9) $(9 - 10i) - (10 - 6i)$<br>$= -1 - 4i$   | (28) $(-2 + 3i) - (4 + 8i)$<br>$= -6 - 5i$     | (47) $(4 + 10i) + (-8 - 2i)$<br>$= -4 + 8i$    |
| (10) $(-2 - 8i) - (-8 - 7i)$<br>$= 6 - i$    | (29) $(-5 - i) - (8 + 8i)$<br>$= -13 - 9i$     | (48) $(7 + i) - (10 + 8i)$<br>$= -3 - 7i$      |
| (11) $(-8 + 3i) - (4 - i)$<br>$= -12 + 4i$   | (30) $(1 + i) + (-6 + 2i)$<br>$= -5 + 3i$      | (49) $(-8 + 3i) - (2 - 2i)$<br>$= -10 + 5i$    |
| (12) $(1 - 8i) + (-7 + 7i)$<br>$= -6 - i$    | (31) $(10 - 7i) - (6 - 6i)$<br>$= 4 - i$       | (50) $(-8 - 2i) + (-5 + 8i)$<br>$= -13 + 6i$   |
| (13) $(5 + 4i) + (-8 + 2i)$<br>$= -3 + 6i$   | (32) $(9 + i) + (10 + 10i)$<br>$= 19 + 11i$    | (51) $(1 + 10i) - (-8 - 4i)$<br>$= 9 + 14i$    |
| (14) $(3 + 4i) + (-1 + 2i)$<br>$= 2 + 6i$    | (33) $(2 - 2i) - (8 + 7i)$<br>$= -6 - 9i$      | (52) $(-6 + i) - (4 + 5i)$<br>$= -10 - 4i$     |
| (15) $(-9 + 9i) + (4 - 8i)$<br>$= -5 + i$    | (34) $(2 - 4i) + (5 - 2i)$<br>$= 7 - 6i$       | (53) $(-3 - 10i) + (-8 - 6i)$<br>$= -11 - 16i$ |
| (16) $(-3 - i) + (6 + 7i)$<br>$= 3 + 6i$     | (35) $(-7 + 7i) + (-10 - 6i)$<br>$= -17 + i$   | (54) $(-1 - 3i) + (6 - 8i)$<br>$= 5 - 11i$     |
| (17) $(4 - 5i) - (-5 + 7i)$<br>$= 9 - 12i$   | (36) $(6 - 2i) + (-3 - 2i)$<br>$= 3 - 4i$      | (55) $(5 - 5i) - (10 + 3i)$<br>$= -5 - 8i$     |
| (18) $(2 - 8i) + (-1 - 3i)$<br>$= 1 - 11i$   | (37) $(-7 - 4i) - (-9 - 5i)$<br>$= 2 + i$      | (56) $(-1 - 10i) - (-2 + 9i)$<br>$= 1 - 19i$   |
| (19) $(-9 - 5i) + (-3 - 2i)$<br>$= -12 - 7i$ | (38) $(-5 + 8i) + (-10 - 10i)$<br>$= -15 - 2i$ | (57) $(1 + i) - (-5 - 8i)$<br>$= 6 + 9i$       |

- |                                               |                                               |                                                |
|-----------------------------------------------|-----------------------------------------------|------------------------------------------------|
| (1) $(-6 - 10i) + (-2 + 3i)$<br>$= -8 - 7i$   | (20) $(-4 + i) + (3 + 9i)$<br>$= -1 + 10i$    | (39) $(5 - 8i) + (9 - i)$<br>$= 14 - 9i$       |
| (2) $(-7 + 4i) - (-5 - 5i)$<br>$= -2 + 9i$    | (21) $(-8 - 3i) - (-3 + 6i)$<br>$= -5 - 9i$   | (40) $(-7 - 10i) + (-10 + 5i)$<br>$= -17 - 5i$ |
| (3) $(2 - 3i) + (-10 - i)$<br>$= -8 - 4i$     | (22) $(2 - 8i) + (4 + 6i)$<br>$= 6 - 2i$      | (41) $(-4 - i) - (4 + 7i)$<br>$= -8 - 8i$      |
| (4) $(7 - 4i) + (-3 + 8i)$<br>$= 4 + 4i$      | (23) $(9 + 7i) + (2 + 2i)$<br>$= 11 + 9i$     | (42) $(8 + 4i) + (5 - 10i)$<br>$= 13 - 6i$     |
| (5) $(-4 - 6i) - (-2 + 9i)$<br>$= -2 - 15i$   | (24) $(6 + 9i) - (-10 - 6i)$<br>$= 16 + 15i$  | (43) $(-9 - 10i) + (10 - 4i)$<br>$= 1 - 14i$   |
| (6) $(-2 - 10i) - (-1 - i)$<br>$= -1 - 9i$    | (25) $(2 + 7i) + (-2 - 5i)$<br>$= 2i$         | (44) $(4 + 6i) + (4 - 5i)$<br>$= 8 + i$        |
| (7) $(-1 - i) + (-5 - 10i)$<br>$= -6 - 11i$   | (26) $(-8 - 9i) + (3 - 10i)$<br>$= -5 - 19i$  | (45) $(-7 + 9i) + (-10 + 6i)$<br>$= -17 + 15i$ |
| (8) $(7 + 3i) + (6 - 4i)$<br>$= 13 - i$       | (27) $(6 + 4i) - (5 + 2i)$<br>$= 1 + 2i$      | (46) $(7 + 3i) + (-8 + 3i)$<br>$= -1 + 6i$     |
| (9) $(1 - 4i) - (-7 + 2i)$<br>$= 8 - 6i$      | (28) $(-1 - 10i) - (-6 + 10i)$<br>$= 5 - 20i$ | (47) $(-2 + 2i) + (6 + 6i)$<br>$= 4 + 8i$      |
| (10) $(-6 - 9i) + (1 - 9i)$<br>$= -5 - 18i$   | (29) $(6 - 6i) + (-8 - 8i)$<br>$= -2 - 14i$   | (48) $(8 + 7i) + (-9 - 2i)$<br>$= -1 + 5i$     |
| (11) $(9 + 10i) + (9 - 3i)$<br>$= 18 + 7i$    | (30) $(9 - 10i) + (1 + 6i)$<br>$= 10 - 4i$    | (49) $(-3 + 6i) + (7 + 5i)$<br>$= 4 + 11i$     |
| (12) $(-3 + 10i) + (5 - 6i)$<br>$= 2 + 4i$    | (31) $(-8 - 7i) - (-6 + 2i)$<br>$= -2 - 9i$   | (50) $(9 + 8i) - (-5 + 6i)$<br>$= 14 + 2i$     |
| (13) $(-1 + i) - (-2 - 7i)$<br>$= 1 + 8i$     | (32) $(-2 - 4i) + (1 - 2i)$<br>$= -1 - 6i$    | (51) $(8 + 7i) - (7 - 9i)$<br>$= 1 + 16i$      |
| (14) $(6 - i) - (-10 + 3i)$<br>$= 16 - 4i$    | (33) $(3 - 2i) + (6 + 7i)$<br>$= 9 + 5i$      | (52) $(3 - 2i) - (-6 - 8i)$<br>$= 9 + 6i$      |
| (15) $(10 + 10i) - (-3 - 3i)$<br>$= 13 + 13i$ | (34) $(5 + 2i) + (-3 + 5i)$<br>$= 2 + 7i$     | (53) $(3 + 6i) + (-7 - 7i)$<br>$= -4 - i$      |
| (16) $(2 - 2i) - (4 + 6i)$<br>$= -2 - 8i$     | (35) $(-5 + 2i) + (-7 - 3i)$<br>$= -12 - i$   | (54) $(-4 + 8i) - (-10 - i)$<br>$= 6 + 9i$     |
| (17) $(10 - 7i) - (-4 + 4i)$<br>$= 14 - 11i$  | (36) $(-8 - 8i) + (5 + 5i)$<br>$= -3 - 3i$    | (55) $(2 - 10i) + (-4 + 9i)$<br>$= -2 - i$     |
| (18) $(-3 + 3i) + (7 + 10i)$<br>$= 4 + 13i$   | (37) $(-8 - 3i) - (2 + 6i)$<br>$= -10 - 9i$   | (56) $(-4 + 10i) - (8 + 7i)$<br>$= -12 + 3i$   |
| (19) $(-1 - 5i) - (1 - i)$<br>$= -2 - 4i$     | (38) $(6 + i) - (-4 + 7i)$<br>$= 10 - 6i$     | (57) $(4 - 8i) + (8 - 4i)$<br>$= 12 - 12i$     |

- |                                               |                                               |                                                   |
|-----------------------------------------------|-----------------------------------------------|---------------------------------------------------|
| (1) $(-1 - 4i) - (-3 - 9i)$<br>$= 2 + 5i$     | (20) $(3 - 8i) + (9 - i)$<br>$= 12 - 9i$      | (39) $(-6 + 10i) + (-6 + 2i)$<br>$= -12 + 12i$    |
| (2) $(-5 + 3i) + (3 - 8i)$<br>$= -2 - 5i$     | (21) $(-3 + 6i) - (-1 - 2i)$<br>$= -2 + 8i$   | (40) $(-10 + 10i) + (-10 + 10i)$<br>$= -20 + 20i$ |
| (3) $(1 + 4i) + (9 + 5i)$<br>$= 10 + 9i$      | (22) $(-3 + 10i) - (-6 - 9i)$<br>$= 3 + 19i$  | (41) $(-1 + 4i) + (6 - 8i)$<br>$= 5 - 4i$         |
| (4) $(-2 + 7i) + (3 - 9i)$<br>$= 1 - 2i$      | (23) $(-6 + 2i) + (-2 + 10i)$<br>$= -8 + 12i$ | (42) $(1 - 3i) + (10 + 3i)$<br>$= 11$             |
| (5) $(3 - 7i) - (-2 + i)$<br>$= 5 - 8i$       | (24) $(-8 - 7i) + (4 - 5i)$<br>$= -4 - 12i$   | (43) $(3 - 6i) - (-4 + 10i)$<br>$= 7 - 16i$       |
| (6) $(8 + 5i) + (-3 + 5i)$<br>$= 5 + 10i$     | (25) $(-8 + 9i) + (6 - 9i)$<br>$= -2$         | (44) $(9 - 4i) + (-10 + 10i)$<br>$= -1 + 6i$      |
| (7) $(10 + 10i) - (-10 - 7i)$<br>$= 20 + 17i$ | (26) $(6 + 7i) - (-6 - 8i)$<br>$= 12 + 15i$   | (45) $(5 + 3i) + (3 - 6i)$<br>$= 8 - 3i$          |
| (8) $(-2 + 9i) + (10 + 3i)$<br>$= 8 + 12i$    | (27) $(-3 - 3i) - (7 - 8i)$<br>$= -10 + 5i$   | (46) $(5 - 8i) - (-4 + 10i)$<br>$= 9 - 18i$       |
| (9) $(7 - 5i) + (-8 - 9i)$<br>$= -1 - 14i$    | (28) $(-7 + 6i) - (3 - 5i)$<br>$= -10 + 11i$  | (47) $(2 - 10i) + (-4 - 10i)$<br>$= -2 - 20i$     |
| (10) $(10 - 7i) + (-7 + 8i)$<br>$= 3 + i$     | (29) $(-1 + i) + (-9 - 4i)$<br>$= -10 - 3i$   | (48) $(-5 - 8i) + (9 - 9i)$<br>$= 4 - 17i$        |
| (11) $(5 + 3i) + (-8 + 2i)$<br>$= -3 + 5i$    | (30) $(-5 + 7i) - (3 + 4i)$<br>$= -8 + 3i$    | (49) $(4 + i) + (7 + 9i)$<br>$= 11 + 10i$         |
| (12) $(7 + 6i) + (5 - 6i)$<br>$= 12$          | (31) $(-7 + 4i) + (-8 - 7i)$<br>$= -15 - 3i$  | (50) $(8 - 9i) - (1 - 8i)$<br>$= 7 - i$           |
| (13) $(-9 - i) - (-10 + 2i)$<br>$= 1 - 3i$    | (32) $(10 + 7i) + (9 - i)$<br>$= 19 + 6i$     | (51) $(-5 - 2i) - (6 - 6i)$<br>$= -11 + 4i$       |
| (14) $(-7 - i) - (-1 - 8i)$<br>$= -6 + 7i$    | (33) $(6 + 5i) + (-9 - 2i)$<br>$= -3 + 3i$    | (52) $(9 - 6i) - (-1 + 2i)$<br>$= 10 - 8i$        |
| (15) $(-10 - 7i) - (1 + 8i)$<br>$= -11 - 15i$ | (34) $(-3 - 9i) - (2 - 9i)$<br>$= -5$         | (53) $(-5 - 9i) - (2 + 2i)$<br>$= -7 - 11i$       |
| (16) $(-10 - i) + (8 - 9i)$<br>$= -2 - 10i$   | (35) $(9 - 3i) + (2 - 4i)$<br>$= 11 - 7i$     | (54) $(-9 - 8i) + (-1 + 5i)$<br>$= -10 - 3i$      |
| (17) $(10 + 3i) + (-2 + i)$<br>$= 8 + 4i$     | (36) $(9 + 3i) + (-5 - 2i)$<br>$= 4 + i$      | (55) $(-8 - 6i) - (-9 + 3i)$<br>$= 1 - 9i$        |
| (18) $(-2 - i) + (5 + 8i)$<br>$= 3 + 7i$      | (37) $(8 + 6i) + (-10 + 5i)$<br>$= -2 + 11i$  | (56) $(-8 - 7i) - (-4 + 7i)$<br>$= -4 - 14i$      |
| (19) $(-8 - 2i) - (-5 - 6i)$<br>$= -3 + 4i$   | (38) $(-7 - i) - (2 + 10i)$<br>$= -9 - 11i$   | (57) $(5 - 10i) - (-8 + 9i)$<br>$= 13 - 19i$      |



## 1.5.2 乗法

複素数の積を求める問題。

(1)  $(-3 - 3i)(-10 + 6i)$

(16)  $(-8 - 7i)(-7 + 5i)$

(31)  $(-10 - 9i)(-6 + 5i)$

(2)  $(-3 - 6i)(-1 - 5i)$

(17)  $(3 + 9i)(4 + 3i)$

(32)  $(-6 + 4i)(4 - 2i)$

(3)  $(10 + 9i)(-4 - 5i)$

(18)  $(-2 - i)(-7 + 4i)$

(33)  $(10 + i)(-4 - i)$

(4)  $(-7 - 8i)(-5 + 2i)$

(19)  $(10 - 7i)(5 + 2i)$

(34)  $(-10 - 8i)(6 + 6i)$

(5)  $(-3 + 8i)(-2 + 7i)$

(20)  $(5 + 4i)(-2 + 7i)$

(35)  $(-3 - 4i)(-6 + 10i)$

(6)  $(4 + 9i)(-6 + 3i)$

(21)  $(1 - i)(1 - 5i)$

(36)  $(10 + 2i)(5 - 8i)$

(7)  $(6 + 7i)(-2 + 8i)$

(22)  $(10 + 10i)(-9 - 9i)$

(37)  $(4 - 6i)(3 - 10i)$

(8)  $(7 + 3i)(-10 - 4i)$

(23)  $(-7 - 4i)(-3 - 8i)$

(38)  $(-7 + 2i)(-10 + 6i)$

(9)  $(5 + 10i)(-2 - 4i)$

(24)  $(10 + i)(7 - 6i)$

(39)  $(2 - 3i)(6 + i)$

(10)  $(-4 - i)(-6 + 5i)$

(25)  $(-4 + 3i)(-5 - 6i)$

(40)  $(-5 - 3i)(-4 - 3i)$

(11)  $(-1 - 4i)(4 - 8i)$

(26)  $(1 - 9i)(-2 + 5i)$

(41)  $(-7 - 6i)(-10 + 6i)$

(12)  $(-10 - 4i)(7 + 7i)$

(27)  $(-9 - 10i)(-7 - 3i)$

(42)  $(-5 - 5i)(-6 - 7i)$

(13)  $(5 - 10i)(5 + 6i)$

(28)  $(-2 + i)(8 + i)$

(43)  $(-10 + 3i)(-3 + 4i)$

(14)  $(5 + 10i)(4 - 7i)$

(29)  $(-6 + 10i)(-7 + 8i)$

(44)  $(4 + 4i)(-4 + 6i)$

(15)  $(9 + 4i)(-4 + 2i)$

(30)  $(-2 - 4i)(2 - 6i)$

(45)  $(8 - i)(-5 - 7i)$

(1)  $(-2 - 6i)(-9 - i)$

(16)  $(-8 - 3i)(-10 - 7i)$

(31)  $(2 - 6i)(-9 - 6i)$

(2)  $(9 - 5i)(4 + 6i)$

(17)  $(6 + 10i)(-4 - 7i)$

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(3)  $(-2 - 3i)(4 - 5i)$

(18)  $(-2 - 8i)(-7 + 8i)$

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(4)  $(10 - i)(-6 - 3i)$

(19)  $(-10 + 2i)(-8 + 4i)$

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(14)  $(6 - i)(6 + i)$

(29)  $(-10 - 9i)(-9 + 9i)$

(44)  $(8 + 2i)(10 + 10i)$

(15)  $(-2 + 10i)(1 + 8i)$

(30)  $(5 - 6i)(-10 + 8i)$

(45)  $(2 + 6i)(4 + 3i)$

(1)  $(-10 - 10i)(-9 - 5i)$

(16)  $(6 + 9i)(6 - i)$

(31)  $(3 - 3i)(-3 + i)$

(2)  $(3 + 7i)(8 + 9i)$

(17)  $(7 + 5i)(9 + 5i)$

(32)  $(-3 + 10i)(-4 - i)$

(3)  $(-5 + i)(2 + 7i)$

(18)  $(-3 + 10i)(2 - 8i)$

(33)  $(1 + 10i)(-4 - 8i)$

(4)  $(9 - i)(1 - 4i)$

(19)  $(-5 - i)(10 - i)$

(34)  $(-2 + 9i)(4 + 8i)$

(5)  $(-1 + i)(-10 + 3i)$

(20)  $(-10 + 8i)(4 + 5i)$

(35)  $(1 + 8i)(-10 - 6i)$

(6)  $(7 - 9i)(-3 - i)$

(21)  $(-8 + 7i)(7 + 3i)$

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(7)  $(10 + i)(9 + 3i)$

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(8)  $(6 - 9i)(10 + 2i)$

(23)  $(1 + 3i)(-4 - 5i)$

(38)  $(7 - 2i)(3 + 3i)$

(9)  $(-10 + 5i)(-1 - i)$

(24)  $(-2 - 8i)(7 - 5i)$

(39)  $(10 - 3i)(-10 - 8i)$

(10)  $(-4 - 9i)(-3 - 4i)$

(25)  $(7 + 3i)(6 + 2i)$

(40)  $(3 - 7i)(2 + 10i)$

(11)  $(-1 - 5i)(5 + 4i)$

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(16)  $(1 + 5i)(1 - 6i)$

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(34)  $(6 - 10i)(2 + i)$

(5)  $(9 - 2i)(8 + i)$

(20)  $(4 + 5i)(1 + 8i)$

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(43)  $(-10 - 3i)(4 + 9i)$

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(30)  $(6 + 5i)(-6 - 4i)$

(45)  $(-3 + 3i)(-10 - 4i)$



(1)  $(-8 + 3i)(10 + 4i)$

(16)  $(-3 - 4i)(10 + 10i)$

(31)  $(2 + 8i)(-2 - 3i)$

(2)  $(5 + 8i)(-7 - i)$

(17)  $(1 - i)(10 + 8i)$

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(27)  $(-9 + 8i)(-10 + 9i)$

(42)  $(7 - 5i)(4 - 8i)$

(13)  $(5 - 2i)(-9 + 4i)$

(28)  $(-7 + 9i)(2 - 7i)$

(43)  $(-1 - 8i)(7 - 5i)$

(14)  $(5 - 8i)(-5 - 5i)$

(29)  $(2 - i)(-9 + 3i)$

(44)  $(-5 - 6i)(1 + 7i)$

(15)  $(-1 - 3i)(10 + 10i)$

(30)  $(-8 + 6i)(-8 - 7i)$

(45)  $(-4 + 7i)(4 + 6i)$

(1)  $(8 - 6i)(5 - 7i)$

(16)  $(7 + 7i)(-10 + 3i)$

(31)  $(9 + 4i)(4 + 10i)$

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(32)  $(-4 - 6i)(8 - 10i)$

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(19)  $(-5 - 3i)(9 + 9i)$

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(35)  $(-2 - 10i)(3 - 10i)$

(6)  $(-6 + 4i)(9 - 8i)$

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(30)  $(5 + 2i)(8 + 3i)$

(45)  $(3 + 8i)(10 + 4i)$

(1)  $(7 - 3i)(5 + 3i)$

(16)  $(2 + 8i)(9 + 7i)$

(31)  $(-2 + 7i)(-1 + i)$

(2)  $(-4 + 7i)(1 + 4i)$

(17)  $(-5 - 6i)(-8 + 10i)$

(32)  $(-5 + 2i)(3 - 4i)$

(3)  $(-7 + 5i)(5 - i)$

(18)  $(-10 + 3i)(1 + 8i)$

(33)  $(9 - 5i)(2 + 4i)$

(4)  $(10 + 3i)(4 - 10i)$

(19)  $(-7 - 9i)(-7 - 3i)$

(34)  $(1 + i)(-1 - i)$

(5)  $(9 + 6i)(8 + i)$

(20)  $(-9 - 9i)(1 - 10i)$

(35)  $(9 - 10i)(-7 + 10i)$

(6)  $(9 + 4i)(-6 - 10i)$

(21)  $(1 + 6i)(6 - 8i)$

(36)  $(4 - 10i)(1 + 2i)$

(7)  $(-5 - 2i)(-2 - 2i)$

(22)  $(-7 - 2i)(-1 + 2i)$

(37)  $(6 + 4i)(-3 + 2i)$

(8)  $(-6 + 6i)(6 + 7i)$

(23)  $(-2 + 7i)(-7 + 6i)$

(38)  $(-10 - 8i)(9 - 9i)$

(9)  $(10 - 7i)(9 - 4i)$

(24)  $(1 + 9i)(6 - 8i)$

(39)  $(3 - 9i)(-4 + 3i)$

(10)  $(3 - 2i)(1 - 10i)$

(25)  $(-2 + 10i)(-8 + 10i)$

(40)  $(5 + 7i)(-8 + 10i)$

(11)  $(-8 + 5i)(-2 + 3i)$

(26)  $(-7 - 4i)(-6 + i)$

(41)  $(-5 - 4i)(-5 - 5i)$

(12)  $(-9 + 6i)(4 - 7i)$

(27)  $(-7 - 9i)(3 + 7i)$

(42)  $(-10 - 7i)(-1 - 8i)$

(13)  $(8 + 9i)(-9 - 3i)$

(28)  $(-5 - 5i)(-2 - 9i)$

(43)  $(10 + 8i)(-9 - 9i)$

(14)  $(4 + 9i)(-1 + 4i)$

(29)  $(-5 + 10i)(1 - 5i)$

(44)  $(-9 + 3i)(-4 + 2i)$

(15)  $(-5 - 8i)(8 - 6i)$

(30)  $(-8 + 9i)(3 + 8i)$

(45)  $(8 - 8i)(4 - 7i)$

$$(1) \quad (-3-3i)(-10+6i) \\ = 48+12i$$

$$(2) \quad (-3-6i)(-1-5i) \\ = -27+21i$$

$$(3) \quad (10+9i)(-4-5i) \\ = 5-86i$$

$$(4) \quad (-7-8i)(-5+2i) \\ = 51+26i$$

$$(5) \quad (-3+8i)(-2+7i) \\ = -50-37i$$

$$(6) \quad (4+9i)(-6+3i) \\ = -51-42i$$

$$(7) \quad (6+7i)(-2+8i) \\ = -68+34i$$

$$(8) \quad (7+3i)(-10-4i) \\ = -58-58i$$

$$(9) \quad (5+10i)(-2-4i) \\ = 30-40i$$

$$(10) \quad (-4-i)(-6+5i) \\ = 29-14i$$

$$(11) \quad (-1-4i)(4-8i) \\ = -36-8i$$

$$(12) \quad (-10-4i)(7+7i) \\ = -42-98i$$

$$(13) \quad (5-10i)(5+6i) \\ = 85-20i$$

$$(14) \quad (5+10i)(4-7i) \\ = 90+5i$$

$$(15) \quad (9+4i)(-4+2i) \\ = -44+2i$$

$$(16) \quad (-8-7i)(-7+5i) \\ = 91+9i$$

$$(17) \quad (3+9i)(4+3i) \\ = -15+45i$$

$$(18) \quad (-2-i)(-7+4i) \\ = 18-i$$

$$(19) \quad (10-7i)(5+2i) \\ = 64-15i$$

$$(20) \quad (5+4i)(-2+7i) \\ = -38+27i$$

$$(21) \quad (1-i)(1-5i) \\ = -4-6i$$

$$(22) \quad (10+10i)(-9-9i) \\ = -180i$$

$$(23) \quad (-7-4i)(-3-8i) \\ = -11+68i$$

$$(24) \quad (10+i)(7-6i) \\ = 76-53i$$

$$(25) \quad (-4+3i)(-5-6i) \\ = 38+9i$$

$$(26) \quad (1-9i)(-2+5i) \\ = 43+23i$$

$$(27) \quad (-9-10i)(-7-3i) \\ = 33+97i$$

$$(28) \quad (-2+i)(8+i) \\ = -17+6i$$

$$(29) \quad (-6+10i)(-7+8i) \\ = -38-118i$$

$$(30) \quad (-2-4i)(2-6i) \\ = -28+4i$$

$$(31) \quad (-10-9i)(-6+5i) \\ = 105+4i$$

$$(32) \quad (-6+4i)(4-2i) \\ = -16+28i$$

$$(33) \quad (10+i)(-4-i) \\ = -39-14i$$

$$(34) \quad (-10-8i)(6+6i) \\ = -12-108i$$

$$(35) \quad (-3-4i)(-6+10i) \\ = 58-6i$$

$$(36) \quad (10+2i)(5-8i) \\ = 66-70i$$

$$(37) \quad (4-6i)(3-10i) \\ = -48-58i$$

$$(38) \quad (-7+2i)(-10+6i) \\ = 58-62i$$

$$(39) \quad (2-3i)(6+i) \\ = 15-16i$$

$$(40) \quad (-5-3i)(-4-3i) \\ = 11+27i$$

$$(41) \quad (-7-6i)(-10+6i) \\ = 106+18i$$

$$(42) \quad (-5-5i)(-6-7i) \\ = -5+65i$$

$$(43) \quad (-10+3i)(-3+4i) \\ = 18-49i$$

$$(44) \quad (4+4i)(-4+6i) \\ = -40+8i$$

$$(45) \quad (8-i)(-5-7i) \\ = -47-51i$$

$$(1) \quad (-2 - 6i)(-9 - i) \\ = 12 + 56i$$

$$(2) \quad (9 - 5i)(4 + 6i) \\ = 66 + 34i$$

$$(3) \quad (-2 - 3i)(4 - 5i) \\ = -23 - 2i$$

$$(4) \quad (10 - i)(-6 - 3i) \\ = -63 - 24i$$

$$(5) \quad (7 - 3i)(10 + 2i) \\ = 76 - 16i$$

$$(6) \quad (-9 + 10i)(-9 - 8i) \\ = 161 - 18i$$

$$(7) \quad (8 - 2i)(-7 - 8i) \\ = -72 - 50i$$

$$(8) \quad (-10 - 10i)(5 + 3i) \\ = -20 - 80i$$

$$(9) \quad (5 + 2i)(6 + 3i) \\ = 24 + 27i$$

$$(10) \quad (-8 - 10i)(-2 - i) \\ = 6 + 28i$$

$$(11) \quad (-4 + 2i)(-9 - 8i) \\ = 52 + 14i$$

$$(12) \quad (1 + 9i)(-7 - 6i) \\ = 47 - 69i$$

$$(13) \quad (-8 + 3i)(1 - 9i) \\ = 19 + 75i$$

$$(14) \quad (6 - i)(6 + i) \\ = 37$$

$$(15) \quad (-2 + 10i)(1 + 8i) \\ = -82 - 6i$$

$$(16) \quad (-8 - 3i)(-10 - 7i) \\ = 59 + 86i$$

$$(17) \quad (6 + 10i)(-4 - 7i) \\ = 46 - 82i$$

$$(18) \quad (-2 - 8i)(-7 + 8i) \\ = 78 + 40i$$

$$(19) \quad (-10 + 2i)(-8 + 4i) \\ = 72 - 56i$$

$$(20) \quad (1 + 9i)(-4 + 7i) \\ = -67 - 29i$$

$$(21) \quad (3 + 8i)(9 - 7i) \\ = 83 + 51i$$

$$(22) \quad (2 - 3i)(-4 + 7i) \\ = 13 + 26i$$

$$(23) \quad (-4 + 3i)(-2 + 5i) \\ = -7 - 26i$$

$$(24) \quad (2 + 8i)(-3 + 7i) \\ = -62 - 10i$$

$$(25) \quad (-4 + 2i)(-1 - 2i) \\ = 8 + 6i$$

$$(26) \quad (9 + 6i)(-2 - 6i) \\ = 18 - 66i$$

$$(27) \quad (-6 + 7i)(5 - 9i) \\ = 33 + 89i$$

$$(28) \quad (-7 + 4i)(-4 - 10i) \\ = 68 + 54i$$

$$(29) \quad (-10 - 9i)(-9 + 9i) \\ = 171 - 9i$$

$$(30) \quad (5 - 6i)(-10 + 8i) \\ = -2 + 100i$$

$$(31) \quad (2 - 6i)(-9 - 6i) \\ = -54 + 42i$$

$$(32) \quad (-8 + 2i)(-6 - 8i) \\ = 64 + 52i$$

$$(33) \quad (2 + 6i)(7 + 2i) \\ = 2 + 46i$$

$$(34) \quad (-5 + 3i)(-5 + 8i) \\ = 1 - 55i$$

$$(35) \quad (-5 - 5i)(10 - 3i) \\ = -65 - 35i$$

$$(36) \quad (1 - 2i)(10 - 2i) \\ = 6 - 22i$$

$$(37) \quad (-4 + 4i)(3 + 3i) \\ = -24$$

$$(38) \quad (-10 + 9i)(8 + 4i) \\ = -116 + 32i$$

$$(39) \quad (2 - 6i)(-7 + i) \\ = -8 + 44i$$

$$(40) \quad (1 - 6i)(-7 + 6i) \\ = 29 + 48i$$

$$(41) \quad (-6 + 2i)(-4 + 6i) \\ = 12 - 44i$$

$$(42) \quad (-6 + 8i)(-7 + 6i) \\ = -6 - 92i$$

$$(43) \quad (-7 + i)(5 - 4i) \\ = -31 + 33i$$

$$(44) \quad (8 + 2i)(10 + 10i) \\ = 60 + 100i$$

$$(45) \quad (2 + 6i)(4 + 3i) \\ = -10 + 30i$$

$$\begin{aligned}(1) \quad & (-10 - 10i)(-9 - 5i) \\ & = 40 + 140i\end{aligned}$$

$$\begin{aligned}(2) \quad & (3 + 7i)(8 + 9i) \\ & = -39 + 83i\end{aligned}$$

$$\begin{aligned}(3) \quad & (-5 + i)(2 + 7i) \\ & = -17 - 33i\end{aligned}$$

$$\begin{aligned}(4) \quad & (9 - i)(1 - 4i) \\ & = 5 - 37i\end{aligned}$$

$$\begin{aligned}(5) \quad & (-1 + i)(-10 + 3i) \\ & = 7 - 13i\end{aligned}$$

$$\begin{aligned}(6) \quad & (7 - 9i)(-3 - i) \\ & = -30 + 20i\end{aligned}$$

$$\begin{aligned}(7) \quad & (10 + i)(9 + 3i) \\ & = 87 + 39i\end{aligned}$$

$$\begin{aligned}(8) \quad & (6 - 9i)(10 + 2i) \\ & = 78 - 78i\end{aligned}$$

$$\begin{aligned}(9) \quad & (-10 + 5i)(-1 - i) \\ & = 15 + 5i\end{aligned}$$

$$\begin{aligned}(10) \quad & (-4 - 9i)(-3 - 4i) \\ & = -24 + 43i\end{aligned}$$

$$\begin{aligned}(11) \quad & (-1 - 5i)(5 + 4i) \\ & = 15 - 29i\end{aligned}$$

$$\begin{aligned}(12) \quad & (-7 - 9i)(10 - 6i) \\ & = -124 - 48i\end{aligned}$$

$$\begin{aligned}(13) \quad & (10 + 2i)(-6 + 10i) \\ & = -80 + 88i\end{aligned}$$

$$\begin{aligned}(14) \quad & (9 + 3i)(-7 - 4i) \\ & = -51 - 57i\end{aligned}$$

$$\begin{aligned}(15) \quad & (3 - 7i)(-2 - 10i) \\ & = -76 - 16i\end{aligned}$$

$$\begin{aligned}(16) \quad & (6 + 9i)(6 - i) \\ & = 45 + 48i\end{aligned}$$

$$\begin{aligned}(17) \quad & (7 + 5i)(9 + 5i) \\ & = 38 + 80i\end{aligned}$$

$$\begin{aligned}(18) \quad & (-3 + 10i)(2 - 8i) \\ & = 74 + 44i\end{aligned}$$

$$\begin{aligned}(19) \quad & (-5 - i)(10 - i) \\ & = -51 - 5i\end{aligned}$$

$$\begin{aligned}(20) \quad & (-10 + 8i)(4 + 5i) \\ & = -80 - 18i\end{aligned}$$

$$\begin{aligned}(21) \quad & (-8 + 7i)(7 + 3i) \\ & = -77 + 25i\end{aligned}$$

$$\begin{aligned}(22) \quad & (-3 - 4i)(-1 + 8i) \\ & = 35 - 20i\end{aligned}$$

$$\begin{aligned}(23) \quad & (1 + 3i)(-4 - 5i) \\ & = 11 - 17i\end{aligned}$$

$$\begin{aligned}(24) \quad & (-2 - 8i)(7 - 5i) \\ & = -54 - 46i\end{aligned}$$

$$\begin{aligned}(25) \quad & (7 + 3i)(6 + 2i) \\ & = 36 + 32i\end{aligned}$$

$$\begin{aligned}(26) \quad & (2 + 10i)(-4 - 2i) \\ & = 12 - 44i\end{aligned}$$

$$\begin{aligned}(27) \quad & (4 - 2i)(-7 - 7i) \\ & = -42 - 14i\end{aligned}$$

$$\begin{aligned}(28) \quad & (-10 + 9i)(-9 + 9i) \\ & = 9 - 171i\end{aligned}$$

$$\begin{aligned}(29) \quad & (2 - 2i)(-6 + 9i) \\ & = 6 + 30i\end{aligned}$$

$$\begin{aligned}(30) \quad & (5 + 2i)(-9 - 3i) \\ & = -39 - 33i\end{aligned}$$

$$\begin{aligned}(31) \quad & (3 - 3i)(-3 + i) \\ & = -6 + 12i\end{aligned}$$

$$\begin{aligned}(32) \quad & (-3 + 10i)(-4 - i) \\ & = 22 - 37i\end{aligned}$$

$$\begin{aligned}(33) \quad & (1 + 10i)(-4 - 8i) \\ & = 76 - 48i\end{aligned}$$

$$\begin{aligned}(34) \quad & (-2 + 9i)(4 + 8i) \\ & = -80 + 20i\end{aligned}$$

$$\begin{aligned}(35) \quad & (1 + 8i)(-10 - 6i) \\ & = 38 - 86i\end{aligned}$$

$$\begin{aligned}(36) \quad & (-5 + i)(-9 - 2i) \\ & = 47 + i\end{aligned}$$

$$\begin{aligned}(37) \quad & (7 - 2i)(-10 + 2i) \\ & = -66 + 34i\end{aligned}$$

$$\begin{aligned}(38) \quad & (7 - 2i)(3 + 3i) \\ & = 27 + 15i\end{aligned}$$

$$\begin{aligned}(39) \quad & (10 - 3i)(-10 - 8i) \\ & = -124 - 50i\end{aligned}$$

$$\begin{aligned}(40) \quad & (3 - 7i)(2 + 10i) \\ & = 76 + 16i\end{aligned}$$

$$\begin{aligned}(41) \quad & (3 + 6i)(5 - 9i) \\ & = 69 + 3i\end{aligned}$$

$$\begin{aligned}(42) \quad & (-1 + 6i)(-9 + 9i) \\ & = -45 - 63i\end{aligned}$$

$$\begin{aligned}(43) \quad & (4 - i)(3 - i) \\ & = 11 - 7i\end{aligned}$$

$$\begin{aligned}(44) \quad & (-9 + 10i)(3 - 3i) \\ & = 3 + 57i\end{aligned}$$

$$\begin{aligned}(45) \quad & (10 - 2i)(9 - 5i) \\ & = 80 - 68i\end{aligned}$$



$$(1) \quad (-8 - 7i)(2 - 4i) \\ = -44 + 18i$$

$$(2) \quad (-8 + 8i)(-1 + 2i) \\ = -8 - 24i$$

$$(3) \quad (-7 - 7i)(8 + 3i) \\ = -35 - 77i$$

$$(4) \quad (-1 - 8i)(-3 + 7i) \\ = 59 + 17i$$

$$(5) \quad (9 - 2i)(8 + i) \\ = 74 - 7i$$

$$(6) \quad (5 + i)(2 - 6i) \\ = 16 - 28i$$

$$(7) \quad (1 + 5i)(1 + 7i) \\ = -34 + 12i$$

$$(8) \quad (2 + 4i)(-9 - i) \\ = -14 - 38i$$

$$(9) \quad (6 + 8i)(-2 + 6i) \\ = -60 + 20i$$

$$(10) \quad (4 + 4i)(8 - 5i) \\ = 52 + 12i$$

$$(11) \quad (-5 + 9i)(-9 - 7i) \\ = 108 - 46i$$

$$(12) \quad (10 - i)(-9 - 7i) \\ = -97 - 61i$$

$$(13) \quad (-7 - 6i)(2 + 3i) \\ = 4 - 33i$$

$$(14) \quad (3 + 9i)(-2 - 6i) \\ = 48 - 36i$$

$$(15) \quad (2 + 3i)(-1 + 3i) \\ = -11 + 3i$$

$$(16) \quad (1 + 5i)(1 - 6i) \\ = 31 - i$$

$$(17) \quad (-10 + 7i)(-9 - 4i) \\ = 118 - 23i$$

$$(18) \quad (6 + 6i)(8 - 7i) \\ = 90 + 6i$$

$$(19) \quad (9 + 4i)(3 - 5i) \\ = 47 - 33i$$

$$(20) \quad (4 + 5i)(1 + 8i) \\ = -36 + 37i$$

$$(21) \quad (-3 + 3i)(-4 + 3i) \\ = 3 - 21i$$

$$(22) \quad (-8 - 3i)(3 + 7i) \\ = -3 - 65i$$

$$(23) \quad (2 + 5i)(3 + i) \\ = 1 + 17i$$

$$(24) \quad (-8 - 4i)(-10 + 2i) \\ = 88 + 24i$$

$$(25) \quad (4 + 8i)(6 - 7i) \\ = 80 + 20i$$

$$(26) \quad (10 - 9i)(3 + 6i) \\ = 84 + 33i$$

$$(27) \quad (-7 + 10i)(6 - 5i) \\ = 8 + 95i$$

$$(28) \quad (10 - 4i)(-4 - 6i) \\ = -64 - 44i$$

$$(29) \quad (-9 + 4i)(7 + 2i) \\ = -71 + 10i$$

$$(30) \quad (-5 + 3i)(-1 + 8i) \\ = -19 - 43i$$

$$(31) \quad (10 + 4i)(-6 + 7i) \\ = -88 + 46i$$

$$(32) \quad (-6 + 10i)(3 - 8i) \\ = 62 + 78i$$

$$(33) \quad (1 - 9i)(-10 + 6i) \\ = 44 + 96i$$

$$(34) \quad (6 - 10i)(2 + i) \\ = 22 - 14i$$

$$(35) \quad (-4 - 2i)(-2 - 3i) \\ = 2 + 16i$$

$$(36) \quad (8 + 8i)(3 + 7i) \\ = -32 + 80i$$

$$(37) \quad (6 - 3i)(-9 - 9i) \\ = -81 - 27i$$

$$(38) \quad (-6 - 10i)(-1 - 7i) \\ = -64 + 52i$$

$$(39) \quad (-7 + i)(-6 + 4i) \\ = 38 - 34i$$

$$(40) \quad (-9 + 4i)(3 + 10i) \\ = -67 - 78i$$

$$(41) \quad (3 + 5i)(1 + 6i) \\ = -27 + 23i$$

$$(42) \quad (9 + i)(-4 + 6i) \\ = -42 + 50i$$

$$(43) \quad (-10 - 3i)(4 + 9i) \\ = -13 - 102i$$

$$(44) \quad (4 - 2i)(-6 + 9i) \\ = -6 + 48i$$

$$(45) \quad (-7 + 3i)(-3 + 2i) \\ = 15 - 23i$$

$$(1) \quad (-1+i)(8-3i) \\ = -5 + 11i$$

$$(2) \quad (8+i)(6-10i) \\ = 58 - 74i$$

$$(3) \quad (-9+5i)(8+9i) \\ = -117 - 41i$$

$$(4) \quad (-8-7i)(-1-10i) \\ = -62 + 87i$$

$$(5) \quad (-3+9i)(6+10i) \\ = -108 + 24i$$

$$(6) \quad (-1+2i)(-2+2i) \\ = -2 - 6i$$

$$(7) \quad (6-4i)(1+8i) \\ = 38 + 44i$$

$$(8) \quad (5-9i)(1+3i) \\ = 32 + 6i$$

$$(9) \quad (4+2i)(-6+i) \\ = -26 - 8i$$

$$(10) \quad (-10+i)(9+10i) \\ = -100 - 91i$$

$$(11) \quad (-10+i)(8+8i) \\ = -88 - 72i$$

$$(12) \quad (10-3i)(5+7i) \\ = 71 + 55i$$

$$(13) \quad (8+9i)(-6-9i) \\ = 33 - 126i$$

$$(14) \quad (-9-5i)(4+5i) \\ = -11 - 65i$$

$$(15) \quad (-4+6i)(4-3i) \\ = 2 + 36i$$

$$(16) \quad (-3-3i)(4-6i) \\ = -30 + 6i$$

$$(17) \quad (9+2i)(-6+i) \\ = -56 - 3i$$

$$(18) \quad (-5-10i)(-7-3i) \\ = 5 + 85i$$

$$(19) \quad (-6-9i)(1+9i) \\ = 75 - 63i$$

$$(20) \quad (10-3i)(-9-10i) \\ = -120 - 73i$$

$$(21) \quad (-5-i)(-6+7i) \\ = 37 - 29i$$

$$(22) \quad (4-8i)(-4-i) \\ = -24 + 28i$$

$$(23) \quad (1-i)(2-6i) \\ = -4 - 8i$$

$$(24) \quad (5+6i)(7-4i) \\ = 59 + 22i$$

$$(25) \quad (4-8i)(-10+8i) \\ = 24 + 112i$$

$$(26) \quad (-5-5i)(9+7i) \\ = -10 - 80i$$

$$(27) \quad (-10+9i)(-7+i) \\ = 61 - 73i$$

$$(28) \quad (-6-10i)(3-4i) \\ = -58 - 6i$$

$$(29) \quad (9+4i)(2+6i) \\ = -6 + 62i$$

$$(30) \quad (-6-9i)(9-3i) \\ = -81 - 63i$$

$$(31) \quad (7-7i)(7-3i) \\ = 28 - 70i$$

$$(32) \quad (3+3i)(-5+5i) \\ = -30$$

$$(33) \quad (4+2i)(-4-7i) \\ = -2 - 36i$$

$$(34) \quad (-3-10i)(9-10i) \\ = -127 - 60i$$

$$(35) \quad (-8+2i)(4+6i) \\ = -44 - 40i$$

$$(36) \quad (-7-i)(-9-5i) \\ = 58 + 44i$$

$$(37) \quad (7+10i)(4-10i) \\ = 128 - 30i$$

$$(38) \quad (-4-2i)(-9-i) \\ = 34 + 22i$$

$$(39) \quad (10+7i)(2+4i) \\ = -8 + 54i$$

$$(40) \quad (-7-7i)(-6+i) \\ = 49 + 35i$$

$$(41) \quad (-7+7i)(-7+i) \\ = 42 - 56i$$

$$(42) \quad (8+7i)(2-6i) \\ = 58 - 34i$$

$$(43) \quad (-2-10i)(7+7i) \\ = 56 - 84i$$

$$(44) \quad (7-i)(4-5i) \\ = 23 - 39i$$

$$(45) \quad (3-5i)(7+3i) \\ = 36 - 26i$$

$$(1) (3 + 6i)(6 + 9i) \\ = -36 + 63i$$

$$(2) (5 + 3i)(-1 - 3i) \\ = 4 - 18i$$

$$(3) (7 + 7i)(4 - 8i) \\ = 84 - 28i$$

$$(4) (-9 + 6i)(-9 + 9i) \\ = 27 - 135i$$

$$(5) (-6 - 2i)(9 + 9i) \\ = -36 - 72i$$

$$(6) (-3 - 3i)(-8 + 8i) \\ = 48$$

$$(7) (8 - 5i)(3 - 6i) \\ = -6 - 63i$$

$$(8) (8 - 3i)(9 + 10i) \\ = 102 + 53i$$

$$(9) (10 - 8i)(-5 + 3i) \\ = -26 + 70i$$

$$(10) (-9 - 8i)(-6 - 9i) \\ = -18 + 129i$$

$$(11) (7 - 10i)(5 + 7i) \\ = 105 - i$$

$$(12) (4 - 2i)(-2 + 5i) \\ = 2 + 24i$$

$$(13) (-5 - 4i)(1 + 2i) \\ = 3 - 14i$$

$$(14) (-9 - 4i)(-4 + i) \\ = 40 + 7i$$

$$(15) (-7 - 4i)(-5 - i) \\ = 31 + 27i$$

$$(16) (-7 - 3i)(-10 - 4i) \\ = 58 + 58i$$

$$(17) (3 + 7i)(-9 + 9i) \\ = -90 - 36i$$

$$(18) (-9 - i)(2 + 5i) \\ = -13 - 47i$$

$$(19) (9 - 2i)(-1 + 4i) \\ = -1 + 38i$$

$$(20) (4 - 4i)(-2 - 7i) \\ = -36 - 20i$$

$$(21) (7 - i)(4 - 6i) \\ = 22 - 46i$$

$$(22) (6 + 4i)(-4 - 5i) \\ = -4 - 46i$$

$$(23) (2 - 7i)(4 + 7i) \\ = 57 - 14i$$

$$(24) (-2 + 3i)(-8 + 2i) \\ = 10 - 28i$$

$$(25) (-1 + 2i)(-3 - 8i) \\ = 19 + 2i$$

$$(26) (7 - 6i)(7 - 7i) \\ = 7 - 91i$$

$$(27) (4 - 7i)(-4 + 6i) \\ = 26 + 52i$$

$$(28) (8 - 6i)(-9 + 10i) \\ = -12 + 134i$$

$$(29) (9 - 6i)(-9 + 7i) \\ = -39 + 117i$$

$$(30) (-7 + 3i)(3 - 2i) \\ = -15 + 23i$$

$$(31) (2 + 2i)(-9 - 5i) \\ = -8 - 28i$$

$$(32) (-5 - 5i)(-10 - 7i) \\ = 15 + 85i$$

$$(33) (1 + 5i)(3 - 2i) \\ = 13 + 13i$$

$$(34) (9 - 3i)(-7 - 7i) \\ = -84 - 42i$$

$$(35) (-3 + 2i)(4 + 5i) \\ = -22 - 7i$$

$$(36) (-5 + 10i)(-3 - 9i) \\ = 105 + 15i$$

$$(37) (1 - 5i)(-2 - 4i) \\ = -22 + 6i$$

$$(38) (-9 - 5i)(9 - 3i) \\ = -96 - 18i$$

$$(39) (3 - i)(10 + 9i) \\ = 39 + 17i$$

$$(40) (-7 - 9i)(7 - 10i) \\ = -139 + 7i$$

$$(41) (3 - 3i)(-8 - 5i) \\ = -39 + 9i$$

$$(42) (6 + 4i)(-5 - 7i) \\ = -2 - 62i$$

$$(43) (8 + 8i)(4 - 5i) \\ = 72 - 8i$$

$$(44) (5 - i)(-7 - 6i) \\ = -41 - 23i$$

$$(45) (-1 + 2i)(1 + 8i) \\ = -17 - 6i$$

$$(1) (5 - 4i)(9 + 2i) \\ = 53 - 26i$$

$$(2) (10 + 9i)(-7 - 7i) \\ = -7 - 133i$$

$$(3) (9 + 6i)(2 + 7i) \\ = -24 + 75i$$

$$(4) (-1 + 2i)(1 + 3i) \\ = -7 - i$$

$$(5) (-6 + 7i)(1 + 5i) \\ = -41 - 23i$$

$$(6) (6 + 5i)(-1 + 6i) \\ = -36 + 31i$$

$$(7) (-5 + i)(-4 - 2i) \\ = 22 + 6i$$

$$(8) (6 - 10i)(-10 + 9i) \\ = 30 + 154i$$

$$(9) (-6 - 4i)(-7 + 3i) \\ = 54 + 10i$$

$$(10) (-9 + 8i)(-7 - 8i) \\ = 127 + 16i$$

$$(11) (-4 - 5i)(2 - 9i) \\ = -53 + 26i$$

$$(12) (10 - 9i)(-8 + 3i) \\ = -53 + 102i$$

$$(13) (6 + 10i)(4 + 8i) \\ = -56 + 88i$$

$$(14) (-3 + i)(-8 - 10i) \\ = 34 + 22i$$

$$(15) (5 - 9i)(8 - 4i) \\ = 4 - 92i$$

$$(16) (-8 + 8i)(-7 + 10i) \\ = -24 - 136i$$

$$(17) (-4 - 3i)(-7 - 8i) \\ = 4 + 53i$$

$$(18) (6 - 5i)(5 - 10i) \\ = -20 - 85i$$

$$(19) (8 + 5i)(10 + 7i) \\ = 45 + 106i$$

$$(20) (-1 - 3i)(-2 - i) \\ = -1 + 7i$$

$$(21) (5 - 4i)(-8 + 8i) \\ = -8 + 72i$$

$$(22) (6 + 8i)(5 + 5i) \\ = -10 + 70i$$

$$(23) (1 - 7i)(-1 - 3i) \\ = -22 + 4i$$

$$(24) (-6 - 8i)(-5 - 4i) \\ = -2 + 64i$$

$$(25) (4 - i)(-10 + 6i) \\ = -34 + 34i$$

$$(26) (2 - 9i)(1 + 10i) \\ = 92 + 11i$$

$$(27) (5 + 5i)(6 - 5i) \\ = 55 + 5i$$

$$(28) (-7 - i)(-5 + 6i) \\ = 41 - 37i$$

$$(29) (5 - 3i)(-1 + 5i) \\ = 10 + 28i$$

$$(30) (6 + 5i)(-6 - 4i) \\ = -16 - 54i$$

$$(31) (5 - 10i)(-5 + 4i) \\ = 15 + 70i$$

$$(32) (-3 + 5i)(6 + 9i) \\ = -63 + 3i$$

$$(33) (5 + 7i)(6 + 3i) \\ = 9 + 57i$$

$$(34) (3 + 3i)(-1 - i) \\ = -6i$$

$$(35) (4 + 4i)(2 + 10i) \\ = -32 + 48i$$

$$(36) (6 - 2i)(1 - 7i) \\ = -8 - 44i$$

$$(37) (7 - 6i)(-5 + i) \\ = -29 + 37i$$

$$(38) (-6 - 4i)(5 - 5i) \\ = -50 + 10i$$

$$(39) (3 - 7i)(-1 + 3i) \\ = 18 + 16i$$

$$(40) (9 - 6i)(4 - 2i) \\ = 24 - 42i$$

$$(41) (3 + 4i)(-6 + 6i) \\ = -42 - 6i$$

$$(42) (10 + 5i)(-4 - 4i) \\ = -20 - 60i$$

$$(43) (-3 - 10i)(-6 - 5i) \\ = -32 + 75i$$

$$(44) (2 - 10i)(-6 + 8i) \\ = 68 + 76i$$

$$(45) (-3 + 3i)(-10 - 4i) \\ = 42 - 18i$$

$$(1) \quad (-8 + 3i)(10 + 4i) \\ = -92 - 2i$$

$$(2) \quad (5 + 8i)(-7 - i) \\ = -27 - 61i$$

$$(3) \quad (-10 - 6i)(-10 + 9i) \\ = 154 - 30i$$

$$(4) \quad (10 + 2i)(7 - 8i) \\ = 86 - 66i$$

$$(5) \quad (-3 + 3i)(1 + i) \\ = -6$$

$$(6) \quad (10 - 7i)(7 - 4i) \\ = 42 - 89i$$

$$(7) \quad (-2 - 2i)(2 - 5i) \\ = -14 + 6i$$

$$(8) \quad (6 + i)(-2 + 4i) \\ = -16 + 22i$$

$$(9) \quad (-1 + 4i)(1 - 3i) \\ = 11 + 7i$$

$$(10) \quad (6 + 5i)(4 + i) \\ = 19 + 26i$$

$$(11) \quad (-7 + i)(6 + 9i) \\ = -51 - 57i$$

$$(12) \quad (-5 - 6i)(-4 - 10i) \\ = -40 + 74i$$

$$(13) \quad (7 - 3i)(-9 - i) \\ = -66 + 20i$$

$$(14) \quad (-4 + 6i)(3 - 6i) \\ = 24 + 42i$$

$$(15) \quad (-10 - 7i)(9 - 6i) \\ = -132 - 3i$$

$$(16) \quad (-3 - 4i)(10 + 10i) \\ = 10 - 70i$$

$$(17) \quad (1 - i)(10 + 8i) \\ = 18 - 2i$$

$$(18) \quad (4 + 4i)(1 + 4i) \\ = -12 + 20i$$

$$(19) \quad (6 - 6i)(-3 - 5i) \\ = -48 - 12i$$

$$(20) \quad (-3 + 3i)(1 - 3i) \\ = 6 + 12i$$

$$(21) \quad (10 + 2i)(-9 - 9i) \\ = -72 - 108i$$

$$(22) \quad (-2 + 4i)(8 + 8i) \\ = -48 + 16i$$

$$(23) \quad (-6 - 6i)(5 + 4i) \\ = -6 - 54i$$

$$(24) \quad (-7 - 7i)(1 - 9i) \\ = -70 + 56i$$

$$(25) \quad (-1 - 8i)(-4 + 4i) \\ = 36 + 28i$$

$$(26) \quad (-7 - i)(-2 - 2i) \\ = 12 + 16i$$

$$(27) \quad (3 + 3i)(2 + 9i) \\ = -21 + 33i$$

$$(28) \quad (-9 - 6i)(-2 + 2i) \\ = 30 - 6i$$

$$(29) \quad (8 - 6i)(-2 - 3i) \\ = -34 - 12i$$

$$(30) \quad (-3 + 6i)(7 + 4i) \\ = -45 + 30i$$

$$(31) \quad (2 + 8i)(-2 - 3i) \\ = 20 - 22i$$

$$(32) \quad (-2 + 4i)(5 + 4i) \\ = -26 + 12i$$

$$(33) \quad (-6 + 9i)(-10 - 10i) \\ = 150 - 30i$$

$$(34) \quad (1 - 5i)(4 + 10i) \\ = 54 - 10i$$

$$(35) \quad (8 - 6i)(6 - 7i) \\ = 6 - 92i$$

$$(36) \quad (-8 + 7i)(-8 + 10i) \\ = -6 - 136i$$

$$(37) \quad (-3 + 6i)(-1 - 6i) \\ = 39 + 12i$$

$$(38) \quad (7 - i)(1 + 5i) \\ = 12 + 34i$$

$$(39) \quad (-10 + 10i)(-9 - 7i) \\ = 160 - 20i$$

$$(40) \quad (8 + 2i)(-3 + 9i) \\ = -42 + 66i$$

$$(41) \quad (5 - 6i)(9 - 9i) \\ = -9 - 99i$$

$$(42) \quad (10 + 6i)(-4 - 8i) \\ = 8 - 104i$$

$$(43) \quad (-8 + 8i)(10 + i) \\ = -88 + 72i$$

$$(44) \quad (-4 - 10i)(-7 + 2i) \\ = 48 + 62i$$

$$(45) \quad (-7 + 9i)(4 - 10i) \\ = 62 + 106i$$

$$(1) \quad (-6 - 4i)(-4 - 6i) \\ = 52i$$

$$(2) \quad (-2 + 9i)(7 + 8i) \\ = -86 + 47i$$

$$(3) \quad (1 + 4i)(-2 + 7i) \\ = -30 - i$$

$$(4) \quad (4 + i)(5 + 5i) \\ = 15 + 25i$$

$$(5) \quad (-10 + 10i)(-8 + 9i) \\ = -10 - 170i$$

$$(6) \quad (-4 + 2i)(-8 - 5i) \\ = 42 + 4i$$

$$(7) \quad (-5 - 8i)(10 - 10i) \\ = -130 - 30i$$

$$(8) \quad (2 - 8i)(6 + 4i) \\ = 44 - 40i$$

$$(9) \quad (-3 - 2i)(9 - 6i) \\ = -39$$

$$(10) \quad (10 + 7i)(-1 + 9i) \\ = -73 + 83i$$

$$(11) \quad (3 - 4i)(8 - 6i) \\ = -50i$$

$$(12) \quad (2 + 6i)(9 + 7i) \\ = -24 + 68i$$

$$(13) \quad (5 - 2i)(-9 + 4i) \\ = -37 + 38i$$

$$(14) \quad (5 - 8i)(-5 - 5i) \\ = -65 + 15i$$

$$(15) \quad (-1 - 3i)(10 + 10i) \\ = 20 - 40i$$

$$(16) \quad (-6 - 7i)(8 + 5i) \\ = -13 - 86i$$

$$(17) \quad (-8 + 10i)(6 + 2i) \\ = -68 + 44i$$

$$(18) \quad (8 + 5i)(9 - 10i) \\ = 122 - 35i$$

$$(19) \quad (-9 + 4i)(-1 - 7i) \\ = 37 + 59i$$

$$(20) \quad (-3 + 8i)(4 - 10i) \\ = 68 + 62i$$

$$(21) \quad (9 + 3i)(10 - 6i) \\ = 108 - 24i$$

$$(22) \quad (7 - 10i)(-4 - 4i) \\ = -68 + 12i$$

$$(23) \quad (-9 - 8i)(-5 + 10i) \\ = 125 - 50i$$

$$(24) \quad (-10 + i)(-6 + 4i) \\ = 56 - 46i$$

$$(25) \quad (9 - 7i)(-3 - 9i) \\ = -90 - 60i$$

$$(26) \quad (2 + 3i)(-4 + 3i) \\ = -17 - 6i$$

$$(27) \quad (-9 + 8i)(-10 + 9i) \\ = 18 - 161i$$

$$(28) \quad (-7 + 9i)(2 - 7i) \\ = 49 + 67i$$

$$(29) \quad (2 - i)(-9 + 3i) \\ = -15 + 15i$$

$$(30) \quad (-8 + 6i)(-8 - 7i) \\ = 106 + 8i$$

$$(31) \quad (7 + 3i)(-1 - 10i) \\ = 23 - 73i$$

$$(32) \quad (2 - 6i)(-6 + 5i) \\ = 18 + 46i$$

$$(33) \quad (-5 - 2i)(5 - 6i) \\ = -37 + 20i$$

$$(34) \quad (-4 - 10i)(-5 + 3i) \\ = 50 + 38i$$

$$(35) \quad (-10 + 8i)(-3 + 3i) \\ = 6 - 54i$$

$$(36) \quad (-5 - 5i)(-5 - 5i) \\ = 50i$$

$$(37) \quad (-1 - 2i)(10 + 3i) \\ = -4 - 23i$$

$$(38) \quad (-2 - 3i)(9 + 4i) \\ = -6 - 35i$$

$$(39) \quad (-6 - i)(-2 - 5i) \\ = 7 + 32i$$

$$(40) \quad (3 + 9i)(-2 + 2i) \\ = -24 - 12i$$

$$(41) \quad (-3 - 2i)(-2 - 7i) \\ = -8 + 25i$$

$$(42) \quad (7 - 5i)(4 - 8i) \\ = -12 - 76i$$

$$(43) \quad (-1 - 8i)(7 - 5i) \\ = -47 - 51i$$

$$(44) \quad (-5 - 6i)(1 + 7i) \\ = 37 - 41i$$

$$(45) \quad (-4 + 7i)(4 + 6i) \\ = -58 + 4i$$

$$(1) (8 - 6i)(5 - 7i) \\ = -2 - 86i$$

$$(2) (-2 + 7i)(-4 - 9i) \\ = 71 - 10i$$

$$(3) (-7 + 4i)(1 + i) \\ = -11 - 3i$$

$$(4) (-4 - 10i)(-5 - 5i) \\ = -30 + 70i$$

$$(5) (-8 + 5i)(-2 + 7i) \\ = -19 - 66i$$

$$(6) (-6 + 4i)(9 - 8i) \\ = -22 + 84i$$

$$(7) (7 + 2i)(-1 - 9i) \\ = 11 - 65i$$

$$(8) (-10 - 2i)(1 - 6i) \\ = -22 + 58i$$

$$(9) (9 + 5i)(3 + 10i) \\ = -23 + 105i$$

$$(10) (-2 + 5i)(-3 - 3i) \\ = 21 - 9i$$

$$(11) (-10 - 5i)(-4 - 5i) \\ = 15 + 70i$$

$$(12) (-5 + 2i)(-3 + 5i) \\ = 5 - 31i$$

$$(13) (-9 - 8i)(-6 - 9i) \\ = -18 + 129i$$

$$(14) (-6 + 7i)(-1 + 6i) \\ = -36 - 43i$$

$$(15) (-4 - 8i)(-2 - 3i) \\ = -16 + 28i$$

$$(16) (7 + 7i)(-10 + 3i) \\ = -91 - 49i$$

$$(17) (7 + 8i)(9 + 4i) \\ = 31 + 100i$$

$$(18) (-8 + 8i)(3 - 9i) \\ = 48 + 96i$$

$$(19) (-5 - 3i)(9 + 9i) \\ = -18 - 72i$$

$$(20) (3 - 3i)(6 - i) \\ = 15 - 21i$$

$$(21) (-9 + 9i)(-8 - 9i) \\ = 153 + 9i$$

$$(22) (-5 + 9i)(5 - 5i) \\ = 20 + 70i$$

$$(23) (-9 + 4i)(-5 + 6i) \\ = 21 - 74i$$

$$(24) (-5 + 6i)(-3 + 7i) \\ = -27 - 53i$$

$$(25) (1 + 7i)(8 - 2i) \\ = 22 + 54i$$

$$(26) (10 + 2i)(6 + 8i) \\ = 44 + 92i$$

$$(27) (-1 + 9i)(8 - 5i) \\ = 37 + 77i$$

$$(28) (-1 - 7i)(7 - 10i) \\ = -77 - 39i$$

$$(29) (-9 + 9i)(10 - 10i) \\ = 180i$$

$$(30) (10 - 5i)(-5 - 5i) \\ = -75 - 25i$$

$$(31) (9 + 4i)(4 + 10i) \\ = -4 + 106i$$

$$(32) (-4 - 6i)(8 - 10i) \\ = -92 - 8i$$

$$(33) (1 + 6i)(8 + 10i) \\ = -52 + 58i$$

$$(34) (10 + 2i)(-3 + 6i) \\ = -42 + 54i$$

$$(35) (-2 - 10i)(3 - 10i) \\ = -106 - 10i$$

$$(36) (-7 - 4i)(7 - 8i) \\ = -81 + 28i$$

$$(37) (-6 - 6i)(9 + 8i) \\ = -6 - 102i$$

$$(38) (10 - 5i)(7 + 10i) \\ = 120 + 65i$$

$$(39) (9 - 6i)(3 + 4i) \\ = 51 + 18i$$

$$(40) (-2 + 10i)(-6 + 3i) \\ = -18 - 66i$$

$$(41) (-7 - 10i)(5 + 5i) \\ = 15 - 85i$$

$$(42) (5 + 8i)(-8 + 10i) \\ = -120 - 14i$$

$$(43) (2 + 4i)(10 - 4i) \\ = 36 + 32i$$

$$(44) (4 + 5i)(10 - 4i) \\ = 60 + 34i$$

$$(45) (9 + 2i)(4 + i) \\ = 34 + 17i$$

$$(1) (10 + 2i)(10 - 9i) \\ = 118 - 70i$$

$$(2) (7 + 5i)(-5 + 2i) \\ = -45 - 11i$$

$$(3) (7 + 3i)(8 + 7i) \\ = 35 + 73i$$

$$(4) (2 + 3i)(-7 + 7i) \\ = -35 - 7i$$

$$(5) (6 + 5i)(7 + 8i) \\ = 2 + 83i$$

$$(6) (4 + 2i)(-2 - i) \\ = -6 - 8i$$

$$(7) (2 - 2i)(-1 - 6i) \\ = -14 - 10i$$

$$(8) (4 - 3i)(2 + 10i) \\ = 38 + 34i$$

$$(9) (8 + 2i)(4 - 6i) \\ = 44 - 40i$$

$$(10) (3 + 6i)(-3 + 5i) \\ = -39 - 3i$$

$$(11) (7 + i)(10 - 5i) \\ = 75 - 25i$$

$$(12) (7 + 5i)(4 - 3i) \\ = 43 - i$$

$$(13) (4 + 10i)(2 - 6i) \\ = 68 - 4i$$

$$(14) (-1 + 6i)(5 + 5i) \\ = -35 + 25i$$

$$(15) (10 + 4i)(-6 + 10i) \\ = -100 + 76i$$

$$(16) (1 - 2i)(2 + 2i) \\ = 6 - 2i$$

$$(17) (3 + 7i)(-10 + 6i) \\ = -72 - 52i$$

$$(18) (1 + 10i)(7 + 3i) \\ = -23 + 73i$$

$$(19) (-7 + 2i)(5 + 4i) \\ = -43 - 18i$$

$$(20) (-3 + 9i)(-8 + 5i) \\ = -21 - 87i$$

$$(21) (-5 + 6i)(-5 + 8i) \\ = -23 - 70i$$

$$(22) (-7 + 10i)(-10 - 5i) \\ = 120 - 65i$$

$$(23) (10 + 6i)(-6 + 10i) \\ = -120 + 64i$$

$$(24) (4 + 10i)(-5 - 10i) \\ = 80 - 90i$$

$$(25) (-6 + 9i)(9 - 6i) \\ = 117i$$

$$(26) (9 - 10i)(-2 + 4i) \\ = 22 + 56i$$

$$(27) (-3 + 8i)(-2 - 8i) \\ = 70 + 8i$$

$$(28) (8 - i)(-7 - 9i) \\ = -65 - 65i$$

$$(29) (9 - 4i)(7 + 6i) \\ = 87 + 26i$$

$$(30) (-9 - 8i)(-9 - 6i) \\ = 33 + 126i$$

$$(31) (5 + 8i)(2 - 10i) \\ = 90 - 34i$$

$$(32) (9 - 10i)(2 - 3i) \\ = -12 - 47i$$

$$(33) (10 - 8i)(8 + i) \\ = 88 - 54i$$

$$(34) (-10 + 4i)(5 - 10i) \\ = -10 + 120i$$

$$(35) (3 - 7i)(8 - 8i) \\ = -32 - 80i$$

$$(36) (8 - 2i)(-4 - 4i) \\ = -40 - 24i$$

$$(37) (5 - 5i)(-10 + 10i) \\ = 100i$$

$$(38) (5 - 8i)(-6 - i) \\ = -38 + 43i$$

$$(39) (-9 + 10i)(-8 + 10i) \\ = -28 - 170i$$

$$(40) (4 - 5i)(-4 + 6i) \\ = 14 + 44i$$

$$(41) (8 - 9i)(-2 + 5i) \\ = 29 + 58i$$

$$(42) (9 - 7i)(3 + 5i) \\ = 62 + 24i$$

$$(43) (8 + 7i)(-8 - 2i) \\ = -50 - 72i$$

$$(44) (-6 + 6i)(8 + 6i) \\ = -84 + 12i$$

$$(45) (-7 - 2i)(-4 - 2i) \\ = 24 + 22i$$



$$(1) \quad (-3-7i)(8+5i) \\ = 11-71i$$

$$(2) \quad (7+7i)(2+8i) \\ = -42+70i$$

$$(3) \quad (9-7i)(-8-5i) \\ = -107+11i$$

$$(4) \quad (9+5i)(6+i) \\ = 49+39i$$

$$(5) \quad (7+5i)(3-7i) \\ = 56-34i$$

$$(6) \quad (10+3i)(-1+4i) \\ = -22+37i$$

$$(7) \quad (-8+i)(3+9i) \\ = -33-69i$$

$$(8) \quad (4-9i)(10-7i) \\ = -23-118i$$

$$(9) \quad (4+i)(-9+5i) \\ = -41+11i$$

$$(10) \quad (6-8i)(5-i) \\ = 22-46i$$

$$(11) \quad (9-5i)(2+3i) \\ = 33+17i$$

$$(12) \quad (5-3i)(10+2i) \\ = 56-20i$$

$$(13) \quad (-3-10i)(-7+8i) \\ = 101+46i$$

$$(14) \quad (10+i)(-6+7i) \\ = -67+64i$$

$$(15) \quad (3-i)(10+i) \\ = 31-7i$$

$$(16) \quad (-1-7i)(-2-9i) \\ = -61+23i$$

$$(17) \quad (4-10i)(2+2i) \\ = 28-12i$$

$$(18) \quad (-8+i)(-3-2i) \\ = 26+13i$$

$$(19) \quad (-6-7i)(-9+i) \\ = 61+57i$$

$$(20) \quad (3-10i)(1+2i) \\ = 23-4i$$

$$(21) \quad (1+6i)(7+7i) \\ = -35+49i$$

$$(22) \quad (2-i)(-8-3i) \\ = -19+2i$$

$$(23) \quad (10-2i)(-10+3i) \\ = -94+50i$$

$$(24) \quad (-6+6i)(3-5i) \\ = 12+48i$$

$$(25) \quad (5+i)(6+3i) \\ = 27+21i$$

$$(26) \quad (-8+i)(-7+5i) \\ = 51-47i$$

$$(27) \quad (2+4i)(2+9i) \\ = -32+26i$$

$$(28) \quad (-6+8i)(10+2i) \\ = -76+68i$$

$$(29) \quad (-9+6i)(4-10i) \\ = 24+114i$$

$$(30) \quad (9+3i)(-9-9i) \\ = -54-108i$$

$$(31) \quad (-6-6i)(-1+8i) \\ = 54-42i$$

$$(32) \quad (-2+10i)(5+5i) \\ = -60+40i$$

$$(33) \quad (6-6i)(-1+2i) \\ = 6+18i$$

$$(34) \quad (10+5i)(9-6i) \\ = 120-15i$$

$$(35) \quad (3+7i)(1+2i) \\ = -11+13i$$

$$(36) \quad (-3+9i)(3+10i) \\ = -99-3i$$

$$(37) \quad (-4-10i)(1-8i) \\ = -84+22i$$

$$(38) \quad (3-7i)(-3-i) \\ = -16+18i$$

$$(39) \quad (-5+10i)(6+9i) \\ = -120+15i$$

$$(40) \quad (8+4i)(7+4i) \\ = 40+60i$$

$$(41) \quad (-2-8i)(6+i) \\ = -4-50i$$

$$(42) \quad (7-7i)(8-i) \\ = 49-63i$$

$$(43) \quad (-2-8i)(-10+10i) \\ = 100+60i$$

$$(44) \quad (7-2i)(8-9i) \\ = 38-79i$$

$$(45) \quad (-8+i)(-3-10i) \\ = 34+77i$$

$$(1) (3 + 8i)(1 - 9i) \\ = 75 - 19i$$

$$(2) (-8 + 8i)(1 - 5i) \\ = 32 + 48i$$

$$(3) (5 - 10i)(-3 - i) \\ = -25 + 25i$$

$$(4) (-1 + 7i)(9 - 10i) \\ = 61 + 73i$$

$$(5) (6 + 8i)(10 - 4i) \\ = 92 + 56i$$

$$(6) (9 + 5i)(-10 - 4i) \\ = -70 - 86i$$

$$(7) (-4 + 6i)(-9 - 2i) \\ = 48 - 46i$$

$$(8) (-5 - 10i)(-3 + 4i) \\ = 55 + 10i$$

$$(9) (-1 - 9i)(1 - 10i) \\ = -91 + i$$

$$(10) (-4 + 9i)(-3 + 3i) \\ = -15 - 39i$$

$$(11) (7 + 5i)(-7 - 3i) \\ = -34 - 56i$$

$$(12) (3 + 3i)(-3 - 2i) \\ = -3 - 15i$$

$$(13) (-9 - 4i)(3 + 10i) \\ = 13 - 102i$$

$$(14) (9 - 10i)(-4 + 8i) \\ = 44 + 112i$$

$$(15) (7 + 6i)(-9 + 9i) \\ = -117 + 9i$$

$$(16) (-10 + 5i)(6 - 4i) \\ = -40 + 70i$$

$$(17) (10 - 3i)(5 - i) \\ = 47 - 25i$$

$$(18) (-1 + 8i)(-1 - 8i) \\ = 65$$

$$(19) (-8 - 8i)(-8 - 6i) \\ = 16 + 112i$$

$$(20) (2 + i)(8 - 10i) \\ = 26 - 12i$$

$$(21) (-7 - 8i)(-3 - 5i) \\ = -19 + 59i$$

$$(22) (8 - i)(3 - 4i) \\ = 20 - 35i$$

$$(23) (5 + 7i)(-5 + i) \\ = -32 - 30i$$

$$(24) (-6 + 7i)(-1 - 5i) \\ = 41 + 23i$$

$$(25) (-3 + 2i)(-6 + 8i) \\ = 2 - 36i$$

$$(26) (-3 + 5i)(1 - 8i) \\ = 37 + 29i$$

$$(27) (-1 + 10i)(-7 - 6i) \\ = 67 - 64i$$

$$(28) (3 - 6i)(10 + 7i) \\ = 72 - 39i$$

$$(29) (4 - 5i)(-7 + 5i) \\ = -3 + 55i$$

$$(30) (2 + 4i)(-8 - 5i) \\ = 4 - 42i$$

$$(31) (-2 + 8i)(1 + 6i) \\ = -50 - 4i$$

$$(32) (-6 + 5i)(1 + 6i) \\ = -36 - 31i$$

$$(33) (-5 + 9i)(9 - 2i) \\ = -27 + 91i$$

$$(34) (-7 + 10i)(3 + 9i) \\ = -111 - 33i$$

$$(35) (-8 - 5i)(4 - 7i) \\ = -67 + 36i$$

$$(36) (6 + 6i)(-8 + 8i) \\ = -96$$

$$(37) (-5 - 8i)(-5 + 3i) \\ = 49 + 25i$$

$$(38) (-6 + 9i)(-7 - 5i) \\ = 87 - 33i$$

$$(39) (-8 - 6i)(-3 + i) \\ = 30 + 10i$$

$$(40) (7 + 5i)(-9 - 10i) \\ = -13 - 115i$$

$$(41) (-2 - 2i)(9 - i) \\ = -20 - 16i$$

$$(42) (10 - 3i)(1 - 8i) \\ = -14 - 83i$$

$$(43) (9 + 5i)(-7 - 7i) \\ = -28 - 98i$$

$$(44) (-3 + 2i)(6 + 2i) \\ = -22 + 6i$$

$$(45) (10 + 9i)(-10 - 8i) \\ = -28 - 170i$$

$$(1) \quad (-9 + 4i)(8 - 8i) \\ = -40 + 104i$$

$$(2) \quad (4 - 7i)(-6 + 4i) \\ = 4 + 58i$$

$$(3) \quad (-7 - 4i)(-2 - 2i) \\ = 6 + 22i$$

$$(4) \quad (9 - 4i)(2 - 5i) \\ = -2 - 53i$$

$$(5) \quad (-5 + 2i)(1 + 5i) \\ = -15 - 23i$$

$$(6) \quad (-6 + 7i)(6 - 3i) \\ = -15 + 60i$$

$$(7) \quad (4 + 9i)(-8 - 6i) \\ = 22 - 96i$$

$$(8) \quad (-1 + 4i)(10 - 10i) \\ = 30 + 50i$$

$$(9) \quad (2 + 2i)(-4 - 8i) \\ = 8 - 24i$$

$$(10) \quad (-2 - 2i)(2 + 8i) \\ = 12 - 20i$$

$$(11) \quad (-7 - 3i)(-5 - 9i) \\ = 8 + 78i$$

$$(12) \quad (-9 - 2i)(-8 + 9i) \\ = 90 - 65i$$

$$(13) \quad (9 - 3i)(-4 - 2i) \\ = -42 - 6i$$

$$(14) \quad (1 - i)(-3 - 6i) \\ = -9 - 3i$$

$$(15) \quad (-3 - 6i)(3 - 6i) \\ = -45$$

$$(16) \quad (-8 + i)(-9 - 8i) \\ = 80 + 55i$$

$$(17) \quad (4 + 3i)(-5 + 9i) \\ = -47 + 21i$$

$$(18) \quad (1 + 4i)(-9 - 3i) \\ = 3 - 39i$$

$$(19) \quad (-2 + 8i)(10 + 2i) \\ = -36 + 76i$$

$$(20) \quad (6 + 9i)(-7 + 3i) \\ = -69 - 45i$$

$$(21) \quad (-4 + 4i)(-4 + 10i) \\ = -24 - 56i$$

$$(22) \quad (-5 + 4i)(5 + 3i) \\ = -37 + 5i$$

$$(23) \quad (8 + 9i)(-9 + 3i) \\ = -99 - 57i$$

$$(24) \quad (-6 + 9i)(2 - 9i) \\ = 69 + 72i$$

$$(25) \quad (-7 + 9i)(6 - 2i) \\ = -24 + 68i$$

$$(26) \quad (-7 + 10i)(8 - 3i) \\ = -26 + 101i$$

$$(27) \quad (4 - i)(-10 + 9i) \\ = -31 + 46i$$

$$(28) \quad (-5 - 7i)(10 - 4i) \\ = -78 - 50i$$

$$(29) \quad (5 + 6i)(-2 - 8i) \\ = 38 - 52i$$

$$(30) \quad (-7 - 9i)(4 + i) \\ = -19 - 43i$$

$$(31) \quad (8 + 5i)(-6 + 4i) \\ = -68 + 2i$$

$$(32) \quad (-8 - 6i)(-5 - 7i) \\ = -2 + 86i$$

$$(33) \quad (3 - 6i)(-5 - 4i) \\ = -39 + 18i$$

$$(34) \quad (4 + i)(-3 + 8i) \\ = -20 + 29i$$

$$(35) \quad (-9 - 7i)(-1 + 9i) \\ = 72 - 74i$$

$$(36) \quad (-3 - 4i)(-2 - 2i) \\ = -2 + 14i$$

$$(37) \quad (4 - 3i)(5 + 9i) \\ = 47 + 21i$$

$$(38) \quad (9 - 7i)(-4 + 6i) \\ = 6 + 82i$$

$$(39) \quad (9 + 8i)(5 - 7i) \\ = 101 - 23i$$

$$(40) \quad (9 - 2i)(3 - 10i) \\ = 7 - 96i$$

$$(41) \quad (2 - i)(1 - 10i) \\ = -8 - 21i$$

$$(42) \quad (7 - 5i)(9 - 7i) \\ = 28 - 94i$$

$$(43) \quad (2 - 9i)(10 - 3i) \\ = -7 - 96i$$

$$(44) \quad (8 + 4i)(-6 + 2i) \\ = -56 - 8i$$

$$(45) \quad (3 - 4i)(8 - 2i) \\ = 16 - 38i$$

$$(1) (2 - 9i)(9 + 9i) \\ = 99 - 63i$$

$$(2) (10 + 4i)(-3 + 6i) \\ = -54 + 48i$$

$$(3) (-10 + 5i)(4 - 6i) \\ = -10 + 80i$$

$$(4) (10 + 8i)(9 + 2i) \\ = 74 + 92i$$

$$(5) (-8 + 6i)(6 - 9i) \\ = 6 + 108i$$

$$(6) (3 - 3i)(-3 + 7i) \\ = 12 + 30i$$

$$(7) (9 + i)(6 + 8i) \\ = 46 + 78i$$

$$(8) (-7 + 2i)(2 - 2i) \\ = -10 + 18i$$

$$(9) (-10 + 8i)(-9 + 3i) \\ = 66 - 102i$$

$$(10) (4 + i)(7 + 2i) \\ = 26 + 15i$$

$$(11) (-10 + 9i)(-4 + 4i) \\ = 4 - 76i$$

$$(12) (-6 - 9i)(-3 - 4i) \\ = -18 + 51i$$

$$(13) (4 + 4i)(10 + 9i) \\ = 4 + 76i$$

$$(14) (5 - 2i)(-10 - 5i) \\ = -60 - 5i$$

$$(15) (-1 - 4i)(-9 + 8i) \\ = 41 + 28i$$

$$(16) (-6 + 8i)(2 - i) \\ = -4 + 22i$$

$$(17) (2 + 9i)(5 + 3i) \\ = -17 + 51i$$

$$(18) (7 - 3i)(-1 - 6i) \\ = -25 - 39i$$

$$(19) (4 + 6i)(-1 + 3i) \\ = -22 + 6i$$

$$(20) (6 - 5i)(7 + 7i) \\ = 77 + 7i$$

$$(21) (-6 - 5i)(7 - 10i) \\ = -92 + 25i$$

$$(22) (1 - 2i)(-9 - 6i) \\ = -21 + 12i$$

$$(23) (5 - 8i)(3 + 6i) \\ = 63 + 6i$$

$$(24) (4 - 9i)(1 + 3i) \\ = 31 + 3i$$

$$(25) (-10 - 3i)(-3 + 2i) \\ = 36 - 11i$$

$$(26) (10 + 7i)(-8 - 10i) \\ = -10 - 156i$$

$$(27) (1 - i)(1 + 10i) \\ = 11 + 9i$$

$$(28) (6 + i)(4 - i) \\ = 25 - 2i$$

$$(29) (3 + 2i)(-4 + 5i) \\ = -22 + 7i$$

$$(30) (3 + 2i)(-2 + 2i) \\ = -10 + 2i$$

$$(31) (-3 + 8i)(-10 - 3i) \\ = 54 - 71i$$

$$(32) (-3 + 3i)(-5 - 7i) \\ = 36 + 6i$$

$$(33) (-7 + 2i)(-10 + 3i) \\ = 64 - 41i$$

$$(34) (-10 - 8i)(-9 + 6i) \\ = 138 + 12i$$

$$(35) (3 + 3i)(8 + i) \\ = 21 + 27i$$

$$(36) (-10 + 10i)(5 - 6i) \\ = 10 + 110i$$

$$(37) (6 - 7i)(1 - 6i) \\ = -36 - 43i$$

$$(38) (8 - 3i)(-8 + 8i) \\ = -40 + 88i$$

$$(39) (-5 - 7i)(10 - 5i) \\ = -85 - 45i$$

$$(40) (3 - 4i)(-1 + 4i) \\ = 13 + 16i$$

$$(41) (8 - 3i)(2 + 7i) \\ = 37 + 50i$$

$$(42) (-9 - 3i)(6 + 10i) \\ = -24 - 108i$$

$$(43) (2 - 2i)(5 + 10i) \\ = 30 + 10i$$

$$(44) (-2 - 7i)(-6 + 9i) \\ = 75 + 24i$$

$$(45) (5 + 7i)(-5 + i) \\ = -32 - 30i$$

$$(1) \quad (-8 + i)(5 + 5i) \\ = -45 - 35i$$

$$(2) \quad (-5 + 5i)(-8 + 7i) \\ = 5 - 75i$$

$$(3) \quad (-8 + 8i)(-2 + 9i) \\ = -56 - 88i$$

$$(4) \quad (-9 - 4i)(-5 - 6i) \\ = 21 + 74i$$

$$(5) \quad (2 - 7i)(8 + 3i) \\ = 37 - 50i$$

$$(6) \quad (-5 + 3i)(-10 - 5i) \\ = 65 - 5i$$

$$(7) \quad (1 - 3i)(-7 + 8i) \\ = 17 + 29i$$

$$(8) \quad (2 - 5i)(-8 + 3i) \\ = -1 + 46i$$

$$(9) \quad (4 - 4i)(9 + 2i) \\ = 44 - 28i$$

$$(10) \quad (-4 - 4i)(-6 - 6i) \\ = 48i$$

$$(11) \quad (-2 + 7i)(-4 + i) \\ = 1 - 30i$$

$$(12) \quad (10 + 3i)(-2 + 7i) \\ = -41 + 64i$$

$$(13) \quad (-6 - 2i)(-5 - 6i) \\ = 18 + 46i$$

$$(14) \quad (-6 + 7i)(10 - 8i) \\ = -4 + 118i$$

$$(15) \quad (1 - 3i)(10 - 4i) \\ = -2 - 34i$$

$$(16) \quad (6 + 4i)(7 - i) \\ = 46 + 22i$$

$$(17) \quad (-9 - 3i)(5 - i) \\ = -48 - 6i$$

$$(18) \quad (3 + 10i)(-6 - 10i) \\ = 82 - 90i$$

$$(19) \quad (2 - 3i)(-10 + 7i) \\ = 1 + 44i$$

$$(20) \quad (2 + 3i)(9 + 6i) \\ = 39i$$

$$(21) \quad (1 + 6i)(1 + 6i) \\ = -35 + 12i$$

$$(22) \quad (9 + 7i)(-10 - i) \\ = -83 - 79i$$

$$(23) \quad (1 + i)(10 + 4i) \\ = 6 + 14i$$

$$(24) \quad (-2 - 10i)(4 + i) \\ = 2 - 42i$$

$$(25) \quad (5 + i)(5 + 2i) \\ = 23 + 15i$$

$$(26) \quad (3 - 5i)(10 + 9i) \\ = 75 - 23i$$

$$(27) \quad (1 + 5i)(-9 - 9i) \\ = 36 - 54i$$

$$(28) \quad (6 - 8i)(-9 - 5i) \\ = -94 + 42i$$

$$(29) \quad (-9 + 2i)(1 + 6i) \\ = -21 - 52i$$

$$(30) \quad (1 - 10i)(8 - 7i) \\ = -62 - 87i$$

$$(31) \quad (6 + 4i)(-5 + 2i) \\ = -38 - 8i$$

$$(32) \quad (-10 - 5i)(-4 - 10i) \\ = -10 + 120i$$

$$(33) \quad (10 - 7i)(9 - 5i) \\ = 55 - 113i$$

$$(34) \quad (-6 + 9i)(-6 + 4i) \\ = -78i$$

$$(35) \quad (6 + 10i)(2 - 2i) \\ = 32 + 8i$$

$$(36) \quad (-1 + 3i)(4 - 7i) \\ = 17 + 19i$$

$$(37) \quad (-1 + 5i)(1 - 2i) \\ = 9 + 7i$$

$$(38) \quad (-1 - 9i)(4 - 10i) \\ = -94 - 26i$$

$$(39) \quad (2 - 5i)(-6 - 2i) \\ = -22 + 26i$$

$$(40) \quad (-5 - 7i)(-7 - 2i) \\ = 21 + 59i$$

$$(41) \quad (9 - 8i)(6 + 2i) \\ = 70 - 30i$$

$$(42) \quad (-5 + 2i)(10 + 6i) \\ = -62 - 10i$$

$$(43) \quad (6 + 6i)(-7 - 4i) \\ = -18 - 66i$$

$$(44) \quad (-9 + 10i)(1 - 6i) \\ = 51 + 64i$$

$$(45) \quad (-2 - 9i)(-1 - i) \\ = -7 + 11i$$

$$(1) \quad (-2-i)(-8-4i) \\ = 12 + 16i$$

$$(2) \quad (5+3i)(-1+i) \\ = -8 + 2i$$

$$(3) \quad (6+5i)(-8-7i) \\ = -13 - 82i$$

$$(4) \quad (-6+10i)(1-4i) \\ = 34 + 34i$$

$$(5) \quad (-3-9i)(2-8i) \\ = -78 + 6i$$

$$(6) \quad (10+4i)(-1+5i) \\ = -30 + 46i$$

$$(7) \quad (-7-3i)(-2-7i) \\ = -7 + 55i$$

$$(8) \quad (2-7i)(-8+10i) \\ = 54 + 76i$$

$$(9) \quad (-3-8i)(5-7i) \\ = -71 - 19i$$

$$(10) \quad (-4+6i)(-2+3i) \\ = -10 - 24i$$

$$(11) \quad (-10+6i)(10+9i) \\ = -154 - 30i$$

$$(12) \quad (-10-3i)(1+3i) \\ = -1 - 33i$$

$$(13) \quad (6-4i)(-7-5i) \\ = -62 - 2i$$

$$(14) \quad (7+4i)(9+8i) \\ = 31 + 92i$$

$$(15) \quad (-9+4i)(-4+6i) \\ = 12 - 70i$$

$$(16) \quad (3+6i)(10+8i) \\ = -18 + 84i$$

$$(17) \quad (2-3i)(4+2i) \\ = 14 - 8i$$

$$(18) \quad (-8-3i)(4-6i) \\ = -50 + 36i$$

$$(19) \quad (-5+9i)(-4+10i) \\ = -70 - 86i$$

$$(20) \quad (-7-i)(-2-10i) \\ = 4 + 72i$$

$$(21) \quad (2-3i)(6-3i) \\ = 3 - 24i$$

$$(22) \quad (-2+3i)(2+4i) \\ = -16 - 2i$$

$$(23) \quad (-7+4i)(-2+4i) \\ = -2 - 36i$$

$$(24) \quad (3+5i)(5-6i) \\ = 45 + 7i$$

$$(25) \quad (3+8i)(-7+10i) \\ = -101 - 26i$$

$$(26) \quad (1+i)(-2-2i) \\ = -4i$$

$$(27) \quad (-7-10i)(9-8i) \\ = -143 - 34i$$

$$(28) \quad (-2-3i)(-4+2i) \\ = 14 + 8i$$

$$(29) \quad (7-3i)(1+3i) \\ = 16 + 18i$$

$$(30) \quad (5-9i)(-1+5i) \\ = 40 + 34i$$

$$(31) \quad (-7+10i)(-3-i) \\ = 31 - 23i$$

$$(32) \quad (-8+6i)(10+2i) \\ = -92 + 44i$$

$$(33) \quad (-6-7i)(-7+10i) \\ = 112 - 11i$$

$$(34) \quad (10+10i)(-8-i) \\ = -70 - 90i$$

$$(35) \quad (-1+i)(-3-3i) \\ = 6$$

$$(36) \quad (1-7i)(-5+5i) \\ = 30 + 40i$$

$$(37) \quad (-2+3i)(-3-7i) \\ = 27 + 5i$$

$$(38) \quad (-4-5i)(6+8i) \\ = 16 - 62i$$

$$(39) \quad (10-6i)(7+4i) \\ = 94 - 2i$$

$$(40) \quad (10-3i)(-6+3i) \\ = -51 + 48i$$

$$(41) \quad (8-10i)(-1+2i) \\ = 12 + 26i$$

$$(42) \quad (2-2i)(7-6i) \\ = 2 - 26i$$

$$(43) \quad (7+i)(-3-6i) \\ = -15 - 45i$$

$$(44) \quad (5+7i)(-4-8i) \\ = 36 - 68i$$

$$(45) \quad (-4+6i)(5-10i) \\ = 40 + 70i$$

$$(1) \quad (-1 - 9i)(-3 - 9i) \\ = -78 + 36i$$

$$(2) \quad (10 - 4i)(-3 + 3i) \\ = -18 + 42i$$

$$(3) \quad (-10 - 9i)(-5 - 4i) \\ = 14 + 85i$$

$$(4) \quad (9 - 3i)(1 + 2i) \\ = 15 + 15i$$

$$(5) \quad (9 - 4i)(6 + 5i) \\ = 74 + 21i$$

$$(6) \quad (7 + 6i)(8 - 4i) \\ = 80 + 20i$$

$$(7) \quad (-1 + 10i)(2 - i) \\ = 8 + 21i$$

$$(8) \quad (8 + 7i)(-1 + 2i) \\ = -22 + 9i$$

$$(9) \quad (-3 - i)(2 + 10i) \\ = 4 - 32i$$

$$(10) \quad (5 + 2i)(-8 + 4i) \\ = -48 + 4i$$

$$(11) \quad (2 - 6i)(-8 - 2i) \\ = -28 + 44i$$

$$(12) \quad (-3 - 4i)(1 - 10i) \\ = -43 + 26i$$

$$(13) \quad (5 + 8i)(5 + 8i) \\ = -39 + 80i$$

$$(14) \quad (5 - 6i)(2 + 7i) \\ = 52 + 23i$$

$$(15) \quad (8 - 6i)(-7 - 3i) \\ = -74 + 18i$$

$$(16) \quad (-2 - 5i)(-2 + 3i) \\ = 19 + 4i$$

$$(17) \quad (8 + 9i)(-10 - i) \\ = -71 - 98i$$

$$(18) \quad (6 - 4i)(-9 + 3i) \\ = -42 + 54i$$

$$(19) \quad (7 - 8i)(-1 + 9i) \\ = 65 + 71i$$

$$(20) \quad (-4 + 4i)(7 + 6i) \\ = -52 + 4i$$

$$(21) \quad (9 - 5i)(10 - 9i) \\ = 45 - 131i$$

$$(22) \quad (9 + 5i)(-8 + 10i) \\ = -122 + 50i$$

$$(23) \quad (-8 - 6i)(8 - 10i) \\ = -124 + 32i$$

$$(24) \quad (-10 - 4i)(-2 - 2i) \\ = 12 + 28i$$

$$(25) \quad (-2 - 7i)(2 - 9i) \\ = -67 + 4i$$

$$(26) \quad (7 + i)(4 + 6i) \\ = 22 + 46i$$

$$(27) \quad (10 - 8i)(-10 - 6i) \\ = -148 + 20i$$

$$(28) \quad (-9 - 4i)(7 - 7i) \\ = -91 + 35i$$

$$(29) \quad (-5 + i)(2 + 5i) \\ = -15 - 23i$$

$$(30) \quad (6 - 8i)(-5 + 3i) \\ = -6 + 58i$$

$$(31) \quad (1 + 5i)(8 - 4i) \\ = 28 + 36i$$

$$(32) \quad (2 + 9i)(-2 + 10i) \\ = -94 + 2i$$

$$(33) \quad (9 - 9i)(8 - 6i) \\ = 18 - 126i$$

$$(34) \quad (1 + 2i)(1 - 4i) \\ = 9 - 2i$$

$$(35) \quad (-6 + 6i)(-2 + 2i) \\ = -24i$$

$$(36) \quad (7 + 2i)(1 + 8i) \\ = -9 + 58i$$

$$(37) \quad (-4 - 2i)(6 - 3i) \\ = -30$$

$$(38) \quad (4 + 10i)(-7 - 8i) \\ = 52 - 102i$$

$$(39) \quad (1 - 6i)(1 + 7i) \\ = 43 + i$$

$$(40) \quad (1 - 7i)(-5 - 10i) \\ = -75 + 25i$$

$$(41) \quad (9 + i)(5 - 7i) \\ = 52 - 58i$$

$$(42) \quad (10 - 6i)(4 + 7i) \\ = 82 + 46i$$

$$(43) \quad (-4 - 7i)(-7 - i) \\ = 21 + 53i$$

$$(44) \quad (-1 + 9i)(-10 + 2i) \\ = -8 - 92i$$

$$(45) \quad (7 - 3i)(6 - 6i) \\ = 24 - 60i$$

$$(1) \quad (-10 - 4i)(8 - 2i) \\ = -88 - 12i$$

$$(2) \quad (9 + 6i)(-5 + 6i) \\ = -81 + 24i$$

$$(3) \quad (2 + 5i)(-2 - 8i) \\ = 36 - 26i$$

$$(4) \quad (9 + 4i)(10 - 6i) \\ = 114 - 14i$$

$$(5) \quad (3 - 7i)(-4 + 6i) \\ = 30 + 46i$$

$$(6) \quad (3 - 10i)(10 - 6i) \\ = -30 - 118i$$

$$(7) \quad (-2 + 4i)(1 + 7i) \\ = -30 - 10i$$

$$(8) \quad (2 - 9i)(10 - 4i) \\ = -16 - 98i$$

$$(9) \quad (-10 - 7i)(9 - 10i) \\ = -160 + 37i$$

$$(10) \quad (5 - 9i)(-6 + 5i) \\ = 15 + 79i$$

$$(11) \quad (-7 + i)(5 - 5i) \\ = -30 + 40i$$

$$(12) \quad (-3 + 8i)(2 + 5i) \\ = -46 + i$$

$$(13) \quad (9 - 6i)(-3 - 8i) \\ = -75 - 54i$$

$$(14) \quad (-7 + 10i)(-10 + 5i) \\ = 20 - 135i$$

$$(15) \quad (9 - 6i)(6 - 10i) \\ = -6 - 126i$$

$$(16) \quad (-5 + 8i)(4 - 6i) \\ = 28 + 62i$$

$$(17) \quad (-8 + 6i)(-3 + 8i) \\ = -24 - 82i$$

$$(18) \quad (-4 - 5i)(-7 - 7i) \\ = -7 + 63i$$

$$(19) \quad (3 + 10i)(3 + i) \\ = -1 + 33i$$

$$(20) \quad (10 - 8i)(3 - 2i) \\ = 14 - 44i$$

$$(21) \quad (-2 - 7i)(9 + 3i) \\ = 3 - 69i$$

$$(22) \quad (4 - 5i)(4 - 9i) \\ = -29 - 56i$$

$$(23) \quad (3 + 3i)(-7 - 2i) \\ = -15 - 27i$$

$$(24) \quad (4 + 10i)(8 + 2i) \\ = 12 + 88i$$

$$(25) \quad (-5 - i)(-6 + 6i) \\ = 36 - 24i$$

$$(26) \quad (-8 - 9i)(5 - 7i) \\ = -103 + 11i$$

$$(27) \quad (2 - 4i)(-8 + 8i) \\ = 16 + 48i$$

$$(28) \quad (-6 + 7i)(10 - 2i) \\ = -46 + 82i$$

$$(29) \quad (9 - 7i)(5 - i) \\ = 38 - 44i$$

$$(30) \quad (5 + 2i)(8 + 3i) \\ = 34 + 31i$$

$$(31) \quad (-3 - 3i)(4 - 5i) \\ = -27 + 3i$$

$$(32) \quad (4 + 7i)(7 + 7i) \\ = -21 + 77i$$

$$(33) \quad (5 + 5i)(6 + 10i) \\ = -20 + 80i$$

$$(34) \quad (5 - 3i)(-7 + i) \\ = -32 + 26i$$

$$(35) \quad (1 + 5i)(8 - 7i) \\ = 43 + 33i$$

$$(36) \quad (-7 + 6i)(9 + 7i) \\ = -105 + 5i$$

$$(37) \quad (-4 + 4i)(-4 - 7i) \\ = 44 + 12i$$

$$(38) \quad (1 + 10i)(-2 - 8i) \\ = 78 - 28i$$

$$(39) \quad (5 - 4i)(-6 - i) \\ = -34 + 19i$$

$$(40) \quad (5 + 4i)(4 - 10i) \\ = 60 - 34i$$

$$(41) \quad (10 + 8i)(-2 + 10i) \\ = -100 + 84i$$

$$(42) \quad (3 - 8i)(-7 + 4i) \\ = 11 + 68i$$

$$(43) \quad (-2 - i)(6 + 6i) \\ = -6 - 18i$$

$$(44) \quad (10 + 4i)(5 - 9i) \\ = 86 - 70i$$

$$(45) \quad (3 + 8i)(10 + 4i) \\ = -2 + 92i$$



$$(1) (7 - 3i)(5 + 3i) \\ = 44 + 6i$$

$$(2) (-4 + 7i)(1 + 4i) \\ = -32 - 9i$$

$$(3) (-7 + 5i)(5 - i) \\ = -30 + 32i$$

$$(4) (10 + 3i)(4 - 10i) \\ = 70 - 88i$$

$$(5) (9 + 6i)(8 + i) \\ = 66 + 57i$$

$$(6) (9 + 4i)(-6 - 10i) \\ = -14 - 114i$$

$$(7) (-5 - 2i)(-2 - 2i) \\ = 6 + 14i$$

$$(8) (-6 + 6i)(6 + 7i) \\ = -78 - 6i$$

$$(9) (10 - 7i)(9 - 4i) \\ = 62 - 103i$$

$$(10) (3 - 2i)(1 - 10i) \\ = -17 - 32i$$

$$(11) (-8 + 5i)(-2 + 3i) \\ = 1 - 34i$$

$$(12) (-9 + 6i)(4 - 7i) \\ = 6 + 87i$$

$$(13) (8 + 9i)(-9 - 3i) \\ = -45 - 105i$$

$$(14) (4 + 9i)(-1 + 4i) \\ = -40 + 7i$$

$$(15) (-5 - 8i)(8 - 6i) \\ = -88 - 34i$$

$$(16) (2 + 8i)(9 + 7i) \\ = -38 + 86i$$

$$(17) (-5 - 6i)(-8 + 10i) \\ = 100 - 2i$$

$$(18) (-10 + 3i)(1 + 8i) \\ = -34 - 77i$$

$$(19) (-7 - 9i)(-7 - 3i) \\ = 22 + 84i$$

$$(20) (-9 - 9i)(1 - 10i) \\ = -99 + 81i$$

$$(21) (1 + 6i)(6 - 8i) \\ = 54 + 28i$$

$$(22) (-7 - 2i)(-1 + 2i) \\ = 11 - 12i$$

$$(23) (-2 + 7i)(-7 + 6i) \\ = -28 - 61i$$

$$(24) (1 + 9i)(6 - 8i) \\ = 78 + 46i$$

$$(25) (-2 + 10i)(-8 + 10i) \\ = -84 - 100i$$

$$(26) (-7 - 4i)(-6 + i) \\ = 46 + 17i$$

$$(27) (-7 - 9i)(3 + 7i) \\ = 42 - 76i$$

$$(28) (-5 - 5i)(-2 - 9i) \\ = -35 + 55i$$

$$(29) (-5 + 10i)(1 - 5i) \\ = 45 + 35i$$

$$(30) (-8 + 9i)(3 + 8i) \\ = -96 - 37i$$

$$(31) (-2 + 7i)(-1 + i) \\ = -5 - 9i$$

$$(32) (-5 + 2i)(3 - 4i) \\ = -7 + 26i$$

$$(33) (9 - 5i)(2 + 4i) \\ = 38 + 26i$$

$$(34) (1 + i)(-1 - i) \\ = -2i$$

$$(35) (9 - 10i)(-7 + 10i) \\ = 37 + 160i$$

$$(36) (4 - 10i)(1 + 2i) \\ = 24 - 2i$$

$$(37) (6 + 4i)(-3 + 2i) \\ = -26$$

$$(38) (-10 - 8i)(9 - 9i) \\ = -162 + 18i$$

$$(39) (3 - 9i)(-4 + 3i) \\ = 15 + 45i$$

$$(40) (5 + 7i)(-8 + 10i) \\ = -110 - 6i$$

$$(41) (-5 - 4i)(-5 - 5i) \\ = 5 + 45i$$

$$(42) (-10 - 7i)(-1 - 8i) \\ = -46 + 87i$$

$$(43) (10 + 8i)(-9 - 9i) \\ = -18 - 162i$$

$$(44) (-9 + 3i)(-4 + 2i) \\ = 30 - 30i$$

$$(45) (8 - 8i)(4 - 7i) \\ = -24 - 88i$$

### 1.5.3 除法

複素数の商を求める問題。

$$(1) (-3 + 2i) \div (4 + i)$$

$$(11) (4 + 3i) \div (5 + i)$$

$$(21) (4 + 2i) \div (-3 + 3i)$$

$$(2) (-4 - 5i) \div (5 - 5i)$$

$$(12) (-1 + 5i) \div (3 - 3i)$$

$$(22) (-4 - 4i) \div (-3 - 5i)$$

$$(3) (-1 - i) \div (-2 - 5i)$$

$$(13) (5 + 3i) \div (-2 + i)$$

$$(23) (1 + 5i) \div (-1 + 4i)$$

$$(4) (2 + 3i) \div (3 + 2i)$$

$$(14) (1 - 5i) \div (5 - 4i)$$

$$(24) (-1 - 2i) \div (-1 - 3i)$$

$$(5) (-2 - 2i) \div (-5 + 4i)$$

$$(15) (5 - i) \div (-2 - 2i)$$

$$(25) (-5 + 4i) \div (5 + i)$$

$$(6) (-4 - i) \div (4 - i)$$

$$(16) (-2 + 4i) \div (1 - 3i)$$

$$(26) (4 + i) \div (-5 - i)$$

$$(7) (-3 + 2i) \div (4 - 2i)$$

$$(17) (-1 - 3i) \div (-4 - 5i)$$

$$(27) (2 + i) \div (2 + 5i)$$

$$(8) (3 - 4i) \div (-2 - 5i)$$

$$(18) (3 + i) \div (-1 + 2i)$$

$$(28) (1 - i) \div (-3 + 2i)$$

$$(9) (-3 - 3i) \div (-4 - 5i)$$

$$(19) (-4 - 5i) \div (1 + 4i)$$

$$(29) (4 + 4i) \div (-5 + 2i)$$

$$(10) (-2 - 3i) \div (3 - 4i)$$

$$(20) (5 - i) \div (-1 + 4i)$$

$$(30) (2 - 5i) \div (1 - i)$$

(1)  $(-2 - 3i) \div (3 + 5i)$

(11)  $(-2 + i) \div (-5 + 3i)$

(21)  $(3 + 5i) \div (-5 + 2i)$

(2)  $(3 + i) \div (-3 + i)$

(12)  $(2 + 4i) \div (-2 + 2i)$

(22)  $(-1 + 4i) \div (-3 - 4i)$

(3)  $(5 - 2i) \div (5 + i)$

(13)  $(-1 + 2i) \div (2 + 3i)$

(23)  $(-5 - 5i) \div (2 + i)$

(4)  $(-1 - 4i) \div (-5 - 5i)$

(14)  $(5 + 5i) \div (1 + 3i)$

(24)  $(1 + i) \div (-3 - 3i)$

(5)  $(-2 + 2i) \div (-4 + i)$

(15)  $(2 + i) \div (1 - 2i)$

(25)  $(-4 + 2i) \div (4 - 4i)$

(6)  $(2 + 4i) \div (-1 + 2i)$

(16)  $(-1 + 5i) \div (-1 + 2i)$

(26)  $(5 - 2i) \div (2 + 2i)$

(7)  $(4 + 4i) \div (-1 - i)$

(17)  $(1 - 4i) \div (-1 - 4i)$

(27)  $(1 + 4i) \div (-1 + 2i)$

(8)  $(1 - 4i) \div (3 - 4i)$

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(19)  $(4 - 3i) \div (-2 - 4i)$

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(10)  $(-5 - i) \div (3 + 5i)$

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(30)  $(3 + 2i) \div (3 - i)$

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$$(22) (5 - 2i) \div (-3 - 4i)$$

$$(3) (2 + i) \div (4 + 2i)$$

$$(13) (3 + i) \div (2 + 4i)$$

$$(23) (2 + 2i) \div (4 + 2i)$$

$$(4) (2 + 2i) \div (-5 + 5i)$$

$$(14) (-2 - 2i) \div (-5 - i)$$

$$(24) (-3 - 5i) \div (4 - 3i)$$

$$(5) (3 + 4i) \div (5 - i)$$

$$(15) (-1 - i) \div (2 + 3i)$$

$$(25) (-5 - i) \div (2 + i)$$

$$(6) (-1 - 4i) \div (2 + 3i)$$

$$(16) (-2 + 4i) \div (-4 + 3i)$$

$$(26) (-4 - 3i) \div (5 - 3i)$$

$$(7) (3 - i) \div (3 + 4i)$$

$$(17) (1 + 4i) \div (-1 + i)$$

$$(27) (4 - 3i) \div (5 + i)$$

$$(8) (2 + 5i) \div (-3 - 4i)$$

$$(18) (5 - 4i) \div (1 - 2i)$$

$$(28) (1 - 3i) \div (5 - 2i)$$

$$(9) (3 + 2i) \div (-3 - 2i)$$

$$(19) (-4 + i) \div (-1 - i)$$

$$(29) (-3 - 2i) \div (3 - 5i)$$

$$(10) (2 - 4i) \div (-3 + 4i)$$

$$(20) (-1 - 4i) \div (-2 + 5i)$$

$$(30) (2 + 5i) \div (4 + 4i)$$



$$(1) (1 + 3i) \div (-5 - 5i)$$

$$(11) (1 - 3i) \div (1 - 4i)$$

$$(21) (-5 + 3i) \div (-1 + i)$$

$$(2) (1 - 3i) \div (-5 + 5i)$$

$$(12) (-5 - 3i) \div (1 + i)$$

$$(22) (-1 + i) \div (-3 - 5i)$$

$$(3) (3 + 4i) \div (4 - 2i)$$

$$(13) (4 - i) \div (4 - 2i)$$

$$(23) (-1 - i) \div (-5 + 2i)$$

$$(4) (5 + 5i) \div (-2 - 2i)$$

$$(14) (1 + i) \div (5 - 5i)$$

$$(24) (-5 + 4i) \div (-4 - 3i)$$

$$(5) (-5 + 5i) \div (1 + 3i)$$

$$(15) (4 - i) \div (-1 + 5i)$$

$$(25) (-2 - 2i) \div (-3 + 3i)$$

$$(6) (-2 + 5i) \div (-1 + i)$$

$$(16) (-3 - 4i) \div (2 + i)$$

$$(26) (1 - i) \div (-3 - 4i)$$

$$(7) (2 - 2i) \div (3 + i)$$

$$(17) (-2 + 2i) \div (-4 - 4i)$$

$$(27) (3 - 4i) \div (-5 - 3i)$$

$$(8) (3 + 3i) \div (4 + 2i)$$

$$(18) (1 + 5i) \div (1 - i)$$

$$(28) (3 - 5i) \div (-1 - 3i)$$

$$(9) (2 + 2i) \div (-3 - 2i)$$

$$(19) (-4 - i) \div (1 - 5i)$$

$$(29) (-1 + 2i) \div (-1 + 3i)$$

$$(10) (-4 + 4i) \div (-3 - 5i)$$

$$(20) (-2 + 4i) \div (4 + 4i)$$

$$(30) (-5 - 2i) \div (2 + 3i)$$

$$(1) (4 - i) \div (1 - 5i)$$

$$(11) (4 + 4i) \div (-5 - 2i)$$

$$(21) (4 + 5i) \div (3 + 2i)$$

$$(2) (1 - 4i) \div (4 + 5i)$$

$$(12) (-5 + i) \div (3 - 4i)$$

$$(22) (2 + 4i) \div (1 - i)$$

$$(3) (-4 + i) \div (3 + 2i)$$

$$(13) (-5 - 3i) \div (4 + 2i)$$

$$(23) (-3 + 2i) \div (2 - 2i)$$

$$(4) (3 + 4i) \div (-3 + i)$$

$$(14) (-1 + 5i) \div (-2 + 2i)$$

$$(24) (1 - 2i) \div (-4 - i)$$

$$(5) (3 - 5i) \div (4 + 2i)$$

$$(15) (-3 - 2i) \div (-5 + 3i)$$

$$(25) (-4 - i) \div (-1 - 4i)$$

$$(6) (2 - 4i) \div (-5 + 2i)$$

$$(16) (-4 - 4i) \div (-1 + 2i)$$

$$(26) (4 + i) \div (1 + 3i)$$

$$(7) (3 + 4i) \div (1 - 2i)$$

$$(17) (5 + 5i) \div (5 - 3i)$$

$$(27) (-3 + 3i) \div (-2 + i)$$

$$(8) (-3 + 2i) \div (3 + 4i)$$

$$(18) (5 + 3i) \div (2 - 3i)$$

$$(28) (5 + 3i) \div (-5 + 5i)$$

$$(9) (1 - 4i) \div (4 + 4i)$$

$$(19) (2 + 5i) \div (5 - 4i)$$

$$(29) (5 - i) \div (2 - 3i)$$

$$(10) (3 - 4i) \div (4 + 3i)$$

$$(20) (5 - 3i) \div (3 + 4i)$$

$$(30) (4 + 5i) \div (3 + 2i)$$

$$(1) (2 + 4i) \div (-5 - 3i)$$

$$(11) (3 - 3i) \div (-1 + 3i)$$

$$(21) (-5 + i) \div (-3 + 2i)$$

$$(2) (1 - i) \div (2 + i)$$

$$(12) (-3 + 2i) \div (-2 - 5i)$$

$$(22) (-3 + 5i) \div (2 + 4i)$$

$$(3) (-5 + 2i) \div (-3 + 2i)$$

$$(13) (3 + 5i) \div (4 - 4i)$$

$$(23) (-3 - i) \div (-4 + i)$$

$$(4) (-1 + 3i) \div (4 + 3i)$$

$$(14) (-3 + 3i) \div (1 - i)$$

$$(24) (-3 + i) \div (-5 - 3i)$$

$$(5) (-3 + 2i) \div (4 + i)$$

$$(15) (-2 - 5i) \div (-1 - 2i)$$

$$(25) (-3 - 2i) \div (1 + 4i)$$

$$(6) (2 - i) \div (-2 - 4i)$$

$$(16) (1 - 3i) \div (-2 + 2i)$$

$$(26) (4 + 5i) \div (4 + 5i)$$

$$(7) (-4 - 4i) \div (1 + 5i)$$

$$(17) (-2 - 4i) \div (4 - 5i)$$

$$(27) (-3 - 3i) \div (-2 + 3i)$$

$$(8) (-1 - i) \div (3 - 4i)$$

$$(18) (-4 + 5i) \div (-1 - 5i)$$

$$(28) (-2 + 3i) \div (-1 - 3i)$$

$$(9) (3 + 2i) \div (-2 - 2i)$$

$$(19) (-4 + 5i) \div (2 + i)$$

$$(29) (5 + i) \div (-5 + 3i)$$

$$(10) (-5 - 5i) \div (1 + 5i)$$

$$(20) (5 + 3i) \div (4 - 3i)$$

$$(30) (-2 - 5i) \div (-5 - 4i)$$

$$(1) (-4 + i) \div (-2 + 2i)$$

$$(11) (-1 - i) \div (3 + i)$$

$$(21) (1 + 2i) \div (-3 - i)$$

$$(2) (-1 - i) \div (-4 - 5i)$$

$$(12) (-1 + 3i) \div (-3 + 3i)$$

$$(22) (-1 + 2i) \div (-5 - i)$$

$$(3) (-4 + i) \div (-4 + i)$$

$$(13) (-2 + 2i) \div (2 - 4i)$$

$$(23) (-1 - i) \div (-1 + i)$$

$$(4) (3 - i) \div (1 - 5i)$$

$$(14) (2 - 2i) \div (5 - 2i)$$

$$(24) (-1 - 5i) \div (5 + 3i)$$

$$(5) (-1 - 2i) \div (1 + 4i)$$

$$(15) (4 + i) \div (-4 - 3i)$$

$$(25) (-1 - i) \div (-4 - i)$$

$$(6) (4 - 3i) \div (-1 - 3i)$$

$$(16) (3 + 3i) \div (-2 - i)$$

$$(26) (3 + 5i) \div (3 - 2i)$$

$$(7) (-1 + 3i) \div (-4 - i)$$

$$(17) (4 - 3i) \div (-2 - 2i)$$

$$(27) (-5 + 5i) \div (4 - 3i)$$

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$$(9) (-2 + 3i) \div (-4 + 4i)$$

$$(19) (-4 + 4i) \div (4 + 4i)$$

$$(29) (-5 - 4i) \div (-1 - i)$$

$$(10) (-1 + i) \div (1 + 2i)$$

$$(20) (-4 - 4i) \div (3 + i)$$

$$(30) (1 - 2i) \div (-4 - 4i)$$

(1)  $(-1 - 3i) \div (-3 + 2i)$

(11)  $(2 - 5i) \div (-4 - 5i)$

(21)  $(-2 - 3i) \div (3 - 2i)$

(2)  $(3 - 2i) \div (5 + i)$

(12)  $(5 + 5i) \div (-5 + 3i)$

(22)  $(3 + 2i) \div (2 - 4i)$

(3)  $(2 + 5i) \div (1 - 5i)$

(13)  $(3 - 5i) \div (5 + 2i)$

(23)  $(5 + i) \div (2 - i)$

(4)  $(-4 + 5i) \div (-5 + 2i)$

(14)  $(5 - i) \div (2 - 5i)$

(24)  $(3 - 5i) \div (-2 - 2i)$

(5)  $(5 - i) \div (-4 - 2i)$

(15)  $(-3 - i) \div (1 - 3i)$

(25)  $(3 - 5i) \div (1 + i)$

(6)  $(-3 + 2i) \div (-5 - i)$

(16)  $(3 - i) \div (-5 + 2i)$

(26)  $(-2 - 3i) \div (1 + 2i)$

(7)  $(-5 - 2i) \div (-2 - 3i)$

(17)  $(1 - 3i) \div (4 + 2i)$

(27)  $(-1 - 3i) \div (5 - i)$

(8)  $(3 + i) \div (1 - i)$

(18)  $(-2 + 5i) \div (-3 - 3i)$

(28)  $(5 + i) \div (4 + 5i)$

(9)  $(-1 - 4i) \div (-4 - 5i)$

(19)  $(3 - 5i) \div (-2 - 5i)$

(29)  $(2 - i) \div (3 + 3i)$

(10)  $(5 - 4i) \div (-4 - 5i)$

(20)  $(4 + 5i) \div (3 + 4i)$

(30)  $(-5 - 5i) \div (1 + 4i)$

(1)  $(2 + i) \div (-3 + 4i)$

(11)  $(5 - 3i) \div (-1 + 5i)$

(21)  $(1 - 3i) \div (-1 - 5i)$

(2)  $(2 + i) \div (2 + 3i)$

(12)  $(-3 - i) \div (2 + 4i)$

(22)  $(-1 - 4i) \div (-2 - 3i)$

(3)  $(-2 + i) \div (-5 + 4i)$

(13)  $(-4 - i) \div (-2 + 3i)$

(23)  $(-1 - 3i) \div (-1 - 3i)$

(4)  $(5 + 3i) \div (-3 + 2i)$

(14)  $(1 + 4i) \div (4 - 5i)$

(24)  $(1 - 2i) \div (3 + 5i)$

(5)  $(-4 + 2i) \div (-2 - 2i)$

(15)  $(2 + i) \div (-2 - 4i)$

(25)  $(4 + 5i) \div (-4 - 4i)$

(6)  $(-1 - 5i) \div (2 - i)$

(16)  $(-1 + 4i) \div (-2 + 4i)$

(26)  $(-1 + 5i) \div (-2 + 5i)$

(7)  $(-5 + 2i) \div (2 + 4i)$

(17)  $(3 - 3i) \div (-1 + i)$

(27)  $(-2 - 4i) \div (1 - 5i)$

(8)  $(-5 - 2i) \div (-3 - 2i)$

(18)  $(-2 + 5i) \div (-1 + 2i)$

(28)  $(2 + 3i) \div (-2 - 5i)$

(9)  $(4 - 2i) \div (2 + 3i)$

(19)  $(-1 - 3i) \div (-2 - 5i)$

(29)  $(2 - 3i) \div (-3 + i)$

(10)  $(-1 + 5i) \div (2 - 5i)$

(20)  $(4 - 5i) \div (1 + 5i)$

(30)  $(1 + 5i) \div (-5 + 4i)$

$$(1) (-3 + 2i) \div (4 + i) \\ = -\frac{10}{17} + \frac{11}{17}i$$

$$(11) (4 + 3i) \div (5 + i) \\ = \frac{23}{26} + \frac{11}{26}i$$

$$(21) (4 + 2i) \div (-3 + 3i) \\ = -\frac{1}{3} - i$$

$$(2) (-4 - 5i) \div (5 - 5i) \\ = \frac{1}{10} - \frac{9}{10}i$$

$$(12) (-1 + 5i) \div (3 - 3i) \\ = -1 + \frac{2}{3}i$$

$$(22) (-4 - 4i) \div (-3 - 5i) \\ = \frac{16}{17} - \frac{4}{17}i$$

$$(3) (-1 - i) \div (-2 - 5i) \\ = \frac{7}{29} - \frac{3}{29}i$$

$$(13) (5 + 3i) \div (-2 + i) \\ = -\frac{7}{5} - \frac{11}{5}i$$

$$(23) (1 + 5i) \div (-1 + 4i) \\ = \frac{19}{17} - \frac{9}{17}i$$

$$(4) (2 + 3i) \div (3 + 2i) \\ = \frac{12}{13} + \frac{5}{13}i$$

$$(14) (1 - 5i) \div (5 - 4i) \\ = \frac{25}{41} - \frac{21}{41}i$$

$$(24) (-1 - 2i) \div (-1 - 3i) \\ = \frac{7}{10} - \frac{1}{10}i$$

$$(5) (-2 - 2i) \div (-5 + 4i) \\ = \frac{2}{41} + \frac{18}{41}i$$

$$(15) (5 - i) \div (-2 - 2i) \\ = -1 + \frac{3}{2}i$$

$$(25) (-5 + 4i) \div (5 + i) \\ = -\frac{21}{26} + \frac{25}{26}i$$

$$(6) (-4 - i) \div (4 - i) \\ = -\frac{15}{17} - \frac{8}{17}i$$

$$(16) (-2 + 4i) \div (1 - 3i) \\ = -\frac{7}{5} - \frac{1}{5}i$$

$$(26) (4 + i) \div (-5 - i) \\ = -\frac{21}{26} - \frac{1}{26}i$$

$$(7) (-3 + 2i) \div (4 - 2i) \\ = -\frac{4}{5} + \frac{1}{10}i$$

$$(17) (-1 - 3i) \div (-4 - 5i) \\ = \frac{19}{41} + \frac{7}{41}i$$

$$(27) (2 + i) \div (2 + 5i) \\ = \frac{9}{29} - \frac{8}{29}i$$

$$(8) (3 - 4i) \div (-2 - 5i) \\ = \frac{14}{29} + \frac{23}{29}i$$

$$(18) (3 + i) \div (-1 + 2i) \\ = -\frac{1}{5} - \frac{7}{5}i$$

$$(28) (1 - i) \div (-3 + 2i) \\ = -\frac{5}{13} + \frac{1}{13}i$$

$$(9) (-3 - 3i) \div (-4 - 5i) \\ = \frac{27}{41} - \frac{3}{41}i$$

$$(19) (-4 - 5i) \div (1 + 4i) \\ = -\frac{24}{17} + \frac{11}{17}i$$

$$(29) (4 + 4i) \div (-5 + 2i) \\ = -\frac{12}{29} - \frac{28}{29}i$$

$$(10) (-2 - 3i) \div (3 - 4i) \\ = \frac{6}{25} - \frac{17}{25}i$$

$$(20) (5 - i) \div (-1 + 4i) \\ = -\frac{9}{17} - \frac{19}{17}i$$

$$(30) (2 - 5i) \div (1 - i) \\ = \frac{7}{2} - \frac{3}{2}i$$

$$(1) \quad (-2-3i) \div (3+5i) \\ = -\frac{21}{34} + \frac{1}{34}i$$

$$(11) \quad (-2+i) \div (-5+3i) \\ = \frac{13}{34} + \frac{1}{34}i$$

$$(21) \quad (3+5i) \div (-5+2i) \\ = -\frac{5}{29} - \frac{31}{29}i$$

$$(2) \quad (3+i) \div (-3+i) \\ = -\frac{4}{5} - \frac{3}{5}i$$

$$(12) \quad (2+4i) \div (-2+2i) \\ = \frac{1}{2} - \frac{3}{2}i$$

$$(22) \quad (-1+4i) \div (-3-4i) \\ = -\frac{13}{25} - \frac{16}{25}i$$

$$(3) \quad (5-2i) \div (5+i) \\ = \frac{23}{26} - \frac{15}{26}i$$

$$(13) \quad (-1+2i) \div (2+3i) \\ = \frac{4}{13} + \frac{7}{13}i$$

$$(23) \quad (-5-5i) \div (2+i) \\ = -3-i$$

$$(4) \quad (-1-4i) \div (-5-5i) \\ = \frac{1}{2} + \frac{3}{10}i$$

$$(14) \quad (5+5i) \div (1+3i) \\ = 2-i$$

$$(24) \quad (1+i) \div (-3-3i) \\ = -\frac{1}{3}$$

$$(5) \quad (-2+2i) \div (-4+i) \\ = \frac{10}{17} - \frac{6}{17}i$$

$$(15) \quad (2+i) \div (1-2i) \\ = i$$

$$(25) \quad (-4+2i) \div (4-4i) \\ = -\frac{3}{4} - \frac{1}{4}i$$

$$(6) \quad (2+4i) \div (-1+2i) \\ = \frac{6}{5} - \frac{8}{5}i$$

$$(16) \quad (-1+5i) \div (-1+2i) \\ = \frac{11}{5} - \frac{3}{5}i$$

$$(26) \quad (5-2i) \div (2+2i) \\ = \frac{3}{4} - \frac{7}{4}i$$

$$(7) \quad (4+4i) \div (-1-i) \\ = -4$$

$$(17) \quad (1-4i) \div (-1-4i) \\ = \frac{15}{17} + \frac{8}{17}i$$

$$(27) \quad (1+4i) \div (-1+2i) \\ = \frac{7}{5} - \frac{6}{5}i$$

$$(8) \quad (1-4i) \div (3-4i) \\ = \frac{19}{25} - \frac{8}{25}i$$

$$(18) \quad (3+5i) \div (4-5i) \\ = -\frac{13}{41} + \frac{35}{41}i$$

$$(28) \quad (-3+i) \div (-1+4i) \\ = \frac{7}{17} + \frac{11}{17}i$$

$$(9) \quad (-5-5i) \div (5-i) \\ = -\frac{10}{13} - \frac{15}{13}i$$

$$(19) \quad (4-3i) \div (-2-4i) \\ = \frac{1}{5} + \frac{11}{10}i$$

$$(29) \quad (-4-3i) \div (-3-i) \\ = \frac{3}{2} + \frac{1}{2}i$$

$$(10) \quad (-5-i) \div (3+5i) \\ = -\frac{10}{17} + \frac{11}{17}i$$

$$(20) \quad (-5+5i) \div (-4+5i) \\ = \frac{45}{41} + \frac{5}{41}i$$

$$(30) \quad (3+2i) \div (3-i) \\ = \frac{7}{10} + \frac{9}{10}i$$



$$(1) (1-i) \div (1-5i) \\ = \frac{3}{13} + \frac{2}{13}i$$

$$(11) (-2-i) \div (5-2i) \\ = -\frac{8}{29} - \frac{9}{29}i$$

$$(21) (3-5i) \div (-4-i) \\ = -\frac{7}{17} + \frac{23}{17}i$$

$$(2) (-4+3i) \div (-4+4i) \\ = \frac{7}{8} + \frac{1}{8}i$$

$$(12) (1+5i) \div (3+2i) \\ = 1+i$$

$$(22) (-3+3i) \div (-3-4i) \\ = -\frac{3}{25} - \frac{21}{25}i$$

$$(3) (-4-4i) \div (2+3i) \\ = -\frac{20}{13} + \frac{4}{13}i$$

$$(13) (2-5i) \div (5+2i) \\ = -i$$

$$(23) (-4-5i) \div (-1-4i) \\ = \frac{24}{17} - \frac{11}{17}i$$

$$(4) (2-2i) \div (1+5i) \\ = -\frac{4}{13} - \frac{6}{13}i$$

$$(14) (-4+3i) \div (-2+4i) \\ = 1 + \frac{1}{2}i$$

$$(24) (-4+3i) \div (-2-5i) \\ = -\frac{7}{29} - \frac{26}{29}i$$

$$(5) (-4-i) \div (-5-2i) \\ = \frac{22}{29} - \frac{3}{29}i$$

$$(15) (2-3i) \div (1+2i) \\ = -\frac{4}{5} - \frac{7}{5}i$$

$$(25) (3-5i) \div (3-5i) \\ = 1$$

$$(6) (-4+i) \div (1-i) \\ = -\frac{5}{2} - \frac{3}{2}i$$

$$(16) (-1-4i) \div (-1-i) \\ = \frac{5}{2} + \frac{3}{2}i$$

$$(26) (-3-2i) \div (-4+5i) \\ = \frac{2}{41} + \frac{23}{41}i$$

$$(7) (-2-3i) \div (1-3i) \\ = \frac{7}{10} - \frac{9}{10}i$$

$$(17) (2+4i) \div (1+2i) \\ = 2$$

$$(27) (-3+i) \div (3-3i) \\ = -\frac{2}{3} - \frac{1}{3}i$$

$$(8) (1-i) \div (-1-3i) \\ = \frac{1}{5} + \frac{2}{5}i$$

$$(18) (-4-i) \div (-3-4i) \\ = \frac{16}{25} - \frac{13}{25}i$$

$$(28) (-1-3i) \div (-2-5i) \\ = \frac{17}{29} + \frac{1}{29}i$$

$$(9) (-5-4i) \div (2-2i) \\ = -\frac{1}{4} - \frac{9}{4}i$$

$$(19) (5+5i) \div (-4-3i) \\ = -\frac{7}{5} - \frac{1}{5}i$$

$$(29) (-2-5i) \div (-2-4i) \\ = \frac{6}{5} + \frac{1}{10}i$$

$$(10) (3-3i) \div (5-5i) \\ = \frac{3}{5}$$

$$(20) (2-5i) \div (-1+5i) \\ = -\frac{27}{26} - \frac{5}{26}i$$

$$(30) (-3-5i) \div (-3+4i) \\ = -\frac{11}{25} + \frac{27}{25}i$$

$$(1) (2+i) \div (5+2i) \\ = \frac{12}{29} + \frac{1}{29}i$$

$$(11) (-1+2i) \div (5+3i) \\ = \frac{1}{34} + \frac{13}{34}i$$

$$(21) (1+2i) \div (4+2i) \\ = \frac{2}{5} + \frac{3}{10}i$$

$$(2) (-5+3i) \div (-5+4i) \\ = \frac{37}{41} + \frac{5}{41}i$$

$$(12) (1-2i) \div (-4+5i) \\ = -\frac{14}{41} + \frac{3}{41}i$$

$$(22) (-1-5i) \div (1+4i) \\ = -\frac{21}{17} - \frac{1}{17}i$$

$$(3) (-2-2i) \div (-5-2i) \\ = \frac{14}{29} + \frac{6}{29}i$$

$$(13) (-2+2i) \div (3-3i) \\ = -\frac{2}{3}$$

$$(23) (-5+3i) \div (-1+2i) \\ = \frac{11}{5} + \frac{7}{5}i$$

$$(4) (3-i) \div (4-i) \\ = \frac{13}{17} - \frac{1}{17}i$$

$$(14) (-1+2i) \div (-1+2i) \\ = 1$$

$$(24) (-3+i) \div (2+2i) \\ = -\frac{1}{2} + i$$

$$(5) (3-i) \div (-2+3i) \\ = -\frac{9}{13} - \frac{7}{13}i$$

$$(15) (-3-5i) \div (2-2i) \\ = \frac{1}{2} - 2i$$

$$(25) (2+4i) \div (-3-5i) \\ = -\frac{13}{17} - \frac{1}{17}i$$

$$(6) (5-i) \div (-5-5i) \\ = -\frac{2}{5} + \frac{3}{5}i$$

$$(16) (2+i) \div (-5-2i) \\ = -\frac{12}{29} - \frac{1}{29}i$$

$$(26) (-4+5i) \div (4+4i) \\ = \frac{1}{8} + \frac{9}{8}i$$

$$(7) (2-4i) \div (-2-4i) \\ = \frac{3}{5} + \frac{4}{5}i$$

$$(17) (1-3i) \div (-4+5i) \\ = -\frac{19}{41} + \frac{7}{41}i$$

$$(27) (2-2i) \div (2-4i) \\ = \frac{3}{5} + \frac{1}{5}i$$

$$(8) (-1-5i) \div (-4+2i) \\ = -\frac{3}{10} + \frac{11}{10}i$$

$$(18) (5+5i) \div (-5-5i) \\ = -1$$

$$(28) (1-5i) \div (-4-5i) \\ = \frac{21}{41} + \frac{25}{41}i$$

$$(9) (4+4i) \div (2+5i) \\ = \frac{28}{29} - \frac{12}{29}i$$

$$(19) (-2-4i) \div (5-5i) \\ = \frac{1}{5} - \frac{3}{5}i$$

$$(29) (3+2i) \div (-4-4i) \\ = -\frac{5}{8} + \frac{1}{8}i$$

$$(10) (-3-i) \div (1-i) \\ = -1 - 2i$$

$$(20) (-5+5i) \div (-2+2i) \\ = \frac{5}{2}$$

$$(30) (1-i) \div (5+3i) \\ = \frac{1}{17} - \frac{4}{17}i$$

$$(1) \quad (-1 + 2i) \div (4 + 5i) \\ = \frac{6}{41} + \frac{13}{41}i$$

$$(11) \quad (1 - 5i) \div (-5 - 4i) \\ = \frac{15}{41} + \frac{29}{41}i$$

$$(21) \quad (1 - i) \div (-5 + 4i) \\ = -\frac{9}{41} + \frac{1}{41}i$$

$$(2) \quad (4 - 3i) \div (1 - 5i) \\ = \frac{19}{26} + \frac{17}{26}i$$

$$(12) \quad (-2 - 3i) \div (-2 + i) \\ = \frac{1}{5} + \frac{8}{5}i$$

$$(22) \quad (3 + 3i) \div (5 + 2i) \\ = \frac{21}{29} + \frac{9}{29}i$$

$$(3) \quad (-4 - 2i) \div (2 - 2i) \\ = -\frac{1}{2} - \frac{3}{2}i$$

$$(13) \quad (3 + 5i) \div (4 - 5i) \\ = -\frac{13}{41} + \frac{35}{41}i$$

$$(23) \quad (5 - 3i) \div (5 + 4i) \\ = \frac{13}{41} - \frac{35}{41}i$$

$$(4) \quad (5 - 2i) \div (3 + 4i) \\ = \frac{7}{25} - \frac{26}{25}i$$

$$(14) \quad (1 + 2i) \div (2 + 5i) \\ = \frac{12}{29} - \frac{1}{29}i$$

$$(24) \quad (-2 - 2i) \div (3 - 3i) \\ = -\frac{2}{3}i$$

$$(5) \quad (2 - 3i) \div (1 - 3i) \\ = \frac{11}{10} + \frac{3}{10}i$$

$$(15) \quad (5 + 4i) \div (4 - 3i) \\ = \frac{8}{25} + \frac{31}{25}i$$

$$(25) \quad (5 - i) \div (2 + 3i) \\ = \frac{7}{13} - \frac{17}{13}i$$

$$(6) \quad (4 + 4i) \div (5 - 2i) \\ = \frac{12}{29} + \frac{28}{29}i$$

$$(16) \quad (2 - 2i) \div (-3 - 2i) \\ = -\frac{2}{13} + \frac{10}{13}i$$

$$(26) \quad (1 + i) \div (-5 - i) \\ = -\frac{3}{13} - \frac{2}{13}i$$

$$(7) \quad (2 - 4i) \div (5 - i) \\ = \frac{7}{13} - \frac{9}{13}i$$

$$(17) \quad (-3 + 5i) \div (4 - 3i) \\ = -\frac{27}{25} + \frac{11}{25}i$$

$$(27) \quad (-2 + 5i) \div (5 - 4i) \\ = -\frac{30}{41} + \frac{17}{41}i$$

$$(8) \quad (2 + 4i) \div (5 + 3i) \\ = \frac{11}{17} + \frac{7}{17}i$$

$$(18) \quad (3 - 3i) \div (3 + 2i) \\ = \frac{3}{13} - \frac{15}{13}i$$

$$(28) \quad (-3 - i) \div (-3 + i) \\ = \frac{4}{5} + \frac{3}{5}i$$

$$(9) \quad (-3 + 3i) \div (4 + i) \\ = -\frac{9}{17} + \frac{15}{17}i$$

$$(19) \quad (5 - 2i) \div (2 - 5i) \\ = \frac{20}{29} + \frac{21}{29}i$$

$$(29) \quad (4 - i) \div (-3 + i) \\ = -\frac{13}{10} - \frac{1}{10}i$$

$$(10) \quad (2 + 5i) \div (-2 - i) \\ = -\frac{9}{5} - \frac{8}{5}i$$

$$(20) \quad (5 - 3i) \div (4 - 3i) \\ = \frac{29}{25} + \frac{3}{25}i$$

$$(30) \quad (5 - 4i) \div (-3 + 4i) \\ = -\frac{31}{25} - \frac{8}{25}i$$

$$(1) (3 + 4i) \div (2 - 5i) \\ = -\frac{14}{29} + \frac{23}{29}i$$

$$(11) (4 + 4i) \div (-4 - 2i) \\ = -\frac{6}{5} - \frac{2}{5}i$$

$$(21) (-5 + 4i) \div (2 - 5i) \\ = -\frac{30}{29} - \frac{17}{29}i$$

$$(2) (2 - 2i) \div (-4 + 5i) \\ = -\frac{18}{41} - \frac{2}{41}i$$

$$(12) (-5 + 3i) \div (1 - i) \\ = -4 - i$$

$$(22) (4 + 4i) \div (-2 - 2i) \\ = -2$$

$$(3) (-4 + 5i) \div (-1 - i) \\ = -\frac{1}{2} - \frac{9}{2}i$$

$$(13) (3 + 3i) \div (1 - i) \\ = 3i$$

$$(23) (4 - 2i) \div (2 - 2i) \\ = \frac{3}{2} + \frac{1}{2}i$$

$$(4) (3 - 2i) \div (-2 + 4i) \\ = -\frac{7}{10} - \frac{2}{5}i$$

$$(14) (3 + 4i) \div (4 - 3i) \\ = i$$

$$(24) (5 + 2i) \div (-5 + 3i) \\ = -\frac{19}{34} - \frac{25}{34}i$$

$$(5) (-4 + 4i) \div (1 - 2i) \\ = -\frac{12}{5} - \frac{4}{5}i$$

$$(15) (-1 + i) \div (-2 - i) \\ = \frac{1}{5} - \frac{3}{5}i$$

$$(25) (3 + i) \div (-2 + 3i) \\ = -\frac{3}{13} - \frac{11}{13}i$$

$$(6) (-2 + 2i) \div (5 + i) \\ = -\frac{4}{13} + \frac{6}{13}i$$

$$(16) (1 + i) \div (-2 + 3i) \\ = \frac{1}{13} - \frac{5}{13}i$$

$$(26) (3 + 5i) \div (1 + 3i) \\ = \frac{9}{5} - \frac{2}{5}i$$

$$(7) (-3 - 5i) \div (5 + 5i) \\ = -\frac{4}{5} - \frac{1}{5}i$$

$$(17) (-4 - 4i) \div (-4 - 4i) \\ = 1$$

$$(27) (-4 - 5i) \div (3 - i) \\ = -\frac{7}{10} - \frac{19}{10}i$$

$$(8) (-5 - i) \div (4 + 5i) \\ = -\frac{25}{41} + \frac{21}{41}i$$

$$(18) (-3 - 2i) \div (-2 - 4i) \\ = \frac{7}{10} - \frac{2}{5}i$$

$$(28) (-5 - 3i) \div (-5 + 2i) \\ = \frac{19}{29} + \frac{25}{29}i$$

$$(9) (4 + 5i) \div (-5 + 4i) \\ = -i$$

$$(19) (-1 + 4i) \div (-3 + 3i) \\ = \frac{5}{6} - \frac{1}{2}i$$

$$(29) (-4 - 3i) \div (-1 - i) \\ = \frac{7}{2} - \frac{1}{2}i$$

$$(10) (4 - 4i) \div (-1 - i) \\ = 4i$$

$$(20) (-3 - 4i) \div (-5 + 3i) \\ = \frac{3}{34} + \frac{29}{34}i$$

$$(30) (2 - 4i) \div (5 + 5i) \\ = -\frac{1}{5} - \frac{3}{5}i$$

$$(1) (4 - 5i) \div (5 + 5i) \\ = -\frac{1}{10} - \frac{9}{10}i$$

$$(11) (2 - 2i) \div (-5 - 3i) \\ = -\frac{2}{17} + \frac{8}{17}i$$

$$(21) (-4 - 4i) \div (-5 + 3i) \\ = \frac{4}{17} + \frac{16}{17}i$$

$$(2) (2 - i) \div (5 - i) \\ = \frac{11}{26} - \frac{3}{26}i$$

$$(12) (-3 + 4i) \div (-1 + i) \\ = \frac{7}{2} - \frac{1}{2}i$$

$$(22) (-4 - i) \div (-3 + 5i) \\ = \frac{7}{34} + \frac{23}{34}i$$

$$(3) (5 - 2i) \div (1 - i) \\ = \frac{7}{2} + \frac{3}{2}i$$

$$(13) (3 - 4i) \div (-2 + i) \\ = -2 + i$$

$$(23) (-5 - 4i) \div (-1 + i) \\ = \frac{1}{2} + \frac{9}{2}i$$

$$(4) (2 - 5i) \div (3 - 3i) \\ = \frac{7}{6} - \frac{1}{2}i$$

$$(14) (-3 + 2i) \div (-2 + 4i) \\ = \frac{7}{10} + \frac{2}{5}i$$

$$(24) (-4 + 5i) \div (-2 + 5i) \\ = \frac{33}{29} + \frac{10}{29}i$$

$$(5) (1 + 2i) \div (1 - i) \\ = -\frac{1}{2} + \frac{3}{2}i$$

$$(15) (-4 - 4i) \div (3 + 4i) \\ = -\frac{28}{25} + \frac{4}{25}i$$

$$(25) (-2 + 3i) \div (5 - i) \\ = -\frac{1}{2} + \frac{1}{2}i$$

$$(6) (3 - 3i) \div (2 + i) \\ = \frac{3}{5} - \frac{9}{5}i$$

$$(16) (3 - 3i) \div (-5 - 2i) \\ = -\frac{9}{29} + \frac{21}{29}i$$

$$(26) (3 + 3i) \div (3 - 5i) \\ = -\frac{3}{17} + \frac{12}{17}i$$

$$(7) (5 + 2i) \div (1 + 4i) \\ = \frac{13}{17} - \frac{18}{17}i$$

$$(17) (-4 + 2i) \div (-4 + 4i) \\ = \frac{3}{4} + \frac{1}{4}i$$

$$(27) (-3 - 3i) \div (1 - 4i) \\ = \frac{9}{17} - \frac{15}{17}i$$

$$(8) (2 - 4i) \div (2 + 3i) \\ = -\frac{8}{13} - \frac{14}{13}i$$

$$(18) (-3 + i) \div (-2 - i) \\ = 1 - i$$

$$(28) (-5 + 3i) \div (4 - 4i) \\ = -1 - \frac{1}{4}i$$

$$(9) (-1 + 2i) \div (2 + 5i) \\ = \frac{8}{29} + \frac{9}{29}i$$

$$(19) (3 + 3i) \div (5 - 2i) \\ = \frac{9}{29} + \frac{21}{29}i$$

$$(29) (4 - 5i) \div (-2 - 4i) \\ = \frac{3}{5} + \frac{13}{10}i$$

$$(10) (5 + 3i) \div (-4 - i) \\ = -\frac{23}{17} - \frac{7}{17}i$$

$$(20) (1 - i) \div (-4 + 3i) \\ = -\frac{7}{25} + \frac{1}{25}i$$

$$(30) (-2 + 4i) \div (5 - 5i) \\ = -\frac{3}{5} + \frac{1}{5}i$$

$$(1) (2 + 4i) \div (-5 + 2i) \\ = -\frac{2}{29} - \frac{24}{29}i$$

$$(11) (-3 - 5i) \div (-5 - 4i) \\ = \frac{35}{41} + \frac{13}{41}i$$

$$(21) (5 - 4i) \div (1 - 3i) \\ = \frac{17}{10} + \frac{11}{10}i$$

$$(2) (-1 + 3i) \div (2 - 4i) \\ = -\frac{7}{10} + \frac{1}{10}i$$

$$(12) (4 + 2i) \div (5 + 3i) \\ = \frac{13}{17} - \frac{1}{17}i$$

$$(22) (-5 + 2i) \div (-3 + 3i) \\ = \frac{7}{6} + \frac{1}{2}i$$

$$(3) (-1 + i) \div (1 - 2i) \\ = -\frac{3}{5} - \frac{1}{5}i$$

$$(13) (3 - 2i) \div (-5 - 2i) \\ = -\frac{11}{29} + \frac{16}{29}i$$

$$(23) (-5 - 5i) \div (5 - 5i) \\ = -i$$

$$(4) (2 + 3i) \div (-1 + 2i) \\ = \frac{4}{5} - \frac{7}{5}i$$

$$(14) (4 - i) \div (-4 - 4i) \\ = -\frac{3}{8} + \frac{5}{8}i$$

$$(24) (1 + 3i) \div (2 + 4i) \\ = \frac{7}{10} + \frac{1}{10}i$$

$$(5) (2 + i) \div (-1 - i) \\ = -\frac{3}{2} + \frac{1}{2}i$$

$$(15) (-3 - 4i) \div (5 - 2i) \\ = -\frac{7}{29} - \frac{26}{29}i$$

$$(25) (-1 - 5i) \div (3 + 4i) \\ = -\frac{23}{25} - \frac{11}{25}i$$

$$(6) (-5 - i) \div (-4 + 4i) \\ = \frac{1}{2} + \frac{3}{4}i$$

$$(16) (-1 + 4i) \div (1 + 2i) \\ = \frac{7}{5} + \frac{6}{5}i$$

$$(26) (-5 - i) \div (2 + 2i) \\ = -\frac{3}{2} + i$$

$$(7) (-4 - i) \div (-1 - 5i) \\ = \frac{9}{26} - \frac{19}{26}i$$

$$(17) (-1 + 4i) \div (4 - 3i) \\ = -\frac{16}{25} + \frac{13}{25}i$$

$$(27) (1 + 4i) \div (-4 - 5i) \\ = -\frac{24}{41} - \frac{11}{41}i$$

$$(8) (3 - 4i) \div (-1 - 4i) \\ = \frac{13}{17} + \frac{16}{17}i$$

$$(18) (-4 - 4i) \div (-5 + i) \\ = \frac{8}{13} + \frac{12}{13}i$$

$$(28) (-2 - 4i) \div (-4 - i) \\ = \frac{12}{17} + \frac{14}{17}i$$

$$(9) (5 - 4i) \div (2 - 4i) \\ = \frac{13}{10} + \frac{3}{5}i$$

$$(19) (-3 - 4i) \div (3 + 3i) \\ = -\frac{7}{6} - \frac{1}{6}i$$

$$(29) (3 - i) \div (5 - 3i) \\ = \frac{9}{17} + \frac{2}{17}i$$

$$(10) (3 + 4i) \div (-4 - 2i) \\ = -1 - \frac{1}{2}i$$

$$(20) (3 - 3i) \div (5 + i) \\ = \frac{6}{13} - \frac{9}{13}i$$

$$(30) (-5 + 4i) \div (4 - 2i) \\ = -\frac{7}{5} + \frac{3}{10}i$$

$$(1) (5+2i) \div (2-3i) \\ = \frac{4}{13} + \frac{19}{13}i$$

$$(11) (-1-i) \div (3-i) \\ = -\frac{1}{5} - \frac{2}{5}i$$

$$(21) (5+2i) \div (5+2i) \\ = 1$$

$$(2) (-1-5i) \div (-2-i) \\ = \frac{7}{5} + \frac{9}{5}i$$

$$(12) (2-3i) \div (2-5i) \\ = \frac{19}{29} + \frac{4}{29}i$$

$$(22) (-2-i) \div (4-5i) \\ = -\frac{3}{41} - \frac{14}{41}i$$

$$(3) (-5+3i) \div (4-4i) \\ = -1 - \frac{1}{4}i$$

$$(13) (-2+5i) \div (1-3i) \\ = -\frac{17}{10} - \frac{1}{10}i$$

$$(23) (-1-3i) \div (5+2i) \\ = -\frac{11}{29} - \frac{13}{29}i$$

$$(4) (-5+2i) \div (-3+2i) \\ = \frac{19}{13} + \frac{4}{13}i$$

$$(14) (2-3i) \div (5+3i) \\ = \frac{1}{34} - \frac{21}{34}i$$

$$(24) (3-2i) \div (5-2i) \\ = \frac{19}{29} - \frac{4}{29}i$$

$$(5) (-3+i) \div (-5+4i) \\ = \frac{19}{41} + \frac{7}{41}i$$

$$(15) (5-i) \div (1-3i) \\ = \frac{4}{5} + \frac{7}{5}i$$

$$(25) (-3+5i) \div (1+4i) \\ = 1 + i$$

$$(6) (3-4i) \div (-3-2i) \\ = -\frac{1}{13} + \frac{18}{13}i$$

$$(16) (-5+3i) \div (-2-5i) \\ = -\frac{5}{29} - \frac{31}{29}i$$

$$(26) (5+5i) \div (-1-3i) \\ = -2 + i$$

$$(7) (-5-i) \div (-3-5i) \\ = \frac{10}{17} - \frac{11}{17}i$$

$$(17) (5-2i) \div (-2+2i) \\ = -\frac{7}{4} - \frac{3}{4}i$$

$$(27) (-2+3i) \div (1+2i) \\ = \frac{4}{5} + \frac{7}{5}i$$

$$(8) (-1+i) \div (-2-5i) \\ = -\frac{3}{29} - \frac{7}{29}i$$

$$(18) (-4-5i) \div (4-i) \\ = -\frac{11}{17} - \frac{24}{17}i$$

$$(28) (3+4i) \div (-4+4i) \\ = \frac{1}{8} - \frac{7}{8}i$$

$$(9) (-4+i) \div (-4+2i) \\ = \frac{9}{10} + \frac{1}{5}i$$

$$(19) (3+5i) \div (-2+5i) \\ = \frac{19}{29} - \frac{25}{29}i$$

$$(29) (5+2i) \div (-5-5i) \\ = -\frac{7}{10} + \frac{3}{10}i$$

$$(10) (1-5i) \div (1+4i) \\ = -\frac{19}{17} - \frac{9}{17}i$$

$$(20) (-4+5i) \div (-1-4i) \\ = -\frac{16}{17} - \frac{21}{17}i$$

$$(30) (4+i) \div (-4+i) \\ = -\frac{15}{17} - \frac{8}{17}i$$

$$(1) (3+5i) \div (-5-4i) \\ = -\frac{35}{41} - \frac{13}{41}i$$

$$(11) (2+5i) \div (-5+i) \\ = -\frac{5}{26} - \frac{27}{26}i$$

$$(21) (-5+i) \div (-2+3i) \\ = 1+i$$

$$(2) (4-2i) \div (1+4i) \\ = -\frac{4}{17} - \frac{18}{17}i$$

$$(12) (1+3i) \div (1-i) \\ = -1+2i$$

$$(22) (-2+4i) \div (-2+4i) \\ = 1$$

$$(3) (-5+4i) \div (-5+5i) \\ = \frac{9}{10} + \frac{1}{10}i$$

$$(13) (5+4i) \div (-5+5i) \\ = -\frac{1}{10} - \frac{9}{10}i$$

$$(23) (2+4i) \div (4+3i) \\ = \frac{4}{5} + \frac{2}{5}i$$

$$(4) (1+4i) \div (-5-4i) \\ = -\frac{21}{41} - \frac{16}{41}i$$

$$(14) (-4+2i) \div (5-5i) \\ = \frac{3}{5} - \frac{1}{5}i$$

$$(24) (3-3i) \div (-1-i) \\ = 3i$$

$$(5) (2+4i) \div (5-2i) \\ = \frac{2}{29} + \frac{24}{29}i$$

$$(15) (1-2i) \div (4-4i) \\ = \frac{3}{8} - \frac{1}{8}i$$

$$(25) (4-5i) \div (4-4i) \\ = \frac{9}{8} - \frac{1}{8}i$$

$$(6) (-2+4i) \div (-3+5i) \\ = \frac{13}{17} - \frac{1}{17}i$$

$$(16) (4-4i) \div (2-i) \\ = \frac{12}{5} - \frac{4}{5}i$$

$$(26) (5+5i) \div (-4-i) \\ = -\frac{25}{17} - \frac{15}{17}i$$

$$(7) (-3+3i) \div (-2-4i) \\ = -\frac{3}{10} - \frac{9}{10}i$$

$$(17) (-3-3i) \div (3-2i) \\ = -\frac{3}{13} - \frac{15}{13}i$$

$$(27) (2+5i) \div (-2+4i) \\ = \frac{4}{5} - \frac{9}{10}i$$

$$(8) (1+4i) \div (3+i) \\ = \frac{7}{10} + \frac{11}{10}i$$

$$(18) (-5-i) \div (-5+i) \\ = \frac{12}{13} + \frac{5}{13}i$$

$$(28) (4+i) \div (-5+3i) \\ = -\frac{1}{2} - \frac{1}{2}i$$

$$(9) (-1-3i) \div (4-i) \\ = -\frac{1}{17} - \frac{13}{17}i$$

$$(19) (1+5i) \div (-3+2i) \\ = \frac{7}{13} - \frac{17}{13}i$$

$$(29) (4-i) \div (-2+2i) \\ = -\frac{5}{4} - \frac{3}{4}i$$

$$(10) (-1+5i) \div (1+i) \\ = 2+3i$$

$$(20) (-1-3i) \div (-5+i) \\ = \frac{1}{13} + \frac{8}{13}i$$

$$(30) (2-5i) \div (-3+3i) \\ = -\frac{7}{6} + \frac{1}{2}i$$



$$(1) (-4 + 3i) \div (5 - i) \\ = -\frac{23}{26} + \frac{11}{26}i$$

$$(11) (1 - i) \div (-2 + 4i) \\ = -\frac{3}{10} - \frac{1}{10}i$$

$$(21) (5 - 4i) \div (2 - 5i) \\ = \frac{30}{29} + \frac{17}{29}i$$

$$(2) (-4 - i) \div (3 + 4i) \\ = -\frac{16}{25} + \frac{13}{25}i$$

$$(12) (2 - 5i) \div (-4 + i) \\ = -\frac{13}{17} + \frac{18}{17}i$$

$$(22) (-5 + 2i) \div (1 + 4i) \\ = \frac{3}{17} + \frac{22}{17}i$$

$$(3) (1 + 5i) \div (-4 + i) \\ = \frac{1}{17} - \frac{21}{17}i$$

$$(13) (-5 + i) \div (5 + 5i) \\ = -\frac{2}{5} + \frac{3}{5}i$$

$$(23) (1 - 4i) \div (1 - 3i) \\ = \frac{13}{10} - \frac{1}{10}i$$

$$(4) (3 - 3i) \div (-1 + 2i) \\ = -\frac{9}{5} - \frac{3}{5}i$$

$$(14) (3 + 4i) \div (-1 + i) \\ = \frac{1}{2} - \frac{7}{2}i$$

$$(24) (-5 - 3i) \div (2 + 4i) \\ = -\frac{11}{10} + \frac{7}{10}i$$

$$(5) (2 + 4i) \div (-5 + 4i) \\ = \frac{6}{41} - \frac{28}{41}i$$

$$(15) (1 + 3i) \div (4 - 4i) \\ = -\frac{1}{4} + \frac{1}{2}i$$

$$(25) (-5 - 2i) \div (3 + 4i) \\ = -\frac{23}{25} + \frac{14}{25}i$$

$$(6) (1 + 5i) \div (5 - 3i) \\ = -\frac{5}{17} + \frac{14}{17}i$$

$$(16) (2 - i) \div (-4 - i) \\ = -\frac{7}{17} + \frac{6}{17}i$$

$$(26) (3 - 5i) \div (2 - 5i) \\ = \frac{31}{29} + \frac{5}{29}i$$

$$(7) (1 - 2i) \div (-3 - 2i) \\ = \frac{1}{13} + \frac{8}{13}i$$

$$(17) (-4 - 4i) \div (-5 + 4i) \\ = \frac{4}{41} + \frac{36}{41}i$$

$$(27) (5 + 3i) \div (-3 - 5i) \\ = -\frac{15}{17} + \frac{8}{17}i$$

$$(8) (1 + i) \div (-3 - 2i) \\ = -\frac{5}{13} - \frac{1}{13}i$$

$$(18) (4 + 3i) \div (-4 - 5i) \\ = -\frac{31}{41} + \frac{8}{41}i$$

$$(28) (3 - i) \div (-3 + 3i) \\ = -\frac{2}{3} - \frac{1}{3}i$$

$$(9) (3 - 5i) \div (-3 - 2i) \\ = \frac{1}{13} + \frac{21}{13}i$$

$$(19) (1 + 5i) \div (5 - i) \\ = i$$

$$(29) (1 + 5i) \div (-3 - 5i) \\ = -\frac{14}{17} - \frac{5}{17}i$$

$$(10) (-3 + 5i) \div (-3 + 4i) \\ = \frac{29}{25} - \frac{3}{25}i$$

$$(20) (4 - 3i) \div (1 + 5i) \\ = -\frac{11}{26} - \frac{23}{26}i$$

$$(30) (2 - 4i) \div (-3 - i) \\ = -\frac{1}{5} + \frac{7}{5}i$$

$$(1) (4 + 3i) \div (3 - i) \\ = \frac{9}{10} + \frac{13}{10}i$$

$$(11) (-5 - 4i) \div (-5 - 5i) \\ = \frac{9}{10} - \frac{1}{10}i$$

$$(21) (1 - 2i) \div (-5 - i) \\ = -\frac{3}{26} + \frac{11}{26}i$$

$$(2) (2 - 4i) \div (-3 - 4i) \\ = \frac{2}{5} + \frac{4}{5}i$$

$$(12) (-5 - 4i) \div (2 + 5i) \\ = -\frac{30}{29} + \frac{17}{29}i$$

$$(22) (-5 + i) \div (3 + 4i) \\ = -\frac{11}{25} + \frac{23}{25}i$$

$$(3) (2 + 3i) \div (-3 - i) \\ = -\frac{9}{10} - \frac{7}{10}i$$

$$(13) (-1 - 5i) \div (-1 + 3i) \\ = -\frac{7}{5} + \frac{4}{5}i$$

$$(23) (-4 + 4i) \div (-4 - 5i) \\ = -\frac{4}{41} - \frac{36}{41}i$$

$$(4) (2 - 4i) \div (-1 - 2i) \\ = \frac{6}{5} + \frac{8}{5}i$$

$$(14) (1 + 5i) \div (-3 - 2i) \\ = -1 - i$$

$$(24) (-5 - i) \div (-3 - 3i) \\ = 1 - \frac{2}{3}i$$

$$(5) (4 + i) \div (1 + 2i) \\ = \frac{6}{5} - \frac{7}{5}i$$

$$(15) (-2 - 2i) \div (4 + i) \\ = -\frac{10}{17} - \frac{6}{17}i$$

$$(25) (3 - i) \div (1 + i) \\ = 1 - 2i$$

$$(6) (-4 - 5i) \div (3 + 2i) \\ = -\frac{22}{13} - \frac{7}{13}i$$

$$(16) (-3 + 4i) \div (-5 + 5i) \\ = \frac{7}{10} - \frac{1}{10}i$$

$$(26) (-1 - 2i) \div (-3 + 3i) \\ = -\frac{1}{6} + \frac{1}{2}i$$

$$(7) (-1 + i) \div (-1 + 3i) \\ = \frac{2}{5} + \frac{1}{5}i$$

$$(17) (-4 + 3i) \div (-1 + 5i) \\ = \frac{19}{26} + \frac{17}{26}i$$

$$(27) (-5 - 3i) \div (-5 - 3i) \\ = 1$$

$$(8) (4 + 3i) \div (-4 - 3i) \\ = -1$$

$$(18) (-5 - 4i) \div (2 + 3i) \\ = -\frac{22}{13} + \frac{7}{13}i$$

$$(28) (4 - 5i) \div (-3 - 4i) \\ = \frac{8}{25} + \frac{31}{25}i$$

$$(9) (1 + 3i) \div (2 + 2i) \\ = 1 + \frac{1}{2}i$$

$$(19) (3 + 4i) \div (5 - 5i) \\ = -\frac{1}{10} + \frac{7}{10}i$$

$$(29) (-2 - 5i) \div (1 - 2i) \\ = \frac{8}{5} - \frac{9}{5}i$$

$$(10) (-5 + 2i) \div (-4 + 3i) \\ = \frac{26}{25} + \frac{7}{25}i$$

$$(20) (-4 - 4i) \div (-4 - 3i) \\ = \frac{28}{25} + \frac{4}{25}i$$

$$(30) (4 + 5i) \div (-4 + i) \\ = -\frac{11}{17} - \frac{24}{17}i$$

$$(1) (-4 - 3i) \div (-5 + 2i) \\ = \frac{14}{29} + \frac{23}{29}i$$

$$(11) (4 + 5i) \div (5 - 4i) \\ = i$$

$$(21) (-4 - i) \div (-4 + 2i) \\ = \frac{7}{10} + \frac{3}{5}i$$

$$(2) (-2 - i) \div (5 - 2i) \\ = -\frac{8}{29} - \frac{9}{29}i$$

$$(12) (-3 - 3i) \div (-5 - 5i) \\ = \frac{3}{5}$$

$$(22) (-4 - 3i) \div (-1 - 5i) \\ = \frac{19}{26} - \frac{17}{26}i$$

$$(3) (-1 + 4i) \div (1 + i) \\ = \frac{3}{2} + \frac{5}{2}i$$

$$(13) (-1 + 2i) \div (-1 - 3i) \\ = -\frac{1}{2} - \frac{1}{2}i$$

$$(23) (-2 - 3i) \div (4 - 2i) \\ = -\frac{1}{10} - \frac{4}{5}i$$

$$(4) (5 - 2i) \div (5 + 4i) \\ = \frac{17}{41} - \frac{30}{41}i$$

$$(14) (-5 + 3i) \div (2 - 3i) \\ = -\frac{19}{13} - \frac{9}{13}i$$

$$(24) (-5 - 4i) \div (-5 + 2i) \\ = \frac{17}{29} + \frac{30}{29}i$$

$$(5) (-2 - 2i) \div (-5 + i) \\ = \frac{4}{13} + \frac{6}{13}i$$

$$(15) (4 + 3i) \div (5 + 3i) \\ = \frac{29}{34} + \frac{3}{34}i$$

$$(25) (2 + 2i) \div (4 - i) \\ = \frac{6}{17} + \frac{10}{17}i$$

$$(6) (-1 - 5i) \div (2 - 4i) \\ = \frac{9}{10} - \frac{7}{10}i$$

$$(16) (2 - i) \div (3 + 5i) \\ = \frac{1}{34} - \frac{13}{34}i$$

$$(26) (3 + 2i) \div (2 + 3i) \\ = \frac{12}{13} - \frac{5}{13}i$$

$$(7) (5 + 2i) \div (-2 + i) \\ = -\frac{8}{5} - \frac{9}{5}i$$

$$(17) (-4 + 4i) \div (5 - 4i) \\ = -\frac{36}{41} + \frac{4}{41}i$$

$$(27) (-5 - 4i) \div (3 + 5i) \\ = -\frac{35}{34} + \frac{13}{34}i$$

$$(8) (4 + i) \div (-5 + 5i) \\ = -\frac{3}{10} - \frac{1}{2}i$$

$$(18) (5 - 4i) \div (4 - 4i) \\ = \frac{9}{8} + \frac{1}{8}i$$

$$(28) (3 + i) \div (-1 + i) \\ = -1 - 2i$$

$$(9) (-5 - 4i) \div (-3 - 5i) \\ = \frac{35}{34} - \frac{13}{34}i$$

$$(19) (1 + 2i) \div (4 + 2i) \\ = \frac{2}{5} + \frac{3}{10}i$$

$$(29) (-5 + 4i) \div (-2 + i) \\ = \frac{14}{5} - \frac{3}{5}i$$

$$(10) (1 + i) \div (4 + i) \\ = \frac{5}{17} + \frac{3}{17}i$$

$$(20) (-5 + 2i) \div (-1 - i) \\ = \frac{3}{2} - \frac{7}{2}i$$

$$(30) (-2 - 5i) \div (-4 + i) \\ = \frac{3}{17} + \frac{22}{17}i$$

$$(1) \quad (-5 + 4i) \div (1 - 5i) \\ = -\frac{25}{26} - \frac{21}{26}i$$

$$(11) \quad (-1 - 5i) \div (1 + i) \\ = -3 - 2i$$

$$(21) \quad (-3 - 2i) \div (5 + 2i) \\ = -\frac{19}{29} - \frac{4}{29}i$$

$$(2) \quad (-3 + 5i) \div (-2 - 2i) \\ = -\frac{1}{2} - 2i$$

$$(12) \quad (4 - i) \div (3 + 3i) \\ = \frac{1}{2} - \frac{5}{6}i$$

$$(22) \quad (5 - 2i) \div (-3 - 4i) \\ = -\frac{7}{25} + \frac{26}{25}i$$

$$(3) \quad (2 + i) \div (4 + 2i) \\ = \frac{1}{2}$$

$$(13) \quad (3 + i) \div (2 + 4i) \\ = \frac{1}{2} - \frac{1}{2}i$$

$$(23) \quad (2 + 2i) \div (4 + 2i) \\ = \frac{3}{5} + \frac{1}{5}i$$

$$(4) \quad (2 + 2i) \div (-5 + 5i) \\ = -\frac{2}{5}i$$

$$(14) \quad (-2 - 2i) \div (-5 - i) \\ = \frac{6}{13} + \frac{4}{13}i$$

$$(24) \quad (-3 - 5i) \div (4 - 3i) \\ = \frac{3}{25} - \frac{29}{25}i$$

$$(5) \quad (3 + 4i) \div (5 - i) \\ = \frac{11}{26} + \frac{23}{26}i$$

$$(15) \quad (-1 - i) \div (2 + 3i) \\ = -\frac{5}{13} + \frac{1}{13}i$$

$$(25) \quad (-5 - i) \div (2 + i) \\ = -\frac{11}{5} + \frac{3}{5}i$$

$$(6) \quad (-1 - 4i) \div (2 + 3i) \\ = -\frac{14}{13} - \frac{5}{13}i$$

$$(16) \quad (-2 + 4i) \div (-4 + 3i) \\ = \frac{4}{5} - \frac{2}{5}i$$

$$(26) \quad (-4 - 3i) \div (5 - 3i) \\ = -\frac{11}{34} - \frac{27}{34}i$$

$$(7) \quad (3 - i) \div (3 + 4i) \\ = \frac{1}{5} - \frac{3}{5}i$$

$$(17) \quad (1 + 4i) \div (-1 + i) \\ = \frac{3}{2} - \frac{5}{2}i$$

$$(27) \quad (4 - 3i) \div (5 + i) \\ = \frac{17}{26} - \frac{19}{26}i$$

$$(8) \quad (2 + 5i) \div (-3 - 4i) \\ = -\frac{26}{25} - \frac{7}{25}i$$

$$(18) \quad (5 - 4i) \div (1 - 2i) \\ = \frac{13}{5} + \frac{6}{5}i$$

$$(28) \quad (1 - 3i) \div (5 - 2i) \\ = \frac{11}{29} - \frac{13}{29}i$$

$$(9) \quad (3 + 2i) \div (-3 - 2i) \\ = -1$$

$$(19) \quad (-4 + i) \div (-1 - i) \\ = \frac{3}{2} - \frac{5}{2}i$$

$$(29) \quad (-3 - 2i) \div (3 - 5i) \\ = \frac{1}{34} - \frac{21}{34}i$$

$$(10) \quad (2 - 4i) \div (-3 + 4i) \\ = -\frac{22}{25} + \frac{4}{25}i$$

$$(20) \quad (-1 - 4i) \div (-2 + 5i) \\ = -\frac{18}{29} + \frac{13}{29}i$$

$$(30) \quad (2 + 5i) \div (4 + 4i) \\ = \frac{7}{8} + \frac{3}{8}i$$

$$(1) (1+3i) \div (-5-5i) \\ = -\frac{2}{5} - \frac{1}{5}i$$

$$(11) (1-3i) \div (1-4i) \\ = \frac{13}{17} + \frac{1}{17}i$$

$$(21) (-5+3i) \div (-1+i) \\ = 4+i$$

$$(2) (1-3i) \div (-5+5i) \\ = -\frac{2}{5} + \frac{1}{5}i$$

$$(12) (-5-3i) \div (1+i) \\ = -4+i$$

$$(22) (-1+i) \div (-3-5i) \\ = -\frac{1}{17} - \frac{4}{17}i$$

$$(3) (3+4i) \div (4-2i) \\ = \frac{1}{5} + \frac{11}{10}i$$

$$(13) (4-i) \div (4-2i) \\ = \frac{9}{10} + \frac{1}{5}i$$

$$(23) (-1-i) \div (-5+2i) \\ = \frac{3}{29} + \frac{7}{29}i$$

$$(4) (5+5i) \div (-2-2i) \\ = -\frac{5}{2}$$

$$(14) (1+i) \div (5-5i) \\ = \frac{1}{5}i$$

$$(24) (-5+4i) \div (-4-3i) \\ = \frac{8}{25} - \frac{31}{25}i$$

$$(5) (-5+5i) \div (1+3i) \\ = 1+2i$$

$$(15) (4-i) \div (-1+5i) \\ = -\frac{9}{26} - \frac{19}{26}i$$

$$(25) (-2-2i) \div (-3+3i) \\ = \frac{2}{3}i$$

$$(6) (-2+5i) \div (-1+i) \\ = \frac{7}{2} - \frac{3}{2}i$$

$$(16) (-3-4i) \div (2+i) \\ = -2-i$$

$$(26) (1-i) \div (-3-4i) \\ = \frac{1}{25} + \frac{7}{25}i$$

$$(7) (2-2i) \div (3+i) \\ = \frac{2}{5} - \frac{4}{5}i$$

$$(17) (-2+2i) \div (-4-4i) \\ = -\frac{1}{2}i$$

$$(27) (3-4i) \div (-5-3i) \\ = -\frac{3}{34} + \frac{29}{34}i$$

$$(8) (3+3i) \div (4+2i) \\ = \frac{9}{10} + \frac{3}{10}i$$

$$(18) (1+5i) \div (1-i) \\ = -2+3i$$

$$(28) (3-5i) \div (-1-3i) \\ = \frac{6}{5} + \frac{7}{5}i$$

$$(9) (2+2i) \div (-3-2i) \\ = -\frac{10}{13} - \frac{2}{13}i$$

$$(19) (-4-i) \div (1-5i) \\ = \frac{1}{26} - \frac{21}{26}i$$

$$(29) (-1+2i) \div (-1+3i) \\ = \frac{7}{10} + \frac{1}{10}i$$

$$(10) (-4+4i) \div (-3-5i) \\ = -\frac{4}{17} - \frac{16}{17}i$$

$$(20) (-2+4i) \div (4+4i) \\ = \frac{1}{4} + \frac{3}{4}i$$

$$(30) (-5-2i) \div (2+3i) \\ = -\frac{16}{13} + \frac{11}{13}i$$

$$(1) (4-i) \div (1-5i) \\ = \frac{9}{26} + \frac{19}{26}i$$

$$(11) (4+4i) \div (-5-2i) \\ = -\frac{28}{29} - \frac{12}{29}i$$

$$(21) (4+5i) \div (3+2i) \\ = \frac{22}{13} + \frac{7}{13}i$$

$$(2) (1-4i) \div (4+5i) \\ = -\frac{16}{41} - \frac{21}{41}i$$

$$(12) (-5+i) \div (3-4i) \\ = -\frac{19}{25} - \frac{17}{25}i$$

$$(22) (2+4i) \div (1-i) \\ = -1 + 3i$$

$$(3) (-4+i) \div (3+2i) \\ = -\frac{10}{13} + \frac{11}{13}i$$

$$(13) (-5-3i) \div (4+2i) \\ = -\frac{13}{10} - \frac{1}{10}i$$

$$(23) (-3+2i) \div (2-2i) \\ = -\frac{5}{4} - \frac{1}{4}i$$

$$(4) (3+4i) \div (-3+i) \\ = -\frac{1}{2} - \frac{3}{2}i$$

$$(14) (-1+5i) \div (-2+2i) \\ = \frac{3}{2} - i$$

$$(24) (1-2i) \div (-4-i) \\ = -\frac{2}{17} + \frac{9}{17}i$$

$$(5) (3-5i) \div (4+2i) \\ = \frac{1}{10} - \frac{13}{10}i$$

$$(15) (-3-2i) \div (-5+3i) \\ = \frac{9}{34} + \frac{19}{34}i$$

$$(25) (-4-i) \div (-1-4i) \\ = \frac{8}{17} - \frac{15}{17}i$$

$$(6) (2-4i) \div (-5+2i) \\ = -\frac{18}{29} + \frac{16}{29}i$$

$$(16) (-4-4i) \div (-1+2i) \\ = -\frac{4}{5} + \frac{12}{5}i$$

$$(26) (4+i) \div (1+3i) \\ = \frac{7}{10} - \frac{11}{10}i$$

$$(7) (3+4i) \div (1-2i) \\ = -1 + 2i$$

$$(17) (5+5i) \div (5-3i) \\ = \frac{5}{17} + \frac{20}{17}i$$

$$(27) (-3+3i) \div (-2+i) \\ = \frac{9}{5} - \frac{3}{5}i$$

$$(8) (-3+2i) \div (3+4i) \\ = -\frac{1}{25} + \frac{18}{25}i$$

$$(18) (5+3i) \div (2-3i) \\ = \frac{1}{13} + \frac{21}{13}i$$

$$(28) (5+3i) \div (-5+5i) \\ = -\frac{1}{5} - \frac{4}{5}i$$

$$(9) (1-4i) \div (4+4i) \\ = -\frac{3}{8} - \frac{5}{8}i$$

$$(19) (2+5i) \div (5-4i) \\ = -\frac{10}{41} + \frac{33}{41}i$$

$$(29) (5-i) \div (2-3i) \\ = 1 + i$$

$$(10) (3-4i) \div (4+3i) \\ = -i$$

$$(20) (5-3i) \div (3+4i) \\ = \frac{3}{25} - \frac{29}{25}i$$

$$(30) (4+5i) \div (3+2i) \\ = \frac{22}{13} + \frac{7}{13}i$$

$$(1) (2+4i) \div (-5-3i) \\ = -\frac{11}{17} - \frac{7}{17}i$$

$$(11) (3-3i) \div (-1+3i) \\ = -\frac{6}{5} - \frac{3}{5}i$$

$$(21) (-5+i) \div (-3+2i) \\ = \frac{17}{13} + \frac{7}{13}i$$

$$(2) (1-i) \div (2+i) \\ = \frac{1}{5} - \frac{3}{5}i$$

$$(12) (-3+2i) \div (-2-5i) \\ = -\frac{4}{29} - \frac{19}{29}i$$

$$(22) (-3+5i) \div (2+4i) \\ = \frac{7}{10} + \frac{11}{10}i$$

$$(3) (-5+2i) \div (-3+2i) \\ = \frac{19}{13} + \frac{4}{13}i$$

$$(13) (3+5i) \div (4-4i) \\ = -\frac{1}{4} + i$$

$$(23) (-3-i) \div (-4+i) \\ = \frac{11}{17} + \frac{7}{17}i$$

$$(4) (-1+3i) \div (4+3i) \\ = \frac{1}{5} + \frac{3}{5}i$$

$$(14) (-3+3i) \div (1-i) \\ = -3$$

$$(24) (-3+i) \div (-5-3i) \\ = \frac{6}{17} - \frac{7}{17}i$$

$$(5) (-3+2i) \div (4+i) \\ = -\frac{10}{17} + \frac{11}{17}i$$

$$(15) (-2-5i) \div (-1-2i) \\ = \frac{12}{5} + \frac{1}{5}i$$

$$(25) (-3-2i) \div (1+4i) \\ = -\frac{11}{17} + \frac{10}{17}i$$

$$(6) (2-i) \div (-2-4i) \\ = \frac{1}{2}i$$

$$(16) (1-3i) \div (-2+2i) \\ = -1 + \frac{1}{2}i$$

$$(26) (4+5i) \div (4+5i) \\ = 1$$

$$(7) (-4-4i) \div (1+5i) \\ = -\frac{12}{13} + \frac{8}{13}i$$

$$(17) (-2-4i) \div (4-5i) \\ = \frac{12}{41} - \frac{26}{41}i$$

$$(27) (-3-3i) \div (-2+3i) \\ = -\frac{3}{13} + \frac{15}{13}i$$

$$(8) (-1-i) \div (3-4i) \\ = \frac{1}{25} - \frac{7}{25}i$$

$$(18) (-4+5i) \div (-1-5i) \\ = -\frac{21}{26} - \frac{25}{26}i$$

$$(28) (-2+3i) \div (-1-3i) \\ = -\frac{7}{10} - \frac{9}{10}i$$

$$(9) (3+2i) \div (-2-2i) \\ = -\frac{5}{4} + \frac{1}{4}i$$

$$(19) (-4+5i) \div (2+i) \\ = -\frac{3}{5} + \frac{14}{5}i$$

$$(29) (5+i) \div (-5+3i) \\ = -\frac{11}{17} - \frac{10}{17}i$$

$$(10) (-5-5i) \div (1+5i) \\ = -\frac{15}{13} + \frac{10}{13}i$$

$$(20) (5+3i) \div (4-3i) \\ = \frac{11}{25} + \frac{27}{25}i$$

$$(30) (-2-5i) \div (-5-4i) \\ = \frac{30}{41} + \frac{17}{41}i$$

$$(1) (-4+i) \div (-2+2i) \\ = \frac{5}{4} + \frac{3}{4}i$$

$$(11) (-1-i) \div (3+i) \\ = -\frac{2}{5} - \frac{1}{5}i$$

$$(21) (1+2i) \div (-3-i) \\ = -\frac{1}{2} - \frac{1}{2}i$$

$$(2) (-1-i) \div (-4-5i) \\ = \frac{9}{41} - \frac{1}{41}i$$

$$(12) (-1+3i) \div (-3+3i) \\ = \frac{2}{3} - \frac{1}{3}i$$

$$(22) (-1+2i) \div (-5-i) \\ = \frac{3}{26} - \frac{11}{26}i$$

$$(3) (-4+i) \div (-4+i) \\ = 1$$

$$(13) (-2+2i) \div (2-4i) \\ = -\frac{3}{5} - \frac{1}{5}i$$

$$(23) (-1-i) \div (-1+i) \\ = i$$

$$(4) (3-i) \div (1-5i) \\ = \frac{4}{13} + \frac{7}{13}i$$

$$(14) (2-2i) \div (5-2i) \\ = \frac{14}{29} - \frac{6}{29}i$$

$$(24) (-1-5i) \div (5+3i) \\ = -\frac{10}{17} - \frac{11}{17}i$$

$$(5) (-1-2i) \div (1+4i) \\ = -\frac{9}{17} + \frac{2}{17}i$$

$$(15) (4+i) \div (-4-3i) \\ = -\frac{19}{25} + \frac{8}{25}i$$

$$(25) (-1-i) \div (-4-i) \\ = \frac{5}{17} + \frac{3}{17}i$$

$$(6) (4-3i) \div (-1-3i) \\ = \frac{1}{2} + \frac{3}{2}i$$

$$(16) (3+3i) \div (-2-i) \\ = -\frac{9}{5} - \frac{3}{5}i$$

$$(26) (3+5i) \div (3-2i) \\ = -\frac{1}{13} + \frac{21}{13}i$$

$$(7) (-1+3i) \div (-4-i) \\ = \frac{1}{17} - \frac{13}{17}i$$

$$(17) (4-3i) \div (-2-2i) \\ = -\frac{1}{4} + \frac{7}{4}i$$

$$(27) (-5+5i) \div (4-3i) \\ = -\frac{7}{5} + \frac{1}{5}i$$

$$(8) (-3+5i) \div (4+3i) \\ = \frac{3}{25} + \frac{29}{25}i$$

$$(18) (1-4i) \div (-1+4i) \\ = -1$$

$$(28) (-2-3i) \div (-1-2i) \\ = \frac{8}{5} - \frac{1}{5}i$$

$$(9) (-2+3i) \div (-4+4i) \\ = \frac{5}{8} - \frac{1}{8}i$$

$$(19) (-4+4i) \div (4+4i) \\ = i$$

$$(29) (-5-4i) \div (-1-i) \\ = \frac{9}{2} - \frac{1}{2}i$$

$$(10) (-1+i) \div (1+2i) \\ = \frac{1}{5} + \frac{3}{5}i$$

$$(20) (-4-4i) \div (3+i) \\ = -\frac{8}{5} - \frac{4}{5}i$$

$$(30) (1-2i) \div (-4-4i) \\ = \frac{1}{8} + \frac{3}{8}i$$



$$(1) \quad (-1 - 3i) \div (-3 + 2i) \\ = -\frac{3}{13} + \frac{11}{13}i$$

$$(11) \quad (2 - 5i) \div (-4 - 5i) \\ = \frac{17}{41} + \frac{30}{41}i$$

$$(21) \quad (-2 - 3i) \div (3 - 2i) \\ = -i$$

$$(2) \quad (3 - 2i) \div (5 + i) \\ = \frac{1}{2} - \frac{1}{2}i$$

$$(12) \quad (5 + 5i) \div (-5 + 3i) \\ = -\frac{5}{17} - \frac{20}{17}i$$

$$(22) \quad (3 + 2i) \div (2 - 4i) \\ = -\frac{1}{10} + \frac{4}{5}i$$

$$(3) \quad (2 + 5i) \div (1 - 5i) \\ = -\frac{23}{26} + \frac{15}{26}i$$

$$(13) \quad (3 - 5i) \div (5 + 2i) \\ = \frac{5}{29} - \frac{31}{29}i$$

$$(23) \quad (5 + i) \div (2 - i) \\ = \frac{9}{5} + \frac{7}{5}i$$

$$(4) \quad (-4 + 5i) \div (-5 + 2i) \\ = \frac{30}{29} - \frac{17}{29}i$$

$$(14) \quad (5 - i) \div (2 - 5i) \\ = \frac{15}{29} + \frac{23}{29}i$$

$$(24) \quad (3 - 5i) \div (-2 - 2i) \\ = \frac{1}{2} + 2i$$

$$(5) \quad (5 - i) \div (-4 - 2i) \\ = -\frac{9}{10} + \frac{7}{10}i$$

$$(15) \quad (-3 - i) \div (1 - 3i) \\ = -i$$

$$(25) \quad (3 - 5i) \div (1 + i) \\ = -1 - 4i$$

$$(6) \quad (-3 + 2i) \div (-5 - i) \\ = \frac{1}{2} - \frac{1}{2}i$$

$$(16) \quad (3 - i) \div (-5 + 2i) \\ = -\frac{17}{29} - \frac{1}{29}i$$

$$(26) \quad (-2 - 3i) \div (1 + 2i) \\ = -\frac{8}{5} + \frac{1}{5}i$$

$$(7) \quad (-5 - 2i) \div (-2 - 3i) \\ = \frac{16}{13} - \frac{11}{13}i$$

$$(17) \quad (1 - 3i) \div (4 + 2i) \\ = -\frac{1}{10} - \frac{7}{10}i$$

$$(27) \quad (-1 - 3i) \div (5 - i) \\ = -\frac{1}{13} - \frac{8}{13}i$$

$$(8) \quad (3 + i) \div (1 - i) \\ = 1 + 2i$$

$$(18) \quad (-2 + 5i) \div (-3 - 3i) \\ = -\frac{1}{2} - \frac{7}{6}i$$

$$(28) \quad (5 + i) \div (4 + 5i) \\ = \frac{25}{41} - \frac{21}{41}i$$

$$(9) \quad (-1 - 4i) \div (-4 - 5i) \\ = \frac{24}{41} + \frac{11}{41}i$$

$$(19) \quad (3 - 5i) \div (-2 - 5i) \\ = \frac{19}{29} + \frac{25}{29}i$$

$$(29) \quad (2 - i) \div (3 + 3i) \\ = \frac{1}{6} - \frac{1}{2}i$$

$$(10) \quad (5 - 4i) \div (-4 - 5i) \\ = i$$

$$(20) \quad (4 + 5i) \div (3 + 4i) \\ = \frac{32}{25} - \frac{1}{25}i$$

$$(30) \quad (-5 - 5i) \div (1 + 4i) \\ = -\frac{25}{17} + \frac{15}{17}i$$

$$(1) (2+i) \div (-3+4i) \\ = -\frac{2}{25} - \frac{11}{25}i$$

$$(11) (5-3i) \div (-1+5i) \\ = -\frac{10}{13} - \frac{11}{13}i$$

$$(21) (1-3i) \div (-1-5i) \\ = \frac{7}{13} + \frac{4}{13}i$$

$$(2) (2+i) \div (2+3i) \\ = \frac{7}{13} - \frac{4}{13}i$$

$$(12) (-3-i) \div (2+4i) \\ = -\frac{1}{2} + \frac{1}{2}i$$

$$(22) (-1-4i) \div (-2-3i) \\ = \frac{14}{13} + \frac{5}{13}i$$

$$(3) (-2+i) \div (-5+4i) \\ = \frac{14}{41} + \frac{3}{41}i$$

$$(13) (-4-i) \div (-2+3i) \\ = \frac{5}{13} + \frac{14}{13}i$$

$$(23) (-1-3i) \div (-1-3i) \\ = 1$$

$$(4) (5+3i) \div (-3+2i) \\ = -\frac{9}{13} - \frac{19}{13}i$$

$$(14) (1+4i) \div (4-5i) \\ = -\frac{16}{41} + \frac{21}{41}i$$

$$(24) (1-2i) \div (3+5i) \\ = -\frac{7}{34} - \frac{11}{34}i$$

$$(5) (-4+2i) \div (-2-2i) \\ = \frac{1}{2} - \frac{3}{2}i$$

$$(15) (2+i) \div (-2-4i) \\ = -\frac{2}{5} + \frac{3}{10}i$$

$$(25) (4+5i) \div (-4-4i) \\ = -\frac{9}{8} - \frac{1}{8}i$$

$$(6) (-1-5i) \div (2-i) \\ = \frac{3}{5} - \frac{11}{5}i$$

$$(16) (-1+4i) \div (-2+4i) \\ = \frac{9}{10} - \frac{1}{5}i$$

$$(26) (-1+5i) \div (-2+5i) \\ = \frac{27}{29} - \frac{5}{29}i$$

$$(7) (-5+2i) \div (2+4i) \\ = -\frac{1}{10} + \frac{6}{5}i$$

$$(17) (3-3i) \div (-1+i) \\ = -3$$

$$(27) (-2-4i) \div (1-5i) \\ = \frac{9}{13} - \frac{7}{13}i$$

$$(8) (-5-2i) \div (-3-2i) \\ = \frac{19}{13} - \frac{4}{13}i$$

$$(18) (-2+5i) \div (-1+2i) \\ = \frac{12}{5} - \frac{1}{5}i$$

$$(28) (2+3i) \div (-2-5i) \\ = -\frac{19}{29} + \frac{4}{29}i$$

$$(9) (4-2i) \div (2+3i) \\ = \frac{2}{13} - \frac{16}{13}i$$

$$(19) (-1-3i) \div (-2-5i) \\ = \frac{17}{29} + \frac{1}{29}i$$

$$(29) (2-3i) \div (-3+i) \\ = -\frac{9}{10} + \frac{7}{10}i$$

$$(10) (-1+5i) \div (2-5i) \\ = -\frac{27}{29} + \frac{5}{29}i$$

$$(20) (4-5i) \div (1+5i) \\ = -\frac{21}{26} - \frac{25}{26}i$$

$$(30) (1+5i) \div (-5+4i) \\ = \frac{15}{41} - \frac{29}{41}i$$

## 1.6 最大公約数

ユークリッドの互除法を利用した最大公約数を求める問題。

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## 第 2 章

# 多項式

### 2.1 加減法

2 次多項式の和や差を求める問題。

(1)  $(9x^2 + 8x - 7) - (7x^2 - 5x + 2)$

(20)  $(-7x^2 + 8x - 6) + (-7x^2 + 3x - 9)$

(2)  $(-8x^2 - 2x - 9) + (6x^2 - 3x + 2)$

(21)  $(4x^2 + x + 6) + (-7x^2 - 3x + 5)$

(3)  $(x^2 - 8x + 7) + (-9x^2 - 2x - 2)$

(22)  $(7x^2 - 4x - 3) + (-3x^2 - 4x - 9)$

(4)  $(3x^2 + 8x + 5) + (-6x^2 - 9x + 7)$

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(31)  $(6x^2 + x - 6) - (7x^2 - 8x + 4)$

(13)  $(-8x^2 + x - 9) - (6x^2 - 7x + 8)$

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(2)  $(8x^2 + 5x - 2) - (2x^2 + 4x + 1)$

(21)  $(8x^2 - 5x + 8) - (3x^2 - 6x)$

(3)  $(-5x^2 - 7x) - (-8x^2 + 9x - 3)$

(22)  $(x^2 + 7x - 9) + (-8x^2 + 8x - 5)$

(4)  $(-5x^2 + 6x + 9) + (-8x^2 + 9x - 4)$

(23)  $(8x^2 + 6) + (2x^2 + 9x - 3)$

(5)  $(6x^2 - 8x + 7) - (3x^2 + 3x - 6)$

(24)  $(-3x^2 - 3x + 8) - (8x^2 + 5x + 5)$

(6)  $(-3x^2 - x - 7) + (4x^2 - 3x - 4)$

(25)  $(-x^2 - 8) + (9x^2 - 7x)$

(7)  $(-7x^2 + 3x) + (4x^2 + x - 8)$

(26)  $(-8x^2 - 2x - 1) + (-7x^2 - 3x + 7)$

(8)  $(9x^2 - 5x - 4) - (-7x^2 - 5x + 2)$

(27)  $(6x^2 + 8x + 1) + (-8x^2 - 4x - 1)$

(9)  $(-2x^2 - 5x) - (-3x^2 + 4x + 5)$

(28)  $(7x^2 - 7x + 5) + (9x^2 + 7x - 8)$

(10)  $(-4x^2 - 2x - 3) + (-7x^2 - x - 1)$

(29)  $(-8x^2 - 9x + 6) + (-4x^2 - 7x + 4)$

(11)  $(5x^2 - x - 6) + (x^2 + 8x - 9)$

(30)  $(9x^2 + 2x - 8) - (-9x^2 + 2x + 2)$

(12)  $(-5x^2 + 7x + 7) + (3x^2 - x - 7)$

(31)  $(3x^2 + 3x + 4) - (x^2 - 7x)$

(13)  $(x^2 + x + 2) - (4x^2 + 8x - 8)$

(32)  $(7x^2 - x + 7) - (4x^2 + 7x + 1)$

(14)  $(3x^2 - 4x + 2) - (-8x^2 + 3x + 9)$

(33)  $(-4x^2 - 8x + 8) + (4x^2 + 2x - 4)$

(15)  $(-4x^2 - 5x + 3) + (3x^2 + 2x - 1)$

(34)  $(7x^2 + 7x - 3) + (-2x^2 - 2x + 3)$

(16)  $(-3x^2 + 7x + 8) - (-2x^2 + 8x - 2)$

(35)  $(-x^2 - x + 5) + (-6x^2 - x - 4)$

(17)  $(4x^2 - 2x - 5) - (-7x^2 - 9x - 8)$

(36)  $(2x^2 + 9x + 7) - (-6x^2 + 6x - 9)$

(18)  $(2x^2 - x + 1) - (7x^2 + 8x + 9)$

(37)  $(5x^2 + 9x - 2) + (8x^2 + 9x + 8)$

(19)  $(4x^2 - 8x - 1) + (-4x^2 - 8x - 8)$

(38)  $(3x^2 - 8x - 4) - (-7x^2 + 4x - 7)$

(1)  $(2x^2 - 4x - 9) - (2x^2 - 2x - 5)$

(20)  $(-3x^2 + 6x + 6) - (-8x^2 + 3x + 9)$

(2)  $(7x^2 - 9x + 1) - (-9x^2 + 3x + 1)$

(21)  $(9x^2 + 6x - 5) + (-8x^2 - 7x + 1)$

(3)  $(-6x^2 - 7x - 2) - (-9x^2 - 6x + 3)$

(22)  $(3x^2 + 3x - 9) - (-2x^2 + 8x - 8)$

(4)  $(6x^2 + 9x) + (7x^2 - 7x + 3)$

(23)  $(x^2 + 7x - 7) - (8x^2 + 6x - 1)$

(5)  $(-5x^2 + 7x + 5) - (-9x^2 - x + 7)$

(24)  $(7x^2 - x + 7) + (4x^2 - 9x - 2)$

(6)  $(9x^2 + 8x) - (2x^2 - 5x + 7)$

(25)  $(-8x^2 + 6x - 9) + (4x^2 - x - 6)$

(7)  $(2x^2 + 9) - (-8x^2 + 4x + 5)$

(26)  $(6x^2 + 6x - 4) + (4x^2 - 4x + 7)$

(8)  $(-9x^2 - 5x - 2) + (9x^2 + 2)$

(27)  $(3x^2 - 8x + 2) - (5x^2 + 9x - 3)$

(9)  $(-3x^2 - 9x + 9) + (3x^2 - 2x - 3)$

(28)  $(7x^2 + 3x + 9) - (-x^2 + 8x - 2)$

(10)  $(5x^2 - 8x - 7) - (2x^2 - 6x + 4)$

(29)  $(-7x^2 + 6x - 4) + (x^2 - 4)$

(11)  $(7x^2 - 2x - 8) + (-8x^2 + 7x + 7)$

(30)  $(3x^2 - 5x - 5) + (-7x^2 + 3x + 1)$

(12)  $(9x^2 + 5x - 3) - (3x^2 - 3x - 3)$

(31)  $(6x^2 + 5x - 5) - (x^2 - 7x + 2)$

(13)  $(-3x^2 - 5x + 4) + (-6x^2 + 4x - 8)$

(32)  $(-7x^2 - 7x - 6) + (-6x^2 + 4x - 9)$

(14)  $(2x^2 - 6x + 8) + (-2x^2 - 2x)$

(33)  $(-7x^2 + 7x + 7) - (-2x^2 + 7x + 1)$

(15)  $(-6x^2 - 2x - 5) - (5x^2 - 9x)$

(34)  $(-7x^2 - 5x + 2) - (7x^2 - 3x - 7)$

(16)  $(-8x^2 - 6x - 6) - (-x^2 + 7x - 8)$

(35)  $(4x^2 + 8x + 3) - (-9x^2 - x + 2)$

(17)  $(6x^2 - 7x + 8) + (x^2 + 4x + 7)$

(36)  $(-3x^2 + x - 2) - (7x^2 + x - 2)$

(18)  $(5x^2 - 3x - 4) - (x^2 + 2x + 6)$

(37)  $(-8x^2 + 3x - 6) - (-8x^2 + 9x - 4)$

(19)  $(3x^2 + 3x - 3) + (-7x^2 + 9x - 1)$

(38)  $(x^2 + 6x - 9) + (-4x^2 - 5x - 5)$

(1)  $(-x^2 + 6x + 3) - (5x^2 - 5x + 5)$

(20)  $(-3x^2 + 9) - (-4x^2 + 4x + 6)$

(2)  $(7x^2 - 8x + 2) + (-8x^2 + 5x + 2)$

(21)  $(-5x^2 - 7x - 7) - (x^2 + 2x - 7)$

(3)  $(-2x^2 + 8x + 9) + (3x^2 + 3x + 6)$

(22)  $(-8x^2 - 9x + 3) + (3x^2 - x + 2)$

(4)  $(-3x^2 + 4x + 1) + (-8x^2 - 8x + 1)$

(23)  $(-8x^2 + 4x) - (-3x^2 - x - 1)$

(5)  $(-8x^2 + 8x + 9) - (7x^2 + 8x + 2)$

(24)  $(-5x^2 - 8x + 8) + (4x^2 - 2x - 1)$

(6)  $(6x^2 - 5x - 7) - (-2x^2 - 3x - 2)$

(25)  $(-4x^2 - 6x) + (7x^2 + x - 2)$

(7)  $(5x^2 + 4x + 2) - (x^2 + 5x + 6)$

(26)  $(5x^2 - 5x - 9) + (-8x^2 + 5)$

(8)  $(-3x^2 - 3x - 8) + (-4x^2 + 8x - 6)$

(27)  $(6x^2 - 4x - 8) - (-x^2 - 9x + 8)$

(9)  $(-9x^2 + 9x + 9) + (8x^2 + 9x - 8)$

(28)  $(-9x^2 - 5x - 1) - (-9x^2 + 6x - 1)$

(10)  $(-8x^2 - 2x - 2) + (-3x^2 + 6x + 5)$

(29)  $(8x^2 + 9x - 8) - (7x^2 + 5)$

(11)  $(-6x^2 + 9x) - (-2x^2 + 3x - 1)$

(30)  $(-5x^2 + 9x - 8) + (-x^2 - 7x + 3)$

(12)  $(3x^2 + 8x - 8) + (-7x^2 + 3x + 6)$

(31)  $(-9x^2 - 6x + 7) + (-8x^2 + 2x - 3)$

(13)  $(3x^2 - 2x - 1) - (7x^2 - x - 9)$

(32)  $(-6x^2 - 6x + 5) + (-6x^2 + 8x + 8)$

(14)  $(-2x^2 - 6x - 3) + (-9x^2 - 9x + 2)$

(33)  $(2x^2 + 3) - (9x^2 - 2x + 8)$

(15)  $(-4x^2 + 3x - 5) + (-4x^2 - 3x - 2)$

(34)  $(-3x^2 - 3x + 7) - (-9x^2 + 4x - 4)$

(16)  $(-8x^2 + 5x - 9) - (2x^2 - 3x + 8)$

(35)  $(-4x^2 + 7x) + (2x^2 - 3x + 5)$

(17)  $(3x^2 + 7) - (3x^2 - 4x + 8)$

(36)  $(5x^2 + 4x - 9) - (5x^2 + 5x + 9)$

(18)  $(8x^2 + 3x + 7) - (8x^2 - 7x - 5)$

(37)  $(2x^2 - 6x + 8) + (9x^2 + 5x)$

(19)  $(4x^2 + 4x - 4) - (-6x^2 - x - 8)$

(38)  $(8x^2 + 4x - 8) - (8x^2 + 9x + 1)$

(1)  $(7x^2 + 8x + 7) - (-x^2 - 3x - 4)$

(20)  $(-4x^2 - 3x - 9) - (-6x^2 + 5x + 4)$

(2)  $(-9x^2 + 4x - 2) - (7x^2 + 3x - 1)$

(21)  $(7x^2 + x) - (-x^2 - 4x - 6)$

(3)  $(3x^2 - 2x) - (-2x^2 - 6x - 8)$

(22)  $(-4x^2 - 4x + 1) - (-4x^2 + 9x - 4)$

(4)  $(-9x^2 + 2x + 9) + (-7x^2 + 2x - 8)$

(23)  $(-2x^2 + 9) - (-2x^2 - 6x + 9)$

(5)  $(x^2 - 7x + 2) - (-7x^2 + 6x - 6)$

(24)  $(-3x^2 - 4x + 5) - (-8x^2 - 2x - 6)$

(6)  $(-x^2 - 5x - 8) + (-8x^2 - 2x - 3)$

(25)  $(-7x^2 + 8x - 3) - (-4x^2 + 6x + 1)$

(7)  $(7x^2 + 7x - 3) + (7x^2 + 5)$

(26)  $(-7x^2 - 2x - 9) + (3x^2)$

(8)  $(x^2 - 9x - 8) - (2x^2 - x - 3)$

(27)  $(-7x^2 + 8x + 5) + (-9x^2 - x + 8)$

(9)  $(6x^2 + 3) + (-9x^2 + 4x - 2)$

(28)  $(-5x^2 - 4x - 8) + (8x^2 + 7x)$

(10)  $(7x^2 - 7x + 3) - (5x^2 - 8x + 2)$

(29)  $(-2x^2 + 6x + 8) - (9x^2 - 7x + 3)$

(11)  $(9x^2 - 4x) + (3x^2 + 6x + 3)$

(30)  $(-7x^2 + x) + (9x^2 + 3x - 5)$

(12)  $(2x^2 - 8x - 2) - (6x^2 - 8x - 1)$

(31)  $(3x^2 - 9x - 2) - (-9x^2 - 2x + 6)$

(13)  $(x^2 - 4x + 8) - (7x^2 - 6)$

(32)  $(-4x^2 - 9x - 8) - (-5x^2 + 6x + 1)$

(14)  $(x^2 - 3x - 5) + (-2x^2 - 2x)$

(33)  $(-x^2 + 3x + 9) - (-7x^2 + 6x)$

(15)  $(-6x^2 - 8x - 1) - (7x^2 - x - 6)$

(34)  $(-8x^2 + 6x - 3) + (-6x^2 - 3x + 2)$

(16)  $(4x^2 - x + 8) + (-3x^2 + 4x - 8)$

(35)  $(9x^2 + 9x - 3) + (5x^2 + 9x)$

(17)  $(x^2 - 8x + 6) - (-5x^2 - 5x + 2)$

(36)  $(-6x^2 - 5x - 2) + (-9x^2 - 8x + 5)$

(18)  $(9x^2 + 8x + 1) + (9x^2 + x)$

(37)  $(-4x^2 + 7x - 3) - (-2x^2 - 5x - 2)$

(19)  $(2x^2 + 9x - 6) + (8x^2 + 4x + 7)$

(38)  $(-7x^2 - 4x - 5) + (-4x^2 + x + 5)$

(1)  $(2x^2 - x + 7) + (5x^2 + 2x + 8)$

(20)  $(-9x^2 - 7x + 1) + (2x^2 + x - 7)$

(2)  $(3x^2 + 9x + 1) - (-9x^2 - 3x + 1)$

(21)  $(7x^2 - 8x) - (2x^2 - 3x - 4)$

(3)  $(-4x^2 + 3x + 3) - (8x^2 - 5x - 4)$

(22)  $(9x^2 + 8x) - (-2x^2 - x + 3)$

(4)  $(6x^2 + 7x - 3) + (2x^2 + 8x + 1)$

(23)  $(-2x^2 - 2x - 1) - (9x^2 - 5x - 5)$

(5)  $(6x^2 - 5x - 6) - (9x^2 + x - 9)$

(24)  $(6x^2 + 4x + 5) - (-3x^2 + 5x + 5)$

(6)  $(-6x^2 - 7x - 5) - (2x^2 - 8x - 5)$

(25)  $(-6x^2 - 2x - 9) - (-5x^2 - 9x + 3)$

(7)  $(-8x^2 + 5x - 5) - (-9x^2 + 8x - 4)$

(26)  $(-9x^2 - 9x + 3) + (-9x^2 + 5x + 8)$

(8)  $(7x^2 - 4) - (-9x^2 - 7x + 9)$

(27)  $(-7x^2 + 9x - 2) + (9x^2 + 6x - 9)$

(9)  $(-6x^2 + 8x + 7) + (-4x^2 - 2x)$

(28)  $(-7x^2 + 5x - 3) + (7x^2 - 5x)$

(10)  $(9x^2 - 7) - (-4x^2 + 8x)$

(29)  $(-6x^2 - 6x - 3) + (3x^2 - 6x - 1)$

(11)  $(9x^2 - 3x - 9) + (-3x^2 - 3x - 6)$

(30)  $(-2x^2 - 5x + 9) - (4x^2 - 6x - 3)$

(12)  $(x^2 - 4x + 4) - (-8x^2 + 7x - 7)$

(31)  $(-9x^2 - 7x + 5) + (8x^2 + 2x - 4)$

(13)  $(-6x^2 + 8x + 1) + (-7x^2 - 4x + 2)$

(32)  $(-3x^2 - 2x + 4) + (4x^2 - 2)$

(14)  $(8x^2 - 4x - 5) - (-3x^2 + 3x - 9)$

(33)  $(-8x^2 + 2) - (3x^2 - 5x - 5)$

(15)  $(-5x^2 + 3x) + (3x^2 - 5x + 3)$

(34)  $(4x^2 - 7x + 2) + (x^2 - 3x - 5)$

(16)  $(-6x^2 + 2x + 3) - (-4x^2 - 4x + 8)$

(35)  $(-5x^2 + 3x - 1) - (5x^2 - 2x - 8)$

(17)  $(-4x^2 - 3x - 5) - (-3x^2 + 9x - 5)$

(36)  $(-x^2 + 8x) + (-8x^2 - 2x - 7)$

(18)  $(7x^2 - 6x - 6) + (-4x^2 - 2x + 7)$

(37)  $(5x^2 + 7x - 5) - (-7x^2 - 4x + 8)$

(19)  $(8x^2 + x + 9) - (x^2 - 8x + 9)$

(38)  $(-x^2 - 7x - 8) - (-9x^2 + 3x - 8)$



- (1)  $(9x^2 + 8x - 7) - (7x^2 - 5x + 2)$   
 $= 2x^2 + 13x - 9$
- (2)  $(-8x^2 - 2x - 9) + (6x^2 - 3x + 2)$   
 $= -2x^2 - 5x - 7$
- (3)  $(x^2 - 8x + 7) + (-9x^2 - 2x - 2)$   
 $= -8x^2 - 10x + 5$
- (4)  $(3x^2 + 8x + 5) + (-6x^2 - 9x + 7)$   
 $= -3x^2 - x + 12$
- (5)  $(8x^2 + 5x - 7) + (x^2 + 2)$   
 $= 9x^2 + 5x - 5$
- (6)  $(7x^2 - 8x + 5) + (-6x^2 + 6x + 9)$   
 $= x^2 - 2x + 14$
- (7)  $(-6x^2 - 9x + 4) + (9x^2 + 3x - 4)$   
 $= 3x^2 - 6x$
- (8)  $(-6x^2 - 5x + 1) - (3x^2 + 5x - 6)$   
 $= -9x^2 - 10x + 7$
- (9)  $(2x^2 - 8x + 5) + (6x^2 - 8x + 1)$   
 $= 8x^2 - 16x + 6$
- (10)  $(-7x^2 - 5x + 7) + (5x^2 + 5x + 8)$   
 $= -2x^2 + 15$
- (11)  $(-9x^2 + 3x + 1) - (-9x^2 + x + 4)$   
 $= 2x - 3$
- (12)  $(-7x^2 + 5) - (-8x^2 + 9x - 2)$   
 $= x^2 - 9x + 7$
- (13)  $(-x^2 - 8x - 3) - (4x^2 + 2x + 4)$   
 $= -5x^2 - 10x - 7$
- (14)  $(7x^2 - 5x + 7) - (-5x^2 - 9x + 8)$   
 $= 12x^2 + 4x - 1$
- (15)  $(-9x^2 - 7x) + (6x^2 + 8x - 8)$   
 $= -3x^2 + x - 8$
- (16)  $(-5x^2 + 9x - 8) + (4x^2 - 5x - 3)$   
 $= -x^2 + 4x - 11$
- (17)  $(-5x^2 - x - 9) - (-8x^2 - x - 4)$   
 $= 3x^2 - 5$
- (18)  $(2x^2 + 5x - 6) - (-6x^2 + 8x + 6)$   
 $= 8x^2 - 3x - 12$
- (19)  $(7x^2 - 8x - 7) + (-x^2 + 2x - 8)$   
 $= 6x^2 - 6x - 15$
- (20)  $(-7x^2 + 8x - 6) + (-7x^2 + 3x - 9)$   
 $= -14x^2 + 11x - 15$
- (21)  $(4x^2 + x + 6) + (-7x^2 - 3x + 5)$   
 $= -3x^2 - 2x + 11$
- (22)  $(7x^2 - 4x - 3) + (-3x^2 - 4x - 9)$   
 $= 4x^2 - 8x - 12$
- (23)  $(6x^2 + 3x + 4) - (-2x^2 - x - 4)$   
 $= 8x^2 + 4x + 8$
- (24)  $(9x^2 + 2x - 1) + (-6x^2 - x - 6)$   
 $= 3x^2 + x - 7$
- (25)  $(7x^2 - 4x + 9) - (x^2 + 4x + 2)$   
 $= 6x^2 - 8x + 7$
- (26)  $(3x^2 - 8x + 8) - (4x^2 + 4x + 9)$   
 $= -x^2 - 12x - 1$
- (27)  $(4x^2 + 3x + 6) + (2x^2 - 7x - 2)$   
 $= 6x^2 - 4x + 4$
- (28)  $(-3x^2 + x + 6) - (7x^2 + 2x - 9)$   
 $= -10x^2 - x + 15$
- (29)  $(3x^2 - x + 1) - (-4x^2 - 3x + 1)$   
 $= 7x^2 + 2x$
- (30)  $(8x^2 - 2x + 2) - (4x^2 + 2x + 2)$   
 $= 4x^2 - 4x$
- (31)  $(9x^2 + 4) - (4x^2 + 5x + 3)$   
 $= 5x^2 - 5x + 1$
- (32)  $(-6x^2 - 8x + 4) + (8x^2 + 2x + 9)$   
 $= 2x^2 - 6x + 13$
- (33)  $(x^2 - 2) + (2x^2 + 5x + 8)$   
 $= 3x^2 + 5x + 6$
- (34)  $(-4x^2 - 8x - 2) - (-3x^2 + x - 4)$   
 $= -x^2 - 9x + 2$
- (35)  $(-8x^2 + 9x + 8) + (8x^2 + 4x + 8)$   
 $= 13x + 16$
- (36)  $(-4x^2 + 9x + 7) + (-3x^2 + x - 5)$   
 $= -7x^2 + 10x + 2$
- (37)  $(-9x^2 + 4x - 7) + (-2x^2 + 8x + 1)$   
 $= -11x^2 + 12x - 6$
- (38)  $(2x^2 + x + 6) + (4x^2 - 7x + 9)$   
 $= 6x^2 - 6x + 15$

$$(1) \quad (-5x^2 - 7x + 2) + (-7x^2 + 2x - 5) \\ = -12x^2 - 5x - 3$$

$$(2) \quad (-3x^2 + 7x + 5) + (8x^2 - 2x - 3) \\ = 5x^2 + 5x + 2$$

$$(3) \quad (-6x^2 + 8x + 3) - (4x^2 - 3x + 8) \\ = -10x^2 + 11x - 5$$

$$(4) \quad (-6x^2 - 5x - 2) + (8x^2 - 3x + 6) \\ = 2x^2 - 8x + 4$$

$$(5) \quad (5x^2 + 6x + 1) - (-x^2 - 9x + 1) \\ = 6x^2 + 15x$$

$$(6) \quad (7x^2 + x + 1) - (x^2 - 2x - 6) \\ = 6x^2 + 3x + 7$$

$$(7) \quad (-x^2 + 7x + 9) + (-9x^2 + 8x + 8) \\ = -10x^2 + 15x + 17$$

$$(8) \quad (x^2 - x - 6) - (-x^2 + 4x - 9) \\ = 2x^2 - 5x + 3$$

$$(9) \quad (3x^2 + 7x - 3) + (-x^2 - 8x + 1) \\ = 2x^2 - x - 2$$

$$(10) \quad (-5x^2 - 6x - 8) + (6x^2 - 4x - 6) \\ = x^2 - 10x - 14$$

$$(11) \quad (4x^2 + 5x + 7) - (-9x^2 - 7x + 9) \\ = 13x^2 + 12x - 2$$

$$(12) \quad (7x^2 + 2x - 9) + (-6x^2 + 4x - 3) \\ = x^2 + 6x - 12$$

$$(13) \quad (x^2 - 4x - 4) + (-4x^2 - 9x + 1) \\ = -3x^2 - 13x - 3$$

$$(14) \quad (-2x^2 + 2x - 1) + (2x^2 + 4x + 6) \\ = 6x + 5$$

$$(15) \quad (-9x^2 - x + 8) - (-7x^2 - 9x - 5) \\ = -2x^2 + 8x + 13$$

$$(16) \quad (-6x^2 - 3x - 5) - (-3x^2 + 5x + 5) \\ = -3x^2 - 8x - 10$$

$$(17) \quad (-6x^2 + 6x + 8) - (8x^2 - x + 2) \\ = -14x^2 + 7x + 6$$

$$(18) \quad (-4x^2 + 8) + (-8x^2 + 6x - 6) \\ = -12x^2 + 6x + 2$$

$$(19) \quad (3x^2 + 3x - 4) + (-8x^2 + 7x + 6) \\ = -5x^2 + 10x + 2$$

$$(20) \quad (5x^2 - x + 6) - (-7x^2 + 2x - 1) \\ = 12x^2 - 3x + 7$$

$$(21) \quad (6x^2 + 7x + 7) - (-5x^2 - 2x - 8) \\ = 11x^2 + 9x + 15$$

$$(22) \quad (-x^2 + x + 7) + (4x^2 - 5x + 1) \\ = 3x^2 - 4x + 8$$

$$(23) \quad (7x^2 + 3x + 5) + (-2x^2 - 7x - 4) \\ = 5x^2 - 4x + 1$$

$$(24) \quad (-6x^2 - 4x + 9) + (-7x^2 + 7x - 7) \\ = -13x^2 + 3x + 2$$

$$(25) \quad (x^2 + 3x + 1) + (4x^2 - 4x - 4) \\ = 5x^2 - x - 3$$

$$(26) \quad (7x^2 - 8x + 4) + (-8x^2 - 8x - 9) \\ = -x^2 - 16x - 5$$

$$(27) \quad (-4x^2 - 4x - 6) - (3x^2 + 9x + 2) \\ = -7x^2 - 13x - 8$$

$$(28) \quad (-5x^2 - 3x - 9) - (-9x^2 + 9x + 2) \\ = 4x^2 - 12x - 11$$

$$(29) \quad (6x^2 + 9x + 9) + (-3x^2 + 7x + 5) \\ = 3x^2 + 16x + 14$$

$$(30) \quad (8x^2 + 2x + 2) - (-8x^2 + 6x - 6) \\ = 16x^2 - 4x + 8$$

$$(31) \quad (-x^2 + 3x + 9) - (x^2 - 6x - 4) \\ = -2x^2 + 9x + 13$$

$$(32) \quad (-9x^2 - 7x + 5) + (4x^2 + 6x - 1) \\ = -5x^2 - x + 4$$

$$(33) \quad (7x^2 + 9x - 6) + (-6x^2 + 4x + 5) \\ = x^2 + 13x - 1$$

$$(34) \quad (-6x^2 + x + 5) - (-9x^2 - 4x - 6) \\ = 3x^2 + 5x + 11$$

$$(35) \quad (9x^2 - 6x) + (-2x^2 - 9x - 7) \\ = 7x^2 - 15x - 7$$

$$(36) \quad (7x^2 + 3x - 3) + (-3x^2 - 7x + 5) \\ = 4x^2 - 4x + 2$$

$$(37) \quad (-8x^2 - 9) + (x^2 - 8x - 6) \\ = -7x^2 - 8x - 15$$

$$(38) \quad (2x^2 - 6x + 7) + (x^2 - 7x - 3) \\ = 3x^2 - 13x + 4$$

$$(1) (4x^2 + 7x - 1) - (-7x^2 + 4x + 4) \\ = 11x^2 + 3x - 5$$

$$(2) (9x^2 + 6x - 1) - (-x^2 - 2x + 7) \\ = 10x^2 + 8x - 8$$

$$(3) (5x^2 - 7x - 6) - (3x^2 + x + 3) \\ = 2x^2 - 8x - 9$$

$$(4) (-7x^2 + 7x - 5) - (-9x^2 - 8x - 3) \\ = 2x^2 + 15x - 2$$

$$(5) (4x^2 + 2x) + (3x^2 - 9x + 2) \\ = 7x^2 - 7x + 2$$

$$(6) (9x^2 + 8x + 4) - (-2x^2 + 8x - 2) \\ = 11x^2 + 6$$

$$(7) (2x^2 + x + 7) + (9x^2 + 2x + 7) \\ = 11x^2 + 3x + 14$$

$$(8) (-3x^2 + 9x + 7) + (2x^2 - 6x - 4) \\ = -x^2 + 3x + 3$$

$$(9) (-6x^2 - x - 1) + (-8x^2 - 9x - 3) \\ = -14x^2 - 10x - 4$$

$$(10) (x^2 + 5x - 3) + (-2x^2 + 2x) \\ = -x^2 + 7x - 3$$

$$(11) (3x^2 - 6x - 5) - (4x^2 - 8x + 2) \\ = -x^2 + 2x - 7$$

$$(12) (-8x^2 - 8) - (-2x^2 - 4x - 2) \\ = -6x^2 + 4x - 6$$

$$(13) (3x^2 - 4x - 2) + (-x^2 - 9x - 8) \\ = 2x^2 - 13x - 10$$

$$(14) (-6x^2 + 5x - 2) - (-7x^2 - 2x - 6) \\ = x^2 + 7x + 4$$

$$(15) (-9x^2 - 6x + 5) + (-4x^2 - 6x - 9) \\ = -13x^2 - 12x - 4$$

$$(16) (-4x^2 - x - 7) + (7x^2 + 5x + 7) \\ = 3x^2 + 4x$$

$$(17) (-8x^2 + 3x + 8) + (-2x^2 - 5x + 2) \\ = -10x^2 - 2x + 10$$

$$(18) (4x^2 - 9) - (-x^2 - 9x - 6) \\ = 5x^2 + 9x - 3$$

$$(19) (-x^2 + 8x + 5) - (8x^2 + 7x - 9) \\ = -9x^2 + x + 14$$

$$(20) (-2x^2 - 3x - 4) - (-4x^2 + 7x - 3) \\ = 2x^2 - 10x - 1$$

$$(21) (3x^2 + 9x + 3) - (6x^2 - 3x - 1) \\ = -3x^2 + 12x + 4$$

$$(22) (-7x^2 - 9x + 5) + (-3x^2 + 7x + 7) \\ = -10x^2 - 2x + 12$$

$$(23) (8x^2 - 5x + 3) + (x^2 - 4x - 3) \\ = 9x^2 - 9x$$

$$(24) (-3x^2 - 3x - 1) - (5x^2 - 5x + 9) \\ = -8x^2 + 2x - 10$$

$$(25) (-8x^2 + 7x - 3) - (-8x^2 + 7x - 7) \\ = 4$$

$$(26) (-3x^2 + 3x + 8) + (3x^2 + 5x - 7) \\ = 8x + 1$$

$$(27) (3x^2 - 3x + 2) + (-6x^2 + x - 8) \\ = -3x^2 - 2x - 6$$

$$(28) (x^2 - 3x - 9) - (9x^2 - 4x + 5) \\ = -8x^2 + x - 14$$

$$(29) (-9x^2 - 9x - 3) - (3x^2 + 5x + 5) \\ = -12x^2 - 14x - 8$$

$$(30) (-4x^2 + x + 8) - (9x^2 + 9x + 9) \\ = -13x^2 - 8x - 1$$

$$(31) (-5x^2 + 6x + 9) + (-6x^2 - 7x - 3) \\ = -11x^2 - x + 6$$

$$(32) (8x^2 + 6x - 9) - (3x^2 - 4x - 9) \\ = 5x^2 + 10x$$

$$(33) (-7x^2 + 2x) - (-8x^2 - 2x + 1) \\ = x^2 + 4x - 1$$

$$(34) (-3x^2 - 2x - 7) - (8x^2 - 6x + 8) \\ = -11x^2 + 4x - 15$$

$$(35) (-x^2 - 9x + 3) + (-2x^2 + 3x - 6) \\ = -3x^2 - 6x - 3$$

$$(36) (-4x^2 - 4x - 7) + (-2x^2 + 7x + 5) \\ = -6x^2 + 3x - 2$$

$$(37) (9x^2 - 5x + 2) + (5x^2 - 5x - 7) \\ = 14x^2 - 10x - 5$$

$$(38) (-4x^2 + 4x + 2) + (2x^2 + 2x + 4) \\ = -2x^2 + 6x + 6$$

- (1)  $(3x^2 - 9x) - (x^2 + 4x - 9)$   
 $= 2x^2 - 13x + 9$
- (2)  $(9x^2 + 9x + 1) + (5x^2 + 8x + 6)$   
 $= 14x^2 + 17x + 7$
- (3)  $(6x^2 + 6x - 6) - (-7x^2 + 8x - 2)$   
 $= 13x^2 - 2x - 4$
- (4)  $(-5x^2 - 2x + 3) + (8x^2 + x - 2)$   
 $= 3x^2 - x + 1$
- (5)  $(4x^2 - 4x + 8) - (-4x^2 + 7x - 6)$   
 $= 8x^2 - 11x + 14$
- (6)  $(-2x^2 - 5x + 8) + (8x^2 - 7x)$   
 $= 6x^2 - 12x + 8$
- (7)  $(4x^2 + 7x - 9) + (x^2 - 5x - 2)$   
 $= 5x^2 + 2x - 11$
- (8)  $(7x^2 - 2x - 1) + (-4x^2 + 4x + 5)$   
 $= 3x^2 + 2x + 4$
- (9)  $(9x^2 + 8x - 3) + (-3x^2 + 8x - 5)$   
 $= 6x^2 + 16x - 8$
- (10)  $(5x^2 + 9x + 5) - (8x^2 + 9x + 4)$   
 $= -3x^2 + 1$
- (11)  $(5x^2 + 5x + 7) - (-4x^2 + 4x - 3)$   
 $= 9x^2 + x + 10$
- (12)  $(9x^2 + 4x - 9) + (-5x^2 - x - 3)$   
 $= 4x^2 + 3x - 12$
- (13)  $(2x^2 - x - 4) - (-5x^2 + x - 1)$   
 $= 7x^2 - 2x - 3$
- (14)  $(7x^2 - x + 4) - (-4x^2 - 8x + 3)$   
 $= 11x^2 + 7x + 1$
- (15)  $(-x^2 - 5x - 6) + (9x^2 + 8x)$   
 $= 8x^2 + 3x - 6$
- (16)  $(2x^2 - x - 7) + (2x^2 - 4x + 4)$   
 $= 4x^2 - 5x - 3$
- (17)  $(4x^2 - 7x + 5) - (-5x^2 - x + 1)$   
 $= 9x^2 - 6x + 4$
- (18)  $(5x^2 - 7x) + (-8x^2 - 6x - 9)$   
 $= -3x^2 - 13x - 9$
- (19)  $(5x^2 + 3x - 4) - (x^2 + 2x + 4)$   
 $= 4x^2 + x - 8$
- (20)  $(-4x^2 - 5x - 1) + (5x^2 + 9x + 4)$   
 $= x^2 + 4x + 3$
- (21)  $(-x^2 - 3x + 1) + (-4x^2 + 7x + 5)$   
 $= -5x^2 + 4x + 6$
- (22)  $(x^2 + 5x + 4) - (-x^2 - 3x + 8)$   
 $= 2x^2 + 8x - 4$
- (23)  $(4x^2 - 8x - 9) - (6x^2 - 6x + 9)$   
 $= -2x^2 - 2x - 18$
- (24)  $(-3x^2 + 8x - 8) - (-3x^2 - 3x + 3)$   
 $= 11x - 11$
- (25)  $(7x^2 + 2x + 6) - (9x^2 - 6x - 1)$   
 $= -2x^2 + 8x + 7$
- (26)  $(8x^2 + 9) - (-5x^2 + 5x - 7)$   
 $= 13x^2 - 5x + 16$
- (27)  $(-6x^2 - 2x + 3) - (-3x^2 + x)$   
 $= -3x^2 - 3x + 3$
- (28)  $(-9x^2 + x - 6) + (3x^2 - 3x + 4)$   
 $= -6x^2 - 2x - 2$
- (29)  $(8x^2 - 9x - 3) + (-6x^2 + 7x + 1)$   
 $= 2x^2 - 2x - 2$
- (30)  $(-3x^2 - 7x + 8) + (-8x^2 + 6x - 8)$   
 $= -11x^2 - x$
- (31)  $(6x^2 + x + 1) + (-7x^2 - x - 5)$   
 $= -x^2 - 4$
- (32)  $(-6x^2 + 3x + 4) - (5x^2 + 7x + 2)$   
 $= -11x^2 - 4x + 2$
- (33)  $(-3x^2 - 6x) + (-8x^2 - 7x - 1)$   
 $= -11x^2 - 13x - 1$
- (34)  $(4x^2 + 3x) - (6x^2 + 5x - 8)$   
 $= -2x^2 - 2x + 8$
- (35)  $(-2x^2 - 4x - 9) - (-8x^2 + 8x - 1)$   
 $= 6x^2 - 12x - 8$
- (36)  $(-7x^2 + x - 1) - (5x^2 - 7x + 9)$   
 $= -12x^2 + 8x - 10$
- (37)  $(6x^2 + 2x - 1) + (-5x^2 + 6x)$   
 $= x^2 + 8x - 1$
- (38)  $(-3x^2 + 4x - 6) + (7x^2 - 6x + 7)$   
 $= 4x^2 - 2x + 1$

$$(1) (-6x^2 + 3x + 7) - (9x^2 + 8x - 9) \\ = -15x^2 - 5x + 16$$

$$(2) (-5x^2 + 5x - 1) - (5x^2 + 6x) \\ = -10x^2 - x - 1$$

$$(3) (-9x^2 - 6x + 4) - (2x^2 - 7) \\ = -11x^2 - 6x + 11$$

$$(4) (4x^2 + 9x - 6) + (-8x^2 - 4x + 2) \\ = -4x^2 + 5x - 4$$

$$(5) (6x^2 + 6x - 2) - (4x^2 - 9x + 4) \\ = 2x^2 + 15x - 6$$

$$(6) (x^2 + 5x + 8) + (6x^2 - 9x - 6) \\ = 7x^2 - 4x + 2$$

$$(7) (x^2 + 8x - 6) - (-6x^2 - 6x + 1) \\ = 7x^2 + 14x - 7$$

$$(8) (2x^2 - 9x - 9) - (-5x^2 + 6x - 4) \\ = 7x^2 - 15x - 5$$

$$(9) (-2x^2 + 5x + 3) - (8x^2 + 7x - 2) \\ = -10x^2 - 2x + 5$$

$$(10) (-7x^2 + x - 1) - (3x^2 + 4x - 1) \\ = -10x^2 - 3x$$

$$(11) (-9x^2 - 3x - 7) + (-8x^2 - 2x - 8) \\ = -17x^2 - 5x - 15$$

$$(12) (-3x^2 + 6x + 6) + (-x^2 - 5x - 6) \\ = -4x^2 + x$$

$$(13) (-3x^2 - 4x + 2) - (-3x^2 + 5x - 7) \\ = -9x + 9$$

$$(14) (2x^2 - 8x + 7) - (3x^2 + 7x) \\ = -x^2 - 15x + 7$$

$$(15) (8x^2 - 5x - 6) + (6x^2 + 7x + 7) \\ = 14x^2 + 2x + 1$$

$$(16) (6x^2 + 7x + 3) + (9x^2 + 3x - 4) \\ = 15x^2 + 10x - 1$$

$$(17) (8x^2 + 5x + 5) + (9x^2 + 7x + 7) \\ = 17x^2 + 12x + 12$$

$$(18) (-7x^2 - 9x + 3) + (-3x^2 + 9x + 8) \\ = -10x^2 + 11$$

$$(19) (-4x^2 - 5x + 1) + (5x^2 - 2x + 2) \\ = x^2 - 7x + 3$$

$$(20) (-2x^2 - 4x - 8) - (6x^2 - 6x - 4) \\ = -8x^2 + 2x - 4$$

$$(21) (7x^2 - 7x + 3) - (-3x^2 - x + 9) \\ = 10x^2 - 6x - 6$$

$$(22) (x^2 + 6x - 9) - (-x^2 - 2x - 3) \\ = 2x^2 + 8x - 6$$

$$(23) (9x^2 + 3x - 2) + (-2x^2 - 4x - 4) \\ = 7x^2 - x - 6$$

$$(24) (-8x^2 - 6x + 9) + (5x^2 + x - 5) \\ = -3x^2 - 5x + 4$$

$$(25) (-x^2 - 6x + 6) - (6x^2 - 9x + 6) \\ = -7x^2 + 3x$$

$$(26) (5x^2 - 2) - (4x^2 + 2x - 7) \\ = x^2 - 2x + 5$$

$$(27) (-8x^2 - 3) - (-3x^2 + 3x - 6) \\ = -5x^2 - 3x + 3$$

$$(28) (-x^2 + 8x) + (-5x^2 + x - 5) \\ = -6x^2 + 9x - 5$$

$$(29) (6x^2 - 9x + 8) + (-3x^2 - 9x - 6) \\ = 3x^2 - 18x + 2$$

$$(30) (6x^2 + 3x + 6) - (-4x^2 - 4x + 1) \\ = 10x^2 + 7x + 5$$

$$(31) (7x^2 + 7) + (5x^2 + 4x + 2) \\ = 12x^2 + 4x + 9$$

$$(32) (-3x^2 + 8x + 3) - (7x^2 - 9x - 4) \\ = -10x^2 + 17x + 7$$

$$(33) (-3x^2 - 8) - (-9x^2 - 6x - 8) \\ = 6x^2 + 6x$$

$$(34) (7x^2 + 4x) + (3x^2 + 4x - 8) \\ = 10x^2 + 8x - 8$$

$$(35) (4x^2 - 4x + 9) - (-2x^2 - 2x + 4) \\ = 6x^2 - 2x + 5$$

$$(36) (-x^2 + 5x + 5) - (-7x^2 + x + 6) \\ = 6x^2 + 4x - 1$$

$$(37) (-9x^2 - 9x) - (-7x^2 + 9x - 3) \\ = -2x^2 - 18x + 3$$

$$(38) (3x^2 - 9x - 5) - (-3x^2 - 9x - 5) \\ = 6x^2$$

- (1)  $(-4x^2 - 9x - 9) - (-4x^2 - x - 8)$   
 $= -8x - 1$
- (2)  $(-4x^2 - 6x + 3) - (9x^2 - 8x + 3)$   
 $= -13x^2 + 2x$
- (3)  $(-3x^2 + 4x + 3) + (3x^2 - 9x - 4)$   
 $= -5x - 1$
- (4)  $(5x^2 + 9x + 6) - (8x^2 + 4x - 2)$   
 $= -3x^2 + 5x + 8$
- (5)  $(4x^2 - x - 6) + (8x^2 - 5x + 4)$   
 $= 12x^2 - 6x - 2$
- (6)  $(-6x^2 - 2x + 8) - (9x^2 + 6x + 3)$   
 $= -15x^2 - 8x + 5$
- (7)  $(-6x^2 - 4x - 1) - (9x^2 + 4x + 6)$   
 $= -15x^2 - 8x - 7$
- (8)  $(9x^2 - 5x + 9) - (2x^2 - 4x + 2)$   
 $= 7x^2 - x + 7$
- (9)  $(-x^2 + 6x - 4) - (-4x^2 - 4x + 1)$   
 $= 3x^2 + 10x - 5$
- (10)  $(-8x^2 - 5x + 2) + (-6x^2 + 6x - 7)$   
 $= -14x^2 + x - 5$
- (11)  $(-5x^2 + 8x + 4) + (-9x^2 + 8x - 7)$   
 $= -14x^2 + 16x - 3$
- (12)  $(9x^2 + x - 3) + (4x^2 - 2x - 4)$   
 $= 13x^2 - x - 7$
- (13)  $(-9x^2 + 8x) + (-6x^2 - 7x - 8)$   
 $= -15x^2 + x - 8$
- (14)  $(7x^2) + (9x^2 - 6)$   
 $= 16x^2 - 6$
- (15)  $(x^2 + 5) + (-7x^2 + 7x + 2)$   
 $= -6x^2 + 7x + 7$
- (16)  $(9x^2 + 8x + 2) - (8x^2 - 3x)$   
 $= x^2 + 11x + 2$
- (17)  $(-5x^2 - 5x - 6) + (-7x^2 - 4x + 7)$   
 $= -12x^2 - 9x + 1$
- (18)  $(-x^2 + 5x - 3) - (-6x^2 + 8x + 2)$   
 $= 5x^2 - 3x - 5$
- (19)  $(-x^2 - 4x - 3) + (2x^2 - 4x - 8)$   
 $= x^2 - 8x - 11$
- (20)  $(-7x^2 - 2x + 3) + (-2x^2 + 6x - 8)$   
 $= -9x^2 + 4x - 5$
- (21)  $(-2x^2 - 9x + 7) - (-9x^2 - 2x + 9)$   
 $= 7x^2 - 7x - 2$
- (22)  $(3x^2 + 9x + 4) - (-5x^2 + 7x - 5)$   
 $= 8x^2 + 2x + 9$
- (23)  $(-4x^2 + 7x + 9) + (-5x^2 + 8x - 3)$   
 $= -9x^2 + 15x + 6$
- (24)  $(9x^2 + x) - (2x^2 + 4x + 9)$   
 $= 7x^2 - 3x - 9$
- (25)  $(-3x^2 - x + 4) - (3x^2 + 3x - 2)$   
 $= -6x^2 - 4x + 6$
- (26)  $(-5x^2 - 3x - 5) + (7x^2 + 4x - 1)$   
 $= 2x^2 + x - 6$
- (27)  $(4x^2 - 5x + 1) + (6x^2 + 4x + 7)$   
 $= 10x^2 - x + 8$
- (28)  $(-4x^2 + 6x + 5) - (-5x^2 - x + 2)$   
 $= x^2 + 7x + 3$
- (29)  $(3x^2 - 4x + 9) - (x^2 + 2x + 5)$   
 $= 2x^2 - 6x + 4$
- (30)  $(x^2 + 7x - 2) + (-x^2 - x - 5)$   
 $= 6x - 7$
- (31)  $(x^2 + 7x - 4) + (2x^2 - 7x - 7)$   
 $= 3x^2 - 11$
- (32)  $(8x^2 - 7x + 6) - (9x^2 + 4x - 4)$   
 $= -x^2 - 11x + 10$
- (33)  $(2x^2 + 7x + 7) - (-x^2 + 9x - 1)$   
 $= 3x^2 - 2x + 8$
- (34)  $(-9x^2 - 3x + 8) + (7x^2 + 8x + 3)$   
 $= -2x^2 + 5x + 11$
- (35)  $(-8x^2 + 2x - 4) + (6x^2 + 8x - 7)$   
 $= -2x^2 + 10x - 11$
- (36)  $(5x^2 + 9x + 4) - (-3x^2 + 8x)$   
 $= 8x^2 + x + 4$
- (37)  $(3x^2 - 9x + 2) + (-6x^2 - 3x + 2)$   
 $= -3x^2 - 12x + 4$
- (38)  $(2x^2 - 3) + (x^2 + x - 1)$   
 $= 3x^2 + x - 4$

- |                                                                   |                                                                  |
|-------------------------------------------------------------------|------------------------------------------------------------------|
| (1) $(3x^2 - 4x - 2) + (2x^2 + 6x + 6)$<br>$= 5x^2 + 2x + 4$      | (20) $(6x^2 + 6x - 9) + (-2x^2 + 4x - 5)$<br>$= 4x^2 + 10x - 14$ |
| (2) $(6x^2 + 2x + 9) - (2x^2 - 2x + 9)$<br>$= 4x^2 + 4x$          | (21) $(7x^2 + 7x - 5) + (9x^2)$<br>$= 16x^2 + 7x - 5$            |
| (3) $(x^2 - 3x + 6) - (x^2 - 7x + 8)$<br>$= 4x - 2$               | (22) $(-3x^2) - (-3x^2 + 5x + 1)$<br>$= -5x - 1$                 |
| (4) $(-9x^2 - 7x - 2) - (3x^2 + 4)$<br>$= -12x^2 - 7x - 6$        | (23) $(-4x^2 + 3x - 8) + (9x^2 - 9x + 4)$<br>$= 5x^2 - 6x - 4$   |
| (5) $(-5x^2 - 4x - 4) - (6x^2 + 4x + 7)$<br>$= -11x^2 - 8x - 11$  | (24) $(5x^2 + 5x - 1) + (5x^2 - 3x + 5)$<br>$= 10x^2 + 2x + 4$   |
| (6) $(6x^2 + 4x) + (-8x^2 - 2x - 4)$<br>$= -2x^2 + 2x - 4$        | (25) $(-9x^2 + 2x - 2) - (6x^2 + 7x - 5)$<br>$= -15x^2 - 5x + 3$ |
| (7) $(9x^2 + 4x + 2) - (3x^2 - 5x - 1)$<br>$= 6x^2 + 9x + 3$      | (26) $(-3x^2 - 7x - 3) - (x^2 - 3x - 8)$<br>$= -4x^2 - 4x + 5$   |
| (8) $(-3x^2 + 5x + 7) + (-9x^2 - 3x - 3)$<br>$= -12x^2 + 2x + 4$  | (27) $(6x^2 + 5x - 9) - (-3x^2 - x + 3)$<br>$= 9x^2 + 6x - 12$   |
| (9) $(-7x^2 + 2x - 2) + (x^2 - 6x + 2)$<br>$= -6x^2 - 4x$         | (28) $(2x^2 - 7x + 2) - (4x^2 - 4x - 8)$<br>$= -2x^2 - 3x + 10$  |
| (10) $(5x^2 + 6x - 7) - (-6x^2 + 4x + 1)$<br>$= 11x^2 + 2x - 8$   | (29) $(4x^2 + x - 3) + (-7x^2 + 9x + 4)$<br>$= -3x^2 + 10x + 1$  |
| (11) $(-4x^2 - 9x - 9) - (-5x^2 - 7x - 4)$<br>$= x^2 - 2x - 5$    | (30) $(-4x^2 + 4x) - (-6x^2 - 9x + 2)$<br>$= 2x^2 + 13x - 2$     |
| (12) $(-6x^2 - 9x + 9) - (-x^2 + 6x - 5)$<br>$= -5x^2 - 15x + 14$ | (31) $(-4x^2 + 3x + 4) + (3x^2 - 7x - 2)$<br>$= -x^2 - 4x + 2$   |
| (13) $(-3x^2 + 9x - 4) + (-7x^2 + x + 8)$<br>$= -10x^2 + 10x + 4$ | (32) $(8x^2 - 5x + 5) - (8x^2 + 5)$<br>$= -5x$                   |
| (14) $(-x^2 - 7x + 2) + (5x^2 - 5x - 8)$<br>$= 4x^2 - 12x - 6$    | (33) $(3x^2 - 7x - 5) + (-6x^2 - 4x + 7)$<br>$= -3x^2 - 11x + 2$ |
| (15) $(-6x^2 - 5x - 5) + (3x^2 - 2x)$<br>$= -3x^2 - 7x - 5$       | (34) $(5x^2 - 2x + 3) - (2x^2 - x - 7)$<br>$= 3x^2 - x + 10$     |
| (16) $(-x^2 - 7x + 2) - (-8x^2 - 3x + 9)$<br>$= 7x^2 - 4x - 7$    | (35) $(-6x^2 + 3x + 1) + (-8x^2 - 3x - 9)$<br>$= -14x^2 - 8$     |
| (17) $(-5x^2 + 8x + 8) + (6x^2 + x - 4)$<br>$= x^2 + 9x + 4$      | (36) $(6x^2 + x - 6) - (-4x^2 - 7x - 6)$<br>$= 10x^2 + 8x$       |
| (18) $(6x^2 - 8x + 9) - (6x^2 - 8x + 2)$<br>$= 7$                 | (37) $(-2x^2 + 4x - 2) - (-8x^2 + 8x + 6)$<br>$= 6x^2 - 4x - 8$  |
| (19) $(3x^2 - 4x - 5) + (-3x^2 + 9x - 4)$<br>$= 5x - 9$           | (38) $(x^2 - 5x + 3) - (-5x^2 + 4x - 7)$<br>$= 6x^2 - 9x + 10$   |

- (1)  $(8x^2 - 9x + 8) + (6x^2 - 6x - 6)$   
 $= 14x^2 - 15x + 2$
- (2)  $(9x^2 - 3x) + (-8x^2 - 4x - 8)$   
 $= x^2 - 7x - 8$
- (3)  $(-2x^2 - 4x - 3) + (-x^2 + 6x + 3)$   
 $= -3x^2 + 2x$
- (4)  $(6x^2 - 7x - 1) + (4x^2 - 3x + 3)$   
 $= 10x^2 - 10x + 2$
- (5)  $(-9x^2 + 5x) - (2x^2 + 6x - 3)$   
 $= -11x^2 - x + 3$
- (6)  $(8x^2 + 9x - 5) - (8x^2 + 9x - 5)$   
 $=$
- (7)  $(-8x^2 - 4x + 3) + (4x^2 + 5x - 7)$   
 $= -4x^2 + x - 4$
- (8)  $(9x^2 - 6x - 4) + (-x^2 - 7x + 4)$   
 $= 8x^2 - 13x$
- (9)  $(-8x^2 + x - 6) + (5x^2 + 4x - 3)$   
 $= -3x^2 + 5x - 9$
- (10)  $(3x^2 - 2x - 6) - (-8x^2 - 6x - 8)$   
 $= 11x^2 + 4x + 2$
- (11)  $(8x^2 - x + 7) + (-9x^2 + 2x - 8)$   
 $= -x^2 + x - 1$
- (12)  $(-9x^2 - 3x + 6) + (-8x^2 + 8x - 7)$   
 $= -17x^2 + 5x - 1$
- (13)  $(-8x^2 + x - 9) - (6x^2 - 7x + 8)$   
 $= -14x^2 + 8x - 17$
- (14)  $(-9x^2 - 2x - 3) + (-2x^2 + 7x + 3)$   
 $= -11x^2 + 5x$
- (15)  $(-x^2 - 7x - 6) - (-9x^2 - 5x - 9)$   
 $= 8x^2 - 2x + 3$
- (16)  $(9x^2 - 3x + 9) + (-x^2 + 4x)$   
 $= 8x^2 + x + 9$
- (17)  $(x^2 + 3x + 4) + (-2x^2 + x - 1)$   
 $= -x^2 + 4x + 3$
- (18)  $(5x^2 - 2x - 3) + (x^2 - 3x - 9)$   
 $= 6x^2 - 5x - 12$
- (19)  $(4x^2 + 3x) + (9x^2 + 7x - 6)$   
 $= 13x^2 + 10x - 6$
- (20)  $(-5x^2 - 9x + 2) + (7x^2 - 2x - 9)$   
 $= 2x^2 - 11x - 7$
- (21)  $(9x^2 + 3x + 7) - (-3x^2 + 3x - 9)$   
 $= 12x^2 + 16$
- (22)  $(-x^2 - 4x - 8) + (4x^2 + 5x - 4)$   
 $= 3x^2 + x - 12$
- (23)  $(4x^2 - 7x + 7) + (2x^2 - 3x + 5)$   
 $= 6x^2 - 10x + 12$
- (24)  $(-9x^2 - 5x + 3) + (-6x^2 + 6x - 1)$   
 $= -15x^2 + x + 2$
- (25)  $(-6x^2 - 2x + 9) + (-3x^2 + 9)$   
 $= -9x^2 - 2x + 18$
- (26)  $(6x^2 - 3x + 6) - (-8x^2 - 8x + 8)$   
 $= 14x^2 + 5x - 2$
- (27)  $(-6x^2 - 4x + 8) - (-5x^2 + 4x + 9)$   
 $= -x^2 - 8x - 1$
- (28)  $(8x^2 - 7x - 4) + (-6x^2 + 7x + 4)$   
 $= 2x^2$
- (29)  $(-2x^2 - 9x + 2) - (6x^2 + x + 2)$   
 $= -8x^2 - 10x$
- (30)  $(-2x^2 + 7x - 8) - (-2x^2 + 7x + 5)$   
 $= -13$
- (31)  $(6x^2 + x - 6) - (7x^2 - 8x + 4)$   
 $= -x^2 + 9x - 10$
- (32)  $(6x^2 - 5x + 6) - (-5x^2 - 6x + 6)$   
 $= 11x^2 + x$
- (33)  $(-9x^2 - x + 8) + (3x^2 + 6x - 8)$   
 $= -6x^2 + 5x$
- (34)  $(-7x^2 + 6x - 4) - (-8x^2 - 5x - 9)$   
 $= x^2 + 11x + 5$
- (35)  $(-7x^2 + 5x + 7) - (-5x^2 + 6x - 4)$   
 $= -2x^2 - x + 11$
- (36)  $(-7x^2 - 6x - 1) - (x^2 - 2x + 9)$   
 $= -8x^2 - 4x - 10$
- (37)  $(8x^2 - 3x - 7) + (-x^2 - 6x)$   
 $= 7x^2 - 9x - 7$
- (38)  $(7x^2 + x - 3) - (-x^2 - x - 5)$   
 $= 8x^2 + 2x + 2$



- (1)  $(7x^2 - 6x + 3) - (-2x^2 + x - 7)$   
 $= 9x^2 - 7x + 10$
- (2)  $(7x^2 + 7x - 3) + (-7x^2 + 8x + 8)$   
 $= 15x + 5$
- (3)  $(-2x^2 + 8x + 8) - (-6x^2 + 9x + 7)$   
 $= 4x^2 - x + 1$
- (4)  $(6x^2 - 4x - 9) - (6x^2 + 8x + 8)$   
 $= -12x - 17$
- (5)  $(-8x^2 + 8x - 8) - (-6x^2 + 9x + 9)$   
 $= -2x^2 - x - 17$
- (6)  $(6x^2 + 7x + 1) + (-9x^2 + 4x + 4)$   
 $= -3x^2 + 11x + 5$
- (7)  $(-5x^2 - 6x + 4) - (-6x^2 - 3x - 2)$   
 $= x^2 - 3x + 6$
- (8)  $(-9x^2 + 4x + 7) - (-4x^2 - 3x + 5)$   
 $= -5x^2 + 7x + 2$
- (9)  $(-8x^2 + 2x + 3) + (-9x^2 - 6)$   
 $= -17x^2 + 2x - 3$
- (10)  $(-7x^2 + 7x - 3) + (7x^2 + 3x - 7)$   
 $= 10x - 10$
- (11)  $(6x^2 - 3x - 2) - (-6x^2 + x)$   
 $= 12x^2 - 4x - 2$
- (12)  $(-4x^2 + 3x + 4) + (6x^2 - 4x)$   
 $= 2x^2 - x + 4$
- (13)  $(-2x^2 - x + 1) + (4x^2 + 7x - 8)$   
 $= 2x^2 + 6x - 7$
- (14)  $(7x^2 + 5x - 9) - (-7x^2 - 7x - 6)$   
 $= 14x^2 + 12x - 3$
- (15)  $(7x^2 - 6x + 6) - (9x^2 - 3x)$   
 $= -2x^2 - 3x + 6$
- (16)  $(3x^2 + 5x - 4) - (5x^2 - 8x + 1)$   
 $= -2x^2 + 13x - 5$
- (17)  $(4x^2 + 3x - 5) + (-7x^2 - 8x + 2)$   
 $= -3x^2 - 5x - 3$
- (18)  $(5x^2 + 9x - 5) + (3x^2 - x + 5)$   
 $= 8x^2 + 8x$
- (19)  $(-7x^2 + 3x - 3) - (-5x^2 - 2x - 8)$   
 $= -2x^2 + 5x + 5$
- (20)  $(8x^2 + 8x - 5) + (5x^2 - 6x + 8)$   
 $= 13x^2 + 2x + 3$
- (21)  $(-x^2 - 7x + 5) - (x^2 - 5x - 7)$   
 $= -2x^2 - 2x + 12$
- (22)  $(7x^2 - 2) - (3x^2 - x - 2)$   
 $= 4x^2 + x$
- (23)  $(-4x^2 + 8x) - (2x^2 - 6x + 9)$   
 $= -6x^2 + 14x - 9$
- (24)  $(9x^2 + 6x + 7) + (-2x^2 + 2x + 4)$   
 $= 7x^2 + 8x + 11$
- (25)  $(-9x^2 - 6x + 5) + (5x^2 - 4x - 3)$   
 $= -4x^2 - 10x + 2$
- (26)  $(-3x^2 - 9x + 6) - (-9x^2 + 9x + 8)$   
 $= 6x^2 - 18x - 2$
- (27)  $(9x^2 + 4) + (-3x^2 + 4x + 7)$   
 $= 6x^2 + 4x + 11$
- (28)  $(6x^2 - 9x) + (-4x^2 - 1)$   
 $= 2x^2 - 9x - 1$
- (29)  $(6x^2 - 5x) + (5x^2 + x + 6)$   
 $= 11x^2 - 4x + 6$
- (30)  $(-2x^2 + 5x - 7) - (-8x^2 - 7x + 8)$   
 $= 6x^2 + 12x - 15$
- (31)  $(-7x^2 - 7x + 2) + (-4x^2 - 3x - 2)$   
 $= -11x^2 - 10x$
- (32)  $(8x^2 + 5x + 4) - (6x^2 - 7x + 5)$   
 $= 2x^2 + 12x - 1$
- (33)  $(-4x^2 - x - 1) - (-x^2 + 7x + 1)$   
 $= -3x^2 - 8x - 2$
- (34)  $(-5x^2 + 3) - (7x^2 + x - 6)$   
 $= -12x^2 - x + 9$
- (35)  $(x^2 + 8) - (3x^2 - 4x + 1)$   
 $= -2x^2 + 4x + 7$
- (36)  $(-7x^2 - 2x - 6) + (2x^2 - 9x + 5)$   
 $= -5x^2 - 11x - 1$
- (37)  $(-6x^2 - 9x + 8) + (3x^2 + 5x + 6)$   
 $= -3x^2 - 4x + 14$
- (38)  $(-3x^2 + 8x - 3) - (-7x^2 + 9x + 7)$   
 $= 4x^2 - x - 10$

- |                                                                   |                                                                  |
|-------------------------------------------------------------------|------------------------------------------------------------------|
| (1) $(-8x^2 - 6x - 1) - (3x^2 - 2)$<br>$= -11x^2 - 6x + 1$        | (20) $(-9x^2 - 8) + (9x^2 - 5x - 6)$<br>$= -5x - 14$             |
| (2) $(-3x^2 + 6) + (2x^2 - 6x)$<br>$= -x^2 - 6x + 6$              | (21) $(2x^2 - 8) - (8x^2 + 7x + 3)$<br>$= -6x^2 - 7x - 11$       |
| (3) $(7x^2 + x) - (-6x^2 - x + 3)$<br>$= 13x^2 + 2x - 3$          | (22) $(9x^2 - 9x + 5) - (-3x^2 - 8x - 4)$<br>$= 12x^2 - x + 9$   |
| (4) $(4x^2 + 8x + 8) + (8x^2 - x + 7)$<br>$= 12x^2 + 7x + 15$     | (23) $(9x^2 + 8x) - (-5x^2 + 6x + 5)$<br>$= 14x^2 + 2x - 5$      |
| (5) $(2x^2 + 3x - 8) + (4x^2 - 2x - 8)$<br>$= 6x^2 + x - 16$      | (24) $(9x^2 - x + 8) - (-7x^2 - 5x - 4)$<br>$= 16x^2 + 4x + 12$  |
| (6) $(4x^2 + 7x + 1) - (-3x^2 - 4x - 7)$<br>$= 7x^2 + 11x + 8$    | (25) $(3x^2 + 8x - 2) + (-x^2 + 8x + 7)$<br>$= 2x^2 + 16x + 5$   |
| (7) $(-3x^2 - 5x - 4) - (-2x^2 + 5x + 4)$<br>$= -x^2 - 10x - 8$   | (26) $(2x^2 - 8x + 5) + (6x^2 - 8x - 9)$<br>$= 8x^2 - 16x - 4$   |
| (8) $(-2x^2 + 3x - 3) + (-3x^2 + 9x - 2)$<br>$= -5x^2 + 12x - 5$  | (27) $(-8x^2 - 2) - (-9x^2 - 6x - 2)$<br>$= x^2 + 6x$            |
| (9) $(-2x^2 + 5x + 9) + (4x^2 - 7x - 8)$<br>$= 2x^2 - 2x + 1$     | (28) $(-x^2 + 8x - 5) - (-4x^2 - 4x + 7)$<br>$= 3x^2 + 12x - 12$ |
| (10) $(7x^2 - 4x + 8) + (-3x^2 + x - 9)$<br>$= 4x^2 - 3x - 1$     | (29) $(-x^2 + 5x) - (3x^2 + x + 1)$<br>$= -4x^2 + 4x - 1$        |
| (11) $(3x^2 - 5x) + (-5x^2 - 7x - 8)$<br>$= -2x^2 - 12x - 8$      | (30) $(-7x^2 + 7x + 6) + (6x^2 + x - 7)$<br>$= -x^2 + 8x - 1$    |
| (12) $(3x^2 - 9x - 6) + (7x^2 + 5)$<br>$= 10x^2 - 9x - 1$         | (31) $(-6x^2 - 2) + (-x^2 + x + 5)$<br>$= -7x^2 + x + 3$         |
| (13) $(-7x^2 + 7x) - (4x^2 + 7x - 3)$<br>$= -11x^2 + 3$           | (32) $(-5x^2 + 4x - 5) + (3x^2 - 3x)$<br>$= -2x^2 + x - 5$       |
| (14) $(-2x^2 + 3x + 9) - (8x^2 + 2x + 4)$<br>$= -10x^2 + x + 5$   | (33) $(8x^2 + 8x - 8) - (-7x^2 - 3x)$<br>$= 15x^2 + 11x - 8$     |
| (15) $(-9x^2 - 4x + 5) + (-6x^2 - x - 8)$<br>$= -15x^2 - 5x - 3$  | (34) $(2x^2 + 4x - 7) - (-6x^2 - 9x + 2)$<br>$= 8x^2 + 13x - 9$  |
| (16) $(-8x^2 - 3x - 9) + (-6x^2 - 5x + 1)$<br>$= -14x^2 - 8x - 8$ | (35) $(6x^2 - 8x + 6) + (-5x^2 + 9x - 3)$<br>$= x^2 + x + 3$     |
| (17) $(-9x^2 + 5) - (4x^2 + 4x + 9)$<br>$= -13x^2 - 4x - 4$       | (36) $(5x^2 - 7x + 6) - (8x^2 - 3x - 5)$<br>$= -3x^2 - 4x + 11$  |
| (18) $(6x^2 + x + 1) - (x^2 + 7x - 5)$<br>$= 5x^2 - 6x + 6$       | (37) $(-9x^2 - 3x) - (-4x^2 + 9x + 7)$<br>$= -5x^2 - 12x - 7$    |
| (19) $(7x^2 + x + 7) - (8x^2 + 7x)$<br>$= -x^2 - 6x + 7$          | (38) $(5x^2 - 6x + 4) + (4x^2 - 4)$<br>$= 9x^2 - 6x$             |

- (1)  $(9x^2 - 5x - 5) - (6x^2 + 6x - 8)$   
 $= 3x^2 - 11x + 3$
- (2)  $(-5x^2 - 8x + 6) + (6x^2 + 5x)$   
 $= x^2 - 3x + 6$
- (3)  $(x^2 - 8x - 6) - (-3x^2 + x - 8)$   
 $= 4x^2 - 9x + 2$
- (4)  $(-7x^2 - 4x + 5) + (-4x^2 - 5x + 7)$   
 $= -11x^2 - 9x + 12$
- (5)  $(8x^2 - 9x - 3) - (9x^2 - 8x - 9)$   
 $= -x^2 - x + 6$
- (6)  $(8x^2 + 4x + 6) - (-9x^2 - 3x)$   
 $= 17x^2 + 7x + 6$
- (7)  $(-4x^2 + 9x - 7) - (-6x^2 + 6x - 5)$   
 $= 2x^2 + 3x - 2$
- (8)  $(4x^2 - 5x - 6) + (-9x^2 - 4x + 9)$   
 $= -5x^2 - 9x + 3$
- (9)  $(x^2 + 2x + 2) - (-6x^2 - 2x + 6)$   
 $= 7x^2 + 4x - 4$
- (10)  $(-7x^2 - 5x - 8) - (-7x^2 - 9x + 1)$   
 $= 4x - 9$
- (11)  $(5x^2 - 2x + 3) + (-8x^2 + 6x)$   
 $= -3x^2 + 4x + 3$
- (12)  $(-4x^2 - 8x + 1) + (-9x^2 + 8x - 9)$   
 $= -13x^2 - 8$
- (13)  $(7x^2 - 3x - 3) + (6x^2 + 4x + 3)$   
 $= 13x^2 + x$
- (14)  $(-2x^2 + 3x - 9) - (6x^2 + 2x + 5)$   
 $= -8x^2 + x - 14$
- (15)  $(-9x^2 + 8x + 4) + (-7x^2 + 3x - 6)$   
 $= -16x^2 + 11x - 2$
- (16)  $(-2x^2 + 6x - 1) + (-x^2 + 2x - 6)$   
 $= -3x^2 + 8x - 7$
- (17)  $(-x^2 - 5x + 4) + (-5x^2 - 4x + 7)$   
 $= -6x^2 - 9x + 11$
- (18)  $(-3x^2 + 6x + 9) - (x^2 + 7x - 5)$   
 $= -4x^2 - x + 14$
- (19)  $(-8x^2 + 4x - 1) + (9x^2 + 8x - 1)$   
 $= x^2 + 12x - 2$
- (20)  $(-8x^2 - 4x) + (7x^2 + 6x)$   
 $= -x^2 + 2x$
- (21)  $(-7x^2 + 8x) + (2x^2 - 9x - 8)$   
 $= -5x^2 - x - 8$
- (22)  $(7x^2 + 7x - 6) + (4x^2 + 4x + 1)$   
 $= 11x^2 + 11x - 5$
- (23)  $(x^2 - 5x - 6) + (-3x^2 - 4x - 8)$   
 $= -2x^2 - 9x - 14$
- (24)  $(5x^2 + 2x - 7) + (-3x^2 + 4x + 3)$   
 $= 2x^2 + 6x - 4$
- (25)  $(7x^2 + x - 2) - (3x^2 - 3x + 8)$   
 $= 4x^2 + 4x - 10$
- (26)  $(9x^2 + 2x + 6) - (-8x^2 + 6x - 6)$   
 $= 17x^2 - 4x + 12$
- (27)  $(-4x^2 - 4x + 3) + (7x^2 - 4x - 5)$   
 $= 3x^2 - 8x - 2$
- (28)  $(4x^2 + 8x) - (-4x^2 + 7x + 4)$   
 $= 8x^2 + x - 4$
- (29)  $(-8x^2 - 4x - 6) - (-7x^2 + 5x - 3)$   
 $= -x^2 - 9x - 3$
- (30)  $(2x^2 - 3x + 9) + (-9x^2 + 3x - 8)$   
 $= -7x^2 + 1$
- (31)  $(-9x^2 - 2x - 5) + (x^2 - 3x + 7)$   
 $= -8x^2 - 5x + 2$
- (32)  $(-3x^2 - 3x + 6) - (-7x^2 + 4x + 5)$   
 $= 4x^2 - 7x + 1$
- (33)  $(2x^2 - 3x + 2) - (3x^2 - 2x + 4)$   
 $= -x^2 - x - 2$
- (34)  $(7x^2 + x - 6) + (-4x^2 - 4x + 9)$   
 $= 3x^2 - 3x + 3$
- (35)  $(-8x^2 + 4x - 6) + (9x^2 - 8x - 4)$   
 $= x^2 - 4x - 10$
- (36)  $(6x^2 + 9x - 3) + (-8x^2 - 6x - 6)$   
 $= -2x^2 + 3x - 9$
- (37)  $(-3x^2 - 2x + 6) - (7x^2 - 2x)$   
 $= -10x^2 + 6$
- (38)  $(-x^2 + 7x - 5) - (2x^2 + 5x + 5)$   
 $= -3x^2 + 2x - 10$

- |                                                                    |                                                                   |
|--------------------------------------------------------------------|-------------------------------------------------------------------|
| (1) $(4x^2 + 8x - 6) + (5x^2 + 7x + 3)$<br>$= 9x^2 + 15x - 3$      | (20) $(-5x^2 - 2x + 1) + (-8x^2 + 4x - 5)$<br>$= -13x^2 + 2x - 4$ |
| (2) $(6x^2 - 8x - 5) + (8x^2 + x - 5)$<br>$= 14x^2 - 7x - 10$      | (21) $(4x^2 - 2x - 8) + (6x^2 + 3x - 9)$<br>$= 10x^2 + x - 17$    |
| (3) $(7x^2 + 6x + 2) + (-x^2 - 6x + 4)$<br>$= 6x^2 + 6$            | (22) $(9x^2 - 4x - 6) - (-3x^2 + 4x - 4)$<br>$= 12x^2 - 8x - 2$   |
| (4) $(7x^2 + 8x - 7) + (-5x^2 + 8x + 2)$<br>$= 2x^2 + 16x - 5$     | (23) $(x^2 - 2x - 5) - (-8x^2 - 7x - 8)$<br>$= 9x^2 + 5x + 3$     |
| (5) $(8x^2) + (-4x^2 - 9x - 2)$<br>$= 4x^2 - 9x - 2$               | (24) $(9x^2 + 7x - 9) - (-3x^2 + 9x + 4)$<br>$= 12x^2 - 2x - 13$  |
| (6) $(-7x^2 - 7x + 4) + (-4x^2 + 9)$<br>$= -11x^2 - 7x + 13$       | (25) $(7x^2 + 7x - 3) - (-x^2 - 6x - 5)$<br>$= 8x^2 + 13x + 2$    |
| (7) $(5x^2 + 4x - 3) + (9x^2 + 2x + 5)$<br>$= 14x^2 + 6x + 2$      | (26) $(2x^2 - 4x) + (-6x^2 + 2x + 8)$<br>$= -4x^2 - 2x + 8$       |
| (8) $(-4x^2 - x + 9) - (-x^2 - 8x + 5)$<br>$= -3x^2 + 7x + 4$      | (27) $(2x^2 + 5x - 1) - (4x^2 + 3x - 9)$<br>$= -2x^2 + 2x + 8$    |
| (9) $(5x^2 - 8x + 1) + (4x^2 - x + 5)$<br>$= 9x^2 - 9x + 6$        | (28) $(3x^2 - 8x + 5) - (3x^2 + 2x - 5)$<br>$= -10x + 10$         |
| (10) $(-5x^2 - 8x + 2) + (-7x^2 - 6x - 7)$<br>$= -12x^2 - 14x - 5$ | (29) $(2x^2 - 4x + 8) + (7x^2 + 5x + 4)$<br>$= 9x^2 + x + 12$     |
| (11) $(5x^2 - 6x + 3) - (3x^2 - 8x + 3)$<br>$= 2x^2 + 2x$          | (30) $(-4x^2 + 9x - 2) - (-x^2 - 3x - 7)$<br>$= -3x^2 + 12x + 5$  |
| (12) $(4x^2 + 2x + 8) + (2x^2 - 8x)$<br>$= 6x^2 - 6x + 8$          | (31) $(9x^2 + 9x - 7) - (7x^2 + 8x + 9)$<br>$= 2x^2 + x - 16$     |
| (13) $(3x^2 - x - 3) + (-7x^2 + 4x - 6)$<br>$= -4x^2 + 3x - 9$     | (32) $(-3x^2 - 7x + 1) - (-7x^2 - x + 3)$<br>$= 4x^2 - 6x - 2$    |
| (14) $(9x^2 + 9x - 1) + (x^2 - 6x + 1)$<br>$= 10x^2 + 3x$          | (33) $(7x^2 - 2x + 2) - (-x^2 - 3x + 5)$<br>$= 8x^2 + x - 3$      |
| (15) $(6x^2 - x + 6) + (2x^2 - 3x + 4)$<br>$= 8x^2 - 4x + 10$      | (34) $(4x^2) - (-7x^2 - 4x - 4)$<br>$= 11x^2 + 4x + 4$            |
| (16) $(4x^2 - 7x - 3) - (-8x^2 - 6)$<br>$= 12x^2 - 7x + 3$         | (35) $(-5x^2 - 4x + 1) - (-2x^2 + x - 8)$<br>$= -3x^2 - 5x + 9$   |
| (17) $(-x^2 + 6x + 9) - (x^2 + 4x + 9)$<br>$= -2x^2 + 2x$          | (36) $(8x^2 + 2x - 3) - (9x^2 - 2x - 6)$<br>$= -x^2 + 4x + 3$     |
| (18) $(2x^2 + 8x + 6) + (4x^2 - x - 7)$<br>$= 6x^2 + 7x - 1$       | (37) $(-9x^2 + 2x - 4) - (-3x^2 + 2x - 3)$<br>$= -6x^2 - 1$       |
| (19) $(2x^2 - 4x - 9) + (-2x^2 - 2x - 1)$<br>$= -6x - 10$          | (38) $(-2x^2 + 5x + 6) + (7x^2 + 9x)$<br>$= 5x^2 + 14x + 6$       |

- (1)  $(-6x^2 - x - 2) - (7x^2 - 5x - 6)$   
 $= -13x^2 + 4x + 4$
- (2)  $(9x^2 + 9x - 8) - (-8x^2 + 2x + 4)$   
 $= 17x^2 + 7x - 12$
- (3)  $(5x^2 - 4x + 1) - (8x^2 - 9x + 2)$   
 $= -3x^2 + 5x - 1$
- (4)  $(-4x^2 + 4x - 5) - (9x^2 + 6x - 5)$   
 $= -13x^2 - 2x$
- (5)  $(-3x^2 + 5x + 7) - (-7x^2 - 6x + 6)$   
 $= 4x^2 + 11x + 1$
- (6)  $(5x^2 + 7x + 7) + (4x^2 - 8x + 9)$   
 $= 9x^2 - x + 16$
- (7)  $(-7x^2 + 6x - 6) - (-3x^2 - 5x - 5)$   
 $= -4x^2 + 11x - 1$
- (8)  $(-3x^2 + 4x - 6) + (-6x^2 - 4x - 7)$   
 $= -9x^2 - 13$
- (9)  $(-2x^2 + 3x - 2) - (-5x^2 - 4x + 6)$   
 $= 3x^2 + 7x - 8$
- (10)  $(2x^2 + 8x - 1) + (5x^2 - x + 8)$   
 $= 7x^2 + 7x + 7$
- (11)  $(8x^2 - 9x + 9) - (-x^2 - 4x + 2)$   
 $= 9x^2 - 5x + 7$
- (12)  $(2x^2 + 3x - 3) - (x^2 - 4x - 4)$   
 $= x^2 + 7x + 1$
- (13)  $(6x^2 - x + 1) + (-9x^2 + 9x - 8)$   
 $= -3x^2 + 8x - 7$
- (14)  $(-3x^2 + 3x + 9) + (-2x^2 - 2x + 2)$   
 $= -5x^2 + x + 11$
- (15)  $(-3x^2 + 5x + 4) - (2x^2 + 9x + 9)$   
 $= -5x^2 - 4x - 5$
- (16)  $(-9x^2 - 5x + 2) + (-x^2 - 6x - 7)$   
 $= -10x^2 - 11x - 5$
- (17)  $(-4x^2 + 8x - 8) - (3x^2 - x + 1)$   
 $= -7x^2 + 9x - 9$
- (18)  $(-4x^2 - 3x - 2) - (-8x^2 + 3x - 2)$   
 $= 4x^2 - 6x$
- (19)  $(-4x^2 + 2x - 5) + (-6x^2 + x + 2)$   
 $= -10x^2 + 3x - 3$
- (20)  $(-6x^2 - 4x - 1) + (6x^2 + x + 5)$   
 $= -3x + 4$
- (21)  $(-5x^2 - 4x - 8) - (x^2 + 8x + 1)$   
 $= -6x^2 - 12x - 9$
- (22)  $(-4x^2 - 8x - 2) - (4x^2 + 5x - 8)$   
 $= -8x^2 - 13x + 6$
- (23)  $(-x^2 + 4x - 8) - (8x^2 - 2x + 3)$   
 $= -9x^2 + 6x - 11$
- (24)  $(-7x^2 + 4x - 5) - (-4x^2 - 4x + 4)$   
 $= -3x^2 + 8x - 9$
- (25)  $(-6x^2 - 5x - 6) + (-7x^2 + 8x - 1)$   
 $= -13x^2 + 3x - 7$
- (26)  $(x^2 + 4x - 8) + (7x^2 + 6x)$   
 $= 8x^2 + 10x - 8$
- (27)  $(-4x^2 + 9x - 1) - (-8x^2 + x - 1)$   
 $= 4x^2 + 8x$
- (28)  $(x^2 + 5x - 3) + (-x^2 - 6x + 4)$   
 $= -x + 1$
- (29)  $(-2x^2 - 8x - 6) + (-4x^2 + 3x + 3)$   
 $= -6x^2 - 5x - 3$
- (30)  $(-5x^2 - 2x + 6) - (2x^2 - 2)$   
 $= -7x^2 - 2x + 8$
- (31)  $(-2x^2 - 3x) - (-2x^2 + 6x + 9)$   
 $= -9x - 9$
- (32)  $(6x^2 - 9) - (x^2 - 5x - 9)$   
 $= 5x^2 + 5x$
- (33)  $(-9x^2 - 2x + 6) - (-8x^2 + 7x - 2)$   
 $= -x^2 - 9x + 8$
- (34)  $(2x^2 - 6x + 9) - (-4x^2 - 2x - 1)$   
 $= 6x^2 - 4x + 10$
- (35)  $(-6x^2 + 5x + 1) + (8x^2 - 5x - 9)$   
 $= 2x^2 - 8$
- (36)  $(5x^2 + 4x + 1) + (8x^2 + 6x + 3)$   
 $= 13x^2 + 10x + 4$
- (37)  $(3x^2 - 6x + 6) - (-2x^2 - 5x - 5)$   
 $= 5x^2 - x + 11$
- (38)  $(-4x^2 + 9x + 3) - (7x^2 + 2x + 5)$   
 $= -11x^2 + 7x - 2$

- (1)  $(3x^2 - 8x) - (-4x^2 + 6x)$   
 $= 7x^2 - 14x$
- (2)  $(-9x^2 - 8x + 7) - (8x^2 - 3x + 8)$   
 $= -17x^2 - 5x - 1$
- (3)  $(-9x^2) - (-6x^2 - 9x + 5)$   
 $= -3x^2 + 9x - 5$
- (4)  $(-9x^2 - 9x - 5) - (-8x^2 + 9x - 5)$   
 $= -x^2 - 18x$
- (5)  $(7x^2 + x - 2) + (-9x^2 - 5x - 1)$   
 $= -2x^2 - 4x - 3$
- (6)  $(-2x^2 + x - 7) + (2x^2 + 4)$   
 $= x - 3$
- (7)  $(-6x^2 + 6x - 9) + (-8x^2 - 6x - 6)$   
 $= -14x^2 - 15$
- (8)  $(-x^2 - 7x - 6) + (-8x^2 - 6x + 6)$   
 $= -9x^2 - 13x$
- (9)  $(6x^2 + 6x - 6) + (-5x^2 - 8x - 8)$   
 $= x^2 - 2x - 14$
- (10)  $(-x^2 + 9x + 3) + (2x^2 - 3x - 2)$   
 $= x^2 + 6x + 1$
- (11)  $(-5x^2 - x - 1) + (-6x^2 + 2)$   
 $= -11x^2 - x + 1$
- (12)  $(-7x^2 - x + 2) + (7x^2 - 4x - 1)$   
 $= -5x + 1$
- (13)  $(-9x^2 - 9x) + (-4x^2 + 5x - 2)$   
 $= -13x^2 - 4x - 2$
- (14)  $(-5x^2 - 5x + 4) + (2x^2 + x - 3)$   
 $= -3x^2 - 4x + 1$
- (15)  $(-2x^2 + 4x + 2) - (-3x^2 - x - 1)$   
 $= x^2 + 5x + 3$
- (16)  $(-7x^2 + 3x - 2) + (7x^2 - 3x - 6)$   
 $= -8$
- (17)  $(9x^2 - 5x + 8) - (4x^2 + 2x + 5)$   
 $= 5x^2 - 7x + 3$
- (18)  $(-6x^2 - 5x - 4) + (-5x^2 + 2x - 5)$   
 $= -11x^2 - 3x - 9$
- (19)  $(9x^2 - 8x + 5) + (8x^2 + 9x + 6)$   
 $= 17x^2 + x + 11$
- (20)  $(6x^2 + x - 3) + (-3x^2 + 4x + 6)$   
 $= 3x^2 + 5x + 3$
- (21)  $(3x^2 + 4x - 4) + (2x^2 - 6)$   
 $= 5x^2 + 4x - 10$
- (22)  $(-2x^2 + 7x + 8) + (2x^2 + 3x - 2)$   
 $= 10x + 6$
- (23)  $(3x^2 - 8x + 2) - (-6x^2 + 6x - 6)$   
 $= 9x^2 - 14x + 8$
- (24)  $(-8x^2 + 2x - 1) - (5x^2 - 9x + 6)$   
 $= -13x^2 + 11x - 7$
- (25)  $(-9x^2 - 8x - 9) - (2x^2 + 5x - 4)$   
 $= -11x^2 - 13x - 5$
- (26)  $(-3x^2 - 7x - 3) + (-5x^2 + 4x + 5)$   
 $= -8x^2 - 3x + 2$
- (27)  $(2x^2 - 8x - 1) + (-6x^2 + 9x - 7)$   
 $= -4x^2 + x - 8$
- (28)  $(-x^2 + 5x + 5) + (-3x^2 - 7x + 3)$   
 $= -4x^2 - 2x + 8$
- (29)  $(2x^2 - 3x - 1) - (-2x^2 - 2x + 6)$   
 $= 4x^2 - x - 7$
- (30)  $(-4x^2 + 9x - 6) - (-9x^2 - 3x + 8)$   
 $= 5x^2 + 12x - 14$
- (31)  $(3x^2 + 2x) + (3x^2 + 4x - 6)$   
 $= 6x^2 + 6x - 6$
- (32)  $(8x^2 + 4x - 2) - (-6x^2 + 8x + 5)$   
 $= 14x^2 - 4x - 7$
- (33)  $(2x^2 + 7x + 6) - (-4x^2 - 8x + 9)$   
 $= 6x^2 + 15x - 3$
- (34)  $(x^2 + 4x - 4) + (4x^2 - 5)$   
 $= 5x^2 + 4x - 9$
- (35)  $(7x^2 - 5x + 7) + (3x^2 - 7x - 2)$   
 $= 10x^2 - 12x + 5$
- (36)  $(-8x^2 - x + 4) + (8x^2 + 6x - 6)$   
 $= 5x - 2$
- (37)  $(-6x^2 - x + 2) + (3x^2 - 4x - 6)$   
 $= -3x^2 - 5x - 4$
- (38)  $(-9x^2 + 5x + 8) + (3x^2 + x + 3)$   
 $= -6x^2 + 6x + 11$

- (1)  $(2x^2 - 7x + 5) + (-7x^2 - 4x)$   
 $= -5x^2 - 11x + 5$
- (2)  $(9x^2 - 9x + 5) - (-5x^2 + 4x - 2)$   
 $= 14x^2 - 13x + 7$
- (3)  $(6x^2 + 7x - 4) - (3x^2 + 9x - 4)$   
 $= 3x^2 - 2x$
- (4)  $(8x^2 - 6x - 3) - (3x^2 + 3x - 8)$   
 $= 5x^2 - 9x + 5$
- (5)  $(-9x^2 - 5x + 4) - (-7x^2 + 5x - 4)$   
 $= -2x^2 - 10x + 8$
- (6)  $(-x^2 + 5x - 2) - (5x^2 - 4x + 1)$   
 $= -6x^2 + 9x - 3$
- (7)  $(9x^2 - 4x - 4) - (-4x^2 - 6x + 8)$   
 $= 13x^2 + 2x - 12$
- (8)  $(-6x^2 - 5x - 1) - (-x^2 - 9x + 3)$   
 $= -5x^2 + 4x - 4$
- (9)  $(x^2 + 9x - 4) + (6x^2 + 8x + 2)$   
 $= 7x^2 + 17x - 2$
- (10)  $(2x^2 + 8x - 5) - (3x^2 - 3x - 7)$   
 $= -x^2 + 11x + 2$
- (11)  $(-3x^2 + 4x + 7) - (3x^2 - 5x + 3)$   
 $= -6x^2 + 9x + 4$
- (12)  $(3x^2 + 5x + 8) + (5x^2 - 3x + 1)$   
 $= 8x^2 + 2x + 9$
- (13)  $(9x^2 - 7x + 8) - (-2x^2 - 1)$   
 $= 11x^2 - 7x + 9$
- (14)  $(-3x^2 - 5x + 1) - (-x^2 - 6)$   
 $= -2x^2 - 5x + 7$
- (15)  $(-7x^2 + x - 6) + (6x^2 + 9x + 3)$   
 $= -x^2 + 10x - 3$
- (16)  $(4x^2 + 6x - 1) - (-x^2 + 5x)$   
 $= 5x^2 + x - 1$
- (17)  $(8x^2 + 7x + 5) - (4x^2 + 4x - 4)$   
 $= 4x^2 + 3x + 9$
- (18)  $(4x^2 + 4x + 6) - (-4x^2 - x + 4)$   
 $= 8x^2 + 5x + 2$
- (19)  $(-4x^2 - 9x + 5) + (6x^2 - 6x + 3)$   
 $= 2x^2 - 15x + 8$
- (20)  $(x^2 + 5x - 9) - (-3x^2 - 7x - 3)$   
 $= 4x^2 + 12x - 6$
- (21)  $(-9x^2 - 8x + 3) + (7x^2 - 3x + 8)$   
 $= -2x^2 - 11x + 11$
- (22)  $(-6x^2 - 7x - 8) - (6x^2 + 9x - 9)$   
 $= -12x^2 - 16x + 1$
- (23)  $(-6x^2 - 8x - 6) + (x^2 + x + 3)$   
 $= -5x^2 - 7x - 3$
- (24)  $(-4x^2 + 6x + 1) - (5x^2 + 7x)$   
 $= -9x^2 - x + 1$
- (25)  $(6x^2 - 4x - 2) - (2x^2 - 8x - 8)$   
 $= 4x^2 + 4x + 6$
- (26)  $(7x^2 - 9x + 6) + (5x^2 + 4x + 6)$   
 $= 12x^2 - 5x + 12$
- (27)  $(-x^2 - 3x - 8) - (-3x^2 + 7x - 4)$   
 $= 2x^2 - 10x - 4$
- (28)  $(-x^2 + 2x - 3) - (-2x^2 + 2x + 4)$   
 $= x^2 - 7$
- (29)  $(-3x^2 - 8x + 6) + (-2x^2 + 5x + 2)$   
 $= -5x^2 - 3x + 8$
- (30)  $(-6x^2 + 8x + 6) - (4x^2 - 6x + 3)$   
 $= -10x^2 + 14x + 3$
- (31)  $(-4x^2 - 5x + 8) - (-8x^2 + 6x + 7)$   
 $= 4x^2 - 11x + 1$
- (32)  $(5x^2) + (9x^2 + 3x)$   
 $= 14x^2 + 3x$
- (33)  $(2x^2 + 3) - (5x^2 + 5x + 5)$   
 $= -3x^2 - 5x - 2$
- (34)  $(7x^2 + 3x - 6) - (-8x^2 + 4x + 4)$   
 $= 15x^2 - x - 10$
- (35)  $(8x^2 - 6x + 1) + (x^2 + 2x - 9)$   
 $= 9x^2 - 4x - 8$
- (36)  $(-x^2 - 5x - 6) - (x^2 + x + 3)$   
 $= -2x^2 - 6x - 9$
- (37)  $(-8x^2 + x + 1) + (3x^2 - x + 8)$   
 $= -5x^2 + 9$
- (38)  $(-4x^2 - 4x + 6) - (2x^2 + 9x - 6)$   
 $= -6x^2 - 13x + 12$

- (1)  $(-x^2 - x - 8) - (3x^2 - 2x + 4)$   
 $= -4x^2 + x - 12$
- (2)  $(8x^2 + 5x - 2) - (2x^2 + 4x + 1)$   
 $= 6x^2 + x - 3$
- (3)  $(-5x^2 - 7x) - (-8x^2 + 9x - 3)$   
 $= 3x^2 - 16x + 3$
- (4)  $(-5x^2 + 6x + 9) + (-8x^2 + 9x - 4)$   
 $= -13x^2 + 15x + 5$
- (5)  $(6x^2 - 8x + 7) - (3x^2 + 3x - 6)$   
 $= 3x^2 - 11x + 13$
- (6)  $(-3x^2 - x - 7) + (4x^2 - 3x - 4)$   
 $= x^2 - 4x - 11$
- (7)  $(-7x^2 + 3x) + (4x^2 + x - 8)$   
 $= -3x^2 + 4x - 8$
- (8)  $(9x^2 - 5x - 4) - (-7x^2 - 5x + 2)$   
 $= 16x^2 - 6$
- (9)  $(-2x^2 - 5x) - (-3x^2 + 4x + 5)$   
 $= x^2 - 9x - 5$
- (10)  $(-4x^2 - 2x - 3) + (-7x^2 - x - 1)$   
 $= -11x^2 - 3x - 4$
- (11)  $(5x^2 - x - 6) + (x^2 + 8x - 9)$   
 $= 6x^2 + 7x - 15$
- (12)  $(-5x^2 + 7x + 7) + (3x^2 - x - 7)$   
 $= -2x^2 + 6x$
- (13)  $(x^2 + x + 2) - (4x^2 + 8x - 8)$   
 $= -3x^2 - 7x + 10$
- (14)  $(3x^2 - 4x + 2) - (-8x^2 + 3x + 9)$   
 $= 11x^2 - 7x - 7$
- (15)  $(-4x^2 - 5x + 3) + (3x^2 + 2x - 1)$   
 $= -x^2 - 3x + 2$
- (16)  $(-3x^2 + 7x + 8) - (-2x^2 + 8x - 2)$   
 $= -x^2 - x + 10$
- (17)  $(4x^2 - 2x - 5) - (-7x^2 - 9x - 8)$   
 $= 11x^2 + 7x + 3$
- (18)  $(2x^2 - x + 1) - (7x^2 + 8x + 9)$   
 $= -5x^2 - 9x - 8$
- (19)  $(4x^2 - 8x - 1) + (-4x^2 - 8x - 8)$   
 $= -16x - 9$
- (20)  $(x^2 + 3x + 6) + (-9x^2 + 9x + 9)$   
 $= -8x^2 + 12x + 15$
- (21)  $(8x^2 - 5x + 8) - (3x^2 - 6x)$   
 $= 5x^2 + x + 8$
- (22)  $(x^2 + 7x - 9) + (-8x^2 + 8x - 5)$   
 $= -7x^2 + 15x - 14$
- (23)  $(8x^2 + 6) + (2x^2 + 9x - 3)$   
 $= 10x^2 + 9x + 3$
- (24)  $(-3x^2 - 3x + 8) - (8x^2 + 5x + 5)$   
 $= -11x^2 - 8x + 3$
- (25)  $(-x^2 - 8) + (9x^2 - 7x)$   
 $= 8x^2 - 7x - 8$
- (26)  $(-8x^2 - 2x - 1) + (-7x^2 - 3x + 7)$   
 $= -15x^2 - 5x + 6$
- (27)  $(6x^2 + 8x + 1) + (-8x^2 - 4x - 1)$   
 $= -2x^2 + 4x$
- (28)  $(7x^2 - 7x + 5) + (9x^2 + 7x - 8)$   
 $= 16x^2 - 3$
- (29)  $(-8x^2 - 9x + 6) + (-4x^2 - 7x + 4)$   
 $= -12x^2 - 16x + 10$
- (30)  $(9x^2 + 2x - 8) - (-9x^2 + 2x + 2)$   
 $= 18x^2 - 10$
- (31)  $(3x^2 + 3x + 4) - (x^2 - 7x)$   
 $= 2x^2 + 10x + 4$
- (32)  $(7x^2 - x + 7) - (4x^2 + 7x + 1)$   
 $= 3x^2 - 8x + 6$
- (33)  $(-4x^2 - 8x + 8) + (4x^2 + 2x - 4)$   
 $= -6x + 4$
- (34)  $(7x^2 + 7x - 3) + (-2x^2 - 2x + 3)$   
 $= 5x^2 + 5x$
- (35)  $(-x^2 - x + 5) + (-6x^2 - x - 4)$   
 $= -7x^2 - 2x + 1$
- (36)  $(2x^2 + 9x + 7) - (-6x^2 + 6x - 9)$   
 $= 8x^2 + 3x + 16$
- (37)  $(5x^2 + 9x - 2) + (8x^2 + 9x + 8)$   
 $= 13x^2 + 18x + 6$
- (38)  $(3x^2 - 8x - 4) - (-7x^2 + 4x - 7)$   
 $= 10x^2 - 12x + 3$



- (1)  $(2x^2 - 4x - 9) - (2x^2 - 2x - 5)$   
 $= -2x - 4$
- (2)  $(7x^2 - 9x + 1) - (-9x^2 + 3x + 1)$   
 $= 16x^2 - 12x$
- (3)  $(-6x^2 - 7x - 2) - (-9x^2 - 6x + 3)$   
 $= 3x^2 - x - 5$
- (4)  $(6x^2 + 9x) + (7x^2 - 7x + 3)$   
 $= 13x^2 + 2x + 3$
- (5)  $(-5x^2 + 7x + 5) - (-9x^2 - x + 7)$   
 $= 4x^2 + 8x - 2$
- (6)  $(9x^2 + 8x) - (2x^2 - 5x + 7)$   
 $= 7x^2 + 13x - 7$
- (7)  $(2x^2 + 9) - (-8x^2 + 4x + 5)$   
 $= 10x^2 - 4x + 4$
- (8)  $(-9x^2 - 5x - 2) + (9x^2 + 2)$   
 $= -5x$
- (9)  $(-3x^2 - 9x + 9) + (3x^2 - 2x - 3)$   
 $= -11x + 6$
- (10)  $(5x^2 - 8x - 7) - (2x^2 - 6x + 4)$   
 $= 3x^2 - 2x - 11$
- (11)  $(7x^2 - 2x - 8) + (-8x^2 + 7x + 7)$   
 $= -x^2 + 5x - 1$
- (12)  $(9x^2 + 5x - 3) - (3x^2 - 3x - 3)$   
 $= 6x^2 + 8x$
- (13)  $(-3x^2 - 5x + 4) + (-6x^2 + 4x - 8)$   
 $= -9x^2 - x - 4$
- (14)  $(2x^2 - 6x + 8) + (-2x^2 - 2x)$   
 $= -8x + 8$
- (15)  $(-6x^2 - 2x - 5) - (5x^2 - 9x)$   
 $= -11x^2 + 7x - 5$
- (16)  $(-8x^2 - 6x - 6) - (-x^2 + 7x - 8)$   
 $= -7x^2 - 13x + 2$
- (17)  $(6x^2 - 7x + 8) + (x^2 + 4x + 7)$   
 $= 7x^2 - 3x + 15$
- (18)  $(5x^2 - 3x - 4) - (x^2 + 2x + 6)$   
 $= 4x^2 - 5x - 10$
- (19)  $(3x^2 + 3x - 3) + (-7x^2 + 9x - 1)$   
 $= -4x^2 + 12x - 4$
- (20)  $(-3x^2 + 6x + 6) - (-8x^2 + 3x + 9)$   
 $= 5x^2 + 3x - 3$
- (21)  $(9x^2 + 6x - 5) + (-8x^2 - 7x + 1)$   
 $= x^2 - x - 4$
- (22)  $(3x^2 + 3x - 9) - (-2x^2 + 8x - 8)$   
 $= 5x^2 - 5x - 1$
- (23)  $(x^2 + 7x - 7) - (8x^2 + 6x - 1)$   
 $= -7x^2 + x - 6$
- (24)  $(7x^2 - x + 7) + (4x^2 - 9x - 2)$   
 $= 11x^2 - 10x + 5$
- (25)  $(-8x^2 + 6x - 9) + (4x^2 - x - 6)$   
 $= -4x^2 + 5x - 15$
- (26)  $(6x^2 + 6x - 4) + (4x^2 - 4x + 7)$   
 $= 10x^2 + 2x + 3$
- (27)  $(3x^2 - 8x + 2) - (5x^2 + 9x - 3)$   
 $= -2x^2 - 17x + 5$
- (28)  $(7x^2 + 3x + 9) - (-x^2 + 8x - 2)$   
 $= 8x^2 - 5x + 11$
- (29)  $(-7x^2 + 6x - 4) + (x^2 - 4)$   
 $= -6x^2 + 6x - 8$
- (30)  $(3x^2 - 5x - 5) + (-7x^2 + 3x + 1)$   
 $= -4x^2 - 2x - 4$
- (31)  $(6x^2 + 5x - 5) - (x^2 - 7x + 2)$   
 $= 5x^2 + 12x - 7$
- (32)  $(-7x^2 - 7x - 6) + (-6x^2 + 4x - 9)$   
 $= -13x^2 - 3x - 15$
- (33)  $(-7x^2 + 7x + 7) - (-2x^2 + 7x + 1)$   
 $= -5x^2 + 6$
- (34)  $(-7x^2 - 5x + 2) - (7x^2 - 3x - 7)$   
 $= -14x^2 - 2x + 9$
- (35)  $(4x^2 + 8x + 3) - (-9x^2 - x + 2)$   
 $= 13x^2 + 9x + 1$
- (36)  $(-3x^2 + x - 2) - (7x^2 + x - 2)$   
 $= -10x^2$
- (37)  $(-8x^2 + 3x - 6) - (-8x^2 + 9x - 4)$   
 $= -6x - 2$
- (38)  $(x^2 + 6x - 9) + (-4x^2 - 5x - 5)$   
 $= -3x^2 + x - 14$

- (1)  $(-x^2 + 6x + 3) - (5x^2 - 5x + 5)$   
 $= -6x^2 + 11x - 2$
- (2)  $(7x^2 - 8x + 2) + (-8x^2 + 5x + 2)$   
 $= -x^2 - 3x + 4$
- (3)  $(-2x^2 + 8x + 9) + (3x^2 + 3x + 6)$   
 $= x^2 + 11x + 15$
- (4)  $(-3x^2 + 4x + 1) + (-8x^2 - 8x + 1)$   
 $= -11x^2 - 4x + 2$
- (5)  $(-8x^2 + 8x + 9) - (7x^2 + 8x + 2)$   
 $= -15x^2 + 7$
- (6)  $(6x^2 - 5x - 7) - (-2x^2 - 3x - 2)$   
 $= 8x^2 - 2x - 5$
- (7)  $(5x^2 + 4x + 2) - (x^2 + 5x + 6)$   
 $= 4x^2 - x - 4$
- (8)  $(-3x^2 - 3x - 8) + (-4x^2 + 8x - 6)$   
 $= -7x^2 + 5x - 14$
- (9)  $(-9x^2 + 9x + 9) + (8x^2 + 9x - 8)$   
 $= -x^2 + 18x + 1$
- (10)  $(-8x^2 - 2x - 2) + (-3x^2 + 6x + 5)$   
 $= -11x^2 + 4x + 3$
- (11)  $(-6x^2 + 9x) - (-2x^2 + 3x - 1)$   
 $= -4x^2 + 6x + 1$
- (12)  $(3x^2 + 8x - 8) + (-7x^2 + 3x + 6)$   
 $= -4x^2 + 11x - 2$
- (13)  $(3x^2 - 2x - 1) - (7x^2 - x - 9)$   
 $= -4x^2 - x + 8$
- (14)  $(-2x^2 - 6x - 3) + (-9x^2 - 9x + 2)$   
 $= -11x^2 - 15x - 1$
- (15)  $(-4x^2 + 3x - 5) + (-4x^2 - 3x - 2)$   
 $= -8x^2 - 7$
- (16)  $(-8x^2 + 5x - 9) - (2x^2 - 3x + 8)$   
 $= -10x^2 + 8x - 17$
- (17)  $(3x^2 + 7) - (3x^2 - 4x + 8)$   
 $= 4x - 1$
- (18)  $(8x^2 + 3x + 7) - (8x^2 - 7x - 5)$   
 $= 10x + 12$
- (19)  $(4x^2 + 4x - 4) - (-6x^2 - x - 8)$   
 $= 10x^2 + 5x + 4$
- (20)  $(-3x^2 + 9) - (-4x^2 + 4x + 6)$   
 $= x^2 - 4x + 3$
- (21)  $(-5x^2 - 7x - 7) - (x^2 + 2x - 7)$   
 $= -6x^2 - 9x$
- (22)  $(-8x^2 - 9x + 3) + (3x^2 - x + 2)$   
 $= -5x^2 - 10x + 5$
- (23)  $(-8x^2 + 4x) - (-3x^2 - x - 1)$   
 $= -5x^2 + 5x + 1$
- (24)  $(-5x^2 - 8x + 8) + (4x^2 - 2x - 1)$   
 $= -x^2 - 10x + 7$
- (25)  $(-4x^2 - 6x) + (7x^2 + x - 2)$   
 $= 3x^2 - 5x - 2$
- (26)  $(5x^2 - 5x - 9) + (-8x^2 + 5)$   
 $= -3x^2 - 5x - 4$
- (27)  $(6x^2 - 4x - 8) - (-x^2 - 9x + 8)$   
 $= 7x^2 + 5x - 16$
- (28)  $(-9x^2 - 5x - 1) - (-9x^2 + 6x - 1)$   
 $= -11x$
- (29)  $(8x^2 + 9x - 8) - (7x^2 + 5)$   
 $= x^2 + 9x - 13$
- (30)  $(-5x^2 + 9x - 8) + (-x^2 - 7x + 3)$   
 $= -6x^2 + 2x - 5$
- (31)  $(-9x^2 - 6x + 7) + (-8x^2 + 2x - 3)$   
 $= -17x^2 - 4x + 4$
- (32)  $(-6x^2 - 6x + 5) + (-6x^2 + 8x + 8)$   
 $= -12x^2 + 2x + 13$
- (33)  $(2x^2 + 3) - (9x^2 - 2x + 8)$   
 $= -7x^2 + 2x - 5$
- (34)  $(-3x^2 - 3x + 7) - (-9x^2 + 4x - 4)$   
 $= 6x^2 - 7x + 11$
- (35)  $(-4x^2 + 7x) + (2x^2 - 3x + 5)$   
 $= -2x^2 + 4x + 5$
- (36)  $(5x^2 + 4x - 9) - (5x^2 + 5x + 9)$   
 $= -x - 18$
- (37)  $(2x^2 - 6x + 8) + (9x^2 + 5x)$   
 $= 11x^2 - x + 8$
- (38)  $(8x^2 + 4x - 8) - (8x^2 + 9x + 1)$   
 $= -5x - 9$

- (1)  $(7x^2 + 8x + 7) - (-x^2 - 3x - 4)$   
 $= 8x^2 + 11x + 11$
- (2)  $(-9x^2 + 4x - 2) - (7x^2 + 3x - 1)$   
 $= -16x^2 + x - 1$
- (3)  $(3x^2 - 2x) - (-2x^2 - 6x - 8)$   
 $= 5x^2 + 4x + 8$
- (4)  $(-9x^2 + 2x + 9) + (-7x^2 + 2x - 8)$   
 $= -16x^2 + 4x + 1$
- (5)  $(x^2 - 7x + 2) - (-7x^2 + 6x - 6)$   
 $= 8x^2 - 13x + 8$
- (6)  $(-x^2 - 5x - 8) + (-8x^2 - 2x - 3)$   
 $= -9x^2 - 7x - 11$
- (7)  $(7x^2 + 7x - 3) + (7x^2 + 5)$   
 $= 14x^2 + 7x + 2$
- (8)  $(x^2 - 9x - 8) - (2x^2 - x - 3)$   
 $= -x^2 - 8x - 5$
- (9)  $(6x^2 + 3) + (-9x^2 + 4x - 2)$   
 $= -3x^2 + 4x + 1$
- (10)  $(7x^2 - 7x + 3) - (5x^2 - 8x + 2)$   
 $= 2x^2 + x + 1$
- (11)  $(9x^2 - 4x) + (3x^2 + 6x + 3)$   
 $= 12x^2 + 2x + 3$
- (12)  $(2x^2 - 8x - 2) - (6x^2 - 8x - 1)$   
 $= -4x^2 - 1$
- (13)  $(x^2 - 4x + 8) - (7x^2 - 6)$   
 $= -6x^2 - 4x + 14$
- (14)  $(x^2 - 3x - 5) + (-2x^2 - 2x)$   
 $= -x^2 - 5x - 5$
- (15)  $(-6x^2 - 8x - 1) - (7x^2 - x - 6)$   
 $= -13x^2 - 7x + 5$
- (16)  $(4x^2 - x + 8) + (-3x^2 + 4x - 8)$   
 $= x^2 + 3x$
- (17)  $(x^2 - 8x + 6) - (-5x^2 - 5x + 2)$   
 $= 6x^2 - 3x + 4$
- (18)  $(9x^2 + 8x + 1) + (9x^2 + x)$   
 $= 18x^2 + 9x + 1$
- (19)  $(2x^2 + 9x - 6) + (8x^2 + 4x + 7)$   
 $= 10x^2 + 13x + 1$
- (20)  $(-4x^2 - 3x - 9) - (-6x^2 + 5x + 4)$   
 $= 2x^2 - 8x - 13$
- (21)  $(7x^2 + x) - (-x^2 - 4x - 6)$   
 $= 8x^2 + 5x + 6$
- (22)  $(-4x^2 - 4x + 1) - (-4x^2 + 9x - 4)$   
 $= -13x + 5$
- (23)  $(-2x^2 + 9) - (-2x^2 - 6x + 9)$   
 $= 6x$
- (24)  $(-3x^2 - 4x + 5) - (-8x^2 - 2x - 6)$   
 $= 5x^2 - 2x + 11$
- (25)  $(-7x^2 + 8x - 3) - (-4x^2 + 6x + 1)$   
 $= -3x^2 + 2x - 4$
- (26)  $(-7x^2 - 2x - 9) + (3x^2)$   
 $= -4x^2 - 2x - 9$
- (27)  $(-7x^2 + 8x + 5) + (-9x^2 - x + 8)$   
 $= -16x^2 + 7x + 13$
- (28)  $(-5x^2 - 4x - 8) + (8x^2 + 7x)$   
 $= 3x^2 + 3x - 8$
- (29)  $(-2x^2 + 6x + 8) - (9x^2 - 7x + 3)$   
 $= -11x^2 + 13x + 5$
- (30)  $(-7x^2 + x) + (9x^2 + 3x - 5)$   
 $= 2x^2 + 4x - 5$
- (31)  $(3x^2 - 9x - 2) - (-9x^2 - 2x + 6)$   
 $= 12x^2 - 7x - 8$
- (32)  $(-4x^2 - 9x - 8) - (-5x^2 + 6x + 1)$   
 $= x^2 - 15x - 9$
- (33)  $(-x^2 + 3x + 9) - (-7x^2 + 6x)$   
 $= 6x^2 - 3x + 9$
- (34)  $(-8x^2 + 6x - 3) + (-6x^2 - 3x + 2)$   
 $= -14x^2 + 3x - 1$
- (35)  $(9x^2 + 9x - 3) + (5x^2 + 9x)$   
 $= 14x^2 + 18x - 3$
- (36)  $(-6x^2 - 5x - 2) + (-9x^2 - 8x + 5)$   
 $= -15x^2 - 13x + 3$
- (37)  $(-4x^2 + 7x - 3) - (-2x^2 - 5x - 2)$   
 $= -2x^2 + 12x - 1$
- (38)  $(-7x^2 - 4x - 5) + (-4x^2 + x + 5)$   
 $= -11x^2 - 3x$

- (1)  $(2x^2 - x + 7) + (5x^2 + 2x + 8)$   
 $= 7x^2 + x + 15$
- (2)  $(3x^2 + 9x + 1) - (-9x^2 - 3x + 1)$   
 $= 12x^2 + 12x$
- (3)  $(-4x^2 + 3x + 3) - (8x^2 - 5x - 4)$   
 $= -12x^2 + 8x + 7$
- (4)  $(6x^2 + 7x - 3) + (2x^2 + 8x + 1)$   
 $= 8x^2 + 15x - 2$
- (5)  $(6x^2 - 5x - 6) - (9x^2 + x - 9)$   
 $= -3x^2 - 6x + 3$
- (6)  $(-6x^2 - 7x - 5) - (2x^2 - 8x - 5)$   
 $= -8x^2 + x$
- (7)  $(-8x^2 + 5x - 5) - (-9x^2 + 8x - 4)$   
 $= x^2 - 3x - 1$
- (8)  $(7x^2 - 4) - (-9x^2 - 7x + 9)$   
 $= 16x^2 + 7x - 13$
- (9)  $(-6x^2 + 8x + 7) + (-4x^2 - 2x)$   
 $= -10x^2 + 6x + 7$
- (10)  $(9x^2 - 7) - (-4x^2 + 8x)$   
 $= 13x^2 - 8x - 7$
- (11)  $(9x^2 - 3x - 9) + (-3x^2 - 3x - 6)$   
 $= 6x^2 - 6x - 15$
- (12)  $(x^2 - 4x + 4) - (-8x^2 + 7x - 7)$   
 $= 9x^2 - 11x + 11$
- (13)  $(-6x^2 + 8x + 1) + (-7x^2 - 4x + 2)$   
 $= -13x^2 + 4x + 3$
- (14)  $(8x^2 - 4x - 5) - (-3x^2 + 3x - 9)$   
 $= 11x^2 - 7x + 4$
- (15)  $(-5x^2 + 3x) + (3x^2 - 5x + 3)$   
 $= -2x^2 - 2x + 3$
- (16)  $(-6x^2 + 2x + 3) - (-4x^2 - 4x + 8)$   
 $= -2x^2 + 6x - 5$
- (17)  $(-4x^2 - 3x - 5) - (-3x^2 + 9x - 5)$   
 $= -x^2 - 12x$
- (18)  $(7x^2 - 6x - 6) + (-4x^2 - 2x + 7)$   
 $= 3x^2 - 8x + 1$
- (19)  $(8x^2 + x + 9) - (x^2 - 8x + 9)$   
 $= 7x^2 + 9x$
- (20)  $(-9x^2 - 7x + 1) + (2x^2 + x - 7)$   
 $= -7x^2 - 6x - 6$
- (21)  $(7x^2 - 8x) - (2x^2 - 3x - 4)$   
 $= 5x^2 - 5x + 4$
- (22)  $(9x^2 + 8x) - (-2x^2 - x + 3)$   
 $= 11x^2 + 9x - 3$
- (23)  $(-2x^2 - 2x - 1) - (9x^2 - 5x - 5)$   
 $= -11x^2 + 3x + 4$
- (24)  $(6x^2 + 4x + 5) - (-3x^2 + 5x + 5)$   
 $= 9x^2 - x$
- (25)  $(-6x^2 - 2x - 9) - (-5x^2 - 9x + 3)$   
 $= -x^2 + 7x - 12$
- (26)  $(-9x^2 - 9x + 3) + (-9x^2 + 5x + 8)$   
 $= -18x^2 - 4x + 11$
- (27)  $(-7x^2 + 9x - 2) + (9x^2 + 6x - 9)$   
 $= 2x^2 + 15x - 11$
- (28)  $(-7x^2 + 5x - 3) + (7x^2 - 5x)$   
 $= -3$
- (29)  $(-6x^2 - 6x - 3) + (3x^2 - 6x - 1)$   
 $= -3x^2 - 12x - 4$
- (30)  $(-2x^2 - 5x + 9) - (4x^2 - 6x - 3)$   
 $= -6x^2 + x + 12$
- (31)  $(-9x^2 - 7x + 5) + (8x^2 + 2x - 4)$   
 $= -x^2 - 5x + 1$
- (32)  $(-3x^2 - 2x + 4) + (4x^2 - 2)$   
 $= x^2 - 2x + 2$
- (33)  $(-8x^2 + 2) - (3x^2 - 5x - 5)$   
 $= -11x^2 + 5x + 7$
- (34)  $(4x^2 - 7x + 2) + (x^2 - 3x - 5)$   
 $= 5x^2 - 10x - 3$
- (35)  $(-5x^2 + 3x - 1) - (5x^2 - 2x - 8)$   
 $= -10x^2 + 5x + 7$
- (36)  $(-x^2 + 8x) + (-8x^2 - 2x - 7)$   
 $= -9x^2 + 6x - 7$
- (37)  $(5x^2 + 7x - 5) - (-7x^2 - 4x + 8)$   
 $= 12x^2 + 11x - 13$
- (38)  $(-x^2 - 7x - 8) - (-9x^2 + 3x - 8)$   
 $= 8x^2 - 10x$

## 2.2 乗法

1 次多項式同士の積を求める問題。

(1)  $(7x + 4)(6x + 3)$

(16)  $(7x + 4)(3x - 8)$

(31)  $(x - 9)(8x - 7)$

(2)  $(9x - 4)(7x + 5)$

(17)  $(x + 9)(3x + 1)$

(32)  $(2x - 8)(3x - 4)$

(3)  $(5x - 2)(7x + 7)$

(18)  $(5x - 4)(x - 9)$

(33)  $(3x + 8)(5x - 2)$

(4)  $(9x - 6)(4x + 5)$

(19)  $(3x - 3)(3x - 6)$

(34)  $(6x + 9)(6x - 8)$

(5)  $(2x - 7)(4x + 3)$

(20)  $(9x + 1)(6x + 7)$

(35)  $(3x + 7)(3x - 1)$

(6)  $(4x - 5)(2x - 7)$

(21)  $(2x + 6)(4x - 4)$

(36)  $(x + 9)(7x - 2)$

(7)  $(9x - 4)(5x + 6)$

(22)  $(3x - 7)(x + 1)$

(37)  $(x + 3)(8x - 5)$

(8)  $(7x - 9)(3x + 5)$

(23)  $(3x - 3)(9x + 4)$

(38)  $(7x + 2)(4x - 1)$

(9)  $(8x + 2)(4x - 5)$

(24)  $(7x - 2)(9x + 9)$

(39)  $(6x - 5)(x + 4)$

(10)  $(5x + 9)(6x + 3)$

(25)  $(2x + 5)(6x - 9)$

(40)  $(9x - 1)(3x - 7)$

(11)  $(2x + 4)(5x + 4)$

(26)  $(9x + 1)(7x + 5)$

(41)  $(3x + 2)(5x + 7)$

(12)  $(6x - 6)(x - 7)$

(27)  $(x + 1)(5x - 4)$

(42)  $(9x - 5)(7x + 1)$

(13)  $(2x + 3)(4x - 3)$

(28)  $(4x - 4)(9x + 3)$

(43)  $(4x - 6)(3x + 5)$

(14)  $(2x + 7)(5x + 3)$

(29)  $(8x + 6)(2x - 9)$

(44)  $(6x + 7)(9x + 8)$

(15)  $(x + 9)(2x + 1)$

(30)  $(9x - 6)(2x + 3)$

(45)  $(2x + 5)(8x + 5)$

(1)  $(9x + 7)(8x + 5)$

(16)  $(7x + 7)(6x - 4)$

(31)  $(8x - 3)(x + 9)$

(2)  $(4x - 3)(x - 9)$

(17)  $(9x + 1)(8x + 6)$

(32)  $(6x - 2)(5x + 9)$

(3)  $(9x - 1)(8x - 3)$

(18)  $(9x - 1)(9x + 9)$

(33)  $(9x + 1)(5x + 7)$

(4)  $(3x - 2)(6x - 4)$

(19)  $(5x + 7)(8x + 7)$

(34)  $(5x - 4)(2x + 2)$

(5)  $(4x + 2)(4x - 4)$

(20)  $(4x + 7)(5x + 4)$

(35)  $(9x - 8)(2x + 7)$

(6)  $(8x + 9)(6x - 4)$

(21)  $(6x + 5)(3x - 9)$

(36)  $(8x + 9)(3x - 5)$

(7)  $(x + 9)(6x + 1)$

(22)  $(7x + 4)(8x + 2)$

(37)  $(x - 4)(7x + 5)$

(8)  $(9x - 5)(8x - 8)$

(23)  $(2x - 9)(x + 4)$

(38)  $(3x - 2)(6x - 7)$

(9)  $(8x - 4)(9x - 2)$

(24)  $(6x + 9)(6x - 8)$

(39)  $(4x + 7)(9x + 2)$

(10)  $(7x - 7)(4x + 9)$

(25)  $(3x - 9)(2x + 5)$

(40)  $(7x + 6)(x + 6)$

(11)  $(6x + 8)(7x - 3)$

(26)  $(5x - 1)(5x - 7)$

(41)  $(8x - 2)(3x + 3)$

(12)  $(8x + 5)(7x - 4)$

(27)  $(x - 1)(2x - 9)$

(42)  $(9x + 7)(4x - 5)$

(13)  $(6x + 2)(6x + 1)$

(28)  $(3x - 4)(2x - 7)$

(43)  $(2x + 3)(9x - 4)$

(14)  $(9x - 9)(5x - 5)$

(29)  $(6x - 7)(x - 5)$

(44)  $(6x - 4)(3x + 4)$

(15)  $(4x + 2)(7x - 5)$

(30)  $(8x + 8)(5x - 8)$

(45)  $(9x - 6)(x - 3)$

(1)  $(8x + 6)(9x + 5)$

(16)  $(6x - 4)(3x - 9)$

(31)  $(8x - 9)(5x - 4)$

(2)  $(4x + 9)(5x + 2)$

(17)  $(x + 3)(2x + 1)$

(32)  $(7x - 5)(8x + 2)$

(3)  $(3x + 4)(2x - 2)$

(18)  $(x - 9)(2x + 5)$

(33)  $(3x - 9)(5x - 5)$

(4)  $(3x + 1)(5x - 2)$

(19)  $(9x - 1)(2x - 5)$

(34)  $(8x - 9)^2$

(5)  $(4x + 6)(5x + 9)$

(20)  $(8x - 9)(6x + 4)$

(35)  $(9x + 8)(7x - 5)$

(6)  $(6x + 2)(2x + 3)$

(21)  $(7x - 4)(4x - 4)$

(36)  $(4x + 5)(9x - 6)$

(7)  $(9x - 7)(7x + 9)$

(22)  $(2x + 9)(6x - 5)$

(37)  $(4x - 4)(x - 7)$

(8)  $(4x - 1)(9x - 6)$

(23)  $(8x + 6)(2x - 1)$

(38)  $(5x + 1)(6x - 5)$

(9)  $(8x + 3)(2x - 1)$

(24)  $(7x + 9)(3x + 8)$

(39)  $(x - 2)(9x - 8)$

(10)  $(7x - 8)(x - 3)$

(25)  $(7x + 2)(2x - 7)$

(40)  $(x - 5)(2x - 5)$

(11)  $(5x + 7)(9x + 8)$

(26)  $(5x - 7)(7x - 1)$

(41)  $(9x + 2)(2x + 1)$

(12)  $(3x - 9)(8x - 5)$

(27)  $(3x - 7)(8x + 2)$

(42)  $(x - 7)(2x - 5)$

(13)  $(5x + 7)(x - 4)$

(28)  $(x + 9)(9x - 3)$

(43)  $(4x + 3)(x + 9)$

(14)  $(7x - 1)(9x - 1)$

(29)  $(8x + 4)(2x + 2)$

(44)  $(x - 5)(3x - 7)$

(15)  $(8x - 8)(3x - 4)$

(30)  $(4x + 5)(6x - 7)$

(45)  $(9x + 8)(3x - 3)$



(1)  $(8x + 5)(9x + 8)$

(16)  $(2x - 5)(2x - 6)$

(31)  $(2x + 7)(7x - 5)$

(2)  $(2x + 9)(7x - 4)$

(17)  $(9x + 4)(7x + 1)$

(32)  $(8x + 4)(8x + 7)$

(3)  $(7x - 6)(9x + 7)$

(18)  $(8x - 8)(5x + 2)$

(33)  $(2x - 8)(x + 6)$

(4)  $(2x + 8)(4x + 1)$

(19)  $(x + 2)(7x - 8)$

(34)  $(3x - 9)(7x - 4)$

(5)  $(8x - 8)(8x + 5)$

(20)  $(x + 8)(9x - 8)$

(35)  $(2x + 1)(5x - 5)$

(6)  $(2x + 4)(6x + 2)$

(21)  $(8x + 4)(9x + 7)$

(36)  $(8x + 5)(3x + 3)$

(7)  $(4x - 9)(2x - 8)$

(22)  $(8x + 9)(8x + 5)$

(37)  $(2x + 1)(5x - 4)$

(8)  $(x - 5)(6x - 8)$

(23)  $(x - 6)(8x - 8)$

(38)  $(9x + 3)(6x + 8)$

(9)  $(6x + 3)(x - 8)$

(24)  $(2x - 7)(8x + 2)$

(39)  $(2x + 6)(7x - 3)$

(10)  $(9x - 5)(2x - 2)$

(25)  $(9x - 6)(2x - 4)$

(40)  $(5x + 6)(7x + 6)$

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(27)  $(2x + 3)(2x + 9)$

(42)  $(3x + 4)(2x + 7)$

(13)  $(8x - 9)(7x + 8)$

(28)  $(x + 7)(9x + 9)$

(43)  $(7x - 2)(7x + 1)$

(14)  $(8x - 1)(8x - 4)$

(29)  $(7x - 8)(4x + 5)$

(44)  $(2x - 4)(6x - 8)$

(15)  $(8x - 6)(6x - 1)$

(30)  $(8x + 2)(4x + 1)$

(45)  $(3x - 1)(2x + 8)$

(1)  $(6x + 3)(6x - 9)$

(16)  $(2x - 2)(4x - 2)$

(31)  $(4x + 6)(5x + 1)$

(2)  $(2x - 6)(6x - 9)$

(17)  $(3x - 9)(9x + 6)$

(32)  $(9x + 1)(3x - 7)$

(3)  $(x + 1)(x - 3)$

(18)  $(2x + 4)(2x + 3)$

(33)  $(5x + 8)(2x + 9)$

(4)  $(2x - 8)(6x - 1)$

(19)  $(x + 2)(4x - 5)$

(34)  $(2x + 2)(3x - 7)$

(5)  $(2x - 6)(6x + 2)$

(20)  $(x - 4)(4x + 2)$

(35)  $(7x + 2)(7x + 4)$

(6)  $(2x - 8)(5x - 1)$

(21)  $(7x + 6)(8x - 9)$

(36)  $(9x - 9)(2x - 1)$

(7)  $(2x + 9)(4x + 5)$

(22)  $(5x - 9)(5x + 6)$

(37)  $(9x - 3)(6x - 8)$

(8)  $(4x - 2)(x + 9)$

(23)  $(8x + 7)(6x + 8)$

(38)  $(4x + 7)(5x + 4)$

(9)  $(5x + 3)(7x - 6)$

(24)  $(7x - 7)(9x - 9)$

(39)  $(9x + 9)(5x - 8)$

(10)  $(4x + 8)(5x + 1)$

(25)  $(6x - 5)(9x - 4)$

(40)  $(x - 1)(3x + 1)$

(11)  $(7x + 5)(7x - 1)$

(26)  $(4x + 1)(5x - 2)$

(41)  $(x - 6)(4x - 6)$

(12)  $(6x + 8)(9x - 5)$

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(13)  $(2x + 1)(x + 4)$

(28)  $(3x + 7)(5x - 6)$

(43)  $(8x + 2)(7x - 3)$

(14)  $(3x - 8)(4x + 5)$

(29)  $(3x - 2)(6x - 3)$

(44)  $(3x - 7)(4x + 1)$

(15)  $(2x + 6)(7x + 6)$

(30)  $(x - 5)(4x - 3)$

(45)  $(8x + 4)(7x - 1)$

(1)  $(4x + 6)(x + 1)$

(16)  $(2x + 9)(7x + 2)$

(31)  $(4x + 7)(3x + 4)$

(2)  $(8x + 6)(8x + 2)$

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(3)  $(6x - 2)(4x - 8)$

(18)  $(8x + 7)(2x + 1)$

(33)  $(9x - 7)(7x - 4)$

(4)  $(3x - 7)(5x + 4)$

(19)  $(6x - 7)(5x + 8)$

(34)  $(x - 5)(5x + 7)$

(5)  $(5x - 2)(7x + 2)$

(20)  $(x + 9)(x - 4)$

(35)  $(3x - 1)(6x + 5)$

(6)  $(5x + 7)(2x + 6)$

(21)  $(3x - 3)(9x - 6)$

(36)  $(x - 6)(7x + 4)$

(7)  $(4x - 3)(7x - 8)$

(22)  $(2x + 2)(4x + 9)$

(37)  $(8x - 2)(7x - 8)$

(8)  $(6x + 6)(7x - 4)$

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(45)  $(x - 1)(6x + 3)$

(1)  $(6x - 6)(5x - 9)$

(16)  $(5x + 6)(8x + 8)$

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(2)  $(7x + 7)(2x + 4)$

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(19)  $(6x - 3)(4x + 4)$

(34)  $(x + 1)(8x - 9)$

(5)  $(x - 3)(4x + 5)$

(20)  $(x - 2)(2x - 1)$

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(24)  $(7x + 1)(x - 3)$

(39)  $(7x + 8)(4x - 6)$

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(1)  $(3x - 7)(4x + 4)$

(16)  $(7x + 7)(5x - 4)$

(31)  $(3x - 1)(x + 5)$

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(35)  $(x - 2)(6x + 7)$

(6)  $(2x - 2)(5x - 6)$

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(9)  $(4x - 7)(6x + 3)$

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(45)  $(4x + 4)(7x + 4)$

(1)  $(2x + 7)(9x - 2)$

(16)  $(9x + 1)(8x - 4)$

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(15)  $(x + 5)(7x - 7)$

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(45)  $(4x - 4)(5x - 4)$

(1)  $(3x + 1)(7x + 7)$

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(4)  $(5x + 5)(8x - 7)$

(19)  $(5x + 1)(7x + 1)$

(34)  $(x - 3)(8x + 4)$

(5)  $(2x - 9)(7x + 9)$

(20)  $(6x - 3)(7x - 4)$

(35)  $(6x - 7)(6x - 6)$

(6)  $(8x - 7)(5x - 5)$

(21)  $(4x - 6)(5x - 6)$

(36)  $(6x + 1)(5x - 7)$

(7)  $(3x + 1)(5x + 6)$

(22)  $(6x + 6)(3x + 3)$

(37)  $(7x - 6)(4x - 3)$

(8)  $(7x - 6)(6x + 7)$

(23)  $(6x - 2)(7x + 1)$

(38)  $(6x - 2)(5x + 6)$

(9)  $(4x + 2)(3x + 8)$

(24)  $(6x - 9)(5x + 3)$

(39)  $(7x - 9)(5x + 3)$

(10)  $(7x + 8)(6x + 5)$

(25)  $(5x + 1)(3x + 1)$

(40)  $(4x + 3)(x + 7)$

(11)  $(8x - 5)(2x + 6)$

(26)  $(3x - 1)(9x - 8)$

(41)  $(x + 3)(7x - 9)$

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(28)  $(6x - 4)(3x + 4)$

(43)  $(4x - 1)(4x + 8)$

(14)  $(4x + 6)(4x + 2)$

(29)  $(5x + 3)(6x - 5)$

(44)  $(8x - 3)(3x - 1)$

(15)  $(8x - 1)(8x + 1)$

(30)  $(6x - 2)(3x - 9)$

(45)  $(7x + 3)(x - 9)$

(1)  $(4x - 2)(7x + 9)$

(16)  $(8x + 7)(2x - 2)$

(31)  $(9x - 5)(4x - 1)$

(2)  $(5x - 2)(3x - 8)$

(17)  $(2x - 9)(x - 6)$

(32)  $(6x - 9)(2x + 3)$

(3)  $(8x + 7)(5x + 1)$

(18)  $(4x + 5)(8x - 8)$

(33)  $(7x - 3)(5x + 7)$

(4)  $(8x - 4)(8x - 8)$

(19)  $(2x + 4)(2x + 5)$

(34)  $(6x + 6)(3x + 8)$

(5)  $(2x + 4)(4x + 1)$

(20)  $(6x - 6)(x - 3)$

(35)  $(3x - 3)(4x - 4)$

(6)  $(8x - 1)(6x - 7)$

(21)  $(5x + 5)(7x + 8)$

(36)  $(2x + 7)(2x + 4)$

(7)  $(4x - 4)(5x + 6)$

(22)  $(5x + 4)(x + 3)$

(37)  $(4x + 3)(2x + 9)$

(8)  $(7x - 6)(8x + 6)$

(23)  $(8x + 3)(x + 5)$

(38)  $(9x + 8)(8x - 2)$

(9)  $(7x + 5)(4x + 4)$

(24)  $(7x + 2)(5x - 3)$

(39)  $(4x + 6)(7x + 3)$

(10)  $(6x + 3)(5x + 6)$

(25)  $(9x - 8)(9x + 9)$

(40)  $(7x - 1)(7x + 6)$

(11)  $(4x - 8)(x + 6)$

(26)  $(x + 8)(4x - 1)$

(41)  $(9x + 2)(x - 9)$

(12)  $(2x - 4)(6x - 6)$

(27)  $(3x + 6)(x - 5)$

(42)  $(6x - 2)(3x - 6)$

(13)  $(8x + 7)(8x - 3)$

(28)  $(5x - 5)(4x + 8)$

(43)  $(3x + 7)(2x + 4)$

(14)  $(2x + 7)(4x - 1)$

(29)  $(2x - 9)(3x + 6)$

(44)  $(9x - 6)(8x - 2)$

(15)  $(2x - 8)(8x + 2)$

(30)  $(9x - 1)(x - 9)$

(45)  $(2x + 7)(5x + 4)$



(1)  $(2x + 4)(x - 1)$

(16)  $(x - 2)(7x + 3)$

(31)  $(6x + 6)(4x + 3)$

(2)  $(7x - 5)(7x + 9)$

(17)  $(7x - 3)(3x - 2)$

(32)  $(4x + 2)(4x + 9)$

(3)  $(x + 8)^2$

(18)  $(2x + 8)(7x - 9)$

(33)  $(8x + 2)(5x - 5)$

(4)  $(9x + 5)(6x - 4)$

(19)  $(4x + 8)(4x + 7)$

(34)  $(8x - 1)(2x + 1)$

(5)  $(9x - 6)(5x + 6)$

(20)  $(6x - 7)(7x + 4)$

(35)  $(4x - 2)(8x + 8)$

(6)  $(4x - 6)(7x + 3)$

(21)  $(6x + 5)(7x + 9)$

(36)  $(x - 8)(4x - 9)$

(7)  $(8x + 4)(3x + 3)$

(22)  $(4x + 8)(9x - 8)$

(37)  $(4x - 5)(x + 9)$

(8)  $(4x + 7)(8x + 9)$

(23)  $(5x + 7)(4x + 6)$

(38)  $(7x + 5)(8x - 3)$

(9)  $(5x - 4)(6x + 6)$

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(45)  $(4x + 9)(6x - 6)$

(1)  $(6x + 3)(4x + 2)$

(16)  $(3x - 4)(7x + 4)$

(31)  $(3x - 3)(x + 2)$

(2)  $(6x + 6)(2x - 5)$

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(22)  $(x - 1)(6x + 4)$

(37)  $(x + 4)(3x - 7)$

(8)  $(3x - 9)(4x + 5)$

(23)  $(3x - 6)(4x + 9)$

(38)  $(x + 2)(6x + 3)$

(9)  $(5x + 7)(x + 1)$

(24)  $(3x - 6)(8x - 2)$

(39)  $(9x + 2)(9x - 7)$

(10)  $(7x - 2)(8x - 6)$

(25)  $(9x - 2)(x + 9)$

(40)  $(5x + 5)(4x + 9)$

(11)  $(3x - 1)(7x - 3)$

(26)  $(9x + 2)(2x + 5)$

(41)  $(5x - 4)(7x + 7)$

(12)  $(9x - 2)(x - 9)$

(27)  $(3x - 5)(x - 3)$

(42)  $(9x + 5)(6x + 7)$

(13)  $(x - 4)(2x - 4)$

(28)  $(7x - 2)(3x + 8)$

(43)  $(8x + 5)(2x - 2)$

(14)  $(4x - 9)(7x + 6)$

(29)  $(3x - 1)(8x + 3)$

(44)  $(x - 7)(6x - 4)$

(15)  $(4x + 9)(7x - 3)$

(30)  $(8x - 9)(6x - 8)$

(45)  $(8x + 6)(2x - 9)$

(1)  $(6x - 6)(7x + 7)$

(16)  $(9x - 6)(5x - 9)$

(31)  $(2x + 6)(8x + 2)$

(2)  $(7x - 2)(4x + 5)$

(17)  $(3x + 2)^2$

(32)  $(3x + 8)(x + 7)$

(3)  $(4x - 7)(6x + 6)$

(18)  $(8x - 3)(5x - 7)$

(33)  $(x + 2)(3x - 9)$

(4)  $(7x - 7)(x - 4)$

(19)  $(2x + 1)(3x + 2)$

(34)  $(9x + 8)(3x - 4)$

(5)  $(8x + 7)(8x - 6)$

(20)  $(6x + 5)(7x + 2)$

(35)  $(5x - 6)(8x - 3)$

(6)  $(3x - 9)(x - 6)$

(21)  $(9x - 7)(7x + 5)$

(36)  $(3x + 3)(8x + 9)$

(7)  $(9x + 3)(x + 5)$

(22)  $(6x - 3)(6x + 3)$

(37)  $(5x + 7)(8x - 3)$

(8)  $(7x + 5)(7x + 1)$

(23)  $(6x - 9)(9x - 1)$

(38)  $(8x - 5)(5x + 8)$

(9)  $(7x - 6)(x - 2)$

(24)  $(5x - 5)(6x - 2)$

(39)  $(4x + 4)(7x - 6)$

(10)  $(5x - 6)(9x - 2)$

(25)  $(2x + 2)(9x + 8)$

(40)  $(9x + 9)(8x + 7)$

(11)  $(4x - 5)(2x - 9)$

(26)  $(9x - 2)(3x - 3)$

(41)  $(3x - 1)(7x - 1)$

(12)  $(3x - 2)(4x + 4)$

(27)  $(6x - 8)(3x + 5)$

(42)  $(x - 8)(9x - 8)$

(13)  $(2x - 3)(5x - 2)$

(28)  $(7x - 7)(3x - 7)$

(43)  $(3x + 5)(8x + 9)$

(14)  $(2x + 1)(9x - 9)$

(29)  $(8x + 1)(5x + 4)$

(44)  $(5x - 3)(5x - 9)$

(15)  $(3x + 1)(7x + 5)$

(30)  $(9x - 5)(x + 2)$

(45)  $(4x - 6)(3x - 5)$

(1)  $(9x + 9)(9x + 6)$

(16)  $(5x + 6)(6x - 1)$

(31)  $(4x + 7)(3x + 6)$

(2)  $(2x + 9)(5x - 6)$

(17)  $(4x - 8)(5x + 4)$

(32)  $(x - 8)(x + 4)$

(3)  $(x - 3)(3x - 6)$

(18)  $(5x - 4)(2x + 7)$

(33)  $(3x - 7)(6x + 9)$

(4)  $(x - 1)(5x - 3)$

(19)  $(4x + 5)(9x + 7)$

(34)  $(x + 8)(3x - 8)$

(5)  $(8x + 9)(3x - 1)$

(20)  $(8x - 3)(x + 3)$

(35)  $(2x + 3)(4x - 7)$

(6)  $(x + 7)(6x - 1)$

(21)  $(5x - 5)(3x + 5)$

(36)  $(5x - 3)(5x - 5)$

(7)  $(7x - 7)(6x - 5)$

(22)  $(x - 3)(3x - 8)$

(37)  $(3x - 2)(4x + 3)$

(8)  $(6x - 7)(3x + 7)$

(23)  $(3x - 8)(x + 6)$

(38)  $(5x + 3)(4x - 3)$

(9)  $(4x + 4)(9x + 5)$

(24)  $(7x - 8)(3x - 1)$

(39)  $(4x + 2)(2x + 7)$

(10)  $(4x + 9)(4x - 2)$

(25)  $(7x - 1)(6x - 7)$

(40)  $(6x + 9)(5x + 5)$

(11)  $(2x - 9)(7x - 5)$

(26)  $(5x - 6)(4x + 4)$

(41)  $(4x + 4)(4x - 6)$

(12)  $(4x - 4)(8x + 4)$

(27)  $(7x - 5)(5x + 2)$

(42)  $(3x + 7)(5x - 8)$

(13)  $(9x - 1)(x + 3)$

(28)  $(6x - 8)(6x + 1)$

(43)  $(8x + 9)(6x - 4)$

(14)  $(8x + 9)(9x + 4)$

(29)  $(9x + 8)(8x + 6)$

(44)  $(9x + 9)(6x - 7)$

(15)  $(7x - 1)(7x + 7)$

(30)  $(3x - 8)(9x - 2)$

(45)  $(9x - 8)^2$

(1)  $(4x - 1)(8x - 9)$

(16)  $(3x - 5)(5x - 1)$

(31)  $(4x - 3)(5x - 8)$

(2)  $(4x + 8)(6x - 5)$

(17)  $(5x + 4)(7x - 9)$

(32)  $(5x + 4)(9x - 8)$

(3)  $(6x - 4)(8x + 3)$

(18)  $(6x - 7)(4x + 4)$

(33)  $(6x + 1)(3x + 9)$

(4)  $(2x - 5)(9x - 5)$

(19)  $(8x - 9)(6x - 2)$

(34)  $(9x + 1)(9x + 9)$

(5)  $(7x - 4)(8x + 9)$

(20)  $(4x - 9)(2x + 9)$

(35)  $(4x - 3)(6x + 5)$

(6)  $(6x - 3)(4x + 7)$

(21)  $(9x - 3)(6x + 6)$

(36)  $(x + 8)(5x - 1)$

(7)  $(6x - 5)(7x + 7)$

(22)  $(2x - 4)(2x + 3)$

(37)  $(6x + 3)(9x - 3)$

(8)  $(5x + 5)(9x - 5)$

(23)  $(7x - 6)(3x + 2)$

(38)  $(8x - 8)(7x + 7)$

(9)  $(6x - 9)(2x - 8)$

(24)  $(9x + 6)(5x - 9)$

(39)  $(3x - 4)(2x + 2)$

(10)  $(8x + 1)(7x - 8)$

(25)  $(6x + 6)(x - 1)$

(40)  $(3x - 2)(6x - 3)$

(11)  $(6x + 4)(8x - 9)$

(26)  $(8x + 3)(6x + 6)$

(41)  $(7x + 1)(3x + 5)$

(12)  $(9x - 9)(3x - 5)$

(27)  $(3x - 9)(8x - 2)$

(42)  $(4x + 5)(3x + 8)$

(13)  $(6x - 9)(8x - 9)$

(28)  $(5x + 7)(6x + 8)$

(43)  $(2x + 2)(8x + 8)$

(14)  $(4x + 6)(3x - 6)$

(29)  $(5x - 8)(x + 3)$

(44)  $(x - 1)(3x - 5)$

(15)  $(4x + 1)(9x - 1)$

(30)  $(2x + 9)(2x + 8)$

(45)  $(4x + 6)(5x + 8)$

(1)  $(5x + 3)(6x - 9)$

(16)  $(x + 1)(2x + 6)$

(31)  $(3x - 3)(4x + 5)$

(2)  $(x + 4)(8x - 5)$

(17)  $(3x - 2)(4x + 9)$

(32)  $(3x - 5)(7x + 8)$

(3)  $(3x + 6)(7x + 4)$

(18)  $(3x - 9)(4x - 9)$

(33)  $(4x + 7)(9x - 1)$

(4)  $(9x + 6)(8x + 7)$

(19)  $(4x + 5)(6x - 5)$

(34)  $(5x + 7)(6x - 7)$

(5)  $(2x - 6)(2x - 1)$

(20)  $(9x - 9)(8x - 4)$

(35)  $(2x + 8)(x + 4)$

(6)  $(x - 8)(x - 1)$

(21)  $(4x + 1)(3x - 1)$

(36)  $(4x - 5)(4x - 8)$

(7)  $(3x - 6)(4x - 6)$

(22)  $(x - 8)(7x - 6)$

(37)  $(8x + 3)(3x + 5)$

(8)  $(7x + 1)(x - 1)$

(23)  $(9x - 1)(5x - 9)$

(38)  $(2x - 1)(6x - 3)$

(9)  $(x + 1)(3x - 4)$

(24)  $(6x - 9)(9x + 4)$

(39)  $(7x + 2)(3x - 9)$

(10)  $(7x - 1)(6x + 3)$

(25)  $(5x - 7)(8x + 8)$

(40)  $(8x + 1)(7x + 5)$

(11)  $(4x + 8)(6x + 7)$

(26)  $(7x + 3)(3x + 4)$

(41)  $(x - 8)(6x - 1)$

(12)  $(5x + 4)(x - 7)$

(27)  $(7x + 7)(9x + 8)$

(42)  $(7x + 3)(5x + 9)$

(13)  $(3x - 4)(5x - 9)$

(28)  $(6x + 9)(2x - 5)$

(43)  $(2x - 7)(2x - 2)$

(14)  $(9x - 4)(8x + 6)$

(29)  $(5x + 1)(7x + 6)$

(44)  $(7x + 9)(5x - 3)$

(15)  $(6x - 9)(2x + 2)$

(30)  $(9x - 6)(7x + 3)$

(45)  $(6x + 1)(6x + 8)$

(1)  $(4x + 7)(3x - 6)$

(16)  $(9x - 4)(4x + 4)$

(31)  $(8x - 2)(9x - 5)$

(2)  $(3x - 8)(9x - 1)$

(17)  $(8x + 1)(x + 4)$

(32)  $(8x - 5)(x + 8)$

(3)  $(7x - 5)(6x - 5)$

(18)  $(9x + 4)(x + 5)$

(33)  $(x - 6)(7x + 3)$

(4)  $(6x + 1)(x + 6)$

(19)  $(x + 5)(9x + 1)$

(34)  $(2x + 4)(7x - 3)$

(5)  $(6x + 9)(5x - 4)$

(20)  $(9x - 2)(4x - 8)$

(35)  $(4x - 5)(6x - 4)$

(6)  $(3x - 1)(x + 9)$

(21)  $(4x + 4)(x - 6)$

(36)  $(3x + 2)(x - 6)$

(7)  $(7x - 3)(5x + 9)$

(22)  $(5x - 7)(4x + 1)$

(37)  $(7x + 8)(x + 9)$

(8)  $(7x - 3)(8x + 1)$

(23)  $(8x - 9)(7x - 1)$

(38)  $(3x - 6)(3x + 2)$

(9)  $(5x + 7)(7x - 5)$

(24)  $(6x + 9)(8x + 7)$

(39)  $(3x - 2)(3x - 8)$

(10)  $(9x + 2)(3x - 3)$

(25)  $(3x + 5)(x + 1)$

(40)  $(x - 9)(7x + 8)$

(11)  $(8x + 3)(4x - 7)$

(26)  $(9x - 6)(2x + 3)$

(41)  $(3x + 2)(8x - 1)$

(12)  $(6x + 1)(4x + 7)$

(27)  $(2x + 3)(6x - 2)$

(42)  $(3x - 1)(7x - 9)$

(13)  $(6x - 7)(x - 1)$

(28)  $(4x - 2)(x - 5)$

(43)  $(3x + 5)(2x + 5)$

(14)  $(2x + 9)(9x + 1)$

(29)  $(9x - 5)(3x + 8)$

(44)  $(x + 8)(9x + 8)$

(15)  $(x - 8)(x + 5)$

(30)  $(x - 7)(7x - 1)$

(45)  $(x - 9)(8x + 3)$



(1)  $(4x - 7)(7x - 2)$

(16)  $(3x + 1)(x + 4)$

(31)  $(9x - 9)(2x + 3)$

(2)  $(x - 2)(4x + 3)$

(17)  $(4x + 1)(x - 8)$

(32)  $(5x - 3)(9x + 8)$

(3)  $(9x + 6)(3x + 1)$

(18)  $(7x - 4)(x + 7)$

(33)  $(8x + 8)(4x - 5)$

(4)  $(2x + 7)(5x - 4)$

(19)  $(6x + 2)(2x - 6)$

(34)  $(8x + 4)(4x - 2)$

(5)  $(5x - 9)(6x - 4)$

(20)  $(3x + 7)(3x + 6)$

(35)  $(8x + 9)(x + 6)$

(6)  $(5x - 1)(9x + 6)$

(21)  $(7x - 8)(2x + 5)$

(36)  $(2x - 3)(x + 8)$

(7)  $(4x + 3)(8x + 3)$

(22)  $(3x + 3)(8x - 5)$

(37)  $(2x + 5)(9x - 8)$

(8)  $(2x - 7)(6x + 7)$

(23)  $(8x + 1)(x + 6)$

(38)  $(2x + 5)(4x - 5)$

(9)  $(2x + 8)(7x + 4)$

(24)  $(9x - 9)(9x + 4)$

(39)  $(6x - 2)(9x + 8)$

(10)  $(4x + 9)(5x - 7)$

(25)  $(7x - 7)(3x - 4)$

(40)  $(7x + 2)(6x - 2)$

(11)  $(7x - 2)(3x - 4)$

(26)  $(5x + 1)(3x - 2)$

(41)  $(2x - 7)(8x + 6)$

(12)  $(2x - 3)(3x + 1)$

(27)  $(2x + 2)(7x + 5)$

(42)  $(x + 2)(6x - 7)$

(13)  $(2x + 4)(3x - 9)$

(28)  $(3x + 1)(x + 1)$

(43)  $(9x - 2)(5x - 4)$

(14)  $(6x - 3)(5x - 7)$

(29)  $(7x + 3)(5x - 8)$

(44)  $(5x - 5)(8x - 3)$

(15)  $(9x + 3)(x - 3)$

(30)  $(2x + 7)(3x - 2)$

(45)  $(3x - 7)(8x - 7)$

$$(1) (7x+4)(6x+3) \\ = 42x^2 + 45x + 12$$

$$(2) (9x-4)(7x+5) \\ = 63x^2 + 17x - 20$$

$$(3) (5x-2)(7x+7) \\ = 35x^2 + 21x - 14$$

$$(4) (9x-6)(4x+5) \\ = 36x^2 + 21x - 30$$

$$(5) (2x-7)(4x+3) \\ = 8x^2 - 22x - 21$$

$$(6) (4x-5)(2x-7) \\ = 8x^2 - 38x + 35$$

$$(7) (9x-4)(5x+6) \\ = 45x^2 + 34x - 24$$

$$(8) (7x-9)(3x+5) \\ = 21x^2 + 8x - 45$$

$$(9) (8x+2)(4x-5) \\ = 32x^2 - 32x - 10$$

$$(10) (5x+9)(6x+3) \\ = 30x^2 + 69x + 27$$

$$(11) (2x+4)(5x+4) \\ = 10x^2 + 28x + 16$$

$$(12) (6x-6)(x-7) \\ = 6x^2 - 48x + 42$$

$$(13) (2x+3)(4x-3) \\ = 8x^2 + 6x - 9$$

$$(14) (2x+7)(5x+3) \\ = 10x^2 + 41x + 21$$

$$(15) (x+9)(2x+1) \\ = 2x^2 + 19x + 9$$

$$(16) (7x+4)(3x-8) \\ = 21x^2 - 44x - 32$$

$$(17) (x+9)(3x+1) \\ = 3x^2 + 28x + 9$$

$$(18) (5x-4)(x-9) \\ = 5x^2 - 49x + 36$$

$$(19) (3x-3)(3x-6) \\ = 9x^2 - 27x + 18$$

$$(20) (9x+1)(6x+7) \\ = 54x^2 + 69x + 7$$

$$(21) (2x+6)(4x-4) \\ = 8x^2 + 16x - 24$$

$$(22) (3x-7)(x+1) \\ = 3x^2 - 4x - 7$$

$$(23) (3x-3)(9x+4) \\ = 27x^2 - 15x - 12$$

$$(24) (7x-2)(9x+9) \\ = 63x^2 + 45x - 18$$

$$(25) (2x+5)(6x-9) \\ = 12x^2 + 12x - 45$$

$$(26) (9x+1)(7x+5) \\ = 63x^2 + 52x + 5$$

$$(27) (x+1)(5x-4) \\ = 5x^2 + x - 4$$

$$(28) (4x-4)(9x+3) \\ = 36x^2 - 24x - 12$$

$$(29) (8x+6)(2x-9) \\ = 16x^2 - 60x - 54$$

$$(30) (9x-6)(2x+3) \\ = 18x^2 + 15x - 18$$

$$(31) (x-9)(8x-7) \\ = 8x^2 - 79x + 63$$

$$(32) (2x-8)(3x-4) \\ = 6x^2 - 32x + 32$$

$$(33) (3x+8)(5x-2) \\ = 15x^2 + 34x - 16$$

$$(34) (6x+9)(6x-8) \\ = 36x^2 + 6x - 72$$

$$(35) (3x+7)(3x-1) \\ = 9x^2 + 18x - 7$$

$$(36) (x+9)(7x-2) \\ = 7x^2 + 61x - 18$$

$$(37) (x+3)(8x-5) \\ = 8x^2 + 19x - 15$$

$$(38) (7x+2)(4x-1) \\ = 28x^2 + x - 2$$

$$(39) (6x-5)(x+4) \\ = 6x^2 + 19x - 20$$

$$(40) (9x-1)(3x-7) \\ = 27x^2 - 66x + 7$$

$$(41) (3x+2)(5x+7) \\ = 15x^2 + 31x + 14$$

$$(42) (9x-5)(7x+1) \\ = 63x^2 - 26x - 5$$

$$(43) (4x-6)(3x+5) \\ = 12x^2 + 2x - 30$$

$$(44) (6x+7)(9x+8) \\ = 54x^2 + 111x + 56$$

$$(45) (2x+5)(8x+5) \\ = 16x^2 + 50x + 25$$

$$(1) (9x+7)(8x+5) \\ = 72x^2 + 101x + 35$$

$$(2) (4x-3)(x-9) \\ = 4x^2 - 39x + 27$$

$$(3) (9x-1)(8x-3) \\ = 72x^2 - 35x + 3$$

$$(4) (3x-2)(6x-4) \\ = 18x^2 - 24x + 8$$

$$(5) (4x+2)(4x-4) \\ = 16x^2 - 8x - 8$$

$$(6) (8x+9)(6x-4) \\ = 48x^2 + 22x - 36$$

$$(7) (x+9)(6x+1) \\ = 6x^2 + 55x + 9$$

$$(8) (9x-5)(8x-8) \\ = 72x^2 - 112x + 40$$

$$(9) (8x-4)(9x-2) \\ = 72x^2 - 52x + 8$$

$$(10) (7x-7)(4x+9) \\ = 28x^2 + 35x - 63$$

$$(11) (6x+8)(7x-3) \\ = 42x^2 + 38x - 24$$

$$(12) (8x+5)(7x-4) \\ = 56x^2 + 3x - 20$$

$$(13) (6x+2)(6x+1) \\ = 36x^2 + 18x + 2$$

$$(14) (9x-9)(5x-5) \\ = 45x^2 - 90x + 45$$

$$(15) (4x+2)(7x-5) \\ = 28x^2 - 6x - 10$$

$$(16) (7x+7)(6x-4) \\ = 42x^2 + 14x - 28$$

$$(17) (9x+1)(8x+6) \\ = 72x^2 + 62x + 6$$

$$(18) (9x-1)(9x+9) \\ = 81x^2 + 72x - 9$$

$$(19) (5x+7)(8x+7) \\ = 40x^2 + 91x + 49$$

$$(20) (4x+7)(5x+4) \\ = 20x^2 + 51x + 28$$

$$(21) (6x+5)(3x-9) \\ = 18x^2 - 39x - 45$$

$$(22) (7x+4)(8x+2) \\ = 56x^2 + 46x + 8$$

$$(23) (2x-9)(x+4) \\ = 2x^2 - x - 36$$

$$(24) (6x+9)(6x-8) \\ = 36x^2 + 6x - 72$$

$$(25) (3x-9)(2x+5) \\ = 6x^2 - 3x - 45$$

$$(26) (5x-1)(5x-7) \\ = 25x^2 - 40x + 7$$

$$(27) (x-1)(2x-9) \\ = 2x^2 - 11x + 9$$

$$(28) (3x-4)(2x-7) \\ = 6x^2 - 29x + 28$$

$$(29) (6x-7)(x-5) \\ = 6x^2 - 37x + 35$$

$$(30) (8x+8)(5x-8) \\ = 40x^2 - 24x - 64$$

$$(31) (8x-3)(x+9) \\ = 8x^2 + 69x - 27$$

$$(32) (6x-2)(5x+9) \\ = 30x^2 + 44x - 18$$

$$(33) (9x+1)(5x+7) \\ = 45x^2 + 68x + 7$$

$$(34) (5x-4)(2x+2) \\ = 10x^2 + 2x - 8$$

$$(35) (9x-8)(2x+7) \\ = 18x^2 + 47x - 56$$

$$(36) (8x+9)(3x-5) \\ = 24x^2 - 13x - 45$$

$$(37) (x-4)(7x+5) \\ = 7x^2 - 23x - 20$$

$$(38) (3x-2)(6x-7) \\ = 18x^2 - 33x + 14$$

$$(39) (4x+7)(9x+2) \\ = 36x^2 + 71x + 14$$

$$(40) (7x+6)(x+6) \\ = 7x^2 + 48x + 36$$

$$(41) (8x-2)(3x+3) \\ = 24x^2 + 18x - 6$$

$$(42) (9x+7)(4x-5) \\ = 36x^2 - 17x - 35$$

$$(43) (2x+3)(9x-4) \\ = 18x^2 + 19x - 12$$

$$(44) (6x-4)(3x+4) \\ = 18x^2 + 12x - 16$$

$$(45) (9x-6)(x-3) \\ = 9x^2 - 33x + 18$$

$$(1) (8x+6)(9x+5) \\ = 72x^2 + 94x + 30$$

$$(16) (6x-4)(3x-9) \\ = 18x^2 - 66x + 36$$

$$(31) (8x-9)(5x-4) \\ = 40x^2 - 77x + 36$$

$$(2) (4x+9)(5x+2) \\ = 20x^2 + 53x + 18$$

$$(17) (x+3)(2x+1) \\ = 2x^2 + 7x + 3$$

$$(32) (7x-5)(8x+2) \\ = 56x^2 - 26x - 10$$

$$(3) (3x+4)(2x-2) \\ = 6x^2 + 2x - 8$$

$$(18) (x-9)(2x+5) \\ = 2x^2 - 13x - 45$$

$$(33) (3x-9)(5x-5) \\ = 15x^2 - 60x + 45$$

$$(4) (3x+1)(5x-2) \\ = 15x^2 - x - 2$$

$$(19) (9x-1)(2x-5) \\ = 18x^2 - 47x + 5$$

$$(34) (8x-9)^2 \\ = 64x^2 - 144x + 81$$

$$(5) (4x+6)(5x+9) \\ = 20x^2 + 66x + 54$$

$$(20) (8x-9)(6x+4) \\ = 48x^2 - 22x - 36$$

$$(35) (9x+8)(7x-5) \\ = 63x^2 + 11x - 40$$

$$(6) (6x+2)(2x+3) \\ = 12x^2 + 22x + 6$$

$$(21) (7x-4)(4x-4) \\ = 28x^2 - 44x + 16$$

$$(36) (4x+5)(9x-6) \\ = 36x^2 + 21x - 30$$

$$(7) (9x-7)(7x+9) \\ = 63x^2 + 32x - 63$$

$$(22) (2x+9)(6x-5) \\ = 12x^2 + 44x - 45$$

$$(37) (4x-4)(x-7) \\ = 4x^2 - 32x + 28$$

$$(8) (4x-1)(9x-6) \\ = 36x^2 - 33x + 6$$

$$(23) (8x+6)(2x-1) \\ = 16x^2 + 4x - 6$$

$$(38) (5x+1)(6x-5) \\ = 30x^2 - 19x - 5$$

$$(9) (8x+3)(2x-1) \\ = 16x^2 - 2x - 3$$

$$(24) (7x+9)(3x+8) \\ = 21x^2 + 83x + 72$$

$$(39) (x-2)(9x-8) \\ = 9x^2 - 26x + 16$$

$$(10) (7x-8)(x-3) \\ = 7x^2 - 29x + 24$$

$$(25) (7x+2)(2x-7) \\ = 14x^2 - 45x - 14$$

$$(40) (x-5)(2x-5) \\ = 2x^2 - 15x + 25$$

$$(11) (5x+7)(9x+8) \\ = 45x^2 + 103x + 56$$

$$(26) (5x-7)(7x-1) \\ = 35x^2 - 54x + 7$$

$$(41) (9x+2)(2x+1) \\ = 18x^2 + 13x + 2$$

$$(12) (3x-9)(8x-5) \\ = 24x^2 - 87x + 45$$

$$(27) (3x-7)(8x+2) \\ = 24x^2 - 50x - 14$$

$$(42) (x-7)(2x-5) \\ = 2x^2 - 19x + 35$$

$$(13) (5x+7)(x-4) \\ = 5x^2 - 13x - 28$$

$$(28) (x+9)(9x-3) \\ = 9x^2 + 78x - 27$$

$$(43) (4x+3)(x+9) \\ = 4x^2 + 39x + 27$$

$$(14) (7x-1)(9x-1) \\ = 63x^2 - 16x + 1$$

$$(29) (8x+4)(2x+2) \\ = 16x^2 + 24x + 8$$

$$(44) (x-5)(3x-7) \\ = 3x^2 - 22x + 35$$

$$(15) (8x-8)(3x-4) \\ = 24x^2 - 56x + 32$$

$$(30) (4x+5)(6x-7) \\ = 24x^2 + 2x - 35$$

$$(45) (9x+8)(3x-3) \\ = 27x^2 - 3x - 24$$

$$(1) (8x+5)(9x+8) \\ = 72x^2 + 109x + 40$$

$$(16) (2x-5)(2x-6) \\ = 4x^2 - 22x + 30$$

$$(31) (2x+7)(7x-5) \\ = 14x^2 + 39x - 35$$

$$(2) (2x+9)(7x-4) \\ = 14x^2 + 55x - 36$$

$$(17) (9x+4)(7x+1) \\ = 63x^2 + 37x + 4$$

$$(32) (8x+4)(8x+7) \\ = 64x^2 + 88x + 28$$

$$(3) (7x-6)(9x+7) \\ = 63x^2 - 5x - 42$$

$$(18) (8x-8)(5x+2) \\ = 40x^2 - 24x - 16$$

$$(33) (2x-8)(x+6) \\ = 2x^2 + 4x - 48$$

$$(4) (2x+8)(4x+1) \\ = 8x^2 + 34x + 8$$

$$(19) (x+2)(7x-8) \\ = 7x^2 + 6x - 16$$

$$(34) (3x-9)(7x-4) \\ = 21x^2 - 75x + 36$$

$$(5) (8x-8)(8x+5) \\ = 64x^2 - 24x - 40$$

$$(20) (x+8)(9x-8) \\ = 9x^2 + 64x - 64$$

$$(35) (2x+1)(5x-5) \\ = 10x^2 - 5x - 5$$

$$(6) (2x+4)(6x+2) \\ = 12x^2 + 28x + 8$$

$$(21) (8x+4)(9x+7) \\ = 72x^2 + 92x + 28$$

$$(36) (8x+5)(3x+3) \\ = 24x^2 + 39x + 15$$

$$(7) (4x-9)(2x-8) \\ = 8x^2 - 50x + 72$$

$$(22) (8x+9)(8x+5) \\ = 64x^2 + 112x + 45$$

$$(37) (2x+1)(5x-4) \\ = 10x^2 - 3x - 4$$

$$(8) (x-5)(6x-8) \\ = 6x^2 - 38x + 40$$

$$(23) (x-6)(8x-8) \\ = 8x^2 - 56x + 48$$

$$(38) (9x+3)(6x+8) \\ = 54x^2 + 90x + 24$$

$$(9) (6x+3)(x-8) \\ = 6x^2 - 45x - 24$$

$$(24) (2x-7)(8x+2) \\ = 16x^2 - 52x - 14$$

$$(39) (2x+6)(7x-3) \\ = 14x^2 + 36x - 18$$

$$(10) (9x-5)(2x-2) \\ = 18x^2 - 28x + 10$$

$$(25) (9x-6)(2x-4) \\ = 18x^2 - 48x + 24$$

$$(40) (5x+6)(7x+6) \\ = 35x^2 + 72x + 36$$

$$(11) (6x+8)(2x+7) \\ = 12x^2 + 58x + 56$$

$$(26) (8x+2)(3x-7) \\ = 24x^2 - 50x - 14$$

$$(41) (7x+6)(5x+9) \\ = 35x^2 + 93x + 54$$

$$(12) (9x-7)(6x-4) \\ = 54x^2 - 78x + 28$$

$$(27) (2x+3)(2x+9) \\ = 4x^2 + 24x + 27$$

$$(42) (3x+4)(2x+7) \\ = 6x^2 + 29x + 28$$

$$(13) (8x-9)(7x+8) \\ = 56x^2 + x - 72$$

$$(28) (x+7)(9x+9) \\ = 9x^2 + 72x + 63$$

$$(43) (7x-2)(7x+1) \\ = 49x^2 - 7x - 2$$

$$(14) (8x-1)(8x-4) \\ = 64x^2 - 40x + 4$$

$$(29) (7x-8)(4x+5) \\ = 28x^2 + 3x - 40$$

$$(44) (2x-4)(6x-8) \\ = 12x^2 - 40x + 32$$

$$(15) (8x-6)(6x-1) \\ = 48x^2 - 44x + 6$$

$$(30) (8x+2)(4x+1) \\ = 32x^2 + 16x + 2$$

$$(45) (3x-1)(2x+8) \\ = 6x^2 + 22x - 8$$

$$(1) (6x+3)(6x-9) \\ = 36x^2 - 36x - 27$$

$$(2) (2x-6)(6x-9) \\ = 12x^2 - 54x + 54$$

$$(3) (x+1)(x-3) \\ = x^2 - 2x - 3$$

$$(4) (2x-8)(6x-1) \\ = 12x^2 - 50x + 8$$

$$(5) (2x-6)(6x+2) \\ = 12x^2 - 32x - 12$$

$$(6) (2x-8)(5x-1) \\ = 10x^2 - 42x + 8$$

$$(7) (2x+9)(4x+5) \\ = 8x^2 + 46x + 45$$

$$(8) (4x-2)(x+9) \\ = 4x^2 + 34x - 18$$

$$(9) (5x+3)(7x-6) \\ = 35x^2 - 9x - 18$$

$$(10) (4x+8)(5x+1) \\ = 20x^2 + 44x + 8$$

$$(11) (7x+5)(7x-1) \\ = 49x^2 + 28x - 5$$

$$(12) (6x+8)(9x-5) \\ = 54x^2 + 42x - 40$$

$$(13) (2x+1)(x+4) \\ = 2x^2 + 9x + 4$$

$$(14) (3x-8)(4x+5) \\ = 12x^2 - 17x - 40$$

$$(15) (2x+6)(7x+6) \\ = 14x^2 + 54x + 36$$

$$(16) (2x-2)(4x-2) \\ = 8x^2 - 12x + 4$$

$$(17) (3x-9)(9x+6) \\ = 27x^2 - 63x - 54$$

$$(18) (2x+4)(2x+3) \\ = 4x^2 + 14x + 12$$

$$(19) (x+2)(4x-5) \\ = 4x^2 + 3x - 10$$

$$(20) (x-4)(4x+2) \\ = 4x^2 - 14x - 8$$

$$(21) (7x+6)(8x-9) \\ = 56x^2 - 15x - 54$$

$$(22) (5x-9)(5x+6) \\ = 25x^2 - 15x - 54$$

$$(23) (8x+7)(6x+8) \\ = 48x^2 + 106x + 56$$

$$(24) (7x-7)(9x-9) \\ = 63x^2 - 126x + 63$$

$$(25) (6x-5)(9x-4) \\ = 54x^2 - 69x + 20$$

$$(26) (4x+1)(5x-2) \\ = 20x^2 - 3x - 2$$

$$(27) (8x-9)(8x-3) \\ = 64x^2 - 96x + 27$$

$$(28) (3x+7)(5x-6) \\ = 15x^2 + 17x - 42$$

$$(29) (3x-2)(6x-3) \\ = 18x^2 - 21x + 6$$

$$(30) (x-5)(4x-3) \\ = 4x^2 - 23x + 15$$

$$(31) (4x+6)(5x+1) \\ = 20x^2 + 34x + 6$$

$$(32) (9x+1)(3x-7) \\ = 27x^2 - 60x - 7$$

$$(33) (5x+8)(2x+9) \\ = 10x^2 + 61x + 72$$

$$(34) (2x+2)(3x-7) \\ = 6x^2 - 8x - 14$$

$$(35) (7x+2)(7x+4) \\ = 49x^2 + 42x + 8$$

$$(36) (9x-9)(2x-1) \\ = 18x^2 - 27x + 9$$

$$(37) (9x-3)(6x-8) \\ = 54x^2 - 90x + 24$$

$$(38) (4x+7)(5x+4) \\ = 20x^2 + 51x + 28$$

$$(39) (9x+9)(5x-8) \\ = 45x^2 - 27x - 72$$

$$(40) (x-1)(3x+1) \\ = 3x^2 - 2x - 1$$

$$(41) (x-6)(4x-6) \\ = 4x^2 - 30x + 36$$

$$(42) (9x+5)(6x-3) \\ = 54x^2 + 3x - 15$$

$$(43) (8x+2)(7x-3) \\ = 56x^2 - 10x - 6$$

$$(44) (3x-7)(4x+1) \\ = 12x^2 - 25x - 7$$

$$(45) (8x+4)(7x-1) \\ = 56x^2 + 20x - 4$$

$$(1) (4x+6)(x+1) \\ = 4x^2 + 10x + 6$$

$$(2) (8x+6)(8x+2) \\ = 64x^2 + 64x + 12$$

$$(3) (6x-2)(4x-8) \\ = 24x^2 - 56x + 16$$

$$(4) (3x-7)(5x+4) \\ = 15x^2 - 23x - 28$$

$$(5) (5x-2)(7x+2) \\ = 35x^2 - 4x - 4$$

$$(6) (5x+7)(2x+6) \\ = 10x^2 + 44x + 42$$

$$(7) (4x-3)(7x-8) \\ = 28x^2 - 53x + 24$$

$$(8) (6x+6)(7x-4) \\ = 42x^2 + 18x - 24$$

$$(9) (9x-9)(8x-7) \\ = 72x^2 - 135x + 63$$

$$(10) (x-8)(7x+3) \\ = 7x^2 - 53x - 24$$

$$(11) (x-8)(5x+8) \\ = 5x^2 - 32x - 64$$

$$(12) (3x-5)(3x-1) \\ = 9x^2 - 18x + 5$$

$$(13) (8x-2)(2x+7) \\ = 16x^2 + 52x - 14$$

$$(14) (4x+4)(7x-1) \\ = 28x^2 + 24x - 4$$

$$(15) (2x-6)(3x+4) \\ = 6x^2 - 10x - 24$$

$$(16) (2x+9)(7x+2) \\ = 14x^2 + 67x + 18$$

$$(17) (5x-9)(9x+2) \\ = 45x^2 - 71x - 18$$

$$(18) (8x+7)(2x+1) \\ = 16x^2 + 22x + 7$$

$$(19) (6x-7)(5x+8) \\ = 30x^2 + 13x - 56$$

$$(20) (x+9)(x-4) \\ = x^2 + 5x - 36$$

$$(21) (3x-3)(9x-6) \\ = 27x^2 - 45x + 18$$

$$(22) (2x+2)(4x+9) \\ = 8x^2 + 26x + 18$$

$$(23) (3x+4)(x-8) \\ = 3x^2 - 20x - 32$$

$$(24) (8x+6)(5x-8) \\ = 40x^2 - 34x - 48$$

$$(25) (6x-7)(5x+4) \\ = 30x^2 - 11x - 28$$

$$(26) (4x+1)(4x-5) \\ = 16x^2 - 16x - 5$$

$$(27) (5x+1)(3x-9) \\ = 15x^2 - 42x - 9$$

$$(28) (5x-7)(8x-8) \\ = 40x^2 - 96x + 56$$

$$(29) (x-5)(2x+6) \\ = 2x^2 - 4x - 30$$

$$(30) (3x+6)(2x+4) \\ = 6x^2 + 24x + 24$$

$$(31) (4x+7)(3x+4) \\ = 12x^2 + 37x + 28$$

$$(32) (6x+8)(8x+6) \\ = 48x^2 + 100x + 48$$

$$(33) (9x-7)(7x-4) \\ = 63x^2 - 85x + 28$$

$$(34) (x-5)(5x+7) \\ = 5x^2 - 18x - 35$$

$$(35) (3x-1)(6x+5) \\ = 18x^2 + 9x - 5$$

$$(36) (x-6)(7x+4) \\ = 7x^2 - 38x - 24$$

$$(37) (8x-2)(7x-8) \\ = 56x^2 - 78x + 16$$

$$(38) (7x-7)(3x+6) \\ = 21x^2 + 21x - 42$$

$$(39) (6x+2)(3x-1) \\ = 18x^2 - 2$$

$$(40) (3x-7)(7x+5) \\ = 21x^2 - 34x - 35$$

$$(41) (9x-6)(6x-6) \\ = 54x^2 - 90x + 36$$

$$(42) (5x-2)(2x-6) \\ = 10x^2 - 34x + 12$$

$$(43) (8x-4)(x-7) \\ = 8x^2 - 60x + 28$$

$$(44) (4x+1)(x+5) \\ = 4x^2 + 21x + 5$$

$$(45) (x-1)(6x+3) \\ = 6x^2 - 3x - 3$$

$$(1) (6x - 6)(5x - 9) \\ = 30x^2 - 84x + 54$$

$$(2) (7x + 7)(2x + 4) \\ = 14x^2 + 42x + 28$$

$$(3) (2x + 4)(3x + 3) \\ = 6x^2 + 18x + 12$$

$$(4) (7x + 6)(5x - 6) \\ = 35x^2 - 12x - 36$$

$$(5) (x - 3)(4x + 5) \\ = 4x^2 - 7x - 15$$

$$(6) (6x - 3)(3x + 1) \\ = 18x^2 - 3x - 3$$

$$(7) (9x + 4)(3x - 4) \\ = 27x^2 - 24x - 16$$

$$(8) (7x - 7)(5x + 4) \\ = 35x^2 - 7x - 28$$

$$(9) (3x + 9)(6x - 4) \\ = 18x^2 + 42x - 36$$

$$(10) (2x - 4)(5x - 3) \\ = 10x^2 - 26x + 12$$

$$(11) (9x + 8)(x - 5) \\ = 9x^2 - 37x - 40$$

$$(12) (3x - 2)(2x + 1) \\ = 6x^2 - x - 2$$

$$(13) (2x - 9)(9x - 1) \\ = 18x^2 - 83x + 9$$

$$(14) (5x - 8)(x + 5) \\ = 5x^2 + 17x - 40$$

$$(15) (7x - 4)(7x - 6) \\ = 49x^2 - 70x + 24$$

$$(16) (5x + 6)(8x + 8) \\ = 40x^2 + 88x + 48$$

$$(17) (4x + 7)(8x + 5) \\ = 32x^2 + 76x + 35$$

$$(18) (2x - 7)(2x + 8) \\ = 4x^2 + 2x - 56$$

$$(19) (6x - 3)(4x + 4) \\ = 24x^2 + 12x - 12$$

$$(20) (x - 2)(2x - 1) \\ = 2x^2 - 5x + 2$$

$$(21) (2x - 4)(8x - 3) \\ = 16x^2 - 38x + 12$$

$$(22) (x - 3)(6x + 2) \\ = 6x^2 - 16x - 6$$

$$(23) (6x - 1)(8x + 1) \\ = 48x^2 - 2x - 1$$

$$(24) (7x + 1)(x - 3) \\ = 7x^2 - 20x - 3$$

$$(25) (6x + 6)(6x + 3) \\ = 36x^2 + 54x + 18$$

$$(26) (7x + 9)(3x - 2) \\ = 21x^2 + 13x - 18$$

$$(27) (9x - 7)(5x + 7) \\ = 45x^2 + 28x - 49$$

$$(28) (9x + 6)(9x + 8) \\ = 81x^2 + 126x + 48$$

$$(29) (4x - 7)(2x + 2) \\ = 8x^2 - 6x - 14$$

$$(30) (9x - 4)(8x - 4) \\ = 72x^2 - 68x + 16$$

$$(31) (2x + 1)(5x - 8) \\ = 10x^2 - 11x - 8$$

$$(32) (7x + 5)(9x - 9) \\ = 63x^2 - 18x - 45$$

$$(33) (7x + 4)(5x + 4) \\ = 35x^2 + 48x + 16$$

$$(34) (x + 1)(8x - 9) \\ = 8x^2 - x - 9$$

$$(35) (9x + 2)(6x + 1) \\ = 54x^2 + 21x + 2$$

$$(36) (3x - 4)(6x + 9) \\ = 18x^2 + 3x - 36$$

$$(37) (2x + 1)(9x - 2) \\ = 18x^2 + 5x - 2$$

$$(38) (8x + 9)(5x - 9) \\ = 40x^2 - 27x - 81$$

$$(39) (7x + 8)(4x - 6) \\ = 28x^2 - 10x - 48$$

$$(40) (9x + 3)(5x - 3) \\ = 45x^2 - 12x - 9$$

$$(41) (7x - 5)(9x + 2) \\ = 63x^2 - 31x - 10$$

$$(42) (6x - 4)(6x + 7) \\ = 36x^2 + 18x - 28$$

$$(43) (8x - 4)(9x - 5) \\ = 72x^2 - 76x + 20$$

$$(44) (x + 7)(x - 5) \\ = x^2 + 2x - 35$$

$$(45) (5x - 6)(5x + 8) \\ = 25x^2 + 10x - 48$$



$$(1) (3x-7)(4x+4) \\ = 12x^2 - 16x - 28$$

$$(16) (7x+7)(5x-4) \\ = 35x^2 + 7x - 28$$

$$(31) (3x-1)(x+5) \\ = 3x^2 + 14x - 5$$

$$(2) (8x-2)(9x+2) \\ = 72x^2 - 2x - 4$$

$$(17) (9x-5)(2x-5) \\ = 18x^2 - 55x + 25$$

$$(32) (5x-4)(4x+8) \\ = 20x^2 + 24x - 32$$

$$(3) (x+4)(2x+2) \\ = 2x^2 + 10x + 8$$

$$(18) (3x-8)(2x-9) \\ = 6x^2 - 43x + 72$$

$$(33) (2x-5)(9x+2) \\ = 18x^2 - 41x - 10$$

$$(4) (x+5)(3x+3) \\ = 3x^2 + 18x + 15$$

$$(19) (x+7)(6x-3) \\ = 6x^2 + 39x - 21$$

$$(34) (9x+3)(8x+5) \\ = 72x^2 + 69x + 15$$

$$(5) (4x+9)(4x-3) \\ = 16x^2 + 24x - 27$$

$$(20) (2x-1)(5x+7) \\ = 10x^2 + 9x - 7$$

$$(35) (x-2)(6x+7) \\ = 6x^2 - 5x - 14$$

$$(6) (2x-2)(5x-6) \\ = 10x^2 - 22x + 12$$

$$(21) (8x+4)(5x+2) \\ = 40x^2 + 36x + 8$$

$$(36) (2x+4)(7x+9) \\ = 14x^2 + 46x + 36$$

$$(7) (4x-9)(3x-3) \\ = 12x^2 - 39x + 27$$

$$(22) (7x+2)(4x-6) \\ = 28x^2 - 34x - 12$$

$$(37) (5x+1)(8x+3) \\ = 40x^2 + 23x + 3$$

$$(8) (6x-8)(4x+3) \\ = 24x^2 - 14x - 24$$

$$(23) (3x-1)(4x+1) \\ = 12x^2 - x - 1$$

$$(38) (8x+8)(2x-5) \\ = 16x^2 - 24x - 40$$

$$(9) (4x-7)(6x+3) \\ = 24x^2 - 30x - 21$$

$$(24) (x+8)(x+5) \\ = x^2 + 13x + 40$$

$$(39) (6x+3)(8x+2) \\ = 48x^2 + 36x + 6$$

$$(10) (8x-7)(5x+9) \\ = 40x^2 + 37x - 63$$

$$(25) (x+6)(4x+9) \\ = 4x^2 + 33x + 54$$

$$(40) (9x+9)(9x-3) \\ = 81x^2 + 54x - 27$$

$$(11) (3x-7)(8x+6) \\ = 24x^2 - 38x - 42$$

$$(26) (4x-3)(9x+3) \\ = 36x^2 - 15x - 9$$

$$(41) (7x-5)(7x-2) \\ = 49x^2 - 49x + 10$$

$$(12) (5x+8)(9x-6) \\ = 45x^2 + 42x - 48$$

$$(27) (6x+7)(7x-5) \\ = 42x^2 + 19x - 35$$

$$(42) (2x+2)(x+6) \\ = 2x^2 + 14x + 12$$

$$(13) (8x-2)(5x+6) \\ = 40x^2 + 38x - 12$$

$$(28) (8x+7)(4x+7) \\ = 32x^2 + 84x + 49$$

$$(43) (2x+1)(8x+3) \\ = 16x^2 + 14x + 3$$

$$(14) (5x-8)(7x-3) \\ = 35x^2 - 71x + 24$$

$$(29) (8x+7)(5x+1) \\ = 40x^2 + 43x + 7$$

$$(44) (2x-4)(3x+1) \\ = 6x^2 - 10x - 4$$

$$(15) (5x+2)(9x-7) \\ = 45x^2 - 17x - 14$$

$$(30) (5x+1)(2x+7) \\ = 10x^2 + 37x + 7$$

$$(45) (4x+4)(7x+4) \\ = 28x^2 + 44x + 16$$

$$(1) (2x+7)(9x-2) \\ = 18x^2 + 59x - 14$$

$$(16) (9x+1)(8x-4) \\ = 72x^2 - 28x - 4$$

$$(31) (4x-9)(7x-7) \\ = 28x^2 - 91x + 63$$

$$(2) (3x+8)(7x-4) \\ = 21x^2 + 44x - 32$$

$$(17) (4x-9)(x-7) \\ = 4x^2 - 37x + 63$$

$$(32) (3x-2)(6x+4) \\ = 18x^2 - 8$$

$$(3) (5x-2)(3x+4) \\ = 15x^2 + 14x - 8$$

$$(18) (5x-8)(8x+5) \\ = 40x^2 - 39x - 40$$

$$(33) (2x-5)(7x-7) \\ = 14x^2 - 49x + 35$$

$$(4) (7x+4)(4x+9) \\ = 28x^2 + 79x + 36$$

$$(19) (x+5)(9x-5) \\ = 9x^2 + 40x - 25$$

$$(34) (2x+7)(8x-1) \\ = 16x^2 + 54x - 7$$

$$(5) (7x+1)(4x+6) \\ = 28x^2 + 46x + 6$$

$$(20) (6x-8)(4x-2) \\ = 24x^2 - 44x + 16$$

$$(35) (8x+8)(7x+8) \\ = 56x^2 + 120x + 64$$

$$(6) (3x-2)(5x-8) \\ = 15x^2 - 34x + 16$$

$$(21) (4x+3)(6x-2) \\ = 24x^2 + 10x - 6$$

$$(36) (8x-3)(2x+5) \\ = 16x^2 + 34x - 15$$

$$(7) (2x+9)(2x-6) \\ = 4x^2 + 6x - 54$$

$$(22) (9x+8)(9x+2) \\ = 81x^2 + 90x + 16$$

$$(37) (3x+5)(9x-1) \\ = 27x^2 + 42x - 5$$

$$(8) (4x+6)(3x-5) \\ = 12x^2 - 2x - 30$$

$$(23) (2x-4)(8x+8) \\ = 16x^2 - 16x - 32$$

$$(38) (9x-2)(x+1) \\ = 9x^2 + 7x - 2$$

$$(9) (6x+2)(4x+1) \\ = 24x^2 + 14x + 2$$

$$(24) (5x+3)(5x-4) \\ = 25x^2 - 5x - 12$$

$$(39) (8x-4)(9x+5) \\ = 72x^2 + 4x - 20$$

$$(10) (x+6)(x+4) \\ = x^2 + 10x + 24$$

$$(25) (2x+3)(7x-7) \\ = 14x^2 + 7x - 21$$

$$(40) (7x-4)(5x-5) \\ = 35x^2 - 55x + 20$$

$$(11) (9x+3)(6x-7) \\ = 54x^2 - 45x - 21$$

$$(26) (3x-2)(3x+2) \\ = 9x^2 - 4$$

$$(41) (3x-3)(4x-9) \\ = 12x^2 - 39x + 27$$

$$(12) (5x+6)(6x-9) \\ = 30x^2 - 9x - 54$$

$$(27) (5x-7)(8x-8) \\ = 40x^2 - 96x + 56$$

$$(42) (8x-5)(9x-2) \\ = 72x^2 - 61x + 10$$

$$(13) (3x+1)(7x-2) \\ = 21x^2 + x - 2$$

$$(28) (2x-7)(3x-9) \\ = 6x^2 - 39x + 63$$

$$(43) (8x-9)(3x+6) \\ = 24x^2 + 21x - 54$$

$$(14) (8x-1)(x+7) \\ = 8x^2 + 55x - 7$$

$$(29) (4x+6)(4x+2) \\ = 16x^2 + 32x + 12$$

$$(44) (6x-4)(7x-7) \\ = 42x^2 - 70x + 28$$

$$(15) (x+5)(7x-7) \\ = 7x^2 + 28x - 35$$

$$(30) (6x+5)(2x-7) \\ = 12x^2 - 32x - 35$$

$$(45) (4x-4)(5x-4) \\ = 20x^2 - 36x + 16$$

$$(1) (3x+1)(7x+7) \\ = 21x^2 + 28x + 7$$

$$(2) (8x+4)(x+9) \\ = 8x^2 + 76x + 36$$

$$(3) (5x-9)(9x+7) \\ = 45x^2 - 46x - 63$$

$$(4) (5x+5)(8x-7) \\ = 40x^2 + 5x - 35$$

$$(5) (2x-9)(7x+9) \\ = 14x^2 - 45x - 81$$

$$(6) (8x-7)(5x-5) \\ = 40x^2 - 75x + 35$$

$$(7) (3x+1)(5x+6) \\ = 15x^2 + 23x + 6$$

$$(8) (7x-6)(6x+7) \\ = 42x^2 + 13x - 42$$

$$(9) (4x+2)(3x+8) \\ = 12x^2 + 38x + 16$$

$$(10) (7x+8)(6x+5) \\ = 42x^2 + 83x + 40$$

$$(11) (8x-5)(2x+6) \\ = 16x^2 + 38x - 30$$

$$(12) (6x-2)(6x+3) \\ = 36x^2 + 6x - 6$$

$$(13) (9x-4)(3x-1) \\ = 27x^2 - 21x + 4$$

$$(14) (4x+6)(4x+2) \\ = 16x^2 + 32x + 12$$

$$(15) (8x-1)(8x+1) \\ = 64x^2 - 1$$

$$(16) (6x-9)(5x+4) \\ = 30x^2 - 21x - 36$$

$$(17) (6x-6)(x+6) \\ = 6x^2 + 30x - 36$$

$$(18) (8x-6)(2x-1) \\ = 16x^2 - 20x + 6$$

$$(19) (5x+1)(7x+1) \\ = 35x^2 + 12x + 1$$

$$(20) (6x-3)(7x-4) \\ = 42x^2 - 45x + 12$$

$$(21) (4x-6)(5x-6) \\ = 20x^2 - 54x + 36$$

$$(22) (6x+6)(3x+3) \\ = 18x^2 + 36x + 18$$

$$(23) (6x-2)(7x+1) \\ = 42x^2 - 8x - 2$$

$$(24) (6x-9)(5x+3) \\ = 30x^2 - 27x - 27$$

$$(25) (5x+1)(3x+1) \\ = 15x^2 + 8x + 1$$

$$(26) (3x-1)(9x-8) \\ = 27x^2 - 33x + 8$$

$$(27) (8x-3)(7x+6) \\ = 56x^2 + 27x - 18$$

$$(28) (6x-4)(3x+4) \\ = 18x^2 + 12x - 16$$

$$(29) (5x+3)(6x-5) \\ = 30x^2 - 7x - 15$$

$$(30) (6x-2)(3x-9) \\ = 18x^2 - 60x + 18$$

$$(31) (x+8)(7x-8) \\ = 7x^2 + 48x - 64$$

$$(32) (8x-9)(x-7) \\ = 8x^2 - 65x + 63$$

$$(33) (2x+3)(x-1) \\ = 2x^2 + x - 3$$

$$(34) (x-3)(8x+4) \\ = 8x^2 - 20x - 12$$

$$(35) (6x-7)(6x-6) \\ = 36x^2 - 78x + 42$$

$$(36) (6x+1)(5x-7) \\ = 30x^2 - 37x - 7$$

$$(37) (7x-6)(4x-3) \\ = 28x^2 - 45x + 18$$

$$(38) (6x-2)(5x+6) \\ = 30x^2 + 26x - 12$$

$$(39) (7x-9)(5x+3) \\ = 35x^2 - 24x - 27$$

$$(40) (4x+3)(x+7) \\ = 4x^2 + 31x + 21$$

$$(41) (x+3)(7x-9) \\ = 7x^2 + 12x - 27$$

$$(42) (4x+8)(8x+8) \\ = 32x^2 + 96x + 64$$

$$(43) (4x-1)(4x+8) \\ = 16x^2 + 28x - 8$$

$$(44) (8x-3)(3x-1) \\ = 24x^2 - 17x + 3$$

$$(45) (7x+3)(x-9) \\ = 7x^2 - 60x - 27$$

$$(1) (4x-2)(7x+9) \\ = 28x^2 + 22x - 18$$

$$(16) (8x+7)(2x-2) \\ = 16x^2 - 2x - 14$$

$$(31) (9x-5)(4x-1) \\ = 36x^2 - 29x + 5$$

$$(2) (5x-2)(3x-8) \\ = 15x^2 - 46x + 16$$

$$(17) (2x-9)(x-6) \\ = 2x^2 - 21x + 54$$

$$(32) (6x-9)(2x+3) \\ = 12x^2 - 27$$

$$(3) (8x+7)(5x+1) \\ = 40x^2 + 43x + 7$$

$$(18) (4x+5)(8x-8) \\ = 32x^2 + 8x - 40$$

$$(33) (7x-3)(5x+7) \\ = 35x^2 + 34x - 21$$

$$(4) (8x-4)(8x-8) \\ = 64x^2 - 96x + 32$$

$$(19) (2x+4)(2x+5) \\ = 4x^2 + 18x + 20$$

$$(34) (6x+6)(3x+8) \\ = 18x^2 + 66x + 48$$

$$(5) (2x+4)(4x+1) \\ = 8x^2 + 18x + 4$$

$$(20) (6x-6)(x-3) \\ = 6x^2 - 24x + 18$$

$$(35) (3x-3)(4x-4) \\ = 12x^2 - 24x + 12$$

$$(6) (8x-1)(6x-7) \\ = 48x^2 - 62x + 7$$

$$(21) (5x+5)(7x+8) \\ = 35x^2 + 75x + 40$$

$$(36) (2x+7)(2x+4) \\ = 4x^2 + 22x + 28$$

$$(7) (4x-4)(5x+6) \\ = 20x^2 + 4x - 24$$

$$(22) (5x+4)(x+3) \\ = 5x^2 + 19x + 12$$

$$(37) (4x+3)(2x+9) \\ = 8x^2 + 42x + 27$$

$$(8) (7x-6)(8x+6) \\ = 56x^2 - 6x - 36$$

$$(23) (8x+3)(x+5) \\ = 8x^2 + 43x + 15$$

$$(38) (9x+8)(8x-2) \\ = 72x^2 + 46x - 16$$

$$(9) (7x+5)(4x+4) \\ = 28x^2 + 48x + 20$$

$$(24) (7x+2)(5x-3) \\ = 35x^2 - 11x - 6$$

$$(39) (4x+6)(7x+3) \\ = 28x^2 + 54x + 18$$

$$(10) (6x+3)(5x+6) \\ = 30x^2 + 51x + 18$$

$$(25) (9x-8)(9x+9) \\ = 81x^2 + 9x - 72$$

$$(40) (7x-1)(7x+6) \\ = 49x^2 + 35x - 6$$

$$(11) (4x-8)(x+6) \\ = 4x^2 + 16x - 48$$

$$(26) (x+8)(4x-1) \\ = 4x^2 + 31x - 8$$

$$(41) (9x+2)(x-9) \\ = 9x^2 - 79x - 18$$

$$(12) (2x-4)(6x-6) \\ = 12x^2 - 36x + 24$$

$$(27) (3x+6)(x-5) \\ = 3x^2 - 9x - 30$$

$$(42) (6x-2)(3x-6) \\ = 18x^2 - 42x + 12$$

$$(13) (8x+7)(8x-3) \\ = 64x^2 + 32x - 21$$

$$(28) (5x-5)(4x+8) \\ = 20x^2 + 20x - 40$$

$$(43) (3x+7)(2x+4) \\ = 6x^2 + 26x + 28$$

$$(14) (2x+7)(4x-1) \\ = 8x^2 + 26x - 7$$

$$(29) (2x-9)(3x+6) \\ = 6x^2 - 15x - 54$$

$$(44) (9x-6)(8x-2) \\ = 72x^2 - 66x + 12$$

$$(15) (2x-8)(8x+2) \\ = 16x^2 - 60x - 16$$

$$(30) (9x-1)(x-9) \\ = 9x^2 - 82x + 9$$

$$(45) (2x+7)(5x+4) \\ = 10x^2 + 43x + 28$$

$$(1) (2x+4)(x-1) \\ = 2x^2 + 2x - 4$$

$$(2) (7x-5)(7x+9) \\ = 49x^2 + 28x - 45$$

$$(3) (x+8)^2 \\ = x^2 + 16x + 64$$

$$(4) (9x+5)(6x-4) \\ = 54x^2 - 6x - 20$$

$$(5) (9x-6)(5x+6) \\ = 45x^2 + 24x - 36$$

$$(6) (4x-6)(7x+3) \\ = 28x^2 - 30x - 18$$

$$(7) (8x+4)(3x+3) \\ = 24x^2 + 36x + 12$$

$$(8) (4x+7)(8x+9) \\ = 32x^2 + 92x + 63$$

$$(9) (5x-4)(6x+6) \\ = 30x^2 + 6x - 24$$

$$(10) (2x-9)(3x-5) \\ = 6x^2 - 37x + 45$$

$$(11) (9x+4)(2x-4) \\ = 18x^2 - 28x - 16$$

$$(12) (5x+6)(x-8) \\ = 5x^2 - 34x - 48$$

$$(13) (7x+1)(x-4) \\ = 7x^2 - 27x - 4$$

$$(14) (4x-8)(5x+8) \\ = 20x^2 - 8x - 64$$

$$(15) (5x-1)(3x-4) \\ = 15x^2 - 23x + 4$$

$$(16) (x-2)(7x+3) \\ = 7x^2 - 11x - 6$$

$$(17) (7x-3)(3x-2) \\ = 21x^2 - 23x + 6$$

$$(18) (2x+8)(7x-9) \\ = 14x^2 + 38x - 72$$

$$(19) (4x+8)(4x+7) \\ = 16x^2 + 60x + 56$$

$$(20) (6x-7)(7x+4) \\ = 42x^2 - 25x - 28$$

$$(21) (6x+5)(7x+9) \\ = 42x^2 + 89x + 45$$

$$(22) (4x+8)(9x-8) \\ = 36x^2 + 40x - 64$$

$$(23) (5x+7)(4x+6) \\ = 20x^2 + 58x + 42$$

$$(24) (4x+1)(2x-2) \\ = 8x^2 - 6x - 2$$

$$(25) (5x+7)(7x-6) \\ = 35x^2 + 19x - 42$$

$$(26) (8x-7)(2x+3) \\ = 16x^2 + 10x - 21$$

$$(27) (9x+6)(8x-7) \\ = 72x^2 - 15x - 42$$

$$(28) (3x+7)(4x+1) \\ = 12x^2 + 31x + 7$$

$$(29) (7x-7)(x+3) \\ = 7x^2 + 14x - 21$$

$$(30) (8x-8)(5x-1) \\ = 40x^2 - 48x + 8$$

$$(31) (6x+6)(4x+3) \\ = 24x^2 + 42x + 18$$

$$(32) (4x+2)(4x+9) \\ = 16x^2 + 44x + 18$$

$$(33) (8x+2)(5x-5) \\ = 40x^2 - 30x - 10$$

$$(34) (8x-1)(2x+1) \\ = 16x^2 + 6x - 1$$

$$(35) (4x-2)(8x+8) \\ = 32x^2 + 16x - 16$$

$$(36) (x-8)(4x-9) \\ = 4x^2 - 41x + 72$$

$$(37) (4x-5)(x+9) \\ = 4x^2 + 31x - 45$$

$$(38) (7x+5)(8x-3) \\ = 56x^2 + 19x - 15$$

$$(39) (2x+9)(3x+9) \\ = 6x^2 + 45x + 81$$

$$(40) (2x+4)(2x-5) \\ = 4x^2 - 2x - 20$$

$$(41) (2x-8)(2x+6) \\ = 4x^2 - 4x - 48$$

$$(42) (x+9)(6x+8) \\ = 6x^2 + 62x + 72$$

$$(43) (6x-9)(3x+7) \\ = 18x^2 + 15x - 63$$

$$(44) (9x-1)(3x-8) \\ = 27x^2 - 75x + 8$$

$$(45) (4x+9)(6x-6) \\ = 24x^2 + 30x - 54$$

$$(1) (6x+3)(4x+2) \\ = 24x^2 + 24x + 6$$

$$(16) (3x-4)(7x+4) \\ = 21x^2 - 16x - 16$$

$$(31) (3x-3)(x+2) \\ = 3x^2 + 3x - 6$$

$$(2) (6x+6)(2x-5) \\ = 12x^2 - 18x - 30$$

$$(17) (x-3)(9x-6) \\ = 9x^2 - 33x + 18$$

$$(32) (6x-8)(2x+9) \\ = 12x^2 + 38x - 72$$

$$(3) (x+4)(7x-3) \\ = 7x^2 + 25x - 12$$

$$(18) (7x+6)(6x-8) \\ = 42x^2 - 20x - 48$$

$$(33) (5x-9)(6x-6) \\ = 30x^2 - 84x + 54$$

$$(4) (3x-2)(5x+9) \\ = 15x^2 + 17x - 18$$

$$(19) (x-5)(2x-3) \\ = 2x^2 - 13x + 15$$

$$(34) (2x-7)(6x+3) \\ = 12x^2 - 36x - 21$$

$$(5) (5x+5)(9x+8) \\ = 45x^2 + 85x + 40$$

$$(20) (4x+5)(6x-5) \\ = 24x^2 + 10x - 25$$

$$(35) (7x-7)(4x-6) \\ = 28x^2 - 70x + 42$$

$$(6) (5x+2)(9x+4) \\ = 45x^2 + 38x + 8$$

$$(21) (4x+4)(6x-6) \\ = 24x^2 - 24$$

$$(36) (8x-5)(9x-6) \\ = 72x^2 - 93x + 30$$

$$(7) (x+5)(9x-7) \\ = 9x^2 + 38x - 35$$

$$(22) (9x+9)(9x-6) \\ = 81x^2 + 27x - 54$$

$$(37) (x-5)(4x+6) \\ = 4x^2 - 14x - 30$$

$$(8) (6x+2)(x+2) \\ = 6x^2 + 14x + 4$$

$$(23) (5x-2)(x-6) \\ = 5x^2 - 32x + 12$$

$$(38) (2x+8)(x-8) \\ = 2x^2 - 8x - 64$$

$$(9) (3x-6)(5x+3) \\ = 15x^2 - 21x - 18$$

$$(24) (9x-7)(3x+1) \\ = 27x^2 - 12x - 7$$

$$(39) (7x-9)(8x+2) \\ = 56x^2 - 58x - 18$$

$$(10) (x+4)(x-5) \\ = x^2 - x - 20$$

$$(25) (8x-7)(5x+1) \\ = 40x^2 - 27x - 7$$

$$(40) (2x+4)(5x-1) \\ = 10x^2 + 18x - 4$$

$$(11) (3x+8)(2x-5) \\ = 6x^2 + x - 40$$

$$(26) (9x-7)(x-9) \\ = 9x^2 - 88x + 63$$

$$(41) (5x+3)(3x+7) \\ = 15x^2 + 44x + 21$$

$$(12) (x+3)(7x-9) \\ = 7x^2 + 12x - 27$$

$$(27) (8x+8)(8x-9) \\ = 64x^2 - 8x - 72$$

$$(42) (5x+1)(3x+8) \\ = 15x^2 + 43x + 8$$

$$(13) (6x+6)(7x+7) \\ = 42x^2 + 84x + 42$$

$$(28) (2x-8)(6x+7) \\ = 12x^2 - 34x - 56$$

$$(43) (9x-6)(2x+1) \\ = 18x^2 - 3x - 6$$

$$(14) (7x+5)(5x-2) \\ = 35x^2 + 11x - 10$$

$$(29) (5x+6)(x-1) \\ = 5x^2 + x - 6$$

$$(44) (6x-7)(x+5) \\ = 6x^2 + 23x - 35$$

$$(15) (5x-5)(3x+5) \\ = 15x^2 + 10x - 25$$

$$(30) (9x-3)(3x+7) \\ = 27x^2 + 54x - 21$$

$$(45) (2x-1)(7x-1) \\ = 14x^2 - 9x + 1$$

$$(1) (2x+9)(5x-8) \\ = 10x^2 + 29x - 72$$

$$(2) (3x-5)(5x+3) \\ = 15x^2 - 16x - 15$$

$$(3) (9x-6)(7x+3) \\ = 63x^2 - 15x - 18$$

$$(4) (9x-7)(2x-3) \\ = 18x^2 - 41x + 21$$

$$(5) (2x+8)(6x-5) \\ = 12x^2 + 38x - 40$$

$$(6) (4x-6)(5x+4) \\ = 20x^2 - 14x - 24$$

$$(7) (7x+7)(8x+7) \\ = 56x^2 + 105x + 49$$

$$(8) (3x-9)(4x+5) \\ = 12x^2 - 21x - 45$$

$$(9) (5x+7)(x+1) \\ = 5x^2 + 12x + 7$$

$$(10) (7x-2)(8x-6) \\ = 56x^2 - 58x + 12$$

$$(11) (3x-1)(7x-3) \\ = 21x^2 - 16x + 3$$

$$(12) (9x-2)(x-9) \\ = 9x^2 - 83x + 18$$

$$(13) (x-4)(2x-4) \\ = 2x^2 - 12x + 16$$

$$(14) (4x-9)(7x+6) \\ = 28x^2 - 39x - 54$$

$$(15) (4x+9)(7x-3) \\ = 28x^2 + 51x - 27$$

$$(16) (4x+3)(7x+2) \\ = 28x^2 + 29x + 6$$

$$(17) (9x-6)(4x-2) \\ = 36x^2 - 42x + 12$$

$$(18) (9x-3)(3x-7) \\ = 27x^2 - 72x + 21$$

$$(19) (2x-2)(9x+9) \\ = 18x^2 - 18$$

$$(20) (7x-7)(3x+9) \\ = 21x^2 + 42x - 63$$

$$(21) (x-5)(8x+9) \\ = 8x^2 - 31x - 45$$

$$(22) (x-1)(6x+4) \\ = 6x^2 - 2x - 4$$

$$(23) (3x-6)(4x+9) \\ = 12x^2 + 3x - 54$$

$$(24) (3x-6)(8x-2) \\ = 24x^2 - 54x + 12$$

$$(25) (9x-2)(x+9) \\ = 9x^2 + 79x - 18$$

$$(26) (9x+2)(2x+5) \\ = 18x^2 + 49x + 10$$

$$(27) (3x-5)(x-3) \\ = 3x^2 - 14x + 15$$

$$(28) (7x-2)(3x+8) \\ = 21x^2 + 50x - 16$$

$$(29) (3x-1)(8x+3) \\ = 24x^2 + x - 3$$

$$(30) (8x-9)(6x-8) \\ = 48x^2 - 118x + 72$$

$$(31) (3x+3)(9x+3) \\ = 27x^2 + 36x + 9$$

$$(32) (9x-6)(7x-9) \\ = 63x^2 - 123x + 54$$

$$(33) (2x-4)(8x-8) \\ = 16x^2 - 48x + 32$$

$$(34) (3x+3)(4x-5) \\ = 12x^2 - 3x - 15$$

$$(35) (3x-9)(x+1) \\ = 3x^2 - 6x - 9$$

$$(36) (5x+8)(9x-9) \\ = 45x^2 + 27x - 72$$

$$(37) (x+4)(3x-7) \\ = 3x^2 + 5x - 28$$

$$(38) (x+2)(6x+3) \\ = 6x^2 + 15x + 6$$

$$(39) (9x+2)(9x-7) \\ = 81x^2 - 45x - 14$$

$$(40) (5x+5)(4x+9) \\ = 20x^2 + 65x + 45$$

$$(41) (5x-4)(7x+7) \\ = 35x^2 + 7x - 28$$

$$(42) (9x+5)(6x+7) \\ = 54x^2 + 93x + 35$$

$$(43) (8x+5)(2x-2) \\ = 16x^2 - 6x - 10$$

$$(44) (x-7)(6x-4) \\ = 6x^2 - 46x + 28$$

$$(45) (8x+6)(2x-9) \\ = 16x^2 - 60x - 54$$

$$(1) (6x - 6)(7x + 7) \\ = 42x^2 - 42$$

$$(2) (7x - 2)(4x + 5) \\ = 28x^2 + 27x - 10$$

$$(3) (4x - 7)(6x + 6) \\ = 24x^2 - 18x - 42$$

$$(4) (7x - 7)(x - 4) \\ = 7x^2 - 35x + 28$$

$$(5) (8x + 7)(8x - 6) \\ = 64x^2 + 8x - 42$$

$$(6) (3x - 9)(x - 6) \\ = 3x^2 - 27x + 54$$

$$(7) (9x + 3)(x + 5) \\ = 9x^2 + 48x + 15$$

$$(8) (7x + 5)(7x + 1) \\ = 49x^2 + 42x + 5$$

$$(9) (7x - 6)(x - 2) \\ = 7x^2 - 20x + 12$$

$$(10) (5x - 6)(9x - 2) \\ = 45x^2 - 64x + 12$$

$$(11) (4x - 5)(2x - 9) \\ = 8x^2 - 46x + 45$$

$$(12) (3x - 2)(4x + 4) \\ = 12x^2 + 4x - 8$$

$$(13) (2x - 3)(5x - 2) \\ = 10x^2 - 19x + 6$$

$$(14) (2x + 1)(9x - 9) \\ = 18x^2 - 9x - 9$$

$$(15) (3x + 1)(7x + 5) \\ = 21x^2 + 22x + 5$$

$$(16) (9x - 6)(5x - 9) \\ = 45x^2 - 111x + 54$$

$$(17) (3x + 2)^2 \\ = 9x^2 + 12x + 4$$

$$(18) (8x - 3)(5x - 7) \\ = 40x^2 - 71x + 21$$

$$(19) (2x + 1)(3x + 2) \\ = 6x^2 + 7x + 2$$

$$(20) (6x + 5)(7x + 2) \\ = 42x^2 + 47x + 10$$

$$(21) (9x - 7)(7x + 5) \\ = 63x^2 - 4x - 35$$

$$(22) (6x - 3)(6x + 3) \\ = 36x^2 - 9$$

$$(23) (6x - 9)(9x - 1) \\ = 54x^2 - 87x + 9$$

$$(24) (5x - 5)(6x - 2) \\ = 30x^2 - 40x + 10$$

$$(25) (2x + 2)(9x + 8) \\ = 18x^2 + 34x + 16$$

$$(26) (9x - 2)(3x - 3) \\ = 27x^2 - 33x + 6$$

$$(27) (6x - 8)(3x + 5) \\ = 18x^2 + 6x - 40$$

$$(28) (7x - 7)(3x - 7) \\ = 21x^2 - 70x + 49$$

$$(29) (8x + 1)(5x + 4) \\ = 40x^2 + 37x + 4$$

$$(30) (9x - 5)(x + 2) \\ = 9x^2 + 13x - 10$$

$$(31) (2x + 6)(8x + 2) \\ = 16x^2 + 52x + 12$$

$$(32) (3x + 8)(x + 7) \\ = 3x^2 + 29x + 56$$

$$(33) (x + 2)(3x - 9) \\ = 3x^2 - 3x - 18$$

$$(34) (9x + 8)(3x - 4) \\ = 27x^2 - 12x - 32$$

$$(35) (5x - 6)(8x - 3) \\ = 40x^2 - 63x + 18$$

$$(36) (3x + 3)(8x + 9) \\ = 24x^2 + 51x + 27$$

$$(37) (5x + 7)(8x - 3) \\ = 40x^2 + 41x - 21$$

$$(38) (8x - 5)(5x + 8) \\ = 40x^2 + 39x - 40$$

$$(39) (4x + 4)(7x - 6) \\ = 28x^2 + 4x - 24$$

$$(40) (9x + 9)(8x + 7) \\ = 72x^2 + 135x + 63$$

$$(41) (3x - 1)(7x - 1) \\ = 21x^2 - 10x + 1$$

$$(42) (x - 8)(9x - 8) \\ = 9x^2 - 80x + 64$$

$$(43) (3x + 5)(8x + 9) \\ = 24x^2 + 67x + 45$$

$$(44) (5x - 3)(5x - 9) \\ = 25x^2 - 60x + 27$$

$$(45) (4x - 6)(3x - 5) \\ = 12x^2 - 38x + 30$$



$$(1) (9x+9)(9x+6) \\ = 81x^2 + 135x + 54$$

$$(16) (5x+6)(6x-1) \\ = 30x^2 + 31x - 6$$

$$(31) (4x+7)(3x+6) \\ = 12x^2 + 45x + 42$$

$$(2) (2x+9)(5x-6) \\ = 10x^2 + 33x - 54$$

$$(17) (4x-8)(5x+4) \\ = 20x^2 - 24x - 32$$

$$(32) (x-8)(x+4) \\ = x^2 - 4x - 32$$

$$(3) (x-3)(3x-6) \\ = 3x^2 - 15x + 18$$

$$(18) (5x-4)(2x+7) \\ = 10x^2 + 27x - 28$$

$$(33) (3x-7)(6x+9) \\ = 18x^2 - 15x - 63$$

$$(4) (x-1)(5x-3) \\ = 5x^2 - 8x + 3$$

$$(19) (4x+5)(9x+7) \\ = 36x^2 + 73x + 35$$

$$(34) (x+8)(3x-8) \\ = 3x^2 + 16x - 64$$

$$(5) (8x+9)(3x-1) \\ = 24x^2 + 19x - 9$$

$$(20) (8x-3)(x+3) \\ = 8x^2 + 21x - 9$$

$$(35) (2x+3)(4x-7) \\ = 8x^2 - 2x - 21$$

$$(6) (x+7)(6x-1) \\ = 6x^2 + 41x - 7$$

$$(21) (5x-5)(3x+5) \\ = 15x^2 + 10x - 25$$

$$(36) (5x-3)(5x-5) \\ = 25x^2 - 40x + 15$$

$$(7) (7x-7)(6x-5) \\ = 42x^2 - 77x + 35$$

$$(22) (x-3)(3x-8) \\ = 3x^2 - 17x + 24$$

$$(37) (3x-2)(4x+3) \\ = 12x^2 + x - 6$$

$$(8) (6x-7)(3x+7) \\ = 18x^2 + 21x - 49$$

$$(23) (3x-8)(x+6) \\ = 3x^2 + 10x - 48$$

$$(38) (5x+3)(4x-3) \\ = 20x^2 - 3x - 9$$

$$(9) (4x+4)(9x+5) \\ = 36x^2 + 56x + 20$$

$$(24) (7x-8)(3x-1) \\ = 21x^2 - 31x + 8$$

$$(39) (4x+2)(2x+7) \\ = 8x^2 + 32x + 14$$

$$(10) (4x+9)(4x-2) \\ = 16x^2 + 28x - 18$$

$$(25) (7x-1)(6x-7) \\ = 42x^2 - 55x + 7$$

$$(40) (6x+9)(5x+5) \\ = 30x^2 + 75x + 45$$

$$(11) (2x-9)(7x-5) \\ = 14x^2 - 73x + 45$$

$$(26) (5x-6)(4x+4) \\ = 20x^2 - 4x - 24$$

$$(41) (4x+4)(4x-6) \\ = 16x^2 - 8x - 24$$

$$(12) (4x-4)(8x+4) \\ = 32x^2 - 16x - 16$$

$$(27) (7x-5)(5x+2) \\ = 35x^2 - 11x - 10$$

$$(42) (3x+7)(5x-8) \\ = 15x^2 + 11x - 56$$

$$(13) (9x-1)(x+3) \\ = 9x^2 + 26x - 3$$

$$(28) (6x-8)(6x+1) \\ = 36x^2 - 42x - 8$$

$$(43) (8x+9)(6x-4) \\ = 48x^2 + 22x - 36$$

$$(14) (8x+9)(9x+4) \\ = 72x^2 + 113x + 36$$

$$(29) (9x+8)(8x+6) \\ = 72x^2 + 118x + 48$$

$$(44) (9x+9)(6x-7) \\ = 54x^2 - 9x - 63$$

$$(15) (7x-1)(7x+7) \\ = 49x^2 + 42x - 7$$

$$(30) (3x-8)(9x-2) \\ = 27x^2 - 78x + 16$$

$$(45) (9x-8)^2 \\ = 81x^2 - 144x + 64$$

$$(1) (4x-1)(8x-9) \\ = 32x^2 - 44x + 9$$

$$(16) (3x-5)(5x-1) \\ = 15x^2 - 28x + 5$$

$$(31) (4x-3)(5x-8) \\ = 20x^2 - 47x + 24$$

$$(2) (4x+8)(6x-5) \\ = 24x^2 + 28x - 40$$

$$(17) (5x+4)(7x-9) \\ = 35x^2 - 17x - 36$$

$$(32) (5x+4)(9x-8) \\ = 45x^2 - 4x - 32$$

$$(3) (6x-4)(8x+3) \\ = 48x^2 - 14x - 12$$

$$(18) (6x-7)(4x+4) \\ = 24x^2 - 4x - 28$$

$$(33) (6x+1)(3x+9) \\ = 18x^2 + 57x + 9$$

$$(4) (2x-5)(9x-5) \\ = 18x^2 - 55x + 25$$

$$(19) (8x-9)(6x-2) \\ = 48x^2 - 70x + 18$$

$$(34) (9x+1)(9x+9) \\ = 81x^2 + 90x + 9$$

$$(5) (7x-4)(8x+9) \\ = 56x^2 + 31x - 36$$

$$(20) (4x-9)(2x+9) \\ = 8x^2 + 18x - 81$$

$$(35) (4x-3)(6x+5) \\ = 24x^2 + 2x - 15$$

$$(6) (6x-3)(4x+7) \\ = 24x^2 + 30x - 21$$

$$(21) (9x-3)(6x+6) \\ = 54x^2 + 36x - 18$$

$$(36) (x+8)(5x-1) \\ = 5x^2 + 39x - 8$$

$$(7) (6x-5)(7x+7) \\ = 42x^2 + 7x - 35$$

$$(22) (2x-4)(2x+3) \\ = 4x^2 - 2x - 12$$

$$(37) (6x+3)(9x-3) \\ = 54x^2 + 9x - 9$$

$$(8) (5x+5)(9x-5) \\ = 45x^2 + 20x - 25$$

$$(23) (7x-6)(3x+2) \\ = 21x^2 - 4x - 12$$

$$(38) (8x-8)(7x+7) \\ = 56x^2 - 56$$

$$(9) (6x-9)(2x-8) \\ = 12x^2 - 66x + 72$$

$$(24) (9x+6)(5x-9) \\ = 45x^2 - 51x - 54$$

$$(39) (3x-4)(2x+2) \\ = 6x^2 - 2x - 8$$

$$(10) (8x+1)(7x-8) \\ = 56x^2 - 57x - 8$$

$$(25) (6x+6)(x-1) \\ = 6x^2 - 6$$

$$(40) (3x-2)(6x-3) \\ = 18x^2 - 21x + 6$$

$$(11) (6x+4)(8x-9) \\ = 48x^2 - 22x - 36$$

$$(26) (8x+3)(6x+6) \\ = 48x^2 + 66x + 18$$

$$(41) (7x+1)(3x+5) \\ = 21x^2 + 38x + 5$$

$$(12) (9x-9)(3x-5) \\ = 27x^2 - 72x + 45$$

$$(27) (3x-9)(8x-2) \\ = 24x^2 - 78x + 18$$

$$(42) (4x+5)(3x+8) \\ = 12x^2 + 47x + 40$$

$$(13) (6x-9)(8x-9) \\ = 48x^2 - 126x + 81$$

$$(28) (5x+7)(6x+8) \\ = 30x^2 + 82x + 56$$

$$(43) (2x+2)(8x+8) \\ = 16x^2 + 32x + 16$$

$$(14) (4x+6)(3x-6) \\ = 12x^2 - 6x - 36$$

$$(29) (5x-8)(x+3) \\ = 5x^2 + 7x - 24$$

$$(44) (x-1)(3x-5) \\ = 3x^2 - 8x + 5$$

$$(15) (4x+1)(9x-1) \\ = 36x^2 + 5x - 1$$

$$(30) (2x+9)(2x+8) \\ = 4x^2 + 34x + 72$$

$$(45) (4x+6)(5x+8) \\ = 20x^2 + 62x + 48$$

$$(1) (5x+3)(6x-9) \\ = 30x^2 - 27x - 27$$

$$(16) (x+1)(2x+6) \\ = 2x^2 + 8x + 6$$

$$(31) (3x-3)(4x+5) \\ = 12x^2 + 3x - 15$$

$$(2) (x+4)(8x-5) \\ = 8x^2 + 27x - 20$$

$$(17) (3x-2)(4x+9) \\ = 12x^2 + 19x - 18$$

$$(32) (3x-5)(7x+8) \\ = 21x^2 - 11x - 40$$

$$(3) (3x+6)(7x+4) \\ = 21x^2 + 54x + 24$$

$$(18) (3x-9)(4x-9) \\ = 12x^2 - 63x + 81$$

$$(33) (4x+7)(9x-1) \\ = 36x^2 + 59x - 7$$

$$(4) (9x+6)(8x+7) \\ = 72x^2 + 111x + 42$$

$$(19) (4x+5)(6x-5) \\ = 24x^2 + 10x - 25$$

$$(34) (5x+7)(6x-7) \\ = 30x^2 + 7x - 49$$

$$(5) (2x-6)(2x-1) \\ = 4x^2 - 14x + 6$$

$$(20) (9x-9)(8x-4) \\ = 72x^2 - 108x + 36$$

$$(35) (2x+8)(x+4) \\ = 2x^2 + 16x + 32$$

$$(6) (x-8)(x-1) \\ = x^2 - 9x + 8$$

$$(21) (4x+1)(3x-1) \\ = 12x^2 - x - 1$$

$$(36) (4x-5)(4x-8) \\ = 16x^2 - 52x + 40$$

$$(7) (3x-6)(4x-6) \\ = 12x^2 - 42x + 36$$

$$(22) (x-8)(7x-6) \\ = 7x^2 - 62x + 48$$

$$(37) (8x+3)(3x+5) \\ = 24x^2 + 49x + 15$$

$$(8) (7x+1)(x-1) \\ = 7x^2 - 6x - 1$$

$$(23) (9x-1)(5x-9) \\ = 45x^2 - 86x + 9$$

$$(38) (2x-1)(6x-3) \\ = 12x^2 - 12x + 3$$

$$(9) (x+1)(3x-4) \\ = 3x^2 - x - 4$$

$$(24) (6x-9)(9x+4) \\ = 54x^2 - 57x - 36$$

$$(39) (7x+2)(3x-9) \\ = 21x^2 - 57x - 18$$

$$(10) (7x-1)(6x+3) \\ = 42x^2 + 15x - 3$$

$$(25) (5x-7)(8x+8) \\ = 40x^2 - 16x - 56$$

$$(40) (8x+1)(7x+5) \\ = 56x^2 + 47x + 5$$

$$(11) (4x+8)(6x+7) \\ = 24x^2 + 76x + 56$$

$$(26) (7x+3)(3x+4) \\ = 21x^2 + 37x + 12$$

$$(41) (x-8)(6x-1) \\ = 6x^2 - 49x + 8$$

$$(12) (5x+4)(x-7) \\ = 5x^2 - 31x - 28$$

$$(27) (7x+7)(9x+8) \\ = 63x^2 + 119x + 56$$

$$(42) (7x+3)(5x+9) \\ = 35x^2 + 78x + 27$$

$$(13) (3x-4)(5x-9) \\ = 15x^2 - 47x + 36$$

$$(28) (6x+9)(2x-5) \\ = 12x^2 - 12x - 45$$

$$(43) (2x-7)(2x-2) \\ = 4x^2 - 18x + 14$$

$$(14) (9x-4)(8x+6) \\ = 72x^2 + 22x - 24$$

$$(29) (5x+1)(7x+6) \\ = 35x^2 + 37x + 6$$

$$(44) (7x+9)(5x-3) \\ = 35x^2 + 24x - 27$$

$$(15) (6x-9)(2x+2) \\ = 12x^2 - 6x - 18$$

$$(30) (9x-6)(7x+3) \\ = 63x^2 - 15x - 18$$

$$(45) (6x+1)(6x+8) \\ = 36x^2 + 54x + 8$$

$$(1) (4x+7)(3x-6) \\ = 12x^2 - 3x - 42$$

$$(16) (9x-4)(4x+4) \\ = 36x^2 + 20x - 16$$

$$(31) (8x-2)(9x-5) \\ = 72x^2 - 58x + 10$$

$$(2) (3x-8)(9x-1) \\ = 27x^2 - 75x + 8$$

$$(17) (8x+1)(x+4) \\ = 8x^2 + 33x + 4$$

$$(32) (8x-5)(x+8) \\ = 8x^2 + 59x - 40$$

$$(3) (7x-5)(6x-5) \\ = 42x^2 - 65x + 25$$

$$(18) (9x+4)(x+5) \\ = 9x^2 + 49x + 20$$

$$(33) (x-6)(7x+3) \\ = 7x^2 - 39x - 18$$

$$(4) (6x+1)(x+6) \\ = 6x^2 + 37x + 6$$

$$(19) (x+5)(9x+1) \\ = 9x^2 + 46x + 5$$

$$(34) (2x+4)(7x-3) \\ = 14x^2 + 22x - 12$$

$$(5) (6x+9)(5x-4) \\ = 30x^2 + 21x - 36$$

$$(20) (9x-2)(4x-8) \\ = 36x^2 - 80x + 16$$

$$(35) (4x-5)(6x-4) \\ = 24x^2 - 46x + 20$$

$$(6) (3x-1)(x+9) \\ = 3x^2 + 26x - 9$$

$$(21) (4x+4)(x-6) \\ = 4x^2 - 20x - 24$$

$$(36) (3x+2)(x-6) \\ = 3x^2 - 16x - 12$$

$$(7) (7x-3)(5x+9) \\ = 35x^2 + 48x - 27$$

$$(22) (5x-7)(4x+1) \\ = 20x^2 - 23x - 7$$

$$(37) (7x+8)(x+9) \\ = 7x^2 + 71x + 72$$

$$(8) (7x-3)(8x+1) \\ = 56x^2 - 17x - 3$$

$$(23) (8x-9)(7x-1) \\ = 56x^2 - 71x + 9$$

$$(38) (3x-6)(3x+2) \\ = 9x^2 - 12x - 12$$

$$(9) (5x+7)(7x-5) \\ = 35x^2 + 24x - 35$$

$$(24) (6x+9)(8x+7) \\ = 48x^2 + 114x + 63$$

$$(39) (3x-2)(3x-8) \\ = 9x^2 - 30x + 16$$

$$(10) (9x+2)(3x-3) \\ = 27x^2 - 21x - 6$$

$$(25) (3x+5)(x+1) \\ = 3x^2 + 8x + 5$$

$$(40) (x-9)(7x+8) \\ = 7x^2 - 55x - 72$$

$$(11) (8x+3)(4x-7) \\ = 32x^2 - 44x - 21$$

$$(26) (9x-6)(2x+3) \\ = 18x^2 + 15x - 18$$

$$(41) (3x+2)(8x-1) \\ = 24x^2 + 13x - 2$$

$$(12) (6x+1)(4x+7) \\ = 24x^2 + 46x + 7$$

$$(27) (2x+3)(6x-2) \\ = 12x^2 + 14x - 6$$

$$(42) (3x-1)(7x-9) \\ = 21x^2 - 34x + 9$$

$$(13) (6x-7)(x-1) \\ = 6x^2 - 13x + 7$$

$$(28) (4x-2)(x-5) \\ = 4x^2 - 22x + 10$$

$$(43) (3x+5)(2x+5) \\ = 6x^2 + 25x + 25$$

$$(14) (2x+9)(9x+1) \\ = 18x^2 + 83x + 9$$

$$(29) (9x-5)(3x+8) \\ = 27x^2 + 57x - 40$$

$$(44) (x+8)(9x+8) \\ = 9x^2 + 80x + 64$$

$$(15) (x-8)(x+5) \\ = x^2 - 3x - 40$$

$$(30) (x-7)(7x-1) \\ = 7x^2 - 50x + 7$$

$$(45) (x-9)(8x+3) \\ = 8x^2 - 69x - 27$$

$$(1) (4x-7)(7x-2) \\ = 28x^2 - 57x + 14$$

$$(16) (3x+1)(x+4) \\ = 3x^2 + 13x + 4$$

$$(31) (9x-9)(2x+3) \\ = 18x^2 + 9x - 27$$

$$(2) (x-2)(4x+3) \\ = 4x^2 - 5x - 6$$

$$(17) (4x+1)(x-8) \\ = 4x^2 - 31x - 8$$

$$(32) (5x-3)(9x+8) \\ = 45x^2 + 13x - 24$$

$$(3) (9x+6)(3x+1) \\ = 27x^2 + 27x + 6$$

$$(18) (7x-4)(x+7) \\ = 7x^2 + 45x - 28$$

$$(33) (8x+8)(4x-5) \\ = 32x^2 - 8x - 40$$

$$(4) (2x+7)(5x-4) \\ = 10x^2 + 27x - 28$$

$$(19) (6x+2)(2x-6) \\ = 12x^2 - 32x - 12$$

$$(34) (8x+4)(4x-2) \\ = 32x^2 - 8$$

$$(5) (5x-9)(6x-4) \\ = 30x^2 - 74x + 36$$

$$(20) (3x+7)(3x+6) \\ = 9x^2 + 39x + 42$$

$$(35) (8x+9)(x+6) \\ = 8x^2 + 57x + 54$$

$$(6) (5x-1)(9x+6) \\ = 45x^2 + 21x - 6$$

$$(21) (7x-8)(2x+5) \\ = 14x^2 + 19x - 40$$

$$(36) (2x-3)(x+8) \\ = 2x^2 + 13x - 24$$

$$(7) (4x+3)(8x+3) \\ = 32x^2 + 36x + 9$$

$$(22) (3x+3)(8x-5) \\ = 24x^2 + 9x - 15$$

$$(37) (2x+5)(9x-8) \\ = 18x^2 + 29x - 40$$

$$(8) (2x-7)(6x+7) \\ = 12x^2 - 28x - 49$$

$$(23) (8x+1)(x+6) \\ = 8x^2 + 49x + 6$$

$$(38) (2x+5)(4x-5) \\ = 8x^2 + 10x - 25$$

$$(9) (2x+8)(7x+4) \\ = 14x^2 + 64x + 32$$

$$(24) (9x-9)(9x+4) \\ = 81x^2 - 45x - 36$$

$$(39) (6x-2)(9x+8) \\ = 54x^2 + 30x - 16$$

$$(10) (4x+9)(5x-7) \\ = 20x^2 + 17x - 63$$

$$(25) (7x-7)(3x-4) \\ = 21x^2 - 49x + 28$$

$$(40) (7x+2)(6x-2) \\ = 42x^2 - 2x - 4$$

$$(11) (7x-2)(3x-4) \\ = 21x^2 - 34x + 8$$

$$(26) (5x+1)(3x-2) \\ = 15x^2 - 7x - 2$$

$$(41) (2x-7)(8x+6) \\ = 16x^2 - 44x - 42$$

$$(12) (2x-3)(3x+1) \\ = 6x^2 - 7x - 3$$

$$(27) (2x+2)(7x+5) \\ = 14x^2 + 24x + 10$$

$$(42) (x+2)(6x-7) \\ = 6x^2 + 5x - 14$$

$$(13) (2x+4)(3x-9) \\ = 6x^2 - 6x - 36$$

$$(28) (3x+1)(x+1) \\ = 3x^2 + 4x + 1$$

$$(43) (9x-2)(5x-4) \\ = 45x^2 - 46x + 8$$

$$(14) (6x-3)(5x-7) \\ = 30x^2 - 57x + 21$$

$$(29) (7x+3)(5x-8) \\ = 35x^2 - 41x - 24$$

$$(44) (5x-5)(8x-3) \\ = 40x^2 - 55x + 15$$

$$(15) (9x+3)(x-3) \\ = 9x^2 - 24x - 9$$

$$(30) (2x+7)(3x-2) \\ = 6x^2 + 17x - 14$$

$$(45) (3x-7)(8x-7) \\ = 24x^2 - 77x + 49$$

## 2.3 除法

3 次多項式を 2 次多項式で割った商と剰余を求める問題。

$$(1) \quad (-12x^3 + 26x^2 + 16x + 2) \div (2x^2 - 5x - 1)$$

$$(6) \quad (-4x^3 - 27x^2 + 8x + 40) \div (x^2 + 8x + 8)$$

$$(2) \quad (45x^3 - 126x^2 + 153x - 72) \div (5x^2 - 9x + 8)$$

$$(7) \quad (-72x^3 + 10x^2 + 34x - 8) \div (9x^2 + x - 4)$$

$$(3) \quad (-28x^3 - 15x^2 - 3x - 35) \div (7x^2 - 5x + 7)$$

$$(8) \quad (18x^3 - 66x^2 + 92x - 48) \div (3x^2 - 7x + 6)$$

$$(4) \quad (-24x^3 - 90x^2 - 97x - 36) \div (6x^2 + 9x + 4)$$

$$(9) \quad (-35x^3 + 69x^2 - 53x + 35) \div (7x^2 - 4x + 5)$$

$$(5) \quad (-14x^3 - 54x^2 - 43x - 6) \div (2x^2 + 6x + 1)$$

$$(10) \quad (-3x^3 + 5x + 14) \div (3x^2 + 6x + 7)$$

(1)  $(-56x^3 - 34x^2 - 34x - 30) \div (7x^2 - x + 5)$

(6)  $(-81x^3 + 45x^2 + 5x - 2) \div (9x^2 - 7x + 1)$

(2)  $(9x^3 + 85x^2 + 34x - 18) \div (9x^2 + 4x - 2)$

(7)  $(-24x^3 - 22x^2 - 16x - 3) \div (6x^2 + 4x + 3)$

(3)  $(64x^3 - 104x^2 + 8x + 20) \div (8x^2 - 8x - 4)$

(8)  $(-8x^3 + 16x^2 + 40x - 48) \div (4x^2 + 4x - 8)$

(4)  $(-21x^3 - 10x^2 + 17x + 6) \div (7x^2 + x - 6)$

(9)  $(35x^3 + 52x^2 + 58x + 63) \div (5x^2 + x + 7)$

(5)  $(-28x^3 + 52x^2 - 20x - 4) \div (7x^2 - 6x - 1)$

(10)  $(-24x^3 + 76x^2 + 8x - 20) \div (3x^2 - 8x - 5)$



(1)  $(3x^3 - 17x^2 - 8x + 12) \div (x^2 - 5x - 6)$

(6)  $(-30x^3 - 48x^2 - 58x - 24) \div (6x^2 + 6x + 8)$

(2)  $(-6x^3 - 39x^2 - 14x + 24) \div (6x^2 + 3x - 4)$

(7)  $(25x^3 - 85x^2 + 37x + 63) \div (5x^2 - 8x - 7)$

(3)  $(-35x^3 - 17x^2 + 12x + 4) \div (5x^2 + x - 2)$

(8)  $(24x^3 + 48x^2) \div (6x^2)$

(4)  $(-32x^3 + 72x^2 - 96x + 56) \div (4x^2 - 5x + 7)$

(9)  $(-30x^3 + 32x^2 + 34x - 28) \div (5x^2 - 2x - 7)$

(5)  $(-28x^3 - 20x^2 + 8x) \div (4x^2 + 4x)$

(10)  $(6x^3 + 59x^2 + 52x + 63) \div (6x^2 + 5x + 7)$

(1)  $(12x^3 + 24x^2 + 15x + 3) \div (4x^2 + 4x + 1)$

(6)  $(8x^3 - 36x^2 + 13x + 12) \div (8x^2 - 4x - 3)$

(2)  $(27x^3 + 63x^2 - 75x - 63) \div (9x^2 - 6x - 7)$

(7)  $(14x^3 + 36x^2 + 20x + 8) \div (7x^2 + 4x + 2)$

(3)  $(32x^3 - 24x^2 - 44x - 12) \div (4x^2 - 5x - 3)$

(8)  $(56x^3 - 4x^2 + 48x + 32) \div (7x^2 - 4x + 8)$

(4)  $(54x^3 + 75x^2 + 48x + 12) \div (9x^2 + 8x + 4)$

(9)  $(9x^3 + 38x^2 - 107x + 56) \div (x^2 + 5x - 8)$

(5)  $(-30x^3 + 12x^2 + 54x) \div (5x^2 - 2x - 9)$

(10)  $(-54x^3 + 57x^2 + 79x + 8) \div (6x^2 - 7x - 8)$

(1)  $(27x^3 - 84x^2 + 90x - 9) \div (3x^2 - 9x + 9)$

(6)  $(9x^3 + 7x^2 + x + 3) \div (9x^2 - 2x + 3)$

(2)  $(-6x^3 + 7x^2 + 15x + 4) \div (2x^2 - 3x - 4)$

(7)  $(45x^3 + 59x^2 + 15x + 1) \div (5x^2 + 6x + 1)$

(3)  $(-27x^3 + 72x^2 - 15x - 14) \div (9x^2 - 3x - 2)$

(8)  $(6x^3 + 53x^2 + 15x - 4) \div (x^2 + 9x + 4)$

(4)  $(12x^3 + 40x^2 + 66x + 18) \div (2x^2 + 6x + 9)$

(9)  $(8x^3 + 27x^2 - 21x - 4) \div (8x^2 - 5x - 1)$

(5)  $(8x^3 - 16x^2 - 14x + 10) \div (4x^2 + 2x - 2)$

(10)  $(-40x^3 - 16x^2 + 64x) \div (5x^2 + 2x - 8)$

$$(1) (35x^3 + 52x^2 - 37x - 14) \div (5x^2 + 6x - 7)$$

$$(6) (-7x^3 - 40x^2 + 67x + 28) \div (7x^2 - 9x - 4)$$

$$(2) (-6x^3 + 18x^2 + 8x - 24) \div (6x^2 - 8)$$

$$(7) (-6x^3 + 34x^2 - 62x + 28) \div (x^2 - 5x + 7)$$

$$(3) (-2x^3 - 12x^2 - 13x + 15) \div (2x^2 + 6x - 5)$$

$$(8) (-45x^3 + 7x^2 + 37x - 15) \div (9x^2 + 4x - 5)$$

$$(4) (8x^3 - 4x^2 + 20x - 24) \div (2x^2 + x + 6)$$

$$(9) (-56x^3 + 121x^2 - 105x + 54) \div (8x^2 - 7x + 6)$$

$$(5) (56x^3 - 99x^2 + 112x - 45) \div (7x^2 - 8x + 9)$$

$$(10) (-9x^3 - 45x^2) \div (x^2 + 5x)$$

$$(1) (18x^3 + 20x^2 - 5x - 24) \div (2x^2 + 4x + 3)$$

$$(6) (12x^3 - 3x^2 - 10x + 56) \div (3x^2 - 6x + 8)$$

$$(2) (21x^3 + 33x^2 - 12) \div (7x^2 + 4x - 4)$$

$$(7) (35x^3 - 94x^2 + 77x - 18) \div (5x^2 - 7x + 2)$$

$$(3) (-6x^3 - 58x^2 - 60x - 16) \div (x^2 + 9x + 4)$$

$$(8) (36x^3 + 39x^2 - 10x + 56) \div (9x^2 - 6x + 8)$$

$$(4) (-36x^3 + 6x^2 + 12x) \div (6x^2 - 4x)$$

$$(9) (-21x^3 - 42x^2 - 36x - 15) \div (7x^2 + 7x + 5)$$

$$(5) (-7x^3 + 43x^2 - 38x - 10) \div (7x^2 - 8x - 2)$$

$$(10) (20x^3 - 57x^2 + 85x - 72) \div (4x^2 - 5x + 9)$$

(1)  $(-63x^3 + 17x^2 + 13x + 5) \div (9x^2 + 4x + 1)$

(6)  $(-30x^3 - 8x^2 + 82x - 24) \div (5x^2 + 8x - 3)$

(2)  $(12x^3 - 11x^2 + 11x - 20) \div (3x^2 + x + 4)$

(7)  $(72x^3 + 46x^2 - 78x + 8) \div (8x^2 + 6x - 8)$

(3)  $(-8x^3 + 26x^2 + 10x + 56) \div (4x^2 + 3x + 7)$

(8)  $(6x^3 - 21x^2 + 18x + 15) \div (x^2 - 4x + 5)$

(4)  $(-5x^3 - 22x^2 + 10x - 25) \div (5x^2 - 3x + 5)$

(9)  $(-36x^3 - 27x^2 + 45x + 18) \div (4x^2 + 7x + 2)$

(5)  $(-49x^3 - 56x^2 - 63x) \div (7x^2 + 8x + 9)$

(10)  $(24x^3 - 38x^2 + 17x - 42) \div (6x^2 + x + 6)$

$$(1) (9x^3 - 4x + 35) \div (3x^2 - 5x + 7)$$

$$(6) (27x^3 + 33x^2 - 78x + 28) \div (3x^2 + 6x - 4)$$

$$(2) (35x^3 + 80x^2 - 4x - 63) \div (5x^2 + 5x - 7)$$

$$(7) (-6x^3 - 13x^2 + 90x - 63) \div (2x^2 + 9x - 9)$$

$$(3) (-20x^3 - 44x^2 + 16) \div (5x^2 + x - 2)$$

$$(8) (-45x^3 + 58x^2 + 7x) \div (9x^2 + x)$$

$$(4) (-18x^3 + 69x^2 - 63x + 15) \div (3x^2 - 9x + 3)$$

$$(9) (42x^3 - 54x^2 + 12x) \div (6x^2 - 6x)$$

$$(5) (-21x^3 - 54x^2 - 39x - 30) \div (7x^2 + 4x + 5)$$

$$(10) (-45x^3 - 21x^2 + 11x - 1) \div (9x^2 + 6x - 1)$$

$$(1) (20x^3 + 13x^2 - 36x - 21) \div (4x^2 - 3x - 3)$$

$$(6) (72x^3 + 37x^2 + 19x + 15) \div (9x^2 - x + 3)$$

$$(2) (32x^3 + 36x^2 - 26x - 12) \div (8x^2 - 3x - 2)$$

$$(7) (54x^3 + 6x^2) \div (6x^2)$$

$$(3) (56x^3 + 45x^2 + 37x + 12) \div (8x^2 + 3x + 4)$$

$$(8) (64x^3 - 64x^2 - 56x) \div (8x^2 - 8x - 7)$$

$$(4) (42x^3 + 90x^2 + 34x - 16) \div (6x^2 + 6x - 2)$$

$$(9) (-6x^3 + 17x^2 + 49) \div (3x^2 + 2x + 7)$$

$$(5) (-16x^3 + 46x^2 + 82x - 40) \div (8x^2 + 9x - 5)$$

$$(10) (14x^3 + 70x^2 + 72x + 64) \div (7x^2 + 7x + 8)$$



(1)  $(45x^3 + 47x^2 + 30x + 8) \div (5x^2 + 3x + 2)$

(6)  $(48x^3 + 70x^2 - 30x - 16) \div (8x^2 + 9x - 8)$

(2)  $(-28x^3 - 27x^2 - 11x + 6) \div (4x^2 + 5x + 3)$

(7)  $(-64x^3 + 88x^2 + 42x - 45) \div (8x^2 - 6x - 9)$

(3)  $(63x^3 - 21x^2 - 69x - 18) \div (7x^2 - 7x - 3)$

(8)  $(8x^3 + 52x^2 + 88x + 32) \div (x^2 + 6x + 8)$

(4)  $(56x^3 - 103x^2 - 18x + 45) \div (8x^2 - 9x - 9)$

(9)  $(8x^3 + 36x^2 + 44x + 8) \div (2x^2 + 5x + 1)$

(5)  $(30x^3 + 46x^2 + 50x + 56) \div (5x^2 + x + 7)$

(10)  $(-32x^3 + 4x^2 + 39x + 15) \div (4x^2 - 3x - 3)$

$$(1) \quad (-5x^3 - 4x^2 - 30x - 24) \div (x^2 + 6)$$

$$(6) \quad (25x^3 + 15x^2 - 74x - 36) \div (5x^2 - 6x - 4)$$

$$(2) \quad (64x^3 - 112x^2 - 3x + 30) \div (8x^2 - 9x - 6)$$

$$(7) \quad (8x^3 + 32x^2 - 8x) \div (x^2 + 4x - 1)$$

$$(3) \quad (6x^3 + 53x^2 - 8x - 40) \div (x^2 + 8x - 8)$$

$$(8) \quad (-6x^3 + 51x + 9) \div (2x^2 + 6x + 1)$$

$$(4) \quad (10x^3 - 15x^2 + 5x) \div (2x^2 - 3x + 1)$$

$$(9) \quad (-15x^3 + 19x^2 + 59x - 8) \div (5x^2 + 7x - 1)$$

$$(5) \quad (-25x^3 - 45x^2 - 10x + 8) \div (5x^2 + 5x - 2)$$

$$(10) \quad (-18x^3 + 34x^2 - 6x + 20) \div (9x^2 + x + 5)$$

(1)  $(-2x^3 + x^2 + 32x + 14) \div (x^2 - 4x - 2)$

(6)  $(63x^3 - 95x^2 + 54x - 8) \div (7x^2 - 9x + 4)$

(2)  $(-5x^3 - 12x^2 + 4x + 16) \div (5x^2 + 2x - 8)$

(7)  $(24x^3 + 84x^2 + 60x + 72) \div (8x^2 + 4x + 8)$

(3)  $(-10x^3 + 13x^2 + 30x) \div (5x^2 + 6x)$

(8)  $(-30x^3 + 29x^2 + 15x - 25) \div (5x^2 - 9x + 5)$

(4)  $(-30x^3 - 15x^2 + 35x) \div (6x^2 + 3x - 7)$

(9)  $(54x^3 + 6x^2 - 13x + 2) \div (6x^2 + 2x - 1)$

(5)  $(-63x^3 - 130x^2 - 63x) \div (7x^2 + 9x)$

(10)  $(-42x^3 + 57x^2 - 21x + 45) \div (7x^2 + x + 5)$

$$(1) (4x^3 - 36x^2 + 20x) \div (x^2 - 9x + 5)$$

$$(6) (-21x^3 - 22x^2 + 58x + 45) \div (3x^2 + x - 9)$$

$$(2) (49x^3 + 14x^2 - 78x + 27) \div (7x^2 + 5x - 9)$$

$$(7) (16x^3 + 50x^2 + 54x + 6) \div (2x^2 + 6x + 6)$$

$$(3) (5x^3 + 40x^2 + 10x) \div (x^2 + 8x + 2)$$

$$(8) (-18x^3 + 60x^2 - 12x - 48) \div (3x^2 - 6x - 6)$$

$$(4) (18x^3 - 30x^2 - 24x + 24) \div (3x^2 - 8x + 4)$$

$$(9) (-72x^3 - 4x^2 + 101x + 45) \div (8x^2 - 4x - 9)$$

$$(5) (36x^3 + 45x^2 - 20x - 25) \div (9x^2 - 5)$$

$$(10) (32x^3 + 20x^2 + 66x + 8) \div (4x^2 + 2x + 8)$$

(1)  $(-40x^3 - 58x^2 - 20x - 2) \div (5x^2 + 6x + 1)$

(6)  $(-63x^3 - 85x^2 - 91x - 36) \div (9x^2 + 7x + 9)$

(2)  $(-48x^3 + 8x^2 + 48x - 20) \div (6x^2 + 2x - 5)$

(7)  $(10x^3 + 20x^2 - 22x + 24) \div (5x^2 - 5x + 4)$

(3)  $(-7x^3 - 2x^2 + 11x + 6) \div (7x^2 - 5x - 6)$

(8)  $(8x^3 - 14x^2 - 52x + 45) \div (2x^2 - 8x + 5)$

(4)  $(-48x^3 + 72x^2 + 16x - 20) \div (6x^2 - 6x - 5)$

(9)  $(12x^3 - 20x^2 - 34x - 42) \div (6x^2 + 8x + 7)$

(5)  $(-2x^3 + x^2 - 6x) \div (2x^2 - x + 6)$

(10)  $(30x^3 - 31x^2 - 60x - 9) \div (5x^2 - 6x - 9)$

(1)  $(-12x^3 - 38x^2 + 64x - 36) \div (6x^2 - 8x + 4)$

(6)  $(10x^3 + 13x^2 + 39x + 28) \div (2x^2 + x + 7)$

(2)  $(-5x^3 - 25x^2 + 5x) \div (x^2 + 5x - 1)$

(7)  $(40x^3 - 31x^2 + 62x - 21) \div (5x^2 - 2x + 7)$

(3)  $(42x^3 - 21x^2 - 49x) \div (6x^2 - 3x - 7)$

(8)  $(12x^3 - 50x^2 - 26x + 36) \div (6x^2 + 2x - 4)$

(4)  $(24x^3 - 52x^2 + 47x - 45) \div (8x^2 - 4x + 9)$

(9)  $(54x^3 - 63x^2 - 54x + 63) \div (9x^2 - 9)$

(5)  $(28x^3 + 22x^2 + 18x + 4) \div (4x^2 + 2x + 2)$

(10)  $(14x^3 + 79x^2 + 100x + 32) \div (2x^2 + 9x + 4)$

(1)  $(36x^3 - 79x^2 - 35x + 28) \div (4x^2 - 7x - 7)$

(6)  $(9x^3 + 12x^2 + 9x + 42) \div (3x^2 - 2x + 7)$

(2)  $(-27x^3 + 15x^2 - 46x + 48) \div (3x^2 + x + 6)$

(7)  $(4x^3 - 16x^2 + 4x - 16) \div (4x^2 + 4)$

(3)  $(-6x^3 + 22x^2 - 34x - 14) \div (x^2 - 4x + 7)$

(8)  $(8x^3 - 4x^2 - 6x + 2) \div (4x^2 + 2x - 1)$

(4)  $(8x^3 - 33x^2 - 31x - 20) \div (8x^2 + 7x + 4)$

(9)  $(48x^3 - 108x^2 + 70x - 12) \div (6x^2 - 9x + 2)$

(5)  $(-24x^3 + 8x^2 - 2x + 2) \div (6x^2 + x + 1)$

(10)  $(-12x^3 - 56x^2 - 53x - 14) \div (6x^2 + 7x + 2)$

(1)  $(-12x^3 - 18x^2 + 48x + 9) \div (2x^2 + 6x + 1)$

(6)  $(-6x^3 - 61x^2 - 56x + 63) \div (6x^2 + 7x - 7)$

(2)  $(24x^3 + 22x^2 - 66x - 8) \div (4x^2 + 9x + 1)$

(7)  $(12x^3 + 6x^2 - 30x - 18) \div (3x^2 - 3x - 3)$

(3)  $(20x^3 - 16x^2 - 32x) \div (5x^2 - 4x - 8)$

(8)  $(15x^3 - 38x^2 + 48x - 32) \div (5x^2 - 6x + 8)$

(4)  $(-4x^3 - 16x^2 - 12x) \div (4x^2 + 4x)$

(9)  $(42x^3 - 30x^2 + 36x - 48) \div (7x^2 + 2x + 8)$

(5)  $(36x^3 + 54x^2 + 74x + 45) \div (6x^2 + 4x + 9)$

(10)  $(-36x^3 + 76x^2 + x - 1) \div (4x^2 - 8x - 1)$



$$(1) (8x^3 - 2x^2 + x + 2) \div (4x^2 - 3x + 2)$$

$$(6) (48x^3 + 56x^2 + 8x) \div (6x^2 + 7x + 1)$$

$$(2) (48x^3 - 28x^2 + 28x - 8) \div (8x^2 - 2x + 4)$$

$$(7) (9x^3 - 45x^2 + 36) \div (x^2 - 4x - 4)$$

$$(3) (-5x^3 + 22x^2 + 16x - 5) \div (5x^2 + 3x - 1)$$

$$(8) (36x^3 - 54x^2 + 56x - 49) \div (6x^2 - 2x + 7)$$

$$(4) (-63x^3 - 36x^2 - 18x - 45) \div (7x^2 - 3x + 5)$$

$$(9) (18x^3 - 6x^2 + 12x - 4) \div (3x^2 + 2)$$

$$(5) (36x^3 + 48x^2 - 5x - 14) \div (6x^2 + x - 2)$$

$$(10) (-20x^3 + 25x^2 + 27x - 8) \div (5x^2 - 5x - 8)$$

(1)  $(-24x^3 + 7x^2 + 3x + 2) \div (8x^2 + 3x + 1)$

(6)  $(5x^3 + x^2 - 6x) \div (x^2 - x)$

(2)  $(14x^3 - 64x^2 + 38x - 24) \div (7x^2 - 4x + 3)$

(7)  $(-15x^3 - 12x^2 - 18x) \div (5x^2 + 4x + 6)$

(3)  $(-8x^3 - 12x^2 + 88x - 64) \div (2x^2 + 7x - 8)$

(8)  $(4x^3 + 8x^2 - 2x) \div (2x^2 + 4x - 1)$

(4)  $(-48x^3 - 24x^2 - 32x - 16) \div (6x^2 + 4)$

(9)  $(24x^3 - 46x^2 - 6x + 4) \div (4x^2 - 9x + 2)$

(5)  $(42x^3 + 30x^2 - 63x - 45) \div (6x^2 - 9)$

(10)  $(18x^3 + 12x^2 - 13x - 2) \div (6x^2 + 8x + 1)$

(1)  $(-12x^3 + 26x^2 + 16x + 2) \div (2x^2 - 5x - 1)$   
 商 :  $-6x - 2$ , 剰余 :  $-x + 2$

(6)  $(-4x^3 - 27x^2 + 8x + 40) \div (x^2 + 8x + 8)$   
 商 :  $-4x + 5$ , 剰余 :  $-6x + 1$

(2)  $(45x^3 - 126x^2 + 153x - 72) \div (5x^2 - 9x + 8)$   
 商 :  $9x - 9$ , 剰余 :  $x - 4$

(7)  $(-72x^3 + 10x^2 + 34x - 8) \div (9x^2 + x - 4)$   
 商 :  $-8x + 2$ , 剰余 :  $-3x$

(3)  $(-28x^3 - 15x^2 - 3x - 35) \div (7x^2 - 5x + 7)$   
 商 :  $-4x - 5$ , 剰余 :  $-4x - 4$

(8)  $(18x^3 - 66x^2 + 92x - 48) \div (3x^2 - 7x + 6)$   
 商 :  $6x - 8$ , 剰余 :  $1$

(4)  $(-24x^3 - 90x^2 - 97x - 36) \div (6x^2 + 9x + 4)$   
 商 :  $-4x - 9$ , 剰余 :  $4x + 5$

(9)  $(-35x^3 + 69x^2 - 53x + 35) \div (7x^2 - 4x + 5)$   
 商 :  $-5x + 7$ , 剰余 :  $-x + 7$

(5)  $(-14x^3 - 54x^2 - 43x - 6) \div (2x^2 + 6x + 1)$   
 商 :  $-7x - 6$ , 剰余 :  $6x - 4$

(10)  $(-3x^3 + 5x + 14) \div (3x^2 + 6x + 7)$   
 商 :  $-x + 2$ , 剰余 :  $5x - 8$

$$(1) (-56x^3 - 34x^2 - 34x - 30) \div (7x^2 - x + 5)$$

商:  $-8x - 6$ , 剰余:  $5x - 2$

$$(6) (-81x^3 + 45x^2 + 5x - 2) \div (9x^2 - 7x + 1)$$

商:  $-9x - 2$ , 剰余:  $-9$

$$(2) (9x^3 + 85x^2 + 34x - 18) \div (9x^2 + 4x - 2)$$

商:  $x + 9$ , 剰余:  $3x - 2$

$$(7) (-24x^3 - 22x^2 - 16x - 3) \div (6x^2 + 4x + 3)$$

商:  $-4x - 1$ , 剰余:  $2x - 2$

$$(3) (64x^3 - 104x^2 + 8x + 20) \div (8x^2 - 8x - 4)$$

商:  $8x - 5$ , 剰余:  $3x - 2$

$$(8) (-8x^3 + 16x^2 + 40x - 48) \div (4x^2 + 4x - 8)$$

商:  $-2x + 6$ , 剰余:  $-7x$

$$(4) (-21x^3 - 10x^2 + 17x + 6) \div (7x^2 + x - 6)$$

商:  $-3x - 1$ , 剰余:  $-4x + 6$

$$(9) (35x^3 + 52x^2 + 58x + 63) \div (5x^2 + x + 7)$$

商:  $7x + 9$ , 剰余:  $4x + 1$

$$(5) (-28x^3 + 52x^2 - 20x - 4) \div (7x^2 - 6x - 1)$$

商:  $-4x + 4$ , 剰余:  $-3$

$$(10) (-24x^3 + 76x^2 + 8x - 20) \div (3x^2 - 8x - 5)$$

商:  $-8x + 4$ , 剰余:  $3x - 8$

$$(1) (3x^3 - 17x^2 - 8x + 12) \div (x^2 - 5x - 6)$$

商 :  $3x - 2$ , 剰余 :  $-3$

$$(6) (-30x^3 - 48x^2 - 58x - 24) \div (6x^2 + 6x + 8)$$

商 :  $-5x - 3$ , 剰余 :  $-9x - 9$

$$(2) (-6x^3 - 39x^2 - 14x + 24) \div (6x^2 + 3x - 4)$$

商 :  $-x - 6$ , 剰余 :  $-5x - 6$

$$(7) (25x^3 - 85x^2 + 37x + 63) \div (5x^2 - 8x - 7)$$

商 :  $5x - 9$ , 剰余 :  $6x - 1$

$$(3) (-35x^3 - 17x^2 + 12x + 4) \div (5x^2 + x - 2)$$

商 :  $-7x - 2$ , 剰余 :  $6x - 7$

$$(8) (24x^3 + 48x^2) \div (6x^2)$$

商 :  $4x + 8$ , 剰余 :  $2$

$$(4) (-32x^3 + 72x^2 - 96x + 56) \div (4x^2 - 5x + 7)$$

商 :  $-8x + 8$ , 剰余 :  $-2x + 3$

$$(9) (-30x^3 + 32x^2 + 34x - 28) \div (5x^2 - 2x - 7)$$

商 :  $-6x + 4$ , 剰余 :  $-5x + 2$

$$(5) (-28x^3 - 20x^2 + 8x) \div (4x^2 + 4x)$$

商 :  $-7x + 2$ , 剰余 :  $9x + 6$

$$(10) (6x^3 + 59x^2 + 52x + 63) \div (6x^2 + 5x + 7)$$

商 :  $x + 9$ , 剰余 :  $-x - 2$

$$(1) (12x^3 + 24x^2 + 15x + 3) \div (4x^2 + 4x + 1)$$

商 :  $3x + 3$ , 剰余 :  $-4x + 4$

$$(6) (8x^3 - 36x^2 + 13x + 12) \div (8x^2 - 4x - 3)$$

商 :  $x - 4$ , 剰余 :  $-5x + 7$

$$(2) (27x^3 + 63x^2 - 75x - 63) \div (9x^2 - 6x - 7)$$

商 :  $3x + 9$ , 剰余 :  $2x + 4$

$$(7) (14x^3 + 36x^2 + 20x + 8) \div (7x^2 + 4x + 2)$$

商 :  $2x + 4$ , 剰余 :  $4x + 4$

$$(3) (32x^3 - 24x^2 - 44x - 12) \div (4x^2 - 5x - 3)$$

商 :  $8x + 4$ , 剰余 :  $-3x + 8$

$$(8) (56x^3 - 4x^2 + 48x + 32) \div (7x^2 - 4x + 8)$$

商 :  $8x + 4$ , 剰余 :  $-x - 8$

$$(4) (54x^3 + 75x^2 + 48x + 12) \div (9x^2 + 8x + 4)$$

商 :  $6x + 3$ , 剰余 :  $-8x + 5$

$$(9) (9x^3 + 38x^2 - 107x + 56) \div (x^2 + 5x - 8)$$

商 :  $9x - 7$ , 剰余 :  $-2x + 7$

$$(5) (-30x^3 + 12x^2 + 54x) \div (5x^2 - 2x - 9)$$

商 :  $-6x$ , 剰余 :  $-2x + 1$

$$(10) (-54x^3 + 57x^2 + 79x + 8) \div (6x^2 - 7x - 8)$$

商 :  $-9x - 1$ , 剰余 :  $-8x + 5$

$$(1) (27x^3 - 84x^2 + 90x - 9) \div (3x^2 - 9x + 9)$$

商 :  $9x - 1$ , 剰余 :  $-9x - 5$

$$(6) (9x^3 + 7x^2 + x + 3) \div (9x^2 - 2x + 3)$$

商 :  $x + 1$ , 剰余 :  $6x - 7$

$$(2) (-6x^3 + 7x^2 + 15x + 4) \div (2x^2 - 3x - 4)$$

商 :  $-3x - 1$ , 剰余 :  $-4x - 2$

$$(7) (45x^3 + 59x^2 + 15x + 1) \div (5x^2 + 6x + 1)$$

商 :  $9x + 1$ , 剰余 :  $9x - 6$

$$(3) (-27x^3 + 72x^2 - 15x - 14) \div (9x^2 - 3x - 2)$$

商 :  $-3x + 7$ , 剰余 :  $-7x + 5$

$$(8) (6x^3 + 53x^2 + 15x - 4) \div (x^2 + 9x + 4)$$

商 :  $6x - 1$ , 剰余 :  $-3x - 7$

$$(4) (12x^3 + 40x^2 + 66x + 18) \div (2x^2 + 6x + 9)$$

商 :  $6x + 2$ , 剰余 :  $-x + 4$

$$(9) (8x^3 + 27x^2 - 21x - 4) \div (8x^2 - 5x - 1)$$

商 :  $x + 4$ , 剰余 :  $x + 3$

$$(5) (8x^3 - 16x^2 - 14x + 10) \div (4x^2 + 2x - 2)$$

商 :  $2x - 5$ , 剰余 :  $9x + 6$

$$(10) (-40x^3 - 16x^2 + 64x) \div (5x^2 + 2x - 8)$$

商 :  $-8x$ , 剰余 :  $-8x - 3$

$$(1) (35x^3 + 52x^2 - 37x - 14) \div (5x^2 + 6x - 7)$$

商 :  $7x + 2$ , 剰余 :  $-2$

$$(6) (-7x^3 - 40x^2 + 67x + 28) \div (7x^2 - 9x - 4)$$

商 :  $-x - 7$ , 剰余 :  $4$

$$(2) (-6x^3 + 18x^2 + 8x - 24) \div (6x^2 - 8)$$

商 :  $-x + 3$ , 剰余 :  $-2x + 3$

$$(7) (-6x^3 + 34x^2 - 62x + 28) \div (x^2 - 5x + 7)$$

商 :  $-6x + 4$ , 剰余 :  $-4x + 9$

$$(3) (-2x^3 - 12x^2 - 13x + 15) \div (2x^2 + 6x - 5)$$

商 :  $-x - 3$ , 剰余 :  $-6x + 9$

$$(8) (-45x^3 + 7x^2 + 37x - 15) \div (9x^2 + 4x - 5)$$

商 :  $-5x + 3$ , 剰余 :  $-2x + 7$

$$(4) (8x^3 - 4x^2 + 20x - 24) \div (2x^2 + x + 6)$$

商 :  $4x - 4$ , 剰余 :  $x + 4$

$$(9) (-56x^3 + 121x^2 - 105x + 54) \div (8x^2 - 7x + 6)$$

商 :  $-7x + 9$ , 剰余 :  $9x + 6$

$$(5) (56x^3 - 99x^2 + 112x - 45) \div (7x^2 - 8x + 9)$$

商 :  $8x - 5$ , 剰余 :  $-x - 2$

$$(10) (-9x^3 - 45x^2) \div (x^2 + 5x)$$

商 :  $-9x$ , 剰余 :  $6x + 3$



$$(1) (18x^3 + 20x^2 - 5x - 24) \div (2x^2 + 4x + 3)$$

商 :  $9x - 8$ , 剰余 :  $4x - 8$

$$(6) (12x^3 - 3x^2 - 10x + 56) \div (3x^2 - 6x + 8)$$

商 :  $4x + 7$ , 剰余 :  $8x - 9$

$$(2) (21x^3 + 33x^2 - 12) \div (7x^2 + 4x - 4)$$

商 :  $3x + 3$ , 剰余 :  $x + 1$

$$(7) (35x^3 - 94x^2 + 77x - 18) \div (5x^2 - 7x + 2)$$

商 :  $7x - 9$ , 剰余 :  $-2x + 8$

$$(3) (-6x^3 - 58x^2 - 60x - 16) \div (x^2 + 9x + 4)$$

商 :  $-6x - 4$ , 剰余 :  $5x + 7$

$$(8) (36x^3 + 39x^2 - 10x + 56) \div (9x^2 - 6x + 8)$$

商 :  $4x + 7$ , 剰余 :  $8x + 1$

$$(4) (-36x^3 + 6x^2 + 12x) \div (6x^2 - 4x)$$

商 :  $-6x - 3$ , 剰余 :  $3$

$$(9) (-21x^3 - 42x^2 - 36x - 15) \div (7x^2 + 7x + 5)$$

商 :  $-3x - 3$ , 剰余 :  $-8x + 5$

$$(5) (-7x^3 + 43x^2 - 38x - 10) \div (7x^2 - 8x - 2)$$

商 :  $-x + 5$ , 剰余 :  $3x + 4$

$$(10) (20x^3 - 57x^2 + 85x - 72) \div (4x^2 - 5x + 9)$$

商 :  $5x - 8$ , 剰余 :  $-x + 5$

$$(1) (-63x^3 + 17x^2 + 13x + 5) \div (9x^2 + 4x + 1)$$

商:  $-7x + 5$ , 剰余:  $-x - 2$

$$(6) (-30x^3 - 8x^2 + 82x - 24) \div (5x^2 + 8x - 3)$$

商:  $-6x + 8$ , 剰余:  $2$

$$(2) (12x^3 - 11x^2 + 11x - 20) \div (3x^2 + x + 4)$$

商:  $4x - 5$ , 剰余:  $-3x + 2$

$$(7) (72x^3 + 46x^2 - 78x + 8) \div (8x^2 + 6x - 8)$$

商:  $9x - 1$ , 剰余:  $-4x - 1$

$$(3) (-8x^3 + 26x^2 + 10x + 56) \div (4x^2 + 3x + 7)$$

商:  $-2x + 8$ , 剰余:  $x + 9$

$$(8) (6x^3 - 21x^2 + 18x + 15) \div (x^2 - 4x + 5)$$

商:  $6x + 3$ , 剰余:  $x + 2$

$$(4) (-5x^3 - 22x^2 + 10x - 25) \div (5x^2 - 3x + 5)$$

商:  $-x - 5$ , 剰余:  $-x + 8$

$$(9) (-36x^3 - 27x^2 + 45x + 18) \div (4x^2 + 7x + 2)$$

商:  $-9x + 9$ , 剰余:  $-6x - 7$

$$(5) (-49x^3 - 56x^2 - 63x) \div (7x^2 + 8x + 9)$$

商:  $-7x$ , 剰余:  $-6x + 7$

$$(10) (24x^3 - 38x^2 + 17x - 42) \div (6x^2 + x + 6)$$

商:  $4x - 7$ , 剰余:  $2x - 4$

$$(1) (9x^3 - 4x + 35) \div (3x^2 - 5x + 7)$$

商 :  $3x + 5$ , 剰余 :  $-6x - 5$

$$(6) (27x^3 + 33x^2 - 78x + 28) \div (3x^2 + 6x - 4)$$

商 :  $9x - 7$ , 剰余 :  $9x + 4$

$$(2) (35x^3 + 80x^2 - 4x - 63) \div (5x^2 + 5x - 7)$$

商 :  $7x + 9$ , 剰余 :  $-6x - 7$

$$(7) (-6x^3 - 13x^2 + 90x - 63) \div (2x^2 + 9x - 9)$$

商 :  $-3x + 7$ , 剰余 :  $-9x - 7$

$$(3) (-20x^3 - 44x^2 + 16) \div (5x^2 + x - 2)$$

商 :  $-4x - 8$ , 剰余 :  $4x + 6$

$$(8) (-45x^3 + 58x^2 + 7x) \div (9x^2 + x)$$

商 :  $-5x + 7$ , 剰余 :  $-3x + 5$

$$(4) (-18x^3 + 69x^2 - 63x + 15) \div (3x^2 - 9x + 3)$$

商 :  $-6x + 5$ , 剰余 :  $4x + 3$

$$(9) (42x^3 - 54x^2 + 12x) \div (6x^2 - 6x)$$

商 :  $7x - 2$ , 剰余 :  $8x + 8$

$$(5) (-21x^3 - 54x^2 - 39x - 30) \div (7x^2 + 4x + 5)$$

商 :  $-3x - 6$ , 剰余 :  $-6x + 4$

$$(10) (-45x^3 - 21x^2 + 11x - 1) \div (9x^2 + 6x - 1)$$

商 :  $-5x + 1$ , 剰余 :  $-x + 1$

$$(1) (20x^3 + 13x^2 - 36x - 21) \div (4x^2 - 3x - 3)$$

商 :  $5x + 7$ , 剰余 :  $8x + 4$

$$(6) (72x^3 + 37x^2 + 19x + 15) \div (9x^2 - x + 3)$$

商 :  $8x + 5$ , 剰余 :  $2x + 7$

$$(2) (32x^3 + 36x^2 - 26x - 12) \div (8x^2 - 3x - 2)$$

商 :  $4x + 6$ , 剰余 :  $9x + 4$

$$(7) (54x^3 + 6x^2) \div (6x^2)$$

商 :  $9x + 1$ , 剰余 :  $x - 7$

$$(3) (56x^3 + 45x^2 + 37x + 12) \div (8x^2 + 3x + 4)$$

商 :  $7x + 3$ , 剰余 :  $-6x - 6$

$$(8) (64x^3 - 64x^2 - 56x) \div (8x^2 - 8x - 7)$$

商 :  $8x$ , 剰余 :  $-3x + 6$

$$(4) (42x^3 + 90x^2 + 34x - 16) \div (6x^2 + 6x - 2)$$

商 :  $7x + 8$ , 剰余 :  $4x + 9$

$$(9) (-6x^3 + 17x^2 + 49) \div (3x^2 + 2x + 7)$$

商 :  $-2x + 7$ , 剰余 :  $3x - 4$

$$(5) (-16x^3 + 46x^2 + 82x - 40) \div (8x^2 + 9x - 5)$$

商 :  $-2x + 8$ , 剰余 :  $-6x + 1$

$$(10) (14x^3 + 70x^2 + 72x + 64) \div (7x^2 + 7x + 8)$$

商 :  $2x + 8$ , 剰余 :  $5x - 9$

$$(1) (45x^3 + 47x^2 + 30x + 8) \div (5x^2 + 3x + 2)$$

商 :  $9x + 4$ , 剰余 :  $5x + 3$

$$(6) (48x^3 + 70x^2 - 30x - 16) \div (8x^2 + 9x - 8)$$

商 :  $6x + 2$ , 剰余 :  $4x - 8$

$$(2) (-28x^3 - 27x^2 - 11x + 6) \div (4x^2 + 5x + 3)$$

商 :  $-7x + 2$ , 剰余 :  $6x - 3$

$$(7) (-64x^3 + 88x^2 + 42x - 45) \div (8x^2 - 6x - 9)$$

商 :  $-8x + 5$ , 剰余 :  $5x + 6$

$$(3) (63x^3 - 21x^2 - 69x - 18) \div (7x^2 - 7x - 3)$$

商 :  $9x + 6$ , 剰余 :  $-7x - 1$

$$(8) (8x^3 + 52x^2 + 88x + 32) \div (x^2 + 6x + 8)$$

商 :  $8x + 4$ , 剰余 :  $8x - 6$

$$(4) (56x^3 - 103x^2 - 18x + 45) \div (8x^2 - 9x - 9)$$

商 :  $7x - 5$ , 剰余 :  $3x + 4$

$$(9) (8x^3 + 36x^2 + 44x + 8) \div (2x^2 + 5x + 1)$$

商 :  $4x + 8$ , 剰余 :  $4x - 5$

$$(5) (30x^3 + 46x^2 + 50x + 56) \div (5x^2 + x + 7)$$

商 :  $6x + 8$ , 剰余 :  $7x - 8$

$$(10) (-32x^3 + 4x^2 + 39x + 15) \div (4x^2 - 3x - 3)$$

商 :  $-8x - 5$ , 剰余 :  $8x - 9$

$$(1) (-5x^3 - 4x^2 - 30x - 24) \div (x^2 + 6)$$

商:  $-5x - 4$ , 剰余:  $6x + 2$

$$(6) (25x^3 + 15x^2 - 74x - 36) \div (5x^2 - 6x - 4)$$

商:  $5x + 9$ , 剰余:  $3x - 2$

$$(2) (64x^3 - 112x^2 - 3x + 30) \div (8x^2 - 9x - 6)$$

商:  $8x - 5$ , 剰余:  $5x + 2$

$$(7) (8x^3 + 32x^2 - 8x) \div (x^2 + 4x - 1)$$

商:  $8x$ , 剰余:  $-7x - 5$

$$(3) (6x^3 + 53x^2 - 8x - 40) \div (x^2 + 8x - 8)$$

商:  $6x + 5$ , 剰余:  $7x - 5$

$$(8) (-6x^3 + 51x + 9) \div (2x^2 + 6x + 1)$$

商:  $-3x + 9$ , 剰余:  $-3x - 4$

$$(4) (10x^3 - 15x^2 + 5x) \div (2x^2 - 3x + 1)$$

商:  $5x$ , 剰余:  $4x - 6$

$$(9) (-15x^3 + 19x^2 + 59x - 8) \div (5x^2 + 7x - 1)$$

商:  $-3x + 8$ , 剰余:  $4x + 1$

$$(5) (-25x^3 - 45x^2 - 10x + 8) \div (5x^2 + 5x - 2)$$

商:  $-5x - 4$ , 剰余:  $2x + 1$

$$(10) (-18x^3 + 34x^2 - 6x + 20) \div (9x^2 + x + 5)$$

商:  $-2x + 4$ , 剰余:  $-6x + 9$

(1)  $(-2x^3 + x^2 + 32x + 14) \div (x^2 - 4x - 2)$   
 商 :  $-2x - 7$ , 剰余 :  $2x + 4$

(6)  $(63x^3 - 95x^2 + 54x - 8) \div (7x^2 - 9x + 4)$   
 商 :  $9x - 2$ , 剰余 :  $-3x - 8$

(2)  $(-5x^3 - 12x^2 + 4x + 16) \div (5x^2 + 2x - 8)$   
 商 :  $-x - 2$ , 剰余 :  $4$

(7)  $(24x^3 + 84x^2 + 60x + 72) \div (8x^2 + 4x + 8)$   
 商 :  $3x + 9$ , 剰余 :  $2x + 7$

(3)  $(-10x^3 + 13x^2 + 30x) \div (5x^2 + 6x)$   
 商 :  $-2x + 5$ , 剰余 :  $-2x + 4$

(8)  $(-30x^3 + 29x^2 + 15x - 25) \div (5x^2 - 9x + 5)$   
 商 :  $-6x - 5$ , 剰余 :  $-5$

(4)  $(-30x^3 - 15x^2 + 35x) \div (6x^2 + 3x - 7)$   
 商 :  $-5x$ , 剰余 :  $-3x - 7$

(9)  $(54x^3 + 6x^2 - 13x + 2) \div (6x^2 + 2x - 1)$   
 商 :  $9x - 2$ , 剰余 :  $7x + 7$

(5)  $(-63x^3 - 130x^2 - 63x) \div (7x^2 + 9x)$   
 商 :  $-9x - 7$ , 剰余 :  $-3x + 6$

(10)  $(-42x^3 + 57x^2 - 21x + 45) \div (7x^2 + x + 5)$   
 商 :  $-6x + 9$ , 剰余 :  $-4x + 6$

$$(1) (4x^3 - 36x^2 + 20x) \div (x^2 - 9x + 5)$$

商 :  $4x$ , 剰余 :  $x + 1$

$$(6) (-21x^3 - 22x^2 + 58x + 45) \div (3x^2 + x - 9)$$

商 :  $-7x - 5$ , 剰余 :  $-2x - 3$

$$(2) (49x^3 + 14x^2 - 78x + 27) \div (7x^2 + 5x - 9)$$

商 :  $7x - 3$ , 剰余 :  $3x + 7$

$$(7) (16x^3 + 50x^2 + 54x + 6) \div (2x^2 + 6x + 6)$$

商 :  $8x + 1$ , 剰余 :  $-6x - 3$

$$(3) (5x^3 + 40x^2 + 10x) \div (x^2 + 8x + 2)$$

商 :  $5x$ , 剰余 :  $-2x + 8$

$$(8) (-18x^3 + 60x^2 - 12x - 48) \div (3x^2 - 6x - 6)$$

商 :  $-6x + 8$ , 剰余 :  $-6$

$$(4) (18x^3 - 30x^2 - 24x + 24) \div (3x^2 - 8x + 4)$$

商 :  $6x + 6$ , 剰余 :  $x + 6$

$$(9) (-72x^3 - 4x^2 + 101x + 45) \div (8x^2 - 4x - 9)$$

商 :  $-9x - 5$ , 剰余 :  $4x - 5$

$$(5) (36x^3 + 45x^2 - 20x - 25) \div (9x^2 - 5)$$

商 :  $4x + 5$ , 剰余 :  $-7x + 5$

$$(10) (32x^3 + 20x^2 + 66x + 8) \div (4x^2 + 2x + 8)$$

商 :  $8x + 1$ , 剰余 :  $-7x + 1$



$$(1) (-40x^3 - 58x^2 - 20x - 2) \div (5x^2 + 6x + 1)$$

商 :  $-8x - 2$ , 剰余 :  $7x + 6$

$$(6) (-63x^3 - 85x^2 - 91x - 36) \div (9x^2 + 7x + 9)$$

商 :  $-7x - 4$ , 剰余 :  $4x - 1$

$$(2) (-48x^3 + 8x^2 + 48x - 20) \div (6x^2 + 2x - 5)$$

商 :  $-8x + 4$ , 剰余 :  $-9x - 3$

$$(7) (10x^3 + 20x^2 - 22x + 24) \div (5x^2 - 5x + 4)$$

商 :  $2x + 6$ , 剰余 :  $-6x - 1$

$$(3) (-7x^3 - 2x^2 + 11x + 6) \div (7x^2 - 5x - 6)$$

商 :  $-x - 1$ , 剰余 :  $8x + 6$

$$(8) (8x^3 - 14x^2 - 52x + 45) \div (2x^2 - 8x + 5)$$

商 :  $4x + 9$ , 剰余 :  $2x + 9$

$$(4) (-48x^3 + 72x^2 + 16x - 20) \div (6x^2 - 6x - 5)$$

商 :  $-8x + 4$ , 剰余 :  $-6x - 4$

$$(9) (12x^3 - 20x^2 - 34x - 42) \div (6x^2 + 8x + 7)$$

商 :  $2x - 6$ , 剰余 :  $8x + 9$

$$(5) (-2x^3 + x^2 - 6x) \div (2x^2 - x + 6)$$

商 :  $-x$ , 剰余 :  $-2x + 9$

$$(10) (30x^3 - 31x^2 - 60x - 9) \div (5x^2 - 6x - 9)$$

商 :  $6x + 1$ , 剰余 :  $-3x - 1$

$$(1) (-12x^3 - 38x^2 + 64x - 36) \div (6x^2 - 8x + 4)$$

商:  $-2x - 9$ , 剰余:  $7x + 5$

$$(6) (10x^3 + 13x^2 + 39x + 28) \div (2x^2 + x + 7)$$

商:  $5x + 4$ , 剰余:  $-7x + 9$

$$(2) (-5x^3 - 25x^2 + 5x) \div (x^2 + 5x - 1)$$

商:  $-5x$ , 剰余:  $-7x + 3$

$$(7) (40x^3 - 31x^2 + 62x - 21) \div (5x^2 - 2x + 7)$$

商:  $8x - 3$ , 剰余:  $-3x - 2$

$$(3) (42x^3 - 21x^2 - 49x) \div (6x^2 - 3x - 7)$$

商:  $7x$ , 剰余:  $2x + 9$

$$(8) (12x^3 - 50x^2 - 26x + 36) \div (6x^2 + 2x - 4)$$

商:  $2x - 9$ , 剰余:  $-7x - 7$

$$(4) (24x^3 - 52x^2 + 47x - 45) \div (8x^2 - 4x + 9)$$

商:  $3x - 5$ , 剰余:  $3x + 2$

$$(9) (54x^3 - 63x^2 - 54x + 63) \div (9x^2 - 9)$$

商:  $6x - 7$ , 剰余:  $x - 5$

$$(5) (28x^3 + 22x^2 + 18x + 4) \div (4x^2 + 2x + 2)$$

商:  $7x + 2$ , 剰余:  $8x + 4$

$$(10) (14x^3 + 79x^2 + 100x + 32) \div (2x^2 + 9x + 4)$$

商:  $7x + 8$ , 剰余:  $-3x + 2$

$$(1) (36x^3 - 79x^2 - 35x + 28) \div (4x^2 - 7x - 7)$$

商 :  $9x - 4$ , 剰余 :  $-8x + 8$

$$(6) (9x^3 + 12x^2 + 9x + 42) \div (3x^2 - 2x + 7)$$

商 :  $3x + 6$ , 剰余 :  $7x - 9$

$$(2) (-27x^3 + 15x^2 - 46x + 48) \div (3x^2 + x + 6)$$

商 :  $-9x + 8$ , 剰余 :  $-2x + 1$

$$(7) (4x^3 - 16x^2 + 4x - 16) \div (4x^2 + 4)$$

商 :  $x - 4$ , 剰余 :  $-6x + 4$

$$(3) (-6x^3 + 22x^2 - 34x - 14) \div (x^2 - 4x + 7)$$

商 :  $-6x - 2$ , 剰余 :  $-8x + 3$

$$(8) (8x^3 - 4x^2 - 6x + 2) \div (4x^2 + 2x - 1)$$

商 :  $2x - 2$ , 剰余 :  $-2x - 9$

$$(4) (8x^3 - 33x^2 - 31x - 20) \div (8x^2 + 7x + 4)$$

商 :  $x - 5$ , 剰余 :  $2x - 3$

$$(9) (48x^3 - 108x^2 + 70x - 12) \div (6x^2 - 9x + 2)$$

商 :  $8x - 6$ , 剰余 :  $9x + 2$

$$(5) (-24x^3 + 8x^2 - 2x + 2) \div (6x^2 + x + 1)$$

商 :  $-4x + 2$ , 剰余 :  $5x + 5$

$$(10) (-12x^3 - 56x^2 - 53x - 14) \div (6x^2 + 7x + 2)$$

商 :  $-2x - 7$ , 剰余 :  $-8x + 6$

$$(1) (-12x^3 - 18x^2 + 48x + 9) \div (2x^2 + 6x + 1)$$

商:  $-6x + 9$ , 剰余:  $-9x - 4$

$$(6) (-6x^3 - 61x^2 - 56x + 63) \div (6x^2 + 7x - 7)$$

商:  $-x - 9$ , 剰余:  $-5$

$$(2) (24x^3 + 22x^2 - 66x - 8) \div (4x^2 + 9x + 1)$$

商:  $6x - 8$ , 剰余:  $-3x - 6$

$$(7) (12x^3 + 6x^2 - 30x - 18) \div (3x^2 - 3x - 3)$$

商:  $4x + 6$ , 剰余:

$$(3) (20x^3 - 16x^2 - 32x) \div (5x^2 - 4x - 8)$$

商:  $4x$ , 剰余:  $-4x + 4$

$$(8) (15x^3 - 38x^2 + 48x - 32) \div (5x^2 - 6x + 8)$$

商:  $3x - 4$ , 剰余:  $-5x + 2$

$$(4) (-4x^3 - 16x^2 - 12x) \div (4x^2 + 4x)$$

商:  $-x - 3$ , 剰余:  $-9x + 8$

$$(9) (42x^3 - 30x^2 + 36x - 48) \div (7x^2 + 2x + 8)$$

商:  $6x - 6$ , 剰余:  $6x + 2$

$$(5) (36x^3 + 54x^2 + 74x + 45) \div (6x^2 + 4x + 9)$$

商:  $6x + 5$ , 剰余:  $-4x + 2$

$$(10) (-36x^3 + 76x^2 + x - 1) \div (4x^2 - 8x - 1)$$

商:  $-9x + 1$ , 剰余:  $-5x + 8$

$$(1) (8x^3 - 2x^2 + x + 2) \div (4x^2 - 3x + 2)$$

商 :  $2x + 1$ , 剰余 :  $8x - 1$

$$(6) (48x^3 + 56x^2 + 8x) \div (6x^2 + 7x + 1)$$

商 :  $8x$ , 剰余 :  $-6x + 7$

$$(2) (48x^3 - 28x^2 + 28x - 8) \div (8x^2 - 2x + 4)$$

商 :  $6x - 2$ , 剰余 :  $-5x - 5$

$$(7) (9x^3 - 45x^2 + 36) \div (x^2 - 4x - 4)$$

商 :  $9x - 9$ , 剰余 :  $-7x + 5$

$$(3) (-5x^3 + 22x^2 + 16x - 5) \div (5x^2 + 3x - 1)$$

商 :  $-x + 5$ , 剰余 :  $-7x - 6$

$$(8) (36x^3 - 54x^2 + 56x - 49) \div (6x^2 - 2x + 7)$$

商 :  $6x - 7$ , 剰余 :  $9x - 6$

$$(4) (-63x^3 - 36x^2 - 18x - 45) \div (7x^2 - 3x + 5)$$

商 :  $-9x - 9$ , 剰余 :  $-7x - 3$

$$(9) (18x^3 - 6x^2 + 12x - 4) \div (3x^2 + 2)$$

商 :  $6x - 2$ , 剰余 :  $-8x - 6$

$$(5) (36x^3 + 48x^2 - 5x - 14) \div (6x^2 + x - 2)$$

商 :  $6x + 7$ , 剰余 :  $x + 4$

$$(10) (-20x^3 + 25x^2 + 27x - 8) \div (5x^2 - 5x - 8)$$

商 :  $-4x + 1$ , 剰余 :  $8x + 8$

$$(1) (-24x^3 + 7x^2 + 3x + 2) \div (8x^2 + 3x + 1)$$

商:  $-3x + 2$ , 剰余:  $8x - 7$

$$(6) (5x^3 + x^2 - 6x) \div (x^2 - x)$$

商:  $5x + 6$ , 剰余:  $-6x$

$$(2) (14x^3 - 64x^2 + 38x - 24) \div (7x^2 - 4x + 3)$$

商:  $2x - 8$ , 剰余:  $-9x + 4$

$$(7) (-15x^3 - 12x^2 - 18x) \div (5x^2 + 4x + 6)$$

商:  $-3x$ , 剰余:  $-9x + 9$

$$(3) (-8x^3 - 12x^2 + 88x - 64) \div (2x^2 + 7x - 8)$$

商:  $-4x + 8$ , 剰余:  $x - 7$

$$(8) (4x^3 + 8x^2 - 2x) \div (2x^2 + 4x - 1)$$

商:  $2x$ , 剰余:  $4x - 2$

$$(4) (-48x^3 - 24x^2 - 32x - 16) \div (6x^2 + 4)$$

商:  $-8x - 4$ , 剰余:  $x + 6$

$$(9) (24x^3 - 46x^2 - 6x + 4) \div (4x^2 - 9x + 2)$$

商:  $6x + 2$ , 剰余:  $8$

$$(5) (42x^3 + 30x^2 - 63x - 45) \div (6x^2 - 9)$$

商:  $7x + 5$ , 剰余:  $-4x - 3$

$$(10) (18x^3 + 12x^2 - 13x - 2) \div (6x^2 + 8x + 1)$$

商:  $3x - 2$ , 剰余:  $2x - 1$

## 2.4 因数分解 (2 次式)

2 次多項式の因数分解をする問題。

(1)  $5x^2 - 4x - 9$

(11)  $5x^2 - 29x + 20$

(21)  $14x^2 - 73x + 45$

(2)  $x^2 - 6x - 7$

(12)  $2x^2 - x - 1$

(22)  $27x^2 - 24x + 5$

(3)  $5x^2 + 6x - 8$

(13)  $2x^2 - 15x + 7$

(23)  $5x^2 + 14x + 9$

(4)  $12x^2 - 13x - 4$

(14)  $7x^2 + 3x - 4$

(24)  $12x^2 - 11x - 5$

(5)  $3x^2 - 11x - 4$

(15)  $x^2 + 4x - 5$

(25)  $2x^2 + 3x - 5$

(6)  $36x^2 - 89x + 18$

(16)  $21x^2 - 2x - 8$

(26)  $2x^2 - 15x + 7$

(7)  $2x^2 + 21x + 54$

(17)  $14x^2 - 19x - 3$

(27)  $10x^2 + 13x - 3$

(8)  $12x^2 - 17x - 5$

(18)  $3x^2 - x - 2$

(28)  $4x^2 + 9x - 28$

(9)  $3x^2 - 4x - 7$

(19)  $2x^2 - 11x + 5$

(29)  $3x^2 - 2x - 16$

(10)  $16x^2 + 32x + 7$

(20)  $5x^2 + 33x + 40$

(30)  $6x^2 + 25x + 21$



(1)  $3x^2 - 7x + 4$

(11)  $18x^2 + 79x - 9$

(21)  $6x^2 + 5x - 6$

(2)  $4x^2 - 33x + 8$

(12)  $14x^2 + 23x - 30$

(22)  $10x^2 + 27x + 18$

(3)  $18x^2 - 21x + 5$

(13)  $2x^2 + 13x - 7$

(23)  $3x^2 - 26x + 16$

(4)  $2x^2 - 23x + 45$

(14)  $21x^2 + 23x + 6$

(24)  $30x^2 + 67x + 35$

(5)  $3x^2 - 7x + 2$

(15)  $7x^2 - 16x + 9$

(25)  $16x^2 - 1$

(6)  $35x^2 + 53x - 18$

(16)  $25x^2 - 55x + 18$

(26)  $2x^2 + 9x + 4$

(7)  $42x^2 + 19x + 2$

(17)  $7x^2 - 29x + 4$

(27)  $4x^2 + 8x - 21$

(8)  $15x^2 + 47x + 28$

(18)  $2x^2 + 3x + 1$

(28)  $2x^2 + 13x - 24$

(9)  $3x^2 + 13x + 14$

(19)  $5x^2 - 12x - 32$

(29)  $x^2 + 9x + 18$

(10)  $2x^2 + 5x + 2$

(20)  $8x^2 + 41x + 36$

(30)  $32x^2 + 84x + 27$

(1)  $24x^2 - 22x - 7$

(11)  $5x^2 - x - 6$

(21)  $3x^2 - x - 2$

(2)  $7x^2 + x - 6$

(12)  $8x^2 + 9x - 14$

(22)  $14x^2 + 45x - 81$

(3)  $7x^2 + 15x + 2$

(13)  $3x^2 + 13x + 4$

(23)  $5x^2 + 3x - 8$

(4)  $2x^2 + 5x - 3$

(14)  $63x^2 + 4x - 35$

(24)  $x^2 - x - 2$

(5)  $25x^2 + 25x + 4$

(15)  $9x^2 - 10x + 1$

(25)  $45x^2 - 14x + 1$

(6)  $21x^2 + 11x - 6$

(16)  $2x^2 + 3x - 9$

(26)  $49x^2 + 49x - 18$

(7)  $6x^2 - 17x + 10$

(17)  $8x^2 - 7x - 18$

(27)  $56x^2 - 45x + 9$

(8)  $2x^2 - 9x + 4$

(18)  $8x^2 + 13x + 5$

(28)  $36x^2 - 13x - 40$

(9)  $12x^2 + 7x + 1$

(19)  $42x^2 + 13x + 1$

(29)  $18x^2 + 37x + 15$

(10)  $5x^2 - 28x - 49$

(20)  $48x^2 + 94x + 45$

(30)  $2x^2 - 3x - 9$

(1)  $5x^2 + 12x - 9$

(11)  $x^2 + 2x + 1$

(21)  $14x^2 + x - 3$

(2)  $5x^2 - 12x + 7$

(12)  $9x^2 + 12x - 32$

(22)  $3x^2 - x - 10$

(3)  $20x^2 + 3x - 9$

(13)  $5x^2 - x - 4$

(23)  $28x^2 - 75x + 27$

(4)  $9x^2 + 6x + 1$

(14)  $10x^2 - 27x + 5$

(24)  $3x^2 + 4x + 1$

(5)  $16x^2 - 16x - 21$

(15)  $10x^2 - 21x + 8$

(25)  $20x^2 + 43x + 21$

(6)  $9x^2 - 3x - 20$

(16)  $40x^2 + 99x + 56$

(26)  $6x^2 + 19x + 3$

(7)  $3x^2 + 2x - 8$

(17)  $5x^2 - 16x + 12$

(27)  $2x^2 - 3x + 1$

(8)  $3x^2 - 8x + 4$

(18)  $5x^2 - 2x - 3$

(28)  $x^2 + 10x + 24$

(9)  $2x^2 + 5x - 3$

(19)  $2x^2 + 3x - 9$

(29)  $3x^2 + 25x - 18$

(10)  $x^2 - 3x - 4$

(20)  $24x^2 + 25x + 6$

(30)  $3x^2 - 10x + 7$

(1)  $21x^2 + 25x - 4$

(11)  $4x^2 - 11x + 7$

(21)  $4x^2 + 5x - 9$

(2)  $x^2 - 6x - 7$

(12)  $3x^2 - 20x + 12$

(22)  $5x^2 - 3x - 2$

(3)  $36x^2 + 23x - 3$

(13)  $18x^2 + 83x + 9$

(23)  $7x^2 - 17x + 6$

(4)  $35x^2 - 31x - 18$

(14)  $25x^2 + 55x + 24$

(24)  $28x^2 - 57x + 27$

(5)  $4x^2 + 3x - 7$

(15)  $14x^2 - 9x - 8$

(25)  $24x^2 + 26x - 63$

(6)  $10x^2 + 9x - 7$

(16)  $24x^2 + 14x - 3$

(26)  $10x^2 + 39x + 35$

(7)  $2x^2 - x - 3$

(17)  $9x^2 + 11x + 2$

(27)  $12x^2 - 7x + 1$

(8)  $27x^2 + 39x + 4$

(18)  $x^2 - 3x + 2$

(28)  $24x^2 - 17x - 20$

(9)  $24x^2 - 29x + 7$

(19)  $49x^2 + 56x + 15$

(29)  $7x^2 + x - 8$

(10)  $15x^2 - 44x + 21$

(20)  $35x^2 + 32x - 12$

(30)  $3x^2 - 11x + 8$

(1)  $49x^2 + 35x - 36$

(11)  $14x^2 - 3x - 2$

(21)  $27x^2 - 30x + 7$

(2)  $45x^2 - 76x + 32$

(12)  $2x^2 + x - 1$

(22)  $14x^2 - 25x + 6$

(3)  $10x^2 + 3x - 1$

(13)  $10x^2 - 23x + 12$

(23)  $4x^2 + 12x - 27$

(4)  $15x^2 - 22x - 9$

(14)  $24x^2 - 17x + 3$

(24)  $9x^2 - 37x + 4$

(5)  $14x^2 + 15x + 4$

(15)  $64x^2 + 80x + 21$

(25)  $4x^2 + 11x - 3$

(6)  $30x^2 + 73x + 40$

(16)  $2x^2 + 11x + 15$

(26)  $15x^2 + 28x - 32$

(7)  $2x^2 + 17x - 9$

(17)  $x^2 - 36$

(27)  $3x^2 - 4x - 7$

(8)  $7x^2 + 26x - 45$

(18)  $x^2 + 3x + 2$

(28)  $7x^2 + 43x - 42$

(9)  $8x^2 - 2x - 1$

(19)  $5x^2 - 31x - 28$

(29)  $56x^2 - 19x - 15$

(10)  $x^2 + 3x + 2$

(20)  $x^2 + 3x + 2$

(30)  $12x^2 - 35x + 25$

(1)  $7x^2 + 2x - 5$

(11)  $3x^2 + 2x - 1$

(21)  $8x^2 - 3x - 5$

(2)  $5x^2 + 6x + 1$

(12)  $2x^2 + 3x + 1$

(22)  $7x^2 - 5x - 2$

(3)  $x^2 - 1$

(13)  $20x^2 - 11x - 4$

(23)  $20x^2 + 9x + 1$

(4)  $56x^2 + 17x - 28$

(14)  $3x^2 - x - 2$

(24)  $3x^2 - 11x + 6$

(5)  $3x^2 + 13x - 10$

(15)  $x^2 - 14x + 48$

(25)  $30x^2 + 31x - 6$

(6)  $42x^2 - 41x + 5$

(16)  $72x^2 + 113x + 36$

(26)  $16x^2 - 9$

(7)  $5x^2 - 27x + 10$

(17)  $56x^2 + 15x - 54$

(27)  $2x^2 + 7x + 5$

(8)  $x^2 + 4x + 3$

(18)  $8x^2 + 6x + 1$

(28)  $3x^2 - 13x + 4$

(9)  $7x^2 + 23x - 20$

(19)  $2x^2 - 3x - 5$

(29)  $25x^2 + 5x - 6$

(10)  $9x^2 + 83x + 18$

(20)  $x^2 - 4x + 3$

(30)  $18x^2 + 95x + 63$

(1)  $9x^2 - 5x - 4$

(11)  $9x^2 + 30x + 16$

(21)  $20x^2 + 17x - 24$

(2)  $6x^2 + 7x - 3$

(12)  $45x^2 + 34x + 5$

(22)  $18x^2 + 51x + 35$

(3)  $21x^2 - 10x - 24$

(13)  $56x^2 - 73x + 21$

(23)  $12x^2 - 29x + 14$

(4)  $3x^2 - 4x + 1$

(14)  $9x^2 + 18x - 16$

(24)  $x^2 - 10x + 16$

(5)  $42x^2 - 25x - 28$

(15)  $24x^2 - 61x - 8$

(25)  $28x^2 - 23x + 4$

(6)  $6x^2 + 13x - 5$

(16)  $4x^2 - 13x + 3$

(26)  $20x^2 - x - 1$

(7)  $x^2 + 4x + 3$

(17)  $14x^2 - 13x + 3$

(27)  $2x^2 + 3x + 1$

(8)  $6x^2 + 17x + 7$

(18)  $x^2 - 5x + 4$

(28)  $x^2 - x - 2$

(9)  $14x^2 - 59x + 35$

(19)  $x^2 + 6x - 27$

(29)  $6x^2 + 23x + 15$

(10)  $4x^2 + 5x - 6$

(20)  $4x^2 + 12x + 5$

(30)  $8x^2 + 15x - 2$

(1)  $2x^2 - x - 1$

(11)  $32x^2 - 28x - 15$

(21)  $21x^2 + 34x + 9$

(2)  $56x^2 + 41x - 7$

(12)  $15x^2 - 22x + 8$

(22)  $8x^2 + 10x - 7$

(3)  $24x^2 + 11x + 1$

(13)  $x^2 + 3x + 2$

(23)  $5x^2 - 11x + 6$

(4)  $15x^2 + 22x + 8$

(14)  $56x^2 + 15x + 1$

(24)  $5x^2 + 12x + 7$

(5)  $x^2 - 3x - 18$

(15)  $10x^2 + 3x - 4$

(25)  $3x^2 - 8x - 3$

(6)  $15x^2 + 49x + 40$

(16)  $81x^2 - 45x - 14$

(26)  $56x^2 - 65x + 14$

(7)  $3x^2 + 2x - 5$

(17)  $6x^2 - 13x + 6$

(27)  $35x^2 - 66x + 27$

(8)  $x^2 - 6x - 16$

(18)  $16x^2 + 2x - 5$

(28)  $18x^2 + 9x + 1$

(9)  $10x^2 + x - 2$

(19)  $x^2 - 3x + 2$

(29)  $24x^2 + 29x - 4$

(10)  $9x^2 - 11x + 2$

(20)  $x^2 + 11x + 24$

(30)  $5x^2 - 4x - 1$



(1)  $18x^2 + 27x + 7$

(11)  $4x^2 + x - 3$

(21)  $45x^2 + 112x + 64$

(2)  $12x^2 + x - 6$

(12)  $3x^2 - 10x + 7$

(22)  $7x^2 + 10x + 3$

(3)  $6x^2 - 47x - 8$

(13)  $40x^2 + 21x - 49$

(23)  $x^2 + 6x + 9$

(4)  $15x^2 + x - 6$

(14)  $16x^2 + 10x - 9$

(24)  $7x^2 + 24x - 16$

(5)  $6x^2 + x - 15$

(15)  $4x^2 + 27x - 40$

(25)  $9x^2 + 11x + 2$

(6)  $6x^2 - 11x - 10$

(16)  $20x^2 - 31x - 7$

(26)  $35x^2 - 24x - 35$

(7)  $x^2 + x - 2$

(17)  $4x^2 + 19x + 12$

(27)  $56x^2 - 59x + 15$

(8)  $2x^2 + 5x + 3$

(18)  $9x^2 - 9x + 2$

(28)  $x^2 - 12x + 27$

(9)  $x^2 - 5x + 6$

(19)  $2x^2 + 7x - 9$

(29)  $x^2 + 3x - 4$

(10)  $72x^2 + 41x - 45$

(20)  $21x^2 + 11x - 2$

(30)  $7x^2 - 27x + 18$

(1)  $10x^2 - 31x + 24$

(11)  $5x^2 + 9x - 18$

(21)  $7x^2 + 27x + 18$

(2)  $9x^2 - 27x + 8$

(12)  $24x^2 + x - 10$

(22)  $4x^2 - 23x - 72$

(3)  $8x^2 - 26x + 21$

(13)  $18x^2 + 23x + 7$

(23)  $21x^2 - 20x + 4$

(4)  $32x^2 + 92x + 63$

(14)  $14x^2 - 27x + 9$

(24)  $6x^2 - 17x - 28$

(5)  $4x^2 - 3x - 1$

(15)  $4x^2 + 3x - 27$

(25)  $x^2 - x - 2$

(6)  $2x^2 - x - 10$

(16)  $4x^2 - 4x - 3$

(26)  $7x^2 + 22x + 16$

(7)  $3x^2 + x - 2$

(17)  $12x^2 - 4x - 1$

(27)  $72x^2 - 137x + 63$

(8)  $27x^2 - 93x + 56$

(18)  $7x^2 + 39x - 18$

(28)  $15x^2 + 4x - 32$

(9)  $7x^2 - 52x + 21$

(19)  $7x^2 + 2x - 5$

(29)  $9x^2 + 9x - 10$

(10)  $9x^2 - 4x - 5$

(20)  $9x^2 - 8x - 1$

(30)  $24x^2 + 2x - 15$

(1)  $18x^2 + 3x - 28$

(11)  $x^2 - 5x + 4$

(21)  $5x^2 - 8x + 3$

(2)  $7x^2 - 58x + 63$

(12)  $49x^2 + 42x - 16$

(22)  $49x^2 - 35x + 6$

(3)  $36x^2 - 25x + 4$

(13)  $x^2 - 4$

(23)  $25x^2 - 25x + 4$

(4)  $12x^2 + 17x + 6$

(14)  $5x^2 + 4x - 1$

(24)  $35x^2 - 44x - 7$

(5)  $7x^2 + 2x - 9$

(15)  $18x^2 + 45x + 28$

(25)  $12x^2 + 7x + 1$

(6)  $x^2 + x - 56$

(16)  $6x^2 - 19x + 8$

(26)  $x^2 - 5x + 6$

(7)  $x^2 + 4x - 21$

(17)  $27x^2 + 48x + 5$

(27)  $40x^2 - x - 15$

(8)  $3x^2 - 17x - 28$

(18)  $4x^2 - 39x + 27$

(28)  $2x^2 - 7x + 5$

(9)  $2x^2 + 17x + 8$

(19)  $x^2 + 10x + 9$

(29)  $8x^2 + 33x + 27$

(10)  $63x^2 - 44x + 5$

(20)  $8x^2 - x - 9$

(30)  $4x^2 - 5x - 6$

(1)  $3x^2 + 5x + 2$

(11)  $3x^2 - 10x + 8$

(21)  $3x^2 - x - 4$

(2)  $14x^2 - 25x + 9$

(12)  $9x^2 + 9x - 4$

(22)  $21x^2 - 29x + 10$

(3)  $56x^2 + 47x - 18$

(13)  $4x^2 + 9x + 5$

(23)  $x^2 - x - 12$

(4)  $2x^2 + 3x - 2$

(14)  $9x^2 + 23x - 12$

(24)  $5x^2 - 8x + 3$

(5)  $3x^2 - 2x - 21$

(15)  $4x^2 - 25x + 6$

(25)  $5x^2 + 7x + 2$

(6)  $3x^2 + 22x + 24$

(16)  $72x^2 + 7x - 49$

(26)  $9x^2 - 85x + 36$

(7)  $25x^2 - 35x - 18$

(17)  $2x^2 - 5x - 7$

(27)  $36x^2 + 79x + 28$

(8)  $5x^2 + 8x - 4$

(18)  $63x^2 - 2x - 48$

(28)  $24x^2 - 55x + 25$

(9)  $12x^2 - 17x + 6$

(19)  $6x^2 + 17x + 12$

(29)  $2x^2 + x - 1$

(10)  $10x^2 - 11x - 6$

(20)  $2x^2 + 15x - 8$

(30)  $4x^2 + 19x + 12$

(1)  $12x^2 + 17x + 6$

(11)  $8x^2 + 39x + 28$

(21)  $21x^2 - 41x + 18$

(2)  $14x^2 - 53x + 45$

(12)  $15x^2 - x - 28$

(22)  $5x^2 - 13x + 8$

(3)  $3x^2 + 2x - 1$

(13)  $14x^2 - 11x - 9$

(23)  $4x^2 - 12x + 5$

(4)  $2x^2 + x - 3$

(14)  $3x^2 + 20x + 12$

(24)  $14x^2 - 3x - 5$

(5)  $16x^2 + 18x + 5$

(15)  $3x^2 + 7x - 6$

(25)  $4x^2 + 19x + 12$

(6)  $4x^2 + 12x + 9$

(16)  $8x^2 - 25x + 18$

(26)  $3x^2 - 11x + 6$

(7)  $9x^2 + 7x - 2$

(17)  $6x^2 + 5x - 1$

(27)  $36x^2 - 47x + 15$

(8)  $3x^2 - 4x - 7$

(18)  $15x^2 - x - 6$

(28)  $35x^2 - 86x + 48$

(9)  $8x^2 - 10x - 25$

(19)  $x^2 - 8x + 7$

(29)  $40x^2 - 77x + 36$

(10)  $7x^2 + 11x + 4$

(20)  $16x^2 + 38x + 21$

(30)  $49x^2 - 21x - 40$

(1)  $4x^2 - 5x + 1$

(11)  $x^2 + 6x - 7$

(21)  $10x^2 - 33x - 54$

(2)  $x^2 + 6x + 5$

(12)  $6x^2 - 17x + 7$

(22)  $14x^2 + 19x - 40$

(3)  $6x^2 - 13x + 7$

(13)  $5x^2 - 3x - 2$

(23)  $x^2 - 5x + 4$

(4)  $12x^2 + 25x + 12$

(14)  $x^2 - 1$

(24)  $18x^2 + 3x - 28$

(5)  $2x^2 + 21x + 49$

(15)  $3x^2 + 7x + 2$

(25)  $x^2 + 2x - 3$

(6)  $7x^2 + 23x + 18$

(16)  $5x^2 + 12x + 4$

(26)  $8x^2 - 22x - 63$

(7)  $8x^2 - 15x + 7$

(17)  $20x^2 + 69x + 54$

(27)  $7x^2 - 16x - 15$

(8)  $4x^2 + 19x + 21$

(18)  $4x^2 - 4x - 3$

(28)  $14x^2 - x - 3$

(9)  $3x^2 - 13x + 14$

(19)  $63x^2 - 62x + 15$

(29)  $3x^2 + 13x + 14$

(10)  $x^2 + 7x - 8$

(20)  $x^2 - x - 12$

(30)  $x^2 + 3x - 4$

(1)  $27x^2 + 12x - 32$

(11)  $27x^2 + 33x + 8$

(21)  $8x^2 + 9x - 14$

(2)  $6x^2 + 31x + 18$

(12)  $10x^2 + 9x - 7$

(22)  $12x^2 + 8x - 7$

(3)  $15x^2 - 29x + 8$

(13)  $5x^2 - 3x - 14$

(23)  $4x^2 + 4x - 15$

(4)  $14x^2 + 23x + 8$

(14)  $18x^2 + 89x + 36$

(24)  $63x^2 - 8x - 15$

(5)  $12x^2 + 25x - 7$

(15)  $3x^2 - 16x + 5$

(25)  $20x^2 + 27x - 14$

(6)  $6x^2 + 5x - 6$

(16)  $81x^2 - 54x - 7$

(26)  $10x^2 - 21x + 8$

(7)  $3x^2 - 16x - 35$

(17)  $7x^2 + 10x - 8$

(27)  $4x^2 - 41x + 72$

(8)  $x^2 - 1$

(18)  $7x^2 + 2x - 5$

(28)  $16x^2 + 6x - 27$

(9)  $15x^2 - 22x - 9$

(19)  $27x^2 - 21x - 20$

(29)  $28x^2 + 19x + 3$

(10)  $2x^2 - 3x + 1$

(20)  $3x^2 + 17x - 28$

(30)  $3x^2 - 20x - 7$

(1)  $21x^2 + 23x + 6$

(11)  $21x^2 - 13x - 20$

(21)  $x^2 - 2x + 1$

(2)  $9x^2 + 6x + 1$

(12)  $4x^2 - x - 5$

(22)  $9x^2 + 38x - 35$

(3)  $3x^2 + 4x - 7$

(13)  $28x^2 + 81x + 56$

(23)  $4x^2 + 5x - 6$

(4)  $14x^2 - 3x - 2$

(14)  $24x^2 - 22x + 3$

(24)  $x^2 - 6x - 7$

(5)  $16x^2 - 22x - 3$

(15)  $2x^2 + x - 3$

(25)  $4x^2 - 27x - 7$

(6)  $4x^2 + 11x - 3$

(16)  $6x^2 - 11x - 2$

(26)  $30x^2 + x - 1$

(7)  $8x^2 - 14x - 9$

(17)  $18x^2 - 11x + 1$

(27)  $48x^2 + 50x + 7$

(8)  $20x^2 - 29x - 36$

(18)  $5x^2 - 31x - 28$

(28)  $20x^2 - 61x + 45$

(9)  $2x^2 - 3x + 1$

(19)  $x^2 - 2x + 1$

(29)  $7x^2 + 12x + 5$

(10)  $4x^2 + 13x + 3$

(20)  $2x^2 + 5x + 3$

(30)  $8x^2 + 33x + 27$



(1)  $18x^2 + 19x - 12$

(11)  $15x^2 + 7x - 4$

(21)  $9x^2 - 17x - 2$

(2)  $9x^2 - 9x + 2$

(12)  $5x^2 + 53x + 72$

(22)  $2x^2 + 5x + 3$

(3)  $2x^2 + x - 3$

(13)  $40x^2 + 49x + 15$

(23)  $9x^2 - 18x + 8$

(4)  $24x^2 + 7x - 6$

(14)  $x^2 - 5x + 4$

(24)  $14x^2 + 45x + 25$

(5)  $x^2 + 3x - 4$

(15)  $10x^2 - 37x + 7$

(25)  $7x^2 - 5x - 2$

(6)  $x^2 + x - 56$

(16)  $24x^2 - 13x - 2$

(26)  $x^2 + 7x - 8$

(7)  $9x^2 + x - 8$

(17)  $32x^2 - 60x - 27$

(27)  $64x^2 - 81$

(8)  $24x^2 + 22x - 7$

(18)  $4x^2 - 19x - 5$

(28)  $2x^2 + 5x - 7$

(9)  $20x^2 + 39x + 7$

(19)  $36x^2 + 5x - 1$

(29)  $4x^2 - 3x - 27$

(10)  $14x^2 + 3x - 2$

(20)  $6x^2 - 29x + 20$

(30)  $7x^2 - 54x - 16$

(1)  $45x^2 - 92x + 32$

(11)  $18x^2 - 17x - 15$

(21)  $5x^2 - 37x - 72$

(2)  $72x^2 - 83x + 21$

(12)  $8x^2 - 3x - 5$

(22)  $32x^2 - 4x - 15$

(3)  $42x^2 - x - 30$

(13)  $8x^2 + 15x + 7$

(23)  $4x^2 + 13x - 12$

(4)  $2x^2 + 11x + 9$

(14)  $x^2 - 4x + 3$

(24)  $28x^2 + 69x + 35$

(5)  $25x^2 + 5x - 6$

(15)  $7x^2 + 16x + 9$

(25)  $28x^2 + 15x + 2$

(6)  $9x^2 - 4x - 5$

(16)  $2x^2 + 9x + 10$

(26)  $32x^2 - 60x + 7$

(7)  $3x^2 - 5x - 8$

(17)  $35x^2 + 9x - 18$

(27)  $16x^2 + 26x + 3$

(8)  $36x^2 + 97x + 36$

(18)  $5x^2 + 13x - 6$

(28)  $3x^2 - 8x + 5$

(9)  $32x^2 - 12x - 9$

(19)  $5x^2 + 19x + 18$

(29)  $36x^2 - 49x + 5$

(10)  $10x^2 - 23x + 12$

(20)  $2x^2 - x - 10$

(30)  $12x^2 - 31x + 7$

(1)  $45x^2 - 53x + 14$

(11)  $6x^2 + 35x - 6$

(21)  $56x^2 - 9x - 35$

(2)  $24x^2 - 11x - 18$

(12)  $4x^2 + 5x - 6$

(22)  $7x^2 - 29x + 24$

(3)  $8x^2 + 14x + 5$

(13)  $2x^2 + 3x - 2$

(23)  $40x^2 - 39x + 9$

(4)  $15x^2 + 4x - 32$

(14)  $3x^2 + 5x - 2$

(24)  $36x^2 + 95x + 56$

(5)  $4x^2 + 21x + 5$

(15)  $6x^2 + 29x + 28$

(25)  $x^2 + 2x - 15$

(6)  $20x^2 + 17x - 10$

(16)  $16x^2 - 74x + 63$

(26)  $3x^2 - 5x - 2$

(7)  $81x^2 + 36x - 32$

(17)  $3x^2 + 32x + 64$

(27)  $2x^2 + x - 21$

(8)  $5x^2 + 14x - 3$

(18)  $16x^2 + 2x - 5$

(28)  $36x^2 + 37x - 10$

(9)  $5x^2 + 4x - 1$

(19)  $4x^2 + 17x + 15$

(29)  $10x^2 + 43x + 45$

(10)  $6x^2 + 13x - 8$

(20)  $24x^2 - 43x + 18$

(30)  $x^2 + 2x + 1$

$$\begin{aligned} (1) \quad & 5x^2 - 4x - 9 \\ &= (x+1)(5x-9) \end{aligned}$$

$$\begin{aligned} (11) \quad & 5x^2 - 29x + 20 \\ &= (x-5)(5x-4) \end{aligned}$$

$$\begin{aligned} (21) \quad & 14x^2 - 73x + 45 \\ &= (2x-9)(7x-5) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^2 - 6x - 7 \\ &= (x+1)(x-7) \end{aligned}$$

$$\begin{aligned} (12) \quad & 2x^2 - x - 1 \\ &= (2x+1)(x-1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 27x^2 - 24x + 5 \\ &= (3x-1)(9x-5) \end{aligned}$$

$$\begin{aligned} (3) \quad & 5x^2 + 6x - 8 \\ &= (5x-4)(x+2) \end{aligned}$$

$$\begin{aligned} (13) \quad & 2x^2 - 15x + 7 \\ &= (2x-1)(x-7) \end{aligned}$$

$$\begin{aligned} (23) \quad & 5x^2 + 14x + 9 \\ &= (5x+9)(x+1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 12x^2 - 13x - 4 \\ &= (4x+1)(3x-4) \end{aligned}$$

$$\begin{aligned} (14) \quad & 7x^2 + 3x - 4 \\ &= (7x-4)(x+1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 12x^2 - 11x - 5 \\ &= (4x-5)(3x+1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^2 - 11x - 4 \\ &= (x-4)(3x+1) \end{aligned}$$

$$\begin{aligned} (15) \quad & x^2 + 4x - 5 \\ &= (x-1)(x+5) \end{aligned}$$

$$\begin{aligned} (25) \quad & 2x^2 + 3x - 5 \\ &= (x-1)(2x+5) \end{aligned}$$

$$\begin{aligned} (6) \quad & 36x^2 - 89x + 18 \\ &= (4x-9)(9x-2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 21x^2 - 2x - 8 \\ &= (3x-2)(7x+4) \end{aligned}$$

$$\begin{aligned} (26) \quad & 2x^2 - 15x + 7 \\ &= (x-7)(2x-1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^2 + 21x + 54 \\ &= (x+6)(2x+9) \end{aligned}$$

$$\begin{aligned} (17) \quad & 14x^2 - 19x - 3 \\ &= (2x-3)(7x+1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 10x^2 + 13x - 3 \\ &= (5x-1)(2x+3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 12x^2 - 17x - 5 \\ &= (4x+1)(3x-5) \end{aligned}$$

$$\begin{aligned} (18) \quad & 3x^2 - x - 2 \\ &= (3x+2)(x-1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 4x^2 + 9x - 28 \\ &= (4x-7)(x+4) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^2 - 4x - 7 \\ &= (3x-7)(x+1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 2x^2 - 11x + 5 \\ &= (2x-1)(x-5) \end{aligned}$$

$$\begin{aligned} (29) \quad & 3x^2 - 2x - 16 \\ &= (x+2)(3x-8) \end{aligned}$$

$$\begin{aligned} (10) \quad & 16x^2 + 32x + 7 \\ &= (4x+7)(4x+1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 5x^2 + 33x + 40 \\ &= (5x+8)(x+5) \end{aligned}$$

$$\begin{aligned} (30) \quad & 6x^2 + 25x + 21 \\ &= (6x+7)(x+3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^2 - 7x + 4 \\ &= (x - 1)(3x - 4) \end{aligned}$$

$$\begin{aligned} (11) \quad & 18x^2 + 79x - 9 \\ &= (2x + 9)(9x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 6x^2 + 5x - 6 \\ &= (3x - 2)(2x + 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 4x^2 - 33x + 8 \\ &= (x - 8)(4x - 1) \end{aligned}$$

$$\begin{aligned} (12) \quad & 14x^2 + 23x - 30 \\ &= (7x - 6)(2x + 5) \end{aligned}$$

$$\begin{aligned} (22) \quad & 10x^2 + 27x + 18 \\ &= (2x + 3)(5x + 6) \end{aligned}$$

$$\begin{aligned} (3) \quad & 18x^2 - 21x + 5 \\ &= (6x - 5)(3x - 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 2x^2 + 13x - 7 \\ &= (2x - 1)(x + 7) \end{aligned}$$

$$\begin{aligned} (23) \quad & 3x^2 - 26x + 16 \\ &= (x - 8)(3x - 2) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^2 - 23x + 45 \\ &= (x - 9)(2x - 5) \end{aligned}$$

$$\begin{aligned} (14) \quad & 21x^2 + 23x + 6 \\ &= (3x + 2)(7x + 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & 30x^2 + 67x + 35 \\ &= (6x + 5)(5x + 7) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^2 - 7x + 2 \\ &= (x - 2)(3x - 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 7x^2 - 16x + 9 \\ &= (7x - 9)(x - 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 16x^2 - 1 \\ &= (4x - 1)(4x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 35x^2 + 53x - 18 \\ &= (5x + 9)(7x - 2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 25x^2 - 55x + 18 \\ &= (5x - 2)(5x - 9) \end{aligned}$$

$$\begin{aligned} (26) \quad & 2x^2 + 9x + 4 \\ &= (x + 4)(2x + 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 42x^2 + 19x + 2 \\ &= (6x + 1)(7x + 2) \end{aligned}$$

$$\begin{aligned} (17) \quad & 7x^2 - 29x + 4 \\ &= (7x - 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (27) \quad & 4x^2 + 8x - 21 \\ &= (2x + 7)(2x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 15x^2 + 47x + 28 \\ &= (5x + 4)(3x + 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & 2x^2 + 3x + 1 \\ &= (x + 1)(2x + 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 2x^2 + 13x - 24 \\ &= (2x - 3)(x + 8) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^2 + 13x + 14 \\ &= (x + 2)(3x + 7) \end{aligned}$$

$$\begin{aligned} (19) \quad & 5x^2 - 12x - 32 \\ &= (5x + 8)(x - 4) \end{aligned}$$

$$\begin{aligned} (29) \quad & x^2 + 9x + 18 \\ &= (x + 6)(x + 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^2 + 5x + 2 \\ &= (2x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (20) \quad & 8x^2 + 41x + 36 \\ &= (x + 4)(8x + 9) \end{aligned}$$

$$\begin{aligned} (30) \quad & 32x^2 + 84x + 27 \\ &= (4x + 9)(8x + 3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 24x^2 - 22x - 7 \\ & = (4x + 1)(6x - 7) \end{aligned}$$

$$\begin{aligned} (11) \quad & 5x^2 - x - 6 \\ & = (5x - 6)(x + 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 3x^2 - x - 2 \\ & = (x - 1)(3x + 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 7x^2 + x - 6 \\ & = (x + 1)(7x - 6) \end{aligned}$$

$$\begin{aligned} (12) \quad & 8x^2 + 9x - 14 \\ & = (x + 2)(8x - 7) \end{aligned}$$

$$\begin{aligned} (22) \quad & 14x^2 + 45x - 81 \\ & = (7x - 9)(2x + 9) \end{aligned}$$

$$\begin{aligned} (3) \quad & 7x^2 + 15x + 2 \\ & = (7x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (13) \quad & 3x^2 + 13x + 4 \\ & = (3x + 1)(x + 4) \end{aligned}$$

$$\begin{aligned} (23) \quad & 5x^2 + 3x - 8 \\ & = (5x + 8)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^2 + 5x - 3 \\ & = (x + 3)(2x - 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 63x^2 + 4x - 35 \\ & = (7x - 5)(9x + 7) \end{aligned}$$

$$\begin{aligned} (24) \quad & x^2 - x - 2 \\ & = (x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 25x^2 + 25x + 4 \\ & = (5x + 4)(5x + 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 9x^2 - 10x + 1 \\ & = (x - 1)(9x - 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 45x^2 - 14x + 1 \\ & = (5x - 1)(9x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 21x^2 + 11x - 6 \\ & = (3x - 1)(7x + 6) \end{aligned}$$

$$\begin{aligned} (16) \quad & 2x^2 + 3x - 9 \\ & = (2x - 3)(x + 3) \end{aligned}$$

$$\begin{aligned} (26) \quad & 49x^2 + 49x - 18 \\ & = (7x - 2)(7x + 9) \end{aligned}$$

$$\begin{aligned} (7) \quad & 6x^2 - 17x + 10 \\ & = (x - 2)(6x - 5) \end{aligned}$$

$$\begin{aligned} (17) \quad & 8x^2 - 7x - 18 \\ & = (8x + 9)(x - 2) \end{aligned}$$

$$\begin{aligned} (27) \quad & 56x^2 - 45x + 9 \\ & = (7x - 3)(8x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 2x^2 - 9x + 4 \\ & = (2x - 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (18) \quad & 8x^2 + 13x + 5 \\ & = (x + 1)(8x + 5) \end{aligned}$$

$$\begin{aligned} (28) \quad & 36x^2 - 13x - 40 \\ & = (9x + 8)(4x - 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 12x^2 + 7x + 1 \\ & = (4x + 1)(3x + 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 42x^2 + 13x + 1 \\ & = (6x + 1)(7x + 1) \end{aligned}$$

$$\begin{aligned} (29) \quad & 18x^2 + 37x + 15 \\ & = (2x + 3)(9x + 5) \end{aligned}$$

$$\begin{aligned} (10) \quad & 5x^2 - 28x - 49 \\ & = (5x + 7)(x - 7) \end{aligned}$$

$$\begin{aligned} (20) \quad & 48x^2 + 94x + 45 \\ & = (8x + 9)(6x + 5) \end{aligned}$$

$$\begin{aligned} (30) \quad & 2x^2 - 3x - 9 \\ & = (x - 3)(2x + 3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 5x^2 + 12x - 9 \\ &= (5x - 3)(x + 3) \end{aligned}$$

$$\begin{aligned} (11) \quad & x^2 + 2x + 1 \\ &= (x + 1)^2 \end{aligned}$$

$$\begin{aligned} (21) \quad & 14x^2 + x - 3 \\ &= (2x + 1)(7x - 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 5x^2 - 12x + 7 \\ &= (5x - 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (12) \quad & 9x^2 + 12x - 32 \\ &= (3x - 4)(3x + 8) \end{aligned}$$

$$\begin{aligned} (22) \quad & 3x^2 - x - 10 \\ &= (x - 2)(3x + 5) \end{aligned}$$

$$\begin{aligned} (3) \quad & 20x^2 + 3x - 9 \\ &= (4x + 3)(5x - 3) \end{aligned}$$

$$\begin{aligned} (13) \quad & 5x^2 - x - 4 \\ &= (5x + 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (23) \quad & 28x^2 - 75x + 27 \\ &= (4x - 9)(7x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 9x^2 + 6x + 1 \\ &= (3x + 1)^2 \end{aligned}$$

$$\begin{aligned} (14) \quad & 10x^2 - 27x + 5 \\ &= (2x - 5)(5x - 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 3x^2 + 4x + 1 \\ &= (x + 1)(3x + 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 16x^2 - 16x - 21 \\ &= (4x - 7)(4x + 3) \end{aligned}$$

$$\begin{aligned} (15) \quad & 10x^2 - 21x + 8 \\ &= (2x - 1)(5x - 8) \end{aligned}$$

$$\begin{aligned} (25) \quad & 20x^2 + 43x + 21 \\ &= (5x + 7)(4x + 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 9x^2 - 3x - 20 \\ &= (3x + 4)(3x - 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 40x^2 + 99x + 56 \\ &= (5x + 8)(8x + 7) \end{aligned}$$

$$\begin{aligned} (26) \quad & 6x^2 + 19x + 3 \\ &= (6x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^2 + 2x - 8 \\ &= (x + 2)(3x - 4) \end{aligned}$$

$$\begin{aligned} (17) \quad & 5x^2 - 16x + 12 \\ &= (x - 2)(5x - 6) \end{aligned}$$

$$\begin{aligned} (27) \quad & 2x^2 - 3x + 1 \\ &= (2x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^2 - 8x + 4 \\ &= (3x - 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (18) \quad & 5x^2 - 2x - 3 \\ &= (5x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & x^2 + 10x + 24 \\ &= (x + 6)(x + 4) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^2 + 5x - 3 \\ &= (x + 3)(2x - 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 2x^2 + 3x - 9 \\ &= (2x - 3)(x + 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 3x^2 + 25x - 18 \\ &= (x + 9)(3x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^2 - 3x - 4 \\ &= (x + 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (20) \quad & 24x^2 + 25x + 6 \\ &= (8x + 3)(3x + 2) \end{aligned}$$

$$\begin{aligned} (30) \quad & 3x^2 - 10x + 7 \\ &= (x - 1)(3x - 7) \end{aligned}$$

$$\begin{aligned} (1) \quad & 21x^2 + 25x - 4 \\ & = (3x + 4)(7x - 1) \end{aligned}$$

$$\begin{aligned} (11) \quad & 4x^2 - 11x + 7 \\ & = (4x - 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 4x^2 + 5x - 9 \\ & = (4x + 9)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^2 - 6x - 7 \\ & = (x + 1)(x - 7) \end{aligned}$$

$$\begin{aligned} (12) \quad & 3x^2 - 20x + 12 \\ & = (x - 6)(3x - 2) \end{aligned}$$

$$\begin{aligned} (22) \quad & 5x^2 - 3x - 2 \\ & = (5x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 36x^2 + 23x - 3 \\ & = (4x + 3)(9x - 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 18x^2 + 83x + 9 \\ & = (9x + 1)(2x + 9) \end{aligned}$$

$$\begin{aligned} (23) \quad & 7x^2 - 17x + 6 \\ & = (x - 2)(7x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 35x^2 - 31x - 18 \\ & = (7x - 9)(5x + 2) \end{aligned}$$

$$\begin{aligned} (14) \quad & 25x^2 + 55x + 24 \\ & = (5x + 8)(5x + 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & 28x^2 - 57x + 27 \\ & = (4x - 3)(7x - 9) \end{aligned}$$

$$\begin{aligned} (5) \quad & 4x^2 + 3x - 7 \\ & = (x - 1)(4x + 7) \end{aligned}$$

$$\begin{aligned} (15) \quad & 14x^2 - 9x - 8 \\ & = (2x + 1)(7x - 8) \end{aligned}$$

$$\begin{aligned} (25) \quad & 24x^2 + 26x - 63 \\ & = (6x - 7)(4x + 9) \end{aligned}$$

$$\begin{aligned} (6) \quad & 10x^2 + 9x - 7 \\ & = (5x + 7)(2x - 1) \end{aligned}$$

$$\begin{aligned} (16) \quad & 24x^2 + 14x - 3 \\ & = (4x + 3)(6x - 1) \end{aligned}$$

$$\begin{aligned} (26) \quad & 10x^2 + 39x + 35 \\ & = (2x + 5)(5x + 7) \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^2 - x - 3 \\ & = (x + 1)(2x - 3) \end{aligned}$$

$$\begin{aligned} (17) \quad & 9x^2 + 11x + 2 \\ & = (9x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 12x^2 - 7x + 1 \\ & = (4x - 1)(3x - 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 27x^2 + 39x + 4 \\ & = (9x + 1)(3x + 4) \end{aligned}$$

$$\begin{aligned} (18) \quad & x^2 - 3x + 2 \\ & = (x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (28) \quad & 24x^2 - 17x - 20 \\ & = (3x - 4)(8x + 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 24x^2 - 29x + 7 \\ & = (8x - 7)(3x - 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 49x^2 + 56x + 15 \\ & = (7x + 5)(7x + 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 7x^2 + x - 8 \\ & = (7x + 8)(x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 15x^2 - 44x + 21 \\ & = (5x - 3)(3x - 7) \end{aligned}$$

$$\begin{aligned} (20) \quad & 35x^2 + 32x - 12 \\ & = (5x + 6)(7x - 2) \end{aligned}$$

$$\begin{aligned} (30) \quad & 3x^2 - 11x + 8 \\ & = (x - 1)(3x - 8) \end{aligned}$$



$$\begin{aligned} (1) \quad & 49x^2 + 35x - 36 \\ & = (7x - 4)(7x + 9) \end{aligned}$$

$$\begin{aligned} (11) \quad & 14x^2 - 3x - 2 \\ & = (2x - 1)(7x + 2) \end{aligned}$$

$$\begin{aligned} (21) \quad & 27x^2 - 30x + 7 \\ & = (9x - 7)(3x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 45x^2 - 76x + 32 \\ & = (5x - 4)(9x - 8) \end{aligned}$$

$$\begin{aligned} (12) \quad & 2x^2 + x - 1 \\ & = (x + 1)(2x - 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 14x^2 - 25x + 6 \\ & = (7x - 2)(2x - 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 10x^2 + 3x - 1 \\ & = (2x + 1)(5x - 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 10x^2 - 23x + 12 \\ & = (2x - 3)(5x - 4) \end{aligned}$$

$$\begin{aligned} (23) \quad & 4x^2 + 12x - 27 \\ & = (2x + 9)(2x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 15x^2 - 22x - 9 \\ & = (5x - 9)(3x + 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 24x^2 - 17x + 3 \\ & = (8x - 3)(3x - 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 9x^2 - 37x + 4 \\ & = (9x - 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (5) \quad & 14x^2 + 15x + 4 \\ & = (2x + 1)(7x + 4) \end{aligned}$$

$$\begin{aligned} (15) \quad & 64x^2 + 80x + 21 \\ & = (8x + 3)(8x + 7) \end{aligned}$$

$$\begin{aligned} (25) \quad & 4x^2 + 11x - 3 \\ & = (4x - 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 30x^2 + 73x + 40 \\ & = (5x + 8)(6x + 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 2x^2 + 11x + 15 \\ & = (x + 3)(2x + 5) \end{aligned}$$

$$\begin{aligned} (26) \quad & 15x^2 + 28x - 32 \\ & = (5x - 4)(3x + 8) \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^2 + 17x - 9 \\ & = (x + 9)(2x - 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & x^2 - 36 \\ & = (x + 6)(x - 6) \end{aligned}$$

$$\begin{aligned} (27) \quad & 3x^2 - 4x - 7 \\ & = (3x - 7)(x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 7x^2 + 26x - 45 \\ & = (7x - 9)(x + 5) \end{aligned}$$

$$\begin{aligned} (18) \quad & x^2 + 3x + 2 \\ & = (x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 7x^2 + 43x - 42 \\ & = (x + 7)(7x - 6) \end{aligned}$$

$$\begin{aligned} (9) \quad & 8x^2 - 2x - 1 \\ & = (2x - 1)(4x + 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 5x^2 - 31x - 28 \\ & = (x - 7)(5x + 4) \end{aligned}$$

$$\begin{aligned} (29) \quad & 56x^2 - 19x - 15 \\ & = (7x - 5)(8x + 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^2 + 3x + 2 \\ & = (x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (20) \quad & x^2 + 3x + 2 \\ & = (x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (30) \quad & 12x^2 - 35x + 25 \\ & = (4x - 5)(3x - 5) \end{aligned}$$

$$\begin{aligned} (1) \quad & 7x^2 + 2x - 5 \\ &= (7x - 5)(x + 1) \end{aligned}$$

$$\begin{aligned} (11) \quad & 3x^2 + 2x - 1 \\ &= (x + 1)(3x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 8x^2 - 3x - 5 \\ &= (8x + 5)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 5x^2 + 6x + 1 \\ &= (x + 1)(5x + 1) \end{aligned}$$

$$\begin{aligned} (12) \quad & 2x^2 + 3x + 1 \\ &= (x + 1)(2x + 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 7x^2 - 5x - 2 \\ &= (x - 1)(7x + 2) \end{aligned}$$

$$\begin{aligned} (3) \quad & x^2 - 1 \\ &= (x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 20x^2 - 11x - 4 \\ &= (5x - 4)(4x + 1) \end{aligned}$$

$$\begin{aligned} (23) \quad & 20x^2 + 9x + 1 \\ &= (4x + 1)(5x + 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 56x^2 + 17x - 28 \\ &= (7x - 4)(8x + 7) \end{aligned}$$

$$\begin{aligned} (14) \quad & 3x^2 - x - 2 \\ &= (3x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 3x^2 - 11x + 6 \\ &= (3x - 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^2 + 13x - 10 \\ &= (3x - 2)(x + 5) \end{aligned}$$

$$\begin{aligned} (15) \quad & x^2 - 14x + 48 \\ &= (x - 6)(x - 8) \end{aligned}$$

$$\begin{aligned} (25) \quad & 30x^2 + 31x - 6 \\ &= (5x + 6)(6x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 42x^2 - 41x + 5 \\ &= (7x - 1)(6x - 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 72x^2 + 113x + 36 \\ &= (8x + 9)(9x + 4) \end{aligned}$$

$$\begin{aligned} (26) \quad & 16x^2 - 9 \\ &= (4x + 3)(4x - 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & 5x^2 - 27x + 10 \\ &= (x - 5)(5x - 2) \end{aligned}$$

$$\begin{aligned} (17) \quad & 56x^2 + 15x - 54 \\ &= (7x - 6)(8x + 9) \end{aligned}$$

$$\begin{aligned} (27) \quad & 2x^2 + 7x + 5 \\ &= (x + 1)(2x + 5) \end{aligned}$$

$$\begin{aligned} (8) \quad & x^2 + 4x + 3 \\ &= (x + 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (18) \quad & 8x^2 + 6x + 1 \\ &= (2x + 1)(4x + 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 3x^2 - 13x + 4 \\ &= (3x - 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (9) \quad & 7x^2 + 23x - 20 \\ &= (x + 4)(7x - 5) \end{aligned}$$

$$\begin{aligned} (19) \quad & 2x^2 - 3x - 5 \\ &= (x + 1)(2x - 5) \end{aligned}$$

$$\begin{aligned} (29) \quad & 25x^2 + 5x - 6 \\ &= (5x + 3)(5x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 9x^2 + 83x + 18 \\ &= (9x + 2)(x + 9) \end{aligned}$$

$$\begin{aligned} (20) \quad & x^2 - 4x + 3 \\ &= (x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (30) \quad & 18x^2 + 95x + 63 \\ &= (9x + 7)(2x + 9) \end{aligned}$$

$$\begin{aligned} (1) \quad & 9x^2 - 5x - 4 \\ &= (x - 1)(9x + 4) \end{aligned}$$

$$\begin{aligned} (11) \quad & 9x^2 + 30x + 16 \\ &= (3x + 8)(3x + 2) \end{aligned}$$

$$\begin{aligned} (21) \quad & 20x^2 + 17x - 24 \\ &= (4x - 3)(5x + 8) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^2 + 7x - 3 \\ &= (3x - 1)(2x + 3) \end{aligned}$$

$$\begin{aligned} (12) \quad & 45x^2 + 34x + 5 \\ &= (9x + 5)(5x + 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 18x^2 + 51x + 35 \\ &= (6x + 7)(3x + 5) \end{aligned}$$

$$\begin{aligned} (3) \quad & 21x^2 - 10x - 24 \\ &= (3x - 4)(7x + 6) \end{aligned}$$

$$\begin{aligned} (13) \quad & 56x^2 - 73x + 21 \\ &= (7x - 3)(8x - 7) \end{aligned}$$

$$\begin{aligned} (23) \quad & 12x^2 - 29x + 14 \\ &= (3x - 2)(4x - 7) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^2 - 4x + 1 \\ &= (3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 9x^2 + 18x - 16 \\ &= (3x - 2)(3x + 8) \end{aligned}$$

$$\begin{aligned} (24) \quad & x^2 - 10x + 16 \\ &= (x - 2)(x - 8) \end{aligned}$$

$$\begin{aligned} (5) \quad & 42x^2 - 25x - 28 \\ &= (7x + 4)(6x - 7) \end{aligned}$$

$$\begin{aligned} (15) \quad & 24x^2 - 61x - 8 \\ &= (3x - 8)(8x + 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 28x^2 - 23x + 4 \\ &= (7x - 4)(4x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 6x^2 + 13x - 5 \\ &= (3x - 1)(2x + 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 4x^2 - 13x + 3 \\ &= (x - 3)(4x - 1) \end{aligned}$$

$$\begin{aligned} (26) \quad & 20x^2 - x - 1 \\ &= (4x - 1)(5x + 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^2 + 4x + 3 \\ &= (x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (17) \quad & 14x^2 - 13x + 3 \\ &= (7x - 3)(2x - 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 2x^2 + 3x + 1 \\ &= (2x + 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^2 + 17x + 7 \\ &= (2x + 1)(3x + 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & x^2 - 5x + 4 \\ &= (x - 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & x^2 - x - 2 \\ &= (x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 14x^2 - 59x + 35 \\ &= (7x - 5)(2x - 7) \end{aligned}$$

$$\begin{aligned} (19) \quad & x^2 + 6x - 27 \\ &= (x + 9)(x - 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 6x^2 + 23x + 15 \\ &= (x + 3)(6x + 5) \end{aligned}$$

$$\begin{aligned} (10) \quad & 4x^2 + 5x - 6 \\ &= (x + 2)(4x - 3) \end{aligned}$$

$$\begin{aligned} (20) \quad & 4x^2 + 12x + 5 \\ &= (2x + 5)(2x + 1) \end{aligned}$$

$$\begin{aligned} (30) \quad & 8x^2 + 15x - 2 \\ &= (8x - 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 2x^2 - x - 1 \\ &= (x - 1)(2x + 1) \end{aligned}$$

$$\begin{aligned} (11) \quad & 32x^2 - 28x - 15 \\ &= (4x - 5)(8x + 3) \end{aligned}$$

$$\begin{aligned} (21) \quad & 21x^2 + 34x + 9 \\ &= (3x + 1)(7x + 9) \end{aligned}$$

$$\begin{aligned} (2) \quad & 56x^2 + 41x - 7 \\ &= (7x - 1)(8x + 7) \end{aligned}$$

$$\begin{aligned} (12) \quad & 15x^2 - 22x + 8 \\ &= (3x - 2)(5x - 4) \end{aligned}$$

$$\begin{aligned} (22) \quad & 8x^2 + 10x - 7 \\ &= (4x + 7)(2x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 24x^2 + 11x + 1 \\ &= (3x + 1)(8x + 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & x^2 + 3x + 2 \\ &= (x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (23) \quad & 5x^2 - 11x + 6 \\ &= (x - 1)(5x - 6) \end{aligned}$$

$$\begin{aligned} (4) \quad & 15x^2 + 22x + 8 \\ &= (3x + 2)(5x + 4) \end{aligned}$$

$$\begin{aligned} (14) \quad & 56x^2 + 15x + 1 \\ &= (8x + 1)(7x + 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 5x^2 + 12x + 7 \\ &= (5x + 7)(x + 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^2 - 3x - 18 \\ &= (x + 3)(x - 6) \end{aligned}$$

$$\begin{aligned} (15) \quad & 10x^2 + 3x - 4 \\ &= (5x + 4)(2x - 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 3x^2 - 8x - 3 \\ &= (3x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 15x^2 + 49x + 40 \\ &= (5x + 8)(3x + 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 81x^2 - 45x - 14 \\ &= (9x + 2)(9x - 7) \end{aligned}$$

$$\begin{aligned} (26) \quad & 56x^2 - 65x + 14 \\ &= (8x - 7)(7x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^2 + 2x - 5 \\ &= (x - 1)(3x + 5) \end{aligned}$$

$$\begin{aligned} (17) \quad & 6x^2 - 13x + 6 \\ &= (2x - 3)(3x - 2) \end{aligned}$$

$$\begin{aligned} (27) \quad & 35x^2 - 66x + 27 \\ &= (7x - 9)(5x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & x^2 - 6x - 16 \\ &= (x - 8)(x + 2) \end{aligned}$$

$$\begin{aligned} (18) \quad & 16x^2 + 2x - 5 \\ &= (8x + 5)(2x - 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 18x^2 + 9x + 1 \\ &= (3x + 1)(6x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 10x^2 + x - 2 \\ &= (5x - 2)(2x + 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & x^2 - 3x + 2 \\ &= (x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (29) \quad & 24x^2 + 29x - 4 \\ &= (8x - 1)(3x + 4) \end{aligned}$$

$$\begin{aligned} (10) \quad & 9x^2 - 11x + 2 \\ &= (9x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & x^2 + 11x + 24 \\ &= (x + 3)(x + 8) \end{aligned}$$

$$\begin{aligned} (30) \quad & 5x^2 - 4x - 1 \\ &= (x - 1)(5x + 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 18x^2 + 27x + 7 \\ & = (3x + 1)(6x + 7) \end{aligned}$$

$$\begin{aligned} (11) \quad & 4x^2 + x - 3 \\ & = (4x - 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 45x^2 + 112x + 64 \\ & = (5x + 8)(9x + 8) \end{aligned}$$

$$\begin{aligned} (2) \quad & 12x^2 + x - 6 \\ & = (3x - 2)(4x + 3) \end{aligned}$$

$$\begin{aligned} (12) \quad & 3x^2 - 10x + 7 \\ & = (3x - 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 7x^2 + 10x + 3 \\ & = (x + 1)(7x + 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 6x^2 - 47x - 8 \\ & = (x - 8)(6x + 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 40x^2 + 21x - 49 \\ & = (8x - 7)(5x + 7) \end{aligned}$$

$$\begin{aligned} (23) \quad & x^2 + 6x + 9 \\ & = (x + 3)^2 \end{aligned}$$

$$\begin{aligned} (4) \quad & 15x^2 + x - 6 \\ & = (5x - 3)(3x + 2) \end{aligned}$$

$$\begin{aligned} (14) \quad & 16x^2 + 10x - 9 \\ & = (2x - 1)(8x + 9) \end{aligned}$$

$$\begin{aligned} (24) \quad & 7x^2 + 24x - 16 \\ & = (x + 4)(7x - 4) \end{aligned}$$

$$\begin{aligned} (5) \quad & 6x^2 + x - 15 \\ & = (3x + 5)(2x - 3) \end{aligned}$$

$$\begin{aligned} (15) \quad & 4x^2 + 27x - 40 \\ & = (4x - 5)(x + 8) \end{aligned}$$

$$\begin{aligned} (25) \quad & 9x^2 + 11x + 2 \\ & = (9x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 6x^2 - 11x - 10 \\ & = (2x - 5)(3x + 2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 20x^2 - 31x - 7 \\ & = (5x + 1)(4x - 7) \end{aligned}$$

$$\begin{aligned} (26) \quad & 35x^2 - 24x - 35 \\ & = (5x - 7)(7x + 5) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^2 + x - 2 \\ & = (x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & 4x^2 + 19x + 12 \\ & = (x + 4)(4x + 3) \end{aligned}$$

$$\begin{aligned} (27) \quad & 56x^2 - 59x + 15 \\ & = (7x - 3)(8x - 5) \end{aligned}$$

$$\begin{aligned} (8) \quad & 2x^2 + 5x + 3 \\ & = (2x + 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (18) \quad & 9x^2 - 9x + 2 \\ & = (3x - 2)(3x - 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & x^2 - 12x + 27 \\ & = (x - 3)(x - 9) \end{aligned}$$

$$\begin{aligned} (9) \quad & x^2 - 5x + 6 \\ & = (x - 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (19) \quad & 2x^2 + 7x - 9 \\ & = (x - 1)(2x + 9) \end{aligned}$$

$$\begin{aligned} (29) \quad & x^2 + 3x - 4 \\ & = (x - 1)(x + 4) \end{aligned}$$

$$\begin{aligned} (10) \quad & 72x^2 + 41x - 45 \\ & = (8x + 9)(9x - 5) \end{aligned}$$

$$\begin{aligned} (20) \quad & 21x^2 + 11x - 2 \\ & = (7x - 1)(3x + 2) \end{aligned}$$

$$\begin{aligned} (30) \quad & 7x^2 - 27x + 18 \\ & = (7x - 6)(x - 3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 10x^2 - 31x + 24 \\ & = (2x - 3)(5x - 8) \end{aligned}$$

$$\begin{aligned} (11) \quad & 5x^2 + 9x - 18 \\ & = (5x - 6)(x + 3) \end{aligned}$$

$$\begin{aligned} (21) \quad & 7x^2 + 27x + 18 \\ & = (7x + 6)(x + 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 9x^2 - 27x + 8 \\ & = (3x - 8)(3x - 1) \end{aligned}$$

$$\begin{aligned} (12) \quad & 24x^2 + x - 10 \\ & = (3x + 2)(8x - 5) \end{aligned}$$

$$\begin{aligned} (22) \quad & 4x^2 - 23x - 72 \\ & = (x - 8)(4x + 9) \end{aligned}$$

$$\begin{aligned} (3) \quad & 8x^2 - 26x + 21 \\ & = (2x - 3)(4x - 7) \end{aligned}$$

$$\begin{aligned} (13) \quad & 18x^2 + 23x + 7 \\ & = (9x + 7)(2x + 1) \end{aligned}$$

$$\begin{aligned} (23) \quad & 21x^2 - 20x + 4 \\ & = (3x - 2)(7x - 2) \end{aligned}$$

$$\begin{aligned} (4) \quad & 32x^2 + 92x + 63 \\ & = (4x + 7)(8x + 9) \end{aligned}$$

$$\begin{aligned} (14) \quad & 14x^2 - 27x + 9 \\ & = (7x - 3)(2x - 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & 6x^2 - 17x - 28 \\ & = (6x + 7)(x - 4) \end{aligned}$$

$$\begin{aligned} (5) \quad & 4x^2 - 3x - 1 \\ & = (4x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 4x^2 + 3x - 27 \\ & = (4x - 9)(x + 3) \end{aligned}$$

$$\begin{aligned} (25) \quad & x^2 - x - 2 \\ & = (x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^2 - x - 10 \\ & = (2x - 5)(x + 2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 4x^2 - 4x - 3 \\ & = (2x - 3)(2x + 1) \end{aligned}$$

$$\begin{aligned} (26) \quad & 7x^2 + 22x + 16 \\ & = (7x + 8)(x + 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^2 + x - 2 \\ & = (x + 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (17) \quad & 12x^2 - 4x - 1 \\ & = (6x + 1)(2x - 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 72x^2 - 137x + 63 \\ & = (8x - 9)(9x - 7) \end{aligned}$$

$$\begin{aligned} (8) \quad & 27x^2 - 93x + 56 \\ & = (9x - 7)(3x - 8) \end{aligned}$$

$$\begin{aligned} (18) \quad & 7x^2 + 39x - 18 \\ & = (7x - 3)(x + 6) \end{aligned}$$

$$\begin{aligned} (28) \quad & 15x^2 + 4x - 32 \\ & = (3x - 4)(5x + 8) \end{aligned}$$

$$\begin{aligned} (9) \quad & 7x^2 - 52x + 21 \\ & = (x - 7)(7x - 3) \end{aligned}$$

$$\begin{aligned} (19) \quad & 7x^2 + 2x - 5 \\ & = (7x - 5)(x + 1) \end{aligned}$$

$$\begin{aligned} (29) \quad & 9x^2 + 9x - 10 \\ & = (3x + 5)(3x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 9x^2 - 4x - 5 \\ & = (9x + 5)(x - 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 9x^2 - 8x - 1 \\ & = (x - 1)(9x + 1) \end{aligned}$$

$$\begin{aligned} (30) \quad & 24x^2 + 2x - 15 \\ & = (4x - 3)(6x + 5) \end{aligned}$$

$$\begin{aligned} (1) \quad & 18x^2 + 3x - 28 \\ & = (3x + 4)(6x - 7) \end{aligned}$$

$$\begin{aligned} (11) \quad & x^2 - 5x + 4 \\ & = (x - 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 5x^2 - 8x + 3 \\ & = (5x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 7x^2 - 58x + 63 \\ & = (7x - 9)(x - 7) \end{aligned}$$

$$\begin{aligned} (12) \quad & 49x^2 + 42x - 16 \\ & = (7x + 8)(7x - 2) \end{aligned}$$

$$\begin{aligned} (22) \quad & 49x^2 - 35x + 6 \\ & = (7x - 3)(7x - 2) \end{aligned}$$

$$\begin{aligned} (3) \quad & 36x^2 - 25x + 4 \\ & = (4x - 1)(9x - 4) \end{aligned}$$

$$\begin{aligned} (13) \quad & x^2 - 4 \\ & = (x + 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (23) \quad & 25x^2 - 25x + 4 \\ & = (5x - 1)(5x - 4) \end{aligned}$$

$$\begin{aligned} (4) \quad & 12x^2 + 17x + 6 \\ & = (3x + 2)(4x + 3) \end{aligned}$$

$$\begin{aligned} (14) \quad & 5x^2 + 4x - 1 \\ & = (x + 1)(5x - 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 35x^2 - 44x - 7 \\ & = (7x + 1)(5x - 7) \end{aligned}$$

$$\begin{aligned} (5) \quad & 7x^2 + 2x - 9 \\ & = (7x + 9)(x - 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 18x^2 + 45x + 28 \\ & = (3x + 4)(6x + 7) \end{aligned}$$

$$\begin{aligned} (25) \quad & 12x^2 + 7x + 1 \\ & = (4x + 1)(3x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & x^2 + x - 56 \\ & = (x + 8)(x - 7) \end{aligned}$$

$$\begin{aligned} (16) \quad & 6x^2 - 19x + 8 \\ & = (2x - 1)(3x - 8) \end{aligned}$$

$$\begin{aligned} (26) \quad & x^2 - 5x + 6 \\ & = (x - 3)(x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^2 + 4x - 21 \\ & = (x + 7)(x - 3) \end{aligned}$$

$$\begin{aligned} (17) \quad & 27x^2 + 48x + 5 \\ & = (3x + 5)(9x + 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 40x^2 - x - 15 \\ & = (8x - 5)(5x + 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^2 - 17x - 28 \\ & = (3x + 4)(x - 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & 4x^2 - 39x + 27 \\ & = (4x - 3)(x - 9) \end{aligned}$$

$$\begin{aligned} (28) \quad & 2x^2 - 7x + 5 \\ & = (x - 1)(2x - 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^2 + 17x + 8 \\ & = (2x + 1)(x + 8) \end{aligned}$$

$$\begin{aligned} (19) \quad & x^2 + 10x + 9 \\ & = (x + 9)(x + 1) \end{aligned}$$

$$\begin{aligned} (29) \quad & 8x^2 + 33x + 27 \\ & = (8x + 9)(x + 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 63x^2 - 44x + 5 \\ & = (7x - 1)(9x - 5) \end{aligned}$$

$$\begin{aligned} (20) \quad & 8x^2 - x - 9 \\ & = (x + 1)(8x - 9) \end{aligned}$$

$$\begin{aligned} (30) \quad & 4x^2 - 5x - 6 \\ & = (4x + 3)(x - 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^2 + 5x + 2 \\ &= (x + 1)(3x + 2) \end{aligned}$$

$$\begin{aligned} (11) \quad & 3x^2 - 10x + 8 \\ &= (x - 2)(3x - 4) \end{aligned}$$

$$\begin{aligned} (21) \quad & 3x^2 - x - 4 \\ &= (x + 1)(3x - 4) \end{aligned}$$

$$\begin{aligned} (2) \quad & 14x^2 - 25x + 9 \\ &= (2x - 1)(7x - 9) \end{aligned}$$

$$\begin{aligned} (12) \quad & 9x^2 + 9x - 4 \\ &= (3x + 4)(3x - 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 21x^2 - 29x + 10 \\ &= (3x - 2)(7x - 5) \end{aligned}$$

$$\begin{aligned} (3) \quad & 56x^2 + 47x - 18 \\ &= (7x - 2)(8x + 9) \end{aligned}$$

$$\begin{aligned} (13) \quad & 4x^2 + 9x + 5 \\ &= (4x + 5)(x + 1) \end{aligned}$$

$$\begin{aligned} (23) \quad & x^2 - x - 12 \\ &= (x + 3)(x - 4) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^2 + 3x - 2 \\ &= (x + 2)(2x - 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 9x^2 + 23x - 12 \\ &= (9x - 4)(x + 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & 5x^2 - 8x + 3 \\ &= (5x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^2 - 2x - 21 \\ &= (x - 3)(3x + 7) \end{aligned}$$

$$\begin{aligned} (15) \quad & 4x^2 - 25x + 6 \\ &= (4x - 1)(x - 6) \end{aligned}$$

$$\begin{aligned} (25) \quad & 5x^2 + 7x + 2 \\ &= (x + 1)(5x + 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 3x^2 + 22x + 24 \\ &= (3x + 4)(x + 6) \end{aligned}$$

$$\begin{aligned} (16) \quad & 72x^2 + 7x - 49 \\ &= (8x + 7)(9x - 7) \end{aligned}$$

$$\begin{aligned} (26) \quad & 9x^2 - 85x + 36 \\ &= (x - 9)(9x - 4) \end{aligned}$$

$$\begin{aligned} (7) \quad & 25x^2 - 35x - 18 \\ &= (5x + 2)(5x - 9) \end{aligned}$$

$$\begin{aligned} (17) \quad & 2x^2 - 5x - 7 \\ &= (2x - 7)(x + 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 36x^2 + 79x + 28 \\ &= (9x + 4)(4x + 7) \end{aligned}$$

$$\begin{aligned} (8) \quad & 5x^2 + 8x - 4 \\ &= (5x - 2)(x + 2) \end{aligned}$$

$$\begin{aligned} (18) \quad & 63x^2 - 2x - 48 \\ &= (7x + 6)(9x - 8) \end{aligned}$$

$$\begin{aligned} (28) \quad & 24x^2 - 55x + 25 \\ &= (8x - 5)(3x - 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 12x^2 - 17x + 6 \\ &= (3x - 2)(4x - 3) \end{aligned}$$

$$\begin{aligned} (19) \quad & 6x^2 + 17x + 12 \\ &= (3x + 4)(2x + 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 2x^2 + x - 1 \\ &= (x + 1)(2x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 10x^2 - 11x - 6 \\ &= (5x + 2)(2x - 3) \end{aligned}$$

$$\begin{aligned} (20) \quad & 2x^2 + 15x - 8 \\ &= (2x - 1)(x + 8) \end{aligned}$$

$$\begin{aligned} (30) \quad & 4x^2 + 19x + 12 \\ &= (4x + 3)(x + 4) \end{aligned}$$



$$\begin{aligned} (1) \quad & 12x^2 + 17x + 6 \\ &= (3x + 2)(4x + 3) \end{aligned}$$

$$\begin{aligned} (11) \quad & 8x^2 + 39x + 28 \\ &= (8x + 7)(x + 4) \end{aligned}$$

$$\begin{aligned} (21) \quad & 21x^2 - 41x + 18 \\ &= (3x - 2)(7x - 9) \end{aligned}$$

$$\begin{aligned} (2) \quad & 14x^2 - 53x + 45 \\ &= (7x - 9)(2x - 5) \end{aligned}$$

$$\begin{aligned} (12) \quad & 15x^2 - x - 28 \\ &= (5x - 7)(3x + 4) \end{aligned}$$

$$\begin{aligned} (22) \quad & 5x^2 - 13x + 8 \\ &= (5x - 8)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 3x^2 + 2x - 1 \\ &= (3x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 14x^2 - 11x - 9 \\ &= (2x + 1)(7x - 9) \end{aligned}$$

$$\begin{aligned} (23) \quad & 4x^2 - 12x + 5 \\ &= (2x - 5)(2x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^2 + x - 3 \\ &= (2x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 3x^2 + 20x + 12 \\ &= (x + 6)(3x + 2) \end{aligned}$$

$$\begin{aligned} (24) \quad & 14x^2 - 3x - 5 \\ &= (7x - 5)(2x + 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 16x^2 + 18x + 5 \\ &= (2x + 1)(8x + 5) \end{aligned}$$

$$\begin{aligned} (15) \quad & 3x^2 + 7x - 6 \\ &= (3x - 2)(x + 3) \end{aligned}$$

$$\begin{aligned} (25) \quad & 4x^2 + 19x + 12 \\ &= (4x + 3)(x + 4) \end{aligned}$$

$$\begin{aligned} (6) \quad & 4x^2 + 12x + 9 \\ &= (2x + 3)^2 \end{aligned}$$

$$\begin{aligned} (16) \quad & 8x^2 - 25x + 18 \\ &= (8x - 9)(x - 2) \end{aligned}$$

$$\begin{aligned} (26) \quad & 3x^2 - 11x + 6 \\ &= (x - 3)(3x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 9x^2 + 7x - 2 \\ &= (9x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & 6x^2 + 5x - 1 \\ &= (x + 1)(6x - 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 36x^2 - 47x + 15 \\ &= (9x - 5)(4x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^2 - 4x - 7 \\ &= (x + 1)(3x - 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & 15x^2 - x - 6 \\ &= (3x - 2)(5x + 3) \end{aligned}$$

$$\begin{aligned} (28) \quad & 35x^2 - 86x + 48 \\ &= (7x - 6)(5x - 8) \end{aligned}$$

$$\begin{aligned} (9) \quad & 8x^2 - 10x - 25 \\ &= (2x - 5)(4x + 5) \end{aligned}$$

$$\begin{aligned} (19) \quad & x^2 - 8x + 7 \\ &= (x - 1)(x - 7) \end{aligned}$$

$$\begin{aligned} (29) \quad & 40x^2 - 77x + 36 \\ &= (8x - 9)(5x - 4) \end{aligned}$$

$$\begin{aligned} (10) \quad & 7x^2 + 11x + 4 \\ &= (7x + 4)(x + 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 16x^2 + 38x + 21 \\ &= (8x + 7)(2x + 3) \end{aligned}$$

$$\begin{aligned} (30) \quad & 49x^2 - 21x - 40 \\ &= (7x + 5)(7x - 8) \end{aligned}$$

$$\begin{aligned} (1) \quad & 4x^2 - 5x + 1 \\ &= (4x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (11) \quad & x^2 + 6x - 7 \\ &= (x + 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 10x^2 - 33x - 54 \\ &= (5x + 6)(2x - 9) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^2 + 6x + 5 \\ &= (x + 1)(x + 5) \end{aligned}$$

$$\begin{aligned} (12) \quad & 6x^2 - 17x + 7 \\ &= (2x - 1)(3x - 7) \end{aligned}$$

$$\begin{aligned} (22) \quad & 14x^2 + 19x - 40 \\ &= (2x + 5)(7x - 8) \end{aligned}$$

$$\begin{aligned} (3) \quad & 6x^2 - 13x + 7 \\ &= (6x - 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 5x^2 - 3x - 2 \\ &= (x - 1)(5x + 2) \end{aligned}$$

$$\begin{aligned} (23) \quad & x^2 - 5x + 4 \\ &= (x - 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 12x^2 + 25x + 12 \\ &= (3x + 4)(4x + 3) \end{aligned}$$

$$\begin{aligned} (14) \quad & x^2 - 1 \\ &= (x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 18x^2 + 3x - 28 \\ &= (6x - 7)(3x + 4) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^2 + 21x + 49 \\ &= (x + 7)(2x + 7) \end{aligned}$$

$$\begin{aligned} (15) \quad & 3x^2 + 7x + 2 \\ &= (x + 2)(3x + 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & x^2 + 2x - 3 \\ &= (x - 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 7x^2 + 23x + 18 \\ &= (7x + 9)(x + 2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 5x^2 + 12x + 4 \\ &= (x + 2)(5x + 2) \end{aligned}$$

$$\begin{aligned} (26) \quad & 8x^2 - 22x - 63 \\ &= (4x + 7)(2x - 9) \end{aligned}$$

$$\begin{aligned} (7) \quad & 8x^2 - 15x + 7 \\ &= (8x - 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & 20x^2 + 69x + 54 \\ &= (5x + 6)(4x + 9) \end{aligned}$$

$$\begin{aligned} (27) \quad & 7x^2 - 16x - 15 \\ &= (7x + 5)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 4x^2 + 19x + 21 \\ &= (x + 3)(4x + 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & 4x^2 - 4x - 3 \\ &= (2x - 3)(2x + 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 14x^2 - x - 3 \\ &= (2x - 1)(7x + 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^2 - 13x + 14 \\ &= (x - 2)(3x - 7) \end{aligned}$$

$$\begin{aligned} (19) \quad & 63x^2 - 62x + 15 \\ &= (9x - 5)(7x - 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 3x^2 + 13x + 14 \\ &= (x + 2)(3x + 7) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^2 + 7x - 8 \\ &= (x + 8)(x - 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & x^2 - x - 12 \\ &= (x + 3)(x - 4) \end{aligned}$$

$$\begin{aligned} (30) \quad & x^2 + 3x - 4 \\ &= (x + 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 27x^2 + 12x - 32 \\ & = (9x - 8)(3x + 4) \end{aligned}$$

$$\begin{aligned} (11) \quad & 27x^2 + 33x + 8 \\ & = (9x + 8)(3x + 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 8x^2 + 9x - 14 \\ & = (8x - 7)(x + 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^2 + 31x + 18 \\ & = (3x + 2)(2x + 9) \end{aligned}$$

$$\begin{aligned} (12) \quad & 10x^2 + 9x - 7 \\ & = (2x - 1)(5x + 7) \end{aligned}$$

$$\begin{aligned} (22) \quad & 12x^2 + 8x - 7 \\ & = (6x + 7)(2x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 15x^2 - 29x + 8 \\ & = (5x - 8)(3x - 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 5x^2 - 3x - 14 \\ & = (x - 2)(5x + 7) \end{aligned}$$

$$\begin{aligned} (23) \quad & 4x^2 + 4x - 15 \\ & = (2x + 5)(2x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 14x^2 + 23x + 8 \\ & = (7x + 8)(2x + 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 18x^2 + 89x + 36 \\ & = (2x + 9)(9x + 4) \end{aligned}$$

$$\begin{aligned} (24) \quad & 63x^2 - 8x - 15 \\ & = (7x + 3)(9x - 5) \end{aligned}$$

$$\begin{aligned} (5) \quad & 12x^2 + 25x - 7 \\ & = (3x + 7)(4x - 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 3x^2 - 16x + 5 \\ & = (x - 5)(3x - 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 20x^2 + 27x - 14 \\ & = (5x - 2)(4x + 7) \end{aligned}$$

$$\begin{aligned} (6) \quad & 6x^2 + 5x - 6 \\ & = (3x - 2)(2x + 3) \end{aligned}$$

$$\begin{aligned} (16) \quad & 81x^2 - 54x - 7 \\ & = (9x - 7)(9x + 1) \end{aligned}$$

$$\begin{aligned} (26) \quad & 10x^2 - 21x + 8 \\ & = (2x - 1)(5x - 8) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^2 - 16x - 35 \\ & = (3x + 5)(x - 7) \end{aligned}$$

$$\begin{aligned} (17) \quad & 7x^2 + 10x - 8 \\ & = (7x - 4)(x + 2) \end{aligned}$$

$$\begin{aligned} (27) \quad & 4x^2 - 41x + 72 \\ & = (x - 8)(4x - 9) \end{aligned}$$

$$\begin{aligned} (8) \quad & x^2 - 1 \\ & = (x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (18) \quad & 7x^2 + 2x - 5 \\ & = (x + 1)(7x - 5) \end{aligned}$$

$$\begin{aligned} (28) \quad & 16x^2 + 6x - 27 \\ & = (8x - 9)(2x + 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 15x^2 - 22x - 9 \\ & = (5x - 9)(3x + 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 27x^2 - 21x - 20 \\ & = (3x - 4)(9x + 5) \end{aligned}$$

$$\begin{aligned} (29) \quad & 28x^2 + 19x + 3 \\ & = (7x + 3)(4x + 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^2 - 3x + 1 \\ & = (2x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 3x^2 + 17x - 28 \\ & = (x + 7)(3x - 4) \end{aligned}$$

$$\begin{aligned} (30) \quad & 3x^2 - 20x - 7 \\ & = (x - 7)(3x + 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 21x^2 + 23x + 6 \\ &= (7x + 3)(3x + 2) \end{aligned}$$

$$\begin{aligned} (11) \quad & 21x^2 - 13x - 20 \\ &= (3x - 4)(7x + 5) \end{aligned}$$

$$\begin{aligned} (21) \quad & x^2 - 2x + 1 \\ &= (x - 1)^2 \end{aligned}$$

$$\begin{aligned} (2) \quad & 9x^2 + 6x + 1 \\ &= (3x + 1)^2 \end{aligned}$$

$$\begin{aligned} (12) \quad & 4x^2 - x - 5 \\ &= (4x - 5)(x + 1) \end{aligned}$$

$$\begin{aligned} (22) \quad & 9x^2 + 38x - 35 \\ &= (x + 5)(9x - 7) \end{aligned}$$

$$\begin{aligned} (3) \quad & 3x^2 + 4x - 7 \\ &= (x - 1)(3x + 7) \end{aligned}$$

$$\begin{aligned} (13) \quad & 28x^2 + 81x + 56 \\ &= (4x + 7)(7x + 8) \end{aligned}$$

$$\begin{aligned} (23) \quad & 4x^2 + 5x - 6 \\ &= (4x - 3)(x + 2) \end{aligned}$$

$$\begin{aligned} (4) \quad & 14x^2 - 3x - 2 \\ &= (7x + 2)(2x - 1) \end{aligned}$$

$$\begin{aligned} (14) \quad & 24x^2 - 22x + 3 \\ &= (6x - 1)(4x - 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & x^2 - 6x - 7 \\ &= (x - 7)(x + 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 16x^2 - 22x - 3 \\ &= (8x + 1)(2x - 3) \end{aligned}$$

$$\begin{aligned} (15) \quad & 2x^2 + x - 3 \\ &= (2x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (25) \quad & 4x^2 - 27x - 7 \\ &= (x - 7)(4x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 4x^2 + 11x - 3 \\ &= (4x - 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (16) \quad & 6x^2 - 11x - 2 \\ &= (x - 2)(6x + 1) \end{aligned}$$

$$\begin{aligned} (26) \quad & 30x^2 + x - 1 \\ &= (5x + 1)(6x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 8x^2 - 14x - 9 \\ &= (4x - 9)(2x + 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & 18x^2 - 11x + 1 \\ &= (2x - 1)(9x - 1) \end{aligned}$$

$$\begin{aligned} (27) \quad & 48x^2 + 50x + 7 \\ &= (8x + 7)(6x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 20x^2 - 29x - 36 \\ &= (5x + 4)(4x - 9) \end{aligned}$$

$$\begin{aligned} (18) \quad & 5x^2 - 31x - 28 \\ &= (5x + 4)(x - 7) \end{aligned}$$

$$\begin{aligned} (28) \quad & 20x^2 - 61x + 45 \\ &= (5x - 9)(4x - 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^2 - 3x + 1 \\ &= (x - 1)(2x - 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & x^2 - 2x + 1 \\ &= (x - 1)^2 \end{aligned}$$

$$\begin{aligned} (29) \quad & 7x^2 + 12x + 5 \\ &= (7x + 5)(x + 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 4x^2 + 13x + 3 \\ &= (x + 3)(4x + 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 2x^2 + 5x + 3 \\ &= (2x + 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (30) \quad & 8x^2 + 33x + 27 \\ &= (8x + 9)(x + 3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 18x^2 + 19x - 12 \\ &= (9x - 4)(2x + 3) \end{aligned}$$

$$\begin{aligned} (11) \quad & 15x^2 + 7x - 4 \\ &= (5x + 4)(3x - 1) \end{aligned}$$

$$\begin{aligned} (21) \quad & 9x^2 - 17x - 2 \\ &= (x - 2)(9x + 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 9x^2 - 9x + 2 \\ &= (3x - 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (12) \quad & 5x^2 + 53x + 72 \\ &= (5x + 8)(x + 9) \end{aligned}$$

$$\begin{aligned} (22) \quad & 2x^2 + 5x + 3 \\ &= (x + 1)(2x + 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 2x^2 + x - 3 \\ &= (x - 1)(2x + 3) \end{aligned}$$

$$\begin{aligned} (13) \quad & 40x^2 + 49x + 15 \\ &= (8x + 5)(5x + 3) \end{aligned}$$

$$\begin{aligned} (23) \quad & 9x^2 - 18x + 8 \\ &= (3x - 2)(3x - 4) \end{aligned}$$

$$\begin{aligned} (4) \quad & 24x^2 + 7x - 6 \\ &= (8x - 3)(3x + 2) \end{aligned}$$

$$\begin{aligned} (14) \quad & x^2 - 5x + 4 \\ &= (x - 1)(x - 4) \end{aligned}$$

$$\begin{aligned} (24) \quad & 14x^2 + 45x + 25 \\ &= (7x + 5)(2x + 5) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^2 + 3x - 4 \\ &= (x + 4)(x - 1) \end{aligned}$$

$$\begin{aligned} (15) \quad & 10x^2 - 37x + 7 \\ &= (5x - 1)(2x - 7) \end{aligned}$$

$$\begin{aligned} (25) \quad & 7x^2 - 5x - 2 \\ &= (x - 1)(7x + 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & x^2 + x - 56 \\ &= (x - 7)(x + 8) \end{aligned}$$

$$\begin{aligned} (16) \quad & 24x^2 - 13x - 2 \\ &= (8x + 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (26) \quad & x^2 + 7x - 8 \\ &= (x - 1)(x + 8) \end{aligned}$$

$$\begin{aligned} (7) \quad & 9x^2 + x - 8 \\ &= (x + 1)(9x - 8) \end{aligned}$$

$$\begin{aligned} (17) \quad & 32x^2 - 60x - 27 \\ &= (4x - 9)(8x + 3) \end{aligned}$$

$$\begin{aligned} (27) \quad & 64x^2 - 81 \\ &= (8x + 9)(8x - 9) \end{aligned}$$

$$\begin{aligned} (8) \quad & 24x^2 + 22x - 7 \\ &= (4x - 1)(6x + 7) \end{aligned}$$

$$\begin{aligned} (18) \quad & 4x^2 - 19x - 5 \\ &= (x - 5)(4x + 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 2x^2 + 5x - 7 \\ &= (2x + 7)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 20x^2 + 39x + 7 \\ &= (4x + 7)(5x + 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 36x^2 + 5x - 1 \\ &= (9x - 1)(4x + 1) \end{aligned}$$

$$\begin{aligned} (29) \quad & 4x^2 - 3x - 27 \\ &= (4x + 9)(x - 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 14x^2 + 3x - 2 \\ &= (7x - 2)(2x + 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 6x^2 - 29x + 20 \\ &= (x - 4)(6x - 5) \end{aligned}$$

$$\begin{aligned} (30) \quad & 7x^2 - 54x - 16 \\ &= (x - 8)(7x + 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 45x^2 - 92x + 32 \\ & = (5x - 8)(9x - 4) \end{aligned}$$

$$\begin{aligned} (11) \quad & 18x^2 - 17x - 15 \\ & = (9x + 5)(2x - 3) \end{aligned}$$

$$\begin{aligned} (21) \quad & 5x^2 - 37x - 72 \\ & = (5x + 8)(x - 9) \end{aligned}$$

$$\begin{aligned} (2) \quad & 72x^2 - 83x + 21 \\ & = (8x - 3)(9x - 7) \end{aligned}$$

$$\begin{aligned} (12) \quad & 8x^2 - 3x - 5 \\ & = (x - 1)(8x + 5) \end{aligned}$$

$$\begin{aligned} (22) \quad & 32x^2 - 4x - 15 \\ & = (4x - 3)(8x + 5) \end{aligned}$$

$$\begin{aligned} (3) \quad & 42x^2 - x - 30 \\ & = (6x + 5)(7x - 6) \end{aligned}$$

$$\begin{aligned} (13) \quad & 8x^2 + 15x + 7 \\ & = (x + 1)(8x + 7) \end{aligned}$$

$$\begin{aligned} (23) \quad & 4x^2 + 13x - 12 \\ & = (x + 4)(4x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^2 + 11x + 9 \\ & = (x + 1)(2x + 9) \end{aligned}$$

$$\begin{aligned} (14) \quad & x^2 - 4x + 3 \\ & = (x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (24) \quad & 28x^2 + 69x + 35 \\ & = (4x + 7)(7x + 5) \end{aligned}$$

$$\begin{aligned} (5) \quad & 25x^2 + 5x - 6 \\ & = (5x + 3)(5x - 2) \end{aligned}$$

$$\begin{aligned} (15) \quad & 7x^2 + 16x + 9 \\ & = (x + 1)(7x + 9) \end{aligned}$$

$$\begin{aligned} (25) \quad & 28x^2 + 15x + 2 \\ & = (4x + 1)(7x + 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 9x^2 - 4x - 5 \\ & = (x - 1)(9x + 5) \end{aligned}$$

$$\begin{aligned} (16) \quad & 2x^2 + 9x + 10 \\ & = (x + 2)(2x + 5) \end{aligned}$$

$$\begin{aligned} (26) \quad & 32x^2 - 60x + 7 \\ & = (4x - 7)(8x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^2 - 5x - 8 \\ & = (3x - 8)(x + 1) \end{aligned}$$

$$\begin{aligned} (17) \quad & 35x^2 + 9x - 18 \\ & = (5x - 3)(7x + 6) \end{aligned}$$

$$\begin{aligned} (27) \quad & 16x^2 + 26x + 3 \\ & = (2x + 3)(8x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 36x^2 + 97x + 36 \\ & = (9x + 4)(4x + 9) \end{aligned}$$

$$\begin{aligned} (18) \quad & 5x^2 + 13x - 6 \\ & = (5x - 2)(x + 3) \end{aligned}$$

$$\begin{aligned} (28) \quad & 3x^2 - 8x + 5 \\ & = (x - 1)(3x - 5) \end{aligned}$$

$$\begin{aligned} (9) \quad & 32x^2 - 12x - 9 \\ & = (4x - 3)(8x + 3) \end{aligned}$$

$$\begin{aligned} (19) \quad & 5x^2 + 19x + 18 \\ & = (x + 2)(5x + 9) \end{aligned}$$

$$\begin{aligned} (29) \quad & 36x^2 - 49x + 5 \\ & = (4x - 5)(9x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 10x^2 - 23x + 12 \\ & = (2x - 3)(5x - 4) \end{aligned}$$

$$\begin{aligned} (20) \quad & 2x^2 - x - 10 \\ & = (x + 2)(2x - 5) \end{aligned}$$

$$\begin{aligned} (30) \quad & 12x^2 - 31x + 7 \\ & = (4x - 1)(3x - 7) \end{aligned}$$

$$\begin{aligned} (1) \quad & 45x^2 - 53x + 14 \\ & = (5x - 2)(9x - 7) \end{aligned}$$

$$\begin{aligned} (11) \quad & 6x^2 + 35x - 6 \\ & = (6x - 1)(x + 6) \end{aligned}$$

$$\begin{aligned} (21) \quad & 56x^2 - 9x - 35 \\ & = (7x + 5)(8x - 7) \end{aligned}$$

$$\begin{aligned} (2) \quad & 24x^2 - 11x - 18 \\ & = (8x - 9)(3x + 2) \end{aligned}$$

$$\begin{aligned} (12) \quad & 4x^2 + 5x - 6 \\ & = (4x - 3)(x + 2) \end{aligned}$$

$$\begin{aligned} (22) \quad & 7x^2 - 29x + 24 \\ & = (7x - 8)(x - 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 8x^2 + 14x + 5 \\ & = (4x + 5)(2x + 1) \end{aligned}$$

$$\begin{aligned} (13) \quad & 2x^2 + 3x - 2 \\ & = (2x - 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (23) \quad & 40x^2 - 39x + 9 \\ & = (8x - 3)(5x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 15x^2 + 4x - 32 \\ & = (5x + 8)(3x - 4) \end{aligned}$$

$$\begin{aligned} (14) \quad & 3x^2 + 5x - 2 \\ & = (x + 2)(3x - 1) \end{aligned}$$

$$\begin{aligned} (24) \quad & 36x^2 + 95x + 56 \\ & = (9x + 8)(4x + 7) \end{aligned}$$

$$\begin{aligned} (5) \quad & 4x^2 + 21x + 5 \\ & = (4x + 1)(x + 5) \end{aligned}$$

$$\begin{aligned} (15) \quad & 6x^2 + 29x + 28 \\ & = (3x + 4)(2x + 7) \end{aligned}$$

$$\begin{aligned} (25) \quad & x^2 + 2x - 15 \\ & = (x - 3)(x + 5) \end{aligned}$$

$$\begin{aligned} (6) \quad & 20x^2 + 17x - 10 \\ & = (4x + 5)(5x - 2) \end{aligned}$$

$$\begin{aligned} (16) \quad & 16x^2 - 74x + 63 \\ & = (2x - 7)(8x - 9) \end{aligned}$$

$$\begin{aligned} (26) \quad & 3x^2 - 5x - 2 \\ & = (x - 2)(3x + 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 81x^2 + 36x - 32 \\ & = (9x - 4)(9x + 8) \end{aligned}$$

$$\begin{aligned} (17) \quad & 3x^2 + 32x + 64 \\ & = (x + 8)(3x + 8) \end{aligned}$$

$$\begin{aligned} (27) \quad & 2x^2 + x - 21 \\ & = (2x + 7)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 5x^2 + 14x - 3 \\ & = (x + 3)(5x - 1) \end{aligned}$$

$$\begin{aligned} (18) \quad & 16x^2 + 2x - 5 \\ & = (8x + 5)(2x - 1) \end{aligned}$$

$$\begin{aligned} (28) \quad & 36x^2 + 37x - 10 \\ & = (4x + 5)(9x - 2) \end{aligned}$$

$$\begin{aligned} (9) \quad & 5x^2 + 4x - 1 \\ & = (x + 1)(5x - 1) \end{aligned}$$

$$\begin{aligned} (19) \quad & 4x^2 + 17x + 15 \\ & = (4x + 5)(x + 3) \end{aligned}$$

$$\begin{aligned} (29) \quad & 10x^2 + 43x + 45 \\ & = (2x + 5)(5x + 9) \end{aligned}$$

$$\begin{aligned} (10) \quad & 6x^2 + 13x - 8 \\ & = (3x + 8)(2x - 1) \end{aligned}$$

$$\begin{aligned} (20) \quad & 24x^2 - 43x + 18 \\ & = (3x - 2)(8x - 9) \end{aligned}$$

$$\begin{aligned} (30) \quad & x^2 + 2x + 1 \\ & = (x + 1)^2 \end{aligned}$$

## 2.5 因数分解 (3 次式)

3 次多項式の因数分解をする問題。



(1)  $3x^3 - 5x^2 + x + 1$

(6)  $2x^3 - x^2 - 2x + 1$

(2)  $x^3 + x^2 - x - 1$

(7)  $x^3 - 2x^2 - 5x + 6$

(3)  $9x^3 - 3x^2 - 5x - 1$

(8)  $6x^3 - 5x^2 - 3x + 2$

(4)  $x^3 + 3x^2 - x - 3$

(9)  $12x^3 + 20x^2 - x - 6$

(5)  $2x^3 - 3x^2 - 11x + 6$

(10)  $2x^3 - 5x^2 - x + 6$

(1)  $4x^3 - 4x^2 - x + 1$

(6)  $18x^3 + 9x^2 - 23x + 6$

(2)  $6x^3 + x^2 - 11x - 6$

(7)  $2x^3 - 7x^2 + 8x - 3$

(3)  $x^3 - 3x^2 - x + 3$

(8)  $6x^3 + 17x^2 - 5x - 6$

(4)  $3x^3 + 7x^2 + 5x + 1$

(9)  $9x^3 + 21x^2 - 17x + 3$

(5)  $2x^3 + 7x^2 + 7x + 2$

(10)  $3x^3 - 4x^2 - x + 2$

(1)  $9x^3 - 21x^2 + 16x - 4$

(6)  $2x^3 + 7x^2 - 9$

(2)  $3x^3 + x^2 - 12x - 4$

(7)  $9x^3 + 21x^2 - 17x + 3$

(3)  $x^3 + x^2 - 9x - 9$

(8)  $3x^3 - 7x^2 + 5x - 1$

(4)  $9x^3 - 3x^2 - 8x + 4$

(9)  $2x^3 - 3x^2 - 18x + 27$

(5)  $x^3 + 3x^2 - x - 3$

(10)  $4x^3 - 4x^2 - x + 1$

(1)  $4x^3 + 4x^2 - 5x - 3$

(6)  $6x^3 + 11x^2 + 6x + 1$

(2)  $x^3 - 5x^2 + 7x - 3$

(7)  $x^3 + 2x^2 - x - 2$

(3)  $27x^3 + 45x^2 + 24x + 4$

(8)  $2x^3 + 5x^2 - 6x - 9$

(4)  $3x^3 + x^2 - 27x - 9$

(9)  $4x^3 - 8x^2 + x + 3$

(5)  $x^3 - x^2 - 4x + 4$

(10)  $9x^3 + 18x^2 + 11x + 2$

(1)  $4x^3 - 4x^2 - 5x + 3$

(6)  $9x^3 + 24x^2 + 13x + 2$

(2)  $2x^3 - 11x^2 + 20x - 12$

(7)  $4x^3 - 8x^2 + x + 3$

(3)  $12x^3 - 4x^2 - 3x + 1$

(8)  $8x^3 - 4x^2 - 18x + 9$

(4)  $3x^3 - 5x^2 - 4x + 4$

(9)  $2x^3 + 3x^2 - 11x - 6$

(5)  $2x^3 - 7x^2 + 4x + 4$

(10)  $2x^3 + x^2 - 2x - 1$

(1)  $x^3 - 7x + 6$

(6)  $6x^3 - 5x^2 - 2x + 1$

(2)  $2x^3 - 5x^2 - x + 6$

(7)  $4x^3 + 12x^2 - 9x - 27$

(3)  $6x^3 - 5x^2 - 33x - 18$

(8)  $2x^3 + 7x^2 - 9$

(4)  $4x^3 - 4x^2 - 5x + 3$

(9)  $4x^3 - 8x^2 + x + 3$

(5)  $9x^3 - 9x^2 - x + 1$

(10)  $2x^3 + 3x^2 - 5x - 6$

(1)  $3x^3 - 8x^2 - 5x + 6$

(6)  $3x^3 + 17x^2 + 21x - 9$

(2)  $2x^3 - 3x^2 - 2x + 3$

(7)  $3x^3 + 10x^2 + x - 6$

(3)  $2x^3 - x^2 - 4x + 3$

(8)  $x^3 - 3x + 2$

(4)  $6x^3 - x^2 - 5x + 2$

(9)  $x^3 - 3x^2 - 9x + 27$

(5)  $9x^3 - 30x^2 + 28x - 8$

(10)  $3x^3 + x^2 - 8x + 4$

(1)  $6x^3 - 5x^2 - 8x + 3$

(6)  $x^3 - 4x^2 + x + 6$

(2)  $2x^3 + 5x^2 + 4x + 1$

(7)  $x^3 - 3x - 2$

(3)  $2x^3 + 3x^2 - 5x - 6$

(8)  $2x^3 + 9x^2 + 13x + 6$

(4)  $3x^3 + 5x^2 - 4x - 4$

(9)  $x^3 - x^2 - 5x - 3$

(5)  $2x^3 + 7x^2 + 8x + 3$

(10)  $3x^3 - 10x^2 + 9x - 2$



(1)  $2x^3 + 7x^2 + 7x + 2$

(6)  $2x^3 + 3x^2 - 8x + 3$

(2)  $18x^3 + 15x^2 - 4x - 4$

(7)  $2x^3 - 5x^2 - x + 6$

(3)  $3x^3 - 19x^2 + 33x - 9$

(8)  $x^3 + x^2 - x - 1$

(4)  $12x^3 - 4x^2 - 3x + 1$

(9)  $6x^3 - 19x^2 + 19x - 6$

(5)  $4x^3 - 4x^2 - 21x - 9$

(10)  $2x^3 - 9x^2 + 13x - 6$

(1)  $3x^3 + 10x^2 + x - 6$

(6)  $12x^3 - 20x^2 - x + 6$

(2)  $6x^3 - 7x^2 - 9x - 2$

(7)  $6x^3 - 19x^2 + 19x - 6$

(3)  $2x^3 - 7x^2 + 2x + 3$

(8)  $6x^3 + x^2 - 4x + 1$

(4)  $6x^3 - 17x^2 - 4x + 3$

(9)  $6x^3 - 11x^2 + 6x - 1$

(5)  $6x^3 + x^2 - 11x - 6$

(10)  $6x^3 - 19x^2 + 19x - 6$

(1)  $3x^3 - 7x^2 - 7x + 3$

(6)  $x^3 - 2x^2 - 5x + 6$

(2)  $8x^3 + 4x^2 - 18x - 9$

(7)  $6x^3 - x^2 - 5x + 2$

(3)  $12x^3 + 28x^2 + 17x + 3$

(8)  $6x^3 - x^2 - 10x - 3$

(4)  $2x^3 + 7x^2 + 2x - 3$

(9)  $3x^3 + x^2 - 12x - 4$

(5)  $3x^3 + 10x^2 + 9x + 2$

(10)  $6x^3 + 13x^2 + x - 2$

(1)  $18x^3 + 21x^2 - 13x - 6$

(6)  $2x^3 - 9x^2 + 13x - 6$

(2)  $3x^3 + 10x^2 + 9x + 2$

(7)  $4x^3 - 4x^2 - x + 1$

(3)  $6x^3 + 17x^2 + 4x - 12$

(8)  $3x^3 - 7x^2 + 5x - 1$

(4)  $4x^3 - 4x^2 - 5x + 3$

(9)  $2x^3 - 3x^2 - 8x - 3$

(5)  $x^3 - 4x^2 + 5x - 2$

(10)  $x^3 - 4x^2 + 5x - 2$

(1)  $9x^3 - 18x^2 - 4x + 8$

(6)  $3x^3 - 8x^2 - 5x + 6$

(2)  $6x^3 - 5x^2 - 3x + 2$

(7)  $2x^3 + 3x^2 - 11x - 6$

(3)  $4x^3 - 13x + 6$

(8)  $9x^3 + 21x^2 + 16x + 4$

(4)  $3x^3 + 4x^2 - 13x + 6$

(9)  $3x^3 + 2x^2 - 3x - 2$

(5)  $3x^3 + 13x^2 + 13x + 3$

(10)  $4x^3 - 3x - 1$

(1)  $x^3 - x^2 - 9x + 9$

(6)  $18x^3 - 27x^2 + 13x - 2$

(2)  $4x^3 + 16x^2 + 9x - 9$

(7)  $3x^3 - x^2 - 3x + 1$

(3)  $4x^3 + 4x^2 - 7x + 2$

(8)  $x^3 + 3x^2 - x - 3$

(4)  $2x^3 + 5x^2 - x - 6$

(9)  $6x^3 - 11x^2 - x + 6$

(5)  $18x^3 + 15x^2 - 16x + 3$

(10)  $2x^3 + 5x^2 - 4x - 3$

(1)  $2x^3 + x^2 - 5x + 2$

(6)  $4x^3 - 8x^2 + 5x - 1$

(2)  $6x^3 - x^2 - 20x + 12$

(7)  $3x^3 + 19x^2 + 33x + 9$

(3)  $6x^3 + 11x^2 - x - 6$

(8)  $6x^3 + 25x^2 + 23x + 6$

(4)  $9x^3 - 18x^2 - x + 2$

(9)  $6x^3 + 17x^2 - 5x - 6$

(5)  $2x^3 + 7x^2 + 7x + 2$

(10)  $x^3 - 3x^2 + 3x - 1$

(1)  $18x^3 + 15x^2 - x - 2$

(6)  $4x^3 + 8x^2 + x - 3$

(2)  $6x^3 + 13x^2 + 9x + 2$

(7)  $3x^3 - 4x^2 - 17x + 6$

(3)  $9x^3 + 18x^2 - x - 2$

(8)  $3x^3 - 2x^2 - 3x + 2$

(4)  $x^3 - 3x^2 + 4$

(9)  $2x^3 + x^2 - 2x - 1$

(5)  $9x^3 + 9x^2 - 4x - 4$

(10)  $9x^3 - 6x^2 - 5x + 2$



(1)  $3x^3 - x^2 - 3x + 1$

(6)  $4x^3 - 8x^2 - 3x + 9$

(2)  $x^3 + 3x^2 + 3x + 1$

(7)  $6x^3 + 7x^2 - 1$

(3)  $9x^3 + 18x^2 - x - 2$

(8)  $6x^3 + 17x^2 + 14x + 3$

(4)  $3x^3 - 7x^2 - 7x + 3$

(9)  $3x^3 + 14x^2 + 17x + 6$

(5)  $3x^3 + x^2 - 3x - 1$

(10)  $x^3 + x^2 - x - 1$

(1)  $x^3 - 2x^2 - x + 2$

(6)  $2x^3 - 3x^2 - 2x + 3$

(2)  $12x^3 - 8x^2 - 13x - 3$

(7)  $x^3 - 2x^2 - 5x + 6$

(3)  $4x^3 + 16x^2 + 19x + 6$

(8)  $4x^3 - 8x^2 + x + 3$

(4)  $x^3 - x^2 - 9x + 9$

(9)  $2x^3 + x^2 - 18x - 9$

(5)  $3x^3 + x^2 - 3x - 1$

(10)  $x^3 + x^2 - x - 1$

(1)  $2x^3 + x^2 - 7x - 6$

(6)  $2x^3 - x^2 - 7x + 6$

(2)  $6x^3 - x^2 - 11x + 6$

(7)  $12x^3 - 8x^2 - 13x - 3$

(3)  $9x^3 - 18x^2 - 4x + 8$

(8)  $9x^3 - 27x^2 - 4x + 12$

(4)  $x^3 + x^2 - 9x - 9$

(9)  $3x^3 - 13x^2 + 8x + 12$

(5)  $x^3 + x^2 - 4x - 4$

(10)  $3x^3 - 11x^2 + 5x + 3$

(1)  $2x^3 - x^2 - 4x + 3$

(6)  $x^3 + 3x^2 - x - 3$

(2)  $3x^3 - 5x^2 - 11x - 3$

(7)  $9x^3 + 3x^2 - 5x + 1$

(3)  $12x^3 - 16x^2 - 7x + 6$

(8)  $3x^3 + 7x^2 + 5x + 1$

(4)  $9x^3 - 9x^2 - x + 1$

(9)  $3x^3 + x^2 - 3x - 1$

(5)  $9x^3 - 9x^2 - 4x + 4$

(10)  $4x^3 - 7x - 3$

$$\begin{aligned} (1) \quad & 3x^3 - 5x^2 + x + 1 \\ &= (3x + 1)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 - x^2 - 2x + 1 \\ &= (2x - 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^3 + x^2 - x - 1 \\ &= (x + 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^3 - 2x^2 - 5x + 6 \\ &= (x + 2)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 9x^3 - 3x^2 - 5x - 1 \\ &= (3x + 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 - 5x^2 - 3x + 2 \\ &= (2x - 1)(3x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & x^3 + 3x^2 - x - 3 \\ &= (x + 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 12x^3 + 20x^2 - x - 6 \\ &= (2x + 3)(2x - 1)(3x + 2) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^3 - 3x^2 - 11x + 6 \\ &= (2x - 1)(x + 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^3 - 5x^2 - x + 6 \\ &= (2x - 3)(x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 4x^3 - 4x^2 - x + 1 \\ &= (2x + 1)(2x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 18x^3 + 9x^2 - 23x + 6 \\ &= (2x + 3)(3x - 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 + x^2 - 11x - 6 \\ &= (2x - 3)(3x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^3 - 7x^2 + 8x - 3 \\ &= (2x - 3)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (3) \quad & x^3 - 3x^2 - x + 3 \\ &= (x + 1)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 + 17x^2 - 5x - 6 \\ &= (2x + 1)(3x - 2)(x + 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 + 7x^2 + 5x + 1 \\ &= (3x + 1)(x + 1)^2 \end{aligned}$$

$$\begin{aligned} (9) \quad & 9x^3 + 21x^2 - 17x + 3 \\ &= (3x - 1)^2(x + 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^3 + 7x^2 + 7x + 2 \\ &= (2x + 1)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 3x^3 - 4x^2 - x + 2 \\ &= (3x + 2)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (1) \quad & 9x^3 - 21x^2 + 16x - 4 \\ &= (3x - 2)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 + 7x^2 - 9 \\ &= (2x + 3)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 3x^3 + x^2 - 12x - 4 \\ &= (3x + 1)(x + 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 9x^3 + 21x^2 - 17x + 3 \\ &= (3x - 1)^2(x + 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & x^3 + x^2 - 9x - 9 \\ &= (x + 1)(x + 3)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^3 - 7x^2 + 5x - 1 \\ &= (3x - 1)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (4) \quad & 9x^3 - 3x^2 - 8x + 4 \\ &= (3x - 2)^2(x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^3 - 3x^2 - 18x + 27 \\ &= (2x - 3)(x + 3)(x - 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^3 + 3x^2 - x - 3 \\ &= (x + 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 4x^3 - 4x^2 - x + 1 \\ &= (2x + 1)(2x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 4x^3 + 4x^2 - 5x - 3 \\ &= (2x + 1)(2x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 6x^3 + 11x^2 + 6x + 1 \\ &= (2x + 1)(3x + 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^3 - 5x^2 + 7x - 3 \\ &= (x - 1)^2(x - 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^3 + 2x^2 - x - 2 \\ &= (x + 1)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 27x^3 + 45x^2 + 24x + 4 \\ &= (3x + 1)(3x + 2)^2 \end{aligned}$$

$$\begin{aligned} (8) \quad & 2x^3 + 5x^2 - 6x - 9 \\ &= (2x - 3)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 + x^2 - 27x - 9 \\ &= (3x + 1)(x + 3)(x - 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 4x^3 - 8x^2 + x + 3 \\ &= (2x + 1)(2x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^3 - x^2 - 4x + 4 \\ &= (x + 2)(x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 9x^3 + 18x^2 + 11x + 2 \\ &= (3x + 1)(3x + 2)(x + 1) \end{aligned}$$



$$\begin{aligned} (1) \quad & 4x^3 - 4x^2 - 5x + 3 \\ &= (2x - 1)(2x - 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 9x^3 + 24x^2 + 13x + 2 \\ &= (3x + 1)^2(x + 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 2x^3 - 11x^2 + 20x - 12 \\ &= (2x - 3)(x - 2)^2 \end{aligned}$$

$$\begin{aligned} (7) \quad & 4x^3 - 8x^2 + x + 3 \\ &= (2x + 1)(2x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 12x^3 - 4x^2 - 3x + 1 \\ &= (2x + 1)(2x - 1)(3x - 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 8x^3 - 4x^2 - 18x + 9 \\ &= (2x + 3)(2x - 1)(2x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 - 5x^2 - 4x + 4 \\ &= (3x - 2)(x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^3 + 3x^2 - 11x - 6 \\ &= (2x + 1)(x + 3)(x - 2) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^3 - 7x^2 + 4x + 4 \\ &= (2x + 1)(x - 2)^2 \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^3 + x^2 - 2x - 1 \\ &= (2x + 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & x^3 - 7x + 6 \\ &= (x + 3)(x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 6x^3 - 5x^2 - 2x + 1 \\ &= (2x + 1)(3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 2x^3 - 5x^2 - x + 6 \\ &= (2x - 3)(x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 4x^3 + 12x^2 - 9x - 27 \\ &= (2x + 3)(2x - 3)(x + 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 6x^3 - 5x^2 - 33x - 18 \\ &= (2x + 3)(3x + 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 2x^3 + 7x^2 - 9 \\ &= (2x + 3)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 4x^3 - 4x^2 - 5x + 3 \\ &= (2x - 1)(2x - 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 4x^3 - 8x^2 + x + 3 \\ &= (2x + 1)(2x - 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 9x^3 - 9x^2 - x + 1 \\ &= (3x + 1)(3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^3 + 3x^2 - 5x - 6 \\ &= (2x - 3)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^3 - 8x^2 - 5x + 6 \\ &= (3x - 2)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 3x^3 + 17x^2 + 21x - 9 \\ &= (3x - 1)(x + 3)^2 \end{aligned}$$

$$\begin{aligned} (2) \quad & 2x^3 - 3x^2 - 2x + 3 \\ &= (2x - 3)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^3 + 10x^2 + x - 6 \\ &= (3x - 2)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 2x^3 - x^2 - 4x + 3 \\ &= (2x + 3)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (8) \quad & x^3 - 3x + 2 \\ &= (x + 2)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (4) \quad & 6x^3 - x^2 - 5x + 2 \\ &= (2x - 1)(3x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & x^3 - 3x^2 - 9x + 27 \\ &= (x + 3)(x - 3)^2 \end{aligned}$$

$$\begin{aligned} (5) \quad & 9x^3 - 30x^2 + 28x - 8 \\ &= (3x - 2)^2(x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 3x^3 + x^2 - 8x + 4 \\ &= (3x - 2)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 6x^3 - 5x^2 - 8x + 3 \\ &= (2x - 3)(3x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & x^3 - 4x^2 + x + 6 \\ &= (x + 1)(x - 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 2x^3 + 5x^2 + 4x + 1 \\ &= (2x + 1)(x + 1)^2 \end{aligned}$$

$$\begin{aligned} (7) \quad & x^3 - 3x - 2 \\ &= (x + 1)^2(x - 2) \end{aligned}$$

$$\begin{aligned} (3) \quad & 2x^3 + 3x^2 - 5x - 6 \\ &= (2x - 3)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 2x^3 + 9x^2 + 13x + 6 \\ &= (2x + 3)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 + 5x^2 - 4x - 4 \\ &= (3x + 2)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & x^3 - x^2 - 5x - 3 \\ &= (x + 1)^2(x - 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^3 + 7x^2 + 8x + 3 \\ &= (2x + 3)(x + 1)^2 \end{aligned}$$

$$\begin{aligned} (10) \quad & 3x^3 - 10x^2 + 9x - 2 \\ &= (3x - 1)(x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 2x^3 + 7x^2 + 7x + 2 \\ &= (2x + 1)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 + 3x^2 - 8x + 3 \\ &= (2x - 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 18x^3 + 15x^2 - 4x - 4 \\ &= (2x - 1)(3x + 2)^2 \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^3 - 5x^2 - x + 6 \\ &= (2x - 3)(x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (3) \quad & 3x^3 - 19x^2 + 33x - 9 \\ &= (3x - 1)(x - 3)^2 \end{aligned}$$

$$\begin{aligned} (8) \quad & x^3 + x^2 - x - 1 \\ &= (x + 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 12x^3 - 4x^2 - 3x + 1 \\ &= (2x + 1)(2x - 1)(3x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 6x^3 - 19x^2 + 19x - 6 \\ &= (2x - 3)(3x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 4x^3 - 4x^2 - 21x - 9 \\ &= (2x + 1)(2x + 3)(x - 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^3 - 9x^2 + 13x - 6 \\ &= (2x - 3)(x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^3 + 10x^2 + x - 6 \\ &= (3x - 2)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 12x^3 - 20x^2 - x + 6 \\ &= (2x + 1)(2x - 3)(3x - 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 - 7x^2 - 9x - 2 \\ &= (2x + 1)(3x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 6x^3 - 19x^2 + 19x - 6 \\ &= (2x - 3)(3x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 2x^3 - 7x^2 + 2x + 3 \\ &= (2x + 1)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 + x^2 - 4x + 1 \\ &= (2x - 1)(3x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 6x^3 - 17x^2 - 4x + 3 \\ &= (2x + 1)(3x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 6x^3 - 11x^2 + 6x - 1 \\ &= (2x - 1)(3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 6x^3 + x^2 - 11x - 6 \\ &= (2x - 3)(3x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 6x^3 - 19x^2 + 19x - 6 \\ &= (2x - 3)(3x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^3 - 7x^2 - 7x + 3 \\ &= (3x - 1)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & x^3 - 2x^2 - 5x + 6 \\ &= (x + 2)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 8x^3 + 4x^2 - 18x - 9 \\ &= (2x + 1)(2x + 3)(2x - 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & 6x^3 - x^2 - 5x + 2 \\ &= (2x - 1)(3x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 12x^3 + 28x^2 + 17x + 3 \\ &= (2x + 1)(2x + 3)(3x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 - x^2 - 10x - 3 \\ &= (2x - 3)(3x + 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^3 + 7x^2 + 2x - 3 \\ &= (2x - 1)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^3 + x^2 - 12x - 4 \\ &= (3x + 1)(x + 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^3 + 10x^2 + 9x + 2 \\ &= (3x + 1)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 6x^3 + 13x^2 + x - 2 \\ &= (2x + 1)(3x - 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (1) \quad & 18x^3 + 21x^2 - 13x - 6 \\ &= (2x + 3)(3x + 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 - 9x^2 + 13x - 6 \\ &= (2x - 3)(x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 3x^3 + 10x^2 + 9x + 2 \\ &= (3x + 1)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 4x^3 - 4x^2 - x + 1 \\ &= (2x + 1)(2x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 6x^3 + 17x^2 + 4x - 12 \\ &= (2x + 3)(3x - 2)(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^3 - 7x^2 + 5x - 1 \\ &= (3x - 1)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (4) \quad & 4x^3 - 4x^2 - 5x + 3 \\ &= (2x - 1)(2x - 3)(x + 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^3 - 3x^2 - 8x - 3 \\ &= (2x + 1)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^3 - 4x^2 + 5x - 2 \\ &= (x - 1)^2(x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^3 - 4x^2 + 5x - 2 \\ &= (x - 1)^2(x - 2) \end{aligned}$$



$$\begin{aligned} (1) \quad & 9x^3 - 18x^2 - 4x + 8 \\ & = (3x + 2)(3x - 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 3x^3 - 8x^2 - 5x + 6 \\ & = (3x - 2)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 - 5x^2 - 3x + 2 \\ & = (2x - 1)(3x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 2x^3 + 3x^2 - 11x - 6 \\ & = (2x + 1)(x + 3)(x - 2) \end{aligned}$$

$$\begin{aligned} (3) \quad & 4x^3 - 13x + 6 \\ & = (2x - 1)(2x - 3)(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 9x^3 + 21x^2 + 16x + 4 \\ & = (3x + 2)^2(x + 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 + 4x^2 - 13x + 6 \\ & = (3x - 2)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^3 + 2x^2 - 3x - 2 \\ & = (3x + 2)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^3 + 13x^2 + 13x + 3 \\ & = (3x + 1)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (10) \quad & 4x^3 - 3x - 1 \\ & = (2x + 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & x^3 - x^2 - 9x + 9 \\ &= (x + 3)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (6) \quad & 18x^3 - 27x^2 + 13x - 2 \\ &= (2x - 1)(3x - 1)(3x - 2) \end{aligned}$$

$$\begin{aligned} (2) \quad & 4x^3 + 16x^2 + 9x - 9 \\ &= (2x + 3)(2x - 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^3 - x^2 - 3x + 1 \\ &= (3x - 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 4x^3 + 4x^2 - 7x + 2 \\ &= (2x - 1)^2(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & x^3 + 3x^2 - x - 3 \\ &= (x + 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 2x^3 + 5x^2 - x - 6 \\ &= (2x + 3)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 6x^3 - 11x^2 - x + 6 \\ &= (2x - 3)(3x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 18x^3 + 15x^2 - 16x + 3 \\ &= (2x + 3)(3x - 1)^2 \end{aligned}$$

$$\begin{aligned} (10) \quad & 2x^3 + 5x^2 - 4x - 3 \\ &= (2x + 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 2x^3 + x^2 - 5x + 2 \\ &= (2x - 1)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 4x^3 - 8x^2 + 5x - 1 \\ &= (2x - 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 - x^2 - 20x + 12 \\ &= (2x - 3)(3x - 2)(x + 2) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^3 + 19x^2 + 33x + 9 \\ &= (3x + 1)(x + 3)^2 \end{aligned}$$

$$\begin{aligned} (3) \quad & 6x^3 + 11x^2 - x - 6 \\ &= (2x + 3)(3x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 + 25x^2 + 23x + 6 \\ &= (2x + 1)(3x + 2)(x + 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & 9x^3 - 18x^2 - x + 2 \\ &= (3x + 1)(3x - 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (9) \quad & 6x^3 + 17x^2 - 5x - 6 \\ &= (2x + 1)(3x - 2)(x + 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 2x^3 + 7x^2 + 7x + 2 \\ &= (2x + 1)(x + 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^3 - 3x^2 + 3x - 1 \\ &= (x - 1)^3 \end{aligned}$$

$$\begin{aligned} (1) \quad & 18x^3 + 15x^2 - x - 2 \\ &= (2x + 1)(3x + 2)(3x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 4x^3 + 8x^2 + x - 3 \\ &= (2x + 3)(2x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 + 13x^2 + 9x + 2 \\ &= (2x + 1)(3x + 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 3x^3 - 4x^2 - 17x + 6 \\ &= (3x - 1)(x + 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 9x^3 + 18x^2 - x - 2 \\ &= (3x + 1)(3x - 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^3 - 2x^2 - 3x + 2 \\ &= (3x - 2)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & x^3 - 3x^2 + 4 \\ &= (x + 1)(x - 2)^2 \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^3 + x^2 - 2x - 1 \\ &= (2x + 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 9x^3 + 9x^2 - 4x - 4 \\ &= (3x + 2)(3x - 2)(x + 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 9x^3 - 6x^2 - 5x + 2 \\ &= (3x + 2)(3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 3x^3 - x^2 - 3x + 1 \\ &= (3x - 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (6) \quad & 4x^3 - 8x^2 - 3x + 9 \\ &= (2x - 3)^2(x + 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & x^3 + 3x^2 + 3x + 1 \\ &= (x + 1)^3 \end{aligned}$$

$$\begin{aligned} (7) \quad & 6x^3 + 7x^2 - 1 \\ &= (2x + 1)(3x - 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 9x^3 + 18x^2 - x - 2 \\ &= (3x + 1)(3x - 1)(x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 6x^3 + 17x^2 + 14x + 3 \\ &= (2x + 3)(3x + 1)(x + 1) \end{aligned}$$

$$\begin{aligned} (4) \quad & 3x^3 - 7x^2 - 7x + 3 \\ &= (3x - 1)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^3 + 14x^2 + 17x + 6 \\ &= (3x + 2)(x + 1)(x + 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^3 + x^2 - 3x - 1 \\ &= (3x + 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^3 + x^2 - x - 1 \\ &= (x + 1)^2(x - 1) \end{aligned}$$

$$\begin{aligned} (1) \quad & x^3 - 2x^2 - x + 2 \\ &= (x+1)(x-1)(x-2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 - 3x^2 - 2x + 3 \\ &= (2x-3)(x+1)(x-1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 12x^3 - 8x^2 - 13x - 3 \\ &= (2x+1)(2x-3)(3x+1) \end{aligned}$$

$$\begin{aligned} (7) \quad & x^3 - 2x^2 - 5x + 6 \\ &= (x+2)(x-1)(x-3) \end{aligned}$$

$$\begin{aligned} (3) \quad & 4x^3 + 16x^2 + 19x + 6 \\ &= (2x+1)(2x+3)(x+2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 4x^3 - 8x^2 + x + 3 \\ &= (2x+1)(2x-3)(x-1) \end{aligned}$$

$$\begin{aligned} (4) \quad & x^3 - x^2 - 9x + 9 \\ &= (x+3)(x-1)(x-3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 2x^3 + x^2 - 18x - 9 \\ &= (2x+1)(x+3)(x-3) \end{aligned}$$

$$\begin{aligned} (5) \quad & 3x^3 + x^2 - 3x - 1 \\ &= (3x+1)(x+1)(x-1) \end{aligned}$$

$$\begin{aligned} (10) \quad & x^3 + x^2 - x - 1 \\ &= (x+1)^2(x-1) \end{aligned}$$

$$\begin{aligned} (1) \quad & 2x^3 + x^2 - 7x - 6 \\ &= (2x + 3)(x + 1)(x - 2) \end{aligned}$$

$$\begin{aligned} (6) \quad & 2x^3 - x^2 - 7x + 6 \\ &= (2x - 3)(x + 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 6x^3 - x^2 - 11x + 6 \\ &= (2x + 3)(3x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (7) \quad & 12x^3 - 8x^2 - 13x - 3 \\ &= (2x + 1)(2x - 3)(3x + 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 9x^3 - 18x^2 - 4x + 8 \\ &= (3x + 2)(3x - 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 9x^3 - 27x^2 - 4x + 12 \\ &= (3x + 2)(3x - 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (4) \quad & x^3 + x^2 - 9x - 9 \\ &= (x + 1)(x + 3)(x - 3) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^3 - 13x^2 + 8x + 12 \\ &= (3x + 2)(x - 2)(x - 3) \end{aligned}$$

$$\begin{aligned} (5) \quad & x^3 + x^2 - 4x - 4 \\ &= (x + 1)(x + 2)(x - 2) \end{aligned}$$

$$\begin{aligned} (10) \quad & 3x^3 - 11x^2 + 5x + 3 \\ &= (3x + 1)(x - 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (1) \quad & 2x^3 - x^2 - 4x + 3 \\ &= (2x + 3)(x - 1)^2 \end{aligned}$$

$$\begin{aligned} (6) \quad & x^3 + 3x^2 - x - 3 \\ &= (x + 1)(x + 3)(x - 1) \end{aligned}$$

$$\begin{aligned} (2) \quad & 3x^3 - 5x^2 - 11x - 3 \\ &= (3x + 1)(x + 1)(x - 3) \end{aligned}$$

$$\begin{aligned} (7) \quad & 9x^3 + 3x^2 - 5x + 1 \\ &= (3x - 1)^2(x + 1) \end{aligned}$$

$$\begin{aligned} (3) \quad & 12x^3 - 16x^2 - 7x + 6 \\ &= (2x - 1)(2x - 3)(3x + 2) \end{aligned}$$

$$\begin{aligned} (8) \quad & 3x^3 + 7x^2 + 5x + 1 \\ &= (3x + 1)(x + 1)^2 \end{aligned}$$

$$\begin{aligned} (4) \quad & 9x^3 - 9x^2 - x + 1 \\ &= (3x + 1)(3x - 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (9) \quad & 3x^3 + x^2 - 3x - 1 \\ &= (3x + 1)(x + 1)(x - 1) \end{aligned}$$

$$\begin{aligned} (5) \quad & 9x^3 - 9x^2 - 4x + 4 \\ &= (3x + 2)(3x - 2)(x - 1) \end{aligned}$$

$$\begin{aligned} (10) \quad & 4x^3 - 7x - 3 \\ &= (2x + 1)(2x - 3)(x + 1) \end{aligned}$$



## 2.6 平方完成 (易しい)

2 次多項式を平方完成する問題。  
式に現れる数は整数となる。

(1)  $2x^2 - 8x + 4$

(11)  $-4x^2 - 40x - 98$

(21)  $-4x^2 - 8x - 6$

(2)  $x^2 - 8x + 19$

(12)  $-5x^2 + 50x - 129$

(22)  $-2x^2 + 20x - 52$

(3)  $x^2 + 6x + 11$

(13)  $-5x^2 - 40x - 85$

(23)  $4x^2 - 40x + 101$

(4)  $-2x^2 - 20x - 45$

(14)  $5x^2 - 30x + 45$

(24)  $-x^2 - 6x - 8$

(5)  $3x^2 + 18x + 22$

(15)  $5x^2 + 50x + 122$

(25)  $-4x^2 + 32x - 65$

(6)  $2x^2 + 16x + 29$

(16)  $-5x^2 - 30x - 43$

(26)  $-5x^2 - 30x - 45$

(7)  $5x^2 + 50x + 130$

(17)  $5x^2 + 30x + 43$

(27)  $2x^2 + 4x + 6$

(8)  $5x^2 + 30x + 48$

(18)  $x^2 - 8x + 14$

(28)  $x^2 - 4x + 4$

(9)  $-2x^2 + 16x - 35$

(19)  $-4x^2 - 16x - 17$

(29)  $2x^2 - 8x + 10$

(10)  $x^2 + 4x + 8$

(20)  $-5x^2 + 10x - 8$

(30)  $-5x^2 - 30x - 46$

(1)  $x^2 - 6x + 8$

(11)  $3x^2 - 6x + 3$

(21)  $2x^2 - 12x + 22$

(2)  $4x^2 + 24x + 36$

(12)  $3x^2 + 24x + 48$

(22)  $-4x^2 - 16x - 20$

(3)  $5x^2 + 30x + 49$

(13)  $-x^2 - 10x - 28$

(23)  $-3x^2 + 12x - 15$

(4)  $4x^2 + 40x + 104$

(14)  $5x^2 + 20x + 21$

(24)  $2x^2 - 20x + 54$

(5)  $5x^2 + 30x + 47$

(15)  $-5x^2 - 50x - 130$

(25)  $-5x^2 + 50x - 123$

(6)  $2x^2 + 4x - 3$

(16)  $-2x^2 + 4x + 3$

(26)  $4x^2 - 24x + 34$

(7)  $-5x^2 - 30x - 45$

(17)  $-3x^2 + 30x - 80$

(27)  $3x^2 - 30x + 71$

(8)  $4x^2 - 16x + 18$

(18)  $2x^2 - 4x + 1$

(28)  $2x^2 + 8x + 8$

(9)  $-4x^2 + 8x + 1$

(19)  $4x^2 + 8x + 6$

(29)  $-2x^2 + 8x - 3$

(10)  $2x^2 + 20x + 53$

(20)  $x^2 - 4x + 4$

(30)  $2x^2 + 4x - 3$

(1)  $x^2 - 10x + 21$

(11)  $5x^2 + 10x + 1$

(21)  $-4x^2 + 40x - 95$

(2)  $2x^2 - 4x + 1$

(12)  $4x^2 - 8x + 1$

(22)  $-x^2 + 6x - 6$

(3)  $-4x^2 + 24x - 41$

(13)  $3x^2 + 18x + 27$

(23)  $2x^2 - 8x + 5$

(4)  $4x^2 + 16x + 14$

(14)  $3x^2 + 24x + 43$

(24)  $3x^2 + 6x - 1$

(5)  $x^2 + 10x + 30$

(15)  $2x^2 + 16x + 27$

(25)  $-5x^2 + 30x - 47$

(6)  $x^2 - 10x + 24$

(16)  $-3x^2 - 6x - 3$

(26)  $-x^2 - 6x - 10$

(7)  $-x^2 + 6x - 7$

(17)  $-5x^2 - 30x - 46$

(27)  $-2x^2 + 4x - 5$

(8)  $x^2 + 2x + 1$

(18)  $-4x^2 - 8x - 9$

(28)  $4x^2 + 8x + 2$

(9)  $4x^2 + 24x + 33$

(19)  $4x^2 + 8x + 4$

(29)  $-x^2 - 10x - 28$

(10)  $-3x^2 + 18x - 23$

(20)  $-3x^2 + 12x - 10$

(30)  $-2x^2 - 12x - 18$

(1)  $x^2 + 6x + 5$

(11)  $-x^2 + 6x - 5$

(21)  $5x^2 - 30x + 49$

(2)  $5x^2 - 40x + 83$

(12)  $-2x^2 - 4x + 2$

(22)  $x^2 + 2x - 2$

(3)  $-3x^2 + 6x - 1$

(13)  $x^2 - 4x + 7$

(23)  $3x^2 - 6x + 7$

(4)  $-3x^2 - 6x + 2$

(14)  $-2x^2 + 16x - 35$

(24)  $-5x^2 - 10x - 4$

(5)  $-4x^2 + 8x - 3$

(15)  $5x^2 + 50x + 125$

(25)  $x^2 + 6x + 13$

(6)  $2x^2 - 8x + 6$

(16)  $4x^2 - 32x + 61$

(26)  $-4x^2 + 32x - 65$

(7)  $-3x^2 + 18x - 31$

(17)  $x^2 - 10x + 20$

(27)  $-2x^2 + 4x - 6$

(8)  $-5x^2 + 30x - 44$

(18)  $-2x^2 + 4x - 3$

(28)  $-4x^2 - 8x$

(9)  $-5x^2 - 40x - 80$

(19)  $5x^2 + 10x + 9$

(29)  $-x^2 - 10x - 20$

(10)  $x^2 - 4x + 2$

(20)  $x^2 - 2x + 3$

(30)  $3x^2 - 18x + 26$

(1)  $-2x^2 + 4x - 3$

(11)  $5x^2 + 30x + 42$

(21)  $-5x^2 - 40x - 75$

(2)  $-3x^2 - 6x - 8$

(12)  $-4x^2 - 16x - 17$

(22)  $-2x^2 + 4x - 5$

(3)  $-3x^2 - 6x$

(13)  $5x^2 - 40x + 82$

(23)  $3x^2 - 24x + 51$

(4)  $-4x^2 + 16x - 18$

(14)  $5x^2 + 20x + 16$

(24)  $3x^2 - 30x + 70$

(5)  $-4x^2 - 32x - 62$

(15)  $3x^2 - 12x + 13$

(25)  $3x^2 + 6x + 5$

(6)  $x^2 + 2x - 3$

(16)  $x^2 + 8x + 17$

(26)  $-5x^2 - 50x - 125$

(7)  $-x^2 + 2x + 3$

(17)  $5x^2 + 40x + 81$

(27)  $-5x^2 - 20x - 21$

(8)  $-3x^2 - 24x - 43$

(18)  $-5x^2 - 30x - 40$

(28)  $2x^2 + 16x + 36$

(9)  $-2x^2 - 4x + 3$

(19)  $3x^2 - 24x + 53$

(29)  $-2x^2 - 12x - 18$

(10)  $-2x^2 + 16x - 34$

(20)  $3x^2 + 24x + 45$

(30)  $-5x^2 - 40x - 81$

(1)  $-4x^2 + 24x - 32$

(11)  $4x^2 + 40x + 105$

(21)  $-3x^2 - 18x - 24$

(2)  $-2x^2 - 4x - 6$

(12)  $5x^2 + 30x + 49$

(22)  $-5x^2 - 20x - 18$

(3)  $2x^2 + 12x + 13$

(13)  $-4x^2 + 40x - 104$

(23)  $-4x^2 + 40x - 97$

(4)  $4x^2 + 40x + 104$

(14)  $-4x^2 - 40x - 103$

(24)  $-5x^2 + 50x - 129$

(5)  $3x^2 - 24x + 45$

(15)  $5x^2 - 40x + 80$

(25)  $-5x^2 + 50x - 123$

(6)  $-x^2 + 6x - 8$

(16)  $5x^2 + 20x + 24$

(26)  $-5x^2 + 30x - 41$

(7)  $x^2 - 4x + 6$

(17)  $-2x^2 - 16x - 28$

(27)  $2x^2 + 8x + 7$

(8)  $-5x^2 + 50x - 124$

(18)  $3x^2 + 18x + 32$

(28)  $-5x^2 + 30x - 46$

(9)  $-x^2 + 2x - 1$

(19)  $5x^2 + 20x + 16$

(29)  $-x^2 + 6x - 6$

(10)  $-3x^2 - 12x - 9$

(20)  $-5x^2 - 10x$

(30)  $x^2 + 4x + 1$

(1)  $3x^2 + 12x + 10$

(11)  $-2x^2 + 12x - 21$

(21)  $-2x^2 + 16x - 31$

(2)  $4x^2 + 40x + 104$

(12)  $3x^2 - 6x + 4$

(22)  $x^2 - 2x - 1$

(3)  $-4x^2 + 40x - 105$

(13)  $4x^2 + 8x + 4$

(23)  $-x^2 + 4x - 5$

(4)  $-2x^2 - 12x - 18$

(14)  $3x^2 + 12x + 16$

(24)  $-3x^2 - 18x - 29$

(5)  $-x^2 - 10x - 30$

(15)  $2x^2 + 4x + 6$

(25)  $4x^2 - 24x + 38$

(6)  $3x^2 + 30x + 75$

(16)  $3x^2 - 6x + 2$

(26)  $-x^2 + 10x - 23$

(7)  $-x^2 + 10x - 25$

(17)  $-4x^2 - 16x - 11$

(27)  $x^2 + 6x + 12$

(8)  $4x^2 + 24x + 38$

(18)  $x^2 - 6x + 13$

(28)  $x^2 - 6x + 11$

(9)  $-2x^2 - 4x - 5$

(19)  $2x^2 - 8x + 6$

(29)  $3x^2 - 18x + 28$

(10)  $-x^2 - 2x + 3$

(20)  $4x^2 - 32x + 67$

(30)  $3x^2 - 30x + 80$



(1)  $x^2 - 10x + 26$

(11)  $-4x^2 - 8x - 1$

(21)  $x^2 - 6x + 12$

(2)  $x^2 + 4x + 2$

(12)  $5x^2 + 10x + 7$

(22)  $x^2 + 6x + 9$

(3)  $-4x^2 + 32x - 61$

(13)  $-2x^2 + 20x - 46$

(23)  $-x^2 - 4x - 1$

(4)  $4x^2 + 8x + 9$

(14)  $-x^2 + 4x - 2$

(24)  $4x^2 + 8x + 3$

(5)  $-2x^2 + 8x - 3$

(15)  $-3x^2 + 30x - 72$

(25)  $5x^2 + 40x + 76$

(6)  $3x^2 + 6x - 2$

(16)  $5x^2 + 20x + 21$

(26)  $x^2 - 4x + 8$

(7)  $x^2 + 10x + 23$

(17)  $2x^2 - 20x + 47$

(27)  $4x^2 + 32x + 62$

(8)  $-4x^2 + 40x - 100$

(18)  $-2x^2 - 16x - 32$

(28)  $-5x^2 - 10x - 2$

(9)  $x^2 - 6x + 6$

(19)  $x^2 + 6x + 14$

(29)  $5x^2 + 50x + 129$

(10)  $x^2 + 8x + 16$

(20)  $-x^2 - 8x - 15$

(30)  $x^2 - 2x + 3$

(1)  $4x^2 + 40x + 103$

(11)  $-3x^2 - 24x - 45$

(21)  $-x^2 + 4x + 1$

(2)  $3x^2 - 24x + 47$

(12)  $-x^2 - 6x - 9$

(22)  $-3x^2 + 12x - 10$

(3)  $3x^2 + 30x + 76$

(13)  $-2x^2 + 16x - 33$

(23)  $x^2 - 4x + 7$

(4)  $2x^2 - 4x + 1$

(14)  $-x^2 - 10x - 28$

(24)  $4x^2 + 32x + 62$

(5)  $-x^2 + 8x - 14$

(15)  $3x^2 + 12x + 7$

(25)  $-4x^2 - 40x - 101$

(6)  $4x^2 - 24x + 40$

(16)  $3x^2 + 24x + 43$

(26)  $5x^2 + 40x + 82$

(7)  $x^2 - 10x + 26$

(17)  $3x^2 + 18x + 32$

(27)  $-5x^2 - 40x - 82$

(8)  $x^2 - 2x + 5$

(18)  $5x^2 + 40x + 80$

(28)  $-5x^2 + 10x - 6$

(9)  $-x^2 - 8x - 18$

(19)  $4x^2 + 8x + 3$

(29)  $x^2 - 2x + 5$

(10)  $x^2 + 10x + 30$

(20)  $4x^2 + 8x + 9$

(30)  $-4x^2 + 32x - 62$

(1)  $-x^2 - 8x - 18$

(11)  $-5x^2 - 10x - 5$

(21)  $2x^2 + 20x + 46$

(2)  $-5x^2 + 40x - 76$

(12)  $-3x^2 - 6x - 4$

(22)  $4x^2 + 40x + 96$

(3)  $-3x^2 + 24x - 46$

(13)  $-5x^2 + 30x - 43$

(23)  $x^2 - 6x + 7$

(4)  $-4x^2 + 16x - 11$

(14)  $5x^2 - 50x + 124$

(24)  $3x^2 - 18x + 29$

(5)  $4x^2 + 8x$

(15)  $-4x^2 - 24x - 39$

(25)  $-4x^2 - 8x - 5$

(6)  $x^2 - 6x + 4$

(16)  $x^2 + 10x + 22$

(26)  $-3x^2 - 30x - 77$

(7)  $-x^2 - 4x + 1$

(17)  $-2x^2 - 8x - 11$

(27)  $5x^2 + 40x + 84$

(8)  $-3x^2 + 30x - 73$

(18)  $-2x^2 - 8x - 5$

(28)  $-x^2 + 10x - 24$

(9)  $-x^2 + 4x - 3$

(19)  $-5x^2 + 30x - 45$

(29)  $5x^2 + 30x + 45$

(10)  $-x^2 + 6x - 14$

(20)  $4x^2 - 32x + 60$

(30)  $4x^2 + 24x + 33$

(1)  $4x^2 + 16x + 17$

(11)  $3x^2 - 24x + 43$

(21)  $-3x^2 - 12x - 7$

(2)  $-5x^2 + 30x - 49$

(12)  $-x^2 + 4x - 3$

(22)  $3x^2 - 30x + 70$

(3)  $5x^2 + 40x + 78$

(13)  $-3x^2 - 12x - 17$

(23)  $-x^2 + 6x - 14$

(4)  $-4x^2 + 24x - 39$

(14)  $-x^2 - 10x - 22$

(24)  $-5x^2 - 20x - 22$

(5)  $5x^2 + 20x + 22$

(15)  $-5x^2 + 40x - 82$

(25)  $-x^2 + 6x - 9$

(6)  $5x^2 + 20x + 24$

(16)  $-x^2 + 8x - 17$

(26)  $-3x^2 - 30x - 71$

(7)  $2x^2 - 4x - 3$

(17)  $-3x^2 + 6x - 8$

(27)  $-2x^2 - 12x - 15$

(8)  $2x^2 - 12x + 16$

(18)  $5x^2 - 10x + 4$

(28)  $-4x^2 - 24x - 35$

(9)  $5x^2 + 30x + 45$

(19)  $-2x^2 - 4x + 1$

(29)  $4x^2 - 40x + 104$

(10)  $2x^2 - 16x + 36$

(20)  $-3x^2 - 24x - 45$

(30)  $-5x^2 - 40x - 79$

(1)  $-3x^2 - 24x - 52$

(11)  $x^2 + 4x - 1$

(21)  $5x^2 + 10x + 3$

(2)  $3x^2 + 6x - 2$

(12)  $-2x^2 + 16x - 33$

(22)  $-2x^2 + 12x - 21$

(3)  $-4x^2 + 40x - 104$

(13)  $4x^2 - 40x + 100$

(23)  $2x^2 + 8x + 12$

(4)  $3x^2 + 6x + 7$

(14)  $x^2 - 4x + 2$

(24)  $x^2 + 2x + 2$

(5)  $-4x^2 + 16x - 16$

(15)  $-4x^2 + 32x - 59$

(25)  $x^2 - 8x + 18$

(6)  $-x^2 - 2x + 1$

(16)  $-x^2 - 2x + 2$

(26)  $x^2 + 10x + 23$

(7)  $-4x^2 + 8x - 1$

(17)  $5x^2 + 40x + 85$

(27)  $-5x^2 + 10x - 8$

(8)  $-5x^2 + 50x - 127$

(18)  $4x^2 - 32x + 69$

(28)  $4x^2 - 32x + 64$

(9)  $-2x^2 - 16x - 27$

(19)  $-5x^2 - 30x - 49$

(29)  $-2x^2 + 20x - 55$

(10)  $-2x^2 + 20x - 46$

(20)  $-5x^2 - 50x - 122$

(30)  $-4x^2 + 40x - 95$

(1)  $x^2 + 2x + 5$

(11)  $5x^2 - 20x + 17$

(21)  $2x^2 + 16x + 31$

(2)  $-3x^2 - 24x - 49$

(12)  $2x^2 - 12x + 15$

(22)  $x^2 - 8x + 18$

(3)  $4x^2 - 16x + 12$

(13)  $3x^2 - 24x + 45$

(23)  $-2x^2 - 4x + 3$

(4)  $-3x^2 + 24x - 47$

(14)  $-2x^2 + 12x - 20$

(24)  $3x^2 - 30x + 74$

(5)  $5x^2 + 20x + 24$

(15)  $-3x^2 + 24x - 47$

(25)  $x^2 + 2x + 5$

(6)  $x^2 + 6x + 11$

(16)  $-3x^2 - 30x - 71$

(26)  $-3x^2 - 30x - 77$

(7)  $-4x^2 - 32x - 68$

(17)  $-x^2 - 2x$

(27)  $x^2 + 6x + 4$

(8)  $4x^2 + 16x + 16$

(18)  $x^2 + 4x - 1$

(28)  $-x^2 + 10x - 22$

(9)  $-4x^2 - 32x - 65$

(19)  $2x^2 + 20x + 48$

(29)  $4x^2 - 24x + 34$

(10)  $2x^2 - 8x + 9$

(20)  $-5x^2 + 10x - 6$

(30)  $-5x^2 - 30x - 49$

(1)  $x^2 + 10x + 24$

(11)  $4x^2 - 24x + 36$

(21)  $-4x^2 + 32x - 60$

(2)  $-5x^2 - 10x$

(12)  $-5x^2 + 50x - 122$

(22)  $-x^2 + 2x + 1$

(3)  $4x^2 - 16x + 15$

(13)  $-2x^2 + 8x - 3$

(23)  $2x^2 - 16x + 31$

(4)  $4x^2 + 32x + 68$

(14)  $x^2 + 2x - 4$

(24)  $-2x^2 - 12x - 23$

(5)  $5x^2 + 10x + 2$

(15)  $-x^2 - 2x + 3$

(25)  $2x^2 - 12x + 16$

(6)  $3x^2 + 12x + 10$

(16)  $4x^2 + 32x + 69$

(26)  $5x^2 + 30x + 50$

(7)  $-5x^2 - 20x - 16$

(17)  $2x^2 + 4x$

(27)  $-5x^2 - 40x - 75$

(8)  $-x^2 + 2x - 1$

(18)  $-3x^2 + 12x - 9$

(28)  $3x^2 - 24x + 50$

(9)  $3x^2 + 18x + 25$

(19)  $-3x^2 + 12x - 8$

(29)  $3x^2 - 18x + 23$

(10)  $-3x^2 - 12x - 17$

(20)  $2x^2 + 4x - 3$

(30)  $x^2 - 6x + 12$

(1)  $5x^2 + 20x + 25$

(11)  $-x^2 - 6x - 11$

(21)  $-2x^2 + 20x - 55$

(2)  $-x^2 + 6x - 11$

(12)  $x^2 + 6x + 8$

(22)  $5x^2 + 20x + 25$

(3)  $5x^2 + 30x + 46$

(13)  $-4x^2 - 8x - 2$

(23)  $2x^2 + 20x + 49$

(4)  $-x^2 + 4x - 8$

(14)  $3x^2 - 12x + 13$

(24)  $4x^2 - 32x + 61$

(5)  $-x^2 + 2x + 3$

(15)  $-4x^2 - 8x - 3$

(25)  $-3x^2 + 12x - 16$

(6)  $-4x^2 + 24x - 35$

(16)  $-4x^2 + 40x - 98$

(26)  $-5x^2 + 10x - 4$

(7)  $5x^2 + 30x + 41$

(17)  $-4x^2 - 16x - 11$

(27)  $-5x^2 - 50x - 129$

(8)  $-4x^2 - 32x - 60$

(18)  $-4x^2 + 24x - 38$

(28)  $-2x^2 + 16x - 37$

(9)  $-2x^2 - 16x - 34$

(19)  $-2x^2 - 20x - 54$

(29)  $2x^2 + 16x + 31$

(10)  $2x^2 - 16x + 32$

(20)  $4x^2 - 32x + 66$

(30)  $-5x^2 + 40x - 83$



(1)  $5x^2 - 20x + 15$

(11)  $-4x^2 - 16x - 12$

(21)  $5x^2 + 50x + 124$

(2)  $5x^2 + 10x + 2$

(12)  $-3x^2 - 30x - 74$

(22)  $3x^2 + 6x + 6$

(3)  $3x^2 - 18x + 26$

(13)  $-x^2 - 4x - 2$

(23)  $4x^2 - 24x + 36$

(4)  $2x^2 + 12x + 23$

(14)  $2x^2 - 12x + 14$

(24)  $-3x^2 - 6x + 1$

(5)  $-x^2 + 10x - 30$

(15)  $4x^2 - 16x + 17$

(25)  $4x^2 + 24x + 36$

(6)  $-5x^2 + 50x - 126$

(16)  $-5x^2 - 10x - 8$

(26)  $-2x^2 + 4x - 3$

(7)  $-4x^2 - 32x - 60$

(17)  $2x^2 - 12x + 19$

(27)  $-2x^2 + 4x - 5$

(8)  $-x^2 + 10x - 23$

(18)  $2x^2 - 8x + 12$

(28)  $4x^2 + 32x + 64$

(9)  $3x^2 - 6x - 2$

(19)  $3x^2 + 30x + 73$

(29)  $3x^2 + 24x + 50$

(10)  $4x^2 + 32x + 63$

(20)  $-x^2 + 10x - 25$

(30)  $4x^2 + 24x + 39$

(1)  $5x^2 + 40x + 77$

(11)  $-2x^2 + 20x - 48$

(21)  $-4x^2 + 40x - 104$

(2)  $-5x^2 + 40x - 78$

(12)  $5x^2 - 50x + 128$

(22)  $2x^2 - 16x + 37$

(3)  $-3x^2 + 6x - 3$

(13)  $-4x^2 + 16x - 20$

(23)  $4x^2 - 24x + 32$

(4)  $5x^2 - 50x + 130$

(14)  $-4x^2 - 16x - 12$

(24)  $-5x^2 - 20x - 18$

(5)  $2x^2 + 20x + 49$

(15)  $2x^2 + 8x + 4$

(25)  $-x^2 + 6x - 9$

(6)  $-3x^2 - 12x - 8$

(16)  $-4x^2 + 40x - 98$

(26)  $x^2 - 4x + 1$

(7)  $5x^2 + 40x + 75$

(17)  $3x^2 + 6x$

(27)  $-x^2 - 10x - 30$

(8)  $2x^2 + 4x - 2$

(18)  $-3x^2 - 12x - 9$

(28)  $-x^2 + 10x - 26$

(9)  $3x^2 + 12x + 9$

(19)  $-2x^2 - 12x - 14$

(29)  $-4x^2 + 32x - 60$

(10)  $-5x^2 + 50x - 129$

(20)  $3x^2 + 24x + 47$

(30)  $x^2 + 2x - 1$

(1)  $-4x^2 - 40x - 101$

(11)  $-3x^2 + 18x - 30$

(21)  $3x^2 - 30x + 73$

(2)  $-4x^2 + 32x - 64$

(12)  $-2x^2 - 16x - 31$

(22)  $-x^2 - 8x - 11$

(3)  $-5x^2 + 30x - 47$

(13)  $-x^2 + 6x - 9$

(23)  $-2x^2 + 16x - 32$

(4)  $3x^2 + 12x + 14$

(14)  $x^2 - 10x + 23$

(24)  $4x^2 - 40x + 98$

(5)  $x^2 + 8x + 16$

(15)  $-4x^2 + 8x$

(25)  $-x^2 - 6x - 8$

(6)  $-5x^2 + 40x - 75$

(16)  $-x^2 + 6x - 5$

(26)  $4x^2 + 40x + 98$

(7)  $-x^2 - 6x - 5$

(17)  $x^2 + 2x + 4$

(27)  $-x^2 - 10x - 27$

(8)  $-5x^2 + 10x - 1$

(18)  $5x^2 - 30x + 46$

(28)  $x^2 - 2x - 1$

(9)  $-5x^2 + 30x - 47$

(19)  $5x^2 - 40x + 79$

(29)  $-5x^2 + 30x - 50$

(10)  $-5x^2 - 30x - 48$

(20)  $x^2 - 6x + 10$

(30)  $5x^2 + 20x + 22$

(1)  $-2x^2 - 16x - 27$

(11)  $-5x^2 + 20x - 15$

(21)  $3x^2 - 12x + 17$

(2)  $2x^2 - 8x + 6$

(12)  $2x^2 - 8x + 3$

(22)  $x^2 + 2x + 2$

(3)  $-x^2 - 8x - 11$

(13)  $-2x^2 - 16x - 35$

(23)  $x^2 + 8x + 15$

(4)  $2x^2 + 4x + 4$

(14)  $-4x^2 - 40x - 105$

(24)  $-x^2 - 6x - 6$

(5)  $2x^2 - 8x + 6$

(15)  $-3x^2 + 30x - 76$

(25)  $5x^2 + 40x + 84$

(6)  $-x^2 - 2x - 5$

(16)  $-4x^2 + 16x - 17$

(26)  $-5x^2 - 50x - 130$

(7)  $-3x^2 - 18x - 24$

(17)  $4x^2 - 24x + 31$

(27)  $-4x^2 - 24x - 32$

(8)  $x^2 + 10x + 23$

(18)  $-x^2 + 8x - 11$

(28)  $5x^2 - 30x + 46$

(9)  $-5x^2 - 20x - 17$

(19)  $2x^2 + 4x + 2$

(29)  $x^2 + 4x + 6$

(10)  $2x^2 + 4x + 2$

(20)  $3x^2 - 30x + 78$

(30)  $2x^2 - 8x + 5$

(1)  $-x^2 + 6x - 8$

(11)  $-5x^2 + 20x - 19$

(21)  $-4x^2 - 32x - 69$

(2)  $x^2 + 10x + 30$

(12)  $3x^2 - 24x + 52$

(22)  $3x^2 + 24x + 47$

(3)  $-3x^2 - 12x - 12$

(13)  $4x^2 + 24x + 35$

(23)  $x^2 + 10x + 27$

(4)  $4x^2 + 8x - 1$

(14)  $-4x^2 - 16x - 15$

(24)  $x^2 - 6x + 10$

(5)  $-2x^2 + 12x - 20$

(15)  $-2x^2 - 8x - 5$

(25)  $-3x^2 + 18x - 32$

(6)  $4x^2 + 24x + 41$

(16)  $-3x^2 - 18x - 25$

(26)  $5x^2 - 50x + 123$

(7)  $-3x^2 + 24x - 46$

(17)  $2x^2 + 12x + 20$

(27)  $-3x^2 + 12x - 12$

(8)  $2x^2 - 20x + 52$

(18)  $4x^2 + 40x + 95$

(28)  $4x^2 - 8x + 8$

(9)  $-3x^2 - 30x - 72$

(19)  $-5x^2 - 40x - 84$

(29)  $-3x^2 - 30x - 70$

(10)  $3x^2 - 18x + 30$

(20)  $x^2 - 6x + 14$

(30)  $3x^2 - 6x + 1$

$$\begin{aligned}(1) \quad & 2x^2 - 8x + 4 \\ &= 2(x-2)^2 - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & -4x^2 - 40x - 98 \\ &= -4(x+5)^2 + 2\end{aligned}$$

$$\begin{aligned}(21) \quad & -4x^2 - 8x - 6 \\ &= -4(x+1)^2 - 2\end{aligned}$$

$$\begin{aligned}(2) \quad & x^2 - 8x + 19 \\ &= (x-4)^2 + 3\end{aligned}$$

$$\begin{aligned}(12) \quad & -5x^2 + 50x - 129 \\ &= -5(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(22) \quad & -2x^2 + 20x - 52 \\ &= -2(x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(3) \quad & x^2 + 6x + 11 \\ &= (x+3)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & -5x^2 - 40x - 85 \\ &= -5(x+4)^2 - 5\end{aligned}$$

$$\begin{aligned}(23) \quad & 4x^2 - 40x + 101 \\ &= 4(x-5)^2 + 1\end{aligned}$$

$$\begin{aligned}(4) \quad & -2x^2 - 20x - 45 \\ &= -2(x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & 5x^2 - 30x + 45 \\ &= 5(x-3)^2\end{aligned}$$

$$\begin{aligned}(24) \quad & -x^2 - 6x - 8 \\ &= -(x+3)^2 + 1\end{aligned}$$

$$\begin{aligned}(5) \quad & 3x^2 + 18x + 22 \\ &= 3(x+3)^2 - 5\end{aligned}$$

$$\begin{aligned}(15) \quad & 5x^2 + 50x + 122 \\ &= 5(x+5)^2 - 3\end{aligned}$$

$$\begin{aligned}(25) \quad & -4x^2 + 32x - 65 \\ &= -4(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(6) \quad & 2x^2 + 16x + 29 \\ &= 2(x+4)^2 - 3\end{aligned}$$

$$\begin{aligned}(16) \quad & -5x^2 - 30x - 43 \\ &= -5(x+3)^2 + 2\end{aligned}$$

$$\begin{aligned}(26) \quad & -5x^2 - 30x - 45 \\ &= -5(x+3)^2\end{aligned}$$

$$\begin{aligned}(7) \quad & 5x^2 + 50x + 130 \\ &= 5(x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(17) \quad & 5x^2 + 30x + 43 \\ &= 5(x+3)^2 - 2\end{aligned}$$

$$\begin{aligned}(27) \quad & 2x^2 + 4x + 6 \\ &= 2(x+1)^2 + 4\end{aligned}$$

$$\begin{aligned}(8) \quad & 5x^2 + 30x + 48 \\ &= 5(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(18) \quad & x^2 - 8x + 14 \\ &= (x-4)^2 - 2\end{aligned}$$

$$\begin{aligned}(28) \quad & x^2 - 4x + 4 \\ &= (x-2)^2\end{aligned}$$

$$\begin{aligned}(9) \quad & -2x^2 + 16x - 35 \\ &= -2(x-4)^2 - 3\end{aligned}$$

$$\begin{aligned}(19) \quad & -4x^2 - 16x - 17 \\ &= -4(x+2)^2 - 1\end{aligned}$$

$$\begin{aligned}(29) \quad & 2x^2 - 8x + 10 \\ &= 2(x-2)^2 + 2\end{aligned}$$

$$\begin{aligned}(10) \quad & x^2 + 4x + 8 \\ &= (x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(20) \quad & -5x^2 + 10x - 8 \\ &= -5(x-1)^2 - 3\end{aligned}$$

$$\begin{aligned}(30) \quad & -5x^2 - 30x - 46 \\ &= -5(x+3)^2 - 1\end{aligned}$$

$$\begin{aligned}(1) \quad & x^2 - 6x + 8 \\ &= (x - 3)^2 - 1\end{aligned}$$

$$\begin{aligned}(11) \quad & 3x^2 - 6x + 3 \\ &= 3(x - 1)^2\end{aligned}$$

$$\begin{aligned}(21) \quad & 2x^2 - 12x + 22 \\ &= 2(x - 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(2) \quad & 4x^2 + 24x + 36 \\ &= 4(x + 3)^2\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 + 24x + 48 \\ &= 3(x + 4)^2\end{aligned}$$

$$\begin{aligned}(22) \quad & -4x^2 - 16x - 20 \\ &= -4(x + 2)^2 - 4\end{aligned}$$

$$\begin{aligned}(3) \quad & 5x^2 + 30x + 49 \\ &= 5(x + 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 - 10x - 28 \\ &= -(x + 5)^2 - 3\end{aligned}$$

$$\begin{aligned}(23) \quad & -3x^2 + 12x - 15 \\ &= -3(x - 2)^2 - 3\end{aligned}$$

$$\begin{aligned}(4) \quad & 4x^2 + 40x + 104 \\ &= 4(x + 5)^2 + 4\end{aligned}$$

$$\begin{aligned}(14) \quad & 5x^2 + 20x + 21 \\ &= 5(x + 2)^2 + 1\end{aligned}$$

$$\begin{aligned}(24) \quad & 2x^2 - 20x + 54 \\ &= 2(x - 5)^2 + 4\end{aligned}$$

$$\begin{aligned}(5) \quad & 5x^2 + 30x + 47 \\ &= 5(x + 3)^2 + 2\end{aligned}$$

$$\begin{aligned}(15) \quad & -5x^2 - 50x - 130 \\ &= -5(x + 5)^2 - 5\end{aligned}$$

$$\begin{aligned}(25) \quad & -5x^2 + 50x - 123 \\ &= -5(x - 5)^2 + 2\end{aligned}$$

$$\begin{aligned}(6) \quad & 2x^2 + 4x - 3 \\ &= 2(x + 1)^2 - 5\end{aligned}$$

$$\begin{aligned}(16) \quad & -2x^2 + 4x + 3 \\ &= -2(x - 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(26) \quad & 4x^2 - 24x + 34 \\ &= 4(x - 3)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -5x^2 - 30x - 45 \\ &= -5(x + 3)^2\end{aligned}$$

$$\begin{aligned}(17) \quad & -3x^2 + 30x - 80 \\ &= -3(x - 5)^2 - 5\end{aligned}$$

$$\begin{aligned}(27) \quad & 3x^2 - 30x + 71 \\ &= 3(x - 5)^2 - 4\end{aligned}$$

$$\begin{aligned}(8) \quad & 4x^2 - 16x + 18 \\ &= 4(x - 2)^2 + 2\end{aligned}$$

$$\begin{aligned}(18) \quad & 2x^2 - 4x + 1 \\ &= 2(x - 1)^2 - 1\end{aligned}$$

$$\begin{aligned}(28) \quad & 2x^2 + 8x + 8 \\ &= 2(x + 2)^2\end{aligned}$$

$$\begin{aligned}(9) \quad & -4x^2 + 8x + 1 \\ &= -4(x - 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(19) \quad & 4x^2 + 8x + 6 \\ &= 4(x + 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(29) \quad & -2x^2 + 8x - 3 \\ &= -2(x - 2)^2 + 5\end{aligned}$$

$$\begin{aligned}(10) \quad & 2x^2 + 20x + 53 \\ &= 2(x + 5)^2 + 3\end{aligned}$$

$$\begin{aligned}(20) \quad & x^2 - 4x + 4 \\ &= (x - 2)^2\end{aligned}$$

$$\begin{aligned}(30) \quad & 2x^2 + 4x - 3 \\ &= 2(x + 1)^2 - 5\end{aligned}$$

$$\begin{aligned}(1) \quad & x^2 - 10x + 21 \\ &= (x - 5)^2 - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & 5x^2 + 10x + 1 \\ &= 5(x + 1)^2 - 4\end{aligned}$$

$$\begin{aligned}(21) \quad & -4x^2 + 40x - 95 \\ &= -4(x - 5)^2 + 5\end{aligned}$$

$$\begin{aligned}(2) \quad & 2x^2 - 4x + 1 \\ &= 2(x - 1)^2 - 1\end{aligned}$$

$$\begin{aligned}(12) \quad & 4x^2 - 8x + 1 \\ &= 4(x - 1)^2 - 3\end{aligned}$$

$$\begin{aligned}(22) \quad & -x^2 + 6x - 6 \\ &= -(x - 3)^2 + 3\end{aligned}$$

$$\begin{aligned}(3) \quad & -4x^2 + 24x - 41 \\ &= -4(x - 3)^2 - 5\end{aligned}$$

$$\begin{aligned}(13) \quad & 3x^2 + 18x + 27 \\ &= 3(x + 3)^2\end{aligned}$$

$$\begin{aligned}(23) \quad & 2x^2 - 8x + 5 \\ &= 2(x - 2)^2 - 3\end{aligned}$$

$$\begin{aligned}(4) \quad & 4x^2 + 16x + 14 \\ &= 4(x + 2)^2 - 2\end{aligned}$$

$$\begin{aligned}(14) \quad & 3x^2 + 24x + 43 \\ &= 3(x + 4)^2 - 5\end{aligned}$$

$$\begin{aligned}(24) \quad & 3x^2 + 6x - 1 \\ &= 3(x + 1)^2 - 4\end{aligned}$$

$$\begin{aligned}(5) \quad & x^2 + 10x + 30 \\ &= (x + 5)^2 + 5\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 + 16x + 27 \\ &= 2(x + 4)^2 - 5\end{aligned}$$

$$\begin{aligned}(25) \quad & -5x^2 + 30x - 47 \\ &= -5(x - 3)^2 - 2\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 - 10x + 24 \\ &= (x - 5)^2 - 1\end{aligned}$$

$$\begin{aligned}(16) \quad & -3x^2 - 6x - 3 \\ &= -3(x + 1)^2\end{aligned}$$

$$\begin{aligned}(26) \quad & -x^2 - 6x - 10 \\ &= -(x + 3)^2 - 1\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 + 6x - 7 \\ &= -(x - 3)^2 + 2\end{aligned}$$

$$\begin{aligned}(17) \quad & -5x^2 - 30x - 46 \\ &= -5(x + 3)^2 - 1\end{aligned}$$

$$\begin{aligned}(27) \quad & -2x^2 + 4x - 5 \\ &= -2(x - 1)^2 - 3\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 + 2x + 1 \\ &= (x + 1)^2\end{aligned}$$

$$\begin{aligned}(18) \quad & -4x^2 - 8x - 9 \\ &= -4(x + 1)^2 - 5\end{aligned}$$

$$\begin{aligned}(28) \quad & 4x^2 + 8x + 2 \\ &= 4(x + 1)^2 - 2\end{aligned}$$

$$\begin{aligned}(9) \quad & 4x^2 + 24x + 33 \\ &= 4(x + 3)^2 - 3\end{aligned}$$

$$\begin{aligned}(19) \quad & 4x^2 + 8x + 4 \\ &= 4(x + 1)^2\end{aligned}$$

$$\begin{aligned}(29) \quad & -x^2 - 10x - 28 \\ &= -(x + 5)^2 - 3\end{aligned}$$

$$\begin{aligned}(10) \quad & -3x^2 + 18x - 23 \\ &= -3(x - 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(20) \quad & -3x^2 + 12x - 10 \\ &= -3(x - 2)^2 + 2\end{aligned}$$

$$\begin{aligned}(30) \quad & -2x^2 - 12x - 18 \\ &= -2(x + 3)^2\end{aligned}$$



$$\begin{aligned}(1) \quad & x^2 + 6x + 5 \\ &= (x + 3)^2 - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & -x^2 + 6x - 5 \\ &= -(x - 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(21) \quad & 5x^2 - 30x + 49 \\ &= 5(x - 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(2) \quad & 5x^2 - 40x + 83 \\ &= 5(x - 4)^2 + 3\end{aligned}$$

$$\begin{aligned}(12) \quad & -2x^2 - 4x + 2 \\ &= -2(x + 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(22) \quad & x^2 + 2x - 2 \\ &= (x + 1)^2 - 3\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 + 6x - 1 \\ &= -3(x - 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 - 4x + 7 \\ &= (x - 2)^2 + 3\end{aligned}$$

$$\begin{aligned}(23) \quad & 3x^2 - 6x + 7 \\ &= 3(x - 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(4) \quad & -3x^2 - 6x + 2 \\ &= -3(x + 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & -2x^2 + 16x - 35 \\ &= -2(x - 4)^2 - 3\end{aligned}$$

$$\begin{aligned}(24) \quad & -5x^2 - 10x - 4 \\ &= -5(x + 1)^2 + 1\end{aligned}$$

$$\begin{aligned}(5) \quad & -4x^2 + 8x - 3 \\ &= -4(x - 1)^2 + 1\end{aligned}$$

$$\begin{aligned}(15) \quad & 5x^2 + 50x + 125 \\ &= 5(x + 5)^2\end{aligned}$$

$$\begin{aligned}(25) \quad & x^2 + 6x + 13 \\ &= (x + 3)^2 + 4\end{aligned}$$

$$\begin{aligned}(6) \quad & 2x^2 - 8x + 6 \\ &= 2(x - 2)^2 - 2\end{aligned}$$

$$\begin{aligned}(16) \quad & 4x^2 - 32x + 61 \\ &= 4(x - 4)^2 - 3\end{aligned}$$

$$\begin{aligned}(26) \quad & -4x^2 + 32x - 65 \\ &= -4(x - 4)^2 - 1\end{aligned}$$

$$\begin{aligned}(7) \quad & -3x^2 + 18x - 31 \\ &= -3(x - 3)^2 - 4\end{aligned}$$

$$\begin{aligned}(17) \quad & x^2 - 10x + 20 \\ &= (x - 5)^2 - 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -2x^2 + 4x - 6 \\ &= -2(x - 1)^2 - 4\end{aligned}$$

$$\begin{aligned}(8) \quad & -5x^2 + 30x - 44 \\ &= -5(x - 3)^2 + 1\end{aligned}$$

$$\begin{aligned}(18) \quad & -2x^2 + 4x - 3 \\ &= -2(x - 1)^2 - 1\end{aligned}$$

$$\begin{aligned}(28) \quad & -4x^2 - 8x \\ &= -4(x + 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(9) \quad & -5x^2 - 40x - 80 \\ &= -5(x + 4)^2\end{aligned}$$

$$\begin{aligned}(19) \quad & 5x^2 + 10x + 9 \\ &= 5(x + 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(29) \quad & -x^2 - 10x - 20 \\ &= -(x + 5)^2 + 5\end{aligned}$$

$$\begin{aligned}(10) \quad & x^2 - 4x + 2 \\ &= (x - 2)^2 - 2\end{aligned}$$

$$\begin{aligned}(20) \quad & x^2 - 2x + 3 \\ &= (x - 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(30) \quad & 3x^2 - 18x + 26 \\ &= 3(x - 3)^2 - 1\end{aligned}$$

$$\begin{aligned}(1) \quad & -2x^2 + 4x - 3 \\ & = -2(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(11) \quad & 5x^2 + 30x + 42 \\ & = 5(x+3)^2 - 3\end{aligned}$$

$$\begin{aligned}(21) \quad & -5x^2 - 40x - 75 \\ & = -5(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(2) \quad & -3x^2 - 6x - 8 \\ & = -3(x+1)^2 - 5\end{aligned}$$

$$\begin{aligned}(12) \quad & -4x^2 - 16x - 17 \\ & = -4(x+2)^2 - 1\end{aligned}$$

$$\begin{aligned}(22) \quad & -2x^2 + 4x - 5 \\ & = -2(x-1)^2 - 3\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 - 6x \\ & = -3(x+1)^2 + 3\end{aligned}$$

$$\begin{aligned}(13) \quad & 5x^2 - 40x + 82 \\ & = 5(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(23) \quad & 3x^2 - 24x + 51 \\ & = 3(x-4)^2 + 3\end{aligned}$$

$$\begin{aligned}(4) \quad & -4x^2 + 16x - 18 \\ & = -4(x-2)^2 - 2\end{aligned}$$

$$\begin{aligned}(14) \quad & 5x^2 + 20x + 16 \\ & = 5(x+2)^2 - 4\end{aligned}$$

$$\begin{aligned}(24) \quad & 3x^2 - 30x + 70 \\ & = 3(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(5) \quad & -4x^2 - 32x - 62 \\ & = -4(x+4)^2 + 2\end{aligned}$$

$$\begin{aligned}(15) \quad & 3x^2 - 12x + 13 \\ & = 3(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(25) \quad & 3x^2 + 6x + 5 \\ & = 3(x+1)^2 + 2\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 + 2x - 3 \\ & = (x+1)^2 - 4\end{aligned}$$

$$\begin{aligned}(16) \quad & x^2 + 8x + 17 \\ & = (x+4)^2 + 1\end{aligned}$$

$$\begin{aligned}(26) \quad & -5x^2 - 50x - 125 \\ & = -5(x+5)^2\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 + 2x + 3 \\ & = -(x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(17) \quad & 5x^2 + 40x + 81 \\ & = 5(x+4)^2 + 1\end{aligned}$$

$$\begin{aligned}(27) \quad & -5x^2 - 20x - 21 \\ & = -5(x+2)^2 - 1\end{aligned}$$

$$\begin{aligned}(8) \quad & -3x^2 - 24x - 43 \\ & = -3(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(18) \quad & -5x^2 - 30x - 40 \\ & = -5(x+3)^2 + 5\end{aligned}$$

$$\begin{aligned}(28) \quad & 2x^2 + 16x + 36 \\ & = 2(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(9) \quad & -2x^2 - 4x + 3 \\ & = -2(x+1)^2 + 5\end{aligned}$$

$$\begin{aligned}(19) \quad & 3x^2 - 24x + 53 \\ & = 3(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(29) \quad & -2x^2 - 12x - 18 \\ & = -2(x+3)^2\end{aligned}$$

$$\begin{aligned}(10) \quad & -2x^2 + 16x - 34 \\ & = -2(x-4)^2 - 2\end{aligned}$$

$$\begin{aligned}(20) \quad & 3x^2 + 24x + 45 \\ & = 3(x+4)^2 - 3\end{aligned}$$

$$\begin{aligned}(30) \quad & -5x^2 - 40x - 81 \\ & = -5(x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(1) \quad & -4x^2 + 24x - 32 \\ & = -4(x-3)^2 + 4\end{aligned}$$

$$\begin{aligned}(11) \quad & 4x^2 + 40x + 105 \\ & = 4(x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(21) \quad & -3x^2 - 18x - 24 \\ & = -3(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(2) \quad & -2x^2 - 4x - 6 \\ & = -2(x+1)^2 - 4\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 + 30x + 49 \\ & = 5(x+3)^2 + 4\end{aligned}$$

$$\begin{aligned}(22) \quad & -5x^2 - 20x - 18 \\ & = -5(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(3) \quad & 2x^2 + 12x + 13 \\ & = 2(x+3)^2 - 5\end{aligned}$$

$$\begin{aligned}(13) \quad & -4x^2 + 40x - 104 \\ & = -4(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(23) \quad & -4x^2 + 40x - 97 \\ & = -4(x-5)^2 + 3\end{aligned}$$

$$\begin{aligned}(4) \quad & 4x^2 + 40x + 104 \\ & = 4(x+5)^2 + 4\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 - 40x - 103 \\ & = -4(x+5)^2 - 3\end{aligned}$$

$$\begin{aligned}(24) \quad & -5x^2 + 50x - 129 \\ & = -5(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(5) \quad & 3x^2 - 24x + 45 \\ & = 3(x-4)^2 - 3\end{aligned}$$

$$\begin{aligned}(15) \quad & 5x^2 - 40x + 80 \\ & = 5(x-4)^2\end{aligned}$$

$$\begin{aligned}(25) \quad & -5x^2 + 50x - 123 \\ & = -5(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(6) \quad & -x^2 + 6x - 8 \\ & = -(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(16) \quad & 5x^2 + 20x + 24 \\ & = 5(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(26) \quad & -5x^2 + 30x - 41 \\ & = -5(x-3)^2 + 4\end{aligned}$$

$$\begin{aligned}(7) \quad & x^2 - 4x + 6 \\ & = (x-2)^2 + 2\end{aligned}$$

$$\begin{aligned}(17) \quad & -2x^2 - 16x - 28 \\ & = -2(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(27) \quad & 2x^2 + 8x + 7 \\ & = 2(x+2)^2 - 1\end{aligned}$$

$$\begin{aligned}(8) \quad & -5x^2 + 50x - 124 \\ & = -5(x-5)^2 + 1\end{aligned}$$

$$\begin{aligned}(18) \quad & 3x^2 + 18x + 32 \\ & = 3(x+3)^2 + 5\end{aligned}$$

$$\begin{aligned}(28) \quad & -5x^2 + 30x - 46 \\ & = -5(x-3)^2 - 1\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 + 2x - 1 \\ & = -(x-1)^2\end{aligned}$$

$$\begin{aligned}(19) \quad & 5x^2 + 20x + 16 \\ & = 5(x+2)^2 - 4\end{aligned}$$

$$\begin{aligned}(29) \quad & -x^2 + 6x - 6 \\ & = -(x-3)^2 + 3\end{aligned}$$

$$\begin{aligned}(10) \quad & -3x^2 - 12x - 9 \\ & = -3(x+2)^2 + 3\end{aligned}$$

$$\begin{aligned}(20) \quad & -5x^2 - 10x \\ & = -5(x+1)^2 + 5\end{aligned}$$

$$\begin{aligned}(30) \quad & x^2 + 4x + 1 \\ & = (x+2)^2 - 3\end{aligned}$$

$$\begin{aligned}(1) \quad & 3x^2 + 12x + 10 \\ &= 3(x+2)^2 - 2\end{aligned}$$

$$\begin{aligned}(11) \quad & -2x^2 + 12x - 21 \\ &= -2(x-3)^2 - 3\end{aligned}$$

$$\begin{aligned}(21) \quad & -2x^2 + 16x - 31 \\ &= -2(x-4)^2 + 1\end{aligned}$$

$$\begin{aligned}(2) \quad & 4x^2 + 40x + 104 \\ &= 4(x+5)^2 + 4\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 - 6x + 4 \\ &= 3(x-1)^2 + 1\end{aligned}$$

$$\begin{aligned}(22) \quad & x^2 - 2x - 1 \\ &= (x-1)^2 - 2\end{aligned}$$

$$\begin{aligned}(3) \quad & -4x^2 + 40x - 105 \\ &= -4(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(13) \quad & 4x^2 + 8x + 4 \\ &= 4(x+1)^2\end{aligned}$$

$$\begin{aligned}(23) \quad & -x^2 + 4x - 5 \\ &= -(x-2)^2 - 1\end{aligned}$$

$$\begin{aligned}(4) \quad & -2x^2 - 12x - 18 \\ &= -2(x+3)^2\end{aligned}$$

$$\begin{aligned}(14) \quad & 3x^2 + 12x + 16 \\ &= 3(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(24) \quad & -3x^2 - 18x - 29 \\ &= -3(x+3)^2 - 2\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 - 10x - 30 \\ &= -(x+5)^2 - 5\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 + 4x + 6 \\ &= 2(x+1)^2 + 4\end{aligned}$$

$$\begin{aligned}(25) \quad & 4x^2 - 24x + 38 \\ &= 4(x-3)^2 + 2\end{aligned}$$

$$\begin{aligned}(6) \quad & 3x^2 + 30x + 75 \\ &= 3(x+5)^2\end{aligned}$$

$$\begin{aligned}(16) \quad & 3x^2 - 6x + 2 \\ &= 3(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(26) \quad & -x^2 + 10x - 23 \\ &= -(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 + 10x - 25 \\ &= -(x-5)^2\end{aligned}$$

$$\begin{aligned}(17) \quad & -4x^2 - 16x - 11 \\ &= -4(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(27) \quad & x^2 + 6x + 12 \\ &= (x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(8) \quad & 4x^2 + 24x + 38 \\ &= 4(x+3)^2 + 2\end{aligned}$$

$$\begin{aligned}(18) \quad & x^2 - 6x + 13 \\ &= (x-3)^2 + 4\end{aligned}$$

$$\begin{aligned}(28) \quad & x^2 - 6x + 11 \\ &= (x-3)^2 + 2\end{aligned}$$

$$\begin{aligned}(9) \quad & -2x^2 - 4x - 5 \\ &= -2(x+1)^2 - 3\end{aligned}$$

$$\begin{aligned}(19) \quad & 2x^2 - 8x + 6 \\ &= 2(x-2)^2 - 2\end{aligned}$$

$$\begin{aligned}(29) \quad & 3x^2 - 18x + 28 \\ &= 3(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(10) \quad & -x^2 - 2x + 3 \\ &= -(x+1)^2 + 4\end{aligned}$$

$$\begin{aligned}(20) \quad & 4x^2 - 32x + 67 \\ &= 4(x-4)^2 + 3\end{aligned}$$

$$\begin{aligned}(30) \quad & 3x^2 - 30x + 80 \\ &= 3(x-5)^2 + 5\end{aligned}$$

$$(1) \ x^2 - 10x + 26 \\ = (x - 5)^2 + 1$$

$$(11) \ -4x^2 - 8x - 1 \\ = -4(x + 1)^2 + 3$$

$$(21) \ x^2 - 6x + 12 \\ = (x - 3)^2 + 3$$

$$(2) \ x^2 + 4x + 2 \\ = (x + 2)^2 - 2$$

$$(12) \ 5x^2 + 10x + 7 \\ = 5(x + 1)^2 + 2$$

$$(22) \ x^2 + 6x + 9 \\ = (x + 3)^2$$

$$(3) \ -4x^2 + 32x - 61 \\ = -4(x - 4)^2 + 3$$

$$(13) \ -2x^2 + 20x - 46 \\ = -2(x - 5)^2 + 4$$

$$(23) \ -x^2 - 4x - 1 \\ = -(x + 2)^2 + 3$$

$$(4) \ 4x^2 + 8x + 9 \\ = 4(x + 1)^2 + 5$$

$$(14) \ -x^2 + 4x - 2 \\ = -(x - 2)^2 + 2$$

$$(24) \ 4x^2 + 8x + 3 \\ = 4(x + 1)^2 - 1$$

$$(5) \ -2x^2 + 8x - 3 \\ = -2(x - 2)^2 + 5$$

$$(15) \ -3x^2 + 30x - 72 \\ = -3(x - 5)^2 + 3$$

$$(25) \ 5x^2 + 40x + 76 \\ = 5(x + 4)^2 - 4$$

$$(6) \ 3x^2 + 6x - 2 \\ = 3(x + 1)^2 - 5$$

$$(16) \ 5x^2 + 20x + 21 \\ = 5(x + 2)^2 + 1$$

$$(26) \ x^2 - 4x + 8 \\ = (x - 2)^2 + 4$$

$$(7) \ x^2 + 10x + 23 \\ = (x + 5)^2 - 2$$

$$(17) \ 2x^2 - 20x + 47 \\ = 2(x - 5)^2 - 3$$

$$(27) \ 4x^2 + 32x + 62 \\ = 4(x + 4)^2 - 2$$

$$(8) \ -4x^2 + 40x - 100 \\ = -4(x - 5)^2$$

$$(18) \ -2x^2 - 16x - 32 \\ = -2(x + 4)^2$$

$$(28) \ -5x^2 - 10x - 2 \\ = -5(x + 1)^2 + 3$$

$$(9) \ x^2 - 6x + 6 \\ = (x - 3)^2 - 3$$

$$(19) \ x^2 + 6x + 14 \\ = (x + 3)^2 + 5$$

$$(29) \ 5x^2 + 50x + 129 \\ = 5(x + 5)^2 + 4$$

$$(10) \ x^2 + 8x + 16 \\ = (x + 4)^2$$

$$(20) \ -x^2 - 8x - 15 \\ = -(x + 4)^2 + 1$$

$$(30) \ x^2 - 2x + 3 \\ = (x - 1)^2 + 2$$

$$\begin{aligned}(1) \quad & 4x^2 + 40x + 103 \\ &= 4(x+5)^2 + 3\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 - 24x - 45 \\ &= -3(x+4)^2 + 3\end{aligned}$$

$$\begin{aligned}(21) \quad & -x^2 + 4x + 1 \\ &= -(x-2)^2 + 5\end{aligned}$$

$$\begin{aligned}(2) \quad & 3x^2 - 24x + 47 \\ &= 3(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(12) \quad & -x^2 - 6x - 9 \\ &= -(x+3)^2\end{aligned}$$

$$\begin{aligned}(22) \quad & -3x^2 + 12x - 10 \\ &= -3(x-2)^2 + 2\end{aligned}$$

$$\begin{aligned}(3) \quad & 3x^2 + 30x + 76 \\ &= 3(x+5)^2 + 1\end{aligned}$$

$$\begin{aligned}(13) \quad & -2x^2 + 16x - 33 \\ &= -2(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(23) \quad & x^2 - 4x + 7 \\ &= (x-2)^2 + 3\end{aligned}$$

$$\begin{aligned}(4) \quad & 2x^2 - 4x + 1 \\ &= 2(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(14) \quad & -x^2 - 10x - 28 \\ &= -(x+5)^2 - 3\end{aligned}$$

$$\begin{aligned}(24) \quad & 4x^2 + 32x + 62 \\ &= 4(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 + 8x - 14 \\ &= -(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(15) \quad & 3x^2 + 12x + 7 \\ &= 3(x+2)^2 - 5\end{aligned}$$

$$\begin{aligned}(25) \quad & -4x^2 - 40x - 101 \\ &= -4(x+5)^2 - 1\end{aligned}$$

$$\begin{aligned}(6) \quad & 4x^2 - 24x + 40 \\ &= 4(x-3)^2 + 4\end{aligned}$$

$$\begin{aligned}(16) \quad & 3x^2 + 24x + 43 \\ &= 3(x+4)^2 - 5\end{aligned}$$

$$\begin{aligned}(26) \quad & 5x^2 + 40x + 82 \\ &= 5(x+4)^2 + 2\end{aligned}$$

$$\begin{aligned}(7) \quad & x^2 - 10x + 26 \\ &= (x-5)^2 + 1\end{aligned}$$

$$\begin{aligned}(17) \quad & 3x^2 + 18x + 32 \\ &= 3(x+3)^2 + 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -5x^2 - 40x - 82 \\ &= -5(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 - 2x + 5 \\ &= (x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(18) \quad & 5x^2 + 40x + 80 \\ &= 5(x+4)^2\end{aligned}$$

$$\begin{aligned}(28) \quad & -5x^2 + 10x - 6 \\ &= -5(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 - 8x - 18 \\ &= -(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(19) \quad & 4x^2 + 8x + 3 \\ &= 4(x+1)^2 - 1\end{aligned}$$

$$\begin{aligned}(29) \quad & x^2 - 2x + 5 \\ &= (x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(10) \quad & x^2 + 10x + 30 \\ &= (x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(20) \quad & 4x^2 + 8x + 9 \\ &= 4(x+1)^2 + 5\end{aligned}$$

$$\begin{aligned}(30) \quad & -4x^2 + 32x - 62 \\ &= -4(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(1) \quad & -x^2 - 8x - 18 \\ & = -(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(11) \quad & -5x^2 - 10x - 5 \\ & = -5(x+1)^2\end{aligned}$$

$$\begin{aligned}(21) \quad & 2x^2 + 20x + 46 \\ & = 2(x+5)^2 - 4\end{aligned}$$

$$\begin{aligned}(2) \quad & -5x^2 + 40x - 76 \\ & = -5(x-4)^2 + 4\end{aligned}$$

$$\begin{aligned}(12) \quad & -3x^2 - 6x - 4 \\ & = -3(x+1)^2 - 1\end{aligned}$$

$$\begin{aligned}(22) \quad & 4x^2 + 40x + 96 \\ & = 4(x+5)^2 - 4\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 + 24x - 46 \\ & = -3(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & -5x^2 + 30x - 43 \\ & = -5(x-3)^2 + 2\end{aligned}$$

$$\begin{aligned}(23) \quad & x^2 - 6x + 7 \\ & = (x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(4) \quad & -4x^2 + 16x - 11 \\ & = -4(x-2)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & 5x^2 - 50x + 124 \\ & = 5(x-5)^2 - 1\end{aligned}$$

$$\begin{aligned}(24) \quad & 3x^2 - 18x + 29 \\ & = 3(x-3)^2 + 2\end{aligned}$$

$$\begin{aligned}(5) \quad & 4x^2 + 8x \\ & = 4(x+1)^2 - 4\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 - 24x - 39 \\ & = -4(x+3)^2 - 3\end{aligned}$$

$$\begin{aligned}(25) \quad & -4x^2 - 8x - 5 \\ & = -4(x+1)^2 - 1\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 - 6x + 4 \\ & = (x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(16) \quad & x^2 + 10x + 22 \\ & = (x+5)^2 - 3\end{aligned}$$

$$\begin{aligned}(26) \quad & -3x^2 - 30x - 77 \\ & = -3(x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 - 4x + 1 \\ & = -(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(17) \quad & -2x^2 - 8x - 11 \\ & = -2(x+2)^2 - 3\end{aligned}$$

$$\begin{aligned}(27) \quad & 5x^2 + 40x + 84 \\ & = 5(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(8) \quad & -3x^2 + 30x - 73 \\ & = -3(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(18) \quad & -2x^2 - 8x - 5 \\ & = -2(x+2)^2 + 3\end{aligned}$$

$$\begin{aligned}(28) \quad & -x^2 + 10x - 24 \\ & = -(x-5)^2 + 1\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 + 4x - 3 \\ & = -(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(19) \quad & -5x^2 + 30x - 45 \\ & = -5(x-3)^2\end{aligned}$$

$$\begin{aligned}(29) \quad & 5x^2 + 30x + 45 \\ & = 5(x+3)^2\end{aligned}$$

$$\begin{aligned}(10) \quad & -x^2 + 6x - 14 \\ & = -(x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(20) \quad & 4x^2 - 32x + 60 \\ & = 4(x-4)^2 - 4\end{aligned}$$

$$\begin{aligned}(30) \quad & 4x^2 + 24x + 33 \\ & = 4(x+3)^2 - 3\end{aligned}$$

$$\begin{aligned}(1) \quad & 4x^2 + 16x + 17 \\ &= 4(x+2)^2 + 1\end{aligned}$$

$$\begin{aligned}(11) \quad & 3x^2 - 24x + 43 \\ &= 3(x-4)^2 - 5\end{aligned}$$

$$\begin{aligned}(21) \quad & -3x^2 - 12x - 7 \\ &= -3(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(2) \quad & -5x^2 + 30x - 49 \\ &= -5(x-3)^2 - 4\end{aligned}$$

$$\begin{aligned}(12) \quad & -x^2 + 4x - 3 \\ &= -(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(22) \quad & 3x^2 - 30x + 70 \\ &= 3(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(3) \quad & 5x^2 + 40x + 78 \\ &= 5(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(13) \quad & -3x^2 - 12x - 17 \\ &= -3(x+2)^2 - 5\end{aligned}$$

$$\begin{aligned}(23) \quad & -x^2 + 6x - 14 \\ &= -(x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(4) \quad & -4x^2 + 24x - 39 \\ &= -4(x-3)^2 - 3\end{aligned}$$

$$\begin{aligned}(14) \quad & -x^2 - 10x - 22 \\ &= -(x+5)^2 + 3\end{aligned}$$

$$\begin{aligned}(24) \quad & -5x^2 - 20x - 22 \\ &= -5(x+2)^2 - 2\end{aligned}$$

$$\begin{aligned}(5) \quad & 5x^2 + 20x + 22 \\ &= 5(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(15) \quad & -5x^2 + 40x - 82 \\ &= -5(x-4)^2 - 2\end{aligned}$$

$$\begin{aligned}(25) \quad & -x^2 + 6x - 9 \\ &= -(x-3)^2\end{aligned}$$

$$\begin{aligned}(6) \quad & 5x^2 + 20x + 24 \\ &= 5(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(16) \quad & -x^2 + 8x - 17 \\ &= -(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(26) \quad & -3x^2 - 30x - 71 \\ &= -3(x+5)^2 + 4\end{aligned}$$

$$\begin{aligned}(7) \quad & 2x^2 - 4x - 3 \\ &= 2(x-1)^2 - 5\end{aligned}$$

$$\begin{aligned}(17) \quad & -3x^2 + 6x - 8 \\ &= -3(x-1)^2 - 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -2x^2 - 12x - 15 \\ &= -2(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(8) \quad & 2x^2 - 12x + 16 \\ &= 2(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(18) \quad & 5x^2 - 10x + 4 \\ &= 5(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(28) \quad & -4x^2 - 24x - 35 \\ &= -4(x+3)^2 + 1\end{aligned}$$

$$\begin{aligned}(9) \quad & 5x^2 + 30x + 45 \\ &= 5(x+3)^2\end{aligned}$$

$$\begin{aligned}(19) \quad & -2x^2 - 4x + 1 \\ &= -2(x+1)^2 + 3\end{aligned}$$

$$\begin{aligned}(29) \quad & 4x^2 - 40x + 104 \\ &= 4(x-5)^2 + 4\end{aligned}$$

$$\begin{aligned}(10) \quad & 2x^2 - 16x + 36 \\ &= 2(x-4)^2 + 4\end{aligned}$$

$$\begin{aligned}(20) \quad & -3x^2 - 24x - 45 \\ &= -3(x+4)^2 + 3\end{aligned}$$

$$\begin{aligned}(30) \quad & -5x^2 - 40x - 79 \\ &= -5(x+4)^2 + 1\end{aligned}$$



$$\begin{aligned}(1) \quad & -3x^2 - 24x - 52 \\ & = -3(x+4)^2 - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & x^2 + 4x - 1 \\ & = (x+2)^2 - 5\end{aligned}$$

$$\begin{aligned}(21) \quad & 5x^2 + 10x + 3 \\ & = 5(x+1)^2 - 2\end{aligned}$$

$$\begin{aligned}(2) \quad & 3x^2 + 6x - 2 \\ & = 3(x+1)^2 - 5\end{aligned}$$

$$\begin{aligned}(12) \quad & -2x^2 + 16x - 33 \\ & = -2(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(22) \quad & -2x^2 + 12x - 21 \\ & = -2(x-3)^2 - 3\end{aligned}$$

$$\begin{aligned}(3) \quad & -4x^2 + 40x - 104 \\ & = -4(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(13) \quad & 4x^2 - 40x + 100 \\ & = 4(x-5)^2\end{aligned}$$

$$\begin{aligned}(23) \quad & 2x^2 + 8x + 12 \\ & = 2(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(4) \quad & 3x^2 + 6x + 7 \\ & = 3(x+1)^2 + 4\end{aligned}$$

$$\begin{aligned}(14) \quad & x^2 - 4x + 2 \\ & = (x-2)^2 - 2\end{aligned}$$

$$\begin{aligned}(24) \quad & x^2 + 2x + 2 \\ & = (x+1)^2 + 1\end{aligned}$$

$$\begin{aligned}(5) \quad & -4x^2 + 16x - 16 \\ & = -4(x-2)^2\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 + 32x - 59 \\ & = -4(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(25) \quad & x^2 - 8x + 18 \\ & = (x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(6) \quad & -x^2 - 2x + 1 \\ & = -(x+1)^2 + 2\end{aligned}$$

$$\begin{aligned}(16) \quad & -x^2 - 2x + 2 \\ & = -(x+1)^2 + 3\end{aligned}$$

$$\begin{aligned}(26) \quad & x^2 + 10x + 23 \\ & = (x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -4x^2 + 8x - 1 \\ & = -4(x-1)^2 + 3\end{aligned}$$

$$\begin{aligned}(17) \quad & 5x^2 + 40x + 85 \\ & = 5(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -5x^2 + 10x - 8 \\ & = -5(x-1)^2 - 3\end{aligned}$$

$$\begin{aligned}(8) \quad & -5x^2 + 50x - 127 \\ & = -5(x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(18) \quad & 4x^2 - 32x + 69 \\ & = 4(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(28) \quad & 4x^2 - 32x + 64 \\ & = 4(x-4)^2\end{aligned}$$

$$\begin{aligned}(9) \quad & -2x^2 - 16x - 27 \\ & = -2(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(19) \quad & -5x^2 - 30x - 49 \\ & = -5(x+3)^2 - 4\end{aligned}$$

$$\begin{aligned}(29) \quad & -2x^2 + 20x - 55 \\ & = -2(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(10) \quad & -2x^2 + 20x - 46 \\ & = -2(x-5)^2 + 4\end{aligned}$$

$$\begin{aligned}(20) \quad & -5x^2 - 50x - 122 \\ & = -5(x+5)^2 + 3\end{aligned}$$

$$\begin{aligned}(30) \quad & -4x^2 + 40x - 95 \\ & = -4(x-5)^2 + 5\end{aligned}$$

$$\begin{aligned}(1) \quad & x^2 + 2x + 5 \\ &= (x + 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(11) \quad & 5x^2 - 20x + 17 \\ &= 5(x - 2)^2 - 3\end{aligned}$$

$$\begin{aligned}(21) \quad & 2x^2 + 16x + 31 \\ &= 2(x + 4)^2 - 1\end{aligned}$$

$$\begin{aligned}(2) \quad & -3x^2 - 24x - 49 \\ &= -3(x + 4)^2 - 1\end{aligned}$$

$$\begin{aligned}(12) \quad & 2x^2 - 12x + 15 \\ &= 2(x - 3)^2 - 3\end{aligned}$$

$$\begin{aligned}(22) \quad & x^2 - 8x + 18 \\ &= (x - 4)^2 + 2\end{aligned}$$

$$\begin{aligned}(3) \quad & 4x^2 - 16x + 12 \\ &= 4(x - 2)^2 - 4\end{aligned}$$

$$\begin{aligned}(13) \quad & 3x^2 - 24x + 45 \\ &= 3(x - 4)^2 - 3\end{aligned}$$

$$\begin{aligned}(23) \quad & -2x^2 - 4x + 3 \\ &= -2(x + 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(4) \quad & -3x^2 + 24x - 47 \\ &= -3(x - 4)^2 + 1\end{aligned}$$

$$\begin{aligned}(14) \quad & -2x^2 + 12x - 20 \\ &= -2(x - 3)^2 - 2\end{aligned}$$

$$\begin{aligned}(24) \quad & 3x^2 - 30x + 74 \\ &= 3(x - 5)^2 - 1\end{aligned}$$

$$\begin{aligned}(5) \quad & 5x^2 + 20x + 24 \\ &= 5(x + 2)^2 + 4\end{aligned}$$

$$\begin{aligned}(15) \quad & -3x^2 + 24x - 47 \\ &= -3(x - 4)^2 + 1\end{aligned}$$

$$\begin{aligned}(25) \quad & x^2 + 2x + 5 \\ &= (x + 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 + 6x + 11 \\ &= (x + 3)^2 + 2\end{aligned}$$

$$\begin{aligned}(16) \quad & -3x^2 - 30x - 71 \\ &= -3(x + 5)^2 + 4\end{aligned}$$

$$\begin{aligned}(26) \quad & -3x^2 - 30x - 77 \\ &= -3(x + 5)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -4x^2 - 32x - 68 \\ &= -4(x + 4)^2 - 4\end{aligned}$$

$$\begin{aligned}(17) \quad & -x^2 - 2x \\ &= -(x + 1)^2 + 1\end{aligned}$$

$$\begin{aligned}(27) \quad & x^2 + 6x + 4 \\ &= (x + 3)^2 - 5\end{aligned}$$

$$\begin{aligned}(8) \quad & 4x^2 + 16x + 16 \\ &= 4(x + 2)^2\end{aligned}$$

$$\begin{aligned}(18) \quad & x^2 + 4x - 1 \\ &= (x + 2)^2 - 5\end{aligned}$$

$$\begin{aligned}(28) \quad & -x^2 + 10x - 22 \\ &= -(x - 5)^2 + 3\end{aligned}$$

$$\begin{aligned}(9) \quad & -4x^2 - 32x - 65 \\ &= -4(x + 4)^2 - 1\end{aligned}$$

$$\begin{aligned}(19) \quad & 2x^2 + 20x + 48 \\ &= 2(x + 5)^2 - 2\end{aligned}$$

$$\begin{aligned}(29) \quad & 4x^2 - 24x + 34 \\ &= 4(x - 3)^2 - 2\end{aligned}$$

$$\begin{aligned}(10) \quad & 2x^2 - 8x + 9 \\ &= 2(x - 2)^2 + 1\end{aligned}$$

$$\begin{aligned}(20) \quad & -5x^2 + 10x - 6 \\ &= -5(x - 1)^2 - 1\end{aligned}$$

$$\begin{aligned}(30) \quad & -5x^2 - 30x - 49 \\ &= -5(x + 3)^2 - 4\end{aligned}$$

$$\begin{aligned}(1) \quad & x^2 + 10x + 24 \\ & = (x + 5)^2 - 1\end{aligned}$$

$$\begin{aligned}(11) \quad & 4x^2 - 24x + 36 \\ & = 4(x - 3)^2\end{aligned}$$

$$\begin{aligned}(21) \quad & -4x^2 + 32x - 60 \\ & = -4(x - 4)^2 + 4\end{aligned}$$

$$\begin{aligned}(2) \quad & -5x^2 - 10x \\ & = -5(x + 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(12) \quad & -5x^2 + 50x - 122 \\ & = -5(x - 5)^2 + 3\end{aligned}$$

$$\begin{aligned}(22) \quad & -x^2 + 2x + 1 \\ & = -(x - 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(3) \quad & 4x^2 - 16x + 15 \\ & = 4(x - 2)^2 - 1\end{aligned}$$

$$\begin{aligned}(13) \quad & -2x^2 + 8x - 3 \\ & = -2(x - 2)^2 + 5\end{aligned}$$

$$\begin{aligned}(23) \quad & 2x^2 - 16x + 31 \\ & = 2(x - 4)^2 - 1\end{aligned}$$

$$\begin{aligned}(4) \quad & 4x^2 + 32x + 68 \\ & = 4(x + 4)^2 + 4\end{aligned}$$

$$\begin{aligned}(14) \quad & x^2 + 2x - 4 \\ & = (x + 1)^2 - 5\end{aligned}$$

$$\begin{aligned}(24) \quad & -2x^2 - 12x - 23 \\ & = -2(x + 3)^2 - 5\end{aligned}$$

$$\begin{aligned}(5) \quad & 5x^2 + 10x + 2 \\ & = 5(x + 1)^2 - 3\end{aligned}$$

$$\begin{aligned}(15) \quad & -x^2 - 2x + 3 \\ & = -(x + 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(25) \quad & 2x^2 - 12x + 16 \\ & = 2(x - 3)^2 - 2\end{aligned}$$

$$\begin{aligned}(6) \quad & 3x^2 + 12x + 10 \\ & = 3(x + 2)^2 - 2\end{aligned}$$

$$\begin{aligned}(16) \quad & 4x^2 + 32x + 69 \\ & = 4(x + 4)^2 + 5\end{aligned}$$

$$\begin{aligned}(26) \quad & 5x^2 + 30x + 50 \\ & = 5(x + 3)^2 + 5\end{aligned}$$

$$\begin{aligned}(7) \quad & -5x^2 - 20x - 16 \\ & = -5(x + 2)^2 + 4\end{aligned}$$

$$\begin{aligned}(17) \quad & 2x^2 + 4x \\ & = 2(x + 1)^2 - 2\end{aligned}$$

$$\begin{aligned}(27) \quad & -5x^2 - 40x - 75 \\ & = -5(x + 4)^2 + 5\end{aligned}$$

$$\begin{aligned}(8) \quad & -x^2 + 2x - 1 \\ & = -(x - 1)^2\end{aligned}$$

$$\begin{aligned}(18) \quad & -3x^2 + 12x - 9 \\ & = -3(x - 2)^2 + 3\end{aligned}$$

$$\begin{aligned}(28) \quad & 3x^2 - 24x + 50 \\ & = 3(x - 4)^2 + 2\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 + 18x + 25 \\ & = 3(x + 3)^2 - 2\end{aligned}$$

$$\begin{aligned}(19) \quad & -3x^2 + 12x - 8 \\ & = -3(x - 2)^2 + 4\end{aligned}$$

$$\begin{aligned}(29) \quad & 3x^2 - 18x + 23 \\ & = 3(x - 3)^2 - 4\end{aligned}$$

$$\begin{aligned}(10) \quad & -3x^2 - 12x - 17 \\ & = -3(x + 2)^2 - 5\end{aligned}$$

$$\begin{aligned}(20) \quad & 2x^2 + 4x - 3 \\ & = 2(x + 1)^2 - 5\end{aligned}$$

$$\begin{aligned}(30) \quad & x^2 - 6x + 12 \\ & = (x - 3)^2 + 3\end{aligned}$$

$$\begin{aligned}(1) \quad & 5x^2 + 20x + 25 \\ &= 5(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(11) \quad & -x^2 - 6x - 11 \\ &= -(x+3)^2 - 2\end{aligned}$$

$$\begin{aligned}(21) \quad & -2x^2 + 20x - 55 \\ &= -2(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(2) \quad & -x^2 + 6x - 11 \\ &= -(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(12) \quad & x^2 + 6x + 8 \\ &= (x+3)^2 - 1\end{aligned}$$

$$\begin{aligned}(22) \quad & 5x^2 + 20x + 25 \\ &= 5(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(3) \quad & 5x^2 + 30x + 46 \\ &= 5(x+3)^2 + 1\end{aligned}$$

$$\begin{aligned}(13) \quad & -4x^2 - 8x - 2 \\ &= -4(x+1)^2 + 2\end{aligned}$$

$$\begin{aligned}(23) \quad & 2x^2 + 20x + 49 \\ &= 2(x+5)^2 - 1\end{aligned}$$

$$\begin{aligned}(4) \quad & -x^2 + 4x - 8 \\ &= -(x-2)^2 - 4\end{aligned}$$

$$\begin{aligned}(14) \quad & 3x^2 - 12x + 13 \\ &= 3(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(24) \quad & 4x^2 - 32x + 61 \\ &= 4(x-4)^2 - 3\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 + 2x + 3 \\ &= -(x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 - 8x - 3 \\ &= -4(x+1)^2 + 1\end{aligned}$$

$$\begin{aligned}(25) \quad & -3x^2 + 12x - 16 \\ &= -3(x-2)^2 - 4\end{aligned}$$

$$\begin{aligned}(6) \quad & -4x^2 + 24x - 35 \\ &= -4(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(16) \quad & -4x^2 + 40x - 98 \\ &= -4(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(26) \quad & -5x^2 + 10x - 4 \\ &= -5(x-1)^2 + 1\end{aligned}$$

$$\begin{aligned}(7) \quad & 5x^2 + 30x + 41 \\ &= 5(x+3)^2 - 4\end{aligned}$$

$$\begin{aligned}(17) \quad & -4x^2 - 16x - 11 \\ &= -4(x+2)^2 + 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -5x^2 - 50x - 129 \\ &= -5(x+5)^2 - 4\end{aligned}$$

$$\begin{aligned}(8) \quad & -4x^2 - 32x - 60 \\ &= -4(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(18) \quad & -4x^2 + 24x - 38 \\ &= -4(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(28) \quad & -2x^2 + 16x - 37 \\ &= -2(x-4)^2 - 5\end{aligned}$$

$$\begin{aligned}(9) \quad & -2x^2 - 16x - 34 \\ &= -2(x+4)^2 - 2\end{aligned}$$

$$\begin{aligned}(19) \quad & -2x^2 - 20x - 54 \\ &= -2(x+5)^2 - 4\end{aligned}$$

$$\begin{aligned}(29) \quad & 2x^2 + 16x + 31 \\ &= 2(x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(10) \quad & 2x^2 - 16x + 32 \\ &= 2(x-4)^2\end{aligned}$$

$$\begin{aligned}(20) \quad & 4x^2 - 32x + 66 \\ &= 4(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(30) \quad & -5x^2 + 40x - 83 \\ &= -5(x-4)^2 - 3\end{aligned}$$

$$\begin{aligned}(1) \quad & 5x^2 - 20x + 15 \\ &= 5(x-2)^2 - 5\end{aligned}$$

$$\begin{aligned}(11) \quad & -4x^2 - 16x - 12 \\ &= -4(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(21) \quad & 5x^2 + 50x + 124 \\ &= 5(x+5)^2 - 1\end{aligned}$$

$$\begin{aligned}(2) \quad & 5x^2 + 10x + 2 \\ &= 5(x+1)^2 - 3\end{aligned}$$

$$\begin{aligned}(12) \quad & -3x^2 - 30x - 74 \\ &= -3(x+5)^2 + 1\end{aligned}$$

$$\begin{aligned}(22) \quad & 3x^2 + 6x + 6 \\ &= 3(x+1)^2 + 3\end{aligned}$$

$$\begin{aligned}(3) \quad & 3x^2 - 18x + 26 \\ &= 3(x-3)^2 - 1\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 - 4x - 2 \\ &= -(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(23) \quad & 4x^2 - 24x + 36 \\ &= 4(x-3)^2\end{aligned}$$

$$\begin{aligned}(4) \quad & 2x^2 + 12x + 23 \\ &= 2(x+3)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & 2x^2 - 12x + 14 \\ &= 2(x-3)^2 - 4\end{aligned}$$

$$\begin{aligned}(24) \quad & -3x^2 - 6x + 1 \\ &= -3(x+1)^2 + 4\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 + 10x - 30 \\ &= -(x-5)^2 - 5\end{aligned}$$

$$\begin{aligned}(15) \quad & 4x^2 - 16x + 17 \\ &= 4(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(25) \quad & 4x^2 + 24x + 36 \\ &= 4(x+3)^2\end{aligned}$$

$$\begin{aligned}(6) \quad & -5x^2 + 50x - 126 \\ &= -5(x-5)^2 - 1\end{aligned}$$

$$\begin{aligned}(16) \quad & -5x^2 - 10x - 8 \\ &= -5(x+1)^2 - 3\end{aligned}$$

$$\begin{aligned}(26) \quad & -2x^2 + 4x - 3 \\ &= -2(x-1)^2 - 1\end{aligned}$$

$$\begin{aligned}(7) \quad & -4x^2 - 32x - 60 \\ &= -4(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(17) \quad & 2x^2 - 12x + 19 \\ &= 2(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(27) \quad & -2x^2 + 4x - 5 \\ &= -2(x-1)^2 - 3\end{aligned}$$

$$\begin{aligned}(8) \quad & -x^2 + 10x - 23 \\ &= -(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(18) \quad & 2x^2 - 8x + 12 \\ &= 2(x-2)^2 + 4\end{aligned}$$

$$\begin{aligned}(28) \quad & 4x^2 + 32x + 64 \\ &= 4(x+4)^2\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 - 6x - 2 \\ &= 3(x-1)^2 - 5\end{aligned}$$

$$\begin{aligned}(19) \quad & 3x^2 + 30x + 73 \\ &= 3(x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(29) \quad & 3x^2 + 24x + 50 \\ &= 3(x+4)^2 + 2\end{aligned}$$

$$\begin{aligned}(10) \quad & 4x^2 + 32x + 63 \\ &= 4(x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(20) \quad & -x^2 + 10x - 25 \\ &= -(x-5)^2\end{aligned}$$

$$\begin{aligned}(30) \quad & 4x^2 + 24x + 39 \\ &= 4(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(1) \quad & 5x^2 + 40x + 77 \\ &= 5(x+4)^2 - 3\end{aligned}$$

$$\begin{aligned}(11) \quad & -2x^2 + 20x - 48 \\ &= -2(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(21) \quad & -4x^2 + 40x - 104 \\ &= -4(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(2) \quad & -5x^2 + 40x - 78 \\ &= -5(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 - 50x + 128 \\ &= 5(x-5)^2 + 3\end{aligned}$$

$$\begin{aligned}(22) \quad & 2x^2 - 16x + 37 \\ &= 2(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 + 6x - 3 \\ &= -3(x-1)^2\end{aligned}$$

$$\begin{aligned}(13) \quad & -4x^2 + 16x - 20 \\ &= -4(x-2)^2 - 4\end{aligned}$$

$$\begin{aligned}(23) \quad & 4x^2 - 24x + 32 \\ &= 4(x-3)^2 - 4\end{aligned}$$

$$\begin{aligned}(4) \quad & 5x^2 - 50x + 130 \\ &= 5(x-5)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 - 16x - 12 \\ &= -4(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(24) \quad & -5x^2 - 20x - 18 \\ &= -5(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(5) \quad & 2x^2 + 20x + 49 \\ &= 2(x+5)^2 - 1\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 + 8x + 4 \\ &= 2(x+2)^2 - 4\end{aligned}$$

$$\begin{aligned}(25) \quad & -x^2 + 6x - 9 \\ &= -(x-3)^2\end{aligned}$$

$$\begin{aligned}(6) \quad & -3x^2 - 12x - 8 \\ &= -3(x+2)^2 + 4\end{aligned}$$

$$\begin{aligned}(16) \quad & -4x^2 + 40x - 98 \\ &= -4(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(26) \quad & x^2 - 4x + 1 \\ &= (x-2)^2 - 3\end{aligned}$$

$$\begin{aligned}(7) \quad & 5x^2 + 40x + 75 \\ &= 5(x+4)^2 - 5\end{aligned}$$

$$\begin{aligned}(17) \quad & 3x^2 + 6x \\ &= 3(x+1)^2 - 3\end{aligned}$$

$$\begin{aligned}(27) \quad & -x^2 - 10x - 30 \\ &= -(x+5)^2 - 5\end{aligned}$$

$$\begin{aligned}(8) \quad & 2x^2 + 4x - 2 \\ &= 2(x+1)^2 - 4\end{aligned}$$

$$\begin{aligned}(18) \quad & -3x^2 - 12x - 9 \\ &= -3(x+2)^2 + 3\end{aligned}$$

$$\begin{aligned}(28) \quad & -x^2 + 10x - 26 \\ &= -(x-5)^2 - 1\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 + 12x + 9 \\ &= 3(x+2)^2 - 3\end{aligned}$$

$$\begin{aligned}(19) \quad & -2x^2 - 12x - 14 \\ &= -2(x+3)^2 + 4\end{aligned}$$

$$\begin{aligned}(29) \quad & -4x^2 + 32x - 60 \\ &= -4(x-4)^2 + 4\end{aligned}$$

$$\begin{aligned}(10) \quad & -5x^2 + 50x - 129 \\ &= -5(x-5)^2 - 4\end{aligned}$$

$$\begin{aligned}(20) \quad & 3x^2 + 24x + 47 \\ &= 3(x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(30) \quad & x^2 + 2x - 1 \\ &= (x+1)^2 - 2\end{aligned}$$

$$\begin{aligned}(1) \quad & -4x^2 - 40x - 101 \\ & = -4(x+5)^2 - 1\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 + 18x - 30 \\ & = -3(x-3)^2 - 3\end{aligned}$$

$$\begin{aligned}(21) \quad & 3x^2 - 30x + 73 \\ & = 3(x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(2) \quad & -4x^2 + 32x - 64 \\ & = -4(x-4)^2\end{aligned}$$

$$\begin{aligned}(12) \quad & -2x^2 - 16x - 31 \\ & = -2(x+4)^2 + 1\end{aligned}$$

$$\begin{aligned}(22) \quad & -x^2 - 8x - 11 \\ & = -(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(3) \quad & -5x^2 + 30x - 47 \\ & = -5(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 + 6x - 9 \\ & = -(x-3)^2\end{aligned}$$

$$\begin{aligned}(23) \quad & -2x^2 + 16x - 32 \\ & = -2(x-4)^2\end{aligned}$$

$$\begin{aligned}(4) \quad & 3x^2 + 12x + 14 \\ & = 3(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(14) \quad & x^2 - 10x + 23 \\ & = (x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(24) \quad & 4x^2 - 40x + 98 \\ & = 4(x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(5) \quad & x^2 + 8x + 16 \\ & = (x+4)^2\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 + 8x \\ & = -4(x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(25) \quad & -x^2 - 6x - 8 \\ & = -(x+3)^2 + 1\end{aligned}$$

$$\begin{aligned}(6) \quad & -5x^2 + 40x - 75 \\ & = -5(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(16) \quad & -x^2 + 6x - 5 \\ & = -(x-3)^2 + 4\end{aligned}$$

$$\begin{aligned}(26) \quad & 4x^2 + 40x + 98 \\ & = 4(x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 - 6x - 5 \\ & = -(x+3)^2 + 4\end{aligned}$$

$$\begin{aligned}(17) \quad & x^2 + 2x + 4 \\ & = (x+1)^2 + 3\end{aligned}$$

$$\begin{aligned}(27) \quad & -x^2 - 10x - 27 \\ & = -(x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(8) \quad & -5x^2 + 10x - 1 \\ & = -5(x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(18) \quad & 5x^2 - 30x + 46 \\ & = 5(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(28) \quad & x^2 - 2x - 1 \\ & = (x-1)^2 - 2\end{aligned}$$

$$\begin{aligned}(9) \quad & -5x^2 + 30x - 47 \\ & = -5(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(19) \quad & 5x^2 - 40x + 79 \\ & = 5(x-4)^2 - 1\end{aligned}$$

$$\begin{aligned}(29) \quad & -5x^2 + 30x - 50 \\ & = -5(x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(10) \quad & -5x^2 - 30x - 48 \\ & = -5(x+3)^2 - 3\end{aligned}$$

$$\begin{aligned}(20) \quad & x^2 - 6x + 10 \\ & = (x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(30) \quad & 5x^2 + 20x + 22 \\ & = 5(x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(1) \quad & -2x^2 - 16x - 27 \\ & = -2(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(11) \quad & -5x^2 + 20x - 15 \\ & = -5(x-2)^2 + 5\end{aligned}$$

$$\begin{aligned}(21) \quad & 3x^2 - 12x + 17 \\ & = 3(x-2)^2 + 5\end{aligned}$$

$$\begin{aligned}(2) \quad & 2x^2 - 8x + 6 \\ & = 2(x-2)^2 - 2\end{aligned}$$

$$\begin{aligned}(12) \quad & 2x^2 - 8x + 3 \\ & = 2(x-2)^2 - 5\end{aligned}$$

$$\begin{aligned}(22) \quad & x^2 + 2x + 2 \\ & = (x+1)^2 + 1\end{aligned}$$

$$\begin{aligned}(3) \quad & -x^2 - 8x - 11 \\ & = -(x+4)^2 + 5\end{aligned}$$

$$\begin{aligned}(13) \quad & -2x^2 - 16x - 35 \\ & = -2(x+4)^2 - 3\end{aligned}$$

$$\begin{aligned}(23) \quad & x^2 + 8x + 15 \\ & = (x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(4) \quad & 2x^2 + 4x + 4 \\ & = 2(x+1)^2 + 2\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 - 40x - 105 \\ & = -4(x+5)^2 - 5\end{aligned}$$

$$\begin{aligned}(24) \quad & -x^2 - 6x - 6 \\ & = -(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(5) \quad & 2x^2 - 8x + 6 \\ & = 2(x-2)^2 - 2\end{aligned}$$

$$\begin{aligned}(15) \quad & -3x^2 + 30x - 76 \\ & = -3(x-5)^2 - 1\end{aligned}$$

$$\begin{aligned}(25) \quad & 5x^2 + 40x + 84 \\ & = 5(x+4)^2 + 4\end{aligned}$$

$$\begin{aligned}(6) \quad & -x^2 - 2x - 5 \\ & = -(x+1)^2 - 4\end{aligned}$$

$$\begin{aligned}(16) \quad & -4x^2 + 16x - 17 \\ & = -4(x-2)^2 - 1\end{aligned}$$

$$\begin{aligned}(26) \quad & -5x^2 - 50x - 130 \\ & = -5(x+5)^2 - 5\end{aligned}$$

$$\begin{aligned}(7) \quad & -3x^2 - 18x - 24 \\ & = -3(x+3)^2 + 3\end{aligned}$$

$$\begin{aligned}(17) \quad & 4x^2 - 24x + 31 \\ & = 4(x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(27) \quad & -4x^2 - 24x - 32 \\ & = -4(x+3)^2 + 4\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 + 10x + 23 \\ & = (x+5)^2 - 2\end{aligned}$$

$$\begin{aligned}(18) \quad & -x^2 + 8x - 11 \\ & = -(x-4)^2 + 5\end{aligned}$$

$$\begin{aligned}(28) \quad & 5x^2 - 30x + 46 \\ & = 5(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(9) \quad & -5x^2 - 20x - 17 \\ & = -5(x+2)^2 + 3\end{aligned}$$

$$\begin{aligned}(19) \quad & 2x^2 + 4x + 2 \\ & = 2(x+1)^2\end{aligned}$$

$$\begin{aligned}(29) \quad & x^2 + 4x + 6 \\ & = (x+2)^2 + 2\end{aligned}$$

$$\begin{aligned}(10) \quad & 2x^2 + 4x + 2 \\ & = 2(x+1)^2\end{aligned}$$

$$\begin{aligned}(20) \quad & 3x^2 - 30x + 78 \\ & = 3(x-5)^2 + 3\end{aligned}$$

$$\begin{aligned}(30) \quad & 2x^2 - 8x + 5 \\ & = 2(x-2)^2 - 3\end{aligned}$$



$$\begin{aligned}(1) \quad & -x^2 + 6x - 8 \\ & = -(x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(11) \quad & -5x^2 + 20x - 19 \\ & = -5(x-2)^2 + 1\end{aligned}$$

$$\begin{aligned}(21) \quad & -4x^2 - 32x - 69 \\ & = -4(x+4)^2 - 5\end{aligned}$$

$$\begin{aligned}(2) \quad & x^2 + 10x + 30 \\ & = (x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 - 24x + 52 \\ & = 3(x-4)^2 + 4\end{aligned}$$

$$\begin{aligned}(22) \quad & 3x^2 + 24x + 47 \\ & = 3(x+4)^2 - 1\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 - 12x - 12 \\ & = -3(x+2)^2\end{aligned}$$

$$\begin{aligned}(13) \quad & 4x^2 + 24x + 35 \\ & = 4(x+3)^2 - 1\end{aligned}$$

$$\begin{aligned}(23) \quad & x^2 + 10x + 27 \\ & = (x+5)^2 + 2\end{aligned}$$

$$\begin{aligned}(4) \quad & 4x^2 + 8x - 1 \\ & = 4(x+1)^2 - 5\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 - 16x - 15 \\ & = -4(x+2)^2 + 1\end{aligned}$$

$$\begin{aligned}(24) \quad & x^2 - 6x + 10 \\ & = (x-3)^2 + 1\end{aligned}$$

$$\begin{aligned}(5) \quad & -2x^2 + 12x - 20 \\ & = -2(x-3)^2 - 2\end{aligned}$$

$$\begin{aligned}(15) \quad & -2x^2 - 8x - 5 \\ & = -2(x+2)^2 + 3\end{aligned}$$

$$\begin{aligned}(25) \quad & -3x^2 + 18x - 32 \\ & = -3(x-3)^2 - 5\end{aligned}$$

$$\begin{aligned}(6) \quad & 4x^2 + 24x + 41 \\ & = 4(x+3)^2 + 5\end{aligned}$$

$$\begin{aligned}(16) \quad & -3x^2 - 18x - 25 \\ & = -3(x+3)^2 + 2\end{aligned}$$

$$\begin{aligned}(26) \quad & 5x^2 - 50x + 123 \\ & = 5(x-5)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & -3x^2 + 24x - 46 \\ & = -3(x-4)^2 + 2\end{aligned}$$

$$\begin{aligned}(17) \quad & 2x^2 + 12x + 20 \\ & = 2(x+3)^2 + 2\end{aligned}$$

$$\begin{aligned}(27) \quad & -3x^2 + 12x - 12 \\ & = -3(x-2)^2\end{aligned}$$

$$\begin{aligned}(8) \quad & 2x^2 - 20x + 52 \\ & = 2(x-5)^2 + 2\end{aligned}$$

$$\begin{aligned}(18) \quad & 4x^2 + 40x + 95 \\ & = 4(x+5)^2 - 5\end{aligned}$$

$$\begin{aligned}(28) \quad & 4x^2 - 8x + 8 \\ & = 4(x-1)^2 + 4\end{aligned}$$

$$\begin{aligned}(9) \quad & -3x^2 - 30x - 72 \\ & = -3(x+5)^2 + 3\end{aligned}$$

$$\begin{aligned}(19) \quad & -5x^2 - 40x - 84 \\ & = -5(x+4)^2 - 4\end{aligned}$$

$$\begin{aligned}(29) \quad & -3x^2 - 30x - 70 \\ & = -3(x+5)^2 + 5\end{aligned}$$

$$\begin{aligned}(10) \quad & 3x^2 - 18x + 30 \\ & = 3(x-3)^2 + 3\end{aligned}$$

$$\begin{aligned}(20) \quad & x^2 - 6x + 14 \\ & = (x-3)^2 + 5\end{aligned}$$

$$\begin{aligned}(30) \quad & 3x^2 - 6x + 1 \\ & = 3(x-1)^2 - 2\end{aligned}$$

## 2.7 平方完成 (難しい)

2 次多項式を平方完成する問題。  
式に現れる数は有理数となる。

(1)  $-5x^2 - 4x - 4$

(6)  $2x^2 + x + 3$

(11)  $-3x^2 - 3x + 2$

(2)  $-2x^2 + 3x - 3$

(7)  $-5x^2 + 4x - 1$

(12)  $-4x^2 - 4x + 4$

(3)  $-x^2 - 2x - 5$

(8)  $-4x^2 + 5x - 1$

(13)  $x^2 - x - 3$

(4)  $-5x^2 + 2x$

(9)  $3x^2 - 5x + 5$

(14)  $-x^2 - 4x - 2$

(5)  $x^2 - 2x + 5$

(10)  $-3x^2 - 4x + 2$

(15)  $3x^2 + 3x - 5$

(1)  $2x^2 + 2x + 5$

(6)  $-x^2 + 4x - 1$

(11)  $-3x^2 - 2x + 1$

(2)  $-x^2 - 2x - 3$

(7)  $4x^2 - 3x - 2$

(12)  $3x^2 - 2x$

(3)  $5x^2 + 2x + 4$

(8)  $4x^2 - 4x + 3$

(13)  $x^2 - x + 3$

(4)  $-5x^2 - 2x - 1$

(9)  $-x^2 + 2x + 4$

(14)  $3x^2 + 3x + 1$

(5)  $-4x^2 - 2x$

(10)  $4x^2 - 3x - 2$

(15)  $-5x^2 + 4x + 2$

(1)  $-3x^2 + 2x + 5$

(6)  $-3x^2 + 2x + 4$

(11)  $-3x^2 - 5x - 4$

(2)  $x^2 - 3x - 1$

(7)  $-4x^2 - 3x - 5$

(12)  $-5x^2 - 3x$

(3)  $-2x^2 + 4x - 1$

(8)  $x^2 - x - 5$

(13)  $x^2 - 3x + 4$

(4)  $-2x^2 + 4x - 5$

(9)  $3x^2 + 5x + 1$

(14)  $-2x^2 - 5x + 3$

(5)  $-5x^2 + x - 2$

(10)  $-5x^2 + 2x - 1$

(15)  $-2x^2 + x + 1$

(1)  $-4x^2 - 4x - 3$

(6)  $x^2 - 2x + 5$

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(2)  $-5x^2 - 4x$

(7)  $x^2 + 3x + 2$

(12)  $5x^2 - 3x - 2$

(3)  $-3x^2 + 3x - 2$

(8)  $2x^2 - 4x - 4$

(13)  $2x^2 + x + 4$

(4)  $-x^2 - x$

(9)  $x^2 + 2x - 5$

(14)  $2x^2 - x$

(10)  $3x^2 + 4x - 5$

(5)  $-5x^2 + 3x$

(15)  $3x^2 - x + 5$

(1)  $5x^2 + x + 3$

(6)  $-4x^2 - 5x - 1$

(11)  $-x^2 + x - 4$

(2)  $5x^2 + x + 1$

(7)  $-3x^2 + 2x + 4$

(12)  $2x^2 - 5x + 3$

(3)  $x^2 - 3x + 1$

(8)  $-4x^2 + x + 1$

(13)  $-2x^2 + x + 2$

(4)  $-2x^2 - 2x + 2$

(9)  $-5x^2 + x$

(14)  $-3x^2 + 4x$

(5)  $-2x^2 - 5x + 4$

(10)  $3x^2 - 2x + 5$

(15)  $3x^2 - 5x + 5$

(1)  $-4x^2 + 4x - 5$

(6)  $3x^2 + 4x + 3$

(11)  $4x^2 - 5x + 5$

(2)  $-3x^2 + x + 4$

(7)  $3x^2 - 2x + 5$

(12)  $5x^2 + 2x - 2$

(3)  $-2x^2 + 2x + 5$

(8)  $5x^2 + 5x$

(13)  $5x^2 - 2x - 5$

(4)  $2x^2 - 3x + 4$

(9)  $2x^2 + x - 5$

(14)  $-2x^2 + 5x + 3$

(5)  $3x^2 + 3x - 3$

(10)  $-4x^2 + 5x - 5$

(15)  $2x^2 + 5x + 3$



(1)  $-5x^2 + 4x + 4$

(6)  $-5x^2 + x - 4$

(11)  $2x^2 + 5x - 3$

(2)  $-4x^2 + 5x + 1$

(7)  $5x^2 - 2x - 4$

(12)  $-2x^2 + x + 5$

(3)  $2x^2 - 4x + 2$

(8)  $-3x^2 + 5x - 4$

(13)  $-x^2 + 4x$

(4)  $2x^2 + x - 1$

(9)  $-3x^2 + 5x + 4$

(14)  $-5x^2 + 3x + 4$

(5)  $4x^2 - 3x + 1$

(10)  $-x^2 + 5x - 1$

(15)  $-5x^2 + 4x$

(1)  $-2x^2 + x + 4$

(6)  $3x^2 - 2x - 5$

(11)  $-4x^2 - 2x - 4$

(2)  $3x^2 + 5x - 5$

(7)  $-x^2 + 5x - 3$

(12)  $2x^2 + 5x - 2$

(3)  $-5x^2 + x - 3$

(8)  $-4x^2 - x - 1$

(13)  $-x^2 - x - 1$

(4)  $3x^2 + 2x + 4$

(9)  $3x^2 + x - 4$

(14)  $4x^2 + 5x - 1$

(5)  $-2x^2 - 4x$

(10)  $-x^2 - 5x - 4$

(15)  $-x^2 - x + 1$

(1)  $-2x^2 + 2x - 2$

(6)  $2x^2 + x + 2$

(11)  $-3x^2 + x + 5$

(2)  $-4x^2 - 5x + 3$

(7)  $-4x^2 - 3x + 1$

(12)  $3x^2 - x + 4$

(3)  $-x^2 - 2x + 4$

(8)  $-3x^2 - 2x - 4$

(13)  $2x^2 + 5x - 4$

(4)  $-3x^2 + 4x - 4$

(9)  $-x^2 - 3x - 2$

(14)  $-2x^2 - x - 1$

(5)  $-x^2 + 2x - 3$

(10)  $-x^2 - 3x + 2$

(15)  $4x^2 + 4x + 3$

(1)  $x^2 + x - 5$

(6)  $-3x^2 - 4x$

(11)  $-3x^2 - 3x + 4$

(2)  $5x^2 + x - 5$

(7)  $2x^2 + 4x - 1$

(12)  $-4x^2 - 2x - 3$

(3)  $-2x^2 - 2x + 1$

(8)  $x^2 + 5x - 2$

(13)  $-2x^2 + 4x - 4$

(4)  $2x^2 - x - 5$

(9)  $-3x^2 - 4x + 1$

(14)  $3x^2 + x + 3$

(5)  $x^2 - 5x - 4$

(10)  $-3x^2 + 3x - 1$

(15)  $5x^2 - 2x - 3$

(1)  $3x^2 + 5x + 2$

(6)  $5x^2 + 3x + 1$

(11)  $-4x^2 - x + 4$

(2)  $x^2 - 3x + 4$

(7)  $-x^2 - 2x - 5$

(12)  $-x^2 - 2x + 1$

(3)  $3x^2 + 4x - 3$

(8)  $-2x^2 + 4x$

(13)  $-x^2 + 4x - 5$

(4)  $-2x^2 + 3x + 3$

(9)  $4x^2 + 2x - 3$

(14)  $5x^2 - x - 5$

(5)  $2x^2 + 2x$

(10)  $x^2 + 5x - 4$

(15)  $-4x^2 + 5x - 4$

(1)  $-3x^2 + 5x - 2$

(6)  $-4x^2 + 4x - 5$

(11)  $5x^2 - 5x + 2$

(2)  $-2x^2 + 3x$

(7)  $5x^2 + x + 5$

(12)  $-4x^2 - x - 3$

(3)  $-2x^2 - 4x - 1$

(8)  $3x^2 - 2x - 2$

(13)  $x^2 + 4x + 4$

(4)  $4x^2 + 4x - 1$

(9)  $2x^2 - 5x - 2$

(14)  $-3x^2 + 5x + 1$

(5)  $-4x^2 - 4x - 5$

(10)  $-3x^2 - 2x + 1$

(15)  $5x^2 - 2x - 4$

(1)  $-4x^2 + 4x + 2$

(6)  $4x^2 - 4x + 2$

(11)  $-x^2 - x$

(2)  $5x^2 + 3x$

(7)  $4x^2 - 2x - 5$

(12)  $-4x^2 - 5x - 2$

(3)  $-5x^2 + 3x - 5$

(8)  $-x^2 + 3x - 5$

(13)  $-3x^2 - x + 5$

(4)  $x^2 + 5x + 5$

(9)  $4x^2 + 3x - 1$

(14)  $-3x^2 - x - 4$

(5)  $2x^2 + 3x - 3$

(10)  $-3x^2 - 3x - 5$

(15)  $x^2 - x + 5$

(1)  $-5x^2 + 4x + 1$

(6)  $5x^2 - x + 1$

(11)  $x^2 + 2x$

(2)  $-2x^2 - 2x - 1$

(7)  $-3x^2 - 3x - 2$

(12)  $5x^2 + 4x + 4$

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(13)  $3x^2 - 2x - 5$

(4)  $-2x^2 + 3x - 3$

(9)  $4x^2 - 4x + 3$

(14)  $2x^2 - 3x + 2$

(5)  $-x^2 - 3x - 4$

(10)  $3x^2 - 4x$

(15)  $-3x^2 + 4x - 3$



(1)  $-5x^2 - 3x + 4$

(6)  $5x^2 - x - 3$

(11)  $3x^2 - x + 5$

(2)  $-x^2 + 3x + 4$

(7)  $-5x^2 - 3x + 5$

(12)  $-3x^2 + 5x + 3$

(3)  $-3x^2 - 3x - 2$

(8)  $x^2 - 3x + 5$

(13)  $5x^2 - 2x + 2$

(4)  $-5x^2 + 3x - 4$

(9)  $-x^2 - 3x - 2$

(14)  $-4x^2 + 5x$

(5)  $5x^2 - x + 1$

(10)  $-2x^2 - x - 2$

(15)  $2x^2 + 3x + 5$

(1)  $3x^2 + 4x + 4$

(6)  $x^2 - 3x + 3$

(11)  $-5x^2 + x + 1$

(2)  $3x^2 - x - 1$

(7)  $-2x^2 + 5x + 2$

(12)  $5x^2 - x - 3$

(3)  $-3x^2 + 5x - 5$

(8)  $2x^2 + 4x + 4$

(13)  $x^2 + 3x + 1$

(4)  $5x^2 - 4x - 5$

(9)  $3x^2 + 5x - 3$

(14)  $-4x^2 + 2x - 2$

(5)  $2x^2 - 2x - 4$

(10)  $-4x^2 + 3x - 3$

(15)  $2x^2 - 2x + 3$

(1)  $3x^2 + 2x + 1$

(6)  $4x^2 - 5x - 3$

(11)  $-5x^2 + x - 4$

(2)  $3x^2 - x - 4$

(7)  $4x^2 - 4x - 4$

(12)  $3x^2 + 3x - 1$

(3)  $-4x^2 + 4x + 5$

(8)  $-4x^2 + 2x$

(13)  $-x^2 + 3x - 4$

(4)  $-4x^2 + 5x$

(9)  $5x^2 + 3x + 1$

(14)  $4x^2 - 3x$

(5)  $3x^2 + x + 3$

(10)  $x^2 - x + 1$

(15)  $-4x^2 - x - 1$

(1)  $2x^2 - 3x$

(6)  $2x^2 + 5x - 1$

(11)  $-x^2 + x - 3$

(2)  $x^2 + 5x - 3$

(7)  $-2x^2 + 2x$

(12)  $2x^2 - x - 4$

(3)  $3x^2 + 2x + 5$

(8)  $-3x^2 + 5x + 3$

(13)  $-4x^2 - 5x - 3$

(4)  $-5x^2 + 4x - 5$

(9)  $-4x^2 - x - 2$

(14)  $-2x^2 + 3x - 3$

(5)  $4x^2 - 3x + 3$

(10)  $3x^2 - 2x - 2$

(15)  $-5x^2 + 4x + 5$

(1)  $2x^2 + 3x + 2$

(6)  $x^2 + 3x + 5$

(11)  $3x^2 + 4x - 5$

(2)  $3x^2 + 5x + 3$

(7)  $5x^2 - 2x + 5$

(12)  $-x^2 - 5x - 4$

(3)  $-x^2 - 3x + 4$

(8)  $-3x^2 + 5x - 3$

(13)  $3x^2 + x$

(4)  $-x^2 - 2x + 1$

(9)  $-3x^2 + 3x + 2$

(14)  $4x^2 + 4x - 4$

(5)  $x^2 + 5x - 2$

(10)  $-2x^2 + 3x + 3$

(15)  $-x^2 + 5x + 5$

(1)  $x^2 + 3x + 3$

(6)  $-x^2 + 5x + 1$

(11)  $-x^2 - 3x - 1$

(2)  $x^2 + x - 5$

(7)  $-3x^2 - 3x$

(12)  $5x^2 - 4x - 1$

(3)  $5x^2 + 4x + 1$

(8)  $3x^2 + 4x - 5$

(13)  $-5x^2 + 5x + 2$

(4)  $-2x^2 - 2x - 1$

(9)  $4x^2 + 4x$

(14)  $-4x^2 - x + 5$

(5)  $2x^2 + 2x + 4$

(10)  $3x^2 - x - 4$

(15)  $-5x^2 + 2x + 2$

$$\begin{aligned}(1) \quad & -5x^2 - 4x - 4 \\ & = -5 \left( x + \frac{2}{5} \right)^2 - \frac{16}{5}\end{aligned}$$

$$\begin{aligned}(6) \quad & 2x^2 + x + 3 \\ & = 2 \left( x + \frac{1}{4} \right)^2 + \frac{23}{8}\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 - 3x + 2 \\ & = -3 \left( x + \frac{1}{2} \right)^2 + \frac{11}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & -2x^2 + 3x - 3 \\ & = -2 \left( x - \frac{3}{4} \right)^2 - \frac{15}{8}\end{aligned}$$

$$\begin{aligned}(7) \quad & -5x^2 + 4x - 1 \\ & = -5 \left( x - \frac{2}{5} \right)^2 - \frac{1}{5}\end{aligned}$$

$$\begin{aligned}(12) \quad & -4x^2 - 4x + 4 \\ & = -4 \left( x + \frac{1}{2} \right)^2 + 5\end{aligned}$$

$$\begin{aligned}(3) \quad & -x^2 - 2x - 5 \\ & = -(x + 1)^2 - 4\end{aligned}$$

$$\begin{aligned}(8) \quad & -4x^2 + 5x - 1 \\ & = -4 \left( x - \frac{5}{8} \right)^2 + \frac{9}{16}\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 - x - 3 \\ & = \left( x - \frac{1}{2} \right)^2 - \frac{13}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & -5x^2 + 2x \\ & = -5 \left( x - \frac{1}{5} \right)^2 + \frac{1}{5}\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 - 5x + 5 \\ & = 3 \left( x - \frac{5}{6} \right)^2 + \frac{35}{12}\end{aligned}$$

$$\begin{aligned}(14) \quad & -x^2 - 4x - 2 \\ & = -(x + 2)^2 + 2\end{aligned}$$

$$\begin{aligned}(5) \quad & x^2 - 2x + 5 \\ & = (x - 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(10) \quad & -3x^2 - 4x + 2 \\ & = -3 \left( x + \frac{2}{3} \right)^2 + \frac{10}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & 3x^2 + 3x - 5 \\ & = 3 \left( x + \frac{1}{2} \right)^2 - \frac{23}{4}\end{aligned}$$

$$\begin{aligned}(1) \quad & 2x^2 + 2x + 5 \\ &= 2 \left( x + \frac{1}{2} \right)^2 + \frac{9}{2}\end{aligned}$$

$$\begin{aligned}(6) \quad & -x^2 + 4x - 1 \\ &= -(x - 2)^2 + 3\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 - 2x + 1 \\ &= -3 \left( x + \frac{1}{3} \right)^2 + \frac{4}{3}\end{aligned}$$

$$\begin{aligned}(2) \quad & -x^2 - 2x - 3 \\ &= -(x + 1)^2 - 2\end{aligned}$$

$$\begin{aligned}(7) \quad & 4x^2 - 3x - 2 \\ &= 4 \left( x - \frac{3}{8} \right)^2 - \frac{41}{16}\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 - 2x \\ &= 3 \left( x - \frac{1}{3} \right)^2 - \frac{1}{3}\end{aligned}$$

$$\begin{aligned}(3) \quad & 5x^2 + 2x + 4 \\ &= 5 \left( x + \frac{1}{5} \right)^2 + \frac{19}{5}\end{aligned}$$

$$\begin{aligned}(8) \quad & 4x^2 - 4x + 3 \\ &= 4 \left( x - \frac{1}{2} \right)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 - x + 3 \\ &= \left( x - \frac{1}{2} \right)^2 + \frac{11}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & -5x^2 - 2x - 1 \\ &= -5 \left( x + \frac{1}{5} \right)^2 - \frac{4}{5}\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 + 2x + 4 \\ &= -(x - 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(14) \quad & 3x^2 + 3x + 1 \\ &= 3 \left( x + \frac{1}{2} \right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(5) \quad & -4x^2 - 2x \\ &= -4 \left( x + \frac{1}{4} \right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(10) \quad & 4x^2 - 3x - 2 \\ &= 4 \left( x - \frac{3}{8} \right)^2 - \frac{41}{16}\end{aligned}$$

$$\begin{aligned}(15) \quad & -5x^2 + 4x + 2 \\ &= -5 \left( x - \frac{2}{5} \right)^2 + \frac{14}{5}\end{aligned}$$



$$\begin{aligned}(1) \quad & -3x^2 + 2x + 5 \\ & = -3\left(x - \frac{1}{3}\right)^2 + \frac{16}{3}\end{aligned}$$

$$\begin{aligned}(6) \quad & -3x^2 + 2x + 4 \\ & = -3\left(x - \frac{1}{3}\right)^2 + \frac{13}{3}\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 - 5x - 4 \\ & = -3\left(x + \frac{5}{6}\right)^2 - \frac{23}{12}\end{aligned}$$

$$\begin{aligned}(2) \quad & x^2 - 3x - 1 \\ & = \left(x - \frac{3}{2}\right)^2 - \frac{13}{4}\end{aligned}$$

$$\begin{aligned}(7) \quad & -4x^2 - 3x - 5 \\ & = -4\left(x + \frac{3}{8}\right)^2 - \frac{71}{16}\end{aligned}$$

$$\begin{aligned}(12) \quad & -5x^2 - 3x \\ & = -5\left(x + \frac{3}{10}\right)^2 + \frac{9}{20}\end{aligned}$$

$$\begin{aligned}(3) \quad & -2x^2 + 4x - 1 \\ & = -2(x - 1)^2 + 1\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 - x - 5 \\ & = \left(x - \frac{1}{2}\right)^2 - \frac{21}{4}\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 - 3x + 4 \\ & = \left(x - \frac{3}{2}\right)^2 + \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & -2x^2 + 4x - 5 \\ & = -2(x - 1)^2 - 3\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 + 5x + 1 \\ & = 3\left(x + \frac{5}{6}\right)^2 - \frac{13}{12}\end{aligned}$$

$$\begin{aligned}(14) \quad & -2x^2 - 5x + 3 \\ & = -2\left(x + \frac{5}{4}\right)^2 + \frac{49}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & -5x^2 + x - 2 \\ & = -5\left(x - \frac{1}{10}\right)^2 - \frac{39}{20}\end{aligned}$$

$$\begin{aligned}(10) \quad & -5x^2 + 2x - 1 \\ & = -5\left(x - \frac{1}{5}\right)^2 - \frac{4}{5}\end{aligned}$$

$$\begin{aligned}(15) \quad & -2x^2 + x + 1 \\ & = -2\left(x - \frac{1}{4}\right)^2 + \frac{9}{8}\end{aligned}$$

$$\begin{aligned}(1) \quad & -4x^2 - 4x - 3 \\ & = -4\left(x + \frac{1}{2}\right)^2 - 2\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 - 2x + 5 \\ & = (x - 1)^2 + 4\end{aligned}$$

$$\begin{aligned}(11) \quad & -2x^2 + 3x - 2 \\ & = -2\left(x - \frac{3}{4}\right)^2 - \frac{7}{8}\end{aligned}$$

$$\begin{aligned}(2) \quad & -5x^2 - 4x \\ & = -5\left(x + \frac{2}{5}\right)^2 + \frac{4}{5}\end{aligned}$$

$$\begin{aligned}(7) \quad & x^2 + 3x + 2 \\ & = \left(x + \frac{3}{2}\right)^2 - \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 - 3x - 2 \\ & = 5\left(x - \frac{3}{10}\right)^2 - \frac{49}{20}\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 + 3x - 2 \\ & = -3\left(x - \frac{1}{2}\right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(8) \quad & 2x^2 - 4x - 4 \\ & = 2(x - 1)^2 - 6\end{aligned}$$

$$\begin{aligned}(13) \quad & 2x^2 + x + 4 \\ & = 2\left(x + \frac{1}{4}\right)^2 + \frac{31}{8}\end{aligned}$$

$$\begin{aligned}(4) \quad & -x^2 - x \\ & = -\left(x + \frac{1}{2}\right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(9) \quad & x^2 + 2x - 5 \\ & = (x + 1)^2 - 6\end{aligned}$$

$$\begin{aligned}(14) \quad & 2x^2 - x \\ & = 2\left(x - \frac{1}{4}\right)^2 - \frac{1}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & -5x^2 + 3x \\ & = -5\left(x - \frac{3}{10}\right)^2 + \frac{9}{20}\end{aligned}$$

$$\begin{aligned}(10) \quad & 3x^2 + 4x - 5 \\ & = 3\left(x + \frac{2}{3}\right)^2 - \frac{19}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & 3x^2 - x + 5 \\ & = 3\left(x - \frac{1}{6}\right)^2 + \frac{59}{12}\end{aligned}$$

$$(1) \quad 5x^2 + x + 3 \\ = 5 \left( x + \frac{1}{10} \right)^2 + \frac{59}{20}$$

$$(6) \quad -4x^2 - 5x - 1 \\ = -4 \left( x + \frac{5}{8} \right)^2 + \frac{9}{16}$$

$$(11) \quad -x^2 + x - 4 \\ = - \left( x - \frac{1}{2} \right)^2 - \frac{15}{4}$$

$$(2) \quad 5x^2 + x + 1 \\ = 5 \left( x + \frac{1}{10} \right)^2 + \frac{19}{20}$$

$$(7) \quad -3x^2 + 2x + 4 \\ = -3 \left( x - \frac{1}{3} \right)^2 + \frac{13}{3}$$

$$(12) \quad 2x^2 - 5x + 3 \\ = 2 \left( x - \frac{5}{4} \right)^2 - \frac{1}{8}$$

$$(3) \quad x^2 - 3x + 1 \\ = \left( x - \frac{3}{2} \right)^2 - \frac{5}{4}$$

$$(8) \quad -4x^2 + x + 1 \\ = -4 \left( x - \frac{1}{8} \right)^2 + \frac{17}{16}$$

$$(13) \quad -2x^2 + x + 2 \\ = -2 \left( x - \frac{1}{4} \right)^2 + \frac{17}{8}$$

$$(4) \quad -2x^2 - 2x + 2 \\ = -2 \left( x + \frac{1}{2} \right)^2 + \frac{5}{2}$$

$$(9) \quad -5x^2 + x \\ = -5 \left( x - \frac{1}{10} \right)^2 + \frac{1}{20}$$

$$(14) \quad -3x^2 + 4x \\ = -3 \left( x - \frac{2}{3} \right)^2 + \frac{4}{3}$$

$$(5) \quad -2x^2 - 5x + 4 \\ = -2 \left( x + \frac{5}{4} \right)^2 + \frac{57}{8}$$

$$(10) \quad 3x^2 - 2x + 5 \\ = 3 \left( x - \frac{1}{3} \right)^2 + \frac{14}{3}$$

$$(15) \quad 3x^2 - 5x + 5 \\ = 3 \left( x - \frac{5}{6} \right)^2 + \frac{35}{12}$$

$$\begin{aligned}(1) \quad & -4x^2 + 4x - 5 \\ & = -4 \left( x - \frac{1}{2} \right)^2 - 4\end{aligned}$$

$$\begin{aligned}(6) \quad & 3x^2 + 4x + 3 \\ & = 3 \left( x + \frac{2}{3} \right)^2 + \frac{5}{3}\end{aligned}$$

$$\begin{aligned}(11) \quad & 4x^2 - 5x + 5 \\ & = 4 \left( x - \frac{5}{8} \right)^2 + \frac{55}{16}\end{aligned}$$

$$\begin{aligned}(2) \quad & -3x^2 + x + 4 \\ & = -3 \left( x - \frac{1}{6} \right)^2 + \frac{49}{12}\end{aligned}$$

$$\begin{aligned}(7) \quad & 3x^2 - 2x + 5 \\ & = 3 \left( x - \frac{1}{3} \right)^2 + \frac{14}{3}\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 + 2x - 2 \\ & = 5 \left( x + \frac{1}{5} \right)^2 - \frac{11}{5}\end{aligned}$$

$$\begin{aligned}(3) \quad & -2x^2 + 2x + 5 \\ & = -2 \left( x - \frac{1}{2} \right)^2 + \frac{11}{2}\end{aligned}$$

$$\begin{aligned}(8) \quad & 5x^2 + 5x \\ & = 5 \left( x + \frac{1}{2} \right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(13) \quad & 5x^2 - 2x - 5 \\ & = 5 \left( x - \frac{1}{5} \right)^2 - \frac{26}{5}\end{aligned}$$

$$\begin{aligned}(4) \quad & 2x^2 - 3x + 4 \\ & = 2 \left( x - \frac{3}{4} \right)^2 + \frac{23}{8}\end{aligned}$$

$$\begin{aligned}(9) \quad & 2x^2 + x - 5 \\ & = 2 \left( x + \frac{1}{4} \right)^2 - \frac{41}{8}\end{aligned}$$

$$\begin{aligned}(14) \quad & -2x^2 + 5x + 3 \\ & = -2 \left( x - \frac{5}{4} \right)^2 + \frac{49}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & 3x^2 + 3x - 3 \\ & = 3 \left( x + \frac{1}{2} \right)^2 - \frac{15}{4}\end{aligned}$$

$$\begin{aligned}(10) \quad & -4x^2 + 5x - 5 \\ & = -4 \left( x - \frac{5}{8} \right)^2 - \frac{55}{16}\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 + 5x + 3 \\ & = 2 \left( x + \frac{5}{4} \right)^2 - \frac{1}{8}\end{aligned}$$

$$\begin{aligned}(1) \quad & -5x^2 + 4x + 4 \\ & = -5\left(x - \frac{2}{5}\right)^2 + \frac{24}{5}\end{aligned}$$

$$\begin{aligned}(6) \quad & -5x^2 + x - 4 \\ & = -5\left(x - \frac{1}{10}\right)^2 - \frac{79}{20}\end{aligned}$$

$$\begin{aligned}(11) \quad & 2x^2 + 5x - 3 \\ & = 2\left(x + \frac{5}{4}\right)^2 - \frac{49}{8}\end{aligned}$$

$$\begin{aligned}(2) \quad & -4x^2 + 5x + 1 \\ & = -4\left(x - \frac{5}{8}\right)^2 + \frac{41}{16}\end{aligned}$$

$$\begin{aligned}(7) \quad & 5x^2 - 2x - 4 \\ & = 5\left(x - \frac{1}{5}\right)^2 - \frac{21}{5}\end{aligned}$$

$$\begin{aligned}(12) \quad & -2x^2 + x + 5 \\ & = -2\left(x - \frac{1}{4}\right)^2 + \frac{41}{8}\end{aligned}$$

$$\begin{aligned}(3) \quad & 2x^2 - 4x + 2 \\ & = 2(x - 1)^2\end{aligned}$$

$$\begin{aligned}(8) \quad & -3x^2 + 5x - 4 \\ & = -3\left(x - \frac{5}{6}\right)^2 - \frac{23}{12}\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 + 4x \\ & = -(x - 2)^2 + 4\end{aligned}$$

$$\begin{aligned}(4) \quad & 2x^2 + x - 1 \\ & = 2\left(x + \frac{1}{4}\right)^2 - \frac{9}{8}\end{aligned}$$

$$\begin{aligned}(9) \quad & -3x^2 + 5x + 4 \\ & = -3\left(x - \frac{5}{6}\right)^2 + \frac{73}{12}\end{aligned}$$

$$\begin{aligned}(14) \quad & -5x^2 + 3x + 4 \\ & = -5\left(x - \frac{3}{10}\right)^2 + \frac{89}{20}\end{aligned}$$

$$\begin{aligned}(5) \quad & 4x^2 - 3x + 1 \\ & = 4\left(x - \frac{3}{8}\right)^2 + \frac{7}{16}\end{aligned}$$

$$\begin{aligned}(10) \quad & -x^2 + 5x - 1 \\ & = -\left(x - \frac{5}{2}\right)^2 + \frac{21}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & -5x^2 + 4x \\ & = -5\left(x - \frac{2}{5}\right)^2 + \frac{4}{5}\end{aligned}$$

$$\begin{aligned}(1) \quad & -2x^2 + x + 4 \\ & = -2\left(x - \frac{1}{4}\right)^2 + \frac{33}{8}\end{aligned}$$

$$\begin{aligned}(6) \quad & 3x^2 - 2x - 5 \\ & = 3\left(x - \frac{1}{3}\right)^2 - \frac{16}{3}\end{aligned}$$

$$\begin{aligned}(11) \quad & -4x^2 - 2x - 4 \\ & = -4\left(x + \frac{1}{4}\right)^2 - \frac{15}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & 3x^2 + 5x - 5 \\ & = 3\left(x + \frac{5}{6}\right)^2 - \frac{85}{12}\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 + 5x - 3 \\ & = -\left(x - \frac{5}{2}\right)^2 + \frac{13}{4}\end{aligned}$$

$$\begin{aligned}(12) \quad & 2x^2 + 5x - 2 \\ & = 2\left(x + \frac{5}{4}\right)^2 - \frac{41}{8}\end{aligned}$$

$$\begin{aligned}(3) \quad & -5x^2 + x - 3 \\ & = -5\left(x - \frac{1}{10}\right)^2 - \frac{59}{20}\end{aligned}$$

$$\begin{aligned}(8) \quad & -4x^2 - x - 1 \\ & = -4\left(x + \frac{1}{8}\right)^2 - \frac{15}{16}\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 - x - 1 \\ & = -\left(x + \frac{1}{2}\right)^2 - \frac{3}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & 3x^2 + 2x + 4 \\ & = 3\left(x + \frac{1}{3}\right)^2 + \frac{11}{3}\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 + x - 4 \\ & = 3\left(x + \frac{1}{6}\right)^2 - \frac{49}{12}\end{aligned}$$

$$\begin{aligned}(14) \quad & 4x^2 + 5x - 1 \\ & = 4\left(x + \frac{5}{8}\right)^2 - \frac{41}{16}\end{aligned}$$

$$\begin{aligned}(5) \quad & -2x^2 - 4x \\ & = -2(x + 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(10) \quad & -x^2 - 5x - 4 \\ & = -\left(x + \frac{5}{2}\right)^2 + \frac{9}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & -x^2 - x + 1 \\ & = -\left(x + \frac{1}{2}\right)^2 + \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(1) \quad & -2x^2 + 2x - 2 \\ & = -2 \left( x - \frac{1}{2} \right)^2 - \frac{3}{2}\end{aligned}$$

$$\begin{aligned}(6) \quad & 2x^2 + x + 2 \\ & = 2 \left( x + \frac{1}{4} \right)^2 + \frac{15}{8}\end{aligned}$$

$$\begin{aligned}(11) \quad & -3x^2 + x + 5 \\ & = -3 \left( x - \frac{1}{6} \right)^2 + \frac{61}{12}\end{aligned}$$

$$\begin{aligned}(2) \quad & -4x^2 - 5x + 3 \\ & = -4 \left( x + \frac{5}{8} \right)^2 + \frac{73}{16}\end{aligned}$$

$$\begin{aligned}(7) \quad & -4x^2 - 3x + 1 \\ & = -4 \left( x + \frac{3}{8} \right)^2 + \frac{25}{16}\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 - x + 4 \\ & = 3 \left( x - \frac{1}{6} \right)^2 + \frac{47}{12}\end{aligned}$$

$$\begin{aligned}(3) \quad & -x^2 - 2x + 4 \\ & = -(x + 1)^2 + 5\end{aligned}$$

$$\begin{aligned}(8) \quad & -3x^2 - 2x - 4 \\ & = -3 \left( x + \frac{1}{3} \right)^2 - \frac{11}{3}\end{aligned}$$

$$\begin{aligned}(13) \quad & 2x^2 + 5x - 4 \\ & = 2 \left( x + \frac{5}{4} \right)^2 - \frac{57}{8}\end{aligned}$$

$$\begin{aligned}(4) \quad & -3x^2 + 4x - 4 \\ & = -3 \left( x - \frac{2}{3} \right)^2 - \frac{8}{3}\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 - 3x - 2 \\ & = - \left( x + \frac{3}{2} \right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(14) \quad & -2x^2 - x - 1 \\ & = -2 \left( x + \frac{1}{4} \right)^2 - \frac{7}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 + 2x - 3 \\ & = -(x - 1)^2 - 2\end{aligned}$$

$$\begin{aligned}(10) \quad & -x^2 - 3x + 2 \\ & = - \left( x + \frac{3}{2} \right)^2 + \frac{17}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & 4x^2 + 4x + 3 \\ & = 4 \left( x + \frac{1}{2} \right)^2 + 2\end{aligned}$$

$$(1) \quad x^2 + x - 5 \\ = \left(x + \frac{1}{2}\right)^2 - \frac{21}{4}$$

$$(6) \quad -3x^2 - 4x \\ = -3\left(x + \frac{2}{3}\right)^2 + \frac{4}{3}$$

$$(11) \quad -3x^2 - 3x + 4 \\ = -3\left(x + \frac{1}{2}\right)^2 + \frac{19}{4}$$

$$(2) \quad 5x^2 + x - 5 \\ = 5\left(x + \frac{1}{10}\right)^2 - \frac{101}{20}$$

$$(7) \quad 2x^2 + 4x - 1 \\ = 2(x + 1)^2 - 3$$

$$(12) \quad -4x^2 - 2x - 3 \\ = -4\left(x + \frac{1}{4}\right)^2 - \frac{11}{4}$$

$$(3) \quad -2x^2 - 2x + 1 \\ = -2\left(x + \frac{1}{2}\right)^2 + \frac{3}{2}$$

$$(8) \quad x^2 + 5x - 2 \\ = \left(x + \frac{5}{2}\right)^2 - \frac{33}{4}$$

$$(13) \quad -2x^2 + 4x - 4 \\ = -2(x - 1)^2 - 2$$

$$(4) \quad 2x^2 - x - 5 \\ = 2\left(x - \frac{1}{4}\right)^2 - \frac{41}{8}$$

$$(9) \quad -3x^2 - 4x + 1 \\ = -3\left(x + \frac{2}{3}\right)^2 + \frac{7}{3}$$

$$(14) \quad 3x^2 + x + 3 \\ = 3\left(x + \frac{1}{6}\right)^2 + \frac{35}{12}$$

$$(5) \quad x^2 - 5x - 4 \\ = \left(x - \frac{5}{2}\right)^2 - \frac{41}{4}$$

$$(10) \quad -3x^2 + 3x - 1 \\ = -3\left(x - \frac{1}{2}\right)^2 - \frac{1}{4}$$

$$(15) \quad 5x^2 - 2x - 3 \\ = 5\left(x - \frac{1}{5}\right)^2 - \frac{16}{5}$$



$$\begin{aligned}(1) \quad & 3x^2 + 5x + 2 \\ &= 3 \left( x + \frac{5}{6} \right)^2 - \frac{1}{12}\end{aligned}$$

$$\begin{aligned}(6) \quad & 5x^2 + 3x + 1 \\ &= 5 \left( x + \frac{3}{10} \right)^2 + \frac{11}{20}\end{aligned}$$

$$\begin{aligned}(11) \quad & -4x^2 - x + 4 \\ &= -4 \left( x + \frac{1}{8} \right)^2 + \frac{65}{16}\end{aligned}$$

$$\begin{aligned}(2) \quad & x^2 - 3x + 4 \\ &= \left( x - \frac{3}{2} \right)^2 + \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(7) \quad & -x^2 - 2x - 5 \\ &= -(x + 1)^2 - 4\end{aligned}$$

$$\begin{aligned}(12) \quad & -x^2 - 2x + 1 \\ &= -(x + 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(3) \quad & 3x^2 + 4x - 3 \\ &= 3 \left( x + \frac{2}{3} \right)^2 - \frac{13}{3}\end{aligned}$$

$$\begin{aligned}(8) \quad & -2x^2 + 4x \\ &= -2(x - 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 + 4x - 5 \\ &= -(x - 2)^2 - 1\end{aligned}$$

$$\begin{aligned}(4) \quad & -2x^2 + 3x + 3 \\ &= -2 \left( x - \frac{3}{4} \right)^2 + \frac{33}{8}\end{aligned}$$

$$\begin{aligned}(9) \quad & 4x^2 + 2x - 3 \\ &= 4 \left( x + \frac{1}{4} \right)^2 - \frac{13}{4}\end{aligned}$$

$$\begin{aligned}(14) \quad & 5x^2 - x - 5 \\ &= 5 \left( x - \frac{1}{10} \right)^2 - \frac{101}{20}\end{aligned}$$

$$\begin{aligned}(5) \quad & 2x^2 + 2x \\ &= 2 \left( x + \frac{1}{2} \right)^2 - \frac{1}{2}\end{aligned}$$

$$\begin{aligned}(10) \quad & x^2 + 5x - 4 \\ &= \left( x + \frac{5}{2} \right)^2 - \frac{41}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 + 5x - 4 \\ &= -4 \left( x - \frac{5}{8} \right)^2 - \frac{39}{16}\end{aligned}$$

$$\begin{aligned}(1) \quad & -3x^2 + 5x - 2 \\ & = -3\left(x - \frac{5}{6}\right)^2 + \frac{1}{12}\end{aligned}$$

$$\begin{aligned}(6) \quad & -4x^2 + 4x - 5 \\ & = -4\left(x - \frac{1}{2}\right)^2 - 4\end{aligned}$$

$$\begin{aligned}(11) \quad & 5x^2 - 5x + 2 \\ & = 5\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & -2x^2 + 3x \\ & = -2\left(x - \frac{3}{4}\right)^2 + \frac{9}{8}\end{aligned}$$

$$\begin{aligned}(7) \quad & 5x^2 + x + 5 \\ & = 5\left(x + \frac{1}{10}\right)^2 + \frac{99}{20}\end{aligned}$$

$$\begin{aligned}(12) \quad & -4x^2 - x - 3 \\ & = -4\left(x + \frac{1}{8}\right)^2 - \frac{47}{16}\end{aligned}$$

$$\begin{aligned}(3) \quad & -2x^2 - 4x - 1 \\ & = -2(x + 1)^2 + 1\end{aligned}$$

$$\begin{aligned}(8) \quad & 3x^2 - 2x - 2 \\ & = 3\left(x - \frac{1}{3}\right)^2 - \frac{7}{3}\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 + 4x + 4 \\ & = (x + 2)^2\end{aligned}$$

$$\begin{aligned}(4) \quad & 4x^2 + 4x - 1 \\ & = 4\left(x + \frac{1}{2}\right)^2 - 2\end{aligned}$$

$$\begin{aligned}(9) \quad & 2x^2 - 5x - 2 \\ & = 2\left(x - \frac{5}{4}\right)^2 - \frac{41}{8}\end{aligned}$$

$$\begin{aligned}(14) \quad & -3x^2 + 5x + 1 \\ & = -3\left(x - \frac{5}{6}\right)^2 + \frac{37}{12}\end{aligned}$$

$$\begin{aligned}(5) \quad & -4x^2 - 4x - 5 \\ & = -4\left(x + \frac{1}{2}\right)^2 - 4\end{aligned}$$

$$\begin{aligned}(10) \quad & -3x^2 - 2x + 1 \\ & = -3\left(x + \frac{1}{3}\right)^2 + \frac{4}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & 5x^2 - 2x - 4 \\ & = 5\left(x - \frac{1}{5}\right)^2 - \frac{21}{5}\end{aligned}$$

$$\begin{aligned}(1) \quad & -4x^2 + 4x + 2 \\ & = -4\left(x - \frac{1}{2}\right)^2 + 3\end{aligned}$$

$$\begin{aligned}(6) \quad & 4x^2 - 4x + 2 \\ & = 4\left(x - \frac{1}{2}\right)^2 + 1\end{aligned}$$

$$\begin{aligned}(11) \quad & -x^2 - x \\ & = -\left(x + \frac{1}{2}\right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & 5x^2 + 3x \\ & = 5\left(x + \frac{3}{10}\right)^2 - \frac{9}{20}\end{aligned}$$

$$\begin{aligned}(7) \quad & 4x^2 - 2x - 5 \\ & = 4\left(x - \frac{1}{4}\right)^2 - \frac{21}{4}\end{aligned}$$

$$\begin{aligned}(12) \quad & -4x^2 - 5x - 2 \\ & = -4\left(x + \frac{5}{8}\right)^2 - \frac{7}{16}\end{aligned}$$

$$\begin{aligned}(3) \quad & -5x^2 + 3x - 5 \\ & = -5\left(x - \frac{3}{10}\right)^2 - \frac{91}{20}\end{aligned}$$

$$\begin{aligned}(8) \quad & -x^2 + 3x - 5 \\ & = -\left(x - \frac{3}{2}\right)^2 - \frac{11}{4}\end{aligned}$$

$$\begin{aligned}(13) \quad & -3x^2 - x + 5 \\ & = -3\left(x + \frac{1}{6}\right)^2 + \frac{61}{12}\end{aligned}$$

$$\begin{aligned}(4) \quad & x^2 + 5x + 5 \\ & = \left(x + \frac{5}{2}\right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(9) \quad & 4x^2 + 3x - 1 \\ & = 4\left(x + \frac{3}{8}\right)^2 - \frac{25}{16}\end{aligned}$$

$$\begin{aligned}(14) \quad & -3x^2 - x - 4 \\ & = -3\left(x + \frac{1}{6}\right)^2 - \frac{47}{12}\end{aligned}$$

$$\begin{aligned}(5) \quad & 2x^2 + 3x - 3 \\ & = 2\left(x + \frac{3}{4}\right)^2 - \frac{33}{8}\end{aligned}$$

$$\begin{aligned}(10) \quad & -3x^2 - 3x - 5 \\ & = -3\left(x + \frac{1}{2}\right)^2 - \frac{17}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & x^2 - x + 5 \\ & = \left(x - \frac{1}{2}\right)^2 + \frac{19}{4}\end{aligned}$$

$$\begin{aligned}(1) \quad & -5x^2 + 4x + 1 \\ & = -5 \left( x - \frac{2}{5} \right)^2 + \frac{9}{5}\end{aligned}$$

$$\begin{aligned}(6) \quad & 5x^2 - x + 1 \\ & = 5 \left( x - \frac{1}{10} \right)^2 + \frac{19}{20}\end{aligned}$$

$$\begin{aligned}(11) \quad & x^2 + 2x \\ & = (x + 1)^2 - 1\end{aligned}$$

$$\begin{aligned}(2) \quad & -2x^2 - 2x - 1 \\ & = -2 \left( x + \frac{1}{2} \right)^2 - \frac{1}{2}\end{aligned}$$

$$\begin{aligned}(7) \quad & -3x^2 - 3x - 2 \\ & = -3 \left( x + \frac{1}{2} \right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 + 4x + 4 \\ & = 5 \left( x + \frac{2}{5} \right)^2 + \frac{16}{5}\end{aligned}$$

$$\begin{aligned}(3) \quad & 3x^2 + 2x \\ & = 3 \left( x + \frac{1}{3} \right)^2 - \frac{1}{3}\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 - 4x \\ & = (x - 2)^2 - 4\end{aligned}$$

$$\begin{aligned}(13) \quad & 3x^2 - 2x - 5 \\ & = 3 \left( x - \frac{1}{3} \right)^2 - \frac{16}{3}\end{aligned}$$

$$\begin{aligned}(4) \quad & -2x^2 + 3x - 3 \\ & = -2 \left( x - \frac{3}{4} \right)^2 - \frac{15}{8}\end{aligned}$$

$$\begin{aligned}(9) \quad & 4x^2 - 4x + 3 \\ & = 4 \left( x - \frac{1}{2} \right)^2 + 2\end{aligned}$$

$$\begin{aligned}(14) \quad & 2x^2 - 3x + 2 \\ & = 2 \left( x - \frac{3}{4} \right)^2 + \frac{7}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & -x^2 - 3x - 4 \\ & = - \left( x + \frac{3}{2} \right)^2 - \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(10) \quad & 3x^2 - 4x \\ & = 3 \left( x - \frac{2}{3} \right)^2 - \frac{4}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & -3x^2 + 4x - 3 \\ & = -3 \left( x - \frac{2}{3} \right)^2 - \frac{5}{3}\end{aligned}$$

$$\begin{aligned}(1) \quad & -5x^2 - 3x + 4 \\ & = -5 \left( x + \frac{3}{10} \right)^2 + \frac{89}{20}\end{aligned}$$

$$\begin{aligned}(6) \quad & 5x^2 - x - 3 \\ & = 5 \left( x - \frac{1}{10} \right)^2 - \frac{61}{20}\end{aligned}$$

$$\begin{aligned}(11) \quad & 3x^2 - x + 5 \\ & = 3 \left( x - \frac{1}{6} \right)^2 + \frac{59}{12}\end{aligned}$$

$$\begin{aligned}(2) \quad & -x^2 + 3x + 4 \\ & = - \left( x - \frac{3}{2} \right)^2 + \frac{25}{4}\end{aligned}$$

$$\begin{aligned}(7) \quad & -5x^2 - 3x + 5 \\ & = -5 \left( x + \frac{3}{10} \right)^2 + \frac{109}{20}\end{aligned}$$

$$\begin{aligned}(12) \quad & -3x^2 + 5x + 3 \\ & = -3 \left( x - \frac{5}{6} \right)^2 + \frac{61}{12}\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 - 3x - 2 \\ & = -3 \left( x + \frac{1}{2} \right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(8) \quad & x^2 - 3x + 5 \\ & = \left( x - \frac{3}{2} \right)^2 + \frac{11}{4}\end{aligned}$$

$$\begin{aligned}(13) \quad & 5x^2 - 2x + 2 \\ & = 5 \left( x - \frac{1}{5} \right)^2 + \frac{9}{5}\end{aligned}$$

$$\begin{aligned}(4) \quad & -5x^2 + 3x - 4 \\ & = -5 \left( x - \frac{3}{10} \right)^2 - \frac{71}{20}\end{aligned}$$

$$\begin{aligned}(9) \quad & -x^2 - 3x - 2 \\ & = - \left( x + \frac{3}{2} \right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 + 5x \\ & = -4 \left( x - \frac{5}{8} \right)^2 + \frac{25}{16}\end{aligned}$$

$$\begin{aligned}(5) \quad & 5x^2 - x + 1 \\ & = 5 \left( x - \frac{1}{10} \right)^2 + \frac{19}{20}\end{aligned}$$

$$\begin{aligned}(10) \quad & -2x^2 - x - 2 \\ & = -2 \left( x + \frac{1}{4} \right)^2 - \frac{15}{8}\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 + 3x + 5 \\ & = 2 \left( x + \frac{3}{4} \right)^2 + \frac{31}{8}\end{aligned}$$

$$\begin{aligned}(1) \quad & 3x^2 + 4x + 4 \\ &= 3\left(x + \frac{2}{3}\right)^2 + \frac{8}{3}\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 - 3x + 3 \\ &= \left(x - \frac{3}{2}\right)^2 + \frac{3}{4}\end{aligned}$$

$$\begin{aligned}(11) \quad & -5x^2 + x + 1 \\ &= -5\left(x - \frac{1}{10}\right)^2 + \frac{21}{20}\end{aligned}$$

$$\begin{aligned}(2) \quad & 3x^2 - x - 1 \\ &= 3\left(x - \frac{1}{6}\right)^2 - \frac{13}{12}\end{aligned}$$

$$\begin{aligned}(7) \quad & -2x^2 + 5x + 2 \\ &= -2\left(x - \frac{5}{4}\right)^2 + \frac{41}{8}\end{aligned}$$

$$\begin{aligned}(12) \quad & 5x^2 - x - 3 \\ &= 5\left(x - \frac{1}{10}\right)^2 - \frac{61}{20}\end{aligned}$$

$$\begin{aligned}(3) \quad & -3x^2 + 5x - 5 \\ &= -3\left(x - \frac{5}{6}\right)^2 - \frac{35}{12}\end{aligned}$$

$$\begin{aligned}(8) \quad & 2x^2 + 4x + 4 \\ &= 2(x + 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(13) \quad & x^2 + 3x + 1 \\ &= \left(x + \frac{3}{2}\right)^2 - \frac{5}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & 5x^2 - 4x - 5 \\ &= 5\left(x - \frac{2}{5}\right)^2 - \frac{29}{5}\end{aligned}$$

$$\begin{aligned}(9) \quad & 3x^2 + 5x - 3 \\ &= 3\left(x + \frac{5}{6}\right)^2 - \frac{61}{12}\end{aligned}$$

$$\begin{aligned}(14) \quad & -4x^2 + 2x - 2 \\ &= -4\left(x - \frac{1}{4}\right)^2 - \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(5) \quad & 2x^2 - 2x - 4 \\ &= 2\left(x - \frac{1}{2}\right)^2 - \frac{9}{2}\end{aligned}$$

$$\begin{aligned}(10) \quad & -4x^2 + 3x - 3 \\ &= -4\left(x - \frac{3}{8}\right)^2 - \frac{39}{16}\end{aligned}$$

$$\begin{aligned}(15) \quad & 2x^2 - 2x + 3 \\ &= 2\left(x - \frac{1}{2}\right)^2 + \frac{5}{2}\end{aligned}$$

$$\begin{aligned}(1) \quad & 3x^2 + 2x + 1 \\ &= 3\left(x + \frac{1}{3}\right)^2 + \frac{2}{3}\end{aligned}$$

$$\begin{aligned}(6) \quad & 4x^2 - 5x - 3 \\ &= 4\left(x - \frac{5}{8}\right)^2 - \frac{73}{16}\end{aligned}$$

$$\begin{aligned}(11) \quad & -5x^2 + x - 4 \\ &= -5\left(x - \frac{1}{10}\right)^2 - \frac{79}{20}\end{aligned}$$

$$\begin{aligned}(2) \quad & 3x^2 - x - 4 \\ &= 3\left(x - \frac{1}{6}\right)^2 - \frac{49}{12}\end{aligned}$$

$$\begin{aligned}(7) \quad & 4x^2 - 4x - 4 \\ &= 4\left(x - \frac{1}{2}\right)^2 - 5\end{aligned}$$

$$\begin{aligned}(12) \quad & 3x^2 + 3x - 1 \\ &= 3\left(x + \frac{1}{2}\right)^2 - \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(3) \quad & -4x^2 + 4x + 5 \\ &= -4\left(x - \frac{1}{2}\right)^2 + 6\end{aligned}$$

$$\begin{aligned}(8) \quad & -4x^2 + 2x \\ &= -4\left(x - \frac{1}{4}\right)^2 + \frac{1}{4}\end{aligned}$$

$$\begin{aligned}(13) \quad & -x^2 + 3x - 4 \\ &= -\left(x - \frac{3}{2}\right)^2 - \frac{7}{4}\end{aligned}$$

$$\begin{aligned}(4) \quad & -4x^2 + 5x \\ &= -4\left(x - \frac{5}{8}\right)^2 + \frac{25}{16}\end{aligned}$$

$$\begin{aligned}(9) \quad & 5x^2 + 3x + 1 \\ &= 5\left(x + \frac{3}{10}\right)^2 + \frac{11}{20}\end{aligned}$$

$$\begin{aligned}(14) \quad & 4x^2 - 3x \\ &= 4\left(x - \frac{3}{8}\right)^2 - \frac{9}{16}\end{aligned}$$

$$\begin{aligned}(5) \quad & 3x^2 + x + 3 \\ &= 3\left(x + \frac{1}{6}\right)^2 + \frac{35}{12}\end{aligned}$$

$$\begin{aligned}(10) \quad & x^2 - x + 1 \\ &= \left(x - \frac{1}{2}\right)^2 + \frac{3}{4}\end{aligned}$$

$$\begin{aligned}(15) \quad & -4x^2 - x - 1 \\ &= -4\left(x + \frac{1}{8}\right)^2 - \frac{15}{16}\end{aligned}$$

$$\begin{aligned}(1) \quad & 2x^2 - 3x \\ &= 2 \left( x - \frac{3}{4} \right)^2 - \frac{9}{8}\end{aligned}$$

$$\begin{aligned}(6) \quad & 2x^2 + 5x - 1 \\ &= 2 \left( x + \frac{5}{4} \right)^2 - \frac{33}{8}\end{aligned}$$

$$\begin{aligned}(11) \quad & -x^2 + x - 3 \\ &= - \left( x - \frac{1}{2} \right)^2 - \frac{11}{4}\end{aligned}$$

$$\begin{aligned}(2) \quad & x^2 + 5x - 3 \\ &= \left( x + \frac{5}{2} \right)^2 - \frac{37}{4}\end{aligned}$$

$$\begin{aligned}(7) \quad & -2x^2 + 2x \\ &= -2 \left( x - \frac{1}{2} \right)^2 + \frac{1}{2}\end{aligned}$$

$$\begin{aligned}(12) \quad & 2x^2 - x - 4 \\ &= 2 \left( x - \frac{1}{4} \right)^2 - \frac{33}{8}\end{aligned}$$

$$\begin{aligned}(3) \quad & 3x^2 + 2x + 5 \\ &= 3 \left( x + \frac{1}{3} \right)^2 + \frac{14}{3}\end{aligned}$$

$$\begin{aligned}(8) \quad & -3x^2 + 5x + 3 \\ &= -3 \left( x - \frac{5}{6} \right)^2 + \frac{61}{12}\end{aligned}$$

$$\begin{aligned}(13) \quad & -4x^2 - 5x - 3 \\ &= -4 \left( x + \frac{5}{8} \right)^2 - \frac{23}{16}\end{aligned}$$

$$\begin{aligned}(4) \quad & -5x^2 + 4x - 5 \\ &= -5 \left( x - \frac{2}{5} \right)^2 - \frac{21}{5}\end{aligned}$$

$$\begin{aligned}(9) \quad & -4x^2 - x - 2 \\ &= -4 \left( x + \frac{1}{8} \right)^2 - \frac{31}{16}\end{aligned}$$

$$\begin{aligned}(14) \quad & -2x^2 + 3x - 3 \\ &= -2 \left( x - \frac{3}{4} \right)^2 - \frac{15}{8}\end{aligned}$$

$$\begin{aligned}(5) \quad & 4x^2 - 3x + 3 \\ &= 4 \left( x - \frac{3}{8} \right)^2 + \frac{39}{16}\end{aligned}$$

$$\begin{aligned}(10) \quad & 3x^2 - 2x - 2 \\ &= 3 \left( x - \frac{1}{3} \right)^2 - \frac{7}{3}\end{aligned}$$

$$\begin{aligned}(15) \quad & -5x^2 + 4x + 5 \\ &= -5 \left( x - \frac{2}{5} \right)^2 + \frac{29}{5}\end{aligned}$$



$$\begin{aligned}(1) \quad & 2x^2 + 3x + 2 \\ &= 2 \left( x + \frac{3}{4} \right)^2 + \frac{7}{8}\end{aligned}$$

$$\begin{aligned}(6) \quad & x^2 + 3x + 5 \\ &= \left( x + \frac{3}{2} \right)^2 + \frac{11}{4}\end{aligned}$$

$$\begin{aligned}(11) \quad & 3x^2 + 4x - 5 \\ &= 3 \left( x + \frac{2}{3} \right)^2 - \frac{19}{3}\end{aligned}$$

$$\begin{aligned}(2) \quad & 3x^2 + 5x + 3 \\ &= 3 \left( x + \frac{5}{6} \right)^2 + \frac{11}{12}\end{aligned}$$

$$\begin{aligned}(7) \quad & 5x^2 - 2x + 5 \\ &= 5 \left( x - \frac{1}{5} \right)^2 + \frac{24}{5}\end{aligned}$$

$$\begin{aligned}(12) \quad & -x^2 - 5x - 4 \\ &= - \left( x + \frac{5}{2} \right)^2 + \frac{9}{4}\end{aligned}$$

$$\begin{aligned}(3) \quad & -x^2 - 3x + 4 \\ &= - \left( x + \frac{3}{2} \right)^2 + \frac{25}{4}\end{aligned}$$

$$\begin{aligned}(8) \quad & -3x^2 + 5x - 3 \\ &= -3 \left( x - \frac{5}{6} \right)^2 - \frac{11}{12}\end{aligned}$$

$$\begin{aligned}(13) \quad & 3x^2 + x \\ &= 3 \left( x + \frac{1}{6} \right)^2 - \frac{1}{12}\end{aligned}$$

$$\begin{aligned}(4) \quad & -x^2 - 2x + 1 \\ &= - (x + 1)^2 + 2\end{aligned}$$

$$\begin{aligned}(9) \quad & -3x^2 + 3x + 2 \\ &= -3 \left( x - \frac{1}{2} \right)^2 + \frac{11}{4}\end{aligned}$$

$$\begin{aligned}(14) \quad & 4x^2 + 4x - 4 \\ &= 4 \left( x + \frac{1}{2} \right)^2 - 5\end{aligned}$$

$$\begin{aligned}(5) \quad & x^2 + 5x - 2 \\ &= \left( x + \frac{5}{2} \right)^2 - \frac{33}{4}\end{aligned}$$

$$\begin{aligned}(10) \quad & -2x^2 + 3x + 3 \\ &= -2 \left( x - \frac{3}{4} \right)^2 + \frac{33}{8}\end{aligned}$$

$$\begin{aligned}(15) \quad & -x^2 + 5x + 5 \\ &= - \left( x - \frac{5}{2} \right)^2 + \frac{45}{4}\end{aligned}$$

$$(1) \quad x^2 + 3x + 3 \\ = \left(x + \frac{3}{2}\right)^2 + \frac{3}{4}$$

$$(6) \quad -x^2 + 5x + 1 \\ = -\left(x - \frac{5}{2}\right)^2 + \frac{29}{4}$$

$$(11) \quad -x^2 - 3x - 1 \\ = -\left(x + \frac{3}{2}\right)^2 + \frac{5}{4}$$

$$(2) \quad x^2 + x - 5 \\ = \left(x + \frac{1}{2}\right)^2 - \frac{21}{4}$$

$$(7) \quad -3x^2 - 3x \\ = -3\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$(12) \quad 5x^2 - 4x - 1 \\ = 5\left(x - \frac{2}{5}\right)^2 - \frac{9}{5}$$

$$(3) \quad 5x^2 + 4x + 1 \\ = 5\left(x + \frac{2}{5}\right)^2 + \frac{1}{5}$$

$$(8) \quad 3x^2 + 4x - 5 \\ = 3\left(x + \frac{2}{3}\right)^2 - \frac{19}{3}$$

$$(13) \quad -5x^2 + 5x + 2 \\ = -5\left(x - \frac{1}{2}\right)^2 + \frac{13}{4}$$

$$(4) \quad -2x^2 - 2x - 1 \\ = -2\left(x + \frac{1}{2}\right)^2 - \frac{1}{2}$$

$$(9) \quad 4x^2 + 4x \\ = 4\left(x + \frac{1}{2}\right)^2 - 1$$

$$(14) \quad -4x^2 - x + 5 \\ = -4\left(x + \frac{1}{8}\right)^2 + \frac{81}{16}$$

$$(5) \quad 2x^2 + 2x + 4 \\ = 2\left(x + \frac{1}{2}\right)^2 + \frac{7}{2}$$

$$(10) \quad 3x^2 - x - 4 \\ = 3\left(x - \frac{1}{6}\right)^2 - \frac{49}{12}$$

$$(15) \quad -5x^2 + 2x + 2 \\ = -5\left(x - \frac{1}{5}\right)^2 + \frac{11}{5}$$

## 2.8 代入 (易しい)

2 次多項式に整数を代入する問題。

$$(1) \begin{aligned} f(x) &= -5x^2 - 2x + 5 \\ f(5) \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 4x^2 - 3x + 4 \\ f(1) \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(5) \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 3x^2 + 3x - 2 \\ f(1) \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 3x^2 - 2x + 3 \\ f(-2) \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 5x^2 - 4x + 4 \\ f(-2) \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4x^2 + 4x - 4 \\ f(3) \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 3x^2 - 2x + 1 \\ f(2) \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -3x^2 - 5x - 1 \\ f(3) \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 - 2x + 2 \\ f(-2) \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 3x^2 + 4x - 2 \\ f(4) \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -x^2 - x + 1 \\ f(2) \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -x^2 - 4x - 5 \\ f(1) \end{aligned}$$

$$(20) \begin{aligned} f(x) &= x^2 - 4x - 1 \\ f(-3) \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -5x^2 + 2x - 2 \\ f(-4) \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3x^2 + 3x - 1 \\ f(-3) \end{aligned}$$

$$(21) \begin{aligned} f(x) &= x^2 + 2x + 4 \\ f(-1) \end{aligned}$$

$$(36) \begin{aligned} f(x) &= 5x^2 + 3x - 4 \\ f(-1) \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -3x^2 + x - 4 \\ f(-4) \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -5x^2 - 3x + 4 \\ f(-5) \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -5x^2 + 3x + 4 \\ f(2) \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 2x^2 - 5x + 2 \\ f(-2) \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 5x^2 - 2x + 2 \\ f(-1) \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 2x^2 - x + 4 \\ f(3) \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -x^2 - 3x + 3 \\ f(3) \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -2x^2 - 2x + 5 \\ f(3) \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -3x^2 + 4x + 4 \\ f(-1) \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -5x^2 - 4x - 1 \\ f(-2) \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 4x^2 + 4x - 2 \\ f(4) \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 5x^2 + 2x + 5 \\ f(1) \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 - 3x + 4 \\ f(3) \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 5x^2 - 4x + 2 \\ f(2) \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -x^2 + 3x - 5 \\ f(4) \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -3x^2 + x + 1 \\ f(-2) \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 2x^2 + 5x - 2 \\ f(-2) \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 3x^2 + 3x - 4 \\ f(1) \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -4x^2 + 3x - 5 \\ f(-3) \end{aligned}$$

$$(28) \begin{aligned} f(x) &= x^2 - 3x - 2 \\ f(3) \end{aligned}$$

$$(43) \begin{aligned} f(x) &= x^2 - 4x - 2 \\ f(-3) \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -4x^2 + 3x - 3 \\ f(-4) \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(-1) \end{aligned}$$

$$(44) \begin{aligned} f(x) &= 5x^2 + 2x + 5 \\ f(3) \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -4x^2 + x - 3 \\ f(3) \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 2x^2 - 5x - 5 \\ f(4) \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 3x^2 - 2x + 5 \\ f(1) \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 5x^2 - 3x + 1 \\ f(2) \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 3x^2 - x + 4 \\ f(-5) \end{aligned}$$

$$(31) \begin{aligned} f(x) &= x^2 - 4x - 1 \\ f(-4) \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -4x^2 - 2x + 4 \\ f(4) \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -3x^2 + 5x - 2 \\ f(3) \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 2x^2 - 4x + 2 \\ f(5) \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 3x^2 - 3x + 1 \\ f(5) \end{aligned}$$

$$(18) \begin{aligned} f(x) &= x^2 + 4x + 5 \\ f(4) \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -2x^2 + 4x + 1 \\ f(-4) \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -x^2 + 4x - 1 \\ f(2) \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -2x^2 - 4x + 1 \\ f(3) \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -3x^2 + 3x - 3 \\ f(-3) \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 5x^2 - 3x - 3 \\ f(1) \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -4x^2 + 2x + 3 \\ f(-4) \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 2x^2 + 2x - 2 \\ f(2) \end{aligned}$$

$$(6) \begin{aligned} f(x) &= x^2 + 4x + 5 \\ f(4) \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -2x^2 + 5x - 2 \\ f(5) \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -2x^2 - 3x - 3 \\ f(-2) \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 4x^2 + 5x - 1 \\ f(4) \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -2x^2 + 2x + 4 \\ f(-1) \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -5x^2 - 2x + 2 \\ f(-1) \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2x^2 - 3x - 3 \\ f(4) \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -4x^2 - 4x + 5 \\ f(-3) \end{aligned}$$

$$(38) \begin{aligned} f(x) &= -2x^2 - 4x + 5 \\ f(-2) \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -3x^2 - 2x + 4 \\ f(1) \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -x^2 + 4x + 2 \\ f(4) \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -2x^2 - 2x + 2 \\ f(4) \end{aligned}$$

$$(10) \begin{aligned} f(x) &= x^2 + 2x + 5 \\ f(-1) \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 2x^2 + x - 1 \\ f(2) \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 2x^2 - 2x + 2 \\ f(3) \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -4x^2 + 3x + 2 \\ f(3) \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 5x^2 + 3x + 4 \\ f(5) \end{aligned}$$

$$(41) \begin{aligned} f(x) &= x^2 + x + 1 \\ f(-2) \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -5x^2 - 4x + 3 \\ f(5) \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -x^2 + 2x + 5 \\ f(2) \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 4x^2 - 4x + 5 \\ f(-1) \end{aligned}$$

$$(13) \begin{aligned} f(x) &= x^2 - 4x + 5 \\ f(3) \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 2x^2 + 2x + 5 \\ f(4) \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -4x^2 + 4x - 2 \\ f(-2) \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 5x^2 - 4x - 4 \\ f(1) \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -2x^2 + 3x - 2 \\ f(4) \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -2x^2 + 5x - 3 \\ f(-2) \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -x^2 - 3x - 2 \\ f(3) \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -x^2 - 5x - 5 \\ f(-4) \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -2x^2 - 3x + 1 \\ f(-4) \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 5x^2 - 4x - 5 \\ f(3) \end{aligned}$$

$$(16) \begin{aligned} f(x) &= -3x^2 - 4x - 2 \\ f(-3) \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 5x^2 - 5x + 5 \\ f(3) \end{aligned}$$

$$(2) \begin{aligned} f(x) &= x^2 + 3x + 2 \\ f(-5) \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -4x^2 - 3x - 3 \\ f(-4) \end{aligned}$$

$$(32) \begin{aligned} f(x) &= x^2 - 5x + 2 \\ f(-2) \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -5x^2 - 5x + 5 \\ f(1) \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -x^2 - 3x - 2 \\ f(2) \end{aligned}$$

$$(33) \begin{aligned} f(x) &= x^2 + 5x + 3 \\ f(-2) \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 2x^2 + x + 2 \\ f(3) \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 5x^2 - 3x + 2 \\ f(1) \end{aligned}$$

$$(34) \begin{aligned} f(x) &= x^2 + 4x - 4 \\ f(-2) \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -5x^2 + 3x - 5 \\ f(3) \end{aligned}$$

$$(20) \begin{aligned} f(x) &= 3x^2 - x + 3 \\ f(2) \end{aligned}$$

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$$(7) \begin{aligned} f(x) &= 5x^2 - 2x - 5 \\ f(1) \end{aligned}$$

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$$(37) \begin{aligned} f(x) &= 3x^2 - 4x - 2 \\ f(-4) \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -5x^2 + 4x + 4 \\ f(5) \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 2x^2 + 5x + 4 \\ f(-5) \end{aligned}$$

$$(38) \begin{aligned} f(x) &= x^2 - x + 2 \\ f(-1) \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -5x^2 + x - 2 \\ f(-1) \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 3x^2 + 5x - 5 \\ f(-1) \end{aligned}$$

$$(39) \begin{aligned} f(x) &= 5x^2 - 2x - 2 \\ f(1) \end{aligned}$$

$$(10) \begin{aligned} f(x) &= x^2 + 3x - 5 \\ f(4) \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -4x^2 - 5x - 4 \\ f(2) \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 4x^2 - 5x - 1 \\ f(4) \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 + 5x + 4 \\ f(-2) \end{aligned}$$

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$$(12) \begin{aligned} f(x) &= -4x^2 + 3x - 2 \\ f(-5) \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 5x^2 + 4x + 5 \\ f(4) \end{aligned}$$

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$$(13) \begin{aligned} f(x) &= -x^2 + 2x - 4 \\ f(-4) \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 2x^2 - 4x - 2 \\ f(-3) \end{aligned}$$

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$$(14) \begin{aligned} f(x) &= -5x^2 - x + 4 \\ f(-3) \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 4x^2 + 5x - 2 \\ f(2) \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -3x^2 + 3x - 5 \\ f(-5) \end{aligned}$$

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$$(40) \quad f(x) = x^2 - 2x + 3 \\ f(1)$$

$$(11) \quad f(x) = 5x^2 - 3x + 4 \\ f(-5)$$

$$(26) \quad f(x) = -5x^2 - 3x - 5 \\ f(1)$$

$$(41) \quad f(x) = -5x^2 - 5x + 2 \\ f(-4)$$

$$(12) \quad f(x) = 3x^2 - 3x + 1 \\ f(-5)$$

$$(27) \quad f(x) = -x^2 + 3x - 5 \\ f(5)$$

$$(42) \quad f(x) = 2x^2 - 4x + 2 \\ f(-5)$$

$$(13) \quad f(x) = 3x^2 - 5x + 4 \\ f(-1)$$

$$(28) \quad f(x) = -x^2 - x + 2 \\ f(4)$$

$$(43) \quad f(x) = -3x^2 + 5x + 2 \\ f(4)$$

$$(14) \quad f(x) = -3x^2 - 3x + 5 \\ f(3)$$

$$(29) \quad f(x) = -5x^2 + 2x - 5 \\ f(-4)$$

$$(44) \quad f(x) = 5x^2 + 5x + 5 \\ f(4)$$

$$(15) \quad f(x) = 3x^2 - 2x - 4 \\ f(-5)$$

$$(30) \quad f(x) = x^2 + 4x - 5 \\ f(3)$$

$$(45) \quad f(x) = -x^2 + 2x + 1 \\ f(2)$$

$$(1) \begin{aligned} f(x) &= -x^2 + 3x + 1 \\ f(-3) \end{aligned}$$

$$(16) \begin{aligned} f(x) &= -3x^2 + x - 2 \\ f(4) \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -2x^2 - 3x + 5 \\ f(-5) \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 2x^2 + x - 4 \\ f(-4) \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 2x^2 + 5x - 5 \\ f(2) \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -x^2 - 4x - 2 \\ f(-4) \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4x^2 + 5x + 5 \\ f(2) \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 5x^2 + 2x - 5 \\ f(-5) \end{aligned}$$

$$(33) \begin{aligned} f(x) &= x^2 - 4x - 2 \\ f(1) \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5x^2 - x + 3 \\ f(5) \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -2x^2 - 3x - 2 \\ f(4) \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -3x^2 - 3x + 3 \\ f(4) \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 3x^2 - 3x - 4 \\ f(-3) \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -2x^2 + x - 1 \\ f(3) \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -x^2 + 4x + 4 \\ f(-3) \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 5x^2 - 4x + 3 \\ f(-5) \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -3x^2 - 2x + 5 \\ f(-1) \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -5x^2 - x - 5 \\ f(-4) \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 3x^2 - x + 1 \\ f(2) \end{aligned}$$

$$(22) \begin{aligned} f(x) &= 5x^2 - 2x - 1 \\ f(-2) \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -2x^2 + 3x - 1 \\ f(-5) \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2x^2 + 4x - 2 \\ f(1) \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -4x^2 - 4x + 2 \\ f(4) \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 5x^2 + x - 3 \\ f(5) \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 4x^2 - x + 4 \\ f(-2) \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 2x^2 - 2x - 4 \\ f(-1) \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -4x^2 - 2x - 1 \\ f(1) \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -3x^2 + 4x + 2 \\ f(1) \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -4x^2 - 4x + 3 \\ f(-2) \end{aligned}$$

$$(40) \begin{aligned} f(x) &= -5x^2 + 5x + 2 \\ f(-5) \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -2x^2 - 4x + 1 \\ f(-3) \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -2x^2 + 2x + 2 \\ f(1) \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 2x^2 - 2x - 3 \\ f(-1) \end{aligned}$$

$$(12) \begin{aligned} f(x) &= 2x^2 + 4x - 1 \\ f(3) \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 4x^2 + 5x + 5 \\ f(5) \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -3x^2 + x + 3 \\ f(-3) \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -2x^2 - 4x + 5 \\ f(-3) \end{aligned}$$

$$(28) \begin{aligned} f(x) &= x^2 - 3x - 3 \\ f(2) \end{aligned}$$

$$(43) \begin{aligned} f(x) &= x^2 - 5x - 1 \\ f(-4) \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -x^2 - 4x - 1 \\ f(2) \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 3x^2 - 3x - 4 \\ f(-5) \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -x^2 - 2x - 1 \\ f(5) \end{aligned}$$

$$(15) \begin{aligned} f(x) &= 5x^2 - x - 5 \\ f(-4) \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 2x^2 - x - 5 \\ f(2) \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 5x^2 + x + 3 \\ f(-2) \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -5x^2 - 2x + 5 \\ f(5) &= -130 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 4x^2 - 3x + 4 \\ f(1) &= 5 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(5) &= -131 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 3x^2 + 3x - 2 \\ f(1) &= 4 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 3x^2 - 2x + 3 \\ f(-2) &= 19 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 5x^2 - 4x + 4 \\ f(-2) &= 32 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4x^2 + 4x - 4 \\ f(3) &= -28 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 3x^2 - 2x + 1 \\ f(2) &= 9 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -3x^2 - 5x - 1 \\ f(3) &= -43 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 - 2x + 2 \\ f(-2) &= 18 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 3x^2 + 4x - 2 \\ f(4) &= 62 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -x^2 - x + 1 \\ f(2) &= -5 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -x^2 - 4x - 5 \\ f(1) &= -10 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= x^2 - 4x - 1 \\ f(-3) &= 20 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -5x^2 + 2x - 2 \\ f(-4) &= -90 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3x^2 + 3x - 1 \\ f(-3) &= 17 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= x^2 + 2x + 4 \\ f(-1) &= 3 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= 5x^2 + 3x - 4 \\ f(-1) &= -2 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -3x^2 + x - 4 \\ f(-4) &= -56 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -5x^2 - 3x + 4 \\ f(-5) &= -106 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -5x^2 + 3x + 4 \\ f(2) &= -10 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 2x^2 - 5x + 2 \\ f(-2) &= 20 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 5x^2 - 2x + 2 \\ f(-1) &= 9 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 2x^2 - x + 4 \\ f(3) &= 19 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -x^2 - 3x + 3 \\ f(3) &= -15 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -2x^2 - 2x + 5 \\ f(3) &= -19 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -3x^2 + 4x + 4 \\ f(-1) &= -3 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -5x^2 - 4x - 1 \\ f(-2) &= -13 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 4x^2 + 4x - 2 \\ f(4) &= 78 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 5x^2 + 2x + 5 \\ f(1) &= 12 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 - 3x + 4 \\ f(3) &= 40 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 5x^2 - 4x + 2 \\ f(2) &= 14 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -x^2 + 3x - 5 \\ f(4) &= -9 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -3x^2 + x + 1 \\ f(-2) &= -13 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 2x^2 + 5x - 2 \\ f(-2) &= -4 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 3x^2 + 3x - 4 \\ f(1) &= 2 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -4x^2 + 3x - 5 \\ f(-3) &= -50 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= x^2 - 3x - 2 \\ f(3) &= -2 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= x^2 - 4x - 2 \\ f(-3) &= 19 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -4x^2 + 3x - 3 \\ f(-4) &= -79 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(-1) &= -5 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= 5x^2 + 2x + 5 \\ f(3) &= 56 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -4x^2 + x - 3 \\ f(3) &= -36 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 2x^2 - 5x - 5 \\ f(4) &= 7 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 3x^2 - 2x + 5 \\ f(1) &= 6 \end{aligned}$$



$$(1) \begin{aligned} f(x) &= 5x^2 - 3x + 1 \\ f(2) &= 15 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 3x^2 - x + 4 \\ f(-5) &= 84 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= x^2 - 4x - 1 \\ f(-4) &= 31 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -4x^2 - 2x + 4 \\ f(4) &= -68 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -3x^2 + 5x - 2 \\ f(3) &= -14 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 2x^2 - 4x + 2 \\ f(5) &= 32 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 3x^2 - 3x + 1 \\ f(5) &= 61 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= x^2 + 4x + 5 \\ f(4) &= 37 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -2x^2 + 4x + 1 \\ f(-4) &= -47 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -x^2 + 4x - 1 \\ f(2) &= 3 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -2x^2 - 4x + 1 \\ f(3) &= -29 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -3x^2 + 3x - 3 \\ f(-3) &= -39 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 5x^2 - 3x - 3 \\ f(1) &= -1 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -4x^2 + 2x + 3 \\ f(-4) &= -69 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 2x^2 + 2x - 2 \\ f(2) &= 10 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= x^2 + 4x + 5 \\ f(4) &= 37 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -2x^2 + 5x - 2 \\ f(5) &= -27 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -2x^2 - 3x - 3 \\ f(-2) &= -5 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 4x^2 + 5x - 1 \\ f(4) &= 83 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -2x^2 + 2x + 4 \\ f(-1) &= 0 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -5x^2 - 2x + 2 \\ f(-1) &= -1 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2x^2 - 3x - 3 \\ f(4) &= -47 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -4x^2 - 4x + 5 \\ f(-3) &= -19 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= -2x^2 - 4x + 5 \\ f(-2) &= 5 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -3x^2 - 2x + 4 \\ f(1) &= -1 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -x^2 + 4x + 2 \\ f(4) &= 2 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -2x^2 - 2x + 2 \\ f(4) &= -38 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= x^2 + 2x + 5 \\ f(-1) &= 4 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 2x^2 + x - 1 \\ f(2) &= 9 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 2x^2 - 2x + 2 \\ f(3) &= 14 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -4x^2 + 3x + 2 \\ f(3) &= -25 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 5x^2 + 3x + 4 \\ f(5) &= 144 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= x^2 + x + 1 \\ f(-2) &= 3 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -5x^2 - 4x + 3 \\ f(5) &= -142 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -x^2 + 2x + 5 \\ f(2) &= 5 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 4x^2 - 4x + 5 \\ f(-1) &= 13 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= x^2 - 4x + 5 \\ f(3) &= 2 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 2x^2 + 2x + 5 \\ f(4) &= 45 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -4x^2 + 4x - 2 \\ f(-2) &= -26 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 5x^2 - 4x - 4 \\ f(1) &= -3 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -2x^2 + 3x - 2 \\ f(4) &= -22 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -2x^2 + 5x - 3 \\ f(-2) &= -21 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -x^2 - 3x - 2 \\ f(3) &= -20 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -x^2 - 5x - 5 \\ f(-4) &= -1 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -2x^2 - 3x + 1 \\ f(-4) &= -19 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 5x^2 - 4x - 5 \\ f(3) &= 28 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= -3x^2 - 4x - 2 \\ f(-3) &= -17 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 5x^2 - 5x + 5 \\ f(3) &= 35 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= x^2 + 3x + 2 \\ f(-5) &= 12 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -4x^2 - 3x - 3 \\ f(-4) &= -55 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= x^2 - 5x + 2 \\ f(-2) &= 16 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -5x^2 - 5x + 5 \\ f(1) &= -5 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -x^2 - 3x - 2 \\ f(2) &= -12 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= x^2 + 5x + 3 \\ f(-2) &= -3 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 2x^2 + x + 2 \\ f(3) &= 23 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 5x^2 - 3x + 2 \\ f(1) &= 4 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= x^2 + 4x - 4 \\ f(-2) &= -8 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -5x^2 + 3x - 5 \\ f(3) &= -41 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= 3x^2 - x + 3 \\ f(2) &= 13 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 2x^2 - 2x + 2 \\ f(-1) &= 6 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -3x^2 + 5x + 3 \\ f(-3) &= -39 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= 2x^2 - x + 2 \\ f(3) &= 17 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -3x^2 - 5x - 1 \\ f(1) &= -9 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 5x^2 - 2x - 5 \\ f(1) &= -2 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -x^2 + 5x - 1 \\ f(2) &= 5 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 3x^2 - 4x - 2 \\ f(-4) &= 62 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -5x^2 + 4x + 4 \\ f(5) &= -101 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 2x^2 + 5x + 4 \\ f(-5) &= 29 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= x^2 - x + 2 \\ f(-1) &= 4 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -5x^2 + x - 2 \\ f(-1) &= -8 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 3x^2 + 5x - 5 \\ f(-1) &= -7 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= 5x^2 - 2x - 2 \\ f(1) &= 1 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= x^2 + 3x - 5 \\ f(4) &= 23 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -4x^2 - 5x - 4 \\ f(2) &= -30 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 4x^2 - 5x - 1 \\ f(4) &= 43 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 + 5x + 4 \\ f(-2) &= 14 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 4x^2 - 2x + 5 \\ f(-3) &= 47 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -4x^2 + 2x - 3 \\ f(4) &= -59 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -4x^2 + 3x - 2 \\ f(-5) &= -117 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 5x^2 + 4x + 5 \\ f(4) &= 101 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 5x^2 - 4x + 1 \\ f(5) &= 106 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -x^2 + 2x - 4 \\ f(-4) &= -28 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 2x^2 - 4x - 2 \\ f(-3) &= 28 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -3x^2 + 4x - 2 \\ f(-3) &= -41 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -5x^2 - x + 4 \\ f(-3) &= -38 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 4x^2 + 5x - 2 \\ f(2) &= 24 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -3x^2 + 3x - 5 \\ f(-5) &= -95 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -4x^2 - x - 2 \\ f(3) &= -41 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -4x^2 - 3x + 3 \\ f(-2) &= -7 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -x^2 - 3x + 3 \\ f(2) &= -7 \end{aligned}$$

$$(1) \quad f(x) = 3x^2 - 2x - 4 \\ f(-5) = 81$$

$$(16) \quad f(x) = 4x^2 - 2x + 2 \\ f(4) = 58$$

$$(31) \quad f(x) = 5x^2 + 2x - 2 \\ f(-5) = 113$$

$$(2) \quad f(x) = -x^2 - 3x + 1 \\ f(5) = -39$$

$$(17) \quad f(x) = 2x^2 + 3x + 5 \\ f(-4) = 25$$

$$(32) \quad f(x) = -2x^2 - 2x - 5 \\ f(-5) = -45$$

$$(3) \quad f(x) = -5x^2 - 4x + 3 \\ f(2) = -25$$

$$(18) \quad f(x) = 3x^2 + x + 2 \\ f(-5) = 72$$

$$(33) \quad f(x) = -2x^2 + 5x + 2 \\ f(-3) = -31$$

$$(4) \quad f(x) = 2x^2 + 3x - 5 \\ f(4) = 39$$

$$(19) \quad f(x) = 3x^2 + 4x + 4 \\ f(2) = 24$$

$$(34) \quad f(x) = -x^2 - 4x - 4 \\ f(3) = -25$$

$$(5) \quad f(x) = 4x^2 + 5x - 3 \\ f(-4) = 41$$

$$(20) \quad f(x) = x^2 - x + 5 \\ f(1) = 5$$

$$(35) \quad f(x) = 3x^2 - 5x + 1 \\ f(1) = -1$$

$$(6) \quad f(x) = 3x^2 - 3x + 4 \\ f(-2) = 22$$

$$(21) \quad f(x) = -3x^2 - x - 3 \\ f(1) = -7$$

$$(36) \quad f(x) = 3x^2 - 5x + 2 \\ f(5) = 52$$

$$(7) \quad f(x) = -3x^2 - 5x - 5 \\ f(3) = -47$$

$$(22) \quad f(x) = -3x^2 - 4x - 5 \\ f(-4) = -37$$

$$(37) \quad f(x) = -2x^2 + 2x - 2 \\ f(-4) = -42$$

$$(8) \quad f(x) = -4x^2 + 3x - 4 \\ f(1) = -5$$

$$(23) \quad f(x) = 4x^2 + x - 2 \\ f(3) = 37$$

$$(38) \quad f(x) = -4x^2 + 5x + 3 \\ f(3) = -18$$

$$(9) \quad f(x) = 2x^2 - 3x - 5 \\ f(-5) = 60$$

$$(24) \quad f(x) = -5x^2 - 5x + 4 \\ f(4) = -96$$

$$(39) \quad f(x) = 2x^2 - 2x - 1 \\ f(-2) = 11$$

$$(10) \quad f(x) = 2x^2 + 5x - 3 \\ f(-4) = 9$$

$$(25) \quad f(x) = -4x^2 + 5x + 1 \\ f(-4) = -83$$

$$(40) \quad f(x) = 2x^2 + 3x + 3 \\ f(-4) = 23$$

$$(11) \quad f(x) = -4x^2 + 2x + 1 \\ f(3) = -29$$

$$(26) \quad f(x) = 4x^2 - x + 5 \\ f(2) = 19$$

$$(41) \quad f(x) = 4x^2 - 5x + 4 \\ f(-3) = 55$$

$$(12) \quad f(x) = -3x^2 + x - 4 \\ f(2) = -14$$

$$(27) \quad f(x) = -3x^2 - 2x + 4 \\ f(-2) = -4$$

$$(42) \quad f(x) = 4x^2 + 3x + 3 \\ f(-2) = 13$$

$$(13) \quad f(x) = -x^2 + x + 2 \\ f(3) = -4$$

$$(28) \quad f(x) = -5x^2 - 2x + 4 \\ f(-2) = -12$$

$$(43) \quad f(x) = -4x^2 - 5x + 1 \\ f(-1) = 2$$

$$(14) \quad f(x) = 5x^2 - x - 3 \\ f(1) = 1$$

$$(29) \quad f(x) = -2x^2 + 4x - 3 \\ f(2) = -3$$

$$(44) \quad f(x) = x^2 + 4x + 4 \\ f(1) = 9$$

$$(15) \quad f(x) = -4x^2 - 2x - 2 \\ f(2) = -22$$

$$(30) \quad f(x) = 2x^2 + 3x + 1 \\ f(3) = 28$$

$$(45) \quad f(x) = -3x^2 + 4x + 1 \\ f(-4) = -63$$

$$(1) \quad f(x) = -2x^2 - 4x - 1 \\ f(1) = -7$$

$$(2) \quad f(x) = 5x^2 + 5x + 3 \\ f(-1) = 3$$

$$(3) \quad f(x) = x^2 + 2x + 3 \\ f(3) = 18$$

$$(4) \quad f(x) = x^2 - 5x + 1 \\ f(4) = -3$$

$$(5) \quad f(x) = x^2 + x - 1 \\ f(-2) = 1$$

$$(6) \quad f(x) = 3x^2 - x + 2 \\ f(-4) = 54$$

$$(7) \quad f(x) = -3x^2 - 4x + 4 \\ f(2) = -16$$

$$(8) \quad f(x) = -x^2 + 5x - 2 \\ f(3) = 4$$

$$(9) \quad f(x) = -5x^2 - 3x - 2 \\ f(5) = -142$$

$$(10) \quad f(x) = 4x^2 - x + 1 \\ f(3) = 34$$

$$(11) \quad f(x) = -3x^2 + 3x + 4 \\ f(1) = 4$$

$$(12) \quad f(x) = 5x^2 - 2x + 3 \\ f(-4) = 91$$

$$(13) \quad f(x) = 2x^2 - 5x - 3 \\ f(4) = 9$$

$$(14) \quad f(x) = 2x^2 + x + 4 \\ f(-4) = 32$$

$$(15) \quad f(x) = -4x^2 - 2x - 1 \\ f(5) = -111$$

$$(16) \quad f(x) = 4x^2 + 5x - 1 \\ f(2) = 25$$

$$(17) \quad f(x) = -2x^2 + x + 2 \\ f(3) = -13$$

$$(18) \quad f(x) = 3x^2 - 4x - 1 \\ f(-2) = 19$$

$$(19) \quad f(x) = -5x^2 + 4x - 5 \\ f(-5) = -150$$

$$(20) \quad f(x) = 4x^2 + x - 4 \\ f(-5) = 91$$

$$(21) \quad f(x) = 3x^2 - 4x + 1 \\ f(-2) = 21$$

$$(22) \quad f(x) = 4x^2 + 4x - 3 \\ f(-1) = -3$$

$$(23) \quad f(x) = -2x^2 - 4x + 4 \\ f(-4) = -12$$

$$(24) \quad f(x) = -3x^2 - 5x + 4 \\ f(4) = -64$$

$$(25) \quad f(x) = 4x^2 + 5x - 2 \\ f(3) = 49$$

$$(26) \quad f(x) = 5x^2 - 3x - 5 \\ f(-3) = 49$$

$$(27) \quad f(x) = 4x^2 - 4x + 2 \\ f(5) = 82$$

$$(28) \quad f(x) = -3x^2 + 5x + 3 \\ f(2) = 1$$

$$(29) \quad f(x) = -5x^2 - x + 2 \\ f(4) = -82$$

$$(30) \quad f(x) = -x^2 + 2x + 2 \\ f(-2) = -6$$

$$(31) \quad f(x) = 2x^2 - 4x - 3 \\ f(-1) = 3$$

$$(32) \quad f(x) = 2x^2 + 3x - 1 \\ f(1) = 4$$

$$(33) \quad f(x) = -5x^2 - 2x + 5 \\ f(3) = -46$$

$$(34) \quad f(x) = -2x^2 + 4x - 5 \\ f(-4) = -53$$

$$(35) \quad f(x) = 2x^2 - x + 2 \\ f(3) = 17$$

$$(36) \quad f(x) = 5x^2 - 5x - 2 \\ f(-4) = 98$$

$$(37) \quad f(x) = 2x^2 + 5x + 3 \\ f(-5) = 28$$

$$(38) \quad f(x) = 3x^2 - 4x - 5 \\ f(-2) = 15$$

$$(39) \quad f(x) = 2x^2 + 3x + 2 \\ f(5) = 67$$

$$(40) \quad f(x) = -4x^2 - 5x + 3 \\ f(-5) = -72$$

$$(41) \quad f(x) = 3x^2 + 5x + 1 \\ f(-1) = -1$$

$$(42) \quad f(x) = -2x^2 - 5x + 1 \\ f(-2) = 3$$

$$(43) \quad f(x) = x^2 + 4x + 1 \\ f(-3) = -2$$

$$(44) \quad f(x) = 5x^2 + x - 2 \\ f(2) = 20$$

$$(45) \quad f(x) = 2x^2 + 3x + 4 \\ f(3) = 31$$

$$(1) \begin{aligned} f(x) &= 3x^2 - x + 4 \\ f(3) &= 28 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= x^2 + x + 3 \\ f(-4) &= 15 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= x^2 - 2x + 4 \\ f(-2) &= 12 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -4x^2 - 2x - 1 \\ f(1) &= -7 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 4x^2 - 5x - 1 \\ f(-3) &= 50 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 4x^2 - 5x - 5 \\ f(2) &= 1 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 3x^2 + 2x - 3 \\ f(1) &= 2 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -x^2 + 4x + 1 \\ f(4) &= 1 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -x^2 + 5x - 1 \\ f(-5) &= -51 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5x^2 - 2x + 2 \\ f(5) &= -133 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 4x^2 + 5x + 4 \\ f(4) &= 88 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= 2x^2 + x - 3 \\ f(-5) &= 42 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 2x^2 + 4x - 5 \\ f(-1) &= -7 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= 5x^2 + 4x - 5 \\ f(-5) &= 100 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -2x^2 + 3x + 4 \\ f(2) &= 2 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3x^2 + 5x - 5 \\ f(-4) &= 23 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= x^2 + 4x + 5 \\ f(-5) &= 10 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -2x^2 - 4x - 3 \\ f(3) &= -33 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -2x^2 - 2x + 2 \\ f(3) &= -22 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -3x^2 - 4x - 2 \\ f(4) &= -66 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -x^2 + 3x - 2 \\ f(4) &= -6 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 2x^2 + 3x + 3 \\ f(-3) &= 12 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -x^2 - x - 2 \\ f(-1) &= -2 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= -3x^2 + 4x + 5 \\ f(3) &= -10 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 3x^2 - x - 1 \\ f(-3) &= 29 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 3x^2 + x - 5 \\ f(-1) &= -3 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= 2x^2 - 4x + 3 \\ f(1) &= 1 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -x^2 + x - 1 \\ f(2) &= -3 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 5x^2 + 4x - 5 \\ f(-1) &= -4 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 3x^2 + 3x + 4 \\ f(4) &= 64 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -2x^2 + 3x + 4 \\ f(-2) &= -10 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -5x^2 + 5x - 5 \\ f(-3) &= -65 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 5x^2 + x + 5 \\ f(2) &= 27 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= x^2 - 3x + 5 \\ f(-1) &= 9 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 3x^2 + 3x + 4 \\ f(4) &= 64 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 4x^2 + 4x - 2 \\ f(-1) &= -2 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -3x^2 + 4x + 2 \\ f(-2) &= -18 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 2x^2 + x - 5 \\ f(-2) &= 1 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -x^2 - x - 1 \\ f(3) &= -13 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= x^2 - x + 2 \\ f(-1) &= 4 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 5x^2 - 5x - 3 \\ f(3) &= 27 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -x^2 - 3x + 5 \\ f(4) &= -23 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= 2x^2 - x + 3 \\ f(-1) &= 6 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -2x^2 + 4x - 4 \\ f(5) &= -34 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -3x^2 - x - 5 \\ f(-2) &= -15 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -2x^2 + 3x + 5 \\ f(2) &= 3 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 4x^2 - 2x + 3 \\ f(5) &= 93 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 3x^2 + 3x + 3 \\ f(-2) &= 9 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5x^2 + 3x - 2 \\ f(4) &= -70 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -x^2 + 4x - 2 \\ f(-5) &= -47 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3x^2 + 5x + 3 \\ f(3) &= 45 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -3x^2 + 4x - 1 \\ f(-5) &= -96 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= x^2 + 5x - 2 \\ f(-5) &= -2 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 3x^2 + 3x + 5 \\ f(1) &= 11 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -3x^2 - 5x + 5 \\ f(-4) &= -23 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 - 2x + 5 \\ f(-2) &= 29 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= 3x^2 + 4x + 4 \\ f(-1) &= 3 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -2x^2 + 3x - 3 \\ f(2) &= -5 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 2x^2 + 2x - 5 \\ f(5) &= 55 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= x^2 + 2x + 3 \\ f(-5) &= 18 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 4x^2 + 4x + 3 \\ f(-4) &= 51 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -4x^2 + 3x - 4 \\ f(2) &= -14 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -4x^2 + 5x - 1 \\ f(1) &= 0 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 5x^2 - 5x - 4 \\ f(-3) &= 56 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= x^2 - 5x - 5 \\ f(-4) &= 31 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= 5x^2 + 3x + 2 \\ f(5) &= 142 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= 3x^2 + 3x - 4 \\ f(3) &= 32 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -4x^2 - x - 5 \\ f(-2) &= -19 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 3x^2 - 2x - 4 \\ f(-3) &= 29 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 3x^2 - 2x - 5 \\ f(1) &= -4 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -3x^2 + 5x - 2 \\ f(1) &= 0 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 4x^2 - x + 4 \\ f(-2) &= 22 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= -3x^2 - 3x + 4 \\ f(-3) &= -14 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -2x^2 - 3x - 5 \\ f(5) &= -70 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -x^2 - 2x - 3 \\ f(3) &= -18 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -4x^2 + 3x + 5 \\ f(5) &= -80 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 5x^2 + 4x + 4 \\ f(-3) &= 37 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -5x^2 + 4x - 4 \\ f(-4) &= -100 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -4x^2 - 2x - 2 \\ f(-4) &= -58 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -4x^2 - x + 3 \\ f(-2) &= -11 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -5x^2 - 2x - 3 \\ f(1) &= -10 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 3x^2 - x + 4 \\ f(-1) &= 8 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 4x^2 + 2x + 1 \\ f(5) &= 111 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -x^2 - 5x + 3 \\ f(3) &= -21 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 5x^2 + 5x - 5 \\ f(4) &= 95 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -3x^2 + 3x + 1 \\ f(-3) &= -35 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 3x^2 - 5x - 5 \\ f(-2) &= 17 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 3x^2 + 5x + 5 \\ f(-1) &= 3 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -5x^2 + 3x - 5 \\ f(2) &= -19 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 2x^2 - 3x + 3 \\ f(-5) &= 68 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 2x^2 - 4x - 3 \\ f(4) &= 13 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 3x^2 + 5x + 3 \\ f(-2) &= 5 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -3x^2 + 3x + 1 \\ f(5) &= -59 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 3x^2 - 4x + 2 \\ f(-5) &= 97 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 5x^2 - 2x - 5 \\ f(2) &= 11 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 2x^2 - 5x - 5 \\ f(-4) &= 47 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -3x^2 + 2x + 4 \\ f(2) &= -4 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -4x^2 + x - 5 \\ f(2) &= -19 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= 5x^2 + 3x + 1 \\ f(-5) &= 111 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5x^2 - 5x - 4 \\ f(2) &= -34 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 3x^2 - 2x + 4 \\ f(3) &= 25 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= 4x^2 + 5x - 4 \\ f(5) &= 121 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -5x^2 + 5x - 2 \\ f(-1) &= -12 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -4x^2 + 5x + 1 \\ f(4) &= -43 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 4x^2 - 3x + 3 \\ f(2) &= 13 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -x^2 + 4x - 3 \\ f(1) &= 0 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -x^2 - 3x + 1 \\ f(3) &= -17 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= 5x^2 + 4x - 1 \\ f(1) &= 8 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 3x^2 + 3x + 1 \\ f(-5) &= 61 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -2x^2 - 3x - 3 \\ f(5) &= -68 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 2x^2 + 2x - 1 \\ f(-3) &= 11 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2x^2 - 5x - 2 \\ f(1) &= -9 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -5x^2 - x + 1 \\ f(-5) &= -119 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= x^2 - x - 3 \\ f(3) &= 3 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4x^2 - 3x - 4 \\ f(4) &= -80 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -2x^2 + 3x - 1 \\ f(2) &= -3 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= 2x^2 + 3x - 1 \\ f(4) &= 43 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 3x^2 - 2x + 2 \\ f(-1) &= 7 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -3x^2 + 4x + 3 \\ f(-3) &= -36 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 3x^2 + 3x - 1 \\ f(3) &= 35 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -x^2 + x + 2 \\ f(3) &= -4 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -2x^2 - 5x + 3 \\ f(-4) &= -9 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 3x^2 + 4x + 4 \\ f(-5) &= 59 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= 4x^2 + x - 5 \\ f(3) &= 34 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -2x^2 - x + 4 \\ f(-3) &= -11 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -3x^2 + 4x + 5 \\ f(3) &= -10 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= 2x^2 - 2x - 3 \\ f(-2) &= 9 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 3x^2 - 2x + 1 \\ f(-3) &= 34 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 2x^2 - 5x - 4 \\ f(1) &= -7 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -5x^2 - 5x - 2 \\ f(-4) &= -62 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -3x^2 - 2x + 2 \\ f(5) &= -83 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -x^2 - 4x + 4 \\ f(-1) &= 7 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= 2x^2 - 4x - 3 \\ f(-5) &= 67 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 5x^2 - 5x + 4 \\ f(5) &= 104 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 5x^2 - 2x - 5 \\ f(-4) &= 83 \end{aligned}$$

$$(1) \quad f(x) = -2x^2 + 3x - 5 \\ f(-4) = -49$$

$$(2) \quad f(x) = x^2 - 3x + 1 \\ f(-3) = 19$$

$$(3) \quad f(x) = -2x^2 - x - 2 \\ f(3) = -23$$

$$(4) \quad f(x) = 4x^2 + 4x - 1 \\ f(-5) = 79$$

$$(5) \quad f(x) = -2x^2 - 2x + 5 \\ f(-3) = -7$$

$$(6) \quad f(x) = 2x^2 + x - 5 \\ f(4) = 31$$

$$(7) \quad f(x) = 4x^2 - x - 5 \\ f(-1) = 0$$

$$(8) \quad f(x) = 3x^2 - 2x - 4 \\ f(1) = -3$$

$$(9) \quad f(x) = 5x^2 + 5x - 4 \\ f(-4) = 56$$

$$(10) \quad f(x) = -x^2 - 5x + 3 \\ f(-4) = 7$$

$$(11) \quad f(x) = 2x^2 + 4x - 3 \\ f(-1) = -5$$

$$(12) \quad f(x) = -2x^2 - 3x + 2 \\ f(-3) = -7$$

$$(13) \quad f(x) = x^2 - 5x + 3 \\ f(2) = -3$$

$$(14) \quad f(x) = -3x^2 + 2x + 5 \\ f(-4) = -51$$

$$(15) \quad f(x) = -5x^2 - 2x + 5 \\ f(5) = -130$$

$$(16) \quad f(x) = -3x^2 + x - 2 \\ f(1) = -4$$

$$(17) \quad f(x) = -2x^2 - 5x + 2 \\ f(2) = -16$$

$$(18) \quad f(x) = 2x^2 + x - 4 \\ f(-3) = 11$$

$$(19) \quad f(x) = -3x^2 - 3x + 2 \\ f(2) = -16$$

$$(20) \quad f(x) = 3x^2 + 2x - 5 \\ f(-1) = -4$$

$$(21) \quad f(x) = 5x^2 + 5x - 1 \\ f(5) = 149$$

$$(22) \quad f(x) = 2x^2 + x - 2 \\ f(-1) = -1$$

$$(23) \quad f(x) = 4x^2 + 3x - 1 \\ f(1) = 6$$

$$(24) \quad f(x) = x^2 - 5x + 3 \\ f(4) = -1$$

$$(25) \quad f(x) = -5x^2 + 4x + 2 \\ f(3) = -31$$

$$(26) \quad f(x) = -x^2 - x + 5 \\ f(5) = -25$$

$$(27) \quad f(x) = 3x^2 - 5x - 1 \\ f(-2) = 21$$

$$(28) \quad f(x) = x^2 - 2x + 3 \\ f(-3) = 18$$

$$(29) \quad f(x) = -2x^2 + 5x + 1 \\ f(3) = -2$$

$$(30) \quad f(x) = -2x^2 - 4x + 5 \\ f(-2) = 5$$

$$(31) \quad f(x) = -x^2 + 3x + 2 \\ f(3) = 2$$

$$(32) \quad f(x) = -3x^2 + x + 3 \\ f(-3) = -27$$

$$(33) \quad f(x) = x^2 - 5x - 1 \\ f(-2) = 13$$

$$(34) \quad f(x) = 2x^2 - 3x + 5 \\ f(2) = 7$$

$$(35) \quad f(x) = -2x^2 + 2x + 2 \\ f(-1) = -2$$

$$(36) \quad f(x) = 3x^2 + 3x + 1 \\ f(5) = 91$$

$$(37) \quad f(x) = -2x^2 - 2x + 5 \\ f(2) = -7$$

$$(38) \quad f(x) = -x^2 + 5x - 5 \\ f(-5) = -55$$

$$(39) \quad f(x) = -2x^2 + 2x + 3 \\ f(-5) = -57$$

$$(40) \quad f(x) = x^2 - 3x + 3 \\ f(-5) = 43$$

$$(41) \quad f(x) = -3x^2 + 2x + 4 \\ f(-2) = -12$$

$$(42) \quad f(x) = -x^2 + 5x + 3 \\ f(2) = 9$$

$$(43) \quad f(x) = -4x^2 - 4x - 1 \\ f(3) = -49$$

$$(44) \quad f(x) = x^2 - 5x + 5 \\ f(-3) = 29$$

$$(45) \quad f(x) = 2x^2 - 2x - 2 \\ f(-3) = 22$$



$$(1) \begin{aligned} f(x) &= -3x^2 - 5x - 2 \\ f(5) &= -102 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 5x^2 + 3x - 1 \\ f(3) &= 53 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -3x^2 + 5x + 3 \\ f(-2) &= -19 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -2x^2 - 2x - 5 \\ f(-2) &= -9 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 4x^2 + 4x + 2 \\ f(-4) &= 50 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 5x^2 + 2x - 2 \\ f(2) &= 22 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 4x^2 + 4x + 2 \\ f(2) &= 26 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 2x^2 - 2x + 3 \\ f(3) &= 15 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -4x^2 - x + 1 \\ f(-5) &= -94 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 - 2x - 1 \\ f(-5) &= 84 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -3x^2 - x + 5 \\ f(-5) &= -65 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -5x^2 + 4x - 2 \\ f(-2) &= -30 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(3) &= -49 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= 4x^2 - 3x - 1 \\ f(-5) &= 114 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 2x^2 - 3x - 2 \\ f(1) &= -3 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -2x^2 + 2x + 1 \\ f(-5) &= -59 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -3x^2 - 4x - 3 \\ f(-4) &= -35 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -x^2 + 3x + 1 \\ f(2) &= 3 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -4x^2 + 4x + 3 \\ f(-5) &= -117 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= x^2 - 4x - 2 \\ f(-1) &= 3 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 2x^2 - 2x + 3 \\ f(2) &= 7 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 3x^2 + 2x - 3 \\ f(1) &= 2 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 2x^2 + x - 2 \\ f(-5) &= 43 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= -2x^2 - 5x - 5 \\ f(4) &= -57 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -5x^2 - x - 5 \\ f(-3) &= -47 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 2x^2 + x - 1 \\ f(-1) &= 0 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= 2x^2 - 4x + 3 \\ f(1) &= 1 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 5x^2 + 4x + 5 \\ f(2) &= 33 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 3x^2 + 3x + 2 \\ f(1) &= 8 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= x^2 + 3x + 4 \\ f(3) &= 22 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 + 2x + 2 \\ f(1) &= 9 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -2x^2 + 5x - 3 \\ f(-5) &= -78 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 5x^2 + 5x + 2 \\ f(-4) &= 62 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= x^2 - 3x + 3 \\ f(-1) &= 7 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -4x^2 + 2x + 5 \\ f(3) &= -25 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= x^2 + 2x - 4 \\ f(-4) &= 4 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -5x^2 + 2x - 1 \\ f(-3) &= -52 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 2x^2 - 5x + 4 \\ f(4) &= 16 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -4x^2 + 5x - 2 \\ f(4) &= -46 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -2x^2 - x - 3 \\ f(3) &= -24 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -x^2 - 4x - 2 \\ f(3) &= -23 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -x^2 + 4x - 5 \\ f(-4) &= -37 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -2x^2 + 2x - 2 \\ f(4) &= -26 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -4x^2 - x - 3 \\ f(-5) &= -98 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 3x^2 - x + 2 \\ f(2) &= 12 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -x^2 + 4x + 4 \\ f(-3) &= -17 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 3x^2 - 3x - 2 \\ f(5) &= 58 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= -4x^2 + x + 3 \\ f(2) &= -11 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -x^2 + 3x + 1 \\ f(-4) &= -27 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 4x^2 - 5x + 3 \\ f(5) &= 78 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 3x^2 - 3x - 4 \\ f(4) &= 32 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -2x^2 - x - 1 \\ f(3) &= -22 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 4x^2 + 2x + 3 \\ f(-3) &= 33 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= x^2 + 3x + 1 \\ f(-1) &= -1 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 + 5x - 3 \\ f(-3) &= 9 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= x^2 + 2x + 3 \\ f(-5) &= 18 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= 3x^2 - 4x + 5 \\ f(4) &= 37 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -5x^2 - 5x - 3 \\ f(4) &= -103 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= 3x^2 + x + 4 \\ f(-2) &= 14 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= x^2 + 3x + 1 \\ f(1) &= 5 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 2x^2 + 2x - 4 \\ f(-3) &= 8 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -x^2 + 2x - 3 \\ f(4) &= -11 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -4x^2 + 5x - 2 \\ f(3) &= -23 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -5x^2 - 4x + 2 \\ f(-1) &= 1 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= 4x^2 + 4x - 4 \\ f(-4) &= 44 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -3x^2 - 5x + 5 \\ f(4) &= -63 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -4x^2 + 4x + 1 \\ f(-5) &= -119 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 4x^2 + 3x - 5 \\ f(3) &= 40 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= x^2 + 5x - 1 \\ f(1) &= 5 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4x^2 - 2x - 5 \\ f(1) &= -11 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -5x^2 - 5x - 2 \\ f(5) &= -152 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -x^2 + 3x - 2 \\ f(-1) &= -6 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 2x^2 - 4x - 2 \\ f(-3) &= 28 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -2x^2 + 2x + 3 \\ f(2) &= -1 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= -3x^2 - x - 1 \\ f(1) &= -5 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 5x^2 + 3x - 4 \\ f(-3) &= 32 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -4x^2 + 5x + 4 \\ f(-4) &= -80 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -2x^2 - 4x - 1 \\ f(1) &= -7 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -5x^2 - 5x - 4 \\ f(4) &= -104 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= x^2 - x - 5 \\ f(5) &= 15 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 3x^2 - 3x - 1 \\ f(-1) &= 5 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= 3x^2 + 3x + 4 \\ f(3) &= 40 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= x^2 - 3x - 2 \\ f(4) &= 2 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 2x^2 + 2x + 2 \\ f(2) &= 14 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -4x^2 - 5x + 2 \\ f(-2) &= -4 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -2x^2 - 3x - 1 \\ f(-5) &= -36 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= 3x^2 - 2x - 3 \\ f(-4) &= 53 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -x^2 + 2x - 4 \\ f(1) &= -3 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 4x^2 + 2x + 2 \\ f(3) &= 44 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 3x^2 + 4x + 5 \\ f(3) &= 44 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -x^2 - 2x + 4 \\ f(3) &= -11 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= -3x^2 + 4x - 3 \\ f(4) &= -35 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 3x^2 + x - 2 \\ f(4) &= 50 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 2x^2 + x + 1 \\ f(1) &= 4 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -2x^2 + x + 1 \\ f(3) &= -14 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -4x^2 + 5x + 5 \\ f(-3) &= -46 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 4x^2 - x - 4 \\ f(-1) &= 1 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -4x^2 + 3x - 5 \\ f(5) &= -90 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -5x^2 - 5x + 3 \\ f(4) &= -97 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5x^2 - x + 2 \\ f(2) &= -20 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -4x^2 + x - 4 \\ f(1) &= -7 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -4x^2 + 3x - 2 \\ f(1) &= -3 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -4x^2 - 4x + 2 \\ f(-5) &= -78 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -x^2 - 5x - 5 \\ f(-5) &= -5 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -4x^2 + 2x + 4 \\ f(-4) &= -68 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 4x^2 - x + 2 \\ f(4) &= 62 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -4x^2 - x + 3 \\ f(-2) &= -11 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -x^2 + 3x - 4 \\ f(-1) &= -8 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= x^2 + 3x - 5 \\ f(3) &= 13 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -5x^2 - x - 3 \\ f(1) &= -9 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 3x^2 + x + 2 \\ f(-3) &= 26 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -5x^2 + 4x - 2 \\ f(2) &= -14 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -3x^2 - 4x - 3 \\ f(-2) &= -7 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 3x^2 + x + 1 \\ f(-1) &= 3 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 4x^2 + 4x - 5 \\ f(-5) &= 75 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -5x^2 - 5x + 1 \\ f(5) &= -149 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -x^2 + 2x - 2 \\ f(3) &= -5 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -3x^2 + 3x - 2 \\ f(3) &= -20 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 2x^2 - 5x + 2 \\ f(1) &= -1 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= -4x^2 + x + 4 \\ f(-5) &= -101 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 3x^2 + x - 2 \\ f(5) &= 78 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= 4x^2 - 4x + 2 \\ f(-5) &= 122 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 4x^2 - 2x + 3 \\ f(4) &= 59 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= 5x^2 + 4x - 4 \\ f(3) &= 53 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -x^2 - 5x + 4 \\ f(4) &= -32 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 3x^2 + 5x + 4 \\ f(-3) &= 16 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -x^2 - 4x + 1 \\ f(3) &= -20 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 5x^2 - 5x + 3 \\ f(-1) &= 13 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 2x^2 - 2x + 1 \\ f(-3) &= 25 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 5x^2 + 2x + 4 \\ f(2) &= 28 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 3x^2 + 3x - 1 \\ f(3) &= 35 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -4x^2 - 3x - 4 \\ f(-3) &= -31 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -4x^2 - 5x + 5 \\ f(-5) &= -70 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -x^2 + 5x + 2 \\ f(-3) &= -22 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= x^2 + 3x + 4 \\ f(-5) &= 14 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 2x^2 + 3x - 2 \\ f(-5) &= 33 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= x^2 - x + 1 \\ f(-3) &= 13 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 2x^2 + 2x + 3 \\ f(4) &= 43 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 5x^2 + 4x - 4 \\ f(5) &= 141 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= 5x^2 - 2x + 5 \\ f(4) &= 77 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -x^2 - x - 1 \\ f(4) &= -21 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -5x^2 + 4x + 1 \\ f(5) &= -104 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -5x^2 + 3x + 5 \\ f(5) &= -105 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= 3x^2 + 4x - 1 \\ f(4) &= 63 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 + 4x - 5 \\ f(-2) &= -1 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -2x^2 + x + 1 \\ f(1) &= 0 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -2x^2 + 4x + 3 \\ f(-3) &= -27 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 3x^2 - x + 5 \\ f(2) &= 15 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= 5x^2 - 3x - 3 \\ f(-2) &= 23 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -4x^2 + 3x + 5 \\ f(-5) &= -110 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= x^2 - 3x - 3 \\ f(-5) &= 37 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -x^2 + 4x + 4 \\ f(2) &= 8 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -5x^2 + 5x + 3 \\ f(5) &= -97 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= x^2 + x - 1 \\ f(-2) &= 1 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -x^2 + 5x - 5 \\ f(5) &= -5 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -x^2 - 3x - 4 \\ f(-5) &= -14 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -4x^2 - x - 1 \\ f(3) &= -40 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 3x^2 + 2x - 2 \\ f(-5) &= 63 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 4x^2 - 2x + 4 \\ f(-4) &= 76 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -2x^2 + 2x - 1 \\ f(-4) &= -41 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -2x^2 + 5x + 4 \\ f(4) &= -8 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -5x^2 + 5x - 4 \\ f(-2) &= -34 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -x^2 - 3x - 1 \\ f(-3) &= -1 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -2x^2 - 2x + 1 \\ f(1) &= -3 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 3x^2 - 2x + 4 \\ f(5) &= 69 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -3x^2 - 4x - 3 \\ f(-4) &= -35 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -5x^2 + 3x - 3 \\ f(1) &= -5 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 2x^2 + 4x + 2 \\ f(1) &= 8 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -5x^2 - x - 1 \\ f(-4) &= -77 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -3x^2 + 4x + 2 \\ f(5) &= -53 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -4x^2 - x + 5 \\ f(-3) &= -28 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -5x^2 + 2x + 4 \\ f(1) &= 1 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= -2x^2 - 4x + 3 \\ f(-1) &= 5 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -4x^2 - 3x - 1 \\ f(-2) &= -11 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 2x^2 + 2x - 1 \\ f(4) &= 39 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 2x^2 + 2x - 3 \\ f(3) &= 21 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -3x^2 + 3x - 1 \\ f(-3) &= -37 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= 5x^2 + 2x - 1 \\ f(-2) &= 15 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -2x^2 + x + 2 \\ f(-3) &= -19 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -4x^2 + x - 4 \\ f(-4) &= -72 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3x^2 - 2x + 5 \\ f(2) &= -11 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 4x^2 + 4x - 3 \\ f(2) &= 21 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 3x^2 + 4x + 4 \\ f(-1) &= 3 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -5x^2 + x + 4 \\ f(-2) &= -18 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -5x^2 - 3x - 1 \\ f(-4) &= -69 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -4x^2 - 3x - 2 \\ f(3) &= -47 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4x^2 + 4x - 5 \\ f(-1) &= -13 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -3x^2 + 4x - 4 \\ f(2) &= -8 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -4x^2 + 2x + 4 \\ f(-4) &= -68 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -4x^2 + x - 4 \\ f(-1) &= -9 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -x^2 + 4x + 4 \\ f(-2) &= -8 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -2x^2 - 4x + 5 \\ f(-5) &= -25 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -4x^2 - 5x + 2 \\ f(4) &= -82 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -3x^2 + 2x + 1 \\ f(5) &= -64 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -4x^2 + 5x + 1 \\ f(3) &= -20 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -x^2 + 4x + 3 \\ f(-2) &= -9 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -x^2 + 5x + 5 \\ f(-5) &= -45 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -2x^2 - x - 5 \\ f(2) &= -15 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -2x^2 - 2x + 5 \\ f(-2) &= 1 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -3x^2 - 4x - 5 \\ f(-5) &= -60 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= x^2 - x - 5 \\ f(-4) &= 15 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 3x^2 - 3x - 1 \\ f(4) &= 35 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 5x^2 - 5x - 5 \\ f(1) &= -5 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 3x^2 - 5x - 3 \\ f(4) &= 25 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -x^2 - 4x + 5 \\ f(1) &= 0 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= x^2 + x + 4 \\ f(-1) &= 4 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -5x^2 + 3x + 4 \\ f(4) &= -64 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -3x^2 + 3x - 1 \\ f(-1) &= -7 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= x^2 - 3x - 3 \\ f(2) &= -5 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 5x^2 - 3x - 3 \\ f(-5) &= 137 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -3x^2 - 4x + 1 \\ f(-2) &= -3 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= x^2 - 2x + 2 \\ f(-1) &= 5 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -2x^2 - x - 2 \\ f(1) &= -5 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= 5x^2 - x - 4 \\ f(-2) &= 18 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -x^2 + 3x - 4 \\ f(2) &= -2 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -x^2 - x - 5 \\ f(5) &= -35 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= 3x^2 - 4x + 5 \\ f(4) &= 37 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 5x^2 - 2x - 3 \\ f(-5) &= 132 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= -x^2 - 4x - 2 \\ f(-2) &= 2 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 5x^2 + x + 5 \\ f(5) &= 135 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= -3x^2 + 3x + 5 \\ f(4) &= -31 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= x^2 - 3x + 3 \\ f(-1) &= 7 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -5x^2 + x + 4 \\ f(2) &= -14 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -4x^2 - 5x - 2 \\ f(-3) &= -23 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -5x^2 - 4x - 4 \\ f(-5) &= -109 \end{aligned}$$

$$(1) \quad f(x) = -4x^2 - 4x - 5 \\ f(1) = -13$$

$$(2) \quad f(x) = 4x^2 - 3x - 2 \\ f(-2) = 20$$

$$(3) \quad f(x) = 5x^2 - 5x + 4 \\ f(-5) = 154$$

$$(4) \quad f(x) = -5x^2 + 4x + 4 \\ f(4) = -60$$

$$(5) \quad f(x) = -x^2 + 5x - 5 \\ f(1) = -1$$

$$(6) \quad f(x) = 4x^2 + 3x - 5 \\ f(4) = 71$$

$$(7) \quad f(x) = -x^2 - 3x + 2 \\ f(3) = -16$$

$$(8) \quad f(x) = -2x^2 - 2x - 2 \\ f(-2) = -6$$

$$(9) \quad f(x) = -3x^2 + 4x - 4 \\ f(5) = -59$$

$$(10) \quad f(x) = -5x^2 - 2x - 5 \\ f(-2) = -21$$

$$(11) \quad f(x) = -5x^2 + 4x - 3 \\ f(3) = -36$$

$$(12) \quad f(x) = 2x^2 + 4x + 5 \\ f(4) = 53$$

$$(13) \quad f(x) = 2x^2 + 2x + 3 \\ f(-5) = 43$$

$$(14) \quad f(x) = x^2 - 4x + 2 \\ f(1) = -1$$

$$(15) \quad f(x) = x^2 - 4x + 1 \\ f(2) = -3$$

$$(16) \quad f(x) = -2x^2 - x - 5 \\ f(5) = -60$$

$$(17) \quad f(x) = -5x^2 - 3x + 3 \\ f(-1) = 1$$

$$(18) \quad f(x) = x^2 - 5x - 2 \\ f(-4) = 34$$

$$(19) \quad f(x) = -2x^2 - 3x - 5 \\ f(5) = -70$$

$$(20) \quad f(x) = -x^2 - 5x + 4 \\ f(-5) = 4$$

$$(21) \quad f(x) = -x^2 - x + 3 \\ f(2) = -3$$

$$(22) \quad f(x) = 5x^2 + x + 4 \\ f(1) = 10$$

$$(23) \quad f(x) = -3x^2 - 5x + 5 \\ f(1) = -3$$

$$(24) \quad f(x) = -5x^2 - 5x - 5 \\ f(2) = -35$$

$$(25) \quad f(x) = -4x^2 + 2x + 1 \\ f(1) = -1$$

$$(26) \quad f(x) = -x^2 + 5x + 3 \\ f(2) = 9$$

$$(27) \quad f(x) = -4x^2 + 5x + 1 \\ f(-4) = -83$$

$$(28) \quad f(x) = 4x^2 + 4x + 1 \\ f(-3) = 25$$

$$(29) \quad f(x) = 5x^2 - 2x + 4 \\ f(5) = 119$$

$$(30) \quad f(x) = -5x^2 - 3x - 4 \\ f(5) = -144$$

$$(31) \quad f(x) = -4x^2 - 4x + 1 \\ f(-4) = -47$$

$$(32) \quad f(x) = -3x^2 + 3x + 4 \\ f(4) = -32$$

$$(33) \quad f(x) = x^2 - x - 3 \\ f(4) = 9$$

$$(34) \quad f(x) = -2x^2 + 2x - 5 \\ f(-2) = -17$$

$$(35) \quad f(x) = x^2 - 3x + 1 \\ f(4) = 5$$

$$(36) \quad f(x) = -2x^2 + x - 5 \\ f(3) = -20$$

$$(37) \quad f(x) = 4x^2 + 5x + 4 \\ f(4) = 88$$

$$(38) \quad f(x) = 3x^2 + 2x - 1 \\ f(1) = 4$$

$$(39) \quad f(x) = -x^2 + 2x + 2 \\ f(-4) = -22$$

$$(40) \quad f(x) = 3x^2 - 2x - 4 \\ f(-4) = 52$$

$$(41) \quad f(x) = 2x^2 - 2x + 3 \\ f(5) = 43$$

$$(42) \quad f(x) = -x^2 - x - 4 \\ f(1) = -6$$

$$(43) \quad f(x) = 2x^2 + 5x - 1 \\ f(-2) = -3$$

$$(44) \quad f(x) = 5x^2 - x + 4 \\ f(-4) = 88$$

$$(45) \quad f(x) = -5x^2 + 2x - 2 \\ f(-1) = -9$$

$$(1) \begin{aligned} f(x) &= -x^2 + 2x + 3 \\ f(-3) &= -12 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= -3x^2 + x + 5 \\ f(-2) &= -9 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 4x^2 - 2x - 4 \\ f(5) &= 86 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -4x^2 + 3x + 4 \\ f(-3) &= -41 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -2x^2 - 3x + 4 \\ f(5) &= -61 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -4x^2 + 5x - 3 \\ f(-4) &= -87 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4x^2 - 4x + 2 \\ f(3) &= -46 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -4x^2 - 2x - 5 \\ f(-3) &= -35 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= -3x^2 + 5x + 1 \\ f(-4) &= -67 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 2x^2 + x + 1 \\ f(-3) &= 16 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= -3x^2 + x + 5 \\ f(1) &= 3 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -3x^2 + 2x + 1 \\ f(-3) &= -32 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 4x^2 + x - 2 \\ f(1) &= 3 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -5x^2 - 5x - 1 \\ f(-2) &= -11 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= 5x^2 - 3x + 2 \\ f(-4) &= 94 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 4x^2 + 2x - 1 \\ f(-4) &= 55 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -3x^2 - x + 3 \\ f(-1) &= 1 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -2x^2 - 4x + 2 \\ f(2) &= -14 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -2x^2 - 3x - 3 \\ f(-4) &= -23 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -5x^2 + 2x + 4 \\ f(-5) &= -131 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -4x^2 - 4x + 4 \\ f(-2) &= -4 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2x^2 + x - 5 \\ f(1) &= -6 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 2x^2 - x - 2 \\ f(3) &= 13 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 3x^2 - x + 5 \\ f(1) &= 7 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4x^2 - 3x - 3 \\ f(-5) &= -88 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= x^2 + 5x + 4 \\ f(-2) &= -2 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= x^2 - x + 4 \\ f(-4) &= 24 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 4x^2 + 4x + 1 \\ f(-2) &= 9 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= 5x^2 - 4x + 3 \\ f(2) &= 15 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= 5x^2 - 2x - 2 \\ f(-3) &= 49 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -5x^2 + 3x - 1 \\ f(-2) &= -27 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -4x^2 - 3x - 2 \\ f(-2) &= -12 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= -2x^2 - 3x - 2 \\ f(4) &= -46 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -5x^2 - x - 4 \\ f(4) &= -88 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -5x^2 - 4x - 1 \\ f(2) &= -29 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -4x^2 - 5x - 5 \\ f(-4) &= -49 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -x^2 + x - 1 \\ f(3) &= -7 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= -x^2 + 3x + 4 \\ f(-4) &= -24 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 3x^2 + 5x - 4 \\ f(5) &= 96 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= 4x^2 + 4x - 5 \\ f(-3) &= 19 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 3x^2 + x + 4 \\ f(5) &= 84 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= 4x^2 - 5x + 2 \\ f(-1) &= 11 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -3x^2 - 4x + 5 \\ f(-3) &= -10 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= -2x^2 - x + 4 \\ f(5) &= -51 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= -4x^2 + 2x - 2 \\ f(-4) &= -74 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3x^2 - 5x + 2 \\ f(3) &= -40 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= -4x^2 + 2x + 2 \\ f(2) &= -10 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= 4x^2 + 3x - 4 \\ f(-4) &= 48 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -x^2 - 4x + 2 \\ f(-4) &= 2 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -5x^2 + x - 1 \\ f(1) &= -5 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= 5x^2 + 5x + 2 \\ f(-4) &= 62 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4x^2 + 2x - 4 \\ f(4) &= -60 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= -4x^2 - 5x - 4 \\ f(-1) &= -3 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= 4x^2 + x + 4 \\ f(-3) &= 37 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3x^2 - x - 3 \\ f(-1) &= 1 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 3x^2 - 4x + 5 \\ f(2) &= 9 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -2x^2 + 4x - 3 \\ f(-5) &= -73 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 4x^2 - 3x - 2 \\ f(-4) &= 74 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -4x^2 + 3x + 2 \\ f(2) &= -8 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -x^2 + 4x - 2 \\ f(2) &= 2 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -3x^2 + 3x - 2 \\ f(1) &= -2 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -5x^2 + 5x + 2 \\ f(-1) &= -8 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= x^2 - 5x + 4 \\ f(-2) &= 18 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 4x^2 - 5x + 5 \\ f(3) &= 26 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -5x^2 + 3x + 4 \\ f(-1) &= -4 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= 5x^2 + 3x + 5 \\ f(-2) &= 19 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 4x^2 - 5x - 5 \\ f(2) &= 1 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= 5x^2 - 3x + 1 \\ f(-2) &= 27 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= -3x^2 + 5x - 3 \\ f(2) &= -5 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -2x^2 + x - 3 \\ f(-4) &= -39 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= 3x^2 - 4x - 3 \\ f(-5) &= 92 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -4x^2 - 4x - 5 \\ f(1) &= -13 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 5x^2 + x - 4 \\ f(2) &= 18 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= x^2 - 2x + 4 \\ f(5) &= 19 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= -3x^2 + 2x - 5 \\ f(4) &= -45 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= -4x^2 - x + 3 \\ f(3) &= -36 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= -4x^2 - x + 2 \\ f(2) &= -16 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 3x^2 - 2x - 2 \\ f(-5) &= 83 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -x^2 + 3x + 2 \\ f(1) &= 4 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= -3x^2 + 2x + 3 \\ f(1) &= 2 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= 3x^2 - 2x + 5 \\ f(-1) &= 10 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= 4x^2 + 4x - 4 \\ f(-2) &= 4 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 3x^2 - x + 2 \\ f(-5) &= 82 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 4x^2 + 4x + 5 \\ f(-4) &= 53 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -2x^2 + 4x - 4 \\ f(-4) &= -52 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 4x^2 - 5x + 3 \\ f(2) &= 9 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= -4x^2 + 4x - 2 \\ f(4) &= -50 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -2x^2 - 3x - 4 \\ f(-2) &= -6 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 2x^2 + x + 2 \\ f(-3) &= 17 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 4x^2 + 5x + 2 \\ f(-4) &= 46 \end{aligned}$$



$$(1) \begin{aligned} f(x) &= 5x^2 - 2x - 3 \\ f(-3) &= 48 \end{aligned}$$

$$(16) \begin{aligned} f(x) &= 3x^2 + 2x + 1 \\ f(-4) &= 41 \end{aligned}$$

$$(31) \begin{aligned} f(x) &= x^2 - 2x - 3 \\ f(-1) &= 0 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -2x^2 + 3x - 1 \\ f(1) &= 0 \end{aligned}$$

$$(17) \begin{aligned} f(x) &= -4x^2 - 4x + 3 \\ f(-3) &= -21 \end{aligned}$$

$$(32) \begin{aligned} f(x) &= -5x^2 + 4x + 2 \\ f(-5) &= -143 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -3x^2 + 5x - 3 \\ f(-3) &= -45 \end{aligned}$$

$$(18) \begin{aligned} f(x) &= 5x^2 + 5x + 3 \\ f(-1) &= 3 \end{aligned}$$

$$(33) \begin{aligned} f(x) &= 2x^2 + 2x - 3 \\ f(4) &= 37 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 4x^2 - x + 2 \\ f(-5) &= 107 \end{aligned}$$

$$(19) \begin{aligned} f(x) &= 2x^2 - 5x + 3 \\ f(-5) &= 78 \end{aligned}$$

$$(34) \begin{aligned} f(x) &= -x^2 - 5x - 2 \\ f(3) &= -26 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -5x^2 + 2x + 4 \\ f(-3) &= -47 \end{aligned}$$

$$(20) \begin{aligned} f(x) &= -4x^2 + 3x + 3 \\ f(-5) &= -112 \end{aligned}$$

$$(35) \begin{aligned} f(x) &= -3x^2 - 5x - 4 \\ f(1) &= -12 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3x^2 + 3x + 4 \\ f(4) &= 64 \end{aligned}$$

$$(21) \begin{aligned} f(x) &= -2x^2 - x - 3 \\ f(-4) &= -31 \end{aligned}$$

$$(36) \begin{aligned} f(x) &= -5x^2 - 3x + 1 \\ f(-1) &= -1 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 3x^2 - 2x + 2 \\ f(5) &= 67 \end{aligned}$$

$$(22) \begin{aligned} f(x) &= -x^2 - 2x + 4 \\ f(-4) &= -4 \end{aligned}$$

$$(37) \begin{aligned} f(x) &= -4x^2 - 3x + 5 \\ f(-4) &= -47 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2x^2 - x + 5 \\ f(5) &= -50 \end{aligned}$$

$$(23) \begin{aligned} f(x) &= -x^2 - 4x - 2 \\ f(-1) &= 1 \end{aligned}$$

$$(38) \begin{aligned} f(x) &= 4x^2 - 2x + 2 \\ f(2) &= 14 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 2x^2 - 3x + 5 \\ f(-1) &= 10 \end{aligned}$$

$$(24) \begin{aligned} f(x) &= -x^2 + x - 4 \\ f(3) &= -10 \end{aligned}$$

$$(39) \begin{aligned} f(x) &= -2x^2 + 2x + 5 \\ f(-4) &= -35 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 4x^2 - 5x - 5 \\ f(-5) &= 120 \end{aligned}$$

$$(25) \begin{aligned} f(x) &= -4x^2 - 2x - 3 \\ f(2) &= -23 \end{aligned}$$

$$(40) \begin{aligned} f(x) &= -x^2 + x + 1 \\ f(2) &= -1 \end{aligned}$$

$$(11) \begin{aligned} f(x) &= 3x^2 + 4x + 5 \\ f(-1) &= 4 \end{aligned}$$

$$(26) \begin{aligned} f(x) &= x^2 - 4x - 3 \\ f(-5) &= 42 \end{aligned}$$

$$(41) \begin{aligned} f(x) &= 4x^2 + 5x + 1 \\ f(-5) &= 76 \end{aligned}$$

$$(12) \begin{aligned} f(x) &= -3x^2 + 4x + 1 \\ f(-4) &= -63 \end{aligned}$$

$$(27) \begin{aligned} f(x) &= 2x^2 + 3x + 1 \\ f(-1) &= 0 \end{aligned}$$

$$(42) \begin{aligned} f(x) &= -2x^2 - 4x - 4 \\ f(1) &= -10 \end{aligned}$$

$$(13) \begin{aligned} f(x) &= -4x^2 - x - 2 \\ f(1) &= -7 \end{aligned}$$

$$(28) \begin{aligned} f(x) &= 4x^2 - 2x - 4 \\ f(-1) &= 2 \end{aligned}$$

$$(43) \begin{aligned} f(x) &= 3x^2 - 2x + 4 \\ f(-4) &= 60 \end{aligned}$$

$$(14) \begin{aligned} f(x) &= -3x^2 + 2x + 3 \\ f(-2) &= -13 \end{aligned}$$

$$(29) \begin{aligned} f(x) &= 3x^2 - 5x + 1 \\ f(3) &= 13 \end{aligned}$$

$$(44) \begin{aligned} f(x) &= 2x^2 + 2x + 5 \\ f(5) &= 65 \end{aligned}$$

$$(15) \begin{aligned} f(x) &= -x^2 + x - 4 \\ f(-2) &= -10 \end{aligned}$$

$$(30) \begin{aligned} f(x) &= 3x^2 + 2x + 2 \\ f(-4) &= 42 \end{aligned}$$

$$(45) \begin{aligned} f(x) &= 4x^2 + x - 2 \\ f(-4) &= 58 \end{aligned}$$

$$(1) \quad f(x) = x^2 + 3x + 3$$

$$f(-3) = 3$$

$$(16) \quad f(x) = 4x^2 + 5x - 2$$

$$f(1) = 7$$

$$(31) \quad f(x) = 2x^2 - 2x - 5$$

$$f(4) = 19$$

$$(2) \quad f(x) = 5x^2 - 2x - 4$$

$$f(-3) = 47$$

$$(17) \quad f(x) = 2x^2 + 4x + 2$$

$$f(-4) = 18$$

$$(32) \quad f(x) = -4x^2 - x - 2$$

$$f(5) = -107$$

$$(3) \quad f(x) = -2x^2 - 3x - 4$$

$$f(1) = -9$$

$$(18) \quad f(x) = -2x^2 + x - 5$$

$$f(4) = -33$$

$$(33) \quad f(x) = -2x^2 + 4x + 3$$

$$f(-4) = -45$$

$$(4) \quad f(x) = 5x^2 + 2x - 4$$

$$f(-4) = 68$$

$$(19) \quad f(x) = 3x^2 + 3x - 1$$

$$f(5) = 89$$

$$(34) \quad f(x) = -x^2 + 2x + 3$$

$$f(-5) = -32$$

$$(5) \quad f(x) = -3x^2 + 3x - 3$$

$$f(-1) = -9$$

$$(20) \quad f(x) = -2x^2 + 4x + 1$$

$$f(1) = 3$$

$$(35) \quad f(x) = -4x^2 + 4x + 1$$

$$f(-3) = -47$$

$$(6) \quad f(x) = -5x^2 + 3x + 2$$

$$f(1) = 0$$

$$(21) \quad f(x) = -3x^2 + 3x - 3$$

$$f(2) = -9$$

$$(36) \quad f(x) = -2x^2 + 4x - 5$$

$$f(-4) = -53$$

$$(7) \quad f(x) = -2x^2 + 3x - 5$$

$$f(4) = -25$$

$$(22) \quad f(x) = 5x^2 - x - 4$$

$$f(5) = 116$$

$$(37) \quad f(x) = 5x^2 + x + 5$$

$$f(-2) = 23$$

$$(8) \quad f(x) = 5x^2 + 5x - 1$$

$$f(4) = 99$$

$$(23) \quad f(x) = -3x^2 + 5x + 4$$

$$f(-5) = -96$$

$$(38) \quad f(x) = -2x^2 + 4x + 4$$

$$f(-5) = -66$$

$$(9) \quad f(x) = -x^2 - x - 2$$

$$f(-5) = -22$$

$$(24) \quad f(x) = 5x^2 - 2x - 4$$

$$f(5) = 111$$

$$(39) \quad f(x) = -3x^2 + 5x - 4$$

$$f(-5) = -104$$

$$(10) \quad f(x) = -x^2 - x + 5$$

$$f(-5) = -15$$

$$(25) \quad f(x) = 5x^2 + 3x + 5$$

$$f(3) = 59$$

$$(40) \quad f(x) = x^2 - 2x + 3$$

$$f(1) = 2$$

$$(11) \quad f(x) = 5x^2 - 3x + 4$$

$$f(-5) = 144$$

$$(26) \quad f(x) = -5x^2 - 3x - 5$$

$$f(1) = -13$$

$$(41) \quad f(x) = -5x^2 - 5x + 2$$

$$f(-4) = -58$$

$$(12) \quad f(x) = 3x^2 - 3x + 1$$

$$f(-5) = 91$$

$$(27) \quad f(x) = -x^2 + 3x - 5$$

$$f(5) = -15$$

$$(42) \quad f(x) = 2x^2 - 4x + 2$$

$$f(-5) = 72$$

$$(13) \quad f(x) = 3x^2 - 5x + 4$$

$$f(-1) = 12$$

$$(28) \quad f(x) = -x^2 - x + 2$$

$$f(4) = -18$$

$$(43) \quad f(x) = -3x^2 + 5x + 2$$

$$f(4) = -26$$

$$(14) \quad f(x) = -3x^2 - 3x + 5$$

$$f(3) = -31$$

$$(29) \quad f(x) = -5x^2 + 2x - 5$$

$$f(-4) = -93$$

$$(44) \quad f(x) = 5x^2 + 5x + 5$$

$$f(4) = 105$$

$$(15) \quad f(x) = 3x^2 - 2x - 4$$

$$f(-5) = 81$$

$$(30) \quad f(x) = x^2 + 4x - 5$$

$$f(3) = 16$$

$$(45) \quad f(x) = -x^2 + 2x + 1$$

$$f(2) = 1$$

$$(1) \quad f(x) = -x^2 + 3x + 1 \\ f(-3) = -17$$

$$(16) \quad f(x) = -3x^2 + x - 2 \\ f(4) = -46$$

$$(31) \quad f(x) = -2x^2 - 3x + 5 \\ f(-5) = -30$$

$$(2) \quad f(x) = 2x^2 + x - 4 \\ f(-4) = 24$$

$$(17) \quad f(x) = 2x^2 + 5x - 5 \\ f(2) = 13$$

$$(32) \quad f(x) = -x^2 - 4x - 2 \\ f(-4) = -2$$

$$(3) \quad f(x) = -4x^2 + 5x + 5 \\ f(2) = -1$$

$$(18) \quad f(x) = 5x^2 + 2x - 5 \\ f(-5) = 110$$

$$(33) \quad f(x) = x^2 - 4x - 2 \\ f(1) = -5$$

$$(4) \quad f(x) = -5x^2 - x + 3 \\ f(5) = -127$$

$$(19) \quad f(x) = -2x^2 - 3x - 2 \\ f(4) = -46$$

$$(34) \quad f(x) = -3x^2 - 3x + 3 \\ f(4) = -57$$

$$(5) \quad f(x) = 3x^2 - 3x - 4 \\ f(-3) = 32$$

$$(20) \quad f(x) = -2x^2 + x - 1 \\ f(3) = -16$$

$$(35) \quad f(x) = -x^2 + 4x + 4 \\ f(-3) = -17$$

$$(6) \quad f(x) = 5x^2 - 4x + 3 \\ f(-5) = 148$$

$$(21) \quad f(x) = -3x^2 - 2x + 5 \\ f(-1) = 4$$

$$(36) \quad f(x) = -5x^2 - x - 5 \\ f(-4) = -81$$

$$(7) \quad f(x) = 3x^2 - x + 1 \\ f(2) = 11$$

$$(22) \quad f(x) = 5x^2 - 2x - 1 \\ f(-2) = 23$$

$$(37) \quad f(x) = -2x^2 + 3x - 1 \\ f(-5) = -66$$

$$(8) \quad f(x) = -2x^2 + 4x - 2 \\ f(1) = 0$$

$$(23) \quad f(x) = -4x^2 - 4x + 2 \\ f(4) = -78$$

$$(38) \quad f(x) = 5x^2 + x - 3 \\ f(5) = 127$$

$$(9) \quad f(x) = 4x^2 - x + 4 \\ f(-2) = 22$$

$$(24) \quad f(x) = 2x^2 - 2x - 4 \\ f(-1) = 0$$

$$(39) \quad f(x) = -4x^2 - 2x - 1 \\ f(1) = -7$$

$$(10) \quad f(x) = -3x^2 + 4x + 2 \\ f(1) = 3$$

$$(25) \quad f(x) = -4x^2 - 4x + 3 \\ f(-2) = -5$$

$$(40) \quad f(x) = -5x^2 + 5x + 2 \\ f(-5) = -148$$

$$(11) \quad f(x) = -2x^2 - 4x + 1 \\ f(-3) = -5$$

$$(26) \quad f(x) = -2x^2 + 2x + 2 \\ f(1) = 2$$

$$(41) \quad f(x) = 2x^2 - 2x - 3 \\ f(-1) = 1$$

$$(12) \quad f(x) = 2x^2 + 4x - 1 \\ f(3) = 29$$

$$(27) \quad f(x) = 4x^2 + 5x + 5 \\ f(5) = 130$$

$$(42) \quad f(x) = -3x^2 + x + 3 \\ f(-3) = -27$$

$$(13) \quad f(x) = -2x^2 - 4x + 5 \\ f(-3) = -1$$

$$(28) \quad f(x) = x^2 - 3x - 3 \\ f(2) = -5$$

$$(43) \quad f(x) = x^2 - 5x - 1 \\ f(-4) = 35$$

$$(14) \quad f(x) = -x^2 - 4x - 1 \\ f(2) = -13$$

$$(29) \quad f(x) = 3x^2 - 3x - 4 \\ f(-5) = 86$$

$$(44) \quad f(x) = -x^2 - 2x - 1 \\ f(5) = -36$$

$$(15) \quad f(x) = 5x^2 - x - 5 \\ f(-4) = 79$$

$$(30) \quad f(x) = 2x^2 - x - 5 \\ f(2) = 1$$

$$(45) \quad f(x) = 5x^2 + x + 3 \\ f(-2) = 21$$

## 2.9 代入 (難しい)

平方完成の式に整数を代入する問題。

$$(1) \begin{array}{l} f(x) = (x+2)^2 - 9 \\ f(4) \end{array}$$

$$(6) \begin{array}{l} f(x) = -5 \left( x + \frac{1}{2} \right)^2 + \frac{9}{4} \\ f(-3) \end{array}$$

$$(2) \begin{array}{l} f(x) = -2 \left( x - \frac{5}{4} \right)^2 - \frac{15}{8} \\ f(-5) \end{array}$$

$$(7) \begin{array}{l} f(x) = -3 \left( x - \frac{2}{3} \right)^2 + \frac{1}{3} \\ f(3) \end{array}$$

$$(3) \begin{array}{l} f(x) = -5 \left( x - \frac{1}{5} \right)^2 - \frac{9}{5} \\ f(-2) \end{array}$$

$$(8) \begin{array}{l} f(x) = (x+2)^2 - 1 \\ f(1) \end{array}$$

$$(4) \begin{array}{l} f(x) = 4 \left( x - \frac{3}{8} \right)^2 - \frac{9}{16} \\ f(-3) \end{array}$$

$$(9) \begin{array}{l} f(x) = 3 \left( x + \frac{1}{2} \right)^2 + \frac{13}{4} \\ f(4) \end{array}$$

$$(5) \begin{array}{l} f(x) = -5 \left( x + \frac{1}{2} \right)^2 - \frac{15}{4} \\ f(5) \end{array}$$

$$(10) \begin{array}{l} f(x) = -2 \left( x - \frac{3}{4} \right)^2 + \frac{9}{8} \\ f(3) \end{array}$$

$$\begin{array}{l} (1) \ f(x) = \left(x - \frac{3}{2}\right)^2 + \frac{3}{4} \\ \quad f(2) \end{array}$$

$$\begin{array}{l} (6) \ f(x) = \left(x + \frac{1}{2}\right)^2 - \frac{21}{4} \\ \quad f(-2) \end{array}$$

$$\begin{array}{l} (2) \ f(x) = 4\left(x + \frac{1}{2}\right)^2 + 4 \\ \quad f(4) \end{array}$$

$$\begin{array}{l} (7) \ f(x) = \left(x - \frac{1}{2}\right)^2 - \frac{17}{4} \\ \quad f(-1) \end{array}$$

$$\begin{array}{l} (3) \ f(x) = -3\left(x + \frac{2}{3}\right)^2 + \frac{4}{3} \\ \quad f(-2) \end{array}$$

$$\begin{array}{l} (8) \ f(x) = -(x + 2)^2 + 7 \\ \quad f(4) \end{array}$$

$$\begin{array}{l} (4) \ f(x) = -4\left(x - \frac{1}{2}\right)^2 + 6 \\ \quad f(-5) \end{array}$$

$$\begin{array}{l} (9) \ f(x) = 3\left(x - \frac{1}{3}\right)^2 + \frac{5}{3} \\ \quad f(0) \end{array}$$

$$\begin{array}{l} (5) \ f(x) = -3\left(x - \frac{1}{2}\right)^2 + \frac{11}{4} \\ \quad f(0) \end{array}$$

$$\begin{array}{l} (10) \ f(x) = 2(x - 1)^2 - 1 \\ \quad f(-5) \end{array}$$

$$(1) \begin{array}{l} f(x) = 3\left(x + \frac{1}{3}\right)^2 - \frac{10}{3} \\ f(-3) \end{array}$$

$$(6) \begin{array}{l} f(x) = -5\left(x + \frac{1}{10}\right)^2 - \frac{99}{20} \\ f(-2) \end{array}$$

$$(2) \begin{array}{l} f(x) = 5\left(x - \frac{1}{2}\right)^2 + \frac{7}{4} \\ f(-5) \end{array}$$

$$(7) \begin{array}{l} f(x) = (x - 1)^2 + 3 \\ f(-3) \end{array}$$

$$(3) \begin{array}{l} f(x) = -\left(x - \frac{5}{2}\right)^2 + \frac{37}{4} \\ f(1) \end{array}$$

$$(8) \begin{array}{l} f(x) = -2\left(x + \frac{3}{4}\right)^2 + \frac{9}{8} \\ f(1) \end{array}$$

$$(4) \begin{array}{l} f(x) = 2\left(x + \frac{1}{4}\right)^2 + \frac{23}{8} \\ f(3) \end{array}$$

$$(9) \begin{array}{l} f(x) = -4\left(x + \frac{1}{2}\right)^2 - 3 \\ f(2) \end{array}$$

$$(5) \begin{array}{l} f(x) = -2\left(x - \frac{5}{4}\right)^2 - \frac{7}{8} \\ f(2) \end{array}$$

$$(10) \begin{array}{l} f(x) = 3\left(x - \frac{2}{3}\right)^2 - \frac{10}{3} \\ f(4) \end{array}$$

$$(1) \begin{array}{l} f(x) = -4 \left( x + \frac{3}{8} \right)^2 - \frac{71}{16} \\ f(-3) \end{array}$$

$$(6) \begin{array}{l} f(x) = \left( x - \frac{3}{2} \right)^2 + \frac{11}{4} \\ f(-4) \end{array}$$

$$(2) \begin{array}{l} f(x) = 5 \left( x - \frac{2}{5} \right)^2 - \frac{19}{5} \\ f(-2) \end{array}$$

$$(7) \begin{array}{l} f(x) = 3 \left( x - \frac{5}{6} \right)^2 + \frac{23}{12} \\ f(5) \end{array}$$

$$(3) \begin{array}{l} f(x) = -3 \left( x - \frac{1}{3} \right)^2 + \frac{7}{3} \\ f(-1) \end{array}$$

$$(8) \begin{array}{l} f(x) = (x + 2)^2 + 1 \\ f(3) \end{array}$$

$$(4) \begin{array}{l} f(x) = -(x - 1)^2 \\ f(0) \end{array}$$

$$(9) \begin{array}{l} f(x) = -4 \left( x + \frac{1}{4} \right)^2 - \frac{19}{4} \\ f(5) \end{array}$$

$$(5) \begin{array}{l} f(x) = -4 \left( x - \frac{1}{2} \right)^2 + 6 \\ f(-4) \end{array}$$

$$(10) \begin{array}{l} f(x) = 2 \left( x - \frac{5}{4} \right)^2 - \frac{49}{8} \\ f(5) \end{array}$$



$$(1) \begin{array}{l} f(x) = -3 \left( x + \frac{5}{6} \right)^2 - \frac{23}{12} \\ f(4) \end{array}$$

$$(6) \begin{array}{l} f(x) = 3 \left( x + \frac{2}{3} \right)^2 - \frac{1}{3} \\ f(5) \end{array}$$

$$(2) \begin{array}{l} f(x) = -4 \left( x + \frac{1}{2} \right)^2 + 4 \\ f(-1) \end{array}$$

$$(7) \begin{array}{l} f(x) = -3 \left( x + \frac{5}{6} \right)^2 + \frac{73}{12} \\ f(0) \end{array}$$

$$(3) \begin{array}{l} f(x) = -2 \left( x - \frac{1}{2} \right)^2 - \frac{1}{2} \\ f(-4) \end{array}$$

$$(8) \begin{array}{l} f(x) = 2(x-1)^2 + 1 \\ f(-3) \end{array}$$

$$(4) \begin{array}{l} f(x) = \left( x - \frac{5}{2} \right)^2 - \frac{13}{4} \\ f(3) \end{array}$$

$$(9) \begin{array}{l} f(x) = -5 \left( x + \frac{1}{2} \right)^2 + \frac{9}{4} \\ f(-4) \end{array}$$

$$(5) \begin{array}{l} f(x) = 5 \left( x - \frac{2}{5} \right)^2 - \frac{14}{5} \\ f(0) \end{array}$$

$$(10) \begin{array}{l} f(x) = -5 \left( x + \frac{2}{5} \right)^2 - \frac{11}{5} \\ f(0) \end{array}$$

$$(1) \begin{array}{l} f(x) = -3\left(x + \frac{1}{6}\right)^2 - \frac{23}{12} \\ f(-1) \end{array}$$

$$(6) \begin{array}{l} f(x) = 4\left(x + \frac{1}{8}\right)^2 + \frac{47}{16} \\ f(4) \end{array}$$

$$(2) \begin{array}{l} f(x) = -\left(x - \frac{5}{2}\right)^2 + \frac{9}{4} \\ f(5) \end{array}$$

$$(7) \begin{array}{l} f(x) = -5\left(x - \frac{3}{10}\right)^2 + \frac{69}{20} \\ f(-3) \end{array}$$

$$(3) \begin{array}{l} f(x) = 2(x - 1)^2 - 7 \\ f(-3) \end{array}$$

$$(8) \begin{array}{l} f(x) = 5\left(x - \frac{2}{5}\right)^2 - \frac{19}{5} \\ f(-3) \end{array}$$

$$(4) \begin{array}{l} f(x) = 4\left(x - \frac{1}{2}\right)^2 + 4 \\ f(0) \end{array}$$

$$(9) \begin{array}{l} f(x) = -5\left(x - \frac{1}{10}\right)^2 - \frac{39}{20} \\ f(-3) \end{array}$$

$$(5) \begin{array}{l} f(x) = -3\left(x - \frac{1}{6}\right)^2 - \frac{35}{12} \\ f(2) \end{array}$$

$$(10) \begin{array}{l} f(x) = 5\left(x + \frac{1}{2}\right)^2 - \frac{9}{4} \\ f(4) \end{array}$$

$$(1) \begin{array}{l} f(x) = -3 \left( x - \frac{1}{2} \right)^2 + \frac{3}{4} \\ f(-2) \end{array}$$

$$(6) \begin{array}{l} f(x) = 5 \left( x + \frac{1}{10} \right)^2 + \frac{59}{20} \\ f(5) \end{array}$$

$$(2) \begin{array}{l} f(x) = 4 \left( x + \frac{3}{8} \right)^2 - \frac{89}{16} \\ f(4) \end{array}$$

$$(7) \begin{array}{l} f(x) = 5 \left( x + \frac{1}{5} \right)^2 + \frac{14}{5} \\ f(2) \end{array}$$

$$(3) \begin{array}{l} f(x) = -4 \left( x - \frac{1}{8} \right)^2 + \frac{1}{16} \\ f(1) \end{array}$$

$$(8) \begin{array}{l} f(x) = 4 \left( x + \frac{1}{4} \right)^2 + \frac{3}{4} \\ f(2) \end{array}$$

$$(4) \begin{array}{l} f(x) = 5 \left( x + \frac{1}{2} \right)^2 + \frac{15}{4} \\ f(-3) \end{array}$$

$$(9) \begin{array}{l} f(x) = -2 \left( x + \frac{1}{2} \right)^2 + \frac{7}{2} \\ f(3) \end{array}$$

$$(5) \begin{array}{l} f(x) = 5 \left( x + \frac{1}{10} \right)^2 - \frac{61}{20} \\ f(-3) \end{array}$$

$$(10) \begin{array}{l} f(x) = -5 \left( x + \frac{2}{5} \right)^2 + \frac{4}{5} \\ f(-3) \end{array}$$

$$(1) \quad f(x) = (x-1)^2 + 2 \\ f(5)$$

$$(6) \quad f(x) = -(x+1)^2 + 1 \\ f(0)$$

$$(2) \quad f(x) = -3\left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \\ f(-1)$$

$$(7) \quad f(x) = -5\left(x - \frac{2}{5}\right)^2 - \frac{1}{5} \\ f(4)$$

$$(3) \quad f(x) = 3\left(x + \frac{2}{3}\right)^2 + \frac{8}{3} \\ f(0)$$

$$(8) \quad f(x) = 4\left(x + \frac{1}{2}\right)^2 \\ f(-4)$$

$$(4) \quad f(x) = 3\left(x - \frac{2}{3}\right)^2 - \frac{10}{3} \\ f(-3)$$

$$(9) \quad f(x) = -3\left(x - \frac{1}{6}\right)^2 + \frac{49}{12} \\ f(-2)$$

$$(5) \quad f(x) = -4\left(x - \frac{1}{2}\right)^2 + 4 \\ f(-4)$$

$$(10) \quad f(x) = 2\left(x + \frac{1}{2}\right)^2 - \frac{5}{2} \\ f(3)$$

$$\begin{array}{l} (1) \ f(x) = 2 \left( x + \frac{1}{4} \right)^2 - \frac{41}{8} \\ f(4) \end{array}$$

$$\begin{array}{l} (6) \ f(x) = -3 \left( x - \frac{5}{6} \right)^2 - \frac{11}{12} \\ f(-1) \end{array}$$

$$\begin{array}{l} (2) \ f(x) = 3 \left( x + \frac{2}{3} \right)^2 + \frac{8}{3} \\ f(4) \end{array}$$

$$\begin{array}{l} (7) \ f(x) = -2 \left( x + \frac{5}{4} \right)^2 + \frac{25}{8} \\ f(-2) \end{array}$$

$$\begin{array}{l} (3) \ f(x) = -5 \left( x + \frac{2}{5} \right)^2 + \frac{14}{5} \\ f(4) \end{array}$$

$$\begin{array}{l} (8) \ f(x) = -4 \left( x + \frac{1}{8} \right)^2 - \frac{63}{16} \\ f(2) \end{array}$$

$$\begin{array}{l} (4) \ f(x) = (x - 1)^2 + 1 \\ f(-5) \end{array}$$

$$\begin{array}{l} (9) \ f(x) = -4 \left( x + \frac{1}{8} \right)^2 + \frac{17}{16} \\ f(-4) \end{array}$$

$$\begin{array}{l} (5) \ f(x) = 4 \left( x + \frac{1}{4} \right)^2 + \frac{3}{4} \\ f(1) \end{array}$$

$$\begin{array}{l} (10) \ f(x) = -4 \left( x + \frac{1}{2} \right)^2 - 3 \\ f(0) \end{array}$$

$$(1) \begin{array}{l} f(x) = -3 \left( x + \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(-1) \end{array}$$

$$(6) \begin{array}{l} f(x) = -4 \left( x - \frac{3}{8} \right)^2 - \frac{55}{16} \\ f(-2) \end{array}$$

$$(2) \begin{array}{l} f(x) = -3 \left( x + \frac{2}{3} \right)^2 + \frac{10}{3} \\ f(0) \end{array}$$

$$(7) \begin{array}{l} f(x) = (x - 2)^2 - 1 \\ f(-3) \end{array}$$

$$(3) \begin{array}{l} f(x) = 3 \left( x - \frac{1}{2} \right)^2 + \frac{1}{4} \\ f(-4) \end{array}$$

$$(8) \begin{array}{l} f(x) = \left( x - \frac{3}{2} \right)^2 - \frac{13}{4} \\ f(-1) \end{array}$$

$$(4) \begin{array}{l} f(x) = -5 \left( x - \frac{3}{10} \right)^2 + \frac{109}{20} \\ f(2) \end{array}$$

$$(9) \begin{array}{l} f(x) = 5 \left( x + \frac{1}{10} \right)^2 + \frac{59}{20} \\ f(3) \end{array}$$

$$(5) \begin{array}{l} f(x) = -3 \left( x + \frac{5}{6} \right)^2 + \frac{61}{12} \\ f(-1) \end{array}$$

$$(10) \begin{array}{l} f(x) = 5 \left( x + \frac{1}{2} \right)^2 - \frac{9}{4} \\ f(-4) \end{array}$$

$$(1) \begin{array}{l} f(x) = 4 \left( x + \frac{5}{8} \right)^2 - \frac{89}{16} \\ f(4) \end{array}$$

$$(6) \begin{array}{l} f(x) = 5 \left( x - \frac{1}{10} \right)^2 - \frac{61}{20} \\ f(-5) \end{array}$$

$$(2) \begin{array}{l} f(x) = -2 \left( x - \frac{5}{4} \right)^2 - \frac{7}{8} \\ f(-3) \end{array}$$

$$(7) \begin{array}{l} f(x) = -2 (x - 1)^2 - 2 \\ f(0) \end{array}$$

$$(3) \begin{array}{l} f(x) = -5 \left( x - \frac{1}{2} \right)^2 - \frac{3}{4} \\ f(5) \end{array}$$

$$(8) \begin{array}{l} f(x) = 3 \left( x + \frac{1}{6} \right)^2 + \frac{23}{12} \\ f(5) \end{array}$$

$$(4) \begin{array}{l} f(x) = 3 \left( x + \frac{1}{6} \right)^2 - \frac{25}{12} \\ f(-3) \end{array}$$

$$(9) \begin{array}{l} f(x) = -2 \left( x + \frac{5}{4} \right)^2 + \frac{65}{8} \\ f(-1) \end{array}$$

$$(5) \begin{array}{l} f(x) = 3 \left( x + \frac{2}{3} \right)^2 - \frac{4}{3} \\ f(5) \end{array}$$

$$(10) \begin{array}{l} f(x) = -5 \left( x + \frac{3}{10} \right)^2 + \frac{69}{20} \\ f(-2) \end{array}$$

$$\begin{array}{l} (1) \ f(x) = \left(x + \frac{3}{2}\right)^2 - \frac{9}{4} \\ \quad f(5) \end{array}$$

$$\begin{array}{l} (6) \ f(x) = -3\left(x - \frac{1}{3}\right)^2 - \frac{5}{3} \\ \quad f(1) \end{array}$$

$$\begin{array}{l} (2) \ f(x) = -2\left(x + \frac{3}{4}\right)^2 + \frac{33}{8} \\ \quad f(-4) \end{array}$$

$$\begin{array}{l} (7) \ f(x) = 5\left(x + \frac{1}{5}\right)^2 + \frac{4}{5} \\ \quad f(4) \end{array}$$

$$\begin{array}{l} (3) \ f(x) = -\left(x + \frac{3}{2}\right)^2 + \frac{1}{4} \\ \quad f(-4) \end{array}$$

$$\begin{array}{l} (8) \ f(x) = 2\left(x + \frac{1}{4}\right)^2 + \frac{31}{8} \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (4) \ f(x) = -5\left(x + \frac{2}{5}\right)^2 - \frac{16}{5} \\ \quad f(5) \end{array}$$

$$\begin{array}{l} (9) \ f(x) = -2(x + 1)^2 + 5 \\ \quad f(5) \end{array}$$

$$\begin{array}{l} (5) \ f(x) = -2\left(x - \frac{1}{2}\right)^2 + \frac{9}{2} \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (10) \ f(x) = -4\left(x + \frac{1}{4}\right)^2 - \frac{19}{4} \\ \quad f(2) \end{array}$$



$$(1) \begin{array}{l} f(x) = (x+2)^2 - 3 \\ f(2) \end{array}$$

$$(6) \begin{array}{l} f(x) = 3\left(x + \frac{1}{2}\right)^2 + \frac{9}{4} \\ f(1) \end{array}$$

$$(2) \begin{array}{l} f(x) = 3\left(x + \frac{5}{6}\right)^2 - \frac{37}{12} \\ f(5) \end{array}$$

$$(7) \begin{array}{l} f(x) = 2(x+1)^2 - 6 \\ f(3) \end{array}$$

$$(3) \begin{array}{l} f(x) = -2\left(x + \frac{5}{4}\right)^2 + \frac{65}{8} \\ f(-1) \end{array}$$

$$(8) \begin{array}{l} f(x) = -4\left(x - \frac{3}{8}\right)^2 + \frac{73}{16} \\ f(5) \end{array}$$

$$(4) \begin{array}{l} f(x) = 2\left(x + \frac{1}{2}\right)^2 - \frac{1}{2} \\ f(3) \end{array}$$

$$(9) \begin{array}{l} f(x) = -2\left(x + \frac{1}{4}\right)^2 - \frac{7}{8} \\ f(-4) \end{array}$$

$$(5) \begin{array}{l} f(x) = 3\left(x + \frac{5}{6}\right)^2 - \frac{85}{12} \\ f(-5) \end{array}$$

$$(10) \begin{array}{l} f(x) = 5\left(x - \frac{1}{2}\right)^2 - \frac{17}{4} \\ f(5) \end{array}$$

$$\begin{array}{l} (1) \ f(x) = 4\left(x + \frac{3}{8}\right)^2 - \frac{89}{16} \\ \quad f(1) \end{array}$$

$$\begin{array}{l} (6) \ f(x) = -3\left(x - \frac{5}{6}\right)^2 + \frac{73}{12} \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (2) \ f(x) = -(x + 1)^2 + 1 \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (7) \ f(x) = -3\left(x - \frac{5}{6}\right)^2 + \frac{25}{12} \\ \quad f(4) \end{array}$$

$$\begin{array}{l} (3) \ f(x) = -5\left(x - \frac{1}{5}\right)^2 + \frac{21}{5} \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (8) \ f(x) = 3\left(x + \frac{1}{3}\right)^2 + \frac{14}{3} \\ \quad f(4) \end{array}$$

$$\begin{array}{l} (4) \ f(x) = 2(x - 1)^2 - 4 \\ \quad f(-3) \end{array}$$

$$\begin{array}{l} (9) \ f(x) = -5\left(x - \frac{1}{5}\right)^2 - \frac{19}{5} \\ \quad f(5) \end{array}$$

$$\begin{array}{l} (5) \ f(x) = -2\left(x + \frac{1}{4}\right)^2 - \frac{7}{8} \\ \quad f(-2) \end{array}$$

$$\begin{array}{l} (10) \ f(x) = 4\left(x + \frac{1}{4}\right)^2 - \frac{9}{4} \\ \quad f(5) \end{array}$$

$$(1) \begin{array}{l} f(x) = \left(x - \frac{1}{2}\right)^2 + \frac{19}{4} \\ f(0) \end{array}$$

$$(6) \begin{array}{l} f(x) = -5\left(x - \frac{2}{5}\right)^2 + \frac{29}{5} \\ f(1) \end{array}$$

$$(2) \begin{array}{l} f(x) = \left(x + \frac{1}{2}\right)^2 - \frac{1}{4} \\ f(-5) \end{array}$$

$$(7) \begin{array}{l} f(x) = 3\left(x + \frac{5}{6}\right)^2 - \frac{25}{12} \\ f(-1) \end{array}$$

$$(3) \begin{array}{l} f(x) = 3\left(x + \frac{2}{3}\right)^2 + \frac{5}{3} \\ f(1) \end{array}$$

$$(8) \begin{array}{l} f(x) = 4\left(x + \frac{1}{4}\right)^2 - \frac{17}{4} \\ f(-4) \end{array}$$

$$(4) \begin{array}{l} f(x) = 4\left(x + \frac{1}{2}\right)^2 - 2 \\ f(-4) \end{array}$$

$$(9) \begin{array}{l} f(x) = -4\left(x - \frac{5}{8}\right)^2 + \frac{41}{16} \\ f(5) \end{array}$$

$$(5) \begin{array}{l} f(x) = 4\left(x - \frac{5}{8}\right)^2 - \frac{41}{16} \\ f(-2) \end{array}$$

$$(10) \begin{array}{l} f(x) = (x + 2)^2 \\ f(1) \end{array}$$

$$(1) \begin{array}{l} f(x) = -4 \left( x - \frac{1}{2} \right)^2 - 4 \\ f(0) \end{array}$$

$$(6) \begin{array}{l} f(x) = -2(x+1)^2 - 2 \\ f(1) \end{array}$$

$$(2) \begin{array}{l} f(x) = -4 \left( x - \frac{3}{8} \right)^2 - \frac{71}{16} \\ f(-2) \end{array}$$

$$(7) \begin{array}{l} f(x) = (x+1)^2 - 4 \\ f(3) \end{array}$$

$$(3) \begin{array}{l} f(x) = -2 \left( x - \frac{5}{4} \right)^2 - \frac{7}{8} \\ f(1) \end{array}$$

$$(8) \begin{array}{l} f(x) = -(x-2)^2 + 5 \\ f(1) \end{array}$$

$$(4) \begin{array}{l} f(x) = \left( x - \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(-5) \end{array}$$

$$(9) \begin{array}{l} f(x) = - \left( x - \frac{3}{2} \right)^2 + \frac{25}{4} \\ f(-2) \end{array}$$

$$(5) \begin{array}{l} f(x) = - \left( x - \frac{3}{2} \right)^2 - \frac{11}{4} \\ f(-2) \end{array}$$

$$(10) \begin{array}{l} f(x) = - \left( x - \frac{1}{2} \right)^2 - \frac{3}{4} \\ f(-1) \end{array}$$

$$(1) \begin{array}{l} f(x) = -5 \left( x + \frac{2}{5} \right)^2 + \frac{9}{5} \\ f(-4) \end{array}$$

$$(6) \begin{array}{l} f(x) = 3 \left( x + \frac{1}{2} \right)^2 - \frac{7}{4} \\ f(-4) \end{array}$$

$$(2) \begin{array}{l} f(x) = -3 \left( x + \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(5) \end{array}$$

$$(7) \begin{array}{l} f(x) = \left( x - \frac{3}{2} \right)^2 - \frac{25}{4} \\ f(0) \end{array}$$

$$(3) \begin{array}{l} f(x) = 5 \left( x - \frac{1}{2} \right)^2 - \frac{9}{4} \\ f(-2) \end{array}$$

$$(8) \begin{array}{l} f(x) = 3 \left( x + \frac{1}{3} \right)^2 + \frac{5}{3} \\ f(5) \end{array}$$

$$(4) \begin{array}{l} f(x) = -5 \left( x - \frac{1}{10} \right)^2 + \frac{41}{20} \\ f(2) \end{array}$$

$$(9) \begin{array}{l} f(x) = 5 \left( x - \frac{2}{5} \right)^2 + \frac{1}{5} \\ f(-5) \end{array}$$

$$(5) \begin{array}{l} f(x) = -2 \left( x - \frac{1}{2} \right)^2 + \frac{3}{2} \\ f(2) \end{array}$$

$$(10) \begin{array}{l} f(x) = 2 \left( x + \frac{1}{2} \right)^2 + \frac{1}{2} \\ f(3) \end{array}$$

$$(1) \begin{array}{l} f(x) = -\left(x + \frac{1}{2}\right)^2 - \frac{15}{4} \\ f(-2) \end{array}$$

$$(6) \begin{array}{l} f(x) = 3\left(x - \frac{5}{6}\right)^2 + \frac{23}{12} \\ f(-1) \end{array}$$

$$(2) \begin{array}{l} f(x) = (x - 1)^2 + 2 \\ f(1) \end{array}$$

$$(7) \begin{array}{l} f(x) = -3\left(x + \frac{5}{6}\right)^2 + \frac{25}{12} \\ f(-4) \end{array}$$

$$(3) \begin{array}{l} f(x) = 4\left(x + \frac{1}{4}\right)^2 + \frac{11}{4} \\ f(-4) \end{array}$$

$$(8) \begin{array}{l} f(x) = -4\left(x + \frac{5}{8}\right)^2 + \frac{25}{16} \\ f(2) \end{array}$$

$$(4) \begin{array}{l} f(x) = 2\left(x + \frac{5}{4}\right)^2 - \frac{25}{8} \\ f(-3) \end{array}$$

$$(9) \begin{array}{l} f(x) = -3\left(x - \frac{2}{3}\right)^2 + \frac{19}{3} \\ f(3) \end{array}$$

$$(5) \begin{array}{l} f(x) = -5\left(x - \frac{1}{5}\right)^2 + \frac{16}{5} \\ f(4) \end{array}$$

$$(10) \begin{array}{l} f(x) = -2(x + 1)^2 + 2 \\ f(-3) \end{array}$$

$$(1) \quad f(x) = -(x-1)^2 + 4$$
$$f(0)$$

$$(6) \quad f(x) = -2(x+1)^2 + 4$$
$$f(-1)$$

$$(2) \quad f(x) = 5\left(x + \frac{1}{2}\right)^2 - \frac{13}{4}$$
$$f(-4)$$

$$(7) \quad f(x) = 5\left(x - \frac{2}{5}\right)^2 - \frac{24}{5}$$
$$f(0)$$

$$(3) \quad f(x) = \left(x - \frac{1}{2}\right)^2 + \frac{15}{4}$$
$$f(3)$$

$$(8) \quad f(x) = 4\left(x + \frac{5}{8}\right)^2 + \frac{39}{16}$$
$$f(-3)$$

$$(4) \quad f(x) = -3\left(x - \frac{1}{3}\right)^2 - \frac{11}{3}$$
$$f(0)$$

$$(9) \quad f(x) = 5\left(x - \frac{1}{5}\right)^2 + \frac{19}{5}$$
$$f(-5)$$

$$(5) \quad f(x) = -2\left(x + \frac{5}{4}\right)^2 - \frac{7}{8}$$
$$f(-3)$$

$$(10) \quad f(x) = -2\left(x + \frac{3}{4}\right)^2 - \frac{23}{8}$$
$$f(-4)$$

$$(1) \begin{array}{l} f(x) = 3\left(x + \frac{5}{6}\right)^2 - \frac{13}{12} \\ f(-4) \end{array}$$

$$(6) \begin{array}{l} f(x) = -2\left(x - \frac{1}{2}\right)^2 + \frac{1}{2} \\ f(5) \end{array}$$

$$(2) \begin{array}{l} f(x) = 3\left(x - \frac{1}{3}\right)^2 + \frac{14}{3} \\ f(0) \end{array}$$

$$(7) \begin{array}{l} f(x) = 2\left(x - \frac{1}{4}\right)^2 + \frac{7}{8} \\ f(0) \end{array}$$

$$(3) \begin{array}{l} f(x) = 4\left(x + \frac{1}{4}\right)^2 - \frac{13}{4} \\ f(3) \end{array}$$

$$(8) \begin{array}{l} f(x) = 3\left(x - \frac{1}{2}\right)^2 - \frac{23}{4} \\ f(-2) \end{array}$$

$$(4) \begin{array}{l} f(x) = 2\left(x - \frac{5}{4}\right)^2 - \frac{57}{8} \\ f(-2) \end{array}$$

$$(9) \begin{array}{l} f(x) = \left(x - \frac{3}{2}\right)^2 - \frac{29}{4} \\ f(4) \end{array}$$

$$(5) \begin{array}{l} f(x) = 5\left(x - \frac{1}{2}\right)^2 + \frac{15}{4} \\ f(3) \end{array}$$

$$(10) \begin{array}{l} f(x) = 4\left(x - \frac{3}{8}\right)^2 - \frac{89}{16} \\ f(2) \end{array}$$



$$(1) \quad f(x) = (x+2)^2 - 9 \\ f(4) = 27$$

$$(6) \quad f(x) = -5 \left( x + \frac{1}{2} \right)^2 + \frac{9}{4} \\ f(-3) = -29$$

$$(2) \quad f(x) = -2 \left( x - \frac{5}{4} \right)^2 - \frac{15}{8} \\ f(-5) = -80$$

$$(7) \quad f(x) = -3 \left( x - \frac{2}{3} \right)^2 + \frac{1}{3} \\ f(3) = -16$$

$$(3) \quad f(x) = -5 \left( x - \frac{1}{5} \right)^2 - \frac{9}{5} \\ f(-2) = -26$$

$$(8) \quad f(x) = (x+2)^2 - 1 \\ f(1) = 8$$

$$(4) \quad f(x) = 4 \left( x - \frac{3}{8} \right)^2 - \frac{9}{16} \\ f(-3) = 45$$

$$(9) \quad f(x) = 3 \left( x + \frac{1}{2} \right)^2 + \frac{13}{4} \\ f(4) = 64$$

$$(5) \quad f(x) = -5 \left( x + \frac{1}{2} \right)^2 - \frac{15}{4} \\ f(5) = -155$$

$$(10) \quad f(x) = -2 \left( x - \frac{3}{4} \right)^2 + \frac{9}{8} \\ f(3) = -9$$

$$(1) \begin{aligned} f(x) &= \left(x - \frac{3}{2}\right)^2 + \frac{3}{4} \\ f(2) &= 1 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= \left(x + \frac{1}{2}\right)^2 - \frac{21}{4} \\ f(-2) &= -3 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 4\left(x + \frac{1}{2}\right)^2 + 4 \\ f(4) &= 85 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= \left(x - \frac{1}{2}\right)^2 - \frac{17}{4} \\ f(-1) &= -2 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -3\left(x + \frac{2}{3}\right)^2 + \frac{4}{3} \\ f(-2) &= -4 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -(x + 2)^2 + 7 \\ f(4) &= -29 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -4\left(x - \frac{1}{2}\right)^2 + 6 \\ f(-5) &= -115 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 3\left(x - \frac{1}{3}\right)^2 + \frac{5}{3} \\ f(0) &= 2 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -3\left(x - \frac{1}{2}\right)^2 + \frac{11}{4} \\ f(0) &= 2 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 2(x - 1)^2 - 1 \\ f(-5) &= 71 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 3\left(x + \frac{1}{3}\right)^2 - \frac{10}{3} \\ f(-3) &= 18 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -5\left(x + \frac{1}{10}\right)^2 - \frac{99}{20} \\ f(-2) &= -23 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 5\left(x - \frac{1}{2}\right)^2 + \frac{7}{4} \\ f(-5) &= 153 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= (x - 1)^2 + 3 \\ f(-3) &= 19 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -\left(x - \frac{5}{2}\right)^2 + \frac{37}{4} \\ f(1) &= 7 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -2\left(x + \frac{3}{4}\right)^2 + \frac{9}{8} \\ f(1) &= -5 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 2\left(x + \frac{1}{4}\right)^2 + \frac{23}{8} \\ f(3) &= 24 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4\left(x + \frac{1}{2}\right)^2 - 3 \\ f(2) &= -28 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -2\left(x - \frac{5}{4}\right)^2 - \frac{7}{8} \\ f(2) &= -2 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 3\left(x - \frac{2}{3}\right)^2 - \frac{10}{3} \\ f(4) &= 30 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -4 \left( x + \frac{3}{8} \right)^2 - \frac{71}{16} \\ f(-3) &= -32 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= \left( x - \frac{3}{2} \right)^2 + \frac{11}{4} \\ f(-4) &= 33 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 5 \left( x - \frac{2}{5} \right)^2 - \frac{19}{5} \\ f(-2) &= 25 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 3 \left( x - \frac{5}{6} \right)^2 + \frac{23}{12} \\ f(5) &= 54 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -3 \left( x - \frac{1}{3} \right)^2 + \frac{7}{3} \\ f(-1) &= -3 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= (x + 2)^2 + 1 \\ f(3) &= 26 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -(x - 1)^2 \\ f(0) &= -1 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4 \left( x + \frac{1}{4} \right)^2 - \frac{19}{4} \\ f(5) &= -115 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -4 \left( x - \frac{1}{2} \right)^2 + 6 \\ f(-4) &= -75 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 2 \left( x - \frac{5}{4} \right)^2 - \frac{49}{8} \\ f(5) &= 22 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3 \left( x + \frac{5}{6} \right)^2 - \frac{23}{12} \\ f(4) &= -72 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3 \left( x + \frac{2}{3} \right)^2 - \frac{1}{3} \\ f(5) &= 96 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -4 \left( x + \frac{1}{2} \right)^2 + 4 \\ f(-1) &= 3 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -3 \left( x + \frac{5}{6} \right)^2 + \frac{73}{12} \\ f(0) &= 4 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -2 \left( x - \frac{1}{2} \right)^2 - \frac{1}{2} \\ f(-4) &= -41 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 2(x-1)^2 + 1 \\ f(-3) &= 33 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= \left( x - \frac{5}{2} \right)^2 - \frac{13}{4} \\ f(3) &= -3 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -5 \left( x + \frac{1}{2} \right)^2 + \frac{9}{4} \\ f(-4) &= -59 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 5 \left( x - \frac{2}{5} \right)^2 - \frac{14}{5} \\ f(0) &= -2 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -5 \left( x + \frac{2}{5} \right)^2 - \frac{11}{5} \\ f(0) &= -3 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3\left(x + \frac{1}{6}\right)^2 - \frac{23}{12} \\ f(-1) &= -4 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 4\left(x + \frac{1}{8}\right)^2 + \frac{47}{16} \\ f(4) &= 71 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -\left(x - \frac{5}{2}\right)^2 + \frac{9}{4} \\ f(5) &= -4 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -5\left(x - \frac{3}{10}\right)^2 + \frac{69}{20} \\ f(-3) &= -51 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 2(x - 1)^2 - 7 \\ f(-3) &= 25 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 5\left(x - \frac{2}{5}\right)^2 - \frac{19}{5} \\ f(-3) &= 54 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 4\left(x - \frac{1}{2}\right)^2 + 4 \\ f(0) &= 5 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -5\left(x - \frac{1}{10}\right)^2 - \frac{39}{20} \\ f(-3) &= -50 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -3\left(x - \frac{1}{6}\right)^2 - \frac{35}{12} \\ f(2) &= -13 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 5\left(x + \frac{1}{2}\right)^2 - \frac{9}{4} \\ f(4) &= 99 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3\left(x - \frac{1}{2}\right)^2 + \frac{3}{4} \\ f(-2) &= -18 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 5\left(x + \frac{1}{10}\right)^2 + \frac{59}{20} \\ f(5) &= 133 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 4\left(x + \frac{3}{8}\right)^2 - \frac{89}{16} \\ f(4) &= 71 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 5\left(x + \frac{1}{5}\right)^2 + \frac{14}{5} \\ f(2) &= 27 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -4\left(x - \frac{1}{8}\right)^2 + \frac{1}{16} \\ f(1) &= -3 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 4\left(x + \frac{1}{4}\right)^2 + \frac{3}{4} \\ f(2) &= 21 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 5\left(x + \frac{1}{2}\right)^2 + \frac{15}{4} \\ f(-3) &= 35 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -2\left(x + \frac{1}{2}\right)^2 + \frac{7}{2} \\ f(3) &= -21 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 5\left(x + \frac{1}{10}\right)^2 - \frac{61}{20} \\ f(-3) &= 39 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -5\left(x + \frac{2}{5}\right)^2 + \frac{4}{5} \\ f(-3) &= -33 \end{aligned}$$

$$(1) \quad f(x) = (x-1)^2 + 2 \\ f(5) = 18$$

$$(6) \quad f(x) = -(x+1)^2 + 1 \\ f(0) = 0$$

$$(2) \quad f(x) = -3\left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \\ f(-1) = 0$$

$$(7) \quad f(x) = -5\left(x - \frac{2}{5}\right)^2 - \frac{1}{5} \\ f(4) = -65$$

$$(3) \quad f(x) = 3\left(x + \frac{2}{3}\right)^2 + \frac{8}{3} \\ f(0) = 4$$

$$(8) \quad f(x) = 4\left(x + \frac{1}{2}\right)^2 \\ f(-4) = 49$$

$$(4) \quad f(x) = 3\left(x - \frac{2}{3}\right)^2 - \frac{10}{3} \\ f(-3) = 37$$

$$(9) \quad f(x) = -3\left(x - \frac{1}{6}\right)^2 + \frac{49}{12} \\ f(-2) = -10$$

$$(5) \quad f(x) = -4\left(x - \frac{1}{2}\right)^2 + 4 \\ f(-4) = -77$$

$$(10) \quad f(x) = 2\left(x + \frac{1}{2}\right)^2 - \frac{5}{2} \\ f(3) = 22$$



$$(1) \quad \begin{aligned} f(x) &= 2 \left( x + \frac{1}{4} \right)^2 - \frac{41}{8} \\ f(4) &= 31 \end{aligned}$$

$$(6) \quad \begin{aligned} f(x) &= -3 \left( x - \frac{5}{6} \right)^2 - \frac{11}{12} \\ f(-1) &= -11 \end{aligned}$$

$$(2) \quad \begin{aligned} f(x) &= 3 \left( x + \frac{2}{3} \right)^2 + \frac{8}{3} \\ f(4) &= 68 \end{aligned}$$

$$(7) \quad \begin{aligned} f(x) &= -2 \left( x + \frac{5}{4} \right)^2 + \frac{25}{8} \\ f(-2) &= 2 \end{aligned}$$

$$(3) \quad \begin{aligned} f(x) &= -5 \left( x + \frac{2}{5} \right)^2 + \frac{14}{5} \\ f(4) &= -94 \end{aligned}$$

$$(8) \quad \begin{aligned} f(x) &= -4 \left( x + \frac{1}{8} \right)^2 - \frac{63}{16} \\ f(2) &= -22 \end{aligned}$$

$$(4) \quad \begin{aligned} f(x) &= (x-1)^2 + 1 \\ f(-5) &= 37 \end{aligned}$$

$$(9) \quad \begin{aligned} f(x) &= -4 \left( x + \frac{1}{8} \right)^2 + \frac{17}{16} \\ f(-4) &= -59 \end{aligned}$$

$$(5) \quad \begin{aligned} f(x) &= 4 \left( x + \frac{1}{4} \right)^2 + \frac{3}{4} \\ f(1) &= 7 \end{aligned}$$

$$(10) \quad \begin{aligned} f(x) &= -4 \left( x + \frac{1}{2} \right)^2 - 3 \\ f(0) &= -4 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= -3 \left( x + \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(-1) &= 1 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -4 \left( x - \frac{3}{8} \right)^2 - \frac{55}{16} \\ f(-2) &= -26 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -3 \left( x + \frac{2}{3} \right)^2 + \frac{10}{3} \\ f(0) &= 2 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= (x - 2)^2 - 1 \\ f(-3) &= 24 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 3 \left( x - \frac{1}{2} \right)^2 + \frac{1}{4} \\ f(-4) &= 61 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= \left( x - \frac{3}{2} \right)^2 - \frac{13}{4} \\ f(-1) &= 3 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5 \left( x - \frac{3}{10} \right)^2 + \frac{109}{20} \\ f(2) &= -9 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 5 \left( x + \frac{1}{10} \right)^2 + \frac{59}{20} \\ f(3) &= 51 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -3 \left( x + \frac{5}{6} \right)^2 + \frac{61}{12} \\ f(-1) &= 5 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 5 \left( x + \frac{1}{2} \right)^2 - \frac{9}{4} \\ f(-4) &= 59 \end{aligned}$$

$$(1) \begin{aligned} f(x) &= 4 \left( x + \frac{5}{8} \right)^2 - \frac{89}{16} \\ f(4) &= 80 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 5 \left( x - \frac{1}{10} \right)^2 - \frac{61}{20} \\ f(-5) &= 127 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -2 \left( x - \frac{5}{4} \right)^2 - \frac{7}{8} \\ f(-3) &= -37 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= -2(x-1)^2 - 2 \\ f(0) &= -4 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -5 \left( x - \frac{1}{2} \right)^2 - \frac{3}{4} \\ f(5) &= -102 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 3 \left( x + \frac{1}{6} \right)^2 + \frac{23}{12} \\ f(5) &= 82 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 3 \left( x + \frac{1}{6} \right)^2 - \frac{25}{12} \\ f(-3) &= 22 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -2 \left( x + \frac{5}{4} \right)^2 + \frac{65}{8} \\ f(-1) &= 8 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 3 \left( x + \frac{2}{3} \right)^2 - \frac{4}{3} \\ f(5) &= 95 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= -5 \left( x + \frac{3}{10} \right)^2 + \frac{69}{20} \\ f(-2) &= -11 \end{aligned}$$

$$(1) \quad f(x) = \left(x + \frac{3}{2}\right)^2 - \frac{9}{4} \\ f(5) = 40$$

$$(6) \quad f(x) = -3\left(x - \frac{1}{3}\right)^2 - \frac{5}{3} \\ f(1) = -3$$

$$(2) \quad f(x) = -2\left(x + \frac{3}{4}\right)^2 + \frac{33}{8} \\ f(-4) = -17$$

$$(7) \quad f(x) = 5\left(x + \frac{1}{5}\right)^2 + \frac{4}{5} \\ f(4) = 89$$

$$(3) \quad f(x) = -\left(x + \frac{3}{2}\right)^2 + \frac{1}{4} \\ f(-4) = -6$$

$$(8) \quad f(x) = 2\left(x + \frac{1}{4}\right)^2 + \frac{31}{8} \\ f(-3) = 19$$

$$(4) \quad f(x) = -5\left(x + \frac{2}{5}\right)^2 - \frac{16}{5} \\ f(5) = -149$$

$$(9) \quad f(x) = -2(x + 1)^2 + 5 \\ f(5) = -67$$

$$(5) \quad f(x) = -2\left(x - \frac{1}{2}\right)^2 + \frac{9}{2} \\ f(-3) = -20$$

$$(10) \quad f(x) = -4\left(x + \frac{1}{4}\right)^2 - \frac{19}{4} \\ f(2) = -25$$

$$(1) \begin{aligned} f(x) &= (x+2)^2 - 3 \\ f(2) &= 13 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3\left(x + \frac{1}{2}\right)^2 + \frac{9}{4} \\ f(1) &= 9 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= 3\left(x + \frac{5}{6}\right)^2 - \frac{37}{12} \\ f(5) &= 99 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 2(x+1)^2 - 6 \\ f(3) &= 26 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= -2\left(x + \frac{5}{4}\right)^2 + \frac{65}{8} \\ f(-1) &= 8 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= -4\left(x - \frac{3}{8}\right)^2 + \frac{73}{16} \\ f(5) &= -81 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 2\left(x + \frac{1}{2}\right)^2 - \frac{1}{2} \\ f(3) &= 24 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -2\left(x + \frac{1}{4}\right)^2 - \frac{7}{8} \\ f(-4) &= -29 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 3\left(x + \frac{5}{6}\right)^2 - \frac{85}{12} \\ f(-5) &= 45 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 5\left(x - \frac{1}{2}\right)^2 - \frac{17}{4} \\ f(5) &= 97 \end{aligned}$$

$$(1) \quad f(x) = 4 \left( x + \frac{3}{8} \right)^2 - \frac{89}{16}$$
$$f(1) = 2$$

$$(6) \quad f(x) = -3 \left( x - \frac{5}{6} \right)^2 + \frac{73}{12}$$
$$f(-3) = -38$$

$$(2) \quad f(x) = -(x+1)^2 + 1$$
$$f(-3) = -3$$

$$(7) \quad f(x) = -3 \left( x - \frac{5}{6} \right)^2 + \frac{25}{12}$$
$$f(4) = -28$$

$$(3) \quad f(x) = -5 \left( x - \frac{1}{5} \right)^2 + \frac{21}{5}$$
$$f(-3) = -47$$

$$(8) \quad f(x) = 3 \left( x + \frac{1}{3} \right)^2 + \frac{14}{3}$$
$$f(4) = 61$$

$$(4) \quad f(x) = 2(x-1)^2 - 4$$
$$f(-3) = 28$$

$$(9) \quad f(x) = -5 \left( x - \frac{1}{5} \right)^2 - \frac{19}{5}$$
$$f(5) = -119$$

$$(5) \quad f(x) = -2 \left( x + \frac{1}{4} \right)^2 - \frac{7}{8}$$
$$f(-2) = -7$$

$$(10) \quad f(x) = 4 \left( x + \frac{1}{4} \right)^2 - \frac{9}{4}$$
$$f(5) = 108$$

$$(1) \begin{aligned} f(x) &= \left(x - \frac{1}{2}\right)^2 + \frac{19}{4} \\ f(0) &= 5 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= -5\left(x - \frac{2}{5}\right)^2 + \frac{29}{5} \\ f(1) &= 4 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= \left(x + \frac{1}{2}\right)^2 - \frac{1}{4} \\ f(-5) &= 20 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= 3\left(x + \frac{5}{6}\right)^2 - \frac{25}{12} \\ f(-1) &= -2 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 3\left(x + \frac{2}{3}\right)^2 + \frac{5}{3} \\ f(1) &= 10 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 4\left(x + \frac{1}{4}\right)^2 - \frac{17}{4} \\ f(-4) &= 52 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= 4\left(x + \frac{1}{2}\right)^2 - 2 \\ f(-4) &= 47 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= -4\left(x - \frac{5}{8}\right)^2 + \frac{41}{16} \\ f(5) &= -74 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= 4\left(x - \frac{5}{8}\right)^2 - \frac{41}{16} \\ f(-2) &= 25 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= (x + 2)^2 \\ f(1) &= 9 \end{aligned}$$

$$(1) \quad f(x) = -4 \left( x - \frac{1}{2} \right)^2 - 4 \\ f(0) = -5$$

$$(6) \quad f(x) = -2(x+1)^2 - 2 \\ f(1) = -10$$

$$(2) \quad f(x) = -4 \left( x - \frac{3}{8} \right)^2 - \frac{71}{16} \\ f(-2) = -27$$

$$(7) \quad f(x) = (x+1)^2 - 4 \\ f(3) = 12$$

$$(3) \quad f(x) = -2 \left( x - \frac{5}{4} \right)^2 - \frac{7}{8} \\ f(1) = -1$$

$$(8) \quad f(x) = -(x-2)^2 + 5 \\ f(1) = 4$$

$$(4) \quad f(x) = \left( x - \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(-5) = 32$$

$$(9) \quad f(x) = - \left( x - \frac{3}{2} \right)^2 + \frac{25}{4} \\ f(-2) = -6$$

$$(5) \quad f(x) = - \left( x - \frac{3}{2} \right)^2 - \frac{11}{4} \\ f(-2) = -15$$

$$(10) \quad f(x) = - \left( x - \frac{1}{2} \right)^2 - \frac{3}{4} \\ f(-1) = -3$$



$$(1) \begin{aligned} f(x) &= -5 \left( x + \frac{2}{5} \right)^2 + \frac{9}{5} \\ f(-4) &= -63 \end{aligned}$$

$$(6) \begin{aligned} f(x) &= 3 \left( x + \frac{1}{2} \right)^2 - \frac{7}{4} \\ f(-4) &= 35 \end{aligned}$$

$$(2) \begin{aligned} f(x) &= -3 \left( x + \frac{1}{2} \right)^2 + \frac{7}{4} \\ f(5) &= -89 \end{aligned}$$

$$(7) \begin{aligned} f(x) &= \left( x - \frac{3}{2} \right)^2 - \frac{25}{4} \\ f(0) &= -4 \end{aligned}$$

$$(3) \begin{aligned} f(x) &= 5 \left( x - \frac{1}{2} \right)^2 - \frac{9}{4} \\ f(-2) &= 29 \end{aligned}$$

$$(8) \begin{aligned} f(x) &= 3 \left( x + \frac{1}{3} \right)^2 + \frac{5}{3} \\ f(5) &= 87 \end{aligned}$$

$$(4) \begin{aligned} f(x) &= -5 \left( x - \frac{1}{10} \right)^2 + \frac{41}{20} \\ f(2) &= -16 \end{aligned}$$

$$(9) \begin{aligned} f(x) &= 5 \left( x - \frac{2}{5} \right)^2 + \frac{1}{5} \\ f(-5) &= 146 \end{aligned}$$

$$(5) \begin{aligned} f(x) &= -2 \left( x - \frac{1}{2} \right)^2 + \frac{3}{2} \\ f(2) &= -3 \end{aligned}$$

$$(10) \begin{aligned} f(x) &= 2 \left( x + \frac{1}{2} \right)^2 + \frac{1}{2} \\ f(3) &= 25 \end{aligned}$$

$$(1) \quad f(x) = -\left(x + \frac{1}{2}\right)^2 - \frac{15}{4}$$
$$f(-2) = -6$$

$$(6) \quad f(x) = 3\left(x - \frac{5}{6}\right)^2 + \frac{23}{12}$$
$$f(-1) = 12$$

$$(2) \quad f(x) = (x - 1)^2 + 2$$
$$f(1) = 2$$

$$(7) \quad f(x) = -3\left(x + \frac{5}{6}\right)^2 + \frac{25}{12}$$
$$f(-4) = -28$$

$$(3) \quad f(x) = 4\left(x + \frac{1}{4}\right)^2 + \frac{11}{4}$$
$$f(-4) = 59$$

$$(8) \quad f(x) = -4\left(x + \frac{5}{8}\right)^2 + \frac{25}{16}$$
$$f(2) = -26$$

$$(4) \quad f(x) = 2\left(x + \frac{5}{4}\right)^2 - \frac{25}{8}$$
$$f(-3) = 3$$

$$(9) \quad f(x) = -3\left(x - \frac{2}{3}\right)^2 + \frac{19}{3}$$
$$f(3) = -10$$

$$(5) \quad f(x) = -5\left(x - \frac{1}{5}\right)^2 + \frac{16}{5}$$
$$f(4) = -69$$

$$(10) \quad f(x) = -2(x + 1)^2 + 2$$
$$f(-3) = -6$$

$$(1) \quad f(x) = -(x-1)^2 + 4 \\ f(0) = 3$$

$$(6) \quad f(x) = -2(x+1)^2 + 4 \\ f(-1) = 4$$

$$(2) \quad f(x) = 5\left(x + \frac{1}{2}\right)^2 - \frac{13}{4} \\ f(-4) = 58$$

$$(7) \quad f(x) = 5\left(x - \frac{2}{5}\right)^2 - \frac{24}{5} \\ f(0) = -4$$

$$(3) \quad f(x) = \left(x - \frac{1}{2}\right)^2 + \frac{15}{4} \\ f(3) = 10$$

$$(8) \quad f(x) = 4\left(x + \frac{5}{8}\right)^2 + \frac{39}{16} \\ f(-3) = 25$$

$$(4) \quad f(x) = -3\left(x - \frac{1}{3}\right)^2 - \frac{11}{3} \\ f(0) = -4$$

$$(9) \quad f(x) = 5\left(x - \frac{1}{5}\right)^2 + \frac{19}{5} \\ f(-5) = 139$$

$$(5) \quad f(x) = -2\left(x + \frac{5}{4}\right)^2 - \frac{7}{8} \\ f(-3) = -7$$

$$(10) \quad f(x) = -2\left(x + \frac{3}{4}\right)^2 - \frac{23}{8} \\ f(-4) = -24$$

$$(1) \quad f(x) = 3 \left( x + \frac{5}{6} \right)^2 - \frac{13}{12}$$
$$f(-4) = 29$$

$$(6) \quad f(x) = -2 \left( x - \frac{1}{2} \right)^2 + \frac{1}{2}$$
$$f(5) = -40$$

$$(2) \quad f(x) = 3 \left( x - \frac{1}{3} \right)^2 + \frac{14}{3}$$
$$f(0) = 5$$

$$(7) \quad f(x) = 2 \left( x - \frac{1}{4} \right)^2 + \frac{7}{8}$$
$$f(0) = 1$$

$$(3) \quad f(x) = 4 \left( x + \frac{1}{4} \right)^2 - \frac{13}{4}$$
$$f(3) = 39$$

$$(8) \quad f(x) = 3 \left( x - \frac{1}{2} \right)^2 - \frac{23}{4}$$
$$f(-2) = 13$$

$$(4) \quad f(x) = 2 \left( x - \frac{5}{4} \right)^2 - \frac{57}{8}$$
$$f(-2) = 14$$

$$(9) \quad f(x) = \left( x - \frac{3}{2} \right)^2 - \frac{29}{4}$$
$$f(4) = -1$$

$$(5) \quad f(x) = 5 \left( x - \frac{1}{2} \right)^2 + \frac{15}{4}$$
$$f(3) = 35$$

$$(10) \quad f(x) = 4 \left( x - \frac{3}{8} \right)^2 - \frac{89}{16}$$
$$f(2) = 5$$

## 2.10 1 次方程式

1 次方程式の問題。

(1)  $-6x - 9 = -5x - 3$

(16)  $8x + 8 = -3x + 8$

(31)  $5x - 4 = 8x + 4$

(2)  $3x + 7 = -4x - 8$

(17)  $-7x + 4 = 9x - 3$

(32)  $2x - 3 = 9x - 1$

(3)  $7x - 8 = -7x - 5$

(18)  $4x + 5 = -7x - 1$

(33)  $-9x - 6 = 8x + 2$

(4)  $-2x + 9 = -4x - 3$

(19)  $-6x - 3 = 6x + 3$

(34)  $3x + 4 = 5x + 1$

(5)  $-4x + 2 = 2x + 6$

(20)  $-6x - 2 = 3x + 8$

(35)  $6x + 4 = -2x - 5$

(6)  $8x - 3 = 3x - 6$

(21)  $-6x + 5 = 4x + 4$

(36)  $7x - 4 = 6x + 2$

(7)  $6x - 9 = -7x - 5$

(22)  $9x - 8 = 5x + 8$

(37)  $5x - 9 = -5x - 6$

(8)  $5x - 9 = x + 1$

(23)  $-5x + 5 = 6x + 9$

(38)  $x - 2 = 2x - 8$

(9)  $-9x + 7 = x - 1$

(24)  $-8x + 5 = 8x - 4$

(39)  $3x + 3 = x + 5$

(10)  $-6x - 7 = 3x - 5$

(25)  $3x - 2 = 7x + 2$

(40)  $9x + 4 = -3x + 7$

(11)  $-7x - 7 = 3x - 5$

(26)  $-x + 8 = 6x + 7$

(41)  $-8x + 3 = -7x - 1$

(12)  $5x + 7 = -4x - 9$

(27)  $-9x - 3 = 8x + 2$

(42)  $-2x - 1 = -3x + 2$

(13)  $6x - 5 = -9x - 3$

(28)  $6x + 1 = 5x + 4$

(43)  $-2x - 7 = 3x - 4$

(14)  $-2x + 5 = -5x - 8$

(29)  $-7x + 8 = -3x + 5$

(44)  $2x + 3 = -4x - 1$

(15)  $-9x - 1 = -5x - 4$

(30)  $-7x + 5 = 7x + 3$

(45)  $4x + 4 = x - 6$

(1)  $-8x + 6 = -3x - 4$

(16)  $-7x - 6 = 4x - 4$

(31)  $7x - 4 = -6x + 6$

(2)  $-3x - 1 = 9x + 3$

(17)  $-5x + 6 = -6x - 7$

(32)  $7x - 6 = 2x - 9$

(3)  $7x + 8 = -8x - 4$

(18)  $2x - 3 = 9x - 2$

(33)  $6x - 4 = -7x - 8$

(4)  $3x - 9 = -x + 9$

(19)  $-x - 9 = -2x + 2$

(34)  $x + 4 = 4x - 7$

(5)  $2x - 5 = x - 4$

(20)  $-5x - 8 = 3x - 7$

(35)  $-6x - 4 = -x + 8$

(6)  $-5x + 8 = -6x - 5$

(21)  $-8x - 9 = 4x + 2$

(36)  $7x + 7 = 5x - 4$

(7)  $7x + 5 = 5x - 1$

(22)  $5x + 3 = 2x - 3$

(37)  $6x + 4 = 5x + 2$

(8)  $-2x - 1 = -7x - 8$

(23)  $3x + 7 = -x + 5$

(38)  $8x - 3 = -4x + 7$

(9)  $5x - 2 = 7x + 1$

(24)  $-4x + 2 = 4x + 3$

(39)  $6x + 7 = -7x + 9$

(10)  $-x - 9 = x + 1$

(25)  $9x - 7 = -2x + 2$

(40)  $4x + 5 = -8x - 1$

(11)  $4x + 7 = -6x - 3$

(26)  $4x - 4 = -4x + 3$

(41)  $-8x + 9 = 2x + 3$

(12)  $7x + 8 = 5x + 6$

(27)  $-3x + 4 = -8x + 4$

(42)  $8x - 9 = 7x + 3$

(13)  $7x - 5 = x - 9$

(28)  $-x - 1 = 3x + 6$

(43)  $7x - 9 = -2x - 2$

(14)  $7x + 9 = -3x - 3$

(29)  $7x - 7 = -9x - 6$

(44)  $6x - 3 = -2x - 6$

(15)  $4x + 5 = -8x - 3$

(30)  $8x + 5 = x + 4$

(45)  $4x + 9 = -x - 3$

(1)  $-7x + 5 = -8x - 8$

(16)  $x - 4 = -x - 2$

(31)  $-5x - 1 = 5x + 1$

(2)  $8x - 9 = -7x - 7$

(17)  $-6x - 5 = 7x - 8$

(32)  $-3x - 1 = -7x + 4$

(3)  $9x - 7 = -6x + 6$

(18)  $3x + 5 = x - 4$

(33)  $x - 5 = 8x - 3$

(4)  $-x - 9 = 2x - 8$

(19)  $8x + 4 = -3x - 6$

(34)  $4x - 5 = -4x - 1$

(5)  $-3x + 2 = -8x - 4$

(20)  $-4x - 7 = -6x - 4$

(35)  $5x + 5 = -4x + 7$

(6)  $3x + 8 = 7x - 3$

(21)  $7x - 8 = 3x - 1$

(36)  $7x - 2 = -9x + 3$

(7)  $5x + 5 = 2x - 2$

(22)  $-7x - 8 = 4x + 8$

(37)  $3x + 3 = -8x + 4$

(8)  $-3x + 3 = 4x - 5$

(23)  $-6x - 5 = 8x - 7$

(38)  $9x - 6 = -x + 5$

(9)  $7x + 8 = -6x + 1$

(24)  $4x - 1 = -2x - 1$

(39)  $2x + 1 = -8x - 2$

(10)  $6x - 3 = -9x - 8$

(25)  $2x - 2 = -x + 1$

(40)  $8x + 3 = 9x + 1$

(11)  $9x - 5 = 4x + 2$

(26)  $-7x + 5 = -6x + 5$

(41)  $-8x - 9 = -2x - 7$

(12)  $-3x - 2 = -6x + 3$

(27)  $3x + 6 = -4x + 9$

(42)  $6x + 9 = -6x - 9$

(13)  $-2x - 5 = 6x + 7$

(28)  $x + 5 = 7x + 6$

(43)  $5x + 2 = 7x + 6$

(14)  $5x + 9 = 3x + 4$

(29)  $-4x - 1 = -x + 7$

(44)  $-5x - 8 = 9x - 9$

(15)  $2x + 6 = 8x - 1$

(30)  $x - 4 = 5x - 3$

(45)  $x + 5 = 8x + 3$



(1)  $6x + 1 = -x - 5$

(16)  $3x + 3 = -8x - 1$

(31)  $-6x + 6 = -x + 9$

(2)  $-9x + 7 = 7x + 2$

(17)  $-x - 6 = 8x + 1$

(32)  $9x - 2 = -8x + 4$

(3)  $4x - 9 = 2x + 6$

(18)  $5x + 1 = -7x + 9$

(33)  $-2x - 5 = 7x + 9$

(4)  $-8x - 4 = 8x + 3$

(19)  $-4x - 1 = -9x - 3$

(34)  $8x - 8 = 6x - 7$

(5)  $x - 6 = 8x + 5$

(20)  $-6x - 1 = x + 7$

(35)  $-7x - 5 = 6x - 2$

(6)  $-8x + 9 = -5x + 7$

(21)  $3x - 1 = -8x - 4$

(36)  $-6x + 8 = -4x + 1$

(7)  $9x - 2 = 2x - 1$

(22)  $-3x - 1 = 3x - 9$

(37)  $-6x + 9 = -3x + 8$

(8)  $-9x - 9 = 2x - 9$

(23)  $-4x + 3 = -2x - 2$

(38)  $5x - 4 = 8x - 4$

(9)  $-x - 6 = 6x + 9$

(24)  $3x - 2 = 5x + 7$

(39)  $4x + 9 = 5x - 9$

(10)  $x + 5 = 9x + 9$

(25)  $-2x - 2 = x - 2$

(40)  $5x + 7 = -5x + 3$

(11)  $5x - 6 = -8x - 3$

(26)  $-x - 2 = -2x - 7$

(41)  $5x + 2 = -9x - 6$

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(16)  $-8x + 7 = x + 1$

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(19)  $-4x + 8 = 2x + 2$

(34)  $3x + 9 = -6x - 7$

(5)  $-5x + 6 = -6x + 8$

(20)  $4x + 4 = 5x - 3$

(35)  $7x + 9 = -6x - 5$

(6)  $x + 7 = 7x - 6$

(21)  $-2x - 3 = -5x + 8$

(36)  $9x + 2 = -3x - 3$

(7)  $-2x - 7 = -x + 6$

(22)  $5x + 3 = -9x + 8$

(37)  $2x - 8 = -7x - 1$

(8)  $x + 4 = 9x + 8$

(23)  $-9x + 1 = -3x - 7$

(38)  $-9x - 2 = -2x - 6$

(9)  $-6x + 4 = -5x + 6$

(24)  $-6x + 8 = -8x + 4$

(39)  $-5x - 9 = 7x - 8$

(10)  $9x - 2 = -4x + 8$

(25)  $-7x + 8 = -3x - 7$

(40)  $5x + 2 = -7x + 3$

(11)  $-6x + 5 = -8x + 6$

(26)  $3x + 5 = 2x - 4$

(41)  $-x - 1 = 6x - 5$

(12)  $-x + 9 = 9x + 2$

(27)  $-9x - 6 = -5x + 4$

(42)  $-8x + 9 = 8x + 9$

(13)  $-7x - 2 = -9x + 4$

(28)  $-2x - 9 = 8x + 9$

(43)  $6x + 4 = 9x + 5$

(14)  $-9x - 6 = x + 5$

(29)  $-9x - 4 = 8x + 4$

(44)  $6x + 3 = 7x - 2$

(15)  $-5x - 3 = 9x + 8$

(30)  $-6x + 4 = 6x - 2$

(45)  $-4x + 3 = -x - 8$

(1)  $2x + 6 = 5x - 5$

(16)  $-9x + 5 = -5x - 2$

(31)  $6x - 4 = 5x + 6$

(2)  $x - 4 = -3x + 1$

(17)  $2x + 1 = x - 7$

(32)  $-9x + 3 = -3x + 8$

(3)  $-7x - 2 = -2x + 8$

(18)  $5x - 6 = 6x - 4$

(33)  $2x - 9 = x + 2$

(4)  $8x + 1 = 2x - 7$

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(16)  $-3x + 7 = 5x - 7$

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(45)  $8x - 6 = -x - 9$

$$(1) -6x - 9 = -5x - 3$$

$$x = -6$$

$$(2) 3x + 7 = -4x - 8$$

$$x = -\frac{15}{7}$$

$$(3) 7x - 8 = -7x - 5$$

$$x = \frac{3}{14}$$

$$(4) -2x + 9 = -4x - 3$$

$$x = -6$$

$$(5) -4x + 2 = 2x + 6$$

$$x = -\frac{2}{3}$$

$$(6) 8x - 3 = 3x - 6$$

$$x = -\frac{3}{5}$$

$$(7) 6x - 9 = -7x - 5$$

$$x = \frac{4}{13}$$

$$(8) 5x - 9 = x + 1$$

$$x = \frac{5}{2}$$

$$(9) -9x + 7 = x - 1$$

$$x = \frac{4}{5}$$

$$(10) -6x - 7 = 3x - 5$$

$$x = -\frac{2}{9}$$

$$(11) -7x - 7 = 3x - 5$$

$$x = -\frac{1}{5}$$

$$(12) 5x + 7 = -4x - 9$$

$$x = -\frac{16}{9}$$

$$(13) 6x - 5 = -9x - 3$$

$$x = \frac{2}{15}$$

$$(14) -2x + 5 = -5x - 8$$

$$x = -\frac{13}{3}$$

$$(15) -9x - 1 = -5x - 4$$

$$x = \frac{3}{4}$$

$$(16) 8x + 8 = -3x + 8$$

$$x = 0$$

$$(17) -7x + 4 = 9x - 3$$

$$x = \frac{7}{16}$$

$$(18) 4x + 5 = -7x - 1$$

$$x = -\frac{6}{11}$$

$$(19) -6x - 3 = 6x + 3$$

$$x = -\frac{1}{2}$$

$$(20) -6x - 2 = 3x + 8$$

$$x = -\frac{10}{9}$$

$$(21) -6x + 5 = 4x + 4$$

$$x = \frac{1}{10}$$

$$(22) 9x - 8 = 5x + 8$$

$$x = 4$$

$$(23) -5x + 5 = 6x + 9$$

$$x = -\frac{4}{11}$$

$$(24) -8x + 5 = 8x - 4$$

$$x = \frac{9}{16}$$

$$(25) 3x - 2 = 7x + 2$$

$$x = -1$$

$$(26) -x + 8 = 6x + 7$$

$$x = \frac{1}{7}$$

$$(27) -9x - 3 = 8x + 2$$

$$x = -\frac{5}{17}$$

$$(28) 6x + 1 = 5x + 4$$

$$x = 3$$

$$(29) -7x + 8 = -3x + 5$$

$$x = \frac{3}{4}$$

$$(30) -7x + 5 = 7x + 3$$

$$x = \frac{1}{7}$$

$$(31) 5x - 4 = 8x + 4$$

$$x = -\frac{8}{3}$$

$$(32) 2x - 3 = 9x - 1$$

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$$(33) -9x - 6 = 8x + 2$$

$$x = -\frac{8}{17}$$

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$$(36) 7x - 4 = 6x + 2$$

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$$(37) 5x - 9 = -5x - 6$$

$$x = \frac{3}{10}$$

$$(38) x - 2 = 2x - 8$$

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$$x = -\frac{2}{3}$$

$$(45) 4x + 4 = x - 6$$

$$x = -\frac{10}{3}$$

$$(1) \quad -8x + 6 = -3x - 4$$

$$x = 2$$

$$(2) \quad -3x - 1 = 9x + 3$$

$$x = -\frac{1}{3}$$

$$(3) \quad 7x + 8 = -8x - 4$$

$$x = -\frac{4}{5}$$

$$(4) \quad 3x - 9 = -x + 9$$

$$x = \frac{9}{2}$$

$$(5) \quad 2x - 5 = x - 4$$

$$x = 1$$

$$(6) \quad -5x + 8 = -6x - 5$$

$$x = -13$$

$$(7) \quad 7x + 5 = 5x - 1$$

$$x = -3$$

$$(8) \quad -2x - 1 = -7x - 8$$

$$x = -\frac{7}{5}$$

$$(9) \quad 5x - 2 = 7x + 1$$

$$x = -\frac{3}{2}$$

$$(10) \quad -x - 9 = x + 1$$

$$x = -5$$

$$(11) \quad 4x + 7 = -6x - 3$$

$$x = -1$$

$$(12) \quad 7x + 8 = 5x + 6$$

$$x = -1$$

$$(13) \quad 7x - 5 = x - 9$$

$$x = -\frac{2}{3}$$

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$$(24) \quad -4x + 2 = 4x + 3$$

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$$(25) \quad 9x - 7 = -2x + 2$$

$$x = \frac{9}{11}$$

$$(26) \quad 4x - 4 = -4x + 3$$

$$x = \frac{7}{8}$$

$$(27) \quad -3x + 4 = -8x + 4$$

$$x = 0$$

$$(28) \quad -x - 1 = 3x + 6$$

$$x = -\frac{7}{4}$$

$$(29) \quad 7x - 7 = -9x - 6$$

$$x = \frac{1}{16}$$

$$(30) \quad 8x + 5 = x + 4$$

$$x = -\frac{1}{7}$$

$$(31) \quad 7x - 4 = -6x + 6$$

$$x = \frac{10}{13}$$

$$(32) \quad 7x - 6 = 2x - 9$$

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$$(45) \quad 4x + 9 = -x - 3$$

$$x = -\frac{12}{5}$$

$$(1) \quad -7x + 5 = -8x - 8$$

$$x = -13$$

$$(2) \quad 8x - 9 = -7x - 7$$

$$x = \frac{2}{15}$$

$$(3) \quad 9x - 7 = -6x + 6$$

$$x = \frac{13}{15}$$

$$(4) \quad -x - 9 = 2x - 8$$

$$x = -\frac{1}{3}$$

$$(5) \quad -3x + 2 = -8x - 4$$

$$x = -\frac{6}{5}$$

$$(6) \quad 3x + 8 = 7x - 3$$

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$$x = -\frac{10}{11}$$

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$$x = \frac{5}{16}$$

$$(37) \quad 3x + 3 = -8x + 4$$

$$x = \frac{1}{11}$$

$$(38) \quad 9x - 6 = -x + 5$$

$$x = \frac{11}{10}$$

$$(39) \quad 2x + 1 = -8x - 2$$

$$x = -\frac{3}{10}$$

$$(40) \quad 8x + 3 = 9x + 1$$

$$x = 2$$

$$(41) \quad -8x - 9 = -2x - 7$$

$$x = -\frac{1}{3}$$

$$(42) \quad 6x + 9 = -6x - 9$$

$$x = -\frac{3}{2}$$

$$(43) \quad 5x + 2 = 7x + 6$$

$$x = -2$$

$$(44) \quad -5x - 8 = 9x - 9$$

$$x = \frac{1}{14}$$

$$(45) \quad x + 5 = 8x + 3$$

$$x = \frac{2}{7}$$

$$(1) \quad 6x + 1 = -x - 5$$

$$x = -\frac{6}{7}$$

$$(2) \quad -9x + 7 = 7x + 2$$

$$x = \frac{5}{16}$$

$$(3) \quad 4x - 9 = 2x + 6$$

$$x = \frac{15}{2}$$

$$(4) \quad -8x - 4 = 8x + 3$$

$$x = -\frac{7}{16}$$

$$(5) \quad x - 6 = 8x + 5$$

$$x = -\frac{11}{7}$$

$$(6) \quad -8x + 9 = -5x + 7$$

$$x = \frac{2}{3}$$

$$(7) \quad 9x - 2 = 2x - 1$$

$$x = \frac{1}{7}$$

$$(8) \quad -9x - 9 = 2x - 9$$

$$x = 0$$

$$(9) \quad -x - 6 = 6x + 9$$

$$x = -\frac{15}{7}$$

$$(10) \quad x + 5 = 9x + 9$$

$$x = -\frac{1}{2}$$

$$(11) \quad 5x - 6 = -8x - 3$$

$$x = \frac{3}{13}$$

$$(12) \quad 7x - 3 = -6x - 4$$

$$x = -\frac{1}{13}$$

$$(13) \quad -7x - 2 = 6x - 6$$

$$x = \frac{4}{13}$$

$$(14) \quad -3x + 4 = -7x - 2$$

$$x = -\frac{3}{2}$$

$$(15) \quad -5x - 5 = -x - 8$$

$$x = \frac{3}{4}$$

$$(16) \quad 3x + 3 = -8x - 1$$

$$x = -\frac{4}{11}$$

$$(17) \quad -x - 6 = 8x + 1$$

$$x = -\frac{7}{9}$$

$$(18) \quad 5x + 1 = -7x + 9$$

$$x = \frac{2}{3}$$

$$(19) \quad -4x - 1 = -9x - 3$$

$$x = -\frac{2}{5}$$

$$(20) \quad -6x - 1 = x + 7$$

$$x = -\frac{8}{7}$$

$$(21) \quad 3x - 1 = -8x - 4$$

$$x = -\frac{3}{11}$$

$$(22) \quad -3x - 1 = 3x - 9$$

$$x = \frac{4}{3}$$

$$(23) \quad -4x + 3 = -2x - 2$$

$$x = \frac{5}{2}$$

$$(24) \quad 3x - 2 = 5x + 7$$

$$x = -\frac{9}{2}$$

$$(25) \quad -2x - 2 = x - 2$$

$$x = 0$$

$$(26) \quad -x - 2 = -2x - 7$$

$$x = -5$$

$$(27) \quad x - 3 = 9x - 7$$

$$x = \frac{1}{2}$$

$$(28) \quad 6x - 1 = 3x + 5$$

$$x = 2$$

$$(29) \quad -2x + 9 = 8x - 8$$

$$x = \frac{17}{10}$$

$$(30) \quad -2x + 6 = -5x - 8$$

$$x = -\frac{14}{3}$$

$$(31) \quad -6x + 6 = -x + 9$$

$$x = -\frac{3}{5}$$

$$(32) \quad 9x - 2 = -8x + 4$$

$$x = \frac{6}{17}$$

$$(33) \quad -2x - 5 = 7x + 9$$

$$x = -\frac{14}{9}$$

$$(34) \quad 8x - 8 = 6x - 7$$

$$x = \frac{1}{2}$$

$$(35) \quad -7x - 5 = 6x - 2$$

$$x = -\frac{3}{13}$$

$$(36) \quad -6x + 8 = -4x + 1$$

$$x = \frac{7}{2}$$

$$(37) \quad -6x + 9 = -3x + 8$$

$$x = \frac{1}{3}$$

$$(38) \quad 5x - 4 = 8x - 4$$

$$x = 0$$

$$(39) \quad 4x + 9 = 5x - 9$$

$$x = 18$$

$$(40) \quad 5x + 7 = -5x + 3$$

$$x = -\frac{2}{5}$$

$$(41) \quad 5x + 2 = -9x - 6$$

$$x = -\frac{4}{7}$$

$$(42) \quad 6x + 1 = 7x - 6$$

$$x = 7$$

$$(43) \quad 7x - 3 = 5x + 1$$

$$x = 2$$

$$(44) \quad -5x + 8 = x + 6$$

$$x = \frac{1}{3}$$

$$(45) \quad -8x + 6 = 4x - 9$$

$$x = \frac{5}{4}$$

$$(1) \quad 8x - 9 = -7x - 3$$

$$x = \frac{2}{5}$$

$$(2) \quad -8x - 5 = -4x - 8$$

$$x = \frac{3}{4}$$

$$(3) \quad -6x + 3 = 5x + 9$$

$$x = -\frac{6}{11}$$

$$(4) \quad 7x + 6 = 5x - 2$$

$$x = -4$$

$$(5) \quad 7x + 6 = -3x + 1$$

$$x = -\frac{1}{2}$$

$$(6) \quad 2x - 7 = -7x - 1$$

$$x = \frac{2}{3}$$

$$(7) \quad 5x - 9 = x + 4$$

$$x = \frac{13}{4}$$

$$(8) \quad -5x + 8 = 6x - 3$$

$$x = 1$$

$$(9) \quad -8x - 6 = 8x - 1$$

$$x = -\frac{5}{16}$$

$$(10) \quad -2x + 7 = 6x + 6$$

$$x = \frac{1}{8}$$

$$(11) \quad -6x + 6 = 4x + 8$$

$$x = -\frac{1}{5}$$

$$(12) \quad 4x - 7 = 2x + 9$$

$$x = 8$$

$$(13) \quad x + 8 = -5x + 3$$

$$x = -\frac{5}{6}$$

$$(14) \quad 2x - 5 = -x + 2$$

$$x = \frac{7}{3}$$

$$(15) \quad 5x + 4 = -8x + 7$$

$$x = \frac{3}{13}$$

$$(16) \quad -8x + 7 = x + 1$$

$$x = \frac{2}{3}$$

$$(17) \quad -4x + 6 = 8x - 4$$

$$x = \frac{5}{6}$$

$$(18) \quad -7x + 7 = 5x - 7$$

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$$(20) \quad 9x + 3 = x + 5$$

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$$(21) \quad 8x - 4 = 9x + 1$$

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$$(22) \quad -4x + 5 = 2x - 9$$

$$x = \frac{7}{3}$$

$$(23) \quad x - 4 = -6x - 6$$

$$x = -\frac{2}{7}$$

$$(24) \quad -5x - 6 = 3x + 5$$

$$x = -\frac{11}{8}$$

$$(25) \quad -5x - 4 = -4x + 1$$

$$x = -5$$

$$(26) \quad 8x + 5 = -5x - 3$$

$$x = -\frac{8}{13}$$

$$(27) \quad 2x + 9 = -9x + 3$$

$$x = -\frac{6}{11}$$

$$(28) \quad 4x + 5 = 7x - 1$$

$$x = 2$$

$$(29) \quad 6x + 1 = 3x - 2$$

$$x = -1$$

$$(30) \quad -7x - 5 = 8x - 3$$

$$x = -\frac{2}{15}$$

$$(31) \quad 5x + 3 = -7x + 4$$

$$x = \frac{1}{12}$$

$$(32) \quad 6x - 6 = -2x - 3$$

$$x = \frac{3}{8}$$

$$(33) \quad -5x + 8 = 8x - 5$$

$$x = 1$$

$$(34) \quad -9x - 3 = 8x - 6$$

$$x = \frac{3}{17}$$

$$(35) \quad 5x - 8 = 4x + 7$$

$$x = 15$$

$$(36) \quad 8x + 2 = 9x - 6$$

$$x = 8$$

$$(37) \quad 5x - 9 = 6x - 4$$

$$x = -5$$

$$(38) \quad 5x + 8 = -7x - 4$$

$$x = -1$$

$$(39) \quad 9x - 7 = -x - 7$$

$$x = 0$$

$$(40) \quad -2x + 3 = 4x - 5$$

$$x = \frac{4}{3}$$

$$(41) \quad 3x - 9 = -x - 2$$

$$x = \frac{7}{4}$$

$$(42) \quad -7x + 8 = -8x + 3$$

$$x = -5$$

$$(43) \quad 4x - 7 = x + 4$$

$$x = \frac{11}{3}$$

$$(44) \quad 3x + 7 = 5x + 6$$

$$x = \frac{1}{2}$$

$$(45) \quad 4x - 9 = -6x - 9$$

$$x = 0$$

$$(1) \quad 8x - 2 = 4x + 3$$

$$x = \frac{5}{4}$$

$$(2) \quad x - 4 = 7x - 4$$

$$x = 0$$

$$(3) \quad -6x - 8 = 8x - 3$$

$$x = -\frac{5}{14}$$

$$(4) \quad -5x - 1 = 2x + 8$$

$$x = -\frac{9}{7}$$

$$(5) \quad 4x + 5 = -3x + 4$$

$$x = -\frac{1}{7}$$

$$(6) \quad 3x - 3 = -5x - 7$$

$$x = -\frac{1}{2}$$

$$(7) \quad -8x + 9 = 3x + 9$$

$$x = 0$$

$$(8) \quad -4x - 9 = 7x - 3$$

$$x = -\frac{6}{11}$$

$$(9) \quad -2x - 2 = 2x + 8$$

$$x = -\frac{5}{2}$$

$$(10) \quad -x - 8 = -3x - 2$$

$$x = 3$$

$$(11) \quad x + 2 = 4x - 8$$

$$x = \frac{10}{3}$$

$$(12) \quad 8x + 5 = -2x - 8$$

$$x = -\frac{13}{10}$$

$$(13) \quad 4x + 6 = 7x + 4$$

$$x = \frac{2}{3}$$

$$(14) \quad 3x - 2 = -6x - 5$$

$$x = -\frac{1}{3}$$

$$(15) \quad -8x + 8 = -9x + 3$$

$$x = -5$$

$$(16) \quad -4x + 2 = 3x - 7$$

$$x = \frac{9}{7}$$

$$(17) \quad -2x - 7 = 9x + 3$$

$$x = -\frac{10}{11}$$

$$(18) \quad -4x - 7 = x + 5$$

$$x = -\frac{12}{5}$$

$$(19) \quad 9x + 3 = 2x - 5$$

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$$x = -\frac{15}{4}$$

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$$(24) \quad -7x - 8 = 5x - 6$$

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$$(25) \quad 4x + 5 = 3x - 6$$

$$x = -11$$

$$(26) \quad 6x - 3 = -6x + 9$$

$$x = 1$$

$$(27) \quad 6x - 9 = -2x - 8$$

$$x = \frac{1}{8}$$

$$(28) \quad 4x + 1 = x - 4$$

$$x = -\frac{5}{3}$$

$$(29) \quad -6x + 3 = 4x + 6$$

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$$(30) \quad -9x + 1 = 6x - 4$$

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$$(31) \quad -3x + 5 = 7x - 2$$

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$$(32) \quad 6x - 8 = 9x - 8$$

$$x = 0$$

$$(33) \quad 9x + 2 = -8x + 7$$

$$x = \frac{5}{17}$$

$$(34) \quad -6x - 6 = -2x + 3$$

$$x = -\frac{9}{4}$$

$$(35) \quad 6x - 7 = 2x - 4$$

$$x = \frac{3}{4}$$

$$(36) \quad 5x + 5 = 2x - 4$$

$$x = -3$$

$$(37) \quad -3x + 5 = 6x - 4$$

$$x = 1$$

$$(38) \quad -3x + 4 = -4x - 1$$

$$x = -5$$

$$(39) \quad 7x + 4 = -8x + 5$$

$$x = \frac{1}{15}$$

$$(40) \quad 5x + 7 = -2x + 5$$

$$x = -\frac{2}{7}$$

$$(41) \quad -6x + 3 = -7x - 8$$

$$x = -11$$

$$(42) \quad -x - 6 = 2x + 7$$

$$x = -\frac{13}{3}$$

$$(43) \quad 7x + 1 = -4x + 9$$

$$x = \frac{8}{11}$$

$$(44) \quad 7x + 4 = -7x - 5$$

$$x = -\frac{9}{14}$$

$$(45) \quad 4x - 6 = -7x + 5$$

$$x = 1$$

$$(1) -7x - 6 = 7x + 9$$

$$x = -\frac{15}{14}$$

$$(2) -2x + 8 = -3x - 1$$

$$x = -9$$

$$(3) 9x + 3 = 2x + 5$$

$$x = \frac{2}{7}$$

$$(4) 3x + 6 = 5x + 5$$

$$x = \frac{1}{2}$$

$$(5) -2x - 3 = -5x - 9$$

$$x = -2$$

$$(6) 2x - 2 = -9x + 2$$

$$x = \frac{4}{11}$$

$$(7) -2x - 3 = -x - 3$$

$$x = 0$$

$$(8) -5x - 8 = 5x - 8$$

$$x = 0$$

$$(9) 7x + 2 = x - 1$$

$$x = -\frac{1}{2}$$

$$(10) 3x - 8 = -3x - 7$$

$$x = \frac{1}{6}$$

$$(11) 2x - 9 = 3x - 8$$

$$x = -1$$

$$(12) -9x + 7 = -2x - 5$$

$$x = \frac{12}{7}$$

$$(13) 6x - 2 = -x + 2$$

$$x = \frac{4}{7}$$

$$(14) 5x - 9 = 4x - 9$$

$$x = 0$$

$$(15) 3x - 6 = 5x - 5$$

$$x = -\frac{1}{2}$$

$$(16) 6x - 1 = 7x + 5$$

$$x = -6$$

$$(17) -7x - 7 = -6x - 3$$

$$x = -4$$

$$(18) -3x - 8 = 8x + 5$$

$$x = -\frac{13}{11}$$

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$$x = -\frac{1}{12}$$

$$(23) -6x + 5 = 6x - 6$$

$$x = \frac{11}{12}$$

$$(24) -8x - 2 = 7x + 3$$

$$x = -\frac{1}{3}$$

$$(25) 6x + 8 = 3x + 4$$

$$x = -\frac{4}{3}$$

$$(26) 7x - 4 = -7x - 8$$

$$x = -\frac{2}{7}$$

$$(27) -4x + 6 = 6x + 6$$

$$x = 0$$

$$(28) 3x + 1 = -9x - 5$$

$$x = -\frac{1}{2}$$

$$(29) -8x - 7 = x - 2$$

$$x = -\frac{5}{9}$$

$$(30) x + 7 = -9x + 1$$

$$x = -\frac{3}{5}$$

$$(31) -5x + 1 = 9x - 9$$

$$x = \frac{5}{7}$$

$$(32) 7x + 8 = 8x - 2$$

$$x = 10$$

$$(33) -3x + 2 = 5x + 1$$

$$x = \frac{1}{8}$$

$$(34) -7x + 9 = -2x - 2$$

$$x = \frac{11}{5}$$

$$(35) -7x + 2 = -x + 5$$

$$x = -\frac{1}{2}$$

$$(36) -8x + 1 = -7x - 9$$

$$x = 10$$

$$(37) 9x - 5 = 3x + 5$$

$$x = \frac{5}{3}$$

$$(38) -6x - 2 = -7x + 4$$

$$x = 6$$

$$(39) 9x + 6 = 2x + 6$$

$$x = 0$$

$$(40) -8x - 6 = -6x + 9$$

$$x = -\frac{15}{2}$$

$$(41) -6x - 9 = 3x + 6$$

$$x = -\frac{5}{3}$$

$$(42) -4x + 6 = 3x - 2$$

$$x = \frac{8}{7}$$

$$(43) -8x + 2 = -2x - 5$$

$$x = \frac{7}{6}$$

$$(44) x + 8 = -4x - 6$$

$$x = -\frac{14}{5}$$

$$(45) -8x - 6 = 9x + 6$$

$$x = -\frac{12}{17}$$



$$(1) \quad 5x - 4 = x + 8$$

$$x = 3$$

$$(2) \quad -7x + 7 = 8x - 1$$

$$x = \frac{8}{15}$$

$$(3) \quad -3x + 8 = -7x - 2$$

$$x = -\frac{5}{2}$$

$$(4) \quad -4x + 4 = -3x + 2$$

$$x = 2$$

$$(5) \quad 9x - 9 = 4x + 5$$

$$x = \frac{14}{5}$$

$$(6) \quad -6x - 7 = 4x - 7$$

$$x = 0$$

$$(7) \quad -4x - 7 = 8x + 3$$

$$x = -\frac{5}{6}$$

$$(8) \quad -9x - 4 = -2x + 2$$

$$x = -\frac{6}{7}$$

$$(9) \quad -x + 3 = -6x - 4$$

$$x = -\frac{7}{5}$$

$$(10) \quad -8x + 8 = -3x - 5$$

$$x = \frac{13}{5}$$

$$(11) \quad 6x + 1 = 7x + 3$$

$$x = -2$$

$$(12) \quad 2x + 4 = -x + 3$$

$$x = -\frac{1}{3}$$

$$(13) \quad -6x - 3 = 5x - 7$$

$$x = \frac{4}{11}$$

$$(14) \quad -9x + 2 = -2x + 5$$

$$x = -\frac{3}{7}$$

$$(15) \quad -8x + 3 = -6x + 4$$

$$x = -\frac{1}{2}$$

$$(16) \quad -8x + 8 = x - 3$$

$$x = \frac{11}{9}$$

$$(17) \quad 6x + 8 = -2x - 6$$

$$x = -\frac{7}{4}$$

$$(18) \quad -7x + 7 = 2x + 3$$

$$x = \frac{4}{9}$$

$$(19) \quad -5x - 2 = 8x + 7$$

$$x = -\frac{9}{13}$$

$$(20) \quad -6x - 5 = -5x + 4$$

$$x = -9$$

$$(21) \quad -6x + 1 = x + 9$$

$$x = -\frac{8}{7}$$

$$(22) \quad 6x + 6 = -x + 5$$

$$x = -\frac{1}{7}$$

$$(23) \quad 3x + 7 = 8x - 2$$

$$x = \frac{9}{5}$$

$$(24) \quad -4x - 5 = -5x - 4$$

$$x = 1$$

$$(25) \quad 8x - 5 = -4x - 3$$

$$x = \frac{1}{6}$$

$$(26) \quad -8x + 6 = -3x - 3$$

$$x = \frac{9}{5}$$

$$(27) \quad 4x + 7 = 3x + 1$$

$$x = -6$$

$$(28) \quad -8x + 1 = -3x + 1$$

$$x = 0$$

$$(29) \quad 7x - 6 = -x - 4$$

$$x = \frac{1}{4}$$

$$(30) \quad 2x - 8 = x - 9$$

$$x = -1$$

$$(31) \quad 8x + 1 = 2x - 7$$

$$x = -\frac{4}{3}$$

$$(32) \quad -x - 9 = 6x - 8$$

$$x = -\frac{1}{7}$$

$$(33) \quad -6x + 3 = -3x - 1$$

$$x = \frac{4}{3}$$

$$(34) \quad 5x - 3 = x - 2$$

$$x = \frac{1}{4}$$

$$(35) \quad -6x + 1 = -4x - 4$$

$$x = \frac{5}{2}$$

$$(36) \quad -9x + 9 = -8x + 9$$

$$x = 0$$

$$(37) \quad -7x - 9 = 4x + 8$$

$$x = -\frac{17}{11}$$

$$(38) \quad -9x + 7 = -6x - 6$$

$$x = \frac{13}{3}$$

$$(39) \quad -5x - 2 = -2x + 9$$

$$x = -\frac{11}{3}$$

$$(40) \quad x - 6 = -5x + 8$$

$$x = \frac{7}{3}$$

$$(41) \quad -2x + 6 = 2x + 8$$

$$x = -\frac{1}{2}$$

$$(42) \quad x + 7 = -7x + 6$$

$$x = -\frac{1}{8}$$

$$(43) \quad 8x + 1 = 9x + 8$$

$$x = -7$$

$$(44) \quad 7x + 6 = -2x - 6$$

$$x = -\frac{4}{3}$$

$$(45) \quad -9x - 1 = -4x - 8$$

$$x = \frac{7}{5}$$

$$(1) \quad 8x + 1 = 9x + 8$$

$$x = -7$$

$$(2) \quad -4x + 3 = -x - 6$$

$$x = 3$$

$$(3) \quad 5x + 9 = -3x + 4$$

$$x = -\frac{5}{8}$$

$$(4) \quad -2x + 8 = 9x + 2$$

$$x = \frac{6}{11}$$

$$(5) \quad 5x + 2 = 4x - 1$$

$$x = -3$$

$$(6) \quad 3x + 3 = -7x - 2$$

$$x = -\frac{1}{2}$$

$$(7) \quad -5x - 4 = 8x + 5$$

$$x = -\frac{9}{13}$$

$$(8) \quad 5x - 1 = -x + 2$$

$$x = \frac{1}{2}$$

$$(9) \quad 9x - 9 = 5x - 4$$

$$x = \frac{5}{4}$$

$$(10) \quad -6x - 9 = -9x - 4$$

$$x = \frac{5}{3}$$

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$$x = \frac{5}{3}$$

$$(20) \quad 7x - 6 = 2x + 6$$

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$$x = \frac{3}{11}$$

$$(22) \quad -3x - 2 = 4x + 3$$

$$x = -\frac{5}{7}$$

$$(23) \quad 7x - 8 = 9x - 4$$

$$x = -2$$

$$(24) \quad -2x + 4 = x + 6$$

$$x = -\frac{2}{3}$$

$$(25) \quad -x - 1 = -9x - 5$$

$$x = -\frac{1}{2}$$

$$(26) \quad -7x - 2 = -9x - 1$$

$$x = \frac{1}{2}$$

$$(27) \quad x - 3 = -3x + 9$$

$$x = 3$$

$$(28) \quad -8x + 8 = 9x - 8$$

$$x = \frac{16}{17}$$

$$(29) \quad 2x + 6 = -x - 7$$

$$x = -\frac{13}{3}$$

$$(30) \quad 5x + 8 = -2x + 9$$

$$x = \frac{1}{7}$$

$$(31) \quad x - 3 = 2x + 5$$

$$x = -8$$

$$(32) \quad -2x + 6 = -7x + 7$$

$$x = \frac{1}{5}$$

$$(33) \quad -2x + 5 = -3x - 1$$

$$x = -6$$

$$(34) \quad -7x - 7 = -3x - 8$$

$$x = \frac{1}{4}$$

$$(35) \quad 8x - 1 = 4x + 5$$

$$x = \frac{3}{2}$$

$$(36) \quad 7x + 9 = -3x - 7$$

$$x = -\frac{8}{5}$$

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$$x = \frac{8}{3}$$

$$(43) \quad 8x - 2 = 2x - 9$$

$$x = -\frac{7}{6}$$

$$(44) \quad 7x + 9 = 8x + 2$$

$$x = 7$$

$$(45) \quad -x - 8 = 7x - 9$$

$$x = \frac{1}{8}$$

$$(1) \quad -7x - 7 = x - 8$$

$$x = \frac{1}{8}$$

$$(2) \quad 5x - 3 = -5x + 5$$

$$x = \frac{4}{5}$$

$$(3) \quad -8x + 7 = 3x - 4$$

$$x = 1$$

$$(4) \quad 5x - 1 = 4x + 4$$

$$x = 5$$

$$(5) \quad 3x + 2 = -3x + 2$$

$$x = 0$$

$$(6) \quad 3x - 8 = 6x + 2$$

$$x = -\frac{10}{3}$$

$$(7) \quad 3x - 2 = -9x + 1$$

$$x = \frac{1}{4}$$

$$(8) \quad x + 9 = -4x + 1$$

$$x = -\frac{8}{5}$$

$$(9) \quad -4x + 1 = 2x + 1$$

$$x = 0$$

$$(10) \quad -5x - 2 = -2x - 5$$

$$x = 1$$

$$(11) \quad 4x - 3 = -6x - 5$$

$$x = -\frac{1}{5}$$

$$(12) \quad -5x - 8 = -x - 5$$

$$x = -\frac{3}{4}$$

$$(13) \quad 4x + 8 = 6x + 1$$

$$x = \frac{7}{2}$$

$$(14) \quad -2x + 3 = 2x - 3$$

$$x = \frac{3}{2}$$

$$(15) \quad -2x + 4 = -5x + 4$$

$$x = 0$$

$$(16) \quad -9x + 3 = x - 9$$

$$x = \frac{6}{5}$$

$$(17) \quad -4x + 5 = 9x - 2$$

$$x = \frac{7}{13}$$

$$(18) \quad 8x - 7 = 9x + 1$$

$$x = -8$$

$$(19) \quad -x - 7 = -6x - 9$$

$$x = -\frac{2}{5}$$

$$(20) \quad 3x - 7 = -8x - 3$$

$$x = \frac{4}{11}$$

$$(21) \quad 8x - 7 = 2x - 3$$

$$x = \frac{2}{3}$$

$$(22) \quad -6x - 9 = 2x + 5$$

$$x = -\frac{7}{4}$$

$$(23) \quad 3x - 5 = -9x + 5$$

$$x = \frac{5}{6}$$

$$(24) \quad -8x - 3 = 4x + 5$$

$$x = -\frac{2}{3}$$

$$(25) \quad -6x + 5 = 4x - 3$$

$$x = \frac{4}{5}$$

$$(26) \quad 7x + 6 = 6x + 3$$

$$x = -3$$

$$(27) \quad x + 6 = 4x + 7$$

$$x = -\frac{1}{3}$$

$$(28) \quad -x + 5 = -5x + 3$$

$$x = -\frac{1}{2}$$

$$(29) \quad x + 4 = 7x - 9$$

$$x = \frac{13}{6}$$

$$(30) \quad x - 3 = -9x - 3$$

$$x = 0$$

$$(31) \quad -4x + 1 = -6x + 1$$

$$x = 0$$

$$(32) \quad -x + 9 = 8x + 6$$

$$x = \frac{1}{3}$$

$$(33) \quad 2x - 3 = -5x - 1$$

$$x = \frac{2}{7}$$

$$(34) \quad -3x + 1 = -6x - 4$$

$$x = -\frac{5}{3}$$

$$(35) \quad 6x - 2 = 3x + 6$$

$$x = \frac{8}{3}$$

$$(36) \quad -2x + 4 = 5x + 3$$

$$x = \frac{1}{7}$$

$$(37) \quad 9x + 4 = 3x - 3$$

$$x = -\frac{7}{6}$$

$$(38) \quad 4x - 7 = 2x + 1$$

$$x = 4$$

$$(39) \quad 9x - 7 = 3x - 9$$

$$x = -\frac{1}{3}$$

$$(40) \quad -5x - 5 = 7x + 1$$

$$x = -\frac{1}{2}$$

$$(41) \quad 7x - 7 = 3x - 7$$

$$x = 0$$

$$(42) \quad -5x + 9 = 5x - 7$$

$$x = \frac{8}{5}$$

$$(43) \quad -9x + 8 = 2x + 5$$

$$x = \frac{3}{11}$$

$$(44) \quad -7x + 2 = 8x - 3$$

$$x = \frac{1}{3}$$

$$(45) \quad 2x - 8 = -7x - 7$$

$$x = \frac{1}{9}$$

$$(1) \quad -8x - 2 = 8x - 4$$

$$x = \frac{1}{8}$$

$$(2) \quad -3x + 5 = -5x + 6$$

$$x = \frac{1}{2}$$

$$(3) \quad 8x + 6 = -8x + 8$$

$$x = \frac{1}{8}$$

$$(4) \quad 9x + 6 = -7x - 5$$

$$x = -\frac{11}{16}$$

$$(5) \quad -x + 9 = 3x + 3$$

$$x = \frac{3}{2}$$

$$(6) \quad -x - 4 = -9x + 8$$

$$x = \frac{3}{2}$$

$$(7) \quad -7x + 9 = 3x - 7$$

$$x = \frac{8}{5}$$

$$(8) \quad 6x + 6 = -6x - 8$$

$$x = -\frac{7}{6}$$

$$(9) \quad -5x - 9 = 4x + 2$$

$$x = -\frac{11}{9}$$

$$(10) \quad 9x - 5 = 6x - 4$$

$$x = \frac{1}{3}$$

$$(11) \quad -5x - 2 = -3x - 1$$

$$x = -\frac{1}{2}$$

$$(12) \quad -2x - 8 = -8x - 2$$

$$x = 1$$

$$(13) \quad 9x - 7 = 2x + 3$$

$$x = \frac{10}{7}$$

$$(14) \quad 8x - 7 = -6x - 8$$

$$x = -\frac{1}{14}$$

$$(15) \quad 4x - 5 = 9x + 6$$

$$x = -\frac{11}{5}$$

$$(16) \quad -x - 9 = 4x - 1$$

$$x = -\frac{8}{5}$$

$$(17) \quad -8x - 7 = 2x - 9$$

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$$x = -4$$

$$(1) \quad -2x - 4 = 5x - 3$$

$$x = -\frac{1}{7}$$

$$(2) \quad 7x + 9 = -5x + 1$$

$$x = -\frac{2}{3}$$

$$(3) \quad -2x + 6 = -6x - 4$$

$$x = -\frac{5}{2}$$

$$(4) \quad -7x + 8 = -4x + 3$$

$$x = \frac{5}{3}$$

$$(5) \quad x - 7 = 5x + 4$$

$$x = -\frac{11}{4}$$

$$(6) \quad 8x - 8 = -9x - 1$$

$$x = \frac{7}{17}$$

$$(7) \quad -5x + 2 = 7x - 3$$

$$x = \frac{5}{12}$$

$$(8) \quad -7x + 2 = -4x - 1$$

$$x = 1$$

$$(9) \quad -8x + 4 = 2x - 7$$

$$x = \frac{11}{10}$$

$$(10) \quad -2x + 5 = -9x + 9$$

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$$(13) \quad -x - 9 = -5x + 2$$

$$x = \frac{11}{4}$$

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$$x = \frac{14}{9}$$

$$(44) \quad -3x - 5 = 6x - 3$$

$$x = -\frac{2}{9}$$

$$(45) \quad 9x + 6 = -7x + 2$$

$$x = -\frac{1}{4}$$

$$(1) \quad 3x - 5 = -x - 1$$

$$x = 1$$

$$(2) \quad -5x - 1 = -6x + 9$$

$$x = 10$$

$$(3) \quad -9x + 4 = -6x + 1$$

$$x = 1$$

$$(4) \quad 9x + 8 = 8x + 7$$

$$x = -1$$

$$(5) \quad 9x + 8 = 6x - 4$$

$$x = -4$$

$$(6) \quad -x - 6 = x + 9$$

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$$(4) \quad -4x - 4 = -5x - 3$$

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$$(5) \quad -5x + 4 = x - 5$$

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$$(9) \quad -3x - 3 = x + 3$$

$$x = -\frac{3}{2}$$

$$(10) \quad -6x - 4 = -4x - 6$$

$$x = 1$$

$$(11) \quad -9x - 5 = x + 5$$

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$$(2) \quad 4x + 5 = -6x + 9$$

$$x = \frac{2}{5}$$

$$(3) \quad -8x - 2 = -9x - 5$$

$$x = -3$$

$$(4) \quad 7x - 5 = 4x + 5$$

$$x = \frac{10}{3}$$

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$$(4) \quad 8x + 1 = 2x - 7$$

$$x = -\frac{4}{3}$$

$$(5) \quad -5x + 7 = -3x + 6$$

$$x = \frac{1}{2}$$

$$(6) \quad 9x + 9 = 2x + 7$$

$$x = -\frac{2}{7}$$

$$(7) \quad 6x + 8 = -5x - 9$$

$$x = -\frac{17}{11}$$

$$(8) \quad 2x + 3 = -6x + 6$$

$$x = \frac{3}{8}$$

$$(9) \quad 3x + 4 = 7x + 8$$

$$x = -1$$

$$(10) \quad -9x + 8 = 3x - 1$$

$$x = \frac{3}{4}$$

$$(11) \quad -2x + 1 = -7x + 3$$

$$x = \frac{2}{5}$$

$$(12) \quad -6x - 4 = 2x + 9$$

$$x = -\frac{13}{8}$$

$$(13) \quad 8x - 7 = 4x - 5$$

$$x = \frac{1}{2}$$

$$(14) \quad 3x + 4 = 4x + 9$$

$$x = -5$$

$$(15) \quad -7x - 5 = -6x - 4$$

$$x = -1$$

$$(16) \quad -9x + 5 = -5x - 2$$

$$x = \frac{7}{4}$$

$$(17) \quad 2x + 1 = x - 7$$

$$x = -8$$

$$(18) \quad 5x - 6 = 6x - 4$$

$$x = -2$$

$$(19) \quad 2x + 6 = -7x - 3$$

$$x = -1$$

$$(20) \quad 7x - 7 = 4x + 8$$

$$x = 5$$

$$(21) \quad 2x - 2 = -8x + 2$$

$$x = \frac{2}{5}$$

$$(22) \quad -x + 7 = 7x + 9$$

$$x = -\frac{1}{4}$$

$$(23) \quad -9x - 3 = 3x + 6$$

$$x = -\frac{3}{4}$$

$$(24) \quad -x + 4 = 4x + 5$$

$$x = -\frac{1}{5}$$

$$(25) \quad -4x + 8 = -3x - 8$$

$$x = 16$$

$$(26) \quad 5x + 5 = 7x + 6$$

$$x = -\frac{1}{2}$$

$$(27) \quad -5x + 8 = -8x - 7$$

$$x = -5$$

$$(28) \quad -3x + 5 = 8x + 3$$

$$x = \frac{2}{11}$$

$$(29) \quad 5x - 6 = -2x - 2$$

$$x = \frac{4}{7}$$

$$(30) \quad 8x - 2 = -2x - 7$$

$$x = -\frac{1}{2}$$

$$(31) \quad 6x - 4 = 5x + 6$$

$$x = 10$$

$$(32) \quad -9x + 3 = -3x + 8$$

$$x = -\frac{5}{6}$$

$$(33) \quad 2x - 9 = x + 2$$

$$x = 11$$

$$(34) \quad 5x - 4 = 4x + 1$$

$$x = 5$$

$$(35) \quad 4x - 3 = -4x - 9$$

$$x = -\frac{3}{4}$$

$$(36) \quad -x - 2 = -3x - 6$$

$$x = -2$$

$$(37) \quad -7x - 2 = 6x - 9$$

$$x = \frac{7}{13}$$

$$(38) \quad 3x - 5 = 9x + 5$$

$$x = -\frac{5}{3}$$

$$(39) \quad -8x + 8 = 9x + 5$$

$$x = \frac{3}{17}$$

$$(40) \quad 9x + 3 = 3x + 9$$

$$x = 1$$

$$(41) \quad -5x + 6 = 4x + 5$$

$$x = \frac{1}{9}$$

$$(42) \quad 5x + 5 = -5x + 3$$

$$x = -\frac{1}{5}$$

$$(43) \quad 5x + 5 = 2x - 8$$

$$x = -\frac{13}{3}$$

$$(44) \quad -9x - 4 = -6x - 8$$

$$x = \frac{4}{3}$$

$$(45) \quad 9x - 5 = -7x + 2$$

$$x = \frac{7}{16}$$

$$(1) \quad -2x + 7 = -9x - 3 \\ x = -\frac{10}{7}$$

$$(2) \quad 9x - 5 = x + 8 \\ x = \frac{13}{8}$$

$$(3) \quad 4x + 6 = -5x - 6 \\ x = -\frac{4}{3}$$

$$(4) \quad -8x - 7 = -6x + 1 \\ x = -4$$

$$(5) \quad 5x + 2 = -9x - 8 \\ x = -\frac{5}{7}$$

$$(6) \quad -5x + 3 = -x - 2 \\ x = \frac{5}{4}$$

$$(7) \quad 9x - 2 = -7x + 6 \\ x = \frac{1}{2}$$

$$(8) \quad -2x - 3 = 7x + 4 \\ x = -\frac{7}{9}$$

$$(9) \quad 5x + 6 = 2x + 8 \\ x = \frac{2}{3}$$

$$(10) \quad x - 5 = -x - 1 \\ x = 2$$

$$(11) \quad -x - 3 = -2x - 3 \\ x = 0$$

$$(12) \quad -7x - 4 = 8x - 8 \\ x = \frac{4}{15}$$

$$(13) \quad -9x - 9 = -6x - 9 \\ x = 0$$

$$(14) \quad 8x + 4 = -2x + 7 \\ x = \frac{3}{10}$$

$$(15) \quad 2x - 4 = x - 6 \\ x = -2$$

$$(16) \quad 6x + 9 = 3x - 5 \\ x = -\frac{14}{3}$$

$$(17) \quad -7x - 5 = 7x - 3 \\ x = -\frac{1}{7}$$

$$(18) \quad 5x + 5 = 3x - 2 \\ x = -\frac{7}{2}$$

$$(19) \quad -3x - 8 = -6x - 5 \\ x = 1$$

$$(20) \quad -9x - 2 = 3x + 7 \\ x = -\frac{3}{4}$$

$$(21) \quad -8x - 1 = 6x - 9 \\ x = \frac{4}{7}$$

$$(22) \quad x + 9 = -7x + 3 \\ x = -\frac{3}{4}$$

$$(23) \quad 4x + 7 = -9x + 2 \\ x = -\frac{5}{13}$$

$$(24) \quad 3x - 9 = x + 3 \\ x = 6$$

$$(25) \quad 2x - 2 = -x - 6 \\ x = -\frac{4}{3}$$

$$(26) \quad 4x + 4 = 7x - 3 \\ x = \frac{7}{3}$$

$$(27) \quad 3x - 3 = -2x + 6 \\ x = \frac{9}{5}$$

$$(28) \quad x - 1 = -4x - 4 \\ x = -\frac{3}{5}$$

$$(29) \quad -6x + 5 = 3x + 3 \\ x = \frac{2}{9}$$

$$(30) \quad x - 7 = -x - 1 \\ x = 3$$

$$(31) \quad x + 1 = 5x - 7 \\ x = 2$$

$$(32) \quad 9x - 5 = x - 2 \\ x = \frac{3}{8}$$

$$(33) \quad -6x - 5 = -2x - 9 \\ x = 1$$

$$(34) \quad 2x + 5 = -4x - 2 \\ x = -\frac{7}{6}$$

$$(35) \quad -3x + 7 = 9x - 6 \\ x = \frac{13}{12}$$

$$(36) \quad -6x + 9 = -7x + 7 \\ x = -2$$

$$(37) \quad 2x + 2 = -5x - 5 \\ x = -1$$

$$(38) \quad -x - 9 = 2x + 1 \\ x = -\frac{10}{3}$$

$$(39) \quad -5x + 6 = -6x - 4 \\ x = -10$$

$$(40) \quad 6x - 3 = 9x + 7 \\ x = -\frac{10}{3}$$

$$(41) \quad x + 6 = 4x - 4 \\ x = \frac{10}{3}$$

$$(42) \quad 6x + 4 = -9x + 9 \\ x = \frac{1}{3}$$

$$(43) \quad -5x - 9 = -8x + 1 \\ x = \frac{10}{3}$$

$$(44) \quad 4x + 4 = 7x + 3 \\ x = \frac{1}{3}$$

$$(45) \quad -2x + 6 = x - 6 \\ x = 4$$

$$(1) \quad x + 5 = -4x + 2$$

$$x = -\frac{3}{5}$$

$$(2) \quad 2x - 9 = -5x + 5$$

$$x = 2$$

$$(3) \quad -8x - 6 = 2x + 4$$

$$x = -1$$

$$(4) \quad 7x + 1 = 4x - 1$$

$$x = -\frac{2}{3}$$

$$(5) \quad 8x + 6 = x - 6$$

$$x = -\frac{12}{7}$$

$$(6) \quad 5x + 2 = -2x + 7$$

$$x = \frac{5}{7}$$

$$(7) \quad 6x - 1 = 5x - 4$$

$$x = -3$$

$$(8) \quad -2x + 7 = -7x + 5$$

$$x = -\frac{2}{5}$$

$$(9) \quad 4x + 9 = -9x - 9$$

$$x = -\frac{18}{13}$$

$$(10) \quad 2x - 3 = 3x + 1$$

$$x = -4$$

$$(11) \quad -8x - 1 = -9x - 8$$

$$x = -7$$

$$(12) \quad 6x + 8 = 4x - 6$$

$$x = -7$$

$$(13) \quad 9x - 6 = -4x + 4$$

$$x = \frac{10}{13}$$

$$(14) \quad -4x - 9 = 2x - 3$$

$$x = -1$$

$$(15) \quad 5x + 5 = 2x + 9$$

$$x = \frac{4}{3}$$

$$(16) \quad 7x - 3 = 5x + 2$$

$$x = \frac{5}{2}$$

$$(17) \quad 4x - 7 = -9x - 8$$

$$x = -\frac{1}{13}$$

$$(18) \quad -7x - 8 = 7x + 7$$

$$x = -\frac{15}{14}$$

$$(19) \quad -x - 6 = 3x + 8$$

$$x = -\frac{7}{2}$$

$$(20) \quad 9x + 1 = 5x - 9$$

$$x = -\frac{5}{2}$$

$$(21) \quad 4x + 9 = -3x + 8$$

$$x = -\frac{1}{7}$$

$$(22) \quad 6x - 2 = -4x - 8$$

$$x = -\frac{3}{5}$$

$$(23) \quad -4x - 2 = -8x - 3$$

$$x = -\frac{1}{4}$$

$$(24) \quad 4x - 1 = 6x + 6$$

$$x = -\frac{7}{2}$$

$$(25) \quad 5x - 7 = 2x - 8$$

$$x = -\frac{1}{3}$$

$$(26) \quad -3x + 4 = 8x - 1$$

$$x = \frac{5}{11}$$

$$(27) \quad 8x - 4 = 5x + 7$$

$$x = \frac{11}{3}$$

$$(28) \quad -6x + 9 = -5x - 1$$

$$x = 10$$

$$(29) \quad 2x - 6 = -2x + 2$$

$$x = 2$$

$$(30) \quad 8x + 4 = -8x - 1$$

$$x = -\frac{5}{16}$$

$$(31) \quad 8x + 6 = 5x + 1$$

$$x = -\frac{5}{3}$$

$$(32) \quad 5x - 4 = -3x - 8$$

$$x = -\frac{1}{2}$$

$$(33) \quad -9x - 8 = -2x - 7$$

$$x = -\frac{1}{7}$$

$$(34) \quad -4x + 2 = 4x - 3$$

$$x = \frac{5}{8}$$

$$(35) \quad -x + 9 = x + 8$$

$$x = \frac{1}{2}$$

$$(36) \quad 9x - 5 = -x + 7$$

$$x = \frac{6}{5}$$

$$(37) \quad -4x - 2 = -7x - 6$$

$$x = -\frac{4}{3}$$

$$(38) \quad -6x + 8 = -9x - 6$$

$$x = -\frac{14}{3}$$

$$(39) \quad 6x + 6 = 9x - 1$$

$$x = \frac{7}{3}$$

$$(40) \quad -8x + 4 = 7x - 9$$

$$x = \frac{13}{15}$$

$$(41) \quad -3x - 5 = 6x + 3$$

$$x = -\frac{8}{9}$$

$$(42) \quad -7x + 6 = 4x + 9$$

$$x = -\frac{3}{11}$$

$$(43) \quad 7x + 9 = 8x + 4$$

$$x = 5$$

$$(44) \quad 9x + 9 = 5x + 2$$

$$x = -\frac{7}{4}$$

$$(45) \quad 6x - 3 = -8x - 5$$

$$x = -\frac{1}{7}$$

$$(1) \quad 6x + 5 = x - 8$$

$$x = -\frac{13}{5}$$

$$(2) \quad 4x + 5 = -7x - 4$$

$$x = -\frac{9}{11}$$

$$(3) \quad 3x - 8 = 6x + 9$$

$$x = -\frac{17}{3}$$

$$(4) \quad 6x + 8 = -8x + 1$$

$$x = -\frac{1}{2}$$

$$(5) \quad 6x - 5 = 8x - 4$$

$$x = -\frac{1}{2}$$

$$(6) \quad -x + 5 = x - 3$$

$$x = 4$$

$$(7) \quad 2x + 2 = -7x + 2$$

$$x = 0$$

$$(8) \quad -9x + 3 = -2x - 9$$

$$x = \frac{12}{7}$$

$$(9) \quad 7x - 1 = -8x - 9$$

$$x = -\frac{8}{15}$$

$$(10) \quad -4x - 6 = -9x + 2$$

$$x = \frac{8}{5}$$

$$(11) \quad -2x + 8 = 6x - 2$$

$$x = \frac{5}{4}$$

$$(12) \quad -7x + 3 = x - 4$$

$$x = \frac{7}{8}$$

$$(13) \quad -3x - 1 = -2x + 6$$

$$x = -7$$

$$(14) \quad 9x - 4 = -x - 6$$

$$x = -\frac{1}{5}$$

$$(15) \quad -6x + 5 = 5x - 2$$

$$x = \frac{7}{11}$$

$$(16) \quad 8x - 3 = -2x - 3$$

$$x = 0$$

$$(17) \quad -8x + 5 = -5x + 3$$

$$x = \frac{2}{3}$$

$$(18) \quad -2x - 1 = -x + 9$$

$$x = -10$$

$$(19) \quad 5x - 4 = 8x - 3$$

$$x = -\frac{1}{3}$$

$$(20) \quad 3x - 5 = 8x - 6$$

$$x = \frac{1}{5}$$

$$(21) \quad -9x - 4 = 6x - 5$$

$$x = \frac{1}{15}$$

$$(22) \quad 6x + 1 = -7x - 5$$

$$x = -\frac{6}{13}$$

$$(23) \quad -3x - 4 = 3x - 1$$

$$x = -\frac{1}{2}$$

$$(24) \quad 3x + 2 = 2x + 5$$

$$x = 3$$

$$(25) \quad 6x - 9 = 3x - 9$$

$$x = 0$$

$$(26) \quad -9x - 3 = -x - 9$$

$$x = \frac{3}{4}$$

$$(27) \quad -4x - 3 = 9x - 9$$

$$x = \frac{6}{13}$$

$$(28) \quad 4x - 9 = -9x - 9$$

$$x = 0$$

$$(29) \quad 3x + 8 = -6x + 4$$

$$x = -\frac{4}{9}$$

$$(30) \quad 7x - 3 = 4x - 1$$

$$x = \frac{2}{3}$$

$$(31) \quad 2x + 5 = x + 1$$

$$x = -4$$

$$(32) \quad -2x - 9 = x + 2$$

$$x = -\frac{11}{3}$$

$$(33) \quad 3x + 7 = -4x + 1$$

$$x = -\frac{6}{7}$$

$$(34) \quad 2x + 9 = -8x + 5$$

$$x = -\frac{2}{5}$$

$$(35) \quad -7x + 7 = -2x + 4$$

$$x = \frac{3}{5}$$

$$(36) \quad -4x + 6 = x + 3$$

$$x = \frac{3}{5}$$

$$(37) \quad 5x + 6 = 9x + 4$$

$$x = \frac{1}{2}$$

$$(38) \quad -6x + 1 = 3x + 2$$

$$x = -\frac{1}{9}$$

$$(39) \quad 3x + 9 = x + 9$$

$$x = 0$$

$$(40) \quad x + 5 = 8x - 6$$

$$x = \frac{11}{7}$$

$$(41) \quad -2x + 3 = 2x + 2$$

$$x = \frac{1}{4}$$

$$(42) \quad 4x - 6 = 8x - 3$$

$$x = -\frac{3}{4}$$

$$(43) \quad -9x + 8 = -2x - 1$$

$$x = \frac{9}{7}$$

$$(44) \quad -7x + 4 = -x - 5$$

$$x = \frac{3}{2}$$

$$(45) \quad -3x + 8 = x + 7$$

$$x = \frac{1}{4}$$

$$(1) \quad -7x - 7 = x - 5$$

$$x = -\frac{1}{4}$$

$$(2) \quad 8x - 1 = 2x - 3$$

$$x = -\frac{1}{3}$$

$$(3) \quad 6x + 7 = 2x - 3$$

$$x = -\frac{5}{2}$$

$$(4) \quad -5x + 4 = -3x + 2$$

$$x = 1$$

$$(5) \quad 9x + 8 = x - 7$$

$$x = -\frac{15}{8}$$

$$(6) \quad x - 4 = 6x - 3$$

$$x = -\frac{1}{5}$$

$$(7) \quad -3x + 5 = 4x + 5$$

$$x = 0$$

$$(8) \quad 5x + 8 = -3x + 9$$

$$x = \frac{1}{8}$$

$$(9) \quad 9x - 2 = -3x - 9$$

$$x = -\frac{7}{12}$$

$$(10) \quad 2x - 8 = -6x + 1$$

$$x = \frac{9}{8}$$

$$(11) \quad -2x + 5 = 2x + 6$$

$$x = -\frac{1}{4}$$

$$(12) \quad -2x + 9 = 2x - 1$$

$$x = \frac{5}{2}$$

$$(13) \quad x + 3 = -9x + 2$$

$$x = -\frac{1}{10}$$

$$(14) \quad 7x + 5 = -2x - 9$$

$$x = -\frac{14}{9}$$

$$(15) \quad -x - 7 = -6x + 3$$

$$x = 2$$

$$(16) \quad -3x + 7 = 5x - 7$$

$$x = \frac{7}{4}$$

$$(17) \quad 6x + 7 = -2x + 3$$

$$x = -\frac{1}{2}$$

$$(18) \quad 7x + 8 = -3x - 1$$

$$x = -\frac{9}{10}$$

$$(19) \quad 8x + 6 = x - 3$$

$$x = -\frac{9}{7}$$

$$(20) \quad -8x - 5 = 2x - 3$$

$$x = -\frac{1}{5}$$

$$(21) \quad 3x - 8 = 6x + 7$$

$$x = -5$$

$$(22) \quad -5x + 6 = -6x + 6$$

$$x = 0$$

$$(23) \quad 7x - 3 = 6x + 7$$

$$x = 10$$

$$(24) \quad 5x - 3 = 9x - 4$$

$$x = \frac{1}{4}$$

$$(25) \quad -7x - 1 = 5x - 6$$

$$x = \frac{5}{12}$$

$$(26) \quad -8x + 8 = 2x - 5$$

$$x = \frac{13}{10}$$

$$(27) \quad 2x - 6 = 8x - 9$$

$$x = \frac{1}{2}$$

$$(28) \quad 2x + 3 = 3x + 8$$

$$x = -5$$

$$(29) \quad -3x + 1 = -4x - 3$$

$$x = -4$$

$$(30) \quad -6x + 4 = -3x - 7$$

$$x = \frac{11}{3}$$

$$(31) \quad 7x + 9 = 8x - 4$$

$$x = 13$$

$$(32) \quad 3x - 9 = -6x + 5$$

$$x = \frac{14}{9}$$

$$(33) \quad 9x - 1 = -4x + 1$$

$$x = \frac{2}{13}$$

$$(34) \quad 7x - 3 = -2x - 9$$

$$x = -\frac{2}{3}$$

$$(35) \quad -9x - 5 = -8x + 2$$

$$x = -7$$

$$(36) \quad -5x + 6 = 5x - 3$$

$$x = \frac{9}{10}$$

$$(37) \quad 5x + 9 = 6x - 1$$

$$x = 10$$

$$(38) \quad -2x + 8 = -9x + 9$$

$$x = \frac{1}{7}$$

$$(39) \quad -2x - 8 = -3x - 5$$

$$x = 3$$

$$(40) \quad 3x + 3 = 9x + 5$$

$$x = -\frac{1}{3}$$

$$(41) \quad -9x + 4 = 3x - 5$$

$$x = \frac{3}{4}$$

$$(42) \quad 9x + 4 = -7x - 5$$

$$x = -\frac{9}{16}$$

$$(43) \quad 5x - 6 = 3x + 9$$

$$x = \frac{15}{2}$$

$$(44) \quad -4x - 3 = 5x + 6$$

$$x = -1$$

$$(45) \quad 8x - 6 = -x - 9$$

$$x = -\frac{1}{3}$$

## 2.11 2 次方程式 因数分解利用

2 次式の因数分解を利用した方程式の問題。

(1)  $6x^2 - 7x - 3 = 0$

(11)  $6x^2 + 35x + 25 = 0$

(21)  $x^2 + x - 56 = 0$

(2)  $12x^2 + 29x + 14 = 0$

(12)  $8x^2 + 22x + 5 = 0$

(22)  $4x^2 + 5x - 9 = 0$

(3)  $45x^2 + 98x + 49 = 0$

(13)  $4x^2 + 3x - 1 = 0$

(23)  $10x^2 - 17x + 6 = 0$

(4)  $16x^2 + 66x + 35 = 0$

(14)  $15x^2 - 4x - 4 = 0$

(24)  $3x^2 - 11x + 6 = 0$

(5)  $18x^2 - 21x + 5 = 0$

(15)  $24x^2 - 14x - 3 = 0$

(25)  $24x^2 - 11x + 1 = 0$

(6)  $20x^2 - 11x - 45 = 0$

(16)  $2x^2 - 9x + 10 = 0$

(26)  $49x^2 + 21x - 54 = 0$

(7)  $5x^2 - 17x + 6 = 0$

(17)  $25x^2 - 30x - 27 = 0$

(27)  $9x^2 - 37x + 4 = 0$

(8)  $x^2 - 5x - 6 = 0$

(18)  $15x^2 - 4x - 3 = 0$

(28)  $16x^2 + 14x - 15 = 0$

(9)  $7x^2 + 13x + 6 = 0$

(19)  $16x^2 + 6x - 1 = 0$

(29)  $5x^2 - 2x - 16 = 0$

(10)  $3x^2 - 26x + 16 = 0$

(20)  $2x^2 - 15x + 18 = 0$

(30)  $4x^2 + 3x - 7 = 0$

(1)  $40x^2 + 31x - 35 = 0$

(11)  $7x^2 - 12x + 5 = 0$

(21)  $9x^2 + 10x + 1 = 0$

(2)  $30x^2 - 43x - 8 = 0$

(12)  $6x^2 - 11x + 3 = 0$

(22)  $10x^2 + 13x + 4 = 0$

(3)  $9x^2 + 46x + 5 = 0$

(13)  $5x^2 + 8x + 3 = 0$

(23)  $54x^2 - 87x + 35 = 0$

(4)  $9x^2 - 28x + 3 = 0$

(14)  $4x^2 - 7x + 3 = 0$

(24)  $18x^2 + 7x - 8 = 0$

(5)  $28x^2 - 41x - 14 = 0$

(15)  $18x^2 + 27x - 56 = 0$

(25)  $32x^2 - 36x - 5 = 0$

(6)  $27x^2 + 15x + 2 = 0$

(16)  $16x^2 + 40x + 9 = 0$

(26)  $56x^2 - 37x + 6 = 0$

(7)  $5x^2 + 19x + 12 = 0$

(17)  $2x^2 - 5x - 12 = 0$

(27)  $8x^2 + 39x + 28 = 0$

(8)  $14x^2 + 47x - 72 = 0$

(18)  $4x^2 - 11x - 20 = 0$

(28)  $8x^2 - 57x + 54 = 0$

(9)  $6x^2 + 13x - 15 = 0$

(19)  $21x^2 + 10x - 16 = 0$

(29)  $36x^2 + 85x + 9 = 0$

(10)  $14x^2 + 17x - 6 = 0$

(20)  $7x^2 + 22x + 3 = 0$

(30)  $63x^2 + 25x + 2 = 0$



(1)  $6x^2 - 5x - 21 = 0$

(11)  $3x^2 - 8x + 4 = 0$

(21)  $25x^2 - 50x + 21 = 0$

(2)  $12x^2 + 35x + 8 = 0$

(12)  $15x^2 - 7x - 36 = 0$

(22)  $45x^2 + 31x - 4 = 0$

(3)  $4x^2 - 11x + 7 = 0$

(13)  $7x^2 - 8x + 1 = 0$

(23)  $6x^2 - 11x + 3 = 0$

(4)  $21x^2 - 64x + 35 = 0$

(14)  $56x^2 + 57x + 7 = 0$

(24)  $x^2 - x - 2 = 0$

(5)  $2x^2 - 13x + 21 = 0$

(15)  $7x^2 + 27x + 18 = 0$

(25)  $8x^2 - 6x + 1 = 0$

(6)  $28x^2 + 29x - 9 = 0$

(16)  $15x^2 - 7x - 2 = 0$

(26)  $16x^2 + 26x + 9 = 0$

(7)  $28x^2 + 57x + 27 = 0$

(17)  $10x^2 - 7x - 12 = 0$

(27)  $64x^2 + 96x + 35 = 0$

(8)  $2x^2 + 17x + 8 = 0$

(18)  $x^2 - 1 = 0$

(28)  $8x^2 - 7x - 1 = 0$

(9)  $3x^2 + 4x + 1 = 0$

(19)  $64x^2 + 32x - 21 = 0$

(29)  $5x^2 - 6x + 1 = 0$

(10)  $63x^2 - 23x + 2 = 0$

(20)  $9x^2 + 29x + 6 = 0$

(30)  $3x^2 + 7x + 4 = 0$

(1)  $7x^2 - 13x - 24 = 0$

(11)  $16x^2 - 6x - 7 = 0$

(21)  $24x^2 - 47x + 20 = 0$

(2)  $x^2 + 7x + 12 = 0$

(12)  $5x^2 + 14x - 3 = 0$

(22)  $x^2 - 8x + 15 = 0$

(3)  $12x^2 + 13x + 3 = 0$

(13)  $72x^2 - 119x + 49 = 0$

(23)  $x^2 - 1 = 0$

(4)  $24x^2 - 23x + 5 = 0$

(14)  $15x^2 - 29x - 14 = 0$

(24)  $20x^2 - 43x + 21 = 0$

(5)  $6x^2 - 25x + 14 = 0$

(15)  $15x^2 - 34x + 15 = 0$

(25)  $49x^2 - 25 = 0$

(6)  $45x^2 + 17x - 14 = 0$

(16)  $6x^2 + 13x - 5 = 0$

(26)  $24x^2 + 25x + 6 = 0$

(7)  $8x^2 - 18x + 9 = 0$

(17)  $10x^2 - 33x - 54 = 0$

(27)  $x^2 - 2x - 8 = 0$

(8)  $18x^2 - 13x + 2 = 0$

(18)  $8x^2 - 27x - 20 = 0$

(28)  $7x^2 - 13x - 2 = 0$

(9)  $5x^2 + 16x + 12 = 0$

(19)  $42x^2 + 5x - 2 = 0$

(29)  $9x^2 - 24x + 7 = 0$

(10)  $2x^2 - 17x + 35 = 0$

(20)  $56x^2 + 53x + 12 = 0$

(30)  $9x^2 + 14x + 5 = 0$

(1)  $8x^2 - 5x - 3 = 0$

(11)  $3x^2 + 22x + 24 = 0$

(21)  $21x^2 + x - 2 = 0$

(2)  $24x^2 - 7x - 6 = 0$

(12)  $8x^2 + 22x + 5 = 0$

(22)  $63x^2 + 82x + 24 = 0$

(3)  $36x^2 - 11x - 12 = 0$

(13)  $2x^2 - 17x - 9 = 0$

(23)  $14x^2 - 31x - 63 = 0$

(4)  $10x^2 + 29x + 10 = 0$

(14)  $63x^2 - 68x + 12 = 0$

(24)  $5x^2 + 13x - 6 = 0$

(5)  $x^2 + 11x + 18 = 0$

(15)  $21x^2 - 2x - 8 = 0$

(25)  $14x^2 - 27x + 9 = 0$

(6)  $2x^2 + 7x - 4 = 0$

(16)  $63x^2 - 73x + 20 = 0$

(26)  $3x^2 - 14x + 8 = 0$

(7)  $4x^2 + 5x + 1 = 0$

(17)  $9x^2 + 10x + 1 = 0$

(27)  $5x^2 + 41x + 42 = 0$

(8)  $4x^2 - 8x + 3 = 0$

(18)  $x^2 - x - 6 = 0$

(28)  $24x^2 - 31x + 10 = 0$

(9)  $6x^2 + 11x + 3 = 0$

(19)  $5x^2 - 6x + 1 = 0$

(29)  $35x^2 + 58x - 9 = 0$

(10)  $12x^2 - 4x - 1 = 0$

(20)  $12x^2 - 17x - 7 = 0$

(30)  $x^2 - 8x + 12 = 0$

(1)  $21x^2 - 46x + 24 = 0$

(11)  $14x^2 - 3x - 27 = 0$

(21)  $x^2 + 6x + 5 = 0$

(2)  $63x^2 - 20x - 3 = 0$

(12)  $56x^2 - 111x + 54 = 0$

(22)  $72x^2 + x - 56 = 0$

(3)  $63x^2 + 53x - 36 = 0$

(13)  $2x^2 + x - 1 = 0$

(23)  $18x^2 + 7x - 1 = 0$

(4)  $35x^2 + 38x - 9 = 0$

(14)  $4x^2 + 21x - 18 = 0$

(24)  $16x^2 - 46x - 35 = 0$

(5)  $40x^2 - 9x - 9 = 0$

(15)  $7x^2 + 12x - 4 = 0$

(25)  $2x^2 + 7x - 72 = 0$

(6)  $x^2 - 7x - 18 = 0$

(16)  $48x^2 + 50x - 7 = 0$

(26)  $8x^2 + 2x - 3 = 0$

(7)  $4x^2 - 3x - 1 = 0$

(17)  $18x^2 + 17x - 15 = 0$

(27)  $21x^2 - 4x - 32 = 0$

(8)  $4x^2 + x - 3 = 0$

(18)  $3x^2 - x - 2 = 0$

(28)  $x^2 - 6x + 8 = 0$

(9)  $64x^2 + 112x + 45 = 0$

(19)  $27x^2 - 3x - 14 = 0$

(29)  $6x^2 - 5x - 4 = 0$

(10)  $x^2 - 2x + 1 = 0$

(20)  $8x^2 - 17x + 9 = 0$

(30)  $12x^2 + 23x - 24 = 0$

(1)  $4x^2 - 15x - 4 = 0$

(11)  $42x^2 + 17x - 4 = 0$

(21)  $14x^2 - 17x - 6 = 0$

(2)  $12x^2 - 13x - 35 = 0$

(12)  $2x^2 + 3x - 5 = 0$

(22)  $3x^2 + 2x - 8 = 0$

(3)  $30x^2 + 13x - 3 = 0$

(13)  $2x^2 - 13x - 24 = 0$

(23)  $15x^2 + 32x + 9 = 0$

(4)  $72x^2 + 23x - 4 = 0$

(14)  $x^2 + 9x + 8 = 0$

(24)  $7x^2 + 11x + 4 = 0$

(5)  $x^2 - 1 = 0$

(15)  $3x^2 - 2x - 1 = 0$

(25)  $10x^2 - 21x + 8 = 0$

(6)  $9x^2 + 30x + 16 = 0$

(16)  $6x^2 + 53x + 40 = 0$

(26)  $2x^2 - 3x + 1 = 0$

(7)  $14x^2 + 37x + 24 = 0$

(17)  $56x^2 + 65x + 14 = 0$

(27)  $8x^2 - 63x - 81 = 0$

(8)  $21x^2 - 76x + 63 = 0$

(18)  $7x^2 + 20x - 32 = 0$

(28)  $5x^2 + 14x - 3 = 0$

(9)  $8x^2 + 13x + 5 = 0$

(19)  $x^2 - 5x + 6 = 0$

(29)  $72x^2 - 31x - 28 = 0$

(10)  $4x^2 - x - 14 = 0$

(20)  $35x^2 - 58x - 9 = 0$

(30)  $4x^2 + x - 18 = 0$

(1)  $12x^2 + x - 1 = 0$

(11)  $2x^2 + 7x + 6 = 0$

(21)  $21x^2 + 31x + 8 = 0$

(2)  $7x^2 + 13x + 6 = 0$

(12)  $6x^2 - 11x + 3 = 0$

(22)  $16x^2 + 34x + 15 = 0$

(3)  $x^2 + 2x - 15 = 0$

(13)  $4x^2 + 13x + 9 = 0$

(23)  $28x^2 + 43x - 45 = 0$

(4)  $3x^2 + 16x + 5 = 0$

(14)  $x^2 + 5x + 6 = 0$

(24)  $3x^2 + 4x + 1 = 0$

(5)  $3x^2 + 2x - 8 = 0$

(15)  $27x^2 - 24x + 4 = 0$

(25)  $x^2 + 6x - 7 = 0$

(6)  $6x^2 + 31x + 28 = 0$

(16)  $5x^2 + 14x + 8 = 0$

(26)  $3x^2 + 7x - 6 = 0$

(7)  $2x^2 + 3x - 2 = 0$

(17)  $8x^2 + 6x - 5 = 0$

(27)  $28x^2 + 11x + 1 = 0$

(8)  $25x^2 + 35x + 6 = 0$

(18)  $3x^2 + 13x + 4 = 0$

(28)  $8x^2 - 34x + 21 = 0$

(9)  $72x^2 + 79x + 14 = 0$

(19)  $18x^2 + 11x - 24 = 0$

(29)  $56x^2 + 57x - 8 = 0$

(10)  $7x^2 + 66x + 27 = 0$

(20)  $2x^2 + 7x + 5 = 0$

(30)  $9x^2 - 4 = 0$

(1)  $2x^2 + 5x + 3 = 0$

(11)  $6x^2 - 17x + 12 = 0$

(21)  $18x^2 - 37x + 15 = 0$

(2)  $8x^2 + 22x - 21 = 0$

(12)  $3x^2 + 20x + 12 = 0$

(22)  $9x^2 + 18x + 5 = 0$

(3)  $6x^2 + 13x + 7 = 0$

(13)  $7x^2 - 11x - 6 = 0$

(23)  $8x^2 + 59x + 21 = 0$

(4)  $14x^2 - 55x + 21 = 0$

(14)  $8x^2 - 10x - 7 = 0$

(24)  $x^2 - 2x + 1 = 0$

(5)  $21x^2 - 11x - 6 = 0$

(15)  $8x^2 - 34x - 9 = 0$

(25)  $15x^2 - 4x - 3 = 0$

(6)  $14x^2 + 9x - 18 = 0$

(16)  $49x^2 + 28x + 3 = 0$

(26)  $2x^2 - 11x - 21 = 0$

(7)  $24x^2 - 13x - 7 = 0$

(17)  $63x^2 - 31x - 10 = 0$

(27)  $14x^2 + 79x + 72 = 0$

(8)  $5x^2 - 21x + 18 = 0$

(18)  $10x^2 + 29x + 21 = 0$

(28)  $3x^2 - 31x + 36 = 0$

(9)  $36x^2 + 37x - 10 = 0$

(19)  $72x^2 - 11x - 6 = 0$

(29)  $8x^2 + x - 7 = 0$

(10)  $16x^2 - 30x + 9 = 0$

(20)  $10x^2 + 9x + 2 = 0$

(30)  $9x^2 - 13x + 4 = 0$

(1)  $2x^2 - 3x - 9 = 0$

(11)  $x^2 - 2x - 8 = 0$

(21)  $x^2 - 7x + 6 = 0$

(2)  $x^2 + 2x + 1 = 0$

(12)  $28x^2 + 3x - 18 = 0$

(22)  $20x^2 - 7x - 6 = 0$

(3)  $12x^2 - 25x + 7 = 0$

(13)  $45x^2 - 79x + 30 = 0$

(23)  $9x^2 - 15x - 14 = 0$

(4)  $63x^2 + 10x - 25 = 0$

(14)  $54x^2 + 15x - 4 = 0$

(24)  $5x^2 + 8x - 21 = 0$

(5)  $x^2 - 6x + 9 = 0$

(15)  $3x^2 - 2x - 1 = 0$

(25)  $7x^2 - 9x + 2 = 0$

(6)  $36x^2 - 29x + 5 = 0$

(16)  $3x^2 - 2x - 1 = 0$

(26)  $7x^2 + 10x + 3 = 0$

(7)  $9x^2 + 6x - 8 = 0$

(17)  $2x^2 - 5x + 3 = 0$

(27)  $32x^2 - 100x + 63 = 0$

(8)  $7x^2 - 6x - 16 = 0$

(18)  $2x^2 + 11x - 21 = 0$

(28)  $12x^2 + x - 20 = 0$

(9)  $2x^2 - 3x + 1 = 0$

(19)  $7x^2 - 8x + 1 = 0$

(29)  $10x^2 + 33x - 7 = 0$

(10)  $8x^2 - 31x + 21 = 0$

(20)  $32x^2 - 28x - 9 = 0$

(30)  $14x^2 + 37x + 24 = 0$



(1)  $5x^2 - 6x - 27 = 0$

(11)  $x^2 - 9 = 0$

(21)  $9x^2 + 25x - 6 = 0$

(2)  $5x^2 - 9x + 4 = 0$

(12)  $7x^2 + 39x + 20 = 0$

(22)  $4x^2 + 15x + 14 = 0$

(3)  $2x^2 - 17x + 36 = 0$

(13)  $2x^2 - 5x + 3 = 0$

(23)  $4x^2 - 4x - 3 = 0$

(4)  $12x^2 + 37x + 21 = 0$

(14)  $40x^2 + 107x + 63 = 0$

(24)  $4x^2 - x - 3 = 0$

(5)  $3x^2 - 8x + 4 = 0$

(15)  $27x^2 + 30x + 7 = 0$

(25)  $6x^2 - 7x - 3 = 0$

(6)  $4x^2 + 15x + 14 = 0$

(16)  $21x^2 + 17x - 30 = 0$

(26)  $x^2 - 1 = 0$

(7)  $10x^2 - 7x + 1 = 0$

(17)  $32x^2 + 4x - 15 = 0$

(27)  $45x^2 + 71x + 28 = 0$

(8)  $5x^2 + 16x + 3 = 0$

(18)  $3x^2 - 4x + 1 = 0$

(28)  $2x^2 + 3x + 1 = 0$

(9)  $12x^2 + 4x - 5 = 0$

(19)  $24x^2 - 17x + 3 = 0$

(29)  $4x^2 + 33x - 27 = 0$

(10)  $10x^2 - 53x + 63 = 0$

(20)  $6x^2 + 13x - 28 = 0$

(30)  $x^2 - 4x + 3 = 0$

(1)  $4x^2 - 3x - 1 = 0$

(11)  $2x^2 - 7x + 6 = 0$

(21)  $4x^2 + 11x + 7 = 0$

(2)  $x^2 + 4x + 3 = 0$

(12)  $16x^2 + 2x - 5 = 0$

(22)  $48x^2 + 22x - 15 = 0$

(3)  $4x^2 - 3x - 1 = 0$

(13)  $40x^2 - 91x + 49 = 0$

(23)  $x^2 - 7x + 12 = 0$

(4)  $28x^2 - 43x + 9 = 0$

(14)  $56x^2 + 15x - 54 = 0$

(24)  $36x^2 + 73x - 18 = 0$

(5)  $16x^2 + 2x - 3 = 0$

(15)  $20x^2 + 37x + 8 = 0$

(25)  $63x^2 - 46x + 8 = 0$

(6)  $21x^2 + 44x - 32 = 0$

(16)  $3x^2 - 8x + 5 = 0$

(26)  $5x^2 - 14x + 8 = 0$

(7)  $3x^2 + 4x + 1 = 0$

(17)  $10x^2 + 3x - 4 = 0$

(27)  $2x^2 + 3x - 2 = 0$

(8)  $x^2 - 2x - 3 = 0$

(18)  $5x^2 - 27x + 28 = 0$

(28)  $28x^2 + 29x - 35 = 0$

(9)  $2x^2 - x - 1 = 0$

(19)  $x^2 + 3x + 2 = 0$

(29)  $3x^2 + 22x - 16 = 0$

(10)  $2x^2 + 5x - 12 = 0$

(20)  $28x^2 + 45x + 18 = 0$

(30)  $9x^2 - 34x + 21 = 0$

(1)  $x^2 + 8x + 7 = 0$

(11)  $3x^2 + 4x + 1 = 0$

(21)  $18x^2 + 33x - 40 = 0$

(2)  $36x^2 + 65x - 36 = 0$

(12)  $7x^2 - 29x + 24 = 0$

(22)  $5x^2 + 13x - 6 = 0$

(3)  $32x^2 - 52x - 7 = 0$

(13)  $9x^2 - 15x - 14 = 0$

(23)  $x^2 - 3x + 2 = 0$

(4)  $7x^2 - 2x - 5 = 0$

(14)  $4x^2 + 20x + 9 = 0$

(24)  $7x^2 + 8x - 12 = 0$

(5)  $16x^2 - 18x + 5 = 0$

(15)  $5x^2 - 29x + 36 = 0$

(25)  $20x^2 - 3x - 35 = 0$

(6)  $16x^2 - 1 = 0$

(16)  $8x^2 + 59x + 21 = 0$

(26)  $56x^2 + 73x + 21 = 0$

(7)  $4x^2 + 11x + 7 = 0$

(17)  $2x^2 + 17x - 9 = 0$

(27)  $7x^2 - 25x + 12 = 0$

(8)  $24x^2 + 22x - 7 = 0$

(18)  $9x^2 - 65x - 56 = 0$

(28)  $6x^2 - 23x - 18 = 0$

(9)  $21x^2 + 17x - 8 = 0$

(19)  $12x^2 + 25x + 12 = 0$

(29)  $24x^2 + 58x + 9 = 0$

(10)  $56x^2 - 57x + 7 = 0$

(20)  $24x^2 + 26x + 5 = 0$

(30)  $32x^2 - 44x - 21 = 0$

(1)  $72x^2 + 29x - 10 = 0$

(11)  $5x^2 - 14x + 9 = 0$

(21)  $28x^2 + 61x + 21 = 0$

(2)  $8x^2 - 17x + 2 = 0$

(12)  $25x^2 - 80x + 64 = 0$

(22)  $40x^2 - 27x - 4 = 0$

(3)  $8x^2 - 9x + 1 = 0$

(13)  $7x^2 - 8x + 1 = 0$

(23)  $10x^2 - 21x + 9 = 0$

(4)  $18x^2 + 21x + 5 = 0$

(14)  $27x^2 + 69x + 14 = 0$

(24)  $6x^2 + 5x - 4 = 0$

(5)  $2x^2 + 5x + 3 = 0$

(15)  $6x^2 - 7x + 1 = 0$

(25)  $16x^2 - 50x + 25 = 0$

(6)  $21x^2 + 31x + 8 = 0$

(16)  $9x^2 - 25 = 0$

(26)  $4x^2 - 12x - 27 = 0$

(7)  $2x^2 + 5x + 2 = 0$

(17)  $40x^2 + 79x + 24 = 0$

(27)  $3x^2 - 4x - 4 = 0$

(8)  $24x^2 - x - 10 = 0$

(18)  $35x^2 - 6x - 8 = 0$

(28)  $x^2 - 1 = 0$

(9)  $35x^2 + 3x - 20 = 0$

(19)  $x^2 - x - 2 = 0$

(29)  $4x^2 + 19x + 12 = 0$

(10)  $5x^2 + 12x + 7 = 0$

(20)  $3x^2 - x - 2 = 0$

(30)  $9x^2 - 43x + 28 = 0$

(1)  $4x^2 + x - 3 = 0$

(11)  $3x^2 + 7x - 6 = 0$

(21)  $3x^2 + 11x + 6 = 0$

(2)  $7x^2 + 13x - 24 = 0$

(12)  $3x^2 + 4x - 4 = 0$

(22)  $45x^2 + 83x + 28 = 0$

(3)  $14x^2 + 13x + 3 = 0$

(13)  $2x^2 + 3x - 9 = 0$

(23)  $35x^2 + 9x - 2 = 0$

(4)  $36x^2 - 84x + 49 = 0$

(14)  $9x^2 + 23x + 10 = 0$

(24)  $12x^2 - 11x - 5 = 0$

(5)  $54x^2 - 39x - 5 = 0$

(15)  $5x^2 - 4x - 1 = 0$

(25)  $9x^2 + 16x + 7 = 0$

(6)  $28x^2 - x - 15 = 0$

(16)  $2x^2 + 5x + 3 = 0$

(26)  $45x^2 - 92x + 32 = 0$

(7)  $15x^2 - 7x - 2 = 0$

(17)  $7x^2 - 25x - 12 = 0$

(27)  $4x^2 + 17x - 15 = 0$

(8)  $2x^2 - 13x + 18 = 0$

(18)  $6x^2 + x - 7 = 0$

(28)  $7x^2 - 22x + 3 = 0$

(9)  $28x^2 + 9x - 9 = 0$

(19)  $14x^2 - 33x - 56 = 0$

(29)  $12x^2 + 41x + 24 = 0$

(10)  $3x^2 - 16x - 12 = 0$

(20)  $4x^2 + 17x + 4 = 0$

(30)  $3x^2 - 4x + 1 = 0$

(1)  $8x^2 - 39x - 54 = 0$

(11)  $3x^2 - 13x + 4 = 0$

(21)  $10x^2 - x - 24 = 0$

(2)  $8x^2 - 15x - 2 = 0$

(12)  $x^2 + 6x - 27 = 0$

(22)  $2x^2 - 3x - 2 = 0$

(3)  $x^2 - 2x + 1 = 0$

(13)  $24x^2 - 31x + 10 = 0$

(23)  $28x^2 - 3x - 18 = 0$

(4)  $7x^2 + 5x - 2 = 0$

(14)  $2x^2 - 9x - 18 = 0$

(24)  $x^2 - 3x + 2 = 0$

(5)  $2x^2 + 13x + 6 = 0$

(15)  $6x^2 - 7x - 10 = 0$

(25)  $27x^2 - 33x + 8 = 0$

(6)  $6x^2 - 23x + 7 = 0$

(16)  $3x^2 + 8x + 5 = 0$

(26)  $28x^2 + 75x + 27 = 0$

(7)  $35x^2 - 31x - 40 = 0$

(17)  $9x^2 + 74x - 63 = 0$

(27)  $4x^2 + 27x - 40 = 0$

(8)  $9x^2 - 3x - 2 = 0$

(18)  $5x^2 + 3x - 8 = 0$

(28)  $5x^2 - 36x - 81 = 0$

(9)  $7x^2 + 5x - 18 = 0$

(19)  $x^2 - 3x - 4 = 0$

(29)  $3x^2 - 8x + 4 = 0$

(10)  $20x^2 - 43x + 14 = 0$

(20)  $2x^2 + 3x - 20 = 0$

(30)  $12x^2 + 11x - 5 = 0$

(1)  $x^2 - 1 = 0$

(11)  $7x^2 + 18x + 8 = 0$

(21)  $x^2 + 7x + 12 = 0$

(2)  $x^2 + x - 2 = 0$

(12)  $12x^2 - 8x - 15 = 0$

(22)  $3x^2 + x - 2 = 0$

(3)  $7x^2 + 5x - 2 = 0$

(13)  $36x^2 - 23x - 8 = 0$

(23)  $4x^2 + 4x - 3 = 0$

(4)  $8x^2 - 2x - 1 = 0$

(14)  $12x^2 - 5x - 2 = 0$

(24)  $8x^2 + 11x - 10 = 0$

(5)  $4x^2 + 23x + 15 = 0$

(15)  $21x^2 - 13x - 18 = 0$

(25)  $8x^2 + 15x + 7 = 0$

(6)  $2x^2 + 11x + 12 = 0$

(16)  $5x^2 + 39x + 54 = 0$

(26)  $30x^2 - 41x + 7 = 0$

(7)  $4x^2 - 13x + 9 = 0$

(17)  $21x^2 - 13x - 20 = 0$

(27)  $12x^2 + 37x + 21 = 0$

(8)  $3x^2 + 22x + 24 = 0$

(18)  $x^2 - 1 = 0$

(28)  $12x^2 - 5x - 25 = 0$

(9)  $40x^2 + 73x + 30 = 0$

(19)  $15x^2 + 7x - 36 = 0$

(29)  $7x^2 + 12x + 5 = 0$

(10)  $6x^2 - 19x + 14 = 0$

(20)  $x^2 + 4x + 3 = 0$

(30)  $49x^2 + 63x + 18 = 0$

(1)  $8x^2 - 14x + 5 = 0$

(11)  $28x^2 + 99x + 81 = 0$

(21)  $2x^2 - 9x + 7 = 0$

(2)  $27x^2 - 96x + 64 = 0$

(12)  $15x^2 - 26x - 21 = 0$

(22)  $5x^2 - 21x + 18 = 0$

(3)  $16x^2 - 18x + 5 = 0$

(13)  $40x^2 + 69x + 8 = 0$

(23)  $6x^2 - 13x - 8 = 0$

(4)  $2x^2 - x - 3 = 0$

(14)  $49x^2 + 42x + 8 = 0$

(24)  $9x^2 + 12x + 4 = 0$

(5)  $9x^2 + 17x + 8 = 0$

(15)  $5x^2 + 2x - 7 = 0$

(25)  $81x^2 + 54x + 8 = 0$

(6)  $7x^2 - 65x + 18 = 0$

(16)  $27x^2 - 12x - 7 = 0$

(26)  $49x^2 - 28x - 12 = 0$

(7)  $16x^2 + 10x - 21 = 0$

(17)  $18x^2 + 31x - 35 = 0$

(27)  $12x^2 + 5x - 2 = 0$

(8)  $10x^2 - x - 3 = 0$

(18)  $3x^2 + 26x + 48 = 0$

(28)  $9x^2 + 61x - 14 = 0$

(9)  $5x^2 + 6x + 1 = 0$

(19)  $9x^2 - 16 = 0$

(29)  $24x^2 + 2x - 1 = 0$

(10)  $2x^2 - 9x + 4 = 0$

(20)  $21x^2 - 73x + 56 = 0$

(30)  $6x^2 - 11x - 7 = 0$



(1)  $x^2 - 4x + 3 = 0$

(11)  $27x^2 - 12x - 32 = 0$

(21)  $63x^2 - 40x - 7 = 0$

(2)  $8x^2 + 47x - 6 = 0$

(12)  $4x^2 - x - 3 = 0$

(22)  $9x^2 + 17x + 8 = 0$

(3)  $9x^2 - 56x - 49 = 0$

(13)  $12x^2 - 7x - 10 = 0$

(23)  $45x^2 + 46x + 8 = 0$

(4)  $x^2 - 2x + 1 = 0$

(14)  $6x^2 - 13x + 6 = 0$

(24)  $2x^2 + 7x + 5 = 0$

(5)  $72x^2 + 13x - 20 = 0$

(15)  $x^2 + 5x + 4 = 0$

(25)  $8x^2 + 2x - 1 = 0$

(6)  $4x^2 + 7x - 15 = 0$

(16)  $18x^2 - 51x + 35 = 0$

(26)  $6x^2 - 11x - 2 = 0$

(7)  $3x^2 - 5x + 2 = 0$

(17)  $5x^2 - 46x + 9 = 0$

(27)  $18x^2 - 29x + 3 = 0$

(8)  $7x^2 - 30x - 25 = 0$

(18)  $8x^2 - 10x - 7 = 0$

(28)  $4x^2 + 5x + 1 = 0$

(9)  $27x^2 + 3x - 10 = 0$

(19)  $7x^2 - 23x + 18 = 0$

(29)  $9x^2 + 44x - 5 = 0$

(10)  $20x^2 - 9x - 18 = 0$

(20)  $14x^2 + 25x + 6 = 0$

(30)  $5x^2 + 29x - 6 = 0$

(1)  $3x^2 + 5x - 12 = 0$

(11)  $25x^2 - 10x - 63 = 0$

(21)  $12x^2 + 5x - 3 = 0$

(2)  $6x^2 + 13x - 5 = 0$

(12)  $2x^2 + 15x - 8 = 0$

(22)  $x^2 + 2x - 3 = 0$

(3)  $35x^2 + 93x + 54 = 0$

(13)  $56x^2 - 55x - 9 = 0$

(23)  $42x^2 + 103x + 63 = 0$

(4)  $2x^2 - 5x - 18 = 0$

(14)  $x^2 - 10x + 16 = 0$

(24)  $2x^2 + x - 21 = 0$

(5)  $4x^2 - 23x + 28 = 0$

(15)  $9x^2 + 9x - 10 = 0$

(25)  $4x^2 - 12x + 5 = 0$

(6)  $9x^2 + 9x + 2 = 0$

(16)  $3x^2 - 29x + 18 = 0$

(26)  $16x^2 + 6x - 1 = 0$

(7)  $36x^2 - 73x + 35 = 0$

(17)  $10x^2 - x - 3 = 0$

(27)  $9x^2 + 25x + 14 = 0$

(8)  $7x^2 - 65x + 18 = 0$

(18)  $49x^2 + 7x - 20 = 0$

(28)  $2x^2 + 5x - 3 = 0$

(9)  $36x^2 + 5x - 24 = 0$

(19)  $45x^2 + 8x - 21 = 0$

(29)  $10x^2 - 11x - 8 = 0$

(10)  $4x^2 - 9x - 28 = 0$

(20)  $16x^2 - 38x + 21 = 0$

(30)  $6x^2 - 13x - 15 = 0$

$$(1) \quad 6x^2 - 7x - 3 = 0$$

$$x = -\frac{1}{3}, \frac{3}{2}$$

$$(11) \quad 6x^2 + 35x + 25 = 0$$

$$x = -5, -\frac{5}{6}$$

$$(21) \quad x^2 + x - 56 = 0$$

$$x = -8, 7$$

$$(2) \quad 12x^2 + 29x + 14 = 0$$

$$x = -\frac{2}{3}, -\frac{7}{4}$$

$$(12) \quad 8x^2 + 22x + 5 = 0$$

$$x = -\frac{1}{4}, -\frac{5}{2}$$

$$(22) \quad 4x^2 + 5x - 9 = 0$$

$$x = -\frac{9}{4}, 1$$

$$(3) \quad 45x^2 + 98x + 49 = 0$$

$$x = -\frac{7}{9}, -\frac{7}{5}$$

$$(13) \quad 4x^2 + 3x - 1 = 0$$

$$x = \frac{1}{4}, -1$$

$$(23) \quad 10x^2 - 17x + 6 = 0$$

$$x = \frac{6}{5}, \frac{1}{2}$$

$$(4) \quad 16x^2 + 66x + 35 = 0$$

$$x = -\frac{5}{8}, -\frac{7}{2}$$

$$(14) \quad 15x^2 - 4x - 4 = 0$$

$$x = \frac{2}{3}, -\frac{2}{5}$$

$$(24) \quad 3x^2 - 11x + 6 = 0$$

$$x = \frac{2}{3}, 3$$

$$(5) \quad 18x^2 - 21x + 5 = 0$$

$$x = \frac{1}{3}, \frac{5}{6}$$

$$(15) \quad 24x^2 - 14x - 3 = 0$$

$$x = -\frac{1}{6}, \frac{3}{4}$$

$$(25) \quad 24x^2 - 11x + 1 = 0$$

$$x = \frac{1}{8}, \frac{1}{3}$$

$$(6) \quad 20x^2 - 11x - 45 = 0$$

$$x = \frac{9}{5}, -\frac{5}{4}$$

$$(16) \quad 2x^2 - 9x + 10 = 0$$

$$x = 2, \frac{5}{2}$$

$$(26) \quad 49x^2 + 21x - 54 = 0$$

$$x = \frac{6}{7}, -\frac{9}{7}$$

$$(7) \quad 5x^2 - 17x + 6 = 0$$

$$x = 3, \frac{2}{5}$$

$$(17) \quad 25x^2 - 30x - 27 = 0$$

$$x = -\frac{3}{5}, \frac{9}{5}$$

$$(27) \quad 9x^2 - 37x + 4 = 0$$

$$x = \frac{1}{9}, 4$$

$$(8) \quad x^2 - 5x - 6 = 0$$

$$x = -1, 6$$

$$(18) \quad 15x^2 - 4x - 3 = 0$$

$$x = \frac{3}{5}, -\frac{1}{3}$$

$$(28) \quad 16x^2 + 14x - 15 = 0$$

$$x = \frac{5}{8}, -\frac{3}{2}$$

$$(9) \quad 7x^2 + 13x + 6 = 0$$

$$x = -\frac{6}{7}, -1$$

$$(19) \quad 16x^2 + 6x - 1 = 0$$

$$x = \frac{1}{8}, -\frac{1}{2}$$

$$(29) \quad 5x^2 - 2x - 16 = 0$$

$$x = -\frac{8}{5}, 2$$

$$(10) \quad 3x^2 - 26x + 16 = 0$$

$$x = \frac{2}{3}, 8$$

$$(20) \quad 2x^2 - 15x + 18 = 0$$

$$x = 6, \frac{3}{2}$$

$$(30) \quad 4x^2 + 3x - 7 = 0$$

$$x = 1, -\frac{7}{4}$$

$$(1) 40x^2 + 31x - 35 = 0$$

$$x = -\frac{7}{5}, \frac{5}{8}$$

$$(11) 7x^2 - 12x + 5 = 0$$

$$x = 1, \frac{5}{7}$$

$$(21) 9x^2 + 10x + 1 = 0$$

$$x = -1, -\frac{1}{9}$$

$$(2) 30x^2 - 43x - 8 = 0$$

$$x = -\frac{1}{6}, \frac{8}{5}$$

$$(12) 6x^2 - 11x + 3 = 0$$

$$x = \frac{1}{3}, \frac{3}{2}$$

$$(22) 10x^2 + 13x + 4 = 0$$

$$x = -\frac{1}{2}, -\frac{4}{5}$$

$$(3) 9x^2 + 46x + 5 = 0$$

$$x = -\frac{1}{9}, -5$$

$$(13) 5x^2 + 8x + 3 = 0$$

$$x = -\frac{3}{5}, -1$$

$$(23) 54x^2 - 87x + 35 = 0$$

$$x = \frac{5}{6}, \frac{7}{9}$$

$$(4) 9x^2 - 28x + 3 = 0$$

$$x = 3, \frac{1}{9}$$

$$(14) 4x^2 - 7x + 3 = 0$$

$$x = 1, \frac{3}{4}$$

$$(24) 18x^2 + 7x - 8 = 0$$

$$x = -\frac{8}{9}, \frac{1}{2}$$

$$(5) 28x^2 - 41x - 14 = 0$$

$$x = \frac{7}{4}, -\frac{2}{7}$$

$$(15) 18x^2 + 27x - 56 = 0$$

$$x = -\frac{8}{3}, \frac{7}{6}$$

$$(25) 32x^2 - 36x - 5 = 0$$

$$x = -\frac{1}{8}, \frac{5}{4}$$

$$(6) 27x^2 + 15x + 2 = 0$$

$$x = -\frac{2}{9}, -\frac{1}{3}$$

$$(16) 16x^2 + 40x + 9 = 0$$

$$x = -\frac{9}{4}, -\frac{1}{4}$$

$$(26) 56x^2 - 37x + 6 = 0$$

$$x = \frac{3}{8}, \frac{2}{7}$$

$$(7) 5x^2 + 19x + 12 = 0$$

$$x = -\frac{4}{5}, -3$$

$$(17) 2x^2 - 5x - 12 = 0$$

$$x = -\frac{3}{2}, 4$$

$$(27) 8x^2 + 39x + 28 = 0$$

$$x = -4, -\frac{7}{8}$$

$$(8) 14x^2 + 47x - 72 = 0$$

$$x = -\frac{9}{2}, \frac{8}{7}$$

$$(18) 4x^2 - 11x - 20 = 0$$

$$x = 4, -\frac{5}{4}$$

$$(28) 8x^2 - 57x + 54 = 0$$

$$x = \frac{9}{8}, 6$$

$$(9) 6x^2 + 13x - 15 = 0$$

$$x = -3, \frac{5}{6}$$

$$(19) 21x^2 + 10x - 16 = 0$$

$$x = \frac{2}{3}, -\frac{8}{7}$$

$$(29) 36x^2 + 85x + 9 = 0$$

$$x = -\frac{9}{4}, -\frac{1}{9}$$

$$(10) 14x^2 + 17x - 6 = 0$$

$$x = \frac{2}{7}, -\frac{3}{2}$$

$$(20) 7x^2 + 22x + 3 = 0$$

$$x = -3, -\frac{1}{7}$$

$$(30) 63x^2 + 25x + 2 = 0$$

$$x = -\frac{2}{7}, -\frac{1}{9}$$

$$(1) \quad 6x^2 - 5x - 21 = 0$$

$$x = -\frac{3}{2}, \frac{7}{3}$$

$$(11) \quad 3x^2 - 8x + 4 = 0$$

$$x = \frac{2}{3}, 2$$

$$(21) \quad 25x^2 - 50x + 21 = 0$$

$$x = \frac{7}{5}, \frac{3}{5}$$

$$(2) \quad 12x^2 + 35x + 8 = 0$$

$$x = -\frac{8}{3}, -\frac{1}{4}$$

$$(12) \quad 15x^2 - 7x - 36 = 0$$

$$x = -\frac{4}{3}, \frac{9}{5}$$

$$(22) \quad 45x^2 + 31x - 4 = 0$$

$$x = \frac{1}{9}, -\frac{4}{5}$$

$$(3) \quad 4x^2 - 11x + 7 = 0$$

$$x = \frac{7}{4}, 1$$

$$(13) \quad 7x^2 - 8x + 1 = 0$$

$$x = 1, \frac{1}{7}$$

$$(23) \quad 6x^2 - 11x + 3 = 0$$

$$x = \frac{1}{3}, \frac{3}{2}$$

$$(4) \quad 21x^2 - 64x + 35 = 0$$

$$x = \frac{5}{7}, \frac{7}{3}$$

$$(14) \quad 56x^2 + 57x + 7 = 0$$

$$x = -\frac{7}{8}, -\frac{1}{7}$$

$$(24) \quad x^2 - x - 2 = 0$$

$$x = -1, 2$$

$$(5) \quad 2x^2 - 13x + 21 = 0$$

$$x = \frac{7}{2}, 3$$

$$(15) \quad 7x^2 + 27x + 18 = 0$$

$$x = -3, -\frac{6}{7}$$

$$(25) \quad 8x^2 - 6x + 1 = 0$$

$$x = \frac{1}{2}, \frac{1}{4}$$

$$(6) \quad 28x^2 + 29x - 9 = 0$$

$$x = \frac{1}{4}, -\frac{9}{7}$$

$$(16) \quad 15x^2 - 7x - 2 = 0$$

$$x = -\frac{1}{5}, \frac{2}{3}$$

$$(26) \quad 16x^2 + 26x + 9 = 0$$

$$x = -\frac{9}{8}, -\frac{1}{2}$$

$$(7) \quad 28x^2 + 57x + 27 = 0$$

$$x = -\frac{9}{7}, -\frac{3}{4}$$

$$(17) \quad 10x^2 - 7x - 12 = 0$$

$$x = -\frac{4}{5}, \frac{3}{2}$$

$$(27) \quad 64x^2 + 96x + 35 = 0$$

$$x = -\frac{5}{8}, -\frac{7}{8}$$

$$(8) \quad 2x^2 + 17x + 8 = 0$$

$$x = -\frac{1}{2}, -8$$

$$(18) \quad x^2 - 1 = 0$$

$$x = 1, -1$$

$$(28) \quad 8x^2 - 7x - 1 = 0$$

$$x = 1, -\frac{1}{8}$$

$$(9) \quad 3x^2 + 4x + 1 = 0$$

$$x = -1, -\frac{1}{3}$$

$$(19) \quad 64x^2 + 32x - 21 = 0$$

$$x = -\frac{7}{8}, \frac{3}{8}$$

$$(29) \quad 5x^2 - 6x + 1 = 0$$

$$x = \frac{1}{5}, 1$$

$$(10) \quad 63x^2 - 23x + 2 = 0$$

$$x = \frac{1}{7}, \frac{2}{9}$$

$$(20) \quad 9x^2 + 29x + 6 = 0$$

$$x = -\frac{2}{9}, -3$$

$$(30) \quad 3x^2 + 7x + 4 = 0$$

$$x = -1, -\frac{4}{3}$$

$$(1) \quad 7x^2 - 13x - 24 = 0$$

$$x = 3, -\frac{8}{7}$$

$$(11) \quad 16x^2 - 6x - 7 = 0$$

$$x = \frac{7}{8}, -\frac{1}{2}$$

$$(21) \quad 24x^2 - 47x + 20 = 0$$

$$x = \frac{4}{3}, \frac{5}{8}$$

$$(2) \quad x^2 + 7x + 12 = 0$$

$$x = -3, -4$$

$$(12) \quad 5x^2 + 14x - 3 = 0$$

$$x = \frac{1}{5}, -3$$

$$(22) \quad x^2 - 8x + 15 = 0$$

$$x = 5, 3$$

$$(3) \quad 12x^2 + 13x + 3 = 0$$

$$x = -\frac{1}{3}, -\frac{3}{4}$$

$$(13) \quad 72x^2 - 119x + 49 = 0$$

$$x = \frac{7}{9}, \frac{7}{8}$$

$$(23) \quad x^2 - 1 = 0$$

$$x = -1, 1$$

$$(4) \quad 24x^2 - 23x + 5 = 0$$

$$x = \frac{1}{3}, \frac{5}{8}$$

$$(14) \quad 15x^2 - 29x - 14 = 0$$

$$x = \frac{7}{3}, -\frac{2}{5}$$

$$(24) \quad 20x^2 - 43x + 21 = 0$$

$$x = \frac{7}{5}, \frac{3}{4}$$

$$(5) \quad 6x^2 - 25x + 14 = 0$$

$$x = \frac{2}{3}, \frac{7}{2}$$

$$(15) \quad 15x^2 - 34x + 15 = 0$$

$$x = \frac{5}{3}, \frac{3}{5}$$

$$(25) \quad 49x^2 - 25 = 0$$

$$x = -\frac{5}{7}, \frac{5}{7}$$

$$(6) \quad 45x^2 + 17x - 14 = 0$$

$$x = \frac{2}{5}, -\frac{7}{9}$$

$$(16) \quad 6x^2 + 13x - 5 = 0$$

$$x = \frac{1}{3}, -\frac{5}{2}$$

$$(26) \quad 24x^2 + 25x + 6 = 0$$

$$x = -\frac{3}{8}, -\frac{2}{3}$$

$$(7) \quad 8x^2 - 18x + 9 = 0$$

$$x = \frac{3}{4}, \frac{3}{2}$$

$$(17) \quad 10x^2 - 33x - 54 = 0$$

$$x = \frac{9}{2}, -\frac{6}{5}$$

$$(27) \quad x^2 - 2x - 8 = 0$$

$$x = 4, -2$$

$$(8) \quad 18x^2 - 13x + 2 = 0$$

$$x = \frac{1}{2}, \frac{2}{9}$$

$$(18) \quad 8x^2 - 27x - 20 = 0$$

$$x = -\frac{5}{8}, 4$$

$$(28) \quad 7x^2 - 13x - 2 = 0$$

$$x = -\frac{1}{7}, 2$$

$$(9) \quad 5x^2 + 16x + 12 = 0$$

$$x = -2, -\frac{6}{5}$$

$$(19) \quad 42x^2 + 5x - 2 = 0$$

$$x = -\frac{2}{7}, \frac{1}{6}$$

$$(29) \quad 9x^2 - 24x + 7 = 0$$

$$x = \frac{7}{3}, \frac{1}{3}$$

$$(10) \quad 2x^2 - 17x + 35 = 0$$

$$x = \frac{7}{2}, 5$$

$$(20) \quad 56x^2 + 53x + 12 = 0$$

$$x = -\frac{3}{8}, -\frac{4}{7}$$

$$(30) \quad 9x^2 + 14x + 5 = 0$$

$$x = -1, -\frac{5}{9}$$

$$(1) \quad 8x^2 - 5x - 3 = 0$$

$$x = 1, -\frac{3}{8}$$

$$(11) \quad 3x^2 + 22x + 24 = 0$$

$$x = -6, -\frac{4}{3}$$

$$(21) \quad 21x^2 + x - 2 = 0$$

$$x = \frac{2}{7}, -\frac{1}{3}$$

$$(2) \quad 24x^2 - 7x - 6 = 0$$

$$x = -\frac{3}{8}, \frac{2}{3}$$

$$(12) \quad 8x^2 + 22x + 5 = 0$$

$$x = -\frac{1}{4}, -\frac{5}{2}$$

$$(22) \quad 63x^2 + 82x + 24 = 0$$

$$x = -\frac{6}{7}, -\frac{4}{9}$$

$$(3) \quad 36x^2 - 11x - 12 = 0$$

$$x = -\frac{4}{9}, \frac{3}{4}$$

$$(13) \quad 2x^2 - 17x - 9 = 0$$

$$x = 9, -\frac{1}{2}$$

$$(23) \quad 14x^2 - 31x - 63 = 0$$

$$x = -\frac{9}{7}, \frac{7}{2}$$

$$(4) \quad 10x^2 + 29x + 10 = 0$$

$$x = -\frac{2}{5}, -\frac{5}{2}$$

$$(14) \quad 63x^2 - 68x + 12 = 0$$

$$x = \frac{2}{9}, \frac{6}{7}$$

$$(24) \quad 5x^2 + 13x - 6 = 0$$

$$x = -3, \frac{2}{5}$$

$$(5) \quad x^2 + 11x + 18 = 0$$

$$x = -9, -2$$

$$(15) \quad 21x^2 - 2x - 8 = 0$$

$$x = -\frac{4}{7}, \frac{2}{3}$$

$$(25) \quad 14x^2 - 27x + 9 = 0$$

$$x = \frac{3}{2}, \frac{3}{7}$$

$$(6) \quad 2x^2 + 7x - 4 = 0$$

$$x = -4, \frac{1}{2}$$

$$(16) \quad 63x^2 - 73x + 20 = 0$$

$$x = \frac{5}{7}, \frac{4}{9}$$

$$(26) \quad 3x^2 - 14x + 8 = 0$$

$$x = \frac{2}{3}, 4$$

$$(7) \quad 4x^2 + 5x + 1 = 0$$

$$x = -\frac{1}{4}, -1$$

$$(17) \quad 9x^2 + 10x + 1 = 0$$

$$x = -1, -\frac{1}{9}$$

$$(27) \quad 5x^2 + 41x + 42 = 0$$

$$x = -\frac{6}{5}, -7$$

$$(8) \quad 4x^2 - 8x + 3 = 0$$

$$x = \frac{1}{2}, \frac{3}{2}$$

$$(18) \quad x^2 - x - 6 = 0$$

$$x = -2, 3$$

$$(28) \quad 24x^2 - 31x + 10 = 0$$

$$x = \frac{5}{8}, \frac{2}{3}$$

$$(9) \quad 6x^2 + 11x + 3 = 0$$

$$x = -\frac{3}{2}, -\frac{1}{3}$$

$$(19) \quad 5x^2 - 6x + 1 = 0$$

$$x = 1, \frac{1}{5}$$

$$(29) \quad 35x^2 + 58x - 9 = 0$$

$$x = -\frac{9}{5}, \frac{1}{7}$$

$$(10) \quad 12x^2 - 4x - 1 = 0$$

$$x = \frac{1}{2}, -\frac{1}{6}$$

$$(20) \quad 12x^2 - 17x - 7 = 0$$

$$x = -\frac{1}{3}, \frac{7}{4}$$

$$(30) \quad x^2 - 8x + 12 = 0$$

$$x = 6, 2$$

$$(1) \quad 21x^2 - 46x + 24 = 0$$
$$x = \frac{6}{7}, \frac{4}{3}$$

$$(11) \quad 14x^2 - 3x - 27 = 0$$
$$x = -\frac{9}{7}, \frac{3}{2}$$

$$(21) \quad x^2 + 6x + 5 = 0$$
$$x = -1, -5$$

$$(2) \quad 63x^2 - 20x - 3 = 0$$
$$x = -\frac{1}{9}, \frac{3}{7}$$

$$(12) \quad 56x^2 - 111x + 54 = 0$$
$$x = \frac{9}{8}, \frac{6}{7}$$

$$(22) \quad 72x^2 + x - 56 = 0$$
$$x = -\frac{8}{9}, \frac{7}{8}$$

$$(3) \quad 63x^2 + 53x - 36 = 0$$
$$x = \frac{4}{9}, -\frac{9}{7}$$

$$(13) \quad 2x^2 + x - 1 = 0$$
$$x = \frac{1}{2}, -1$$

$$(23) \quad 18x^2 + 7x - 1 = 0$$
$$x = -\frac{1}{2}, \frac{1}{9}$$

$$(4) \quad 35x^2 + 38x - 9 = 0$$
$$x = \frac{1}{5}, -\frac{9}{7}$$

$$(14) \quad 4x^2 + 21x - 18 = 0$$
$$x = \frac{3}{4}, -6$$

$$(24) \quad 16x^2 - 46x - 35 = 0$$
$$x = -\frac{5}{8}, \frac{7}{2}$$

$$(5) \quad 40x^2 - 9x - 9 = 0$$
$$x = \frac{3}{5}, -\frac{3}{8}$$

$$(15) \quad 7x^2 + 12x - 4 = 0$$
$$x = -2, \frac{2}{7}$$

$$(25) \quad 2x^2 + 7x - 72 = 0$$
$$x = -8, \frac{9}{2}$$

$$(6) \quad x^2 - 7x - 18 = 0$$
$$x = 9, -2$$

$$(16) \quad 48x^2 + 50x - 7 = 0$$
$$x = -\frac{7}{6}, \frac{1}{8}$$

$$(26) \quad 8x^2 + 2x - 3 = 0$$
$$x = -\frac{3}{4}, \frac{1}{2}$$

$$(7) \quad 4x^2 - 3x - 1 = 0$$
$$x = -\frac{1}{4}, 1$$

$$(17) \quad 18x^2 + 17x - 15 = 0$$
$$x = \frac{5}{9}, -\frac{3}{2}$$

$$(27) \quad 21x^2 - 4x - 32 = 0$$
$$x = \frac{4}{3}, -\frac{8}{7}$$

$$(8) \quad 4x^2 + x - 3 = 0$$
$$x = -1, \frac{3}{4}$$

$$(18) \quad 3x^2 - x - 2 = 0$$
$$x = 1, -\frac{2}{3}$$

$$(28) \quad x^2 - 6x + 8 = 0$$
$$x = 2, 4$$

$$(9) \quad 64x^2 + 112x + 45 = 0$$
$$x = -\frac{9}{8}, -\frac{5}{8}$$

$$(19) \quad 27x^2 - 3x - 14 = 0$$
$$x = \frac{7}{9}, -\frac{2}{3}$$

$$(29) \quad 6x^2 - 5x - 4 = 0$$
$$x = \frac{4}{3}, -\frac{1}{2}$$

$$(10) \quad x^2 - 2x + 1 = 0$$
$$x = 1$$

$$(20) \quad 8x^2 - 17x + 9 = 0$$
$$x = \frac{9}{8}, 1$$

$$(30) \quad 12x^2 + 23x - 24 = 0$$
$$x = \frac{3}{4}, -\frac{8}{3}$$



$$(1) \quad 4x^2 - 15x - 4 = 0$$
$$x = -\frac{1}{4}, 4$$

$$(11) \quad 42x^2 + 17x - 4 = 0$$
$$x = -\frac{4}{7}, \frac{1}{6}$$

$$(21) \quad 14x^2 - 17x - 6 = 0$$
$$x = \frac{3}{2}, -\frac{2}{7}$$

$$(2) \quad 12x^2 - 13x - 35 = 0$$
$$x = -\frac{5}{4}, \frac{7}{3}$$

$$(12) \quad 2x^2 + 3x - 5 = 0$$
$$x = -\frac{5}{2}, 1$$

$$(22) \quad 3x^2 + 2x - 8 = 0$$
$$x = -2, \frac{4}{3}$$

$$(3) \quad 30x^2 + 13x - 3 = 0$$
$$x = \frac{1}{6}, -\frac{3}{5}$$

$$(13) \quad 2x^2 - 13x - 24 = 0$$
$$x = 8, -\frac{3}{2}$$

$$(23) \quad 15x^2 + 32x + 9 = 0$$
$$x = -\frac{9}{5}, -\frac{1}{3}$$

$$(4) \quad 72x^2 + 23x - 4 = 0$$
$$x = -\frac{4}{9}, \frac{1}{8}$$

$$(14) \quad x^2 + 9x + 8 = 0$$
$$x = -8, -1$$

$$(24) \quad 7x^2 + 11x + 4 = 0$$
$$x = -\frac{4}{7}, -1$$

$$(5) \quad x^2 - 1 = 0$$
$$x = -1, 1$$

$$(15) \quad 3x^2 - 2x - 1 = 0$$
$$x = -\frac{1}{3}, 1$$

$$(25) \quad 10x^2 - 21x + 8 = 0$$
$$x = \frac{8}{5}, \frac{1}{2}$$

$$(6) \quad 9x^2 + 30x + 16 = 0$$
$$x = -\frac{8}{3}, -\frac{2}{3}$$

$$(16) \quad 6x^2 + 53x + 40 = 0$$
$$x = -8, -\frac{5}{6}$$

$$(26) \quad 2x^2 - 3x + 1 = 0$$
$$x = 1, \frac{1}{2}$$

$$(7) \quad 14x^2 + 37x + 24 = 0$$
$$x = -\frac{8}{7}, -\frac{3}{2}$$

$$(17) \quad 56x^2 + 65x + 14 = 0$$
$$x = -\frac{7}{8}, -\frac{2}{7}$$

$$(27) \quad 8x^2 - 63x - 81 = 0$$
$$x = 9, -\frac{9}{8}$$

$$(8) \quad 21x^2 - 76x + 63 = 0$$
$$x = \frac{9}{7}, \frac{7}{3}$$

$$(18) \quad 7x^2 + 20x - 32 = 0$$
$$x = \frac{8}{7}, -4$$

$$(28) \quad 5x^2 + 14x - 3 = 0$$
$$x = \frac{1}{5}, -3$$

$$(9) \quad 8x^2 + 13x + 5 = 0$$
$$x = -1, -\frac{5}{8}$$

$$(19) \quad x^2 - 5x + 6 = 0$$
$$x = 2, 3$$

$$(29) \quad 72x^2 - 31x - 28 = 0$$
$$x = \frac{7}{8}, -\frac{4}{9}$$

$$(10) \quad 4x^2 - x - 14 = 0$$
$$x = 2, -\frac{7}{4}$$

$$(20) \quad 35x^2 - 58x - 9 = 0$$
$$x = \frac{9}{5}, -\frac{1}{7}$$

$$(30) \quad 4x^2 + x - 18 = 0$$
$$x = 2, -\frac{9}{4}$$

$$(1) \quad 12x^2 + x - 1 = 0$$

$$x = \frac{1}{4}, -\frac{1}{3}$$

$$(11) \quad 2x^2 + 7x + 6 = 0$$

$$x = -2, -\frac{3}{2}$$

$$(21) \quad 21x^2 + 31x + 8 = 0$$

$$x = -\frac{1}{3}, -\frac{8}{7}$$

$$(2) \quad 7x^2 + 13x + 6 = 0$$

$$x = -1, -\frac{6}{7}$$

$$(12) \quad 6x^2 - 11x + 3 = 0$$

$$x = \frac{3}{2}, \frac{1}{3}$$

$$(22) \quad 16x^2 + 34x + 15 = 0$$

$$x = -\frac{3}{2}, -\frac{5}{8}$$

$$(3) \quad x^2 + 2x - 15 = 0$$

$$x = -5, 3$$

$$(13) \quad 4x^2 + 13x + 9 = 0$$

$$x = -1, -\frac{9}{4}$$

$$(23) \quad 28x^2 + 43x - 45 = 0$$

$$x = \frac{5}{7}, -\frac{9}{4}$$

$$(4) \quad 3x^2 + 16x + 5 = 0$$

$$x = -\frac{1}{3}, -5$$

$$(14) \quad x^2 + 5x + 6 = 0$$

$$x = -3, -2$$

$$(24) \quad 3x^2 + 4x + 1 = 0$$

$$x = -\frac{1}{3}, -1$$

$$(5) \quad 3x^2 + 2x - 8 = 0$$

$$x = \frac{4}{3}, -2$$

$$(15) \quad 27x^2 - 24x + 4 = 0$$

$$x = \frac{2}{9}, \frac{2}{3}$$

$$(25) \quad x^2 + 6x - 7 = 0$$

$$x = 1, -7$$

$$(6) \quad 6x^2 + 31x + 28 = 0$$

$$x = -4, -\frac{7}{6}$$

$$(16) \quad 5x^2 + 14x + 8 = 0$$

$$x = -2, -\frac{4}{5}$$

$$(26) \quad 3x^2 + 7x - 6 = 0$$

$$x = -3, \frac{2}{3}$$

$$(7) \quad 2x^2 + 3x - 2 = 0$$

$$x = \frac{1}{2}, -2$$

$$(17) \quad 8x^2 + 6x - 5 = 0$$

$$x = \frac{1}{2}, -\frac{5}{4}$$

$$(27) \quad 28x^2 + 11x + 1 = 0$$

$$x = -\frac{1}{4}, -\frac{1}{7}$$

$$(8) \quad 25x^2 + 35x + 6 = 0$$

$$x = -\frac{6}{5}, -\frac{1}{5}$$

$$(18) \quad 3x^2 + 13x + 4 = 0$$

$$x = -4, -\frac{1}{3}$$

$$(28) \quad 8x^2 - 34x + 21 = 0$$

$$x = \frac{7}{2}, \frac{3}{4}$$

$$(9) \quad 72x^2 + 79x + 14 = 0$$

$$x = -\frac{7}{8}, -\frac{2}{9}$$

$$(19) \quad 18x^2 + 11x - 24 = 0$$

$$x = -\frac{3}{2}, \frac{8}{9}$$

$$(29) \quad 56x^2 + 57x - 8 = 0$$

$$x = -\frac{8}{7}, \frac{1}{8}$$

$$(10) \quad 7x^2 + 66x + 27 = 0$$

$$x = -\frac{3}{7}, -9$$

$$(20) \quad 2x^2 + 7x + 5 = 0$$

$$x = -1, -\frac{5}{2}$$

$$(30) \quad 9x^2 - 4 = 0$$

$$x = \frac{2}{3}, -\frac{2}{3}$$

$$(1) \quad 2x^2 + 5x + 3 = 0$$

$$x = -1, -\frac{3}{2}$$

$$(11) \quad 6x^2 - 17x + 12 = 0$$

$$x = \frac{4}{3}, \frac{3}{2}$$

$$(21) \quad 18x^2 - 37x + 15 = 0$$

$$x = \frac{3}{2}, \frac{5}{9}$$

$$(2) \quad 8x^2 + 22x - 21 = 0$$

$$x = \frac{3}{4}, -\frac{7}{2}$$

$$(12) \quad 3x^2 + 20x + 12 = 0$$

$$x = -\frac{2}{3}, -6$$

$$(22) \quad 9x^2 + 18x + 5 = 0$$

$$x = -\frac{5}{3}, -\frac{1}{3}$$

$$(3) \quad 6x^2 + 13x + 7 = 0$$

$$x = -1, -\frac{7}{6}$$

$$(13) \quad 7x^2 - 11x - 6 = 0$$

$$x = 2, -\frac{3}{7}$$

$$(23) \quad 8x^2 + 59x + 21 = 0$$

$$x = -\frac{3}{8}, -7$$

$$(4) \quad 14x^2 - 55x + 21 = 0$$

$$x = \frac{7}{2}, \frac{3}{7}$$

$$(14) \quad 8x^2 - 10x - 7 = 0$$

$$x = \frac{7}{4}, -\frac{1}{2}$$

$$(24) \quad x^2 - 2x + 1 = 0$$

$$x = 1$$

$$(5) \quad 21x^2 - 11x - 6 = 0$$

$$x = -\frac{1}{3}, \frac{6}{7}$$

$$(15) \quad 8x^2 - 34x - 9 = 0$$

$$x = \frac{9}{2}, -\frac{1}{4}$$

$$(25) \quad 15x^2 - 4x - 3 = 0$$

$$x = -\frac{1}{3}, \frac{3}{5}$$

$$(6) \quad 14x^2 + 9x - 18 = 0$$

$$x = \frac{6}{7}, -\frac{3}{2}$$

$$(16) \quad 49x^2 + 28x + 3 = 0$$

$$x = -\frac{1}{7}, -\frac{3}{7}$$

$$(26) \quad 2x^2 - 11x - 21 = 0$$

$$x = -\frac{3}{2}, 7$$

$$(7) \quad 24x^2 - 13x - 7 = 0$$

$$x = -\frac{1}{3}, \frac{7}{8}$$

$$(17) \quad 63x^2 - 31x - 10 = 0$$

$$x = \frac{5}{7}, -\frac{2}{9}$$

$$(27) \quad 14x^2 + 79x + 72 = 0$$

$$x = -\frac{9}{2}, -\frac{8}{7}$$

$$(8) \quad 5x^2 - 21x + 18 = 0$$

$$x = \frac{6}{5}, 3$$

$$(18) \quad 10x^2 + 29x + 21 = 0$$

$$x = -\frac{7}{5}, -\frac{3}{2}$$

$$(28) \quad 3x^2 - 31x + 36 = 0$$

$$x = 9, \frac{4}{3}$$

$$(9) \quad 36x^2 + 37x - 10 = 0$$

$$x = \frac{2}{9}, -\frac{5}{4}$$

$$(19) \quad 72x^2 - 11x - 6 = 0$$

$$x = -\frac{2}{9}, \frac{3}{8}$$

$$(29) \quad 8x^2 + x - 7 = 0$$

$$x = -1, \frac{7}{8}$$

$$(10) \quad 16x^2 - 30x + 9 = 0$$

$$x = \frac{3}{2}, \frac{3}{8}$$

$$(20) \quad 10x^2 + 9x + 2 = 0$$

$$x = -\frac{1}{2}, -\frac{2}{5}$$

$$(30) \quad 9x^2 - 13x + 4 = 0$$

$$x = 1, \frac{4}{9}$$

$$(1) \quad 2x^2 - 3x - 9 = 0$$

$$x = 3, -\frac{3}{2}$$

$$(11) \quad x^2 - 2x - 8 = 0$$

$$x = -2, 4$$

$$(21) \quad x^2 - 7x + 6 = 0$$

$$x = 6, 1$$

$$(2) \quad x^2 + 2x + 1 = 0$$

$$x = -1$$

$$(12) \quad 28x^2 + 3x - 18 = 0$$

$$x = -\frac{6}{7}, \frac{3}{4}$$

$$(22) \quad 20x^2 - 7x - 6 = 0$$

$$x = -\frac{2}{5}, \frac{3}{4}$$

$$(3) \quad 12x^2 - 25x + 7 = 0$$

$$x = \frac{7}{4}, \frac{1}{3}$$

$$(13) \quad 45x^2 - 79x + 30 = 0$$

$$x = \frac{5}{9}, \frac{6}{5}$$

$$(23) \quad 9x^2 - 15x - 14 = 0$$

$$x = -\frac{2}{3}, \frac{7}{3}$$

$$(4) \quad 63x^2 + 10x - 25 = 0$$

$$x = \frac{5}{9}, -\frac{5}{7}$$

$$(14) \quad 54x^2 + 15x - 4 = 0$$

$$x = \frac{1}{6}, -\frac{4}{9}$$

$$(24) \quad 5x^2 + 8x - 21 = 0$$

$$x = \frac{7}{5}, -3$$

$$(5) \quad x^2 - 6x + 9 = 0$$

$$x = 3$$

$$(15) \quad 3x^2 - 2x - 1 = 0$$

$$x = 1, -\frac{1}{3}$$

$$(25) \quad 7x^2 - 9x + 2 = 0$$

$$x = 1, \frac{2}{7}$$

$$(6) \quad 36x^2 - 29x + 5 = 0$$

$$x = \frac{5}{9}, \frac{1}{4}$$

$$(16) \quad 3x^2 - 2x - 1 = 0$$

$$x = 1, -\frac{1}{3}$$

$$(26) \quad 7x^2 + 10x + 3 = 0$$

$$x = -1, -\frac{3}{7}$$

$$(7) \quad 9x^2 + 6x - 8 = 0$$

$$x = \frac{2}{3}, -\frac{4}{3}$$

$$(17) \quad 2x^2 - 5x + 3 = 0$$

$$x = \frac{3}{2}, 1$$

$$(27) \quad 32x^2 - 100x + 63 = 0$$

$$x = \frac{9}{4}, \frac{7}{8}$$

$$(8) \quad 7x^2 - 6x - 16 = 0$$

$$x = -\frac{8}{7}, 2$$

$$(18) \quad 2x^2 + 11x - 21 = 0$$

$$x = -7, \frac{3}{2}$$

$$(28) \quad 12x^2 + x - 20 = 0$$

$$x = \frac{5}{4}, -\frac{4}{3}$$

$$(9) \quad 2x^2 - 3x + 1 = 0$$

$$x = 1, \frac{1}{2}$$

$$(19) \quad 7x^2 - 8x + 1 = 0$$

$$x = \frac{1}{7}, 1$$

$$(29) \quad 10x^2 + 33x - 7 = 0$$

$$x = \frac{1}{5}, -\frac{7}{2}$$

$$(10) \quad 8x^2 - 31x + 21 = 0$$

$$x = \frac{7}{8}, 3$$

$$(20) \quad 32x^2 - 28x - 9 = 0$$

$$x = \frac{9}{8}, -\frac{1}{4}$$

$$(30) \quad 14x^2 + 37x + 24 = 0$$

$$x = -\frac{3}{2}, -\frac{8}{7}$$

$$(1) \quad 5x^2 - 6x - 27 = 0$$

$$x = 3, -\frac{9}{5}$$

$$(11) \quad x^2 - 9 = 0$$

$$x = -3, 3$$

$$(21) \quad 9x^2 + 25x - 6 = 0$$

$$x = -3, \frac{2}{9}$$

$$(2) \quad 5x^2 - 9x + 4 = 0$$

$$x = \frac{4}{5}, 1$$

$$(12) \quad 7x^2 + 39x + 20 = 0$$

$$x = -\frac{4}{7}, -5$$

$$(22) \quad 4x^2 + 15x + 14 = 0$$

$$x = -2, -\frac{7}{4}$$

$$(3) \quad 2x^2 - 17x + 36 = 0$$

$$x = \frac{9}{2}, 4$$

$$(13) \quad 2x^2 - 5x + 3 = 0$$

$$x = 1, \frac{3}{2}$$

$$(23) \quad 4x^2 - 4x - 3 = 0$$

$$x = \frac{3}{2}, -\frac{1}{2}$$

$$(4) \quad 12x^2 + 37x + 21 = 0$$

$$x = -\frac{7}{3}, -\frac{3}{4}$$

$$(14) \quad 40x^2 + 107x + 63 = 0$$

$$x = -\frac{9}{5}, -\frac{7}{8}$$

$$(24) \quad 4x^2 - x - 3 = 0$$

$$x = 1, -\frac{3}{4}$$

$$(5) \quad 3x^2 - 8x + 4 = 0$$

$$x = 2, \frac{2}{3}$$

$$(15) \quad 27x^2 + 30x + 7 = 0$$

$$x = -\frac{1}{3}, -\frac{7}{9}$$

$$(25) \quad 6x^2 - 7x - 3 = 0$$

$$x = -\frac{1}{3}, \frac{3}{2}$$

$$(6) \quad 4x^2 + 15x + 14 = 0$$

$$x = -\frac{7}{4}, -2$$

$$(16) \quad 21x^2 + 17x - 30 = 0$$

$$x = \frac{6}{7}, -\frac{5}{3}$$

$$(26) \quad x^2 - 1 = 0$$

$$x = 1, -1$$

$$(7) \quad 10x^2 - 7x + 1 = 0$$

$$x = \frac{1}{2}, \frac{1}{5}$$

$$(17) \quad 32x^2 + 4x - 15 = 0$$

$$x = \frac{5}{8}, -\frac{3}{4}$$

$$(27) \quad 45x^2 + 71x + 28 = 0$$

$$x = -\frac{4}{5}, -\frac{7}{9}$$

$$(8) \quad 5x^2 + 16x + 3 = 0$$

$$x = -3, -\frac{1}{5}$$

$$(18) \quad 3x^2 - 4x + 1 = 0$$

$$x = 1, \frac{1}{3}$$

$$(28) \quad 2x^2 + 3x + 1 = 0$$

$$x = -\frac{1}{2}, -1$$

$$(9) \quad 12x^2 + 4x - 5 = 0$$

$$x = \frac{1}{2}, -\frac{5}{6}$$

$$(19) \quad 24x^2 - 17x + 3 = 0$$

$$x = \frac{3}{8}, \frac{1}{3}$$

$$(29) \quad 4x^2 + 33x - 27 = 0$$

$$x = -9, \frac{3}{4}$$

$$(10) \quad 10x^2 - 53x + 63 = 0$$

$$x = \frac{9}{5}, \frac{7}{2}$$

$$(20) \quad 6x^2 + 13x - 28 = 0$$

$$x = -\frac{7}{2}, \frac{4}{3}$$

$$(30) \quad x^2 - 4x + 3 = 0$$

$$x = 1, 3$$

$$(1) \quad 4x^2 - 3x - 1 = 0$$

$$x = -\frac{1}{4}, 1$$

$$(11) \quad 2x^2 - 7x + 6 = 0$$

$$x = \frac{3}{2}, 2$$

$$(21) \quad 4x^2 + 11x + 7 = 0$$

$$x = -1, -\frac{7}{4}$$

$$(2) \quad x^2 + 4x + 3 = 0$$

$$x = -1, -3$$

$$(12) \quad 16x^2 + 2x - 5 = 0$$

$$x = -\frac{5}{8}, \frac{1}{2}$$

$$(22) \quad 48x^2 + 22x - 15 = 0$$

$$x = -\frac{5}{6}, \frac{3}{8}$$

$$(3) \quad 4x^2 - 3x - 1 = 0$$

$$x = -\frac{1}{4}, 1$$

$$(13) \quad 40x^2 - 91x + 49 = 0$$

$$x = \frac{7}{8}, \frac{7}{5}$$

$$(23) \quad x^2 - 7x + 12 = 0$$

$$x = 4, 3$$

$$(4) \quad 28x^2 - 43x + 9 = 0$$

$$x = \frac{9}{7}, \frac{1}{4}$$

$$(14) \quad 56x^2 + 15x - 54 = 0$$

$$x = \frac{6}{7}, -\frac{9}{8}$$

$$(24) \quad 36x^2 + 73x - 18 = 0$$

$$x = \frac{2}{9}, -\frac{9}{4}$$

$$(5) \quad 16x^2 + 2x - 3 = 0$$

$$x = -\frac{1}{2}, \frac{3}{8}$$

$$(15) \quad 20x^2 + 37x + 8 = 0$$

$$x = -\frac{8}{5}, -\frac{1}{4}$$

$$(25) \quad 63x^2 - 46x + 8 = 0$$

$$x = \frac{2}{7}, \frac{4}{9}$$

$$(6) \quad 21x^2 + 44x - 32 = 0$$

$$x = \frac{4}{7}, -\frac{8}{3}$$

$$(16) \quad 3x^2 - 8x + 5 = 0$$

$$x = 1, \frac{5}{3}$$

$$(26) \quad 5x^2 - 14x + 8 = 0$$

$$x = \frac{4}{5}, 2$$

$$(7) \quad 3x^2 + 4x + 1 = 0$$

$$x = -1, -\frac{1}{3}$$

$$(17) \quad 10x^2 + 3x - 4 = 0$$

$$x = -\frac{4}{5}, \frac{1}{2}$$

$$(27) \quad 2x^2 + 3x - 2 = 0$$

$$x = \frac{1}{2}, -2$$

$$(8) \quad x^2 - 2x - 3 = 0$$

$$x = -1, 3$$

$$(18) \quad 5x^2 - 27x + 28 = 0$$

$$x = \frac{7}{5}, 4$$

$$(28) \quad 28x^2 + 29x - 35 = 0$$

$$x = \frac{5}{7}, -\frac{7}{4}$$

$$(9) \quad 2x^2 - x - 1 = 0$$

$$x = -\frac{1}{2}, 1$$

$$(19) \quad x^2 + 3x + 2 = 0$$

$$x = -2, -1$$

$$(29) \quad 3x^2 + 22x - 16 = 0$$

$$x = -8, \frac{2}{3}$$

$$(10) \quad 2x^2 + 5x - 12 = 0$$

$$x = \frac{3}{2}, -4$$

$$(20) \quad 28x^2 + 45x + 18 = 0$$

$$x = -\frac{3}{4}, -\frac{6}{7}$$

$$(30) \quad 9x^2 - 34x + 21 = 0$$

$$x = 3, \frac{7}{9}$$

$$(1) \quad x^2 + 8x + 7 = 0$$

$$x = -7, -1$$

$$(11) \quad 3x^2 + 4x + 1 = 0$$

$$x = -\frac{1}{3}, -1$$

$$(21) \quad 18x^2 + 33x - 40 = 0$$

$$x = -\frac{8}{3}, \frac{5}{6}$$

$$(2) \quad 36x^2 + 65x - 36 = 0$$

$$x = \frac{4}{9}, -\frac{9}{4}$$

$$(12) \quad 7x^2 - 29x + 24 = 0$$

$$x = 3, \frac{8}{7}$$

$$(22) \quad 5x^2 + 13x - 6 = 0$$

$$x = -3, \frac{2}{5}$$

$$(3) \quad 32x^2 - 52x - 7 = 0$$

$$x = -\frac{1}{8}, \frac{7}{4}$$

$$(13) \quad 9x^2 - 15x - 14 = 0$$

$$x = -\frac{2}{3}, \frac{7}{3}$$

$$(23) \quad x^2 - 3x + 2 = 0$$

$$x = 1, 2$$

$$(4) \quad 7x^2 - 2x - 5 = 0$$

$$x = -\frac{5}{7}, 1$$

$$(14) \quad 4x^2 + 20x + 9 = 0$$

$$x = -\frac{9}{2}, -\frac{1}{2}$$

$$(24) \quad 7x^2 + 8x - 12 = 0$$

$$x = -2, \frac{6}{7}$$

$$(5) \quad 16x^2 - 18x + 5 = 0$$

$$x = \frac{1}{2}, \frac{5}{8}$$

$$(15) \quad 5x^2 - 29x + 36 = 0$$

$$x = \frac{9}{5}, 4$$

$$(25) \quad 20x^2 - 3x - 35 = 0$$

$$x = \frac{7}{5}, -\frac{5}{4}$$

$$(6) \quad 16x^2 - 1 = 0$$

$$x = -\frac{1}{4}, \frac{1}{4}$$

$$(16) \quad 8x^2 + 59x + 21 = 0$$

$$x = -7, -\frac{3}{8}$$

$$(26) \quad 56x^2 + 73x + 21 = 0$$

$$x = -\frac{7}{8}, -\frac{3}{7}$$

$$(7) \quad 4x^2 + 11x + 7 = 0$$

$$x = -\frac{7}{4}, -1$$

$$(17) \quad 2x^2 + 17x - 9 = 0$$

$$x = \frac{1}{2}, -9$$

$$(27) \quad 7x^2 - 25x + 12 = 0$$

$$x = 3, \frac{4}{7}$$

$$(8) \quad 24x^2 + 22x - 7 = 0$$

$$x = -\frac{7}{6}, \frac{1}{4}$$

$$(18) \quad 9x^2 - 65x - 56 = 0$$

$$x = -\frac{7}{9}, 8$$

$$(28) \quad 6x^2 - 23x - 18 = 0$$

$$x = \frac{9}{2}, -\frac{2}{3}$$

$$(9) \quad 21x^2 + 17x - 8 = 0$$

$$x = \frac{1}{3}, -\frac{8}{7}$$

$$(19) \quad 12x^2 + 25x + 12 = 0$$

$$x = -\frac{3}{4}, -\frac{4}{3}$$

$$(29) \quad 24x^2 + 58x + 9 = 0$$

$$x = -\frac{1}{6}, -\frac{9}{4}$$

$$(10) \quad 56x^2 - 57x + 7 = 0$$

$$x = \frac{1}{7}, \frac{7}{8}$$

$$(20) \quad 24x^2 + 26x + 5 = 0$$

$$x = -\frac{5}{6}, -\frac{1}{4}$$

$$(30) \quad 32x^2 - 44x - 21 = 0$$

$$x = -\frac{3}{8}, \frac{7}{4}$$

$$(1) \quad 72x^2 + 29x - 10 = 0 \\ x = -\frac{5}{8}, \frac{2}{9}$$

$$(11) \quad 5x^2 - 14x + 9 = 0 \\ x = \frac{9}{5}, 1$$

$$(21) \quad 28x^2 + 61x + 21 = 0 \\ x = -\frac{7}{4}, -\frac{3}{7}$$

$$(2) \quad 8x^2 - 17x + 2 = 0 \\ x = \frac{1}{8}, 2$$

$$(12) \quad 25x^2 - 80x + 64 = 0 \\ x = \frac{8}{5}$$

$$(22) \quad 40x^2 - 27x - 4 = 0 \\ x = \frac{4}{5}, -\frac{1}{8}$$

$$(3) \quad 8x^2 - 9x + 1 = 0 \\ x = \frac{1}{8}, 1$$

$$(13) \quad 7x^2 - 8x + 1 = 0 \\ x = 1, \frac{1}{7}$$

$$(23) \quad 10x^2 - 21x + 9 = 0 \\ x = \frac{3}{2}, \frac{3}{5}$$

$$(4) \quad 18x^2 + 21x + 5 = 0 \\ x = -\frac{5}{6}, -\frac{1}{3}$$

$$(14) \quad 27x^2 + 69x + 14 = 0 \\ x = -\frac{7}{3}, -\frac{2}{9}$$

$$(24) \quad 6x^2 + 5x - 4 = 0 \\ x = -\frac{4}{3}, \frac{1}{2}$$

$$(5) \quad 2x^2 + 5x + 3 = 0 \\ x = -1, -\frac{3}{2}$$

$$(15) \quad 6x^2 - 7x + 1 = 0 \\ x = 1, \frac{1}{6}$$

$$(25) \quad 16x^2 - 50x + 25 = 0 \\ x = \frac{5}{8}, \frac{5}{2}$$

$$(6) \quad 21x^2 + 31x + 8 = 0 \\ x = -\frac{1}{3}, -\frac{8}{7}$$

$$(16) \quad 9x^2 - 25 = 0 \\ x = \frac{5}{3}, -\frac{5}{3}$$

$$(26) \quad 4x^2 - 12x - 27 = 0 \\ x = -\frac{3}{2}, \frac{9}{2}$$

$$(7) \quad 2x^2 + 5x + 2 = 0 \\ x = -2, -\frac{1}{2}$$

$$(17) \quad 40x^2 + 79x + 24 = 0 \\ x = -\frac{8}{5}, -\frac{3}{8}$$

$$(27) \quad 3x^2 - 4x - 4 = 0 \\ x = -\frac{2}{3}, 2$$

$$(8) \quad 24x^2 - x - 10 = 0 \\ x = \frac{2}{3}, -\frac{5}{8}$$

$$(18) \quad 35x^2 - 6x - 8 = 0 \\ x = \frac{4}{7}, -\frac{2}{5}$$

$$(28) \quad x^2 - 1 = 0 \\ x = 1, -1$$

$$(9) \quad 35x^2 + 3x - 20 = 0 \\ x = \frac{5}{7}, -\frac{4}{5}$$

$$(19) \quad x^2 - x - 2 = 0 \\ x = 2, -1$$

$$(29) \quad 4x^2 + 19x + 12 = 0 \\ x = -4, -\frac{3}{4}$$

$$(10) \quad 5x^2 + 12x + 7 = 0 \\ x = -\frac{7}{5}, -1$$

$$(20) \quad 3x^2 - x - 2 = 0 \\ x = 1, -\frac{2}{3}$$

$$(30) \quad 9x^2 - 43x + 28 = 0 \\ x = \frac{7}{9}, 4$$



$$(1) 4x^2 + x - 3 = 0$$

$$x = -1, \frac{3}{4}$$

$$(11) 3x^2 + 7x - 6 = 0$$

$$x = \frac{2}{3}, -3$$

$$(21) 3x^2 + 11x + 6 = 0$$

$$x = -\frac{2}{3}, -3$$

$$(2) 7x^2 + 13x - 24 = 0$$

$$x = -3, \frac{8}{7}$$

$$(12) 3x^2 + 4x - 4 = 0$$

$$x = -2, \frac{2}{3}$$

$$(22) 45x^2 + 83x + 28 = 0$$

$$x = -\frac{4}{9}, -\frac{7}{5}$$

$$(3) 14x^2 + 13x + 3 = 0$$

$$x = -\frac{3}{7}, -\frac{1}{2}$$

$$(13) 2x^2 + 3x - 9 = 0$$

$$x = -3, \frac{3}{2}$$

$$(23) 35x^2 + 9x - 2 = 0$$

$$x = \frac{1}{7}, -\frac{2}{5}$$

$$(4) 36x^2 - 84x + 49 = 0$$

$$x = \frac{7}{6}$$

$$(14) 9x^2 + 23x + 10 = 0$$

$$x = -2, -\frac{5}{9}$$

$$(24) 12x^2 - 11x - 5 = 0$$

$$x = \frac{5}{4}, -\frac{1}{3}$$

$$(5) 54x^2 - 39x - 5 = 0$$

$$x = \frac{5}{6}, -\frac{1}{9}$$

$$(15) 5x^2 - 4x - 1 = 0$$

$$x = -\frac{1}{5}, 1$$

$$(25) 9x^2 + 16x + 7 = 0$$

$$x = -1, -\frac{7}{9}$$

$$(6) 28x^2 - x - 15 = 0$$

$$x = \frac{3}{4}, -\frac{5}{7}$$

$$(16) 2x^2 + 5x + 3 = 0$$

$$x = -\frac{3}{2}, -1$$

$$(26) 45x^2 - 92x + 32 = 0$$

$$x = \frac{8}{5}, \frac{4}{9}$$

$$(7) 15x^2 - 7x - 2 = 0$$

$$x = \frac{2}{3}, -\frac{1}{5}$$

$$(17) 7x^2 - 25x - 12 = 0$$

$$x = 4, -\frac{3}{7}$$

$$(27) 4x^2 + 17x - 15 = 0$$

$$x = -5, \frac{3}{4}$$

$$(8) 2x^2 - 13x + 18 = 0$$

$$x = \frac{9}{2}, 2$$

$$(18) 6x^2 + x - 7 = 0$$

$$x = -\frac{7}{6}, 1$$

$$(28) 7x^2 - 22x + 3 = 0$$

$$x = 3, \frac{1}{7}$$

$$(9) 28x^2 + 9x - 9 = 0$$

$$x = -\frac{3}{4}, \frac{3}{7}$$

$$(19) 14x^2 - 33x - 56 = 0$$

$$x = \frac{7}{2}, -\frac{8}{7}$$

$$(29) 12x^2 + 41x + 24 = 0$$

$$x = -\frac{8}{3}, -\frac{3}{4}$$

$$(10) 3x^2 - 16x - 12 = 0$$

$$x = -\frac{2}{3}, 6$$

$$(20) 4x^2 + 17x + 4 = 0$$

$$x = -\frac{1}{4}, -4$$

$$(30) 3x^2 - 4x + 1 = 0$$

$$x = \frac{1}{3}, 1$$

$$(1) \quad 8x^2 - 39x - 54 = 0$$

$$x = -\frac{9}{8}, 6$$

$$(11) \quad 3x^2 - 13x + 4 = 0$$

$$x = \frac{1}{3}, 4$$

$$(21) \quad 10x^2 - x - 24 = 0$$

$$x = -\frac{3}{2}, \frac{8}{5}$$

$$(2) \quad 8x^2 - 15x - 2 = 0$$

$$x = -\frac{1}{8}, 2$$

$$(12) \quad x^2 + 6x - 27 = 0$$

$$x = -9, 3$$

$$(22) \quad 2x^2 - 3x - 2 = 0$$

$$x = 2, -\frac{1}{2}$$

$$(3) \quad x^2 - 2x + 1 = 0$$

$$x = 1$$

$$(13) \quad 24x^2 - 31x + 10 = 0$$

$$x = \frac{5}{8}, \frac{2}{3}$$

$$(23) \quad 28x^2 - 3x - 18 = 0$$

$$x = -\frac{3}{4}, \frac{6}{7}$$

$$(4) \quad 7x^2 + 5x - 2 = 0$$

$$x = \frac{2}{7}, -1$$

$$(14) \quad 2x^2 - 9x - 18 = 0$$

$$x = 6, -\frac{3}{2}$$

$$(24) \quad x^2 - 3x + 2 = 0$$

$$x = 2, 1$$

$$(5) \quad 2x^2 + 13x + 6 = 0$$

$$x = -6, -\frac{1}{2}$$

$$(15) \quad 6x^2 - 7x - 10 = 0$$

$$x = -\frac{5}{6}, 2$$

$$(25) \quad 27x^2 - 33x + 8 = 0$$

$$x = \frac{1}{3}, \frac{8}{9}$$

$$(6) \quad 6x^2 - 23x + 7 = 0$$

$$x = \frac{7}{2}, \frac{1}{3}$$

$$(16) \quad 3x^2 + 8x + 5 = 0$$

$$x = -\frac{5}{3}, -1$$

$$(26) \quad 28x^2 + 75x + 27 = 0$$

$$x = -\frac{9}{4}, -\frac{3}{7}$$

$$(7) \quad 35x^2 - 31x - 40 = 0$$

$$x = -\frac{5}{7}, \frac{8}{5}$$

$$(17) \quad 9x^2 + 74x - 63 = 0$$

$$x = \frac{7}{9}, -9$$

$$(27) \quad 4x^2 + 27x - 40 = 0$$

$$x = \frac{5}{4}, -8$$

$$(8) \quad 9x^2 - 3x - 2 = 0$$

$$x = -\frac{1}{3}, \frac{2}{3}$$

$$(18) \quad 5x^2 + 3x - 8 = 0$$

$$x = 1, -\frac{8}{5}$$

$$(28) \quad 5x^2 - 36x - 81 = 0$$

$$x = -\frac{9}{5}, 9$$

$$(9) \quad 7x^2 + 5x - 18 = 0$$

$$x = \frac{9}{7}, -2$$

$$(19) \quad x^2 - 3x - 4 = 0$$

$$x = -1, 4$$

$$(29) \quad 3x^2 - 8x + 4 = 0$$

$$x = \frac{2}{3}, 2$$

$$(10) \quad 20x^2 - 43x + 14 = 0$$

$$x = \frac{2}{5}, \frac{7}{4}$$

$$(20) \quad 2x^2 + 3x - 20 = 0$$

$$x = -4, \frac{5}{2}$$

$$(30) \quad 12x^2 + 11x - 5 = 0$$

$$x = \frac{1}{3}, -\frac{5}{4}$$

$$(1) \quad x^2 - 1 = 0$$

$$x = 1, -1$$

$$(11) \quad 7x^2 + 18x + 8 = 0$$

$$x = -\frac{4}{7}, -2$$

$$(21) \quad x^2 + 7x + 12 = 0$$

$$x = -3, -4$$

$$(2) \quad x^2 + x - 2 = 0$$

$$x = 1, -2$$

$$(12) \quad 12x^2 - 8x - 15 = 0$$

$$x = \frac{3}{2}, -\frac{5}{6}$$

$$(22) \quad 3x^2 + x - 2 = 0$$

$$x = -1, \frac{2}{3}$$

$$(3) \quad 7x^2 + 5x - 2 = 0$$

$$x = \frac{2}{7}, -1$$

$$(13) \quad 36x^2 - 23x - 8 = 0$$

$$x = \frac{8}{9}, -\frac{1}{4}$$

$$(23) \quad 4x^2 + 4x - 3 = 0$$

$$x = -\frac{3}{2}, \frac{1}{2}$$

$$(4) \quad 8x^2 - 2x - 1 = 0$$

$$x = -\frac{1}{4}, \frac{1}{2}$$

$$(14) \quad 12x^2 - 5x - 2 = 0$$

$$x = -\frac{1}{4}, \frac{2}{3}$$

$$(24) \quad 8x^2 + 11x - 10 = 0$$

$$x = \frac{5}{8}, -2$$

$$(5) \quad 4x^2 + 23x + 15 = 0$$

$$x = -5, -\frac{3}{4}$$

$$(15) \quad 21x^2 - 13x - 18 = 0$$

$$x = -\frac{2}{3}, \frac{9}{7}$$

$$(25) \quad 8x^2 + 15x + 7 = 0$$

$$x = -\frac{7}{8}, -1$$

$$(6) \quad 2x^2 + 11x + 12 = 0$$

$$x = -4, -\frac{3}{2}$$

$$(16) \quad 5x^2 + 39x + 54 = 0$$

$$x = -6, -\frac{9}{5}$$

$$(26) \quad 30x^2 - 41x + 7 = 0$$

$$x = \frac{7}{6}, \frac{1}{5}$$

$$(7) \quad 4x^2 - 13x + 9 = 0$$

$$x = 1, \frac{9}{4}$$

$$(17) \quad 21x^2 - 13x - 20 = 0$$

$$x = -\frac{5}{7}, \frac{4}{3}$$

$$(27) \quad 12x^2 + 37x + 21 = 0$$

$$x = -\frac{3}{4}, -\frac{7}{3}$$

$$(8) \quad 3x^2 + 22x + 24 = 0$$

$$x = -\frac{4}{3}, -6$$

$$(18) \quad x^2 - 1 = 0$$

$$x = 1, -1$$

$$(28) \quad 12x^2 - 5x - 25 = 0$$

$$x = -\frac{5}{4}, \frac{5}{3}$$

$$(9) \quad 40x^2 + 73x + 30 = 0$$

$$x = -\frac{6}{5}, -\frac{5}{8}$$

$$(19) \quad 15x^2 + 7x - 36 = 0$$

$$x = \frac{4}{3}, -\frac{9}{5}$$

$$(29) \quad 7x^2 + 12x + 5 = 0$$

$$x = -1, -\frac{5}{7}$$

$$(10) \quad 6x^2 - 19x + 14 = 0$$

$$x = \frac{7}{6}, 2$$

$$(20) \quad x^2 + 4x + 3 = 0$$

$$x = -1, -3$$

$$(30) \quad 49x^2 + 63x + 18 = 0$$

$$x = -\frac{6}{7}, -\frac{3}{7}$$

$$(1) \quad 8x^2 - 14x + 5 = 0$$

$$x = \frac{1}{2}, \frac{5}{4}$$

$$(11) \quad 28x^2 + 99x + 81 = 0$$

$$x = -\frac{9}{7}, -\frac{9}{4}$$

$$(21) \quad 2x^2 - 9x + 7 = 0$$

$$x = \frac{7}{2}, 1$$

$$(2) \quad 27x^2 - 96x + 64 = 0$$

$$x = \frac{8}{9}, \frac{8}{3}$$

$$(12) \quad 15x^2 - 26x - 21 = 0$$

$$x = \frac{7}{3}, -\frac{3}{5}$$

$$(22) \quad 5x^2 - 21x + 18 = 0$$

$$x = 3, \frac{6}{5}$$

$$(3) \quad 16x^2 - 18x + 5 = 0$$

$$x = \frac{1}{2}, \frac{5}{8}$$

$$(13) \quad 40x^2 + 69x + 8 = 0$$

$$x = -\frac{1}{8}, -\frac{8}{5}$$

$$(23) \quad 6x^2 - 13x - 8 = 0$$

$$x = \frac{8}{3}, -\frac{1}{2}$$

$$(4) \quad 2x^2 - x - 3 = 0$$

$$x = -1, \frac{3}{2}$$

$$(14) \quad 49x^2 + 42x + 8 = 0$$

$$x = -\frac{4}{7}, -\frac{2}{7}$$

$$(24) \quad 9x^2 + 12x + 4 = 0$$

$$x = -\frac{2}{3}$$

$$(5) \quad 9x^2 + 17x + 8 = 0$$

$$x = -1, -\frac{8}{9}$$

$$(15) \quad 5x^2 + 2x - 7 = 0$$

$$x = -\frac{7}{5}, 1$$

$$(25) \quad 81x^2 + 54x + 8 = 0$$

$$x = -\frac{2}{9}, -\frac{4}{9}$$

$$(6) \quad 7x^2 - 65x + 18 = 0$$

$$x = \frac{2}{7}, 9$$

$$(16) \quad 27x^2 - 12x - 7 = 0$$

$$x = \frac{7}{9}, -\frac{1}{3}$$

$$(26) \quad 49x^2 - 28x - 12 = 0$$

$$x = \frac{6}{7}, -\frac{2}{7}$$

$$(7) \quad 16x^2 + 10x - 21 = 0$$

$$x = \frac{7}{8}, -\frac{3}{2}$$

$$(17) \quad 18x^2 + 31x - 35 = 0$$

$$x = \frac{7}{9}, -\frac{5}{2}$$

$$(27) \quad 12x^2 + 5x - 2 = 0$$

$$x = \frac{1}{4}, -\frac{2}{3}$$

$$(8) \quad 10x^2 - x - 3 = 0$$

$$x = -\frac{1}{2}, \frac{3}{5}$$

$$(18) \quad 3x^2 + 26x + 48 = 0$$

$$x = -\frac{8}{3}, -6$$

$$(28) \quad 9x^2 + 61x - 14 = 0$$

$$x = -7, \frac{2}{9}$$

$$(9) \quad 5x^2 + 6x + 1 = 0$$

$$x = -\frac{1}{5}, -1$$

$$(19) \quad 9x^2 - 16 = 0$$

$$x = \frac{4}{3}, \frac{4}{3}$$

$$(29) \quad 24x^2 + 2x - 1 = 0$$

$$x = -\frac{1}{4}, \frac{1}{6}$$

$$(10) \quad 2x^2 - 9x + 4 = 0$$

$$x = \frac{1}{2}, 4$$

$$(20) \quad 21x^2 - 73x + 56 = 0$$

$$x = \frac{8}{7}, \frac{7}{3}$$

$$(30) \quad 6x^2 - 11x - 7 = 0$$

$$x = -\frac{1}{2}, \frac{7}{3}$$

$$(1) \quad x^2 - 4x + 3 = 0$$

$$x = 1, 3$$

$$(11) \quad 27x^2 - 12x - 32 = 0$$

$$x = \frac{4}{3}, -\frac{8}{9}$$

$$(21) \quad 63x^2 - 40x - 7 = 0$$

$$x = \frac{7}{9}, -\frac{1}{7}$$

$$(2) \quad 8x^2 + 47x - 6 = 0$$

$$x = \frac{1}{8}, -6$$

$$(12) \quad 4x^2 - x - 3 = 0$$

$$x = -\frac{3}{4}, 1$$

$$(22) \quad 9x^2 + 17x + 8 = 0$$

$$x = -1, -\frac{8}{9}$$

$$(3) \quad 9x^2 - 56x - 49 = 0$$

$$x = -\frac{7}{9}, 7$$

$$(13) \quad 12x^2 - 7x - 10 = 0$$

$$x = -\frac{2}{3}, \frac{5}{4}$$

$$(23) \quad 45x^2 + 46x + 8 = 0$$

$$x = -\frac{2}{9}, -\frac{4}{5}$$

$$(4) \quad x^2 - 2x + 1 = 0$$

$$x = 1$$

$$(14) \quad 6x^2 - 13x + 6 = 0$$

$$x = \frac{2}{3}, \frac{3}{2}$$

$$(24) \quad 2x^2 + 7x + 5 = 0$$

$$x = -1, -\frac{5}{2}$$

$$(5) \quad 72x^2 + 13x - 20 = 0$$

$$x = \frac{4}{9}, -\frac{5}{8}$$

$$(15) \quad x^2 + 5x + 4 = 0$$

$$x = -1, -4$$

$$(25) \quad 8x^2 + 2x - 1 = 0$$

$$x = \frac{1}{4}, -\frac{1}{2}$$

$$(6) \quad 4x^2 + 7x - 15 = 0$$

$$x = -3, \frac{5}{4}$$

$$(16) \quad 18x^2 - 51x + 35 = 0$$

$$x = \frac{5}{3}, \frac{7}{6}$$

$$(26) \quad 6x^2 - 11x - 2 = 0$$

$$x = -\frac{1}{6}, 2$$

$$(7) \quad 3x^2 - 5x + 2 = 0$$

$$x = 1, \frac{2}{3}$$

$$(17) \quad 5x^2 - 46x + 9 = 0$$

$$x = 9, \frac{1}{5}$$

$$(27) \quad 18x^2 - 29x + 3 = 0$$

$$x = \frac{3}{2}, \frac{1}{9}$$

$$(8) \quad 7x^2 - 30x - 25 = 0$$

$$x = -\frac{5}{7}, 5$$

$$(18) \quad 8x^2 - 10x - 7 = 0$$

$$x = \frac{7}{4}, -\frac{1}{2}$$

$$(28) \quad 4x^2 + 5x + 1 = 0$$

$$x = -1, -\frac{1}{4}$$

$$(9) \quad 27x^2 + 3x - 10 = 0$$

$$x = \frac{5}{9}, -\frac{2}{3}$$

$$(19) \quad 7x^2 - 23x + 18 = 0$$

$$x = \frac{9}{7}, 2$$

$$(29) \quad 9x^2 + 44x - 5 = 0$$

$$x = -5, \frac{1}{9}$$

$$(10) \quad 20x^2 - 9x - 18 = 0$$

$$x = -\frac{3}{4}, \frac{6}{5}$$

$$(20) \quad 14x^2 + 25x + 6 = 0$$

$$x = -\frac{3}{2}, -\frac{2}{7}$$

$$(30) \quad 5x^2 + 29x - 6 = 0$$

$$x = \frac{1}{5}, -6$$

$$(1) \quad 3x^2 + 5x - 12 = 0 \\ x = \frac{4}{3}, -3$$

$$(11) \quad 25x^2 - 10x - 63 = 0 \\ x = \frac{9}{5}, -\frac{7}{5}$$

$$(21) \quad 12x^2 + 5x - 3 = 0 \\ x = -\frac{3}{4}, \frac{1}{3}$$

$$(2) \quad 6x^2 + 13x - 5 = 0 \\ x = \frac{1}{3}, -\frac{5}{2}$$

$$(12) \quad 2x^2 + 15x - 8 = 0 \\ x = -8, \frac{1}{2}$$

$$(22) \quad x^2 + 2x - 3 = 0 \\ x = 1, -3$$

$$(3) \quad 35x^2 + 93x + 54 = 0 \\ x = -\frac{6}{7}, -\frac{9}{5}$$

$$(13) \quad 56x^2 - 55x - 9 = 0 \\ x = \frac{9}{8}, -\frac{1}{7}$$

$$(23) \quad 42x^2 + 103x + 63 = 0 \\ x = -\frac{7}{6}, -\frac{9}{7}$$

$$(4) \quad 2x^2 - 5x - 18 = 0 \\ x = -2, \frac{9}{2}$$

$$(14) \quad x^2 - 10x + 16 = 0 \\ x = 8, 2$$

$$(24) \quad 2x^2 + x - 21 = 0 \\ x = -\frac{7}{2}, 3$$

$$(5) \quad 4x^2 - 23x + 28 = 0 \\ x = 4, \frac{7}{4}$$

$$(15) \quad 9x^2 + 9x - 10 = 0 \\ x = \frac{2}{3}, -\frac{5}{3}$$

$$(25) \quad 4x^2 - 12x + 5 = 0 \\ x = \frac{1}{2}, \frac{5}{2}$$

$$(6) \quad 9x^2 + 9x + 2 = 0 \\ x = -\frac{1}{3}, -\frac{2}{3}$$

$$(16) \quad 3x^2 - 29x + 18 = 0 \\ x = 9, \frac{2}{3}$$

$$(26) \quad 16x^2 + 6x - 1 = 0 \\ x = \frac{1}{8}, -\frac{1}{2}$$

$$(7) \quad 36x^2 - 73x + 35 = 0 \\ x = \frac{7}{9}, \frac{5}{4}$$

$$(17) \quad 10x^2 - x - 3 = 0 \\ x = \frac{3}{5}, -\frac{1}{2}$$

$$(27) \quad 9x^2 + 25x + 14 = 0 \\ x = -2, -\frac{7}{9}$$

$$(8) \quad 7x^2 - 65x + 18 = 0 \\ x = 9, \frac{2}{7}$$

$$(18) \quad 49x^2 + 7x - 20 = 0 \\ x = \frac{4}{7}, -\frac{5}{7}$$

$$(28) \quad 2x^2 + 5x - 3 = 0 \\ x = -3, \frac{1}{2}$$

$$(9) \quad 36x^2 + 5x - 24 = 0 \\ x = -\frac{8}{9}, \frac{3}{4}$$

$$(19) \quad 45x^2 + 8x - 21 = 0 \\ x = -\frac{7}{9}, \frac{3}{5}$$

$$(29) \quad 10x^2 - 11x - 8 = 0 \\ x = -\frac{1}{2}, \frac{8}{5}$$

$$(10) \quad 4x^2 - 9x - 28 = 0 \\ x = -\frac{7}{4}, 4$$

$$(20) \quad 16x^2 - 38x + 21 = 0 \\ x = \frac{3}{2}, \frac{7}{8}$$

$$(30) \quad 6x^2 - 13x - 15 = 0 \\ x = 3, -\frac{5}{6}$$

## 2.12 1 次不等式

1 次不等式の問題。

(1)  $2x - 7 \geq -8x + 8$

(16)  $2x - 3 \geq 9x - 8$

(31)  $4x + 6 < -x + 7$

(2)  $-6x - 2 \geq 4x - 5$

(17)  $7x - 8 \geq 2x + 3$

(32)  $-4x + 1 \leq 7x + 9$

(3)  $3x + 9 < -3x + 5$

(18)  $x - 5 \leq 6x - 3$

(33)  $5x + 6 \geq -5x + 4$

(4)  $9x + 3 \geq -5x - 9$

(19)  $-8x - 2 > -3x - 4$

(34)  $-9x + 3 > -8x - 2$

(5)  $-6x + 8 < -4x - 9$

(20)  $x + 3 \geq -4x + 8$

(35)  $5x + 9 \geq -2x - 5$

(6)  $-6x - 7 \leq 3x + 7$

(21)  $5x - 9 > x + 9$

(36)  $-5x - 5 > -6x + 7$

(7)  $x - 9 \leq 2x - 7$

(22)  $3x - 9 < 8x + 2$

(37)  $3x + 8 \leq -2x - 9$

(8)  $-6x + 3 \leq 9x + 4$

(23)  $2x + 7 \geq -4x - 4$

(38)  $5x + 6 > 3x - 6$

(9)  $-2x - 8 > 5x + 8$

(24)  $-8x - 3 > 2x - 9$

(39)  $3x + 7 < 9x - 5$

(10)  $9x + 6 \leq -5x + 9$

(25)  $-7x - 9 < -2x - 1$

(40)  $-6x - 6 \geq -8x - 7$

(11)  $-2x - 8 < 6x - 6$

(26)  $9x - 4 < -x + 7$

(41)  $-7x - 3 > 5x + 9$

(12)  $x - 5 \geq 3x - 8$

(27)  $-5x - 8 \geq -4x - 9$

(42)  $6x + 5 \leq 2x - 4$

(13)  $-6x + 7 < -4x - 7$

(28)  $-8x - 3 \geq 2x + 4$

(43)  $-x + 2 \geq -2x - 2$

(14)  $-2x - 5 > -9x + 8$

(29)  $9x - 8 > -5x - 5$

(44)  $-2x - 3 \leq x + 7$

(15)  $-6x - 9 \leq 9x + 9$

(30)  $5x - 8 \leq -3x - 1$

(45)  $-9x + 8 < 2x + 7$



(1)  $6x - 8 > 9x + 8$

(16)  $4x - 7 < -7x + 5$

(31)  $6x + 5 < 8x + 8$

(2)  $5x - 4 \leq -3x + 1$

(17)  $-2x + 2 > -3x + 1$

(32)  $5x + 3 < -7x + 7$

(3)  $-x + 8 \geq -4x + 4$

(18)  $-x - 8 < 2x + 9$

(33)  $7x - 7 < 9x - 3$

(4)  $2x + 2 > 9x + 8$

(19)  $-3x + 2 \leq 4x - 6$

(34)  $-4x + 9 > 7x - 6$

(5)  $5x + 5 \leq 3x - 8$

(20)  $-3x - 5 \leq -4x - 3$

(35)  $-5x + 7 \geq -7x + 3$

(6)  $-7x - 3 \geq -4x - 9$

(21)  $4x + 8 > -7x + 9$

(36)  $2x + 5 < -8x + 2$

(7)  $-4x - 6 \leq 3x - 5$

(22)  $-x - 1 > -9x - 2$

(37)  $4x + 9 \geq 7x - 3$

(8)  $-4x + 1 > -2x + 8$

(23)  $8x - 9 < 4x + 2$

(38)  $7x - 4 \geq -9x + 3$

(9)  $-4x - 2 > 5x + 8$

(24)  $-4x - 7 \geq -3x + 9$

(39)  $x - 4 \geq 2x - 6$

(10)  $-3x + 5 > -5x + 9$

(25)  $9x + 2 \geq 4x - 6$

(40)  $-4x + 8 > -3x - 8$

(11)  $-4x + 3 \geq 2x + 2$

(26)  $-5x + 1 \geq -x - 3$

(41)  $-2x + 3 \geq 9x + 4$

(12)  $8x + 4 < x - 8$

(27)  $-2x - 6 \geq -5x - 1$

(42)  $8x + 4 > -7x - 2$

(13)  $7x + 3 < 3x + 8$

(28)  $3x - 3 < x - 6$

(43)  $-x + 9 > -2x + 7$

(14)  $-8x + 2 < -5x - 3$

(29)  $-6x + 5 \leq -2x + 6$

(44)  $-4x - 6 \leq -8x - 3$

(15)  $-5x + 5 > -7x - 1$

(30)  $-3x - 7 \geq -7x - 5$

(45)  $-3x + 1 > 8x + 4$

(1)  $x + 9 \geq 6x - 4$

(16)  $4x + 6 \leq 7x + 6$

(31)  $6x - 8 \geq -9x + 6$

(2)  $-x + 9 \geq 9x + 4$

(17)  $2x + 2 > -4x + 2$

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(21)  $-6x + 2 \leq 7x + 1$

(36)  $-9x - 9 < 7x + 3$

(7)  $-3x - 5 \geq 2x - 4$

(22)  $2x + 8 < -2x + 8$

(37)  $2x - 4 < -5x - 2$

(8)  $5x - 6 < -4x + 5$

(23)  $8x + 9 > 5x + 5$

(38)  $5x + 2 < -4x - 7$

(9)  $-8x - 4 \geq 3x + 4$

(24)  $-8x + 9 \geq -2x - 9$

(39)  $7x + 5 < -x + 3$

(10)  $-9x - 8 > 5x + 1$

(25)  $-8x + 2 > 7x + 4$

(40)  $7x - 4 < 8x + 8$

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(26)  $-9x + 5 \leq 6x + 5$

(41)  $4x + 6 \leq -2x - 5$

(12)  $-5x - 2 > -4x + 1$

(27)  $4x + 1 > 3x - 4$

(42)  $-8x + 1 \leq 2x - 6$

(13)  $-8x - 2 < -3x + 6$

(28)  $8x - 4 > -4x - 7$

(43)  $-8x - 7 \geq -9x + 5$

(14)  $x + 2 \geq -4x - 7$

(29)  $-x - 8 > -8x + 8$

(44)  $6x - 7 < 9x - 8$

(15)  $7x - 7 > 9x - 7$

(30)  $-4x - 7 \leq 3x + 9$

(45)  $9x + 8 > x + 1$

(1)  $7x - 3 \geq 5x + 7$

(16)  $7x + 1 < 8x + 4$

(31)  $-7x - 8 \geq 7x - 4$

(2)  $7x - 6 \leq 6x - 6$

(17)  $9x + 9 < 2x - 4$

(32)  $-7x + 4 < -x - 5$

(3)  $6x + 8 \geq 5x + 5$

(18)  $-2x + 8 \leq x - 9$

(33)  $2x + 7 < 3x - 1$

(4)  $-2x + 8 < -6x + 4$

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(17)  $-6x - 3 \leq -x - 1$

(32)  $8x - 1 \leq 7x + 2$

(3)  $4x - 3 \leq -x + 9$

(18)  $7x - 6 \geq -8x - 9$

(33)  $-6x + 4 \leq 5x + 6$

(4)  $7x + 4 \geq 2x + 6$

(19)  $-7x + 9 \geq 6x - 6$

(34)  $-4x - 1 < -6x + 6$

(5)  $-3x - 8 \geq -x + 2$

(20)  $7x + 8 \geq 5x - 2$

(35)  $-9x - 6 > 4x + 9$

(6)  $-8x + 8 < -6x + 3$

(21)  $5x + 1 \geq 4x - 5$

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(8)  $6x - 4 < -4x - 8$

(23)  $9x - 3 \geq 3x + 6$

(38)  $-6x - 4 \leq 5x - 7$

(9)  $-5x - 7 \geq 6x - 6$

(24)  $5x + 6 < 4x - 9$

(39)  $8x - 7 \leq -7x + 5$

(10)  $2x + 8 < -x - 1$

(25)  $-5x + 5 \geq 6x - 6$

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(29)  $5x - 4 \geq x + 2$

(44)  $-3x - 5 > -7x + 5$

(15)  $6x - 3 < 4x - 9$

(30)  $-6x - 5 < -9x + 3$

(45)  $9x + 8 > -6x - 3$

$$(1) \quad 2x - 7 \geq -8x + 8$$

$$x \geq \frac{3}{2}$$

$$(2) \quad -6x - 2 \geq 4x - 5$$

$$x \leq \frac{3}{10}$$

$$(3) \quad 3x + 9 < -3x + 5$$

$$x < -\frac{2}{3}$$

$$(4) \quad 9x + 3 \geq -5x - 9$$

$$x \geq -\frac{6}{7}$$

$$(5) \quad -6x + 8 < -4x - 9$$

$$x > \frac{17}{2}$$

$$(6) \quad -6x - 7 \leq 3x + 7$$

$$x \geq -\frac{14}{9}$$

$$(7) \quad x - 9 \leq 2x - 7$$

$$x \geq -2$$

$$(8) \quad -6x + 3 \leq 9x + 4$$

$$x \geq -\frac{1}{15}$$

$$(9) \quad -2x - 8 > 5x + 8$$

$$x < -\frac{16}{7}$$

$$(10) \quad 9x + 6 \leq -5x + 9$$

$$x \leq \frac{3}{14}$$

$$(11) \quad -2x - 8 < 6x - 6$$

$$x > -\frac{1}{4}$$

$$(12) \quad x - 5 \geq 3x - 8$$

$$x \leq \frac{3}{2}$$

$$(13) \quad -6x + 7 < -4x - 7$$

$$x > 7$$

$$(14) \quad -2x - 5 > -9x + 8$$

$$x > \frac{13}{7}$$

$$(15) \quad -6x - 9 \leq 9x + 9$$

$$x \geq -\frac{6}{5}$$

$$(16) \quad 2x - 3 \geq 9x - 8$$

$$x \leq \frac{5}{7}$$

$$(17) \quad 7x - 8 \geq 2x + 3$$

$$x \geq \frac{11}{5}$$

$$(18) \quad x - 5 \leq 6x - 3$$

$$x \geq -\frac{2}{5}$$

$$(19) \quad -8x - 2 > -3x - 4$$

$$x < \frac{2}{5}$$

$$(20) \quad x + 3 \geq -4x + 8$$

$$x \geq 1$$

$$(21) \quad 5x - 9 > x + 9$$

$$x > \frac{9}{2}$$

$$(22) \quad 3x - 9 < 8x + 2$$

$$x > -\frac{11}{5}$$

$$(23) \quad 2x + 7 \geq -4x - 4$$

$$x \geq -\frac{11}{6}$$

$$(24) \quad -8x - 3 > 2x - 9$$

$$x < \frac{3}{5}$$

$$(25) \quad -7x - 9 < -2x - 1$$

$$x > -\frac{8}{5}$$

$$(26) \quad 9x - 4 < -x + 7$$

$$x < \frac{11}{10}$$

$$(27) \quad -5x - 8 \geq -4x - 9$$

$$x \leq 1$$

$$(28) \quad -8x - 3 \geq 2x + 4$$

$$x \leq -\frac{7}{10}$$

$$(29) \quad 9x - 8 > -5x - 5$$

$$x > \frac{3}{14}$$

$$(30) \quad 5x - 8 \leq -3x - 1$$

$$x \leq \frac{7}{8}$$

$$(31) \quad 4x + 6 < -x + 7$$

$$x < \frac{1}{5}$$

$$(32) \quad -4x + 1 \leq 7x + 9$$

$$x \geq -\frac{8}{11}$$

$$(33) \quad 5x + 6 \geq -5x + 4$$

$$x \geq -\frac{1}{5}$$

$$(34) \quad -9x + 3 > -8x - 2$$

$$x < 5$$

$$(35) \quad 5x + 9 \geq -2x - 5$$

$$x \geq -2$$

$$(36) \quad -5x - 5 > -6x + 7$$

$$x > 12$$

$$(37) \quad 3x + 8 \leq -2x - 9$$

$$x \leq -\frac{17}{5}$$

$$(38) \quad 5x + 6 > 3x - 6$$

$$x > -6$$

$$(39) \quad 3x + 7 < 9x - 5$$

$$x > 2$$

$$(40) \quad -6x - 6 \geq -8x - 7$$

$$x \geq -\frac{1}{2}$$

$$(41) \quad -7x - 3 > 5x + 9$$

$$x < -1$$

$$(42) \quad 6x + 5 \leq 2x - 4$$

$$x \leq -\frac{9}{4}$$

$$(43) \quad -x + 2 \geq -2x - 2$$

$$x \geq -4$$

$$(44) \quad -2x - 3 \leq x + 7$$

$$x \geq -\frac{10}{3}$$

$$(45) \quad -9x + 8 < 2x + 7$$

$$x > \frac{1}{11}$$

$$(1) \quad 6x - 8 > 9x + 8$$

$$x < -\frac{16}{3}$$

$$(2) \quad 5x - 4 \leq -3x + 1$$

$$x \leq \frac{5}{8}$$

$$(3) \quad -x + 8 \geq -4x + 4$$

$$x \geq -\frac{4}{3}$$

$$(4) \quad 2x + 2 > 9x + 8$$

$$x < -\frac{6}{7}$$

$$(5) \quad 5x + 5 \leq 3x - 8$$

$$x \leq -\frac{13}{2}$$

$$(6) \quad -7x - 3 \geq -4x - 9$$

$$x \leq 2$$

$$(7) \quad -4x - 6 \leq 3x - 5$$

$$x \geq -\frac{1}{7}$$

$$(8) \quad -4x + 1 > -2x + 8$$

$$x < -\frac{7}{2}$$

$$(9) \quad -4x - 2 > 5x + 8$$

$$x < -\frac{10}{9}$$

$$(10) \quad -3x + 5 > -5x + 9$$

$$x > 2$$

$$(11) \quad -4x + 3 \geq 2x + 2$$

$$x \leq \frac{1}{6}$$

$$(12) \quad 8x + 4 < x - 8$$

$$x < -\frac{12}{7}$$

$$(13) \quad 7x + 3 < 3x + 8$$

$$x < \frac{5}{4}$$

$$(14) \quad -8x + 2 < -5x - 3$$

$$x > \frac{5}{3}$$

$$(15) \quad -5x + 5 > -7x - 1$$

$$x > -3$$

$$(16) \quad 4x - 7 < -7x + 5$$

$$x < \frac{12}{11}$$

$$(17) \quad -2x + 2 > -3x + 1$$

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$$(21) \quad 4x + 8 > -7x + 9$$

$$x > \frac{1}{11}$$

$$(22) \quad -x - 1 > -9x - 2$$

$$x > -\frac{1}{8}$$

$$(23) \quad 8x - 9 < 4x + 2$$

$$x < \frac{11}{4}$$

$$(24) \quad -4x - 7 \geq -3x + 9$$

$$x \leq -16$$

$$(25) \quad 9x + 2 \geq 4x - 6$$

$$x \geq -\frac{8}{5}$$

$$(26) \quad -5x + 1 \geq -x - 3$$

$$x \leq 1$$

$$(27) \quad -2x - 6 \geq -5x - 1$$

$$x \geq \frac{5}{3}$$

$$(28) \quad 3x - 3 < x - 6$$

$$x < -\frac{3}{2}$$

$$(29) \quad -6x + 5 \leq -2x + 6$$

$$x \geq -\frac{1}{4}$$

$$(30) \quad -3x - 7 \geq -7x - 5$$

$$x \geq \frac{1}{2}$$

$$(31) \quad 6x + 5 < 8x + 8$$

$$x > -\frac{3}{2}$$

$$(32) \quad 5x + 3 < -7x + 7$$

$$x < \frac{1}{3}$$

$$(33) \quad 7x - 7 < 9x - 3$$

$$x > -2$$

$$(34) \quad -4x + 9 > 7x - 6$$

$$x < \frac{15}{11}$$

$$(35) \quad -5x + 7 \geq -7x + 3$$

$$x \geq -2$$

$$(36) \quad 2x + 5 < -8x + 2$$

$$x < -\frac{3}{10}$$

$$(37) \quad 4x + 9 \geq 7x - 3$$

$$x \leq 4$$

$$(38) \quad 7x - 4 \geq -9x + 3$$

$$x \geq \frac{7}{16}$$

$$(39) \quad x - 4 \geq 2x - 6$$

$$x \leq 2$$

$$(40) \quad -4x + 8 > -3x - 8$$

$$x < 16$$

$$(41) \quad -2x + 3 \geq 9x + 4$$

$$x \leq -\frac{1}{11}$$

$$(42) \quad 8x + 4 > -7x - 2$$

$$x > -\frac{2}{5}$$

$$(43) \quad -x + 9 > -2x + 7$$

$$x > -2$$

$$(44) \quad -4x - 6 \leq -8x - 3$$

$$x \leq \frac{3}{4}$$

$$(45) \quad -3x + 1 > 8x + 4$$

$$x < -\frac{3}{11}$$

$$(1) \quad x + 9 \geq 6x - 4$$

$$x \leq \frac{13}{5}$$

$$(2) \quad -x + 9 \geq 9x + 4$$

$$x \leq \frac{1}{2}$$

$$(3) \quad 5x - 4 \geq -x - 4$$

$$x \geq 0$$

$$(4) \quad 9x + 3 > 5x + 4$$

$$x > \frac{1}{4}$$

$$(5) \quad -9x + 3 \leq -2x + 5$$

$$x \geq -\frac{2}{7}$$

$$(6) \quad x + 7 > 8x + 3$$

$$x < \frac{4}{7}$$

$$(7) \quad 6x + 4 \geq 5x + 2$$

$$x \geq -2$$

$$(8) \quad 8x - 3 \geq 2x + 6$$

$$x \geq \frac{3}{2}$$

$$(9) \quad -7x + 9 > -2x - 7$$

$$x < \frac{16}{5}$$

$$(10) \quad x - 8 \geq -7x - 5$$

$$x \geq \frac{3}{8}$$

$$(11) \quad 5x - 9 \geq x + 3$$

$$x \geq 3$$

$$(12) \quad 7x + 4 > 5x + 1$$

$$x > -\frac{3}{2}$$

$$(13) \quad 3x - 3 < -8x - 9$$

$$x < -\frac{6}{11}$$

$$(14) \quad -8x - 4 > -2x + 4$$

$$x < -\frac{4}{3}$$

$$(15) \quad -5x + 7 > 6x + 5$$

$$x < \frac{2}{11}$$

$$(16) \quad 4x + 6 \leq 7x + 6$$

$$x \geq 0$$

$$(17) \quad 2x + 2 > -4x + 2$$

$$x > 0$$

$$(18) \quad 9x - 8 \leq -9x - 8$$

$$x \leq 0$$

$$(19) \quad -x - 3 > -8x + 5$$

$$x > \frac{8}{7}$$

$$(20) \quad -3x - 2 > -9x - 8$$

$$x > -1$$

$$(21) \quad -9x - 1 > 8x - 7$$

$$x < \frac{6}{17}$$

$$(22) \quad -2x + 7 < 7x + 3$$

$$x > \frac{4}{9}$$

$$(23) \quad -8x + 8 < 4x - 6$$

$$x > \frac{7}{6}$$

$$(24) \quad 7x + 9 \leq -2x + 3$$

$$x \leq -\frac{2}{3}$$

$$(25) \quad -9x + 2 \leq -6x - 4$$

$$x \geq 2$$

$$(26) \quad 8x - 3 > -x - 8$$

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$$(27) \quad -x - 7 > -8x - 7$$

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$$(28) \quad 7x + 2 \leq -8x - 1$$

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$$(29) \quad 9x + 3 \geq 8x + 5$$

$$x \geq 2$$

$$(30) \quad 9x + 2 \leq 2x - 2$$

$$x \leq -\frac{4}{7}$$

$$(31) \quad 6x - 8 \geq -9x + 6$$

$$x \geq \frac{14}{15}$$

$$(32) \quad -9x + 7 < -2x - 7$$

$$x > 2$$

$$(33) \quad x - 8 > 9x - 2$$

$$x < -\frac{3}{4}$$

$$(34) \quad -6x - 3 \geq x - 9$$

$$x \leq \frac{6}{7}$$

$$(35) \quad 6x + 6 < -2x - 3$$

$$x < -\frac{9}{8}$$

$$(36) \quad -9x + 1 < x - 5$$

$$x > \frac{3}{5}$$

$$(37) \quad -8x + 5 \geq -4x - 7$$

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$$(38) \quad -8x + 4 > -3x + 6$$

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$$(40) \quad x + 8 < 3x - 1$$

$$x > \frac{9}{2}$$

$$(41) \quad -2x + 9 > -7x - 7$$

$$x > -\frac{16}{5}$$

$$(42) \quad 2x + 9 \geq 9x - 6$$

$$x \leq \frac{15}{7}$$

$$(43) \quad 4x - 3 < 7x - 4$$

$$x > \frac{1}{3}$$

$$(44) \quad 7x - 5 < -5x + 5$$

$$x < \frac{5}{6}$$

$$(45) \quad 9x - 3 < -2x + 5$$

$$x < \frac{8}{11}$$

$$(1) -8x - 3 \geq 3x - 6$$

$$x \leq \frac{3}{11}$$

$$(2) 7x - 7 > -6x - 6$$

$$x > \frac{1}{13}$$

$$(3) 4x - 7 > -8x - 8$$

$$x > -\frac{1}{12}$$

$$(4) -8x + 5 \geq -x + 4$$

$$x \leq \frac{1}{7}$$

$$(5) -8x + 1 > 9x + 4$$

$$x < -\frac{3}{17}$$

$$(6) 4x + 2 > -8x + 3$$

$$x > \frac{1}{12}$$

$$(7) x + 9 \geq 9x - 3$$

$$x \leq \frac{3}{2}$$

$$(8) 6x - 8 < -2x - 4$$

$$x < \frac{1}{2}$$

$$(9) -7x + 6 \leq 3x - 2$$

$$x \geq \frac{4}{5}$$

$$(10) -2x - 8 \leq 5x - 1$$

$$x \geq -1$$

$$(11) 9x - 5 \leq 8x - 8$$

$$x \leq -3$$

$$(12) 7x + 2 < 6x - 2$$

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$$(13) -3x + 1 \geq -x + 7$$

$$x \leq -3$$

$$(14) 2x - 3 > 8x + 1$$

$$x < -\frac{2}{3}$$

$$(15) -8x - 2 \geq -2x - 4$$

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$$(16) 3x + 8 > -7x + 9$$

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$$(17) 2x - 3 < x + 7$$

$$x < 10$$

$$(18) -2x - 5 \geq -7x + 7$$

$$x \geq \frac{12}{5}$$

$$(19) -2x + 9 \geq -8x - 3$$

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$$(20) -x + 4 > 6x + 6$$

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$$(22) -2x - 3 \leq 9x + 1$$

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$$(23) 2x + 5 \geq -9x - 1$$

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$$(24) -8x - 7 \geq 4x + 1$$

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$$(28) -4x - 9 < -5x - 2$$

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$$(29) -3x + 1 \geq 8x - 5$$

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$$(30) -3x - 7 \leq -x - 7$$

$$x \geq 0$$

$$(31) -7x + 4 < 3x + 4$$

$$x > 0$$

$$(32) -8x - 4 \geq 7x + 9$$

$$x \leq -\frac{13}{15}$$

$$(33) -3x + 6 \geq -6x - 7$$

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$$(34) 7x - 2 > -3x - 9$$

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$$(35) -x + 3 \leq -7x + 4$$

$$x \leq \frac{1}{6}$$

$$(36) -5x + 5 > 5x + 4$$

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$$(38) 5x + 7 < -6x + 8$$

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$$(40) -6x - 6 \leq 9x - 9$$

$$x \geq \frac{1}{5}$$

$$(41) -8x + 1 \leq -4x + 1$$

$$x \geq 0$$

$$(42) -5x - 8 \geq -7x + 8$$

$$x \geq 8$$

$$(43) -x - 1 > -2x - 9$$

$$x > -8$$

$$(44) -9x - 6 < 4x + 5$$

$$x > -\frac{11}{13}$$

$$(45) -9x - 3 \geq x - 8$$

$$x \leq \frac{1}{2}$$

$$(1) -5x + 7 < -x - 4$$

$$x > \frac{11}{4}$$

$$(2) 8x + 1 > 2x - 3$$

$$x > -\frac{2}{3}$$

$$(3) -7x - 3 > 8x + 9$$

$$x < -\frac{4}{5}$$

$$(4) 7x - 5 \leq 8x + 3$$

$$x \geq -8$$

$$(5) x - 4 \geq 2x - 4$$

$$x \leq 0$$

$$(6) 6x - 5 \geq 4x + 3$$

$$x \geq 4$$

$$(7) x - 1 \leq 8x - 2$$

$$x \geq \frac{1}{7}$$

$$(8) 2x + 6 \leq -3x + 3$$

$$x \leq -\frac{3}{5}$$

$$(9) -x + 5 > -8x - 7$$

$$x > -\frac{12}{7}$$

$$(10) -9x - 8 \geq x - 1$$

$$x \leq -\frac{7}{10}$$

$$(11) 8x - 3 < 4x + 9$$

$$x < 3$$

$$(12) 9x + 1 < 7x + 5$$

$$x < 2$$

$$(13) -8x + 9 > -5x + 5$$

$$x < \frac{4}{3}$$

$$(14) 7x - 8 < -5x + 6$$

$$x < \frac{7}{6}$$

$$(15) 3x - 6 \geq -3x - 9$$

$$x \geq -\frac{1}{2}$$

$$(16) 5x - 9 < 3x + 8$$

$$x < \frac{17}{2}$$

$$(17) 7x - 3 < 3x - 7$$

$$x < -1$$

$$(18) -2x - 8 > -6x + 7$$

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$$(22) 8x - 3 > 3x - 3$$

$$x > 0$$

$$(23) 3x + 9 \geq 8x + 1$$

$$x \leq \frac{8}{5}$$

$$(24) -x - 2 \leq -4x + 7$$

$$x \leq 3$$

$$(25) 7x + 7 \geq 8x + 2$$

$$x \leq 5$$

$$(26) -8x + 9 \leq -6x - 5$$

$$x \geq 7$$

$$(27) -8x - 4 \leq -3x - 7$$

$$x \geq \frac{3}{5}$$

$$(28) -5x + 4 \leq 8x - 4$$

$$x \geq \frac{8}{13}$$

$$(29) -x + 8 < -2x + 7$$

$$x < -1$$

$$(30) 3x + 3 > x + 5$$

$$x > 1$$

$$(31) 2x + 5 > 8x + 1$$

$$x < \frac{2}{3}$$

$$(32) 9x - 1 \geq -2x + 3$$

$$x \geq \frac{4}{11}$$

$$(33) 3x - 3 \leq -8x + 2$$

$$x \leq \frac{5}{11}$$

$$(34) -8x + 8 \leq x + 7$$

$$x \geq \frac{1}{9}$$

$$(35) -8x - 8 < -6x + 7$$

$$x > -\frac{15}{2}$$

$$(36) 9x + 4 < -x + 3$$

$$x < -\frac{1}{10}$$

$$(37) -6x - 7 < -5x + 8$$

$$x > -15$$

$$(38) -6x - 2 \leq 9x + 3$$

$$x \geq -\frac{1}{3}$$

$$(39) -5x + 6 > 7x + 7$$

$$x < -\frac{1}{12}$$

$$(40) 4x + 1 < 2x - 7$$

$$x < -4$$

$$(41) -5x - 3 \geq -9x + 8$$

$$x \geq \frac{11}{4}$$

$$(42) -7x + 2 \geq 2x + 9$$

$$x \leq -\frac{7}{9}$$

$$(43) 5x + 9 < -5x + 2$$

$$x < -\frac{7}{10}$$

$$(44) 8x - 6 < -6x - 2$$

$$x < \frac{2}{7}$$

$$(45) -2x + 6 < 4x - 6$$

$$x > 2$$



$$(1) \quad 9x + 7 > 4x + 8 \\ x > \frac{1}{5}$$

$$(2) \quad -3x + 4 > -8x - 2 \\ x > -\frac{6}{5}$$

$$(3) \quad 4x + 8 < 9x + 9 \\ x > -\frac{1}{5}$$

$$(4) \quad -5x + 2 < -6x + 4 \\ x < 2$$

$$(5) \quad 2x + 4 > 4x + 2 \\ x < 1$$

$$(6) \quad 8x + 4 \leq -4x + 5 \\ x \leq \frac{1}{12}$$

$$(7) \quad -4x - 6 < 8x + 6 \\ x > -1$$

$$(8) \quad -8x - 5 > -4x - 7 \\ x < \frac{1}{2}$$

$$(9) \quad 5x + 5 \geq 4x + 4 \\ x \geq -1$$

$$(10) \quad -5x - 3 \leq -6x + 4 \\ x \leq 7$$

$$(11) \quad 5x - 8 \geq 4x + 4 \\ x \geq 12$$

$$(12) \quad 6x + 2 > 4x + 7 \\ x > \frac{5}{2}$$

$$(13) \quad 8x + 2 \leq 4x - 7 \\ x \leq -\frac{9}{4}$$

$$(14) \quad 8x - 1 \geq -8x + 5 \\ x \geq \frac{3}{8}$$

$$(15) \quad -8x + 4 < -9x - 6 \\ x < -10$$

$$(16) \quad -8x - 9 \geq -6x + 4 \\ x \leq -\frac{13}{2}$$

$$(17) \quad -3x - 5 \geq x + 7 \\ x \leq -3$$

$$(18) \quad 4x - 1 < -5x + 8 \\ x < 1$$

$$(19) \quad 2x + 1 > 7x - 6 \\ x < \frac{7}{5}$$

$$(20) \quad -4x + 4 < -7x - 5 \\ x < -3$$

$$(21) \quad 4x - 4 \leq 9x - 9 \\ x \geq 1$$

$$(22) \quad 7x + 5 > x - 4 \\ x > -\frac{3}{2}$$

$$(23) \quad 8x - 6 \geq x - 2 \\ x \geq \frac{4}{7}$$

$$(24) \quad x + 3 \geq -5x - 8 \\ x \geq -\frac{11}{6}$$

$$(25) \quad -3x - 3 > 2x + 8 \\ x < -\frac{11}{5}$$

$$(26) \quad 9x - 3 > 5x + 3 \\ x > \frac{3}{2}$$

$$(27) \quad 6x - 2 < 2x + 1 \\ x < \frac{3}{4}$$

$$(28) \quad 5x + 7 \geq -6x - 3 \\ x \geq -\frac{10}{11}$$

$$(29) \quad 6x + 2 > -4x - 6 \\ x > -\frac{4}{5}$$

$$(30) \quad -5x + 1 \leq 4x - 3 \\ x \geq \frac{4}{9}$$

$$(31) \quad 2x - 9 \leq -7x + 3 \\ x \leq \frac{4}{3}$$

$$(32) \quad 6x + 7 > -5x + 8 \\ x > \frac{1}{11}$$

$$(33) \quad -6x - 3 \geq 6x - 1 \\ x \leq -\frac{1}{6}$$

$$(34) \quad -6x + 9 \leq 7x + 9 \\ x \geq 0$$

$$(35) \quad 2x - 6 < -7x - 2 \\ x < \frac{4}{9}$$

$$(36) \quad -9x + 1 \geq -x + 2 \\ x \leq -\frac{1}{8}$$

$$(37) \quad -3x + 3 \leq 2x + 8 \\ x \geq -1$$

$$(38) \quad x - 9 > 8x + 4 \\ x < -\frac{13}{7}$$

$$(39) \quad -7x + 2 > -6x + 9 \\ x < -7$$

$$(40) \quad -8x - 9 < 7x + 4 \\ x > -\frac{13}{15}$$

$$(41) \quad 2x - 2 < -2x + 3 \\ x < \frac{5}{4}$$

$$(42) \quad 3x + 7 \leq -5x + 7 \\ x \leq 0$$

$$(43) \quad -x - 2 \leq -3x - 7 \\ x \leq -\frac{5}{2}$$

$$(44) \quad 6x - 8 > -4x + 4 \\ x > \frac{6}{5}$$

$$(45) \quad 7x - 3 > x + 5 \\ x > \frac{4}{3}$$

$$(1) \quad -6x - 9 > -2x - 4$$

$$x < -\frac{5}{4}$$

$$(2) \quad -4x + 4 < -7x - 5$$

$$x < -3$$

$$(3) \quad -6x - 7 < -9x + 2$$

$$x < 3$$

$$(4) \quad 8x + 6 < 4x - 9$$

$$x < -\frac{15}{4}$$

$$(5) \quad -9x - 4 \leq -7x + 6$$

$$x \geq -5$$

$$(6) \quad 9x + 7 < -4x + 3$$

$$x < -\frac{4}{13}$$

$$(7) \quad -7x + 8 < -3x - 3$$

$$x > \frac{11}{4}$$

$$(8) \quad 7x - 8 > -4x - 6$$

$$x > \frac{2}{11}$$

$$(9) \quad -9x - 7 < 3x - 6$$

$$x > -\frac{1}{12}$$

$$(10) \quad x + 9 \leq -4x - 2$$

$$x \leq -\frac{11}{5}$$

$$(11) \quad -8x - 4 > 3x + 8$$

$$x < -\frac{12}{11}$$

$$(12) \quad 2x + 9 < 3x - 5$$

$$x > 14$$

$$(13) \quad -3x + 7 > x + 3$$

$$x < 1$$

$$(14) \quad 2x - 9 > 4x - 2$$

$$x < -\frac{7}{2}$$

$$(15) \quad -9x - 6 > 6x + 5$$

$$x < -\frac{11}{15}$$

$$(16) \quad -2x + 1 > 4x - 7$$

$$x < \frac{4}{3}$$

$$(17) \quad x - 9 \geq -3x - 4$$

$$x \geq \frac{5}{4}$$

$$(18) \quad x - 3 \geq -x - 7$$

$$x \geq -2$$

$$(19) \quad 8x - 4 \geq 4x + 8$$

$$x \geq 3$$

$$(20) \quad -8x - 2 > -2x - 7$$

$$x < \frac{5}{6}$$

$$(21) \quad -7x + 6 < -9x + 6$$

$$x < 0$$

$$(22) \quad -8x + 1 \leq -3x + 3$$

$$x \geq -\frac{2}{5}$$

$$(23) \quad -4x - 7 > 3x - 2$$

$$x < -\frac{5}{7}$$

$$(24) \quad 2x - 7 < -4x - 4$$

$$x < \frac{1}{2}$$

$$(25) \quad -x - 7 > 5x - 4$$

$$x < -\frac{1}{2}$$

$$(26) \quad 6x - 7 > 9x - 9$$

$$x < \frac{2}{3}$$

$$(27) \quad -5x - 8 \geq -7x + 9$$

$$x \geq \frac{17}{2}$$

$$(28) \quad 4x + 9 < 7x - 1$$

$$x > \frac{10}{3}$$

$$(29) \quad 7x - 4 \leq 6x - 1$$

$$x \leq 3$$

$$(30) \quad 9x + 5 \geq 6x - 8$$

$$x \geq -\frac{13}{3}$$

$$(31) \quad x - 7 \geq -5x - 2$$

$$x \geq \frac{5}{6}$$

$$(32) \quad 7x - 8 \geq 4x + 9$$

$$x \geq \frac{17}{3}$$

$$(33) \quad -x - 7 \leq -5x + 7$$

$$x \leq \frac{7}{2}$$

$$(34) \quad 8x + 1 > 4x - 5$$

$$x > -\frac{3}{2}$$

$$(35) \quad -4x + 9 \geq 2x - 3$$

$$x \leq 2$$

$$(36) \quad -6x + 4 > 3x + 2$$

$$x < \frac{2}{9}$$

$$(37) \quad -x + 4 > 8x + 9$$

$$x < -\frac{5}{9}$$

$$(38) \quad -2x - 4 < 7x + 3$$

$$x > -\frac{7}{9}$$

$$(39) \quad -x + 9 < -4x - 9$$

$$x < -6$$

$$(40) \quad 7x + 6 \leq -x + 6$$

$$x \leq 0$$

$$(41) \quad 4x + 3 < -4x - 9$$

$$x < -\frac{3}{2}$$

$$(42) \quad -8x + 8 < 8x + 6$$

$$x > \frac{1}{8}$$

$$(43) \quad -x + 5 \leq 4x - 5$$

$$x \geq 2$$

$$(44) \quad 3x + 3 < 7x - 5$$

$$x > 2$$

$$(45) \quad 2x - 1 > 8x + 9$$

$$x < -\frac{5}{3}$$

$$(1) \quad -5x + 5 < 2x + 1$$

$$x > \frac{4}{7}$$

$$(2) \quad -8x + 1 \leq 5x - 6$$

$$x \geq \frac{7}{13}$$

$$(3) \quad 5x + 2 \leq -4x + 5$$

$$x \leq \frac{1}{3}$$

$$(4) \quad x - 1 \leq 8x - 2$$

$$x \geq \frac{1}{7}$$

$$(5) \quad 2x + 7 \leq -8x - 2$$

$$x \leq -\frac{9}{10}$$

$$(6) \quad -5x + 5 > 5x - 4$$

$$x < \frac{9}{10}$$

$$(7) \quad 3x + 6 \geq -9x - 7$$

$$x \geq -\frac{13}{12}$$

$$(8) \quad -9x - 3 \leq -4x + 5$$

$$x \geq -\frac{8}{5}$$

$$(9) \quad -9x - 7 \leq -2x - 8$$

$$x \geq \frac{1}{7}$$

$$(10) \quad -3x - 6 \leq -8x + 4$$

$$x \leq 2$$

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$$(12) \quad 9x - 9 < 3x - 7$$

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$$(24) \quad 7x - 7 \geq 2x + 5$$

$$x \geq \frac{12}{5}$$

$$(25) \quad -7x + 3 < 7x + 4$$

$$x > -\frac{1}{14}$$

$$(26) \quad -9x - 6 \leq 5x - 1$$

$$x \geq -\frac{5}{14}$$

$$(27) \quad 9x - 1 > 2x - 5$$

$$x > -\frac{4}{7}$$

$$(28) \quad -x - 9 \geq -6x + 5$$

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$$(29) \quad 3x - 3 \leq -5x + 8$$

$$x \leq \frac{11}{8}$$

$$(30) \quad 7x - 2 \leq 5x + 7$$

$$x \leq \frac{9}{2}$$

$$(31) \quad 9x - 9 > -4x - 7$$

$$x > \frac{2}{13}$$

$$(32) \quad -7x + 7 < 8x - 4$$

$$x > \frac{11}{15}$$

$$(33) \quad -x - 7 \geq 3x - 6$$

$$x \leq -\frac{1}{4}$$

$$(34) \quad 4x + 1 \leq x + 8$$

$$x \leq \frac{7}{3}$$

$$(35) \quad -5x + 7 \geq 4x - 3$$

$$x \leq \frac{10}{9}$$

$$(36) \quad -7x + 7 \leq -6x - 3$$

$$x \geq 10$$

$$(37) \quad -8x + 8 \geq 5x + 8$$

$$x \leq 0$$

$$(38) \quad -x + 7 < -4x - 1$$

$$x < -\frac{8}{3}$$

$$(39) \quad -6x - 3 \geq -5x + 6$$

$$x \leq -9$$

$$(40) \quad 3x + 2 < 9x - 7$$

$$x > \frac{3}{2}$$

$$(41) \quad 8x - 6 \geq x + 7$$

$$x \geq \frac{13}{7}$$

$$(42) \quad 6x + 9 > 7x + 3$$

$$x < 6$$

$$(43) \quad 8x + 6 \geq x + 9$$

$$x \geq \frac{3}{7}$$

$$(44) \quad -9x - 1 \leq 5x + 8$$

$$x \geq -\frac{9}{14}$$

$$(45) \quad -5x + 4 < -x + 6$$

$$x > -\frac{1}{2}$$

$$(1) -9x - 8 \leq -8x - 2$$

$$x \geq -6$$

$$(2) 5x + 8 < -x + 9$$

$$x < \frac{1}{6}$$

$$(3) x - 1 > 2x - 5$$

$$x < 4$$

$$(4) 6x + 4 \leq 3x - 5$$

$$x \leq -3$$

$$(5) -6x + 7 > x + 8$$

$$x < -\frac{1}{7}$$

$$(6) 6x - 2 < x + 6$$

$$x < \frac{8}{5}$$

$$(7) x - 2 \leq -9x - 7$$

$$x \leq -\frac{1}{2}$$

$$(8) 5x + 2 \leq -x + 7$$

$$x \leq \frac{5}{6}$$

$$(9) -2x - 8 \geq 7x + 5$$

$$x \leq -\frac{13}{9}$$

$$(10) 3x - 8 \geq 2x + 3$$

$$x \geq 11$$

$$(11) 5x + 7 < -8x + 2$$

$$x < -\frac{5}{13}$$

$$(12) 3x + 6 \leq -8x - 4$$

$$x \leq -\frac{10}{11}$$

$$(13) 5x - 9 < -2x + 5$$

$$x < 2$$

$$(14) 9x - 7 < 3x - 9$$

$$x < -\frac{1}{3}$$

$$(15) -x + 5 \geq 6x - 4$$

$$x \leq \frac{9}{7}$$

$$(16) 6x - 9 < -7x + 2$$

$$x < \frac{11}{13}$$

$$(17) 3x + 7 < 6x + 6$$

$$x > \frac{1}{3}$$

$$(18) -7x + 4 > 6x - 2$$

$$x < \frac{6}{13}$$

$$(19) 3x - 6 > -x - 6$$

$$x > 0$$

$$(20) -5x - 6 \leq -7x + 5$$

$$x \leq \frac{11}{2}$$

$$(21) 7x + 2 \leq -9x + 7$$

$$x \leq \frac{5}{16}$$

$$(22) -2x - 6 < 4x + 9$$

$$x > -\frac{5}{2}$$

$$(23) -6x - 1 \geq 4x + 7$$

$$x \leq -\frac{4}{5}$$

$$(24) 8x + 4 \leq -7x - 3$$

$$x \leq -\frac{7}{15}$$

$$(25) 7x - 3 > x - 2$$

$$x > \frac{1}{6}$$

$$(26) -3x + 8 \leq -x - 2$$

$$x \geq 5$$

$$(27) -5x + 2 \leq 6x + 7$$

$$x \geq -\frac{5}{11}$$

$$(28) -7x - 5 \geq -2x + 2$$

$$x \leq -\frac{7}{5}$$

$$(29) -7x - 5 > 4x + 3$$

$$x < -\frac{8}{11}$$

$$(30) 6x - 4 > -6x + 9$$

$$x > \frac{13}{12}$$

$$(31) -4x - 8 < 9x - 4$$

$$x > -\frac{4}{13}$$

$$(32) 2x + 5 > 5x + 7$$

$$x < -\frac{2}{3}$$

$$(33) -2x - 5 \geq 2x + 1$$

$$x \leq -\frac{3}{2}$$

$$(34) 7x + 7 \geq 9x - 8$$

$$x \leq \frac{15}{2}$$

$$(35) 9x - 1 \leq 7x - 4$$

$$x \leq -\frac{3}{2}$$

$$(36) x + 1 < -x - 8$$

$$x < -\frac{9}{2}$$

$$(37) x + 2 > 6x - 9$$

$$x < \frac{11}{5}$$

$$(38) -9x + 5 < 3x + 3$$

$$x > \frac{1}{6}$$

$$(39) -5x - 8 < 7x + 8$$

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$$(41) 5x + 9 > x + 4$$

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$$(42) 3x + 1 > 5x - 4$$

$$x < \frac{5}{2}$$

$$(43) -5x + 8 > -2x + 7$$

$$x < \frac{1}{3}$$

$$(44) -6x - 7 < 4x + 6$$

$$x > -\frac{13}{10}$$

$$(45) 7x - 3 < -4x - 3$$

$$x < 0$$

$$(1) \quad 4x + 5 < -x - 6$$

$$x < -\frac{11}{5}$$

$$(2) \quad 8x + 3 \geq 4x + 6$$

$$x \geq \frac{3}{4}$$

$$(3) \quad -2x + 9 > -7x - 3$$

$$x > -\frac{12}{5}$$

$$(4) \quad -x + 9 < 3x - 2$$

$$x > \frac{11}{4}$$

$$(5) \quad x - 2 \geq 4x + 5$$

$$x \leq -\frac{7}{3}$$

$$(6) \quad 9x + 1 < -4x - 3$$

$$x < -\frac{4}{13}$$

$$(7) \quad 5x + 1 > 9x + 2$$

$$x < -\frac{1}{4}$$

$$(8) \quad 6x - 3 \geq -5x + 9$$

$$x \geq \frac{12}{11}$$

$$(9) \quad -6x - 3 < 8x - 1$$

$$x > -\frac{1}{7}$$

$$(10) \quad x + 6 \leq -6x + 5$$

$$x \leq -\frac{1}{7}$$

$$(11) \quad -9x + 9 \leq x - 4$$

$$x \geq \frac{13}{10}$$

$$(12) \quad 6x + 2 > x + 6$$

$$x > \frac{4}{5}$$

$$(13) \quad -6x + 5 \leq 3x + 8$$

$$x \geq -\frac{1}{3}$$

$$(14) \quad -8x + 3 < x + 8$$

$$x > -\frac{5}{9}$$

$$(15) \quad 7x + 9 > -5x + 1$$

$$x > -\frac{2}{3}$$

$$(16) \quad -2x + 3 \geq -3x + 2$$

$$x \geq -1$$

$$(17) \quad -x - 7 \leq 8x - 5$$

$$x \geq -\frac{2}{9}$$

$$(18) \quad 4x - 6 \geq -9x + 7$$

$$x \geq 1$$

$$(19) \quad 7x - 1 \geq -8x + 7$$

$$x \geq \frac{8}{15}$$

$$(20) \quad -6x + 8 < -9x + 1$$

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$$(21) \quad -5x + 4 \geq 7x - 7$$

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$$(24) \quad x + 6 < 3x + 9$$

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$$(28) \quad 6x + 8 < -4x + 5$$

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$$(31) \quad 7x - 7 < 3x + 8$$

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$$(32) \quad 5x + 6 < -4x - 9$$

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$$(33) \quad 3x - 5 < x + 2$$

$$x < \frac{7}{2}$$

$$(34) \quad 7x + 7 < 8x + 6$$

$$x > 1$$

$$(35) \quad x - 9 \leq -7x - 3$$

$$x \leq \frac{3}{4}$$

$$(36) \quad 6x - 6 < -x + 3$$

$$x < \frac{9}{7}$$

$$(37) \quad 8x + 1 \leq -2x + 1$$

$$x \leq 0$$

$$(38) \quad -6x - 1 \leq -2x - 4$$

$$x \geq \frac{3}{4}$$

$$(39) \quad 5x + 5 \geq 3x + 3$$

$$x \geq -1$$

$$(40) \quad -3x + 1 \leq -5x + 4$$

$$x \leq \frac{3}{2}$$

$$(41) \quad -7x + 2 \leq 7x + 7$$

$$x \geq -\frac{5}{14}$$

$$(42) \quad -4x - 2 < 2x + 3$$

$$x > -\frac{5}{6}$$

$$(43) \quad -4x + 7 < 3x - 7$$

$$x > 2$$

$$(44) \quad -3x + 1 \leq -4x - 4$$

$$x \leq -5$$

$$(45) \quad -3x - 3 > 3x + 9$$

$$x < -2$$

$$(1) \quad 6x - 4 > 2x + 1$$

$$x > \frac{5}{4}$$

$$(2) \quad 8x - 3 < 2x - 7$$

$$x < -\frac{2}{3}$$

$$(3) \quad x + 7 > -9x - 2$$

$$x > -\frac{9}{10}$$

$$(4) \quad -5x + 5 \geq -9x - 6$$

$$x \geq -\frac{11}{4}$$

$$(5) \quad 4x - 2 > 2x + 7$$

$$x > \frac{9}{2}$$

$$(6) \quad -4x + 1 > 6x + 5$$

$$x < -\frac{2}{5}$$

$$(7) \quad x - 2 < -x - 8$$

$$x < -3$$

$$(8) \quad 3x - 8 \leq 5x + 3$$

$$x \geq -\frac{11}{2}$$

$$(9) \quad 8x - 7 \geq -8x - 8$$

$$x \geq -\frac{1}{16}$$

$$(10) \quad -8x + 2 \geq -7x - 3$$

$$x \leq 5$$

$$(11) \quad 8x - 9 < 9x + 7$$

$$x > -16$$

$$(12) \quad 7x - 2 < 2x - 2$$

$$x < 0$$

$$(13) \quad 4x - 9 \leq 2x + 5$$

$$x \leq 7$$

$$(14) \quad 2x + 9 < -6x + 8$$

$$x < -\frac{1}{8}$$

$$(15) \quad 2x + 8 > -6x + 8$$

$$x > 0$$

$$(16) \quad -7x + 2 > 8x + 5$$

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$$(17) \quad -4x + 6 \geq -5x - 5$$

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$$(18) \quad -8x - 1 \geq -x + 1$$

$$x \leq -\frac{2}{7}$$

$$(19) \quad -6x - 9 \leq -x - 5$$

$$x \geq -\frac{4}{5}$$

$$(20) \quad 4x - 5 \geq -5x - 2$$

$$x \geq \frac{1}{3}$$

$$(21) \quad 4x - 8 \geq 7x + 7$$

$$x \leq -5$$

$$(22) \quad 4x - 1 < 3x - 4$$

$$x < -3$$

$$(23) \quad -x + 7 \leq -8x - 1$$

$$x \leq -\frac{8}{7}$$

$$(24) \quad -8x - 2 \geq 9x + 6$$

$$x \leq -\frac{8}{17}$$

$$(25) \quad 3x + 3 < 2x + 8$$

$$x < 5$$

$$(26) \quad -3x - 6 \geq -6x - 3$$

$$x \geq 1$$

$$(27) \quad 9x + 1 > -2x + 3$$

$$x > \frac{2}{11}$$

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$$x \geq 0$$

$$(33) \quad 3x + 9 < -x + 6$$

$$x < -\frac{3}{4}$$

$$(34) \quad 9x - 6 \leq 4x + 7$$

$$x \leq \frac{13}{5}$$

$$(35) \quad 2x + 2 \geq 5x - 5$$

$$x \leq \frac{7}{3}$$

$$(36) \quad -2x + 3 < -x - 7$$

$$x > 10$$

$$(37) \quad 6x - 7 > -9x - 8$$

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$$(40) \quad -4x - 5 < -x - 1$$

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$$(42) \quad -9x - 3 < -x + 8$$

$$x > -\frac{11}{8}$$

$$(43) \quad 3x + 7 \leq -8x + 9$$

$$x \leq \frac{2}{11}$$

$$(44) \quad -x + 9 \leq 8x - 2$$

$$x \geq \frac{11}{9}$$

$$(45) \quad 4x - 1 \leq -6x - 1$$

$$x \leq 0$$

$$(1) \quad 9x - 3 \leq 7x + 6$$

$$x \leq \frac{9}{2}$$

$$(2) \quad 2x + 4 \leq -x + 5$$

$$x \leq \frac{1}{3}$$

$$(3) \quad x - 5 > 7x + 7$$

$$x < -2$$

$$(4) \quad 4x - 5 > -9x + 3$$

$$x > \frac{8}{13}$$

$$(5) \quad -x - 3 \leq -3x + 4$$

$$x \leq \frac{7}{2}$$

$$(6) \quad 2x - 9 < -6x + 2$$

$$x < \frac{11}{8}$$

$$(7) \quad 3x - 2 \geq 9x + 6$$

$$x \leq -\frac{4}{3}$$

$$(8) \quad 4x - 9 \geq -9x - 1$$

$$x \geq \frac{8}{13}$$

$$(9) \quad -6x + 9 > 9x + 9$$

$$x < 0$$

$$(10) \quad -6x - 1 \leq 2x - 5$$

$$x \geq \frac{1}{2}$$

$$(11) \quad 8x - 4 \leq -2x + 9$$

$$x \leq \frac{13}{10}$$

$$(12) \quad 9x + 2 \geq -6x + 2$$

$$x \geq 0$$

$$(13) \quad -3x - 6 < x + 5$$

$$x > -\frac{11}{4}$$

$$(14) \quad 4x - 3 > -7x + 9$$

$$x > \frac{12}{11}$$

$$(15) \quad -9x + 9 < 9x + 5$$

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$$(16) \quad -9x - 9 \leq -3x + 6$$

$$x \geq -\frac{5}{2}$$

$$(17) \quad 8x - 8 > -5x - 7$$

$$x > \frac{1}{13}$$

$$(18) \quad 4x + 5 \leq -2x - 9$$

$$x \leq -\frac{7}{3}$$

$$(19) \quad -2x + 7 < 4x + 8$$

$$x > -\frac{1}{6}$$

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$$(21) \quad 4x - 5 \geq -7x + 6$$

$$x \geq 1$$

$$(22) \quad -5x + 6 > -2x + 3$$

$$x < 1$$

$$(23) \quad -3x + 5 < 4x + 3$$

$$x > \frac{2}{7}$$

$$(24) \quad -3x - 5 < -8x + 5$$

$$x < 2$$

$$(25) \quad 9x + 2 > 7x + 6$$

$$x > 2$$

$$(26) \quad -4x - 1 < x + 9$$

$$x > -2$$

$$(27) \quad 7x + 9 \leq 3x + 3$$

$$x \leq -\frac{3}{2}$$

$$(28) \quad x + 4 \geq 8x - 3$$

$$x \leq 1$$

$$(29) \quad 8x - 7 > x - 3$$

$$x > \frac{4}{7}$$

$$(30) \quad -9x - 5 \leq -3x - 8$$

$$x \geq \frac{1}{2}$$

$$(31) \quad -5x - 9 > -6x + 4$$

$$x > 13$$

$$(32) \quad 2x + 4 \geq -4x - 6$$

$$x \geq -\frac{5}{3}$$

$$(33) \quad -9x - 4 < -5x + 8$$

$$x > -3$$

$$(34) \quad -9x - 1 \geq x + 9$$

$$x \leq -1$$

$$(35) \quad -2x - 8 < 4x - 6$$

$$x > -\frac{1}{3}$$

$$(36) \quad -x - 9 > 8x - 4$$

$$x < -\frac{5}{9}$$

$$(37) \quad 8x - 6 > x - 4$$

$$x > \frac{2}{7}$$

$$(38) \quad -x + 1 < 7x - 5$$

$$x > \frac{3}{4}$$

$$(39) \quad -8x + 4 \geq 5x + 1$$

$$x \leq \frac{3}{13}$$

$$(40) \quad 9x + 6 \geq x - 7$$

$$x \geq -\frac{13}{8}$$

$$(41) \quad -x + 5 > 6x - 8$$

$$x < \frac{13}{7}$$

$$(42) \quad 5x + 2 \geq 6x + 1$$

$$x \leq 1$$

$$(43) \quad 9x + 2 \leq -x + 3$$

$$x \leq \frac{1}{10}$$

$$(44) \quad -8x - 6 \leq -5x - 5$$

$$x \geq -\frac{1}{3}$$

$$(45) \quad -5x - 5 > -7x - 5$$

$$x > 0$$

$$(1) \quad 5x - 3 > 2x - 6 \\ x > -1$$

$$(2) \quad -2x - 1 \leq -7x + 4 \\ x \leq 1$$

$$(3) \quad 3x + 9 < 5x - 4 \\ x > \frac{13}{2}$$

$$(4) \quad 7x - 1 > 2x - 3 \\ x > -\frac{2}{5}$$

$$(5) \quad -9x - 5 < -6x - 2 \\ x > -1$$

$$(6) \quad -9x + 1 < 4x + 6 \\ x > -\frac{5}{13}$$

$$(7) \quad -3x - 5 \geq 2x - 4 \\ x \leq -\frac{1}{5}$$

$$(8) \quad 5x - 6 < -4x + 5 \\ x < \frac{11}{9}$$

$$(9) \quad -8x - 4 \geq 3x + 4 \\ x \leq -\frac{8}{11}$$

$$(10) \quad -9x - 8 > 5x + 1 \\ x < -\frac{9}{14}$$

$$(11) \quad -2x + 5 \leq -8x - 7 \\ x \leq -2$$

$$(12) \quad -5x - 2 > -4x + 1 \\ x < -3$$

$$(13) \quad -8x - 2 < -3x + 6 \\ x > -\frac{8}{5}$$

$$(14) \quad x + 2 \geq -4x - 7 \\ x \geq -\frac{9}{5}$$

$$(15) \quad 7x - 7 > 9x - 7 \\ x < 0$$

$$(16) \quad -2x + 6 < 5x - 3 \\ x > \frac{9}{7}$$

$$(17) \quad -5x + 5 < 2x - 4 \\ x > \frac{9}{7}$$

$$(18) \quad -5x - 6 > -2x - 4 \\ x < -\frac{2}{3}$$

$$(19) \quad -3x - 5 > 8x + 5 \\ x < -\frac{10}{11}$$

$$(20) \quad -x + 3 > 7x + 9 \\ x < -\frac{3}{4}$$

$$(21) \quad -6x + 2 \leq 7x + 1 \\ x \geq \frac{1}{13}$$

$$(22) \quad 2x + 8 < -2x + 8 \\ x < 0$$

$$(23) \quad 8x + 9 > 5x + 5 \\ x > -\frac{4}{3}$$

$$(24) \quad -8x + 9 \geq -2x - 9 \\ x \leq 3$$

$$(25) \quad -8x + 2 > 7x + 4 \\ x < -\frac{2}{15}$$

$$(26) \quad -9x + 5 \leq 6x + 5 \\ x \geq 0$$

$$(27) \quad 4x + 1 > 3x - 4 \\ x > -5$$

$$(28) \quad 8x - 4 > -4x - 7 \\ x > -\frac{1}{4}$$

$$(29) \quad -x - 8 > -8x + 8 \\ x > \frac{16}{7}$$

$$(30) \quad -4x - 7 \leq 3x + 9 \\ x \geq -\frac{16}{7}$$

$$(31) \quad x + 8 > -3x - 6 \\ x > -\frac{7}{2}$$

$$(32) \quad -x - 5 \leq 2x + 6 \\ x \geq -\frac{11}{3}$$

$$(33) \quad 5x - 8 \leq 8x + 1 \\ x \geq -3$$

$$(34) \quad 4x - 2 > 8x - 5 \\ x < \frac{3}{4}$$

$$(35) \quad -7x - 5 \leq -8x + 9 \\ x \leq 14$$

$$(36) \quad -9x - 9 < 7x + 3 \\ x > -\frac{3}{4}$$

$$(37) \quad 2x - 4 < -5x - 2 \\ x < \frac{2}{7}$$

$$(38) \quad 5x + 2 < -4x - 7 \\ x < -1$$

$$(39) \quad 7x + 5 < -x + 3 \\ x < -\frac{1}{4}$$

$$(40) \quad 7x - 4 < 8x + 8 \\ x > -12$$

$$(41) \quad 4x + 6 \leq -2x - 5 \\ x \leq -\frac{11}{6}$$

$$(42) \quad -8x + 1 \leq 2x - 6 \\ x \geq \frac{7}{10}$$

$$(43) \quad -8x - 7 \geq -9x + 5 \\ x \geq 12$$

$$(44) \quad 6x - 7 < 9x - 8 \\ x > \frac{1}{3}$$

$$(45) \quad 9x + 8 > x + 1 \\ x > -\frac{7}{8}$$



$$(1) \quad 7x - 3 \geq 5x + 7$$

$$x \geq 5$$

$$(2) \quad 7x - 6 \leq 6x - 6$$

$$x \leq 0$$

$$(3) \quad 6x + 8 \geq 5x + 5$$

$$x \geq -3$$

$$(4) \quad -2x + 8 < -6x + 4$$

$$x < -1$$

$$(5) \quad 9x - 8 > -7x + 8$$

$$x > 1$$

$$(6) \quad -9x + 5 \geq 3x + 8$$

$$x \leq -\frac{1}{4}$$

$$(7) \quad -5x - 1 \geq -4x - 4$$

$$x \leq 3$$

$$(8) \quad -7x + 6 < -9x - 5$$

$$x < -\frac{11}{2}$$

$$(9) \quad 3x - 5 > 7x - 4$$

$$x < -\frac{1}{4}$$

$$(10) \quad 8x + 2 \geq -6x - 2$$

$$x \geq -\frac{2}{7}$$

$$(11) \quad -8x + 6 \geq x - 8$$

$$x \leq \frac{14}{9}$$

$$(12) \quad -2x - 3 > -4x + 3$$

$$x > 3$$

$$(13) \quad x + 1 > 6x + 1$$

$$x < 0$$

$$(14) \quad -3x - 9 > -x + 9$$

$$x < -9$$

$$(15) \quad -3x + 6 \geq 2x - 2$$

$$x \leq \frac{8}{5}$$

$$(16) \quad 7x + 1 < 8x + 4$$

$$x > -3$$

$$(17) \quad 9x + 9 < 2x - 4$$

$$x < -\frac{13}{7}$$

$$(18) \quad -2x + 8 \leq x - 9$$

$$x \geq \frac{17}{3}$$

$$(19) \quad 6x - 2 > 4x - 2$$

$$x > 0$$

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$$(21) \quad 5x - 8 \geq -3x + 6$$

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$$x \geq 3$$

$$(23) \quad -7x + 3 > -5x - 1$$

$$x < 2$$

$$(24) \quad -7x - 3 \geq -6x - 3$$

$$x \leq 0$$

$$(25) \quad -5x - 7 \geq 4x - 6$$

$$x \leq -\frac{1}{9}$$

$$(26) \quad -3x - 2 > 7x + 5$$

$$x < -\frac{7}{10}$$

$$(27) \quad -7x + 7 > 6x - 5$$

$$x < \frac{12}{13}$$

$$(28) \quad 7x - 3 > 8x + 9$$

$$x < -12$$

$$(29) \quad -4x + 9 < 9x - 8$$

$$x > \frac{17}{13}$$

$$(30) \quad 5x - 9 \geq -x + 9$$

$$x \geq 3$$

$$(31) \quad -7x - 8 \geq 7x - 4$$

$$x \leq -\frac{2}{7}$$

$$(32) \quad -7x + 4 < -x - 5$$

$$x > \frac{3}{2}$$

$$(33) \quad 2x + 7 < 3x - 1$$

$$x > 8$$

$$(34) \quad 2x - 4 < -6x + 4$$

$$x < 1$$

$$(35) \quad x - 6 \leq -6x - 5$$

$$x \leq \frac{1}{7}$$

$$(36) \quad -5x + 6 < -3x + 5$$

$$x > \frac{1}{2}$$

$$(37) \quad 9x - 2 \leq -x - 7$$

$$x \leq -\frac{1}{2}$$

$$(38) \quad 4x + 9 > -7x - 9$$

$$x > -\frac{18}{11}$$

$$(39) \quad 5x - 3 \geq -7x - 6$$

$$x \geq -\frac{1}{4}$$

$$(40) \quad -6x - 7 \geq 5x + 2$$

$$x \leq -\frac{9}{11}$$

$$(41) \quad 5x + 6 \geq 3x - 7$$

$$x \geq -\frac{13}{2}$$

$$(42) \quad 4x - 4 \geq -6x - 7$$

$$x \geq -\frac{3}{10}$$

$$(43) \quad 4x + 1 \leq -9x - 5$$

$$x \leq -\frac{6}{13}$$

$$(44) \quad -2x + 9 > -4x - 2$$

$$x > -\frac{11}{2}$$

$$(45) \quad -x - 6 \leq -9x - 1$$

$$x \leq \frac{5}{8}$$

$$(1) \quad 2x - 5 < -8x - 1$$

$$x < \frac{2}{5}$$

$$(2) \quad 2x + 6 > 3x - 8$$

$$x < 14$$

$$(3) \quad -7x + 1 > -2x - 6$$

$$x < \frac{7}{5}$$

$$(4) \quad 9x - 2 > -6x + 7$$

$$x > \frac{3}{5}$$

$$(5) \quad 8x - 1 < -x + 8$$

$$x < 1$$

$$(6) \quad 4x + 7 \geq 2x - 1$$

$$x \geq -4$$

$$(7) \quad 5x + 1 \leq 6x + 1$$

$$x \geq 0$$

$$(8) \quad -5x + 8 \leq 5x - 9$$

$$x \geq \frac{17}{10}$$

$$(9) \quad -7x + 5 < -5x + 4$$

$$x > \frac{1}{2}$$

$$(10) \quad 9x - 9 > x + 8$$

$$x > \frac{17}{8}$$

$$(11) \quad 8x - 1 \geq 4x - 6$$

$$x \geq -\frac{5}{4}$$

$$(12) \quad x + 1 \leq -6x - 9$$

$$x \leq -\frac{10}{7}$$

$$(13) \quad 2x + 7 < 3x + 6$$

$$x > 1$$

$$(14) \quad -x - 5 < 6x + 4$$

$$x > -\frac{9}{7}$$

$$(15) \quad -4x + 1 > 8x + 6$$

$$x < -\frac{5}{12}$$

$$(16) \quad -8x - 6 > 5x - 4$$

$$x < -\frac{2}{13}$$

$$(17) \quad 5x - 2 \leq -7x + 2$$

$$x \leq \frac{1}{3}$$

$$(18) \quad -9x + 3 \leq 2x - 5$$

$$x \geq \frac{8}{11}$$

$$(19) \quad -9x - 1 \leq -6x - 2$$

$$x \geq \frac{1}{3}$$

$$(20) \quad 6x + 6 > 4x - 2$$

$$x > -4$$

$$(21) \quad 6x + 5 \geq -6x - 7$$

$$x \geq -1$$

$$(22) \quad 3x - 8 \leq 9x - 9$$

$$x \geq \frac{1}{6}$$

$$(23) \quad 2x + 1 < 6x - 7$$

$$x > 2$$

$$(24) \quad -6x - 2 \geq 8x - 3$$

$$x \leq \frac{1}{14}$$

$$(25) \quad 3x + 9 > x + 9$$

$$x > 0$$

$$(26) \quad -9x + 2 > 7x - 3$$

$$x < \frac{5}{16}$$

$$(27) \quad -2x - 4 < -5x + 8$$

$$x < 4$$

$$(28) \quad -8x - 7 > -5x - 6$$

$$x < -\frac{1}{3}$$

$$(29) \quad 7x - 9 > -x + 5$$

$$x > \frac{7}{4}$$

$$(30) \quad 6x - 2 \leq -9x + 5$$

$$x \leq \frac{7}{15}$$

$$(31) \quad 2x - 8 > -4x - 9$$

$$x > -\frac{1}{6}$$

$$(32) \quad -8x - 3 > 2x - 8$$

$$x < \frac{1}{2}$$

$$(33) \quad 7x - 4 < 8x - 5$$

$$x > 1$$

$$(34) \quad 4x - 3 \geq 5x + 9$$

$$x \leq -12$$

$$(35) \quad -9x + 3 \geq -5x + 9$$

$$x \leq -\frac{3}{2}$$

$$(36) \quad 4x + 3 \leq 9x + 6$$

$$x \geq -\frac{3}{5}$$

$$(37) \quad -5x + 9 \geq 6x - 8$$

$$x \leq \frac{17}{11}$$

$$(38) \quad 3x + 9 < 4x + 4$$

$$x > 5$$

$$(39) \quad -4x - 5 \leq x + 5$$

$$x \geq -2$$

$$(40) \quad 4x + 1 > 9x + 5$$

$$x < -\frac{4}{5}$$

$$(41) \quad -8x - 2 > 7x + 2$$

$$x < -\frac{4}{15}$$

$$(42) \quad 9x + 7 < 6x - 4$$

$$x < -\frac{11}{3}$$

$$(43) \quad -7x + 3 < -6x + 3$$

$$x > 0$$

$$(44) \quad 8x + 5 > -5x - 8$$

$$x > -1$$

$$(45) \quad -2x + 4 \leq -7x - 8$$

$$x \leq -\frac{12}{5}$$

$$(1) \quad 2x - 1 \geq -4x + 3$$

$$x \geq \frac{2}{3}$$

$$(2) \quad 7x + 1 > 4x - 8$$

$$x > -3$$

$$(3) \quad x - 7 > -8x + 9$$

$$x > \frac{16}{9}$$

$$(4) \quad 3x + 3 > -2x + 8$$

$$x > 1$$

$$(5) \quad 3x - 2 < -x - 1$$

$$x < \frac{1}{4}$$

$$(6) \quad 3x + 7 > -2x + 4$$

$$x > -\frac{3}{5}$$

$$(7) \quad 4x - 5 \geq 7x + 3$$

$$x \leq -\frac{8}{3}$$

$$(8) \quad 5x - 3 < 7x + 4$$

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$$(9) \quad -2x + 5 \leq -7x - 1$$

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$$x \geq \frac{6}{5}$$

$$(45) \quad x - 2 \geq 2x + 8$$

$$x \leq -10$$

$$(1) \quad 3x + 7 \leq 7x + 1$$

$$x \geq \frac{3}{2}$$

$$(2) \quad -5x - 4 \geq 9x - 3$$

$$x \leq -\frac{1}{14}$$

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$$(11) \quad 8x + 6 < x - 9$$

$$x < -\frac{15}{7}$$

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$$x \geq -\frac{3}{13}$$

$$(1) \quad -9x - 9 \leq 9x + 2$$

$$x \geq -\frac{11}{18}$$

$$(2) \quad -3x + 8 \geq -6x - 9$$

$$x \geq -\frac{17}{3}$$

$$(3) \quad -5x - 3 > x - 9$$

$$x < 1$$

$$(4) \quad x - 8 > 6x + 5$$

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$$(30) \quad x + 1 > 8x + 1$$

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$$(36) \quad -3x - 9 \leq -4x - 6$$

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$$(43) \quad 7x + 8 \geq 2x + 4$$

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$$(44) \quad 3x - 4 \geq 9x + 1$$

$$x \leq -\frac{5}{6}$$

$$(45) \quad -7x + 2 > -3x + 3$$

$$x < -\frac{1}{4}$$

$$(1) \quad -5x - 8 \geq -x + 2$$

$$x \leq -\frac{5}{2}$$

$$(2) \quad -5x + 7 \geq -3x - 1$$

$$x \leq 4$$

$$(3) \quad -4x + 7 \leq x + 3$$

$$x \geq \frac{4}{5}$$

$$(4) \quad -5x + 7 < 9x - 6$$

$$x > \frac{13}{14}$$

$$(5) \quad -x + 3 < -5x + 8$$

$$x < \frac{5}{4}$$

$$(6) \quad 3x - 9 \geq 6x - 8$$

$$x \leq -\frac{1}{3}$$

$$(7) \quad 8x + 1 \geq 9x + 6$$

$$x \leq -5$$

$$(8) \quad -x - 1 \geq x - 1$$

$$x \leq 0$$

$$(9) \quad -2x - 4 \leq 8x - 9$$

$$x \geq \frac{1}{2}$$

$$(10) \quad 7x - 5 < 3x + 4$$

$$x < \frac{9}{4}$$

$$(11) \quad x - 8 < -9x - 1$$

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$$(12) \quad -8x - 5 > 9x - 4$$

$$x < -\frac{1}{17}$$

$$(13) \quad x + 2 < -2x - 5$$

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$$x \leq -\frac{10}{11}$$

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$$x > \frac{4}{5}$$

$$(38) -6x - 4 \leq 5x - 7$$

$$x \geq \frac{3}{11}$$

$$(39) 8x - 7 \leq -7x + 5$$

$$x \leq \frac{4}{5}$$

$$(40) 9x - 7 < 3x - 5$$

$$x < \frac{1}{3}$$

$$(41) -7x + 2 \geq -x - 6$$

$$x \leq \frac{4}{3}$$

$$(42) -3x + 7 < -4x - 7$$

$$x < -14$$

$$(43) -3x + 7 < -x - 3$$

$$x > 5$$

$$(44) -3x - 5 > -7x + 5$$

$$x > \frac{5}{2}$$

$$(45) 9x + 8 > -6x - 3$$

$$x > -\frac{11}{15}$$

## 2.13 微分 多項式

多項式の微分。



(1)  $f(x) = 2x^3 + 6x^2 + 2x + 5$

(16)  $f(x) = 5x^3 + 7x^2 + 6x + 2$

(31)  $f(x) = 4x^3 + 6x^2 + 3x + 7$

(2)  $f(x) = 8x^3 - 7x^2 - 8x - 1$

(17)  $f(x) = 6x^3 + 5x^2 - x + 8$

(32)  $f(x) = 7x^3 + 8x + 9$

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- |                                                                 |                                                                 |                                                                 |
|-----------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|
| (1) $f(x) = 2x^3 + 6x^2 + 2x + 5$<br>$f'(x) = 6x^2 + 12x + 2$   | (16) $f(x) = 5x^3 + 7x^2 + 6x + 2$<br>$f'(x) = 15x^2 + 14x + 6$ | (31) $f(x) = 4x^3 + 6x^2 + 3x + 7$<br>$f'(x) = 12x^2 + 12x + 3$ |
| (2) $f(x) = 8x^3 - 7x^2 - 8x - 1$<br>$f'(x) = 24x^2 - 14x - 8$  | (17) $f(x) = 6x^3 + 5x^2 - x + 8$<br>$f'(x) = 18x^2 + 10x - 1$  | (32) $f(x) = 7x^3 + 8x + 9$<br>$f'(x) = 21x^2 + 8$              |
| (3) $f(x) = 7x^3 - 2x^2 + x - 3$<br>$f'(x) = 21x^2 - 4x + 1$    | (18) $f(x) = 8x^3 - 2x^2 - 5x + 2$<br>$f'(x) = 24x^2 - 4x - 5$  | (33) $f(x) = 7x^3 + 8x^2 + 6x + 1$<br>$f'(x) = 21x^2 + 16x + 6$ |
| (4) $f(x) = 6x^3 + 7x^2 + x - 6$<br>$f'(x) = 18x^2 + 14x + 1$   | (19) $f(x) = 3x^3 - x - 2$<br>$f'(x) = 9x^2 - 1$                | (34) $f(x) = 8x^3 + 2x^2 - 5x - 2$<br>$f'(x) = 24x^2 + 4x - 5$  |
| (5) $f(x) = 2x^3 - 9x^2 - 3x - 1$<br>$f'(x) = 6x^2 - 18x - 3$   | (20) $f(x) = 3x^3 + x^2 - 6x - 8$<br>$f'(x) = 9x^2 + 2x - 6$    | (35) $f(x) = 8x^3 + 5x^2 - 8x - 3$<br>$f'(x) = 24x^2 + 10x - 8$ |
| (6) $f(x) = 4x^3 + x^2 + 8x - 8$<br>$f'(x) = 12x^2 + 2x + 8$    | (21) $f(x) = 8x^3 - x^2 - 5x - 9$<br>$f'(x) = 24x^2 - 2x - 5$   | (36) $f(x) = 3x^3 + 3x^2 - 5x + 2$<br>$f'(x) = 9x^2 + 6x - 5$   |
| (7) $f(x) = 9x^3 - 5x^2 + 7x + 1$<br>$f'(x) = 27x^2 - 10x + 7$  | (22) $f(x) = 8x^3 - 7x + 2$<br>$f'(x) = 24x^2 - 7$              | (37) $f(x) = 3x^3 - 8x^2 - 3x - 9$<br>$f'(x) = 9x^2 - 16x - 3$  |
| (8) $f(x) = 5x^3 - 3x^2 + 9x$<br>$f'(x) = 15x^2 - 6x + 9$       | (23) $f(x) = x^3 + 8x^2 + 8x - 1$<br>$f'(x) = 3x^2 + 16x + 8$   | (38) $f(x) = 4x^3 + x^2 + 3x - 9$<br>$f'(x) = 12x^2 + 2x + 3$   |
| (9) $f(x) = 6x^3 - 2x^2 + 5x - 5$<br>$f'(x) = 18x^2 - 4x + 5$   | (24) $f(x) = 2x^3 - 8x^2 - 5x + 5$<br>$f'(x) = 6x^2 - 16x - 5$  | (39) $f(x) = 8x^3 - 5x^2 + 6x - 1$<br>$f'(x) = 24x^2 - 10x + 6$ |
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